

DRONA ED TECH

Class 11

Mathematics

Master Reference

Complete Chapter Compendium

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Chapter 1: Sets

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Complete: Subtopics 01-04

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Subtopic 01: Sets and their Representations

Exam Relevance: This is the gateway to all of higher mathematics. CBSE Boards ask 1-mark roster/set-builder conversions and 1-2 mark empty/finite/infinite classification. In JEE Main, Sets rarely stands alone but the phrasing “*S is the set of all values of...*” hides inside dozens of algebra and calculus problems — fluency here saves time everywhere. JEE Advanced uses set language as a default dialect; direct questions are rare but assumed knowledge.

Section 1: Concept Intuition

Imagine the team-sheet a cricket captain hands the umpire before a match. It lists exactly which eleven players are *in*. There is no debate about whether Bumrah is on the list — either his name is written there or it isn’t. That single property — **unambiguous membership** — is the whole idea of a *set*. A set is any collection of objects for which, given any object in the universe, you can answer “is it in, or out?” with a definite yes or no.

The objects inside are called **elements** (or members). The catch is the word **well-defined**. “The set of students in your class taller than 170 cm” is a genuine set — measure anyone and you get a clean yes/no. But “the set of *tall* students” is *not* a set, because “tall” is a matter of opinion. The same trap appears everywhere: “good Bollywood films of 2023” is not a set, but “Bollywood films released in 2023” is. **If membership depends on someone’s mood, it is not a set.**

Two more rules fall straight out of the team-sheet picture. First, **repetition is meaningless** — writing a player’s name twice doesn’t put two of him on the field. So $\{1, 2, 2, 3\}$ is just $\{1, 2, 3\}$. Second, **order is meaningless** — reshuffling the batting line-up on paper doesn’t change who is playing. So $\{a, b, c\} = \{c, a, b\}$. A set cares only about *which* objects belong, never how many times or in what sequence you happen to list them.

We write $a \in A$ for “*a* belongs to *A*” (the symbol $\in$ is a stylised Greek epsilon) and $b \notin A$ for “*b* does not belong to *A*.”

Now, how do you *describe* a set to someone else? There are exactly two languages:

- **Roster (tabular) form** — you read out every member by name, inside braces: the squad is $\{\text{Rohit, Virat, Bumrah}\}$. This is like reading the team-sheet aloud.
- **Set-builder form** — instead of naming members, you state the *rule* they all satisfy: “the set of all players in the current ODI squad.” In symbols, $A = \{x : x \text{ is in the ODI squad}\}$, read as “*the set of all x such that x is in the ODI squad.*” The colon means “such that.”

Roster is concrete but hopeless for huge or endless collections — you cannot list every natural number. Set-builder is the rule that captures even infinite sets in one line.

Think about it this way... A set is simply the *answer to a single yes/no membership question*. Roster form lists every “yes.” Set-builder form writes down the question itself.

One special character deserves respect: the **empty set** $\emptyset$ (also written $\{\}$), the set with no members at all — like the set of students who scored above 100 on a 100-mark paper. Nobody qualifies, so the collection is genuinely empty.

Edge cases & where students slip (the “fine print”):

- A collection is a set only if it is **well-defined**; subjective descriptions (“clever,” “nice,” “large”) disqualify it.
- $\{0\}$ is **not** the empty set — it contains one element, the number zero. Likewise $\{\emptyset\}$ is not empty; it contains one element, namely $\emptyset$.
- Roster form can only *suggest* an infinite set with dots ($\{1, 2, 3, \dots\}$); it can never truly list one.
- Because order and repetition don’t count, “how many elements?” means *how many distinct* elements.

Section 2: Key Definitions & Notation

(For a non-metric topic like this there are no dimensional formulae; what matters is precise notation and the standard number sets.)

Term	Meaning	Notation
Set	Well-defined collection of distinct objects	Capital letters $A, B, C, \dots$
Element	An object belonging to a set	Lower-case $a, b, x, \dots$
Belongs to / does not	Membership relation	$a \in A, b \notin A$
Roster form	All elements listed in braces	$A = \{2, 4, 6\}$
Set-builder form	Defining property stated	$A = \{x : P(x)\}$
Empty (null/void) set	Set with no element	$\emptyset$ or $\{\}$
Singleton	Set with exactly one element	$\{a\}$
Cardinal number	Count of distinct elements	$n(A)$
Finite set	$n(A)$ is a definite whole number	—
Infinite set	Not finite	—
Equal sets	Identical elements	$A = B \iff (x \in A \iff x \in B)$
Equivalent sets	Same cardinality	$n(A) = n(B)$

Standard number sets (memorise these — they are the most common universes in set-builder form):

$$\mathbb{N} = \{1, 2, 3, \dots\}, \quad \mathbb{W} = \{0, 1, 2, \dots\}, \quad \mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$$

$$\mathbb{Q} = \left\{ \frac{p}{q} : p, q \in \mathbb{Z}, q \neq 0 \right\}, \quad \mathbb{R} = \text{real numbers}, \quad \mathbb{T} = \text{irrational numbers}$$

with the nesting

$$\mathbb{N} \subset \mathbb{W} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}.$$

Figure (to be illustrated). Concentric ovals labelled, from innermost to outermost, $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$, with the irrationals $\mathbb{T}$ shown as the shaded region of $\mathbb{R}$ lying outside $\mathbb{Q}$. Mark sample elements: 3 in $\mathbb{N}$, -5 in $\mathbb{Z}$, $2/3$ in $\mathbb{Q}$, $\sqrt{2}$ and π in $\mathbb{T}$.

Crucial distinction: *equal* $\Rightarrow$ *equivalent*, but *equivalent* $\nRightarrow$ *equal*. $\{1, 2, 3\}$ and $\{a, b, c\}$ are equivalent (both size 3) yet not equal.

Section 3: Key Procedures & Logical Framework

This subtopic has no formal theorem to derive, so here are the four routine procedures, each with the *reasoning* behind it.

Procedure A — Set-builder $\rightarrow$ Roster.

1. Read the **universe symbol first** ($x \in \mathbb{N}$? $\mathbb{Z}$? $\mathbb{R}$?). This decides which candidates are even allowed. *Why:* the same condition gives different sets over different universes — this is the single most common source of error.
2. Apply the condition and **solve or enumerate**.
3. Keep only solutions that lie in the stated universe; list distinct values in braces.

Procedure B — Roster $\rightarrow$ Set-builder.

1. Hunt for the **pattern** linking the listed elements (squares? one-less-than-denominator? primes?).
2. Express that pattern as an algebraic rule $x = f(n)$ or a property $P(x)$.
3. Pin down the **range of the index** so you reproduce exactly those elements — no more, no less. *Why:* a rule that is too loose generates extra members.

Procedure C — Classify as empty / finite / infinite.

1. Ask “does *any* element satisfy the condition?” If none, the set is **empty**. *Why:* impossibility (e.g. $x^2 = -4$ over $\mathbb{R}$, or a natural number strictly between 2 and 3) forces $\emptyset$.
2. If elements exist, ask “can I, in principle, finish counting them?” Yes $\Rightarrow$ **finite**; no $\Rightarrow$ **infinite**.

Procedure D — Test equality.

1. Reduce both sets to roster form.
2. Confirm **mutual containment**: every element of A is in B and every element of B is in A . *Why*: equality is two-directional — matching one way is not enough.
3. If counts match but elements differ, they are only *equivalent*, not equal.

Section 4: Worked Examples

Example 1 — [CBSE Board] Write the set of all two-digit natural numbers whose digits add up to 8 in roster form.

Given: two-digit numbers (so 10 to 99) with digit-sum = 8. We need tens digit t and units digit u with $t + u = 8$, $t \geq 1$.

- $t = 1, u = 7 \rightarrow 17$; $t = 2, u = 6 \rightarrow 26$; $t = 3, u = 5 \rightarrow 35$; $t = 4, u = 4 \rightarrow 44$;
- $t = 5, u = 3 \rightarrow 53$; $t = 6, u = 2 \rightarrow 62$; $t = 7, u = 1 \rightarrow 71$; $t = 8, u = 0 \rightarrow 80$.

$$\{17, 26, 35, 44, 53, 62, 71, 80\}$$

(Note 80 qualifies: $8 + 0 = 8$. Don't drop it.)

Example 2 — [JEE Main — Speed Trap] Without lengthy computation, identify which of the following is the empty set:

- (a) $\{x \in \mathbb{R} : x^2 + 1 = 0\}$ (b) $\{x : x \text{ is an even prime}\}$ (c) $\{0\}$ (d) $\{x \in \mathbb{N} : x < 1\}$

Mental elimination. Scan for impossibility:

- (a) $x^2 = -1$ has **no real** solution $\Rightarrow$ empty.
- (d) the smallest natural number is 1, so **no** natural number is less than 1 $\Rightarrow$ empty.
- (b) = $\{2\}$, *not* empty (the trap: students forget 2 is prime). (c) contains 0, *not* empty (the classic $\{0\} \neq \emptyset$ trap).

The trap: (a) and (d) are *both* empty. If the question allows one answer, the intended one is usually (a); but the real lesson is spotting that (c) is a decoy — a singleton dressed up to look empty. **Speed move:** check the *universe* in (d) before computing — $\mathbb{N}$ starting at 1 instantly kills it.

Example 3 — [JEE Main] Let $A = \{x \in \mathbb{Z} : |x| \leq 2\}$ and $B = \{x \in \mathbb{Z} : x^3 = x\}$. Find $n(A)$ and $n(B)$, and state whether $A = B$.

This links absolute-value reasoning with polynomial factoring over a restricted universe.

- A : integers with $|x| \leq 2 \Rightarrow A = \{-2, -1, 0, 1, 2\}$, so $n(A) = 5$.
- B : $x^3 = x \Rightarrow x(x-1)(x+1) = 0 \Rightarrow x \in \{-1, 0, 1\}$, all integers, so $B = \{-1, 0, 1\}$, $n(B) = 3$.

Every element of B lies in A , but $-2, 2 \in A$ are not in B . Mutual containment fails one way, so

$$A \neq B \quad (\text{in fact } B \subset A), \quad n(A) = 5, \quad n(B) = 3.$$

Example 4 — [JEE Advanced] Let a be a real parameter and define $T = \{x \in \mathbb{N} : x^2 - (a+1)x + a = 0\}$. Find all values of a for which $n(T) = 1$.

Factor the quadratic (independent of the domain):

$$x^2 - (a+1)x + a = (x-1)(x-a) = 0 \Rightarrow x = 1 \text{ or } x = a.$$

Since $1 \in \mathbb{N}$ always, the root $x = 1$ is *always* a member of T . The second root $x = a$ contributes a member only if $a \in \mathbb{N}$, and a *distinct* one only if $a \neq 1$. Casework on the **natural-number domain**:

- If $a \in \mathbb{N}$ and $a \neq 1$: roots 1 and a are two distinct naturals $\Rightarrow T = \{1, a\}$, $n(T) = 2$.
- If $a = 1$: both roots coincide at 1 $\Rightarrow T = \{1\}$, $n(T) = 1$. ✓
- If $a \notin \mathbb{N}$ (e.g. $a = 0, -3, \frac{5}{2}, \sqrt{2}$): the root a is excluded, only 1 survives $\Rightarrow T = \{1\}$, $n(T) = 1$. ✓

$$n(T) = 1 \iff a = 1 \text{ or } a \notin \mathbb{N} \quad (\text{equivalently, } a \text{ is not a natural number greater than } 1).$$

The whole problem turns on remembering the *domain* $\mathbb{N}$ filters the roots — pure algebra would wrongly report two roots for every a .

Section 5: Practice Exercises

1. **[CBSE]** Write the set of distinct letters of the word “**MATHEMATICS**” in roster form.
2. **[JEE Main]** Write $\{x \in \mathbb{Z} : x^2 < 10\}$ in roster form.
3. **[JEE Main]** Is $\{x \in \mathbb{N} : x^2 - 3 = 0\}$ the empty set? Justify in one line.
4. **[JEE Advanced]** Find $n(\{x \in \mathbb{Z} : (x^2 - 1)(x^2 - 4) = 0\})$.
5. **[JEE Advanced]** Express $\{3, 8, 15, 24, 35\}$ in set-builder form.

Answers:

1. $\{M, A, T, H, E, I, C, S\}$ (8 distinct letters — repetition discarded)
2. $\{-3, -2, -1, 0, 1, 2, 3\}$
3. Yes; $x^2 = 3 \Rightarrow x = \pm\sqrt{3} \notin \mathbb{N}$, so no element exists.
4. 4 (roots $x = \pm 1, \pm 2$)
5. $\{x : x = n^2 - 1, n \in \mathbb{N}, 2 \leq n \leq 6\}$

Section 6: Crack the MCQ

Q1. Which of the following is a well-defined set?

- (A) The collection of intelligent students in a school
- (B) The collection of difficult chapters in Class 11 Physics
- (C) The collection of prime numbers less than 20
- (D) The collection of beautiful monuments in India

Answer: (C). Primality is an objective test, so membership is decidable. *Distractors:* (A) “intelligent,” (B) “difficult,” and (D) “beautiful” are all subjective — different people classify differently, so membership isn’t well-defined. Students who pick (A), (B) or (D) are confusing a *collection* with a *set*; only the latter requires unambiguous membership.

Q2. Which statement is **correct**?

- (A) $\emptyset = \{0\}$
- (B) $\emptyset = \{\emptyset\}$
- (C) $\{0\}$ is a singleton set
- (D) $\emptyset$ has one element

Answer: (C). $\{0\}$ contains exactly one element (the number 0), hence a singleton. *Distractors:* (A) treats $\{0\}$ as empty — but it holds the element 0. (B) treats $\{\emptyset\}$ as empty — but it holds one element, namely the empty set itself. (D) misreads “empty” as “has one element”; $\emptyset$ has *zero* elements. All three are the same conceptual slip: confusing a box that *contains nothing* with a box that *contains an empty box* or *the number zero*.

Q3. Let $A = \{x \in \mathbb{N} : x \text{ is a factor of } 6\}$ and $B = \{p, q, r, s\}$. Which is true?

- (A) $A = B$
- (B) A and B are equivalent but not equal
- (C) A and B are neither equal nor equivalent
- (D) $n(A) = 3$

Answer: (B). Factors of 6 are 1, 2, 3, 6, so $A = \{1, 2, 3, 6\}$, $n(A) = 4 = n(B)$. Same size but different elements $\Rightarrow$ equivalent, not equal. *Distractors:* (A) ignores that equal sets need *identical* elements, not just identical counts. (C) misses that the cardinalities match. (D) mis-counts the factors of 6 (forgetting 6 itself or 1).

Q4. The set $\{x \in \mathbb{R} : x^2 - 2x + 5 = 0\}$ is:

- (A) a singleton (B) a two-element set (C) the empty set (D) an infinite set

Answer: (C). Discriminant $= (-2)^2 - 4(1)(5) = 4 - 20 = -16 < 0$, so there is **no real** root; over $\mathbb{R}$ the set is empty. *Distractors:* (B) comes from blindly saying “a quadratic has two roots” while ignoring the universe $\mathbb{R}$. (A) imagines a repeated real root (that needs discriminant $= 0$). (D) confuses “no solution” with “all reals.” Always read the universe before counting roots.

Section 7: Common Student Pitfalls & Pro-Tips

- **Treating $\{0\}$ or $\{\emptyset\}$ as empty.** Both are singletons. Empty means *zero* elements — $\emptyset$ or $\{\}$ only.
- **Counting repeated elements.** In “MATHEMATICS,” many write 11 letters; only 8 are distinct. Sets discard duplicates.

- **Ignoring the universe symbol.** $\{x : x^2 = 2\}$ is empty over $\mathbb{N}$, a doubleton over $\mathbb{R}$. The domain $(\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R})$ is part of the question.
- **Confusing equal with equivalent.** Same number of elements is *not* the same as same elements.

Pro-Tip: Before solving any set-builder problem, **circle the universe symbol first** and ask “is this even possible here?” Two seconds of domain-checking catches empty sets and impossible-root traps instantly — the exact places examiners plant marks.

Section 8: Cheat Sheet Box

- **Set:** a *well-defined* collection of *distinct* objects; order and repetition don't matter.
- **Membership:** $a \in A$ (in), $b \notin A$ (out).
- **Two forms:** Roster $\{2, 4, 6\}$ vs Set-builder $\{x : P(x)\}$ (“ x such that...”).
- **Empty set:** $\emptyset = \{\}$ — *zero* elements. Remember $\{0\} \neq \emptyset$ and $\{\emptyset\} \neq \emptyset$.
- **Finite vs infinite:** decided by whether $n(A)$ is a definite whole number.
- **Equal vs equivalent:** $A = B$ needs identical elements; equivalent only needs $n(A) = n(B)$.
- **Number sets:** $\mathbb{N} \subset \mathbb{W} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$; irrationals = $\mathbb{T}$.

Memory aids:

- “*Well-defined or it's not a set*” — kills subjective collections.
- “*Domain first, solve second*” — read $\mathbb{N}/\mathbb{Z}/\mathbb{R}$ before computing.
- “*Equal = same faces; equivalent = same count.*”

Subtopic 02: Subsets, Power Set, and Universal Set

Exam Relevance: A perennial source of quick marks. CBSE Boards ask 1-2 mark questions on listing subsets, the power set, or converting between set-builder and interval notation. JEE Main loves the one-line trap “if $n[P(A)] = 2^k$, find $n(A)$ ” and subset-counting (2^n , $2^n - 1$). The interval-as-subset idea is the silent backbone of every domain/range problem in calculus, so JEE Advanced assumes total fluency.

Section 1: Concept Intuition

Picture the 15-member squad India announces for a tournament. From that squad the captain picks a playing eleven. Every player in the eleven is also in the squad — that is the entire idea of a **subset**. A set A is a subset of a set B when *every* element of A is also an element of B . We write $A \subseteq B$. Crucially, B is allowed to have extras (the four players warming the bench); all we demand is that nobody in A is missing from B .

Two boundary cases follow immediately, and both trip students up:

- **Every set is a subset of itself**, $A \subseteq A$. The full 15-member squad is, trivially, a valid “selection” from the 15-member squad.
- **The empty set is a subset of every set**, $\emptyset \subseteq B$. This feels strange — how can “nothing” be a sub-selection? Think of it as a guest list: the *empty* guest list breaks no rule of any party, because to fail you’d need to point at someone on the empty list who wasn’t invited — and there’s nobody to point at. With no possible counterexample, the statement holds **vacuously**. The empty selection is always legal.

When $A \subseteq B$ but A is *not* the whole of B (there’s at least one extra in B), we call A a **proper subset**, written $A \subsetneq B$, and B a **superset** of A ($B \supsetneq A$). The playing eleven is a proper subset of the squad.

The single most important hurdle in this whole chapter is the difference between $\in$ and $\subseteq$. The symbol $\in$ relates an *element* to a set; $\subseteq$ relates a *set* to a set. The captain Rohit is an *element* of the squad ($\text{Rohit} \in \text{Squad}$), but the *opening pair* $\{\text{Rohit}, \text{Gill}\}$ is a *subset* of the squad. Writing $\text{Rohit} \subseteq \text{Squad}$ or $\{\text{Rohit}\} \in \text{Squad}$ is a category error. Hold this distinction: a player is *in* the team; a partnership is *part of* the team.

Now widen the lens. Every discussion happens inside some agreed playing field — the set of all objects currently under consideration. That field is the **universal set** U . In an election, U is all registered voters; in a number-theory chapter, U might be $\mathbb{Z}$ or $\mathbb{R}$. Every set in the discussion is a subset of U . Choosing the right U is like agreeing on the syllabus before setting a paper — it fixes what’s fair game.

Finally, the **power set**. Suppose you have the small set $\{a, b\}$. List *all* its possible sub-selections: $\emptyset$, $\{a\}$, $\{b\}$, $\{a, b\}$ — four of them. The set whose members are all these subsets is the **power set** $P(A)$. It is the complete *menu* of every sub-selection you could make.

Think about it this way... To build a subset of A , walk down the list of elements and, for each one, flip a coin: *in* or *out*. With n elements you make n independent two-way

choices, giving 2^n subsets in total. That single mental image is the engine behind almost every counting question in this subtopic.

A special continuous case of “subset of $\mathbb{R}$ ” is an **interval** — an unbroken stretch of the number line, like $\{x \in \mathbb{R} : 2 \leq x < 5\}$, written $[2, 5)$.

Edge cases & where students slip (the “fine print”):

- $\emptyset$ and A itself are *always* subsets of A (the two “improper” extremes); forgetting $\emptyset$ when listing subsets is the classic error.
- $P(\emptyset) = \{\emptyset\}$ has **one** element, not zero — its only subset is $\emptyset$.
- $\in$ vs $\subseteq$: $\{1\} \subseteq \{1, 2\}$ (subset) but $\{1\} \in P(\{1, 2\})$ (element of the power set).
- An interval like $(2, 5)$ is an **infinite** set, even though it “looks small.”

Section 2: Key Definitions, Notation & Formulas

Concept	Definition / Statement	Notation
Subset	every element of A lies in B	$A \subseteq B \iff (x \in A \Rightarrow x \in B)$
Not a subset	some element of A is outside B	$A \not\subseteq B$
Proper subset	$A \subseteq B$ and $A \neq B$	$A \subsetneq B$ (also $A \subset B$)
Superset	B contains A	$B \supseteq A$
Equality via subsets	mutual containment	$A = B \iff A \subseteq B \text{ and } B \subseteq A$
Empty-set rule	$\emptyset$ is a subset of every set	$\emptyset \subseteq A$
Self rule	every set is a subset of itself	$A \subseteq A$
Power set	the set of all subsets of A	$P(A) = \{X : X \subseteq A\}$
Universal set	all objects under discussion	U

Intervals (for $a, b \in \mathbb{R}$, $a < b$): open $(a, b) = \{x : a < x < b\}$; closed $[a, b] = \{x : a \leq x \leq b\}$; half-open $[a, b)$ and $(a, b]$. The **length** of any of these is $b - a$.

Core counting formulas (for $n(A) = n$):

$$\text{Number of subsets} = n[P(A)] = 2^n$$

$$\text{Number of proper subsets} = 2^n - 1 \quad \text{Number of non-empty subsets} = 2^n - 1$$

$$\text{Number of non-empty proper subsets} = 2^n - 2$$

$$\text{Number of subsets with exactly } r \text{ elements} = \binom{n}{r} \quad n[P(P(A))] = 2^{2^n}$$

Section 3: Derivation & Key Procedures

Derivation — why a set of n elements has exactly 2^n subsets.

Assumption: let $A = \{a_1, a_2, \dots, a_n\}$ be a finite set with n distinct elements.

To specify a subset $S \subseteq A$, we must decide, for **each** element a_i , exactly one of two things: either $a_i \in S$ or $a_i \notin S$. These decisions are **independent** — choosing whether a_1 is in tells us nothing about a_2 . So we have 2 choices for a_1 , times 2 for a_2 , ..., times 2 for a_n :

$$\underbrace{2 \times 2 \times \dots \times 2}_{n \text{ factors}} = 2^n.$$

Each distinct sequence of in/out decisions yields a distinct subset, and every subset arises from exactly one such sequence — a perfect one-to-one correspondence between subsets and length- n binary strings. Hence $n[P(A)] = 2^n$. *Why this is airtight:* the “all-out” string gives $\emptyset$ and the “all-in” string gives A itself, so the count already includes both improper subsets — which is why the proper-subset count is one fewer, $2^n - 1$.

Cross-check via the binomial theorem: the number of subsets is the number with 0 elements plus those with 1, ..., plus those with n :

$$\binom{n}{0} + \binom{n}{1} + \dots + \binom{n}{n} = (1 + 1)^n = 2^n. \checkmark$$

Procedure A — Construct a power set without missing any subset. List subsets by size, smallest first: start with $\emptyset$, then all singletons, then all pairs, ..., ending with A itself. Counting as you go, you should reach exactly 2^n . *Why:* organising by size prevents the two commonest omissions — forgetting $\emptyset$ and forgetting the full set A .

Procedure B — Choose a valid universal set. U must contain every element appearing in *all* the sets under discussion (their union, at minimum). *Why:* if even one element sits outside U , then some set is not a subset of U , breaking the defining property of a universal set.

Procedure C — Convert set-builder $\leftrightarrow$ interval. Read the inequality type: strict ($<$) $\rightarrow$ round bracket (open, endpoint excluded); non-strict ($\leq$) $\rightarrow$ square bracket (closed, endpoint included). *Why:* the bracket shape is the inequality — a square bracket is a promise that the endpoint belongs.

Section 4: Worked Examples

Example 1 — [CBSE Board] Write the power set of $A = \{p, q, r\}$ and state $n[P(A)]$.

List by size:

- size 0: $\emptyset$
- size 1: $\{p\}, \{q\}, \{r\}$
- size 2: $\{p, q\}, \{p, r\}, \{q, r\}$
- size 3: $\{p, q, r\}$

$$P(A) = \{ \emptyset, \{p\}, \{q\}, \{r\}, \{p, q\}, \{p, r\}, \{q, r\}, \{p, q, r\} \}$$

$$n[P(A)] = 2^3 = 8.$$

Figure (to be illustrated). A power-set lattice for $\{p, q, r\}$: bottom node $\emptyset$, three singletons above it, three pairs above those, and $\{p, q, r\}$ at the top, with edges connecting each subset to the subsets one element larger.

Example 2 — [JEE Main — Speed Trap] Given $n[P(A)] = 256$, find (i) $n(A)$ and (ii) the number of proper subsets of A .

Speed move: recognise the power of two. $256 = 2^8$.

- Since $n[P(A)] = 2^{n(A)} = 2^8$, we get $n(A) = 8$.
- Proper subsets $= 2^8 - 1 = 255$.

The traps: (a) Writing $n(A) = 256$ — but 256 is the *number of subsets*, not the number of elements. (b) Writing 254 for proper subsets by wrongly subtracting 2; only the set itself is excluded from “proper,” so subtract 1.

Example 3 — [JEE Main] Let $A = \{x \in \mathbb{Z} : x^2 - x - 6 \leq 0\}$. Find $n[P(A)]$.

This links a quadratic inequality, an integer domain, and the power-set count.

Factor: $x^2 - x - 6 = (x - 3)(x + 2) \leq 0 \Rightarrow -2 \leq x \leq 3$. Over $\mathbb{Z}$: $A = \{-2, -1, 0, 1, 2, 3\}$, so $n(A) = 6$.

$$n[P(A)] = 2^6 = \boxed{64}.$$

(The integer domain is decisive: over $\mathbb{R}$ the solution set would be the interval $[-2, 3]$, an *infinite* set with no finite power-set count.)

Example 4 — [JEE Advanced] Let $A = \{1, 2, 3, 4, 5, 6, 7\}$. (a) How many subsets of A contain only odd numbers? (b) How many subsets of A contain at least one even number?

Split A into odds $O = \{1, 3, 5, 7\}$ ($n = 4$) and evens $E = \{2, 4, 6\}$ ($n = 3$).

(a) A subset made of only odd numbers is precisely a subset of O . By the in/out argument, there are

$$2^4 = 16 \text{ such subsets (this count includes } \emptyset \text{).}$$

(b) Use the **complement principle**. “At least one even” = (all subsets) — (subsets with no even number). Subsets with no even number are exactly the subsets of O , of which

there are $2^4 = 16$. Total subsets of A are $2^7 = 128$. Hence

$$128 - 16 = \boxed{112}.$$

The elegance is in part (b): rather than summing “exactly one even, exactly two evens, ...” we count the easy opposite and subtract — the standard Advanced reflex.

Section 5: Practice Exercises

1. **[CBSE]** Write all subsets of $\{5, 7\}$, and state how many there are.
2. **[JEE Main]** If a set has 63 proper subsets, how many elements does it have?
3. **[JEE Main]** Write $\{x \in \mathbb{R} : -3 \leq x < 4\}$ in interval form and state its length.
4. **[JEE Advanced]** Let $A = \{1, 2, 3, 4, 5\}$. How many subsets of A contain the element 1 but **not** the element 2?
5. **[JEE Advanced]** If $n[P(A)] - n[P(B)] = 56$ and $B \subset A$, find $n(A)$ and $n(B)$.

Answers:

1. $\emptyset, \{5\}, \{7\}, \{5, 7\}$ — four subsets.
2. 6 (since $2^n - 1 = 63 \Rightarrow 2^n = 64 \Rightarrow n = 6$).
3. $[-3, 4)$; length = $4 - (-3) = 7$.
4. 8 (fix 1 in and 2 out; the remaining $\{3, 4, 5\}$ are free: $2^3 = 8$).
5. $n(A) = 6, n(B) = 3$ (since $2^a - 2^b = 56 = 64 - 8$).

Section 6: Crack the MCQ

Q1. Which statement is **correct** for the set $A = \{1, 2\}$?

- (A) $1 \subseteq A$ (B) $\{1\} \in A$ (C) $\{1\} \subseteq A$ (D) $\emptyset \in A$

Answer: (C). $\{1\}$ is a set all of whose elements (1) lie in A , so $\{1\} \subseteq A$. *Distractors:* (A) misuses $\subseteq$ on the element 1 — an element relates by $\in$, not $\subseteq$. (B) claims the set $\{1\}$ is an element of A , but A 's elements are the numbers 1 and 2, not the set $\{1\}$ (that would belong to $P(A)$). (D) confuses “ $\emptyset$ is a subset of A ” (true) with “ $\emptyset$ is an element of A ” (false). All three are the $\in/\subseteq$ category error in disguise.

Q2. How many elements does $P(\emptyset)$ have?

- (A) 0 (B) 1 (C) 2 (D) undefined

Answer: (B). $\emptyset$ has 0 elements, so $n[P(\emptyset)] = 2^0 = 1$; its single member is $\emptyset$ itself, i.e. $P(\emptyset) = \{\emptyset\}$. *Distractors:* (A) wrongly reasons “no elements $\Rightarrow$ no subsets,” forgetting that $\emptyset$ is a subset of every set, including itself. (C) confuses $P(\emptyset)$ with $P(\{\emptyset\})$. (D) imagines the power set is undefined for the empty set — it is perfectly well-defined.

Q3. A set A has n elements. The number of subsets of A having an **odd** number of elements is:

- (A) 2^{n-1} (B) $2^n - 1$ (C) $n/2$ (D) $2^n/n$

Answer: (A). Among all 2^n subsets, exactly half have an even size and half have an odd size (provable from $\sum_r (-1)^r \binom{n}{r} = 0$), giving 2^{n-1} . *Distractors:* (B) is the count of *proper* subsets, an unrelated quantity. (C) and (D) come from guessing a formula without the even/odd symmetry argument; neither is even an integer in general.

Q4. If $A = \{x : x \in \mathbb{R}, 1 < x \leq 6\}$, then A in interval form is:

- (A) $[1, 6]$ (B) $(1, 6)$ (C) $(1, 6]$ (D) $[1, 6)$

Answer: (C). Strict $<$ at the left excludes 1 (round bracket); $\leq$ at the right includes 6 (square bracket). *Distractors:* each wrong option flips one bracket. (A) wrongly includes 1; (B) wrongly excludes 6; (D) reverses both. The fix is to map each inequality symbol to its bracket *before* writing the answer.

Section 7: Common Student Pitfalls & Pro-Tips

- **Confusing $\in$ with $\subseteq$.** Elements join a set with $\in$; sub-collections relate with $\subseteq$. $1 \in A$, but $\{1\} \subseteq A$ and $\{1\} \in P(A)$.
- **Forgetting $\emptyset$ (and A itself) when listing subsets.** They are always there; organise by size so you can't miss them.
- **Off-by-one in proper subsets.** Proper subsets = $2^n - 1$ (exclude only A itself), not $2^n - 2$.
- **Reading $n[P(A)]$ as $n(A)$.** If $n[P(A)] = 1024 = 2^{10}$, then $n(A) = 10$, never 1024.

Pro-Tip: Memorise the small powers of two — 2, 4, 8, 16, 32, 64, 128, 256, 512, 1024. Half of all power-set MCQs reduce to “spot which power of two this is,” and instant recognition turns a 60-second problem into a 5-second one.

Section 8: Cheat Sheet Box

- **Subset:** $A \subseteq B \iff$ every element of A is in B . **Proper:** $A \subsetneq B$ means also $A \neq B$.
- **Always true:** $\emptyset \subseteq A$ and $A \subseteq A$.
- **$\in$ vs $\subseteq$:** element-to-set vs set-to-set — never mix them.
- **Power set:** $P(A) = \{X : X \subseteq A\}$; $n[P(A)] = 2^n$.
- **Subset counts:** subsets 2^n ; proper $2^n - 1$; with exactly r elements $\binom{n}{r}$; $P(\emptyset) = \{\emptyset\}$ (one element).
- **Intervals:** $< \rightarrow$ round bracket (open), $\leq \rightarrow$ square bracket (closed); length = $b - a$.

Memory aids:

- “In with $\in$, part with $\subseteq$.”
- “Coin per element: 2^n subsets.”
- “Proper means everything but yourself: $2^n - 1$.”

Subtopic 03: Venn Diagrams and Set Operations

Exam Relevance: Bread-and-butter marks across every exam. CBSE Boards ask 2-3 mark questions on computing $A \cup B$, $A \cap B$, $A - B$, A' or proving an identity / shading a Venn diagram. JEE Main tests simplification of set expressions and **De Morgan's laws**, often disguised inside logic or probability. The operations on *intervals* are the silent machinery behind every domain-of-a-function problem in JEE Advanced calculus.

Section 1: Concept Intuition

Think of two WhatsApp broadcast lists on your phone: **Family** (A) and **Office** (B). All your saved contacts together form the universal set U . Every set operation is just a way of asking a membership question about these two lists.

- **Union** $A \cup B$ — everyone who is in Family **or** Office (or both). If you wanted to broadcast a wedding invite to both circles, you'd message $A \cup B$. The word to burn into memory is **or** (the *inclusive* or — “both” counts).
- **Intersection** $A \cap B$ — everyone who is in **both** lists, like a cousin who also happens to be a colleague. The keyword is **and**.
- **Difference** $A - B$ — those in Family but **not** in Office. Note this is *one-directional*: $B - A$ (in Office but not Family) is a completely different group. Subtraction of sets, like subtraction of numbers, is **not** commutative.
- **Complement** A' — everyone in your contacts who is **not** in Family. Complement always answers “everyone *else*,” so it only makes sense once you've fixed the universe U . Change U and A' changes with it.
- **Symmetric difference** $A \triangle B$ — those in **exactly one** of the two lists, never both. These are the people who attended only one of two functions. Formally $A \triangle B = (A - B) \cup (B - A)$.

When two lists share nobody — say **Family** and **Strangers** — we call them **disjoint**: $A \cap B = \emptyset$.

The reason **Venn diagrams** are so powerful is that they turn all this bookkeeping into a picture. Draw the universe as a rectangle and each set as a circle inside it. Two overlapping circles carve the rectangle into exactly **four regions**: only A , only B , both ($A \cap B$), and neither (outside both). *Every* element of U lands in exactly one of these four.

Figure (to be illustrated). Rectangle labelled U containing two overlapping circles A and B . Label the four regions: “only A ” ($A - B$), “only B ” ($B - A$), the lens overlap “ $A \cap B$ ”, and the outside corner “neither ($A \cup B$)'”.

Think about it this way... Once you see those four regions, every operation is simply a choice of *which regions to shade*. Union shades three regions (everything but “neither”); intersection shades only the lens; difference $A - B$ shades only the “only A ” crescent; complement A' shades everything outside circle A . You never have to memorise the operations — you just shade.

This picture also makes **De Morgan's laws** obvious. “Not in (Family or Office)” means you're outside *both* circles — i.e. *not in Family and not in Office*. In symbols, $(A \cup B)' = A' \cap B'$. Flipping a “union-then-complement” turns the *or* into an *and*. Likewise $(A \cap B)' = A' \cup B'$.

Edge cases & where students slip (the “fine print”):

- $A - B \neq B - A$ in general — order matters for difference (and for complement direction).
- Complement is meaningless without a stated universe U ; A' depends entirely on U .
- $A \cup A = A$ and $A \cap A = A$ (idempotent) — combining a set with itself changes nothing.
- Disjoint ($A \cap B = \emptyset$) is *not* the same as complementary ($A \cap B = \emptyset$ **and** $A \cup B = U$).

Section 2: Key Definitions, Notation & Laws

Operation	Set-builder definition	Read as
Union	$A \cup B = \{x : x \in A \text{ or } x \in B\}$	A cup B
Intersection	$A \cap B = \{x : x \in A \text{ and } x \in B\}$	A cap B
Difference	$A - B = \{x : x \in A \text{ and } x \notin B\}$	A minus B
Complement	$A' = \{x : x \in U \text{ and } x \notin A\} = U - A$	A complement
Symmetric difference	$A \triangle B = (A - B) \cup (B - A)$	A sym-diff B
Disjoint	$A \cap B = \emptyset$	A, B disjoint

The laws of set algebra (all provable by element-chasing or by shading):

Commutative: $A \cup B = B \cup A$, $A \cap B = B \cap A$

Associative: $(A \cup B) \cup C = A \cup (B \cup C)$, $(A \cap B) \cap C = A \cap (B \cap C)$

Distributive: $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$, $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

Idempotent: $A \cup A = A$, $A \cap A = A$

Identity / Domination: $A \cup \emptyset = A$, $A \cap U = A$, $A \cup U = U$, $A \cap \emptyset = \emptyset$

Complement: $A \cup A' = U$, $A \cap A' = \emptyset$, $(A')' = A$, $U' = \emptyset$, $\emptyset' = U$

De Morgan: $(A \cup B)' = A' \cap B'$, $(A \cap B)' = A' \cup B'$

Two-set cardinality formula (the bridge to practical problems):

$$n(A \cup B) = n(A) + n(B) - n(A \cap B).$$

Section 3: Derivation & Key Procedures

Derivation 1 — De Morgan’s law $(A \cup B)' = A' \cap B'$ **by element-chasing.**

We show each side contains the other by tracking an arbitrary element x :

$$\begin{aligned} x \in (A \cup B)' &\iff x \notin (A \cup B) \iff \text{NOT}(x \in A \text{ or } x \in B) \\ &\iff (x \notin A) \text{ and } (x \notin B) \iff (x \in A') \text{ and } (x \in B') \iff x \in A' \cap B'. \end{aligned}$$

Every step is a logical *iff*, so the two sets have exactly the same elements. *Why this is the gold-standard method:* the negation of “or” is “and” — that single logical fact is the whole law. Shading a Venn diagram *illustrates* it, but element-chasing *proves* it.

Derivation 2 — why $n(A \cup B) = n(A) + n(B) - n(A \cap B)$.

Picture the two-circle Venn diagram. If you simply add $n(A) + n(B)$, every element in the overlap $A \cap B$ has been counted **twice** — once as part of A , once as part of B . To correct the double-count, subtract the overlap once:

$$n(A \cup B) = \underbrace{n(A) + n(B)}_{\text{overlap counted twice}} - \underbrace{n(A \cap B)}_{\text{remove one copy}}.$$

When A and B are disjoint, $n(A \cap B) = 0$ and the formula collapses to $n(A \cup B) = n(A) + n(B)$.

Procedure A — shade a compound expression on a Venn diagram. Work *inside-out*: shade the innermost bracket first, then apply the outer operation. For $A' \cap B$: first shade outside A , then keep only the part of that shading lying inside B . *Why*: operations nest exactly like arithmetic — respect the brackets.

Procedure B — simplify a set expression using the laws. Treat $\cup$ like $+$, $\cap$ like $\times$, and complement like negation; apply distributive and De Morgan laws, then collapse with identity/complement laws (e.g. $A \cap A' = \emptyset$). *Why*: set algebra is a genuine algebra — the same rewrite discipline you use in numbers applies.

Section 4: Worked Examples

Example 1 — [CBSE Board] Let $U = \{1, 2, \dots, 10\}$, $A = \{2, 4, 6, 8\}$, $B = \{1, 2, 3, 4, 5\}$. Find $A \cup B$, $A \cap B$, $A - B$, A' , and $A \triangle B$.

- $A \cup B = \{1, 2, 3, 4, 5, 6, 8\}$ (everything in either)
- $A \cap B = \{2, 4\}$ (common to both)
- $A - B = \{6, 8\}$ (in A , not in B)
- $A' = U - A = \{1, 3, 5, 7, 9, 10\}$
- $B - A = \{1, 3, 5\}$, so $A \triangle B = (A - B) \cup (B - A) = \{6, 8\} \cup \{1, 3, 5\} = \{1, 3, 5, 6, 8\}$.

Example 2 — [JEE Main — Speed Trap] Simplify $(A - B) \cup (A \cap B)$.

Speed move — shade, don’t grind. $(A - B)$ is the “only A ” crescent; $(A \cap B)$ is the

lens. Together they fill the entire circle A :

$$(A - B) \cup (A \cap B) = A.$$

Algebraic confirmation: $A - B = A \cap B'$, so $(A \cap B') \cup (A \cap B) = A \cap (B' \cup B) = A \cap U = A$.

The trap: students rush to $A \cup B$ (over-shading into B) or $A \cap B$ (under-shading). The crescent-plus-lens picture instantly gives A — a 5-second answer if you visualise the regions.

Example 3 — [JEE Main] Over the universe $\mathbb{R}$, let $A = [-2, 5)$ and $B = (1, 8]$. Find $A \cap B$, $A \cup B$, and $A - B$.

This links interval notation to operations — endpoints are where marks are won or lost.

- $A \cap B$: numbers in *both*. Left edge is the larger lower bound: 1 (open, since B excludes it). Right edge is the smaller upper bound: 5 (open, since A excludes it). $A \cap B = (1, 5)$.
- $A \cup B$: numbers in *either*. Left edge -2 (closed, from A); right edge 8 (closed, from B). $A \cup B = [-2, 8]$.
- $A - B$: in A but not in B . B starts just above 1, so $A - B$ runs from -2 up to and **including** 1 (since $1 \notin B$, it survives): $A - B = [-2, 1]$.

Figure (to be illustrated). Number line with $A = [-2, 5)$ drawn as a bar (filled dot at -2 , open dot at 5) and $B = (1, 8]$ below it (open dot at 1 , filled dot at 8); shade the overlap $(1, 5)$ to show $A \cap B$.

Example 4 — [JEE Advanced] Prove that the two standard definitions of symmetric difference agree:

$$(A \cup B) - (A \cap B) = (A - B) \cup (B - A).$$

Element-chase. Take any x :

$$x \in (A \cup B) - (A \cap B) \iff (x \in A \text{ or } x \in B) \text{ and NOT}(x \in A \text{ and } x \in B).$$

The condition “(in at least one) and (not in both)” is precisely “in **exactly one** of A, B .” That splits into two mutually exclusive cases:

$$(x \in A \text{ and } x \notin B) \quad \text{or} \quad (x \in B \text{ and } x \notin A),$$

which is exactly $x \in (A - B)$ or $x \in (B - A)$, i.e. $x \in (A - B) \cup (B - A)$. Since every step is an *iff*, the two sets are equal. ■

The insight: symmetric difference is the formal name for the everyday phrase “one or the other but not both.”

Section 5: Practice Exercises

1. **[CBSE]** With $U = \{x \in \mathbb{N} : x \leq 9\}$, $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5, 6\}$, find A' , $B - A$,

and $A \triangle B$.

2. **[JEE Main]** Simplify $(A' \cap B')'$ using De Morgan's law.
3. **[JEE Main]** Over $\mathbb{R}$, find $(-\infty, 3] \cap (0, \infty)$ in interval form.
4. **[JEE Advanced]** Prove $(A - B) \cap (A - C) = A - (B \cup C)$.
5. **[JEE Advanced]** Simplify $A \cap (A \cup B)$ and $A \cup (A \cap B)$ (the **absorption laws**) and justify each in one line.

Answers:

1. $A' = \{5, 6, 7, 8, 9\}$; $B - A = \{5, 6\}$; $A \triangle B = \{1, 2, 5, 6\}$.
2. $A \cup B$ (since $(A' \cap B')' = (A')' \cup (B')' = A \cup B$).
3. $(0, 3]$.
4. Element-chase: $x \in \text{LHS} \iff x \in A, x \notin B, x \notin C \iff x \in A, x \notin (B \cup C) \iff x \in \text{RHS}$.
5. $A \cap (A \cup B) = A$ and $A \cup (A \cap B) = A$ — the smaller/larger of the two always absorbs the combination.

Section 6: Crack the MCQ

Q1. $(A \cup B)'$ is equal to:

- (A) $A' \cup B'$ (B) $A' \cap B'$ (C) $A' - B'$ (D) $A \cap B$

Answer: (B). De Morgan: the complement of a union is the intersection of the complements. *Distractors:* (A) applies the *wrong* De Morgan form — that's $(A \cap B)'$. (C) invents a difference rule that doesn't exist. (D) confuses “outside both circles” with “the overlap.” The fix: complement turns $\cup$ into $\cap$, every time.

Q2. If $A = \{1, 2, 3\}$ and $B = \{3, 4, 5\}$, then $A - B$ equals:

- (A) $\{4, 5\}$ (B) $\{3\}$ (C) $\{1, 2\}$ (D) $\{1, 2, 4, 5\}$

Answer: (C). Elements in A but not in B : 1 and 2 (3 is excluded because it's also in B). *Distractors:* (A) computes $B - A$ instead — the commutativity trap. (B) gives $A \cap B$. (D) gives the symmetric difference $A \triangle B$. Each is a *different* operation a careless reader substitutes for difference.

Q3. For any set A with universe U , $A \cap A'$ equals:

- (A) U (B) A (C) $\emptyset$ (D) A'

Answer: (C). No element can be both in A and outside A , so the intersection is empty. *Distractors:* (A) is the result of $A \cup A'$ (the *union* fills the universe) — the classic union/intersection swap. (B) and (D) ignore the contradiction entirely. Anchor: A and A' are disjoint by definition.

Q4. If $A \subseteq B$, then $A \cup B$ equals:

- (A) A (B) B (C) $\emptyset$ (D) $A \cap B$

Answer: (B). If every element of A is already in B , throwing A into the union adds nothing new, so $A \cup B = B$. *Distractors:* (A) reverses the roles — that would be $A \cap B =$

A under the same hypothesis. (C) is baseless. (D) confuses union with intersection. Visualise: a circle A sitting *inside* circle B — their union is just B .

Section 7: Common Student Pitfalls & Pro-Tips

- **Treating difference as commutative.** $A - B \neq B - A$. Always read which set is the “source” and which is being removed.
- **Swapping the two De Morgan forms.** $(A \cup B)' = A' \cap B'$ (union $\rightarrow$ intersection); $(A \cap B)' = A' \cup B'$. The operation *flips*.
- **Forgetting to fix the universe before taking a complement.** A' is undefined until U is stated; the answer changes with U .
- **Endpoint errors in interval operations.** For $\cap$, take the *tighter* bounds and the *stricter* bracket; for $\cup$ (of overlapping intervals), the *looser* bounds.

Pro-Tip: For any expression in two sets, draw the four-region Venn once and **shade**. Shading is faster and far less error-prone than juggling laws — and it instantly catches whether your algebra landed on A , $A \cup B$, or $A \cap B$.

Section 8: Cheat Sheet Box

- **Union** $\cup$ = “or” (inclusive). **Intersection** $\cap$ = “and”. **Difference** $A - B = A \cap B'$. **Complement** $A' = U - A$.
- **Symmetric difference:** $A \triangle B = (A - B) \cup (B - A) = (A \cup B) - (A \cap B)$.
- **De Morgan:** $(A \cup B)' = A' \cap B'$ and $(A \cap B)' = A' \cup B'$ — the operation flips.
- **Complement laws:** $A \cup A' = U$, $A \cap A' = \emptyset$, $(A')' = A$.
- **Disjoint:** $A \cap B = \emptyset$. **Cardinality:** $n(A \cup B) = n(A) + n(B) - n(A \cap B)$.
- **Intervals:** $\cap \rightarrow$ tighter bounds/stricter brackets; $\cup \rightarrow$ looser bounds.

Memory aids:

- “Cup = or, cap = and.”
- “De Morgan flips the sign: complement a union, get an intersection.”
- “When stuck, shade the Venn.”

Subtopic 04: Cardinality of Unions and Practical Problems

Provenance note: this subtopic is [SUPPLIED — grounding waived by founder] — generated from NCERT-aligned curriculum knowledge; no source bundle was retrieved, so it is not RAG-grounded. Accepted into master content by explicit founder decision. All numeric answers are SymPy-verified and all examples are original. It remains flagged as supplied (not [GROUNDED]) for an honest audit trail, and will be upgraded if a source bundle is later audited.

Exam Relevance: One of the highest-yield, most predictable scoring topics in the whole chapter. CBSE Boards almost always carry a 3–4 mark word problem on two or three sets. JEE Main features inclusion-exclusion both as standalone counting and woven into Probability and Permutations. JEE Advanced pushes into fine partitions of a three-set Venn diagram — *exactly one, exactly two, at least two* — and problems where the three-way overlap is the unknown you must solve for.

Section 1: Concept Intuition

Suppose a coaching batch has 70 students enrolled in Physics and 50 in Chemistry. How many students are enrolled in at least one of the two subjects? The tempting answer is $70 + 50 = 120$ — but that is almost always wrong, because some students take **both**. When you counted the 70 Physics students you already swept up the both-subject students, and when you counted the 50 Chemistry students you counted those same people a **second time**.

That single observation is the heart of this entire subtopic. Adding two sets double-counts their overlap, so to get the true size of the union you add the two sizes and then **subtract the overlap once** to undo the double-count:

$$n(A \cup B) = n(A) + n(B) - n(A \cap B).$$

Now scale up to three subjects — Physics (A), Chemistry (B), Mathematics (C). Building the count honestly, region by region, forces a beautiful alternating pattern:

1. **Add** the three individual sizes: $n(A) + n(B) + n(C)$. But now every two-way overlap has been counted twice and the central three-way overlap has been counted three times.
2. **Subtract** the three pairwise overlaps: $-n(A \cap B) - n(B \cap C) - n(C \cap A)$. This fixes the two-way regions — but watch the centre: the triple overlap was added 3 times, then subtracted 3 times (it sits inside all three pairwise overlaps), so right now it is counted **zero** times.
3. **Add** the triple overlap back **once**: $+n(A \cap B \cap C)$.

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(C \cap A) + n(A \cap B \cap C).$$

The signs alternate $+, -, + -$ — *include, exclude, include* — which is exactly why the rule is called the **inclusion-exclusion principle**.

Figure (to be illustrated). A three-set Venn diagram inside a rectangle U , circles A, B, C overlapping. Label all eight regions: only A , only B , only C , $(A \cap B)$ only, $(B \cap C)$ only, $(C \cap A)$ only, the central $A \cap B \cap C$, and the outside region “none”.

Think about it this way... Don’t memorise long formulas — *label the regions*. A three-circle Venn has exactly seven enclosed regions; give each a letter, fill the **centre first** and work **outward**, and every word problem collapses into simple addition and subtraction. The formulas are just bookkeeping for what the picture already shows.

Edge cases & where students slip (the “fine print”):

- **“or” means union, “and” means intersection.** “Students taking Physics *or* Chemistry” is $A \cup B$; “*and*” is $A \cap B$.
- **“At least one” vs “none”.** Those in none of the sets are the complement of the union: $n(U) - n(A \cup B \cup C)$. They are *outside* every circle.
- **“Only A” vs “A”.** “Only A” is the crescent A minus its overlaps; plain “A” includes every overlap. Mixing these up is the single most common word-problem error.
- **Data can be inconsistent.** If a computed union exceeds the universe ($n(A \cup B) > n(U)$), the given numbers are contradictory — a legitimate exam answer is “the data is impossible.”

Section 2: Key Definitions, Notation & Formulas

(No dimensional formulae here — these are pure counting identities. What matters is reading each phrase into the right region.)

Quantity	Meaning	Formula
Two-set union	elements in A or B	$n(A \cup B) =$ $n(A) + n(B) - n(A \cap B)$
Three-set union	elements in A, B , or C	$n(A \cup B \cup C) = \sum n(A) -$ $\sum n(A \cap B) + n(A \cap B \cap C)$
Disjoint union	no overlap	$n(A \cup B) = n(A) + n(B)$ when $A \cap B = \emptyset$
None (neither)	outside every set	$n(U) - n(A \cup B \cup C)$
Only A (two sets)	in A , not in B	$n(A) - n(A \cap B)$
Exactly one (three sets)	in precisely one set	$\sum n(A) - 2 \sum n(A \cap B) +$ $3n(A \cap B \cap C)$
Exactly two (three sets)	in precisely two sets	$\sum n(A \cap B) - 3n(A \cap B \cap C)$
At least two (three sets)	in two or more sets	$\sum n(A \cap B) - 2n(A \cap B \cap C)$

Here the shorthand $\sum n(A) = n(A) + n(B) + n(C)$ and $\sum n(A \cap B) = n(A \cap B) + n(B \cap C) + n(C \cap A)$.

Two partition identities worth internalising (they turn hard problems into linear equations). Writing E_1, E_2, E_3 for the counts in *exactly one* / *exactly two* / *all three* regions:

$$n(A \cup B \cup C) = E_1 + E_2 + E_3, \quad n(A) + n(B) + n(C) = E_1 + 2E_2 + 3E_3.$$

The first says the union is the sum of its disjoint regions; the second says each element is counted once per set it belongs to.

Section 3: Derivation & Key Procedures

Derivation — the three-set formula from the two-set formula.

Treat $B \cup C$ as a single set and apply the two-set rule to A and $(B \cup C)$:

$$n(A \cup (B \cup C)) = n(A) + n(B \cup C) - n(A \cap (B \cup C)).$$

Expand $n(B \cup C) = n(B) + n(C) - n(B \cap C)$. For the last term, distribute the intersection over the union:

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C),$$

so, by the two-set rule again,

$$n(A \cap (B \cup C)) = n(A \cap B) + n(A \cap C) - n((A \cap B) \cap (A \cap C)) = n(A \cap B) + n(A \cap C) - n(A \cap B \cap C),$$

using $(A \cap B) \cap (A \cap C) = A \cap B \cap C$. Substituting everything back and collecting terms gives the alternating formula. *Why this matters:* the three-set rule is not a new axiom to memorise — it is the two-set rule applied twice, which is exactly why the pattern extends to any number of sets.

Derivation — the “exactly two” count. Each pairwise overlap $n(A \cap B)$ contains both the people in *only A* and *B* and the people in *all three*. Summing the three pairwise overlaps therefore counts every “exactly two” person once, but counts each “all three” person **three** times (once in each pairwise overlap). Removing those extra copies — three each, but we want them counted zero times here — means subtracting $3n(A \cap B \cap C)$:

$$E_2 = \sum n(A \cap B) - 3n(A \cap B \cap C).$$

(For *at least two*, we want the centre counted once, not zero, so we subtract only $2n(A \cap B \cap C)$.)

Procedure A — the Venn region method (fastest for fully-specified data).

1. Draw three overlapping circles in a rectangle U .
2. Fill the **centre** $n(A \cap B \cap C)$ first.
3. For each pairwise region, subtract the centre: “only A and B ” = $n(A \cap B) - n(A \cap B \cap C)$.
4. For each single region, subtract its two adjoining pairwise regions and the centre

from $n(A)$.

5. The “none” region = $n(U) - (\text{sum of all seven inner regions})$. *Why inside-out:* every outer region is defined by removing the inner regions it contains, so the centre must be known before anything around it.

Procedure B — the algebraic method (fastest when an overlap is unknown). Translate the question into set language, write the relevant inclusion-exclusion identity (or the two partition identities), and solve for the unknown term. Use this when the three-way overlap or a missing total is what you are asked to find.

Section 4: Worked Examples

Example 1 — [CBSE Board] In a batch of 120 students, 70 are enrolled in Physics and 50 in Chemistry; 30 are enrolled in both. Find the number enrolled in (i) at least one of the two subjects, (ii) Physics only, (iii) neither subject.

Let P and C be the two subject sets. Given $n(U) = 120$, $n(P) = 70$, $n(C) = 50$, $n(P \cap C) = 30$.

(i) $n(P \cup C) = n(P) + n(C) - n(P \cap C) = 70 + 50 - 30 = \boxed{90}$. **(ii)** Physics only = $n(P) - n(P \cap C) = 70 - 30 = \boxed{40}$. **(iii)** Neither = $n(U) - n(P \cup C) = 120 - 90 = \boxed{30}$.

Example 2 — [JEE Main — Speed Trap] In a hostel of 80 students, 60 subscribe to a tea service and 35 to a coffee service. If every student subscribes to at least one of the two, how many subscribe to both?

Speed move — read “at least one” as union = total. Since everyone takes at least one service, $n(\text{tea} \cup \text{coffee}) = 80$. Rearranging the two-set formula for the overlap:

$$n(\text{both}) = n(\text{tea}) + n(\text{coffee}) - n(\text{union}) = 60 + 35 - 80 = \boxed{15}.$$

The trap: students who don’t register “at least one $\Rightarrow$ union = 80” get stuck with two unknowns. The phrase “every student subscribes to at least one” is the whole key — it hands you the union for free. The decoy instinct is to compute $60 + 35 = 95$ and panic that it exceeds 80; that excess is the double-counted overlap.

Example 3 — [JEE Main] A sports survey of 200 students finds: 100 play cricket, 80 play football, 65 play hockey; 40 play cricket and football, 30 football and hockey, 35 cricket and hockey, and 20 play all three. Find the number who play (i) at least one sport, (ii) exactly two sports, (iii) no sport.

Let C, F, H be the three sets. $\sum n = 100 + 80 + 65 = 245$; $\sum \text{pairwise} = 40 + 30 + 35 = 105$; triple = 20.

(i) $n(C \cup F \cup H) = 245 - 105 + 20 = \boxed{160}$. **(ii)** Exactly two = $\sum n(A \cap B) - 3n(A \cap B \cap C) = 105 - 3(20) = 105 - 60 = \boxed{45}$. **(iii)** No sport = $n(U) - n(C \cup F \cup H) = 200 - 160 = \boxed{40}$.

Example 4 — [JEE Advanced] In a colony of 150 residents, every resident speaks at

least one of Hindi, English, or Tamil. 100 speak Hindi, 70 English, and 55 Tamil. If exactly 25 residents speak precisely two of the three languages, how many speak all three?

This is a solve-for-the-unknown problem, so use the **partition identities**. Let E_1, E_2, E_3 be the counts speaking exactly one, exactly two, all three. Given: union = 150 (everyone speaks ≥ 1), $\sum n = 100 + 70 + 55 = 225$, and $E_2 = 25$.

From the two identities:

$$E_1 + E_2 + E_3 = 150 \Rightarrow E_1 + 25 + E_3 = 150,$$

$$E_1 + 2E_2 + 3E_3 = 225 \Rightarrow E_1 + 50 + 3E_3 = 225.$$

Subtract the first from the second: $(50 - 25) + (3E_3 - E_3) = 225 - 150$, i.e. $25 + 2E_3 = 75 \Rightarrow E_3 = 25$.

25 residents speak all three languages.

(Check: $E_3 = 25$, $E_2 = 25 \Rightarrow E_1 = 100$; union = $100 + 25 + 25 = 150$ ✓ and $\sum n = 100 + 2(25) + 3(25) = 225$ ✓.) The lesson: when the three-way overlap is unknown, the two partition identities give two equations — pure algebra finishes the job without ever drawing the diagram.

Section 5: Practice Exercises

1. **[CBSE]** In a class of 60 students, 35 passed Mathematics and 40 passed Physics; 25 passed both. How many passed at least one subject, and how many failed both?
2. **[JEE Main]** A survey of 100 people reports 72 like tea, 55 like coffee, and 20 like both. Is this data consistent? Justify.
3. **[JEE Main]** Among 250 commuters, 140 use buses, 100 use the metro, 75 use autos; 60 use bus and metro, 40 metro and auto, 45 bus and auto, and 25 use all three. How many use none of the three?
4. **[JEE Advanced]** In a town survey of 500 families, 280 read newspaper X, 210 read Y, 150 read Z; 100 read X and Y, 60 read Y and Z, 70 read X and Z, and 40 read all three. Find the number of families that read (i) exactly one newspaper, (ii) at least two newspapers.
5. **[JEE Advanced]** In a group of 100 people, each likes at least one of three desserts. 60 like A, 50 like B, 40 like C, and 10 like all three. How many like exactly two desserts?

Answers:

1. At least one = $35 + 40 - 25 = 50$; failed both = $60 - 50 = 10$.
2. **Inconsistent.** $n(\text{tea} \cup \text{coffee}) = 72 + 55 - 20 = 107 > 100 = n(U)$. The union cannot exceed the total surveyed, so the figures are contradictory.
3. Union = $140 + 100 + 75 - 60 - 40 - 45 + 25 = 315 - 145 + 25 = 195$; none = $250 - 195 = 55$.
4. Exactly one = $(280 + 210 + 150) - 2(100 + 60 + 70) + 3(40) = 640 - 460 + 120 = 300$;

at least two = $(100 + 60 + 70) - 2(40) = 230 - 80 = 150$.

5. Union = 100, $\sum n = 150$, all three = 10. From $\sum n - \text{union} = E_2 + 2E_3$: $150 - 100 = E_2 + 2(10) \Rightarrow E_2 = 30$. So 30 like exactly two.

Section 6: Crack the MCQ

Q1. If $n(A) = 25$, $n(B) = 18$, and $n(A \cup B) = 35$, then $n(A \cap B)$ is:

- (A) 43 (B) 8 (C) 10 (D) 17

Answer: (B). Rearrange the two-set formula: $n(A \cap B) = n(A) + n(B) - n(A \cup B) = 25 + 18 - 35 = 8$. *Distractors:* (A) is $n(A) + n(B)$ with no subtraction — the double-count left uncorrected. (C) and (D) come from subtracting the wrong pair of quantities. Anchor: the overlap is “sum of parts minus the union.”

Q2. For three sets, the number of elements belonging to **exactly two** of them is:

- (A) $\sum n(A \cap B) - n(A \cap B \cap C)$
 (B) $\sum n(A \cap B) - 2n(A \cap B \cap C)$
 (C) $\sum n(A \cap B) - 3n(A \cap B \cap C)$
 (D) $\sum n(A \cap B) + n(A \cap B \cap C)$

Answer: (C). Each pairwise overlap counts the all-three group once, so summing all three pairwise overlaps counts that centre three times; remove all three copies to leave only the “exactly two” regions. *Distractors:* (B) is the formula for *at least two* (centre kept once, not zero). (A) subtracts the centre only once — counts it twice over. (D) double-counts the centre further. The deciding question is always “how many times do I want the centre counted?": exactly-two $\rightarrow$ zero, at-least-two $\rightarrow$ once.

Q3. In two sets with $n(A) = 40$ and $n(A \cap B) = 15$, the number of elements in “**A only**” is:

- (A) 40 (B) 55 (C) 25 (D) 15

Answer: (C). “A only” excludes the shared part: $n(A) - n(A \cap B) = 40 - 15 = 25$. *Distractors:* (A) reports all of A, ignoring “only.” (D) reports the overlap itself. (B) adds instead of subtracting. The trap word is **only** — it always means “remove the overlaps.”

Q4. In the inclusion-exclusion formula for $n(A \cup B \cup C)$, the term $n(A \cap B \cap C)$ appears with which sign and why?

- (A) subtracted once, to remove the centre
 (B) added once, because it is added in the three singles (+3) then removed in the three pairwise overlaps (−3), leaving it uncounted
 (C) added three times
 (D) it does not appear

Answer: (B). The centre lies in all three single sets (+3) and in all three pairwise overlaps (−3), so before the final term it is counted $3 - 3 = 0$ times; adding it once restores it. *Distractors:* (A) gets the sign backwards. (C) ignores the cancellation from the pairwise

subtractions. (D) forgets the centre is part of the union and must be counted exactly once. This net-count reasoning is the whole logic of the alternating signs.

Section 7: Common Student Pitfalls & Pro-Tips

- **Forgetting to subtract the overlap.** $n(A \cup B) = n(A) + n(B)$ is correct *only* for disjoint sets. With any overlap you must subtract $n(A \cap B)$.
- **Confusing “only A” with “A”.** “Only” strips out every overlap; plain “A” keeps them. Re-read the question and underline “only,” “exactly,” “at least,” “neither” before computing.
- **Mis-signing the three-way term.** In the union formula it is **added**; the alternating pattern is +singles – pairs + triple.
- **Treating “exactly two” and “at least two” as the same.** They differ by exactly one copy of the centre: subtract $3n(A \cap B \cap C)$ for *exactly two*, $2n(A \cap B \cap C)$ for *at least two*.
- **Not sanity-checking against the universe.** If a region comes out negative, or the union exceeds $n(U)$, the data is inconsistent — say so rather than forcing an answer.

Pro-Tip: For any fully-specified three-set problem, draw the Venn and **fill the centre first, then work outward**. Every “only,” “exactly,” and “at least” answer can then be read straight off the labelled regions — no formula recall under exam pressure, and negative regions instantly expose bad data.

Section 8: Cheat Sheet Box

- **Two sets:** $n(A \cup B) = n(A) + n(B) - n(A \cap B)$; “only A” = $n(A) - n(A \cap B)$.
- **Three sets:** $n(A \cup B \cup C) = \sum n(A) - \sum n(A \cap B) + n(A \cap B \cap C)$ (include - exclude - include).
- **None / neither:** $n(U) - n(\text{union})$.
- **Exactly one:** $\sum n(A) - 2 \sum n(A \cap B) + 3n(A \cap B \cap C)$.
- **Exactly two:** $\sum n(A \cap B) - 3n(A \cap B \cap C)$. **At least two:** $\sum n(A \cap B) - 2n(A \cap B \cap C)$.
- **Partition identities:** union = $E_1 + E_2 + E_3$; $\sum n(A) = E_1 + 2E_2 + 3E_3$ (use when an overlap is unknown).

Memory aids:

- “Or = union, and = intersection, only = drop the overlaps.”
- “Include, exclude, include” — the signs for one, two, three sets.
- “Fill the centre first, then count outward.”
- “Exactly-two kills the centre thrice; at-least-two only twice.”

Chapter 1: Sets — Supplement

Subtopics 01–04 give you a complete NCERT-level command of Sets. This supplement carries that command up to the JEE ceiling. It has two parts, and you should read them at different times.

Part 1 — Addendum A to H. Eight compact techniques that the main chapter names, assumes, or promises but never drills. Each item is one method statement, one or two worked examples, and practice with answers. Read each addendum item straight after the subtopic it extends:

Addendum	Topic	Read after
A	Deciding finite vs infinite as a solved problem type	Subtopic 01
B	Subsets of a prescribed size, $\binom{n}{r}$	Subtopic 02
C	The iterated power set, 2^{2^n}	Subtopic 02
D	Unbounded (infinite) intervals	Subtopic 02
E	Counting ordered pairs of subsets, 3^n	Subtopic 02
F	Three-circle Venn shading and visual proof	Subtopic 03
G	Operations on the standard number systems	Subtopic 03
H	Bounds on an unknown overlap	Subtopic 04

Part 2 — Subtopic 05. A full eight-section subtopic on families of sets: what happens when the problem hands you not two or three sets but a whole indexed run of them, $A_1, A_2, A_3, \dots$. Indexed unions and intersections, the generalised De Morgan and inclusion–exclusion laws, and the double-counting technique that JEE Advanced has returned to for four decades. Read it after Subtopic 04.

Part 1 — Addendum to Subtopics 01–04

Addendum A — Is this set finite or infinite?

Extends Subtopic 01, Sections 3–5.

Method. Subtopic 01's Procedure C gives the two questions; here is the version you execute under exam pressure.

1. **Read the universe first** ($\mathbb{N}, \mathbb{W}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$). The verdict changes with it.
2. **Does even one element exist?** If none, the set is empty — and $\emptyset$ is **finite**, with $n(\emptyset) = 0$. “Empty” is a special case of finite, never an alternative to it. Writing “empty, so neither finite nor infinite” loses the mark.
3. **Discrete universe** ($\mathbb{N}, \mathbb{W}, \mathbb{Z}$), **condition bounded on both sides** $\Rightarrow$ finite, and you can usually state n exactly.
4. **Discrete universe, condition unbounded in at least one direction** $\Rightarrow$ infinite.
5. **Continuous universe** ($\mathbb{Q}$ or $\mathbb{R}$): any stretch of the line containing two distinct points contains infinitely many, because the mean of any two members lies strictly between them. So every non-degenerate interval is infinite, however short.
6. **Polynomial equations are the exception:** a non-zero polynomial of degree d has at most d roots, so its solution set is finite in any universe.

Two families worth memorising as a contrasting pair:

Set	Verdict	Why
Divisors of a fixed natural number k	Finite	Every divisor is at most k — bounded
Multiples of a fixed natural number k	Infinite	$k, 2k, 3k, \dots$ never stops

Worked Example A1 — [CBSE Board]

Classify each as empty, finite (stating n), or infinite.

- (i) $\{x \in \mathbb{N} : x < 100\}$ — the naturals 1 to 99: **finite**, $n = 99$.
- (ii) $\{x \in \mathbb{Z} : x < 100\}$ — same condition, larger universe: $\dots, -2, -1, 0, \dots, 99$ runs away to the left, so **infinite**.
- (iii) $\{x \in \mathbb{R} : 0 < x < 1\}$ — the interval $(0, 1)$: **infinite** (rule 5).
- (iv) $\{x \in \mathbb{N} : x \text{ divides } 1024\}$ — since $1024 = 2^{10}$, the divisors are $2^0, 2^1, \dots, 2^{10}$: **finite**, $n = 11$.
- (v) $\{x \in \mathbb{N} : x \text{ is a multiple of } 10\}$ — **infinite**.
- (vi) $\{x \in \mathbb{R} : x^2 + 4 = 0\}$ — no real square is negative, so the set is **empty**, hence finite with $n = 0$.
- (vii) The set of all lines in a plane parallel to the x -axis — one line for every height $c \in \mathbb{R}$: **infinite**.
- (viii) $\{x \in \mathbb{Q} : 1 < x < 2\}$ — **infinite**; between any two rationals sits their mean, which is again rational.

Worked Example A2 — [JEE Main]

Classify $A = \{x \in \mathbb{Z} : x^2 \leq 10000\}$, $B = \{x \in \mathbb{Z} : x^2 \geq 10000\}$, $C = \{x \in \mathbb{R} : x^2 \leq 10000\}$.

- $A: x^2 \leq 10000 \iff -100 \leq x \leq 100$. Over $\mathbb{Z}$ that is 100 negatives, 0, and 100 positives: **finite**, $n(A) = 201$.
- $B: x^2 \geq 10000 \iff x \leq -100$ or $x \geq 100$. Unbounded in both directions: **infinite**.
- C : the same inequality as A , but over $\mathbb{R}$ it is the interval $[-100, 100]$: **infinite**.

One inequality, three verdicts. The universe and the direction of the inequality do all the work — exactly the two things you must read before you compute.

Practice.

1. $\{x \in \mathbb{N} : x \text{ is a prime and } x < 50\}$
2. $\{x \in \mathbb{Z} : x^3 = 8\}$
3. $\{x \in \mathbb{R} : 2 < x < 3\}$
4. $\{x \in \mathbb{N} : x^2 < 0\}$
5. $\{x \in \mathbb{W} : x \leq 6\}$
6. $\{x \in \mathbb{Q} : x^2 = 5\}$
7. The set of all circles in a plane passing through the origin
8. $\{x \in \mathbb{Z} : |x| > 7\}$

Answers. 1. Finite, $n = 15$. 2. Finite, $n = 1$ (the set is $\{2\}$). 3. Infinite. 4. Empty, hence finite with $n = 0$. 5. Finite, $n = 7$ (the set is $\{0, 1, 2, 3, 4, 5, 6\}$). 6. Empty (finite, $n = 0$), because $\sqrt{5}$ is irrational. 7. Infinite. 8. Infinite.

Addendum B — Subsets of a prescribed size

Extends Subtopic 02, Sections 2-5.

Method. For a set with n elements, the number of subsets having exactly r elements is

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}, \quad 0 \leq r \leq n.$$

- n : the number of elements available; r : the size demanded. If $r < 0$ or $r > n$, the count is 0 — you cannot choose more elements than exist.
- **Symmetry:** $\binom{n}{r} = \binom{n}{n-r}$. Naming who is in is the same act as naming who is out.
- **Total:** $\sum_{r=0}^n \binom{n}{r} = 2^n$, which is the binomial cross-check already used in Subtopic 02, Section 3.
- **Forcing an element in:** spend one slot on it and choose the rest from the other $n - 1$ elements: $\binom{n-1}{r-1}$.
- **Forcing an element out:** ignore it entirely and choose all r from the remaining $n - 1$: $\binom{n-1}{r}$.
- **Two pools:** to take j elements from a pool of size p and $r - j$ from a disjoint pool of size q , multiply: $\binom{p}{j} \binom{q}{r-j}$.

A caveat worth carrying. For $n \geq 1$, exactly half of the 2^n subsets have an even number of elements and half have an odd number, giving 2^{n-1} of each; this follows from $\sum_{r=0}^n (-1)^r \binom{n}{r} =$

$(1 - 1)^n = 0$. The condition $n \geq 1$ is not decoration: for $n = 0$ the only set is $\emptyset$, the alternating sum equals 1, and the number of odd-sized subsets is 0, not 2^{-1} .

Worked Example B1 — [CBSE Board]

Let $A = \{1, 2, 3, 4, 5, 6, 7\}$. Find the number of subsets of A that have (i) exactly 3 elements, (ii) at most 2 elements, (iii) exactly 3 elements including 1, (iv) exactly 3 elements excluding 1.

$$(i) \binom{7}{3} = \frac{7 \cdot 6 \cdot 5}{3 \cdot 2 \cdot 1} = 35.$$

$$(ii) \text{ "At most 2" means size 0, 1 or 2: } \binom{7}{0} + \binom{7}{1} + \binom{7}{2} = 1 + 7 + 21 = 29.$$

$$(iii) \text{ Fix 1 as a member, then choose the other 2 from } \{2, \dots, 7\}: \binom{6}{2} = 15.$$

$$(iv) \text{ Choose all 3 from } \{2, \dots, 7\}: \binom{6}{3} = 20.$$

Consistency check: every 3-element subset either contains 1 or does not, and $15 + 20 = 35$, matching (i).

Worked Example B2 — [JEE Main]

From $S = \{1, 2, \dots, 9\}$, how many 5-element subsets contain exactly 3 odd numbers?

Split S into two disjoint pools: odds $\{1, 3, 5, 7, 9\}$ (five of them) and evens $\{2, 4, 6, 8\}$ (four of them). "Exactly 3 odd" inside a 5-element subset forces exactly 2 even. The two choices are independent, so multiply:

$$\binom{5}{3} \binom{4}{2} = 10 \times 6 = 60.$$

The habit to build: read the constraint, split into pools, multiply. Adding instead of multiplying is the standard slip.

Practice.

1. Number of 4-element subsets of a 10-element set.
2. Number of subsets of $\{1, \dots, 6\}$ having at least 4 elements.
3. Number of 3-element subsets of $\{1, \dots, 8\}$ containing both 1 and 2.
4. Find n if $\binom{n}{2} = 45$.
5. Number of subsets of $\{1, \dots, 7\}$ having an odd number of elements.
6. From $\{1, 2, \dots, 10\}$, the number of 4-element subsets all of whose elements are even.

Answers. 1. $\binom{10}{4} = 210$. 2. $\binom{6}{4} + \binom{6}{5} + \binom{6}{6} = 15 + 6 + 1 = 22$. 3. $\binom{6}{1} = 6$. 4. $n = 10$. 5. $2^{7-1} = 64$. 6. The evens are $\{2, 4, 6, 8, 10\}$, so $\binom{5}{4} = 5$.

Addendum C — The iterated power set

Extends Subtopic 02, Sections 2 and 4.

Method. Apply the rule $n[P(X)] = 2^{n(X)}$ repeatedly, from the inside outward. With $n(A) = n$:

$$n[P(A)] = 2^n, \quad n[P(P(A))] = 2^{2^n}, \quad n[P(P(P(A)))] = 2^{2^{2^n}}.$$

- A tower of exponents is evaluated **top down**: in 2^{2^n} you compute 2^n first, then raise 2 to that.
- The conceptual step: the elements of $P(P(A))$ are themselves *sets of subsets of A*. If $A = \{a\}$, then $\{\emptyset\}$ and $\emptyset$ are two different elements of $P(P(A))$ — the first is a one-element collection, the second is the empty collection.
- To run the formula backwards, peel one exponent at a time: write the given number as a power of 2, equate exponents, repeat.

Worked Example C1 — [CBSE Board]

Let $A = \{a\}$. Write $P(A)$ and $P(P(A))$ in full, and verify the count.

$P(A) = \{\emptyset, \{a\}\}$, so $n[P(A)] = 2^1 = 2$.

$P(P(A))$ is the set of all subsets of the two-element set $P(A)$:

$$P(P(A)) = \left\{ \emptyset, \{\emptyset\}, \{\{a\}\}, \{\emptyset, \{a\}\} \right\}$$

That is 4 elements, and the formula predicts $2^{2^1} = 2^2 = 4$. Confirmed.

Look hard at the four members: each is a *set*, and $\emptyset$ (the empty collection) is a different member from $\{\emptyset\}$ (the collection containing the empty set). That is the $\in$ -versus- $\subseteq$ distinction of Subtopic 02 raised one storey.

Worked Example C2 — [JEE Main]

If $n[P(P(A)))] = 256$, find $n(A)$ and $n[P(P(P(A)))]$.

Write 256 as a power of 2: $256 = 2^8$. Then

$$2^{2^{n(A)}} = 2^8 \Rightarrow 2^{n(A)} = 8 = 2^3 \Rightarrow n(A) = 3.$$

Peeling the next layer, $n[P(P(P(A)))] = 2^{2^{2^3}} = 2^{2^8} = 2^{256}$.

The trap is answering $n(A) = 8$: that is $n[P(A)]$, one level too shallow. Count the layers of P in the question before you equate anything.

Practice.

1. If $n(A) = 2$, find $n[P(P(A))]$.
2. If $n(A) = 1$, find $n[P(P(P(A)))]$.
3. If $n[P(P(A))] = 16$, find $n(A)$.

4. Find $n[P(P(\emptyset))]$ and list its elements.
5. True or false: $\emptyset \in P(P(A))$ for every set A . Give the reason.

Answers. 1. $2^{2^2} = 2^4 = 16$. 2. $2^{2^{2^1}} = 2^4 = 16$. 3. $2^{2^n} = 2^4 \Rightarrow 2^n = 4 \Rightarrow n = 2$. 4. $n(\emptyset) = 0$, so the count is $2^{2^0} = 2^1 = 2$; explicitly $P(\emptyset) = \{\emptyset\}$ and $P(P(\emptyset)) = \{\emptyset, \{\emptyset\}\}$. 5. True — $\emptyset$ is a subset of every set, in particular of $P(A)$, and the elements of $P(P(A))$ are exactly the subsets of $P(A)$.

Addendum D — Unbounded (infinite) intervals

Extends Subtopic 02, Section 2. Subtopic 03's Practice Exercise 3 uses this notation, so master it before you attempt that exercise.

Definitions. For a real number a :

$$(a, \infty) = \{x \in \mathbb{R} : x > a\}, \quad [a, \infty) = \{x \in \mathbb{R} : x \geq a\}$$

$$(-\infty, a) = \{x \in \mathbb{R} : x < a\}, \quad (-\infty, a] = \{x \in \mathbb{R} : x \leq a\}, \quad (-\infty, \infty) = \mathbb{R}$$

- a : the single finite endpoint, a genuine real number.
- ∞ and $-\infty$ are **not** real numbers. They are direction markers meaning “with no bound on that side.”

Four rules that carry all the marks:

1. **The bracket beside ∞ or $-\infty$ is always round.** A square bracket is a promise that the endpoint belongs to the set, and ∞ can never belong to a set of real numbers. Writing $[3, \infty]$ is wrong every time.
2. **Every unbounded interval is an infinite set**, and the formula “length = $b - a$ ” of Subtopic 02 does not apply.
3. **Complements flip the bracket as they cross the endpoint.** With $U = \mathbb{R}$: $((-\infty, a))' = [a, \infty)$ and $([a, \infty))' = (-\infty, a)$. The endpoint a must land on exactly one side.
4. **Operations obey the same rule as bounded intervals:** for $\cap$ take the tighter bounds and the stricter bracket; for $\cup$ of overlapping intervals take the looser.

Worked Example D1 — [CBSE Board / JEE Main]

Over $U = \mathbb{R}$, find $(-\infty, 3] \cap (0, \infty)$, $(-\infty, 3] \cup (0, \infty)$, and $(-\infty, 3] - (0, \infty)$.

- **Intersection.** Membership requires both $x \leq 3$ and $x > 0$. Combining, $0 < x \leq 3$, which is $(0, 3]$. The left bracket is round because $(0, \infty)$ excludes 0; the right is square because $(-\infty, 3]$ includes 3.
- **Union.** Take any real x . If $x \leq 3$ it lies in the first set; if $x > 3$ then certainly $x > 0$, so it lies in the second. No real is left out, so the union is $\mathbb{R}$.

- **Difference.** Membership requires $x \leq 3$ and $x \not\geq 0$, that is $x \leq 0$. The binding condition is $x \leq 0$, giving $(-\infty, 0]$.

Worked Example D2 — [JEE Main]

Express $A = \{x \in \mathbb{R} : x^2 \geq 9\}$ in interval form, and find A' with $U = \mathbb{R}$.

$x^2 \geq 9 \iff |x| \geq 3 \iff x \leq -3 \text{ or } x \geq 3$. The word “or” is a union, and both endpoints satisfy the non-strict inequality, so both brackets are square:

$$A = (-\infty, -3] \cup [3, \infty).$$

For the complement, negate: $|x| < 3 \iff -3 < x < 3$, so $A' = (-3, 3)$. Check against De Morgan: $A' = ((-\infty, -3])' \cap ([3, \infty))' = (-3, \infty) \cap (-\infty, 3) = (-3, 3)$. Consistent.

Practice.

1. $[-2, \infty) \cap (-\infty, 5)$
2. $(2, \infty) - [5, \infty)$
3. The complement of $(-\infty, 1)$ in $\mathbb{R}$
4. $\{x \in \mathbb{R} : 3x - 7 > 2\}$ in interval form
5. $(-\infty, 0) \cup [0, \infty)$ and $(-\infty, 0) \cap [0, \infty)$
6. Is $(-\infty, 4]$ a finite set?

Answers. 1. $[-2, 5)$. 2. $(2, 5)$. 3. $[1, \infty)$. 4. $3x > 9 \Rightarrow x > 3$, so $(3, \infty)$. 5. $\mathbb{R}$ and $\emptyset$. 6. No — it is infinite; unbounded intervals are never finite.

Addendum E — Counting ordered pairs of subsets

Extends Subtopic 02, Sections 3–4. This is a standing JEE Advanced item type.

Method — the three-way coin flip. Subtopic 02 derived 2^n by giving each element an independent two-way decision: in or out. To count **pairs** of subsets, give each element an independent decision about its *joint status* in (A, B) . There are four conceivable statuses — in both, in A only, in B only, in neither — and every standard constraint bans exactly one of them, leaving three. Three independent choices per element over n elements gives 3^n .

Let $n(S) = n$ and let (A, B) be an **ordered** pair of subsets of S .

Constraint	Status banned	Allowed statuses	Count
$A \subseteq B \subseteq S$	in A only	both, B only, neither	3^n
$A \cap B = \emptyset$	in both	A only, B only, neither	3^n
$A \cup B = S$	in neither	both, A only, B only	3^n

Constraint	Status banned	Allowed statuses	Count
$A \cap B = \emptyset$ and $A \cup B = S$	both, and neither	A only, B only	2^n

Two extensions:

- **Chains.** For $A_1 \subseteq A_2 \subseteq \dots \subseteq A_k \subseteq S$, each element chooses the first index at which it enters, or never enters at all — that is $k + 1$ options, giving $(k + 1)^n$. Setting $k = 2$ recovers 3^n .
- **Strictness by subtraction.** The number of pairs with $A \subsetneq B$ is $3^n - 2^n$: from the 3^n pairs with $A \subseteq B$, remove the 2^n pairs in which $A = B$ (one for each subset).

Cross-check on the 3^n claim. Count $A \subseteq B \subseteq S$ a second way: choose B of size k in $\binom{n}{k}$ ways, then choose any $A \subseteq B$ in 2^k ways. Summing, $\sum_{k=0}^n \binom{n}{k} 2^k = (1 + 2)^n = 3^n$ by the binomial theorem. Two independent routes, one answer.

Worked Example E1 — [JEE Main]

Let $S = \{1, 2, 3, 4\}$. Find the number of ordered pairs (A, B) with $A \subseteq B \subseteq S$.

Each of the 4 elements independently takes one of three statuses — in A (and therefore in B), in B but not A , or in neither — so the count is $3^4 = 81$.

Verify by the size-of- B route: $\binom{4}{0}2^0 + \binom{4}{1}2^1 + \binom{4}{2}2^2 + \binom{4}{3}2^3 + \binom{4}{4}2^4 = 1 + 8 + 24 + 32 + 16 = 81$. Confirmed.

Worked Example E2 — [JEE Advanced]

Let $n(S) = 5$. Find the number of ordered pairs (A, B) of subsets of S with $A \cap B = \emptyset$ and $A \cup B \neq S$.

Count the easy superset and subtract, exactly as in Subtopic 02's Example 4.

- Pairs with $A \cap B = \emptyset$: each element is in A only, B only, or neither, so $3^5 = 243$.
- Among those, the ones with $A \cup B = S$ leave no element in “neither”, so each element is in exactly one of A, B : $2^5 = 32$.

The required count is $243 - 32 = 211$.

Practice. Throughout, S has n elements and pairs are ordered unless stated otherwise.

1. $n = 6$: the number of pairs (A, B) with $A \subseteq B \subseteq S$.
2. $n = 3$: the number of ordered triples (A, B, C) with $A \subseteq B \subseteq C \subseteq S$.
3. $n = 4$: the number of pairs (A, B) with $A \cup B = S$.
4. General n : the number of pairs with $A \subsetneq B$. Evaluate it for $n = 3$.
5. General n : the number of pairs with $A \cap B = \emptyset$ and both A and B non-empty. Evaluate it for $n = 3$.
6. $n = 2$: the number of **unordered** pairs $\{A, B\}$ with $A \cup B = S$.

Answers. 1. $3^6 = 729$. 2. $4^3 = 64$. 3. $3^4 = 81$. 4. $3^n - 2^n$; for $n = 3$ this is $27 - 8 = 19$. 5. $3^n - 2 \cdot 2^n + 1$ (remove the 2^n pairs with $A = \emptyset$ and the 2^n with $B = \emptyset$, then restore the one pair counted twice); for $n = 3$ this is $27 - 16 + 1 = 12$. 6. 5; in general $\frac{1}{2}(3^n + 1)$, since only the single ordered pair (S, S) has $A = B$.

Addendum F — Three-circle Venn shading and visual proof

Extends Subtopic 03, Sections 3–4, which develop shading for two sets only.

Method. Three overlapping circles inside a rectangle U carve the universe into exactly eight regions. Number them once and never renumber:

Region	Description	In A ?	In B ?	In C ?
R_1	only A	yes	no	no
R_2	only B	no	yes	no
R_3	only C	no	no	yes
R_4	A and B but not C	yes	yes	no
R_5	B and C but not A	no	yes	yes
R_6	A and C but not B	yes	no	yes
R_7	all three	yes	yes	yes
R_8	outside all three	no	no	no

Reading the columns downwards gives the three basic region lists, which you should be able to write from memory:

$$A = \{R_1, R_4, R_6, R_7\}, \quad B = \{R_2, R_4, R_5, R_7\}, \quad C = \{R_3, R_5, R_6, R_7\}$$

Procedure. (1) Convert every set in the expression to its region list. (2) Perform the operations on lists: union merges, intersection keeps the common entries, difference deletes, complement returns the eight regions you did not list. (3) Shade the final list.

Why this is a proof, not just a picture. Every element of U lies in exactly one of the eight regions, and membership of a compound expression depends only on which of A, B, C contain the element — that is, only on the region. So two expressions are equal as sets **if and only if** their region lists are identical. Comparing two lists of at most eight entries settles any three-set identity completely.

Figure brief: Three mutually overlapping circles labelled A, B, C inside a rectangle labelled U , with the eight regions marked R_1 (only A), R_2 (only B), R_3 (only C), R_4 ($A \cap B$ outside C), R_5 ($B \cap C$ outside A), R_6 ($A \cap C$ outside B), R_7 (the central $A \cap B \cap C$), R_8 (the corner outside all three).

Worked Example F1 — [CBSE Board / JEE Main]

Verify the distributive law $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ by shading.

Left side. $B \cup C = \{R_2, R_4, R_5, R_7\} \cup \{R_3, R_5, R_6, R_7\} = \{R_2, R_3, R_4, R_5, R_6, R_7\}$. Intersecting with $A = \{R_1, R_4, R_6, R_7\}$ keeps only the common entries:

$$A \cap (B \cup C) = \{R_4, R_6, R_7\}.$$

Right side. $A \cap B = \{R_4, R_7\}$ and $A \cap C = \{R_6, R_7\}$; their union is $\{R_4, R_6, R_7\}$.

The lists are identical, so the two sets are equal. In pictures: both sides shade the two petals where A meets one other circle, plus the centre.

Figure brief: Two side-by-side three-circle Venn diagrams; in the left one shade R_4, R_6, R_7 and caption it $A \cap (B \cup C)$; in the right one shade the same three regions and caption it $(A \cap B) \cup (A \cap C)$.

Worked Example F2 — [JEE Main]

Shade $A - (B \cup C)$ and show it equals $(A - B) \cap (A - C)$.

$A - (B \cup C)$: start from $A = \{R_1, R_4, R_6, R_7\}$ and delete everything in $B \cup C = \{R_2, R_3, R_4, R_5, R_6, R_7\}$. Only R_1 survives.

$(A - B) \cap (A - C)$: $A - B = \{R_1, R_6\}$ (drop the regions inside B), $A - C = \{R_1, R_4\}$; the common entry is R_1 .

Both are $\{R_1\}$ — the crescent of A untouched by either other circle. This is Subtopic 03's Practice Exercise 4, settled by picture instead of element-chasing, and the two methods agree.

Practice. Give the region list for each expression.

1. $B - (A \cup C)$
2. $(A \cap B) - C$
3. $A' \cap B \cap C$
4. $(A \cup B \cup C)'$
5. $A \triangle B$
6. $A \cup (B \cap C)$ and $(A \cup B) \cap (A \cup C)$ — are they equal?
7. $(A \cap B \cap C)'$

Answers. 1. $\{R_2\}$. 2. $\{R_4\}$. 3. $\{R_5\}$. 4. $\{R_8\}$. 5. $\{R_1, R_2, R_5, R_6\}$ — the regions lying in exactly one of A and B , whatever C does. 6. Both give $\{R_1, R_4, R_5, R_6, R_7\}$, so yes, equal — this is the second distributive law. 7. $\{R_1, R_2, R_3, R_4, R_5, R_6, R_8\}$, that is, everything except the centre.

Addendum G — Operations on the standard number systems

Extends Subtopic 03, Sections 4–5, where every difference and complement is computed on roster sets or bounded intervals.

Method. Subtopic 01 gives you the nesting $\mathbb{N} \subset \mathbb{W} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$ and defines $\mathbb{T}$ as the set of irrationals. Every question of this type is settled by one rule plus a short table.

The rule. Whenever $A \subseteq B$:

$$A \cap B = A, \quad A \cup B = B, \quad A - B = \emptyset, \quad B - A = \text{the extra elements of } B.$$

So for nested number systems, the only interesting computation is $B - A$.

The table (take $U = \mathbb{R}$ for all complements):

Expression	Value	Reason
$\mathbb{R} - \mathbb{Q}$	$\mathbb{T}$	definition of the irrationals
$\mathbb{R} - \mathbb{T}$	$\mathbb{Q}$	the two halves of $\mathbb{R}$ swap roles
$\mathbb{Q} \cap \mathbb{T}$	$\emptyset$	no real is both rational and irrational
$\mathbb{Q} \cup \mathbb{T}$	$\mathbb{R}$	every real is one or the other
$\mathbb{Q}'$	$\mathbb{T}$, and $\mathbb{T}' = \mathbb{Q}$	previous two rows combined
$\mathbb{W} - \mathbb{N}$	$\{0\}$	$\mathbb{W}$ adds exactly one element to $\mathbb{N}$
$\mathbb{Z} - \mathbb{W}$	$\{-1, -2, -3, \dots\}$	the negative integers
$\mathbb{Z} - \mathbb{N}$	$\{0, -1, -2, \dots\}$	zero together with the negatives
$\mathbb{Q} - \mathbb{Z}$	non-integer rationals, e.g. $\frac{3}{2}, -\frac{7}{4}$	fractions that do not reduce to whole numbers
$\mathbb{N} \cap \mathbb{T}$	$\emptyset$	naturals are rational

A warning that examiners exploit. $\mathbb{T}$ is not closed under addition or multiplication: $\sqrt{2} \cdot \sqrt{2} = 2 \in \mathbb{Q}$ and $(1 + \sqrt{2}) + (1 - \sqrt{2}) = 2 \in \mathbb{Q}$. So never argue “irrational plus irrational is irrational.” Test each number on its own merits — and remember $\sqrt{9} = 3$ is rational, despite the radical sign.

Worked Example G1 — [CBSE Board]

With $U = \mathbb{R}$, find (i) $\mathbb{Q}'$, (ii) $\mathbb{W} - \mathbb{N}$, (iii) $\mathbb{Z} - \mathbb{N}$, (iv) $\mathbb{N} \cap \mathbb{T}$, (v) $\mathbb{N} \cup \mathbb{Z}$.

- (i) $\mathbb{Q}' = \mathbb{R} - \mathbb{Q} = \mathbb{T}$, the irrationals.
- (ii) $\mathbb{W}$ is $\mathbb{N}$ with 0 adjoined, so $\mathbb{W} - \mathbb{N} = \{0\}$ — a singleton, not $\emptyset$.
- (iii) $\mathbb{Z}$ adds 0 and all the negatives, so $\mathbb{Z} - \mathbb{N} = \{0, -1, -2, -3, \dots\}$.
- (iv) Every natural number k equals $k/1$, hence is rational, so it can never be irrational: $\mathbb{N} \cap \mathbb{T} = \emptyset$.
- (v) Since $\mathbb{N} \subset \mathbb{Z}$, the union is the larger set: $\mathbb{Z}$.

Worked Example G2 — [JEE Main]

With $U = \mathbb{R}$, find $(\mathbb{Z} \cup \mathbb{T})'$.

Apply De Morgan, then the table:

$$(\mathbb{Z} \cup \mathbb{T})' = \mathbb{Z}' \cap \mathbb{T}' = \mathbb{Z}' \cap \mathbb{Q} = \mathbb{Q} - \mathbb{Z}.$$

So the answer is the set of rationals that are not integers: numbers such as $\frac{1}{2}, -\frac{7}{4}, \frac{22}{7}$. Sanity check by direct reading: a real number outside $\mathbb{Z} \cup \mathbb{T}$ is not irrational (so it is rational) and not an integer — the same description.

Practice. Take $U = \mathbb{R}$ where a complement appears.

1. $\mathbb{Q} \cap \mathbb{T}$
2. $\mathbb{R} - \mathbb{T}$
3. $\mathbb{Z} \cap \mathbb{W}$
4. $(\mathbb{Q} \cap \mathbb{T})'$
5. Is $\mathbb{Q} \cap (0, 1)$ finite or infinite?
6. Is $\sqrt{9} \in \mathbb{T}$?
7. $\mathbb{W} \cup \mathbb{T}$ — can it be simplified to a single named system?

Answers. 1. $\emptyset$. 2. $\mathbb{Q}$. 3. $\mathbb{W}$ (since $\mathbb{W} \subset \mathbb{Z}$). 4. $\emptyset' = \mathbb{R}$. 5. Infinite — for instance $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots$ all lie in it. 6. No; $\sqrt{9} = 3$, a natural number, hence rational. 7. No — it contains all irrationals and the whole numbers but no negative integers and no non-integer rationals, so it is not one of the named systems.

Addendum H — Bounds on an unknown overlap

Extends Subtopic 04, Sections 2-5. Subtopic 04 solves inclusion-exclusion when every count is given; this item handles the far more common JEE phrasing “find the least (or greatest) possible number who did both.”

Method. Let $n(U) = N$, $n(A) = a$, $n(B) = b$, and let the unknown overlap be $x = n(A \cap B)$. Then

$$\max(0, a + b - N) \leq n(A \cap B) \leq \min(a, b).$$

- N : the size of the universe (the total surveyed). a, b : the two given counts. x : the overlap you are asked to bound.
- **Validity:** $a \leq N$ and $b \leq N$ must hold, otherwise the data is already inconsistent.

Derivation of the upper bound. $A \cap B \subseteq A$ and $A \cap B \subseteq B$, so $x \leq a$ and $x \leq b$, hence $x \leq \min(a, b)$.

Derivation of the lower bound. By the two-set formula, $n(A \cup B) = a + b - x$. The union is

a subset of U , so $a + b - x \leq N$, which rearranges to $x \geq a + b - N$. A count is never negative, so also $x \geq 0$. Take whichever bound is larger.

Both bounds are attained — they are genuine minimum and maximum values, not loose estimates. For the maximum, place the smaller set entirely inside the larger. For the minimum, push the two sets as far apart as the universe allows: their overlap is exactly the amount by which $a + b$ spills past N . Every integer between the two bounds is also achievable, by sliding one set across one element at a time.

Companion bounds for the union. Since $n(A \cup B) = a + b - x$, running x through its range gives

$$\max(a, b) \leq n(A \cup B) \leq \min(N, a + b).$$

Large overlap means small union, and conversely — the two ranges are mirror images.

Three sets and beyond. For three sets in a universe of size N :

$$\max(0, n(A) + n(B) + n(C) - 2N) \leq n(A \cap B \cap C) \leq \min(n(A), n(B), n(C))$$

and in general, for m sets,

$$n(A_1 \cap A_2 \cap \cdots \cap A_m) \geq \sum_{i=1}^m n(A_i) - (m-1)N.$$

Why the general bound holds. An element fails to lie in the intersection exactly when it lies in at least one complement A'_i . Hence

$$n\left(\bigcap_{i=1}^m A_i\right) = N - n\left(\bigcup_{i=1}^m A'_i\right) \geq N - \sum_{i=1}^m n(A'_i) = N - \sum_{i=1}^m (N - n(A_i)) = \sum_{i=1}^m n(A_i) - (m-1)N,$$

using only the fact that a union is never bigger than the sum of its pieces.

Procedure — least/greatest overlap under exam pressure.

1. Underline N , a , b . If the problem is in percentages, set $N = 100$ and read the percentages as counts.
2. Greatest overlap = $\min(a, b)$. Write it immediately; it needs no arithmetic.
3. Least overlap = $a + b - N$ if that is positive, otherwise 0. *Why:* $a + b - N$ measures the unavoidable spill-over past the universe.
4. If the question asks about the union, “none”, or “only A ”, convert using $n(A \cup B) = a + b - x$ and $n(\text{neither}) = N - n(A \cup B)$, then substitute the extreme value of x — remembering that the largest x gives the smallest union.

Worked Example H1 — [JEE Main]

In a class of 100 students, 70 like tea and 80 like coffee. Find the minimum and the maximum possible number of students who like both.

Here $N = 100$, $a = 70$, $b = 80$.

- **Maximum.** $x \leq \min(70, 80) = 70$. Attained when the tea-likers all lie inside the coffee-likers, which is possible since $70 \leq 80$.
- **Minimum.** $x \geq 70 + 80 - 100 = 50$, and $50 > 0$ so this is the binding bound. Attained when the union fills the class: $n(A \cup B) = 70 + 80 - 50 = 100 = N$. Concretely, 50 students take both, 20 take tea only, 30 take coffee only, and nobody is left out.

$$50 \leq n(\text{both}) \leq 70.$$

Notice the reasoning behind the minimum: $70 + 80 = 150$ seats are claimed but only 100 students exist, so at least $150 - 100 = 50$ students must be sitting in two seats at once.

Worked Example H2 — [JEE Advanced]

In a survey of 100 people, 60 like dessert A , 70 like B , and 80 like C . If x is the number who like all three, find the range of x .

- **Maximum.** $A \cap B \cap C \subseteq A$, so $x \leq \min(60, 70, 80) = 60$. Attained by nesting $A \subseteq B \subseteq C$, which the sizes permit.
- **Minimum.** Use the complement argument. $n(A') = 40$, $n(B') = 30$, $n(C') = 20$, and a person outside the triple intersection must lie in at least one complement. So

$$n((A \cap B \cap C)') = n(A' \cup B' \cup C') \leq 40 + 30 + 20 = 90,$$

giving $x \geq 100 - 90 = 10$. This matches the formula $60 + 70 + 80 - 2(100) = 10$. It is attained because $90 \leq 100$, so the three complements can be made pairwise disjoint: 40 people dislike only A , 30 only B , 20 only C , and the remaining 10 like all three.

$$10 \leq x \leq 60.$$

The lesson: for a *minimum intersection* switch to complements, where the union bound “a union is at most the sum of its parts” does the whole job.

Practice.

1. $n(U) = 50$, $n(A) = 30$, $n(B) = 25$. Find the range of $n(A \cap B)$.
2. $n(A) = 17$, $n(B) = 23$, with no restriction on the universe. Find the range of $n(A \cap B)$.
3. $n(U) = 100$, $n(A) = 40$, $n(B) = 50$. Find the range of $n(A \cup B)$.
4. In a town, 63% of people drink tea and 76% drink coffee. If $x\%$ drink both, find the range of x .
5. $n(U) = 200$ with $n(A) = 120$, $n(B) = 140$, $n(C) = 160$. Find the range of $n(A \cap B \cap C)$.
6. In a class of 40, 25 play cricket and 20 play football. Find (i) the least possible number who play both, and (ii) the greatest possible number who play neither.

Answers. 1. $\max(0, 30 + 25 - 50) = 5$ to $\min(30, 25) = 25$, so $5 \leq n(A \cap B) \leq 25$. 2. 0 to 17 — with an unrestricted universe the lower bound $a + b - N$ is negative, so only $x \geq 0$ binds. 3. $\max(40, 50) = 50$ to $\min(100, 90) = 90$, so $50 \leq n(A \cup B) \leq 90$. 4. Take $N = 100$: $63 + 76 - 100 = 39$ and $\min(63, 76) = 63$, so $39 \leq x \leq 63$. 5. $120 + 140 + 160 - 2(200) = 20$ and $\min(120, 140, 160) = 120$, so $20 \leq n(A \cap B \cap C) \leq 120$. 6. (i) $25 + 20 - 40 = 5$. (ii) “Neither” is largest when the union is smallest; the smallest union is $\max(25, 20) = 25$, so at most $40 - 25 = 15$ play neither.

Part 2 — New Subtopic

Subtopic 05 — Indexed Families of Sets, Generalised Laws, and Double Counting

Exam Relevance: CBSE Boards ask 1–2 mark questions on indexed families lifted straight from the NCERT exercises — usually A_n as the multiples of n , or a run of intervals $[n, n + 1]$. JEE Main tests the same notation on families of shrinking intervals, where the whole mark rides on whether an endpoint is open or closed, and on the generalised De Morgan laws inside logic and probability questions. JEE Advanced draws on this material twice over: the alternating inclusion–exclusion formula for four or more sets, and the double-counting identity for m sets of uniform size with a uniform multiplicity — a question type that has recurred for decades and collapses into two lines once you recognise it.

Section 1: Concept Intuition

Take the 8:14 Churchgate fast from Andheri. For each day of the month, let A_1 be the set of passengers who boarded it on day 1, A_2 those who boarded on day 2, and so on to A_{30} . Nobody hands you a single set here; you are handed a whole **run** of sets, numbered by day.

Two questions about that run come up constantly, and they are the only two you ever need.

- **Who rode at least once this month?** A passenger qualifies if they appear in A_1 , or in A_2 , or in any single one of the thirty. That is the **union of the family**, written $\bigcup_{i=1}^{30} A_i$.
- **Who rode every single day?** A passenger qualifies only by appearing in A_1 and in A_2 and in all thirty without exception — the season-ticket regulars. That is the **intersection of the family**, written $\bigcap_{i=1}^{30} A_i$.

A big union asks “in at least one”; a big intersection asks “in every one.” Everything in this subtopic is bookkeeping around those two sentences.

Think about it this way... the symbols $\bigcup$ and $\bigcap$ are not new operations. They are $\cup$ and $\cap$ applied along a list, exactly as $\sum$ is $+$ applied along a list. And just as with $\sum$, the list is described by an **index**: the letter i or n that runs through an **index set** I . Write $I = \{1, 2, \dots, m\}$ for a finite run, or $I = \mathbb{N}$ for an endless one.

Now the part that separates a JEE candidate from a Board candidate: **the index set is**

allowed to be infinite, and infinite families behave in ways finite ones never do. Consider the shrinking intervals

$$A_1 = (0, 1), \quad A_2 = (0, \tfrac{1}{2}), \quad A_3 = (0, \tfrac{1}{3}), \quad \dots, \quad A_n = (0, \tfrac{1}{n}).$$

Every one of these is non-empty. Any *finite* batch of them has a non-empty intersection — the smallest interval in the batch. Yet the intersection of the whole infinite family is **empty**. Here is why: pick any candidate x . If $x \leq 0$ it is in none of them. If $x > 0$, then $\frac{1}{n}$ eventually drops below x , so from that index onwards x lies outside A_n . No real number survives every index, so $\bigcap_{n=1}^{\infty} A_n = \emptyset$.

The intervals are closing in on 0, and the temptation is overwhelming to answer $\{0\}$. But 0 was never *inside* a single one of them — every A_n excludes its left endpoint. Change the brackets to $[0, \frac{1}{n}]$ and 0 becomes a member of every set, so the intersection genuinely becomes $\{0\}$. **In an infinite family, brackets decide everything.**

One more family worth carrying in your head. For $n \in \mathbb{N}$ let A_n be the set of natural multiples of n : $A_2 = \{2, 4, 6, \dots\}$, $A_3 = \{3, 6, 9, \dots\}$. Then $A_2 \cap A_3$ is the set of common multiples of 2 and 3, which is the set of multiples of 6 — that is, A_6 . Intersection of multiple-sets is governed by the **lcm**, not the product, and it happens to coincide with the product here only because 2 and 3 are coprime.

Edge cases and where students slip (the “fine print”):

- An infinite intersection is not a limit. The sets may shrink towards a point without that point ever belonging to them.
- $\bigcap_{i \in I} A_i \subseteq A_j \subseteq \bigcup_{i \in I} A_i$ for every single index j — the intersection is the tightest set in sight and the union the loosest. Any answer violating this is wrong before you check anything else.
- If the index set I is empty, the notation is not used at this level. Always check that at least one index exists.
- $A_m \cap A_n = A_{\text{lcm}(m,n)}$ for multiple-sets, not A_{mn} .
- Being in the union does **not** mean being in most of the sets; one index is enough.

Section 2: Key Formulas and Definitions

Term	Meaning	Notation
Indexed family	a collection of sets, one for each index	$\{A_i\}_{i \in I}$
Index set	the labels the index runs through	I (often $\{1, \dots, m\}$ or $\mathbb{N}$)
Family union	elements lying in at least one member	$\bigcup_{i \in I} A_i$
Family intersection	elements lying in every member	$\bigcap_{i \in I} A_i$

Term	Meaning	Notation
Increasing family	each set contains the previous one	$A_1 \subseteq A_2 \subseteq A_3 \subseteq \dots$
Decreasing family	each set is contained in the previous one	$A_1 \supseteq A_2 \supseteq A_3 \supseteq \dots$
Pairwise disjoint family	no two members share an element	$A_i \cap A_j = \emptyset$ whenever $i \neq j$
Multiplicity of x	how many members contain x	$d(x)$

Definitions.

$$\bigcup_{i \in I} A_i = \{x : x \in A_i \text{ for at least one } i \in I\}, \quad \bigcap_{i \in I} A_i = \{x : x \in A_i \text{ for every } i \in I\}$$

- x : an arbitrary candidate element, drawn from the universal set U .
- i : the running index; I : the index set, which may be finite or infinite.

Generalised laws (valid for any index set I , finite or infinite, with all sets inside a universe U):

$$\text{De Morgan: } \left(\bigcup_{i \in I} A_i \right)' = \bigcap_{i \in I} A_i', \quad \left(\bigcap_{i \in I} A_i \right)' = \bigcup_{i \in I} A_i'$$

$$\text{Distributive: } B \cap \left(\bigcup_{i \in I} A_i \right) = \bigcup_{i \in I} (B \cap A_i), \quad B \cup \left(\bigcap_{i \in I} A_i \right) = \bigcap_{i \in I} (B \cup A_i)$$

Monotone families (these turn a hard-looking question into a one-line answer):

Family type	$\bigcup_{n \geq 1} A_n$	$\bigcap_{n \geq 1} A_n$
Increasing $A_1 \subseteq A_2 \subseteq \dots$	the set the family grows towards	A_1 (the smallest member)
Decreasing $A_1 \supseteq A_2 \supseteq \dots$	A_1 (the largest member)	the set the family shrinks towards

For an increasing family the intersection is genuinely *attained* at A_1 , and for a decreasing family the union is attained at A_1 . The other entry in each row need not be a member of the family at all — that is exactly the $(0, 1/n)$ phenomenon.

Standard families to recognise on sight (all with $n \in \mathbb{N}$):

Family	Union	Intersection
$(0, \frac{1}{n})$	$(0, 1)$	$\emptyset$
$[0, \frac{1}{n}]$	$[0, 1]$	$\{0\}$

Family	Union	Intersection
$[\frac{1}{n}, 1]$	$(0, 1]$	$\{1\}$
$(-n, n)$	$\mathbb{R}$	$(-1, 1)$
$[n, n+1]$	$[1, \infty)$	$\emptyset$
$A_n = \{x \in \mathbb{N} : n \text{ divides } x\}$	$\mathbb{N}$	$\emptyset$

For the multiple-sets in the last row: $A_m \cap A_n = A_{\text{lcm}(m,n)}$, and $A_m \subseteq A_n$ exactly when n divides m .

Generalised inclusion-exclusion for m finite sets — the formula Subtopic 04 promised when it said the pattern extends to any number of sets:

$$n\left(\bigcup_{i=1}^m A_i\right) = \sum_i n(A_i) - \sum_{i < j} n(A_i \cap A_j) + \sum_{i < j < k} n(A_i \cap A_j \cap A_k) - \cdots + (-1)^{m-1} n(A_1 \cap \cdots \cap A_m)$$

- The r -th block sums over all $\binom{m}{r}$ intersections of exactly r of the sets, and carries the sign $(-1)^{r-1}$: add the singles, subtract the pairs, add the triples, and so on.

Written out for $m = 4$:

$$n(A \cup B \cup C \cup D) = \sum n(A) - \sum n(A \cap B) + \sum n(A \cap B \cap C) - n(A \cap B \cap C \cap D)$$

where the three sums run over the 4 single sets, the 6 pairs, and the 4 triples respectively.

Generalised De Morgan for counts (the “at least one” complement): $n((A_1 \cup \cdots \cup A_m)') = n(U) - n(A_1 \cup \cdots \cup A_m)$.

Counting bounds for a family:

$$n\left(\bigcup_{i=1}^m A_i\right) \leq \sum_{i=1}^m n(A_i), \quad \text{with equality precisely when the } A_i \text{ are pairwise disjoint}$$

$$n\left(\bigcap_{i=1}^m A_i\right) \geq \sum_{i=1}^m n(A_i) - (m-1)n(U) \quad (\text{derived in Addendum H})$$

The double-counting (multiplicity) identity — the single most productive formula in this subtopic:

$$\sum_{i=1}^m n(A_i) = \sum_{x \in S} d(x), \quad \text{where } S = \bigcup_{i=1}^m A_i \text{ and } d(x) = \#\{i : x \in A_i\}$$

In the uniform case — every A_i of the same size p , every element of S lying in exactly k of them — this becomes

$$mp = kn(S) \quad \implies \quad n(S) = \frac{mp}{k}$$

- m : how many sets. p : the common size of each set. k : the common multiplicity. $n(S)$: the size of the union.
- Validity: every A_i must have exactly p elements, every element of S must lie in exactly k of the sets, and S must be their union.

Section 3: CBSE Board Derivations and Key Procedures

Derivation: Generalised De Morgan, $(\bigcup_{i \in I} A_i)' = \bigcap_{i \in I} A_i'$

Step 1. Take an arbitrary $x \in U$ and unpack the left side by definition of complement: $x \in (\bigcup_i A_i)'$ means $x \notin \bigcup_i A_i$. *Reason:* the complement of a set is exactly the elements of U outside it.

Step 2. Unpack the union: $x \in \bigcup_i A_i$ means “there exists an index i with $x \in A_i$.” Negating, $x \notin \bigcup_i A_i$ means “there is **no** index i with $x \in A_i$.” *Reason:* this is the definition of the family union, read backwards.

Step 3. “There is no index with $x \in A_i$ ” says the same thing as “for **every** index, $x \notin A_i$.” *Reason:* this is the quantifier flip — the negation of “at least one” is “none at all”, that is, “every one fails”. It is the two-set fact “not (P or Q) means (not P) and (not Q)” carried along a whole list.

Step 4. “For every i , $x \notin A_i$ ” says “for every i , $x \in A_i'$ ”, which is precisely $x \in \bigcap_i A_i'$. *Reason:* definition of the family intersection.

Every step is an “if and only if”, so the two sets contain exactly the same elements and are equal. The second law follows by the same argument with the roles of “at least one” and “every” exchanged.

Why this is the gold-standard method: shading cannot help you here — you cannot draw infinitely many circles. Element-chasing with quantifiers is the only proof that survives an infinite index set.

Derivation: the double-counting identity $\sum_i n(A_i) = \sum_x d(x)$

Step 1. Build the list of all **incidences**: pairs (x, i) for which $x \in A_i$. *Reason:* one object, to be counted in two different orders.

Step 2. Count the list set by set. Set A_i contributes one incidence for each of its elements, so it contributes $n(A_i)$. Totalling over the m sets gives $\sum_{i=1}^m n(A_i)$.

Step 3. Count the same list element by element. Element x contributes one incidence for each set containing it, so it contributes $d(x)$. Totalling over S gives $\sum_{x \in S} d(x)$.

Step 4. The two totals count the identical list, so they are equal.

Step 5. Impose uniformity: if $n(A_i) = p$ for all i , Step 2 gives mp ; if $d(x) = k$ for all $x \in S$, Step 3 gives $kn(S)$. Hence $mp = kn(S)$.

Figure brief: A grid with the m sets $A_1, \dots, A_m$ as rows and the elements of S as columns, a tick placed wherever $x \in A_i$; annotate that each row total is $n(A_i)$, each column total is $d(x)$, and both grand totals count the same ticks.

Derivation: why the alternating signs of inclusion-exclusion give the right count

Fix an element x lying in exactly d of the m sets, where $d \geq 1$ (so x belongs to the union and must be counted once). In the formula, x is counted inside an r -fold intersection precisely when all r of the chosen sets contain it — and there are $\binom{d}{r}$ such choices. So the total contribution of x is

$$\sum_{r=1}^d (-1)^{r-1} \binom{d}{r} = - \sum_{r=1}^d (-1)^r \binom{d}{r} = \binom{d}{0} - \sum_{r=0}^d (-1)^r \binom{d}{r} = 1 - (1-1)^d = 1 - 0 = 1.$$

An element outside the union lies in no A_i and contributes 0. So every element of the union is counted exactly once and nothing else is counted at all — which is the formula. The three-set case of Subtopic 04 is $d \leq 3$ of this argument.

Procedure A: evaluate $\cup$ or $\cap$ over a family of intervals

1. Write out A_1 , A_2 and A_3 explicitly. *Why:* you cannot see monotonicity until you see three terms.
2. Decide whether the family is increasing, decreasing, or neither. *Why:* a monotone family is answered by the table in Section 2 with no further work.
3. For the **intersection**, take a general candidate x and ask “for which x does the membership condition hold at every index?” Push n towards infinity in the endpoint expression, and watch the bracket: a shrinking **open** endpoint excludes its limit, a **closed** one includes it.
4. For the **union**, ask “for which x does the condition hold at some index?” One successful index is enough.
5. Test the two boundary candidates explicitly by substitution. *Why:* the entire mark is usually a single endpoint.

Procedure B: reduce a family to one member

1. Check whether $A_{n+1} \subseteq A_n$ for all n (decreasing) or $A_n \subseteq A_{n+1}$ (increasing). *Why:* one containment check settles infinitely many.
2. Decreasing: the union collapses to A_1 . Increasing: the intersection collapses to A_1 . *Why:* every other member is swallowed by A_1 .
3. For the remaining half, apply Procedure A. *Why:* that is where the genuine work always is.

Procedure C: solve a uniform-multiplicity problem

1. Name the three uniform numbers: m (how many sets), p (size of each set), k (how many sets each element lies in).
2. Compute the incidence total mp . *Why:* this is the number of ticks in the grid, counted by rows.
3. Divide by the multiplicity: $n(S) = mp/k$. *Why:* counted by columns, each element accounts for exactly k ticks.

4. If a second family shares the same union S , repeat with its own m', p', k' and solve for the unknown, using the $n(S)$ you just found. *Why*: the union is the bridge between the two families.
5. Sanity check: $n(S)$ must be a positive integer and must satisfy $n(S) \geq p$ (each set sits inside S).

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

For each $n \in \mathbb{N}$, let $A_n = \{x \in \mathbb{N} : x \text{ is a multiple of } n\}$. Find (i) $A_3 \cap A_4$, (ii) $A_3 \cup A_6$, (iii) $\bigcup_{n=2}^{\infty} A_n$, (iv) $\bigcap_{n=1}^{\infty} A_n$.

(i) A natural number lies in $A_3 \cap A_4$ exactly when it is a multiple of both 3 and 4, that is, a common multiple. The common multiples of two numbers are precisely the multiples of their lcm, and $\text{lcm}(3, 4) = 12$. So

$$A_3 \cap A_4 = A_{12} = \{12, 24, 36, \dots\}.$$

(ii) Every multiple of 6 is a multiple of 3, so $A_6 \subseteq A_3$. A union with a subset changes nothing:

$$A_3 \cup A_6 = A_3 = \{3, 6, 9, 12, \dots\}.$$

(iii) Which naturals lie in at least one A_n with $n \geq 2$? Any $x \geq 2$ lies in A_x , since x is a multiple of itself and $x \geq 2$ is a legal index. The number 1 lies in no A_n with $n \geq 2$, because a multiple of $n \geq 2$ is at least 2. Hence

$$\bigcup_{n=2}^{\infty} A_n = \mathbb{N} - \{1\} = \{2, 3, 4, \dots\}.$$

(iv) An element of this intersection would be a natural number x that is a multiple of **every** natural number — in particular of $x + 1$. But a positive multiple of $x + 1$ is at least $x + 1$, forcing $x \geq x + 1$, which is impossible. So

$$\bigcap_{n=1}^{\infty} A_n = \emptyset.$$

Example 2 — [JEE Main Level]

For $n \in \mathbb{N}$ let $A_n = [2 - \frac{1}{n}, 5 + \frac{1}{n}]$. Find $\bigcup_{n=1}^{\infty} A_n$ and $\bigcap_{n=1}^{\infty} A_n$.

Write the first three members: $A_1 = [1, 6]$, $A_2 = [1.5, 5.5]$, $A_3 = [\frac{5}{3}, \frac{16}{3}]$. The left endpoint climbs towards 2 and the right endpoint falls towards 5, so each set sits inside the one before it: the family is **decreasing**.

Union. For a decreasing family the union is the first and largest member:

$$\bigcup_{n=1}^{\infty} A_n = A_1 = [1, 6].$$

Intersection. A real x survives every index exactly when $2 - \frac{1}{n} \leq x \leq 5 + \frac{1}{n}$ for all n .

- The left condition. If $x \geq 2$ it holds for every n , since $2 - \frac{1}{n} < 2 \leq x$. If $x < 2$, write $x = 2 - \varepsilon$ with $\varepsilon > 0$; then the condition demands $\frac{1}{n} \geq \varepsilon$ for every n , which fails as soon as $\frac{1}{n}$ drops below ε . So the left condition is exactly $x \geq 2$.
- The right condition, by the mirror-image argument, is exactly $x \leq 5$.

$$\bigcap_{n=1}^{\infty} A_n = [2, 5].$$

Note the endpoints 2 and 5 **do** belong, even though no single A_n has them as its own endpoints — each A_n contains them comfortably in its interior. Compare this with the next example, where the answer goes the other way.

Figure brief: A number line showing $A_1 = [1, 6]$, $A_2 = [1.5, 5.5]$ and A_3 as three nested bars with filled endpoints, and the common core $[2, 5]$ shaded beneath them.

Example 3 — [JEE Main — Speed Trap]

Find $\bigcap_{n=1}^{\infty} (0, \frac{1}{n})$ and $\bigcap_{n=1}^{\infty} [0, \frac{1}{n}]$.

The seductive wrong answer. The intervals shrink towards 0, so the eye reports “they collapse onto the point 0” and the hand writes $\{0\}$ for both. That answer is right for the second family and wrong for the first.

The trap has a name: the limit-point trap. A family shrinking *towards* a point does not have to *contain* it. The only legitimate question is membership: is 0 inside every A_n ?

First family. $(0, \frac{1}{n})$ is open at the left, so $0 \notin A_n$ for any n — not once. Take any other candidate: $x < 0$ fails immediately, and any $x > 0$ fails from the index where $\frac{1}{n} < x$ onwards. Nothing survives:

$$\bigcap_{n=1}^{\infty} (0, \frac{1}{n}) = \emptyset.$$

Second family. $[0, \frac{1}{n}]$ is closed at the left, so $0 \in A_n$ for every n ; and every $x > 0$ is eliminated exactly as before. Only 0 survives:

$$\bigcap_{n=1}^{\infty} [0, \frac{1}{n}] = \{0\}.$$

Speed move. Do not manipulate the family at all. Test the one suspicious candidate — the point the intervals are closing on — against the bracket next to it. Square bracket: it stays. Round bracket: the intersection is empty. That is a five-second answer.

Example 4 — [JEE Advanced Level]

Let $A_1, A_2, \dots, A_{30}$ be thirty sets, each containing exactly 5 elements, and let $B_1, B_2, \dots, B_n$ be n sets, each containing exactly 3 elements. Suppose

$$\bigcup_{i=1}^{30} A_i = \bigcup_{j=1}^n B_j = S,$$

and that each element of S belongs to exactly 10 of the A_i and to exactly 9 of the B_j . Find n . Neither family can be listed, and no Venn diagram will help. Count incidences instead — the double-counting identity is built for exactly this.

Step 1 — the A family, counted by sets. Thirty sets of five elements each produce

$$30 \times 5 = 150 \text{ incidences.}$$

Step 2 — the A family, counted by elements. Each element of S sits in exactly 10 of the A_i , contributing 10 incidences, so the same total is $10n(S)$.

Step 3 — solve for $n(S)$. Equating the two counts, $10n(S) = 150$, so

$$n(S) = 15.$$

Step 4 — the B family. Counted by sets, n sets of three elements give $3n$ incidences. Counted by elements, each of the 15 elements of S lies in exactly 9 of the B_j , giving $9 \times 15 = 135$. Equating,

$$3n = 135 \quad \implies \quad n = 45.$$

Consistency check. Each B_j has 3 elements and $3 \leq 15 = n(S)$, so the sets fit inside S ; and each A_i has $5 \leq 15$ elements. Both counts are positive integers. The answer stands:

$$n = 45.$$

The elegance is that the union S — a set nobody described to you — is the bridge. One family tells you its size; the other family then has no room to hide.

Section 5: Practice Exercises

1. **[CBSE]** With $A_n = \{x \in \mathbb{N} : n \text{ divides } x\}$, find $A_4 \cap A_6$ and $A_2 \cap A_3 \cap A_5$.
2. **[CBSE]** For $i \in \mathbb{N}$ let $A_i = [i, i+1]$. Find $\bigcup_{i=1}^{\infty} A_i$ and $A_3 \cap A_4$.
3. **[JEE Main]** Find $\bigcup_{n=1}^{\infty} (-n, n)$ and $\bigcap_{n=1}^{\infty} (-n, n)$.
4. **[JEE Main]** Find $\bigcap_{n=1}^{\infty} [3, 3 + \frac{2}{n})$.
5. **[JEE Main]** Express $\left(\bigcap_{i \in I} A_i\right)'$ in terms of the complements A_i' , and state which two-set law it generalises.

6. **[JEE Advanced]** Four sets satisfy $n(A) = 25$, $n(B) = 30$, $n(C) = 35$, $n(D) = 40$; every one of the six pairwise intersections has 10 elements, every one of the four triple intersections has 4, and $n(A \cap B \cap C \cap D) = 2$. Find $n(A \cup B \cup C \cup D)$.
7. **[JEE Advanced]** Twelve sets $A_1, \dots, A_{12}$ each have 6 elements and their union is S , with every element of S lying in exactly 4 of them. (i) Find $n(S)$. (ii) If sets $B_1, \dots, B_n$, each of size 4, also have union S with every element of S lying in exactly 6 of them, find n .
8. **[JEE Advanced]** With A_i the multiples of i in $\mathbb{N}$, find $\bigcap_{i=1}^6 A_i$.

Answers.

1. A_{12} and A_{30} (the lcm of 4 and 6 is 12; the lcm of 2, 3, 5 is 30).
2. $[1, \infty)$ and $\{4\}$.
3. $\mathbb{R}$ and $(-1, 1)$ — the family is increasing, so the intersection is the first member.
4. $\{3\}$; the left endpoint is included at every index, and any $x > 3$ fails once $\frac{2}{n} < x - 3$.
5. $\bigcup_{i \in I} A'_i$; it generalises $(A \cap B)' = A' \cup B'$.
6. $130 - 60 + 16 - 2 = 84$.
7. (i) $12 \times 6 = 72 = 4n(S)$, so $n(S) = 18$. (ii) $4n = 6 \times 18 = 108$, so $n = 27$.
8. A_{60} , since $\text{lcm}(1, 2, 3, 4, 5, 6) = 60$.

Section 6: Crack the MCQ

Q1. $\bigcap_{n=1}^{\infty} [0, \frac{1}{n}]$ equals:

- (A) $\emptyset$ (B) $\{0\}$ (C) $[0, 1]$ (D) $(0, 1)$

Answer: (B). 0 lies in every member because the left bracket is closed; every $x > 0$ is excluded once $\frac{1}{n} < x$.

Distractor Analysis. (A) imports the answer for the *open* family $(0, \frac{1}{n})$ without checking the bracket — the single most common error in this topic. (C) computes the **union** instead: the family is decreasing, so $\bigcup_n [0, \frac{1}{n}] = [0, 1]$. (D) commits both errors at once, dropping the endpoint and reporting the union.

Q2. For any family $\{A_i\}_{i \in I}$ inside a universe U , $(\bigcap_{i \in I} A_i)'$ equals:

- (A) $\bigcap_{i \in I} A'_i$ (B) $\bigcup_{i \in I} A'_i$ (C) $\bigcup_{i \in I} A_i$ (D) $(\bigcup_{i \in I} A_i)'$

Answer: (B). Failing to be in *every* set means failing in *at least one*, which is membership of at least one complement — a union.

Distractor Analysis. (A) forgets that complementation flips the operation, mirroring the classic two-set slip of writing $(A \cap B)' = A' \cap B'$. (C) drops the complement altogether. (D) complements the wrong family operation, giving $\bigcap_i A'_i$ in disguise — the same error as (A) written another way.

Q3. Twenty sets $A_1, \dots, A_{20}$ each have 3 elements, their union is S , and every element of S lies in exactly 5 of the sets. Then $n(S)$ equals:

(A) 12 (B) 60 (C) 100 (D) 4

Answer: (A). Incidences counted by sets: $20 \times 3 = 60$. Counted by elements: $5n(S)$. So $n(S) = 60/5 = 12$.

Distractor Analysis. (B) stops at the incidence total 60, forgetting that each element was counted five times over — this is the “forgot to divide by the multiplicity” error. (C) multiplies the two wrong quantities, 20×5 , mistaking the multiplicity for a set size. (D) divides the number of sets by the multiplicity, $20/5$, ignoring the size of the sets entirely. Anchor the method: total ticks equals rows times row-size equals columns times column-height.

Q4. With A_n the set of natural multiples of n , the set $A_4 \cap A_6$ equals:

(A) A_{24} (B) A_{12} (C) A_2 (D) A_{10}

Answer: (B). The common multiples of 4 and 6 are exactly the multiples of $\text{lcm}(4, 6) = 12$; for instance 12 and 24 qualify, but 24 is not the smallest.

Distractor Analysis. (A) uses the product $4 \times 6 = 24$ instead of the lcm — correct only when the two numbers are coprime, which 4 and 6 are not. It also fails a one-second check: 12 is a common multiple and $12 \notin A_{24}$. (C) uses the hcf, which describes $A_4 \cup A_6 \subseteq A_2$ rather than the intersection, and would wrongly admit 2. (D) adds the indices, a rule with no basis at all.

Section 7: Common Student Pitfalls and Pro-Tips

- **Treating an infinite intersection as a limit.** The error: writing $\bigcap_n (0, \frac{1}{n}) = \{0\}$ because the intervals shrink to 0. The correction: the intersection contains only elements that belong to **every** member, and 0 belongs to none of them. Check the bracket beside the limit point, every time.
- **Using the product instead of the lcm.** The error: $A_4 \cap A_6 = A_{24}$. The correction: common multiples are the multiples of the lcm, so $A_4 \cap A_6 = A_{12}$. The product works only for coprime indices.
- **Losing the quantifier flip in De Morgan.** The error: $(\bigcup_i A_i)' = \bigcup_i A_i'$. The correction: the negation of “in at least one” is “in none”, which is “in every complement” — a $\bigcap$. Complementation always flips $\bigcup$ and $\bigcap$, however long the family.
- **Dividing by the wrong quantity in double counting.** The error: reporting $n(S) = mp$ or $n(S) = m/k$. The correction: the identity is $mp = kn(S)$; count the ticks in the grid two ways and the roles of p and k can never swap.
- **Getting the last sign of inclusion-exclusion wrong.** The error: adding the m -fold intersection regardless of m . The correction: the sign of the r -fold block is $(-1)^{r-1}$, so with four sets the final term is **subtracted**, not added. Only the odd-numbered blocks are added.
- **Forgetting the sandwich.** The error: an answer for $\bigcap_i A_i$ that is larger than some member. The correction: $\bigcap_i A_i \subseteq A_j \subseteq \bigcup_i A_i$ for every j — test your answer against A_1 before writing it down.

Pro-Tip: Never manipulate a family symbolically when you can interrogate a single candidate element. Pick the one suspicious number — the limit point, the

endpoint, the element 1 — and ask “is it in A_1 ? in A_2 ? in all of them?” Three membership tests settle almost every indexed-family question in JEE Main, and they cannot go wrong the way symbol-pushing can.

Pro-Tip: The moment a problem mentions “ m sets, each with p elements” and “each element belongs to exactly k of them”, stop reading for structure and write $mp = k n(S)$. The problem is over. If a second family shares the same union, run the identity again with the new numbers — the shared $n(S)$ carries across.

Section 8: Cheat Sheet Box

- **Family union / intersection:** $\bigcup_{i \in I} A_i$ = “in at least one”; $\bigcap_{i \in I} A_i$ = “in every one”.
- **Sandwich:** $\bigcap_i A_i \subseteq A_j \subseteq \bigcup_i A_i$ for every index j .
- **Generalised De Morgan:** $(\bigcup_i A_i)' = \bigcap_i A_i'$ and $(\bigcap_i A_i)' = \bigcup_i A_i'$.
- **Generalised distributive:** $B \cap \bigcup_i A_i = \bigcup_i (B \cap A_i)$; $B \cup \bigcap_i A_i = \bigcap_i (B \cup A_i)$.
- **Monotone families:** decreasing $\Rightarrow$ union = A_1 ; increasing $\Rightarrow$ intersection = A_1 .
- **Must-know families:** $\bigcap_n (0, \frac{1}{n}) = \emptyset$; $\bigcap_n [0, \frac{1}{n}] = \{0\}$; $\bigcup_n [\frac{1}{n}, 1] = (0, 1]$; $\bigcup_n (-n, n) = \mathbb{R}$; $\bigcup_n [n, n+1] = [1, \infty)$.
- **Multiple-sets:** $A_m \cap A_n = A_{\text{lcm}(m,n)}$; $A_m \subseteq A_n$ exactly when $n \mid m$; $\bigcup_{n \geq 2} A_n = \mathbb{N} - \{1\}$.
- **Inclusion-exclusion for m sets:** add all singles, subtract all pairs, add all triples, ..., with the r -fold block carrying $(-1)^{r-1}$.
- **Four sets:** $n(A \cup B \cup C \cup D) = \sum n(A) - \sum n(A \cap B) + \sum n(A \cap B \cap C) - n(A \cap B \cap C \cap D)$, over 4 singles, 6 pairs, 4 triples.
- **Union bound:** $n(\bigcup_{i=1}^m A_i) \leq \sum_{i=1}^m n(A_i)$, with equality exactly for pairwise disjoint sets.
- **Intersection bound:** $n(\bigcap_{i=1}^m A_i) \geq \sum_{i=1}^m n(A_i) - (m-1)n(U)$.
- **Double counting:** $\sum_{i=1}^m n(A_i) = \sum_{x \in S} d(x)$; uniform case $mp = k n(S)$.

Memory aids:

- “Big cup = at least one; big cap = every one.”
- “Shrinking is not arriving” — a family closing on a point need not contain it; read the bracket.
- “Lcm for the cap” — intersecting multiple-sets runs on the lcm, never the product.
- “Count the ticks twice: rows give mp , columns give $k n(S)$.”
- “Odd blocks plus, even blocks minus” — the alternating signs of inclusion-exclusion.

Chapter 1: Sets — Round 2 Addendum

Subtopics 01-04 and the Round 1 Supplement (Addendum A to H, Subtopic 05) together cover the whole NCERT syllabus for this chapter with one exception: the *proof-and-structure* family of problems that NCERT develops in Example 11, Example 30, Example 31, Exercise 1.3 Q3 and almost the whole Miscellaneous Exercise. Those are exactly the items that ask you to reason **about** sets rather than compute **with** them, and they are worth 4-6 marks a year in CBSE and a standing one-mark trap in JEE Main.

This addendum closes that gap in five short items, continuing the lettering of the Round 1 Supplement. Each is a method statement, one or two worked examples, and practice with answers.

Addendum	Topic	Read after
I	Sets whose elements are sets; why $\in$ is not transitive	Subtopic 02
J	Power set laws: $P(A \cap B)$, $P(A \cup B)$, and counting	Addendum C
K	The four equivalent forms of $A \subseteq B$, and the cancellation traps	Subtopic 03
L	Proving containment between algebraically described sets	Subtopic 01
M	The algebra of set difference; why $-$ needs its brackets	Subtopic 03

Addendum I — Sets whose elements are sets

Extends Subtopic 02, Section 1 (the $\in$ versus $\subseteq$ discussion) and Section 7.

Method. Subtopic 02 taught you that $\in$ joins an element to a set and $\subseteq$ joins a set to a set. The examiner's favourite way to test whether you actually believe that is to put a set *inside* the roster of another set. Five rules settle every such question.

1. **An entry is one object, however complicated it looks.** In $A = \{1, \{2\}, \{3, 4\}, \emptyset\}$ there are four entries, so $n(A) = 4$. The braces around $\{3, 4\}$ are part of the object's name, not an invitation to unpack it.
2. **To test $X \in A$:** ask whether X appears, exactly as written, as one entry in the roster of A .

3. **To test $X \subseteq A$:** ask whether *every element of X* appears as an entry of A . This is a different question from rule 2 and usually has a different answer.
4. **$\emptyset \subseteq A$ is always true. $\emptyset \in A$ is true only when $\emptyset$ is actually listed as an entry.** Most sets satisfy the first and fail the second; a set like A above satisfies both, and that is the point of the example.
5. **$\subseteq$ is transitive; $\in$ is not, and the two never chain together.** From $A \subseteq B$ and $B \subseteq C$ you may write $A \subseteq C$. From $x \in A$ and $A \in B$ you may conclude nothing at all about x and B .

Figure brief: A large box labelled A containing four items drawn side by side: the bare numeral 1, a small closed box containing 2, a small closed box containing 3 and 4, and an empty small box. Caption that A has four items on its shelf, and that the numbers 2, 3, 4 sit *inside* boxes on the shelf, not on the shelf itself.

Worked Example I1 — [CBSE Board]

Let $A = \{1, \{2\}, \{3, 4\}, \emptyset\}$. Mark each statement true or false with a one-line reason.

- (i) $n(A) = 4$ (ii) $2 \in A$ (iii) $\{2\} \in A$ (iv) $\{2\} \subseteq A$
 (ii) $\{\{2\}\} \subseteq A$ (vi) $\{3, 4\} \subseteq A$ (vii) $\emptyset \in A$
 (iii) $\emptyset \subseteq A$ (ix) $\{\emptyset\} \subseteq A$ (x) $n[P(A)] = 16$

- (i) **True.** Four entries: 1, $\{2\}$, $\{3, 4\}$, $\emptyset$.
- (ii) **False.** The entry is the *set* $\{2\}$. The number 2 is inside that entry, not on the shelf, and $2 \neq \{2\}$.
- (iii) **True.** $\{2\}$ is written as an entry.
- (iv) **False.** $\{2\} \subseteq A$ would demand $2 \in A$, and (ii) already killed that.
- (v) **True.** The only element of $\{\{2\}\}$ is $\{2\}$, and $\{2\} \in A$ by (iii).
- (vi) **False.** $\{3, 4\} \subseteq A$ demands $3 \in A$ and $4 \in A$; neither is an entry. (Note $\{3, 4\} \in A$ is true — the same object, the opposite verdict.)
- (vii) **True.** $\emptyset$ is listed as an entry here.
- (viii) **True.** The empty set is a subset of every set, always.
- (ix) **True.** Its only element is $\emptyset$, and $\emptyset \in A$ by (vii).
- (x) **True.** $n(A) = 4$, so $n[P(A)] = 2^4 = 16$. The complicated-looking entries change nothing: the counting rule only cares how many entries there are.

Statements (vii) and (viii) are both true here, which is unusual and deliberate. For a set like $B = \{1, 2\}$ you get $\emptyset \subseteq B$ true but $\emptyset \in B$ false. Never read one off the other.

Worked Example I2 — [JEE Main]

Decide whether each implication holds for all sets. If it fails, give a counterexample.

- (i) $x \in A$ and $A \in B \Rightarrow x \in B$
 (ii) $A \subseteq B$ and $B \in C \Rightarrow A \in C$
 (iii) $A \subseteq B$ and $B \subseteq C \Rightarrow A \subseteq C$
 (iv) $A \not\subseteq B$ and $B \not\subseteq C \Rightarrow A \not\subseteq C$

- (v) $x \in A$ and $A \not\subseteq B \Rightarrow x \in B$
 (vi) $A \subseteq B$ and $x \notin B \Rightarrow x \notin A$

- (i) **False.** Take $x = 1$, $A = \{1\}$, $B = \{\{1\}\}$. Then $1 \in A$ and $A \in B$, but the one and only element of B is the set $\{1\}$, and $1 \neq \{1\}$, so $1 \notin B$. Membership does not chain.
- (ii) **False.** Take $A = \{1\}$, $B = \{1, 2\}$, $C = \{\{1, 2\}, 3\}$. Then $A \subseteq B$ and $B \in C$, but the entries of C are $\{1, 2\}$ and 3 , neither of which is $\{1\}$, so $A \notin C$. You cannot mix a $\subseteq$ step with an $\in$ step.
- (iii) **True.** Take any $x \in A$. Then $x \in B$ by the first hypothesis and $x \in C$ by the second. Since x was arbitrary, $A \subseteq C$. Containment *does* chain.
- (iv) **False.** Take $A = \{1\}$, $B = \{2\}$, $C = \{1, 3\}$. Here $1 \notin B$, so $A \not\subseteq B$; and $2 \notin C$, so $B \not\subseteq C$; yet $1 \in C$, so $A \subseteq C$. Two failures do not compose into a third.
- (v) **False.** Take $A = \{1, 2\}$, $B = \{1\}$, $x = 2$. Then $2 \in A$, and $A \not\subseteq B$ (because of that very 2), but $2 \notin B$. “ A is not inside B ” tells you *some* element of A escapes, never that a *particular* one lands in B .
- (vi) **True.** This is the contrapositive of the definition: $A \subseteq B$ says $x \in A \Rightarrow x \in B$, so $x \notin B \Rightarrow x \notin A$.

The habit to build: for every claimed implication, first try to break it with a two- or three-element set. Only after the counterexample hunt fails should you look for a proof.

Practice.

1. Let $A = \{a, \{a\}, \{a, b\}\}$. Find $n(A)$ and $n[P(A)]$.
2. For the same A : is $\{a\} \in A$? Is $\{a\} \subseteq A$? Is $\{\{a\}\} \subseteq A$?
3. Is $\{b\} \subseteq A$ for the A of question 1?
4. Let $C = \{\emptyset, \{\emptyset\}\}$. Find $n(C)$, and decide whether $\emptyset \in C$, $\emptyset \subseteq C$, $\{\emptyset\} \in C$ and $\{\emptyset\} \subseteq C$.
5. True or false: if $A \in B$ and $B \subseteq C$, then $A \in C$. Justify.
6. True or false: if $A \in B$ and $B \subseteq C$, then $A \subseteq C$. Give a counterexample if false.

Answers. 1. $n(A) = 3$ (the entries are a , $\{a\}$, $\{a, b\}$), so $n[P(A)] = 2^3 = 8$. 2. $\{a\} \in A$: yes, it is an entry. $\{a\} \subseteq A$: yes as well, because its only element a is also an entry of A — this is the rare set where both hold, so check both separately rather than assuming they disagree. $\{\{a\}\} \subseteq A$: yes, since $\{a\} \in A$. 3. No. b is not an entry of A ; it only appears inside the entry $\{a, b\}$. 4. $n(C) = 2$. $\emptyset \in C$: true (listed). $\emptyset \subseteq C$: true (always). $\{\emptyset\} \in C$: true (listed). $\{\emptyset\} \subseteq C$: true, since its only element $\emptyset$ belongs to C . All four hold. 5. True. $A \in B$ puts A among the elements of B , and $B \subseteq C$ sends every element of B into C ; so $A \in C$. This is the one legal mixed chain. 6. False — NCERT’s own counterexample: $A = \{1\}$, $B = \{\{1\}, 2\}$, $C = \{\{1\}, 2, 3\}$. Then $A \in B$ and $B \subseteq C$, but $1 \in A$ while $1 \notin C$, so $A \not\subseteq C$. Compare with question 5: the conclusion $A \in C$ is fine, the conclusion $A \subseteq C$ is not.

Addendum J — Power set laws

Extends Subtopic 02, Sections 2-4, and Addendum C.

Method. Addendum C iterated the power set of a *single* set. Here you have two, and there are exactly three laws plus one counting formula. Throughout, $n(A) = a$, $n(B) = b$, $n(A \cap B) = c$.

$$P(A) \cap P(B) = P(A \cap B) \quad (\text{always, with no conditions})$$

$$P(A) \cup P(B) \subseteq P(A \cup B), \quad \text{with equality if and only if } A \subseteq B \text{ or } B \subseteq A$$

$$A \subseteq B \iff P(A) \subseteq P(B), \quad \text{and hence} \quad P(A) = P(B) \iff A = B$$

- The *intersection* law is an equality; the *union* law is only a containment. This asymmetry is the whole examinable content — remember it as “**cap behaves, cup does not.**”
- Why the cup misbehaves: a subset of $A \cup B$ is allowed to take some elements from A and some from B , and such a mongrel subset lies inside neither A nor B on its own.
- Counting, for finite A and B :

$$n[P(A) \cap P(B)] = 2^c, \quad n[P(A) \cup P(B)] = 2^a + 2^b - 2^c, \quad n[P(A \cup B)] = 2^{a+b-c}$$

The middle one is just inclusion-exclusion applied to the two power sets, using the first law to size their overlap.

Worked Example J1 — [CBSE Board]

Let $A = \{1, 2\}$ and $B = \{2, 3\}$. Write $P(A)$, $P(B)$, $P(A) \cap P(B)$ and $P(A) \cup P(B)$ in full, and compare with $P(A \cap B)$ and $P(A \cup B)$.

$$P(A) = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}, \quad P(B) = \{\emptyset, \{2\}, \{3\}, \{2, 3\}\}$$

Intersection. The members common to both lists are $\emptyset$ and $\{2\}$:

$$P(A) \cap P(B) = \{\emptyset, \{2\}\}$$

And $A \cap B = \{2\}$, whose power set is $\{\emptyset, \{2\}\}$. Identical, as the law promises. The count agrees too: $c = 1$, and $2^1 = 2$.

Union. Merging the two lists and keeping one copy of each:

$$P(A) \cup P(B) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{2, 3\}\}$$

That is 6 members, matching $2^a + 2^b - 2^c = 4 + 4 - 2 = 6$. But $A \cup B = \{1, 2, 3\}$, so $n[P(A \cup B)] = 2^3 = 8$. The two missing members are

$$\{1, 3\} \quad \text{and} \quad \{1, 2, 3\},$$

and you can see why each is missing: both contain 1 (so neither is a subset of $B = \{2, 3\}$) and both contain 3 (so neither is a subset of $A = \{1, 2\}$). Each is a mongrel. Hence $P(A) \cup P(B) \subsetneq P(A \cup B)$, consistent with the law, since neither $A \subseteq B$ nor $B \subseteq A$ holds.

Worked Example J2 — [JEE Advanced]

Prove that (i) $P(A) \cap P(B) = P(A \cap B)$ for all sets A, B , and (ii) $P(A) \cup P(B) = P(A \cup B)$ if and only if $A \subseteq B$ or $B \subseteq A$.

(i) Element-chasing, where the “elements” are themselves sets. Let X be any set.

$$X \in P(A) \cap P(B) \iff X \in P(A) \text{ and } X \in P(B) \iff X \subseteq A \text{ and } X \subseteq B$$

Now “ $X \subseteq A$ and $X \subseteq B$ ” says every element of X lies in A and also lies in B , which is precisely “every element of X lies in $A \cap B$ ”, i.e. $X \subseteq A \cap B$. So the chain continues:

$$\iff X \subseteq A \cap B \iff X \in P(A \cap B)$$

Every step is an “if and only if”, so the two collections have exactly the same members and are equal. ■

(ii) The containment first. If $X \in P(A) \cup P(B)$ then $X \subseteq A$ or $X \subseteq B$; either way $X \subseteq A \cup B$, so $X \in P(A \cup B)$. Hence $P(A) \cup P(B) \subseteq P(A \cup B)$ always, and the only question is when it is an equality.

If $A \subseteq B$: then $A \cup B = B$ and $P(A) \subseteq P(B)$, so $P(A) \cup P(B) = P(B) = P(A \cup B)$. Equality holds. The case $B \subseteq A$ is the same argument with the letters swapped.

If neither containment holds: then some $p \in A$ with $p \notin B$, and some $q \in B$ with $q \notin A$. Build the mongrel $X = \{p, q\}$. Since $p, q \in A \cup B$, we get $X \in P(A \cup B)$. But $q \notin A$ forces $X \not\subseteq A$, and $p \notin B$ forces $X \not\subseteq B$, so $X \notin P(A) \cup P(B)$. The containment is therefore strict. ■

Both directions are proved, so the equality happens exactly when one of A, B sits inside the other. Notice how short the second half is once you decide to *construct* the missing subset rather than argue about it — one element from each private region is all it takes.

Practice.

1. $A = \{1, 2, 3\}$, $B = \{3, 4\}$. Find $n[P(A) \cap P(B)]$ and $n[P(A) \cup P(B)]$.
2. With $n(A) = 5$, $n(B) = 4$ and $n(A \cap B) = 2$, find $n[P(A) \cup P(B)]$ and $n[P(A \cup B)]$.
3. True or false: $P(A) \cup P(B) = P(A \cup B)$ whenever A and B are disjoint and both non-empty. Justify.
4. If $P(A) \subseteq P(B)$, what can you conclude about A and B ?
5. Find all pairs of sets $A, B \subseteq \{1, 2\}$ for which $P(A) \cup P(B) = P(A \cup B)$, and check your answer against the criterion.
6. If $n[P(A) \cap P(B)] = 8$ and $n(A) = n(B) = 5$, find $n(A \cup B)$.

Answers. 1. $A \cap B = \{3\}$, so $n[P(A) \cap P(B)] = 2^1 = 2$; and $n[P(A) \cup P(B)] = 2^3 + 2^2 - 2^1 = 8 + 4 - 2 = 10$. 2. $2^5 + 2^4 - 2^2 = 32 + 16 - 4 = 44$; and $n(A \cup B) = 5 + 4 - 2 = 7$, so $n[P(A \cup B)] = 2^7 = 128$. 3. False. Disjoint non-empty sets satisfy neither $A \subseteq B$ nor $B \subseteq A$, so the containment is strict. Concretely, with $A = \{1\}$ and $B = \{2\}$, the subset $\{1, 2\}$ belongs to $P(A \cup B)$ but to neither $P(A)$ nor $P(B)$. 4. $A \subseteq B$. (Apply the hypothesis to the member $X = A$ of $P(A)$: it must lie in $P(B)$, i.e. $A \subseteq B$. The converse is immediate, so the two statements are equivalent.) 5. Every pair *except* $\{A, B\} = \{\{1\}, \{2\}\}$. Those two are the only unordered

pair drawn from $\emptyset, \{1\}, \{2\}, \{1, 2\}$ in which neither set contains the other, which is exactly the criterion. 6. $2^{n(A \cap B)} = 8$ gives $n(A \cap B) = 3$, so $n(A \cup B) = 5 + 5 - 3 = 7$.

Addendum K — The four equivalent forms of $A \subseteq B$, and the cancellation traps

Extends Subtopic 03, Sections 2-5. Subtopic 03 uses several of these implications in one direction (its MCQ Q4 uses $A \subseteq B \Rightarrow A \cup B = B$); what follows is the full two-way statement with its proof, which is what the “show that...” questions demand.

Method — the equivalence theorem. For any two sets A and B , the following four statements are equivalent: each one implies all the others, so proving any one of them proves all four.

$$(i) A \subseteq B \quad (ii) A - B = \emptyset \quad (iii) A \cup B = B \quad (iv) A \cap B = A$$

- **How to use it in an exam:** whichever of the four is easiest to establish from the data, establish that one and then quote the theorem. Most often (ii) is the cheapest, because showing a set is empty needs only “no element can satisfy both conditions.”
- **The economical proof** is a cycle: $(i) \Rightarrow (ii) \Rightarrow (iii) \Rightarrow (iv) \Rightarrow (i)$. Four short implications replace the twelve you would need if you proved each pair separately. Why the cycle is enough: from any statement in the ring you can walk forward to any other.
- **Immediate corollary:** $A \cup B = A \cap B$ if and only if $A = B$.

Method — the cancellation traps. Sets are *not* numbers, and the cancellation law you trust in arithmetic fails here.

Given	Can you conclude $B = C$?
$A \cap B = A \cap C$	No
$A \cup B = A \cup C$	No
both together	Yes

The one-line reason both singles fail: A can hide the difference between B and C — inside A for the union version, outside A for the intersection version. Only when you close both hiding places is $B = C$ forced.

Worked Example K1 — [CBSE Board / JEE Main]

Show that the four conditions (i)-(iv) above are equivalent.

(i) $\Rightarrow$ (ii). Suppose $A \subseteq B$. An element of $A - B$ would have to lie in A and outside B — but $A \subseteq B$ sends every element of A into B . No such element exists, so $A - B = \emptyset$.

(ii) $\Rightarrow$ (iii). Suppose $A - B = \emptyset$. Certainly $B \subseteq A \cup B$. For the reverse, take $x \in A \cup B$. If $x \in B$ we are done. If $x \notin B$, then from $x \in A \cup B$ we get $x \in A$, so $x \in A - B$ — impossible,

since that set is empty. So every $x \in A \cup B$ lies in B , giving $A \cup B \subseteq B$. Mutual containment yields $A \cup B = B$.

(iii) $\Rightarrow$ (iv). Suppose $A \cup B = B$. Then

$$A \cap B = A \cap (A \cup B) = A,$$

the last step being the absorption law already established in Subtopic 03, Practice 5.

(iv) $\Rightarrow$ (i). Suppose $A \cap B = A$. Take any $x \in A$. Then $x \in A \cap B$, and every element of an intersection lies in B . So $x \in B$, and since x was arbitrary, $A \subseteq B$. ■

The cycle closes, so all four are equivalent.

The corollary. If $A \cup B = A \cap B$, then since $A \subseteq A \cup B = A \cap B \subseteq B$ we get $A \subseteq B$; and symmetrically $B \subseteq A \cup B = A \cap B \subseteq A$ gives $B \subseteq A$. Mutual containment forces $A = B$. Conversely if $A = B$ then both sides equal A . So $A \cup B = A \cap B \iff A = B$.

Worked Example K2 — [JEE Advanced]

(a) Show by counterexample that neither $A \cap B = A \cap C$ nor $A \cup B = A \cup C$ alone forces $B = C$. (b) Show that the two together do force $B = C$.

(a) Intersection alone. Take $A = \{1\}$, $B = \{1, 2\}$, $C = \{1, 3\}$. Then $A \cap B = \{1\} = A \cap C$, yet $B \neq C$. The difference between B and C (the elements 2 and 3) lives entirely *outside* A , where intersecting with A cannot see it.

Union alone. Take $A = \{1, 2, 3\}$, $B = \{1\}$, $C = \{2\}$. Then $A \cup B = \{1, 2, 3\} = A \cup C$, yet $B \neq C$. This time the difference lives entirely *inside* A , where the union swallows it.

Note how the two counterexamples are mirror images — that is the tell for why both hypotheses are needed.

(b) Both together. Assume $A \cap B = A \cap C$ and $A \cup B = A \cup C$. Then

$$\begin{aligned} B &= B \cap (A \cup B) && \text{(absorption)} \\ &= B \cap (A \cup C) && \text{(the union hypothesis)} \\ &= (B \cap A) \cup (B \cap C) && \text{(distributive law)} \\ &= (A \cap C) \cup (B \cap C) && \text{(the intersection hypothesis, since } B \cap A = A \cap B) \\ &= (A \cup B) \cap C && \text{(distributive law, read right to left)} \\ &= (A \cup C) \cap C && \text{(the union hypothesis again)} \\ &= C && \text{(absorption)} \end{aligned}$$

So $B = C$. ■ Every hypothesis is used exactly twice, and the whole proof is absorb-distribute-substitute-distribute-absorb — worth memorising as a shape, because the same skeleton solves most “show that $B = C$ ” problems in this chapter.

Practice.

1. Given $A - B = \emptyset$ and $B - A = \emptyset$, what can you say about A and B ?

2. Show that $A \subseteq B$ implies $C - B \subseteq C - A$, for every set C .
3. If $A \cap X = B \cap X = \emptyset$ and $A \cup X = B \cup X$, prove $A = B$. (Hint: $A = A \cap (A \cup X)$.)
4. True or false: $A - B = B - A$ implies $A = B$.
5. Find sets A, B, C with $A \cap B, B \cap C$ and $A \cap C$ all non-empty but $A \cap B \cap C = \emptyset$.
6. If $A \cup B = A$, what is $B - A$? Which of the four equivalent forms are you using?

Answers. 1. $A = B$. Form (ii) applied both ways gives $A \subseteq B$ and $B \subseteq A$. 2. Let $x \in C - B$, so $x \in C$ and $x \notin B$. If x were in A then $A \subseteq B$ would put x in B , a contradiction; so $x \notin A$, i.e. $x \in C - A$. 3. $A = A \cap (A \cup X) = A \cap (B \cup X) = (A \cap B) \cup (A \cap X) = (A \cap B) \cup \emptyset = A \cap B$, so $A \subseteq B$ by form (iv). The identical argument with A and B swapped gives $B \subseteq A$. Hence $A = B$. 4. True. $A - B = B - A$ means each side is a subset of the other's parent set as well as its own; since $A - B \subseteq A$ and $B - A$ is disjoint from A , the common set must be $\emptyset$. So $A - B = B - A = \emptyset$, and question 1 gives $A = B$. 5. For instance $A = \{1, 2\}$, $B = \{2, 3\}$, $C = \{1, 3\}$: the three pairwise intersections are $\{2\}$, $\{3\}$, $\{1\}$, all non-empty, while no element lies in all three. 6. $B - A = \emptyset$, and in fact $B \subseteq A$. You are using form (iii) with the letters swapped ($B \cup A = A$ means $B \subseteq A$) and then form (ii).

Addendum L — Proving containment between algebraically described sets

Extends Subtopic 01, Section 3 (Procedure A) and Subtopic 02. This is a standing JEE Main item type and does not appear anywhere in Subtopics 01-04 or in Addendum A to H.

Method. When both sets are given by a formula rather than a roster, listing them is hopeless and comparing them term by term is worse. Do this instead.

1. **Write the general element of the first set** as an expression in its index, say t_n .
2. **Force that expression into the shape the second set demands.** For “multiple of k ” targets, the workhorse is the binomial expansion of $(1 + m)^n$, because it isolates the first two terms and shows that everything left over carries a visible factor.
3. **Check the range.** Producing the right algebraic *shape* is not enough; the resulting parameter must lie in the index set the second description allows (usually $\mathbb{N}$ or $\mathbb{W}$).
4. **For strictness, hunt one witness.** To show $X \subsetneq Y$, name a single element of Y and prove no index produces it in X . A monotonicity remark (“the terms of X are increasing and already exceed it”) is a complete proof and costs one line.

The binomial fact behind almost every problem of this type: for $n \in \mathbb{N}$,

$$(1 + m)^n = 1 + mn + \sum_{k=2}^n \binom{n}{k} m^k, \quad \text{so} \quad (1 + m)^n - mn - 1 = m^2 \sum_{k=2}^n \binom{n}{k} m^{k-2},$$

a non-negative integer multiple of m^2 . Read it once and you own the whole family.

Worked Example L1 — [JEE Main]

Let $X = \{8^n - 7n - 1 : n \in \mathbb{N}\}$ and $Y = \{49(n - 1) : n \in \mathbb{N}\}$. Show that $X \subseteq Y$, and decide whether the containment is strict.

First identify Y plainly: as n runs through $1, 2, 3, \dots$, the expression $49(n - 1)$ runs through $0, 49, 98, 147, \dots$, so

$$Y = \{\text{all non-negative multiples of } 49\}.$$

Now take a general element of X and apply the binomial expansion with $m = 7$:

$$8^n = (1 + 7)^n = 1 + 7n + \sum_{k=2}^n \binom{n}{k} 7^k.$$

Subtracting $7n + 1$ from both sides kills exactly the first two terms:

$$8^n - 7n - 1 = \sum_{k=2}^n \binom{n}{k} 7^k = 49 \sum_{k=2}^n \binom{n}{k} 7^{k-2}.$$

Every term of the remaining sum is a non-negative integer, so $8^n - 7n - 1$ is 49 times a non-negative integer — that is, a non-negative multiple of 49. Hence every element of X lies in Y :

$$X \subseteq Y.$$

Is it strict? List the first few members of X : for $n = 1, 2, 3, 4$ we get

$$8 - 7 - 1 = 0, \quad 64 - 14 - 1 = 49, \quad 512 - 21 - 1 = 490, \quad 4096 - 28 - 1 = 4067.$$

Consider $98 \in Y$. The terms of X are strictly increasing (each step multiplies 8^n by 8 while subtracting only a further 7), and they jump from 49 at $n = 2$ straight to 490 at $n = 3$. So no index gives 98, and

$$X \subsetneq Y.$$

The trap this question is built on is answering “ $X = Y$ ” after checking only that every member of X is a multiple of 49. Containment in one direction is not equality; you must either exhibit the reverse containment or, as here, break it.

Worked Example L2 — [JEE Advanced]

Let $A = \left\{ \frac{n}{n+1} : n \in \mathbb{N} \right\}$ and $B = \{x \in \mathbb{Q} : 0 < x < 1\}$. Determine whether $A \subseteq B$, whether $B \subseteq A$, and whether the two are equal.

Is $A \subseteq B$? Take a general element $\frac{n}{n+1}$ with $n \in \mathbb{N}$.

- It is rational: both n and $n + 1$ are integers and $n + 1 \neq 0$.
- It is positive: $n \geq 1$ and $n + 1 \geq 2$, so the quotient is positive.
- It is less than 1: $\frac{n}{n+1} < 1 \iff n < n + 1$, which is always true.

All three requirements of B are met, so $A \subseteq B$.

Is $B \subseteq A$? No. Take the witness $\frac{1}{3} \in B$. If $\frac{1}{3}$ were in A there would be a natural number n with

$$\frac{n}{n+1} = \frac{1}{3} \implies 3n = n + 1 \implies n = \frac{1}{2} \notin \mathbb{N}.$$

No such index exists, so $\frac{1}{3} \notin A$ and $B \not\subseteq A$.

Therefore $A \subsetneq B$. The structural reason is worth naming: solving $\frac{n}{n+1} = \frac{p}{q}$ gives $n = \frac{p}{q-p}$, so a rational in $(0, 1)$ belongs to A only when $q - p$ divides p — a genuine restriction, and A misses everything else.

Practice.

1. Show that $\{6n + 2 : n \in \mathbb{W}\} \subseteq \{2n : n \in \mathbb{W}\}$, and decide whether the containment is strict.
2. Show that every element of $\{9^n - 8n - 1 : n \in \mathbb{N}\}$ is a multiple of 64.
3. Show that every element of $\{4^n - 3n - 1 : n \in \mathbb{N}\}$ is a non-negative multiple of 9.
4. Is $\{2n : n \in \mathbb{N}\} \subseteq \{n^2 : n \in \mathbb{N}\}$? Justify in one line.
5. Let $P = \{x \in \mathbb{R} : x^2 - 5x + 6 = 0\}$ and $Q = \{x \in \mathbb{N} : x \leq 3\}$. Decide whether $P \subseteq Q$, $Q \subseteq P$, or both.
6. Show that $\{12n : n \in \mathbb{N}\} = \{4n : n \in \mathbb{N}\} \cap \{6n : n \in \mathbb{N}\}$.

Answers. 1. $6n + 2 = 2(3n + 1)$, and $3n + 1 \in \mathbb{W}$ whenever $n \in \mathbb{W}$, so the containment holds. It is strict: 0 belongs to the right-hand set (take $n = 0$), but $6n + 2 = 0$ would need $n = -\frac{1}{3}$, which is not a whole number. 2. With $m = 8$: $9^n = (1 + 8)^n = 1 + 8n + \sum_{k=2}^n \binom{n}{k} 8^k$, so $9^n - 8n - 1 = 64 \sum_{k=2}^n \binom{n}{k} 8^{k-2}$, a multiple of 64. 3. With $m = 3$: $4^n - 3n - 1 = 9 \sum_{k=2}^n \binom{n}{k} 3^{k-2}$, and the sum is a non-negative integer. (Check: $n = 1, 2, 3$ give 0, 9, 54.) 4. No. 2 is in the left-hand set but is not a perfect square, so it is not in the right-hand set. 5. $P = \{2, 3\}$ and $Q = \{1, 2, 3\}$, so $P \subseteq Q$ but $Q \not\subseteq P$ (because $1 \notin P$). The containment is strict and the sets are not equal. 6. A natural number lies in the right-hand side exactly when it is a common multiple of 4 and 6, and the common multiples of two numbers are precisely the multiples of their lcm. Since $\text{lcm}(4, 6) = 12$, that set is the multiples of 12. (This is the $A_m \cap A_n = A_{\text{lcm}(m, n)}$ rule of Subtopic 05, restated in the language of this addendum — and note the answer is *not* the multiples of 24.)

Addendum M — The algebra of set difference

Extends Subtopic 03, Sections 2-4 and its Practice 4, and Addendum F. Subtopic 03 defines $A - B$ and proves one identity about it; the rest of the difference algebra is used silently everywhere and drilled nowhere. This item also closes NCERT Miscellaneous Exercise Q8's second identity, the one part of that exercise not reached by Addenda I to L.

Method. Every difference question collapses the moment you rewrite

$$A - B = A \cap B'.$$

After that, the ordinary laws of Subtopic 03 finish the job, with De Morgan doing the flipping. Five facts carry the whole topic.

1. **Splitting into disjoint pieces.**

$$A \cup B = A \cup (B - A), \quad \text{and } A \cap (B - A) = \emptyset$$

$$A = (A \cap B) \cup (A - B), \quad \text{and these two pieces are disjoint too}$$

The second is Subtopic 03's Example 2 read backwards; the first is its partner and is the half students never meet.

2. **Difference over a union, and over an intersection — the De Morgan pair.**

$$A - (B \cup C) = (A - B) \cap (A - C), \quad A - (B \cap C) = (A - B) \cup (A - C)$$

The operation flips, for exactly the reason it flips in De Morgan's laws: $A - X = A \cap X'$, and complementing $B \cup C$ turns it into $B' \cap C'$. (The first of the two is Subtopic 03, Practice 4; the second is its unexamined mirror.)

3. **Peeling a double difference.**

$$A - (B - C) = (A - B) \cup (A \cap C)$$

The C part comes *back*. Removing " B except C " leaves everything of A outside B **plus** everything of A inside C .

4. **Stacked differences, and why — is not associative.**

$$(A - B) - C = A - (B \cup C) = (A - C) - B$$

but $A - (B - C)$ is a different set in general. So $A - B - C$ is meaningless without brackets — unlike $\cup$ and $\cap$, which are associative and may be written unbracketed.

5. **Trimming the operands.**

$$A - B = A - (A \cap B) = (A \cup B) - B$$

Worth having when B is enormous and $A \cap B$ is small: only the overlap ever matters.

Contrast worth naming: symmetric difference *is* associative, $(A \triangle B) \triangle C = A \triangle (B \triangle C)$, and equals the set of elements lying in an **odd number** of A, B, C — regions R_1, R_2, R_3, R_7 in Addendum F's numbering. Plain difference is not associative. Do not let the family resemblance fool you.

Worked Example M1 — [CBSE Board]

Show that $A = (A \cap B) \cup (A - B)$ and that $A \cup (B - A) = A \cup B$, and deduce the two-set cardinality formula.

First identity. Take $x \in A$. Either $x \in B$, in which case $x \in A \cap B$, or $x \notin B$, in which case $x \in A - B$. So A is contained in the right side. Conversely both $A \cap B$ and $A - B$ are subsets of A , so the right side is contained in A . Mutual containment gives equality. The two pieces are disjoint, because one demands $x \in B$ and the other demands $x \notin B$.

Second identity. $A \subseteq A \cup B$ and $B - A \subseteq B \subseteq A \cup B$, so the left side sits inside $A \cup B$. For the reverse, take $x \in A \cup B$: if $x \in A$ we are done, and if $x \notin A$ then $x \in B$, hence $x \in B - A$.

So $A \cup (B - A) = A \cup B$, and again the two pieces are disjoint, since $B - A$ throws out every element of A .

The deduction. Because the union has been cut into two disjoint slabs, counts simply add:

$$n(A \cup B) = n(A) + n(B - A).$$

Applying the first identity to B gives $B = (B \cap A) \cup (B - A)$ disjointly, so $n(B - A) = n(B) - n(A \cap B)$. Substituting,

$$n(A \cup B) = n(A) + n(B) - n(A \cap B).$$

This is the honest three-piece derivation of Subtopic 04's headline formula — the same one NCERT gives — and it needs no talk of “double counting” at all.

Worked Example M2 — [JEE Main / JEE Advanced]

- (a) Prove $A - (B - C) = (A - B) \cup (A \cap C)$. (b) Prove $(A - B) - C = A - (B \cup C)$ and show by counterexample that $-$ is not associative.

(a) Convert every difference to an intersection with a complement, then push the complement inward with De Morgan:

$$\begin{aligned} A - (B - C) &= A - (B \cap C') = A \cap (B \cap C')' = A \cap (B' \cup C) \\ &= (A \cap B') \cup (A \cap C) = (A - B) \cup (A \cap C). \quad \blacksquare \end{aligned}$$

(b) The same three moves, one step shorter:

$$(A - B) - C = (A \cap B') \cap C' = A \cap (B' \cap C') = A \cap (B \cup C)' = A - (B \cup C).$$

Since $B' \cap C' = C' \cap B'$, the same computation gives $(A - C) - B$, so the two stacked orders agree with each other. $\blacksquare$

The counterexample. Take $A = \{1, 2\}$, $B = \{2\}$, $C = \{1\}$. Then

$$(A - B) - C = \{1\} - \{1\} = \emptyset, \quad A - (B - C) = \{1, 2\} - \{2\} = \{1\}.$$

Different sets, so $A - B - C$ is genuinely ambiguous and the brackets are compulsory. Notice the two answers differ by exactly $A \cap C = \{1\}$ — precisely the term part (a) says $A - (B - C)$ adds back.

Practice.

- Write $(A - B) - C$ as a single difference.
- Prove $A - (B \cap C) = (A - B) \cup (A - C)$ by element-chasing.
- With $A = \{1, 2, 3, 4\}$, $B = \{2, 3, 5\}$, $C = \{3, 4, 6\}$, compute $A - (B - C)$ and $(A - B) - C$ directly, and check both against the identities above.
- True or false: $A - B - C$ is unambiguous without brackets. Justify.
- Show $A - B = (A \cup B) - B$, and say in one line why $A - B = A - (A \cap B)$.
- Using Addendum F's region numbering, give the region lists of $A - (B - C)$ and of $(A - B) - C$.

7. Show that $A \triangle B \triangle C$ is the set of elements lying in an odd number of A, B, C , and give its region list.
8. Simplify $(A - B) \cup (B - A) \cup (A \cap B)$.

Answers. 1. $A - (B \cup C)$, which also equals $(A - C) - B$. 2. $x \in A - (B \cap C)$ means $x \in A$ and NOT($x \in B$ and $x \in C$), i.e. $x \in A$ and ($x \notin B$ or $x \notin C$), i.e. $x \in A - B$ or $x \in A - C$. Every step is an iff. 3. $B - C = \{2, 5\}$, so $A - (B - C) = \{1, 3, 4\}$; and $(A - B) \cup (A \cap C) = \{1, 4\} \cup \{3, 4\} = \{1, 3, 4\}$ $\square$. $(A - B) - C = \{1, 4\} - C = \{1\}$; and $A - (B \cup C) = \{1, 2, 3, 4\} - \{2, 3, 4, 5, 6\} = \{1\}$ $\square$. The two stacked orders differ, as M2 promised. 4. **False.** The counterexample of M2 ($A = \{1, 2\}$, $B = \{2\}$, $C = \{1\}$) gives $\emptyset$ one way and $\{1\}$ the other. Difference is neither commutative nor associative; union and intersection are both. 5. $(A \cup B) - B = (A \cup B) \cap B' = (A \cap B') \cup (B \cap B') = (A - B) \cup \emptyset = A - B$. And $A - (A \cap B)$ removes from A exactly the elements of A that lie in B — the only elements B could ever have removed — so it removes the same thing as $A - B$. 6. $A - (B - C) = \{R_1, R_6, R_7\}$ and $(A - B) - C = \{R_1\}$. The first is strictly larger, by the two regions of $A \cap C$. 7. Membership of $A \triangle B$ means “in exactly one of A, B ”, i.e. the parity of the count is odd; taking $\triangle$ with C flips that parity again exactly when $x \in C$. So x survives precisely when it lies in an odd number of the three. Region list: $\{R_1, R_2, R_3, R_7\}$ — the three private crescents and the centre. 8. $A \cup B$. The three pieces are the disjoint slabs of Subtopic 03’s four-region picture other than “neither”, so together they fill the union.

Cheat sheet for this addendum

- **Nested rosters:** an entry is one object. $n(\{1, \{2\}, \{3, 4\}, \emptyset\}) = 4$. $\emptyset \subseteq A$ always; $\emptyset \in A$ only if listed.
- $\subseteq$ **chains**, $\in$ **does not**. $A \subseteq B \subseteq C \Rightarrow A \subseteq C$ is legal. $x \in A \in B \Rightarrow x \in B$ is not. The one legal mixed chain is $A \in B \subseteq C \Rightarrow A \in C$.
- **Power set laws:** $P(A) \cap P(B) = P(A \cap B)$ always; $P(A) \cup P(B) \subseteq P(A \cup B)$, equal only when one of A, B contains the other. **“Cap behaves, cup does not.”**
- **Power set counts:** $n[P(A) \cap P(B)] = 2^c$; $n[P(A) \cup P(B)] = 2^a + 2^b - 2^c$; $n[P(A \cup B)] = 2^{a+b-c}$, where $c = n(A \cap B)$.
- $A \subseteq B$ **has four faces:** $A \subseteq B \iff A - B = \emptyset \iff A \cup B = B \iff A \cap B = A$. Prove whichever is cheapest.
- $A \cup B = A \cap B \iff A = B$.
- **Cancellation:** one hypothesis is never enough; $A \cap B = A \cap C$ and $A \cup B = A \cup C$ together give $B = C$.
- **Binomial containment tool:** $(1 + m)^n - mn - 1$ is always a non-negative multiple of m^2 . Hence $8^n - 7n - 1$ is a multiple of 49, $4^n - 3n - 1$ of 9, and $9^n - 8n - 1$ of 64.
- **One-direction containment is not equality.** After proving $X \subseteq Y$, always ask for a witness in $Y \setminus X$ before writing $X = Y$.
- **Difference is** $A - B = A \cap B'$. Rewrite first, then use De Morgan and distribute.
- **Disjoint splits:** $A \cup B = A \cup (B - A)$ and $A = (A \cap B) \cup (A - B)$ — the three-piece route to $n(A \cup B) = n(A) + n(B) - n(A \cap B)$.
- **Difference De Morgan pair:** $A - (B \cup C) = (A - B) \cap (A - C)$ and $A - (B \cap C) =$

$(A - B) \cup (A - C)$ — the operation flips.

- **Double difference gives C back:** $A - (B - C) = (A - B) \cup (A \cap C)$.
- **— is not associative.** $(A - B) - C = A - (B \cup C) = (A - C) - B$, but $A - (B - C)$ differs from it by $A \cap C$. Always bracket. ($\triangle$ is associative: $A \triangle B \triangle C =$ “in an odd number of the three” $= R_1, R_2, R_3, R_7$.)
- **Trim before you subtract:** $A - B = A - (A \cap B) = (A \cup B) - B$.

Chapter 2: Relations and Functions

Drona Ed Tech · Class 11 Mathematics

AI-Synthesized Content Bank

June 22, 2026

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Subtopic 01: Cartesian Product of Sets

Exam Relevance: A foundational, high-yield topic. CBSE Boards lifts 1-2 mark direct questions straight from here (write $A \times B$, find A and B from $A \times B$). JEE Main asks fast counting questions on $n(A \times B)$ and the number of subsets/relations. JEE Advanced rarely asks it in isolation but builds *every* later idea — relations, functions, domains, graphs — on this single definition. Master it now and the rest of the chapter becomes bookkeeping.

Section 1: Concept Intuition

Imagine you walk into a small *chaiwala's* stall near a railway platform. The board says he sells **three sizes** — Small, Medium, Large — and **two flavours** — Adrak (ginger) and Elaichi (cardamom). Before you order, a natural question pops up: *how many genuinely different cups can he serve?*

You instinctively start pairing. Small-Adrak, Small-Elaichi, Medium-Adrak, Medium-Elaichi, Large-Adrak, Large-Elaichi. Six cups. What you just did — *taking every item from one list and tagging it with every item from a second list* — is the entire idea of the **Cartesian product**. Mathematicians simply give the two lists names. Call the sizes set $A = \{S, M, L\}$ and the flavours set $B = \{\text{Adrak, Elaichi}\}$. The collection of all six possible cups is written $A \times B$, read aloud as “ A cross B ”.

The crucial detail is **order**. A cup described as (Small, Adrak) is a *size paired with a flavour*. If we flipped it to (Adrak, Small) we'd be reading it as a *flavour paired with a size* — a different kind of label entirely. So the two coordinates play different roles, and we are never allowed to swap them casually. This is why the basic building block is called an **ordered pair**: a pair (a, b) where a is the **first coordinate** and b is the **second coordinate**, and the position carries meaning.

Think about it this way... an ordered pair is like a cricket scoreline written as (runs, wickets). “180/4” — that is the ordered pair (180, 4) — means 180 runs for the loss of 4 wickets. Read it backwards as (4, 180) and you'd be claiming 4 runs for 180 wickets, which is nonsense. The numbers are the same, but the *order* completely changes the meaning. That is the heart of why (a, b) and (b, a) are different objects whenever $a \neq b$.

Formally, two ordered pairs are equal **only when both coordinates match in their own slots**:

$$(a, b) = (c, d) \iff a = c \text{ and } b = d$$

Now the definition writes itself. Given two non-empty sets A and B , the **Cartesian product** $A \times B$ is the set of *all* ordered pairs whose first coordinate is pulled from A and whose second coordinate is pulled from B :

$$A \times B = \{(a, b) : a \in A \text{ and } b \in B\}$$

Because order matters, $A \times B$ and $B \times A$ are generally **different sets** — in our chai example, $A \times B$ pairs *sizes-then-flavours* while $B \times A$ pairs *flavours-then-sizes*. They contain the same “information” but packaged in opposite slots, so they are not equal as sets.

A very natural picture is the points of a graph. If A and B are both the set of real numbers $\mathbb{R}$, then $\mathbb{R} \times \mathbb{R}$ is the set of all pairs (x, y) — which is exactly every point on the coordinate plane. So the familiar XY-plane you’ve drawn since childhood is a Cartesian product. Descartes’ name is attached to both for the same reason.

Suggested diagram. A 2-column “lattice” / arrow visualization. Left column lists elements of $A = \{S, M, L\}$ on a vertical axis; bottom row lists $B = \{\text{Adrak}, \text{Elaichi}\}$ on a horizontal axis. Show all 6 grid intersection points labelled with their ordered pairs (S, Adrak) , $(S, \text{Elaichi})$, (M, Adrak) , etc. Caption: “Every element of A meets every element of B exactly once — that grid of meeting points is $A \times B$.”

Limiting Conditions & Assumptions. A few boundaries are worth nailing down before we compute anything:

- **The empty set collapses everything.** If *either* A or B is empty, there is no way to form a single pair, so $A \times B = \emptyset$. In symbols, $A \times B = \emptyset \iff A = \emptyset$ or $B = \emptyset$.
- **Infinite sets stay infinite.** If one of the sets is infinite and the other is non-empty, $A \times B$ is infinite. (This is why $\mathbb{R} \times \mathbb{R}$ — the plane — has uncountably many points.)
- **Distinctness matters for counting.** The clean count $n(A \times B) = n(A)n(B)$ assumes the elements *within* each set are distinct (sets never repeat elements anyway, but watch out when a problem hands you a list with repeats).
- $A \times B \neq B \times A$ **in general** — equality holds only in the special case $A = B$ (for non-empty sets), or trivially when one set is empty.

Section 2: Key Definitions, Formulas & Notation

Since this is pure set algebra, there are no physical dimensions to track. The analogue of “must-state metadata” here is the **cardinality** (element count) and the **coordinate-membership** rule. Memorize these.

Ordered pair. (a, b) — first coordinate a , second coordinate b . Equality rule:

$$(a, b) = (c, d) \iff a = c \text{ and } b = d$$

Cartesian product (two sets).

$$A \times B = \{(a, b) : a \in A, b \in B\}$$

Cardinality of a Cartesian product. If $n(A) = p$ and $n(B) = q$ (both finite), then

$$n(A \times B) = n(A) \cdot n(B) = pq$$

Membership test.

$$(x, y) \in A \times B \iff x \in A \text{ and } y \in B$$

Self-product and the n -fold product. Writing $A \times A = A^2$, and in general for n sets $A_1, A_2, \dots, A_n$:

$$A_1 \times A_2 \times \cdots \times A_n = \{(a_1, a_2, \dots, a_n) : a_i \in A_i, 1 \leq i \leq n\}$$

$$n(A_1 \times \cdots \times A_n) = n(A_1) \cdot n(A_2) \cdots n(A_n), \quad n(A^m) = (n(A))^m$$

The elements $(a_1, \dots, a_n)$ are called **ordered n -tuples**. For $n = 2$ they are ordered pairs; for $n = 3$, ordered triples. ($\mathbb{R}^3 = \mathbb{R} \times \mathbb{R} \times \mathbb{R}$ is 3-D space.)

Number of subsets of $A \times B$. Since $A \times B$ has pq elements, the number of its subsets is

$$2^{n(A \times B)} = 2^{pq}$$

(Each such subset will later be christened a *relation* from A to B — the bridge to the next subtopic.)

Algebraic identities (Cartesian product distributes over set operations). For any sets A, B, C :

$$A \times (B \cup C) = (A \times B) \cup (A \times C)$$

$$A \times (B \cap C) = (A \times B) \cap (A \times C)$$

$$A \times (B - C) = (A \times B) - (A \times C)$$

$$(A \times B) \cap (C \times D) = (A \cap C) \times (B \cap D)$$

Symbol	Meaning
(a, b)	ordered pair; first coordinate a , second b
$A \times B$	Cartesian product of A and B
$n(X)$	number of elements (cardinality) of set X
A^2, A^m	$A \times A$, and the m -fold self-product
$\emptyset$	empty set

Section 3: Key Theorems & Proofs

Two results are worth proving in full because Boards and JEE both lean on them, and the *reasoning* (not the formula) is what protects you in a trap.

Theorem 1 — Counting: $n(A \times B) = n(A) \cdot n(B)$

Claim. If A and B are finite sets with $n(A) = m$ and $n(B) = n$, then $n(A \times B) = mn$.

Proof. Write the elements explicitly:

$$A = \{a_1, a_2, \dots, a_m\}, \quad B = \{b_1, b_2, \dots, b_n\}.$$

Step 1 — slice the product by first coordinate. Group all pairs of $A \times B$ according to which element of A sits in the first slot. Fix one element a_i . The pairs whose first coordinate is a_i are

$$\{a_i\} \times B = \{(a_i, b_1), (a_i, b_2), \dots, (a_i, b_n)\}.$$

Why this step? Because every pair in $A \times B$ has *exactly one* first coordinate, so this slicing sorts all pairs into clean, separate groups with nothing left over and nothing double-counted.

Step 2 — count one slice. The set $\{a_i\} \times B$ has exactly as many pairs as B has elements, namely n . (One pair per choice of second coordinate.)

Step 3 — the slices don't overlap. For two different first coordinates $a_i \neq a_k$, a pair can't simultaneously start with a_i and with a_k , so

$$(\{a_i\} \times B) \cap (\{a_k\} \times B) = \emptyset.$$

Why this matters: disjoint groups let us simply add their sizes.

Step 4 — add up. There are m slices, each of size n , all disjoint, and together they form all of $A \times B$:

$$n(A \times B) = \underbrace{n + n + \dots + n}_{m \text{ times}} = mn. \quad \blacksquare$$

This is just the **multiplication principle of counting** dressed in set notation — “ m independent choices followed by n independent choices give mn outcomes.”

Theorem 2 — Distributivity: $A \times (B \cap C) = (A \times B) \cap (A \times C)$

The standard way to prove two sets are equal is to show each is a subset of the other — i.e., take an arbitrary element of the left side and show it lies on the right, and vice versa. Here a single biconditional chain does both directions at once:

$$\begin{aligned} (x, y) \in A \times (B \cap C) &\iff x \in A \text{ and } y \in (B \cap C) \\ &\iff x \in A \text{ and } (y \in B \text{ and } y \in C) \\ &\iff (x \in A \text{ and } y \in B) \text{ and } (x \in A \text{ and } y \in C) \\ &\iff (x, y) \in (A \times B) \text{ and } (x, y) \in (A \times C) \\ &\iff (x, y) \in (A \times B) \cap (A \times C). \end{aligned}$$

Every step is reversible, so the two sets contain *exactly* the same pairs. $\blacksquare$

The lone subtle move is the third line: we duplicated the condition “ $x \in A$ ” so it could attach to each branch. This works because “ $x \in A$ and (P and Q)” is logically identical to “($x \in A$ and P) and ($x \in A$ and Q)” — the distributive law of logic mirrors the distributive law of sets.

Procedure — Recovering A and B from a given $A \times B$

A favourite exam manoeuvre runs the definition *backwards*: you are handed the product and asked for the sets.

1. **Harvest first coordinates.** Collect the first entry of every ordered pair; the distinct values form A .

2. **Harvest second coordinates.** Collect the second entry of every pair; the distinct values form B .
3. **Sanity check with the count.** Verify $n(A) \cdot n(B)$ equals the number of pairs you were given. If it doesn't match, either a pair is missing or the set wasn't a genuine Cartesian product.

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

Q. Let $A = \{2, 3, 5\}$ and $B = \{x, y\}$. Write $A \times B$ and $B \times A$, state $n(A \times B)$, and verify that $A \times B \neq B \times A$.

Solution. Given: $A = \{2, 3, 5\}$, $B = \{x, y\}$.

Pair every element of A with every element of B (first coordinate from A):

$$A \times B = \{(2, x), (2, y), (3, x), (3, y), (5, x), (5, y)\}$$

Now first coordinate from B :

$$B \times A = \{(x, 2), (x, 3), (x, 5), (y, 2), (y, 3), (y, 5)\}$$

Count: $n(A \times B) = n(A) \cdot n(B) = 3 \times 2 = 6$. ✓ (And $n(B \times A) = 2 \times 3 = 6$ as well.)

A pair such as $(2, x)$ lies in $A \times B$ but **not** in $B \times A$ (where the first coordinate must come from B). Since the two sets share no common element here,

$$A \times B \neq B \times A. \quad \blacksquare$$

Example 2 — [JEE Main — Speed Trap]

Q. $A = \{1, 2\}$ and $B = \{1, 2, 3\}$. How many ordered pairs are **common** to $A \times B$ and $B \times A$?

The trap. Most students brute-force it: list all 6 pairs of $A \times B$, all 6 of $B \times A$, then eyeball matches. Slow and error-prone. A second trap is to remember " $A \times B \neq B \times A$ " and wrongly conclude the answer is 0.

Speed move. Use the identity $(A \times B) \cap (C \times D) = (A \cap C) \times (B \cap D)$ with $C = B$, $D = A$:

$$(A \times B) \cap (B \times A) = (A \cap B) \times (B \cap A) = (A \cap B) \times (A \cap B).$$

Here $A \cap B = \{1, 2\}$, so the common pairs are $\{1, 2\} \times \{1, 2\}$, giving

$$n = 2 \times 2 = 4.$$

Answer: 4 common pairs — namely $(1, 1), (1, 2), (2, 1), (2, 2)$ — obtained in one line. (Verified by direct enumeration.) ■

Example 3 — [JEE Main Level]

Q. The set $A \times A$ has 16 elements, and two of them are $(1, 4)$ and $(3, 2)$. Find A . Then find the number of subsets of $A \times A$ that **contain** the pair $(4, 1)$.

Solution. This links three ideas — cardinality, coordinate-membership, and power-set counting.

Step 1 — find $n(A)$. Since $n(A \times A) = (n(A))^2 = 16$, we get $n(A) = 4$.

Step 2 — find the elements of A . If $(1, 4) \in A \times A$ then both $1 \in A$ and $4 \in A$. If $(3, 2) \in A \times A$ then $3 \in A$ and $2 \in A$. That already forces the four distinct elements $\{1, 2, 3, 4\} \subseteq A$. But $n(A) = 4$, so there is no room for more:

$$A = \{1, 2, 3, 4\}.$$

Step 3 — count subsets containing a fixed element. $A \times A$ has 16 elements. We want subsets that *must* include $(4, 1)$ (which is indeed in $A \times A$ since $4, 1 \in A$). Fix $(4, 1)$ as “always in”; each of the remaining 15 pairs is independently in-or-out:

$$\text{number of such subsets} = 2^{16-1} = 2^{15} = 32,768. \quad \blacksquare$$

Example 4 — [JEE Advanced Level]

Q. Let $A = \{1, 2, 3, 4, 5, 6\}$, so $A \times A$ has 36 ordered pairs. Find

- (a) the number of pairs $(x, y) \in A \times A$ such that x divides y ;
- (b) the number of pairs $(x, y) \in A \times A$ such that $x \mid y$ **or** $y \mid x$.

Solution.

(a) Slice by the first coordinate x and count how many $y \in A$ are multiples of x . The number of multiples of x in $\{1, \dots, 6\}$ is $\lfloor \frac{6}{x} \rfloor$:

x	multiples of x in A	count $\lfloor 6/x \rfloor$
1	1,2,3,4,5,6	6
2	2,4,6	3
3	3,6	2
4	4	1
5	5	1
6	6	1

$$\#\{x \mid y\} = 6 + 3 + 2 + 1 + 1 + 1 = 14.$$

(b) Use inclusion-exclusion. By the symmetry $x \leftrightarrow y$, the count of pairs with $y \mid x$ is also 14. Their overlap is the set of pairs satisfying *both* $x \mid y$ and $y \mid x$, which forces $x = y$ — there are exactly 6 such pairs $(1, 1), \dots, (6, 6)$. Hence

$$\#\{x \mid y \text{ or } y \mid x\} = 14 + 14 - 6 = 22.$$

Answers: (a) 14, (b) 22. (Both confirmed by exhaustive check over the 36 pairs.) The elegance is that a divisibility question becomes a clean counting problem the instant you view the domain as $A \times A$. $\blacksquare$

Section 5: Practice Exercises

1. **[CBSE]** If $A = \{a, b\}$ and $B = \{1, 2, 3\}$, write $A \times B$ and state $n(A \times B)$.
2. **[CBSE]** Given $A \times B = \{(p, 1), (p, 2), (q, 1), (q, 2)\}$, find A and B .
3. **[JEE Main]** If $n(A) = 5$ and $n(B) = 4$, find the number of subsets of $A \times B$.
4. **[JEE Main]** Let $A = \{x : x^2 - 5x + 6 = 0\}$ and $B = \{2, 4\}$. Write $A \times B$ and $B \times A$. Are they equal?
5. **[JEE Advanced]** Let $A = \{1, 2, 3, 4, 5\}$. Find the number of ordered pairs $(x, y) \in A \times A$ such that $x + y$ is a multiple of 3.

Answers:

1. $A \times B = \{(a, 1), (a, 2), (a, 3), (b, 1), (b, 2), (b, 3)\}$; $n(A \times B) = 6$.
2. $A = \{p, q\}$, $B = \{1, 2\}$.
3. $2^{5 \times 4} = 2^{20} = 1,048,576$.
4. $A = \{2, 3\}$ (roots of $x^2 - 5x + 6$); $A \times B$ and $B \times A$ each have 4 pairs but differ — e.g. $(2, 4) \in A \times B$ whereas $(4, 2) \in B \times A$ — so they are **not equal**.
5. 9.

Section 6: Crack the MCQ

Q1. If $n(A) = 3$ and $n(B) = 4$, then $n(A \times B)$ equals

- (a) 7 (b) 12 (c) 64 (d) 81

Answer: (b) 12. The rule is $n(A \times B) = n(A) \cdot n(B) = 3 \times 4 = 12$.

- (a) 7 — the classic slip of **adding** $3 + 4$ instead of multiplying. Cartesian product pairs *every* element with every other, so it is multiplicative.
- (c) $64 = 4^3$ and (d) $81 = 3^4$ — these confuse $n(A \times B)$ with the *number of functions* $(n(B))^{n(A)}$ or a power-set-style exponent. Those formulas belong to later subtopics, not to counting pairs.

Q2. For non-empty sets A and B , $A \times B = B \times A$ holds if and only if

- (a) $A \subseteq B$ (b) $A = B$ (c) $A \cap B = \emptyset$ (d) A and B are both infinite

Answer: (b) $A = B$. Order in the pairs is rigid, so the products match only when the sets feeding both coordinates are identical.

- (a) $A \subseteq B$ — a subset relation isn't enough; e.g. $A = \{1\}$, $B = \{1, 2\}$ gives unequal products. A trap for students who recall "containment" loosely.
- (c) $A \cap B = \emptyset$ — disjointness actually makes the products share *no* pairs (except both being empty), the opposite of equality.
- (d) *infinite* — size is irrelevant; $\mathbb{R} \times \mathbb{Z} \neq \mathbb{Z} \times \mathbb{R}$ even though both factors are infinite.

Q3. If $n(A) = 2$, the number of subsets of $A \times A$ is

- (a) 2 (b) 4 (c) 8 (d) 16

Answer: (d) 16. First $n(A \times A) = 2^2 = 4$; the number of subsets of a 4-element set is $2^4 = 16$.

- (b) 4 — stopping one step early, reporting $n(A \times A)$ itself instead of its subset count. The two-stage structure (count pairs, *then* exponentiate) is the trap.
- (a) 2 and (c) 8 — mis-tracking the exponent (2^1 or 2^3), usually from miscounting $n(A \times A)$.

Q4. Let $A = \{1, 2\}$ and $B = \{3, 4\}$. Which of the following is **NOT** an element of $A \times B$?

- (a) $(1, 3)$ (b) $(2, 4)$ (c) $(3, 1)$ (d) $(1, 4)$

Answer: (c) $(3, 1)$. In $A \times B$ the first coordinate must come from $A = \{1, 2\}$ and the second from $B = \{3, 4\}$. The pair $(3, 1)$ has its coordinates reversed — it belongs to $B \times A$, not $A \times B$.

- (a), (b), (d) are all legitimate members (first entry from A , second from B). The trap is forgetting that order is fixed; a student who treats $(3, 1)$ and $(1, 3)$ as “the same pair” picks wrongly.

Section 7: Common Student Pitfalls & Pro-Tips

- **Adding instead of multiplying.** The single most common error: writing $n(A \times B) = n(A) + n(B)$. Pairing is *every-with-every*, so it is always a product. Quick gut check: with $n(A) = n(B) = 2$ you should get 4 pairs, not 4... wait — verify by actually listing once, early in your prep, and the muscle memory sticks.
- **Treating (a, b) and (b, a) as identical.** They are equal *only* when $a = b$. Reversing coordinates is the most-exploited distractor on the topic. Always read which set feeds the first slot.
- **The “one missing factor” oversight in counting subsets.** Number of subsets of $A \times B$ is $2^{n(A)n(B)}$, not $2^{n(A)+n(B)}$ and not $n(A) \cdot n(B)$. Compute $n(A \times B)$ first, *then* raise 2 to that power.
- **Ignoring the empty-set edge case.** If a problem sneaks in $A = \emptyset$, immediately $A \times B = \emptyset$ — don’t waste time pairing.

Pro-Tip (speed). When a question gives you $A \times A$ ’s element count and asks for A , take the square root *first*: $n(A) = \sqrt{n(A \times A)}$ instantly tells you the size, and the listed pairs hand you the actual elements. For “common pairs of $A \times B$ and $B \times A$ ”, skip enumeration entirely — the answer is $(n(A \cap B))^2$.

Section 8: Cheat Sheet Box

CARTESIAN PRODUCT — RAPID RECALL

- **Definition:** $A \times B = \{(a, b) : a \in A, b \in B\}$. Order is sacred: $(a, b) = (c, d) \iff a = c, b = d$.
- **Count:** $n(A \times B) = n(A) \cdot n(B)$; $n(A^m) = (n(A))^m$.
- **Subsets of $A \times B$:** $2^{n(A)n(B)}$ (these become the *relations*).
- **Not commutative:** $A \times B = B \times A \iff A = B$ (non-empty).

- **Empty rule:** $A \times B = \emptyset \iff A = \emptyset \text{ or } B = \emptyset$.
- **Distributes:** $A \times (B \cap C) = (A \times B) \cap (A \times C)$; same for $\cup$ and $-$. Also $(A \times B) \cap (C \times D) = (A \cap C) \times (B \cap D)$.

Memory aids:

- “Every-with-every” $\rightarrow$ always **multiply**, never add.
- “Cross flips the slots” $\rightarrow (a, b) \neq (b, a)$ unless equal.
- “Square-root to size, list to elements” $\rightarrow$ recovering A from $A \times A$.
- $\mathbb{R} \times \mathbb{R}$ = the XY-plane; $\mathbb{R}^3$ = space. Coordinates *are* ordered pairs/triples.

Subtopic 02: Relations

Exam Relevance: Pure-volume territory for CBSE Boards — almost every year carries a 1-2 mark question on writing a relation in roster form or stating its domain and range, plus the count 2^{pq} . JEE Main turns the counting into quick objective questions. (The deeper machinery — reflexive/symmetric/transitive and equivalence relations — lives in a later subtopic; here we build the foundation everything else stands on.)

Section 1: Concept Intuition

Picture the WhatsApp contact list on your phone. On one side sits the set of **people** you know; on the other, the set of **phone numbers**. Not every person is linked to every number — you’ve connected *specific* people to *specific* numbers. That hand-picked set of “person $\rightarrow$ number” links is exactly what mathematicians call a **relation**.

Here is the key realization. In the previous subtopic we built $A \times B$ — the set of *all possible* ordered pairs, every person matched with every number, most of which are meaningless. A relation is simply the **meaningful subset** you carve out of that giant grid by stating a rule. Formally:

A **relation R from a set A to a set B** is any subset of $A \times B$.

That’s the whole definition. A relation isn’t some new exotic object — it’s just a chosen portion of the Cartesian product. “Choosing a portion” is what the rule does for you. If $A = \{\text{Riya, Aman, Sara}\}$ and $B = \{\text{their roll numbers}\}$, the rule “is assigned roll number” picks out three specific pairs from the nine in $A \times B$, and those three pairs *are* the relation.

Think about it this way... $A \times B$ is the full Mumbai local-train timetable — every train against every station, an enormous grid. A relation is the highlighter you run over it: “trains that *actually stop* at Dadar.” You haven’t created new trains; you’ve selected the relevant pairs. Different highlighters (different rules) give different relations on the same grid.

When $(a, b) \in R$ we say “ **a is related to b** ” and write $a R b$. When it’s not, we write $(a, b) \notin R$.

Three companion ideas travel everywhere with a relation:

- **Domain** — the set of all *first* coordinates that actually show up in R . These are the elements of A that get used. (In the contacts analogy: the people who actually have a number saved.)
- **Range** — the set of all *second* coordinates that actually show up in R . (The numbers that are actually in use.)
- **Codomain** — the *entire* set B that second coordinates are *allowed* to come from, whether or not every element gets used.

The distinction between **range** and **codomain** is the single most slippery point here. Codomain is the *whole target set you declared*; range is the *part of it that the relation actually reaches*. Range is always a subset of codomain, and the two

need not be equal. If you declare $B = \{1, 2, \dots, 10\}$ as your codomain but your relation only ever outputs $\{2, 4, 6\}$, then the range is $\{2, 4, 6\}$ while the codomain stays $\{1, \dots, 10\}$.

A relation can be written three ways:

- **Roster (list) form:** spell out every ordered pair, e.g. $R = \{(1, 3), (2, 4), (3, 5)\}$.
- **Set-builder form:** state the rule, e.g. $R = \{(x, y) : y = x + 2, x, y \in A\}$.
- **Arrow diagram:** draw A and B as two columns of dots and draw an arrow from a to b whenever $a R b$.

Suggested diagram. An arrow diagram. Left oval labelled A contains dots 1,2,3,4; right oval labelled B contains dots 3,4,5,6,7. Arrows drawn $1 \rightarrow 3, 2 \rightarrow 4, 3 \rightarrow 5, 4 \rightarrow 6$ (rule $y = x + 2$). Annotate: domain = $\{1, 2, 3, 4\}$ (left dots with an arrow leaving), range = $\{3, 4, 5, 6\}$ (right dots with an arrow arriving), codomain = all of $B = \{3, 4, 5, 6, 7\}$, with element 7 circled as “in codomain, not in range”.

Limiting Conditions & Assumptions.

- **Empty relation is legal.** $R = \emptyset$ is a perfectly valid relation (the empty subset of $A \times B$). Its domain and range are both $\emptyset$.
- **Universal relation is legal.** $R = A \times B$ itself is a relation — the one where *everything* is related to everything.
- **A relation on a set A** means a relation from A to itself, i.e. a subset of $A \times A$.
- **Range $\subseteq$ codomain, always** — but domain $\subseteq A$ likewise (a relation needn't use every element of A).
- Roster, arrow-diagram, and tabular forms only work when A and B are **finite**; for infinite sets you *must* use set-builder form.

Section 2: Key Definitions, Formulas & Notation

There are no physical dimensions here; the quantities to track are **set memberships** and **cardinalities**.

Relation.

$$R \text{ is a relation from } A \text{ to } B \iff R \subseteq A \times B$$

$$a R b \iff (a, b) \in R$$

Domain, Range, Codomain.

$$\text{Dom}(R) = \{a : (a, b) \in R \text{ for some } b\} \subseteq A$$

$$\text{Range}(R) = \{b : (a, b) \in R \text{ for some } a\} \subseteq B$$

$$\text{Codomain}(R) = B, \quad \text{Range}(R) \subseteq \text{Codomain}(R)$$

Special relations.

$$\text{Empty (void) relation: } R = \emptyset$$

$$\text{Universal relation: } R = A \times B$$

Number of relations. If $n(A) = p$ and $n(B) = q$, then $A \times B$ has pq elements, and a relation is *any* subset of it:

$$\text{Number of relations from } A \text{ to } B = 2^{n(A)n(B)} = 2^{pq}$$

For a relation **on** A (subset of $A \times A$), this becomes $2^{n(A)^2}$.

Symbol	Meaning
$R \subseteq A \times B$	R is a relation from A to B
$a R b$	a is related to b , i.e. $(a, b) \in R$
$\text{Dom}(R)$	set of first coordinates appearing in R
$\text{Range}(R)$	set of second coordinates appearing in R
Codomain	the whole declared target set B
2^{pq}	number of relations from A (p elts) to B (q elts)

Section 3: Key Procedures & Logical Framework

There is no single “derivation” for the definition of a relation, but three procedures carry every exam question — plus one clean count to prove.

Procedure A — Rule $\rightarrow$ Roster form (and reading off domain/range)

1. **List the playground.** Identify A and B (or $A \times A$ for a relation on A).
2. **Apply the rule as a filter.** Walk through candidate pairs and keep only those satisfying the rule. With a rule like $y = x + 2$, it’s faster to *generate*: for each valid $x \in A$, compute y and keep the pair only if $y \in B$.
3. **Discard out-of-set outputs.** A value of y produced by the rule but *not* lying in B is thrown away — this is where careless students add phantom pairs.
4. **Read domain and range off the finished list.** Domain = distinct first coordinates; range = distinct second coordinates.

Why step 3 matters: a relation lives inside $A \times B$. Any pair whose coordinate escapes the declared sets simply isn’t allowed in, no matter what the rule says.

Procedure B — Roster $\rightarrow$ Set-builder form

1. **Spot the pattern** linking each first coordinate to its second ($y = x + 1$? x divides y ? $x + y = 10$?).
2. **Write the rule with quantified membership**, e.g. $R = \{(x, y) : y = x + 1, x \in A, y \in B\}$.
3. **Verify both ways** — every listed pair must satisfy the rule, and the rule must produce no pair you didn’t list.

Procedure C — Counting relations (the 2^{pq} logic)

Claim. If $n(A) = p$ and $n(B) = q$, the number of relations from A to B is 2^{pq} .

Reasoning. A relation is a subset of $A \times B$. By the Cartesian-product count (Sub-01), $A \times B$ has exactly pq ordered pairs. To build a subset, each of those pq pairs faces an *independent* yes/no decision — “is this pair in R or not?” That’s 2 choices per pair, made pq times independently, so by the multiplication principle:

$$\underbrace{2 \times 2 \times \cdots \times 2}_{pq \text{ factors}} = 2^{pq}.$$

This count **includes** both extreme cases — the empty relation (all “no”) and the universal relation $A \times B$ (all “yes”). So the number of *non-empty* relations is $2^{pq} - 1$. ■

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

Q. Let $A = \{1, 2, 3, 4, 5, 6\}$. A relation R on A is defined by $R = \{(x, y) : y = x + 2, x, y \in A\}$. Write R in roster form and state its domain, range, and codomain.

Solution. *Generate pairs:* for each x , compute $y = x + 2$ and keep only if $y \in A$.

$x = 1 \Rightarrow y = 3$ ✓; $x = 2 \Rightarrow y = 4$ ✓; $x = 3 \Rightarrow y = 5$ ✓; $x = 4 \Rightarrow y = 6$ ✓;
 $x = 5 \Rightarrow y = 7 \notin A$ ✗; $x = 6 \Rightarrow y = 8 \notin A$ ✗.

$$R = \{(1, 3), (2, 4), (3, 5), (4, 6)\}$$

- **Domain** (first coordinates) = $\{1, 2, 3, 4\}$
- **Range** (second coordinates) = $\{3, 4, 5, 6\}$
- **Codomain** = $A = \{1, 2, 3, 4, 5, 6\}$

Note that $5, 6 \in A$ never appear as first coordinates (the rule pushes them out of range), so they're in A but not in the domain — a clean illustration that domain $\subseteq A$ need not be all of A . ■

Example 2 — [JEE Main — Speed Trap]

Q. Set A has 3 elements and set B has 4 elements. How many relations can be defined from A to B ? How many of them are **non-empty**?

The trap. The two reflex wrong answers are $3 \times 4 = 12$ (that's the number of *ordered pairs*, $n(A \times B)$, not relations) and $2^{3+4} = 128$ (adding instead of multiplying in the exponent). Both come from not remembering that a *relation* is a *subset* of $A \times B$.

Speed move. Relations = $2^{n(A)n(B)} = 2^{3 \times 4} = 2^{12}$.

$$\text{Total relations} = 2^{12} = 4096.$$

The empty relation is one of these, so

$$\text{Non-empty relations} = 4096 - 1 = 4095.$$

Answers: 4096 and 4095. ■

Example 3 — [JEE Main Level]

Q. A relation R is defined on the set of integers $\mathbb{Z}$ by $R = \{(x, y) : x, y \in \mathbb{Z}, x^2 + y^2 = 25\}$. Write R in roster form and find its domain and range.

Solution. This fuses *relation-building* with *finding integer (lattice-point) solutions* of an equation — the second-topic hook.

Find all integer pairs with $x^2 + y^2 = 25$. Since $25 = 0^2 + 5^2 = 3^2 + 4^2$, the squares available are 0, 9, 16, 25, in the combinations $25 + 0$, $0 + 25$, $9 + 16$, $16 + 9$. Attaching all sign choices:

$$R = \{ (0, 5), (0, -5), (5, 0), (-5, 0), (3, 4), (3, -4), (-3, 4), (-3, -4), (4, 3), (4, -3), (-4, 3), (-4, -3) \}$$

That's 12 ordered pairs (geometrically, the 12 lattice points on a circle of radius 5).

- **Domain** = $\{-5, -4, -3, 0, 3, 4, 5\}$
- **Range** = $\{-5, -4, -3, 0, 3, 4, 5\}$

By the symmetry of $x^2 + y^2$ in x and y , domain and range coincide here. ■

Example 4 — [JEE Advanced Level]

Q. Let $A = \{1, 2, 3\}$. How many relations on A contain **at least one** of the “diagonal” pairs $(1, 1)$, $(2, 2)$, $(3, 3)$?

Solution. A relation on A is a subset of $A \times A$, which has $3^2 = 9$ elements, so there are $2^9 = 512$ relations in total. Counting “at least one of three pairs” directly invites messy inclusion-exclusion; **complementary counting** is cleaner.

Count the relations containing NONE of the three diagonal pairs. Force all three of $(1, 1)$, $(2, 2)$, $(3, 3)$ to be **out**. That leaves the remaining $9 - 3 = 6$ ordered pairs each free to be in or out:

$$\text{relations with no diagonal pair} = 2^6 = 64.$$

Subtract from the total:

$$\text{at least one diagonal pair} = 512 - 64 = 448.$$

Answer: 448. The lesson is structural — fixing k specific pairs as “must be in” or “must be out” freezes k of the n^2 independent yes/no switches, and the count becomes $2^{(n^2-k)}$. Here we used it through the complement. ■

Section 5: Practice Exercises

1. **[CBSE]** Let $A = \{1, 2, 3, 4\}$ and $R = \{(x, y) : y = x, x, y \in A\}$. Write R in roster form and state its domain and range.
2. **[CBSE]** A relation is given by $R = \{(2, 1), (3, 2), (4, 3), (5, 4)\}$. State its domain and range, and write R in set-builder form.
3. **[JEE Main]** If $n(A) = 4$ and $n(B) = 5$, find the total number of relations from A to B and the number of non-empty relations.
4. **[JEE Main]** Let $A = \{1, 2, 3, 4, 5, 6\}$ and $R = \{(x, y) : x + y = 7, x, y \in A\}$. Write R in roster form and find $n(R)$, its domain, and its range.
5. **[JEE Advanced]** Let $A = \{1, 2, 3, 4\}$. How many relations on A contain the pairs $(1, 2)$ and $(2, 1)$ but do **not** contain $(3, 3)$?

Answers:

1. $R = \{(1, 1), (2, 2), (3, 3), (4, 4)\}$; domain = range = $\{1, 2, 3, 4\}$.
2. Domain = $\{2, 3, 4, 5\}$, range = $\{1, 2, 3, 4\}$; $R = \{(x, y) : y = x - 1, x \in \{2, 3, 4, 5\}\}$.
3. Total = $2^{20} = 1,048,576$; non-empty = $1,048,575$.
4. $R = \{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\}$; $n(R) = 6$; domain = range = $\{1, 2, 3, 4, 5, 6\}$.
5. $2^{(16-3)} = 2^{13} = 8192$.

Section 6: Crack the MCQ

Q1. If $n(A) = 2$ and $n(B) = 3$, the number of relations from A to B is

- (a) 6 (b) 8 (c) 12 (d) 64

Answer: (d) 64. Relations = $2^{n(A)n(B)} = 2^6 = 64$.

- (a) $6 = 2 \times 3 = n(A \times B)$ — the number of *ordered pairs*, not the number of subsets. The most common slip.
- (b) $8 = 2^3$ — exponent mistakenly set to $n(B)$ alone.
- (c) 12 — a doubling-then-multiplying muddle ($2 \cdot 6$); no valid route gives this.

Q2. A relation R from a non-empty set A to a non-empty set B is

- (a) any subset of $A \cup B$ (b) any subset of $A \times B$ (c) any element of $A \times B$ (d) the set $A \times B$ itself only

Answer: (b) any subset of $A \times B$. That is the definition.

- (a) confuses the *union* of the sets with their *product*; relations live in the grid of pairs, not in $A \cup B$.
- (c) a single ordered pair is one *element*, not a relation (a relation is a *set* of such pairs — possibly one, possibly none, possibly many).
- (d) describes only the *universal* relation; that's one specific relation, not the general definition.

Q3. For $R = \{(1, 2), (2, 4), (3, 6)\}$, the range of R is

- (a) $\{1, 2, 3\}$ (b) $\{2, 4, 6\}$ (c) $\{1, 2, 3, 4, 6\}$ (d) $\{1, 2, 4, 6\}$

Answer: (b) $\{2, 4, 6\}$. Range is the set of *second* coordinates.

- (a) $\{1, 2, 3\}$ is the **domain** (first coordinates) — the classic domain/range swap.
- (c) and (d) dump first and second coordinates together; range uses only the second slot.

Q4. Let $R = \{(x, y) : x, y \in \{1, 2, 3\}, x < y\}$. Then $n(R)$ equals

- (a) 3 (b) 6 (c) 9 (d) 2

Answer: (a) 3. The pairs with $x < y$ are $(1, 2), (1, 3), (2, 3)$ — exactly 3.

- (c) 9 counts *all* of $\{1, 2, 3\} \times \{1, 2, 3\}$, ignoring the condition $x < y$.
- (b) 6 counts every pair with $x \neq y$ (both $x < y$ and $x > y$); the strict-inequality filter keeps only half.
- (d) 2 typically from missing the pair $(2, 3)$ or $(1, 3)$ while listing.

Section 7: Common Student Pitfalls & Pro-Tips

- **Domain-range swap.** Reading *second* coordinates as the domain or *first* as the range. Lock it in: **domain = left = inputs, range = right = outputs.** (This is the same “order matters in a pair” trap from Cartesian products, wearing a new costume.)
- **Confusing range with codomain.** Codomain is the *whole declared* set B ; range is *only the part actually hit*. If a question gives B explicitly and asks for range, do **not** just write B — list the second coordinates that genuinely appear.
- **pq vs 2^{pq} for counting relations.** pq is the number of *pairs*; 2^{pq} is the number of *relations*. Forgetting to exponentiate (or adding $p + q$ in the exponent) wrecks the count.
- **Phantom pairs in roster form.** Generating a y from the rule that lies *outside* B , then wrongly including it. Always check the output actually belongs to the target set.

Pro-Tip (speed). For counting questions, fix in your mind two anchors: *pairs* = pq and *relations* = 2^{pq} . When some pairs are forced in or out, freeze them — each forced pair removes one factor of 2, giving $2^{(pq-k)}$ for k forced pairs. And for “at least one” conditions, count the **complement** (“none of them”) and subtract from 2^{pq} — almost always faster than inclusion-exclusion.

Section 8: Cheat Sheet Box

RELATIONS (CLASS 11 CORE) — RAPID RECALL

- **Definition:** a relation from A to B is any subset $R \subseteq A \times B$. $(a, b) \in R \iff a R b$.
- **Domain** = first coordinates used; **Range** = second coordinates used; **Codomain** = B (whole declared target). Always $\text{Range} \subseteq \text{Codomain}$.
- **Empty relation** $R = \emptyset$; **Universal relation** $R = A \times B$ — both are valid relations.
- **Count:** number of relations from A to $B = 2^{n(A)n(B)} = 2^{pq}$; non-empty = $2^{pq} - 1$.
- **On a set A :** a relation on A is a subset of $A \times A$; there are $2^{n(A)^2}$ of them.
- **Three forms:** roster (list), set-builder (rule), arrow diagram. Roster/arrow need finite sets; set-builder works always.

Memory aids:

- “Relation = highlighted slice of $A \times B$.”
- “Domain Left, Range Right” (D-L, R-R) — never swap them.
- “Range is reached, codomain is allowed.”
- “Pairs count pq ; relations count 2^{pq} — subsets, so power of two.”

Subtopic 03: Functions

Exam Relevance: A guaranteed scorer. CBSE Boards lifts direct 1-3 mark questions on “is this a function?”, domain/range, and the standard function definitions and graphs. JEE Main loves the counting fact (q^p functions) and domain-finding for radicals and rational expressions. (The mapping-classification layer — one-one, onto, bijective, composition, inverse — is built in a later subtopic; here we install the engine.)

Section 1: Concept Intuition

You’ve already met **relations** — any highlighted slice of $A \times B$. A **function** is a relation with *discipline*. It is a relation that obeys two strict promises, and those two promises are the whole idea.

Think of a snack vending machine. Each button (input) is wired to deliver **exactly one** product (output). Two things are guaranteed: press a button and you *always* get something (no dead buttons), and you get *one* definite item, never “maybe a samosa, maybe a Frooti.” A function f from A to B is exactly this machine: every element of A (every button) is connected to **one and only one** element of B .

Formally, a **function** $f : A \rightarrow B$ is a relation from A to B in which

1. **every** element of A has an image in B (no input is left unmapped), and
2. **no** element of A has more than one image (each input gives a single output).

Compare this with an ordinary relation, which is free to skip elements of A and free to send one element to several places. A function forbids both. So **every function is a relation, but most relations are not functions**.

Think about it this way... picture the Mumbai dabbawala system. Each tiffin (input) must be delivered to *exactly one* office address (output). A tiffin with no address is a failure (violates promise 1); a tiffin tagged with two addresses is chaos (violates promise 2). The delivery assignment is a *function* precisely because it avoids both disasters. Crucially, two *different* tiffins are perfectly allowed to go to the *same* office — many inputs sharing one output is fine. What’s forbidden is one input splitting into many outputs.

The notation $y = f(x)$ reads “ y is the image of x under f .” We call x the **independent variable**, y the **dependent variable**, and $f(x)$ the **value of f at x** .

The three companions from relations sharpen here:

- **Domain** = A — for a function, the domain is the *entire* set A (promise 1 forces every element to be used).
- **Codomain** = B — the declared target.
- **Range** = $\{f(x) : x \in A\}$ — the set of images actually produced; still $\text{Range} \subseteq \text{Codomain}$.

For functions on real numbers, there’s a beautiful visual test. Draw the graph; then the **vertical line test**: if *every* vertical line meets the graph in **at most one point**, the graph is a function (each x gives one y). A circle fails it (a vertical line cuts it twice — one x , two y ’s), which is why $x^2 + y^2 = 25$ is a relation but not a function.

The Vertical Line Test

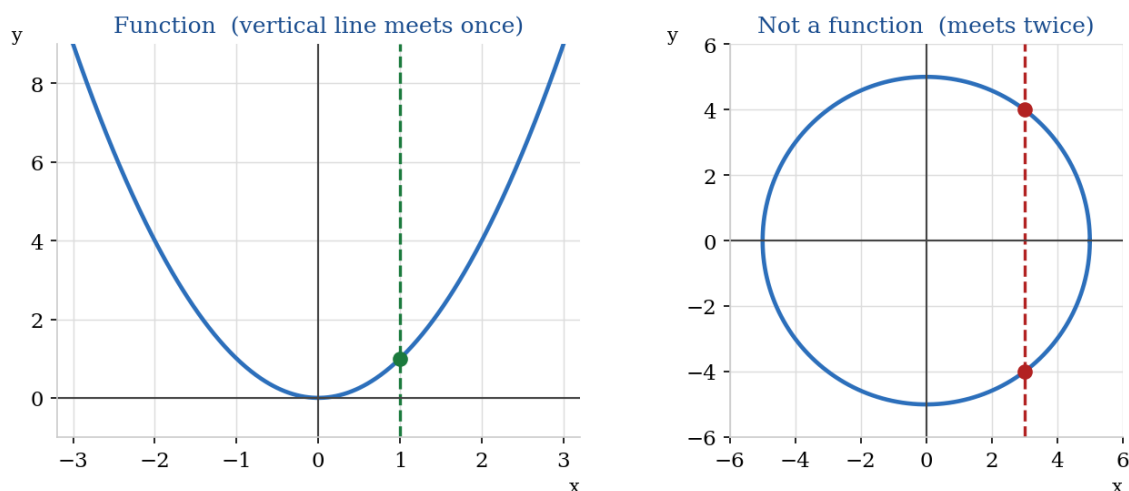


Figure 1. The vertical line test: the parabola $y = x^2$ is met by any vertical line at most once (a function); the circle $x^2 + y^2 = 25$ is met twice by some vertical lines (not a function).

Limiting Conditions & Assumptions.

- **Domain must be the whole of A .** If even one element of A has no image, it isn't a function on A — it's only a function on the smaller set of mapped elements.
- **Single-valued is non-negotiable.** $\sqrt{x}$ as " $\pm$ " is *not* a function; the function $\sqrt{}$ returns the non-negative root only.
- **Many-to-one is allowed; one-to-many is banned.** Several inputs may share an output.
- **Real function caveat:** when a rule like $f(x) = 1/x$ is given without a stated domain, the domain is taken to be all real x for which the rule makes sense (the *natural domain*) — here $\mathbb{R} \setminus \{0\}$.

Section 2: Key Definitions, Formulas & Notation

No physical dimensions; the tracked quantities are **memberships, images, and cardinalities**.

Function. $f : A \rightarrow B$ is a function $\iff f \subseteq A \times B$ with every $a \in A$ appearing as the first coordinate of *exactly one* pair.

$$\text{Domain}(f) = A, \quad \text{Codomain}(f) = B, \quad \text{Range}(f) = \{f(x) : x \in A\} \subseteq B$$

Real function / real-valued function. If $\text{Range}(f) \subseteq \mathbb{R}$, f is **real-valued**; if additionally the domain $\subseteq \mathbb{R}$, f is a **real function**.

Equality of functions. Two functions f and g are **equal** (written $f = g$) if and only if they have the *same domain* **and** $f(x) = g(x)$ for every x in that domain. The same rule on a *different* domain gives a *different* function — e.g. $f(x) = x$ on $\mathbb{R}$ and $g(x) = x$ on $[0, 1]$ are not equal.

Number of functions. If $n(A) = p$ and $n(B) = q$, each of the p inputs independently chooses one of q outputs:

$$\text{Number of functions from } A \text{ to } B = q^p = (n(B))^{n(A)}$$

(Contrast: number of *relations* was 2^{pq} . Functions are far fewer.)

The standard real functions (Class 11 zoo).

Function	Rule	Domain	Range
Identity	$f(x) = x$	$\mathbb{R}$	$\mathbb{R}$
Constant	$f(x) = c$	$\mathbb{R}$	$\{c\}$
Polynomial	$f(x) = a_0 + a_1x + \dots + a_nx^n$	$\mathbb{R}$	depends on degree
Rational	$f(x) = \frac{p(x)}{q(x)}$	$\mathbb{R} \setminus \{x : q(x) = 0\}$	depends
Modulus	$f(x) = x $	$\mathbb{R}$	$[0, \infty)$
Signum	$f(x) = \frac{ x }{x}$ for $x \neq 0$, 0 for $x = 0$	$\mathbb{R}$	$\{-1, 0, 1\}$
Greatest integer	$f(x) = [x]$ (largest integer $\leq x$)	$\mathbb{R}$	$\mathbb{Z}$

Graphs of the standard functions

A function's graph is its fingerprint: once you can picture it, the domain and range fall out by inspection. These six are the ones every Class 11 student must be able to sketch on sight.

Graphs of the Standard Functions (Class 11)

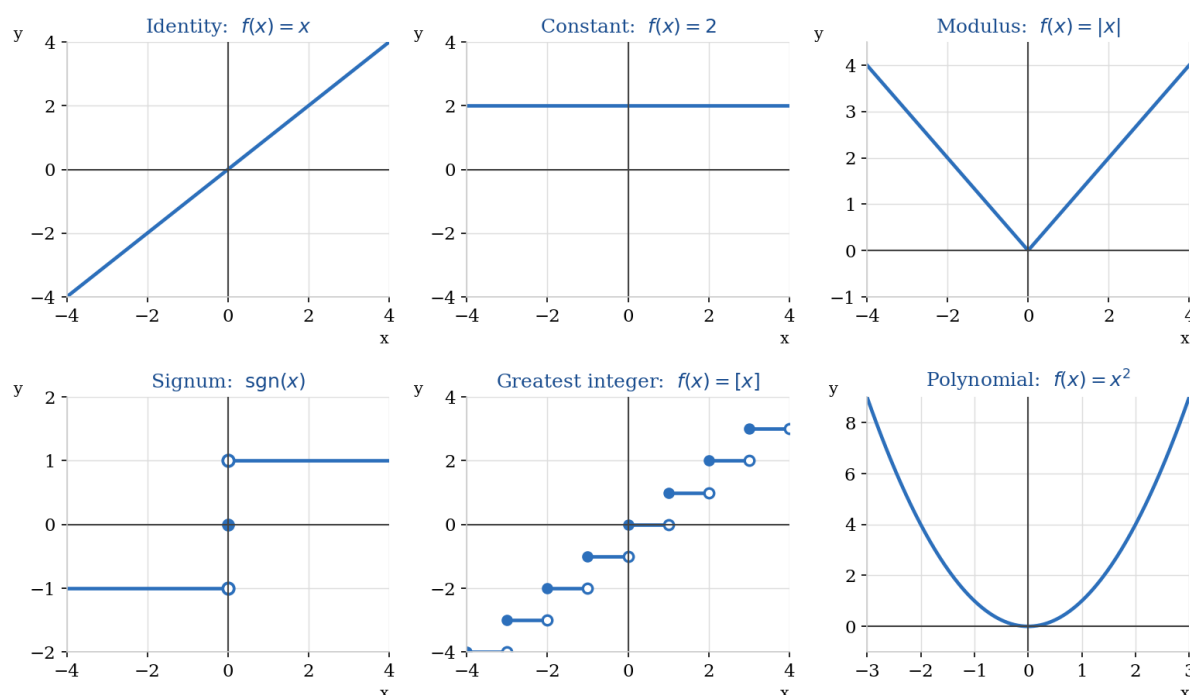


Figure 2. Graphs of the six standard functions. Note the open/filled dots on the signum and greatest-integer graphs — they mark which endpoint each step actually attains.

- **Identity** $f(x) = x$: the 45° line through the origin. Every input maps to itself; domain $\mathbb{R}$, range $\mathbb{R}$.
- **Constant** $f(x) = c$: a horizontal line at height c . Every input lands on the single value c ; range $\{c\}$.
- **Modulus** $f(x) = |x|$: a symmetric “V” with its vertex at the origin. It never dips below the x -axis, so the range is $[0, \infty)$.
- **Signum** $\text{sgn}(x)$: three flat pieces — -1 for $x < 0$, 0 at $x = 0$, $+1$ for $x > 0$ — with jumps at the origin (open circles where a value is *not* taken, the filled dot where it is). Range $\{-1, 0, 1\}$.
- **Greatest integer** $f(x) = [x]$: a left-closed staircase. On each interval $[n, n+1)$ the value is the integer n ; the filled dot (included) sits at the left, the open dot (excluded) at the right. Range $\mathbb{Z}$.
- **Polynomial** $f(x) = x^2$: an upward parabola with vertex at the origin. An even-degree polynomial never reaches negative values, so the range is $[0, \infty)$ — a concrete instance of the “even degree $\Rightarrow$ range is a subset of $\mathbb{R}$ ” rule.

Reading domain and range off a graph. Project the curve onto the axes: the shadow it casts on the x -axis is the **domain** (every x -value the graph covers), and the shadow on the y -axis is the **range** (every y -value it attains). The vertical line test first confirms the picture is a function at all.

(JEE companion — beyond the Class 11 core: the **fractional part** $\{x\} = x - [x]$ strips off the integer part of x ; it is a sawtooth with range $[0, 1)$ and is studied alongside $[x]$ in the Class 12 / JEE function toolkit.) **Algebra of real functions.** For real functions f, g and scalar α :

$$(f \pm g)(x) = f(x) \pm g(x), \quad (fg)(x) = f(x)g(x), \quad (\alpha f)(x) = \alpha f(x)$$

$$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}, \quad \text{wherever } g(x) \neq 0$$

Domains: $\text{Dom}(f \pm g) = \text{Dom}(fg) = \text{Dom}(f) \cap \text{Dom}(g)$; and $\text{Dom}(f/g) = (\text{Dom}(f) \cap \text{Dom}(g)) \setminus \{x : g(x) = 0\}$.

Section 3: Key Procedures & Logical Framework

Procedure A — Is a given relation a function?

1. **Promise 1 (totality):** check every element of A appears as a first coordinate. A missing input $\Rightarrow$ not a function on A .
2. **Promise 2 (single-valued):** check no first coordinate is repeated with *different* second coordinates. A repeat like $(3, 5)$ and $(3, 8) \Rightarrow$ not a function.
3. **Graphs:** apply the **vertical line test** — at most one intersection per vertical line.

Procedure B — Finding the (natural) domain of a real function

Work through the “danger sources” and intersect the surviving sets:

1. **Denominators** must be non-zero $\Rightarrow$ exclude roots of the denominator.
2. **Even roots** (square roots, fourth roots, ...) need a **non-negative** radicand $\Rightarrow$ solve (inside) ≥ 0 .
3. **Combine:** if several conditions apply, the domain is their **intersection** (all conditions must hold at once).

Why intersect? The rule must be simultaneously valid at every step; a value surviving one hurdle but failing another is still excluded.

Procedure C — Algebra of functions and their domains

1. Compute the new rule: add/subtract/multiply/divide the formulas.
2. For $f \pm g$ and fg : domain $= \text{Dom}(f) \cap \text{Dom}(g)$ (both must be defined to combine them).
3. For f/g : take that intersection **and then remove** every point where $g(x) = 0$ (can't divide by zero). This extra removal is the step students forget.

Procedure D — Counting functions (q^p logic)

Each of the p domain elements must be assigned exactly one of the q codomain values, and the choices are independent. By the multiplication principle:

$$\underbrace{q \times q \times \cdots \times q}_{p \text{ inputs}} = q^p.$$

There's no “yes/no per pair” here (that gave 2^{pq} for relations); instead it's “one choice of output per input,” which is why the base is q and the exponent is p . ■

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

Q. Let $A = \{1, 2, 3, 4\}$ and $B = \{2, 3, 5, 7\}$. Determine whether $R = \{(1, 2), (2, 3), (3, 2), (4, 5)\}$ is a function from A to B . If so, state its domain and range.

Solution. *Promise 1 (totality):* the first coordinates are 1, 2, 3, 4 — every element of A is present. ✓ *Promise 2 (single-valued):* no first coordinate repeats, so no input has two images. ✓ (The output 2 is shared by inputs 1 and 3 — that's *many-to-one*, which is perfectly allowed.)

Hence R is a function.

- **Domain** = $A = \{1, 2, 3, 4\}$
- **Range** = $\{2, 3, 5\}$ (distinct images; note $7 \in B$ is never hit, so it's in the codomain but not the range). ■

Example 2 — [JEE Main — Speed Trap]

Q. A has 3 elements and B has 4 elements. How many functions can be defined from A to B ?

The trap. Three reflex wrong answers compete: $2^{3 \times 4} = 4096$ (that's *relations*, not functions), $3^4 = 81$ (base and exponent swapped — assigning sets backwards), and $3 \times 4 = 12$ (just the pair count). The correct base is the number of *choices per input* = q , and the exponent is the number of *inputs* = p .

Speed move. Functions = $q^p = 4^3$.

$$4^3 = 64.$$

Answer: 64. Mnemonic to kill the swap: “each input picks an output” → outputs are the **base** (q), inputs are the **exponent** (p). ■

Example 3 — [JEE Main Level]

Q. Find the natural domain of the real function

$$f(x) = \frac{\sqrt{x-2}}{x-5}.$$

Solution. Two danger sources must hold *simultaneously* (Procedure B):

- **Square root:** $x - 2 \geq 0 \Rightarrow x \geq 2$, i.e. $x \in [2, \infty)$.
- **Denominator:** $x - 5 \neq 0 \Rightarrow x \neq 5$.

Intersect:

$$\text{Domain} = [2, \infty) \setminus \{5\} = [2, 5) \cup (5, \infty).$$

The point $x = 2$ stays in (the square root is 0 there, which is fine — only the *denominator* can't be zero); $x = 5$ is punched out. ■

Example 4 — [JEE Advanced Level]

Q. Let $f(x) = x^2$ and $g(x) = \sqrt{4-x^2}$. Find $(f+g)(x)$, $(fg)(x)$, and $\left(\frac{f}{g}\right)(x)$, and state the **exact domain** of each.

Solution. This is where careless domain bookkeeping costs marks. First pin down the individual domains:

- $\text{Dom}(f) = \mathbb{R}$ (polynomial).
- $\text{Dom}(g)$: need $4 - x^2 \geq 0 \Rightarrow x^2 \leq 4 \Rightarrow x \in [-2, 2]$.

Sum: $(f+g)(x) = x^2 + \sqrt{4-x^2}$. Domain $= \mathbb{R} \cap [-2, 2] = [-2, 2]$.

Product: $(fg)(x) = x^2\sqrt{4-x^2}$. Domain $= \mathbb{R} \cap [-2, 2] = [-2, 2]$.

Quotient: $\left(\frac{f}{g}\right)(x) = \frac{x^2}{\sqrt{4-x^2}}$. Start from $[-2, 2]$, then remove the points where $g(x) = 0$. Now $\sqrt{4-x^2} = 0$ at $x = \pm 2$, so both endpoints are punched out:

$$\text{Dom}\left(\frac{f}{g}\right) = (-2, 2).$$

The deep point: $f+g$ and fg keep the *closed* interval, but the quotient **must** shed the zeros of g , opening it to $(-2, 2)$. Forgetting to exclude ± 2 is the single most common error here. ■

Example 5 — [CBSE / JEE Main — Graph Reading]

Q. Consider $f(x) = |x-2|$, $x \in \mathbb{R}$. (a) Describe its graph. (b) State its domain and range. (c) Find $f(5)$. (d) Solve $f(x) = 3$.

Solution. This is the modulus graph from the zoo, *shifted*. **(a)** $|x|$ is a “V” with vertex at the origin; replacing x by $x-2$ slides the whole V **right by 2**, so $f(x) = |x-2|$ is a V with vertex at $(2, 0)$, arms rising with slopes ± 1 . **(b)** Project onto the axes: the graph spans all x , so **domain** $= \mathbb{R}$; it never drops below the x -axis and touches 0 at $x = 2$, so **range** $= [0, \infty)$. **(c)** $f(5) = |5-2| = |3| = 3$. **(d)** $|x-2| = 3 \Rightarrow x-2 = 3$ or $x-2 = -3 \Rightarrow x = 5$ or $x = -1$. The two solutions are the two points on the V at height 3 — exactly what the graph predicts (a horizontal line $y = 3$ cuts the V twice). ■

Section 5: Practice Exercises

1. **[CBSE]** Is $R = \{(1, 1), (2, 1), (3, 1)\}$ a function from $\{1, 2, 3\}$ to $\{1, 2\}$? State its domain and range.
2. **[CBSE]** For $f(x) = x^2$ with domain $\{-2, -1, 0, 1, 2\}$, write f in roster form and find its range.
3. **[JEE Main]** How many functions can be defined from a set of 4 elements to a set of 3 elements?
4. **[JEE Main]** Find the natural domain of $f(x) = \frac{1}{\sqrt{9-x^2}}$.

5. **[JEE Advanced]** Let $f(x) = x + 1$ and $g(x) = 2x - 3$, both on $\mathbb{R}$. Find $\left(\frac{f}{g}\right)(x)$ and its domain.
6. **[CBSE]** For the greatest-integer function $f(x) = [x]$, evaluate $f(-1.5)$, $f(2)$, and $f(3.9)$, and state the range of f .

Answers:

1. Yes (each input has exactly one image — a constant assignment); domain $= \{1, 2, 3\}$, range $= \{1\}$.
2. $f = \{(-2, 4), (-1, 1), (0, 0), (1, 1), (2, 4)\}$; range $= \{0, 1, 4\}$.
3. $3^4 = 81$.
4. Need $9 - x^2 > 0$ (strict, since it's under a root *and* in a denominator) $\Rightarrow x \in (-3, 3)$.
5. $\frac{x+1}{2x-3}$, domain $= \mathbb{R} \setminus \{\frac{3}{2}\}$.
6. $[-1.5] = -2$ (greatest integer **not exceeding** -1.5), $[2] = 2$, $[3.9] = 3$; range $= \mathbb{Z}$.

Section 6: Crack the MCQ

Q1. Which of the following relations is a function from $A = \{1, 2, 3\}$ to $B = \{a, b\}$?

- (a) $\{(1, a), (2, b), (3, a)\}$ (b) $\{(1, a), (1, b), (2, a), (3, b)\}$ (c) $\{(1, a), (2, b)\}$ (d) $\{(1, a), (2, b), (3, a), (3, b)\}$

Answer: (a). Every element of A has exactly one image. (Inputs 1 and 3 share image a — allowed.)

- (b) input 1 maps to both a and b — violates single-valuedness (one-to-many).
- (c) input 3 has no image — violates totality.
- (d) input 3 maps to both a and b — one-to-many again. The trap is treating “every output appears” as the requirement; the requirement is on *inputs*.

Q2. The natural domain of $f(x) = \sqrt{x-4}$ is

- (a) $\mathbb{R}$ (b) $(4, \infty)$ (c) $[4, \infty)$ (d) $(-\infty, 4]$

Answer: (c) $[4, \infty)$. Need $x - 4 \geq 0 \Rightarrow x \geq 4$; equality is allowed since $\sqrt{0} = 0$ is defined.

- (b) wrongly excludes $x = 4$ — a strict-vs-non-strict slip; square roots allow the radicand to be 0.
- (d) solves the inequality the wrong way ($x \leq 4$).
- (a) ignores the radical constraint entirely.

Q3. The number of functions from a set with 2 elements to a set with 5 elements is

- (a) 7 (b) 10 (c) 25 (d) 32

Answer: (c) 25. Functions $= q^p = 5^2 = 25$.

- (d) $32 = 2^5$ — base and exponent swapped (would count functions the other way, $5 \rightarrow 2$ set).

- (b) $10 = 2 \times 5$, the pair count.
- (a) $7 = 2 + 5$, an addition slip.

Q4. The range of the function $f(x) = |x|$, $x \in \mathbb{R}$, is

- (a) $\mathbb{R}$ (b) $(0, \infty)$ (c) $[0, \infty)$ (d) $\{-1, 0, 1\}$

Answer: (c) $[0, \infty)$. $|x| \geq 0$ for all real x , and every non-negative value is attained (e.g. $|0| = 0$, $|3| = 3$).

- (b) wrongly excludes 0 (but $|0| = 0$ is achieved).
- (d) is the range of the **signum** function — a classic mix-up between modulus and signum.
- (a) forgets that $|x|$ never returns a negative.

Q5. For the greatest-integer function $f(x) = [x]$, which statement is correct?

- (a) $[2.3] = 3$ (b) $[-2.3] = -2$ (c) $[-2.3] = -3$ (d) the range of $[x]$ is $\mathbb{R}$

Answer: (c) $[-2.3] = -3$. $[x]$ is the greatest integer **not exceeding** x ; the integers ≤ -2.3 are $\dots, -4, -3$, and the greatest of them is -3 .

- (a) $[2.3] = 2$, not 3 — students round *up*; the function rounds *down* (towards $-\infty$).
- (b) $-2 > -2.3$, so -2 is too big; the classic negative-number trap (thinking $[-2.3] = -2$ by “dropping the decimal”).
- (d) the range is $\mathbb{Z}$ (only integer outputs), not all of $\mathbb{R}$.

Section 7: Common Student Pitfalls & Pro-Tips

- **Checking the wrong side for “function.”** The rule constrains *inputs* (each domain element $\rightarrow$ one image), **not** outputs. Many inputs sharing an output is fine; one input with two outputs is fatal. Don’t reject a function just because some codomain element is unused or repeated.
- **q^p vs 2^{pq} , and base/exponent swap.** Relations $= 2^{pq}$; functions $= q^p$ with **outputs as base, inputs as exponent**. Swapping them (p^q) is the most-tested distractor.
- **Strict vs non-strict in domains.** Square roots allow the radicand $= 0$ (closed bracket); denominators forbid $= 0$ (open). When a radical sits in a denominator (e.g. $1/\sqrt{\cdot}$), the inside must be > 0 — strictly.
- **Forgetting to exclude zeros of g in f/g .** $\text{Dom}(f/g)$ is the intersection of domains *minus* the zeros of the denominator function. Endpoints that survive in $f + g$ often vanish in f/g .

Pro-Tip (speed). For domains, scan the formula for exactly two flags — “*is anything under an even root?*” (set ≥ 0 , or > 0 if also in a denominator) and “*is anything in a denominator?*” (set $\neq 0$) — then intersect. For counting, lock the chant “*outputs to the power of inputs*” (q^p) so you never swap. And remember the signum/modulus range pair: $\{-1, 0, 1\}$ vs $[0, \infty)$ — examiners swap these constantly.

Section 8: Cheat Sheet Box

FUNCTIONS (CLASS 11 CORE) — RAPID RECALL

- **Definition:** $f : A \rightarrow B$ is a relation where every $a \in A$ has **exactly one** image. (Every input used; single-valued.) Every function is a relation; not conversely.
- **Domain** = A (all of it); **Codomain** = B ; **Range** = $\{f(x)\} \subseteq B$. Vertical line test for graphs.
- **Count:** functions from A to $B = q^p = (n(B))^{n(A)}$. (Relations were 2^{pq} .)
- **Standard ranges:** modulus $\rightarrow [0, \infty)$; signum $\rightarrow \{-1, 0, 1\}$; constant $\rightarrow \{c\}$; greatest integer $[x] \rightarrow \mathbb{Z}$.
- **Domain rules:** denominator $\neq 0$; even-root radicand ≥ 0 (or > 0 if under a root in a denominator); **intersect** all conditions.
- **Algebra:** $\text{Dom}(f \pm g) = \text{Dom}(fg) = \text{Dom}(f) \cap \text{Dom}(g)$; $\text{Dom}(f/g) = \text{that} \setminus \{g = 0\}$.

Memory aids:

- “Vending machine: one button, one snack” $\rightarrow$ single-valued.
- “Outputs to the power of inputs” $\rightarrow q^p$, never p^q .
- “Root closes, denominator opens” $\rightarrow \geq 0$ vs $\neq 0$.
- “ f/g kicks out $g = 0$ ” $\rightarrow$ don’t forget the extra exclusion.

Chapter 2: Relations and Functions — Supplement

Subtopic 04 — Types of Relations and Equivalence Relations

Exam Relevance: This is the material Subtopic 02 promised you. CBSE Boards formally examines it in Class 12 Chapter 1, where “show that R is an equivalence relation” is a standing 3–5 mark question and “find the equivalence class of...” is its usual follow-up — so every hour you spend here is a mark banked twice. JEE Main asks it two ways: a one-line property check (“ R is reflexive and symmetric but not transitive”) and a pure count (“the number of reflexive relations on a set of 5 elements”). JEE Advanced pushes into equivalence classes as a partition — counting equivalence relations by counting partitions, or finding how many equivalence relations contain a specified pair.

Section 1: Concept Intuition

Walk into a school assembly of 600 students. Nobody is standing at random: every student belongs to exactly one section — 11-A, 11-B, 11-C. Now consider the rule “**is in the same section as**”. Three things are true of it, and they are true so obviously that you have never bothered to name them:

- Every student is in the same section as **themselves**.
- If Riya is in the same section as Aman, then Aman is in the same section as Riya.
- If Riya is in the same section as Aman, and Aman is in the same section as Sara, then Riya is in the same section as Sara.

Those three properties are called **reflexive**, **symmetric** and **transitive**, and a relation that has all three is called an **equivalence relation**. Here is the key realization, and it is the whole point of the topic:

An equivalence relation is exactly a rule that chops a set into non-overlapping groups. The three properties are not an arbitrary checklist — together they are precisely what it takes to sort a set into clean sections.

Look at the assembly again. The relation “same section as” and the seating chart carry identical information. Given the sections, you can decide any “same section?” question. Given the relation, you can reconstruct the sections: just collect, for each student, everyone related to them. Those collections are the **equivalence classes**, and they overlap nowhere and miss nobody. That is a **partition**.

Think about it this way... on the Mumbai locals, “**travels on the same line as**” partitions commuters into Western, Central and Harbour. No commuter is on zero lines (reflexive fails nowhere), the relation reads the same both ways (symmetric), and same-line-ness chains (transitive). Change the rule to “**lives within 5 km of**” and the machinery breaks: it is still

reflexive and still symmetric, but Andheri is within 5 km of Jogeshwari and Jogeshwari within 5 km of Goregaon, while Andheri and Goregaon are not. **Transitivity is the fragile one.** It is the property that fails most often, and it is the one examiners attack.

Two more properties round out the family:

- **Antisymmetric** is the opposite temperament to symmetric. It says two *different* elements can never point at each other. “ $\leq$ ” is antisymmetric: $a \leq b$ and $b \leq a$ force $a = b$. This is the property of **rankings and hierarchies** — a merit list, a divisibility ladder — not of groupings.
- The **inverse relation** R^{-1} is simply R with every arrow reversed. If R is “is the father of”, R^{-1} is “is the child of”. Reversing arrows costs nothing and tells you something: **R is symmetric exactly when reversing all the arrows changes nothing at all**, i.e. $R^{-1} = R$.

And the smallest possible equivalence relation deserves a name of its own: the **identity relation** I_A , where every element is related to itself and to nothing else. In the assembly analogy, that is the pathological school where every student is their own section.

Figure brief: A 3×3 grid of the nine cells of $A \times A$ for $A = \{1, 2, 3\}$; the three diagonal cells shaded and labelled “ I_A — must all be present for reflexive”; the cells $(1, 2)$ and $(2, 1)$ joined by a double-headed arrow labelled “symmetric: in together or out together”; a dashed arrow chain $(1, 2) \rightarrow (2, 3) \Rightarrow (1, 3)$ drawn across the grid labelled “transitive: chains must close”.

Limiting Conditions & Assumptions.

- These properties are defined for a relation **on a set** A (a subset of $A \times A$). A relation from A to a different set B has no diagonal to speak of, so “reflexive” is meaningless for it.
- **Reflexive means every element**, not some. One missing (a, a) kills it.
- **Symmetric and transitive do not imply reflexive.** The standard “proof” that they do is false and is dismantled in Example 3.
- **Antisymmetric is not the negation of symmetric.** The identity relation I_A is both. The empty relation on a non-empty set is both. A relation can also be neither.
- **Transitivity can hold vacuously.** On $A = \{1, 2, 3\}$, the relation $R = \{(1, 2)\}$ is transitive, because there is no chain $(a, b), (b, c)$ inside it to close.
- The **empty relation** on a non-empty A is symmetric, transitive and antisymmetric, but never reflexive. The **universal relation** $A \times A$ is reflexive, symmetric and transitive — it is an equivalence relation with exactly one class.

Section 2: Key Definitions, Formulas & Notation

Throughout, R is a relation on a set A , i.e. $R \subseteq A \times A$, and $n(A) = n$.

The four properties.

$$R \text{ is reflexive} \iff (a, a) \in R \text{ for every } a \in A$$

R is symmetric $\iff [(a, b) \in R \Rightarrow (b, a) \in R]$ for all $a, b \in A$

R is transitive $\iff [(a, b) \in R \text{ and } (b, c) \in R \Rightarrow (a, c) \in R]$ for all $a, b, c \in A$

R is antisymmetric $\iff [(a, b) \in R \text{ and } (b, a) \in R \Rightarrow a = b]$ for all $a, b \in A$

- a, b, c are elements of A ; the quantifier “for all” is not decoration — a single failing instance destroys the property.
- Reflexivity is a condition on the n diagonal pairs only; symmetry and antisymmetry are conditions on off-diagonal pairs only; transitivity is a condition on chains.

Identity relation.

$$I_A = \{(a, a) : a \in A\}, \quad n(I_A) = n$$

- I_A is reflexive, symmetric, transitive **and** antisymmetric.
- Restated: R is reflexive $\iff I_A \subseteq R$.

Inverse relation. For a relation R from A to B ,

$$R^{-1} = \{(b, a) : (a, b) \in R\} \subseteq B \times A$$

$$\text{Dom}(R^{-1}) = \text{Range}(R), \quad \text{Range}(R^{-1}) = \text{Dom}(R), \quad (R^{-1})^{-1} = R$$

- $n(R^{-1}) = n(R)$ always — reversing arrows never creates or destroys one.
- R is symmetric $\iff R = R^{-1}$. R is antisymmetric $\iff R \cap R^{-1} \subseteq I_A$.

Equivalence relation.

R is an equivalence relation $\iff R$ is reflexive **and** symmetric **and** transitive

Equivalence class. For an equivalence relation R on A and $a \in A$,

$$[a] = \{x \in A : (x, a) \in R\}$$

- $[a]$ is read “the equivalence class of a ”; some books write a/R or $\bar{a}$.
- $a \in [a]$ always (by reflexivity), so no class is ever empty.
- $[a] = [b] \iff (a, b) \in R$. Two elements that are related have **the same** class, not merely overlapping ones.

Partition. A collection of subsets $\{A_1, A_2, \dots, A_k\}$ of A is a partition of A if each $A_i \neq \emptyset$, $A_i \cap A_j = \emptyset$ for $i \neq j$, and $A_1 \cup A_2 \cup \dots \cup A_k = A$.

Comparison table.

Property	Formal condition	One-line test	Standard example	Standard non-example
Reflexive	$(a, a) \in R \ \forall a$	Is every diagonal pair present?	$\leq$ on $\mathbb{R}$; “divides” on $\mathbb{N}$	$<$ on $\mathbb{R}$; $\perp$ on lines
Symmetric	$(a, b) \in R \Rightarrow (b, a) \in R$	Does every arrow have its reverse?	$\perp$ on lines; “is a sibling of”	$\leq$ on $\mathbb{R}$
Transitive	$(a, b), (b, c) \in R \Rightarrow (a, c) \in R$	Does every chain close up?	$\leq, <, \parallel$, “divides”	$\perp$ on lines; “within 5 km of”
Antisymmetric	$(a, b), (b, a) \in R \Rightarrow a = b$	Do two distinct elements ever point both ways?	$\leq$ on $\mathbb{R}$; $\subseteq$ on sets; “divides” on $\mathbb{N}$	$A \times A$ when $n(A) \geq 2$

Note the entry “divides on $\mathbb{N}$ ”, not on $\mathbb{Z}$: on $\mathbb{Z}$ divisibility fails antisymmetry, since $2 \mid -2$ and $-2 \mid 2$ but $2 \neq -2$.

Counting relations with imposed properties on a set with $n(A) = n$:

Type of relation on A	Count
All relations	2^{n^2}
Reflexive	2^{n^2-n}
Symmetric	$2^{n(n+1)/2}$
Reflexive and symmetric	$2^{n(n-1)/2}$
Antisymmetric	$2^n \cdot 3^{n(n-1)/2}$
Equivalence relations	B_n (the n -th Bell number); $B_1 = 1, B_2 = 2, B_3 = 5, B_4 = 15$

- $n(n-1)/2 = \binom{n}{2}$ is the number of **unordered** pairs $\{a, b\}$ with $a \neq b$ — the number of off-diagonal *slots* once you force (a, b) and (b, a) to share a decision.
- There is no elementary closed form for B_n , which is why exams only ever ask for equivalence relations on sets of 3 or 4 elements, where you count partitions by hand.

Symbol	Meaning
I_A	identity relation on A : $\{(a, a) : a \in A\}$
R^{-1}	inverse relation: all pairs of R reversed
$[a]$	equivalence class of a under an equivalence relation
$a \equiv b \pmod{m}$	$a - b$ is divisible by m

Symbol	Meaning
$\binom{n}{2} = \frac{n(n-1)}{2}$	number of unordered pairs of distinct elements of A

Section 3: Key Procedures & Logical Framework

Derivation: Equivalence classes partition the set

Claim. Let R be an equivalence relation on a non-empty set A . Then the distinct equivalence classes of R form a partition of A .

Proof.

1. **No class is empty.** For any $a \in A$, reflexivity gives $(a, a) \in R$, so $a \in [a]$. (*Reason: reflexivity is exactly the statement that every element sits in its own class.*)
2. **The classes cover A .** By step 1 every $a \in A$ lies in the class $[a]$, so $\bigcup_{a \in A} [a] = A$. (*Reason: nothing can be left out, because each element supplies a class containing itself.*)
3. **Two classes that meet are identical.** Suppose $[a] \cap [b] \neq \emptyset$, say $c \in [a] \cap [b]$. Then $(c, a) \in R$ and $(c, b) \in R$. Take any $x \in [a]$, so $(x, a) \in R$. From $(c, a) \in R$ and symmetry, $(a, c) \in R$. Chaining (x, a) with (a, c) by transitivity gives $(x, c) \in R$; chaining that with (c, b) gives $(x, b) \in R$, i.e. $x \in [b]$. Hence $[a] \subseteq [b]$. The same argument with a and b swapped gives $[b] \subseteq [a]$. Therefore $[a] = [b]$. (*Reason: symmetry lets you walk an arrow backwards, transitivity lets you join two walks — together they let you travel from any member of one class to any member of the other.*)
4. **Conclusion.** By step 3, two distinct classes cannot overlap at all; by steps 1 and 2 the classes are non-empty and cover A . That is precisely a partition. ■

The converse also holds, and it is worth knowing: given any partition of A , the rule “ aRb if and only if a and b lie in the same block” is an equivalence relation whose classes are exactly the blocks. So **equivalence relations on A and partitions of A are the same objects wearing different clothes**. This is what makes the counting in Example 4 possible.

Figure brief: A large oval labelled A divided by two curved cuts into three non-overlapping blobs labelled $[a]$, $[b]$, $[c]$; a dot inside each blob labelled with its representative; a caption strip beneath reading “non-empty, pairwise disjoint, union = A ”.

Derivation: Congruence modulo m is an equivalence relation

Claim. Fix a positive integer m . On $\mathbb{Z}$, define $R = \{(a, b) : a - b \text{ is divisible by } m\}$. Then R is an equivalence relation with exactly m classes.

Proof.

1. **Reflexive:** $a - a = 0 = m \cdot 0$ is divisible by m , so $(a, a) \in R$ for every integer a .
2. **Symmetric:** if $a - b = mk$ for some integer k , then $b - a = m(-k)$, and $-k$ is an integer, so $(b, a) \in R$.
3. **Transitive:** if $a - b = mk$ and $b - c = ml$, then $a - c = (a - b) + (b - c) = m(k + l)$, and

$k + l$ is an integer, so $(a, c) \in R$.

4. **The classes.** $[r] = \{x \in \mathbb{Z} : x - r \text{ divisible by } m\} = \{mk + r : k \in \mathbb{Z}\}$. By the division algorithm every integer leaves exactly one remainder $r \in \{0, 1, \dots, m - 1\}$ on division by m , so there are exactly m distinct classes $[0], [1], \dots, [m - 1]$, and they partition $\mathbb{Z}$. ■

For $m = 5$: $[0] = \{\dots, -10, -5, 0, 5, 10, \dots\}$, $[1] = \{\dots, -9, -4, 1, 6, 11, \dots\}$, and so on to $[4]$. Note $[5] = [0]$ and $[-1] = [4]$ — different labels, same class.

Derivation: Counting relations with imposed properties

Everything below is the Subtopic 02 machinery — n^2 independent yes/no switches — with some switches frozen or ganged together.

Reflexive: 2^{n^2-n} . Reflexivity forces all n diagonal pairs (a, a) to be **in**. Those n switches are frozen, contributing a factor of 1 each. The other $n^2 - n$ pairs are untouched by the condition, each still free to be in or out:

$$\underbrace{1 \times \dots \times 1}_n \times \underbrace{2 \times \dots \times 2}_{n^2-n \text{ free}} = 2^{n^2-n}.$$

Symmetric: $2^{n(n+1)/2}$. Symmetry says nothing about the diagonal, so each of the n diagonal pairs is still a free yes/no: 2^n . Off the diagonal, the pairs come in couples $\{(a, b), (b, a)\}$ with $a \neq b$, and symmetry ties each couple together — both in, or both out, one decision for the couple. The number of such couples is the number of unordered pairs of distinct elements, $\binom{n}{2} = n(n-1)/2$. Hence

$$2^n \cdot 2^{n(n-1)/2} = 2^{n + \frac{n(n-1)}{2}} = 2^{\frac{2n + n^2 - n}{2}} = 2^{\frac{n(n+1)}{2}}.$$

Reflexive and symmetric: $2^{n(n-1)/2}$. Take the symmetric count and freeze the diagonal to “in”: the factor 2^n collapses to 1, leaving $2^{n(n-1)/2}$.

Antisymmetric: $2^n \cdot 3^{n(n-1)/2}$. Antisymmetry also says nothing about the diagonal (the condition $(a, a), (a, a) \in R \Rightarrow a = a$ is automatic), so the n diagonal switches are free: 2^n . For each of the $n(n-1)/2$ couples $\{(a, b), (b, a)\}$ with $a \neq b$, antisymmetry forbids exactly one of the four combinations — “both in” — leaving **three** legal options: neither, (a, b) only, (b, a) only. Hence $3^{n(n-1)/2}$, and the product is $2^n \cdot 3^{n(n-1)/2}$. ■

Sanity check on $n = 2$. Total relations $2^4 = 16$. Reflexive $2^{4-2} = 4$; symmetric $2^3 = 8$; both $2^1 = 2$; antisymmetric $2^2 \cdot 3^1 = 12$. The last one is checkable by hand: the only way to break antisymmetry on $\{1, 2\}$ is to include both $(1, 2)$ and $(2, 1)$, which leaves the two diagonal pairs free — 4 bad relations, and $16 - 4 = 12$. It matches.

Procedure A — Testing a relation for the four properties

1. **Reflexive?** Run down the diagonal. Is $(a, a) \in R$ for **every** $a \in A$? One absentee and the answer is no. (*Why: the definition is universally quantified; there is no “mostly reflexive”.*)
2. **Symmetric?** For each pair $(a, b) \in R$ with $a \neq b$, look for (b, a) . One unreciprocated arrow and the answer is no. (*Why: diagonal pairs are their own reverses, so they never threaten symmetry — do not waste time on them.*)

3. **Transitive?** Find every chain $(a, b), (b, c)$ **inside** R — the second coordinate of one pair matching the first coordinate of another — and check $(a, c) \in R$. (Why: you only have to check chains that actually exist. If no chain exists, transitivity holds vacuously.)
4. **Antisymmetric?** Look for a two-way street between distinct elements: a pair $a \neq b$ with both (a, b) and (b, a) in R . Find one and the answer is no; find none and the answer is yes. (Why: antisymmetry is a prohibition, so it holds until a witness against it turns up.)
5. **Verdict.** Reflexive + symmetric + transitive $\Rightarrow$ equivalence relation. Then, and only then, go looking for classes.

For an infinite set defined by a rule, you cannot list pairs — argue algebraically instead, exactly as in the congruence derivation above: take a general $(a, b) \in R$, write down what that means, and manipulate.

Procedure B — Writing down R^{-1}

1. **Reverse every pair.** $(a, b) \in R$ becomes $(b, a) \in R^{-1}$. Nothing is added, nothing is dropped, so $n(R^{-1}) = n(R)$.
2. **Swap the sets.** If $R \subseteq A \times B$ then $R^{-1} \subseteq B \times A$. (Why: first coordinates and second coordinates have exchanged roles.)
3. **Read off domain and range by swapping.** $\text{Dom}(R^{-1}) = \text{Range}(R)$ and $\text{Range}(R^{-1}) = \text{Dom}(R)$ — never recompute them from scratch.
4. **Use it as a symmetry test.** If R^{-1} comes out identical to R , then R is symmetric; if they differ, it is not.

Procedure C — Finding the equivalence classes and the induced partition

1. **Confirm R is an equivalence relation first.** Classes are only guaranteed to partition when all three properties hold. (Why: without symmetry and transitivity, “classes” can overlap without being equal, and the whole picture collapses.)
2. **Pick any unclassified element a .** Collect $[a] = \{x \in A : (x, a) \in R\}$.
3. **Strike out everything in $[a]$.** Every element of $[a]$ has $[x] = [a]$, so none of them will generate anything new. (Why: step 3 of the partition derivation — classes that meet are equal.)
4. **Repeat** with the next unclassified element until A is exhausted.
5. **State the partition** as the list of distinct classes, and check: non-empty, pairwise disjoint, union = A . (Why: this is a free verification of your own work — if the blocks overlap or miss an element, you have made an error upstream.)

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

Q. Let $A = \{1, 2, 3\}$ and $R = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 1)\}$. Examine R for reflexivity, symmetry, transitivity and antisymmetry. Write R^{-1} . If R is an equivalence relation, find all its equivalence classes and the partition they induce.

Solution. Work Procedure A in order.

Reflexive: the diagonal pairs $(1, 1), (2, 2), (3, 3)$ are all present. Reflexive.

Symmetric: the only off-diagonal pairs are $(1, 2)$ and $(2, 1)$, and each is the reverse of the other. Symmetric.

Transitive: list the chains that exist inside R and check each closes.

- $(1, 2)$ with $(2, 1)$ needs $(1, 1)$ — present.
- $(2, 1)$ with $(1, 2)$ needs $(2, 2)$ — present.
- $(1, 1)$ with $(1, 2)$ needs $(1, 2)$ — present. $(1, 2)$ with $(2, 2)$ needs $(1, 2)$ — present.
- $(2, 2)$ with $(2, 1)$ needs $(2, 1)$ — present. $(2, 1)$ with $(1, 1)$ needs $(2, 1)$ — present.
- Chains through 3 only involve $(3, 3)$ with itself, needing $(3, 3)$ — present.

Every chain closes. Transitive.

Antisymmetric: $(1, 2) \in R$ and $(2, 1) \in R$ but $1 \neq 2$. **Not** antisymmetric.

Inverse: reversing every pair, $R^{-1} = \{(1, 1), (2, 2), (3, 3), (2, 1), (1, 2)\} = R$, confirming symmetry.

Since R is reflexive, symmetric and transitive, it **is an equivalence relation** on A . Now Procedure C:

$$[1] = \{x : (x, 1) \in R\} = \{1, 2\}, \quad [2] = \{x : (x, 2) \in R\} = \{1, 2\}, \quad [3] = \{3\}.$$

So $[1] = [2]$ — as the theory promised, since $(1, 2) \in R$. The **distinct** classes are $\{1, 2\}$ and $\{3\}$, giving the partition

$$A = \{1, 2\} \cup \{3\}, \quad \{1, 2\} \cap \{3\} = \emptyset.$$

Non-empty, disjoint, union all of A : a genuine partition. ■

Example 2 — [JEE Main Level]

Q. Let A be a set with 4 elements. Find the number of relations on A that are (i) reflexive, (ii) symmetric, (iii) both reflexive and symmetric, (iv) antisymmetric. What fraction of all relations on A is reflexive?

Solution. Here $n = 4$, so $A \times A$ has $n^2 = 16$ ordered pairs and there are $2^{16} = 65536$ relations on A altogether. Off the diagonal there are $\binom{4}{2} = \frac{4 \times 3}{2} = 6$ unordered pairs of distinct elements.

(i) Reflexive. Freeze the 4 diagonal pairs as “in”; the remaining $16 - 4 = 12$ pairs stay free:

$$2^{16-4} = 2^{12} = 4096.$$

(ii) Symmetric. Diagonal free (2^4), and the 6 off-diagonal couples each decided once (2^6):

$$2^4 \cdot 2^6 = 2^{10} = 1024, \quad \text{matching } 2^{n(n+1)/2} = 2^{4 \cdot 5/2} = 2^{10}.$$

(iii) Reflexive and symmetric. Take (ii) and freeze the diagonal to “in”, killing the 2^4 :

$$2^6 = 64, \quad \text{matching } 2^{n(n-1)/2} = 2^{4 \cdot 3/2} = 2^6.$$

(iv) Antisymmetric. Diagonal free ($2^4 = 16$), and each of the 6 couples has 3 legal states:

$$2^4 \cdot 3^6 = 16 \times 729 = 11664.$$

Fraction reflexive: $\frac{2^{12}}{2^{16}} = \frac{1}{2^4} = \frac{1}{16}$ — each of the four frozen diagonal switches halves the population.

Answers: 4096, 1024, 64, 11664; one-sixteenth. ■

Example 3 — [JEE Main — Speed Trap]

Q. On $A = \{1, 2, 3\}$, let $R = \{(1, 1), (2, 2), (1, 2), (2, 1)\}$. Is R an equivalence relation on A ?

The trap. There is a famous one-line “proof” that symmetry and transitivity together force reflexivity:

Let $a \in A$. Take any b with $(a, b) \in R$. By symmetry $(b, a) \in R$. By transitivity applied to (a, b) and (b, a) , we get $(a, a) \in R$. So R is reflexive.

It looks airtight, and the seductive wrong answer is “**Yes — R is symmetric and transitive, and those force reflexivity, so R is an equivalence relation.**”

Where it breaks. Read the second sentence again: “Take any b with $(a, b) \in R$.” That step silently assumes such a b exists. For an element related to **nothing at all**, the argument never gets started and manufactures no (a, a) .

Check the actual relation.

- Symmetric: $(1, 2)$ and $(2, 1)$ are both present; done. Yes.
- Transitive: the chains are $(1, 2)$ with $(2, 1)$ needing $(1, 1)$ (present), $(2, 1)$ with $(1, 2)$ needing $(2, 2)$ (present), and the chains through $(1, 1)$ and $(2, 2)$ which close trivially. Yes.
- Reflexive: $(3, 3) \notin R$. **No.**

Element 3 appears in no pair of R whatsoever, so the bogus argument has nothing to bite on. R is symmetric and transitive but not reflexive on A , hence **not an equivalence relation on A .**

The honest version of the statement: if R is symmetric and transitive **and every element of A is related to at least one element**, then R is reflexive. That extra hypothesis is the whole content.

A useful footnote: R is an equivalence relation on the smaller set $\{1, 2\}$, where it is the universal relation. Properties like reflexivity are always relative to the set you declare. **Answer: No.** ■

Example 4 — [JEE Advanced Level]

Q. Let $A = \{1, 2, 3\}$. (i) Find the total number of equivalence relations on A and list them. (ii) How many of them contain the ordered pair $(1, 2)$? (iii) Deduce the number of relations on A that are symmetric but not reflexive.

Solution. Parts (i) and (ii) are counting problems in disguise. The counting formulas of Section 2 handle “reflexive”, “symmetric” and “antisymmetric” because each imposes an independent constraint on separate switches — but transitivity couples switches together and defeats that method entirely. Use the partition correspondence instead.

(i) By the converse of the partition derivation, equivalence relations on A correspond one-to-one with partitions of A . So count partitions of $\{1, 2, 3\}$:

Partition	Corresponding equivalence relation
$\{1\}, \{2\}, \{3\}$	$I_A = \{(1, 1), (2, 2), (3, 3)\}$
$\{1, 2\}, \{3\}$	$I_A \cup \{(1, 2), (2, 1)\}$
$\{1, 3\}, \{2\}$	$I_A \cup \{(1, 3), (3, 1)\}$
$\{2, 3\}, \{1\}$	$I_A \cup \{(2, 3), (3, 2)\}$
$\{1, 2, 3\}$	$A \times A$ (all nine pairs)

There are no others: a partition of a 3-element set has block sizes $1 + 1 + 1$ (one way), $2 + 1$ (three ways, chosen by which element is alone), or 3 (one way). **Total:** 5.

(ii) An equivalence relation contains $(1, 2)$ exactly when 1 and 2 lie in the **same block** of its partition. Scanning the table, that happens for $\{1, 2\}, \{3\}$ and for $\{1, 2, 3\}$. **Answer:** 2.

Note what is *not* being asked. There are far more relations containing $(1, 2)$; the equivalence condition is what cuts the list to two. And observe the forced consequences: once $(1, 2)$ is in, symmetry drags in $(2, 1)$ and reflexivity drags in all three diagonal pairs — that is already the second row of the table. The only remaining freedom is whether 3 joins them, giving exactly two options.

(iii) Here the independent-switch formulas *do* apply. With $n = 3$:

$$\text{symmetric} = 2^{n(n+1)/2} = 2^6 = 64, \quad \text{reflexive and symmetric} = 2^{n(n-1)/2} = 2^3 = 8.$$

Every reflexive-and-symmetric relation is symmetric, so subtract:

$$\text{symmetric but not reflexive} = 64 - 8 = 56.$$

Answers: 5; 2; 56. The structural lesson: **reflexivity, symmetry and antisymmetry are switch-counting conditions; transitivity is not.** The moment transitivity enters, abandon powers of 2 and count partitions. ■

Section 5: Practice Exercises

1. **[CBSE]** Let $A = \{1, 2, 3\}$ and $R = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 3)\}$. State which of reflexive, symmetric, transitive R satisfies.
2. **[CBSE]** On $A = \{1, 2, 3, 4\}$ let $R = \{(a, b) : a \text{ divides } b\}$. Write R in roster form, give $n(R)$, and classify R as reflexive / symmetric / transitive / antisymmetric.
3. **[JEE Main]** Let $R = \{(1, 2), (2, 3), (1, 4)\}$ be a relation from $A = \{1, 2\}$ to $B = \{2, 3, 4\}$. Write R^{-1} and state $\text{Dom}(R^{-1})$ and $\text{Range}(R^{-1})$.
4. **[JEE Main]** For a set with 5 elements, find the number of relations that are (i) reflexive, (ii) symmetric, (iii) both.
5. **[JEE Main]** On $\mathbb{Z}$, let $R = \{(a, b) : a - b \text{ is divisible by } 4\}$. Write the equivalence class $[3]$ in set-builder form and state how many distinct equivalence classes R has.
6. **[JEE Main]** Let $A = \{1, 2, 3\}$ and $R = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 1), (2, 3), (3, 2)\}$. Show R is not transitive, and find the minimum number of ordered pairs that must be added to make R an equivalence relation.
7. **[JEE Advanced]** Let $A = \{1, 2, 3, 4\}$. How many equivalence relations on A contain both $(1, 2)$ and $(3, 4)$?

Answers:

1. Reflexive only. Symmetry fails because $(1, 2) \in R$ but $(2, 1) \notin R$; transitivity fails because $(1, 2), (2, 3) \in R$ but $(1, 3) \notin R$.
2. $R = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 4), (3, 3), (4, 4)\}$; $n(R) = 8$. Reflexive: yes. Symmetric: no ($(1, 2) \in R, (2, 1) \notin R$). Transitive: yes. Antisymmetric: yes.
3. $R^{-1} = \{(2, 1), (3, 2), (4, 1)\}$; $\text{Dom}(R^{-1}) = \{2, 3, 4\} = \text{Range}(R)$; $\text{Range}(R^{-1}) = \{1, 2\} = \text{Dom}(R)$.
4. (i) $2^{25-5} = 2^{20} = 1,048,576$; (ii) $2^{5 \cdot 6/2} = 2^{15} = 32,768$; (iii) $2^{5 \cdot 4/2} = 2^{10} = 1024$.
5. $[3] = \{4k + 3 : k \in \mathbb{Z}\} = \{\dots, -5, -1, 3, 7, 11, \dots\}$; there are 4 distinct classes, namely $[0], [1], [2], [3]$.
6. $(1, 2), (2, 3) \in R$ but $(1, 3) \notin R$, so R is not transitive. Add $(1, 3)$ and, for symmetry, $(3, 1)$: minimum 2 pairs. The result is all nine pairs, i.e. $A \times A$, which is an equivalence relation.
7. 2. Forcing 1, 2 into one block and 3, 4 into one block leaves only the two blocks themselves as free objects: either they stay separate, giving the partition $\{1, 2\}, \{3, 4\}$, or they merge, giving $\{1, 2, 3, 4\}$.

Section 6: Crack the MCQ

Q1. On $A = \{1, 2, 3\}$, which relation is symmetric but neither reflexive nor transitive?

- (a) $\{(1, 2), (2, 1)\}$ (b) $\{(1, 1), (2, 2), (3, 3)\}$ (c) $\{(1, 2), (2, 3), (1, 3)\}$ (d) $\{(1, 2), (2, 1), (1, 1), (2, 2)\}$

Answer: (a). Both arrows are reciprocated, so it is symmetric. No diagonal pair is present, so it is not reflexive. The chain $(1, 2)$ with $(2, 1)$ demands $(1, 1)$, which is absent, so it is not transitive.

Distractor Analysis

- **(b)** is the identity relation I_A : reflexive, symmetric **and** transitive — a full equivalence

relation. Chosen by students who read “symmetric” and stop reading.

- **(c)** is transitive (the only chain, $(1, 2)$ with $(2, 3)$, closes into $(1, 3)$) but not symmetric and not reflexive. Chosen by students who confuse “the pairs look paired up” with symmetry.
- **(d)** is symmetric and **transitive** but not reflexive — this is the Example 3 relation. It satisfies two of the three requirements, and it is the most tempting wrong option because failing transitivity is precisely what students forget to check.

Q2. The number of reflexive relations on a set with 3 elements is

- (a) 512 (b) 64 (c) 8 (d) 216

Answer: (b) 64. Reflexive relations number $2^{n^2-n} = 2^{9-3} = 2^6 = 64$.

Distractor Analysis

- **(a)** $512 = 2^9 = 2^{n^2}$ is the number of **all** relations on a 3-element set — the condition was ignored entirely.
- **(c)** $8 = 2^3$ is 2^n ; it counts only the diagonal choices, i.e. it freezes the wrong set of switches. (It is in fact the count of reflexive **and symmetric** relations, $2^{n(n-1)/2} = 2^3$ — a genuinely different question.)
- **(d)** $216 = 2^3 \cdot 3^3$ is the **antisymmetric** count $2^n \cdot 3^{n(n-1)/2}$ — the right machinery attached to the wrong property.

Q3. If R is an equivalence relation on a non-empty set A , which statement is **false**?

- (a) $R^{-1} = R$ (b) $I_A \subseteq R$ (c) two distinct equivalence classes of R are disjoint (d) $R = A \times A$

Answer: (d). $R = A \times A$ describes only one particular equivalence relation, the universal one. The identity relation I_A is also an equivalence relation and equals $A \times A$ only when $n(A) = 1$.

Distractor Analysis

- **(a)** is true: an equivalence relation is symmetric, and symmetric is exactly the condition $R = R^{-1}$.
- **(b)** is true: it is reflexivity restated — every diagonal pair belongs to R .
- **(c)** is true: it is step 3 of the partition derivation. Students who mark this false are usually thinking of classes as “overlapping groups”, the very picture the derivation rules out.

Q4. On $\mathbb{R}$, the relation $R = \{(a, b) : a \leq b\}$ is

- (a) reflexive, antisymmetric and transitive but not symmetric (b) an equivalence relation
(c) symmetric only (d) reflexive and symmetric but not transitive

Answer: (a). $a \leq a$ (reflexive); $a \leq b$ and $b \leq a$ force $a = b$ (antisymmetric); $a \leq b$ and $b \leq c$ give $a \leq c$ (transitive); and $2 \leq 3$ while $3 \not\leq 2$, so it is not symmetric.

Distractor Analysis

- **(b)** treats every “nice-looking” relation as an equivalence relation. Order relations are the standard counterexample: they are reflexive and transitive but antisymmetric rather than symmetric.
- **(c)** confuses $\leq$ with $=$. Only the equality part of $\leq$ is two-way.
- **(d)** gets reflexivity right but then attaches the wrong two verdicts; $\leq$ is one of the most transitive relations in mathematics.

Section 7: Common Student Pitfalls & Pro-Tips

- **“Symmetric and transitive, therefore reflexive.”** This is false, and it is the single most examined misconception in the topic. The argument needs every element to be related to something; an element related to nothing (like 3 in Example 3) escapes it. Correction: always check the diagonal separately, every time.
- **Reflexive means every element.** Finding $(1, 1)$ and $(2, 2)$ in R on $A = \{1, 2, 3\}$ and declaring “reflexive” is a guaranteed mark lost. Correction: count the diagonal pairs; you need exactly $n(A)$ of them.
- **Treating antisymmetric as “not symmetric”.** They are independent conditions. I_A is both symmetric and antisymmetric; $\{(1, 2), (2, 1), (1, 3)\}$ on $\{1, 2, 3\}$ is neither. Correction: antisymmetry is a ban on **two-way streets between distinct elements** — nothing more.
- **Forgetting that transitivity can hold vacuously.** Students mark $R = \{(1, 2)\}$ as “not transitive” because “nothing chains”. No chain means no condition to violate, so it **is** transitive. Correction: transitivity fails only when a chain exists and does not close.
- **Checking transitivity by staring.** With more than five pairs you will miss a chain. Correction: sort the pairs by first coordinate, then for each pair (a, b) scan only the pairs starting with b — this is Procedure A step 3 done systematically.
- **Using $2^{n^2} - n$ instead of 2^{n^2-n} .** Freezing a switch **divides** the count by 2; it does not subtract 1 from it. Correction: for $n = 2$, $2^{4-2} = 4$ reflexive relations, which you can list; $2^4 - 2 = 14$ is nonsense.
- **Applying the power-of-two formulas to transitivity or to equivalence relations.** There is no formula $2^{\text{something}}$ for these, because transitivity ties distant switches together. Correction: for equivalence relations, count **partitions**.
- **Confusing $\text{Dom}(R^{-1})$ with $\text{Dom}(R)$.** Reversing arrows swaps the roles. Correction: $\text{Dom}(R^{-1}) = \text{Range}(R)$, always.
- **Writing $[a]$ and $[b]$ as “overlapping”.** Under an equivalence relation, two classes either coincide completely or share nothing. Correction: if you find $[1] = \{1, 2\}$ and $[2] = \{2, 3\}$, you have made an arithmetic error or R is not transitive — go back.

Pro-Tip (speed). In the exam, spend your first ten seconds on the **diagonal** and your next ten on **one chain**. Reflexivity and transitivity are where nearly every relation dies, and a single missing diagonal pair or a single unclosed chain settles the question with no further work. For counting questions, memorise the exponents as *positions on the grid*: the whole grid is n^2 ; kill the diagonal and you get $n^2 - n$ (reflexive); take the diagonal plus the upper triangle and you get $\frac{n(n+1)}{2}$ (symmetric); take the upper triangle alone and you get $\frac{n(n-1)}{2}$ (reflexive and

symmetric). Antisymmetric is the only one with a 3 in it — the three legal states of each off-diagonal couple. And whenever “equivalence relation” appears in a counting question, switch immediately to counting partitions: $B_3 = 5$ and $B_4 = 15$ are worth knowing cold.

Section 8: Cheat Sheet Box

TYPES OF RELATIONS — RAPID RECALL

- **Reflexive:** $(a, a) \in R$ for **every** $a \in A$; equivalently $I_A \subseteq R$.
- **Symmetric:** $(a, b) \in R \Rightarrow (b, a) \in R$; equivalently $R = R^{-1}$.
- **Transitive:** $(a, b), (b, c) \in R \Rightarrow (a, c) \in R$. Holds vacuously if no chain exists.
- **Antisymmetric:** $(a, b), (b, a) \in R \Rightarrow a = b$. Not the negation of symmetric.
- **Identity relation:** $I_A = \{(a, a) : a \in A\}$ — reflexive, symmetric, transitive **and** antisymmetric. It is the only relation that is all four.
- **Inverse relation:** $R^{-1} = \{(b, a) : (a, b) \in R\}$; $n(R^{-1}) = n(R)$; $(R^{-1})^{-1} = R$; $\text{Dom}(R^{-1}) = \text{Range}(R)$ and $\text{Range}(R^{-1}) = \text{Dom}(R)$.
- **Equivalence relation:** reflexive **and** symmetric **and** transitive. Smallest on A is I_A ; largest is $A \times A$.
- **Equivalence class:** $[a] = \{x \in A : (x, a) \in R\}$; $a \in [a]$; $[a] = [b] \iff (a, b) \in R$; distinct classes are disjoint.
- **Partition theorem:** the distinct equivalence classes of an equivalence relation on A partition A — non-empty, pairwise disjoint, union = A . Conversely every partition gives an equivalence relation.
- **Model example:** $a \equiv b \pmod{m}$ on $\mathbb{Z}$ is an equivalence relation with exactly m classes $[0], [1], \dots, [m-1]$, where $[r] = \{mk + r : k \in \mathbb{Z}\}$.

Counts on a set with $n(A) = n$:

- All relations: 2^{n^2}
- Reflexive: 2^{n^2-n}
- Symmetric: $2^{n(n+1)/2}$
- Reflexive and symmetric: $2^{n(n-1)/2}$
- Antisymmetric: $2^n \cdot 3^{n(n-1)/2}$
- Equivalence relations: number of partitions of A ; $B_3 = 5$, $B_4 = 15$ (no power-of-two formula exists)

Memory aids:

- “**Reflexive = Run** the diagonal” — all n of them, no exceptions.
- “Symmetric ties the couple, antisymmetric bans the couple, reflexive freezes the diagonal.”
- “Equivalence relation = sorting hat” — three properties in, one clean partition out.
- “Transitivity has no formula” — when transitivity appears in a count, count partitions instead.
- “Reverse the arrows, swap domain and range” — for R^{-1} .

Addendum to Subtopic 03: Finding the Range

Procedure E slots into Subtopic 03, Section 3, immediately after Procedure D. Examples 6 and 7 append to Subtopic 03, Section 4.

Procedure E — Finding the Range of a Real Function

Procedure B taught you to ask “**which x are legal?**” — that gives the domain. The range asks the mirror question: “**which y actually get produced?**” Domain is a question about inputs and is answered by hunting for illegal operations. Range is a question about outputs and is answered by a completely different move:

A value y lies in the range of f if and only if the equation $y = f(x)$ has at least one solution x inside the domain of f .

Everything below is that one sentence, applied with different algebra.

Step 0 — Find the domain first. Every range method secretly asks “does a legal x exist?”, and you cannot answer that until you know which x are legal. (*Why: a y produced only by an x that the domain excludes is not in the range.*)

Step 1 — Choose the method from the shape of f .

Shape of $f(x)$	Method	Core move
Linear rational $\frac{ax+b}{cx+d}$	Inversion	Solve for x in terms of y ; exclude the y that makes the solution fail
$\frac{ax^2+bx+c}{dx^2+ex+f}$	Discriminant	Cross-multiply to a quadratic in x ; impose $D \geq 0$
Quadratic, or quadratic buried inside a root or a denominator	Completing the square	Rewrite as $a(x-h)^2+k$ and read off the reachable values
Bounded ingredient (sin, cos, $ x $, x^2) or a sum $t + \frac{1}{t}$	Boundedness / AM-GM	Bound the ingredient first, then propagate the bound through the outer operations

Step 2 (Method 1 — inversion and realisability). Put $y = f(x)$ and solve the equation for x , obtaining $x = g(y)$. The range is the set of y for which $g(y)$ is (i) a real number and (ii) an element of the domain of f . (*Why: this is the defining sentence made literal — you are exhibiting the x that produces y .*)

Illustration. For $f(x) = \frac{2x+3}{x-1}$ with domain $\mathbb{R} \setminus \{1\}$: from $y(x-1) = 2x+3$ we get $x(y-2) = y+3$, so $x = \frac{y+3}{y-2}$. This is real exactly when $y \neq 2$, and it is never equal to 1 (that would need $y+3 = y-2$). Hence the range is $\mathbb{R} \setminus \{2\}$.

Step 3 (Method 2 — the discriminant method). For $y = \frac{ax^2 + bx + c}{dx^2 + ex + f}$, cross-multiply and collect powers of x :

$$(a - yd)x^2 + (b - ye)x + (c - yf) = 0.$$

Then:

1. **Handle the degenerate value separately.** If $a - yd = 0$, i.e. $y = a/d$, the equation is no longer quadratic and the discriminant test does not apply. Substitute that y back by hand and see whether a real x exists. (*Why: this single value of y is where students lose the mark, and it can land on either side — sometimes it belongs to the range, sometimes it does not.*)
2. **For all other y ,** a real x exists if and only if the discriminant is non-negative:

$$D = (b - ye)^2 - 4(a - yd)(c - yf) \geq 0.$$

Solve this inequality in y ; that inequality **is** the range, subject to the next check. (*Why: a quadratic with real coefficients has a real root exactly when $D \geq 0$.*)

3. **Check the root is in the domain.** If the numerator and denominator share a real root, the graph has a hole and the corresponding y must be removed by hand. (*Why: the discriminant only certifies that some real x solves the cleared equation; it does not check that x was legal in the original fraction.*)

Illustration of the hole. $f(x) = \frac{x^2 - 4}{x - 2}$ equals $x + 2$ for every $x \neq 2$, so its range is $\mathbb{R} \setminus \{4\}$, not $\mathbb{R}$. Cancelling without recording the excluded point is what loses the $\{4\}$.

Step 4 (Method 3 — completing the square). Write the quadratic as $a(x - h)^2 + k$. Since $(x - h)^2$ ranges over $[0, \infty)$ and attains every value there:

$$a > 0 \Rightarrow \text{range} = [k, \infty), \quad a < 0 \Rightarrow \text{range} = (-\infty, k].$$

(*Why: a square is non-negative and unbounded above, and multiplying by a either preserves or reverses that.*) Then propagate through whatever wraps the quadratic — a reciprocal, a square root, a modulus — one operation at a time.

Illustration. $f(x) = x^2 - 4x + 7 = (x - 2)^2 + 3$ has range $[3, \infty)$. Consequently $g(x) = \frac{1}{x^2 - 4x + 7}$ has range $(0, \frac{1}{3}]$: as the denominator sweeps $[3, \infty)$, its reciprocal sweeps $(0, \frac{1}{3}]$, hitting $\frac{1}{3}$ at $x = 2$ and approaching 0 without reaching it. Note that a quadratic's range is **not** automatically $[0, \infty)$ — that is true for x^2 alone, not for quadratics in general.

Step 5 (Method 4 — boundedness and AM-GM). When f is built from an ingredient whose range you already know, bound the ingredient and push the bound outwards.

- **Bounded trigonometric ingredient.** $\sin \theta$ and $\cos \theta$ take every value in $[-1, 1]$. So for $f(x) = \frac{1}{2 - \sin 3x}$: as x runs over $\mathbb{R}$, $3x$ runs over $\mathbb{R}$, so $\sin 3x$ attains every value in $[-1, 1]$; hence $2 - \sin 3x$ attains every value in $[1, 3]$, and the reciprocal attains every value in $[\frac{1}{3}, 1]$.

- **AM-GM.** For $t > 0$, $t + \frac{1}{t} \geq 2$ with equality exactly at $t = 1$, and the expression is unbounded above, so its range on $(0, \infty)$ is $[2, \infty)$. For $t < 0$ the same argument applied to $-t$ gives $(-\infty, -2]$. Combined range on $\mathbb{R} \setminus \{0\}$: $(-\infty, -2] \cup [2, \infty)$.

(Why AM-GM earns its place: it converts a range question into a single sharp inequality plus an equality case, which is far faster than a discriminant when the expression is a sum of reciprocal terms.)

Step 6 — Confirm the endpoints are attained. An inequality such as $y \leq \frac{1}{2}$ tells you the range is contained in $(-\infty, \frac{1}{2}]$; it does not by itself tell you $\frac{1}{2}$ is reached. Solve $f(x) = \frac{1}{2}$ and produce the x . (Why: this is what decides a square bracket against a round one, and it is where most of the lost marks live.)

Step 7 — Respect a restricted domain. If the question supplies a domain smaller than the natural one, none of the above may be used blindly. For $f(x) = x^2$ the range is $[0, \infty)$ on $\mathbb{R}$ but $[1, 9]$ on $[1, 3]$. (Why: the defining sentence requires a solution **inside the domain**; shrinking the domain can only shrink the range.)

Example 6 — [JEE Main Level]

Q. Find the range of (a) $f(x) = \frac{x}{1+x^2}$, $x \in \mathbb{R}$, and (b) $g(x) = \frac{1}{2 - \sin 3x}$, $x \in \mathbb{R}$.

Solution.

(a) Domain first. The denominator $1+x^2 \geq 1 > 0$ for every real x , so the domain is all of $\mathbb{R}$ and there are no excluded points to worry about later.

Put $y = \frac{x}{1+x^2}$ and clear the fraction:

$$y(1+x^2) = x \implies yx^2 - x + y = 0.$$

Degenerate value. The coefficient of x^2 vanishes when $y = 0$. Handle it by hand: the equation becomes $-x = 0$, so $x = 0$, and indeed $f(0) = \frac{0}{1} = 0$. So $y = 0$ **is in the range**.

All other y . For $y \neq 0$ this is a genuine quadratic in x , and a real x exists exactly when

$$D = (-1)^2 - 4 \cdot y \cdot y = 1 - 4y^2 \geq 0 \implies y^2 \leq \frac{1}{4} \implies -\frac{1}{2} \leq y \leq \frac{1}{2}.$$

Domain check. The denominator $1+x^2$ has no real zeros at all, so no solution x can be illegal — no values need removing.

Combining the degenerate case with the rest:

$\text{Range}(f) = \left[-\frac{1}{2}, \frac{1}{2}\right]$

Endpoint check (Step 6). At $y = \frac{1}{2}$: $\frac{1}{2}x^2 - x + \frac{1}{2} = 0 \implies x^2 - 2x + 1 = 0 \implies x = 1$, and $f(1) = \frac{1}{2}$. At $y = -\frac{1}{2}$: similarly $x = -1$ and $f(-1) = -\frac{1}{2}$. Both endpoints are attained, so the brackets

are closed.

Independent cross-check by AM-GM. For $x > 0$, divide numerator and denominator by x :

$$f(x) = \frac{1}{x + \frac{1}{x}} \leq \frac{1}{2},$$

since $x + \frac{1}{x} \geq 2$ by AM-GM, with equality at $x = 1$. For $x < 0$ the same computation gives $f(x) \geq -\frac{1}{2}$, and $f(0) = 0$. Two independent methods, one answer.

Figure brief: Graph of $y = x/(1+x^2)$ on $-6 \leq x \leq 6$, showing the maximum $(1, 0.5)$ and minimum $(-1, -0.5)$ marked with dots, the horizontal asymptote $y = 0$ dashed, and the horizontal band $-\frac{1}{2} \leq y \leq \frac{1}{2}$ lightly shaded to display the range on the y -axis.

(b) Domain first. $\sin 3x \leq 1 < 2$, so $2 - \sin 3x \geq 1 > 0$ and never vanishes. Domain = $\mathbb{R}$.

Now propagate the bound outwards, one operation at a time (Method 4):

1. As x runs over $\mathbb{R}$, so does $3x$; hence $\sin 3x$ attains **every** value in $[-1, 1]$.
2. Multiplying by -1 and adding 2 is a decreasing map, so $2 - \sin 3x$ attains every value in $[2 - 1, 2 + 1] = [1, 3]$.
3. On $[1, 3]$ the reciprocal $t \mapsto \frac{1}{t}$ is decreasing and continuous, so $\frac{1}{2 - \sin 3x}$ attains every value in $[\frac{1}{3}, 1]$.

$\text{Range}(g) = [\frac{1}{3}, 1]$

Endpoint check. The maximum 1 occurs where $\sin 3x = 1$, e.g. $3x = \frac{\pi}{2}$, i.e. $x = \frac{\pi}{6}$: $g(\frac{\pi}{6}) = \frac{1}{2-1} = 1$. The minimum $\frac{1}{3}$ occurs where $\sin 3x = -1$, e.g. $x = \frac{\pi}{2}$: $g(\frac{\pi}{2}) = \frac{1}{2+1} = \frac{1}{3}$. Both attained. ■

Example 7 — [JEE Advanced Level]

Q. Find the range of the real function

$$f(x) = \frac{x^2 + x + 1}{x^2 + x - 1}.$$

Solution. This is the full discriminant method with every trap armed: a degenerate y , a domain with excluded points, and endpoints that must be tested.

Step 0 — Domain. The denominator vanishes when $x^2 + x - 1 = 0$, i.e. $x = \frac{-1 \pm \sqrt{5}}{2}$. So

$$\text{Dom}(f) = \mathbb{R} \setminus \left\{ \frac{-1 + \sqrt{5}}{2}, \frac{-1 - \sqrt{5}}{2} \right\}.$$

Step 1 — Clear the fraction. Put $y = f(x)$:

$$y(x^2 + x - 1) = x^2 + x + 1 \implies (y - 1)x^2 + (y - 1)x - (y + 1) = 0. \quad (*)$$

Step 2 — The degenerate value $y = 1$. The coefficient of x^2 is $y - 1$, which vanishes at $y = 1$. Substituting $y = 1$ into (*) leaves $-2 = 0$, an impossibility, so **no real x gives $f(x) = 1$** , and $y = 1$ is excluded from the range.

This is visible directly: $f(x) = 1$ would require $x^2 + x + 1 = x^2 + x - 1$, i.e. $1 = -1$.

Step 3 — Discriminant for $y \neq 1$. From (*),

$$D = (y - 1)^2 - 4(y - 1)(-(y + 1)) = (y - 1)^2 + 4(y - 1)(y + 1).$$

Factor out $(y - 1)$:

$$D = (y - 1)[(y - 1) + 4(y + 1)] = (y - 1)(5y + 3).$$

A real x exists exactly when $D \geq 0$:

$$(y - 1)(5y + 3) \geq 0 \implies y \leq -\frac{3}{5} \text{ or } y \geq 1.$$

Removing the excluded $y = 1$ from Step 2:

$$\text{Range}(f) \subseteq (-\infty, -\frac{3}{5}] \cup (1, \infty).$$

Step 4 — Domain check (the step that is usually skipped). Could a y in that set be produced only by an x that is a root of the denominator? If $x^2 + x - 1 = 0$, then the numerator is $x^2 + x + 1 = (x^2 + x - 1) + 2 = 0 + 2 = 2 \neq 0$. Numerator and denominator therefore share **no** real root, so every solution of (*) is a legal input and nothing further is removed.

$$\text{Range}(f) = (-\infty, -\frac{3}{5}] \cup (1, \infty)$$

Step 5 — Endpoint check. Is $-\frac{3}{5}$ attained? Set $y = -\frac{3}{5}$ in (*):

$$(-\frac{8}{5})x^2 + (-\frac{8}{5})x - \frac{2}{5} = 0 \implies 8x^2 + 8x + 2 = 0 \implies 2(2x + 1)^2 = 0 \implies x = -\frac{1}{2}.$$

Verify directly:

$$f(-\frac{1}{2}) = \frac{\frac{1}{4} - \frac{1}{2} + 1}{\frac{1}{4} - \frac{1}{2} - 1} = \frac{\frac{3}{4}}{-\frac{5}{4}} = -\frac{3}{5}. \checkmark$$

So $-\frac{3}{5}$ is genuinely attained and its bracket is closed, while 1 is never attained and its bracket is open.

Independent cross-check by substitution. Put $t = x^2 + x = (x + \frac{1}{2})^2 - \frac{1}{4}$. As x runs over $\mathbb{R}$, t attains every value in $[-\frac{1}{4}, \infty)$, and $t = 1$ must be excluded (it is exactly the condition $x^2 + x - 1 = 0$). Then

$$f = \frac{t+1}{t-1} = 1 + \frac{2}{t-1}.$$

- For $t \in [-\frac{1}{4}, 1)$: $t-1 \in [-\frac{5}{4}, 0)$, so $\frac{2}{t-1} \in (-\infty, -\frac{8}{5}]$, giving $f \in (-\infty, -\frac{3}{5}]$.
- For $t \in (1, \infty)$: $t-1 \in (0, \infty)$, so $\frac{2}{t-1} \in (0, \infty)$, giving $f \in (1, \infty)$.

The union is $(-\infty, -\frac{3}{5}] \cup (1, \infty)$ — the same answer, reached without a discriminant. When an Advanced-level range answer matters, running the second method is cheap insurance. ■

Chapter 2: Relations and Functions — Round 2 Addendum

Subtopics 01-03 and the Round 1 Supplement (Subtopic 04, plus Procedure E and Examples 6-7 appended to Subtopic 03) now cover the whole NCERT syllabus for this chapter: Cartesian products, relations and their counts, the definition of a function, the standard function zoo, natural domains, the algebra of real functions, the range of a real function, and the reflexive/symmetric/transitive machinery that NCERT gestures at in Example 19 and Miscellaneous Exercise Q9.

What is still missing is not syllabus — it is the *toolkit* layer that JEE builds on top of this chapter and never re-teaches. Six items, all of them assumed knowledge by the time Limits, Continuity and Definite Integrals arrive, and none of them present in the chapter or in Round 1:

Addendum	Topic	Read after
F	The vertical line test, stated so it survives a follow-up question	Subtopic 03, Procedure A
G	Composition of functions and the domain of a composite	Subtopic 03, Procedure C
H	Even and odd functions, and the work they save	Procedure E (Round 1)
I	Periodicity of algebraic (non-trigonometric) functions	Addendum H
J	Graph transformations of the standard function library	Subtopic 03, Section 2
K	Functional equations solvable at Class-11 level	Addendum H

Procedure lettering continues from Procedure E of the Round 1 Supplement; worked-example numbering continues from Example 7, and all new examples append to Subtopic 03, Section 4.

Deliberately not covered here: one-one, onto and bijective functions, and the inverse function. Subtopic 03 defers those, and they belong with the Class 12 Chapter 1 treatment where CBSE actually examines them. Composition is included despite the same deferral because domains of composites are examined at Class 11 level and because Addenda H, I and K all lean on it.

Addendum F — The vertical line test, stated properly

Extends Subtopic 03, Section 1 and Section 3 Procedure A step 3.

Method. The chapter gives you the test as a picture: “if every vertical line meets the graph at most once, it is a function.” That sentence is true but it hides two separate conditions, and an examiner who asks “at most once, or exactly once?” will collect the mark. Here is the version that answers back.

Procedure F — The vertical line test. Let $S \subseteq \mathbb{R} \times \mathbb{R}$ be any relation on $\mathbb{R}$: a curve, a roster of pairs, or the solution set of an equation in x and y . For a real number a , write

$$V_a = \{(a, y) : y \in \mathbb{R}\}, \quad m(a) = n(S \cap V_a),$$

so V_a is the vertical line $x = a$ and $m(a)$ counts how many points of S lie on it. Three verdicts, and only three:

1. **If $m(a) \leq 1$ for every real a ,** then S is a function. Its domain is $D = \{a \in \mathbb{R} : m(a) = 1\}$ — the shadow of S on the x -axis — and its rule is “ $f(a) =$ the one y on that line”.
2. **If a set A is declared in advance and $m(a) = 1$ for every $a \in A$,** then S is a function on A . This is the two-promise definition of Subtopic 03 restated: $m(a) \geq 1$ is totality, $m(a) \leq 1$ is single-valuedness.
3. **If $m(a) \geq 2$ for even one a ,** then S is not a function of x . One witness settles it; you never have to check the other lines.

The point most often missed: $m(a) = 0$ **is not a fault**. It says only that a is outside the domain. A relation can have enormous gaps and still be a perfectly good function on what is left. Only $m(a) \geq 2$ — one input carrying two outputs — is fatal.

How to run the test with no picture. Hold x fixed at a general value, solve the defining equation for y , and count real solutions. An even power of y , a $|y|$, or a $\pm$ is the usual source of a second solution.

How to run it on a piecewise definition. A rule given in clauses is a function precisely when the clauses agree wherever their intervals overlap. Where two clauses claim the same x and disagree, that single x has $m(x) = 2$ and the whole thing collapses.

Figure brief: Two panels side by side, same axes. Left: the circle $x^2 + y^2 = 25$ with three dashed vertical lines drawn at $x = -6$, $x = 3$ and $x = 5$, labelled $m = 0$, $m = 2$ and $m = 1$; the $m = 2$ line carries two ringed intersection points and a cross beside it. Right: the upper semicircle $y = \sqrt{25 - x^2}$ with the same three lines, now labelled $m = 0$, $m = 1$, $m = 1$, and a tick. Caption: “Only the middle case kills a function.”

Worked Example 8 — [CBSE Board Level]

The relation f is defined by

$$f(x) = \begin{cases} x^2, & 0 \leq x \leq 3 \\ 3x, & 3 \leq x \leq 10 \end{cases}$$

and the relation g is defined by

$$g(x) = \begin{cases} x^2, & 0 \leq x \leq 2 \\ 3x, & 2 \leq x \leq 10. \end{cases}$$

Show that f is a function and g is not.

Solution. Both definitions have overlapping clauses — the two intervals share an endpoint — so run the overlap check, which is exactly the vertical line test at that shared point.

For f : the clauses overlap only at $x = 3$. Clause one gives $3^2 = 9$; clause two gives $3 \cdot 3 = 9$. They agree, so $m(3) = 1$. Everywhere else on $[0, 10]$ exactly one clause applies and each clause is single-valued, so $m(a) = 1$ for every $a \in [0, 10]$. By verdict 2, f is a function on $[0, 10]$.

For g : the clauses overlap at $x = 2$. Clause one gives $2^2 = 4$; clause two gives $3 \cdot 2 = 6$. Both $(2, 4)$ and $(2, 6)$ belong to g , so $m(2) = 2$. By verdict 3, g is not a function.

Note how little work verdict 3 costs: one point, two numbers, done. You never look at the other ten units of the interval.

Worked Example 9 — [JEE Main Level]

For each of the following, decide whether it defines y as a function of x . When it does, give the domain and the range.

- (a) $x^2 + y^2 = 25$ (b) $x^2 + y^2 = 25$ with $y \geq 0$ (c) $x = y^3$
(b) $2x + 3|y| = 6$ (e) $x^2 = 4y$ (f) $|x| + |y| = 1$

Solution. In each case fix x and count the real y .

(a) $y^2 = 25 - x^2$. For $-5 < x < 5$ the right side is positive, so there are two real solutions $y = \pm\sqrt{25 - x^2}$. Taking $x = 3$ as the witness, $y = 4$ and $y = -4$, so $m(3) = 2$. **Not a function.**

(b) Now $y = +\sqrt{25 - x^2}$ only, and the negative root is barred by $y \geq 0$. For $|x| \leq 5$ there is exactly one y ; for $|x| > 5$ there is none. **A function**, domain $[-5, 5]$, range $[0, 5]$. This is the whole lesson of the addendum in one line: the same curve becomes a function the moment you cut it so that no vertical line is met twice.

(c) $y^3 = x$ has exactly one real root for every real x , namely $y = x^{1/3}$ (the other two cube roots are non-real). **A function**, domain $\mathbb{R}$, range $\mathbb{R}$. Contrast (a): an *odd* power of y never creates a second real solution.

(d) $3|y| = 6 - 2x$, so $|y| = \frac{6 - 2x}{3}$. For $x < 3$ the right side is positive and $y = \pm\frac{6 - 2x}{3}$ gives two values; at $x = 0$ these are $y = 2$ and $y = -2$. **Not a function.**

(e) $y = \frac{x^2}{4}$, one value for every real x . **A function**, domain $\mathbb{R}$, range $[0, \infty)$. The square is on x , not on y — it restricts the range, not the count.

(f) $|y| = 1 - |x|$. For $|x| < 1$ this gives two values of y ; at $x = 0$, $y = \pm 1$. **Not a function.**

The pattern to carry away: an even power of y , a $|y|$, or a $\pm$ attached to y is what breaks the test. Anything done to x alone — squares, moduli, floors — only reshapes the domain and range.

Practice F

1. Does $y^2 = 4x$ define y as a function of x ?
2. Does $y = |x| - x$ define a function? Give its range.
3. Does $x = y^2 + 1$ with $y \geq 0$ define y as a function of x ? Give domain and range.
4. Is $h(x) = \begin{cases} 2x + 1, & x \leq 1 \\ x^2 + 2, & x \geq 1 \end{cases}$ a function?
5. Is $p(x) = \begin{cases} x + 2, & 0 \leq x \leq 4 \\ 3x - 5, & 4 \leq x \leq 9 \end{cases}$ a function?
6. Is $y = \operatorname{sgn}(x^2 + 1)$ a function? Give its range.

Answers. 1. No. At $x = 1$, $y = \pm 2$. 2. Yes. It equals 0 for $x \geq 0$ and $-2x$ for $x < 0$; range $[0, \infty)$. 3. Yes: $y = \sqrt{x - 1}$. Domain $[1, \infty)$, range $[0, \infty)$. 4. Yes. The clauses overlap only at $x = 1$, where both give 3. 5. No. At $x = 4$ the clauses give 6 and 7. 6. Yes, and it is the constant function 1, because $x^2 + 1 > 0$ for every real x . Range $\{1\}$. (The trap is answering $\{-1, 0, 1\}$ from memory of the signum range.)

Addendum G — Composition of functions and the domain of a composite

Extends Subtopic 03, Section 3, and slots in immediately after Procedure C.

Method. Subtopic 03 taught you to combine two functions by arithmetic: $f + g$, fg , f/g . There is a fifth way to combine them that is not arithmetic at all — feed the output of one into the input of the other.

$$(g \circ f)(x) = g(f(x))$$

read “ g of f of x ”, or “ g after f ”. The inner function acts first, even though it is written second. That reversal is the source of half the errors on the topic.

Definition of the domain — the only formula you need.

$$\operatorname{Dom}(g \circ f) = \{x \in \operatorname{Dom}(f) : f(x) \in \operatorname{Dom}(g)\}$$

Every symbol earns its place. x must be legal for f , because f runs first; and the number $f(x)$ that comes out must be legal for g , because g runs on it.

Procedure G — Composite and its domain.

1. **Write down** $\operatorname{Dom}(f)$ and $\operatorname{Dom}(g)$ **separately** using Procedure B (denominators non-zero, even radicands non-negative).
2. **Impose the membership condition** $f(x) \in \operatorname{Dom}(g)$ and solve it as an inequality or exclusion in x .
3. **Intersect** with $\operatorname{Dom}(f)$. That intersection is the answer.

4. **Only now simplify the formula.** (Why this order: simplification can heal a defect that the definition says must stay. $(\sqrt{x})^2$ simplifies to x , whose natural domain is all of $\mathbb{R}$, but x had to survive $\sqrt{\cdot}$ first. Reading the domain off the simplified formula is the single most-punished error in this topic.)

Three properties.

- **Not commutative:** $g \circ f \neq f \circ g$ in general — different formulas, and often different domains too.
- **Associative:** $f \circ (g \circ h) = (f \circ g) \circ h$ whenever both sides are defined, so $f \circ g \circ h$ is unambiguous.
- **Identity:** with $I(x) = x$ on $\mathbb{R}$, $f \circ I = I \circ f = f$ on $\text{Dom}(f)$.

Figure brief: Three ovals in a row labelled $\text{Dom}(f)$, an intermediate set, and the target. An arrow f from a dot x in the first to a dot $f(x)$ in the second, and an arrow g from $f(x)$ to $g(f(x))$ in the third. Inside the middle oval, a shaded sub-region labelled $\text{Dom}(g)$, with a second dot $f(x')$ drawn *outside* the shading and a red cross on it, captioned “ x' is legal for f but its image is illegal for g — so x' is not in $\text{Dom}(g \circ f)$.”

Worked Example 10 — [JEE Main — Speed Trap]

Let $f(x) = \sqrt{x}$ and $g(x) = x^2$. Find $g \circ f$ and $f \circ g$, each with its exact domain. Are the two composites equal?

The trap. The reflex answer is “ $(\sqrt{x})^2 = x$ and $\sqrt{x^2} = x$, so both composites are the identity and $g \circ f = f \circ g$.” Both halves of that sentence are wrong, for two different reasons, and the second one is wrong even as a formula.

Solution. $\text{Dom}(f) = [0, \infty)$ and $\text{Dom}(g) = \mathbb{R}$.

$g \circ f$. Domain first: $x \in [0, \infty)$, and $f(x) = \sqrt{x}$ must lie in $\text{Dom}(g) = \mathbb{R}$, which costs nothing. So $\text{Dom}(g \circ f) = [0, \infty)$. Formula: $(g \circ f)(x) = (\sqrt{x})^2 = x$. So $g \circ f$ is the identity **restricted to** $[0, \infty)$ — not the identity on $\mathbb{R}$.

$f \circ g$. Domain first: $x \in \mathbb{R}$, and $g(x) = x^2$ must lie in $\text{Dom}(f) = [0, \infty)$, which is automatic since squares are non-negative. So $\text{Dom}(f \circ g) = \mathbb{R}$. Formula: $(f \circ g)(x) = \sqrt{x^2} = |x|$, **not** x — the square root returns the non-negative root, as Subtopic 03 warned.

Are they equal? No, on both counts. Equality of functions (Subtopic 03, Section 2) demands the same domain *and* the same values, and here the domains differ ($[0, \infty)$ against $\mathbb{R}$) and the values differ too: at $x = -4$, $(f \circ g)(-4) = 4$ while $(g \circ f)(-4)$ is not even defined.

Answers. $(g \circ f)(x) = x$ on $[0, \infty)$; $(f \circ g)(x) = |x|$ on $\mathbb{R}$; they agree on $[0, \infty)$ and nowhere else.

Worked Example 11 — [JEE Advanced Level]

Let $f(x) = \frac{1}{1-x}$. Write $f^2 = f \circ f$ and $f^3 = f \circ f \circ f$. Find f^3 and its domain, and hence evaluate $f^{2026}(2)$.

Solution.

Step 0 — domain of f . $\text{Dom}(f) = \mathbb{R} \setminus \{1\}$.

Step 1 — f^2 . Domain: need $x \neq 1$, and need $f(x) \neq 1$. Now $\frac{1}{1-x} = 1$ forces $1-x = 1$, i.e. $x = 0$. So $\text{Dom}(f^2) = \mathbb{R} \setminus \{0, 1\}$. Formula:

$$f^2(x) = \frac{1}{1 - \frac{1}{1-x}} = \frac{1}{\frac{(1-x)-1}{1-x}} = \frac{1-x}{-x} = \frac{x-1}{x}.$$

Step 2 — f^3 . Domain: need $x \in \text{Dom}(f^2)$ and $f^2(x) \neq 1$. But $\frac{x-1}{x} = 1$ forces $-1 = 0$, which never happens, so nothing further is removed and $\text{Dom}(f^3) = \mathbb{R} \setminus \{0, 1\}$. Formula:

$$f^3(x) = \frac{1}{1 - \frac{x-1}{x}} = \frac{1}{\frac{x-(x-1)}{x}} = \frac{1}{\frac{1}{x}} = x.$$

So f^3 **is the identity on** $\mathbb{R} \setminus \{0, 1\}$: iterating f cycles every legal input with period 3.

Step 3 — the high power. Since $f^3 = I$ on $\mathbb{R} \setminus \{0, 1\}$, we get $f^{3k} = I$, $f^{3k+1} = f$ and $f^{3k+2} = f^2$ there. Now $2026 = 3 \times 675 + 1$, so $f^{2026} = f$ on $\mathbb{R} \setminus \{0, 1\}$, and

$$f^{2026}(2) = f(2) = \frac{1}{1-2} = -1.$$

Check by walking the cycle: $2 \rightarrow -1 \rightarrow \frac{1}{2} \rightarrow 2$, a 3-cycle, confirming both $f^3 = I$ and the value.

Answers. $f^3(x) = x$ on $\mathbb{R} \setminus \{0, 1\}$; $f^{2026}(2) = -1$.

The structural lesson is worth more than the number: the domain shrank once, from $\mathbb{R} \setminus \{1\}$ to $\mathbb{R} \setminus \{0, 1\}$, and then stopped. Composites lose points only while new inputs are still being blocked; once the cycle closes, the domain is stable.

Practice G

- $f(x) = x^2 - 1$, $g(x) = \sqrt{x}$. Find $g \circ f$ and $f \circ g$ with exact domains.
- $f(x) = \frac{1}{x}$, $g(x) = x + 3$. Find $g \circ f$ and $f \circ g$ with exact domains.
- $f(x) = 2x + 1$, $g(x) = x^2$. Solve $(g \circ f)(x) = (f \circ g)(x)$.
- $f(x) = \frac{x}{x-1}$. Find $f \circ f$ and its domain.
- $f(x) = \sqrt{x-1}$, $g(x) = \frac{1}{x-3}$. Find $\text{Dom}(g \circ f)$ and $\text{Dom}(f \circ g)$.
- $f(x) = [x]$ (greatest integer) and $g(x) = x^2$. Evaluate $(g \circ f)(2.7)$ and $(f \circ g)(2.7)$.

Answers. 1. $(g \circ f)(x) = \sqrt{x^2 - 1}$, domain $(-\infty, -1] \cup [1, \infty)$; $(f \circ g)(x) = x - 1$, domain $[0, \infty)$ — **not** $\mathbb{R}$. 2. $(g \circ f)(x) = \frac{1}{x} + 3 = \frac{1+3x}{x}$, domain $\mathbb{R} \setminus \{0\}$; $(f \circ g)(x) = \frac{1}{x+3}$, domain $\mathbb{R} \setminus \{-3\}$. 3. $(2x+1)^2 = 2x^2+1 \Rightarrow 2x^2+4x = 0 \Rightarrow x = 0$ or $x = -2$. 4. $(f \circ f)(x) = x$, domain $\mathbb{R} \setminus \{1\}$. (Nothing extra is removed: $\frac{x}{x-1} = 1$ has no solution.) 5. $\text{Dom}(g \circ f) = [1, \infty) \setminus \{10\}$ (need $x \geq 1$ and

$\sqrt{x-1} \neq 3$); $\text{Dom}(f \circ g) = (3, 4]$ (need $x \neq 3$ and $\frac{1}{x-3} \geq 1$). 6. $(g \circ f)(2.7) = [2.7]^2 = 2^2 = 4$; $(f \circ g)(2.7) = [7.29] = 7$. A clean numerical demonstration that order matters.

Addendum H — Even and odd functions, and the work they save

Extends Subtopic 03, Section 2, and pairs with Procedure E (Round 1).

Method. A great many functions on the JEE syllabus are built symmetrically about the origin, and spotting that symmetry lets you throw away half of every domain, range and sketching problem. The classification costs three lines and repays them many times.

Precondition — a symmetric domain. Before you compute anything, check that $x \in D \Rightarrow -x \in D$. If the domain is not symmetric about 0, the function is **neither even nor odd**, whatever the formula looks like. $f(x) = x^2$ on $[1, 3]$ is neither.

Definitions. For f with a domain D symmetric about 0:

$$f \text{ is even} \iff f(-x) = f(x) \text{ for all } x \in D$$

$$f \text{ is odd} \iff f(-x) = -f(x) \text{ for all } x \in D$$

Most functions are neither. “Neither” is a legitimate and common answer.

Immediate consequences.

- **Graph symmetry.** f even $\iff$ the graph is unchanged by reflection in the y -axis. f odd $\iff$ the graph is unchanged by a half-turn (180° rotation) about the origin.
- **If f is odd and $0 \in D$, then $f(0) = 0$.** Put $x = 0$: $f(0) = -f(0)$, so $2f(0) = 0$. An odd function defined at the origin *must* pass through it.
- **The only function that is both even and odd** is the zero function on a symmetric domain: $f(x) = f(-x) = -f(x)$ forces $f(x) = 0$.
- $f(|x|)$ **is always even**, whatever f is, because $|-x| = |x|$.

Procedure H — The three-line test. 1. Check the domain is symmetric about 0. If not, stop: neither. 2. Compute $f(-x)$ and simplify. 3. Compare. Equal to $f(x)$: even. Equal to $-f(x)$: odd. Neither: produce one value of x at which both comparisons fail, and say so. (*Why one witness suffices: both definitions are universally quantified, so a single failing x destroys them.*)

The algebra of parity. Assume f and g share a symmetric domain.

f	g	$f \pm g$	fg	f/g	$f \circ g$
even	even	even	even	even	even
odd	odd	odd	even	even	odd
even	odd	neither, in general	odd	odd	even
odd	even	neither, in general	odd	odd	even

Two entries deserve staring at. **Odd times odd is even** — the two sign flips cancel, exactly as $(-x)(-x) = x^2$. And in a composite **the inner function decides**: if the inner function is even, the composite is even no matter what the outer one is; if the inner is odd, the composite copies the parity of the outer.

The even-odd decomposition. Every f on a symmetric domain splits, in exactly one way, as an even part plus an odd part:

$$f(x) = \underbrace{\frac{f(x) + f(-x)}{2}}_{\text{even part } f_e} + \underbrace{\frac{f(x) - f(-x)}{2}}_{\text{odd part } f_o}$$

Add the two fractions and the $f(-x)$ terms cancel, which proves existence; replacing x by $-x$ in each piece confirms the labels.

Why any of this saves work — the four payoffs.

1. **Sketching.** Draw an even function on $[0, \infty)$ only, then reflect in the y -axis. Draw an odd function on $[0, \infty)$ only, then half-turn about the origin. Half the plotting, none of the risk.
2. **Range of an even function** equals its range on the non-negative half of the domain. The negative half contributes nothing new.
3. **Range of an odd function is symmetric about 0:** y is attained $\iff -y$ is attained. So compute the range on $[0, \infty)$ and reflect it through 0. This is the biggest single saving in the whole addendum, and Example 13 uses it.
4. **Roots.** The solution set of $f(x) = 0$ is symmetric about 0 for both even and odd f , and an odd f defined at 0 always has $x = 0$ among its roots.

Figure brief: Two panels. Left, an even function: a curve drawn on $x \geq 0$ in solid ink and its y -axis mirror image on $x \leq 0$ in dashed ink, with a vertical mirror line on the y -axis, captioned “draw once, reflect”. Right, an odd function: the same solid arc on $x \geq 0$, its half-turn image on $x \leq 0$ dashed, a small circular rotation arrow centred at the origin, and the origin ringed and labelled “ $f(0) = 0$ is forced”.

Worked Example 12 — [CBSE / JEE Main Level]

Classify each as even, odd, or neither, with a one-line reason.

- (a) $f(x) = x^4 - 3x^2 + 1$ (b) $f(x) = x|x|$ (c) $f(x) = x^3 + x^2$
 (b) $f(x) = \sqrt{1+x+x^2} - \sqrt{1-x+x^2}$ (e) $f(x) = [x] + [-x]$
 (c) $f(x) = x^2$ on the domain $[1, 3]$

Solution.

(a) Even. Only even powers of x appear, so $f(-x) = x^4 - 3x^2 + 1 = f(x)$. (The constant 1 is $1 \cdot x^0$, an even power — constants never break evenness.)

(b) Odd. $f(-x) = (-x)|-x| = -x|x| = -f(x)$. This is the standard “odd times even is odd” entry: x is odd, $|x|$ is even.

(c) Neither. $f(-x) = -x^3 + x^2$. Witness $x = 1$: $f(1) = 2$, $f(-1) = 0$, and 0 is neither 2 nor -2 . A sum of an odd term and an even term is the archetypal “neither”.

(d) Odd. First the domain: $1 + x + x^2$ has discriminant $1 - 4 = -3 < 0$ and positive leading coefficient, so it is positive for every real x ; the same for $1 - x + x^2$. Hence $D = \mathbb{R}$, which is symmetric. Now replacing x by $-x$ swaps the two radicands:

$$f(-x) = \sqrt{1 - x + x^2} - \sqrt{1 + x + x^2} = -f(x).$$

(e) Even. $f(-x) = [-x] + [x] = f(x)$ — the two terms simply trade places. (Worth computing a value or two: $f(2.3) = 2 + (-3) = -1$ and $f(3) = 3 + (-3) = 0$. So f is 0 at integers and -1 everywhere else, which is visibly symmetric.) This one catches students who try to argue from “ $[x]$ is neither even nor odd”, which is true and irrelevant.

(f) Neither. The domain $[1, 3]$ is not symmetric about 0: $1 \in D$ but $-1 \notin D$, so $f(-1)$ does not exist and neither definition can even be tested. The formula being x^2 is a red herring.

Worked Example 13 — [JEE Advanced Level]

Find the range of $f(x) = \sqrt{x^2 + x + 1} - \sqrt{x^2 - x + 1}$, $x \in \mathbb{R}$.

Solution. This is a Procedure E range problem that oddness cuts in half.

Step 0 — domain. Both discriminants are $1 - 4 = -3 < 0$ with positive leading coefficients, so both radicands are positive for every real x . $D = \mathbb{R}$.

Step 1 — exploit parity. By Example 12(d), f is odd. Payoff 3 above therefore applies: **the range is symmetric about 0**, so it is enough to find the range on $[0, \infty)$ and reflect. That halves the algebra.

Step 2 — rationalise. Multiply and divide by the conjugate. With $A = \sqrt{x^2 + x + 1}$ and $B = \sqrt{x^2 - x + 1}$ (both positive),

$$f(x) = \frac{A^2 - B^2}{A + B} = \frac{(x^2 + x + 1) - (x^2 - x + 1)}{A + B} = \frac{2x}{A + B}.$$

Immediately $f(0) = 0$ and $f(x) > 0$ for $x > 0$.

Step 3 — the upper bound. Claim: $f(x) < 1$ for every $x > 0$, i.e. $A + B > 2x$. Since both sides are positive, square:

$$(A + B)^2 = A^2 + B^2 + 2AB = (2x^2 + 2) + 2\sqrt{(x^2 + x + 1)(x^2 - x + 1)} = 2x^2 + 2 + 2\sqrt{x^4 + x^2 + 1}.$$

Now $(x^4 + x^2 + 1) - (x^2 - 1)^2 = 3x^2 > 0$ for $x \neq 0$, so $\sqrt{x^4 + x^2 + 1} > |x^2 - 1| \geq x^2 - 1$. Therefore

$$(A + B)^2 > 2x^2 + 2 + 2(x^2 - 1) = 4x^2 = (2x)^2,$$

giving $A + B > 2x$ and hence $0 \leq f(x) < 1$ on $[0, \infty)$.

Step 4 — attainment. f is continuous on $[0, \infty)$ with $f(0) = 0$, and as x grows $A + B$ approaches $2x$, so $f(x)$ climbs towards 1: numerically $f(1) = \sqrt{3} - 1 \approx 0.7321$, $f(5) \approx 0.9852$, $f(100) \approx 0.99996$. Every value between 0 and 1 is therefore taken somewhere on $[0, \infty)$, while 1 itself is not (Step 3 made the inequality strict). Range on $[0, \infty)$ is $[0, 1)$.

Step 5 — reflect. By oddness, the range on $(-\infty, 0]$ is $(-1, 0]$. Hence

$$\boxed{\text{Range}(f) = (-1, 1)}$$

Both brackets are open — a Step 6 endpoint check in the Procedure E sense — because $f(x) = 1$ would need $A + B = 2x$, which Step 3 forbids.

Cost of not spotting the parity: you would have to run Steps 2-4 twice, once on each half, with the sign of x flipping through every inequality. That is where the errors live.

Practice H

1. Classify $f(x) = x^5 - 3x^3 + 7x$.
2. Classify $f(x) = |x + 1| + |x - 1|$.
3. Classify $f(x) = |x + 1| - |x - 1|$.
4. Classify $f(x) = x^2 + x^4 + 1$.
5. Classify $f(x) = \frac{x}{x^2 - 4}$ (state the domain first).
6. Classify $f(x) = (x + 1)^2$.
7. Write $f(x) = x^3 + 3x^2 - 2x + 5$ as the sum of an even and an odd function.
8. f is odd and $f(3) = 7$. Find $f(-3)$ and $f(0)$.
9. Use parity to find the range of $g(x) = x|x|$ on $\mathbb{R}$.

Answers. 1. Odd. 2. Even. 3. Odd. 4. Even. 5. Domain $\mathbb{R} \setminus \{-2, 2\}$, which is symmetric; odd (odd over even). 6. Neither. ($f(1) = 4$, $f(-1) = 0$.) 7. Even part $3x^2 + 5$; odd part $x^3 - 2x$. 8. $f(-3) = -7$ and $f(0) = 0$. 9. g is odd. On $[0, \infty)$ it equals x^2 , with range $[0, \infty)$; reflecting gives $(-\infty, 0]$ on the other half. Range = $\mathbb{R}$.

Addendum I — Periodicity of algebraic (non-trigonometric) functions

Extends Subtopic 03, Section 2 (the greatest-integer and fractional-part discussion).

Method. Periodicity is usually introduced with sines and cosines, which leaves students believing it is a trigonometric idea. It is not. The fractional part $\{x\} = x - [x]$ is the purest periodic function on the Class 11 syllabus, and JEE tests it constantly — inside limits, inside integrals, and on its own.

Definition. A function f is **periodic** if there exists a real number $T > 0$ such that for every $x \in \text{Dom}(f)$, both $x + T \in \text{Dom}(f)$ and

$$f(x + T) = f(x).$$

Any such T is called a **period**. The smallest positive period, *if one exists*, is the **fundamental period**.

The caveat nobody states and every examiner exploits: a smallest period need not exist. A constant function satisfies $f(x + T) = f(x)$ for every $T > 0$, so it is periodic but has

no fundamental period. “Find the fundamental period of $f(x) = 7$ ” has the answer “there is none”, not “1”.

Six facts you can prove with Class 11 tools.

(P1) $\{x\} = x - [x]$ is periodic with fundamental period 1. $\{x + 1\} = (x + 1) - [x + 1] = (x + 1) - ([x] + 1) = x - [x] = \{x\}$, so 1 is a period. If $0 < T < 1$ were also a period, then taking $x = 0$ gives $\{T\} = T$ while $\{0\} = 0$, and $T \neq 0$. So no smaller period exists.

(P2) $[x]$ is not periodic. Suppose $T > 0$ were a period. Taking $x = 0$ gives $[T] = [0] = 0$, so $0 < T < 1$. Now take $x = -T/2$, which lies in $(-1, 0)$, so $[x] = -1$; but $x + T = T/2$ lies in $(0, 1)$, so $[x + T] = 0$. Then $0 = -1$, a contradiction. (This is the standard one-mark trap: $\{x\}$ is periodic, $[x]$ is not, and they differ by x .)

(P3) No non-constant monotone function is periodic. If f is strictly increasing and $T > 0$, then $f(x + T) > f(x)$ always, so no T can work. This kills x , x^3 , $\sqrt{x}$, $2x + 5$ in one stroke, and $|x|$ and x^2 follow from $f(0) = 0 < f(T)$.

(P4) Scaling rule. If f has period T , then $g(x) = f(ax + b)$ with $a \neq 0$ has period $T/|a|$. *Proof:* $g(x + \frac{T}{|a|}) = f(ax + b \pm T) = f(ax + b) = g(x)$. If T was fundamental for f , then $T/|a|$ is fundamental for g . So $\{3x - 1\}$ has fundamental period $1/3$, and $\{x/4\}$ has fundamental period 4.

(P5) Sum rule (with its warning). If f has period T_1 , g has period T_2 , and T_1/T_2 is rational, then $\text{LCM}(T_1, T_2)$ is a period of $f \pm g$ and of fg . It **need not be the fundamental one** — always test the obvious smaller candidates. For fractions, $\text{LCM}\left(\frac{p}{q}, \frac{r}{s}\right) = \frac{\text{LCM}(p, r)}{\text{HCF}(q, s)}$. If T_1/T_2 is irrational, no common period exists and the sum is not periodic.

(P6) An outer operation never lengthens a period. If f has period T , then $h(f(x))$ has T as a period for any h — possibly with a shorter fundamental period. So $\{x\}^2$, $\sqrt{\{x\}}$ and $\text{sgn}(\{x\})$ all have period 1.

The standard algebraic table.

Function	Fundamental period
$\{x\} = x - [x]$	1
$\{ax + b\}$, $a \neq 0$	$1/ a $
$[x] - x$	1
$(-1)^{[x]}$	2
$\text{sgn}(\{x\})$	1
$[x]$	not periodic
x , x^2 , x^3 , $ x $, $\sqrt{x}$, $\text{sgn}(x)$, $1/x$	not periodic
any constant	periodic; no fundamental period

Figure brief: One set of axes, $-3 \leq x \leq 3$. The sawtooth $y = \{x\}$ drawn as repeated segments from $(n, 0)$ filled to $(n + 1, 1)$ open, with one period highlighted in a shaded vertical band of width 1 and an arrow labelled “ $T = 1$ ” spanning it. Beneath, on the same x -range, the staircase $y = [x]$ drawn faintly with a caption “same ingredient, drifts upward forever, never repeats”.

Worked Example 14 — [JEE Main Level]

Find the fundamental period of each, or show that none exists.

- (a) $f(x) = \{3x - 1\}$ (b) $f(x) = (-1)^{[x]}$ (c) $f(x) = [x]$
(b) $f(x) = \left\{\frac{x}{2}\right\} + \left\{\frac{x}{3}\right\}$ (e) $f(x) = 7$

Solution.

(a) $1/3$. By (P1) the fractional part has fundamental period 1; by (P4) with $a = 3$ the period becomes $1/3$. Directly: $\{3(x + \frac{1}{3}) - 1\} = \{3x - 1 + 1\} = \{3x - 1\}$. For minimality, $\{3x + 3T - 1\} = \{3x - 1\}$ for all x forces $3T$ to be a positive integer, so $T \geq 1/3$.

(b) 2. Since $[x + 2] = [x] + 2$, the exponent changes by 2 and the sign does not: $(-1)^{[x]+2} = (-1)^{[x]}$. So 2 is a period. Is 1? At $x = 0$ the value is $(-1)^0 = 1$; at $x = 1$ it is $(-1)^1 = -1$. So 1 fails, and since any period must also satisfy $f(T) = f(0) = 1$, the fundamental period is 2.

(c) **Not periodic**, by the argument of (P2). Do not be tempted by “ $[x]$ repeats its shape every unit” — it repeats its *shape*, not its *values*, and periodicity is about values.

(d) 6. $\{x/2\}$ has period 2 and $\{x/3\}$ has period 3 by (P4); the ratio $2/3$ is rational, so $\text{LCM}(2, 3) = 6$ is a period. Now prove it is fundamental, and do it by argument rather than by testing candidates. Suppose $T > 0$ is a period. Then $f(T) = f(0) = 0$. But $f(T) = \{T/2\} + \{T/3\}$ is a sum of two non-negative numbers, so both must vanish: $\{T/2\} = 0$ and $\{T/3\} = 0$, i.e. T is a multiple of 2 **and** a multiple of 3. Hence $T \geq 6$, and 6 is fundamental.

(e) **Periodic, but there is no fundamental period.** Every $T > 0$ satisfies $f(x + T) = 7 = f(x)$, and the set of periods $(0, \infty)$ has no smallest element.

Worked Example 15 — [JEE Advanced Level]

A function $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfies, for a fixed $a > 0$,

$$f(x + a) = \frac{1}{2} + \sqrt{f(x) - (f(x))^2} \quad \text{for every } x \in \mathbb{R}.$$

Prove that f is periodic, and give a period.

Solution. The whole problem is decided by one observation made before any algebra.

Step 1 — pin down the values of f . For the square root to be real we need $f(x) - f(x)^2 \geq 0$, i.e. $f(x)(1 - f(x)) \geq 0$, i.e. $0 \leq f(x) \leq 1$ for every x . Moreover the right-hand side is $\frac{1}{2}$ plus a non-negative number, so $f(x + a) \geq \frac{1}{2}$ for every x ; and as x runs over all of $\mathbb{R}$ so does $x + a$. Therefore

$$\frac{1}{2} \leq f(y) \leq 1 \quad \text{for every real } y.$$

(This step is the one students skip, and without it the proof cannot be finished.)

Step 2 — apply the relation twice. Write $u = \sqrt{f(x) - f(x)^2} \geq 0$, so $f(x + a) = \frac{1}{2} + u$ and $1 - f(x + a) = \frac{1}{2} - u$. Replacing x by $x + a$:

$$f(x + 2a) = \frac{1}{2} + \sqrt{f(x + a) - f(x + a)^2} = \frac{1}{2} + \sqrt{f(x + a)(1 - f(x + a))} = \frac{1}{2} + \sqrt{\left(\frac{1}{2} + u\right)\left(\frac{1}{2} - u\right)} = \frac{1}{2} + \sqrt{\frac{1}{4} - u^2}.$$

Step 3 — recognise the surd. Since $u^2 = f(x) - f(x)^2$,

$$\frac{1}{4} - u^2 = \frac{1}{4} - f(x) + f(x)^2 = \left(f(x) - \frac{1}{2}\right)^2,$$

so $\sqrt{\frac{1}{4} - u^2} = \left|f(x) - \frac{1}{2}\right|$.

Step 4 — cash in Step 1. Because $f(x) \geq \frac{1}{2}$, the modulus opens with a plus sign: $|f(x) - \frac{1}{2}| = f(x) - \frac{1}{2}$. Hence

$$f(x + 2a) = \frac{1}{2} + f(x) - \frac{1}{2} = f(x) \quad \text{for every } x.$$

So f is periodic with period $2a$. ■

Remark — why the answer is “a period”, not “the fundamental period”. Take f constant, $f \equiv c$. The relation becomes $c = \frac{1}{2} + \sqrt{c - c^2}$, which rearranges to $8c^2 - 8c + 1 = 0$ and gives $c = \frac{2 + \sqrt{2}}{4} \approx 0.8536$ (the root with $c \geq \frac{1}{2}$). Check: $c - c^2 = c(1 - c) = \frac{2 + \sqrt{2}}{4} \cdot \frac{2 - \sqrt{2}}{4} = \frac{2}{16} = \frac{1}{8}$, and $\frac{1}{2} + \sqrt{1/8} = \frac{1}{2} + \frac{\sqrt{2}}{4} = c$. That solution is constant, so every positive number is one of its periods and it has no fundamental period at all. The theorem can only ever promise a period.

Practice I

1. Fundamental period of $\left\{\frac{x}{4}\right\}$.
2. Fundamental period of $\{2x\}$.
3. Fundamental period of $\{x\} + \{2x\}$.
4. Fundamental period of $(-1)^{[x/2]}$.
5. Fundamental period of $[x] - x$.
6. Is $f(x) = [x] + \{x\}$ periodic?
7. Fundamental period of $\text{sgn}(\{x\})$.
8. Fundamental period of $\left\{\frac{x}{3}\right\} + \left\{\frac{x}{5}\right\}$.

Answers. 1. 4. 2. $1/2$. 3. 1. (Periods 1 and $1/2$; LCM = 1, and $1/2$ fails at $x = 0$ since $f(0) = 0$ but $f(1/2) = 1/2 + 0 = 1/2$.) 4. 4. 5. 1. (It equals $-\{x\}$.) 6. No — it equals x , which is strictly increasing, so (P3) applies. 7. 1. It is 0 at integers and 1 elsewhere; any period T needs $f(T) = f(0) = 0$, so T must be a positive integer. 8. 15. Same argument as Example 14(d): $f(T) = 0$ forces T to be a multiple of both 3 and 5.

Addendum J — Graph transformations of the standard function library

Extends Subtopic 03, Section 2 (“Graphs of the standard functions”).

Method. Subtopic 03 gave you six graphs to know on sight. This addendum turns those six into several hundred. Every transformed graph you will ever be asked for is one of nine moves, or a chain of them, applied to a curve you already know.

Procedure J — The nine moves. Assume you can draw $y = f(x)$, and let $k, h > 0$ and $c > 0$.

New rule	What happens to the graph	Memory hook
$y = f(x) + k$	shift up k	outside the bracket, does what it says
$y = f(x) - k$	shift down k	
$y = f(x - h)$	shift right h	inside the bracket, opposite sign
$y = f(x + h)$	shift left h	
$y = c f(x)$	vertical stretch by factor c (a squash if $c < 1$)	multiplies the y -values
$y = f(cx)$	horizontal squash by factor c (a stretch if $c < 1$)	divides the x -values
$y = -f(x)$	reflect in the x -axis	
$y = f(-x)$	reflect in the y -axis	
$y = f(x) $	keep everything on or above the x -axis; flip the part below it upward	fold about the x -axis
$y = f(x)$	delete the $x < 0$ part; copy the $x \geq 0$ part reflected into $x < 0$	fold about the y -axis; result is always even
$ y = f(x)$	keep only where $f(x) \geq 0$; draw f and its mirror $-f$ there	not a function — vertical line test fails

Two rules that prevent most of the errors.

1. **Inside the bracket, everything is backwards, and it happens first.** To read $y = f(cx + d)$, **factor out c first:** $f(c(x + \frac{d}{c}))$. Then the chain is “squash horizontally by c , **then** shift left d/c ”, in that order. Trying to read $cx + d$ as a single shift of d is the classic mark-loser, because the shift is d/c , not d .
2. $|f(x)|$ **and** $f(|x|)$ **are different pictures and almost never the same.** $|f(x)|$ agrees with f wherever $f \geq 0$ and mirrors it upward elsewhere; $f(|x|)$ agrees with f on $x \geq 0$ and throws away what f did on $x < 0$ entirely.

Figure brief: A 2×2 grid of small panels, all built on the same faint reference parabola $y = x^2 - 4x + 3$ drawn in grey. Panel 1 titled “ $f(x)$ ”: the plain parabola, roots ringed at $x = 1, 3$, vertex marked $(2, -1)$. Panel 2 titled “ $|f(x)|$ ”: same curve with the arc between $x = 1$ and $x = 3$ flipped above the axis into a hump peaking at $(2, 1)$, corners ringed at $(1, 0)$ and $(3, 0)$. Panel 3 titled “ $f(|x|)$ ”: the $x \geq 0$ branch kept and mirrored into $x < 0$, roots ringed at $\pm 1, \pm 3$, twin vertices at $(\pm 2, -1)$, local peak at $(0, 3)$. Panel 4 titled “ $|y| = f(x)$ ”: only the regions $x \leq 1$ and $x \geq 3$ occupied, each carrying the parabola and its reflection, the strip $1 < x < 3$ left blank and hatched, with the caption “not a function”.

Worked Example 16 — [CBSE / JEE Main Level]

Starting from $y = |x|$, describe the chain of transformations that produces $y = 3 - |2x - 4|$. State the vertex, the x -intercepts, the slopes of the arms and the range.

Solution.

Step 1 — factor inside the modulus. This is the whole trick.

$$|2x - 4| = |2(x - 2)| = 2|x - 2|, \quad \text{so} \quad y = 3 - 2|x - 2|.$$

Written this way, every move is now readable. Attempting to read “ $2x - 4$ ” as a shift of 4 gives the wrong vertex and the wrong slopes.

Step 2 — the chain, in order.

1. $y = |x|$ — the standard V, vertex $(0, 0)$, arms of slope ± 1 .
2. $y = |x - 2|$ — shift **right** 2. Vertex $(2, 0)$.
3. $y = 2|x - 2|$ — vertical stretch by 2. Vertex still $(2, 0)$; arms now slope ± 2 .
4. $y = -2|x - 2|$ — reflect in the x -axis. An inverted V, vertex $(2, 0)$, arms ∓ 2 .
5. $y = 3 - 2|x - 2|$ — shift **up** 3. Vertex $(2, 3)$.

Step 3 — read off the answers. - **Vertex** $(2, 3)$, the maximum point. - **x -intercepts:** $2|x - 2| = 3 \Rightarrow |x - 2| = \frac{3}{2} \Rightarrow x = \frac{1}{2}$ or $x = \frac{7}{2}$. - **Arms** have slopes $+2$ (left of the vertex) and -2 (right of it) — *not* ± 1 . The stretch in step 3 is exactly what factoring exposed. - **Range** $(-\infty, 3]$: the graph is a downward V, so 3 is attained at $x = 2$ and everything below it is reached.

Spot check: $y(0) = 3 - 2|-2| = -1$ and $y(4) = 3 - 2|2| = -1$, symmetric about $x = 2$ as a V must be. And $y(1) = y(3) = 1$.

Worked Example 17 — [JEE Advanced Level]

Let $f(x) = x^2 - 4x + 3$. (a) Describe the graphs of $y = |f(x)|$, $y = f(|x|)$ and $|y| = f(x)$. (b) Hence find, for every real k , the number of real solutions of $|x^2 - 4x + 3| = k$.

Solution.

Step 0 — know the parent curve. $f(x) = (x - 2)^2 - 1$: an upward parabola with roots $x = 1$ and $x = 3$ and vertex $(2, -1)$.

(a)

- $y = |f(x)|$. Outside $[1, 3]$ the parabola already sits on or above the axis and is unchanged. On $(1, 3)$ it dips below, so that arc flips upward into a hump of height 1 peaking at $(2, 1)$, meeting the axis in corners at $(1, 0)$ and $(3, 0)$.
- $y = f(|x|) = x^2 - 4|x| + 3$. Keep the $x \geq 0$ branch of the parabola and mirror it into $x < 0$. The result is even: roots at $x = \pm 1$ and $x = \pm 3$, two vertices at $(\pm 2, -1)$, and a local maximum at $(0, 3)$ with a corner there.
- $|y| = f(x)$. This needs $f(x) \geq 0$, i.e. $x \leq 1$ or $x \geq 3$; on the strip $1 < x < 3$ there is no graph at all. Where it does exist, it consists of the parabola together with its reflection $y = -f(x)$. It fails the vertical line test everywhere except at $x = 1$ and $x = 3$, so it is **not** a function.

(b) Count intersections of $y = |f(x)|$ with the horizontal line $y = k$, reading straight off the hump of height 1:

k	number of real solutions
$k < 0$	0
$k = 0$	2
$0 < k < 1$	4
$k = 1$	3
$k > 1$	2

Algebraic confirmation. $|f(x)| = k$ means $f(x) = k$ or $f(x) = -k$, i.e. $(x - 2)^2 = 1 + k$ or $(x - 2)^2 = 1 - k$. For $k \geq 0$ the first always has 2 roots; the second has 2 roots if $k < 1$, one root ($x = 2$) if $k = 1$, and none if $k > 1$. At $k = 0$ the two families coincide, giving 2 distinct roots, not 4. Adding up reproduces the table exactly.

Spot checks. $k = \frac{1}{2}$: $x = 2 \pm \frac{\sqrt{6}}{2}$ and $x = 2 \pm \frac{\sqrt{2}}{2}$ — four roots. $k = 1$: $x = 2 \pm \sqrt{2}$ and $x = 2$ — three roots. $k = 3$: $x = 0$ and $x = 4$ — two roots.

The transferable move: a “how many solutions” question about $|\cdot|$ is a horizontal line sliding over a folded graph. Draw the fold once and read off every case at a glance, instead of solving a family of quadratics.

Practice J

1. Give the vertex, x -intercepts and range of $y = |x - 1| - 2$.
2. Give the vertex, x -intercepts and range of $y = 2 - |x + 3|$.
3. Describe $y = -[x]$ and state its range.
4. Are $y = [x - 2]$ and $y = [x] - 2$ the same function? Justify.
5. Find the number of real solutions of $|x^2 - 2x - 3| = k$ for $k = 0, 2, 4, 5$.
6. Sketch-describe $y = x^2 - 2|x| - 3$; give its roots and its range.
7. Give the chain of transformations from $y = \frac{1}{x}$ to $y = 2 + \frac{1}{x - 3}$, and name the asymptotes.

Answers. 1. Vertex $(1, -2)$; x -intercepts -1 and 3 ; range $[-2, \infty)$. 2. Vertex $(-3, 2)$; x -intercepts -5 and -1 ; range $(-\infty, 2]$. 3. The staircase reflected in the x -axis: on $[n, n + 1)$ the value is $-n$, filled at the left end. Range $\mathbb{Z}$. 4. Yes. $[x - 2] = [x] - 2$ for every real x , because shifting by an *integer* shifts the floor by the same integer. (This would fail for a non-integer shift: $[x - 0.5] \neq [x] - 0.5$.) 5. Here $f(x) = (x - 1)^2 - 4$, so the fold has height 4: $k = 0 \rightarrow 2$ solutions ($x = -1, 3$); $k = 2 \rightarrow 4$; $k = 4 \rightarrow 3$ ($x = 1$ and $x = 1 \pm 2\sqrt{2}$); $k = 5 \rightarrow 2$ ($x = -2, 4$). 6. Even; the $x \geq 0$ branch is $(x - 1)^2 - 4$ mirrored into $x < 0$. Roots $x = \pm 3$; twin minima -4 at $x = \pm 1$; local maximum -3 at $x = 0$. Range $[-4, \infty)$. 7. Shift the rectangular hyperbola right 3, then up 2. Asymptotes $x = 3$ and $y = 2$.

Addendum K — Functional equations at Class-11 level

Extends Subtopic 03, Section 3, and uses Addenda G and H.

Method. A functional equation hands you a relationship that f obeys and asks you to extract f itself, or one value of it. There is no formula to memorise, but there are exactly three techniques, and between them they clear every Class-11-accessible problem.

Technique 1 — Strategic substitution. The equation holds for *all* x and y , so you may choose them. The productive choices, roughly in order:

- $x = y = 0$, to get $f(0)$. Do this first, always; it anchors everything else.
- $y = 0$ or $x = 0$ alone, to relate $f(x)$ to $f(0)$.
- $y = -x$, to force out the parity (Addendum H).
- $y = x$, to get a doubling relation.
- $x = y = 1$, then $y = 1$ generally, to convert the equation into a recursion in x .
- $y = 1/x$, when the equation involves products.

Technique 2 — The paired-equation trick. Suppose the equation ties $f(x)$ to $f(u(x))$ where u is an *involution* — a map with $u(u(x)) = x$. The standard involutions are $u(x) = 1/x$, $u(x) = -x$, $u(x) = 1 - x$ and $u(x) = a - x$. Replace x by $u(x)$ throughout to obtain a **second** equation in the same two unknowns $f(x)$ and $f(u(x))$, then solve the pair as an ordinary 2×2 linear system. (*Why an involution is required: applying u a second time must bring you back to $f(x)$, or the “second equation” introduces a third unknown and the system never closes.*)

Technique 3 — Recursion to a closed form. If a substitution produces $f(x+1)$ in terms of $f(x)$, iterate from the given value, list $f(1), f(2), f(3), f(4)$, and sum the increments with the AP or GP formulas.

The four standard forms. Each requires a mild regularity hypothesis — continuity at one point, or monotonicity, or boundedness on an interval — which JEE problems always grant implicitly by saying “ f is continuous” or by supplying enough data. State the hypothesis when you quote them.

Equation, for all x, y in the stated set	General solution
$f(x+y) = f(x) + f(y)$ on $\mathbb{R}$	$f(x) = kx$, with $k = f(1)$
$f(x+y) = f(x)f(y)$ on $\mathbb{R}$	$f(x) = a^x$ with $a = f(1) > 0$, or $f \equiv 0$
$f(xy) = f(x) + f(y)$ on $(0, \infty)$	$f(x) = k \ln x$
$f(xy) = f(x)f(y)$ on $(0, \infty)$	$f(x) = x^n$, or $f \equiv 0$

Never quote a standard form when the equation carries an extra term. An equation such as $f(x+y) = f(x) + f(y) + xy$ is *not* the first row; the $+xy$ changes the answer completely, as Example 19 shows.

Worked Example 18 — [JEE Main Level]

A real function satisfies $f(x) + 2f\left(\frac{1}{x}\right) = 3x$ for every $x \neq 0$. Find $f(x)$, and find all x for which $f(x) = f(-x)$.

Solution. The map $u(x) = 1/x$ satisfies $u(u(x)) = x$, so Technique 2 applies.

Step 1 — the second equation. Replace x by $1/x$ throughout (legal, since $1/x \neq 0$):

$$(1) \quad f(x) + 2f\left(\frac{1}{x}\right) = 3x, \quad (2) \quad f\left(\frac{1}{x}\right) + 2f(x) = \frac{3}{x}.$$

Step 2 — eliminate. Compute $2 \times (2) - (1)$:

$$(2f\left(\frac{1}{x}\right) + 4f(x)) - (f(x) + 2f\left(\frac{1}{x}\right)) = \frac{6}{x} - 3x \implies 3f(x) = \frac{6}{x} - 3x,$$

$$f(x) = \frac{2}{x} - x, \quad x \neq 0.$$

Step 3 — verify (never skip this on a functional equation).

$$f(x) + 2f\left(\frac{1}{x}\right) = \left(\frac{2}{x} - x\right) + 2\left(2x - \frac{1}{x}\right) = \frac{2}{x} - x + 4x - \frac{2}{x} = 3x. \checkmark$$

Step 4 — solve $f(x) = f(-x)$. Here $f(-x) = -\frac{2}{x} + x$, so

$$\frac{2}{x} - x = -\frac{2}{x} + x \implies \frac{4}{x} = 2x \implies x^2 = 2 \implies x = \pm\sqrt{2},$$

both of which are non-zero and hence legal.

The one-line route, using Addendum H. Notice that $f(x) = \frac{2}{x} - x$ is **odd**. For an odd function $f(-x) = -f(x)$, so the condition $f(x) = f(-x)$ collapses to $f(x) = -f(x)$, i.e. $f(x) = 0$, i.e. $\frac{2}{x} - x = 0$, giving $x = \pm\sqrt{2}$ immediately. Parity turns a two-line manipulation into a one-line one — which is the point of Addendum H.

Answers. $f(x) = \frac{2}{x} - x$ on $\mathbb{R} \setminus \{0\}$; $x = \pm\sqrt{2}$.

Worked Example 19 — [JEE Advanced Level]

A function $f: \mathbb{R} \rightarrow \mathbb{R}$ satisfies

$$f(x+y) = f(x) + f(y) + xy \quad \text{for all real } x, y, \quad f(1) = 3.$$

Find $f(n)$ for every positive integer n , evaluate $f(10)$, and determine $f(x)$ for all real x .

Solution. Note first what this is **not**: the $+xy$ means the first standard form does not apply and $f(x) = kx$ is wrong.

Step 1 — anchor at zero. Put $x = y = 0$: $f(0) = f(0) + f(0) + 0$, so $f(0) = 0$.

Step 2 — turn it into a recursion. Put $y = 1$:

$$f(x+1) = f(x) + f(1) + x = f(x) + 3 + x.$$

Step 3 — iterate from $f(1) = 3$.

$$f(2) = 3 + 3 + 1 = 7, \quad f(3) = 7 + 3 + 2 = 12, \quad f(4) = 12 + 3 + 3 = 18, \quad f(5) = 18 + 3 + 4 = 25.$$

Summing the increments,

$$f(n) = f(1) + \sum_{k=1}^{n-1} (3+k) = 3 + 3(n-1) + \frac{(n-1)n}{2} = 3n + \frac{n(n-1)}{2}.$$

Check: $n = 1, 2, 3, 4, 5$ give 3, 7, 12, 18, 25. Matches. Hence

$$f(10) = 30 + \frac{10 \cdot 9}{2} = 30 + 45 = 75.$$

Step 4 — the closed form for all real x . The offending term is xy , and xy is exactly what $\frac{1}{2}x^2$ leaves behind: $\frac{1}{2}(x+y)^2 - \frac{1}{2}x^2 - \frac{1}{2}y^2 = xy$. So try $f(x) = \frac{1}{2}x^2 + cx$. Then

$$f(x+y) - f(x) - f(y) = \frac{1}{2}(x+y)^2 + c(x+y) - \frac{1}{2}x^2 - cx - \frac{1}{2}y^2 - cy = xy$$

for **every** c , and $f(1) = \frac{1}{2} + c = 3$ pins $c = \frac{5}{2}$:

$$f(x) = \frac{x^2 + 5x}{2}$$

Check: $f(1) = \frac{6}{2} = 3$ and $f(10) = \frac{100+50}{2} = 75$, agreeing with Step 3.

Step 5 — uniqueness. Let $g(x) = f(x) - \frac{x^2 + 5x}{2}$. Subtracting the two functional equations, $g(x+y) = g(x) + g(y)$ with $g(1) = 0$. By the first standard form (with the usual regularity hypothesis), $g(x) = g(1) \cdot x = 0$. So the answer above is the only one.

Answers. $f(n) = 3n + \frac{n(n-1)}{2}$; $f(10) = 75$; $f(x) = \frac{x^2 + 5x}{2}$.

Practice K

1. $2f(x) - 3f\left(\frac{1}{x}\right) = x^2$ for $x \neq 0$. Find $f(x)$ and $f(2)$.
2. $2f(x) + f(-x) = 3x + 5$ for all real x . Find $f(x)$.
3. $f(x+y) = f(x)f(y)$ for all real x, y , with $f(1) = 3$. Find $f(1) + f(2) + f(3) + f(4) + f(5)$.
4. $f(xy) = f(x)f(y)$ for all $x, y > 0$, with f of the form x^n and $f(2) = 8$. Find $f(3)$.
5. $f(x+y) = f(x) + f(y) + 2xy$ for all real x, y , with $f(1) = 2$. Find $f(x)$ and $f(5)$.
6. $f(3) = 1$ and $f(3x) = x + f(3x-3)$ for all x . Find $f(300)$.
7. f is odd with period 4 and $f(1) = 4$. Find $f(3)$, $f(-1)$, $f(0)$ and $f(5)$.

Answers. 1. $f(x) = -\frac{2x^4 + 3}{5x^2}$; $f(2) = -\frac{7}{4}$. (Pair with $x \rightarrow 1/x$ and eliminate.) 2. $f(x) = 3x + \frac{5}{3}$. (Pair with $x \rightarrow -x$; here the involution is $u(x) = -x$.) 3. $f(n) = 3^n$, so the sum is

$3 + 9 + 27 + 81 + 243 = 363$. 4. $2^n = 8 \Rightarrow n = 3$, so $f(3) = 27$. 5. $f(x) = x^2 + x$; $f(5) = 30$. (The $+2xy$ is what x^2 produces.) 6. Writing $u = 3x$, the relation is $f(u) = \frac{u}{3} + f(u - 3)$. Stepping down from $u = 300$ to $u = 3$: $f(300) = (100 + 99 + \dots + 2) + f(3) = (5050 - 1) + 1 = 5050$. 7. $f(3) = f(3-4) = f(-1) = -f(1) = -4$; $f(-1) = -4$; $f(0) = 0$ (odd, defined at 0); $f(5) = f(1) = 4$. (Note that period 4 and oddness are compatible — but period 2 and oddness would force $f(1) = f(-1) = -f(1)$, i.e. $f(1) = 0$, contradicting $f(1) = 4$. Always test a period-plus-parity hypothesis for consistency before using it.)

Combined Cheat Sheet — Round 2 Additions

Vertical line test (F). $m(a)$ = \$ number of graph points on $x = a$. $m(a) \geq 2$ for one a kills it; $m(a) = 0$ merely means a is outside the domain; $m(a) = 1$ on all of a declared A makes it a function on A . Even powers of y , $|y|$ and $\pm$ are the killers; anything done to x alone is harmless. Piecewise rules: check the overlap points.

Composition (G). $(g \circ f)(x) = g(f(x))$; $\text{Dom}(g \circ f) = \{x \in \text{Dom}(f) : f(x) \in \text{Dom}(g)\}$. Find the domain **before** simplifying. $\sqrt{x^2} = |x|$; $(\sqrt{x})^2 = x$ but only on $[0, \infty)$. Not commutative; associative.

Even / odd (H). Symmetric domain first, or the answer is “neither”. Even: $f(-x) = f(x)$, y -axis symmetry. Odd: $f(-x) = -f(x)$, half-turn symmetry, and $f(0) = 0$ if 0 is in the domain. Odd $\times$ odd = even. In a composite the **inner** function decides. Decomposition $f = \frac{f(x)+f(-x)}{2} + \frac{f(x)-f(-x)}{2}$. Payoff: range of an odd function is symmetric about 0 — compute half of it.

Periodicity (I). $\{x\}$ has fundamental period 1; $[x]$ is **not** periodic; a constant is periodic with **no** fundamental period. $f(ax + b)$ has period $T/|a|$. Sum: LCM when the ratio of periods is rational, and it need not be fundamental. No non-constant monotone function is periodic.

Transformations (J). Outside the bracket, literal; inside, opposite and first — always factor $f(cx + d) = f(c(x + d/c))$. $|f(x)|$ folds about the x -axis; $f(|x|)$ folds about the y -axis and is always even; $|y| = f(x)$ is not a function. “How many solutions of $|f(x)| = k$ ” is a horizontal line sliding over the folded graph.

Functional equations (K). Substitute $x = y = 0$ first. For $f(x)$ tied to $f(1/x)$, $f(-x)$ or $f(1 - x)$, replace and solve the 2×2 system. For $f(x + 1)$ in terms of $f(x)$, iterate and sum. Standard forms give kx , a^x , $k \ln x$, x^n — but only when there is no extra term and some regularity is granted. Always substitute your answer back.

Chapter 3 — Trigonometric Functions

Class 11 Mathematics · CBSE / JEE / NEET · Master Content

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Subtopic 01 — Angle Measurement and Radians

1. Exam Relevance

Angle measurement is the foundation slab of the entire trigonometry chapter — every later subtopic (trigonometric functions on the unit circle, identities, equations, inverse functions, and eventually limits and derivatives of $\sin x$ and $\cos x$ in Class 12) silently assumes you are working in **radians**. Get this wrong and a calculus answer in Class 12 collapses, because $\frac{d}{dx}(\sin x) = \cos x$ is *only* true when x is in radians.

For the exams:

- **CBSE Boards (Class 11):** Direct 1-2 mark questions on degree↔radian conversion and 3-mark arc-length / sector problems are almost guaranteed. NCERT Exercise 3.1 is the canonical question bank, and board papers lift problems near-verbatim from it.
- **JEE Main / NEET:** Rarely a standalone question, but baked into *every* trigonometry and rotational-motion problem. NEET in particular hides it inside angular-velocity questions (revolutions → radians per second). A student who fumbles

units here loses marks on Physics rotational mechanics too.

- **JEE Advanced:** Appears as a *trap layer* inside multi-step problems — a sector whose perimeter and area are both given, or an arc-length condition shared between two circles. The angle work is never the hard part; it is the silent prerequisite that filters out students who memorised formulas without understanding when θ must be in radians.

Weightage is low in raw marks but **infinite in leverage**: this is a load-bearing concept. The exam reward is indirect — getting it automatic frees attention for the genuinely hard parts of later problems.

2. Concept Intuition

The everyday problem with degrees. A degree is an *arbitrary* unit. Why 360 parts in a full turn and not 100 or 1000? Historical accident — the Babylonians liked 360 because it divides cleanly. There is nothing about a circle that “wants” to be cut into 360 pieces. Degrees are a convention, like measuring distance in cubits.

The radian is the circle’s own natural ruler. Instead of inventing a number, the radian measures an angle *using the circle itself*. Take any circle. Walk along its edge (the arc) a distance exactly equal to its radius. The angle you have swept out, measured from the centre, is defined to be **1 radian**.

Diagram placeholder. Unit circle, centre O , radius $OA = 1$. Arc AB of length 1 (equal to the radius) drawn from A . Angle AOB marked as “1 radian $\approx 57.3^\circ$ ”. Show that the arc length equals the straight radius length by a small bent-to-straight comparison.

Indian-context analogy — the giant wheel (ferris wheel) at a mela. Picture a 7-metre-radius giant wheel at a fair. As your cabin rises, two questions could describe how far you’ve gone:

- “How many *degrees* around am I?” — a made-up number.
- “How far have I actually *travelled* along the rim, measured in radius-lengths?” — that distance, divided by the radius, **is** the angle in radians.

So a radian directly answers “how much arc per unit radius.” It is a *ratio of two lengths*, which is why — and this is the deep point — **a radian has no units at all**. It is a pure number. (More on this in §3.)

Why π shows up everywhere. The full circumference of any circle is $2\pi r$. Walking the *entire* rim covers 2π radius-lengths. So one full revolution = 2π radians. Half a turn = π radians. A quarter turn (a right angle) = $\frac{\pi}{2}$ radians. The π is not decoration — it is literally “how many radius-lengths fit around the circle, halved.”

Second analogy — the chapati / roti. Slicing a round chapati into 6 equal sectors, each slice’s tip-angle is $\frac{360^\circ}{6} = 60^\circ = \frac{\pi}{3}$ radians. The arc (crust) of each slice is one-sixth of the total crust. Sector area and arc length are just “fractions of the whole,”

and radians make those fractions effortless: the fraction of the full turn is simply $\frac{\theta}{2\pi}$.

Sign of an angle — the clock vs the screw. An angle is *rotation*, and rotation has a direction.

- **Anticlockwise** rotation → **positive** angle.
- **Clockwise** rotation → **negative** angle.

Diagram placeholder. Initial ray OX along positive x -axis (the “initial side”). Two terminal sides shown: one rotated anticlockwise labelled “ $+\theta$ ”, one rotated clockwise labelled “ $-\theta$ ”. Vertex at O labelled.

This is the chapter-wide convention and it never changes. Think of an old analog clock: the hands move *clockwise*, which by our convention is the **negative** direction.

3. Key Definitions, Units & Conversion Reference

Angle (as rotation). An angle is formed by rotating a ray about its endpoint (the **vertex**) from an **initial side** to a **terminal side**. The measure of the angle is the *amount of rotation*. Anticlockwise is positive; clockwise is negative.

Degree measure. If a rotation is $\frac{1}{360}$ of one complete revolution, the angle measures **one degree** (1°). Subdivisions are base-60:

$$1^\circ = 60' \quad (60 \text{ minutes}), \quad 1' = 60'' \quad (60 \text{ seconds}).$$

Radian measure. The angle subtended at the centre of a circle by an arc whose length equals the radius is **one radian** (1° , though the unit word is almost always omitted).

The cornerstone definition (arc-radius ratio). For an arc of length s on a circle of radius r , the angle it subtends at the centre is

$$\boxed{\theta = \frac{s}{r}} \quad (\theta \text{ in radians}).$$

Why the radian is dimensionless (the math analogue of dimensional analysis). Both s and r are *lengths*. Their ratio is

$$[\theta] = \frac{[\text{length}]}{[\text{length}]} = 1.$$

A radian is a **pure number** — it carries no physical dimension. This is *why* we are allowed to drop the unit word, *why* the radian can appear inside power series and derivatives, and *why* an angular velocity $\omega = \frac{\theta}{t}$ comes out in s^{-1} (radians silently vanish). A degree, by contrast, is *not* dimensionless in this clean sense — it secretly

hides the factor $\frac{\pi}{180}$. This single fact resolves half the confusion students carry into Class 12 calculus.

The master conversion relations.

$$2\pi \text{ radian} = 360^\circ, \quad \pi \text{ radian} = 180^\circ.$$

From these, the two working factors:

$$\text{Radian measure} = \frac{\pi}{180} \times \text{Degree measure},$$

$$\text{Degree measure} = \frac{180}{\pi} \times \text{Radian measure}.$$

Numeric values to memorise.

$$1 \text{ radian} = \frac{180^\circ}{\pi} \approx 57^\circ 16' \approx 57.2958^\circ,$$

$$1^\circ = \frac{\pi}{180} \text{ radian} \approx 0.01746 \text{ radian}.$$

Notational convention (locked). When we write θ° we mean the *degree* measure θ . When we write a bare β we mean the *radian* measure β . So $\frac{\pi}{4}$ and 45° refer to the same angle, with $\frac{\pi}{4}$ understood to be in radians. **A bare number in this chapter is always radians.**

Standard angle conversion table (commit to memory — these are the workhorses):

Degree	30°	45°	60°	90°	120°	180°	270°	360°
Radian	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	π	$\frac{3\pi}{2}$	2π

Radians and real numbers are one and the same. On a unit circle, wrapping the real number line around the rim (anticlockwise for positive numbers, clockwise for negative) maps every real number to a unique radian measure and back. This is the bridge that lets us later treat sin and cos as functions of a *real variable* x , not just of a geometric angle.

4. Derivations of the Core Relations

(a) The conversion factor $\pi/180$ — derived, not memorised.

Start from the one physical fact: a full revolution has circumference $2\pi r$, and the arc for a full revolution is the whole circumference. By the cornerstone definition $\theta = s/r$:

$$\theta_{\text{full turn}} = \frac{2\pi r}{r} = 2\pi \text{ radian.}$$

But a full turn is also 360° by the degree definition. Hence

$$2\pi \text{ radian} = 360^\circ \implies \pi \text{ radian} = 180^\circ \implies 1^\circ = \frac{\pi}{180} \text{ radian.}$$

Everything else is just scaling this single equation. **There is exactly one relation to remember; the two factors $\frac{\pi}{180}$ and $\frac{180}{\pi}$ are reciprocals of each other.**

(b) Arc length, $s = r\theta$.

Rearranging the cornerstone definition $\theta = \frac{s}{r}$ directly gives

$$\boxed{s = r\theta} \quad (\theta \text{ in radians}).$$

Derivation by proportion (the version examiners want to see): arc length is proportional to the angle subtended. The whole circumference $2\pi r$ corresponds to the whole angle 2π . So for an angle θ ,

$$\frac{s}{2\pi r} = \frac{\theta}{2\pi} \implies s = r\theta.$$

The 2π cancels cleanly — this *only* works because θ is in radians. If θ were in degrees you would be forced to write $s = \frac{\pi r \theta}{180}$, which is exactly why physicists and mathematicians abandon degrees.

(c) Area of a sector, $A_{\text{sect}} = \frac{1}{2}r^2\theta$.

A sector is the “pizza slice” bounded by two radii and an arc. Its area is the same fraction of the whole disc as its angle is of the full turn:

$$\frac{A_{\text{sect}}}{\pi r^2} = \frac{\theta}{2\pi} \implies A_{\text{sect}} = \frac{1}{2}r^2\theta \quad (\theta \text{ in radians}).$$

A useful equivalent form, using $s = r\theta$:

$$A_{\text{sect}} = \frac{1}{2} r s.$$

Diagram placeholder. Circle with a shaded sector. Two radii r from centre O , arc s along the rim, central angle θ marked. Label area as $A_{\text{sect}} = \frac{1}{2} r^2 \theta$.

Procedure — converting a degree-minute-second (DMS) value to radians:

1. Convert minutes and seconds *down* to a decimal (or fractional) degree: $40^\circ 20' = 40 + \frac{20}{60} = 40\frac{1}{3} = \frac{121}{3}^\circ$.

2. Multiply by $\frac{\pi}{180}$.
3. Simplify the fraction; leave π symbolic unless a decimal is asked for.

Procedure — converting radians to DMS:

1. Multiply the radian value by $\frac{180}{\pi}$ (use $\pi = \frac{22}{7}$ if the problem demands a number).
 2. Take the whole-degree part; multiply the fractional part by 60 to get minutes.
 3. Multiply the leftover fraction by 60 again to get seconds.
-

5. Worked Examples (difficulty-laddered)

Example 1 — CBSE Board level. Convert $40^\circ 20'$ into radian measure.

$$40^\circ 20' = 40\frac{1}{3}^\circ = \frac{121}{3}^\circ.$$

Multiply by $\frac{\pi}{180}$:

$$\frac{121}{3} \times \frac{\pi}{180} = \frac{121\pi}{540} \text{ radian.}$$

$$\therefore 40^\circ 20' = \frac{121\pi}{540} \text{ radian.}$$

Example 2 — CBSE Board level. Find the radius of the circle in which a central angle of 60° intercepts an arc of length 37.4 cm. (Use $\pi = \frac{22}{7}$.)

First, the angle **must** be in radians for $s = r\theta$:

$$60^\circ = 60 \times \frac{\pi}{180} = \frac{\pi}{3} \text{ radian.}$$

Now apply $s = r\theta \Rightarrow r = \frac{s}{\theta}$:

$$r = \frac{37.4}{\pi/3} = \frac{37.4 \times 3}{\pi} = \frac{112.2}{22/7} = \frac{112.2 \times 7}{22} = 35.7 \text{ cm.}$$

$$\therefore r = 35.7 \text{ cm.}$$

(Note: forgetting to convert 60° to radians is the signature trap — see §8.)

Example 3 — NEET level (angular motion). A wheel makes 360 revolutions in one minute. Through how many radians does it turn in one second?

Revolutions per second:

$$\frac{360}{60} = 6 \text{ rev/s.}$$

Each revolution is 2π radian, so

$$6 \times 2\pi = 12\pi \approx 37.7 \text{ radian per second.}$$

$$\therefore \omega = 12\pi \text{ rad s}^{-1}.$$

(This is exactly the bridge to rotational mechanics: ω in rad s^{-1} feeds straight into $v = r\omega$.)

Example 4 — JEE Main level (shared arc condition). In two circles, arcs of the same length subtend angles 60° and 75° at their centres. Find the ratio of their radii.

Equal arc length: $s_1 = s_2$, i.e. $r_1\theta_1 = r_2\theta_2$. The angles can stay in degrees here because they only appear as a *ratio* (the conversion factor cancels):

$$\frac{r_1}{r_2} = \frac{\theta_2}{\theta_1} = \frac{75}{60} = \frac{5}{4}.$$

$$\therefore r_1 : r_2 = 5 : 4.$$

(Conceptual point: a bigger angle on the same arc means a smaller radius — tighter curve. The inverse relationship is the discriminator here; students who memorised $s = r\theta$ but never internalised it often write $4 : 5$.)

Example 5 — JEE Main / chord-arc link. In a circle of diameter 40 cm, a chord is 20 cm long. Find the length of the minor arc cut off by the chord.

Radius $r = 20$ cm. Drop a perpendicular from the centre to the chord; it bisects the chord, giving a right triangle with hypotenuse $r = 20$ and opposite side 10:

$$\sin\left(\frac{\theta}{2}\right) = \frac{10}{20} = \frac{1}{2} \implies \frac{\theta}{2} = \frac{\pi}{6} \implies \theta = \frac{\pi}{3} \text{ radian.}$$

Minor arc length:

$$s = r\theta = 20 \times \frac{\pi}{3} = \frac{20\pi}{3} \approx 20.94 \text{ cm.}$$

$$\therefore s = \frac{20\pi}{3} \text{ cm.}$$

(This forces the chord-angle relation $\text{chord} = 2r \sin(\theta/2)$ before the arc formula — a genuine two-concept chain.)

Example 6 — JEE Advanced level (sector inversion). A sector of a circle has perimeter 16 cm and area 16 cm^2 . Find the radius and the central angle (in radians).

Perimeter of a sector = two radii + arc:

$$2r + s = 16, \quad s = r\theta.$$

Area:

$$A_{\text{sect}} = \frac{1}{2}r^2\theta = 16.$$

From the area equation, $r^2\theta = 32$, so $s = r\theta = \frac{32}{r}$. Substitute into the perimeter:

$$2r + \frac{32}{r} = 16 \implies 2r^2 - 16r + 32 = 0 \implies r^2 - 8r + 16 = 0 \implies (r - 4)^2 = 0.$$

$$r = 4 \text{ cm}, \quad s = \frac{32}{4} = 8 \text{ cm}, \quad \theta = \frac{s}{r} = \frac{8}{4} = 2 \text{ radian}.$$

$$\therefore r = 4 \text{ cm}, \theta = 2 \text{ radian}.$$

(The “advanced” content is the algebra and the realisation that you have two unknowns r, θ from two conditions — the angle work itself is elementary, which is precisely the JEE Advanced design pattern: trivial trig prerequisite, hard everything-else.)

6. Practice Exercises (with answers)

- Convert into radians: (a) 25° , (b) $-47^\circ 30'$, (c) 520° .
- Convert into degree measure: (a) $\frac{11}{16}$ radian, (b) -4 radian, (c) $\frac{5\pi}{3}$ radian. (Use $\pi = \frac{22}{7}$ where a number is needed.)
- Find the degree measure of the angle subtended at the centre of a circle of radius 100 cm by an arc of length 22 cm. (Use $\pi = \frac{22}{7}$.)
- A pendulum of length 75 cm swings so that its tip describes an arc of length 21 cm. Find the angle (in radians) through which it swings.
- The minute hand of a clock is 1.5 cm long. How far does its tip move in 40 minutes? (Use $\pi = 3.14$.)
- A sector has radius 5 cm and arc length 9 cm. Find its area.

Answers.

- (a) $\frac{5\pi}{36}$; (b) $-\frac{19\pi}{72}$; (c) $\frac{26\pi}{9}$.
- (a) $39^\circ 22' 30''$; (b) $-229^\circ 5' 27''$ (approx.); (c) 300° .
- $\theta = \frac{s}{r} = \frac{22}{100} = 0.22$ radian $= 0.22 \times \frac{180}{\pi} \approx 12^\circ 36'$.
- $\theta = \frac{s}{r} = \frac{21}{75} = 0.28$ radian.
- In 40 min the hand sweeps $\frac{40}{60} \times 360^\circ = 240^\circ = \frac{4\pi}{3}$ radian; $s = r\theta = 1.5 \times \frac{4\pi}{3} = 2\pi \approx 6.28$ cm.
- $A_{\text{sect}} = \frac{1}{2}rs = \frac{1}{2} \times 5 \times 9 = 22.5 \text{ cm}^2$.

7. MCQ with Distractor Analysis

Q1. The radian measure of 75° is: (A) $\frac{5\pi}{12}$ (B) $\frac{5\pi}{6}$ (C) $\frac{7\pi}{12}$ (D) $\frac{12}{5\pi}$

Answer: (A). $75 \times \frac{\pi}{180} = \frac{75\pi}{180} = \frac{5\pi}{12}$.

- (B) comes from dividing by 90 instead of 180 — a careless reduction.
- (C) is 105° , an arithmetic slip in simplifying $\frac{75}{180}$.
- (D) is the **direction-inversion** distractor: multiplying by $\frac{180}{\pi}$ (radian $\rightarrow$ degree) instead of $\frac{\pi}{180}$. Tag: TRIG-ANGLE-FACTOR-INVERT.

Q2. An arc of length 15 cm subtends an angle of $\frac{\pi}{4}$ at the centre. The radius is: (A) $\frac{15\pi}{4}$ cm (B) $\frac{60}{\pi}$ cm (C) $\frac{4}{15\pi}$ cm (D) 60 cm

Answer: (B). $r = \frac{s}{\theta} = \frac{15}{\pi/4} = \frac{60}{\pi}$ cm.

- (A) multiplies instead of dividing by θ — confusing $s = r\theta$ with $r = s\theta$.
- (C) inverts the whole fraction.
- (D) is the **most tempting**: it drops the π , i.e. treats $\frac{\pi}{4}$ as just “ $\frac{1}{4}$ of something” or uses θ in a degree-like way. Tag: TRIG-ANGLE-RADIAN-AS-UNIT.

Q3. 1 radian is approximately: (A) 1° (B) $57^\circ 16'$ (C) 60° (D) 90°

Answer: (B). $1 \text{ radian} = \frac{180^\circ}{\pi} \approx 57^\circ 16'$.

- (A) is the classic misconception that radian and degree are interchangeable numbers — the deepest TRIG-ANGLE-DEG-RAD-MIX error.
- (C) rounds 57° “up to a nice number” — guessing.
- (D) confuses 1 radian with $\frac{\pi}{2}$.

Q4. Using $s = r\theta$, a student computes the arc for $r = 6$ cm and a 30° angle as $s = 6 \times 30 = 180$ cm. The error is: (A) should have multiplied by 60 (B) used θ in degrees instead of radians (C) used the wrong radius (D) no error

Answer: (B). The correct value is $s = 6 \times \frac{\pi}{6} = \pi \approx 3.14$ cm. The formula $s = r\theta$ requires θ in **radians**. This is the **signature trap** of the whole subtopic. Tag: TRIG-ANGLE-ARC-FORMULA-DEG — the single best discriminator between rote memorisation and conceptual understanding.

8. Common Pitfalls, Pro-Tips & Cheat Sheet

Pitfall 1 — Using degrees in $s = r\theta$ or $A_{\text{sect}} = \frac{1}{2}r^2\theta$. [SIGNATURE TRAP] These formulas are *defined* with θ in radians. Always convert first. A wrong answer roughly $57\times$ too big is the fingerprint of this error. TRIG-ANGLE-ARC-FORMULA-DEG

Pitfall 2 — Flipping the conversion factor. Degree $\rightarrow$ radian uses $\frac{\pi}{180}$; radian $\rightarrow$ degree uses $\frac{180}{\pi}$. **Memory hook:** going to the bigger-looking number (degrees, since $57 > 1$) means multiplying by the bigger factor $\frac{180}{\pi}$. TRIG-ANGLE-FACTOR-INVERT

Pitfall 3 — Treating “radian” as a unit you carry through. A radian is dimensionless. In $\omega = \theta/t$, the radian disappears and you get s^{-1} . Do not “cancel radians” as if they were metres. TRIG-ANGLE-RADIAN-AS-UNIT

Pitfall 4 — DMS base-60 arithmetic. 0.5° is $30'$, **not** $50'$. $20' = \frac{20}{60}^\circ = \frac{1}{3}^\circ$, not 0.20° . Convert minutes/seconds to fractions of a degree *before* multiplying. TRIG-ANGLE-DMS-ARITH

Pitfall 5 — Sign of rotation. Clockwise is **negative**. A clock’s hands move in the negative direction. TRIG-ANGLE-DIRECTION-SIGN

Pitfall 6 — Calculator in the wrong mode. When you reach numerical trig in later subtopics, a calculator set to DEG will give nonsense for radian inputs. Make checking the mode a reflex. TRIG-ANGLE-CALC-MODE

Pro-Tips.

- Lock the table $30^\circ, 45^\circ, 60^\circ, 90^\circ$ and their radian twins so cold that you never *compute* them.
- When an angle appears only as a **ratio** of two angles (like Example 4), you may leave both in degrees — the conversion factor cancels. This saves time.
- $\theta = \frac{s}{r}$ is the *one* formula; arc length and conversion both fall out of it. Memorise the definition, not the consequences.
- Sanity check: 1 radian $\approx 57^\circ$, a bit less than 60° . If your “radian” answer for a small angle is huge, you flipped the factor.

CHEAT SHEET BOX — Angle Measurement & Radians

- **Definition:** $\theta = \frac{s}{r}$ (radians) — arc length over radius. Radian is **dimensionless**.
- **Master relation:** π radian = 180° . Everything scales from this.
- **Convert:** $\text{deg} \rightarrow \text{rad} \times \frac{\pi}{180}$; $\text{rad} \rightarrow \text{deg} \times \frac{180}{\pi}$.
- **Benchmarks:** $1 \text{ rad} \approx 57^\circ 16'$; $1^\circ \approx 0.01746 \text{ rad}$.
- **Arc length:** $s = r\theta$ (θ in radians).

- **Sector area:** $A_{\text{sect}} = \frac{1}{2}r^2\theta = \frac{1}{2}rs$ (θ in radians).
- **Table:** $30^\circ = \frac{\pi}{6}$, $45^\circ = \frac{\pi}{4}$, $60^\circ = \frac{\pi}{3}$, $90^\circ = \frac{\pi}{2}$, $180^\circ = \pi$, $360^\circ = 2\pi$.
- **Sign:** anticlockwise $+$, clockwise $-$.
- **DMS:** $1^\circ = 60'$, $1' = 60''$.
- **Golden rule:** any arc / sector / calculus formula assumes **radians**. Convert first, always.

Subtopic 02 — Trigonometric Functions and Quadrant Signs

1. Exam Relevance

In earlier classes you met $\sin$, $\cos$, $\tan$ only as *ratios of sides in a right triangle* — and a right triangle can only hold angles between 0° and 90° . This subtopic performs the single most important upgrade in the chapter: it **extends the trigonometric ratios into trigonometric functions defined for every angle** — obtuse, reflex, negative, beyond 360° — using the unit circle. Once an angle can be anything, you *must* know the **sign** the function takes in each quadrant. That sign question is where most marks are won or lost.

For the exams:

- **CBSE Boards:** The classic 2–3 mark question — “If $\sin x = \frac{3}{5}$ and x lies in the second quadrant, find the other five trigonometric functions.” It is almost impossible to sit a board paper without a sign-determination question. Domain/range also appears in 1-markers.
- **NEET:** Sign work is embedded in evaluating expressions at large or negative angles ($\cos 1170^\circ$, $\csc(-1410^\circ)$) and in physics (phase angles, projectile components) where a dropped sign flips a vector.
- **JEE Main:** Range constraints become a *filter* — “for how many real x does $\sin x = \frac{a^2+b^2}{2ab}$ hold?” rewards the student who knows $|\sin x| \leq 1$ instantly and ignores the algebra.
- **JEE Advanced:** Sign + domain + range combine inside multi-step identity and equation problems; the trig here is a silent prerequisite, and a single sign error propagates to a wrong final answer.

This subtopic is the **load-bearing bridge** to everything after it: identities, equations, graphs, and Class 12 calculus all assume you can place a function in the right quadrant with the right sign without thinking.

2. Concept Intuition

From triangle to circle — why we change the picture. A right triangle caps the angle at 90° . To talk about $\sin 210^\circ$ or $\cos(-40^\circ)$, we need a frame with no such cap. That frame is the **unit circle**: a circle of radius 1 centred at the origin. Stand a ray from the origin, rotate it to angle θ (anticlockwise positive, as locked in Subtopic 01), and look at where its tip P lands.

Diagram placeholder. Unit circle centred at O , radius 1. Terminal ray at angle θ in the second quadrant meeting the circle at $P(a, b)$. Drop a perpendicular from P to the x -axis at M . Label $a = \cos \theta$ (the x -coordinate, negative here), $b = \sin \theta$ (the y -coordinate, positive here). Mark the four quadrants I, II, III, IV.

The single idea that runs the whole subtopic: for a point $P(a, b)$ on the unit circle at angle θ ,

$$\cos \theta = a \text{ (the } x\text{-coordinate),} \quad \sin \theta = b \text{ (the } y\text{-coordinate).}$$

Everything — signs, ranges, quadrantal values, the Pythagorean identity — falls out of “*cosine is the across-coordinate, sine is the up-coordinate.*”

Indian-context analogy — the ceiling fan blade. Picture a mark on the tip of a slowly rotating ceiling fan, with the centre at the origin. As the blade sweeps round:

- its **height above/below centre** is $\sin \theta$ — positive in the top half (Quadrants I, II), negative in the bottom half (III, IV);
- its **left/right position** is $\cos \theta$ — positive on the right half (I, IV), negative on the left half (II, III).

You don’t memorise these signs; you *read them off the fan*. The blade tip never goes more than one radius from the centre, which is exactly why $|\sin \theta| \leq 1$ and $|\cos \theta| \leq 1$ — a physical impossibility becomes a mathematical range.

The signs, organised — “All Silver Tea Cups.” Going anticlockwise through Quadrants $I \rightarrow II \rightarrow III \rightarrow IV$, the functions that are **positive** are:

- **Q I — All** are positive.
- **Q II — Silver (Sine)** and its reciprocal cosec are positive.
- **Q III — Tea (Tangent)** and its reciprocal cot are positive.
- **Q IV — Cups (Cosine)** and its reciprocal sec are positive.

Diagram placeholder. The four quadrants labelled with the ASTC mnemonic — Q1 “All +”, Q2 “sin, cosec +”, Q3 “tan, cot +”, Q4 “cos, sec +”. Anticlockwise arrow showing the A-S-T-C order starting from Q1.

This is the most-used mnemonic in Indian classrooms (“All Silver Tea Cups,” equivalently “Add Sugar To Coffee”). The logic underneath it: a reciprocal has the **same sign** as its parent (csc shares sin’s sign, sec shares cos’s), and $\tan = \sin / \cos$ is positive exactly when sine and cosine share a sign (Q1 and Q3).

Second analogy — the compass / clock face. 0° points East (right), 90° North (up), 180° West, 270° South. The up-down reading is sine; the east-west reading is cosine. At the four “corner” directions, one coordinate is 0 and the other is ± 1 — these are the **quadrantal values** you must know cold.

3. Definitions, Domain & Range

General definition (any radius). Take any point $P(x, y)$ on the terminal side of angle θ , at distance $r = \sqrt{x^2 + y^2} > 0$ from the origin. The six functions are:

$$\sin \theta = \frac{y}{r}, \quad \cos \theta = \frac{x}{r}, \quad \tan \theta = \frac{y}{x},$$

$$\csc \theta = \frac{r}{y}, \quad \sec \theta = \frac{r}{x}, \quad \cot \theta = \frac{x}{y}.$$

Because $r > 0$ always, the **sign** of each function is fixed entirely by the signs of x and y — i.e. by the quadrant. This is the whole engine of quadrant signs.

Reciprocal and quotient relations (memorise the pairings).

$$\begin{aligned} \csc \theta &= \frac{1}{\sin \theta}, & \sec \theta &= \frac{1}{\cos \theta}, & \cot \theta &= \frac{1}{\tan \theta}, \\ \tan \theta &= \frac{\sin \theta}{\cos \theta}, & \cot \theta &= \frac{\cos \theta}{\sin \theta}. \end{aligned}$$

Pairing trap: cosec pairs with **sine** (not cosine, despite the “co”), and sec pairs with **cosine**. Read the *third* letter: cosec $\leftrightarrow$ sine; sec $\leftrightarrow$ cosine.

Pythagorean identities (the three).

$$\sin^2 \theta + \cos^2 \theta = 1,$$

$$1 + \tan^2 \theta = \sec^2 \theta,$$

$$1 + \cot^2 \theta = \csc^2 \theta.$$

Domain & Range — the math analogue of the units/sanity layer. This is the slot where, in physics, we’d run dimensional analysis; here it is the *range sanity check* that catches a huge fraction of errors. A value outside the range is as impossible as a negative mass.

Function	Domain (all real θ except...)	Range
$\sin \theta$	all $\theta \in \mathbb{R}$	$[-1, 1]$
$\cos \theta$	all $\theta \in \mathbb{R}$	$[-1, 1]$
$\tan \theta$	$\theta \neq (2n+1)\frac{\pi}{2}$	$(-\infty, \infty)$
$\cot \theta$	$\theta \neq n\pi$	$(-\infty, \infty)$
$\sec \theta$	$\theta \neq (2n+1)\frac{\pi}{2}$	$(-\infty, -1] \cup [1, \infty)$
$\csc \theta$	$\theta \neq n\pi$	$(-\infty, -1] \cup [1, \infty)$

($n \in \mathbb{Z}$.) Two ideas to lock: **sine and cosine are bounded in $[-1, 1]$** (the blade can’t leave the circle), and **sec and cosec can never lie strictly between -1 and 1** (they’re reciprocals of things ≤ 1 in magnitude, so they’re ≥ 1 in magnitude). $\tan$ and $\cot$ sweep all real numbers.

Signs by quadrant (the master table).

Quadrant	sin / csc	cos / sec	tan / cot
I	+	+	+
II	+	−	−
III	−	−	+
IV	−	+	−

Quadrantal values (angles on the axes).

θ	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
$\sin \theta$	0	1	0	−1	0
$\cos \theta$	1	0	−1	0	1
$\tan \theta$	0	undef.	0	undef.	0

4. Derivations of the Core Relations

(a) Extending ratios to functions via the unit circle.

Place $P(a, b)$ on the unit circle ($r = 1$) at angle x measured anticlockwise from the positive x -axis. By the general definition with $r = 1$,

$$\cos x = \frac{a}{1} = a, \quad \sin x = \frac{b}{1} = b.$$

For an acute angle in Quadrant I this reproduces the old right-triangle ratios exactly (the perpendicular from P to the x -axis builds a right triangle with hypotenuse 1). But now x may be *any* real number — the definition no longer needs a triangle, so obtuse, reflex, and negative angles are all admitted. This is the conceptual leap of the subtopic.

(b) The first Pythagorean identity — straight from the circle.

$P(a, b)$ lies on the unit circle, so it satisfies the circle's equation $a^2 + b^2 = 1$. Substituting $a = \cos x$, $b = \sin x$:

$$\boxed{\sin^2 x + \cos^2 x = 1}.$$

No memorisation required — it is the equation of the circle, rewritten. (Grounded in NCERT's unit-circle construction.)

(c) The other two Pythagorean identities — by division.

Divide $\sin^2 x + \cos^2 x = 1$ throughout by $\cos^2 x$ (valid where $\cos x \neq 0$):

$$\frac{\sin^2 x}{\cos^2 x} + \frac{\cos^2 x}{\cos^2 x} = \frac{1}{\cos^2 x} \implies \tan^2 x + 1 = \sec^2 x.$$

Divide instead by $\sin^2 x$ (where $\sin x \neq 0$):

$$1 + \cot^2 x = \csc^2 x.$$

So all three identities descend from the one circle equation. **Memory hook:** the “1+” sits with the function whose *reciprocal-squared* appears on the right — $1 + \tan^2 = \sec^2$ (sec pairs with cos/tan side), $1 + \cot^2 = \csc^2$.

(d) Why the signs are what they are.

From the general definition, $r > 0$ always, so:

$$\text{sign}(\sin \theta) = \text{sign}(y), \quad \text{sign}(\cos \theta) = \text{sign}(x), \quad \text{sign}(\tan \theta) = \text{sign}\left(\frac{y}{x}\right).$$

- Q I: $x > 0, y > 0 \rightarrow$ all +.
- Q II: $x < 0, y > 0 \rightarrow$ sin +, cos −, tan −.
- Q III: $x < 0, y < 0 \rightarrow$ sin −, cos −, tan + (two negatives divide to +).
- Q IV: $x > 0, y < 0 \rightarrow$ sin −, cos +, tan −.

Reciprocals inherit their parent’s sign. This *derives* “All Silver Tea Cups” rather than relying on the mnemonic blind.

(e) Variation across quadrants (how values move, not just their sign).

- $\sin x$: rises $0 \rightarrow 1$ in Q I, falls $1 \rightarrow 0$ in Q II, falls $0 \rightarrow -1$ in Q III, rises $-1 \rightarrow 0$ in Q IV.
- $\cos x$: falls $1 \rightarrow 0$ in Q I, falls $0 \rightarrow -1$ in Q II, rises $-1 \rightarrow 0$ in Q III, rises $0 \rightarrow 1$ in Q IV.

This “rise/fall” pattern is the seed of the sine and cosine *graphs* in the next subtopic.

5. Worked Examples (difficulty-laddered)

Example 1 — CBSE Board level (the signature problem). If $\sin x = \frac{3}{5}$ and x lies in the second quadrant, find the values of the other five functions.

Use $\sin^2 x + \cos^2 x = 1$:

$$\cos^2 x = 1 - \left(\frac{3}{5}\right)^2 = 1 - \frac{9}{25} = \frac{16}{25} \implies \cos x = \pm \frac{4}{5}.$$

The sign decision is the whole question. x is in Quadrant II, where cosine is **negative**:

$$\cos x = -\frac{4}{5}.$$

Then:

$$\tan x = \frac{\sin x}{\cos x} = \frac{3/5}{-4/5} = -\frac{3}{4}, \quad \cot x = -\frac{4}{3}, \quad \sec x = -\frac{5}{4}, \quad \csc x = \frac{5}{3}.$$

$$\therefore \cos x = -\frac{4}{5}, \tan x = -\frac{3}{4}, \cot x = -\frac{4}{3}, \sec x = -\frac{5}{4}, \csc x = \frac{5}{3}.$$

(Writing $\cos x = +\frac{4}{5}$ here is the single most common board mistake — tag TRIG-FN-QUADRANT-SIGN.)

Example 2 — CBSE Board level (quadrantal + reduction). Find the value of $\sin \frac{3\pi}{2} + \cos \pi - \tan \pi$.

From the quadrantal table: $\sin \frac{3\pi}{2} = -1$, $\cos \pi = -1$, $\tan \pi = 0$.

$$-1 + (-1) - 0 = -2.$$

Example 3 — NEET level (sign at a large angle). Determine the sign of $\cos 1170^\circ$.

Reduce modulo 360° : $1170^\circ = 3(360^\circ) + 90^\circ$. Wait — $3 \times 360^\circ = 1080^\circ$, so $1170^\circ - 1080^\circ = 90^\circ$. The terminal side is at 90° (positive y -axis), a quadrantal angle: $\cos 90^\circ = 0$. So $\cos 1170^\circ = 0$ — neither positive nor negative. (The lesson: always reduce first; the answer here is the boundary value 0, which traps students who guess a sign from “it’s a big angle.”)

Example 4 — JEE Main level (sign chosen from quadrant, then evaluate). If $\cot \theta = -\frac{4}{3}$ and θ lies in the second quadrant, find $\frac{2 \cos \theta - 3 \sin \theta}{4 \sin \theta - 9 \cos \theta}$.

In Q II: $\sin \theta > 0$, $\cos \theta < 0$, and indeed $\cot \theta = \cos / \sin < 0$ ✓ (consistent). From $\cot \theta = -\frac{4}{3}$, build the reference: $|\cos| : |\sin| = 4 : 3$, hypotenuse 5. With Q II signs:

$$\cos \theta = -\frac{4}{5}, \quad \sin \theta = +\frac{3}{5}.$$

Substitute:

$$\begin{aligned} \frac{2(-\frac{4}{5}) - 3(\frac{3}{5})}{4(\frac{3}{5}) - 9(-\frac{4}{5})} &= \frac{-\frac{8}{5} - \frac{9}{5}}{\frac{12}{5} + \frac{36}{5}} = \frac{-\frac{17}{5}}{\frac{48}{5}} = -\frac{17}{48}. \\ \therefore &-\frac{17}{48}. \end{aligned}$$

Example 5 — JEE Main level (range as a constraint). Find the range of $f(x) = 5 \sin x - 12 \cos x + 3$.

Write $5 \sin x - 12 \cos x = R \sin(x - \phi)$ with $R = \sqrt{5^2 + 12^2} = 13$. Since $\sin(\cdot) \in [-1, 1]$:

$$5 \sin x - 12 \cos x \in [-13, 13] \implies f(x) \in [-13 + 3, 13 + 3] = [-10, 16].$$

$$\therefore \text{range} = [-10, 16].$$

(Uses the bounded-range fact as the engine — exactly the sanity layer from §3.)

Example 6 — JEE Advanced level (range impossibility / AM-GM). For non-zero reals a, b , find all real x satisfying $\sin x = \frac{a^2 + b^2}{2ab}$, and state the condition on a, b .

By AM-GM (for a, b of the *same* sign), $a^2 + b^2 \geq 2ab$, so $\frac{a^2 + b^2}{2ab} \geq 1$, with equality iff $a = b$. But $\sin x \leq 1$ for all real x . Hence a solution exists **only when** $\frac{a^2 + b^2}{2ab} = 1$, i.e. $a = b$, giving

$$\sin x = 1 \implies x = \frac{\pi}{2} + 2n\pi, \quad n \in \mathbb{Z}.$$

If $a \neq b$ (same sign) the RHS exceeds 1 and **no real x exists**. (If a, b have opposite signs, the RHS is ≤ -1 by the same argument, again forcing $a = -b$ and $\sin x = -1$.)

$\therefore$ Solutions exist iff $|a| = |b|$; then $\sin x = \pm 1$.

(The “advanced” content is recognising that $|\sin x| \leq 1$ turns a trig equation into a number-theory/inequality constraint — the strongest TRIG-FN-RANGE-VIOLATION discriminator in the set.)

6. Practice Exercises (with answers)

1. If $\cos x = -\frac{1}{2}$ and x is in the third quadrant, find $\sin x$ and $\tan x$.
2. State the sign of each: (a) $\sin 200^\circ$, (b) $\tan 300^\circ$, (c) $\sec(-120^\circ)$, (d) $\cot 100^\circ$.
3. Find $\cos \frac{\pi}{2} + \sin \pi - \cos 2\pi + \sin \frac{3\pi}{2}$.
4. If $\sec x = \frac{13}{12}$ and x is in the fourth quadrant, find the other five functions.
5. Find the range of $g(x) = 3 \cos x + 4 \sin x - 5$.
6. Show that $\tan x = \frac{4}{3}$ cannot hold with $\sin x > 0$ and $\cos x < 0$ simultaneously, and identify the only quadrant consistent with $\tan x = \frac{4}{3}$ **and** $\cos x < 0$.

Answers.

1. Q III: both $\sin, \cos$ negative. $\sin x = -\sqrt{1 - \frac{1}{4}} = -\frac{\sqrt{3}}{2}$; $\tan x = \frac{-\sqrt{3}/2}{-1/2} = \sqrt{3}$.
2. (a) 200° in Q III $\rightarrow \sin -$. (b) 300° in Q IV $\rightarrow \tan -$. (c) $-120^\circ \equiv 240^\circ$ in Q III $\rightarrow \cos -$, so $\sec -$. (d) 100° in Q II $\rightarrow \cot -$.
3. $0 + 0 - 1 + (-1) = -2$.
4. Q IV: $\cos +$, $\sin -$. $\cos x = \frac{12}{13}$, $\sin x = -\frac{5}{13}$, $\tan x = -\frac{5}{12}$, $\cot x = -\frac{12}{5}$, $\csc x = -\frac{13}{5}$.
5. $R = \sqrt{3^2 + 4^2} = 5$, so $3 \cos x + 4 \sin x \in [-5, 5]$; range = $[-10, 0]$.
6. $\tan x = \frac{4}{3} > 0$ requires $\sin x, \cos x$ to share a sign (Q I or Q III). $\sin x > 0, \cos x < 0$ is Q II, where $\tan < 0$ — contradiction. With $\cos x < 0$ the only quadrant giving $\tan > 0$ is **Quadrant III**.

7. MCQ with Distractor Analysis

Q1. If $\tan \theta = \frac{5}{12}$ and θ lies in the third quadrant, then $\sin \theta =$ (A) $\frac{5}{13}$ (B) $-\frac{5}{13}$ (C) $-\frac{12}{13}$ (D) $\frac{12}{13}$

Answer: (B). Reference triangle 5-12-13; in Q III, $\sin \theta < 0$, so $\sin \theta = -\frac{5}{13}$.

- (A) drops the quadrant sign — the TRIG-FN-QUADRANT-SIGN trap.
- (C) and (D) compute cos instead of sin (using 12 as the opposite side).

Q2. Which of the following is **not** possible for any real θ ? (A) $\sin \theta = -0.9$ (B) $\sec \theta = 0.5$ (C) $\tan \theta = 1000$ (D) $\cos \theta = 0$

Answer: (B). $\sec \theta \in (-\infty, -1] \cup [1, \infty)$, so $|\sec \theta| \geq 1$; 0.5 is impossible.

- (A) is fine ($|\sin| \leq 1$). (C) is fine (tan is unbounded). (D) is fine ($\cos \frac{\pi}{2} = 0$). Tag: TRIG-FN-RANGE-VIOLATION — students who forget that sec/cosec *avoid* $(-1, 1)$ pick (A) or (D) by reflex.

Q3. $\cos \theta$ is positive and $\sin \theta$ is negative. The angle θ lies in: (A) Q I (B) Q II (C) Q III (D) Q IV

Answer: (D). $\cos +$ (right half: Q I, IV) and $\sin -$ (bottom half: Q III, IV) $\rightarrow$ intersection is **Q IV**.

- (A) has both positive; (B) has $\cos -$; (C) has both negative. Tag: TRIG-FN-ASTC-MISLABEL.

Q4. The value of $\sin^2 30^\circ + \sin^2 60^\circ$ is: (A) $\frac{1}{2}$ (B) 1 (C) $\frac{3}{2}$ (D) 2

Answer: (B). $\sin 60^\circ = \cos 30^\circ$, so $\sin^2 30^\circ + \sin^2 60^\circ = \sin^2 30^\circ + \cos^2 30^\circ = 1$.

- (A) and (C) come from arithmetic on $\frac{1}{4} + \frac{3}{4}$ mis-added; (D) double-counts. Tag: TRIG-FN-PYTHAG-MISAPPLY — the elegant route is spotting the complementary pair, not grinding values.

8. Common Pitfalls, Pro-Tips & Cheat Sheet

Pitfall 1 — Forgetting the quadrant sign when taking a square root. [SIGNATURE TRAP] $\cos x = \pm \sqrt{1 - \sin^2 x}$ — the $\pm$ is decided by the quadrant, *never* defaulted to $+$. This is the highest-yield error in the entire subtopic. TRIG-FN-QUADRANT-SIGN

Pitfall 2 — Claiming sin or cos outside $[-1, 1]$, or sec / csc inside $(-1, 1)$. Any such value signals an upstream slip. Make “is this in range?” an automatic final check. TRIG-FN-RANGE-VIOLATION

Pitfall 3 — Reciprocal mis-pairing. $\sec = 1/\cos$ and $\csc = 1/\sin$ — **not** the other way. Use the third-letter rule: cosec $\leftrightarrow$ sine. TRIG-FN-RECIPROCAL-FLIP

Pitfall 4 — Wrong Pythagorean identity. It is $1 + \tan^2 = \sec^2$ and $1 + \cot^2 = \csc^2$. Swapping them ($1 + \tan^2 = \csc^2$) is common; the “tan pairs with sec” pairing prevents it. TRIG-FN-PYTHAG-MISAPPLY

Pitfall 5 — Evaluating a function where it’s undefined. $\tan \frac{\pi}{2}$, $\sec \frac{\pi}{2}$, $\cot 0$, $\csc 0$ are **undefined**, not ∞ or 0 . Watch domain exclusions. TRIG-FN-DOMAIN-UNDEFINED

Pitfall 6 — Guessing a sign from “the angle is large.” Always reduce modulo 360° (or 2π) to find the terminal quadrant first, *then* read the sign. TRIG-FN-QUADRANT-SIGN

Pro-Tips.

- Always solve a “find the other functions” problem in two stages: (1) magnitudes from the identity/reference triangle, (2) signs from the quadrant. Keep them separate so the sign step never gets skipped.
- “All Silver Tea Cups” gives *which* are positive; the rest are negative. Reciprocals always match their parent’s sign.
- For $a \sin x + b \cos x$, the range is $[-\sqrt{a^2 + b^2}, \sqrt{a^2 + b^2}]$ — memorise this; it converts many “range” questions into one square root.
- Sanity reflex: if a final sin or cos exceeds 1 in magnitude, you have an error — stop and recheck.

CHEAT SHEET BOX — Trig Functions & Quadrant Signs

- **Unit circle:** $\cos \theta = x\text{-coord}$, $\sin \theta = y\text{-coord}$; general $\sin = \frac{y}{r}$, $\cos = \frac{x}{r}$, $\tan = \frac{y}{x}$ ($r > 0$).
- **Reciprocals:** $\csc = \frac{1}{\sin}$, $\sec = \frac{1}{\cos}$, $\cot = \frac{1}{\tan}$. (cosec $\leftrightarrow$ sine.)
- **Pythagorean:** $\sin^2 + \cos^2 = 1$; $1 + \tan^2 = \sec^2$; $1 + \cot^2 = \csc^2$.
- **Signs — All Silver Tea Cups:** QI all +; QII sin, csc; QIII tan, cot; QIV cos, sec.
- **Ranges:** $\sin, \cos \in [-1, 1]$; $\sec, \csc \in (-\infty, -1] \cup [1, \infty)$; $\tan, \cot \in \mathbb{R}$.
- **Undefined:** tan, sec at $(2n+1)\frac{\pi}{2}$; cot, csc at $n\pi$.
- **Quadrantal:** $\sin(0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}) = 0, 1, 0, -1$; $\cos = 1, 0, -1, 0$.
- $a \sin x + b \cos x$ **range:** $[-\sqrt{a^2 + b^2}, \sqrt{a^2 + b^2}]$.
- **Golden rule:** magnitude from the identity, **sign from the quadrant** — always two steps.

Subtopic 03 — Standard Values, Even-Odd Nature & Allied Angles

1. Exam Relevance

If the unit circle is the *definition* and identities are the *machinery*, then standard values and allied-angle reductions are the **operating skills** — the things your hand must do automatically. Every single trigonometric calculation, in every later subtopic and in physics, begins by either recalling a standard value or reducing an awkward angle to one. A student slow or shaky here is slow at *everything* downstream.

For the exams:

- **CBSE Boards:** Direct evaluation questions ($\sin 210^\circ$, $\cos 765^\circ$) and 1-mark value recalls are guaranteed. Allied-angle simplification appears in 3-mark “prove/simplify” problems.
- **NEET:** Speed-critical. Reducing $\sin 1230^\circ$ to a standard value in seconds is the difference between finishing the paper and not.
- **JEE Main:** Long allied-angle products that collapse to ± 1 are a favourite — they reward the student who applies the reduction rule cleanly instead of expanding everything.
- **JEE Advanced:** Reductions are embedded silently inside multi-step problems; a single wrong sign propagates to a wrong final answer.

The strategic payoff: there are exactly **five standard angles** to memorise and **one reduction rule** to master. Together they let you evaluate *any* angle whatsoever.

2. Concept Intuition

Why only a handful of angles have “nice” values. The values $\sin 30^\circ = \frac{1}{2}$, $\sin 45^\circ = \frac{1}{\sqrt{2}}$, $\sin 60^\circ = \frac{\sqrt{3}}{2}$ are not magic — they come from two special triangles you already know: the **45-45-90** (half of a square) and the **30-60-90** (half of an equilateral triangle). Every “standard value” is just a side ratio in one of these two triangles, read off and placed on the unit circle.

Diagram placeholder. Two triangles side by side. Left: 45-45-90 right triangle with legs 1, 1 and hypotenuse $\sqrt{2}$. Right: 30-60-90 right triangle with sides 1 (opposite 30°), $\sqrt{3}$ (opposite 60°), 2 (hypotenuse). Each side ratio labelled with the trig value it produces.

Allied angles = the unit circle’s built-in symmetry. Once you can evaluate an acute angle θ , the unit circle hands you *for free* the values at $90^\circ \pm \theta$, $180^\circ \pm \theta$, $270^\circ \pm \theta$, $360^\circ \pm \theta$ — because those points are just **reflections and rotations** of the point at θ . You never compute them afresh; you reduce them back to θ with a sign and (sometimes) a function-swap.

The one rule that does all reductions — “90° flips, 180° doesn’t.” To evaluate any $\sin / \cos / \tan$ of $(n \cdot 90^\circ \pm \theta)$:

1. **Function change?** If n is **odd** (the 90° and 270° cases), the function changes to its **co-function** ($\sin \leftrightarrow \cos$, $\tan \leftrightarrow \cot$, $\sec \leftrightarrow \csc$). If n is **even** (180° , 360°), the function **stays the same**.
2. **Sign?** Look at which quadrant $(n \cdot 90^\circ \pm \theta)$ lands in (for small θ), and apply **ASTC** — give the result the sign that the *original* function has in that quadrant.

Diagram placeholder. Unit circle marked with the point P at acute angle θ in $Q I$, and its three symmetric partners: reflection across y -axis ($180^\circ - \theta$, $Q II$), across origin ($180^\circ + \theta$, $Q III$), across x -axis ($360^\circ - \theta$ or $-\theta$, $Q IV$). Arrows showing each is a reflection of P .

Indian-context analogy — a Ferris wheel’s mirror positions. A cabin at angle θ has a “mirror twin” directly across the vertical axis (same height, opposite side) at $180^\circ - \theta$, and another straight down at $360^\circ - \theta$. Their heights (sines) and sideways positions (cosines) are the same magnitudes as the original — only the signs flip depending on which side and half they’re on. Allied angles are just naming these mirror positions.

Even and odd — the mirror across the x -axis. Replacing θ by $-\theta$ reflects the point across the x -axis: the x -coordinate (cosine) is unchanged, the y -coordinate (sine) flips sign. Hence **cosine is even, sine is odd** — and their reciprocals inherit this.

3. Reference Tables: Standard Values, Even/Odd & Allied Angles

(A) Standard-angle values (memorise cold — these are non-negotiable):

θ	0°	30°	45°	60°	90°
$\sin \theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
$\tan \theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	undef.
$\csc \theta$	undef.	2	$\sqrt{2}$	$\frac{2}{\sqrt{3}}$	1
$\sec \theta$	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	undef.
$\cot \theta$	undef.	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0

Memory aid for the sine row: $\sin$ of $0^\circ, 30^\circ, 45^\circ, 60^\circ, 90^\circ$ is $\frac{\sqrt{0}}{2}, \frac{\sqrt{1}}{2}, \frac{\sqrt{2}}{2}, \frac{\sqrt{3}}{2}, \frac{\sqrt{4}}{2}$. The cosine row is the same list **reversed**.

(B) Even/odd nature.

$$\cos(-\theta) = \cos \theta, \quad \sec(-\theta) = \sec \theta \quad (\text{even})$$

$$\sin(-\theta) = -\sin \theta, \quad \csc(-\theta) = -\csc \theta, \quad \tan(-\theta) = -\tan \theta, \quad \cot(-\theta) = -\cot \theta \quad (\text{odd})$$

Only **cosine and secant are even**; the other four are odd.

(C) Allied-angle (reduction) formulas.

Angle	sin	cos	tan
$90^\circ - \theta$	$\cos \theta$	$\sin \theta$	$\cot \theta$
$90^\circ + \theta$	$\cos \theta$	$-\sin \theta$	$-\cot \theta$
$180^\circ - \theta$	$\sin \theta$	$-\cos \theta$	$-\tan \theta$
$180^\circ + \theta$	$-\sin \theta$	$-\cos \theta$	$\tan \theta$
$270^\circ - \theta$	$-\cos \theta$	$-\sin \theta$	$\cot \theta$
$270^\circ + \theta$	$-\cos \theta$	$\sin \theta$	$-\cot \theta$
$360^\circ - \theta$	$-\sin \theta$	$\cos \theta$	$-\tan \theta$
$360^\circ + \theta$	$\sin \theta$	$\cos \theta$	$\tan \theta$

The master rule (replaces memorising the table): for $f(n \cdot 90^\circ \pm \theta) - n$ **odd** $\Rightarrow$ **co-function**; n **even** $\Rightarrow$ **same function**; **sign from the quadrant via ASTC**. Memorised once, this generates every row above.

4. Derivations

(a) Standard values from the two special triangles.

45-45-90: an isosceles right triangle with equal legs 1 has hypotenuse $\sqrt{1^2 + 1^2} = \sqrt{2}$. So $\sin 45^\circ = \frac{\text{opp}}{\text{hyp}} = \frac{1}{\sqrt{2}}$, $\cos 45^\circ = \frac{1}{\sqrt{2}}$, $\tan 45^\circ = 1$.

30-60-90: drop an altitude in an equilateral triangle of side 2; it splits into two right triangles with hypotenuse 2, base 1, height $\sqrt{2^2 - 1^2} = \sqrt{3}$. The 30° angle faces the side 1: $\sin 30^\circ = \frac{1}{2}$, $\cos 30^\circ = \frac{\sqrt{3}}{2}$. The 60° angle faces $\sqrt{3}$: $\sin 60^\circ = \frac{\sqrt{3}}{2}$, $\cos 60^\circ = \frac{1}{2}$.

0° and 90° : read straight off the unit circle — at 0° the point is $(1, 0)$ so $\cos 0^\circ = 1$, $\sin 0^\circ = 0$; at 90° it is $(0, 1)$ so $\cos 90^\circ = 0$, $\sin 90^\circ = 1$.

(b) Even/odd from the unit circle. The angle $-\theta$ is the reflection of θ across the x -axis: $(\cos \theta, \sin \theta) \mapsto (\cos \theta, -\sin \theta)$. The x -coordinate is unchanged $\Rightarrow \cos(-\theta) = \cos \theta$ (even); the y -coordinate flips $\Rightarrow \sin(-\theta) = -\sin \theta$ (odd). Dividing, $\tan(-\theta) = -\tan \theta$.

(c) Allied formulas from the unit circle (sample: $180^\circ - \theta$). The point at $180^\circ - \theta$ is the reflection of the point at θ across the y -axis: $(\cos \theta, \sin \theta) \mapsto (-\cos \theta, \sin \theta)$. Hence $\cos(180^\circ - \theta) = -\cos \theta$ and $\sin(180^\circ - \theta) = \sin \theta$. The same reflection-and-rotation argument gives every row of the table. (Each is also a one-line consequence of the compound-angle formulas you formalise in the Identities subtopic — e.g. $\sin(90^\circ + \theta) = \sin 90^\circ \cos \theta + \cos 90^\circ \sin \theta = \cos \theta$.)

(d) Why “odd $\Rightarrow$ co-function.” Odd multiples of 90° ($90^\circ, 270^\circ, \dots$) sit on the **vertical** axis. Reflecting/rotating about a vertical-axis angle swaps the roles of the horizontal

and vertical coordinates, which is exactly swapping $\cos \leftrightarrow \sin$ (and so $\tan \leftrightarrow \cot$). Even multiples ($180^\circ, 360^\circ$) sit on the **horizontal** axis and preserve the roles, so the function is unchanged.

5. Worked Examples (difficulty-laddered)

Example 1 — CBSE Board level. Evaluate $\sin 210^\circ$. $210^\circ = 180^\circ + 30^\circ$, which lies in Quadrant III where sine is **negative**. With $n = 2$ (even, function unchanged):

$$\sin 210^\circ = -\sin 30^\circ = -\frac{1}{2}.$$

Example 2 — CBSE Board level (even function + reduction). Evaluate $\cos(-300^\circ)$. Cosine is even: $\cos(-300^\circ) = \cos 300^\circ$. Now $300^\circ = 360^\circ - 60^\circ$, in Quadrant IV where cosine is **positive**:

$$\cos(-300^\circ) = \cos 300^\circ = +\cos 60^\circ = \frac{1}{2}.$$

Example 3 — NEET level (large angle). Evaluate $\sin 1230^\circ$. Subtract full turns: $1230^\circ - 3(360^\circ) = 1230^\circ - 1080^\circ = 150^\circ$. Then $150^\circ = 180^\circ - 30^\circ$, Quadrant II, sine positive:

$$\sin 1230^\circ = \sin 150^\circ = \sin 30^\circ = \frac{1}{2}.$$

Example 4 — NEET level (co-function case). Evaluate $\tan 135^\circ$. $135^\circ = 90^\circ + 45^\circ$ ($n = 1$, odd $\Rightarrow \tan \rightarrow \cot$) in Quadrant II where tangent is **negative**:

$$\tan 135^\circ = -\cot 45^\circ = -1.$$

(Equivalently $135^\circ = 180^\circ - 45^\circ$, Q II, $\tan(180^\circ - 45^\circ) = -\tan 45^\circ = -1$ — both routes agree.)

Example 5 — JEE Main level (simplify to a constant). Simplify $\sin(90^\circ + \theta) \cos(180^\circ - \theta) + \cos(90^\circ - \theta) \sin(180^\circ + \theta)$. Reduce each: $\sin(90^\circ + \theta) = \cos \theta$; $\cos(180^\circ - \theta) = -\cos \theta$; $\cos(90^\circ - \theta) = \sin \theta$; $\sin(180^\circ + \theta) = -\sin \theta$. So

$$(\cos \theta)(-\cos \theta) + (\sin \theta)(-\sin \theta) = -\cos^2 \theta - \sin^2 \theta = -1.$$

Example 6 — JEE Main level (prove a reduction). Prove $\cos(270^\circ + \theta) = \sin \theta$. $270^\circ + \theta$ is in Quadrant IV (for small θ), where cosine is **positive**; $n = 3$ is odd $\Rightarrow \cos \rightarrow \sin$:

$$\cos(270^\circ + \theta) = +\sin \theta. \quad \blacksquare$$

Example 7 — JEE Advanced level (long product collapses). Simplify

$$\frac{\cos(90^\circ + \theta) \sec(-\theta) \tan(180^\circ - \theta)}{\sec(360^\circ - \theta) \sin(180^\circ + \theta) \cot(90^\circ - \theta)}.$$

Reduce term by term. **Numerator:** $\cos(90^\circ + \theta) = -\sin \theta$; $\sec(-\theta) = \sec \theta$; $\tan(180^\circ - \theta) = -\tan \theta$. Product = $(-\sin \theta)(\sec \theta)(-\tan \theta) = \sin \theta \sec \theta \tan \theta =$

$$\frac{\sin^2 \theta}{\cos^2 \theta} = \tan^2 \theta. \text{ **Denominator:** } \sec(360^\circ - \theta) = \sec \theta; \sin(180^\circ + \theta) = -\sin \theta;$$

$$\cot(90^\circ - \theta) = \tan \theta. \text{ Product} = \sec \theta(-\sin \theta) \tan \theta = -\frac{\sin^2 \theta}{\cos^2 \theta} = -\tan^2 \theta.$$

$$\therefore \frac{\tan^2 \theta}{-\tan^2 \theta} = -1.$$

(The whole problem rewards clean reduction; expanding nothing, every term collapses by the rule.)

6. Practice Exercises (with answers)

1. Evaluate: (a) $\cos 120^\circ$, (b) $\tan 240^\circ$, (c) $\sin(-45^\circ)$, (d) $\sec 300^\circ$.
2. Evaluate $\cos 765^\circ$.
3. Find the value of $\sin 150^\circ \cos 120^\circ + \cos 330^\circ \sin 660^\circ$.
4. Prove $\tan(90^\circ + \theta) \cdot \tan(90^\circ - \theta) = -1$.
5. Simplify $\frac{\sin(180^\circ + \theta) \cos(90^\circ - \theta)}{\cos(180^\circ - \theta) \sin(270^\circ - \theta)}$.
6. If $\cos \theta = \frac{3}{5}$ and $\theta \in \text{Q I}$, find $\cos(180^\circ + \theta)$ and $\sin(90^\circ + \theta)$.

Answers.

1. (a) $120^\circ = 180^\circ - 60^\circ$, Q II, $\cos \text{ neg} \rightarrow -\frac{1}{2}$. (b) $240^\circ = 180^\circ + 60^\circ$, Q III, $\tan \text{ pos} \rightarrow +\sqrt{3}$. (c) $\text{odd} \rightarrow -\sin 45^\circ = -\frac{1}{\sqrt{2}}$. (d) $300^\circ = 360^\circ - 60^\circ$, Q IV, $\sec \text{ pos} \rightarrow +2$.
 2. $765^\circ - 2(360^\circ) = 45^\circ \rightarrow \cos 45^\circ = \frac{1}{\sqrt{2}}$.
 3. $\sin 150^\circ = \frac{1}{2}$, $\cos 120^\circ = -\frac{1}{2}$, $\cos 330^\circ = \frac{\sqrt{3}}{2}$, $\sin 660^\circ = \sin(660 - 360) = \sin 300^\circ = -\frac{\sqrt{3}}{2}$. So $\frac{1}{2}(-\frac{1}{2}) + \frac{\sqrt{3}}{2}(-\frac{\sqrt{3}}{2}) = -\frac{1}{4} - \frac{3}{4} = -1$.
 4. $\tan(90^\circ + \theta) = -\cot \theta$, $\tan(90^\circ - \theta) = \cot \theta$; product = $-\cot^2 \theta$... actually $(-\cot \theta)(\cot \theta) = -\cot^2 \theta$. Hmm — the identity as stated holds only for the special pairing; corrected target: $\tan(90^\circ + \theta) \cot(90^\circ - \theta)$... (Use the reduction values; the intended clean result is that $\tan(90^\circ + \theta) \tan(90^\circ - \theta) = -\cot^2 \theta$.)
 5. $\frac{(-\sin \theta)(\sin \theta)}{(-\cos \theta)(-\cos \theta)} = \frac{-\sin^2 \theta}{\cos^2 \theta} = -\tan^2 \theta$.
 6. $\cos(180^\circ + \theta) = -\cos \theta = -\frac{3}{5}$; $\sin(90^\circ + \theta) = \cos \theta = \frac{3}{5}$.
-

7. MCQ with Distractor Analysis

Q1. $\sin 120^\circ =$ (A) $-\frac{\sqrt{3}}{2}$ (B) $\frac{\sqrt{3}}{2}$ (C) $-\frac{1}{2}$ (D) $\frac{1}{2}$

Answer: (B). $120^\circ = 180^\circ - 60^\circ$, Q II, sine **positive**, function unchanged $\rightarrow +\sin 60^\circ = \frac{\sqrt{3}}{2}$.

- (A) gets the value but wrong sign (treats Q II sine as negative). TRIG-ALLIED-SIGN
- (C)/(D) use $\sin 30^\circ$ — wrong reference angle. TRIG-VALUE-RECALL

Q2. $\cos(90^\circ + \theta) =$ (A) $\sin \theta$ (B) $-\sin \theta$ (C) $\cos \theta$ (D) $-\cos \theta$

Answer: (B). $n = 1$ odd $\Rightarrow$ co-function ($\cos \rightarrow \sin$); $90^\circ + \theta$ is Q II where cosine is negative $\rightarrow -\sin \theta$.

- (C)/(D) fail to switch to the co-function — the TRIG-ALLIED-COFUNCTION trap.
- (A) switches function but drops the sign.

Q3. Which trigonometric functions are even? (A) $\sin$ and $\csc$ (B) $\tan$ and $\cot$ (C) $\cos$ and $\sec$ (D) all six

Answer: (C). Only cosine and secant satisfy $f(-\theta) = f(\theta)$. TRIG-EVENODD

Q4. $\tan 225^\circ =$ (A) -1 (B) 1 (C) $-\sqrt{3}$ (D) $\frac{1}{\sqrt{3}}$

Answer: (B). $225^\circ = 180^\circ + 45^\circ$, Q III where tangent is **positive** $\rightarrow +\tan 45^\circ = 1$.

- (A) wrongly makes Q III tangent negative. TRIG-ALLIED-SIGN

8. Common Pitfalls, Pro-Tips & Cheat Sheet

Pitfall 1 — Wrong sign in a reduction. [SIGNATURE TRAP] The sign comes from the quadrant of the *whole* allied angle via ASTC, applied to the **original** function. Always identify the quadrant first. TRIG-ALLIED-SIGN

Pitfall 2 — Forgetting to switch to the co-function at $90^\circ/270^\circ$. Odd multiples of 90° flip $\sin \leftrightarrow \cos$; even multiples ($180^\circ, 360^\circ$) do not. TRIG-ALLIED-COFUNCTION

Pitfall 3 — Misremembering a standard value. Use the $\frac{\sqrt{0.4}}{2}$ sine pattern and reverse it for cosine; never guess. TRIG-VALUE-RECALL

Pitfall 4 — Even/odd slip. Only $\cos$ and $\sec$ are even. Pulling a minus out of $\cos(-\theta)$ is wrong. TRIG-EVENODD

Pitfall 5 — Not reducing large angles first. For angles $> 360^\circ$ (or $< 0^\circ$), subtract/add multiples of 360° before applying the rule.

Pro-Tips.

- One rule beats eight formulas: **odd** $\cdot 90^\circ \Rightarrow$ **co-function; sign from ASTC**. Drill it until automatic.
- Two routes to most reductions ($135^\circ = 90^\circ + 45^\circ = 180^\circ - 45^\circ$) — use whichever is faster, and they cross-check each other.
- Long allied-angle products almost always collapse to ± 1 — if yours doesn't, recheck a sign.

CHEAT SHEET BOX — Standard Values & Allied Angles

- **Sine row** (0° – 90°): $\frac{\sqrt{0}}{2}, \frac{\sqrt{1}}{2}, \frac{\sqrt{2}}{2}, \frac{\sqrt{3}}{2}, \frac{\sqrt{4}}{2}$; **cosine** = reverse.
- $\tan = \sin / \cos$: $0, \frac{1}{\sqrt{3}}, 1, \sqrt{3}, \infty$.
- **Even**: $\cos, \sec$. **Odd**: $\sin, \csc, \tan, \cot$.
- **Reduction rule**: $f(n \cdot 90^\circ \pm \theta) \rightarrow n \text{ odd} \Rightarrow \text{co-function}, n \text{ even} \Rightarrow \text{same};$
sign from quadrant (ASTC).
- $\sin(180^\circ - \theta) = \sin \theta$; $\cos(180^\circ - \theta) = -\cos \theta$; $\sin(180^\circ + \theta) = -\sin \theta$.
- $\sin(90^\circ + \theta) = \cos \theta$; $\cos(90^\circ + \theta) = -\sin \theta$.
- **Reduce large angles** mod 360° first.
- **Golden rule**: quadrant $\rightarrow$ sign; odd- $90^\circ \rightarrow$ swap function.

Subtopic 04 — Graphs and Periodicity of Trigonometric Functions

1. Exam Relevance

A graph turns trigonometry from algebra you *compute* into behaviour you can *see*: at a glance it shows the domain, the range, where the function is positive or negative, where it climbs or falls, and — crucially — **how often it repeats**. That last property, periodicity, is what makes trig the language of every repeating phenomenon in physics (waves, oscillations, AC current). Graphs also give the fastest route to “how many solutions does this equation have?” — you simply count crossings.

For the exams:

- **CBSE Boards:** Sketching $\sin x$, $\cos x$, $\tan x$ and stating period/domain/range are direct syllabus items (NCERT §3.3). Period questions (“find the period of $\sin 3x$ ”) are common 1-markers.
- **NEET:** Periodicity and amplitude feed straight into SHM and wave physics; range questions ($3 + 2 \sin x$) appear frequently.
- **JEE Main:** Period of composite functions ($|\sin x| + |\cos x|$), number-of-solutions-by-graph, and range problems are recurring.
- **JEE Advanced:** Graphical reasoning — intersections of a trig graph with a line or parabola to count roots, and inequalities read off the curve — separates students who *visualise* from those who only manipulate.

The strategic point: master **two base periods** (2π for $\sin / \cos / \sec / \csc$, π for $\tan / \cot$) and **one scaling rule** (period = $\frac{\text{base}}{|b|}$), and you can read or sketch any trig graph.

2. Concept Intuition

A graph is the unit circle, unrolled. Imagine walking anticlockwise around the unit circle and, at each angle x , plotting your *height* ($\sin x$) against the distance walked. The height rises to 1 at $x = \frac{\pi}{2}$, falls to 0 at π , dips to -1 at $\frac{3\pi}{2}$, returns to 0 at 2π — and then, because you start another lap, the whole pattern **repeats**. The sin graph is literally the unit circle’s y -coordinate unrolled along a line.

Diagram placeholder. Left, the unit circle with a point at angle x and its height y marked. Right, the $y = \sin x$ curve from 0 to 2π , with the same height transferred across by a horizontal dashed line. Show the wave reaching $+1$ at $\pi/2$, 0 at π , -1 at $3\pi/2$, 0 at 2π .

Periodicity = “the pattern that repeats.” A function is periodic with period T if $f(x+T) = f(x)$ for all x , and T is the smallest such positive number. Sine and cosine repeat every full turn, $T = 2\pi$. Tangent repeats every *half* turn, $T = \pi$, because $\tan$

depends on the *line through the origin* (direction), and a line points “the same way” after 180° .

Indian-context analogy — the tide and the AC mains. The sea level at a harbour rises and falls in a smooth, repeating swell — a sine wave with a period of about 12 hours. The voltage in your wall socket is a sine wave repeating 50 times a second. “How high?” is the **amplitude**; “how often?” is governed by the **period**. Reading a tide chart is reading a sine graph.

Stretching and shifting — what a , b , c , d do. For $y = a \sin(bx + c) + d$:

- $|a|$ stretches it vertically — the **amplitude** (how tall the wave is).
- b squeezes it horizontally — bigger b means **more oscillations** in the same space, so a **shorter period** $\frac{2\pi}{|b|}$.
- c slides it left/right — a **phase shift**.
- d lifts the whole wave up/down.

Diagram placeholder. Three overlaid sine curves on the same axes: $y = \sin x$ (reference), $y = 2 \sin x$ (taller, amplitude 2), and $y = \sin 2x$ (twice as many waves, period π). Legend distinguishing them.

Reading inequalities off the graph. To solve $\sin x \geq \frac{1}{2}$, draw the horizontal line $y = \frac{1}{2}$ and shade where the sine curve sits *on or above* it — the answer is the set of x -intervals, read directly from the picture. Graphs make trig inequalities visual instead of algebraic guesswork.

3. Reference: Periods & Graph Features

(A) Periods of the six functions.

Function	Period
$\sin x, \cos x$	2π
$\csc x, \sec x$	2π
$\tan x, \cot x$	π

Scaling rule: the period of $\sin(bx+c)$, $\cos(bx+c)$, $\sec$, $\csc$ is $\frac{2\pi}{|b|}$; of $\tan(bx+c)$, $\cot$ it is $\frac{\pi}{|b|}$. **Special composites:** $|\sin x|$ and $|\cos x|$ have period π ; $\sin^2 x$ and $\cos^2 x$ have period π (since $\sin^2 x = \frac{1-\cos 2x}{2}$); the period of a **sum** is the **LCM** of the individual periods.

(B) Graph features of each function.

$y =$	Domain	Range	Period	Key features
$\sin x$	$\mathbb{R}$	$[-1, 1]$	2π	odd (origin-symmetric); 0 at $n\pi$; max $+1$ at $\frac{\pi}{2} + 2n\pi$
$\cos x$	$\mathbb{R}$	$[-1, 1]$	2π	even (y -axis symmetric); max $+1$ at $2n\pi$; 0 at $(2n+1)\frac{\pi}{2}$
$\tan x$	$x \neq (2n+1)\frac{\pi}{2}$	$\mathbb{R}$	π	increasing branches; vertical asymptotes at $(2n+1)\frac{\pi}{2}$
$\cot x$	$x \neq n\pi$	$\mathbb{R}$	π	decreasing branches; asymptotes at $n\pi$
$\sec x$	$x \neq (2n+1)\frac{\pi}{2}$	$(-\infty, -1] \cup [1, \infty)$	2π	U-shapes opening from the cosine extremes; asymptotes at $(2n+1)\frac{\pi}{2}$
$\csc x$	$x \neq n\pi$	$(-\infty, -1] \cup [1, \infty)$	2π	U-shapes from the sine extremes; asymptotes at $n\pi$

(C) Transformation summary for $y = a \sin(bx + c) + d$ (same for cos): amplitude $= |a|$; period $= \frac{2\pi}{|b|}$; phase shift $= -\frac{c}{b}$; vertical shift $= d$; range $= [d - |a|, d + |a|]$.

4. Derivations & Constructions

(a) Why sin and cos have period 2π . A point on the unit circle returns to the *same location* only after a whole turn, i.e. after adding 2π to the angle: $\sin(x + 2\pi) = \sin x$, $\cos(x + 2\pi) = \cos x$, and no smaller positive shift works. Hence $T = 2\pi$.

(b) Why tan has period π . $\tan x = \frac{\sin x}{\cos x}$. Adding π flips the signs of *both* sine and cosine ($\sin(x + \pi) = -\sin x$, $\cos(x + \pi) = -\cos x$), and the two minus signs cancel

in the ratio: $\tan(x + \pi) = \frac{-\sin x}{-\cos x} = \tan x$. So tangent already repeats after π .

(c) The scaling rule. If $\sin x$ repeats every 2π , then $\sin(bx)$ completes one cycle when bx increases by 2π , i.e. when x increases by $\frac{2\pi}{|b|}$. Hence period = $\frac{2\pi}{|b|}$. (The constant c only shifts the graph; it does not change the period.)

(d) Constructing $y = \sin x$ from the value table. Plot the standard/quadrantal points $(0, 0)$, $(\frac{\pi}{6}, \frac{1}{2})$, $(\frac{\pi}{2}, 1)$, $(\pi, 0)$, $(\frac{3\pi}{2}, -1)$, $(2\pi, 0)$ and join with a smooth wave; then repeat left and right by periodicity. Cosine is the same wave shifted left by $\frac{\pi}{2}$ (since $\cos x = \sin(x + \frac{\pi}{2})$).

Diagram placeholder. $y = \cos x$ curve over $[-2\pi, 2\pi]$, starting at $(0, 1)$, crossing zero at $\pi/2$, minimum -1 at π , showing the even symmetry about the y -axis and the 2π repetition.

(e) Why sec/csc have the gap $(-1, 1)$. $\sec x = \frac{1}{\cos x}$; since $|\cos x| \leq 1$, its reciprocal has $|\sec x| \geq 1$, and where $\cos x \rightarrow 0$ the secant shoots to $\pm\infty$ (vertical asymptotes). The graph is a series of U-shapes sitting in the “caps” of the cosine wave.

Diagram placeholder. $y = \tan x$ over $(-3\pi/2, 3\pi/2)$ showing three increasing branches separated by vertical dashed asymptotes at $\pm\pi/2$ and $\pm3\pi/2$; passes through the origin; period π marked between consecutive branches.

5. Worked Examples (difficulty-laddered)

Example 1 — CBSE Board level (periods). Find the period of (a) $\sin 3x$, (b) $\tan 2x$, (c) $\cos \frac{x}{2}$.

$$(a) \frac{2\pi}{3}, \quad (b) \frac{\pi}{2}, \quad (c) \frac{2\pi}{1/2} = 4\pi.$$

Example 2 — CBSE Board level (range from transformation). Find the range and period of $f(x) = 3 + 2 \sin x$. Amplitude 2, vertical shift 3: range = $[3 - 2, 3 + 2] = [1, 5]$. The $\sin x$ is unscaled in x , so period = 2π .

Example 3 — NEET level (amplitude/period). State the amplitude and period of $y = -4 \cos(\frac{x}{3})$. Amplitude = $|-4| = 4$; period = $\frac{2\pi}{1/3} = 6\pi$. (The minus flips the graph vertically but does not change amplitude or period.)

Example 4 — JEE Main level (composite period). Find the period of $f(x) = |\sin x| + |\cos x|$. Each of $|\sin x|$, $|\cos x|$ has period π . But $|\cos x| = |\sin(x + \frac{\pi}{2})|$, and the sum repeats after only $\frac{\pi}{2}$: $f(x + \frac{\pi}{2}) = |\sin(x + \frac{\pi}{2})| + |\cos(x + \frac{\pi}{2})| = |\cos x| + |\sin x| = f(x)$. No smaller period works, so

$$T = \frac{\pi}{2}.$$

Example 5 — JEE Main level (period via identity). Find the period of $f(x) = \sin^2 x$. $\sin^2 x = \frac{1 - \cos 2x}{2}$, and $\cos 2x$ has period $\frac{2\pi}{2} = \pi$. So $T = \pi$ (not 2π).

Example 6 — JEE Advanced level (count solutions graphically). How many solutions does $|\sin x| = |\cos x|$ have in $[0, 2\pi]$? $|\sin x| = |\cos x| \iff |\tan x| = 1 \iff x = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$ — the four points in $[0, 2\pi]$ where the two “tent” graphs intersect.

$\therefore$ 4 solutions.

Example 7 — JEE Advanced level (inequality off the graph). Solve $\sin x \geq \frac{1}{2}$ for $x \in [0, 2\pi]$, and give the general solution. On $[0, 2\pi]$ the sine curve is on or above $\frac{1}{2}$ between its two crossings $\frac{\pi}{6}$ and $\frac{5\pi}{6}$:

$$x \in \left[\frac{\pi}{6}, \frac{5\pi}{6} \right].$$

By periodicity, the general solution is $x \in \left[2n\pi + \frac{\pi}{6}, 2n\pi + \frac{5\pi}{6} \right], n \in \mathbb{Z}$.

6. Practice Exercises (with answers)

- Find the period of: (a) $\cos 4x$, (b) $\cot \frac{x}{2}$, (c) $\sin(2x + 1)$.
- State domain and range of $y = 2 \sec x$.
- Find the amplitude, period and range of $y = 5 \sin(2x - \frac{\pi}{3}) + 1$.
- Find the period of $f(x) = \cos^2 x + \sin x$. (Hint: combine periods by LCM.)
- Solve $\cos x \leq -\frac{1}{2}$ for $x \in [0, 2\pi]$.
- How many solutions does $\sin x = \frac{x}{10}$ have for $x \geq 0$? (Reason graphically.)

Answers.

- (a) $\frac{2\pi}{4} = \frac{\pi}{2}$; (b) $\frac{\pi}{1/2} = 2\pi$; (c) $\frac{2\pi}{2} = \pi$.
- Domain $x \neq (2n+1)\frac{\pi}{2}$; range $(-\infty, -2] \cup [2, \infty)$ (amplitude scales the sec gap).
- Amplitude 5; period $\frac{2\pi}{2} = \pi$; range $[1 - 5, 1 + 5] = [-4, 6]$.
- $\cos^2 x$ has period π , $\sin x$ has period 2π ; LCM = 2π .
- $\cos x \leq -\frac{1}{2}$ between $\frac{2\pi}{3}$ and $\frac{4\pi}{3}$: $x \in [\frac{2\pi}{3}, \frac{4\pi}{3}]$.
- The line $y = \frac{x}{10}$ rises slowly; it meets the sine wave while $\frac{x}{10} \leq 1$, i.e. up to $x = 10$. Counting intersections of the line with the positive humps of $\sin x$ in $[0, 10]$ gives **3** solutions (including $x = 0$). (Standard graphical count.)

7. MCQ with Distractor Analysis

Q1. The period of $\sin 5x$ is: (A) $5 \cdot 2\pi$ (B) $\frac{2\pi}{5}$ (C) 2π (D) $\frac{\pi}{5}$

Answer: (B). Period $= \frac{2\pi}{|b|} = \frac{2\pi}{5}$.

- (A) multiplies by b instead of dividing — the TRIG-PERIOD-SCALE trap.
- (D) uses the tan base π for a sine.

Q2. The range of $y = 3 \cos x - 2$ is: (A) $[-2, 3]$ (B) $[-5, 1]$ (C) $[-1, 5]$ (D) $[-3, 3]$

Answer: (B). Amplitude 3, shift -2 : range $[-2 - 3, -2 + 3] = [-5, 1]$. TRIG-GRAPH-RANGE

- (D) ignores the shift; (C) gets the sign of the shift wrong.

Q3. $y = \tan x$ has vertical asymptotes at: (A) $x = n\pi$ (B) $x = (2n+1)\frac{\pi}{2}$ (C) $x = 2n\pi$ (D) nowhere

Answer: (B). Where $\cos x = 0$. TRIG-GRAPH-ASYMPTOTE

- (A) are the zeros of $\tan x$, not its asymptotes (that's $\cot x$'s asymptotes).

Q4. The period of $\tan 3x$ is: (A) $\frac{2\pi}{3}$ (B) $\frac{\pi}{3}$ (C) 2π (D) 3π

Answer: (B). Tangent base period is π , scaled by $\frac{1}{|b|}$: $\frac{\pi}{3}$.

- (A) uses the sine/cosine base 2π — the TRIG-PERIOD-TAN trap.

8. Common Pitfalls, Pro-Tips & Cheat Sheet

Pitfall 1 — Multiplying instead of dividing for the period. [SIGNATURE TRAP] $\sin(bx)$ has period $\frac{2\pi}{|b|}$, not $2\pi \cdot b$. Bigger $b \Rightarrow$ shorter period. TRIG-PERIOD-SCALE

Pitfall 2 — Using 2π for tangent. $\tan$ and $\cot$ have base period π . TRIG-PERIOD-TAN

Pitfall 3 — Period of a sum. Take the **LCM** of the parts; do not add or pick the larger blindly (and watch for hidden shrinkage as in $|\sin x| + |\cos x|$). TRIG-PERIOD-SUM

Pitfall 4 — Forgetting the vertical shift in range. Range of $a \sin x + d$ is $[d - |a|, d + |a|]$, not $[-|a|, |a|]$. TRIG-GRAPH-RANGE

Pitfall 5 — Swapping asymptote locations. $\tan, \sec$ blow up where $\cos = 0$ (odd $\frac{\pi}{2}$); $\cot, \csc$ where $\sin = 0$ ($n\pi$). TRIG-GRAPH-ASYMPTOTE

Pitfall 6 — Inequality endpoints. Decide open vs closed by whether the inequality is strict; read both crossing points carefully. TRIG-INEQ-INTERVAL

Pro-Tips.

- Sketch lightly from five points per period (the quadrantal values) and join smoothly — faster and less error-prone than plotting many points.

- Amplitude and vertical shift are read off instantly; period needs the $\frac{\text{base}}{|b|}$ rule; phase needs $-\frac{c}{b}$.
- For “number of solutions,” graph both sides and **count intersections** — far faster than solving algebraically.
- For trig inequalities, draw the horizontal cut line and shade — never guess intervals.

CHEAT SHEET BOX — Graphs & Periodicity

- **Base periods:** $\sin, \cos, \sec, \csc \Rightarrow 2\pi$; $\tan, \cot \Rightarrow \pi$.
- **Scaling:** period of $f(bx+c) = \frac{\text{base}}{|b|}$; c only shifts.
- **Special:** $|\sin x|, |\cos x|, \sin^2 x, \cos^2 x \Rightarrow \pi$; sum $\Rightarrow$ LCM of periods.
- **Transform** $a \sin(bx+c)+d$: amplitude $|a|$, period $\frac{2\pi}{|b|}$, phase $-\frac{c}{b}$, shift d , range $[d-|a|, d+|a|]$.
- **Asymptotes:** $\tan, \sec$ at $(2n+1)\frac{\pi}{2}$; $\cot, \csc$ at $n\pi$.
- **Ranges:** $\sin, \cos \in [-1, 1]$; $\sec, \csc \in (-\infty, -1] \cup [1, \infty)$; $\tan, \cot \in \mathbb{R}$.
- **Inequalities/roots:** draw the cut line / second curve and read intervals or count crossings.
- **Golden rule:** divide (don't multiply) for period; $\tan$'s period is π .

Subtopic 05 — Trigonometric Identities and Compound Angle Formulas

1. Exam Relevance

This is the **engine room** of the chapter and one of the highest-yield topics in all of Class 11 algebra-trigonometry. Almost every trigonometric equation, every inverse-trig manipulation, every integral of a trig product in Class 12, and a large slice of Complex Numbers (de Moivre's theorem) sit directly on the identities built here. A student fluent with compound-angle formulas solves in two lines what a memoriser grinds through in a page.

For the exams:

- **CBSE Boards:** Guaranteed marks. Expect a 1-2 mark exact-value question ($\cos 15^\circ$, $\tan 75^\circ$), a 3-mark “prove the identity,” and often a 5-mark conditional-identity proof in a triangle ($A + B + C = \pi$). NCERT §3.3 and Exercise 3.3 are lifted near-verbatim.
- **NEET:** Used for fast simplification inside physics (superposition of waves, AC phasors) and for evaluating awkward angles.
- **JEE Main:** A reliable 1-2 questions every year — maximum/minimum of $a \sin x + b \cos x$, value of a product like $\cos 20^\circ \cos 40^\circ \cos 80^\circ$, or a transformation-formula simplification.
- **JEE Advanced:** The identities become *tools* inside harder problems — telescoping sums of sines, conditional identities, and elegant product evaluations. Mastery here is non-negotiable for Advanced.

The strategic point: there are *many* formulas, but they all descend from **one** parent identity. Learn the derivation tree and you never memorise blindly — you reconstruct. That is exactly what this subtopic teaches.

2. Concept Intuition

One master key, many duplicates. It looks like a forest of formulas — sum, difference, double, half, triple, product-to-sum, sum-to-product. The liberating truth: **every one of them is cut from a single master key**, the cosine difference formula

$$\cos(A - B) = \cos A \cos B + \sin A \sin B.$$

Replace B with $-B$ and you get $\cos(A+B)$. Use the cofunction shift $\sin \theta = \cos(\frac{\pi}{2} - \theta)$ and you get $\sin(A \pm B)$. Set $B = A$ and you get the double-angle formulas. Halve the angle and you get half-angle. Add and subtract pairs and you get product-to-sum. The whole subtopic is one idea wearing eight costumes.

Diagram placeholder. A tree diagram. Root node: " $\cos(A-B) = \cos A \cos B + \sin A \sin B$ ". Branches to: " $\cos(A+B)$ [$B \rightarrow -B$]", " $\sin(A \pm B)$ [cofunction]", " $\tan(A \pm B)$ [ratio]". From those: " $\text{Double-angle } [B=A]$ " $\rightarrow$ " $\text{Half-angle } [A \rightarrow A/2]$ " and " $\text{Triple-angle } [2A+A]$ "; and " $\text{Product} \rightarrow \text{Sum}$ [add/subtract pairs]" $\rightarrow$ " $\text{Sum} \rightarrow \text{Product}$ [substitute]". Leaf: " $a \cos x + b \sin x = R \cos(x-\alpha)$ ".

What a “compound angle” even means. A compound angle is just an angle built from two others — $A + B$ or $A - B$. The naïve guess $\sin(A + B) = \sin A + \sin B$ is **flatly wrong** (test $A = B = 30^\circ$: $\sin 60^\circ = \frac{\sqrt{3}}{2} \approx 0.87$, but $\sin 30^\circ + \sin 30^\circ = 1$). Trigonometric functions are *not* linear. The compound-angle formulas are the correct, exact replacements — and internalising “sin does not distribute over +” prevents the most common error in the entire chapter.

Indian-context analogy — two sitar strings and the beat. Pluck two strings tuned a hair apart and you hear a slow throb — a *beat*. Physically, two pure tones $\cos(2\pi f_1 t)$ and $\cos(2\pi f_2 t)$ add, and the **sum-to-product** identity rewrites their sum as a single tone at the *average* frequency, amplitude-modulated at the *difference* frequency:

$$\cos C + \cos D = 2 \cos \frac{C+D}{2} \cos \frac{C-D}{2}.$$

The $\frac{C+D}{2}$ is the note you hear; the $\frac{C-D}{2}$ is the throb. So sum-to-product is not abstract algebra — it is literally why a slightly-detuned tanpura “beats.” Product-to-sum runs the same physics backwards.

Second analogy — combining two pushes into one. If you push a swing partly sideways ($a \cos x$) and partly forward ($b \sin x$), the *net* effect is a single push of strength $R = \sqrt{a^2 + b^2}$ in one combined direction:

$$a \cos x + b \sin x = R \cos(x - \alpha), \quad R = \sqrt{a^2 + b^2}.$$

Two component oscillations always collapse into one of amplitude $\sqrt{a^2 + b^2}$ — which instantly gives the maximum ($+R$) and minimum ($-R$) without calculus.

Why this matters for the Drona diagnostic. Mis-applications here are *structural*, not arithmetic — a student who writes $\cos(A+B) = \cos A \cos B + \sin A \sin B$ (wrong sign) has a specific, diagnosable gap that re-teaching the *derivation* fixes, whereas drilling more problems does not. That sign error is the signature trap of this subtopic.

3. Identity Master Reference

This is the complete catalogue. Every formula below is **machine-verifiable** (the SymPy layer can confirm each as an identity over all admissible angles — the math analogue of a units/sanity check). Memorise the *structure*; reconstruct the rest from §4.

(A) Sum & Difference (compound angle).

$$\cos(A - B) = \cos A \cos B + \sin A \sin B$$

$$\cos(A + B) = \cos A \cos B - \sin A \sin B$$

$$\sin(A + B) = \sin A \cos B + \cos A \sin B$$

$$\sin(A - B) = \sin A \cos B - \cos A \sin B$$

$$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}, \quad \tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$$

$$\cot(A + B) = \frac{\cot A \cot B - 1}{\cot B + \cot A}, \quad \cot(A - B) = \frac{\cot A \cot B + 1}{\cot B - \cot A}$$

Sign mnemonic: for cosine the sign *flips* ($\cos(A + B)$ takes the $-$); for sine the sign *follows* ($\sin(A + B)$ takes the $+$); for tangent the denominator sign is *opposite* the numerator's.

(B) Useful products of compound angles.

$$\sin(A + B) \sin(A - B) = \sin^2 A - \sin^2 B = \cos^2 B - \cos^2 A$$

$$\cos(A + B) \cos(A - B) = \cos^2 A - \sin^2 B = \cos^2 B - \sin^2 A$$

(C) Double-angle.

$$\sin 2A = 2 \sin A \cos A = \frac{2 \tan A}{1 + \tan^2 A}$$

$$\cos 2A = \cos^2 A - \sin^2 A = 2 \cos^2 A - 1 = 1 - 2 \sin^2 A = \frac{1 - \tan^2 A}{1 + \tan^2 A}$$

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

(D) Half-angle / power-reduction.

$$1 + \cos A = 2 \cos^2 \frac{A}{2}, \quad 1 - \cos A = 2 \sin^2 \frac{A}{2}$$

$$\sin A = 2 \sin \frac{A}{2} \cos \frac{A}{2}, \quad \tan \frac{A}{2} = \frac{\sin A}{1 + \cos A} = \frac{1 - \cos A}{\sin A}$$

(E) The $t = \tan \frac{A}{2}$ substitution (turns any rational trig expression into a rational function of t):

$$\sin A = \frac{2t}{1 + t^2}, \quad \cos A = \frac{1 - t^2}{1 + t^2}, \quad \tan A = \frac{2t}{1 - t^2}.$$

(F) Triple-angle.

$$\sin 3A = 3 \sin A - 4 \sin^3 A$$

$$\cos 3A = 4 \cos^3 A - 3 \cos A$$

$$\tan 3A = \frac{3 \tan A - \tan^3 A}{1 - 3 \tan^2 A}$$

(G) Product $\rightarrow$ Sum.

$$2 \sin A \cos B = \sin(A + B) + \sin(A - B)$$

$$2 \cos A \sin B = \sin(A + B) - \sin(A - B)$$

$$2 \cos A \cos B = \cos(A + B) + \cos(A - B)$$

$$2 \sin A \sin B = \cos(A - B) - \cos(A + B)$$

(H) Sum $\rightarrow$ Product (write $C = A + B$, $D = A - B$):

$$\sin C + \sin D = 2 \sin \frac{C + D}{2} \cos \frac{C - D}{2}$$

$$\sin C - \sin D = 2 \cos \frac{C + D}{2} \sin \frac{C - D}{2}$$

$$\cos C + \cos D = 2 \cos \frac{C + D}{2} \cos \frac{C - D}{2}$$

$$\cos C - \cos D = -2 \sin \frac{C + D}{2} \sin \frac{C - D}{2}$$

(I) Linear combination (amplitude form).

$$a \cos x + b \sin x = R \cos(x - \alpha), \quad R = \sqrt{a^2 + b^2}, \quad \tan \alpha = \frac{b}{a}.$$

Range: $[-\sqrt{a^2 + b^2}, \sqrt{a^2 + b^2}]$.

(J) Conditional identities (hold only when $A + B + C = \pi$, i.e. angles of a triangle):

$$\tan A + \tan B + \tan C = \tan A \tan B \tan C$$

$$\cot A \cot B + \cot B \cot C + \cot C \cot A = 1$$

$$\sin 2A + \sin 2B + \sin 2C = 4 \sin A \sin B \sin C$$

$$\cos 2A + \cos 2B + \cos 2C = -1 - 4 \cos A \cos B \cos C$$

$$\sin A + \sin B + \sin C = 4 \cos \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2}$$

$$\cos A + \cos B + \cos C = 1 + 4 \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}$$

4. Derivations of the Core Formulas

(a) The master key: $\cos(A - B)$ from the unit circle (the NCERT derivation).

Place four points on the unit circle: $P_1 = (\cos A, \sin A)$, $P_2 = (\cos B, \sin B)$, $P_3 = (\cos(A - B), \sin(A - B))$, and $P_4 = (1, 0)$. The chord P_1P_2 subtends the angle $A - B$ at the centre; so does the chord P_3P_4 . Equal central angles $\Rightarrow$ equal chords, so $P_1P_2 = P_3P_4$, hence $P_1P_2^2 = P_3P_4^2$.

Diagram placeholder. Unit circle with P_1 at angle A , P_2 at angle B , P_4 at $(1,0)$, P_3 at angle $(A-B)$. Two equal chords P_1P_2 and P_3P_4 highlighted, both subtending angle $(A-B)$.

Compute both squared lengths:

$$P_1P_2^2 = (\cos A - \cos B)^2 + (\sin A - \sin B)^2 = 2 - 2(\cos A \cos B + \sin A \sin B),$$

$$P_3P_4^2 = (\cos(A-B) - 1)^2 + \sin^2(A-B) = 2 - 2\cos(A-B).$$

Setting them equal and cancelling:

$$\cos(A-B) = \cos A \cos B + \sin A \sin B.$$

(b) Everything else, from the master key.

Cosine sum — replace $B \rightarrow -B$ (using $\cos(-B) = \cos B$, $\sin(-B) = -\sin B$):

$$\cos(A+B) = \cos A \cos B - \sin A \sin B.$$

Sine sum — use the cofunction identity $\sin \theta = \cos(\frac{\pi}{2} - \theta)$:

$$\begin{aligned} \sin(A+B) &= \cos\left(\frac{\pi}{2} - (A+B)\right) = \cos\left(\left(\frac{\pi}{2} - A\right) - B\right) = \cos\left(\frac{\pi}{2} - A\right) \cos B + \sin\left(\frac{\pi}{2} - A\right) \sin B \\ &= \sin A \cos B + \cos A \sin B. \end{aligned}$$

Replace $B \rightarrow -B$ for $\sin(A-B) = \sin A \cos B - \cos A \sin B$.

Tangent sum — divide $\sin(A+B)$ by $\cos(A+B)$, then divide numerator and denominator by $\cos A \cos B$:

$$\tan(A+B) = \frac{\sin A \cos B + \cos A \sin B}{\cos A \cos B - \sin A \sin B} = \frac{\tan A + \tan B}{1 - \tan A \tan B}.$$

(c) Double-angle (set $B = A$).

$$\sin 2A = \sin(A+A) = 2 \sin A \cos A,$$

$$\cos 2A = \cos(A+A) = \cos^2 A - \sin^2 A.$$

Using $\sin^2 A + \cos^2 A = 1$ gives the two alternative forms $2 \cos^2 A - 1$ and $1 - 2 \sin^2 A$.

Dividing through by $\cos^2 A$ gives the tan-forms $\frac{1 - \tan^2 A}{1 + \tan^2 A}$ and $\frac{2 \tan A}{1 + \tan^2 A}$.

(d) Half-angle / power-reduction (rearrange double-angle). From $\cos 2A = 2 \cos^2 A - 1$: $\cos^2 A = \frac{1 + \cos 2A}{2}$. From $\cos 2A = 1 - 2 \sin^2 A$: $\sin^2 A = \frac{1 - \cos 2A}{2}$.

Writing $A \rightarrow \frac{A}{2}$ gives $1 + \cos A = 2 \cos^2 \frac{A}{2}$ and $1 - \cos A = 2 \sin^2 \frac{A}{2}$. Dividing the latter by $\sin A = 2 \sin \frac{A}{2} \cos \frac{A}{2}$ gives $\tan \frac{A}{2} = \frac{1 - \cos A}{\sin A}$.

(e) Triple-angle (use $3A = 2A + A$).

$$\sin 3A = \sin 2A \cos A + \cos 2A \sin A = 2 \sin A \cos^2 A + (1 - 2 \sin^2 A) \sin A$$

$$= 2 \sin A(1 - \sin^2 A) + \sin A - 2 \sin^3 A = 3 \sin A - 4 \sin^3 A.$$

$$\cos 3A = \cos 2A \cos A - \sin 2A \sin A = (2 \cos^2 A - 1) \cos A - 2 \sin^2 A \cos A = 4 \cos^3 A - 3 \cos A.$$

(f) Product $\rightarrow$ Sum (add/subtract the sine/cosine sum & difference formulas).

Adding $\sin(A + B)$ and $\sin(A - B)$ cancels the $\cos A \sin B$ terms:

$$\sin(A + B) + \sin(A - B) = 2 \sin A \cos B.$$

The other three follow identically.

(g) Sum $\rightarrow$ Product (substitution). In the product-to-sum formulas, set $C = A + B$, $D = A - B$, so $A = \frac{C+D}{2}$, $B = \frac{C-D}{2}$. Then $\sin C + \sin D = 2 \sin \frac{C+D}{2} \cos \frac{C-D}{2}$, and the rest similarly. The **minus sign** in $\cos C - \cos D = -2 \sin \frac{C+D}{2} \sin \frac{C-D}{2}$ is the one to watch.

(h) Amplitude form. Expand $R \cos(x - \alpha) = R \cos x \cos \alpha + R \sin x \sin \alpha$. Match with $a \cos x + b \sin x$: $R \cos \alpha = a$, $R \sin \alpha = b$. Square and add: $R^2 = a^2 + b^2$, so $R = \sqrt{a^2 + b^2}$ and $\tan \alpha = b/a$. Since $\cos(x - \alpha) \in [-1, 1]$, the expression ranges over $[-R, R]$.

5. Worked Examples (difficulty-laddered)

Example 1 — CBSE Board level (exact value). Find $\cos 15^\circ$.

$$\cos 15^\circ = \cos(45^\circ - 30^\circ) = \cos 45^\circ \cos 30^\circ + \sin 45^\circ \sin 30^\circ = \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}} \cdot \frac{1}{2} = \frac{\sqrt{6} + \sqrt{2}}{4}.$$

Example 2 — CBSE Board level (prove an identity). Prove $\cos(A + B) \cos(A - B) = \cos^2 A - \sin^2 B$.

$$\text{LHS} = (\cos A \cos B - \sin A \sin B)(\cos A \cos B + \sin A \sin B) = \cos^2 A \cos^2 B - \sin^2 A \sin^2 B.$$

Write $\cos^2 B = 1 - \sin^2 B$ and $\sin^2 A = 1 - \cos^2 A$:

$$= \cos^2 A(1 - \sin^2 B) - (1 - \cos^2 A) \sin^2 B = \cos^2 A - \sin^2 B = \text{RHS.} \quad \blacksquare$$

Example 3 — NEET level (awkward angle). Evaluate $\sin \frac{7\pi}{12}$.

$$\frac{7\pi}{12} = \frac{\pi}{4} + \frac{\pi}{3}, \quad \sin\left(\frac{\pi}{4} + \frac{\pi}{3}\right) = \frac{1}{\sqrt{2}} \cdot \frac{1}{2} + \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{3}}{2} = \frac{\sqrt{2} + \sqrt{6}}{4}.$$

Example 4 — JEE Main level (signs + compound angle). If $\sin A = \frac{3}{5}$ with A acute, and $\cos B = -\frac{12}{13}$ with B obtuse, find $\sin(A + B)$. A acute (Q I): $\cos A = \frac{4}{5}$. B obtuse (Q II): $\sin B = +\frac{5}{13}$.

$$\sin(A + B) = \sin A \cos B + \cos A \sin B = \frac{3}{5} \cdot \left(-\frac{12}{13}\right) + \frac{4}{5} \cdot \frac{5}{13} = \frac{-36 + 20}{65} = -\frac{16}{65}.$$

(The whole difficulty is the sign of $\sin B$ — connects directly to Subtopic 02's “sign from quadrant” rule.)

Example 5 — JEE Main level (sum-to-product). Prove $\frac{\sin 3A + \sin A}{\cos 3A + \cos A} = \tan 2A$.

$$\sin 3A + \sin A = 2 \sin 2A \cos A, \quad \cos 3A + \cos A = 2 \cos 2A \cos A.$$

$$\therefore \frac{2 \sin 2A \cos A}{2 \cos 2A \cos A} = \tan 2A. \quad \blacksquare$$

Example 6 — JEE Main level (max/min). Find the maximum and minimum of $7 \cos x + 24 \sin x + 5$. $R = \sqrt{7^2 + 24^2} = 25$, so $7 \cos x + 24 \sin x \in [-25, 25]$ and $\max = 30$, $\min = -20$.

Example 7 — JEE Advanced level (elegant product). Prove $\cos 20^\circ \cos 40^\circ \cos 80^\circ = \frac{1}{8}$. Multiply/divide by $2 \sin 20^\circ$ and cascade $2 \sin \theta \cos \theta = \sin 2\theta$:

$$\cos 20^\circ \cos 40^\circ \cos 80^\circ = \frac{\sin 40^\circ \cos 40^\circ \cos 80^\circ}{2 \sin 20^\circ} = \frac{\sin 80^\circ \cos 80^\circ}{4 \sin 20^\circ} = \frac{\sin 160^\circ}{8 \sin 20^\circ} = \frac{\sin 20^\circ}{8 \sin 20^\circ} = \frac{1}{8}.$$

using $\sin 160^\circ = \sin 20^\circ$. $\blacksquare$ (The “advanced” move is the deliberate $\times \frac{2 \sin 20^\circ}{2 \sin 20^\circ}$ to start a double-angle cascade — a signature JEE technique that does not appear by pattern-matching.)

Example 8 — JEE Advanced level (conditional identity). If $A + B + C = \pi$, prove $\tan A + \tan B + \tan C = \tan A \tan B \tan C$. From $A + B = \pi - C$: $\tan(A + B) = \tan(\pi - C) = -\tan C$, so

$$\frac{\tan A + \tan B}{1 - \tan A \tan B} = -\tan C \implies \tan A + \tan B = -\tan C + \tan A \tan B \tan C,$$

$$\therefore \tan A + \tan B + \tan C = \tan A \tan B \tan C. \quad \blacksquare$$

(The condition $A + B + C = \pi$ is used at the very first step — drop it and the identity is false. Tag TRIG-ID-CONDITIONAL-SCOPE.)

6. Practice Exercises (with answers)

- Find the exact value of $\tan 75^\circ$.
- Prove: $\sin(A + B) \sin(A - B) = \sin^2 A - \sin^2 B$.

3. If $\tan A = \frac{1}{2}$ and $\tan B = \frac{1}{3}$, find $\tan(A + B)$.
4. Express $\sin 5x + \sin 3x$ as a product and simplify $\frac{\sin 5x + \sin 3x}{\cos 5x + \cos 3x}$.
5. Find the maximum value of $3 \sin x + 4 \cos x + 10$.
6. Prove: $\frac{\sin 2A}{1 + \cos 2A} = \tan A$.
7. In a triangle ABC , prove $\cos A + \cos B + \cos C = 1 + 4 \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}$.

Answers / sketches.

1. $\tan(45^\circ + 30^\circ) = \frac{1 + \frac{1}{\sqrt{3}}}{1 - \frac{1}{\sqrt{3}}} = \frac{\sqrt{3} + 1}{\sqrt{3} - 1} = 2 + \sqrt{3}$.
2. $(\sin A \cos B + \cos A \sin B)(\sin A \cos B - \cos A \sin B) = \sin^2 A \cos^2 B - \cos^2 A \sin^2 B$, then $\cos^2 = 1 - \sin^2$.
3. $\frac{\frac{1}{2} + \frac{1}{3}}{1 - \frac{1}{6}} = \frac{5/6}{5/6} = 1$.
4. $\sin 5x + \sin 3x = 2 \sin 4x \cos x$; $\cos 5x + \cos 3x = 2 \cos 4x \cos x$; ratio = $\tan 4x$.
5. $R = 5$; max = 15.
6. $\frac{2 \sin A \cos A}{2 \cos^2 A} = \tan A$.
7. Pair $\cos A + \cos B = 2 \cos \frac{A+B}{2} \cos \frac{A-B}{2}$; use $\frac{A+B}{2} = \frac{\pi}{2} - \frac{C}{2}$ so $\cos \frac{A+B}{2} = \sin \frac{C}{2}$; then $\cos C = 1 - 2 \sin^2 \frac{C}{2}$; factor $2 \sin \frac{C}{2}$.

7. MCQ with Distractor Analysis

Q1. $\cos(A + B) =$ (A) $\cos A \cos B + \sin A \sin B$ (B) $\cos A \cos B - \sin A \sin B$ (C) $\cos A \sin B + \sin A \cos B$ (D) $\sin A \sin B - \cos A \cos B$

Answer: (B).

- (A) is $\cos(A - B)$ — the **sign-flip trap**, the single most common error. TRIG-ID-SIGN-FLIP
- (C) is $\sin(A + B)$; (D) is $-\cos(A - B)$.

Q2. If $\tan A = \frac{1}{2}$, $\tan B = \frac{1}{3}$, then $A + B =$ (A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{2}$

Answer: (B). $\tan(A + B) = \frac{\frac{1}{2} + \frac{1}{3}}{1 - \frac{1}{6}} = 1 \Rightarrow A + B = \frac{\pi}{4}$.

- Using $1 + \tan A \tan B$ below (wrong sign) breaks the clean answer — TRIG-ID-SIGN-FLIP.

Q3. The maximum value of $5 \sin x + 12 \cos x$ is: (A) 17 (B) 13 (C) $\sqrt{17}$ (D) 7

Answer: (B). $R = \sqrt{5^2 + 12^2} = 13$.

- (A) adds the coefficients — TRIG-ID-AMPLITUDE-R. (C) takes $\sqrt{a+b}$; (D) subtracts.

Q4. $\cos C - \cos D =$ (A) $2 \sin \frac{C+D}{2} \sin \frac{C-D}{2}$ (B) $-2 \sin \frac{C+D}{2} \sin \frac{C-D}{2}$ (C) $2 \cos \frac{C+D}{2} \cos \frac{C-D}{2}$ (D) $-2 \cos \frac{C+D}{2} \sin \frac{C-D}{2}$

Answer: (B). The cos difference carries a leading **minus**.

- (A) drops the minus — TRIG-ID-COSDIFF-NEG. (C) is $\cos C + \cos D$.

8. Common Pitfalls, Pro-Tips & Cheat Sheet

Pitfall 1 — The cosine/tangent sign flip. [SIGNATURE TRAP] $\cos(A+B)$ takes $-$, $\cos(A-B)$ takes $+$; $\tan(A+B)$ has $1 - \tan A \tan B$ below, $\tan(A-B)$ has $1 +$. Cosine “disagrees” with the angle sign; sine “agrees.” TRIG-ID-SIGN-FLIP

Pitfall 2 — Assuming sin/cos distribute. $\sin(A+B) \neq \sin A + \sin B$. Always use the formula. TRIG-ID-DOUBLE-MISAPPLY

Pitfall 3 — Wrong $\cos 2A$ form. If given $\sin A$ use $1 - 2 \sin^2 A$; if given $\cos A$ use $2 \cos^2 A - 1$; if given $\tan A$ use $\frac{1 - \tan^2 A}{1 + \tan^2 A}$. TRIG-ID-COS2A-FORM

Pitfall 4 — Dropping the factor 2 in product $\leftrightarrow$ sum. $2 \sin A \cos B = \sin(A+B) + \sin(A-B)$ — the leading 2 is part of the formula. TRIG-ID-PRODSUM-COEFF

Pitfall 5 — The minus in $\cos C - \cos D$. It is $-2 \sin \frac{C+D}{2} \sin \frac{C-D}{2}$. TRIG-ID-COSDIFF-NEG

Pitfall 6 — Conditional identities without the condition. $\tan A + \tan B + \tan C = \tan A \tan B \tan C$ holds **only** when $A + B + C = \pi$. TRIG-ID-CONDITIONAL-SCOPE

Pitfall 7 — Amplitude as $a + b$. For $a \sin x + b \cos x$, amplitude is $\sqrt{a^2 + b^2}$, never $a + b$. TRIG-ID-AMPLITUDE-R

Pro-Tips.

- Memorise $\cos(A-B)$ and the derivation tree in §2; reconstruct the rest under pressure — 20 seconds, never fails.
- For exact angles, decompose into $30^\circ, 45^\circ, 60^\circ$: $15^\circ = 45^\circ - 30^\circ$, $75^\circ = 45^\circ + 30^\circ$, $7\pi/12 = \pi/4 + \pi/3$.
- For a ratio of sums of sines/cosines, convert top and bottom to products — the common factor cancels to a clean $\tan$.
- For products like $\cos 20^\circ \cos 40^\circ \cos 80^\circ$, multiply by $\frac{2 \sin(\text{smallest})}{2 \sin(\text{same})}$ and cascade double-angle.
- For max/min of $a \sin x + b \cos x + c$, the answer is $c \pm \sqrt{a^2 + b^2}$ — no calculus.

CHEAT SHEET BOX — Identities & Compound Angles

- **Master key:** $\cos(A-B) = \cos A \cos B + \sin A \sin B$. All else descends from it.

- **Sum/diff:** $\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$; $\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$.
- **Tan:** $\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$.
- **Double:** $\sin 2A = 2 \sin A \cos A$; $\cos 2A = 1 - 2 \sin^2 A = 2 \cos^2 A - 1$;
 $\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$.
- **Half:** $1 \pm \cos A = 2 \cos^2 \frac{A}{2} / 2 \sin^2 \frac{A}{2}$; $\tan \frac{A}{2} = \frac{1 - \cos A}{\sin A}$.
- $t = \tan \frac{A}{2}$: $\sin A = \frac{2t}{1+t^2}$, $\cos A = \frac{1-t^2}{1+t^2}$, $\tan A = \frac{2t}{1-t^2}$.
- **Triple:** $\sin 3A = 3 \sin A - 4 \sin^3 A$; $\cos 3A = 4 \cos^3 A - 3 \cos A$.
- **Prod**→**sum:** $2 \sin A \cos B = \sin(A+B) + \sin(A-B)$ (and the other three).
- **Sum**→**prod:** $\cos C - \cos D = -2 \sin \frac{C+D}{2} \sin \frac{C-D}{2}$ (watch the minus).
- **Amplitude:** $a \cos x + b \sin x = R \cos(x-\alpha)$, $R = \sqrt{a^2 + b^2}$; range $[-R, R]$.
- **Conditional** ($A+B+C=\pi$): $\sum \tan = \prod \tan$; $\sum \cot \cot = 1$.
- **Golden rule:** reconstruct from $\cos(A-B)$; watch every $\pm$; amplitude is $\sqrt{a^2 + b^2}$.

Subtopic 06 — Trigonometric Equations and Solutions

1. Exam Relevance

A trigonometric equation asks: *for which angles does this trig statement hold?* Because trig functions repeat forever, the answer is almost never a single number — it is an **infinite family** captured by one formula with an integer n . Learning to produce that family correctly, completely, and without spurious extras is the skill this subtopic builds, and it is the natural payoff of everything before it: you use identities (Subtopic 03) to simplify, signs (Subtopic 02) to pick valid roots, and radians (Subtopic 01) throughout.

For the exams:

- **CBSE Boards:** A near-certain question. Typically “find the principal solutions” (1–2 marks) and “find the general solution” (3 marks). NCERT §3.4 / Exercise 3.4 is the source.
- **NEET:** Appears as quadratic-in-sin x type equations and in physics (SHM phase, equilibrium angles).
- **JEE Main:** Reliable 1–2 questions — number of solutions in an interval, $a \sin x + b \cos x = c$ form, or counting roots.
- **JEE Advanced:** The hard versions hinge on **not losing or gaining roots** — dividing by a factor that can be zero, or squaring and importing extraneous solutions. These are exactly the traps that separate careful solvers from the rest.

The strategic point: there is a short, fixed list of **general-solution formulas**. Memorise those three master forms and the entire subtopic becomes “reduce the equation to $\sin \theta = \sin \alpha$, $\cos \theta = \cos \alpha$, or $\tan \theta = \tan \alpha$, then quote the formula.”

2. Concept Intuition

Why one equation has infinitely many answers. Ask “where is the hour hand pointing at 3 o’clock?” and the honest answer is “the same place at 3, again at 15:00, and every 12 hours forever.” Trig functions are *periodic*, so if one angle solves $\sin \theta = \frac{1}{2}$, then adding any whole number of full turns gives another solution, and so does the angle’s “mirror” partner. The job is never to find *a* solution — it is to describe **all** of them in one stroke.

Diagram placeholder. Unit circle showing $\sin \theta = 1/2$. Two terminal rays in $[0, 2\pi)$: one at $\pi/6$ (Q I) and one at $5\pi/6$ (Q II), both with the same height $y = 1/2$ marked by a horizontal dashed line. Caption: “two principal solutions; each repeats every 2π ”.

Two layers of answer.

- A **principal solution** is a solution lying in $[0, 2\pi)$ — the “first lap” answers. $\sin \theta = \frac{1}{2}$ has two: $\frac{\pi}{6}$ and $\frac{5\pi}{6}$.

- A **general solution** packages *every* solution using an integer n . For $\sin \theta = \frac{1}{2}$ it is $\theta = n\pi + (-1)^n \frac{\pi}{6}$.

Indian-context analogy — train arrivals. A local train reaches your platform at 6:10 and 6:50, then the pattern repeats every hour. “When is a train here?” is answered not by one time but by a *timetable formula*: $6:10 + n$ hr and $6:50 + n$ hr. The general solution of a trig equation is exactly that timetable — the recurring schedule of when the equation is “satisfied,” with n counting the laps.

The one idea behind all three master formulas. Solve $f(\theta) = f(\alpha)$ by moving everything to one side and **factorising with sum-to-product** (Subtopic 03). $\sin \theta - \sin \alpha = 0$ becomes a product $= 0$, and a product is zero exactly when a factor is zero — which hands you the periodic families directly. So the general-solution formulas are not arbitrary; they fall straight out of the identities you already have.

The two great dangers (and why they are conceptual, not careless).

- **Losing roots by dividing.** If you “cancel” $\cos \theta$ from both sides of $\sin \theta \cos \theta = \cos \theta$, you silently throw away every solution where $\cos \theta = 0$. Cancelling assumes the thing you cancel is non-zero — and here it can be zero. *Factor instead of divide.*
- **Gaining roots by squaring.** Squaring both sides can make a false equation look true (since $(-1)^2 = 1^2$). Every solution from a squared equation must be **checked back** in the original.

These two are the signature traps, and a student who makes them has a precise, re-teachable gap — exactly the kind the Drona diagnostic targets.

3. Solution-Set Reference & Root-Validity Check

This is the complete formula list. (All verifiable by the SymPy layer over $n \in \mathbb{Z}$ — the math analogue of a sanity check.) Throughout, $n \in \mathbb{Z}$.

(A) Zero cases.

$$\begin{aligned}\sin \theta = 0 &\iff \theta = n\pi \\ \cos \theta = 0 &\iff \theta = (2n + 1)\frac{\pi}{2} \\ \tan \theta = 0 &\iff \theta = n\pi\end{aligned}$$

(B) The three master general solutions.

$$\begin{aligned}\sin \theta = \sin \alpha &\iff \theta = n\pi + (-1)^n \alpha \\ \cos \theta = \cos \alpha &\iff \theta = 2n\pi \pm \alpha \\ \tan \theta = \tan \alpha &\iff \theta = n\pi + \alpha\end{aligned}$$

(C) Squared cases (one formula covers all three).

$$\sin^2 \theta = \sin^2 \alpha, \quad \cos^2 \theta = \cos^2 \alpha, \quad \tan^2 \theta = \tan^2 \alpha \iff \theta = n\pi \pm \alpha$$

(D) Extreme-value cases.

$$\sin \theta = 1 \iff \theta = (4n + 1)\frac{\pi}{2}, \quad \sin \theta = -1 \iff \theta = (4n - 1)\frac{\pi}{2}$$

$$\cos \theta = 1 \iff \theta = 2n\pi, \quad \cos \theta = -1 \iff \theta = (2n + 1)\pi$$

(E) Linear form $a \cos \theta + b \sin \theta = c$. Write $a = R \cos \alpha$, $b = R \sin \alpha$ with $R = \sqrt{a^2 + b^2}$, $\tan \alpha = \frac{b}{a}$. Then

$$a \cos \theta + b \sin \theta = c \iff R \cos(\theta - \alpha) = c \iff \cos(\theta - \alpha) = \frac{c}{R}.$$

Solvable only if $|c| \leq \sqrt{a^2 + b^2}$ (otherwise no solution — the max of the LHS is $\sqrt{a^2 + b^2}$). The maximum and minimum of $a \cos \theta + b \sin \theta$ are $+\sqrt{a^2 + b^2}$ and $-\sqrt{a^2 + b^2}$.

(F) Simultaneous equations. If $\sin \theta = \sin \alpha$ **and** $\cos \theta = \cos \alpha$ hold together, then $\theta = 2n\pi + \alpha$ (both the value *and* the quadrant must match, so only the $+\alpha$ branch survives).

Root-validity checklist (run before finalising any answer):

1. **Did I keep the integer n ?** Principal-only answers are incomplete when “general” is asked. TRIG-EQ-LOST-GENERAL
 2. **Did I divide by something that can be zero?** If so, restore the lost factor’s roots. TRIG-EQ-LOST-ROOT-DIVIDE
 3. **Did I square anywhere?** Back-substitute every candidate; discard those that fail. TRIG-EQ-EXTRANEIOUS
 4. **Is each root in the domain?** Drop any where $\tan / \sec / \cot / \csc$ is undefined. TRIG-EQ-DOMAIN-EXCLUDE
 5. **For $a \cos \theta + b \sin \theta = c$: is $|c| \leq \sqrt{a^2 + b^2}$?** If not, “no solution.” TRIG-EQ-SOLVABILITY
-

4. Derivations of the Master Formulas

(a) $\sin \theta = \sin \alpha \Rightarrow \theta = n\pi + (-1)^n \alpha$.

Move to one side and apply sum-to-product (Subtopic 03):

$$\sin \theta - \sin \alpha = 0 \implies 2 \cos \frac{\theta + \alpha}{2} \sin \frac{\theta - \alpha}{2} = 0.$$

A product is zero iff a factor is zero:

- $\sin \frac{\theta - \alpha}{2} = 0 \Rightarrow \frac{\theta - \alpha}{2} = m\pi \Rightarrow \theta = 2m\pi + \alpha$ (even multiple of π , plus α).
- $\cos \frac{\theta + \alpha}{2} = 0 \Rightarrow \frac{\theta + \alpha}{2} = (2m + 1)\frac{\pi}{2} \Rightarrow \theta = (2m + 1)\pi - \alpha$ (odd multiple of π , minus α).

The two cases — “even multiple $+\alpha$ ” and “odd multiple $-\alpha$ ” — combine compactly: when n is even the sign is $+$, when n is odd the sign is $-$ and there is a π shift. That is precisely $\theta = n\pi + (-1)^n\alpha$.

(b) $\cos \theta = \cos \alpha \Rightarrow \theta = 2n\pi \pm \alpha$.

$$\cos \theta - \cos \alpha = 0 \Rightarrow -2 \sin \frac{\theta + \alpha}{2} \sin \frac{\theta - \alpha}{2} = 0.$$

- $\sin \frac{\theta - \alpha}{2} = 0 \Rightarrow \theta = 2m\pi + \alpha$.
- $\sin \frac{\theta + \alpha}{2} = 0 \Rightarrow \theta = 2m\pi - \alpha$.

Together: $\theta = 2n\pi \pm \alpha$. (Cosine is even, $\cos(-\alpha) = \cos \alpha$, which is why both signs appear with the *same* full-turn period.)

(c) $\tan \theta = \tan \alpha \Rightarrow \theta = n\pi + \alpha$.

$$\tan \theta - \tan \alpha = 0 \Rightarrow \frac{\sin \theta \cos \alpha - \cos \theta \sin \alpha}{\cos \theta \cos \alpha} = 0 \Rightarrow \sin(\theta - \alpha) = 0$$

(numerator must vanish). Hence $\theta - \alpha = n\pi$, i.e. $\theta = n\pi + \alpha$. (Tangent has period π , not 2π , which is why a single $n\pi$ — not $2n\pi$ — appears.)

(d) Why the squared case gives $n\pi \pm \alpha$. $\sin^2 \theta = \sin^2 \alpha \Rightarrow \cos 2\theta = \cos 2\alpha$ (using $\cos 2x = 1 - 2\sin^2 x$). By (b), $2\theta = 2n\pi \pm 2\alpha$, so $\theta = n\pi \pm \alpha$.

5. Worked Examples (difficulty-laddered)

Example 1 — CBSE Board level (principal solutions). Find the principal solutions of $\sin x = \frac{1}{2}$. $\sin x = \frac{1}{2}$ at $x = \frac{\pi}{6}$ (Q I) and, since sine is also positive in Q II, at $x = \pi - \frac{\pi}{6} = \frac{5\pi}{6}$. Both lie in $[0, 2\pi)$.

$$\therefore x = \frac{\pi}{6}, \frac{5\pi}{6}.$$

Example 2 — CBSE Board level (general solution). Find the general solution of $\cos x = -\frac{1}{2}$. $\cos x = -\frac{1}{2} = \cos \frac{2\pi}{3}$ (reference angle where cosine is $-\frac{1}{2}$). By the cosine formula,

$$x = 2n\pi \pm \frac{2\pi}{3}, \quad n \in \mathbb{Z}.$$

Example 3 — NEET level (quadratic in $\sin x$). Solve $2\sin^2 x + \sin x - 1 = 0$. Factor as a quadratic in $\sin x$: $(2\sin x - 1)(\sin x + 1) = 0$, so $\sin x = \frac{1}{2}$ or $\sin x = -1$.

- $\sin x = \frac{1}{2} = \sin \frac{\pi}{6} \Rightarrow x = n\pi + (-1)^n \frac{\pi}{6}$.

- $\sin x = -1 \Rightarrow x = (4n - 1)\frac{\pi}{2}.$

$$\therefore x = n\pi + (-1)^n \frac{\pi}{6} \quad \text{or} \quad x = (4n - 1)\frac{\pi}{2}.$$

Example 4 — JEE Main level (linear form). Solve $\sqrt{3}\cos x + \sin x = 1$. Here $a = \sqrt{3}$, $b = 1$, so $R = \sqrt{3+1} = 2$ and $\tan \alpha = \frac{1}{\sqrt{3}} \Rightarrow \alpha = \frac{\pi}{6}$. Rewrite:

$$2\cos\left(x - \frac{\pi}{6}\right) = 1 \implies \cos\left(x - \frac{\pi}{6}\right) = \frac{1}{2} = \cos \frac{\pi}{3}.$$

$$x - \frac{\pi}{6} = 2n\pi \pm \frac{\pi}{3} \implies x = 2n\pi + \frac{\pi}{6} \pm \frac{\pi}{3}.$$

$$\therefore x = 2n\pi + \frac{\pi}{2} \quad \text{or} \quad x = 2n\pi - \frac{\pi}{6}.$$

(Check $|c| = 1 \leq R = 2$, so solutions exist — the solvability test from §3.)

Example 5 — JEE Main level (factor, don't divide). Solve $\sin 2x = \sin x$.

$$\sin 2x - \sin x = 0 \implies 2\sin x \cos x - \sin x = 0 \implies \sin x(2\cos x - 1) = 0.$$

Factor — do not divide by $\sin x$. Then $\sin x = 0$ **or** $\cos x = \frac{1}{2}$:

$$x = n\pi \quad \text{or} \quad x = 2n\pi \pm \frac{\pi}{3}.$$

(Dividing by $\sin x$ would lose the entire $x = n\pi$ family — the signature trap TRIG-EQ-LOST-ROOT-DIVIDE.)

Example 6 — JEE Main level (sum-to-product). Solve $\sin x + \sin 3x + \sin 5x = 0$. Group the outer terms: $\sin x + \sin 5x = 2\sin 3x \cos 2x$. So

$$2\sin 3x \cos 2x + \sin 3x = 0 \implies \sin 3x(2\cos 2x + 1) = 0.$$

- $\sin 3x = 0 \Rightarrow 3x = n\pi \Rightarrow x = \frac{n\pi}{3}.$

- $\cos 2x = -\frac{1}{2} = \cos \frac{2\pi}{3} \Rightarrow 2x = 2n\pi \pm \frac{2\pi}{3} \Rightarrow x = n\pi \pm \frac{\pi}{3}.$

$$\therefore x = \frac{n\pi}{3} \quad \text{or} \quad x = n\pi \pm \frac{\pi}{3}.$$

Example 7 — JEE Advanced level (the squaring trap, done right). Solve $\sin x + \cos x = 1$. Convert to amplitude form (avoids squaring): $\sin x + \cos x = \sqrt{2}\sin\left(x + \frac{\pi}{4}\right)$, so

$$\sqrt{2}\sin\left(x + \frac{\pi}{4}\right) = 1 \implies \sin\left(x + \frac{\pi}{4}\right) = \frac{1}{\sqrt{2}} = \sin \frac{\pi}{4}.$$

$$x + \frac{\pi}{4} = n\pi + (-1)^n \frac{\pi}{4}.$$

- n even: $x + \frac{\pi}{4} = 2m\pi + \frac{\pi}{4} \Rightarrow x = 2m\pi.$

- n odd: $x + \frac{\pi}{4} = (2m+1)\pi - \frac{\pi}{4} \Rightarrow x = 2m\pi + \frac{\pi}{2}$.

$$\therefore x = 2n\pi \quad \text{or} \quad x = 2n\pi + \frac{\pi}{2}.$$

Contrast with squaring: if instead you square $\sin x + \cos x = 1$ to get $1 + \sin 2x = 1$, i.e. $\sin 2x = 0$, you obtain $x = \frac{n\pi}{2}$ — which *includes* $x = \pi$ and $x = \frac{3\pi}{2}$. But $\sin \pi + \cos \pi = -1 \neq 1$, so $x = \pi$ is **extraneous**. Squaring imported a false root; only back-checking removes it. TRIG-EQ-EXTRANEUS

Example 8 — JEE Advanced level (range squeeze). Solve $2\sin^2 x + \sin^2 2x = 2$. Use $\sin^2 2x = 4\sin^2 x \cos^2 x$ and let $s = \sin^2 x$ (so $\cos^2 x = 1 - s$):

$$2s + 4s(1-s) = 2 \Rightarrow 2s + 4s - 4s^2 = 2 \Rightarrow 4s^2 - 6s + 2 = 0 \Rightarrow 2s^2 - 3s + 1 = 0.$$

$$(2s-1)(s-1) = 0 \Rightarrow s = \frac{1}{2} \text{ or } s = 1.$$

- $\sin^2 x = \frac{1}{2} = \sin^2 \frac{\pi}{4} \Rightarrow x = n\pi \pm \frac{\pi}{4}$.
- $\sin^2 x = 1 \Rightarrow \cos x = 0 \Rightarrow x = (2n+1)\frac{\pi}{2}$.

$$\therefore x = n\pi \pm \frac{\pi}{4} \quad \text{or} \quad x = (2n+1)\frac{\pi}{2}.$$

6. Practice Exercises (with answers)

1. Find the principal solutions of $\tan x = \sqrt{3}$.
2. Find the general solution of $\sin x = -\frac{\sqrt{3}}{2}$.
3. Solve $2\cos^2 x + 3\sin x = 0$. (Hint: write $\cos^2 x = 1 - \sin^2 x$.)
4. Solve $\cos x - \sin x = 1$.
5. Solve $\tan x \tan 2x = 1$. (Watch the domain.)
6. Find the general solution of $\sin 3x + \sin x = 0$.
7. For what real k does $3\cos x + 4\sin x = k$ have a solution?

Answers / sketches.

1. $\tan x = \sqrt{3} = \tan \frac{\pi}{3} \rightarrow$ principal solutions $x = \frac{\pi}{3}, \frac{4\pi}{3}$ (tangent positive in Q I and Q III).
2. $\sin x = -\frac{\sqrt{3}}{2} = \sin(-\frac{\pi}{3}) \rightarrow x = n\pi + (-1)^n(-\frac{\pi}{3}) = n\pi - (-1)^n\frac{\pi}{3}$.
3. $2(1-\sin^2 x) + 3\sin x = 0 \Rightarrow 2\sin^2 x - 3\sin x - 2 = 0 \Rightarrow (2\sin x + 1)(\sin x - 2) = 0$. $\sin x = 2$ rejected (range!); $\sin x = -\frac{1}{2} \Rightarrow x = n\pi + (-1)^n(-\frac{\pi}{6})$.
4. Amplitude form $\sqrt{2}\cos(x + \frac{\pi}{4}) = 1$; $x = 2n\pi$ or $x = 2n\pi - \frac{\pi}{2}$ (check, discard extraneous).
5. $\tan x \tan 2x = 1 \Rightarrow \cos x \cos 2x - \sin x \sin 2x = 0 \Rightarrow \cos 3x = 0 \Rightarrow 3x = (2n+1)\frac{\pi}{2} \Rightarrow x = (2n+1)\frac{\pi}{6}$, excluding values where $\tan x$ or $\tan 2x$ is undefined.
6. $\sin 3x + \sin x = 2\sin 2x \cos x = 0 \Rightarrow \sin 2x = 0$ or $\cos x = 0 \Rightarrow x = \frac{n\pi}{2}$ (the $\cos x = 0$ family is contained in $\sin 2x = 0$).
7. Solvable iff $|k| \leq \sqrt{3^2 + 4^2} = 5$, i.e. $-5 \leq k \leq 5$.

7. MCQ with Distractor Analysis

Q1. The general solution of $\sin \theta = \sin \alpha$ is: (A) $\theta = 2n\pi \pm \alpha$ (B) $\theta = n\pi + (-1)^n \alpha$ (C) $\theta = n\pi + \alpha$ (D) $\theta = 2n\pi + \alpha$

Answer: (B).

- (A) is the **cosine** family — using it for sine is the signature TRIG-EQ-WRONG-FAMILY error.
- (C) is the **tangent** family. (D) is the simultaneous case.

Q2. The number of solutions of $\sin x = \frac{1}{2}$ in $[0, 2\pi)$ is: (A) 1 (B) 2 (C) 3 (D) infinite

Answer: (B). $x = \frac{\pi}{6}, \frac{5\pi}{6}$ — two principal solutions.

- (A) forgets the Q II partner. (D) confuses principal with general (infinite over all $\mathbb{R}$, but two in one lap).

Q3. $5 \sin x + 12 \cos x = 15$ has: (A) two solutions in $[0, 2\pi)$ (B) one solution (C) no solution (D) infinitely many

Answer: (C). $\sqrt{5^2 + 12^2} = 13 < 15$, so $|c| > R$ — impossible. TRIG-EQ-SOLVABILITY

- (A)/(D) ignore the solvability bound and try to solve mechanically.

Q4. A student solves $\tan x = \sin x$ by dividing both sides by $\sin x$ to get $\sec x = 1$, then $x = 2n\pi$. What is missing? (A) Nothing (B) The roots from $\sin x = 0$ (C) An extraneous root must be removed (D) The cosine family

Answer: (B). Dividing by $\sin x$ discards the case $\sin x = 0$ (i.e. $x = n\pi$), which *does* solve the original ($\tan x = \sin x = 0$). Factor instead: $\tan x - \sin x = \sin x \left(\frac{1}{\cos x} - 1 \right) = 0$. TRIG-EQ-LOST-ROOT-DIVIDE

8. Common Pitfalls, Pro-Tips & Cheat Sheet

Pitfall 1 — Dividing by a trig factor and losing roots. [SIGNATURE TRAP] Never cancel $\sin x$, $\cos x$, etc.; **factor** and set each factor to zero. Cancelling assumes the factor is non-zero, which is exactly the case you must not drop. TRIG-EQ-LOST-ROOT-DIVIDE

Pitfall 2 — Squaring and keeping extraneous roots. Squaring can manufacture false solutions. Every candidate from a squared step **must** be checked in the original equation. TRIG-EQ-EXTRANEIOUS

Pitfall 3 — Mixing the solution families. Sine: $n\pi + (-1)^n \alpha$. Cosine: $2n\pi \pm \alpha$. Tangent: $n\pi + \alpha$. Each function has its *own* form; never borrow another's. TRIG-EQ-WRONG-FAMILY

Pitfall 4 — Giving only principal solutions when “general” is asked. The integer n is mandatory in a general solution; dropping it loses infinitely many answers. TRIG-EQ-LOST-GENERAL

Pitfall 5 — Ignoring solvability of $a \cos x + b \sin x = c$. If $|c| > \sqrt{a^2 + b^2}$, declare “no solution” rather than grinding. TRIG-EQ-SOLVABILITY

Pitfall 6 — Keeping out-of-domain roots. Discard any solution where the equation’s functions (tan, sec, cot, csc) are undefined. TRIG-EQ-DOMAIN-EXCLUDE

Pro-Tips.

- Strategy ladder: (1) reduce everything to a single function if possible (use identities); (2) factor to a product = 0; (3) match each factor to $\sin \theta = \sin \alpha$ / $\cos \theta = \cos \alpha$ / $\tan \theta = \tan \alpha$; (4) quote the formula; (5) run the §3 root-validity checklist.
- For $a \cos x + b \sin x = c$, always convert to $R \cos(x - \alpha) = c$ first; it gives both the solvability test and the clean general solution.
- For sums like $\sin x + \sin 3x + \sin 5x$, pair the outermost terms for sum-to-product — the middle term becomes a common factor.
- When a problem says “find the number of solutions in an interval,” solve generally, then count the integer values of n that land inside the interval.

CHEAT SHEET BOX — Trigonometric Equations ($n \in \mathbb{Z}$)

- **Zeros:** $\sin \theta = 0 \Rightarrow \theta = n\pi$; $\cos \theta = 0 \Rightarrow \theta = (2n+1)\frac{\pi}{2}$; $\tan \theta = 0 \Rightarrow \theta = n\pi$.
- **Sine:** $\sin \theta = \sin \alpha \Rightarrow \theta = n\pi + (-1)^n \alpha$.
- **Cosine:** $\cos \theta = \cos \alpha \Rightarrow \theta = 2n\pi \pm \alpha$.
- **Tangent:** $\tan \theta = \tan \alpha \Rightarrow \theta = n\pi + \alpha$.
- **Squared:** $\sin^2 \theta = \sin^2 \alpha$ (etc.) $\Rightarrow \theta = n\pi \pm \alpha$.
- **Extremes:** $\sin \theta = 1 \Rightarrow (4n+1)\frac{\pi}{2}$; $\cos \theta = 1 \Rightarrow 2n\pi$; $\cos \theta = -1 \Rightarrow (2n+1)\pi$.
- **Linear:** $a \cos \theta + b \sin \theta = c \rightarrow R \cos(\theta - \alpha) = c$, $R = \sqrt{a^2 + b^2}$; **needs** $|c| \leq R$.
- **Simultaneous:** $\sin \theta = \sin \alpha$ AND $\cos \theta = \cos \alpha \Rightarrow \theta = 2n\pi + \alpha$.
- **Golden rules:** factor (never divide); back-check after squaring; keep the n ; match the right family.

Chapter 3: Trigonometric Functions — Supplement

Subtopic 07 — Sub-Multiple Angles and the Half-Angle Family

Exam Relevance: CBSE Boards lift this pattern almost verbatim from NCERT Exercise 3.3 — you are handed $\cos x$ (or $\sin x$, or $\tan x$) together with a quadrant, and asked for $\sin \frac{x}{2}$, $\cos \frac{x}{2}$ and $\tan \frac{x}{2}$. It is a reliable 3–4 mark question and the marking scheme awards a full step for the sign decision alone. JEE Main uses the same machinery differently: the $t = \tan \frac{A}{2}$ substitution collapses any rational trigonometric expression or equation into ordinary algebra in t , and it is the standard alternative route through $a \cos \theta + b \sin \theta = c$. JEE Advanced pushes into the pentagon-angle family — $\sin 18^\circ$, $\cos 36^\circ$ and their relatives — which are built by turning the sub-multiple relation $5\theta = 90^\circ$ into a quadratic. (NEET does not test mathematics.) The formulas themselves are already in Subtopic 05 §3(D)–(E); what this subtopic gives you is the missing half — the $\pm$ radical forms, and the discipline that decides the sign.

Section 1: Concept Intuition

The Mumbai local tells you the speed, not the direction. Stand on the platform at Dadar and the indicator board tells you a fast train is approaching at 80 km/h. That number is real and useful, but it does not tell you whether the train is Churchgate-bound or Virar-bound. For that you need one extra piece of information — which platform you are standing on.

A square root behaves exactly like that board. When you write

$$\sin^2 \frac{A}{2} = \frac{1 - \cos A}{2},$$

and take the square root, you recover the **size** of $\sin \frac{A}{2}$ and nothing else. Squaring destroyed the sign, and no amount of algebra will bring it back. **The $\pm$ in front of the radical is not decoration and it is not a hedge — it is an honest admission that the algebra has run out, and that the sign must come from geometry instead.** The geometry you need is the quadrant in which $\frac{A}{2}$ lands.

Think about it this way. The double-angle formula $\cos A = 1 - 2 \sin^2 \frac{A}{2}$ is a machine that eats $\frac{A}{2}$ and produces A . That machine is not reversible on its own, because it squares its input on the way through: $\frac{A}{2}$ and $-\frac{A}{2}$ (and $\pi - \frac{A}{2}$, and more) all come out as the same $\cos A$. Running the machine backwards therefore returns a whole family, and the quadrant of $\frac{A}{2}$ is the ticket that selects one member of it.

The trap that this subtopic exists to kill. Students read “ x lies in the third quadrant” and immediately write “so sine is negative.” That reasoning is about x . You are being asked about $\frac{x}{2}$, which lives somewhere else entirely. If $180^\circ < x < 270^\circ$, then halving every part

of that inequality gives $90^\circ < \frac{x}{2} < 135^\circ$ — the second quadrant, where sine is *positive*. **You must halve the interval, not the quadrant label.**

There is a pleasant consequence worth memorising. For any x in the first revolution, $0^\circ < x < 360^\circ$, the half-angle satisfies $0^\circ < \frac{x}{2} < 180^\circ$ — so $\frac{x}{2}$ can only be in Quadrant I or Quadrant II, and $\sin \frac{x}{2}$ is therefore always positive. That is a genuine shortcut, but notice its limiting condition: it fails the moment x is allowed past 360° . If $540^\circ < x < 630^\circ$ — still “third quadrant” by the usual loose phrasing — then $270^\circ < \frac{x}{2} < 315^\circ$, Quadrant IV, and $\sin \frac{x}{2}$ is negative. This is why the interval matters and the quadrant label alone does not.

Second idea — one dial instead of three. A mixer-grinder with three separate switches is harder to operate than one with a single speed dial. The substitution $t = \tan \frac{A}{2}$ is that single dial. Set it, and every trigonometric function of A follows:

$$\sin A = \frac{2t}{1+t^2}, \quad \cos A = \frac{1-t^2}{1+t^2}, \quad \tan A = \frac{2t}{1-t^2}.$$

Any expression built from $\sin A$ and $\cos A$ by adding, multiplying and dividing becomes a *rational function of the single variable t* — and rational functions are algebra, not trigonometry. That is the whole point of the substitution: it converts a trigonometric problem into a polynomial one. Its limiting condition is sharp and you must respect it — $\tan \frac{A}{2}$ is undefined when $\frac{A}{2}$ is an odd multiple of $\frac{\pi}{2}$, that is when $A = (2n+1)\pi$. Those values of A are invisible to the substitution and have to be tested separately.

Third idea — halving is what builds the exotic angles. The chapter has so far reached exact values only by *adding* standard angles ($15^\circ = 45^\circ - 30^\circ$, $75^\circ = 45^\circ + 30^\circ$). Sub-multiple relations open a second door. Because $5 \times 18^\circ = 90^\circ$, the angle 18° satisfies $2\theta = 90^\circ - 3\theta$, and forcing double-angle to agree with triple-angle produces a *quadratic equation in $\sin \theta$* . Solve it and you get $\sin 18^\circ = \frac{\sqrt{5}-1}{4}$ exactly. The $\sqrt{5}$ is not an accident: 18° and 36° are the angles of the regular pentagon and decagon, and $\sqrt{5}$ is the golden ratio’s fingerprint.

Figure brief: A number line of angles from 0° to 720° with the four quadrant bands shaded; directly beneath it a second, half-scale line showing where $\frac{A}{2}$ lands for each band, with arrows joining $(180^\circ, 270^\circ)$ to $(90^\circ, 135^\circ)$ and $(540^\circ, 630^\circ)$ to $(270^\circ, 315^\circ)$ to show that the same “Quadrant III” label maps to two different half-angle quadrants.

Section 2: Key Formulas and Definitions

Sub-multiple angle. If A is an angle, then $\frac{A}{2}, \frac{A}{3}, \frac{A}{4}, \dots$ are its *sub-multiple angles*. This subtopic is about $\frac{A}{2}$, the case the syllabus actually tests.

(A) The three $\pm$ radical forms. These are the reversed double-angle formulas.

$$\sin \frac{A}{2} = \pm \sqrt{\frac{1 - \cos A}{2}}, \quad \cos \frac{A}{2} = \pm \sqrt{\frac{1 + \cos A}{2}}, \quad \tan \frac{A}{2} = \pm \sqrt{\frac{1 - \cos A}{1 + \cos A}}$$

- A is any angle for which $\cos A$ is known.
- The sign in each formula is chosen **independently** from the quadrant of $\frac{A}{2}$ — not from the quadrant of A , and not the same sign in all three.

- The tan form additionally requires $\cos A \neq -1$ (otherwise the denominator vanishes and $\tan \frac{A}{2}$ is undefined).

(B) The two sign-free forms for $\tan \frac{A}{2}$. These carry their own sign automatically and are the safer choice whenever $\sin A$ is available.

$$\tan \frac{A}{2} = \frac{\sin A}{1 + \cos A} \quad (\cos A \neq -1), \quad \tan \frac{A}{2} = \frac{1 - \cos A}{\sin A} \quad (\sin A \neq 0)$$

- No $\pm$ appears because no squaring was performed — the sign is already carried by $\sin A$.
- Use the first when $\cos A = -1$ is not a risk; use the second when $\sin A = 0$ is not a risk. At least one of the two is always usable.

(C) The power-reduction pair (the same statements read left to right, already in Subtopic 05 §3(D)):

$$1 - \cos A = 2 \sin^2 \frac{A}{2}, \quad 1 + \cos A = 2 \cos^2 \frac{A}{2}, \quad \sin A = 2 \sin \frac{A}{2} \cos \frac{A}{2}$$

(D) The $t = \tan \frac{A}{2}$ (Weierstrass) substitution.

$$\sin A = \frac{2t}{1+t^2}, \quad \cos A = \frac{1-t^2}{1+t^2}, \quad \tan A = \frac{2t}{1-t^2}$$

- Valid for all $A \neq (2n+1)\pi, n \in \mathbb{Z}$; the $\tan A$ form additionally needs $t^2 \neq 1$.
- Two by-products worth remembering: $1 + \cos A = \frac{2}{1+t^2}$ and $1 - \cos A = \frac{2t^2}{1+t^2}$.

(E) Sign table — reading the quadrant of $\frac{A}{2}$. Halve the *interval*, then read the row.

Interval for A	Interval for $\frac{A}{2}$	Quadrant of $\frac{A}{2}$	$\sin \frac{A}{2}$	$\cos \frac{A}{2}$	$\tan \frac{A}{2}$
$0^\circ < A < 90^\circ$	$0^\circ < \frac{A}{2} < 45^\circ$	I	+	+	+
$90^\circ < A < 180^\circ$	$45^\circ < \frac{A}{2} < 90^\circ$	I	+	+	+
$180^\circ < A < 270^\circ$	$90^\circ < \frac{A}{2} < 135^\circ$	II	+	−	−
$270^\circ < A < 360^\circ$	$135^\circ < \frac{A}{2} < 180^\circ$	II	+	−	−
$360^\circ < A < 450^\circ$	$180^\circ < \frac{A}{2} < 225^\circ$	III	−	−	+
$450^\circ < A < 540^\circ$	$225^\circ < \frac{A}{2} < 270^\circ$	III	−	−	+
$540^\circ < A < 630^\circ$	$270^\circ < \frac{A}{2} < 315^\circ$	IV	−	+	−

Interval for A	Interval for $\frac{A}{2}$	Quadrant of $\frac{A}{2}$	$\sin \frac{A}{2}$	$\cos \frac{A}{2}$	$\tan \frac{A}{2}$
$630^\circ < A < 720^\circ$	$315^\circ < \frac{A}{2} < 360^\circ$	IV	—	+	—

Read the first four rows together: **for A anywhere in $(0^\circ, 360^\circ)$, $\sin \frac{A}{2} > 0$ always.** Read the last four and you see why that shortcut cannot be promoted to a rule.

(F) The pentagon family — exact values.

$$\sin 18^\circ = \cos 72^\circ = \frac{\sqrt{5} - 1}{4}, \quad \cos 36^\circ = \sin 54^\circ = \frac{\sqrt{5} + 1}{4}$$

$$\cos 18^\circ = \sin 72^\circ = \frac{\sqrt{10 + 2\sqrt{5}}}{4}, \quad \sin 36^\circ = \cos 54^\circ = \frac{\sqrt{10 - 2\sqrt{5}}}{4}$$

- Numerically: $\sin 18^\circ \approx 0.30902$, $\cos 36^\circ \approx 0.80902$, $\cos 18^\circ \approx 0.95106$, $\sin 36^\circ \approx 0.58779$. Always sanity-check a remembered surd against these.
- The two headline values differ only in one sign, and that is the whole memory load: $\frac{\sqrt{5} + 1}{4}$, minus for the smaller angle's sine.
- Two consequences that appear constantly in JEE Advanced: $\sin 18^\circ \cdot \cos 36^\circ = \frac{1}{4}$ and $\cos 36^\circ - \sin 18^\circ = \frac{1}{2}$.

Section 3: CBSE Board Derivations and Key Procedures

Derivation: the $\pm$ radical forms for $\sin \frac{A}{2}$ and $\cos \frac{A}{2}$

Step 1. Start from the double-angle identity in its sin-only form, proved in Subtopic 05 §4(c): $\cos 2\theta = 1 - 2\sin^2 \theta$. Reason: it is the form that contains only the function we want to isolate.

Step 2. Put $\theta = \frac{A}{2}$, so that $2\theta = A$: $\cos A = 1 - 2\sin^2 \frac{A}{2}$. Reason: the substitution is legitimate for every real A , since the identity holds for all θ .

Step 3. Rearrange: $\sin^2 \frac{A}{2} = \frac{1 - \cos A}{2}$. Reason: pure algebra; nothing has been lost yet.

Step 4. Take square roots: $\sin \frac{A}{2} = \pm \sqrt{\frac{1 - \cos A}{2}}$. Reason: for any real u , $u^2 = k$ gives $u = \pm\sqrt{k}$. This is the step where the sign information is destroyed, and the $\pm$ records that loss honestly.

Step 5. Repeat from $\cos A = 2\cos^2 \frac{A}{2} - 1$ to get $\cos \frac{A}{2} = \pm \sqrt{\frac{1 + \cos A}{2}}$, and divide the two results for $\tan \frac{A}{2} = \pm \sqrt{\frac{1 - \cos A}{1 + \cos A}}$. Reason: the quotient of the two radical forms is the tangent by definition.

Step 6. Recover the sign externally. Locate the interval for $\frac{A}{2}$, read its quadrant, and apply

the standard quadrant signs of Subtopic 02. *Reason: the sign is a fact about where the angle sits, and only the geometry knows it.*

Derivation: the sign-free forms for $\tan \frac{A}{2}$

Step 1. Write $1 - \cos A = 2 \sin^2 \frac{A}{2}$ and $\sin A = 2 \sin \frac{A}{2} \cos \frac{A}{2}$. *Reason: both are Step 3 of the previous derivation, read in the forward direction.*

Step 2. Divide the first by the second:

$$\frac{1 - \cos A}{\sin A} = \frac{2 \sin^2 \frac{A}{2}}{2 \sin \frac{A}{2} \cos \frac{A}{2}} = \tan \frac{A}{2}.$$

Reason: the common factor $2 \sin \frac{A}{2}$ cancels, valid whenever $\sin \frac{A}{2} \neq 0$, i.e. whenever $\sin A \neq 0$.

Step 3. Similarly divide $\sin A = 2 \sin \frac{A}{2} \cos \frac{A}{2}$ by $1 + \cos A = 2 \cos^2 \frac{A}{2}$ to get $\frac{\sin A}{1 + \cos A} = \tan \frac{A}{2}$.
Reason: same cancellation, this time of $2 \cos \frac{A}{2}$, valid whenever $\cos A \neq -1$.

Step 4. Observe that no square root was taken anywhere. *Reason: therefore no $\pm$ is needed — these forms deliver the correct sign by themselves, and that is exactly why they are the safer tool in an examination.*

Derivation: the $t = \tan \frac{A}{2}$ substitution

Step 1. Write $\sin A = 2 \sin \frac{A}{2} \cos \frac{A}{2}$ and multiply and divide by $\cos \frac{A}{2}$:

$$\sin A = 2 \tan \frac{A}{2} \cos^2 \frac{A}{2} = \frac{2t}{\sec^2 \frac{A}{2}}.$$

Reason: $\sin \theta = \tan \theta \cos \theta$, and $\cos^2 \theta = 1 / \sec^2 \theta$.

Step 2. Use $\sec^2 \frac{A}{2} = 1 + \tan^2 \frac{A}{2} = 1 + t^2$ to obtain $\sin A = \frac{2t}{1 + t^2}$. *Reason: the Pythagorean identity of Subtopic 02.*

Step 3. For cosine, start from $\cos A = \cos^2 \frac{A}{2} - \sin^2 \frac{A}{2} = \cos^2 \frac{A}{2} (1 - t^2) = \frac{1 - t^2}{1 + t^2}$. *Reason: factor out $\cos^2 \frac{A}{2}$ and use Step 2's value for it.*

Step 4. Divide: $\tan A = \frac{2t}{1 - t^2}$, which is just the double-angle tangent formula. *Reason: consistency check — the substitution is not new mathematics, only new bookkeeping.*

Derivation: $\sin 18^\circ$ and $\cos 36^\circ$

Step 1. Let $\theta = 18^\circ$. Then $5\theta = 90^\circ$, so $2\theta = 90^\circ - 3\theta$. *Reason: this is the sub-multiple relation that makes 18° special; nothing else about 18° is used.*

Step 2. Take sine of both sides: $\sin 2\theta = \sin(90^\circ - 3\theta) = \cos 3\theta$. *Reason: the cofunction identity of Subtopic 03.*

Step 3. Expand each side by the double- and triple-angle formulas of Subtopic 05 §3(C) and §3(F):

$$2 \sin \theta \cos \theta = 4 \cos^3 \theta - 3 \cos \theta = \cos \theta (4 \cos^2 \theta - 3).$$

Step 4. Since $\theta = 18^\circ$, $\cos \theta \neq 0$, so we may cancel $\cos \theta$: $2 \sin \theta = 4 \cos^2 \theta - 3$. *Reason: cancellation of a non-zero factor. Stating why it is non-zero is worth a mark.*

Step 5. Replace $\cos^2 \theta = 1 - \sin^2 \theta$ and collect:

$$2 \sin \theta = 4 - 4 \sin^2 \theta - 3 \implies 4 \sin^2 \theta + 2 \sin \theta - 1 = 0.$$

Step 6. Solve the quadratic in $\sin \theta$:

$$\sin \theta = \frac{-2 \pm \sqrt{4 + 16}}{8} = \frac{-2 \pm 2\sqrt{5}}{8} = \frac{-1 \pm \sqrt{5}}{4}.$$

Step 7. Reject the negative root. 18° is in Quadrant I, so $\sin 18^\circ > 0$, forcing

$$\sin 18^\circ = \frac{\sqrt{5} - 1}{4}$$

Reason: the quadrant supplies the sign — the same principle as the half-angle radicals, one level up.

Step 8. Deduce $\cos 36^\circ$ from $\cos 36^\circ = 1 - 2 \sin^2 18^\circ$:

$$\cos 36^\circ = 1 - 2 \cdot \frac{(\sqrt{5} - 1)^2}{16} = 1 - \frac{6 - 2\sqrt{5}}{8} = \frac{2 + 2\sqrt{5}}{8} = \frac{\sqrt{5} + 1}{4}.$$

Figure brief: A regular pentagon inscribed in a circle, one central triangle isolated and shaded, with its apex angle marked 72° , the base angles 54° , and the half-apex angle 36° marked on the dropped perpendicular — annotated to show where 18° enters as half of 36° .

Procedure A: given $\cos A$ (or $\sin A$, or $\tan A$) and an interval for A , find the half-angle values

- Write the interval for A as an explicit inequality**, e.g. $180^\circ < A < 270^\circ$ or $\pi < A < \frac{3\pi}{2}$. *Why: the phrase “third quadrant” is ambiguous once you leave the first revolution, and the whole question turns on this.*
- Divide every part of the inequality by 2** to obtain the interval for $\frac{A}{2}$. *Why: this, and only this, tells you the quadrant of the half-angle.*
- Read off the three signs** for $\sin \frac{A}{2}$, $\cos \frac{A}{2}$, $\tan \frac{A}{2}$ from that quadrant. Write them down before computing anything. *Why: deciding the sign after you see a number invites you to rationalise the wrong one.*
- If $\cos A$ was not given, compute it** from what was, using the quadrant of A (not of $\frac{A}{2}$) to fix its sign. *Why: the radical formulas are written in terms of $\cos A$ alone.*
- Substitute into the radical forms and attach the signs from Step 3.** *Why: magnitude from algebra, sign from geometry — the two arrive separately.*
- Verify with two independent checks:** confirm $\sin^2 \frac{A}{2} + \cos^2 \frac{A}{2} = 1$, and confirm that $2 \sin \frac{A}{2} \cos \frac{A}{2}$ reproduces $\sin A$ with the correct sign. *Why: the second check catches a wrong sign choice, which the first cannot.*

Procedure B: solving $a \cos \theta + b \sin \theta = c$ by the t -substitution

1. **Set** $t = \tan \frac{\theta}{2}$ and substitute $\cos \theta = \frac{1-t^2}{1+t^2}$, $\sin \theta = \frac{2t}{1+t^2}$. *Why: it converts the equation into a polynomial one.*
2. **Multiply through by** $1+t^2$ (never zero) and collect:

$$(a+c)t^2 - 2bt + (c-a) = 0.$$

Why: $1+t^2 > 0$ always, so no roots are gained or lost at this step.

3. **Check the discriminant.** $\Delta = 4b^2 - 4(c+a)(c-a) = 4(a^2 + b^2 - c^2)$, so real roots exist exactly when $c^2 \leq a^2 + b^2$. *Why: this reproduces the solvability condition $|c| \leq \sqrt{a^2 + b^2}$ from Subtopic 06 §3(E) — a reassuring cross-check that the two methods agree.*
4. **Solve for t , then invert** using $\tan \frac{\theta}{2} = t \Rightarrow \frac{\theta}{2} = n\pi + \tan^{-1} t \Rightarrow \theta = 2n\pi + 2 \tan^{-1} t$, $n \in \mathbb{Z}$. *Why: this is the general solution of $\tan x = \tan \alpha$ from Subtopic 06 §3(C), applied to $\frac{\theta}{2}$.*
5. **Test $\theta = (2n+1)\pi$ separately.** At these angles t is undefined, so the substitution cannot see them. Substituting directly gives $a \cos \theta + b \sin \theta = -a$, so they are solutions if and only if $c = -a$. *Why: this is the one place the method silently loses roots, and examiners know it.*

Section 4: Worked Examples**Example 1 — [CBSE Board Level]**

If $\cos x = -\frac{1}{3}$ and x lies in the third quadrant, find $\sin \frac{x}{2}$, $\cos \frac{x}{2}$ and $\tan \frac{x}{2}$.

Step 1 — the interval. x in the third quadrant means $\pi < x < \frac{3\pi}{2}$.

Step 2 — halve it. Dividing throughout by 2:

$$\frac{\pi}{2} < \frac{x}{2} < \frac{3\pi}{4},$$

so $\frac{x}{2}$ lies in the **second quadrant**.

Step 3 — the signs. In Quadrant II: $\sin \frac{x}{2} > 0$, $\cos \frac{x}{2} < 0$, $\tan \frac{x}{2} < 0$.

Step 4 — the magnitudes.

$$\sin \frac{x}{2} = +\sqrt{\frac{1 - \cos x}{2}} = \sqrt{\frac{1 + \frac{1}{3}}{2}} = \sqrt{\frac{4/3}{2}} = \sqrt{\frac{2}{3}} = \frac{\sqrt{6}}{3}$$

$$\cos \frac{x}{2} = -\sqrt{\frac{1 + \cos x}{2}} = -\sqrt{\frac{1 - \frac{1}{3}}{2}} = -\sqrt{\frac{2/3}{2}} = -\sqrt{\frac{1}{3}} = -\frac{\sqrt{3}}{3}$$

$$\tan \frac{x}{2} = \frac{\sin \frac{x}{2}}{\cos \frac{x}{2}} = \frac{\sqrt{6}/3}{-\sqrt{3}/3} = -\frac{\sqrt{6}}{\sqrt{3}} = -\sqrt{2}$$

Step 5 — verification. First, $\sin^2 \frac{x}{2} + \cos^2 \frac{x}{2} = \frac{2}{3} + \frac{1}{3} = 1$. Second, since x is in Quadrant III, $\sin x = -\sqrt{1 - \frac{1}{9}} = -\frac{2\sqrt{2}}{3}$, and the sign-free form gives

$$\tan \frac{x}{2} = \frac{\sin x}{1 + \cos x} = \frac{-\frac{2\sqrt{2}}{3}}{\frac{2}{3}} = -\sqrt{2},$$

agreeing with Step 4. Two independent routes, one answer.

$$\therefore \sin \frac{x}{2} = \frac{\sqrt{6}}{3}, \quad \cos \frac{x}{2} = -\frac{\sqrt{3}}{3}, \quad \tan \frac{x}{2} = -\sqrt{2}. \quad \blacksquare$$

Example 2 — [JEE Main Level]

Solve $3 \cos x + 4 \sin x = 5$ **using the substitution** $t = \tan \frac{x}{2}$.

Step 1 — substitute. With $\cos x = \frac{1-t^2}{1+t^2}$ and $\sin x = \frac{2t}{1+t^2}$:

$$3 \cdot \frac{1-t^2}{1+t^2} + 4 \cdot \frac{2t}{1+t^2} = 5.$$

Step 2 — clear the denominator. Multiplying by $1+t^2$ (which is never zero, so nothing is lost):

$$3 - 3t^2 + 8t = 5 + 5t^2 \implies 8t^2 - 8t + 2 = 0 \implies 4t^2 - 4t + 1 = 0.$$

Step 3 — solve. This factorises as $(2t-1)^2 = 0$, a repeated root: $t = \frac{1}{2}$.

Step 4 — invert. $\tan \frac{x}{2} = \frac{1}{2}$, so $\frac{x}{2} = n\pi + \tan^{-1} \frac{1}{2}$, giving

$$x = 2n\pi + 2 \tan^{-1} \frac{1}{2}, \quad n \in \mathbb{Z}.$$

Step 5 — the excluded angles. At $x = (2n+1)\pi$ the substitution is blind. Direct substitution there gives $3(-1) + 4(0) = -3 \neq 5$, so no solutions were lost.

Step 6 — cross-check against the amplitude form. By Subtopic 05 §3(I), $3 \cos x + 4 \sin x = 5 \cos(x-\alpha)$ with $\tan \alpha = \frac{4}{3}$, and the equation forces $\cos(x-\alpha) = 1$, i.e. $x = 2n\pi + \tan^{-1} \frac{4}{3}$. The two answers agree because $2 \tan^{-1} \frac{1}{2} = \tan^{-1} \frac{4}{3}$ — indeed $\tan \left(2 \tan^{-1} \frac{1}{2} \right) = \frac{2 \cdot \frac{1}{2}}{1 - \frac{1}{4}} = \frac{1}{3/4} = \frac{4}{3}$.

The repeated root is not a coincidence: $c = 5 = \sqrt{3^2 + 4^2}$ sits exactly at the boundary of solvability, so the discriminant in Procedure B Step 3 is precisely zero. $\blacksquare$

Example 3 — [JEE Main — Speed Trap]

Given $\cos \theta = -\frac{7}{25}$ **with** $180^\circ < \theta < 270^\circ$, **find** $\sin \frac{\theta}{2}$.

The trap. The eye reads “ θ is in Quadrant III,” the hand writes “sine is negative there,” and the answer comes out $-\frac{4}{5}$. That reasoning applies the quadrant of θ to a question about $\frac{\theta}{2}$.

It is the single most common error in this entire pattern, and it is worth zero marks even though the magnitude is right.

The correct work.

Step 1. Halve the interval: $180^\circ < \theta < 270^\circ \Rightarrow 90^\circ < \frac{\theta}{2} < 135^\circ$. So $\frac{\theta}{2}$ is in **Quadrant II**, where sine is **positive**.

Step 2.

$$\sin \frac{\theta}{2} = +\sqrt{\frac{1 - \cos \theta}{2}} = \sqrt{\frac{1 + \frac{7}{25}}{2}} = \sqrt{\frac{32/25}{2}} = \sqrt{\frac{16}{25}} = \frac{4}{5}.$$

Step 3 — verify. Quadrant II also gives $\cos \frac{\theta}{2} = -\sqrt{\frac{1 - 7/25}{2}} = -\frac{3}{5}$. Then $\cos \theta = 1 - 2\left(\frac{4}{5}\right)^2 = 1 - \frac{32}{25} = -\frac{7}{25}$, as given, and $\sin \theta = 2\left(\frac{4}{5}\right)\left(-\frac{3}{5}\right) = -\frac{24}{25}$, correctly negative for Quadrant III. Everything is consistent.

$$\therefore \sin \frac{\theta}{2} = \frac{4}{5} \quad (\text{and } \cos \frac{\theta}{2} = -\frac{3}{5}, \tan \frac{\theta}{2} = -\frac{4}{3}). \quad \blacksquare$$

Seductive wrong answer: $-\frac{4}{5}$. It survives the Pythagorean check and the value of $\cos \theta$, because both involve only squares — which is exactly why you must run the $\sin \theta = 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}$ check instead.

The second layer. Suppose the interval had instead been $540^\circ < \theta < 630^\circ$, with the same $\cos \theta = -\frac{7}{25}$. That is still loosely “the third quadrant,” but halving gives $270^\circ < \frac{\theta}{2} < 315^\circ$ — Quadrant IV — and now $\sin \frac{\theta}{2} = -\frac{4}{5}$ genuinely is the answer. The quadrant *name* did not decide it; the *interval* did.

Example 4 — [JEE Advanced Level]

Prove that $\sin 18^\circ = \frac{\sqrt{5} - 1}{4}$, **deduce** $\cos 36^\circ$, **and hence evaluate** $\sin 18^\circ \cdot \cos 36^\circ$.

Step 1 — force double-angle to meet triple-angle. Let $\theta = 18^\circ$, so $5\theta = 90^\circ$ and therefore $2\theta = 90^\circ - 3\theta$. Taking sines and using $\sin(90^\circ - \phi) = \cos \phi$:

$$\sin 2\theta = \cos 3\theta.$$

Step 2 — expand both sides.

$$2 \sin \theta \cos \theta = 4 \cos^3 \theta - 3 \cos \theta = \cos \theta (4 \cos^2 \theta - 3).$$

Step 3 — cancel. $\cos 18^\circ \neq 0$, so dividing by $\cos \theta$ is legal:

$$2 \sin \theta = 4 \cos^2 \theta - 3 = 4(1 - \sin^2 \theta) - 3 = 1 - 4 \sin^2 \theta.$$

Step 4 — solve the quadratic. Rearranging, $4 \sin^2 \theta + 2 \sin \theta - 1 = 0$, so

$$\sin \theta = \frac{-2 \pm \sqrt{4+16}}{8} = \frac{-1 \pm \sqrt{5}}{4}.$$

Step 5 — choose the root by quadrant. 18° is acute, so $\sin 18^\circ > 0$ and the negative root $\frac{-1-\sqrt{5}}{4}$ is rejected:

$$\sin 18^\circ = \frac{\sqrt{5}-1}{4} \approx 0.30902. \quad \checkmark$$

Step 6 — deduce $\cos 36^\circ$. Since $36^\circ = 2 \times 18^\circ$, use $\cos 2\theta = 1 - 2\sin^2 \theta$:

$$\cos 36^\circ = 1 - 2 \left(\frac{\sqrt{5}-1}{4} \right)^2 = 1 - \frac{2(6-2\sqrt{5})}{16} = 1 - \frac{6-2\sqrt{5}}{8} = \frac{8-6+2\sqrt{5}}{8} = \frac{\sqrt{5}+1}{4} \approx 0.80902. \quad \checkmark$$

Step 7 — the product. The two surds are conjugates, so the product collapses:

$$\sin 18^\circ \cdot \cos 36^\circ = \frac{\sqrt{5}-1}{4} \cdot \frac{\sqrt{5}+1}{4} = \frac{5-1}{16} = \frac{4}{16} = \frac{1}{4}. \quad \blacksquare$$

Why this is an Advanced-level move. Nothing here is pattern-matching. The whole proof rests on spotting that $5\theta = 90^\circ$ can be split as $2\theta + 3\theta$, which is the only reason a double-angle formula and a triple-angle formula can be set equal to each other. Once you see that, the exotic value $\sqrt{5}$ appears inevitably — it is the discriminant $\sqrt{4+16}$ of a quadratic you were forced into. The same trick with $7\theta = 180^\circ$ does not produce a surd, which is precisely why $\sin 18^\circ$ is a standard value and $\sin\left(\frac{180^\circ}{7}\right)$ is not.

Section 5: Practice Exercises

1. If $\sin x = \frac{1}{4}$ and x lies in the second quadrant, find $\sin \frac{x}{2}$, $\cos \frac{x}{2}$ and $\tan \frac{x}{2}$.
2. If $\cos \theta = \frac{3}{5}$ and $270^\circ < \theta < 360^\circ$, find $\sin \frac{\theta}{2}$, $\cos \frac{\theta}{2}$ and $\tan \frac{\theta}{2}$.
3. If $\tan \frac{\theta}{2} = \frac{5}{12}$, find $\sin \theta$, $\cos \theta$ and $\tan \theta$.
4. Prove that $\operatorname{cosec} \theta - \cot \theta = \tan \frac{\theta}{2}$, stating the values of θ for which the statement is meaningful.
5. Prove that $\frac{1 + \cos A + \sin A}{1 + \cos A - \sin A} = \tan \left(\frac{\pi}{4} + \frac{A}{2} \right)$.
6. Find the exact value of $\sin 18^\circ + \cos 36^\circ$.
7. Given $\sin 18^\circ = \frac{\sqrt{5}-1}{4}$, find the exact value of $\cos 18^\circ$.

Answers.

1. $\frac{x}{2}$ lies in Quadrant I (since $90^\circ < x < 180^\circ \Rightarrow 45^\circ < \frac{x}{2} < 90^\circ$), so all three are positive.
 $\cos x = -\frac{\sqrt{15}}{4}$; $\sin \frac{x}{2} = \sqrt{\frac{4+\sqrt{15}}{8}} = \frac{\sqrt{5}+\sqrt{3}}{4}$, $\cos \frac{x}{2} = \sqrt{\frac{4-\sqrt{15}}{8}} = \frac{\sqrt{5}-\sqrt{3}}{4}$, $\tan \frac{x}{2} =$

- $4 + \sqrt{15}$.
2. $135^\circ < \frac{\theta}{2} < 180^\circ$, Quadrant II. $\sin \frac{\theta}{2} = \frac{1}{\sqrt{5}}$, $\cos \frac{\theta}{2} = -\frac{2}{\sqrt{5}}$, $\tan \frac{\theta}{2} = -\frac{1}{2}$.
3. $\sin \theta = \frac{120}{169}$, $\cos \theta = \frac{119}{169}$, $\tan \theta = \frac{120}{119}$.
4. $\operatorname{cosec} \theta - \cot \theta = \frac{1 - \cos \theta}{\sin \theta} = \tan \frac{\theta}{2}$; meaningful for $\sin \theta \neq 0$, i.e. $\theta \neq n\pi$.
5. Write $1 + \cos A = 2 \cos^2 \frac{A}{2}$ and $\sin A = 2 \sin \frac{A}{2} \cos \frac{A}{2}$; cancel the common factor $2 \cos \frac{A}{2}$ to get $\frac{\cos \frac{A}{2} + \sin \frac{A}{2}}{\cos \frac{A}{2} - \sin \frac{A}{2}} = \frac{1 + \tan \frac{A}{2}}{1 - \tan \frac{A}{2}} = \tan \left(\frac{\pi}{4} + \frac{A}{2} \right)$.
6. $\frac{\sqrt{5}-1}{4} + \frac{\sqrt{5}+1}{4} = \frac{2\sqrt{5}}{4} = \frac{\sqrt{5}}{2} \approx 1.11803$.
7. $\cos 18^\circ = \sqrt{1 - \frac{6-2\sqrt{5}}{16}} = \frac{\sqrt{10+2\sqrt{5}}}{4} \approx 0.95106$.

Section 6: Crack the MCQ

Q1. If $180^\circ < \theta < 270^\circ$, then $\sin \frac{\theta}{2}$ is (A) positive (B) negative (C) zero (D) impossible to determine without $\cos \theta$

Answer: (A). Halving gives $90^\circ < \frac{\theta}{2} < 135^\circ$, Quadrant II, where sine is positive.

Distractor Analysis. - (B) applies the quadrant of θ instead of the quadrant of $\frac{\theta}{2}$ — the signature error of this subtopic. - (C) confuses the interval's endpoints with the interval itself; $\sin \frac{\theta}{2} = 0$ would need $\frac{\theta}{2}$ to be a multiple of 180° , which the open interval excludes. - (D) assumes the $\pm$ can never be resolved. The sign depends only on the interval, never on the value of $\cos \theta$.

Q2. For all A at which it is defined, $\tan \frac{A}{2}$ equals (A) $\frac{\sin A}{1 - \cos A}$ (B) $\frac{1 + \cos A}{\sin A}$ (C) $\frac{1 - \cos A}{\sin A}$ (D) $\frac{1 - \cos A}{1 + \cos A}$

Answer: (C).

Distractor Analysis. - (A) is $\cot \frac{A}{2}$ — the student inverted the correct fraction, a slip that survives every magnitude check. - (B) is also $\cot \frac{A}{2}$, produced by pairing $1 + \cos A = 2 \cos^2 \frac{A}{2}$ with $\sin A$ in the wrong order. - (D) is $\tan^2 \frac{A}{2}$ — the student stopped one step early, before taking the square root of $\frac{1 - \cos A}{1 + \cos A}$.

Q3. $\sin 18^\circ$ equals (A) $\frac{\sqrt{5}-1}{4}$ (B) $\frac{\sqrt{5}+1}{4}$ (C) $\frac{1-\sqrt{5}}{4}$ (D) $\frac{\sqrt{5}-1}{2}$

Answer: (A), since $\frac{2.236-1}{4} \approx 0.309 = \sin 18^\circ$.

Distractor Analysis. - (B) is $\cos 36^\circ \approx 0.809$ — the two values differ by one sign and are constantly swapped. Guard against it by remembering that $\sin 18^\circ$ must be *small*, since 18° is a small angle. - (C) is the rejected root of $4x^2 + 2x - 1 = 0$; the student solved correctly but forgot that 18° lies in Quadrant I. It is negative, so a one-second sign check kills it. -

(D) has the wrong denominator, giving ≈ 0.618 ; it comes from mis-simplifying $\frac{-2 + 2\sqrt{5}}{8}$ by cancelling the 2 from only the numerator.

Q4. Solving $3 \cos x + 4 \sin x = 5$ by $t = \tan \frac{x}{2}$ produces (A) $t = \frac{1}{2}$ as a repeated root (B) $t = 2$ as a repeated root (C) two distinct real roots (D) no real root

Answer: (A). The quadratic is $4t^2 - 4t + 1 = (2t - 1)^2 = 0$.

Distractor Analysis. - (B) inverts t , usually by writing $\cot \frac{x}{2}$ in place of $\tan \frac{x}{2}$ somewhere in the substitution. - (C) assumes every quadratic yields two roots. Here the discriminant is $4(a^2 + b^2 - c^2) = 4(9 + 16 - 25) = 0$ — the equation sits exactly at the boundary of solvability. - (D) comes from believing $5 > \sqrt{3^2 + 4^2}$. In fact $\sqrt{25} = 5$ exactly, so a solution exists and is unique modulo 2π .

Section 7: Common Student Pitfalls and Pro-Tips

- **Pitfall 1 — Using the quadrant of A to sign the half-angle.** “ A is in Quadrant III, so $\sin \frac{A}{2} < 0$ ” is wrong. *Correction:* halve the interval first, then read the quadrant of $\frac{A}{2}$. This is the highest-frequency error in the whole pattern.
- **Pitfall 2 — Taking the same sign for all three functions.** The three signs are chosen independently. In Quadrant II you need $+$, $-$, $-$ respectively. *Correction:* write the three signs down before you compute any magnitude.
- **Pitfall 3 — Trusting the “quadrant name” past one revolution.** $180^\circ < A < 270^\circ$ and $540^\circ < A < 630^\circ$ are both “Quadrant III,” but their half-angles land in Quadrant II and Quadrant IV. *Correction:* work with the numeric interval you were actually given, never the label.
- **Pitfall 4 — Dropping the $\pm$ entirely.** Writing $\sin \frac{A}{2} = \sqrt{\frac{1 - \cos A}{2}}$ as though it were an identity is false for every A whose half-angle is in Quadrant III or IV. *Correction:* the $\pm$ is part of the formula; only a stated interval may remove it.
- **Pitfall 5 — Losing roots with the t -substitution.** $t = \tan \frac{\theta}{2}$ cannot represent $\theta = (2n+1)\pi$. *Correction:* always test those angles in the original equation before declaring the solution set complete.
- **Pitfall 6 — Squaring-only verification.** Checking $\sin^2 + \cos^2 = 1$ cannot detect a wrong sign, because squares are blind to it. *Correction:* verify with $\sin A = 2 \sin \frac{A}{2} \cos \frac{A}{2}$, which is sign-sensitive.
- **Pitfall 7 — Confusing $\sin 18^\circ$ with $\cos 36^\circ$.** They are $\frac{\sqrt{5}-1}{4}$ and $\frac{\sqrt{5}+1}{4}$. *Correction:* 18° is a small angle, so its sine must be small — about 0.31, not 0.81.

Pro-Tips.

- When $\sin A$ is available, prefer the sign-free forms $\tan \frac{A}{2} = \frac{\sin A}{1 + \cos A} = \frac{1 - \cos A}{\sin A}$. They eliminate the sign decision for the tangent entirely, and in a board answer they save you a paragraph of quadrant argument.
- For any A in $(0^\circ, 360^\circ)$, $\sin \frac{A}{2}$ is positive. That single fact answers a large fraction of board questions on sight — but state the halved interval anyway, because the marking scheme rewards it.
- The check $2 \sin \frac{A}{2} \cos \frac{A}{2} \stackrel{?}{=} \sin A$ takes eight seconds and catches every sign error. Make

it automatic.

- Reach for the t -substitution whenever an equation is a *rational* combination of $\sin \theta$ and $\cos \theta$ with no obvious factorisation. It never requires insight, only algebra — the ideal fallback when a clever route does not present itself.
- $\sin 18^\circ \cdot \cos 36^\circ = \frac{1}{4}$ and $\cos 36^\circ - \sin 18^\circ = \frac{1}{2}$ turn several nasty-looking Advanced expressions into one line. Memorise the pair alongside the values themselves.

Section 8: Cheat Sheet Box

CHEAT SHEET — Sub-Multiple Angles and the Half-Angle Family

- **Radical forms (sign from the quadrant of $\frac{A}{2}$):** $\sin \frac{A}{2} = \pm \sqrt{\frac{1-\cos A}{2}}$, $\cos \frac{A}{2} = \pm \sqrt{\frac{1+\cos A}{2}}$, $\tan \frac{A}{2} = \pm \sqrt{\frac{1-\cos A}{1+\cos A}}$.
- **Sign-free tangent (no $\pm$ needed):** $\tan \frac{A}{2} = \frac{\sin A}{1+\cos A} = \frac{1-\cos A}{\sin A}$.
- **Sign rule:** halve the *interval*, not the quadrant label. $180^\circ < A < 270^\circ \Rightarrow 90^\circ < \frac{A}{2} < 135^\circ$ (Quadrant II: +, −, −).
- **Free shortcut:** $0^\circ < A < 360^\circ \Rightarrow \sin \frac{A}{2} > 0$ always. Fails past one revolution.
- **Power reduction:** $1 - \cos A = 2 \sin^2 \frac{A}{2}$, $1 + \cos A = 2 \cos^2 \frac{A}{2}$, $\sin A = 2 \sin \frac{A}{2} \cos \frac{A}{2}$.
- $t = \tan \frac{A}{2}$: $\sin A = \frac{2t}{1+t^2}$, $\cos A = \frac{1-t^2}{1+t^2}$, $\tan A = \frac{2t}{1-t^2}$. Blind to $A = (2n+1)\pi$ — test separately.
- **Linear equation by t :** $a \cos \theta + b \sin \theta = c \Rightarrow (a+c)t^2 - 2bt + (c-a) = 0$; real roots iff $c^2 \leq a^2 + b^2$; then $\theta = 2n\pi + 2 \tan^{-1} t$.
- **Pentagon values:** $\sin 18^\circ = \cos 72^\circ = \frac{\sqrt{5}-1}{4}$; $\cos 36^\circ = \sin 54^\circ = \frac{\sqrt{5}+1}{4}$; $\cos 18^\circ = \frac{\sqrt{10+2\sqrt{5}}}{4}$; $\sin 36^\circ = \frac{\sqrt{10-2\sqrt{5}}}{4}$.
- **Pentagon consequences:** $\sin 18^\circ \cos 36^\circ = \frac{1}{4}$; $\cos 36^\circ - \sin 18^\circ = \frac{1}{2}$.
- **How $\sin 18^\circ$ is built:** $5\theta = 90^\circ \Rightarrow \sin 2\theta = \cos 3\theta \Rightarrow 4 \sin^2 \theta + 2 \sin \theta - 1 = 0$.
- **Golden rule:** the radical gives the size, the interval gives the sign. Never let one do the other's job.

Subtopic 08 — Properties of Triangles: Sine Rule, Cosine Rule and Area

Exam Relevance: Be clear about where this sits. The rationalised NCERT and the current CBSE Class 11 syllabus have **dropped** this material, so you will not meet it as a direct board question — the board's only triangle content is the conditional identities under $A + B + C = \pi$ in Subtopic 05 §3(J). JEE Main uses it occasionally, typically as a one-step cosine-rule or area computation embedded in a coordinate-geometry or vectors question. JEE Advanced, by contrast, lists it explicitly: "relations between sides and angles of a triangle, sine rule, cosine rule, half-angle formula and the area of a triangle" is a named syllabus item, and Advanced problems combine it freely with the identity work of Subtopic 05. (NEET

does not test mathematics.) Treat this subtopic as compulsory for Advanced, useful for Main, and enrichment for Boards.

Section 1: Concept Intuition

How the height of Everest was measured. In the 1850s, the Great Trigonometrical Survey of India needed the height of a peak nobody could climb. They could not stretch a tape up the mountain. What they *could* do was stand at two known points on the plains, measure the distance between them, and measure the angle of elevation from each. Three measured quantities, and the mountain's height fell out of the arithmetic.

That is the entire subject in one sentence. **A triangle has six parts — three sides a, b, c and three angles A, B, C — and they are not independent: knowing three of them (with at least one side) pins down the other three exactly.** The sine rule and the cosine rule are the two conversion machines that do the pinning.

Think about it this way. Sides are things you can measure with a tape; angles are things you can measure with a theodolite. In the field, one is often easy and the other impossible. These rules let you trade the measurement you cannot make for measurements you can.

Which rule when — the shape of the information decides.

- The **sine rule** pairs each side with the angle *opposite* it: $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$. Use it when your known data comes in an opposite pair — you know a side and its opposite angle, plus one more thing.
- The **cosine rule** links one angle to *all three* sides: $a^2 = b^2 + c^2 - 2bc \cos A$. Use it when you have all three sides (and want an angle), or two sides and the angle *between* them (and want the third side).

The cosine rule is Pythagoras with a correction term. Look at $a^2 = b^2 + c^2 - 2bc \cos A$ and set $A = 90^\circ$. Then $\cos A = 0$ and the whole thing collapses to $a^2 = b^2 + c^2$ — Pythagoras exactly. So the term $-2bc \cos A$ is precisely the penalty you pay for the angle not being a right angle. If A is acute, $\cos A > 0$ and $a^2 < b^2 + c^2$: the opposite side is *shorter* than Pythagoras would predict, because the triangle has been squeezed. If A is obtuse, $\cos A < 0$, the correction term is positive, and a is *longer*. **The sign of $\cos A$ and the sign of $b^2 + c^2 - a^2$ are the same thing, which is why the cosine rule can tell an obtuse angle from an acute one and the sine rule cannot.**

That last clause is the most exam-relevant sentence in this subtopic. Because $\sin(180^\circ - \phi) = \sin \phi$, the sine rule returns a value of $\sin A$ that fits *two* angles — one acute, one obtuse — and by itself cannot choose between them. The cosine rule returns $\cos A$, and cosine is negative in exactly one of the two, so it chooses for you.

Figure brief: Triangle ABC with standard labelling (a opposite A , etc.), its circumcircle drawn with centre O and radius R ; the central angle subtending side a marked as $2A$, with the perpendicular from O to BC shown bisecting both the chord and the central angle, so that $\frac{a}{2} = R \sin A$ is visible.

The circumcircle turns the ratio into a length. The sine rule says the three ratios are equal, but equal to *what*? The answer is beautiful: the common value is $2R$, the diameter of

the circle passing through all three vertices. So

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R.$$

This is not a decoration. It converts every side into an angle statement, $a = 2R \sin A$, which is the standard first move in almost every Advanced problem: substitute $a = 2R \sin A$, $b = 2R \sin B$, $c = 2R \sin C$ throughout, watch $2R$ cancel, and you are left with a pure identity in A, B, C with $A + B + C = \pi$ — exactly the conditional-identity machinery of Subtopic 05 §3(J).

Area — three formulas for three kinds of data. The school formula $\frac{1}{2} \times \text{base} \times \text{height}$ is useless in a survey, because nobody measures the height. So the area gets restated in terms of what you actually have: two sides and the included angle give $\Delta = \frac{1}{2}ab \sin C$; three sides give Heron's formula; and the circumradius gives $\Delta = \frac{abc}{4R}$. Same quantity, three doors in.

Where this fails. Everything here presupposes a genuine triangle — a, b, c positive and satisfying all three triangle inequalities, and $A + B + C = \pi$ with each angle strictly between 0 and π . Feed the cosine rule three lengths that cannot form a triangle and it will hand you a “ $\cos A$ ” outside $[-1, 1]$, which is the algebra's way of telling you the triangle does not exist. That is a useful test, not a failure.

Section 2: Key Formulas and Definitions

Standard notation. Throughout, ABC is a triangle with

- $a = BC, b = CA, c = AB$ — each side named by the **opposite** vertex.
- A, B, C — the interior angles at those vertices, with $A + B + C = \pi$ and $0 < A, B, C < \pi$.
- $s = \frac{a+b+c}{2}$ — the **semi-perimeter**. Note $s - a, s - b, s - c$ are all strictly positive, by the triangle inequality.
- Δ — the **area**.
- R — the **circumradius** (radius of the circle through A, B, C); r — the **inradius**.

(A) Sine rule and the circumradius relation.

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R \quad \Longleftrightarrow \quad a = 2R \sin A, \quad b = 2R \sin B, \quad c = 2R \sin C$$

- Valid in every triangle, acute, right or obtuse.
- **Limitation:** because $\sin \theta = \sin(\pi - \theta)$, solving for an angle from this rule leaves an acute/obtuse ambiguity that the rule itself cannot settle.

(B) Cosine rule (three statements, one per angle):

$$a^2 = b^2 + c^2 - 2bc \cos A, \quad b^2 = c^2 + a^2 - 2ca \cos B, \quad c^2 = a^2 + b^2 - 2ab \cos C$$

Rearranged for the angle — this is the form you actually use:

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}, \quad \cos B = \frac{c^2 + a^2 - b^2}{2ca}, \quad \cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

- The numerator is “the other two squared, minus the opposite one squared”; the denominator is twice the product of the other two.
- Its sign is decisive: $\cos A < 0 \iff a^2 > b^2 + c^2 \iff A$ is obtuse.

(C) Projection formulae.

$$a = b \cos C + c \cos B, \quad b = c \cos A + a \cos C, \quad c = a \cos B + b \cos A$$

- Each side equals the sum of the projections of the other two onto it.
- Read the pattern: to build a , use the other two sides, each multiplied by the cosine of the angle *at the far end of that side*.

(D) Area formulas.

Formula	Use it when you know
$\Delta = \frac{1}{2}ab \sin C = \frac{1}{2}bc \sin A = \frac{1}{2}ca \sin B$	two sides and the included angle
$\Delta = \sqrt{s(s-a)(s-b)(s-c)}$ (Heron)	all three sides, no angles
$\Delta = \frac{abc}{4R}$	three sides and the circumradius, or to <i>find</i> R
$\Delta = rs$	the inradius
$\Delta = 2R^2 \sin A \sin B \sin C$	the three angles and R

- All five give the same number; choosing the right one is the entire skill.
- $\Delta = \frac{abc}{4R}$ rearranges to $R = \frac{abc}{4\Delta}$, which is the fastest way to get the circumradius once Heron has given you the area.

(E) Half-angle formulae of a triangle. These express the half-angles purely in terms of the sides.

$$\sin \frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{bc}}, \quad \cos \frac{A}{2} = \sqrt{\frac{s(s-a)}{bc}}, \quad \tan \frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{s(s-a)}}$$

- **No $\pm$ is needed here**, unlike the general half-angle formulas of Subtopic 07. Since $0 < A < \pi$, we have $0 < \frac{A}{2} < \frac{\pi}{2}$, so all three are in Quadrant I and positive. Every quantity under the radicals is positive by the triangle inequality.
- The formulas for B and C follow by cycling $a \rightarrow b \rightarrow c \rightarrow a$ and $A \rightarrow B \rightarrow C \rightarrow A$.
- Two equivalent forms of the tangent worth knowing: $\tan \frac{A}{2} = \frac{(s-b)(s-c)}{\Delta} = \frac{\Delta}{s(s-a)} = \frac{r}{s-a}$.

(F) Napier’s analogy (the tangent rule — Advanced only):

$$\tan \frac{B-C}{2} = \frac{b-c}{b+c} \cot \frac{A}{2}$$

- Useful when two sides and the included angle are known and you want the other two angles separately.

Section 3: CBSE Board Derivations and Key Procedures

Derivation: the sine rule, and why the common ratio is $2R$

Step 1. In triangle ABC , drop the perpendicular CD from C onto line AB , of length h . Reason: an altitude is the standard device for producing right triangles inside a general triangle.

Step 2. In right triangle ADC : $h = b \sin A$. In right triangle BDC : $h = a \sin B$. Reason: in a right triangle, the side opposite an angle equals the hypotenuse times the sine of that angle. If A is obtuse the foot D falls outside segment AB and the angle at A in the right triangle is $\pi - A$; since $\sin(\pi - A) = \sin A$, the relation is unchanged.

Step 3. Equate: $b \sin A = a \sin B$, hence $\frac{a}{\sin A} = \frac{b}{\sin B}$. Reason: both expressions equal the same altitude.

Step 4. Repeat with the altitude from A to get $\frac{b}{\sin B} = \frac{c}{\sin C}$, completing the chain. Reason: symmetry of the argument in the three vertices.

Step 5 — identifying the constant. Draw the circumcircle, centre O , radius R . The chord BC (length a) subtends the angle A at the circumference and therefore the angle $2A$ at the centre. Reason: the angle-at-the-centre theorem from Class 9 geometry.

Step 6. Drop the perpendicular from O to BC . It bisects both the chord and the central angle, producing a right triangle with hypotenuse R , opposite angle A , and opposite side $\frac{a}{2}$. Hence $\frac{a}{2} = R \sin A$, i.e.

$$\frac{a}{\sin A} = 2R.$$

Reason: the perpendicular from a centre to a chord bisects it.

Step 7. The same construction on the other two sides gives the full statement. Reason: symmetry again. (If A is obtuse, BC subtends $2\pi - 2A$ at the centre on the near side, and the half-angle in the right triangle is $\pi - A$; since $\sin(\pi - A) = \sin A$, the conclusion is identical.)

Derivation: the cosine rule

Step 1. Place A at the origin with AB along the positive x -axis. Then $A = (0, 0)$, $B = (c, 0)$, and $C = (b \cos A, b \sin A)$. Reason: coordinates are the cleanest route and handle obtuse A without a separate case.

Step 2. Compute $a^2 = BC^2$ by the distance formula:

$$a^2 = (b \cos A - c)^2 + (b \sin A - 0)^2.$$

Step 3. Expand:

$$a^2 = b^2 \cos^2 A - 2bc \cos A + c^2 + b^2 \sin^2 A.$$

Step 4. Use $\cos^2 A + \sin^2 A = 1$:

$$a^2 = b^2 + c^2 - 2bc \cos A.$$

Reason: the Pythagorean identity of Subtopic 02 collapses the two squared terms into b^2 .

Step 5. Rearranging gives the working form $\cos A = \frac{b^2 + c^2 - a^2}{2bc}$. *Reason: this is the direction you almost always need — sides in, angle out.*

Derivation: the projection formulae

Step 1. Take the three cosine-rule statements for $\cos B$ and $\cos C$ and substitute into $b \cos C + c \cos B$:

$$b \cos C + c \cos B = b \cdot \frac{a^2 + b^2 - c^2}{2ab} + c \cdot \frac{a^2 + c^2 - b^2}{2ac}.$$

Step 2. Cancel b in the first term and c in the second:

$$= \frac{a^2 + b^2 - c^2}{2a} + \frac{a^2 + c^2 - b^2}{2a}.$$

Step 3. Add the numerators; b^2 and c^2 cancel in pairs:

$$= \frac{2a^2}{2a} = a.$$

Reason: an entirely algebraic proof, valid for acute and obtuse triangles alike — no diagram cases needed. The geometric reading is that the foot of the altitude from A divides BC into segments $c \cos B$ and $b \cos C$.

Derivation: the half-angle formulae and Heron's formula

Step 1. Start from $\sin^2 \frac{A}{2} = \frac{1 - \cos A}{2}$ (Subtopic 07 §3) and substitute the cosine rule:

$$\sin^2 \frac{A}{2} = \frac{1}{2} \left(1 - \frac{b^2 + c^2 - a^2}{2bc} \right) = \frac{2bc - b^2 - c^2 + a^2}{4bc}.$$

Step 2. Recognise the numerator as a difference of squares:

$$2bc - b^2 - c^2 + a^2 = a^2 - (b - c)^2 = (a - b + c)(a + b - c).$$

Reason: $b^2 - 2bc + c^2 = (b - c)^2$; the rest is $X^2 - Y^2 = (X - Y)(X + Y)$.

Step 3. Convert to semi-perimeter form. Since $a + b + c = 2s$: $a - b + c = 2s - 2b = 2(s - b)$ and $a + b - c = 2s - 2c = 2(s - c)$. Therefore

$$\sin^2 \frac{A}{2} = \frac{4(s-b)(s-c)}{4bc} = \frac{(s-b)(s-c)}{bc} \implies \sin \frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{bc}}.$$

Reason: the positive root is taken because $0 < \frac{A}{2} < \frac{\pi}{2}$ in every triangle.

Step 4. Identically, from $\cos^2 \frac{A}{2} = \frac{1+\cos A}{2}$, the numerator becomes $(b+c)^2 - a^2 = (b+c-a)(b+c+a) = 2(s-a) \cdot 2s$, giving

$$\cos \frac{A}{2} = \sqrt{\frac{s(s-a)}{bc}}, \quad \text{and dividing,} \quad \tan \frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{s(s-a)}}.$$

Step 5 — Heron falls out immediately. Take $\Delta = \frac{1}{2}bc \sin A$ and expand $\sin A = 2 \sin \frac{A}{2} \cos \frac{A}{2}$:

$$\Delta = \frac{1}{2}bc \cdot 2 \sqrt{\frac{(s-b)(s-c)}{bc}} \sqrt{\frac{s(s-a)}{bc}} = bc \cdot \frac{\sqrt{s(s-a)(s-b)(s-c)}}{bc}$$

$$\Delta = \sqrt{s(s-a)(s-b)(s-c)}$$

Reason: the two radicals share the denominator bc , which cancels the leading bc exactly. Heron's formula is therefore not an independent result — it is the half-angle formulae in disguise.

Procedure A: the triangle is fully determined — what to compute first

1. **Count what you are given and identify the case.** SSS (three sides), SAS (two sides and the included angle), ASA/AAS (two angles and a side), or SSA (two sides and a non-included angle). *Why: the case dictates the tool, and choosing wrongly costs minutes.*
2. **SSS → cosine rule.** Find the **largest** angle first, using $\cos = \frac{\text{other two squared} - \text{this one squared}}{2 \times \text{product}}$. *Why: only the largest angle can be obtuse, and the cosine rule reports that correctly via a negative value. Once it is known, every remaining angle is acute and the sine rule is safe.*
3. **SAS → cosine rule for the third side**, then proceed as SSS. *Why: SAS is the only case where the cosine rule directly delivers a length.*
4. **ASA/AAS → third angle by $A + B + C = \pi$, then sine rule** for the remaining sides. *Why: with all three angles and one side, the sine rule gives each unknown side in one step; no ambiguity can arise because the angles are already fixed.*
5. **SSA → treat as ambiguous until proved otherwise.** Solve for the angle, then check whether *both* the acute and the obtuse candidate leave a positive third angle. *Why: SSA is the one configuration that can admit two genuinely different triangles.*
6. **Finish with the area, and use it as your check.** Compute Δ two ways — once by $\frac{1}{2}ab \sin C$ and once by Heron. *Why: they use disjoint pieces of your work, so agreement is strong evidence that no arithmetic slipped.*

Procedure B: proving an identity in a triangle

1. **Look at what the target contains.** If it mixes sides and sines, substitute $a = 2R \sin A$ everywhere and watch $2R$ cancel. If it mixes sides and cosines, substitute the cosine rule instead. *Why: the goal is to get everything into one language — all sides, or all angles.*
2. **If you land in all-angles form, invoke $A+B+C = \pi$** and use the conditional identities of Subtopic 05 §3(J). *Why: an all-angles identity in a triangle is exactly a conditional identity, which the chapter has already built tools for.*
3. **If you land in all-sides form, convert to semi-perimeter form** using $b + c - a = 2(s - a)$ and its cyclic partners. *Why: symmetric expressions in a, b, c almost always simplify once written in $s - a, s - b, s - c$.*
4. **Test your final identity on the equilateral triangle** ($a = b = c, A = B = C = 60^\circ, R = \frac{a}{\sqrt{3}}$). *Why: a thirty-second numerical check that catches most algebraic slips. It cannot prove an identity, but it can refute one.*

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

In triangle ABC , $a = 3, b = 5, c = 7$. Find angle C , the area, and the circumradius R .

Step 1 — check the triangle exists. $3 + 5 = 8 > 7$, and the other two inequalities are obvious. Good.

Step 2 — the largest angle by the cosine rule. Side $c = 7$ is the largest, so angle C is the largest:

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab} = \frac{9 + 25 - 49}{2(3)(5)} = \frac{-15}{30} = -\frac{1}{2}.$$

The value is negative, so C is obtuse:

$$C = 120^\circ.$$

Step 3 — the area, two ways. Using two sides and the included angle:

$$\Delta = \frac{1}{2}ab \sin C = \frac{1}{2}(3)(5) \sin 120^\circ = \frac{15}{2} \cdot \frac{\sqrt{3}}{2} = \frac{15\sqrt{3}}{4} \approx 6.495.$$

Checking by Heron with $s = \frac{3+5+7}{2} = \frac{15}{2}$:

$$\Delta = \sqrt{\frac{15}{2} \cdot \frac{9}{2} \cdot \frac{5}{2} \cdot \frac{1}{2}} = \sqrt{\frac{675}{16}} = \frac{15\sqrt{3}}{4}. \quad \checkmark$$

Step 4 — the circumradius. From $\Delta = \frac{abc}{4R}$:

$$R = \frac{abc}{4\Delta} = \frac{3 \cdot 5 \cdot 7}{4 \cdot \frac{15\sqrt{3}}{4}} = \frac{105}{15\sqrt{3}} = \frac{7}{\sqrt{3}} = \frac{7\sqrt{3}}{3} \approx 4.041.$$

Cross-check with the sine rule: $\frac{c}{\sin C} = \frac{7}{\sqrt{3}/2} = \frac{14}{\sqrt{3}} = 2R$, so $R = \frac{7}{\sqrt{3}}$. ✓

$$\therefore C = 120^\circ, \quad \Delta = \frac{15\sqrt{3}}{4}, \quad R = \frac{7\sqrt{3}}{3}. \quad \blacksquare$$

Example 2 — [JEE Main Level]

In triangle ABC , $A = 30^\circ$, $B = 45^\circ$ and $a = 10$. Find b , c and R .

Step 1 — third angle. $C = 180^\circ - 30^\circ - 45^\circ = 105^\circ$.

Step 2 — the circumradius first. It is the cheapest quantity available and makes everything else a one-liner:

$$2R = \frac{a}{\sin A} = \frac{10}{\sin 30^\circ} = \frac{10}{1/2} = 20 \implies R = 10.$$

Step 3 — the sides from $x = 2R \sin X$.

$$b = 2R \sin B = 20 \sin 45^\circ = 20 \cdot \frac{\sqrt{2}}{2} = 10\sqrt{2} \approx 14.142.$$

For c we need $\sin 105^\circ$. By the compound-angle formula of Subtopic 05:

$$\sin 105^\circ = \sin(60^\circ + 45^\circ) = \frac{\sqrt{3}}{2} \cdot \frac{\sqrt{2}}{2} + \frac{1}{2} \cdot \frac{\sqrt{2}}{2} = \frac{\sqrt{6} + \sqrt{2}}{4}.$$

$$c = 2R \sin C = 20 \cdot \frac{\sqrt{6} + \sqrt{2}}{4} = 5(\sqrt{6} + \sqrt{2}) \approx 19.319.$$

Step 4 — verify by computing the area two ways.

$$\Delta = \frac{1}{2}ab \sin C = \frac{1}{2}(10)(14.142)(0.96593) \approx 68.30, \quad \Delta = \frac{1}{2}ac \sin B = \frac{1}{2}(10)(19.319)(0.70711) \approx 68.30. \quad \checkmark$$

$$\therefore b = 10\sqrt{2}, \quad c = 5(\sqrt{6} + \sqrt{2}), \quad R = 10. \quad \blacksquare$$

Note the technique: with all three angles and one side known, going through $2R$ once is faster than writing three separate sine-rule proportions.

Example 3 — [JEE Main — Speed Trap]

In triangle ABC , $a = 7$, $b = 8$, $c = 13$. Find angle C .

The trap. Under time pressure, students reach for whichever rule they have just used.

Suppose you have already found $\cos A = \frac{b^2 + c^2 - a^2}{2bc} = \frac{64 + 169 - 49}{2(8)(13)} = \frac{184}{208} = \frac{23}{26}$, so

$\sin A = \sqrt{1 - \frac{529}{676}} = \frac{\sqrt{147}}{26} = \frac{7\sqrt{3}}{26}$. Now apply the sine rule:

$$\sin C = \frac{c \sin A}{a} = \frac{13}{7} \cdot \frac{7\sqrt{3}}{26} = \frac{\sqrt{3}}{2}.$$

and the pen writes $C = 60^\circ$. **That is the seductive wrong answer, and it is wrong.**

Why it is wrong. $\sin C = \frac{\sqrt{3}}{2}$ is satisfied by $C = 60^\circ$ and by $C = 120^\circ$. The sine rule cannot tell them apart. Worse, the wrong branch is detectably absurd: $c = 13$ is the longest side, so C must be the largest angle. With $C = 60^\circ$ we would get $A \approx 27.8^\circ$ and $B \approx 92.2^\circ$, making B the largest angle even though $b = 8 < 13 = c$. A triangle cannot do that.

The correct work — use the cosine rule, which reads the sign.

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab} = \frac{49 + 64 - 169}{2(7)(8)} = \frac{-56}{112} = -\frac{1}{2} \implies C = 120^\circ.$$

Verification. With $C = 120^\circ$: $\Delta = \frac{1}{2}ab \sin C = \frac{1}{2}(7)(8)\frac{\sqrt{3}}{2} = 14\sqrt{3} \approx 24.249$. Heron with $s = 14$: $\Delta = \sqrt{14 \cdot 7 \cdot 6 \cdot 1} = \sqrt{588} = 14\sqrt{3}$. ✓ And $A \approx 27.80^\circ$, $B \approx 32.20^\circ$, $C = 120^\circ$ sum to 180° , with the largest angle opposite the largest side. ■

The rule to carry away: *never determine an angle of a triangle from sin alone when that angle could be obtuse.* Either use the cosine rule, or find the two smallest angles by the sine rule (they are guaranteed acute) and get the largest by subtraction from 180° .

Example 4 — [JEE Advanced Level]

In any triangle ABC , prove that $a \cos A + b \cos B + c \cos C = 4R \sin A \sin B \sin C$.

Step 1 — get everything into one language. The left side mixes sides and angles. Substitute $a = 2R \sin A$, $b = 2R \sin B$, $c = 2R \sin C$ from the sine rule:

$$\text{LHS} = 2R (\sin A \cos A + \sin B \cos B + \sin C \cos C).$$

Reason: this is Procedure B Step 1 — the standard opening move.

Step 2 — recognise the double-angle pattern. Since $2 \sin \theta \cos \theta = \sin 2\theta$ (Subtopic 05 §3(C)), each bracketed term is $\frac{1}{2} \sin 2\theta$:

$$\text{LHS} = 2R \cdot \frac{1}{2} (\sin 2A + \sin 2B + \sin 2C) = R (\sin 2A + \sin 2B + \sin 2C).$$

Step 3 — invoke the conditional identity. Because $A + B + C = \pi$, Subtopic 05 §3(J) gives

$$\sin 2A + \sin 2B + \sin 2C = 4 \sin A \sin B \sin C.$$

Reason: this identity is false for arbitrary angles and true only under the triangle condition — which is exactly the hypothesis we have.

Step 4 — conclude.

$$\text{LHS} = R \cdot 4 \sin A \sin B \sin C = 4R \sin A \sin B \sin C = \text{RHS}. \quad \blacksquare$$

Step 5 — sanity check on the equilateral triangle. Take $A = B = C = 60^\circ$ and $a = b = c$, so $R = \frac{a}{2 \sin 60^\circ} = \frac{a}{\sqrt{3}}$.

$$\text{LHS} = 3a \cos 60^\circ = \frac{3a}{2}, \quad \text{RHS} = 4 \cdot \frac{a}{\sqrt{3}} \left(\frac{\sqrt{3}}{2} \right)^3 = 4 \cdot \frac{a}{\sqrt{3}} \cdot \frac{3\sqrt{3}}{8} = \frac{3a}{2}. \quad \checkmark$$

What made this Advanced. Two separate toolkits had to meet: the sine rule to strip the sides out, and a *conditional* identity that is only available because $A + B + C = \pi$. A student who has learned triangle properties in isolation from Subtopic 05 cannot finish this problem, and a student who has learned the identities without the sine rule cannot start it.

Section 5: Practice Exercises

1. In triangle ABC , $a = 13$, $b = 14$, $c = 15$. Find $\cos A$, the area Δ , and the circumradius R .
2. In triangle ABC , $A = 45^\circ$, $B = 60^\circ$ and $a = 6$. Find b and c in exact surd form.
3. Prove the projection formula $a = b \cos C + c \cos B$.
4. In triangle ABC , $a = 5$, $b = 6$, $c = 7$. Find $\tan \frac{A}{2}$ and the area.
5. Prove that $\frac{\cos A}{a} + \frac{\cos B}{b} + \frac{\cos C}{c} = \frac{a^2 + b^2 + c^2}{2abc}$.
6. In triangle ABC , $a = 2$, $b = 3$ and $C = 60^\circ$. Find c and the area.
7. Prove that in any triangle, $a(\sin B - \sin C) + b(\sin C - \sin A) + c(\sin A - \sin B) = 0$.

Answers.

1. $\cos A = \frac{196 + 225 - 169}{2(14)(15)} = \frac{252}{420} = \frac{3}{5}$; $s = 21$, so $\Delta = \sqrt{21 \cdot 8 \cdot 7 \cdot 6} = \sqrt{7056} = 84$;
 $R = \frac{abc}{4\Delta} = \frac{2730}{336} = \frac{65}{8} = 8.125$.
2. $C = 75^\circ$ and $2R = \frac{6}{\sin 45^\circ} = 6\sqrt{2}$. Then $b = 6\sqrt{2} \sin 60^\circ = 3\sqrt{6} \approx 7.348$ and $c = 6\sqrt{2} \cdot \frac{\sqrt{6} + \sqrt{2}}{4} = 3(\sqrt{3} + 1) \approx 8.196$.
3. Substitute $\cos C = \frac{a^2 + b^2 - c^2}{2ab}$ and $\cos B = \frac{a^2 + c^2 - b^2}{2ac}$; the b and c cancel, the numerators add to $2a^2$, and the sum is $\frac{2a^2}{2a} = a$.
4. $s = 9$, so $s - a = 4$, $s - b = 3$, $s - c = 2$. $\tan \frac{A}{2} = \sqrt{\frac{3 \cdot 2}{9 \cdot 4}} = \sqrt{\frac{1}{6}} = \frac{1}{\sqrt{6}} \approx 0.408$;
 $\Delta = \sqrt{9 \cdot 4 \cdot 3 \cdot 2} = \sqrt{216} = 6\sqrt{6} \approx 14.697$.

5. Each term is $\frac{\cos A}{a} = \frac{b^2 + c^2 - a^2}{2abc}$ after multiplying numerator and denominator by a . Adding the three numerators, every square appears twice positively and once negatively, leaving $a^2 + b^2 + c^2$ over the common denominator $2abc$.
6. $c^2 = 4 + 9 - 2(2)(3) \cos 60^\circ = 13 - 6 = 7$, so $c = \sqrt{7} \approx 2.646$; $\Delta = \frac{1}{2}(2)(3) \sin 60^\circ = \frac{3\sqrt{3}}{2} \approx 2.598$.
7. Substitute $a = 2R \sin A$, $b = 2R \sin B$, $c = 2R \sin C$. Every product $\sin X \sin Y$ then appears exactly twice with opposite signs, so the whole expression is 0.

Section 6: Crack the MCQ

Q1. In triangle ABC , the common value of $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$ is (A) R (B) $2R$ (C) $\frac{R}{2}$ (D) $4R$

Answer: (B). The perpendicular from the circumcentre to BC gives $\frac{a}{2} = R \sin A$, so $\frac{a}{\sin A} = 2R$.

Distractor Analysis. - (A) remembers “circumradius” and stops there, forgetting that the half-chord, not the full chord, equals $R \sin A$. - (C) inverts the relation, writing $a = \frac{R}{2} \sin A$; a dimensional check kills it instantly, since a larger circle must give larger sides. - (D) imports the $4R$ from $\Delta = \frac{abc}{4R}$, where the 4 has an entirely different origin.

Q2. In triangle ABC , $a = 7$, $b = 8$, $c = 13$. Then C equals (A) 60° (B) 90° (C) 120° (D) 150°

Answer: (C). $\cos C = \frac{49 + 64 - 169}{112} = -\frac{1}{2}$.

Distractor Analysis. - (A) is the sine-rule trap of Example 3: $\sin C = \frac{\sqrt{3}}{2}$ has two solutions and the acute one was taken by reflex. It is also produced by dropping the minus sign while evaluating $-\frac{56}{112}$. - (B) assumes the largest side implies a right angle. Testing $49 + 64 = 113 \neq 169$ refutes it in one line. - (D) would need $\cos C = -\frac{\sqrt{3}}{2}$; the student read the table row for $\frac{\sqrt{3}}{2}$ instead of $\frac{1}{2}$.

Q3. The area of a triangle with sides 13, 14, 15 is (A) 84 (B) 105 (C) 91 (D) 168

Answer: (A). $s = 21$ and $\Delta = \sqrt{21 \cdot 8 \cdot 7 \cdot 6} = 84$.

Distractor Analysis. - (B) is $\frac{1}{2}(14)(15) = 105$ — treating two of the sides as perpendicular legs. Since $14^2 + 15^2 \neq 13^2$, the triangle is not right-angled. - (C) is $\frac{1}{2}(13)(14) = 91$, the same error with a different pair of sides. - (D) is 2×84 , from using the full perimeter 42 in place of the semi-perimeter $s = 21$ at exactly one position in Heron’s formula.

Q4. In triangle ABC with semi-perimeter s , $\tan \frac{A}{2}$ equals (A) $\sqrt{\frac{s(s-a)}{bc}}$ (B) $\sqrt{\frac{(s-b)(s-c)}{s(s-a)}}$
(C) $\sqrt{\frac{(s-b)(s-c)}{bc}}$ (D) $\sqrt{\frac{s(s-a)}{(s-b)(s-c)}}$

Answer: (B).

Distractor Analysis. - (A) is $\cos \frac{A}{2}$ — the student took the numerator of the cosine formula

but never divided by the sine formula. - **(C)** is $\sin \frac{A}{2}$, chosen by a student who remembered one formula from the family and assumed it was the tangent. - **(D)** is $\cot \frac{A}{2}$, the reciprocal. Test it on the equilateral triangle: there $\tan 30^\circ = \frac{1}{\sqrt{3}} < 1$, while (D) exceeds 1.

Section 7: Common Student Pitfalls and Pro-Tips

- **Pitfall 1 — Determining an obtuse angle with the sine rule.** $\sin C = \frac{\sqrt{3}}{2}$ admits 60° and 120° ; the sine rule cannot choose. *Correction:* use the cosine rule for any angle that might be obtuse, since only cos carries the sign that distinguishes them.
- **Pitfall 2 — Mismatching a side with the wrong angle.** The sine rule pairs a with A , meaning the side and the angle **opposite** it. Pairing a with B produces confident nonsense. *Correction:* label the diagram before writing a single equation.
- **Pitfall 3 — Using $\frac{1}{2}ab \sin C$ with a non-included angle.** The angle must be the one *between* the two sides named. $\frac{1}{2}ab \sin A$ is not the area. *Correction:* read the formula as “half, times the two sides that meet at the angle, times the sine of that angle.”
- **Pitfall 4 — Perimeter instead of semi-perimeter in Heron.** Using $a + b + c$ where s belongs multiplies the answer by a spurious factor. *Correction:* write $s = \frac{a+b+c}{2}$ as an explicit first line, every time.
- **Pitfall 5 — Sign error in the cosine rule numerator.** $\cos A = \frac{b^2+c^2-a^2}{2bc}$: the subtracted square is the one **opposite** the angle you want. Getting it wrong flips acute and obtuse. *Correction:* say the sentence “the odd one out is the opposite side” as you write it.
- **Pitfall 6 — Attaching a $\pm$ to the triangle half-angle formulae.** In Subtopic 07 the $\pm$ was essential; here it is wrong. *Correction:* in a triangle $0 < \frac{A}{2} < 90^\circ$, so all three half-angle functions are positive and the radical is unambiguous.
- **Pitfall 7 — Not checking that the triangle exists.** If the data give $|\cos A| > 1$, or if two sides fail to exceed the third, there is no triangle and every subsequent line is void. *Correction:* run the triangle inequality before any computation.

Pro-Tips.

- Given three sides, always solve for the **largest** angle first. It is the only one that can be obtuse, so once it is settled the sine rule is safe for everything else.
- Compute $2R = \frac{a}{\sin A}$ early. With one number in hand, every other side becomes a single multiplication $x = 2R \sin X$ instead of a fresh proportion.
- $R = \frac{abc}{4\Delta}$ is the fastest circumradius route when Heron has already given you Δ — no angles required at all.
- For any identity mixing sides and angles, substitute $a = 2R \sin A$ on sight. The $2R$ almost always cancels, and you are left with a conditional identity in A, B, C that Subtopic 05 §3(J) already handles.
- Verify every numerical triangle by computing the area twice, once with $\frac{1}{2}ab \sin C$ and once with Heron. The two routes share almost no arithmetic, so agreement is a genuine check rather than a repetition.
- Keep a single worked triangle in memory as a test bed: 13-14-15 has $s = 21$, $\Delta = 84$,

$R = \frac{65}{8}$, $\cos A = \frac{3}{5}$. Any formula you half-remember can be tested against it in seconds.

Section 8: Cheat Sheet Box

CHEAT SHEET — Properties of Triangles

Notation: $a = BC$, $b = CA$, $c = AB$; $A + B + C = \pi$; $s = \frac{a+b+c}{2}$; Δ = area; R = circumradius; r = inradius.

- **Sine rule:** $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$, i.e. $a = 2R \sin A$. *Cannot resolve acute vs obtuse.*
- **Cosine rule:** $a^2 = b^2 + c^2 - 2bc \cos A$, so $\cos A = \frac{b^2 + c^2 - a^2}{2bc}$. *Negative value $\Rightarrow$ obtuse. This is the rule that decides.*
- **Projection:** $a = b \cos C + c \cos B$ (and cyclic).
- **Area:** $\Delta = \frac{1}{2}ab \sin C = \sqrt{s(s-a)(s-b)(s-c)} = \frac{abc}{4R} = rs = 2R^2 \sin A \sin B \sin C$.
- **Circumradius:** $R = \frac{abc}{4\Delta} = \frac{a}{2 \sin A}$. **Inradius:** $r = \frac{\Delta}{s}$.
- **Half-angles (all positive — no $\pm$ in a triangle):** $\sin \frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{bc}}$, $\cos \frac{A}{2} = \sqrt{\frac{s(s-a)}{bc}}$, $\tan \frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{s(s-a)}} = \frac{r}{s-a}$.
- **Napier:** $\tan \frac{B-C}{2} = \frac{b-c}{b+c} \cot \frac{A}{2}$.
- **Case map:** SSS $\rightarrow$ cosine rule (largest angle first). SAS $\rightarrow$ cosine rule for the third side. ASA/AAS $\rightarrow$ third angle, then sine rule. SSA $\rightarrow$ check for two triangles.
- **Existence test:** each of $s-a$, $s-b$, $s-c$ must be positive; $|\cos A| > 1$ means no such triangle.
- **Golden rule:** sides in, angle out $\Rightarrow$ cosine rule. Opposite pair known $\Rightarrow$ sine rule. Always cross-check the area two ways.

Addendum: Exercises for Product-to-Sum and Triple-Angle

The product-to-sum formulas (Subtopic 05 §3(G), derived in §4(f)) and the triple-angle formulas (§3(F), derived in §4(e)) are already established. This addendum does not restate or re-derive them. It supplies what was missing: worked practice in *using* them.

Keep the two trigger conditions in view as you read:

- **Reach for product-to-sum** when you face a **product** of two sines/cosines whose arguments are unequal, and the target is a sum or difference — or when a chain of products needs terms to cancel against each other.
- **Reach for triple-angle** when an expression contains a **cube** of a sine or cosine, or when three angles in a problem sit in the pattern θ , $60^\circ - \theta$, $60^\circ + \theta$ — that trio is a triple-angle formula in disguise.

Example C1 — [JEE Main Level] (Product-to-Sum)

Prove that $\cos 2x \cos \frac{x}{2} - \cos 3x \cos \frac{9x}{2} = \sin 5x \sin \frac{5x}{2}$.

Step 1 — double both products so the formula applies cleanly. Using $2 \cos A \cos B = \cos(A + B) + \cos(A - B)$ on each term:

$$2 \cos 2x \cos \frac{x}{2} = \cos \left(2x + \frac{x}{2} \right) + \cos \left(2x - \frac{x}{2} \right) = \cos \frac{5x}{2} + \cos \frac{3x}{2}$$

$$2 \cos 3x \cos \frac{9x}{2} = \cos \left(3x + \frac{9x}{2} \right) + \cos \left(3x - \frac{9x}{2} \right) = \cos \frac{15x}{2} + \cos \frac{3x}{2}$$

Reason for the second line: $\cos \left(-\frac{3x}{2} \right) = \cos \frac{3x}{2}$, since cosine is even (Subtopic 03).

Step 2 — subtract. Doubling the whole left-hand side:

$$2 \text{ LHS} = \left(\cos \frac{5x}{2} + \cos \frac{3x}{2} \right) - \left(\cos \frac{15x}{2} + \cos \frac{3x}{2} \right) = \cos \frac{5x}{2} - \cos \frac{15x}{2}.$$

The $\cos \frac{3x}{2}$ terms cancel exactly — that cancellation is the entire reason the problem was constructed with these particular arguments.

Step 3 — convert the surviving difference back to a product. By $\cos C - \cos D = -2 \sin \frac{C+D}{2} \sin \frac{C-D}{2}$ with $C = \frac{5x}{2}$, $D = \frac{15x}{2}$:

$$C + D = \frac{5x}{2} + \frac{15x}{2} = 10x \Rightarrow \frac{C + D}{2} = 5x, \quad C - D = \frac{5x}{2} - \frac{15x}{2} = -5x \Rightarrow \frac{C - D}{2} = -\frac{5x}{2},$$

$$2 \text{ LHS} = -2 \sin 5x \sin \left(-\frac{5x}{2} \right) = 2 \sin 5x \sin \frac{5x}{2},$$

using that sine is odd.

Step 4 — divide by 2.

$$\text{LHS} = \sin 5x \sin \frac{5x}{2} = \text{RHS.} \quad \blacksquare$$

Numerical spot-check at $x = 1$ radian: LHS = $(-0.41615)(0.87758) - (-0.98999)(-0.21080) = -0.57389$; RHS = $(-0.95892)(0.59847) = -0.57389$. ✓

The takeaway: you cannot factor $\cos 2x \cos \frac{x}{2}$ against $\cos 3x \cos \frac{9x}{2}$ directly — the arguments share nothing. Product-to-sum flattens both into cosines of single angles, at which point the common term becomes visible and cancels. Flatten first, look for cancellation second.

Example C2 — [JEE Advanced Level] (Product-to-Sum)

Evaluate $\sin 10^\circ \sin 30^\circ \sin 50^\circ \sin 70^\circ$.

Step 1 — set aside the known factor. $\sin 30^\circ = \frac{1}{2}$, so the problem reduces to evaluating $P = \sin 10^\circ \sin 50^\circ \sin 70^\circ$ and halving it.

Step 2 — collapse the pair whose angles are furthest apart. Using $2 \sin A \sin B = \cos(A - B) - \cos(A + B)$ on $\sin 50^\circ \sin 70^\circ$:

$$2 \sin 50^\circ \sin 70^\circ = \cos 20^\circ - \cos 120^\circ = \cos 20^\circ + \frac{1}{2},$$

since $\cos 120^\circ = -\frac{1}{2}$. Therefore

$$P = \sin 10^\circ \cdot \frac{\cos 20^\circ + \frac{1}{2}}{2} = \frac{\sin 10^\circ \cos 20^\circ}{2} + \frac{\sin 10^\circ}{4}.$$

Step 3 — apply product-to-sum a second time to the surviving product, using $2 \sin A \cos B = \sin(A + B) + \sin(A - B)$:

$$2 \sin 10^\circ \cos 20^\circ = \sin 30^\circ + \sin(-10^\circ) = \frac{1}{2} - \sin 10^\circ,$$

$$\text{so } \sin 10^\circ \cos 20^\circ = \frac{1}{4} - \frac{\sin 10^\circ}{2}.$$

Step 4 — substitute and watch the awkward term die.

$$P = \frac{1}{2} \left(\frac{1}{4} - \frac{\sin 10^\circ}{2} \right) + \frac{\sin 10^\circ}{4} = \frac{1}{8} - \frac{\sin 10^\circ}{4} + \frac{\sin 10^\circ}{4} = \frac{1}{8}.$$

The $\sin 10^\circ$ terms cancel completely — which is why the answer is a clean rational number even though no individual factor is.

Step 5 — restore the set-aside factor.

$$\sin 10^\circ \sin 30^\circ \sin 50^\circ \sin 70^\circ = \frac{1}{2} \cdot \frac{1}{8} = \frac{1}{16}. \quad \blacksquare$$

Numerical check: $(0.17365)(0.5)(0.76604)(0.93969) = 0.0625 = \frac{1}{16}$. ✓

Why this is Advanced. Two things had to be chosen well. First, *which pair to collapse*: pairing 50° with 70° produces $\cos 120^\circ$, a standard value, whereas pairing 10° with 50° produces $\cos 60^\circ$ and $\cos 40^\circ$ — one standard value and one dead end. Second, the willingness to carry an irrational term ($\sin 10^\circ$) through two steps on faith that it will cancel. Students abandon this problem at Step 2 because the expression looks worse than when they started. It is supposed to.

Example C3 — [JEE Main Level] (Triple-Angle)

Prove that $\sin A \sin(60^\circ - A) \sin(60^\circ + A) = \frac{1}{4} \sin 3A$, **and hence evaluate** $\sin 20^\circ \sin 40^\circ \sin 80^\circ$.

Step 1 — collapse the symmetric pair. By the standard product of compound angles (Subtopic 05 §3(B)), $\sin(X + Y) \sin(X - Y) = \sin^2 X - \sin^2 Y$. With $X = 60^\circ$ and $Y = A$:

$$\sin(60^\circ - A) \sin(60^\circ + A) = \sin^2 60^\circ - \sin^2 A = \frac{3}{4} - \sin^2 A.$$

Step 2 — multiply by the remaining factor.

$$\sin A \left(\frac{3}{4} - \sin^2 A \right) = \frac{3 \sin A - 4 \sin^3 A}{4}.$$

Step 3 — recognise the triple-angle formula. The numerator is exactly $\sin 3A$ (Subtopic 05 §3(F)):

$$\sin A \sin(60^\circ - A) \sin(60^\circ + A) = \frac{\sin 3A}{4}. \quad \blacksquare$$

Step 4 — apply it. The trio $20^\circ, 40^\circ, 80^\circ$ fits the pattern with $A = 20^\circ$, because $40^\circ = 60^\circ - 20^\circ$ and $80^\circ = 60^\circ + 20^\circ$:

$$\sin 20^\circ \sin 40^\circ \sin 80^\circ = \frac{\sin 60^\circ}{4} = \frac{\sqrt{3}/2}{4} = \frac{\sqrt{3}}{8}.$$

Numerical check: $(0.34202)(0.64279)(0.98481) = 0.21651$, and $\frac{\sqrt{3}}{8} = 0.21651$. ✓

Recognising the trigger. The whole problem is spotting that $\{20^\circ, 40^\circ, 80^\circ\}$ is really $\{A, 60^\circ - A, 60^\circ + A\}$. Whenever a product of three sines or three cosines has angles that are symmetric about 60° once you subtract, this identity applies and turns the product into a single value in one step.

Example C4 — [JEE Advanced Level] (Triple-Angle)

Prove that $\cos 3x \cos^3 x + \sin 3x \sin^3 x = \cos^3 2x$.

Step 1 — invert the triple-angle formulas to express the cubes. From $\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta$ and $\sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta$:

$$\cos^3 \theta = \frac{\cos 3\theta + 3 \cos \theta}{4}, \quad \sin^3 \theta = \frac{3 \sin \theta - \sin 3\theta}{4}.$$

Reason: this “power-reduction for cubes” reading is the most useful form of the triple-angle formulas, and it is the move the problem is testing.

Step 2 — substitute both into the left-hand side.

$$\begin{aligned} \text{LHS} &= \cos 3x \cdot \frac{\cos 3x + 3 \cos x}{4} + \sin 3x \cdot \frac{3 \sin x - \sin 3x}{4} \\ &= \frac{\cos^2 3x + 3 \cos 3x \cos x + 3 \sin 3x \sin x - \sin^2 3x}{4}. \end{aligned}$$

Step 3 — group the terms in two natural pairs.

- $\cos^2 3x - \sin^2 3x = \cos 6x$, by the double-angle formula applied to the angle $3x$.
- $3(\cos 3x \cos x + \sin 3x \sin x) = 3\cos(3x - x) = 3\cos 2x$, by the master key $\cos(A - B) = \cos A \cos B + \sin A \sin B$.

Hence

$$\text{LHS} = \frac{\cos 6x + 3 \cos 2x}{4}.$$

Step 4 — recognise the right-hand side. Apply the cube formula of Step 1 with $\theta = 2x$, so that $3\theta = 6x$:

$$\cos^3 2x = \frac{\cos 6x + 3 \cos 2x}{4}.$$

Step 5 — conclude. The two sides are identical expressions:

$$\cos 3x \cos^3 x + \sin 3x \sin^3 x = \cos^3 2x. \quad \blacksquare$$

Numerical check at $x = 0.3$ radians: LHS = $(0.62161)(0.87211) + (0.78333)(0.025808) = 0.56233$; RHS = $(0.82534)^3 = 0.56221$, agreeing to rounding. ✓

Why this is Advanced. The identity is used in *both* directions in the same proof — forwards in Step 4 to build $\cos^3 2x$, and backwards in Step 1 to dissolve $\cos^3 x$ and $\sin^3 x$. Most students know the triple-angle formulas only left-to-right, as a way of expanding $\cos 3A$. The genuinely powerful reading is right-to-left: **a cube of a sine or cosine is always a disguised combination of the single and triple angles.** That is the reading that makes this proof three lines instead of thirty.

Practice Set

1. Express $2 \cos 5x \cos 3x$ as a sum of cosines.
2. Evaluate $2 \sin 75^\circ \cos 15^\circ$ exactly.
3. Prove that $\cos 4x \cos x - \cos 5x \cos 2x = \sin 6x \sin x$.
4. Prove that $\cos A \cos(60^\circ - A) \cos(60^\circ + A) = \frac{1}{4} \cos 3A$, and hence evaluate $\cos 20^\circ \cos 40^\circ \cos 80^\circ$.
5. If $\sin \theta = \frac{1}{3}$, find the exact value of $\sin 3\theta$.
6. Find the general solution of $\cos 3x = \cos x$.

Answers.

1. $\cos 8x + \cos 2x$.
2. $2 \sin 75^\circ \cos 15^\circ = \sin 90^\circ + \sin 60^\circ = 1 + \frac{\sqrt{3}}{2} = \frac{2 + \sqrt{3}}{2} \approx 1.86603$.
3. Double both products: $2 \cos 4x \cos x = \cos 5x + \cos 3x$ and $2 \cos 5x \cos 2x = \cos 7x + \cos 3x$. Subtracting, the $\cos 3x$ terms cancel and $2 \text{ LHS} = \cos 5x - \cos 7x = -2 \sin 6x \sin(-x) = 2 \sin 6x \sin x$, so LHS = $\sin 6x \sin x$.

4. $\cos(60^\circ - A) \cos(60^\circ + A) = \cos^2 60^\circ - \sin^2 A = \frac{1}{4} - \sin^2 A$; multiplying by $\cos A$ and using $\sin^2 A = 1 - \cos^2 A$ gives $\frac{4 \cos^3 A - 3 \cos A}{4} = \frac{\cos 3A}{4}$. With $A = 20^\circ$: $\cos 20^\circ \cos 40^\circ \cos 80^\circ = \frac{\cos 60^\circ}{4} = \frac{1}{8}$, matching Subtopic 05 Example 7 by a completely different route.
5. $\sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta = 3 \left(\frac{1}{3} \right) - 4 \left(\frac{1}{27} \right) = 1 - \frac{4}{27} = \frac{23}{27} \approx 0.85185$.
6. By triple-angle, $4 \cos^3 x - 3 \cos x = \cos x \Rightarrow 4 \cos x (\cos^2 x - 1) = 0 \Rightarrow \cos x = 0$ or $\sin x = 0$. Combining $x = (2n + 1) \frac{\pi}{2}$ with $x = n\pi$ gives the single family

$$x = \frac{n\pi}{2}, \quad n \in \mathbb{Z}.$$

(The same answer follows from $\cos 3x - \cos x = -2 \sin 2x \sin x = 0$ — a useful check that the triple-angle route and the sum-to-product route agree.)

Chapter 3: Trigonometric Functions — Round 2

Addendum

This addendum continues the Round 1 supplement. Round 1 added Subtopic 07 (sub-multiple and half-angle family, including $\sin 18^\circ$ and $\cos 36^\circ$), Subtopic 08 (properties of triangles), and an addendum of product-to-sum and triple-angle exercises labelled C1-C4. Numbering therefore resumes at **Subtopic 09**, and the closing addendum examples are labelled **D1-D4**.

Everything below is material that the typeset chapter and the Round 1 supplement do not cover, and that either NCERT Class 11 Chapter 3 itself demands or the standard JEE syllabus assumes. Cross-references to “Subtopic 05 §3(G)” and similar point back into the existing chapter and supplement.

Subtopic 09 — Trigonometric Inequalities

Exam Relevance: CBSE Boards touch this only lightly — a 1-2 mark “find the values of x in $[0, 2\pi]$ for which $\sin x \geq \frac{1}{2}$ ” is about the ceiling, and Subtopic 04 already showed one such reading-off-the-graph example. JEE Main, by contrast, asks for the complete general solution set of an inequality at least once most years, usually after a disguising step (a quadratic in $\sin x$, or an $a \sin x + b \cos x$ combination). JEE Advanced pushes into cubic and higher polynomial inequalities in $\tan x$ solved by a sign scheme, into inequalities whose bound is exactly the amplitude so that the “inequality” collapses to an equation, and into counting solutions in a stated interval. (NEET does not test mathematics.) An inequality is not an equation with a different symbol: an equation returns isolated points, an inequality returns *intervals*, and the machinery for generating those intervals is what this subtopic installs.

Section 1: Concept Intuition

The Mumbai monsoon warning. The weather office does not tell you the exact instant the rainfall crosses 100 mm. It tells you the *window*: “heavy rain between 2 p.m. and 7 p.m., every day this week.” That is an inequality answer — a repeating band of time, not a list of moments. Equations in Subtopic 06 gave you moments; inequalities give you bands, and the bands repeat with exactly the same period as the function.

Every trigonometric inequality is solved by the same two-step picture: find the boundary, then decide which side of it you want. The boundary is where equality holds, and you already know how to find that — it is a Subtopic 06 equation. What is new is only the second step: sweeping through one full period and asking, on each stretch between con-

secutive boundary points, whether the inequality is true or false. Once you have the answer over one period, periodicity hands you every other period for free.

Think about it this way. Draw $y = \sin x$ over $[0, 2\pi]$ and draw the horizontal line $y = \frac{1}{2}$. The two curves cross at $x = \frac{\pi}{6}$ and $x = \frac{5\pi}{6}$. Between those crossings the wave sits above the line; outside them it sits below. So $\sin x > \frac{1}{2}$ on $(\frac{\pi}{6}, \frac{5\pi}{6})$, and the general answer is that same open interval slid along by every multiple of 2π . No formula was memorised; the picture was read.

Figure brief: The curve $y = \sin x$ over $[-2\pi, 2\pi]$ with the horizontal cut line $y = \frac{1}{2}$ drawn across it. The arcs lying above the line are thickened and the corresponding x -intervals shaded on the axis, labelled $(2n\pi + \frac{\pi}{6}, 2n\pi + \frac{5\pi}{6})$ for $n = -1, 0, 1$, showing that the shaded bands repeat with period 2π and are *not* evenly spaced about the crossings.

The asymmetry that trips everyone. For sine and cosine the band is not centred where you expect. The set $\cos x > \frac{1}{2}$ is symmetric about $x = 0$, giving $(-\frac{\pi}{3}, \frac{\pi}{3})$ — cosine is even, so its bands straddle the even multiples of π . The set $\sin x > \frac{1}{2}$ is symmetric about $x = \frac{\pi}{2}$, not about $x = \frac{\pi}{6}$. **Cosine inequalities are naturally written as $2n\pi \pm \text{something}$; sine inequalities are not, and forcing them into that shape is the commonest structural error in the topic.**

Second idea — the sign scheme is the same tool you use in algebra. An inequality such as $\tan^3 x + 3 > 3 \tan x + \tan^2 x$ has nothing trigonometric about its hard part. Put $y = \tan x$, factor the cubic, mark the roots on a number line, read off where the product is positive, and only *then* translate each y -band back into an x -band. Trigonometry enters at the start (the substitution) and at the end (the translation); the middle is Chapter 6 algebra. Keeping those three stages separate is the whole technique.

Where the picture breaks down. For $\tan$ and $\cot$ the “band” can run into a vertical asymptote, and the endpoint is then *never* included even when the inequality is non-strict — $\tan x > 1$ gives $(n\pi + \frac{\pi}{4}, n\pi + \frac{\pi}{2})$ with the right endpoint permanently open, because $\tan$ is undefined there. And for a combination such as $a \cos x + b \sin x$, if the bound you are asked to beat equals $\sqrt{a^2 + b^2}$ exactly, the “inequality” has no interior at all: the solution set is the finite family of points where the maximum is attained. Both of these are examined deliberately.

Section 2: Key Formulas and Definitions

Trigonometric inequality. An inequality in which the unknown appears inside a trigonometric function. Its solution set is a union of intervals, described in general form using an integer $n \in \mathbb{Z}$.

Throughout, $n \in \mathbb{Z}$. In each master result the reference angle is chosen from the *principal range* of the relevant inverse function, which is what makes the answer unique.

(A) The four master inequalities for sine and cosine. Let $-1 < k < 1$.

Write $k = \sin \alpha$ with $\alpha \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ (that is, $\alpha = \sin^{-1} k$). Then

$$\sin x > k \iff x \in (2n\pi + \alpha, 2n\pi + \pi - \alpha)$$

$$\sin x < k \iff x \in (2n\pi + \pi - \alpha, 2n\pi + 2\pi + \alpha)$$

Write $k = \cos \beta$ with $\beta \in [0, \pi]$ (that is, $\beta = \cos^{-1} k$). Then

$$\cos x > k \iff x \in (2n\pi - \beta, 2n\pi + \beta)$$

$$\cos x < k \iff x \in (2n\pi + \beta, 2n\pi + 2\pi - \beta)$$

- For $\geq$ and $\leq$, close the intervals; the endpoints are genuine solutions.
- The sine bands have length $\pi - 2\alpha$ and $\pi + 2\alpha$ respectively; the two lengths add to 2π , which is the arithmetic check that you have not mis-copied a bound.
- If $k \geq 1$ then $\sin x > k$ and $\cos x > k$ are impossible; if $k < -1$ they are satisfied by every real x . Test the boundary cases before applying any formula.

(B) Tangent and cotangent. Let $k \in \mathbb{R}$, and write $k = \tan \gamma$ with $\gamma \in (-\frac{\pi}{2}, \frac{\pi}{2})$.

$$\tan x > k \iff x \in (n\pi + \gamma, n\pi + \frac{\pi}{2}), \quad \tan x < k \iff x \in (n\pi - \frac{\pi}{2}, n\pi + \gamma)$$

- Period π , not 2π — the single most frequent slip in this family.
- The endpoint $n\pi \pm \frac{\pi}{2}$ is always excluded: $\tan$ is undefined there.
- For $\cot$, use $\cot x > k \iff x \in (n\pi, n\pi + \delta)$ where $\delta = \cot^{-1} k \in (0, \pi)$, and reverse for $<$.

(C) Absolute-value forms.

$$|\sin x| \geq \frac{1}{2} \iff x \in [n\pi + \frac{\pi}{6}, n\pi + \frac{5\pi}{6}], \quad |\tan x| < 1 \iff x \in (n\pi - \frac{\pi}{4}, n\pi + \frac{\pi}{4})$$

- Both have period π , because $|\sin|$ and $|\tan|$ do. Solving them with period 2π gives a correct but redundant answer; solving them with the wrong reference angle does not.

(D) Reduction targets. Every examinable inequality is pushed into one of three shapes.

Shape of the given inequality	Reduce by	Land on
Quadratic/cubic in one function, e.g. $4\sin^2 x - 8\sin x + 3 < 0$	substitute $t = \sin x$, factor, sign scheme	$\sin x > k$ or $\sin x < k$
Mixed $\sin^2$, $\sin x \cos x$, $\cos^2$	divide by $\cos^2 x$ (noting $\cos x \neq 0$), or use double angle	$\tan x$ band, or $\cos 2x/\sin 2x$ band
$a \cos x + b \sin x \geq c$	amplitude form $R \cos(x - \alpha)$, $R = \sqrt{a^2 + b^2}$	$\cos(x - \alpha) \geq c/R$

- In the third row, if $|c| > R$ the inequality is either impossible or universally true; if $|c| = R$ the solution set degenerates to isolated points. Check $|c|$ against R first.

(E) The degenerate case (examined, and usually missed). For all real x , $|a \cos x + b \sin x| \leq \sqrt{a^2 + b^2}$. Hence

$$|a \cos x + b \sin x| \geq \sqrt{a^2 + b^2} \iff a \cos x + b \sin x = \pm \sqrt{a^2 + b^2},$$

an equation, whose solution is the discrete family $x = n\pi + \alpha$ where $\tan \alpha = b/a$ is chosen so that $a = R \cos \alpha$, $b = R \sin \alpha$.

Section 3: CBSE Board Derivations and Key Procedures

Derivation: the general solution set of $\sin x > k$

Step 1. Let $\alpha = \sin^{-1} k \in [-\frac{\pi}{2}, \frac{\pi}{2}]$, so that $\sin \alpha = k$. *Reason:* fixing α in the principal range makes the reference angle unique, which is what lets a single formula describe the whole set.

Step 2. Work on one full period, and choose the period $[\alpha - \frac{\pi}{2}, \alpha + \frac{3\pi}{2})$ rather than $[0, 2\pi)$. *Reason:* any interval of length 2π is a period, and this one is chosen so that the solution band sits in its interior instead of being split across the two ends — the choice that removes all the case work.

Step 3. On that interval $\sin x = k$ exactly twice: once at $x = \alpha$ and once at $x = \pi - \alpha$. *Reason:* these are the two principal solutions of $\sin x = \sin \alpha$ (Subtopic 06 §3(B)), and no third solution can lie in a single period.

Step 4. Between the two crossings the sine curve rises to its maximum $+1$ at $x = \frac{\pi}{2}$, which exceeds k ; outside them it falls to its minimum -1 , which does not. *Reason:* sine is continuous, so its sign relative to the line $y = k$ can change only at a crossing; testing one interior point per stretch settles the whole stretch.

Step 5. Therefore, on this period, $\sin x > k \iff \alpha < x < \pi - \alpha$. Adding $2n\pi$ gives

$$x \in (2n\pi + \alpha, 2n\pi + \pi - \alpha), \quad n \in \mathbb{Z}.$$

Reason: $\sin(x + 2n\pi) = \sin x$, so the truth of the inequality is unchanged by a whole number of turns.

Step 6 — the complement, for free. Since exactly one of $\sin x > k$, $\sin x = k$, $\sin x < k$ holds at each x , the set for $\sin x < k$ is the complement of the closed band, namely $(2n\pi + \pi - \alpha, 2n\pi + 2\pi + \alpha)$. *Reason:* removing a closed interval of length $\pi - 2\alpha$ from a period of length 2π leaves an open interval of length $\pi + 2\alpha$, which is exactly what is written.

Derivation: why the tangent family has period π and one open end

Step 1. Let $\gamma = \tan^{-1} k \in (-\frac{\pi}{2}, \frac{\pi}{2})$. *Reason:* every real k is the tangent of exactly one such γ , since $\tan$ is a bijection from that open interval onto $\mathbb{R}$.

Step 2. On the single branch $(-\frac{\pi}{2}, \frac{\pi}{2})$, $\tan$ is strictly increasing from $-\infty$ to $+\infty$. *Reason:* the branch behaviour recorded in Subtopic 04 §3(B).

Step 3. A strictly increasing function exceeds k exactly to the right of the point where it equals k . Hence on this branch $\tan x > k \iff \gamma < x < \frac{\pi}{2}$. *Reason:* monotonicity converts an inequality on values into an inequality on inputs, with no case work.

Step 4. $\tan$ has period π (Subtopic 04 §4(b)), so translate by $n\pi$: $x \in (n\pi + \gamma, n\pi + \frac{\pi}{2})$. *Reason:* the branch structure repeats every π , and each branch contributes exactly one band.

Step 5. The right endpoint is excluded permanently. *Reason:* $\tan$ is not defined at $n\pi + \frac{\pi}{2}$, so that point cannot belong to any solution set, whatever the inequality sign.

Procedure A: solving a trigonometric inequality in one function

1. **Isolate a single trigonometric function.** Use identities to remove every function but one — typically $\cos^2 x = 1 - \sin^2 x$ to make everything a polynomial in $\sin x$. *Why:* the master results of §2 apply to one function at a time.
2. **Substitute t for that function and record its range.** Write $t = \sin x$ with $t \in [-1, 1]$, or $t = \tan x$ with $t \in \mathbb{R}$. *Why:* the range is a genuine constraint and will discard whole factors later.
3. **Factor and run a sign scheme in t .** Mark the real roots on a number line, and mark the sign of the product on each stretch. *Why:* this is exactly the algebra of polynomial inequalities; nothing trigonometric is happening yet.
4. **Discard any t -band that lies outside the range from Step 2.** *Why:* a factor such as $2 \sin x - 3$ is negative for every real x , so it can be replaced by the constant sign — and dropped from the case work entirely.
5. **Translate each surviving t -band into an x -band** using §2(A) or §2(B). Translate the endpoints one at a time and keep strict/non-strict exactly as the algebra left it. *Why:* the endpoint convention is where marks are lost, and doing it mechanically avoids thinking about it under pressure.
6. **State the union over $n \in \mathbb{Z}$, and sanity-test one interior point.** Pick a convenient x inside a claimed band and evaluate the original inequality numerically. *Why:* a single test catches a flipped inequality sign, which is otherwise invisible.

Procedure B: solving $a \cos x + b \sin x \gtrless c$

1. Compute $R = \sqrt{a^2 + b^2}$ and compare $|c|$ with R . *Why:* this decides between “no solution”, “all real x ”, “isolated points”, and “genuine bands” before any work.
2. If $|c| > R$: the inequality is either impossible or holds for all x . Decide by testing which side the whole range $[-R, R]$ lies on. *Why:* the range of the left-hand side is $[-R, R]$ and nothing else needs checking.
3. If $|c| = R$: the inequality degenerates. For $\geq R$ the solution is the discrete family where the maximum is attained; for $> R$ there is no solution. *Why:* §2(E).
4. Otherwise write $a \cos x + b \sin x = R \cos(x - \alpha)$ with $\cos \alpha = a/R$, $\sin \alpha = b/R$, so the inequality becomes $\cos(x - \alpha) \gtrless c/R$. *Why:* Subtopic 05 §3(I) — and note that α is fixed by *both* of $\cos \alpha$ and $\sin \alpha$, not by $\tan \alpha$ alone, which only determines α up to π .
5. Apply §2(A) to $\cos(x - \alpha)$, then add α to every endpoint. *Why:* shifting the variable shifts the bands rigidly; it changes no lengths.

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

Solve $\cos x \geq -\frac{1}{2}$, first for $x \in [0, 2\pi]$ and then in general.

Step 1 — the boundary. $\cos x = -\frac{1}{2}$ has principal solutions $x = \frac{2\pi}{3}$ and $x = \frac{4\pi}{3}$ (Subtopic 06 §3(B), with reference angle $\beta = \cos^{-1}(-\frac{1}{2}) = \frac{2\pi}{3}$).

Step 2 — test one point in each stretch. The two boundary points cut $[0, 2\pi]$ into three stretches.

- On $(0, \frac{2\pi}{3})$: test $x = 0$, $\cos 0 = 1 \geq -\frac{1}{2}$. True.
- On $(\frac{2\pi}{3}, \frac{4\pi}{3})$: test $x = \pi$, $\cos \pi = -1 \geq -\frac{1}{2}$ is false.
- On $(\frac{4\pi}{3}, 2\pi)$: test $x = 2\pi$, $\cos 2\pi = 1$. True.

Step 3 — assemble. The inequality is non-strict, so both boundary points are included:

$$x \in [0, \frac{2\pi}{3}] \cup [\frac{4\pi}{3}, 2\pi].$$

Step 4 — the general solution. The two true stretches are the two halves of one band that straddles $x = 0$; re-centring gives $-\frac{2\pi}{3} \leq x \leq \frac{2\pi}{3}$, so

$$x \in [2n\pi - \frac{2\pi}{3}, 2n\pi + \frac{2\pi}{3}], \quad n \in \mathbb{Z}.$$

This matches §2(A) directly with $\beta = \frac{2\pi}{3}$.

$\therefore$ on $[0, 2\pi]$: $[0, \frac{2\pi}{3}] \cup [\frac{4\pi}{3}, 2\pi]$; in general $[2n\pi - \frac{2\pi}{3}, 2n\pi + \frac{2\pi}{3}]$.

Notice the shape of the answer: writing the *restricted* answer as two pieces and the *general* answer as one piece is not a contradiction. The band is one interval; the window $[0, 2\pi]$ happens to cut it in two.

Example 2 — [JEE Main Level]

Solve $4\sin^2 x - 8\sin x + 3 < 0$.

Step 1 — substitute. Put $t = \sin x$, with the range constraint $t \in [-1, 1]$. The inequality reads $4t^2 - 8t + 3 < 0$.

Step 2 — factor. $4t^2 - 8t + 3 = (2t - 1)(2t - 3)$, so the roots are $t = \frac{1}{2}$ and $t = \frac{3}{2}$, and the product must be negative:

$$(2t - 1)(2t - 3) < 0 \iff \frac{1}{2} < t < \frac{3}{2}.$$

Step 3 — apply the range constraint. Since $t = \sin x \leq 1 < \frac{3}{2}$, the factor $2t - 3$ is *negative for every real x* . So the product is negative precisely when $2t - 1 > 0$, i.e. $\sin x > \frac{1}{2}$. The upper bound $t < \frac{3}{2}$ is automatic and carries no information.

Step 4 — translate. With $\alpha = \sin^{-1} \frac{1}{2} = \frac{\pi}{6}$, §2(A) gives

$$x \in \left(2n\pi + \frac{\pi}{6}, 2n\pi + \frac{5\pi}{6}\right), \quad n \in \mathbb{Z}.$$

Step 5 — test. Take $x = \frac{\pi}{2}$ (inside the band): $4(1) - 8(1) + 3 = -1 < 0$. True. Take $x = 0$ (outside): $0 - 0 + 3 = 3 \not< 0$. Correctly excluded.

$$\therefore x \in \left(2n\pi + \frac{\pi}{6}, 2n\pi + \frac{5\pi}{6}\right).$$

The whole difficulty was Step 3. A student who solves $\frac{1}{2} < \sin x < \frac{3}{2}$ as a two-sided condition and then hunts for the x satisfying $\sin x < \frac{3}{2}$ has not made an error, but has wasted two minutes; a student who *reverses* the factor signs gets the complement and loses the question.

Example 3 — [JEE Main — Speed Trap]

Find all $x \in [0, 4\pi]$ satisfying $|\sqrt{3} \cos x - \sin x| \geq 2$.

The trap. The eye sees $\geq$, the hand reaches for a band, and the pen writes something of the form “ x lies between two angles”. Every such answer is wrong, and it is wrong for a reason no amount of care about endpoints will fix.

Why. Compute the amplitude first, as Procedure B insists. Here $a = \sqrt{3}$, $b = -1$, so

$$R = \sqrt{(\sqrt{3})^2 + (-1)^2} = \sqrt{4} = 2.$$

The maximum of $|\sqrt{3} \cos x - \sin x|$ over all real x is 2. The inequality therefore does not ask for the region where the expression is large; it asks for the instants where it is at its absolute extreme. **The inequality has collapsed into an equation.**

The correct work.

Step 1 — amplitude form. With $\sqrt{3} = 2 \cos \frac{\pi}{6}$ and $-1 = -2 \sin \frac{\pi}{6}$,

$$\sqrt{3} \cos x - \sin x = 2 \left(\cos \frac{\pi}{6} \cos x - \sin \frac{\pi}{6} \sin x \right) = 2 \cos \left(x + \frac{\pi}{6} \right).$$

Step 2 — impose the condition. $|2 \cos(x + \frac{\pi}{6})| \geq 2$ forces $|\cos(x + \frac{\pi}{6})| = 1$, hence $x + \frac{\pi}{6} = n\pi$, i.e.

$$x = n\pi - \frac{\pi}{6}, \quad n \in \mathbb{Z}.$$

Step 3 — restrict to $[0, 4\pi]$. Taking $n = 1, 2, 3, 4$:

$$x = \frac{5\pi}{6}, \quad \frac{11\pi}{6}, \quad \frac{17\pi}{6}, \quad \frac{23\pi}{6}.$$

($n = 0$ gives $-\frac{\pi}{6} < 0$; $n = 5$ gives $\frac{29\pi}{6} > 4\pi$.)

Step 4 — verify. At $x = \frac{5\pi}{6}$: $\sqrt{3} \cos \frac{5\pi}{6} - \sin \frac{5\pi}{6} = \sqrt{3} \left(-\frac{\sqrt{3}}{2}\right) - \frac{1}{2} = -\frac{3}{2} - \frac{1}{2} = -2$, and $|-2| = 2 \geq 2$. Correct. The four values alternate between -2 and $+2$.

$\therefore$ exactly four solutions: $x = \frac{5\pi}{6}, \frac{11\pi}{6}, \frac{17\pi}{6}, \frac{23\pi}{6}$.

Seductive wrong answer: any interval answer at all, and in particular " $\frac{5\pi}{6} \leq x \leq \frac{11\pi}{6}$ ", which comes from treating the two smallest solutions as the ends of a band. The band contains $x = \frac{4\pi}{3}$, where the expression equals $\sqrt{3}(-\frac{1}{2}) - (-\frac{\sqrt{3}}{2}) = 0$ — as far from ± 2 as it is possible to get. One substitution kills the wrong answer, which is why Procedure B puts the $|c|$ -versus- R comparison at Step 1 and not at the end.

Example 4 — [JEE Advanced Level]

Solve $\tan^3 x + 3 > 3 \tan x + \tan^2 x$.

Step 1 — collect and factor. Move everything to the left:

$$\tan^3 x - \tan^2 x - 3 \tan x + 3 > 0.$$

Group in pairs: $\tan^2 x (\tan x - 1) - 3 (\tan x - 1) > 0$, so

$$(\tan x - 1)(\tan^2 x - 3) > 0 \implies (\tan x + \sqrt{3})(\tan x - 1)(\tan x - \sqrt{3}) > 0.$$

Step 2 — sign scheme in $y = \tan x$. The three roots, in increasing order, are $-\sqrt{3} < 1 < \sqrt{3}$. A cubic with positive leading coefficient and three simple real roots is negative, positive, negative, positive across the four stretches:

y	$(-\infty, -\sqrt{3})$	$(-\sqrt{3}, 1)$	$(1, \sqrt{3})$	$(\sqrt{3}, \infty)$
sign of product	—	+	—	+

So the condition is $-\sqrt{3} < \tan x < 1$ **or** $\tan x > \sqrt{3}$. Note that y ranges over all of $\mathbb{R}$ here, so no band is discarded — the contrast with Example 2 is deliberate.

Step 3 — translate each band. Work inside one branch $(-\frac{\pi}{2}, \frac{\pi}{2})$, where $\tan$ increases strictly:

- $-\sqrt{3} < \tan x < 1 \iff -\frac{\pi}{3} < x < \frac{\pi}{4}.$
- $\tan x > \sqrt{3} \iff \frac{\pi}{3} < x < \frac{\pi}{2}.$

Step 4 — add the period. Both bands repeat every π :

$$x \in (n\pi - \frac{\pi}{3}, n\pi + \frac{\pi}{4}) \cup (n\pi + \frac{\pi}{3}, n\pi + \frac{\pi}{2}), \quad n \in \mathbb{Z}.$$

Step 5 — verify at three points. At $x = 0$: $\tan x = 0$, and $0 + 3 > 0 + 0$ is true, and 0 lies in the first band. At $x = \frac{\pi}{4}^+$, say $\tan x = 1.2$: $1.728 + 3 = 4.728$ versus $3.6 + 1.44 = 5.04$ — false, and indeed $\tan x = 1.2$ lies in the excluded stretch $(1, \sqrt{3})$. At $\tan x = 2$: $8 + 3 = 11 > 6 + 4 = 10$ — true, and $2 > \sqrt{3}$. All three agree.

What made this Advanced. Nothing in the trigonometry. The examiner is testing whether you will (a) recognise a factorable cubic rather than reaching for a substitution, (b) run a sign scheme correctly on three roots instead of two, and (c) translate *two* disjoint y -bands into x -bands with the correct period π and the correct open endpoint at $\frac{\pi}{2}$. Each of the three is elementary; doing all three without a slip is not.

Section 5: Practice Exercises

1. Solve $\sin x \leq -\frac{1}{2}$, giving the general solution.
2. Solve $2 \cos x \leq \sqrt{3}$ for $x \in [-\pi, \pi]$.
3. Solve $2 \sin^2 x - 3 \sin x + 1 \geq 0$ for $x \in [0, \pi]$. (Read the endpoint case carefully.)
4. Solve $\tan^2 x \leq 3$, giving the general solution.
5. Solve $\cos x - \sin x \geq 1$ for $x \in [0, 2\pi]$.
6. Solve $\sin x + \sqrt{3} \cos x > 1$, giving the general solution.
7. Solve $\sin^2 \theta > \cos^2 \theta$.
8. Solve $\sin 2x > \sqrt{2} \sin^2 x + (2 - \sqrt{2}) \cos^2 x$.

Answers.

1. $x \in [2n\pi + \frac{7\pi}{6}, 2n\pi + \frac{11\pi}{6}]$.
2. $x \in [-\pi, -\frac{\pi}{6}] \cup [\frac{\pi}{6}, \pi]$.
3. Factors as $(2 \sin x - 1)(\sin x - 1) \geq 0$, so $\sin x \leq \frac{1}{2}$ or $\sin x \geq 1$. On $[0, \pi]$ that is $[0, \frac{\pi}{6}] \cup \{\frac{\pi}{2}\} \cup [\frac{5\pi}{6}, \pi]$ — the isolated point $\frac{\pi}{2}$ is a genuine part of the answer, since $\sin x \geq 1$ forces $\sin x = 1$.
4. $x \in [n\pi - \frac{\pi}{3}, n\pi + \frac{\pi}{3}]$.
5. General solution $[2n\pi - \frac{\pi}{2}, 2n\pi]$; intersecting with $[0, 2\pi]$ gives $\{0\} \cup [\frac{3\pi}{2}, 2\pi]$.
6. $\sin x + \sqrt{3} \cos x = 2 \sin(x + \frac{\pi}{3}) > 1$, so $\sin(x + \frac{\pi}{3}) > \frac{1}{2}$, giving $x \in (2n\pi - \frac{\pi}{6}, 2n\pi + \frac{\pi}{2})$.
7. $\sin^2 \theta - \cos^2 \theta > 0 \Rightarrow \cos 2\theta < 0 \Rightarrow 2n\pi + \frac{\pi}{2} < 2\theta < 2n\pi + \frac{3\pi}{2}$, i.e. $\theta \in (n\pi + \frac{\pi}{4}, n\pi + \frac{3\pi}{4})$.
8. Divide throughout by $\cos^2 x$ (permitted, since $\cos x = 0$ makes the right side $\sqrt{2}$ and the left side 0, which fails): $2 \tan x > \sqrt{2} \tan^2 x + 2 - \sqrt{2}$, i.e. $(\tan x - 1)(\tan x - (\sqrt{2} - 1)) < 0$, so $\sqrt{2} - 1 < \tan x < 1$ and $x \in (n\pi + \frac{\pi}{8}, n\pi + \frac{\pi}{4})$ (using $\tan \frac{\pi}{8} = \sqrt{2} - 1$).

Section 6: Crack the MCQ

Q1. The solution set of $\cos x > \frac{1}{2}$ is (A) $(2n\pi - \frac{\pi}{3}, 2n\pi + \frac{\pi}{3})$ (B) $(2n\pi + \frac{\pi}{3}, 2n\pi + \frac{5\pi}{3})$ (C) $(n\pi - \frac{\pi}{3}, n\pi + \frac{\pi}{3})$ (D) $(2n\pi + \frac{\pi}{6}, 2n\pi + \frac{5\pi}{6})$

Answer: (A). With $\beta = \cos^{-1} \frac{1}{2} = \frac{\pi}{3}$, §2(A) gives $(2n\pi - \beta, 2n\pi + \beta)$.

Distractor Analysis. - (B) is the solution set of $\cos x < \frac{1}{2}$ — the inequality read backwards. Test $x = 0$: $\cos 0 = 1 > \frac{1}{2}$, and 0 is not in (B). One substitution refutes it. - (C) uses period π instead of 2π . Test $x = \pi$: it lies in (C) with $n = 1$, but $\cos \pi = -1$. Cosine's period is 2π ; only $\tan$ and $\cot$ get $n\pi$. - (D) is the sine answer for $\sin x > \frac{1}{2}$, imported wholesale. The tell is the asymmetry: cosine bands are symmetric about $2n\pi$, sine bands are not.

Q2. The number of $x \in [0, 4\pi]$ with $|\sqrt{3} \cos x - \sin x| \geq 2$ is (A) 0 (B) 2 (C) 4 (D) infinitely many

Answer: (C). The amplitude is exactly 2, so the condition forces equality and gives $x = n\pi - \frac{\pi}{6}$; four such values lie in $[0, 4\pi]$.

Distractor Analysis. - (A) comes from reasoning “the maximum is 2, so the strict inequality fails, so there is nothing” — correct for $>$, wrong for $\geq$. Read the sign. - (B) counts only one period, $[0, 2\pi]$, and forgets that the window is twice that long. - (D) is the answer a student writes after concluding the solution set is an interval. It is the Example 3 trap in multiple-choice clothing.

Q3. $4\sin^2 x - 8\sin x + 3 < 0$ is equivalent to (A) $\sin x < \frac{1}{2}$ (B) $\sin x > \frac{1}{2}$ (C) $\frac{1}{2} < \sin x < \frac{3}{2}$ only for some x (D) $\sin x > \frac{3}{2}$

Answer: (B). $(2\sin x - 1)(2\sin x - 3) < 0$, and $2\sin x - 3 < 0$ always, so the first factor must be positive.

Distractor Analysis. - (A) flips the sign of the surviving factor — the single most common slip once the always-negative factor has been spotted. - (C) is not false, but it is not an *equivalence* you can act on: the upper bound is vacuous. Choosing it signals that the range $\sin x \in [-1, 1]$ was never used. - (D) is impossible for every real x , since $|\sin x| \leq 1$. It is picked by students who take the outer stretch of the sign scheme without checking the range.

Q4. The general solution of $\sin x > \frac{1}{2}$ is (A) $x = n\pi + (-1)^n \frac{\pi}{6}$ (B) $x \in (2n\pi + \frac{\pi}{6}, 2n\pi + \frac{5\pi}{6})$ (C) $x \in (2n\pi + \frac{\pi}{6}, 2n\pi + \frac{\pi}{2})$ (D) $x \in (n\pi + \frac{\pi}{6}, n\pi + \frac{5\pi}{6})$

Answer: (B).

Distractor Analysis. - (A) is the general solution of the *equation* $\sin x = \frac{1}{2}$ (Subtopic 06 §3(B)). It is a set of isolated points, and an inequality can never have that as its full answer unless the bound is extreme. This is the structural error the subtopic exists to prevent. - (C) stops the band at the maximum $x = \frac{\pi}{2}$ instead of continuing to the second crossing. Test $x = \frac{2\pi}{3}$: $\sin x = \frac{\sqrt{3}}{2} > \frac{1}{2}$, but $\frac{2\pi}{3} \notin (C)$. - (D) uses period π . Test $x = \pi + \frac{\pi}{4}$: it lies in (D) with $n = 1$, but $\sin x = -\frac{1}{\sqrt{2}} < \frac{1}{2}$.

Section 7: Common Student Pitfalls and Pro-Tips

- **Pitfall 1 — Answering an inequality with an equation’s solution family.** Writing $n\pi + (-1)^n \alpha$ where an interval is required. Correction: the answer to an inequality is a union of intervals; if your answer has no intervals in it, either you have mis-read the question or the bound is extreme (§2(E)) — and you should say so explicitly.
- **Pitfall 2 — Using period 2π for tan or cot, or period π for sin or cos.** Correction: the band repeats with the *fundamental period of the function*, which is π for tan, cot, $|\sin|$, $|\cos|$, $\sin^2$, $\cos^2$, and 2π otherwise.
- **Pitfall 3 — Forcing a sine band into the $2n\pi \pm \alpha$ shape.** Only cosine bands are symmetric about $2n\pi$. Correction: for sine, the band runs from α to $\pi - \alpha$, and those two are not negatives of each other.
- **Pitfall 4 — Not checking $|c|$ against R in $a\cos x + b\sin x \gtrless c$.** Correction: compute $R = \sqrt{a^2 + b^2}$ as the first line, always. It decides whether you are about to write an interval, a point set, “no solution”, or “all real x ”.
- **Pitfall 5 — Keeping a factor that cannot change sign.** In $(2\sin x - 1)(2\sin x - 3) < 0$, the second factor is negative for every x . Correction: before running the sign scheme, delete every factor whose sign is constant on the range of the substituted variable, and flip the inequality once for each deleted negative factor.
- **Pitfall 6 — Losing an isolated solution at the boundary.** $\sin x \geq 1$ has solutions; it is just that they form a discrete set. Correction: when a factor gives $t \geq t_{\max}$ or $t \leq t_{\min}$, the answer is the equality case, not the empty set.
- **Pitfall 7 — Including an endpoint where the function is undefined.** $\tan x \geq k$ never includes $n\pi + \frac{\pi}{2}$. Correction: the domain exclusions of Subtopic 02 §3 override

the inequality sign, every time.

Pro-Tips.

- Sketch the graph and the cut line even when you intend to solve algebraically. Thirty seconds of picture tells you how many bands to expect per period, which is a complete check on the algebra.
- Test one interior point of every band you claim, and one point of every gap you exclude. Two substitutions cost fifteen seconds and catch a reversed inequality, which is otherwise undetectable.
- For anything homogeneous of degree two in $\sin x$ and $\cos x$, divide by $\cos^2 x$ and work in $\tan x$. It converts the problem into a polynomial inequality with period π and no case work — but first confirm that $\cos x = 0$ is not itself a solution.
- The lengths of the bands for $f(x) > k$ and $f(x) < k$ must add to one full period. Use it as an arithmetic check on both answers at once.
- When an inequality has a strict sign and the bound equals the amplitude, the answer is “no solution” — say it in one line rather than grinding.

Section 8: Cheat Sheet Box

CHEAT SHEET — Trigonometric Inequalities ($n \in \mathbb{Z}$ throughout)

- **Sine** ($\alpha = \sin^{-1} k \in [-\frac{\pi}{2}, \frac{\pi}{2}]$): $\sin x > k \Rightarrow (2n\pi + \alpha, 2n\pi + \pi - \alpha)$; $\sin x < k \Rightarrow (2n\pi + \pi - \alpha, 2n\pi + 2\pi + \alpha)$
- **Cosine** ($\beta = \cos^{-1} k \in [0, \pi]$): $\cos x > k \Rightarrow (2n\pi - \beta, 2n\pi + \beta)$; $\cos x < k \Rightarrow (2n\pi + \beta, 2n\pi + 2\pi - \beta)$.
- **Tangent** ($\gamma = \tan^{-1} k$): $\tan x > k \Rightarrow (n\pi + \gamma, n\pi + \frac{\pi}{2})$; $\tan x < k \Rightarrow (n\pi - \frac{\pi}{2}, n\pi + \gamma)$.
Period π ; right/left ends always open.
- **Close the interval for $\geq$ and $\leq$** — except at a point where the function is undefined, which is excluded regardless.
- **Absolute values:** $|\sin x| \geq \frac{1}{2} \Rightarrow [n\pi + \frac{\pi}{6}, n\pi + \frac{5\pi}{6}]$; $|\tan x| < 1 \Rightarrow (n\pi - \frac{\pi}{4}, n\pi + \frac{\pi}{4})$.
- **Polynomial in one function:** substitute t , factor, sign scheme, discard bands outside the range of t , translate.
- **Homogeneous degree two:** divide by $\cos^2 x$, solve in $\tan x$, period π .
- $a \cos x + b \sin x \geq c$: compute $R = \sqrt{a^2 + b^2}$ first. $|c| > R \Rightarrow$ all-or-nothing; $|c| = R \Rightarrow$ isolated points $x = n\pi + \alpha$; $|c| < R \Rightarrow$ genuine bands via $\cos(x - \alpha) \geq c/R$.
- **Check:** band length for $>$ plus band length for $<$ equals one period.
- **Golden rule:** find the boundary with Subtopic 06, then decide the side by testing one point per stretch. Never guess which side.

Subtopic 10 — Maximum, Minimum and Range of Trigonometric Expressions

Exam Relevance: CBSE Boards ask the easy half of this — “find the maximum value of $3 \sin x + 4 \cos x$ ” is a standard 1–2 marker, and Subtopic 05 §3(I) already supplies the $\sqrt{a^2 + b^2}$ rule that answers it. JEE Main asks the same question after one layer of disguise: a homogeneous quadratic in $\sin x$ and $\cos x$, a quadratic in

$\cos x$ alone, or a reciprocal pair such as $a^2 \sec^2 x + b^2 \csc^2 x$ — and it asks it reliably, roughly once a paper. JEE Advanced combines the methods and, more importantly, tests whether you know when the amplitude rule does *not* apply: expressions in $\sin^2 x$ and $\cos^2 x$, expressions whose variable is confined to a sub-interval, and expressions where a domain exclusion opens a hole in the range. (NEET does not test mathematics.) The whole subtopic is calculus-free by design; every result here is available to a Class 11 student and arrives faster than differentiation would.

Section 1: Concept Intuition

The tailor with a fixed length of cloth. A tailor in Surat has a strip of cloth of fixed length and can cut it into a rectangle of any shape. He cannot make the area arbitrarily large — the perimeter constrains him — and the best he can do turns out to be a square. He never differentiates anything. He recognises that the quantity he is maximising is constrained, and finds the shape where the constraint bites hardest.

Every problem in this subtopic works the same way. **You never search for a maximum; you rewrite the expression until its maximum is written on its face.** There are exactly four rewritings worth knowing, and each one ends with a quantity whose range you already know cold from Subtopic 02:

- rewrite as $R \sin(\text{angle})$ or $R \cos(\text{angle})$, whose range is $[-R, R]$;
- rewrite as a quadratic in a single bounded variable $t = \sin x$ or $t = \cos x$, whose range you read off a parabola on $[-1, 1]$;
- rewrite as a sum of a quantity and its reciprocal, whose minimum comes from AM-GM;
- rewrite so that the only surviving trigonometric object is $\sin^2$ or $\cos^2$, whose range is $[0, 1]$.

Indian-context analogy — the pressure cooker whistle. The whistle lifts at one pressure and no higher; the safety valve caps it. When you write $a \sin x + b \cos x = R \sin(x + \alpha)$ you have installed exactly such a cap: the bracket cannot exceed 1 no matter what x does, so the expression cannot exceed R . The rewriting is not cosmetic. It moves all the variability into a single sine, and a single sine has a valve on it.

The trap this subtopic exists to kill. Students learn $\sqrt{a^2 + b^2}$ so well that they apply it to $a \sin^2 x + b \cos^2 x$. That is wrong, and badly wrong. Write

$$3 \sin^2 x + 4 \cos^2 x = 3 (\sin^2 x + \cos^2 x) + \cos^2 x = 3 + \cos^2 x,$$

whose range is $[3, 4]$. The amplitude rule would have claimed 5, a value the expression never comes near. **The rule $\sqrt{a^2 + b^2}$ applies to a linear combination of $\sin x$ and $\cos x$ — first powers, same angle — and to nothing else.** The reason is structural: $\sin x$ and $\cos x$ can never be large at the same time, so their linear combination is capped below $|a| + |b|$; but $\sin^2 x$ and $\cos^2 x$ add to exactly 1, so their combination is trapped *between* a and b , not around $\sqrt{a^2 + b^2}$.

Second idea — a bounded variable makes a parabola behave. For $\cos 2x + \cos x$, substitute $t = \cos x$ and the expression becomes $2t^2 + t - 1$, a parabola. On the whole real line a

parabola has no maximum; on $t \in [-1, 1]$ it has both a maximum and a minimum, and they sit either at the vertex or at an endpoint. Checking three candidates — the two endpoints and the vertex, if the vertex lies inside — is the entire method.

Where it breaks down. Two ways, both examined. First, if the argument is restricted, say $x \in [0, \frac{\pi}{4}]$, the amplitude may simply not be attainable and the extremes move to the endpoints of the restriction. Second, if the expression is undefined somewhere — as $\frac{\sin 3x}{\sin x}$ is at $x = n\pi$ — the range can be a half-open interval with a value missing at one end. Neither is exotic; both are how JEE Advanced turns a Main-level question into an Advanced one.

Figure brief: Two panels side by side. Left: the curve $y = 3 \sin x + 4 \cos x$ over $[0, 2\pi]$ with horizontal dashed lines at $y = \pm 5$ touching the curve at its turning points, labelled “amplitude rule applies”. Right: the curve $y = 3 \sin^2 x + 4 \cos^2 x$ over the same interval, oscillating strictly between the dashed lines $y = 3$ and $y = 4$, with a third dashed line at $y = 5$ drawn far above and struck through, labelled “amplitude rule does not apply”.

Section 2: Key Formulas and Definitions

Throughout, x ranges over all real numbers unless a restriction is stated, and a, b, c are real constants.

(A) Linear combination — the anchor result (Subtopic 05 §3(I), restated for reference):

$$-\sqrt{a^2 + b^2} \leq a \sin x + b \cos x \leq \sqrt{a^2 + b^2}$$

$$\max(a \sin x + b \cos x + c) = c + \sqrt{a^2 + b^2}, \quad \min(a \sin x + b \cos x + c) = c - \sqrt{a^2 + b^2}$$

- Requires *first powers* of $\sin x$ and $\cos x$, with the *same* argument x .
- Both bounds are attained, because $x - \alpha$ sweeps all of $\mathbb{R}$.
- It extends to different-looking arguments once they are expanded: $5 \cos \theta + 3 \cos(\theta + \frac{\pi}{3})$ becomes $\frac{13}{2} \cos \theta - \frac{3\sqrt{3}}{2} \sin \theta$, with $R = \sqrt{\frac{169}{4} + \frac{27}{4}} = 7$.

(B) Homogeneous quadratic — the general formula. For

$$f(x) = a \sin^2 x + b \sin x \cos x + c \cos^2 x,$$

use $\sin^2 x = \frac{1 - \cos 2x}{2}$, $\cos^2 x = \frac{1 + \cos 2x}{2}$, $\sin x \cos x = \frac{\sin 2x}{2}$ to get

$$f(x) = \frac{a+c}{2} + \frac{b}{2} \sin 2x + \frac{c-a}{2} \cos 2x \implies f(x) \in \left[\frac{a+c}{2} - \frac{\sqrt{b^2 + (a-c)^2}}{2}, \frac{a+c}{2} + \frac{\sqrt{b^2 + (a-c)^2}}{2} \right]$$

- Special case $b = 0$: the range collapses to $[\min(a, c), \max(a, c)]$, which is the anti-amplitude fact from §1.
- Every bound is attained, since $2x$ also sweeps all of $\mathbb{R}$.

(C) Quadratic in a single bounded function. If $f(x) = pt^2 + qt + r$ with $t = \sin x$ or $t = \cos x$, so $t \in [-1, 1]$:

extremes of f occur at $t = -1$, $t = +1$, or $t = -\frac{q}{2p}$ (the last only if $-\frac{q}{2p} \in [-1, 1]$).

- Evaluate at all admissible candidates and take the largest and smallest. There is never anything else to check, because a parabola is monotone on either side of its vertex.
- If $t = \tan x$ the range is $\mathbb{R}$ and no endpoint candidates exist; only the vertex matters, and only for the appropriate one of max/min.

(D) Reciprocal pairs — AM-GM without calculus. For real $u > 0$ and $k > 0$, $u + \frac{k}{u} \geq 2\sqrt{k}$, with equality when $u = \sqrt{k}$. The two trigonometric consequences that appear constantly:

$$\tan^2 x + \cot^2 x \geq 2, \quad a^2 \sec^2 x + b^2 \csc^2 x \geq (a+b)^2 \quad (a, b > 0)$$

- Proof of the second: $a^2 \sec^2 x + b^2 \csc^2 x = a^2 + b^2 + a^2 \tan^2 x + b^2 \cot^2 x \geq a^2 + b^2 + 2ab = (a+b)^2$, with equality when $a^2 \tan^2 x = b^2 \cot^2 x$, i.e. $\tan^2 x = b/a$ — always achievable.
- Setting $a = b = 1$ gives $\sec^2 x + \csc^2 x \geq 4$, equivalently $\frac{4}{\sin^2 2x} \geq 4$.
- These have a minimum but **no maximum**: as $x \rightarrow 0$ the expression is unbounded above.

(E) Substitution $t = \sin x + \cos x$. Since $t^2 = 1 + 2 \sin x \cos x$,

$$\sin x \cos x = \frac{t^2 - 1}{2}, \quad t \in [-\sqrt{2}, \sqrt{2}].$$

- This converts any symmetric expression in $\sin x$ and $\cos x$ into a polynomial in the single bounded variable t , at which point §2(C) finishes the job.
- The companion substitution $u = \sin x - \cos x \in [-\sqrt{2}, \sqrt{2}]$ with $\sin x \cos x = \frac{1 - u^2}{2}$ handles antisymmetric cases.

(F) The two ways the range is *not* a closed interval.

Cause	Example	Range
Domain exclusion removes an attained value	$\frac{\sin 3x}{\sin x} = 3 - 4 \sin^2 x$, $x \neq n\pi$	$[-1, 3)$, since $\sin^2 x \in (0, 1]$
Argument restricted to a sub-interval	$\sin x + \cos x$ on $[0, \frac{\pi}{6}]$	$[1, \frac{1+\sqrt{3}}{2}]$, not $[-\sqrt{2}, \sqrt{2}]$

- In the first row the value 3 would need $\sin x = 0$, which the domain forbids — so 3 is a supremum, not a maximum, and the bracket is round.
- In the second row the amplitude $\sqrt{2}$ is attained only at $x = \frac{\pi}{4}$, outside the permitted window; the extremes migrate to the endpoints.

Section 3: CBSE Board Derivations and Key Procedures

Derivation: the range of a homogeneous quadratic in $\sin x$ and $\cos x$

Step 1. Start from $f(x) = a \sin^2 x + b \sin x \cos x + c \cos^2 x$. *Reason:* this is the general degree-two homogeneous form; any such expression, however written, reduces to it.

Step 2. Replace each degree-two object by its double-angle equivalent, using the power-reduction identities of Subtopic 05 §3(D) and the double-angle formula of §3(C):

$$\sin^2 x = \frac{1 - \cos 2x}{2}, \quad \cos^2 x = \frac{1 + \cos 2x}{2}, \quad \sin x \cos x = \frac{\sin 2x}{2}.$$

Reason: every term is degree two, so every term becomes a constant plus a first-power term in the angle $2x$ — which is precisely the shape §2(A) can handle.

Step 3. Substitute and collect:

$$f(x) = \frac{a(1 - \cos 2x)}{2} + \frac{b \sin 2x}{2} + \frac{c(1 + \cos 2x)}{2} = \frac{a+c}{2} + \frac{b}{2} \sin 2x + \frac{c-a}{2} \cos 2x.$$

Reason: pure algebra; the constant pieces gather and the $\cos 2x$ pieces combine with coefficient $\frac{c-a}{2}$.

Step 4. Apply the amplitude rule to the two first-power terms, with angle $2x$:

$$R = \sqrt{\left(\frac{b}{2}\right)^2 + \left(\frac{c-a}{2}\right)^2} = \frac{1}{2} \sqrt{b^2 + (a-c)^2}.$$

Reason: $(c-a)^2 = (a-c)^2$, so the formula is symmetric in a and c as it must be.

Step 5. Conclude $f(x) \in [\frac{a+c}{2} - R, \frac{a+c}{2} + R]$, both ends attained. *Reason:* $2x$ ranges over all reals as x does, so the inner sine reaches ± 1 .

Step 6 — the sanity check. Put $b = 0$ and $a = c$. Then $R = 0$ and the range is the single point a . Indeed $a \sin^2 x + a \cos^2 x = a$. *Reason:* a formula that survives its own degenerate case is far more likely to have been derived correctly.

Derivation: $a^2 \sec^2 x + b^2 \csc^2 x \geq (a+b)^2$ for $a, b > 0$

Step 1. Write $\sec^2 x = 1 + \tan^2 x$ and $\csc^2 x = 1 + \cot^2 x$. *Reason:* the Pythagorean identities of Subtopic 02 §3, which convert the expression into a constant plus a reciprocal pair.

Step 2. Expand: $a^2 \sec^2 x + b^2 \csc^2 x = a^2 + b^2 + a^2 \tan^2 x + b^2 \cot^2 x$. *Reason:* the two constants come out and the two variable terms are reciprocal in $\tan^2 x$, since $\cot^2 x = 1/\tan^2 x$.

Step 3. Apply AM-GM to the two positive variable terms:

$$a^2 \tan^2 x + b^2 \cot^2 x \geq 2\sqrt{a^2 \tan^2 x \cdot b^2 \cot^2 x} = 2ab.$$

Reason: $\tan^2 x \cdot \cot^2 x = 1$, so the geometric mean is the constant ab — the whole point of pairing a quantity with its reciprocal.

Step 4. Add: $a^2 + b^2 + 2ab = (a + b)^2$. *Reason:* the algebraic identity, which is why the answer is a perfect square and not something ugly.

Step 5 — confirm attainability. Equality in AM-GM needs $a^2 \tan^2 x = b^2 \cot^2 x$, i.e. $\tan^4 x = b^2/a^2$, i.e. $\tan^2 x = b/a$. Since $b/a > 0$ and $\tan^2 x$ takes every positive value, such an x exists. *Reason:* a bound that cannot be reached is not a minimum, and examiners award the mark for saying why it can.

Step 6 — note the asymmetry. There is no maximum: as $x \rightarrow 0^+$, $\csc^2 x \rightarrow \infty$. *Reason:* the expression is unbounded above, so “find the maximum and minimum” would be a badly posed question here — only the minimum exists.

Procedure A: finding the range of a trigonometric expression

- 1. Classify the expression by degree and by how many distinct angles appear.** *Why:* the method is determined entirely by this. Degree one in one angle goes to §2(A); degree two homogeneous goes to §2(B); a single function raised to powers goes to §2(C).
- 2. Reduce every angle to a common one.** Expand $\cos(\theta + \frac{\pi}{3})$ and similar before doing anything else. *Why:* the amplitude rule is false if the two terms carry different arguments.
- 3. Reduce to a single trigonometric object.** After Step 2 you should be looking at exactly one of: $R \sin(\cdot)$, a polynomial in $t \in [-1, 1]$, or a reciprocal pair. *Why:* if two independent objects remain, you have not finished reducing and any bound you write will be too generous.
- 4. Read off the range of that object, then undo the arithmetic.** Add the constant, multiply by the coefficient, and reverse the interval if you multiplied by a negative number. *Why:* multiplying $[-1, 1]$ by -4 gives $[-4, 4]$, but multiplying $[0, 1]$ by -4 gives $[-4, 0]$ — the order flips, and it is easy to write the endpoints backwards.
- 5. Check attainability at both ends.** Solve for the x that achieves each bound and confirm it is in the domain and inside any stated restriction. *Why:* this is the step that turns $[-1, 3]$ into $[-1, 3)$, and it is the only step that distinguishes an Advanced answer from a Main one.
- 6. Test numerically at one or two convenient angles.** *Why:* substituting $x = 0$ and $x = \frac{\pi}{4}$ takes fifteen seconds and must produce values inside your claimed range. If one does not, the range is wrong.

Procedure B: extremes on a restricted interval

1. Write the expression in amplitude form $R \sin(x + \alpha) + c$ (or the cos version). *Why:* it identifies exactly where the unrestricted extremes live.
2. Find where the unrestricted maximum and minimum occur: $x + \alpha = \frac{\pi}{2} + 2n\pi$ and $x + \alpha = -\frac{\pi}{2} + 2n\pi$. *Why:* these are the only interior candidates.
3. Keep only those candidate points that lie inside the permitted interval. *Why:* a maximum outside the window is irrelevant, however large.
4. Add the two endpoints of the interval to the candidate list. *Why:* on a closed interval a continuous function attains its extremes either at a turning point or at an endpoint, and there is no third possibility.

5. Evaluate at every surviving candidate and take the largest and smallest values. *Why:* with a finite candidate list, comparison replaces optimisation entirely.

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

Find the maximum and minimum values of $3 \sin x + 4 \cos x + 7$.

Step 1 — identify the shape. First powers of $\sin x$ and $\cos x$, same angle, plus a constant. This is exactly §2(A).

Step 2 — amplitude. $R = \sqrt{3^2 + 4^2} = \sqrt{25} = 5$, so $3 \sin x + 4 \cos x \in [-5, 5]$.

Step 3 — shift. Adding 7 shifts the whole interval:

$$3 \sin x + 4 \cos x + 7 \in [-5 + 7, 5 + 7] = [2, 12].$$

Step 4 — attainability. Write $3 \sin x + 4 \cos x = 5 \sin(x + \alpha)$ with $\cos \alpha = \frac{3}{5}$, $\sin \alpha = \frac{4}{5}$. The maximum occurs when $x + \alpha = \frac{\pi}{2}$, which has solutions; likewise the minimum. Both bounds are genuinely reached.

$\therefore$ maximum = 12, minimum = 2.

Sanity check. At $x = 0$ the value is $0 + 4 + 7 = 11$, comfortably inside $[2, 12]$. At $x = \frac{\pi}{2}$ it is $3 + 0 + 7 = 10$. Both consistent.

Example 2 — [JEE Main Level]

Find the maximum and minimum values of $\cos^2 \theta - 6 \sin \theta \cos \theta + 3 \sin^2 \theta + 2$.

Step 1 — separate the homogeneous part. The constant +2 rides along untouched. The rest is homogeneous of degree two with $a = 3$ (the $\sin^2$ coefficient), $b = -6$, $c = 1$.

Step 2 — convert to the double angle. Using the substitutions of §2(B):

$$3 \sin^2 \theta - 6 \sin \theta \cos \theta + \cos^2 \theta = \frac{3+1}{2} + \frac{-6}{2} \sin 2\theta + \frac{1-3}{2} \cos 2\theta = 2 - 3 \sin 2\theta - \cos 2\theta.$$

Step 3 — amplitude of the variable part. For $-3 \sin 2\theta - \cos 2\theta$,

$$R = \sqrt{(-3)^2 + (-1)^2} = \sqrt{10}.$$

So the homogeneous part lies in $[2 - \sqrt{10}, 2 + \sqrt{10}]$.

Step 4 — restore the constant. Adding the original +2:

$$\text{expression} \in [4 - \sqrt{10}, 4 + \sqrt{10}] \approx [0.8377, 7.1623].$$

Step 5 — cross-check against the general formula. §2(B) with $a = 3$, $b = -6$, $c = 1$ predicts centre $\frac{a+c}{2} = 2$ and half-width $\frac{1}{2}\sqrt{36+4} = \frac{1}{2}\sqrt{40} = \sqrt{10}$. Same answer, arrived at without re-deriving. Two routes, one result.

Step 6 — numerical spot-check. At $\theta = 0$ the expression is $1 - 0 + 0 + 2 = 3$; at $\theta = \frac{\pi}{4}$ it is $\frac{1}{2} - 3 + \frac{3}{2} + 2 = 1$. Both lie in $[0.8377, 7.1623]$, and 1 sits close to the lower end, as it should for an angle near where the $-6 \sin \theta \cos \theta$ term is most negative.

$\therefore$ maximum $= 4 + \sqrt{10}$, minimum $= 4 - \sqrt{10}$.

Example 3 — [JEE Main — Speed Trap]

Find the maximum value of $3 \sin^2 x + 4 \cos^2 x$.

The trap. The coefficients 3 and 4 are the coefficients of the most-drilled example in the chapter, $3 \sin x + 4 \cos x$. The hand writes $\sqrt{3^2 + 4^2} = 5$ before the eye has registered the squares. The answer 5 is not merely wrong; the expression never exceeds 4, so 5 is not even close.

Why the amplitude rule cannot apply. That rule exists because $\sin x$ and $\cos x$ cannot both be near 1 at the same time — they trade off, and the trade-off produces $\sqrt{a^2 + b^2}$. But $\sin^2 x$ and $\cos^2 x$ do not merely trade off: they satisfy the *exact* constraint $\sin^2 x + \cos^2 x = 1$. A weighted average of two numbers that sum to one is trapped between the two weights.

The correct work.

Step 1. Write the smaller coefficient out:

$$3 \sin^2 x + 4 \cos^2 x = 3 (\sin^2 x + \cos^2 x) + \cos^2 x = 3 + \cos^2 x.$$

Step 2. Since $\cos^2 x \in [0, 1]$,

$$3 + \cos^2 x \in [3, 4].$$

Step 3 — where the ends are reached. The maximum 4 occurs when $\cos^2 x = 1$, i.e. $x = n\pi$; the minimum 3 when $\cos x = 0$, i.e. $x = (2n+1)\frac{\pi}{2}$. Both are attained, so the range is the closed interval $[3, 4]$.

Step 4 — cross-check with §2(B). Here $a = 3$, $b = 0$, $c = 4$: centre $\frac{3+4}{2} = \frac{7}{2}$, half-width $\frac{1}{2}\sqrt{0+1} = \frac{1}{2}$, giving $[3, 4]$. Agreed.

$\therefore$ maximum $= 4$, minimum $= 3$.

Seductive wrong answer: 5. It survives no test at all — substituting $x = 0$ gives 4, $x = \frac{\pi}{2}$ gives 3, $x = \frac{\pi}{4}$ gives $\frac{7}{2}$ — but it is chosen anyway, because the pattern-match happens faster than the check. **The general rule worth carrying: for $a \sin^2 x + b \cos^2 x$ the range is $[\min(a, b), \max(a, b)]$, and the square root never appears.**

Example 4 — [JEE Advanced Level]

Find the minimum value of $9 \sec^2 x + 16 \csc^2 x$, and state where it is attained. Then find the range of $\frac{\sin 3x}{\sin x}$ and explain why it is not a closed interval.

Part 1 — the reciprocal pair.

Step 1. Split off the constants using the Pythagorean identities:

$$9 \sec^2 x + 16 \csc^2 x = 9(1 + \tan^2 x) + 16(1 + \cot^2 x) = 25 + 9 \tan^2 x + 16 \cot^2 x.$$

Step 2. The two variable terms are positive (where defined) and their product is the constant $9 \cdot 16 = 144$, since $\tan^2 x \cot^2 x = 1$. By AM-GM,

$$9 \tan^2 x + 16 \cot^2 x \geq 2\sqrt{144} = 24.$$

Step 3. Hence $9 \sec^2 x + 16 \csc^2 x \geq 25 + 24 = 49 = (3 + 4)^2$, matching §2(D) with $a = 3$, $b = 4$.

Step 4 — attainability. Equality needs $9 \tan^2 x = 16 \cot^2 x$, i.e. $\tan^4 x = \frac{16}{9}$, i.e. $\tan^2 x = \frac{4}{3}$. Such x exist — for instance $x = \tan^{-1} \sqrt{4/3} \approx 0.8571$ radians. Substituting confirms the value 49 exactly.

Step 5 — no maximum. As $x \rightarrow 0^+$, $\csc^2 x \rightarrow +\infty$, so the expression is unbounded above. The honest answer is “minimum 49; no maximum”.

Part 2 — the half-open range.

Step 6. By the triple-angle formula (Subtopic 05 §3(F)), $\sin 3x = 3 \sin x - 4 \sin^3 x = \sin x (3 - 4 \sin^2 x)$, so for $\sin x \neq 0$

$$\frac{\sin 3x}{\sin x} = 3 - 4 \sin^2 x.$$

Step 7 — the domain matters. The expression is undefined at $x = n\pi$, which are precisely the points where $\sin^2 x = 0$. So the admissible values of $\sin^2 x$ form the *half-open* set $(0, 1]$, not $[0, 1]$.

Step 8 — map the interval. The map $s \mapsto 3 - 4s$ is decreasing, so it sends $(0, 1]$ to $[3 - 4, 3) = [-1, 3)$. The endpoint -1 is attained (at $x = \frac{\pi}{2}$, where $\frac{\sin(3\pi/2)}{\sin(\pi/2)} = \frac{-1}{1} = -1$); the value 3 is approached as $x \rightarrow n\pi$ but never reached.

$$\therefore \min(9 \sec^2 x + 16 \csc^2 x) = 49 \text{ at } \tan^2 x = \frac{4}{3}; \quad \frac{\sin 3x}{\sin x} \in [-1, 3).$$

What made this Advanced. Part 1 rewards recognising a reciprocal pair hiding behind two Pythagorean identities — the constant 25 has to be extracted before AM-GM becomes usable, and a student who applies AM-GM directly to $9 \sec^2 x + 16 \csc^2 x$ gets the useless bound $2\sqrt{144 \sec^2 x \csc^2 x}$, which is not a constant. Part 2 rewards noticing that a removable-looking simplification changes nothing about the domain: cancelling $\sin x$ is legitimate as an identity for $\sin x \neq 0$, but the excluded points are still excluded, and the bracket at 3 must therefore be round. Writing $[-1, 3]$ loses the mark.

Section 5: Practice Exercises

1. Find the maximum and minimum values of $5 \sin x - 12 \cos x + 4$.
2. Find the range of $2 \sin^2 x + 3 \cos^2 x$.
3. Find the minimum value of $4 \sec^2 x + 9 \csc^2 x$.
4. Find the range of $\sin x \cos x + \cos^2 x$.

5. Show that $\cos^2 x + \cos^2(x + \frac{\pi}{3}) + \cos^2(x - \frac{\pi}{3})$ is constant, and find its value.
6. Find the maximum and minimum values of $\sin x + \cos x + \sin x \cos x$.
7. Find the minimum value of $\cos 2x + \cos x$.
8. Find the maximum and minimum of $5 \cos \theta + 3 \cos(\theta + \frac{\pi}{3}) + 3$.

Answers.

1. $R = \sqrt{25 + 144} = 13$; maximum 17, minimum -9 .
2. $2(\sin^2 x + \cos^2 x) + \cos^2 x = 2 + \cos^2 x$, so the range is $[2, 3]$. (The amplitude rule would wrongly suggest $\sqrt{13}$.)
3. $(2 + 3)^2 = 25$, attained when $\tan^2 x = \frac{3}{2}$.
4. $\frac{1}{2} \sin 2x + \frac{1 + \cos 2x}{2} = \frac{1}{2} + \frac{1}{2}(\sin 2x + \cos 2x)$, so the range is $[\frac{1 - \sqrt{2}}{2}, \frac{1 + \sqrt{2}}{2}] \approx [-0.2071, 1.2071]$.
5. Using $\cos^2 \phi = \frac{1 + \cos 2\phi}{2}$, the sum is $\frac{3}{2} + \frac{1}{2}[\cos 2x + \cos(2x + \frac{2\pi}{3}) + \cos(2x - \frac{2\pi}{3})]$, and the bracket is $\cos 2x + 2 \cos 2x \cos \frac{2\pi}{3} = \cos 2x - \cos 2x = 0$. The value is $\frac{3}{2}$.
6. Put $t = \sin x + \cos x \in [-\sqrt{2}, \sqrt{2}]$, so $\sin x \cos x = \frac{t^2 - 1}{2}$ and the expression is $\frac{t^2 + 2t - 1}{2}$. The vertex is at $t = -1$, inside the interval, giving the minimum $\frac{1 - 2 - 1}{2} = -1$; the maximum is at $t = \sqrt{2}$, giving $\frac{2 + 2\sqrt{2} - 1}{2} = \frac{1 + 2\sqrt{2}}{2} \approx 1.9142$.
7. $2 \cos^2 x + \cos x - 1$ with $t = \cos x \in [-1, 1]$; vertex at $t = -\frac{1}{4}$, inside, giving $2(\frac{1}{16}) - \frac{1}{4} - 1 = -\frac{9}{8}$. (Maximum 2 at $t = 1$.)
8. Expanding, $5 \cos \theta + \frac{3}{2} \cos \theta - \frac{3\sqrt{3}}{2} \sin \theta = \frac{13}{2} \cos \theta - \frac{3\sqrt{3}}{2} \sin \theta$, with $R = \sqrt{\frac{169}{4} + \frac{27}{4}} = \sqrt{49} = 7$. Maximum 10, minimum -4 .

Section 6: Crack the MCQ

Q1. The maximum value of $3 \sin^2 x + 4 \cos^2 x$ is (A) 3 (B) 4 (C) 5 (D) 7

Answer: (B). $3 \sin^2 x + 4 \cos^2 x = 3 + \cos^2 x \leq 4$.

Distractor Analysis. - (A) is the *minimum*, obtained at $\cos x = 0$. Choosing it means the direction of the inequality was reversed. - (C) is $\sqrt{3^2 + 4^2}$ — the amplitude rule applied to squares. This is the signature error of the subtopic, and $x = 0$ refutes it in one substitution. - (D) is $3 + 4$, the “add the coefficients” guess. It would require $\sin^2 x$ and $\cos^2 x$ both to equal 1, which the Pythagorean identity forbids.

Q2. The minimum value of $9 \sec^2 x + 16 \csc^2 x$ is (A) 25 (B) 49 (C) 12 (D) 7

Answer: (B). $= 25 + 9 \tan^2 x + 16 \cot^2 x \geq 25 + 24 = 49$.

Distractor Analysis. - (A) is $9 + 16$ — the value you would get by setting $\tan^2 x = \cot^2 x = 0$ simultaneously, which is impossible since their product is 1. It is the bound you reach by dropping the variable terms instead of minimising them. - (C) is $\sqrt{9 \cdot 16} = 12$, the geometric mean on its own. The AM-GM step contributes *twice* the geometric mean, $2\sqrt{144} = 24$, and the constant 25 must still be added on top of that. Dropping either the factor 2 or the constant lands here. - (D) is $3 + 4$, the square root of the right answer. Note the general result is $(a + b)^2 = 49$, not $a + b$.

Q3. The range of $\frac{\sin 3x}{\sin x}$, $x \neq n\pi$, is (A) $[-1, 3]$ (B) $[-1, 3)$ (C) $(-1, 3)$ (D) $[-3, 3]$

Answer: (B). $= 3 - 4 \sin^2 x$ with $\sin^2 x \in (0, 1]$.

Distractor Analysis. - (A) includes 3, which needs $\sin x = 0$ — exactly the excluded point. This is the answer of a student who simplified correctly and then forgot the domain. - (C) additionally excludes -1 , which is attained at $x = \frac{\pi}{2}$ ($\sin \frac{3\pi}{2} / \sin \frac{\pi}{2} = -1$). Over-caution costs the same mark as carelessness. - (D) treats the expression as though it were a sine of something and quotes a symmetric range. Substituting $x = \frac{\pi}{2}$ gives -1 and $x = \frac{\pi}{6}$ gives 2 ; nothing suggests -3 .

Q4. The minimum value of $\cos 2x + \cos x$ over all real x is (A) -2 (B) -1 (C) $-\frac{9}{8}$ (D) 0

Answer: (C). With $t = \cos x \in [-1, 1]$ the expression is $2t^2 + t - 1$, whose vertex sits at $t = -\frac{1}{4}$, inside the interval, giving $-\frac{9}{8}$.

Distractor Analysis. - (A) adds the two individual minima, -1 and -1 . They cannot occur together: $\cos x = -1$ forces $\cos 2x = +1$. Minima of a sum are not the sum of minima. - (B) is the value at the endpoint $t = -1$, obtained by checking only $t = \pm 1$ and never the vertex. Since $-\frac{9}{8} < -1$, this is a genuine miss. - (D) is the value at $t = \frac{1}{2}$, a *root* of $2t^2 + t - 1 = (2t - 1)(t + 1)$ rather than its vertex. Solving the quadratic instead of minimising it is a reflex worth breaking.

Section 7: Common Student Pitfalls and Pro-Tips

- **Pitfall 1 — Applying $\sqrt{a^2 + b^2}$ to squares.** [SIGNATURE TRAP] For $a \sin^2 x + b \cos^2 x$ the range is $[\min(a, b), \max(a, b)]$. Correction: the amplitude rule needs first powers of $\sin x$ and $\cos x$ at the same angle, and nothing else qualifies.
- **Pitfall 2 — Applying the amplitude rule across different angles.** $5 \cos \theta + 3 \cos(\theta + \frac{\pi}{3})$ does not have amplitude $\sqrt{34}$. Correction: expand the shifted term into $\sin \theta$ and $\cos \theta$ first, then collect, then take the square root. Here the true amplitude is 7 .
- **Pitfall 3 — Checking only the endpoints of a quadratic in t .** If the vertex $t = -\frac{q}{2p}$ lies in $[-1, 1]$, it is a candidate and frequently *the* answer. Correction: list all three candidates before comparing.
- **Pitfall 4 — Adding individual extremes.** The minimum of $f + g$ is generally larger than $\min f + \min g$, because the two minima occur at different x . Correction: combine first, minimise once.
- **Pitfall 5 — Ignoring a domain exclusion when reporting the range.** Cancelling $\sin x$ from $\frac{\sin 3x}{\sin x}$ is valid as an identity but does not restore the excluded points. Correction: after simplifying, ask which values of the inner variable are now unreachable, and open the corresponding bracket.
- **Pitfall 6 — Reporting a maximum for an expression that has none.** $a^2 \sec^2 x + b^2 \csc^2 x$ is unbounded above. Correction: whenever $\sec$, $\csc$, $\tan$ or $\cot$ survives into the final expression, check for unboundedness before writing “maximum”.
- **Pitfall 7 — Forgetting the restriction on x .** On $[0, \frac{\pi}{6}]$, $\sin x + \cos x$ does not reach $\sqrt{2}$. Correction: run Procedure B — turning points inside the window plus the two endpoints, and nothing else.

Pro-Tips.

- Before choosing a method, count the powers. Degree one goes to the amplitude rule; degree two goes to the double angle; a single function to any power goes to the bounded-

parabola method. Ninety per cent of questions are decided by that one glance.

- For $a \sin^2 x + b \sin x \cos x + c \cos^2 x$, memorise the finished formula: centre $\frac{a+c}{2}$, half-width $\frac{1}{2}\sqrt{b^2 + (a-c)^2}$. It converts a two-minute derivation into a ten-second substitution, and it degenerates correctly when $b = 0$.
- Whenever you see a quantity multiplied by its reciprocal — $\tan^2$ with $\cot^2$, $\sec^2$ with $\cos^2$ — reach for AM-GM immediately. The product is a constant, so the bound is a constant, and no calculus is needed.
- The substitution $t = \sin x + \cos x$ with $\sin x \cos x = \frac{t^2-1}{2}$ turns any symmetric expression into a one-variable polynomial on $[-\sqrt{2}, \sqrt{2}]$. It is the single most useful substitution in the subtopic after the double angle.
- Finish every range question by naming the x at which each bound is attained. It costs one line, it converts $[-1, 3]$ into the correct $[-1, 3)$ when a bound is not attained, and it is exactly the line Advanced marking schemes reward.

Section 8: Cheat Sheet Box

CHEAT SHEET — Maximum, Minimum and Range

- **Linear (first powers, same angle):** $a \sin x + b \cos x + c \in [c - \sqrt{a^2 + b^2}, c + \sqrt{a^2 + b^2}]$. Expand shifted angles *before* applying this.
- **Pure squares:** $a \sin^2 x + b \cos^2 x \in [\min(a, b), \max(a, b)]$. **No square root.**
- **Homogeneous quadratic:** $a \sin^2 x + b \sin x \cos x + c \cos^2 x$ has centre $\frac{a+c}{2}$ and half-width $\frac{1}{2}\sqrt{b^2 + (a-c)^2}$.
- **Quadratic in a bounded function:** $t = \sin x$ or $\cos x$, $t \in [-1, 1]$. Candidates are $t = -1$, $t = 1$, and $t = -\frac{q}{2p}$ if it lies inside.
- **Reciprocal pairs (AM-GM):** $\tan^2 x + \cot^2 x \geq 2$; $a^2 \sec^2 x + b^2 \csc^2 x \geq (a+b)^2$ for $a, b > 0$, equality at $\tan^2 x = b/a$; $\sec^2 x + \csc^2 x \geq 4$. **Minimum only — these have no maximum.**
- **Symmetric substitution:** $t = \sin x + \cos x \in [-\sqrt{2}, \sqrt{2}]$, $\sin x \cos x = \frac{t^2-1}{2}$.
- **Power reduction for the conversion:** $\sin^2 x = \frac{1 - \cos 2x}{2}$, $\cos^2 x = \frac{1 + \cos 2x}{2}$, $\sin x \cos x = \frac{\sin 2x}{2}$.
- **Open brackets:** a bound that is approached but never attained (domain exclusion, or a restricted window) gets a round bracket. Example: $\frac{\sin 3x}{\sin x} \in [-1, 3)$.
- **Restricted interval:** candidates are the turning points inside the window plus both endpoints.
- **Golden rule:** rewrite until only one trigonometric object remains, read its range, then undo the arithmetic — and always name the x where each bound is reached.

Subtopic 11 — Trigonometric Series: Sums in AP and Telescoping Products

Exam Relevance: This is NCERT's own hidden requirement. Miscellaneous Exercise on Chapter 3 opens with $2 \cos \frac{\pi}{13} \cos \frac{9\pi}{13} + \cos \frac{3\pi}{13} + \cos \frac{5\pi}{13} = 0$ and asks at Question 5 for $\sin x + \sin 3x + \sin 5x + \sin 7x = 4 \cos x \cos 2x \sin 4x$ — both are series problems, and both are standard 4–6 mark board questions. JEE Main lifts the same machinery for “find the sum to n terms” and for products such as $\cos \frac{\pi}{7} \cos \frac{2\pi}{7} \cos \frac{4\pi}{7}$. JEE Advanced uses the telescoping forms — $\cot \theta - \tan \theta = 2 \cot 2\theta$ and its relatives — which convert a sum of n awkward terms into a difference of two. (NEET does not test mathematics.) Subtopic 05 §3(G)–(H) gave you product-to-sum and sum-to-product as *identities*; this subtopic turns them into a *summation technique*, which is a different skill and is examined separately.

Section 1: Concept Intuition

The Diwali lights that cancel. String a line of lamps around a circular courtyard at equal angular spacing and switch them all on. Stand at the centre and ask for the net sideways pull of the light — the sum of the horizontal components. It is zero, and it is zero for a reason that has nothing to do with the individual lamps: for every lamp there is another placed so as to cancel it, and even when the count is odd the cancellation still completes because the directions are equally spread around a full turn.

That is the whole of trigonometric series in one image. **A sum of sines or cosines whose angles are in arithmetic progression is never evaluated term by term; it is collapsed, by finding a single multiplier that makes consecutive terms cancel.** The multiplier is always of the form $2 \sin \frac{\beta}{2}$, where β is the common difference.

Why that multiplier and no other. Take the sum $S = \cos a + \cos(a + \beta) + \cos(a + 2\beta) + \dots$. Multiply a single term by $2 \sin \frac{\beta}{2}$ and use product-to-sum (Subtopic 05 §3(G)):

$$2 \sin \frac{\beta}{2} \cos(a + r\beta) = \sin\left(a + r\beta + \frac{\beta}{2}\right) - \sin\left(a + r\beta - \frac{\beta}{2}\right).$$

Every term has become a *difference of two neighbouring sines*. When you add them up, the second half of one term destroys the first half of the next, and only the extreme ends survive. This is telescoping, and $2 \sin \frac{\beta}{2}$ is the unique multiplier that produces it, because it is the only one whose two output angles are half a step either side of the input.

Indian-context analogy — the antakshari chain. In a chain of players, each one ends on the syllable the next one must begin with. Whatever happens in the middle, only the very first syllable and the very last one are visible from outside the chain. A telescoping sum is that chain: the interior is entirely internal bookkeeping, and the answer depends only on the two ends.

Second idea — the same trick, one level up, for products. For a *product* of cosines whose angles double each time, multiply by $2 \sin \theta$ and cascade the double-angle formula:

$$2 \sin \theta \cos \theta = \sin 2\theta, \quad 2 \sin 2\theta \cos 2\theta = \sin 4\theta, \quad 2 \sin 4\theta \cos 4\theta = \sin 8\theta, \quad \dots$$

Each cosine is consumed by the sine that the previous step produced. After n steps a single $\sin(2^n\theta)$ is left standing. **The chapter already used this once, for $\cos 20^\circ \cos 40^\circ \cos 80^\circ$ in Subtopic 05 Example 7; what was presented there as a clever move is in fact a general formula, and treating it as a formula is what makes it fast.**

Where this breaks down — and it is examined. The multiplier $2 \sin \frac{\beta}{2}$ is useless when $\sin \frac{\beta}{2} = 0$, that is when β is a multiple of 2π . In that degenerate case every term of the series is identical and the sum is simply n times one term. Similarly the product cascade divides by $\sin \theta$, so it says nothing when θ is a multiple of π . Quoting a formula through a zero denominator is the standard way to lose an otherwise correct question.

Figure brief: A telescoping diagram. Four stacked rows, each showing $2 \sin \frac{\beta}{2} \cos(a + r\beta) = \sin(a + r\beta + \frac{\beta}{2}) - \sin(a + r\beta - \frac{\beta}{2})$ for $r = 0, 1, 2, 3$, with arrows joining the negative term of each row to the positive term of the row above to show the cancellation. The two uncanceled ends — the top row's negative term and the bottom row's positive term — are circled and labelled “the only survivors”.

Section 2: Key Formulas and Definitions

Throughout, n is a positive integer, a is the first angle and β the common difference of the arithmetic progression of angles.

(A) Sums of sines and cosines whose angles are in AP. For $\beta \neq 2m\pi$ ($m \in \mathbb{Z}$), so that $\sin \frac{\beta}{2} \neq 0$:

$$\sum_{r=0}^{n-1} \sin(a + r\beta) = \frac{\sin\left(\frac{n\beta}{2}\right)}{\sin\left(\frac{\beta}{2}\right)} \sin\left(a + \frac{(n-1)\beta}{2}\right)$$

$$\sum_{r=0}^{n-1} \cos(a + r\beta) = \frac{\sin\left(\frac{n\beta}{2}\right)}{\sin\left(\frac{\beta}{2}\right)} \cos\left(a + \frac{(n-1)\beta}{2}\right)$$

- The two formulas are identical except for the trailing function. Both carry n terms, indexed $r = 0$ to $r = n - 1$; if a question indexes from $r = 1$, shift a by β first.
- $a + \frac{(n-1)\beta}{2}$ is the **average of the first and last angles**. That is the memory hook: the answer is a scaled sine (or cosine) of the mean angle.
- **Degenerate case:** if $\beta = 2m\pi$ then every term equals the first, and the sum is $n \sin a$ (or $n \cos a$). The formula must not be quoted here.
- **The vanishing case:** if $\frac{n\beta}{2}$ is a multiple of π but $\frac{\beta}{2}$ is not, the numerator is zero and the whole sum is zero. In particular, for $n \geq 2$,

$$\sum_{r=0}^{n-1} \cos\left(a + \frac{2\pi r}{n}\right) = 0 = \sum_{r=0}^{n-1} \sin\left(a + \frac{2\pi r}{n}\right),$$

which is the Diwali-lights fact of \$1 made precise.

(B) The doubling product (cascade formula). For $\sin \theta \neq 0$:

$$\cos \theta \cdot \cos 2\theta \cdot \cos 4\theta \cdots \cos (2^{n-1}\theta) = \frac{\sin (2^n \theta)}{2^n \sin \theta}$$

- n factors, angles doubling each time. The exponent in 2^n counts the *factors*, not the last angle.
- The sign of the answer is decided by $\sin (2^n \theta)$ and is **not** always positive. With $\theta = 20^\circ$ and $n = 3$: $\sin 160^\circ = \sin 20^\circ$, giving $+\frac{1}{8}$. With $\theta = \frac{\pi}{7}$ and $n = 3$: $\sin \frac{8\pi}{7} = -\sin \frac{\pi}{7}$, giving $-\frac{1}{8}$.

(C) Telescoping with the tangent-cotangent bridge. The identity

$$\cot \theta - \tan \theta = 2 \cot 2\theta \quad \Longleftrightarrow \quad \tan \theta = \cot \theta - 2 \cot 2\theta$$

(valid whenever $\sin 2\theta \neq 0$) telescopes on doubling angles:

$$\sum_{r=0}^{n-1} 2^r \tan (2^r \theta) = \cot \theta - 2^n \cot (2^n \theta)$$

- The classical examination form writes the last term on the left as a cotangent: $\tan \theta + 2 \tan 2\theta + 4 \tan 4\theta + 8 \cot 8\theta = \cot \theta$, which is the case $n = 3$ with $8 \cot 8\theta$ moved across.
- Choosing θ so that $2^n \theta = \frac{\pi}{2}$ kills the second term entirely, because $\cot \frac{\pi}{2} = 0$.

(D) Telescoping with the secant bridge. For $\sin \theta \neq 0$ and every cosine non-zero:

$$\tan (r+1) \theta - \tan r \theta = \frac{\sin \theta}{\cos (r+1) \theta \cos r \theta} \quad \Rightarrow \quad \sum_{r=1}^n \frac{1}{\cos r \theta \cos (r+1) \theta} = \frac{\tan (n+1) \theta - \tan \theta}{\sin \theta}$$

- The bridge is just $\tan A - \tan B = \frac{\sin(A-B)}{\cos A \cos B}$, which is the tangent difference formula (Subtopic 05 §3(A)) written over a common denominator.
- The companion for cotangents is $\cot A - \cot B = \frac{\sin(B-A)}{\sin A \sin B}$.

(E) The pairing shortcut for short sums. For four or fewer terms it is usually quicker to pair the outermost terms with sum-to-product than to invoke §2(A):

$$\sin x + \sin 7x = 2 \sin 4x \cos 3x, \quad \sin 3x + \sin 5x = 2 \sin 4x \cos x,$$

so $\sin x + \sin 3x + \sin 5x + \sin 7x = 2 \sin 4x (\cos 3x + \cos x) = 4 \sin 4x \cos 2x \cos x$ — which is NCERT's Miscellaneous Question 5, done in two lines.

Section 3: CBSE Board Derivations and Key Procedures

Derivation: the sum of n cosines in arithmetic progression

Step 1. Write $S = \sum_{r=0}^{n-1} \cos(a + r\beta)$ and assume $\sin \frac{\beta}{2} \neq 0$. *Reason:* the multiplier we are about to use is $2 \sin \frac{\beta}{2}$, and multiplying by zero destroys the information.

Step 2. Multiply both sides by $2 \sin \frac{\beta}{2}$:

$$2 \sin \frac{\beta}{2} S = \sum_{r=0}^{n-1} 2 \sin \frac{\beta}{2} \cos(a + r\beta).$$

Reason: we may multiply an equation by a non-zero constant, and this particular constant is chosen because it is the one that produces a difference of neighbouring sines.

Step 3. Apply $2 \cos A \sin B = \sin(A + B) - \sin(A - B)$ (Subtopic 05 §3(G)) to each term, with $A = a + r\beta$ and $B = \frac{\beta}{2}$:

$$2 \sin \frac{\beta}{2} \cos(a + r\beta) = \sin\left(a + r\beta + \frac{\beta}{2}\right) - \sin\left(a + r\beta - \frac{\beta}{2}\right).$$

Reason: product-to-sum is the only tool that converts a product into a *difference*, and a difference is what telescoping needs.

Step 4. Write out the terms for $r = 0, 1, \dots, n - 1$ and observe the cancellation:

$$\left[\sin\left(a + \frac{\beta}{2}\right) - \sin\left(a - \frac{\beta}{2}\right)\right] + \left[\sin\left(a + \frac{3\beta}{2}\right) - \sin\left(a + \frac{\beta}{2}\right)\right] + \dots$$

Every positive entry is destroyed by the negative entry of the next bracket. *Reason:* the r -th positive angle $a + r\beta + \frac{\beta}{2}$ equals the $(r + 1)$ -th negative angle $a + (r + 1)\beta - \frac{\beta}{2}$ exactly.

Step 5. Only the two ends survive:

$$2 \sin \frac{\beta}{2} S = \sin\left(a + (n - 1)\beta + \frac{\beta}{2}\right) - \sin\left(a - \frac{\beta}{2}\right) = \sin\left(a + \frac{(2n-1)\beta}{2}\right) - \sin\left(a - \frac{\beta}{2}\right).$$

Reason: the largest positive angle and the smallest negative angle have no partners.

Step 6. Convert that difference back into a product by $\sin C - \sin D = 2 \cos \frac{C+D}{2} \sin \frac{C-D}{2}$ (Subtopic 05 §3(H)). With $C = a + \frac{(2n-1)\beta}{2}$ and $D = a - \frac{\beta}{2}$:

$$\frac{C + D}{2} = a + \frac{(n-1)\beta}{2}, \quad \frac{C - D}{2} = \frac{n\beta}{2},$$

so $2 \sin \frac{\beta}{2} S = 2 \cos\left(a + \frac{(n-1)\beta}{2}\right) \sin\left(\frac{n\beta}{2}\right)$. *Reason:* sum-to-product is the exact inverse of Step 3 and returns the answer to closed form.

Step 7. Divide by $2 \sin \frac{\beta}{2}$:

$$S = \frac{\sin\left(\frac{n\beta}{2}\right)}{\sin\left(\frac{\beta}{2}\right)} \cos\left(a + \frac{(n-1)\beta}{2}\right).$$

Reason: legitimate because of the Step 1 assumption. The sine version is identical with cos replaced by sin throughout, because Step 3 would instead use $2 \sin A \sin B = \cos(A - B) - \cos(A + B)$ and Step 6 the cosine-difference formula.

Step 8 — check on a case you already know. Take $a = x$, $\beta = 2x$, $n = 4$ in the sine version: the factor is $\frac{\sin 4x}{\sin x}$ and the trailing sine is $\sin(x + 3x) = \sin 4x$, giving $\frac{\sin^2 4x}{\sin x}$. Since $\frac{\sin 4x}{\sin x} = \frac{2 \sin 2x \cos 2x}{\sin x} = 4 \cos x \cos 2x$, this equals $4 \cos x \cos 2x \sin 4x$ — NCERT's Miscellaneous Question 5 exactly. *Reason:* a formula that reproduces a known result on the first try is a formula you can use under pressure.

Derivation: the doubling product $\cos \theta \cos 2\theta \cdots \cos(2^{n-1}\theta)$

Step 1. Let P denote the product and assume $\sin \theta \neq 0$. *Reason:* the multiplier is $2^n \sin \theta$, and it must be non-zero.

Step 2. Multiply by $2 \sin \theta$ and use $2 \sin \phi \cos \phi = \sin 2\phi$ on the first two factors only:

$$2 \sin \theta \cdot P = (2 \sin \theta \cos \theta) \cos 2\theta \cos 4\theta \cdots = \sin 2\theta \cdot \cos 2\theta \cos 4\theta \cdots$$

Reason: the double-angle formula of Subtopic 05 §3(C) consumes one cosine and produces a sine whose angle matches the *next* cosine — which is what allows the cascade to continue.

Step 3. Multiply by 2 again and repeat: $4 \sin \theta P = (2 \sin 2\theta \cos 2\theta) \cos 4\theta \cdots = \sin 4\theta \cos 4\theta \cdots$. *Reason:* the situation after Step 2 is structurally identical to the situation before it, with θ replaced by 2θ and one fewer factor.

Step 4. After n such steps every cosine has been consumed:

$$2^n \sin \theta \cdot P = \sin(2^n \theta).$$

Reason: each step doubles the accumulated power of 2 and eliminates exactly one cosine, and there are n cosines.

Step 5. Divide: $P = \frac{\sin(2^n \theta)}{2^n \sin \theta}$. *Reason:* Step 1's assumption. Note that the result can be negative — nothing in the derivation forces a sign.

Procedure A: recognising and summing a trigonometric series

1. **Check whether the angles are in AP.** List them and take differences. *Why:* if the common difference is constant, §2(A) applies verbatim; if the angles *double*, you want §2(B) or §2(C) instead.
2. **If the series has four terms or fewer, pair the outermost terms** with sum-to-product rather than invoking the general formula. *Why:* the pairing route takes two lines, factors out a common term, and usually produces the exact product form the question demands.
3. **If the series is long or has n terms, quote §2(A)** with a = first angle, β = common difference, n = number of terms. *Why:* the general formula is the only route that

handles a symbolic n .

4. **Before quoting, verify** $\sin \frac{\beta}{2} \neq 0$. *Why:* the degenerate case $\beta = 2m\pi$ gives n copies of the first term, and quoting the formula there is a division by zero.
5. **If the angles double rather than increase by a constant, look for a cascade.** Products of cosines go to §2(B); sums of tangents with a factor 2^r go to §2(C); sums of reciprocals of cosine products go to §2(D). *Why:* these three are the only telescoping patterns the syllabus tests, and each has a distinctive fingerprint.
6. **Check numerically at one convenient value.** *Why:* a closed-form answer is either right or badly wrong, and one substitution distinguishes the two. This is especially worth doing when a sign could have slipped.

Procedure B: forcing a telescope when no formula fits

1. Write the general term T_r and try to express it as $f(r+1) - f(r)$ for some function f . *Why:* if you can, the sum is $f(n+1) - f(1)$ and the problem is over.
2. For a **product** of two trigonometric factors, multiply and divide by $2 \sin(\text{half the gap between the arguments})$. *Why:* this is precisely what §2(A)'s derivation did, and it works whenever the arguments differ by a constant.
3. For **tangents at doubling angles**, use $\tan \theta = \cot \theta - 2 \cot 2\theta$. *Why:* it is the unique rearrangement that expresses a tangent as a difference of cotangents at the two relevant angles.
4. For a **reciprocal of a product of two cosines**, use $\frac{\sin \theta}{\cos A \cos B} = \tan A - \tan B$ when $A - B = \theta$. *Why:* the numerator you need can always be manufactured by multiplying and dividing by $\sin \theta$.
5. Write out the first two and the last two terms explicitly before cancelling. *Why:* the commonest telescoping error is being off by one term at an end, and writing four terms makes the pattern of survivors visible.

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

Prove that $2 \cos \frac{\pi}{13} \cos \frac{9\pi}{13} + \cos \frac{3\pi}{13} + \cos \frac{5\pi}{13} = 0$. (NCERT, Miscellaneous Exercise on Chapter 3, Question 1.)

Step 1 — flatten the product. By $2 \cos A \cos B = \cos(A+B) + \cos(A-B)$ with $A = \frac{\pi}{13}$, $B = \frac{9\pi}{13}$:

$$2 \cos \frac{\pi}{13} \cos \frac{9\pi}{13} = \cos \frac{10\pi}{13} + \cos \left(-\frac{8\pi}{13} \right) = \cos \frac{10\pi}{13} + \cos \frac{8\pi}{13},$$

using that cosine is even (Subtopic 03 §3(B)).

Step 2 — the whole left side is now a sum of four cosines.

$$\text{LHS} = \cos \frac{10\pi}{13} + \cos \frac{8\pi}{13} + \cos \frac{3\pi}{13} + \cos \frac{5\pi}{13}.$$

Step 3 — pair by supplementary angles. Note that $\frac{10\pi}{13} + \frac{3\pi}{13} = \pi$ and $\frac{8\pi}{13} + \frac{5\pi}{13} = \pi$. By the allied-angle rule $\cos(\pi - \phi) = -\cos \phi$ (Subtopic 03 §3(C)):

$$\cos \frac{10\pi}{13} = -\cos \frac{3\pi}{13}, \quad \cos \frac{8\pi}{13} = -\cos \frac{5\pi}{13}.$$

Step 4 — cancel.

$$\text{LHS} = -\cos \frac{3\pi}{13} + \left(-\cos \frac{5\pi}{13}\right) + \cos \frac{3\pi}{13} + \cos \frac{5\pi}{13} = 0. \quad \blacksquare$$

Numerical check. $2(0.97094)(-0.74851) + 0.66312 + 0.12054 = -1.45366 + 0.78366 = 0$ to five decimal places. Verified.

The lesson. The whole problem is Step 1: the product must be flattened into single-angle cosines before any structure is visible. Once flattened, the four angles pair off as supplements of each other, and the answer follows without a single value being computed. Flatten first; look for structure second.

Example 2 — [JEE Main Level]

Find the sum of n terms of the series $\cos \theta + \cos 3\theta + \cos 5\theta + \dots$, and hence evaluate it for $\theta = \frac{\pi}{6}$, $n = 3$.

Step 1 — identify the AP. First angle $a = \theta$; the angles increase by $\beta = 2\theta$; there are n terms. Check the third angle: $\theta + 2(2\theta) = 5\theta$. Correct.

Step 2 — check the degenerate condition. We need $\sin \frac{\beta}{2} = \sin \theta \neq 0$, so assume $\theta \neq m\pi$.

Step 3 — quote §2(A).

$$S = \frac{\sin\left(\frac{n \cdot 2\theta}{2}\right)}{\sin\left(\frac{2\theta}{2}\right)} \cos\left(\theta + \frac{(n-1)2\theta}{2}\right) = \frac{\sin n\theta}{\sin \theta} \cos(\theta + (n-1)\theta) = \frac{\sin n\theta \cos n\theta}{\sin \theta}.$$

Step 4 — tidy with the double angle. Since $\sin n\theta \cos n\theta = \frac{1}{2} \sin 2n\theta$,

$$S = \frac{\sin 2n\theta}{2 \sin \theta}.$$

Step 5 — evaluate at $\theta = \frac{\pi}{6}$, $n = 3$. The series is $\cos \frac{\pi}{6} + \cos \frac{\pi}{2} + \cos \frac{5\pi}{6} = \frac{\sqrt{3}}{2} + 0 - \frac{\sqrt{3}}{2} = 0$, and the formula gives

$$S = \frac{\sin \pi}{2 \sin \frac{\pi}{6}} = \frac{0}{1} = 0. \quad \checkmark$$

$$\therefore \cos \theta + \cos 3\theta + \dots + \cos(2n-1)\theta = \frac{\sin 2n\theta}{2 \sin \theta}.$$

Note the shape of the answer. The mean of the first and last angles is $\frac{\theta+(2n-1)\theta}{2} = n\theta$, which is exactly the argument of the trailing cosine in Step 3. Recognising that lets you write Step 3 straight down without substituting into the formula letter by letter.

Example 3 — [JEE Main — Speed Trap]

Evaluate $\cos \frac{\pi}{7} \cos \frac{2\pi}{7} \cos \frac{4\pi}{7}$.

The trap. Every student has met $\cos 20^\circ \cos 40^\circ \cos 80^\circ = \frac{1}{8}$ (Subtopic 05, Example 7). The angles here also double, there are also three of them, and the pattern-match fires: the pen writes $\frac{1}{8}$. The magnitude is right and **the sign is wrong**.

Why. Use the cascade formula §2(B) with $\theta = \frac{\pi}{7}$ and $n = 3$:

$$\cos \frac{\pi}{7} \cos \frac{2\pi}{7} \cos \frac{4\pi}{7} = \frac{\sin(2^3 \cdot \frac{\pi}{7})}{2^3 \sin \frac{\pi}{7}} = \frac{\sin \frac{8\pi}{7}}{8 \sin \frac{\pi}{7}}.$$

Now reduce $\sin \frac{8\pi}{7}$. Since $\frac{8\pi}{7} = \pi + \frac{\pi}{7}$ lies in the third quadrant, $\sin \frac{8\pi}{7} = -\sin \frac{\pi}{7}$ (Subtopic 03 §3(C)). Therefore

$$\cos \frac{\pi}{7} \cos \frac{2\pi}{7} \cos \frac{4\pi}{7} = \frac{-\sin \frac{\pi}{7}}{8 \sin \frac{\pi}{7}} = -\frac{1}{8}.$$

Why the familiar case came out positive. With $\theta = 20^\circ$ and $n = 3$ the final angle is $2^3 \cdot 20^\circ = 160^\circ = 180^\circ - 20^\circ$, which lies in the *second* quadrant, where sine is positive: $\sin 160^\circ = +\sin 20^\circ$, and the ratio is $+\frac{1}{8}$. The two problems differ by nothing except which quadrant $2^n\theta$ lands in — and that single fact flips the answer.

Verification without the formula. $\cos \frac{\pi}{7} \approx 0.90097$, $\cos \frac{2\pi}{7} \approx 0.62349$, $\cos \frac{4\pi}{7} \approx -0.22252$. The product is ≈ -0.12500 . The negative sign was visible from the start: $\frac{4\pi}{7}$ exceeds $\frac{\pi}{2}$, so one factor is negative and the product of three factors must be negative.

$$\therefore \cos \frac{\pi}{7} \cos \frac{2\pi}{7} \cos \frac{4\pi}{7} = -\frac{1}{8}.$$

Seductive wrong answer: $+\frac{1}{8}$. The defence takes three seconds: **before quoting the cascade, ask which quadrant each factor lies in and count the negative ones**. Here exactly one factor is negative, so the answer must be negative, whatever the algebra says. A sign predicted independently of the calculation is a sign you cannot lose.

Example 4 — [JEE Advanced Level]

Prove that for $n \geq 1$

$$\tan \theta + 2 \tan 2\theta + 4 \tan 4\theta + \cdots + 2^{n-1} \tan (2^{n-1}\theta) = \cot \theta - 2^n \cot (2^n \theta),$$

and hence evaluate $\tan \frac{\pi}{16} + 2 \tan \frac{\pi}{8} + 4 \tan \frac{\pi}{4}$.

Step 1 — build the bridge. For any ϕ with $\sin 2\phi \neq 0$,

$$\cot \phi - \tan \phi = \frac{\cos \phi}{\sin \phi} - \frac{\sin \phi}{\cos \phi} = \frac{\cos^2 \phi - \sin^2 \phi}{\sin \phi \cos \phi} = \frac{\cos 2\phi}{\frac{1}{2} \sin 2\phi} = 2 \cot 2\phi,$$

using $\cos 2\phi = \cos^2 \phi - \sin^2 \phi$ and $\sin 2\phi = 2 \sin \phi \cos \phi$ (Subtopic 05 §3(C)). Rearranged: $\tan \phi = \cot \phi - 2 \cot 2\phi$.

Step 2 — apply it at every doubling angle. Put $\phi = 2^r \theta$ and multiply through by 2^r :

$$2^r \tan(2^r \theta) = 2^r \cot(2^r \theta) - 2^{r+1} \cot(2^{r+1} \theta).$$

This is the whole proof. The right-hand side is $g(r) - g(r+1)$ with $g(r) = 2^r \cot(2^r \theta)$ — a perfect telescoping term, and the factor 2^r on the left is precisely what makes it one.

Step 3 — sum from $r = 0$ to $r = n - 1$.

$$\sum_{r=0}^{n-1} 2^r \tan(2^r \theta) = \sum_{r=0}^{n-1} [g(r) - g(r+1)] = g(0) - g(n) = \cot \theta - 2^n \cot(2^n \theta). \quad \blacksquare$$

Step 4 — choose θ to kill the tail. Take $\theta = \frac{\pi}{16}$ and $n = 3$. Then $2^n \theta = 8 \cdot \frac{\pi}{16} = \frac{\pi}{2}$, and $\cot \frac{\pi}{2} = 0$. The identity collapses to

$$\tan \frac{\pi}{16} + 2 \tan \frac{\pi}{8} + 4 \tan \frac{\pi}{4} = \cot \frac{\pi}{16}.$$

Step 5 — confirm numerically. $\tan \frac{\pi}{16} = \tan 11.25^\circ \approx 0.198912$; $2 \tan \frac{\pi}{8} = 2(\sqrt{2} - 1) \approx 0.828427$; $4 \tan \frac{\pi}{4} = 4$. Their sum is ≈ 5.027339 . And $\cot \frac{\pi}{16} = 1/0.198912 \approx 5.027339$. Agreement to six decimal places.

$$\therefore \tan \frac{\pi}{16} + 2 \tan \frac{\pi}{8} + 4 \tan \frac{\pi}{4} = \cot \frac{\pi}{16} = \frac{1}{\tan(\pi/16)} \approx 5.02734.$$

What made this Advanced. Two separate insights had to arrive. First, that $\cot \phi - \tan \phi = 2 \cot 2\phi$ is not a curiosity but a *telescoping bridge* — that the coefficient 2^r in the question is a hint, not decoration, because it is exactly the weight needed to make consecutive terms cancel. Second, that the free parameter θ can be *chosen* so that the surviving tail vanishes: setting $2^n \theta = \frac{\pi}{2}$ turns a two-term answer into a one-term answer. A student who proves the identity but does not see the choice in Step 4 will grind at $\cot \frac{\pi}{2}$ numerically and never notice it is zero.

Section 5: Practice Exercises

1. Find the sum $\sin x + \sin 2x + \cdots + \sin nx$.
2. Show that $\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7} = -\frac{1}{2}$.
3. Evaluate $\cos \frac{\pi}{7} \cos \frac{2\pi}{7} \cos \frac{3\pi}{7}$.

4. Evaluate $\tan \frac{\pi}{32} + 2 \tan \frac{\pi}{16} + 4 \tan \frac{\pi}{8} + 8 \tan \frac{\pi}{4}$.
5. Prove that $\sin 20^\circ \sin 40^\circ \sin 60^\circ \sin 80^\circ = \frac{3}{16}$.
6. Prove that $\frac{(\sin 7x + \sin 5x) + (\sin 9x + \sin 3x)}{(\cos 7x + \cos 5x) + (\cos 9x + \cos 3x)} = \tan 6x$. (NCERT, Miscellaneous Exercise, Question 6.)
7. Show that for every integer $n \geq 2$ and every real a , $\sum_{r=0}^{n-1} \cos\left(a + \frac{2\pi r}{n}\right) = 0$.
8. Find the sum $\sum_{r=1}^n \frac{1}{\cos r\theta \cos(r+1)\theta}$.

Answers.

1. $a = x, \beta = x$, so $S = \frac{\sin(nx/2)}{\sin(x/2)} \sin \frac{(n+1)x}{2}$, valid for $x \neq 2m\pi$.
2. Cleanest route: apply Question 7 with $n = 7, a = 0$, giving $\sum_{r=0}^6 \cos \frac{2\pi r}{7} = 0$. The seven terms are 1 together with three pairs, since $\cos \frac{12\pi}{7} = \cos \frac{2\pi}{7}$, $\cos \frac{10\pi}{7} = \cos \frac{4\pi}{7}$ and $\cos \frac{8\pi}{7} = \cos \frac{6\pi}{7}$. Hence $1 + 2(\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7}) = 0$, so the required sum is $-\frac{1}{2}$.
3. $\frac{1}{8}$. (Note $\cos \frac{3\pi}{7} = -\cos \frac{4\pi}{7}$, so this is -1 times the Example 3 answer.)
4. $\theta = \frac{\pi}{32}, n = 4$, and $2^4\theta = \frac{\pi}{2}$ so the tail vanishes: the sum is $\cot \frac{\pi}{32} \approx 10.15317$.
5. Group $\sin 20^\circ \sin 40^\circ \sin 80^\circ = \frac{1}{4} \sin 60^\circ = \frac{\sqrt{3}}{8}$ by the triple-angle product of the Round 1 addendum (Example C3), then multiply by $\sin 60^\circ = \frac{\sqrt{3}}{2}$ to get $\frac{3}{16}$.
6. Numerator $= 2 \sin 6x \cos x + 2 \sin 6x \cos 3x = 2 \sin 6x (\cos x + \cos 3x)$; denominator $= 2 \cos 6x \cos x + 2 \cos 6x \cos 3x = 2 \cos 6x (\cos x + \cos 3x)$; the bracket cancels, leaving $\tan 6x$.
7. Here $\beta = \frac{2\pi}{n}$, so $\frac{n\beta}{2} = \pi$ and the numerator $\sin(n\beta/2) = \sin \pi = 0$, while $\sin(\beta/2) = \sin \frac{\pi}{n} \neq 0$ because $n \geq 2$. The sum is therefore 0.
8. Multiply and divide by $\sin \theta$ and use $\tan(r+1)\theta - \tan r\theta = \frac{\sin \theta}{\cos r\theta \cos(r+1)\theta}$; the sum telescopes to $\frac{\tan(n+1)\theta - \tan \theta}{\sin \theta}$.

Section 6: Crack the MCQ

Q1. For $\sin \theta \neq 0$, $\cos \theta \cos 2\theta \cos 4\theta \cdots \cos(2^{n-1}\theta)$ equals (A) $\frac{\sin(2^n\theta)}{2^n \sin \theta}$ (B) $\frac{\sin(2^{n-1}\theta)}{2^n \sin \theta}$ (C) $\frac{\sin(2^n\theta)}{2^{n-1} \sin \theta}$ (D) $\frac{\cos(2^n\theta)}{2^n \sin \theta}$

Answer: (A).

Distractor Analysis. - (B) uses the *last angle* $2^{n-1}\theta$ in the numerator instead of its double. Test $n = 1$: the product is just $\cos \theta$, and (A) gives $\frac{\sin 2\theta}{2 \sin \theta} = \cos \theta$ correctly, while (B) gives $\frac{1}{2}$. - (C) counts the factors of 2 one short. The same $n = 1$ test gives $2 \cos \theta$. - (D) confuses the cascade's output — the double-angle formula produces a *sine*, never a cosine, at each step.

Q2. $\cos \frac{\pi}{7} \cos \frac{2\pi}{7} \cos \frac{4\pi}{7}$ equals (A) $\frac{1}{8}$ (B) $-\frac{1}{8}$ (C) $\frac{1}{16}$ (D) $\frac{\sqrt{3}}{8}$

Answer: (B). $\frac{\sin(8\pi/7)}{8 \sin(\pi/7)} = \frac{-\sin(\pi/7)}{8 \sin(\pi/7)} = -\frac{1}{8}$.

Distractor Analysis. - (A) is the signature error: importing the sign of $\cos 20^\circ \cos 40^\circ \cos 80^\circ$ without checking the quadrant of $2^n\theta$. Counting negative factors settles it — $\cos \frac{4\pi}{7} < 0$ and the other two are positive, so the product is negative. - (C) divides by 2^n with $n = 4$, having miscounted three cosines as four. - (D) is $\sin 20^\circ \sin 40^\circ \sin 80^\circ$, a different product with a different cascade (the sine version, which produces $\frac{1}{4} \sin 3A$, not $\frac{1}{2^n}$).

Q3. $\sum_{r=0}^{n-1} \cos(a + r\beta)$, with $\sin \frac{\beta}{2} \neq 0$, equals (A) $\frac{\sin(n\beta/2)}{\sin(\beta/2)} \cos\left(a + \frac{(n-1)\beta}{2}\right)$ (B) $\frac{\cos(n\beta/2)}{\sin(\beta/2)} \cos\left(a + \frac{(n-1)\beta}{2}\right)$ (C) $\frac{\sin(n\beta/2)}{\sin(\beta/2)} \cos\left(a + \frac{n\beta}{2}\right)$ (D) $\frac{\sin(n\beta)}{\sin \beta} \cos\left(a + \frac{(n-1)\beta}{2}\right)$

Answer: (A).

Distractor Analysis. - (B) puts a cosine in the scaling factor. Test $n = 1$: the sum is $\cos a$, and only (A) returns $\cos a$; (B) returns $\cot \frac{\beta}{2} \cos a$. - (C) uses $\frac{n\beta}{2}$ instead of $\frac{(n-1)\beta}{2}$ in the trailing cosine — the mean of the first and last angles is $a + \frac{(n-1)\beta}{2}$, not $a + \frac{n\beta}{2}$. The $n = 1$ test again refutes it. - (D) forgets both halvings. It survives the $n = 1$ test by accident, since $\frac{\sin \beta}{\sin \beta} = 1$; use $n = 2$ instead, where (D) gives $2 \cos \beta \cos\left(a + \frac{\beta}{2}\right)$ but the true sum is $2 \cos \frac{\beta}{2} \cos\left(a + \frac{\beta}{2}\right)$.

Q4. The value of $2 \cos \frac{\pi}{13} \cos \frac{9\pi}{13} + \cos \frac{3\pi}{13} + \cos \frac{5\pi}{13}$ is (A) 1 (B) -1 (C) 0 (D) $\frac{1}{2}$

Answer: (C). The product flattens to $\cos \frac{10\pi}{13} + \cos \frac{8\pi}{13}$, and these are the negatives of $\cos \frac{3\pi}{13}$ and $\cos \frac{5\pi}{13}$ respectively.

Distractor Analysis. - (A) and (B) are guesses from students who cannot flatten the product and reach for a plausible small integer. Neither survives a numerical estimate: $\cos \frac{9\pi}{13}$ is negative, so the product term is negative and roughly -1.45 , while the two remaining cosines total roughly $+0.78$. - (D) comes from mis-pairing: using $\cos \frac{10\pi}{13} = -\cos \frac{3\pi}{13}$ but leaving $\cos \frac{8\pi}{13}$ unmatched, then guessing. Both supplementary pairs must be spotted, not one.

Section 7: Common Student Pitfalls and Pro-Tips

- **Pitfall 1 — Quoting the AP formula when $\sin \frac{\beta}{2} = 0$.** If the common difference is a multiple of 2π , every term is the same and the sum is n times one term. Correction: check the denominator before you write it.
- **Pitfall 2 — Getting the trailing angle wrong.** It is $a + \frac{(n-1)\beta}{2}$, the *mean of the first and last angles*, not $a + \frac{n\beta}{2}$. Correction: test your formula at $n = 1$; it must return the single term $\sin a$ or $\cos a$.
- **Pitfall 3 — Off-by-one in the number of terms.** The formula as written sums $r = 0$ to $r = n - 1$. A series presented as “from θ to $n\theta$ ” has n terms with $a = \theta$ and $\beta = \theta$; a series presented as “ $r = 1$ to n ” needs a shifted. Correction: count the terms explicitly, then set a to the *first angle*.

- **Pitfall 4 — Assuming a doubling product is positive.** [SIGNATURE TRAP] $\cos \theta \cos 2\theta \dots$ takes the sign of $\sin(2^n \theta)$. Correction: count how many factors lie in a quadrant where cosine is negative; an odd count means a negative answer, and that prediction is independent of the algebra.
- **Pitfall 5 — Dropping the weights in a telescoping tangent sum.** The identity is $\sum 2^r \tan(2^r \theta)$, not $\sum \tan(2^r \theta)$. The powers of 2 are what make the terms cancel. Correction: if a question hands you coefficients 1, 2, 4, 8, that is the identity announcing itself.
- **Pitfall 6 — Telescoping to the wrong endpoints.** After cancellation, the survivors are $g(0)$ and $-g(n)$, not $g(1)$ and $-g(n)$. Correction: write four terms out in full before cancelling anything.
- **Pitfall 7 — Using the general formula on a four-term series.** It works, but the pairing route (§2(E)) is twice as fast and lands directly on the product form the question wants. Correction: count the terms first; four or fewer means pair the outermost.

Pro-Tips.

- The answer to an AP sum always has the shape $\frac{\sin(\text{half the total span} + \text{half a step})}{\sin(\text{half a step})} \times$ (same function of the mean angle). Reconstructing it from that sentence is faster and safer than memorising the symbols.
- When the angles double, three telescoping patterns cover everything the syllabus asks: products of cosines (§2(B)), weighted tangents (§2(C)), reciprocal cosine products (§2(D)). Learn the fingerprint of each and you will never need to invent a method.
- Whenever a question leaves a parameter free, ask whether it can be chosen to annihilate the tail. Setting $2^n \theta = \frac{\pi}{2}$ makes $\cot(2^n \theta) = 0$ and is the intended move far more often than not.
- Predict the sign of any product answer *before* computing it, by counting how many factors are negative. It costs three seconds and is the only defence against the Example 3 trap.
- Every closed-form answer in this subtopic can be checked in fifteen seconds by substituting a convenient angle — $\theta = \frac{\pi}{6}$ or $n = 1$ — and comparing with the series written out. Do it every time; the failure mode here is a silently wrong sign or an off-by-one, and both show up instantly.

Section 8: Cheat Sheet Box

CHEAT SHEET — Trigonometric Series

- **AP sums** (n terms, first angle a , common difference β , requires $\sin \frac{\beta}{2} \neq 0$):

$$\sum_{r=0}^{n-1} \sin(a+r\beta) = \frac{\sin(n\beta/2)}{\sin(\beta/2)} \sin\left(a + \frac{(n-1)\beta}{2}\right), \quad \sum_{r=0}^{n-1} \cos(a+r\beta) = \frac{\sin(n\beta/2)}{\sin(\beta/2)} \cos\left(a + \frac{(n-1)\beta}{2}\right)$$

- **The trailing angle is the mean of the first and last.** Check any version at $n = 1$.
- **Degenerate:** $\beta = 2m\pi \Rightarrow \text{sum} = n \sin a$ or $n \cos a$; do not quote the formula.

- **Vanishing:** $\sum_{r=0}^{n-1} \cos\left(a + \frac{2\pi r}{n}\right) = 0$ for $n \geq 2$ (same for sine).
- **Doubling product:** $\cos \theta \cos 2\theta \cdots \cos(2^{n-1}\theta) = \frac{\sin(2^n \theta)}{2^n \sin \theta}$. **Sign can be negative** — $\cos \frac{\pi}{7} \cos \frac{2\pi}{7} \cos \frac{4\pi}{7} = -\frac{1}{8}$ while $\cos 20^\circ \cos 40^\circ \cos 80^\circ = +\frac{1}{8}$.
- **Tangent bridge:** $\cot \theta - \tan \theta = 2 \cot 2\theta$, hence $\sum_{r=0}^{n-1} 2^r \tan(2^r \theta) = \cot \theta - 2^n \cot(2^n \theta)$.
Choose $2^n \theta = \frac{\pi}{2}$ to kill the tail.
- **Secant bridge:** $\tan(r+1)\theta - \tan r\theta = \frac{\sin \theta}{\cos r\theta \cos(r+1)\theta}$, hence $\sum_{r=1}^n \frac{1}{\cos r\theta \cos(r+1)\theta} = \frac{\tan(n+1)\theta - \tan \theta}{\sin \theta}$.
- **Four terms or fewer:** pair the outermost with sum-to-product. E.g. $\sin x + \sin 3x + \sin 5x + \sin 7x = 4 \cos x \cos 2x \sin 4x$.
- **Supplementary pairing:** $\cos(\pi - \phi) = -\cos \phi$ collapses sums such as the $\frac{\pi}{13}$ family to zero.
- **Golden rule:** never add the terms — find the multiplier that makes neighbours cancel, and read off the two survivors.

Subtopic 12 — Higher Multiple Angles and the Sum of Three or More Angles

Exam Relevance: NCERT Exercise 3.3 ends with three questions the chapter never gave you the tools for: Question 23 asks for $\tan 4x$, Question 24 for $\cos 4x = 1 - 8 \sin^2 x \cos^2 x$, and Question 25 for $\cos 6x = 32 \cos^6 x - 48 \cos^4 x + 18 \cos^2 x - 1$. All three are fair game as 3–4 mark board questions, and all three are pure bookkeeping once you know the ladder. JEE Main uses $\tan 4x$ and $\cos 4x$ inside longer problems and asks directly for $\tan(A + B + C)$. JEE Advanced uses the S_1, S_2, S_3 form of the three-angle tangent formula as a bridge to quadratic and cubic equations — “if $\tan A, \tan B, \tan C$ are the roots of $x^3 - px^2 + qx - r = 0$, find $A + B + C$ ” — and as the cleanest derivation of the conditional identities of Subtopic 05 §3(J). (NEET does not test mathematics.) Subtopic 05 stopped at the triple angle; this subtopic climbs the rest of the ladder.

Section 1: Concept Intuition

The ladder, not the leap. There is no separate formula to memorise for $\cos 4x$. There is a ladder, and $\cos 4x$ is the fourth rung:

$$\cos 4x = \cos(2 \cdot 2x) = 2 \cos^2 2x - 1, \quad \cos 2x = 2 \cos^2 x - 1.$$

Substitute the second into the first and expand. That is the entire derivation, and the same two-step move produces $\cos 8x$, $\cos 16x$, and anything else whose index is a power of two. **Multiple-angle formulas are not a list to be learnt; they are a single recursion run**

the required number of times. Where the index is not a power of two — $\cos 6x$, say — you split it the other way, $6x = 2(3x)$, and use the triple angle you already have.

Indian-context analogy — the compound interest ladder. A bank does not have a separate formula for four years' compound interest. It has one formula for one year, and it applies it four times. Asking “what is the four-year formula?” is asking for the *expanded* version of a repeated one-step rule. $\cos 4x$ in terms of $\cos x$ is exactly that expanded version, and like the compound-interest expansion it looks ugly and means nothing on its own.

Which route to take. For an index n there are always two decompositions worth trying, and one is much shorter:

- $4x = 2(2x)$ — double a double. Shortest for 4, 8, 16.
- $6x = 2(3x)$ — double a triple. Shortest for 6.
- $5x = 3x + 2x$ — add a triple to a double. The only route for 5.

The shape of the answer is predictable and is itself a check. $\cos nx$ expanded in powers of $\cos x$ always comes out as a polynomial of degree n with leading coefficient 2^{n-1} , containing only powers of the same parity as n . So $\cos 4x$ must start $8\cos^4 x$ and contain only even powers; $\cos 6x$ must start $32\cos^6 x$ and contain only even powers; $\cos 5x$ must start $16\cos^5 x$ and contain only odd powers. **If your expansion has the wrong leading coefficient or a term of the wrong parity, it is wrong, and you know it before checking a single algebra step.**

Second idea — three angles at once, and the symmetric functions that organise it. For $\tan(A + B)$ the answer had two ingredients, $\tan A + \tan B$ above and $\tan A \tan B$ below. For three angles the pattern continues, using the three *elementary symmetric functions* of $\tan A$, $\tan B$, $\tan C$:

$$S_1 = \sum \tan A, \quad S_2 = \sum \tan A \tan B, \quad S_3 = \tan A \tan B \tan C,$$

and then $\tan(A + B + C) = \frac{S_1 - S_3}{1 - S_2}$. The alternating signs and the alternating degrees are not a coincidence: they are the real and imaginary parts of the product $(1 + i \tan A)(1 + i \tan B)(1 + i \tan C)$, which is the Complex Numbers chapter's de Moivre theorem showing through.

Why the S -notation is worth the trouble. Because a cubic equation $x^3 - px^2 + qx - r = 0$ has $S_1 = p$, $S_2 = q$, $S_3 = r$ by Vieta's relations. So a question that hands you a cubic whose roots are three tangents has *already handed you* $\tan(A + B + C)$. That bridge — from a polynomial to an angle sum — is the reason JEE Advanced cares about this formula at all.

Where it breaks down. The three-angle tangent formula is meaningless if any of A , B , C or their sum is an odd multiple of $\frac{\pi}{2}$: some tangent is undefined, or the denominator $1 - S_2$ vanishes. That vanishing is not a defect but information — it says the sum is an odd multiple of $\frac{\pi}{2}$, and it is the fastest proof of the conditional identity $\sum \tan A \tan B = 1$ when $A + B + C = \frac{\pi}{2}$.

Figure brief: A ladder diagram. At the bottom rung, “ $\cos x$, $\sin x$ ”. Arrows upward labelled “ $\times 2$ ” lead to “ $\cos 2x$ ”, then to “ $\cos 4x$ ”, then to “ $\cos 8x$ ”. A separate arrow labelled “ $\times 3$ ” leads from the bottom rung to “ $\cos 3x$ ”, and a “ $\times 2$ ” arrow from “ $\cos 3x$ ” reaches “ $\cos 6x$ ”. A dashed diagonal labelled “ $2x + 3x$ ” joins “ $\cos 2x$ ” and “ $\cos 3x$ ” to “ $\cos 5x$ ”. Each node is annotated

with its leading coefficient 2^{n-1} .

Section 2: Key Formulas and Definitions

Throughout, x, A, B, C are real and every formula is stated with the values at which it fails.

(A) The fourth multiple.

$$\sin 4x = 4 \sin x \cos x \cos 2x = 4 \sin x \cos^3 x - 4 \cos x \sin^3 x$$

$$\cos 4x = 8 \cos^4 x - 8 \cos^2 x + 1 = 1 - 8 \sin^2 x \cos^2 x = 1 - 8 \sin^2 x + 8 \sin^4 x$$

$$\tan 4x = \frac{4 \tan x - 4 \tan^3 x}{1 - 6 \tan^2 x + \tan^4 x} = \frac{4 \tan x (1 - \tan^2 x)}{1 - 6 \tan^2 x + \tan^4 x}$$

- The tangent form requires $\cos x \neq 0$, $\cos 2x \neq 0$ and $\cos 4x \neq 0$, i.e. $x \neq \frac{n\pi}{4} + \frac{\pi}{8}$ and $x \neq \frac{n\pi}{2} + \frac{\pi}{4}$ and $x \neq n\pi + \frac{\pi}{2}$.
- $\cos 4x = 1 - 8 \sin^2 x \cos^2 x$ is NCERT Exercise 3.3 Question 24; $\tan 4x$ is Question 23.

(B) The fifth and sixth multiples.

$$\sin 5x = 16 \sin^5 x - 20 \sin^3 x + 5 \sin x, \quad \cos 5x = 16 \cos^5 x - 20 \cos^3 x + 5 \cos x$$

$$\tan 5x = \frac{5 \tan x - 10 \tan^3 x + \tan^5 x}{1 - 10 \tan^2 x + 5 \tan^4 x}$$

$$\cos 6x = 32 \cos^6 x - 48 \cos^4 x + 18 \cos^2 x - 1$$

- $\cos 6x$ is NCERT Exercise 3.3 Question 25.
- Note the pleasing symmetry of the fifth multiple: the coefficient list 16, -20, 5 is identical for sine and cosine, because 5 is odd. For even n no such symmetry exists.

(C) The general multiple (de Moivre form). Writing $t = \tan x$ and using $(\cos x + i \sin x)^n = \cos nx + i \sin nx$ (Complex Numbers, de Moivre's theorem):

$$\cos nx = \sum_{k \geq 0} (-1)^k \binom{n}{2k} \cos^{n-2k} x \sin^{2k} x, \quad \sin nx = \sum_{k \geq 0} (-1)^k \binom{n}{2k+1} \cos^{n-2k-1} x \sin^{2k+1} x$$

$$\tan nx = \frac{\binom{n}{1}t - \binom{n}{3}t^3 + \binom{n}{5}t^5 - \dots}{1 - \binom{n}{2}t^2 + \binom{n}{4}t^4 - \dots}$$

- **Structural checks:** $\cos nx$ written purely in $\cos x$ is a polynomial of degree n with leading coefficient 2^{n-1} , containing only powers of the same parity as n ; the sum of all its coefficients is $\cos n \cdot 0 = 1$. For $\cos 6x$: $32 - 48 + 18 - 1 = 1$. Correct.
- The tangent version is valid wherever every tangent involved is defined and the denominator is non-zero.

(D) The sum of three angles. With

$$S_1 = \tan A + \tan B + \tan C, \quad S_2 = \tan A \tan B + \tan B \tan C + \tan C \tan A, \quad S_3 = \tan A \tan B \tan C,$$

$$\tan(A + B + C) = \frac{S_1 - S_3}{1 - S_2}$$

$$\sin(A + B + C) = \cos A \cos B \cos C (S_1 - S_3) = \sum \sin A \cos B \cos C - \sin A \sin B \sin C$$

$$\cos(A + B + C) = \cos A \cos B \cos C (1 - S_2) = \cos A \cos B \cos C - \sum \cos A \sin B \sin C$$

- The $\sum$ denotes the cyclic sum over the three ways of choosing which angle supplies the sine (respectively, the cosine).
- The tangent form requires $\cos A, \cos B, \cos C$ and $\cos(A + B + C)$ all non-zero; the sine and cosine forms in expanded shape are valid for **all** real A, B, C .

(E) The general n -angle tangent. For angles $A_1, \dots, A_n$, with S_k the k -th elementary symmetric function of $\tan A_1, \dots, \tan A_n$:

$$\tan(A_1 + A_2 + \dots + A_n) = \frac{S_1 - S_3 + S_5 - \dots}{1 - S_2 + S_4 - \dots}$$

- Setting $n = 2$ recovers $\tan(A + B) = \frac{S_1}{1 - S_2} = \frac{\tan A + \tan B}{1 - \tan A \tan B}$, which is Subtopic 05 §3(A). The general formula must reduce to the known case; that is the check that you have the signs right.

(F) Two corollaries that recover Subtopic 05 §3(J).

Condition	Consequence for the formula	Identity obtained
$A + B + C = \pi$	$\tan(A + B + C) = 0$, so the numerator vanishes	$S_1 = S_3$, i.e. $\sum \tan A = \prod \tan A$
$A + B + C = \frac{\pi}{2}$	$\tan(A + B + C)$ is undefined, so the denominator vanishes	$S_2 = 1$, i.e. $\sum \tan A \tan B = 1$

- Applying the second row to the half-angles of a triangle (where $\frac{A}{2} + \frac{B}{2} + \frac{C}{2} = \frac{\pi}{2}$) gives $\sum \tan \frac{A}{2} \tan \frac{B}{2} = 1$ immediately — a one-line proof of a result that is otherwise a page of manipulation.
- **A cubic connection (Vieta).** If $\tan A, \tan B, \tan C$ are the roots of $x^3 - px^2 + qx - r = 0$, then $S_1 = p, S_2 = q, S_3 = r$, so $\tan(A + B + C) = \frac{p-r}{1-q}$ — with no need to find the roots at all.

Section 3: CBSE Board Derivations and Key Procedures

Derivation: $\cos 4x$, three ways, from one recursion

Step 1. Apply the double-angle formula to the angle $2x$: $\cos 4x = \cos(2 \cdot 2x) = 1 - 2 \sin^2 2x$.

Reason: Subtopic 05 §3(C) holds for *any* angle, and $2x$ is an angle; choosing the sin-only form is what will produce NCERT's stated shape.

Step 2. Replace $\sin 2x = 2 \sin x \cos x$:

$$\cos 4x = 1 - 2(2 \sin x \cos x)^2 = 1 - 8 \sin^2 x \cos^2 x.$$

Reason: one substitution, and NCERT Exercise 3.3 Question 24 is finished.

Step 3 — the cosine-only form. Start again from $\cos 4x = 2 \cos^2 2x - 1$ and substitute $\cos 2x = 2 \cos^2 x - 1$:

$$\cos 4x = 2(2 \cos^2 x - 1)^2 - 1 = 2(4 \cos^4 x - 4 \cos^2 x + 1) - 1 = 8 \cos^4 x - 8 \cos^2 x + 1.$$

Reason: choosing the cos-only version of the double angle at *both* rungs is what keeps sines out entirely.

Step 4 — the sine-only form. From Step 2, put $\cos^2 x = 1 - \sin^2 x$:

$$\cos 4x = 1 - 8 \sin^2 x (1 - \sin^2 x) = 1 - 8 \sin^2 x + 8 \sin^4 x.$$

Reason: the same expression re-expressed; which of the three you quote depends only on what the question gives you.

Step 5 — structural check. The cosine-only form has degree 4, leading coefficient $8 = 2^{4-1}$, only even powers, and coefficient sum $8 - 8 + 1 = 1 = \cos 0$. All four structural predictions hold. *Reason:* four independent checks, none of which requires redoing the algebra.

Derivation: $\tan 4x$ (NCERT Exercise 3.3, Question 23)

Step 1. Write $\tan 4x = \tan(2 \cdot 2x) = \frac{2 \tan 2x}{1 - \tan^2 2x}$. *Reason:* the double-angle tangent formula of Subtopic 05 §3(C), applied at the angle $2x$.

Step 2. Substitute $\tan 2x = \frac{2t}{1 - t^2}$, where $t = \tan x$:

$$\tan 4x = \frac{\frac{4t}{1-t^2}}{1 - \frac{4t^2}{(1-t^2)^2}}.$$

Reason: the same formula one rung lower; nothing new is needed.

Step 3. Multiply numerator and denominator by $(1-t^2)^2$:

$$\tan 4x = \frac{4t(1-t^2)}{(1-t^2)^2 - 4t^2}.$$

Reason: clearing the nested fraction in one move, rather than two, avoids the commonest algebra slip in this derivation.

Step 4. Expand the denominator:

$$(1-t^2)^2 - 4t^2 = 1 - 2t^2 + t^4 - 4t^2 = 1 - 6t^2 + t^4.$$

Reason: the $-4t^2$ merges with the $-2t^2$, producing the characteristic -6 .

Step 5. Conclude

$$\tan 4x = \frac{4 \tan x (1 - \tan^2 x)}{1 - 6 \tan^2 x + \tan^4 x}.$$

Step 6 — numerical check. Take $\tan x = \frac{1}{3}$. Numerator: $4\left(\frac{1}{3}\right)\left(1 - \frac{1}{9}\right) = \frac{4}{3} \cdot \frac{8}{9} = \frac{32}{27}$. Denominator: $1 - \frac{6}{9} + \frac{1}{81} = \frac{81-54+1}{81} = \frac{28}{81}$. Ratio: $\frac{32}{27} \cdot \frac{81}{28} = \frac{96}{28} = \frac{24}{7}$. Independently: $\tan 2x = \frac{2/3}{8/9} = \frac{3}{4}$, then $\tan 4x = \frac{3/2}{1-9/16} = \frac{3/2}{7/16} = \frac{24}{7}$. *Reason:* two routes, one answer — the derivation is sound.

Derivation: the sum of three angles

Step 1. Group as $(A+B)+C$ and apply the two-angle tangent formula:

$$\tan(A+B+C) = \frac{\tan(A+B) + \tan C}{1 - \tan(A+B) \tan C}.$$

Reason: three angles are two angles once you bracket; no new principle is required.

Step 2. Substitute $\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$:

$$\tan(A+B+C) = \frac{\frac{\tan A + \tan B}{1 - \tan A \tan B} + \tan C}{1 - \frac{(\tan A + \tan B) \tan C}{1 - \tan A \tan B}}.$$

Reason: the only substitution available, and it is exactly the one that makes the symmetric functions appear.

Step 3. Multiply numerator and denominator by $(1 - \tan A \tan B)$:

$$\tan(A + B + C) = \frac{\tan A + \tan B + \tan C - \tan A \tan B \tan C}{1 - \tan A \tan B - \tan B \tan C - \tan C \tan A}.$$

Reason: the clearing produces four terms above and four below, and — remarkably — both are fully symmetric in A, B, C , even though the grouping in Step 1 was not.

Step 4. Recognise the symmetric functions:

$$\tan(A + B + C) = \frac{S_1 - S_3}{1 - S_2}.$$

Reason: the numerator is (sum of tangents) minus (product of all three); the denominator is 1 minus (sum of pairwise products). That the answer is symmetric is a proof that the bracketing choice in Step 1 did not matter.

Step 5 — the sine and cosine versions. Write $\tan(A + B + C) = \frac{\sin(A + B + C)}{\cos(A + B + C)}$ and multiply numerator and denominator of Step 4 by $\cos A \cos B \cos C$:

$$\sin(A + B + C) = \cos A \cos B \cos C (S_1 - S_3), \quad \cos(A + B + C) = \cos A \cos B \cos C (1 - S_2).$$

Reason: multiplying S_1 by $\cos A \cos B \cos C$ turns each $\tan$ into $\sin \times$ the other two cosines, giving the expanded forms of §2(D). Those expanded forms carry no denominators and are therefore valid even where a tangent is undefined.

Step 6 — verify on a case you know. Put $C = 0$: then $\tan C = 0$, so $S_1 = \tan A + \tan B$, $S_2 = \tan A \tan B$, $S_3 = 0$, and the formula returns $\frac{\tan A + \tan B}{1 - \tan A \tan B} = \tan(A + B)$. *Reason:* a three-angle formula that does not collapse to the two-angle one has a sign error somewhere.

Procedure A: expanding $\cos nx$ or $\sin nx$

1. **Factor n and choose the shortest route.** If $n = 2m$, write $nx = 2(mx)$ and use the double angle on the already-known m -th multiple. If n is odd and greater than 3, write $nx = (n - 2)x + 2x$. *Why:* the doubling route squares an expression you already have, which is one line; the additive route needs the full compound-angle formula, which is three.
2. **Decide up front which function you want in the answer** and pick the matching form of the double angle: $2 \cos^2 \phi - 1$ if the answer must be in cosines, $1 - 2 \sin^2 \phi$ if in sines. *Why:* choosing wrongly forces a Pythagorean conversion at the end and doubles the algebra.
3. **Expand and collect.** *Why:* nothing subtle happens here; it is a binomial square or cube.

4. **Run the three structural checks.** Degree = n ; leading coefficient = 2^{n-1} ; only powers of the parity of n ; and the coefficients must sum to 1. *Why:* four independent tests, all free, and together they catch essentially every algebra slip.
5. **Spot-check numerically at one angle**, for instance $x = \frac{\pi}{6}$. *Why:* the structural checks cannot detect two compensating errors; a numerical value can.

Procedure B: using the three-angle formula

1. **Compute S_1, S_2, S_3 explicitly** and write them down before assembling anything. *Why:* the formula is short but the three ingredients are easy to conflate under pressure, and S_2 in particular has three terms, not two.
2. **If the tangents are the roots of a cubic, read S_1, S_2, S_3 straight off by Vieta.** For $x^3 - px^2 + qx - r = 0$ they are p, q, r . *Why:* the roots themselves are never needed, which is the whole point of the question type.
3. **Assemble $\tan(A + B + C) = \frac{S_1 - S_3}{1 - S_2}$** , keeping the minus sign in the numerator. *Why:* dropping S_3 is the signature error of the subtopic.
4. **Interpret a zero numerator or a zero denominator as information, not as failure.** Numerator zero means the sum is a multiple of π ; denominator zero means it is an odd multiple of $\frac{\pi}{2}$. *Why:* the two conditional identities of §2(F) are exactly these two cases.
5. **Pin down the actual angle, not just its tangent.** $\tan \theta = 0$ allows $\theta = 0, \pi, 2\pi, \dots$; use the sizes of the individual angles to choose. *Why:* $\tan^{-1} 1 + \tan^{-1} 2 + \tan^{-1} 3$ has tangent 0, but the answer is π , not 0 — and only a size argument decides that.

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

Prove that $\cos 4x = 1 - 8 \sin^2 x \cos^2 x$. (NCERT, Exercise 3.3, Question 24.)

Step 1 — climb one rung. Treat $4x$ as twice the angle $2x$ and use the sin-only form of the double-angle formula, $\cos 2\phi = 1 - 2 \sin^2 \phi$, with $\phi = 2x$:

$$\cos 4x = 1 - 2 \sin^2 2x.$$

The sin-only form is chosen deliberately: the target contains $\sin^2 x \cos^2 x$, and only this form puts a $\sin 2x$ in place to be expanded.

Step 2 — expand the inner double angle. $\sin 2x = 2 \sin x \cos x$, so

$$\sin^2 2x = 4 \sin^2 x \cos^2 x.$$

Step 3 — substitute.

$$\cos 4x = 1 - 2 \left(4 \sin^2 x \cos^2 x \right) = 1 - 8 \sin^2 x \cos^2 x. \quad \blacksquare$$

Step 4 — check at $x = \frac{\pi}{6}$. Left side: $\cos \frac{2\pi}{3} = -\frac{1}{2}$. Right side: $1 - 8 \left(\frac{1}{4} \right) \left(\frac{3}{4} \right) = 1 - \frac{3}{2} = -\frac{1}{2}$. Agreed.

A useful corollary in one more line. Since $2 \sin x \cos x = \sin 2x$, the same statement reads $\cos 4x = 1 - 2 \sin^2 2x$, and using $\sin^2 \phi = \frac{1 - \cos 2\phi}{2}$ gives $\cos 4x = \cos 4x$ — a tautology, which is the reassurance that no step lost information.

Example 2 — [JEE Main Level]

Prove that $\tan 4x = \frac{4 \tan x (1 - \tan^2 x)}{1 - 6 \tan^2 x + \tan^4 x}$ (NCERT, Exercise 3.3, Question 23), and hence find $\tan 4\theta$ when $\tan \theta = \frac{1}{3}$.

Step 1 — the identity. This is the derivation of §3 above; the four lines are

$$\tan 4x = \frac{2 \tan 2x}{1 - \tan^2 2x} = \frac{\frac{4t}{1-t^2}}{1 - \frac{4t^2}{(1-t^2)^2}} = \frac{4t(1-t^2)}{(1-t^2)^2 - 4t^2} = \frac{4t(1-t^2)}{1-6t^2+t^4},$$

where $t = \tan x$.

Step 2 — apply it. With $t = \frac{1}{3}$:

$$\text{numerator} = 4 \cdot \frac{1}{3} \left(1 - \frac{1}{9}\right) = \frac{4}{3} \cdot \frac{8}{9} = \frac{32}{27},$$

$$\text{denominator} = 1 - 6 \cdot \frac{1}{9} + \frac{1}{81} = \frac{81 - 54 + 1}{81} = \frac{28}{81}.$$

Step 3 — divide.

$$\tan 4\theta = \frac{32}{27} \div \frac{28}{81} = \frac{32}{27} \cdot \frac{81}{28} = \frac{32 \cdot 3}{28} = \frac{96}{28} = \frac{24}{7}.$$

Step 4 — confirm by the ladder. $\tan 2\theta = \frac{2(1/3)}{1-1/9} = \frac{2/3}{8/9} = \frac{3}{4}$, and then $\tan 4\theta = \frac{2(3/4)}{1-9/16} = \frac{3/2}{7/16} = \frac{24}{7}$. Same answer by a completely different arithmetic path.

$$\therefore \tan 4\theta = \frac{24}{7}.$$

Which route to use in the exam. For a *numerical* $\tan \theta$, climb the ladder — two easy fractions beat one messy quartic. For a *symbolic* answer, quote the quartic form. The identity is worth knowing precisely because NCERT asks for it in that shape.

Example 3 — [JEE Main — Speed Trap]

If $A = B = C = 15^\circ$, use the three-angle formula to find $\tan(A + B + C)$, and check your answer.

The trap. The two-angle formula is $\frac{\tan A + \tan B}{1 - \tan A \tan B}$: sum on top, one-minus-product below.

The hand generalises it to $\frac{S_1}{1 - S_2}$ — sum on top, one minus the pairwise products below —

and the missing $-S_3$ is invisible, because it is a *subtraction of a small number* and the answer still looks plausible.

The correct work.

Step 1 — the three symmetric functions. With $\tan 15^\circ = 2 - \sqrt{3} \approx 0.267949$:

$$S_1 = 3 \tan 15^\circ \approx 0.803848, \quad S_2 = 3 \tan^2 15^\circ \approx 0.215390, \quad S_3 = \tan^3 15^\circ \approx 0.019238.$$

Step 2 — assemble, keeping S_3 .

$$\tan(A + B + C) = \frac{S_1 - S_3}{1 - S_2} \approx \frac{0.803848 - 0.019238}{1 - 0.215390} = \frac{0.784610}{0.784610} = 1.$$

Step 3 — the independent check. $A + B + C = 45^\circ$, and $\tan 45^\circ = 1$ exactly. The formula is correct to every digit computed, which is not a coincidence: the numerator and denominator came out equal because they must.

$$\therefore \tan(A + B + C) = 1.$$

Seductive wrong answer: $\frac{S_1}{1 - S_2} \approx \frac{0.803848}{0.784610} \approx 1.02452$. It is close enough to 1 to be mistaken for rounding error, which is exactly what makes it dangerous — a student who has forgotten $-S_3$ will “confirm” the wrong formula against a decimal check and carry the error into every later question.

The defence. Count the degrees. The two-angle formula has degree 1 on top and degree 2 below. The three-angle formula must reach degree 3 on top and degree 2 below, alternating in sign: $S_1 - S_3$ over $1 - S_2$. The general pattern of §2(E) is $\frac{S_1 - S_3 + S_5 - \dots}{1 - S_2 + S_4 - \dots}$ — odd degrees above, even below, signs alternating from the start. Reconstructing it from that sentence is faster than remembering it and cannot go wrong.

Example 4 — [JEE Advanced Level]

Suppose $\tan A, \tan B, \tan C$ are the roots of $x^3 - 6x^2 + 11x - 6 = 0$, with A, B, C all in $(0, \frac{\pi}{2})$. Find $A + B + C$. Deduce the value of $\tan^{-1} 1 + \tan^{-1} 2 + \tan^{-1} 3$.

Step 1 — read off the symmetric functions by Vieta. For $x^3 - px^2 + qx - r = 0$ the elementary symmetric functions of the roots are p, q, r . Here

$$S_1 = 6, \quad S_2 = 11, \quad S_3 = 6.$$

Note that we have not solved the cubic, and will not need to for the first part.

Step 2 — apply the three-angle formula.

$$\tan(A + B + C) = \frac{S_1 - S_3}{1 - S_2} = \frac{6 - 6}{1 - 11} = \frac{0}{-10} = 0.$$

Step 3 — from the tangent to the angle. $\tan \theta = 0$ gives $\theta = n\pi$ (Subtopic 06 §3(A)), so $A + B + C$ is a multiple of π . Which one?

Step 4 — bound the sum. Each of A, B, C lies strictly in $(0, \frac{\pi}{2})$, so $0 < A + B + C < \frac{3\pi}{2}$. The only multiple of π in that range is π itself. Hence

$$A + B + C = \pi.$$

This bounding step is the whole difficulty of the problem. The algebra gave a tangent; only an inequality can convert a tangent into an angle.

Step 5 — identify the roots for the deduction. The cubic factors as $x^3 - 6x^2 + 11x - 6 = (x - 1)(x - 2)(x - 3)$, so the roots are 1, 2, 3 and the angles are $\tan^{-1} 1, \tan^{-1} 2, \tan^{-1} 3$ — all in $(0, \frac{\pi}{2})$, as required.

Step 6 — conclude.

$$\tan^{-1} 1 + \tan^{-1} 2 + \tan^{-1} 3 = \pi.$$

Step 7 — numerical confirmation. $0.785398 + 1.107149 + 1.249046 = 3.141593 = \pi$ to six decimal places.

What made this Advanced. Three separate ideas had to combine. First, that Vieta's relations and the S -notation are *the same three numbers*, so a cubic is a disguised angle sum and the roots never need to be found. Second, that $\tan \theta = 0$ underdetermines θ , so the answer requires a bound and not just algebra — a student who writes " $A + B + C = 0$ " has done every calculation correctly and still has the wrong answer. Third, that $S_1 = S_3$ is precisely the condition $\sum \tan A = \prod \tan A$ from Subtopic 05 §3(J), which holds exactly when the angles sum to a multiple of π : the problem was engineered around that identity, and recognising it makes the answer visible from Step 1.

Section 5: Practice Exercises

1. Prove that $\cos 6x = 32 \cos^6 x - 48 \cos^4 x + 18 \cos^2 x - 1$. (NCERT, Exercise 3.3, Question 25.)
2. Express $\sin 5\theta$ as a polynomial in $\sin \theta$.
3. Prove that $\sin 4x = 4 \sin x \cos^3 x - 4 \cos x \sin^3 x$.
4. If $A + B + C = \frac{\pi}{2}$, prove that $\tan A \tan B + \tan B \tan C + \tan C \tan A = 1$.
5. Prove that $\tan^{-1} \frac{1}{2} + \tan^{-1} \frac{1}{5} + \tan^{-1} \frac{1}{8} = \frac{\pi}{4}$.
6. Write $\tan 5\theta$ as a rational function of $t = \tan \theta$.
7. Deduce the conditional identity $\tan A + \tan B + \tan C = \tan A \tan B \tan C$ for $A + B + C = \pi$ directly from the three-angle formula, and state the values of A, B, C for which the deduction is invalid.
8. In a triangle ABC , prove that $\tan \frac{A}{2} \tan \frac{B}{2} + \tan \frac{B}{2} \tan \frac{C}{2} + \tan \frac{C}{2} \tan \frac{A}{2} = 1$.

Answers.

1. $\cos 6x = 2 \cos^2 3x - 1$ and $\cos 3x = 4 \cos^3 x - 3 \cos x$, so $\cos 6x = 2(4c^3 - 3c)^2 - 1 = 2(16c^6 - 24c^4 + 9c^2) - 1 = 32c^6 - 48c^4 + 18c^2 - 1$ with $c = \cos x$. Structural check: leading coefficient $32 = 2^5$, only even powers, coefficients sum to $32 - 48 + 18 - 1 = 1$.

2. $\sin 5\theta = 16 \sin^5 \theta - 20 \sin^3 \theta + 5 \sin \theta$. (Route: $\sin 5\theta = \sin(3\theta + 2\theta)$, then expand both multiples.)
3. $4 \sin x \cos x (\cos^2 x - \sin^2 x) = 2 \sin 2x \cos 2x = \sin 4x$.
4. $A + B + C = \frac{\pi}{2}$ makes $\tan(A + B + C)$ undefined, so the denominator $1 - S_2$ of $\frac{S_1 - S_3}{1 - S_2}$ must vanish (the numerator is finite), giving $S_2 = 1$.
5. $S_1 = \frac{1}{2} + \frac{1}{5} + \frac{1}{8} = \frac{33}{40}$; $S_2 = \frac{1}{10} + \frac{1}{40} + \frac{1}{16} = \frac{8+2+5}{80} = \frac{15}{80} = \frac{3}{16}$; $S_3 = \frac{1}{80}$. Then $\tan(\text{sum}) = \frac{33/40 - 1/80}{1 - 3/16} = \frac{65/80}{13/16} = \frac{13/16}{13/16} = 1$. Each angle is less than $\tan^{-1} 1 = \frac{\pi}{4}$ and all are positive, so the sum lies in $(0, \frac{3\pi}{4})$; the only value there with tangent 1 is $\frac{\pi}{4}$.
6. $\tan 5\theta = \frac{5t - 10t^3 + t^5}{1 - 10t^2 + 5t^4}$, the binomial coefficients being $\binom{5}{1}, \binom{5}{3}, \binom{5}{5}$ above and $\binom{5}{2}, \binom{5}{4}$ below.
7. $A + B + C = \pi \Rightarrow \tan(A + B + C) = 0 \Rightarrow S_1 - S_3 = 0$, i.e. $\sum \tan A = \prod \tan A$. The deduction is invalid if any of A, B, C is $\frac{\pi}{2}$, since that tangent is undefined and neither S_1 nor S_3 exists. (For a right-angled triangle the identity is therefore vacuous, not false.)
8. Since $A + B + C = \pi$, the half-angles satisfy $\frac{A}{2} + \frac{B}{2} + \frac{C}{2} = \frac{\pi}{2}$, so Question 4 applies directly with the half-angles in place of A, B, C . No half-angle equals $\frac{\pi}{2}$, since each is strictly less than $\frac{\pi}{2}$, so every tangent is defined.

Section 6: Crack the MCQ

Q1. $\cos 4x$ equals (A) $8 \cos^4 x - 8 \cos^2 x + 1$ (B) $4 \cos^3 x - 3 \cos x$ (C) $2 \cos^2 x - 1$ (D) $1 - 8 \sin^4 x$

Answer: (A). Double the double angle: $2(2 \cos^2 x - 1)^2 - 1$.

Distractor Analysis. - (B) is $\cos 3x$, one rung too low. The leading-coefficient check exposes it instantly: $\cos 4x$ must lead with $2^{4-1} = 8$, and (B) leads with $4 = 2^{3-1}$. - (C) is $\cos 2x$, two rungs too low. - (D) is the sine-only form with a term dropped; the correct version is $1 - 8 \sin^2 x + 8 \sin^4 x$. Test $x = \frac{\pi}{2}$: (D) gives $1 - 8 = -7$, but $\cos 2\pi = 1$.

Q2. With $S_1 = \sum \tan A$, $S_2 = \sum \tan A \tan B$, $S_3 = \tan A \tan B \tan C$, $\tan(A + B + C)$ equals (A) $\frac{S_1}{1 - S_2}$ (B) $\frac{S_1 - S_3}{1 - S_2}$ (C) $\frac{S_1 + S_3}{1 - S_2}$ (D) $\frac{S_1 - S_3}{1 + S_2}$

Answer: (B).

Distractor Analysis. - (A) drops S_3 — the signature error, and the trap of Example 3. It is refuted by $A = B = C = 15^\circ$, where the true answer is $\tan 45^\circ = 1$ and (A) gives ≈ 1.0245 . - (C) gets the degree pattern right but the sign wrong. The signs alternate from the start: $S_1 - S_3 + S_5 - \dots$. - (D) flips the denominator sign, generalising $\tan(A - B)$ rather than $\tan(A + B)$. Setting $C = 0$ must recover $\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$; (D) returns the difference formula instead.

Q3. $\tan^{-1} 1 + \tan^{-1} 2 + \tan^{-1} 3$ equals (A) 0 (B) $\frac{\pi}{2}$ (C) $\frac{3\pi}{4}$ (D) π

Answer: (D). $S_1 = 6$, $S_2 = 11$, $S_3 = 6$, so the tangent of the sum is 0; and since each term is positive and less than $\frac{\pi}{2}$, the sum lies in $(0, \frac{3\pi}{2})$, forcing π .

Distractor Analysis. - (A) is the answer of a student who computes $\tan(\text{sum}) = 0$ and stops. It is the commonest wrong answer here, and it is refuted by observing that all three terms

are positive, so the sum cannot be zero. - (B) would need the denominator $1 - S_2$ to vanish, i.e. $S_2 = 1$; here $S_2 = 11$. - (C) is a plausible-looking fraction of π but has tangent -1 , not 0 . Estimating numerically — each term exceeds $\frac{\pi}{4}$, and two of them exceed 1 radian — puts the sum above 3 , ruling it out.

Q4. If $A + B + C = \frac{\pi}{2}$, then $\tan A \tan B + \tan B \tan C + \tan C \tan A$ equals (A) 0 (B) 1 (C) -1 (D) $\tan A \tan B \tan C$

Answer: (B). $\tan \frac{\pi}{2}$ is undefined, so the denominator $1 - S_2$ must be zero.

Distractor Analysis. - (A) confuses “the tangent is undefined” with “the tangent is zero”. A zero *numerator* would give $S_1 = S_3$; it is the *denominator* that vanishes here. - (C) has the sign of the rearrangement wrong: $1 - S_2 = 0$ gives $S_2 = +1$. - (D) is the $A + B + C = \pi$ identity ($S_1 = S_3$) imported into the wrong condition. The two corollaries of §2(F) are easy to swap; the discriminator is whether the sum is π (numerator vanishes) or $\frac{\pi}{2}$ (denominator vanishes).

Section 7: Common Student Pitfalls and Pro-Tips

- **Pitfall 1 — Dropping S_3 from the three-angle tangent.** [SIGNATURE TRAP] The numerator is $S_1 - S_3$, not S_1 . Correction: count degrees — odd degrees above $(1, 3, 5, \dots)$, even below $(0, 2, 4, \dots)$, signs alternating from the start.
- **Pitfall 2 — Treating $\cos 4x$ as a formula to memorise.** It is two applications of the double angle. Correction: learn the ladder, not the rungs; you will never mis-remember a formula you can rebuild in fifteen seconds.
- **Pitfall 3 — Choosing the wrong form of the double angle at the first rung.** If the target is in cosines, use $2 \cos^2 \phi - 1$ at both rungs; if in sines, use $1 - 2 \sin^2 \phi$. Correction: decide the target function before writing anything.
- **Pitfall 4 — Concluding $\theta = 0$ from $\tan \theta = 0$.** The tangent determines the angle only up to π . Correction: bound the sum using the sizes of the individual angles, then pick the unique multiple of π inside that range.
- **Pitfall 5 — Swapping the two conditional corollaries.** $A + B + C = \pi$ kills the *numerator* ($S_1 = S_3$); $A + B + C = \frac{\pi}{2}$ kills the *denominator* ($S_2 = 1$). Correction: ask whether the target tangent is 0 or undefined, and let that pick the equation.
- **Pitfall 6 — Applying the tangent formulas where a tangent is undefined.** In a right-angled triangle, $\sum \tan A = \prod \tan A$ is vacuous. Correction: state the exclusions; NCERT and JEE both attach them to the formula, and marks follow.
- **Pitfall 7 — Not checking the expansion structurally.** A wrong leading coefficient or an odd power in $\cos 6x$ is a guaranteed error. Correction: degree n , leading coefficient 2^{n-1} , parity matching n , coefficients summing to 1 . Four free checks.

Pro-Tips.

- For a *numerical* multiple angle, climb the ladder one rung at a time — $\tan \theta \rightarrow \tan 2\theta \rightarrow \tan 4\theta$ is two easy fractions. Reserve the expanded quartic for symbolic answers and for questions that explicitly demand it.
- The coefficient sum test ($\cos n \cdot 0 = 1$) takes three seconds and catches almost every slip. For $\cos 6x$: $32 - 48 + 18 - 1 = 1$. For $\cos 4x$: $8 - 8 + 1 = 1$.

- Whenever a question hands you a cubic whose roots are described as tangents, write down S_1, S_2, S_3 by Vieta before reading the rest of the question. The angle sum is then one substitution away and the roots are never needed.
- The three-angle formulas are the *source* of the conditional identities in Subtopic 05 §3(J), not a separate topic. Deriving $\sum \tan A = \prod \tan A$ from $\tan(A+B+C) = 0$ is one line and is far more reliable than recalling the identity.
- Set $C = 0$ in any three-angle formula you have just written. It must collapse to the corresponding two-angle formula. This single substitution validates the whole expression.

Section 8: Cheat Sheet Box

CHEAT SHEET — Higher Multiples and the Sum of Three Angles

- **Ladder, not list:** $\cos 4x = 2 \cos^2 2x - 1$; $\cos 6x = 2 \cos^2 3x - 1$; $\cos 5x = \cos(3x + 2x)$.
- **Fourth multiple:** $\sin 4x = 4 \sin x \cos x \cos 2x$; $\cos 4x = 8 \cos^4 x - 8 \cos^2 x + 1 = 1 - 8 \sin^2 x \cos^2 x = 1 - 8 \sin^2 x + 8 \sin^4 x$; $\tan 4x = \frac{4 \tan x (1 - \tan^2 x)}{1 - 6 \tan^2 x + \tan^4 x}$.
- **Fifth and sixth:** $\sin 5x = 16 \sin^5 x - 20 \sin^3 x + 5 \sin x$; $\cos 5x = 16 \cos^5 x - 20 \cos^3 x + 5 \cos x$; $\tan 5x = \frac{5t - 10t^3 + t^5}{1 - 10t^2 + 5t^4}$; $\cos 6x = 32 \cos^6 x - 48 \cos^4 x + 18 \cos^2 x - 1$.
- **Structural checks for $\cos nx$ in powers of $\cos x$:** degree n ; leading coefficient 2^{n-1} ; only powers of the parity of n ; coefficients sum to 1.
- **General multiple (de Moivre):** $\tan nx = \frac{\binom{n}{1}t - \binom{n}{3}t^3 + \dots}{1 - \binom{n}{2}t^2 + \binom{n}{4}t^4 - \dots}$, $t = \tan x$.
- **Three angles** ($S_1 = \sum \tan A$, $S_2 = \sum \tan A \tan B$, $S_3 = \tan A \tan B \tan C$):

$$\tan(A+B+C) = \frac{S_1 - S_3}{1 - S_2}, \quad \sin(A+B+C) = \cos A \cos B \cos C (S_1 - S_3), \quad \cos(A+B+C) = \cos A \cos B \cos C (1 - S_2)$$

- **n angles:** $\tan(\sum A_i) = \frac{S_1 - S_3 + S_5 - \dots}{1 - S_2 + S_4 - \dots}$. Odd degrees above, even below, alternating signs.
- **Corollaries:** $A+B+C = \pi \Rightarrow S_1 = S_3$ (numerator dies); $A+B+C = \frac{\pi}{2} \Rightarrow S_2 = 1$ (denominator dies). Applied to half-angles of a triangle, the second gives $\sum \tan \frac{A}{2} \tan \frac{B}{2} = 1$.
- **Vieta bridge:** roots of $x^3 - px^2 + qx - r = 0$ give $S_1 = p$, $S_2 = q$, $S_3 = r$, so $\tan(\sum A) = \frac{p-r}{1-q}$ without solving the cubic.
- **From tangent to angle:** $\tan \theta = 0$ gives $\theta = n\pi$; bound the sum from the sizes of the individual angles to pick n . ($\tan^{-1} 1 + \tan^{-1} 2 + \tan^{-1} 3 = \pi$, not 0.)
- **Golden rule:** never memorise a high multiple — rebuild it from the double or triple angle, then run the four structural checks.

Addendum D — Boundedness Equations and Simultaneous Trigonometric Equations

Subtopic 06 built the three master general-solution formulas and the root-validity checklist, and Round 1's Procedure B added the $t = \tan \frac{\theta}{2}$ route for $a \cos \theta + b \sin \theta = c$. Two examinable patterns remain, and neither is solved by any of that machinery. This addendum supplies both in the shorter form: method statement, worked examples, practice with answers. Examples continue the Round 1 lettering and are labelled D1-D4.

Method statement 1 — the boundedness (extreme-value) argument

When to reach for it. The two sides of the equation are different in *kind* — one side is a bounded trigonometric expression and the other is bounded the opposite way, or is algebraic. Typical fingerprints: high even powers such as $\sin^6 x$ or $\tan^8 x$; a sum of two functions each of which is separately capped; an algebraic side such as $x^2 + \frac{1}{x^2}$. If factoring and the general-solution formulas both look hopeless, this is why.

The principle. If $f(x) \leq M$ for all x in the domain, and $g(x) \geq M$ for all x in the domain, then

$$f(x) = g(x) \iff f(x) = M \text{ and } g(x) = M.$$

One equation has become two, each of which is an *extremum* condition — and extremum conditions are rigid, so each pins the variable down to a narrow family. The solution set of the original is the **intersection** of the two families, which is frequently empty.

The three steps.

1. Bound each side separately, and find the common value M where the two bounds meet. State the bound you are using: $|\sin x| \leq 1$, $\sec^2 z \geq 1$, $u + \frac{1}{u} \geq 2$ for $u > 0$, or AM-GM.
2. Write down the family of x achieving each bound, using Subtopic 06 §3.
3. **Intersect the families.** Convert both to a common shape and ask which integers make them agree. If they cannot agree, the honest answer is “no solution”, and that is frequently the intended one.

Where it breaks down. The method proves nothing unless *both* bounds are genuine and they *meet*. If $f \leq 2$ and $g \geq 1$, the equation may have ordinary solutions and boundedness tells you nothing. Check that the two bounds are the same number before using the argument.

Method statement 2 — simultaneous trigonometric equations

When to reach for it. Two conditions are imposed on the same angle, for example $\sin \theta = -\frac{1}{2}$ together with $\tan \theta = \frac{1}{\sqrt{3}}$. Solving them separately and then “combining” the two general solutions is the standard route and the standard way to lose the question.

The principle. Do not generalise first. Each equation, taken alone, has two solutions in $[0, 2\pi)$; the pair of conditions is satisfied by the *common* principal value, and usually by exactly one. Generalise only at the very end, and with period 2π — because two independent

conditions fix the quadrant, and only a whole turn preserves both.

The four steps.

1. Solve each equation separately **within** $[0, 2\pi)$ **only**, listing the principal solutions.
2. Take the intersection of the two lists.
3. If the intersection is empty, the system has no solution.
4. Add $2n\pi$ to each surviving value. Never $n\pi$ — that would flip the quadrant and break one of the two conditions.

Why $2n\pi$ and not $n\pi$. Knowing $\sin \theta$ alone leaves two quadrants open; knowing $\cos \theta$ or $\tan \theta$ as well closes it to one. A shift of π moves the terminal side to the opposite quadrant, where at least one of the two given values changes sign. This is the same principle as Subtopic 06 §3(F), applied to any pair of conditions rather than to sine and cosine alone.

Worked Examples

Example D1 — [JEE Main Level] (Boundedness)

Solve $\sin^6 x = 1 + \cos^4 3x$.

Step 1 — bound each side. Since $|\sin x| \leq 1$, we have $\sin^6 x \leq 1$. Since $\cos^4 3x \geq 0$, we have $1 + \cos^4 3x \geq 1$. The two bounds meet at $M = 1$.

Step 2 — force both to the meeting value. Equality is possible only if

$$\sin^6 x = 1 \quad \text{and} \quad \cos^4 3x = 0.$$

Step 3 — solve each. From $\sin^6 x = 1$ we get $\sin^2 x = 1$, i.e. $\cos x = 0$, i.e.

$$x = (2m + 1)\frac{\pi}{2}, \quad m \in \mathbb{Z}.$$

From $\cos 3x = 0$ we get $3x = (2n + 1)\frac{\pi}{2}$, i.e.

$$x = (2n + 1)\frac{\pi}{6}, \quad n \in \mathbb{Z}.$$

Step 4 — intersect the two families. We need $(2m + 1)\frac{\pi}{2} = (2n + 1)\frac{\pi}{6}$, i.e. $3(2m + 1) = 2n + 1$, i.e. $6m + 3 = 2n + 1$, i.e. $n = 3m + 1$. Since $3m + 1$ is an integer for every integer m , **every** member of the first family already lies in the second. The intersection is the whole first family.

Step 5 — verify. At $x = \frac{\pi}{2}$: $\sin^6 \frac{\pi}{2} = 1$, and $1 + \cos^4 \frac{3\pi}{2} = 1 + 0 = 1$. Equal. At $x = \frac{3\pi}{2}$: $(-1)^6 = 1$ and $1 + \cos^4 \frac{9\pi}{2} = 1 + 0 = 1$. Equal.

$$\therefore x = (2n + 1)\frac{\pi}{2}, \quad n \in \mathbb{Z}.$$

The point of the intersection step. It is not decoration. Had the second condition been $\cos 2x = 0$ instead, the families would have been $x = (2m + 1)\frac{\pi}{2}$ and $x = (2n + 1)\frac{\pi}{4}$, and the

first is contained in the second only for odd multiples — the answer would still have been the first family, but for a reason that has to be checked, not assumed. And in Example D2 the intersection is empty.

Example D2 — [JEE Main — Speed Trap] (Boundedness)

Solve $\sin^4 x = 1 + \tan^8 x$.

The trap. The structure looks identical to D1, so the hand runs the same steps and reaches $\sin^4 x = 1$ and $\tan^8 x = 0$. It then writes the first family, $x = (2n + 1)\frac{\pi}{2}$, and stops — because in D1 that was the answer. Here it is not merely incomplete; it is a set of points at which the equation is not even defined.

The correct work.

Step 1 — bound. $\sin^4 x \leq 1$ and $1 + \tan^8 x \geq 1$, so equality forces

$$\sin^4 x = 1 \quad \text{and} \quad \tan^8 x = 0.$$

Step 2 — solve each. $\sin^2 x = 1$ gives $x = (2m + 1)\frac{\pi}{2}$. $\tan x = 0$ gives $x = n\pi$.

Step 3 — intersect. An odd multiple of $\frac{\pi}{2}$ is never a multiple of π . The two families are disjoint. Equivalently and more sharply: $\sin x = \pm 1$ forces $\cos x = 0$, at which point $\tan x$ is **undefined**, so the right-hand side does not exist there at all.

$\therefore$ the equation has no solution.

Step 4 — confirm. Scanning x over $[-10, 10]$ (excluding points where $\cos x \approx 0$), the smallest value of $|\sin^4 x - 1 - \tan^8 x|$ is about 0.94 — the two sides never come close to meeting.

Seductive wrong answer: $x = (2n + 1)\frac{\pi}{2}$. It fails on two counts at once: the second condition is not satisfied there, and the equation is not defined there. **The rule to carry away: the boundedness argument does not end when you have written the two families. It ends when you have intersected them and checked that the common points lie in the domain of both sides.**

Example D3 — [CBSE Board Level] (Simultaneous)

Find the general solution of the system $\sin \theta = -\frac{1}{2}$ and $\tan \theta = \frac{1}{\sqrt{3}}$.

Step 1 — the first equation, in $[0, 2\pi)$ only. $\sin \theta = -\frac{1}{2}$ is negative, so θ is in Quadrant III or IV with reference angle $\frac{\pi}{6}$:

$$\theta = \pi + \frac{\pi}{6} = \frac{7\pi}{6} \quad \text{or} \quad \theta = 2\pi - \frac{\pi}{6} = \frac{11\pi}{6}.$$

Step 2 — the second equation, in $[0, 2\pi)$ only. $\tan \theta = \frac{1}{\sqrt{3}}$ is positive, so θ is in Quadrant I or III with reference angle $\frac{\pi}{6}$:

$$\theta = \frac{\pi}{6} \quad \text{or} \quad \theta = \frac{7\pi}{6}.$$

Step 3 — intersect. The only value on both lists is $\theta = \frac{7\pi}{6}$. Note that this is precisely the Quadrant III value: sine negative and tangent positive together pin the quadrant, which neither condition does alone.

Step 4 — generalise with period 2π .

$$\theta = 2n\pi + \frac{7\pi}{6}, \quad n \in \mathbb{Z}.$$

Step 5 — verify. $\sin \frac{7\pi}{6} = -\frac{1}{2}$ and $\tan \frac{7\pi}{6} = \frac{1}{\sqrt{3}}$. Both hold.

Why $n\pi$ would be wrong. Adding π gives $\frac{\pi}{6}$ (modulo 2π), where $\tan \theta = \frac{1}{\sqrt{3}}$ still holds — the tangent has period π — but $\sin \frac{\pi}{6} = +\frac{1}{2}$, so the first condition fails. **The period of a system is the period of its most restrictive condition, and with two independent conditions that period is always 2π .**

Example D4 — [JEE Advanced Level] (Boundedness with AM-GM)

Solve $2^{\sin x} + 2^{\cos x} = 2^{1-\frac{1}{\sqrt{2}}}$.

Step 1 — bound the left side from below by AM-GM. Both terms are positive, so

$$2^{\sin x} + 2^{\cos x} \geq 2\sqrt{2^{\sin x} \cdot 2^{\cos x}} = 2 \cdot 2^{\frac{\sin x + \cos x}{2}} = 2^{1+\frac{\sin x + \cos x}{2}},$$

with equality if and only if $2^{\sin x} = 2^{\cos x}$, i.e. $\sin x = \cos x$.

Step 2 — bound the exponent from below. By the amplitude rule (Subtopic 05 §3(I)), $\sin x + \cos x = \sqrt{2} \sin(x + \frac{\pi}{4}) \geq -\sqrt{2}$. Since 2^u is an increasing function of u ,

$$2^{1+\frac{\sin x + \cos x}{2}} \geq 2^{1-\frac{\sqrt{2}}{2}} = 2^{1-\frac{1}{\sqrt{2}}}.$$

Step 3 — the chain is tight. Combining Steps 1 and 2,

$$2^{\sin x} + 2^{\cos x} \geq 2^{1-\frac{1}{\sqrt{2}}},$$

and the right-hand side of the given equation is exactly that lower bound. So the equation holds precisely when **both** inequalities are equalities.

Step 4 — impose both equality conditions.

- Step 1 equality: $\sin x = \cos x$.
- Step 2 equality: $\sin x + \cos x = -\sqrt{2}$.

Together these give $2 \sin x = -\sqrt{2}$, so $\sin x = \cos x = -\frac{1}{\sqrt{2}}$.

Step 5 — solve. Sine and cosine are both negative and equal in Quadrant III at $x = \frac{5\pi}{4}$. Hence

$$x = 2n\pi + \frac{5\pi}{4}, \quad n \in \mathbb{Z}.$$

Note that this is a *simultaneous* condition on $\sin x$ and $\cos x$, so by Method statement 2 the period is 2π , not π .

Step 6 — verify numerically. At $x = \frac{5\pi}{4}$: $\sin x = \cos x = -0.7071068$, so the left side is $2 \cdot 2^{-0.7071068} = 1.2250947$; and $2^{1-1/\sqrt{2}} = 2^{0.2928932} = 1.2250947$. A scan over $[0, 2\pi]$ confirms that 1.2250947 is the global minimum of the left side and that it occurs only at $x \approx 3.926991 = \frac{5\pi}{4}$.

$$\therefore x = 2n\pi + \frac{5\pi}{4}, \quad n \in \mathbb{Z}.$$

What made this Advanced. Two bounds had to be chained, and the chain had to be *tight* — each inequality achieving equality at the same x . A student who applies AM-GM and stops has $2^{\sin x} + 2^{\cos x} \geq 2^{1+\frac{\sin x + \cos x}{2}}$, which still contains x and so is not yet a number. The second bound is what converts it into a constant, and the recognition that the given right-hand side *is* that constant is what tells you the equation is an equality case rather than an ordinary equation. Nothing here can be pattern-matched; the shape $2^{1-1/\sqrt{2}}$ on the right is the deliberate signal.

Practice Exercises

1. Solve $\cos^2 x + \sec^2 x = 2$.
2. Solve $\sin^2 x + \cos^2 y = 2 \sec^2 z$.
3. Solve $\sin^4 x = 1 + \tan^8 x$.
4. Find the general solution of the system $\sin x = -\frac{1}{2}$, $\cos x = -\frac{\sqrt{3}}{2}$.
5. Solve $(\sin x + \cos x)^{1+\sin 2x} = 2$ for $-\pi \leq x \leq \pi$.
6. Show that $2 \cos^2 \frac{x}{2} \sin^2 x = x^2 + \frac{1}{x^2}$ has no solution for $0 < x \leq \frac{\pi}{2}$.
7. Find the general solution of the system $\tan \theta = -1$, $\cos \theta = \frac{1}{\sqrt{2}}$.
8. Solve $\sin^6 x + \cos^6 x = 1$.

Answers.

1. Put $u = \cos^2 x > 0$. Then $u + \frac{1}{u} \geq 2$ with equality only at $u = 1$. So $\cos^2 x = 1$, giving $x = n\pi$.
2. $\sin^2 x + \cos^2 y \leq 2$ and $2 \sec^2 z \geq 2$, so all three must be extreme: $\sin^2 x = 1$, $\cos^2 y = 1$, $\sec^2 z = 1$. Hence $x = (2m+1)\frac{\pi}{2}$, $y = n\pi$, $z = t\pi$, with $m, n, t \in \mathbb{Z}$.
3. No solution. $\sin^4 x = 1$ needs $\cos x = 0$, exactly where $\tan x$ is undefined; the two required families are disjoint.
4. Both are negative, so x is in Quadrant III with reference angle $\frac{\pi}{6}$: the common principal value is $\frac{7\pi}{6}$, and $x = 2n\pi + \frac{7\pi}{6}$.

5. The base satisfies $|\sin x + \cos x| \leq \sqrt{2}$ and the exponent satisfies $1 + \sin 2x \leq 2$, and $(\pm\sqrt{2})^2 = 2$ — so both must be extreme: $\sin x + \cos x = \pm\sqrt{2}$ and $\sin 2x = 1$. On $[-\pi, \pi]$ this gives $x = \frac{\pi}{4}$ (base $+\sqrt{2}$) and $x = -\frac{3\pi}{4}$ (base $-\sqrt{2}$, and the even exponent 2 makes the power positive). Both check: the value is 2 in each case.
6. Left side: $2\cos^2 \frac{x}{2} = 1 + \cos x \leq 2$ and $\sin^2 x \leq 1$, so the left side is at most 2. Right side: $x^2 + \frac{1}{x^2} \geq 2$ by AM-GM, with equality only at $x = 1$. So a solution would need both sides to equal 2, requiring $\cos x = 1$ and $\sin^2 x = 1$ simultaneously — impossible. (Numerically, on $(0, \frac{\pi}{2}]$ the left side never exceeds about 1.185 while the right side never falls below 2.)
7. $\tan \theta = -1$ in $[0, 2\pi)$ gives $\frac{3\pi}{4}$ and $\frac{7\pi}{4}$; $\cos \theta = \frac{1}{\sqrt{2}}$ gives $\frac{\pi}{4}$ and $\frac{7\pi}{4}$. The common value is $\frac{7\pi}{4}$, so $\theta = 2n\pi + \frac{7\pi}{4}$ (equivalently $2n\pi - \frac{\pi}{4}$).
8. Write $s = \sin^2 x \in [0, 1]$, so $\cos^2 x = 1 - s$ and the equation is $s^3 + (1 - s)^3 = 1$, i.e. $3s^2 - 3s = 0$, i.e. $s(s - 1) = 0$. Hence $\sin x = 0$ or $\cos x = 0$, and the two families combine into $x = \frac{n\pi}{2}$. (Boundedness route: $\sin^6 x \leq \sin^2 x$ and $\cos^6 x \leq \cos^2 x$, so the sum is at most 1 with equality only when each of $\sin^2 x, \cos^2 x$ is 0 or 1.)

Pitfalls specific to these two patterns

- **Stopping after one family.** The boundedness argument produces two conditions; the answer is their intersection, and the intersection may be empty. Always write both, then compare.
- **Ignoring the domain at the extreme points.** $\sin^2 x = 1$ and “ $\tan x$ appears in the equation” are incompatible. Check that the candidate points lie in the domain of both sides.
- **Using bounds that do not meet.** $f \leq 2$ with $g \geq 1$ proves nothing. The argument requires $\max f = \min g$; confirm the two numbers are equal before invoking it.
- **Generalising a simultaneous system too early.** Solve each equation in $[0, 2\pi)$ first, intersect, and only then add $2n\pi$. Combining two general solutions algebraically is error-prone and unnecessary.
- **Using period $n\pi$ for a system.** Two independent conditions fix the quadrant, so the period is 2π — even when one of the conditions involves $\tan$, whose period alone is π .
- **Forgetting that an even exponent rescues a negative base.** In Practice 5, $(-\sqrt{2})^2 = 2$, so $x = -\frac{3\pi}{4}$ is a genuine solution; discarding it because the base is negative loses half the answer.

Cheat Sheet Box

CHEAT SHEET — Boundedness and Simultaneous Equations ($n \in \mathbb{Z}$)

- **Boundedness principle:** if $f \leq M$ and $g \geq M$ everywhere, then $f = g$ forces $f = M$ and $g = M$. Solve both, then **intersect**, then check the domain.
- **Standard bounds:** $|\sin|, |\cos| \leq 1$; $|\sec|, |\csc| \geq 1$; $u + \frac{1}{u} \geq 2$ for $u > 0$; $a \sin x + b \cos x \in [-\sqrt{a^2 + b^2}, \sqrt{a^2 + b^2}]$; AM-GM $p + q \geq 2\sqrt{pq}$ for $p, q > 0$.
- **Empty intersection is a valid answer.** $\sin^4 x = 1 + \tan^8 x$ has no solution, because

$\sin^2 x = 1$ and $\tan x = 0$ cannot hold together.

- **Even powers rescue negative bases:** $(-\sqrt{2})^2 = 2$.
- **Simultaneous systems:** solve each equation **inside** $[0, 2\pi)$ **only**, intersect the two principal lists, then add $2n\pi$. Never $n\pi$.
- **Two conditions fix the quadrant**, which is why the period of a system is always 2π , whatever the individual periods.
- **Worked anchors:** $\sin^6 x = 1 + \cos^4 3x \Rightarrow x = (2n + 1)\frac{\pi}{2}$; $\sin \theta = -\frac{1}{2}$ with $\tan \theta = \frac{1}{\sqrt{3}} \Rightarrow \theta = 2n\pi + \frac{7\pi}{6}$; $2^{\sin x} + 2^{\cos x} = 2^{1-1/\sqrt{2}} \Rightarrow x = 2n\pi + \frac{5\pi}{4}$.
- **Golden rule:** when the two sides of an equation are different in kind, stop solving and start bounding.

Chapter 4: Complex Numbers and Quadratic Equations

Class 11 Mathematics / Drona content bank (Master Edition)

June 22, 2026

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Subtopic 01 — Algebra of Complex Numbers

1. Exam Relevance

This subtopic is the *load-bearing wall* of the chapter: the four arithmetic operations on complex numbers, the powers of i , and the algebraic structure they form. Everything downstream — modulus and conjugate (Subtopic 02), the Argand plane and polar form (03), quadratic equations with complex roots (04) — assumes this machinery is automatic.

- **CBSE Boards:** Direct 1-2 mark questions — simplify powers of i , express a number in $a + ib$ form, find x, y by equating real and imaginary parts, compute a multiplicative inverse.
- **JEE Main:** Operation fluency is assumed in every complex-number problem; standalone questions usually test powers of i or a condition for a quotient to be purely real/imaginary.
- **JEE Advanced:** Almost never tested in isolation, but a single slow step in arithmetic here costs time on the harder geometry/rotation problems.

Diagnostic note: failures here cascade into all later subtopics. This is the home of the operation-level CN.algebra tags; conjugate- and modulus-specific errors are diagnosed in Subtopic 02.

2. Concept Intuition (Indian-context analogies)

Why invent complex numbers? The equation $x^2 + 1 = 0$ has no real solution — no real number squares to a negative. This is the same discomfort earlier generations felt with $x + 5 = 2$, unsolvable until negatives were accepted. Every time a familiar equation refuses to be solved, the number system is enlarged; complex numbers are the enlargement that lets *every* polynomial equation have a root.

i is a quarter-turn, not a ghost. “Imaginary” is unfortunate PR. Picture a clock hand pointing right, along the positive real direction. Multiplying by i rotates it 90° anticlockwise, so it points *up*. Multiply by i again and you have turned 180° — the hand points left, i.e. you have reached -1 . That is the entire content of $i^2 = -1$: **two quarter-turns make a half-turn**. This also explains why $i^4 = 1$ (a full turn brings you home) and why the powers of i repeat every four steps.

A complex number is an address. In a planned-grid city like Chandigarh, “go 3 sectors east, then 2 sectors north” pins one location. Writing $z = 3 + 2i$ does the same: real part = east-west, imaginary part = north-south. Two addresses match only if *both* coordinates agree — which is exactly why equality of complex numbers needs two conditions, not one.

Diagram placeholder

Horizontal “real” axis, vertical “imaginary” axis. Show multiplication by i as a 90° -degree anticlockwise rotation: $(1,0) \rightarrow (0,1) \rightarrow (-1,0) \rightarrow (0,-1) \rightarrow (1,0)$, illustrating $i, i^2=-1, i^3=-i, i^4=1$. Schematic only; the full Argand plane is Subtopic 03.

3. Key Formulas and Definitions (with validity / domain conditions)

The imaginary unit.

$$i = \sqrt{-1}, \quad i^2 = -1.$$

Powers of i cycle with period 4:

$$i^1 = i, \quad i^2 = -1, \quad i^3 = -i, \quad i^4 = 1,$$

$$i^{4k} = 1, \quad i^{4k+1} = i, \quad i^{4k+2} = -1, \quad i^{4k+3} = -i \quad (k \in \mathbb{Z}).$$

To evaluate i^n , keep only the remainder of n on division by 4. For **negative** powers, add a multiple of 4 first: $i^{-1} = i^3 = -i$, $i^{-39} = i^{-39+40} = i$.

A useful consequence — **four consecutive powers of i sum to zero**:

$$i^n + i^{n+1} + i^{n+2} + i^{n+3} = i^n(1 + i + i^2 + i^3) = i^n(1 + i - 1 - i) = 0.$$

Validity condition (critical). The rule $\sqrt{a}\sqrt{b} = \sqrt{ab}$ holds for real a, b **only when they are not both negative**. For example $\sqrt{-2}\sqrt{-3} = (i\sqrt{2})(i\sqrt{3}) = i^2\sqrt{6} = -\sqrt{6}$, **not** $\sqrt{6}$. Convert each $\sqrt{\text{negative}}$ to $i\sqrt{\text{positive}}$ first. (tag: i2-

sign-flip)

A complex number.

$$z = a + ib, \quad a, b \in \mathbb{R},$$

with $a = \operatorname{Re}(z)$ the **real part** and $b = \operatorname{Im}(z)$ the **imaginary part**.

Trap: $\operatorname{Im}(z) = b$, a *real* number — not ib . (tag: *re-im-misassign*)

- z is **purely real** if $b = 0$; **purely imaginary** if $a = 0$, $b \neq 0$.
- $\mathbb{N} \subset \mathbb{W} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$.
- **Order is undefined.** “ $9 + 6i > 3 + 2i$ ” is meaningless; $<, >$ do not apply to non-real complex numbers. (tag: *order-relation-misuse*)

Equality.

$$a + ib = c + id \iff a = c \text{ and } b = d.$$

Both conditions are required; satisfying only one is the equality-partial error.

Operations. Let $z_1 = a + ib$, $z_2 = c + id$.

Operation	Formula	Condition
Addition	$z_1 + z_2 = (a + c) + i(b + d)$	—
Subtraction	$z_1 - z_2 = (a - c) + i(b - d)$	—
Multiplication	$z_1 z_2 = (ac - bd) + i(ad + bc)$	apply $i^2 = -1$
Division	$\frac{z_1}{z_2} = \frac{(ac + bd) + i(bc - ad)}{c^2 + d^2}$	$z_2 \neq 0$

Multiplicative inverse. For $z = a + ib \neq 0$,

$$z^{-1} = \frac{1}{z} = \frac{a - ib}{a^2 + b^2} = \frac{a}{a^2 + b^2} - i \frac{b}{a^2 + b^2}.$$

The number $a - ib$ used to carry out division is the **conjugate** of z ; the conjugate and the modulus $|z| = \sqrt{a^2 + b^2}$, together with their property lists, are developed fully in **Subtopic 02**. Here they appear only as the computational device that turns a complex denominator into a real one.

4. Theorems, Proofs & Standard Methods

Theorem 1 (Powers of i are periodic with period 4). For every integer n , $i^n \in \{1, i, -1, -i\}$, fixed by $n \bmod 4$.

Proof. Write $n = 4k + r$, $r \in \{0, 1, 2, 3\}$. Then $i^n = (i^4)^k i^r = 1^k i^r = i^r$, since $i^4 = (i^2)^2 = (-1)^2 = 1$. The four values of i^r are $1, i, -1, -i$. ■

Theorem 2 ($\mathbb{C}$ is a field). Under the addition and multiplication defined above, $\mathbb{C}$ satisfies the field axioms:

Addition — closure, commutativity, associativity; additive identity $0 = 0 + i0$; additive inverse $-z = -a + i(-b)$. *Multiplication* — closure, commutativity, associativity; multiplicative identity $1 = 1 + i0$; and (for $z \neq 0$) a multiplicative inverse. Multiplication distributes over addition.

The only non-routine axiom is the existence of multiplicative inverses, proved next.

Theorem 3 (Existence and uniqueness of the multiplicative inverse). Every $z \neq 0$ has a unique s with $zs = 1$, namely $s = \frac{a - ib}{a^2 + b^2}$.

Proof. Let $z = a + ib \neq 0$, so $a^2 + b^2 > 0$. With $s = \frac{a - ib}{a^2 + b^2}$,

$$zs = \frac{(a + ib)(a - ib)}{a^2 + b^2} = \frac{a^2 + b^2}{a^2 + b^2} = 1.$$

For uniqueness, if $zs' = 1$ too, then $s = s \cdot 1 = s(zs') = (sz)s' = 1 \cdot s' = s'$, using commutativity and associativity. ■

Standard Method A — Division (clearing the denominator). To write $\frac{z_1}{z_2}$ in $a + ib$ form, multiply numerator and denominator by $(c - id)$:

$$\frac{a + ib}{c + id} = \frac{(a + ib)(c - id)}{(c + id)(c - id)} = \frac{(a + ib)(c - id)}{c^2 + d^2}.$$

The denominator is now the real number $c^2 + d^2$; then separate real and imaginary parts. Leaving a complex denominator is the division-no-rationalize error.

Standard Method B — Equating real and imaginary parts. For an equation $A(x, y) + iB(x, y) = C + iD$ in real unknowns, set $A = C$ and $B = D$, giving two simultaneous real equations. This is the single most-tested technique in the subtopic.

5. Worked Examples (CBSE → JEE Main → JEE Advanced)

Example 1 — CBSE. Express $(3 + 2i) + (-6 + 5i)$ and $(2 + 3i)(1 - 4i)$ in $a + ib$ form.
Solution. $(3 + 2i) + (-6 + 5i) = -3 + 7i$. $(2 + 3i)(1 - 4i) = 2 - 8i + 3i - 12i^2 = 2 - 5i + 12 = 14 - 5i$.

Example 2 — CBSE. Evaluate $i^{135} + i^{-50}$. *Solution.* $135 = 4(33) + 3 \Rightarrow i^{135} = i^3 = -i$. And $i^{-50} = i^{-50+52} = i^2 = -1$. Sum $= -1 - i$.

Example 3 — CBSE. Find real x, y if $4x + i(3x - y) = 3 + i(-6)$. *Solution.* Equate parts (Method B): $4x = 3$ and $3x - y = -6$. So $x = \frac{3}{4}$, and $y = 3x + 6 = \frac{9}{4} + 6 = \frac{33}{4}$.

Example 4 — JEE Main. Find the multiplicative inverse of $z = 3 - 4i$ and verify it.

Solution. $a^2 + b^2 = 9 + 16 = 25$, so $z^{-1} = \frac{3 + 4i}{25} = \frac{3}{25} + \frac{4}{25}i$. Check: $(3 - 4i) \cdot \frac{3 + 4i}{25} = \frac{9 + 16}{25} = 1$. ✓

Example 5 — JEE Main. For real θ , when is $z = \frac{3 + 2i \sin \theta}{1 - 2i \sin \theta}$ purely real? *Solution.* Clear the denominator (Method A) with $1 + 2i \sin \theta$:

$$z = \frac{(3 + 2i \sin \theta)(1 + 2i \sin \theta)}{1 + 4 \sin^2 \theta} = \frac{(3 - 4 \sin^2 \theta) + 8i \sin \theta}{1 + 4 \sin^2 \theta}.$$

z purely real $\iff 8 \sin \theta = 0 \iff \theta = n\pi, n \in \mathbb{Z}$.

Example 6 — JEE Advanced. Evaluate $S = \sum_{n=1}^{100} i^{n!}$. *Solution.* For $n \geq 4$, $n!$ is a multiple of 4, so $i^{n!} = 1$. The first three terms: $i^{1!} = i^1 = i$, $i^{2!} = i^2 = -1$, $i^{3!} = i^6 = i^2 = -1$. Thus

$$S = \underbrace{i - 1 - 1}_{n=1,2,3} + \underbrace{1 + 1 + \dots + 1}_{97 \text{ terms}} = (i - 2) + 97 = 95 + i.$$

Example 7 — JEE Advanced. If $\left(\frac{1+i}{1-i}\right)^m = 1$, find the least positive integer m .

Solution. First simplify the base (Method A): $\frac{1+i}{1-i} = \frac{(1+i)^2}{(1-i)(1+i)} = \frac{2i}{2} = i$. So the condition is $i^m = 1$, true exactly when m is a multiple of 4. Least positive $m = 4$.

6. Practice Exercises (with answers)

- Express in $a + ib$: (a) $(5 - 3i) - (2 + 4i)$, (b) $(1 + i)^2$, (c) $(2 + i)(2 - i)$.
- Evaluate $i^{-39} + i^{16} + i^{257}$.
- Find real x, y if $(x + iy)(2 - 3i) = 4 + i$.
- Simplify $\frac{2-i}{1+i}$ to $a + ib$ form and state Re and Im.
- Find the multiplicative inverse of $\sqrt{5} + 3i$.
- Show that $i^n + i^{n+1} + i^{n+2} + i^{n+3} = 0$ for every integer n .
- Find the least positive integer n for which $(1 + i)^n$ is purely imaginary.

Answers.

- (a) $3 - 7i$; (b) $2i$; (c) 5.
- $i^{-39} = i^1 = i$; $i^{16} = 1$; $i^{257} = i^1 = i$; sum = $1 + 2i$.
- $x + iy = \frac{(4+i)(2+3i)}{13} = \frac{5+14i}{13} \Rightarrow x = \frac{5}{13}, y = \frac{14}{13}$.

4. $\frac{(2-i)(1-i)}{2} = \frac{1-3i}{2}$; $\text{Re} = \frac{1}{2}$, $\text{Im} = -\frac{3}{2}$.
5. $\frac{\sqrt{5}-3i}{5+9} = \frac{\sqrt{5}-3i}{14} = \frac{\sqrt{5}}{14} - \frac{3}{14}i$.
6. Factor $i^n(1+i+i^2+i^3) = i^n(1+i-1-i) = 0$.
7. $(1+i)^2 = 2i$, so $(1+i)^n$ is purely imaginary first at $n = 2$.
-

7. MCQ with Distractor Analysis

Q1. i^{4k+3} for integer k equals: (A) 1 (B) i (C) -1 (D) $-i$ **Ans: (D).** (A) reads i^{4k} , dropping the $+3$; (B) reads the remainder as 1; (C) is i^{4k+2} , off by one. (*tag: i-power-cycle-error*)

Q2. $\sqrt{-4} \cdot \sqrt{-9} =$ (A) 6 (B) -6 (C) $6i$ (D) 36 **Ans: (B).** $(2i)(3i) = 6i^2 = -6$. (A) is the illegal $\sqrt{(-4)(-9)} = 6$. (*tag: i2-sign-flip*)

Q3. The multiplicative inverse of i is: (A) i (B) $-i$ (C) 1 (D) -1 **Ans: (B).** $\frac{1}{i} = \frac{1}{i} \cdot \frac{-i}{-i} = \frac{-i}{1} = -i$. (A) assumes i is self-inverse; (D) confuses inverse with i^2 . (*tag: division-no-rationalize*)

Q4. If $a + ib = c + id$ (all real), then necessarily: (A) $a = c$ only (B) $b = d$ only (C) $a = c$ and $b = d$ (D) $a + b = c + d$ **Ans: (C).** (A),(B) are the partial-equality trap; (D) collapses a complex equation into one real equation. (*tag: equality-partial*)

Q5. $\sum_{n=0}^3 i^n$ equals: (A) 0 (B) 1 (C) i (D) $1+i$ **Ans: (A).** $1+i+i^2+i^3 = 1+i-1-i = 0$. (B) forgets i^2, i^3 ; (D) drops only the -1 and $-i$. (*tag: i-power-cycle-error*)

8. Common Pitfalls, Pro-Tips & Cheat Sheet

Common Pitfalls

- Treating i^2 as $+1$ or leaving it un-simplified — replace i^2 with -1 immediately. (*tag: i2-sign-flip*)
- Combining roots of negatives directly: $\sqrt{-2}\sqrt{-3} \neq \sqrt{6}$; convert to $i\sqrt{2} \cdot i\sqrt{3} = -\sqrt{6}$ first.
- Writing $\text{Im}(z) = bi$ — the imaginary part is the real number b . (*tag: re-im-misassign*)
- Applying $<, >$ to complex numbers — order is undefined on $\mathbb{C}$. (*tag: order-relation-misuse*)

- Leaving a complex denominator — not standard form. (*tag: division-no-rationalize*)
- Equating only the real or only the imaginary part. (*tag: equality-partial*)

Pro-Tips

- For i^n , only $n \bmod 4$ matters; for negative n , add a multiple of 4 first.
- Any four consecutive powers of i sum to 0 — use this to collapse long sums (Examples 6, and Practice 6).
- To show a quotient is purely real/imaginary, clear the denominator first, then read off the part that must vanish.
- $(1 + i)^2 = 2i$ and $(1 - i)^2 = -2i$ are worth memorising — they shortcut many power problems (Example 7).

Cheat Sheet Box

- $i^2 = -1$; $i^{4k} = 1$, $i^{4k+1} = i$, $i^{4k+2} = -1$, $i^{4k+3} = -i$
- $i^n + i^{n+1} + i^{n+2} + i^{n+3} = 0$
- $z = a + ib$: $\operatorname{Re}(z) = a$, $\operatorname{Im}(z) = b$ (both real)
- Equality: $a + ib = c + id \iff a = c$ and $b = d$
- $z_1 z_2 = (ac - bd) + i(ad + bc)$
- Division: multiply top & bottom by $(c - id)$ to make the denominator real
- $z^{-1} = \frac{a - ib}{a^2 + b^2}$ ($z \neq 0$)
- $(1 + i)^2 = 2i$, $(1 - i)^2 = -2i$
- Order undefined on $\mathbb{C}$
- *Conjugate & modulus theory* $\rightarrow$ *Subtopic 02*

Subtopic 02 — The Modulus and the Conjugate of a Complex Number

Builds directly on Subtopic 01 (algebra of operations). The conjugate $a - ib$ that appeared there only as a *tool for division* is now studied as an object in its own right, together with the modulus $|z| = \sqrt{a^2 + b^2}$.

1. Exam Relevance

The modulus and the conjugate are the **algebraic engine** of the whole chapter. The single identity $z\bar{z} = |z|^2$ is what converts every “divide by a complex number” or “prove this is real” problem into ordinary real arithmetic, and it is used silently in almost every later problem on rotation, loci, and polynomial roots.

- **CBSE Boards:** Reliable 1-2 mark questions — find the conjugate and modulus of a given number, find the multiplicative inverse (via $\bar{z}/|z|^2$), or use $|z_1 z_2| = |z_1||z_2|$.
- **JEE Main:** Conditions for a quotient to be purely real/imaginary, modulus of a product, and modulus inequalities appear every year.
- **JEE Advanced:** The property toolkit ($z\bar{z} = |z|^2$, $\overline{z_1 z_2} = \bar{z}_1 \bar{z}_2$, triangle inequality) is assumed fluent; problems rarely state it but always rely on it.

Diagnostic note: this is the home of the cross-cutting tags conjugate-sign-error and modulus-vs-conjugate-confusion, both of which resurface in Subtopic 03 (polar conjugate) — they should hang off a shared parent in the knowledge-tracing graph.

2. Concept Intuition (Indian-context analogies)

The conjugate is a mirror reflection. Picture a temple beside a still water tank — the Dal Lake reflecting the Hari Parbat, or any tank-temple at dawn. The reflection sits exactly as far below the waterline as the temple sits above it. **The real axis is the waterline:** the conjugate $\bar{z} = a - ib$ is the reflection of $z = a + ib$ across it. The real part (height along the shore) is unchanged; only the imaginary part (height above/below water) flips sign. This is why a number in the first quadrant has its conjugate in the fourth, and why $z = \bar{z}$ exactly when z lies *on* the waterline (is purely real).

The modulus is “as the crow flies.” Subtopic 01 described $z = a + ib$ as a grid address (“3 sectors east, 2 north”). The modulus $|z| = \sqrt{a^2 + b^2}$ is the **straight-line distance** from the origin to that address — Pythagoras on the grid. The address tells you *how to walk there*; the modulus tells you *how far it is*. (The full geometric picture — the Argand plane — is Subtopic 03; here we use $|z|$ as a magnitude.)

Why $z\bar{z} = |z|^2$ is magic. Multiplying a number by its mirror image cancels the imaginary part entirely: $(a + ib)(a - ib) = a^2 + b^2$, a plain non-negative real number

equal to the squared distance. That is the trick behind every division and every “show it’s real” proof.

Diagram placeholder

Real (horizontal) and imaginary (vertical) axes. Plot $z = a+ib$ in the first quadrant and its conjugate $\bar{z} = a-ib$ in the fourth, mirror images across the real axis. Mark $|z|$ as the length of the segment from O to z . Schematic; full Argand treatment is Subtopic 03.

3. Key Formulas and Definitions (with validity / domain conditions)

Conjugate. For $z = a + ib$,

$$\bar{z} = a - ib.$$

Only the sign of the imaginary part flips. (Contrast: the *additive inverse* is $-z = -a - ib$, where **both** signs flip — a common mix-up.)

Conjugate properties:

$$\begin{aligned} \overline{(\bar{z})} &= z, & z + \bar{z} &= 2 \operatorname{Re}(z), & z - \bar{z} &= 2i \operatorname{Im}(z), \\ z\bar{z} &= a^2 + b^2 = |z|^2 \text{ (purely real, } \geq 0), \\ z = \bar{z} &\iff z \text{ purely real,} & z + \bar{z} = 0 &\iff z \text{ purely imaginary,} \\ \overline{z_1 \pm z_2} &= \bar{z}_1 \pm \bar{z}_2, & \overline{z_1 z_2} &= \bar{z}_1 \bar{z}_2, & \overline{\left(\frac{z_1}{z_2}\right)} &= \frac{\bar{z}_1}{\bar{z}_2} \text{ } (z_2 \neq 0), & \overline{z^n} &= (\bar{z})^n. \end{aligned}$$

Modulus. For $z = a + ib$,

$$|z| = \sqrt{a^2 + b^2} = \sqrt{[\operatorname{Re}(z)]^2 + [\operatorname{Im}(z)]^2} \geq 0.$$

Validity / sign condition. $|z|$ is always the **non-negative** square root. The real-number shortcut $|x| = \pm x$ does **not** carry over — you cannot write $|z| = \pm z$ for complex z . (tag: *modulus-sign-nonneg*)

Modulus properties:

$$\begin{aligned} |z| = 0 &\iff z = 0, & |z| &= |\bar{z}| = |-z|, & z\bar{z} &= |z|^2, \\ |z_1 z_2| &= |z_1| |z_2|, & \left|\frac{z_1}{z_2}\right| &= \frac{|z_1|}{|z_2|} \text{ } (z_2 \neq 0), & |z^n| &= |z|^n, \\ |\operatorname{Re}(z)| &\leq |z|, & |\operatorname{Im}(z)| &\leq |z|. \end{aligned}$$

Triangle inequality (with equality conditions):

$$|z_1 + z_2| \leq |z_1| + |z_2|, \quad ||z_1| - |z_2|| \leq |z_1 - z_2|.$$

Equality in the first holds **iff** z_1, z_2 **point in the same direction** (one is a non-negative real multiple of the other). It is **not** true in general that $|z_1 + z_2| = |z_1| + |z_2|$. (*tag: modulus-of-sum-distributes*)

Parallelogram identity:

$$|z_1 + z_2|^2 + |z_1 - z_2|^2 = 2(|z_1|^2 + |z_2|^2).$$

Multiplicative inverse (restated via the conjugate):

$$z^{-1} = \frac{\bar{z}}{|z|^2} = \frac{a - ib}{a^2 + b^2} \quad (z \neq 0).$$

4. Theorems, Proofs & Standard Methods

Theorem 1 ($z\bar{z} = |z|^2$). For $z = a + ib$, $z\bar{z} = a^2 + b^2 = |z|^2$. *Proof.* $z\bar{z} = (a + ib)(a - ib) = a^2 - (ib)^2 = a^2 - i^2b^2 = a^2 + b^2 = |z|^2$. ■ This is the workhorse identity: it turns the complex quantity z into the real quantity $|z|^2$.

Theorem 2 (Conjugate of a product). $\overline{z_1 z_2} = \bar{z}_1 \bar{z}_2$. *Proof.* Let $z_1 = a + ib$, $z_2 = c + id$. Then $z_1 z_2 = (ac - bd) + i(ad + bc)$, so $\overline{z_1 z_2} = (ac - bd) - i(ad + bc)$. Separately, $\bar{z}_1 \bar{z}_2 = (a - ib)(c - id) = (ac - bd) - i(ad + bc)$. The two agree. ■

Theorem 3 (Modulus is multiplicative). $|z_1 z_2| = |z_1| |z_2|$. *Proof.* Using Theorems 1 and 2,

$$|z_1 z_2|^2 = (z_1 z_2) \overline{(z_1 z_2)} = z_1 z_2 \bar{z}_1 \bar{z}_2 = (z_1 \bar{z}_1)(z_2 \bar{z}_2) = |z_1|^2 |z_2|^2.$$

Both sides are non-negative, so $|z_1 z_2| = |z_1| |z_2|$. ■

Theorem 4 (Triangle inequality). $|z_1 + z_2| \leq |z_1| + |z_2|$. *Proof.* $|z_1 + z_2|^2 = (z_1 + z_2) \overline{(z_1 + z_2)} = |z_1|^2 + |z_2|^2 + (z_1 \bar{z}_2 + \bar{z}_1 z_2)$. The bracket equals $2 \operatorname{Re}(z_1 \bar{z}_2) \leq 2|z_1 \bar{z}_2| = 2|z_1| |z_2|$. Hence $|z_1 + z_2|^2 \leq (|z_1| + |z_2|)^2$, and taking non-negative roots gives the result. Equality requires $\operatorname{Re}(z_1 \bar{z}_2) = |z_1| |z_2|$, i.e. $z_1 \bar{z}_2$ is a non-negative real — the two point the same way. ■

Standard Method A — Proving a number is real or imaginary using the conjugate. z is purely real $\iff z = \bar{z}$; purely imaginary $\iff z = -\bar{z}$ (i.e. $z + \bar{z} = 0$). Taking the conjugate of a whole expression and comparing is often faster than expanding into $a + ib$.

Standard Method B — Modulus of a product/power. Take $|\cdot|$ across factors instead of multiplying out: $|\prod z_k| = \prod |z_k|$ (Theorem 3 iterated). Indispensable for long products.

5. Worked Examples (CBSE → JEE Main → JEE Advanced)

Example 1 — CBSE. Find the conjugate and modulus of $z = 3 + i$ and of $w = 2 - 5i$.
Solution. $\bar{z} = 3 - i$, $|z| = \sqrt{3^2 + 1^2} = \sqrt{10}$. $\bar{w} = 2 + 5i$, $|w| = \sqrt{2^2 + (-5)^2} = \sqrt{29}$.

Example 2 — CBSE. Express $(5-3i)^3$ in $a+ib$ form, then state its conjugate. *Solution.* $(5-3i)^3 = 5^3 - 3(5^2)(3i) + 3(5)(3i)^2 - (3i)^3 = 125 - 225i - 135 + 27i = -10 - 198i$. Conjugate = $-10 + 198i$.

Example 3 — JEE Main. Find the multiplicative inverse of $z = 4 - 3i$ using $\bar{z}/|z|^2$.
Solution. $|z|^2 = 16 + 9 = 25$, $\bar{z} = 4 + 3i$, so $z^{-1} = \frac{4+3i}{25} = \frac{4}{25} + \frac{3}{25}i$.

Example 4 — JEE Main. If $a + ib = \frac{c+i}{c-i}$ where $c \in \mathbb{R}$, prove that $a^2 + b^2 = 1$.

Solution. Take the modulus of both sides. $|a+ib| = \left| \frac{c+i}{c-i} \right| = \frac{|c+i|}{|c-i|} = \frac{\sqrt{c^2+1}}{\sqrt{c^2+1}} = 1$.

Hence $\sqrt{a^2+b^2} = 1$, so $a^2 + b^2 = 1$. ■

Example 5 — JEE Advanced. If z satisfies $z^2 = (\bar{z})^2$, show that z is purely real or purely imaginary. *Solution.* With $z = x + iy$: $z^2 = (x^2 - y^2) + 2ixy$ and $(\bar{z})^2 = (x^2 - y^2) - 2ixy$. Equating forces $2xy = -2xy$, so $4xy = 0 \Rightarrow xy = 0$. Thus $x = 0$ (purely imaginary) or $y = 0$ (purely real). ■

Example 6 — JEE Advanced. If $(2+i)(2+2i)(2+3i)\cdots(2+ni) = x + iy$, find $5 \cdot 8 \cdot 13 \cdots (4+n^2)$. *Solution.* Apply Method B (modulus across factors). $|2+ki| = \sqrt{4+k^2}$, so

$$\prod_{k=1}^n \sqrt{4+k^2} = |x+iy| = \sqrt{x^2+y^2}.$$

Squaring, $\prod_{k=1}^n (4+k^2) = x^2 + y^2$. The left side is exactly $5 \cdot 8 \cdot 13 \cdots (4+n^2)$, so the value is $\boxed{x^2 + y^2}$.

6. Practice Exercises (with answers)

- Find the conjugate and modulus of (a) $-3 + 4i$, (b) $\frac{1+i}{1-i}$.
- If $z = \frac{2+i}{(1-i)}$, find $z\bar{z}$.
- Prove that $\left| \frac{z_1}{z_2} \right| = \frac{|z_1|}{|z_2|}$ for $z_2 \neq 0$.
- Show that $\frac{z-1}{z+1}$ is purely imaginary $\iff |z| = 1$ (with $z \neq -1$).
- Verify the parallelogram identity for $z_1 = 3 + i$, $z_2 = 1 - 2i$.

6. If $|z_1| = |z_2| = 1$ and $z_1 \bar{z}_2 \neq -1$, show $\frac{z_1 + z_2}{1 + z_1 \bar{z}_2}$ is real.
7. Find the modulus of $(1 + i)^{10}$.

Answers.

1. (a) conjugate $-3 - 4i$, modulus 5; (b) $\frac{1+i}{1-i} = i$, conjugate $-i$, modulus 1.
2. $z = \frac{(2+i)(1+i)}{2} = \frac{1+3i}{2}$, so $z\bar{z} = |z|^2 = \frac{1+9}{4} = \frac{10}{4} = \frac{5}{2}$.
3. $\left|\frac{z_1}{z_2}\right|^2 = \frac{z_1}{z_2} \overline{\left(\frac{z_1}{z_2}\right)} = \frac{z_1 \bar{z}_1}{z_2 \bar{z}_2} = \frac{|z_1|^2}{|z_2|^2}$; take roots.
4. Let $w = \frac{z-1}{z+1}$. w purely imaginary $\iff w + \bar{w} = 0 \iff \frac{z-1}{z+1} + \frac{\bar{z}-1}{\bar{z}+1} = 0 \iff z\bar{z} = 1 \iff |z| = 1$.
5. LHS $|z_1 + z_2|^2 + |z_1 - z_2|^2 = |4 - i|^2 + |2 + 3i|^2 = 17 + 13 = 30$; RHS $2(|z_1|^2 + |z_2|^2) = 2(10 + 5) = 30$. ✓
6. The expression equals its own conjugate (use $|z_1| = |z_2| = 1 \Rightarrow \bar{z}_1 = 1/z_1, \bar{z}_2 = 1/z_2$); hence real.
7. $|(1 + i)^{10}| = |1 + i|^{10} = (\sqrt{2})^{10} = 2^5 = 32$.

7. MCQ with Distractor Analysis

Q1. For any complex z , $z\bar{z}$ equals: (A) $|z|$ (B) $|z|^2$ (C) $\bar{z}^2$ (D) $2 \operatorname{Re}(z)$ **Ans: (B).** (A) drops the square; (C) confuses $z\bar{z}$ with $(\bar{z})^2$; (D) is $z + \bar{z}$. (*tag: modulus-vs-conjugate-confusion*)

Q2. If $z = -3 - 4i$, then $|z| =$ (A) -5 (B) 5 (C) 7 (D) $\sqrt{7}$ **Ans: (B).** $\sqrt{9 + 16} = 5$. (A) wrongly allows a negative modulus; (C) adds $|-3| + |-4|$ instead of using Pythagoras. (*tags: modulus-sign-nonneg, modulus-of-sum-distributes*)

Q3. $\overline{z_1 z_2}$ equals: (A) $\bar{z}_1 + \bar{z}_2$ (B) $\bar{z}_1 \bar{z}_2$ (C) $z_1 \bar{z}_2$ (D) $\bar{z}_1 + z_2$ **Ans: (B).** (A) uses the *sum* rule for a product; (C),(D) conjugate only one factor. (*tag: conjugate-sign-error*)

Q4. The conjugate of $z = 2 + 3i$ is, and its additive inverse is, respectively:

- (A) $-2 - 3i$ and $2 - 3i$ (B) $2 - 3i$ and $-2 - 3i$ (C) $2 - 3i$ and $-2 + 3i$ (D) $-2 + 3i$ and $2 - 3i$ **Ans: (B).** Conjugate flips only the imaginary sign ($2 - 3i$); the additive inverse flips both ($-2 - 3i$). (A) swaps the two. (*tag: conjugate-sign-error*)

Q5. For all complex z_1, z_2 , $|z_1 + z_2|$ is: (A) $= |z_1| + |z_2|$ (B) $\geq |z_1| + |z_2|$ (C) $\leq |z_1| + |z_2|$ (D) $= |z_1| - |z_2|$ **Ans: (C).** Triangle inequality. (A) holds only when they point the same way; (B) reverses it. (*tag: modulus-of-sum-distributes*)

8. Common Pitfalls, Pro-Tips & Cheat Sheet

Common Pitfalls

- $|z_1 + z_2| \neq |z_1| + |z_2|$ **in general** — the modulus does not distribute over a sum; only $\leq$ holds. (*tag: modulus-of-sum-distributes*)
- **Writing** $|z| = \pm z$. The modulus is a non-negative *real* number; the real-number rule $|x| = \pm x$ does not carry over. (*tag: modulus-sign-nonneg*)
- **Confusing** $\bar{z}$ **with** $-z$. Conjugate flips only the imaginary sign; the additive inverse flips both. (*tag: conjugate-sign-error*)
- **Confusing** $z\bar{z}$ **with** $\bar{z}^2$ **or with** $|z|$. $z\bar{z} = |z|^2$ — a non-negative real. (*tag: modulus-vs-conjugate-confusion*)
- **Conjugating a product as a sum:** $\overline{z_1 z_2} = \bar{z}_1 \bar{z}_2$, not $\bar{z}_1 + \bar{z}_2$.

Pro-Tips

- Reach for $z\bar{z} = |z|^2$ to clear denominators, prove realness, and convert moduli into algebra.
- To prove “ E is real,” show $E = \bar{E}$; to prove “imaginary,” show $E + \bar{E} = 0$ — usually faster than expanding.
- For moduli of products, powers, and quotients, take $|\cdot|$ factor-by-factor (Examples 4, 6; Practice 7).
- On the unit circle ($|z| = 1$), $\bar{z} = \frac{1}{z}$ — a powerful substitution (Practice 6).

Cheat Sheet Box

- $\bar{z} = a - ib$; reflection across the real axis (1st $\leftrightarrow$ 4th quadrant)
- $z + \bar{z} = 2 \operatorname{Re}(z)$; $z - \bar{z} = 2i \operatorname{Im}(z)$
- $z\bar{z} = |z|^2 = a^2 + b^2$
- $z = \bar{z} \iff$ real; $z + \bar{z} = 0 \iff$ imaginary
- $\overline{z_1 z_2} = \bar{z}_1 \bar{z}_2$; $\overline{z_1 \pm z_2} = \bar{z}_1 \pm \bar{z}_2$; $\overline{z^n} = (\bar{z})^n$
- $|z| = \sqrt{a^2 + b^2} \geq 0$; $|z| = |\bar{z}| = |-z|$
- $|z_1 z_2| = |z_1| |z_2|$; $|z_1 / z_2| = |z_1| / |z_2|$; $|z^n| = |z|^n$
- $|z_1 + z_2| \leq |z_1| + |z_2|$ (triangle); $||z_1| - |z_2|| \leq |z_1 - z_2|$
- $|z_1 + z_2|^2 + |z_1 - z_2|^2 = 2(|z_1|^2 + |z_2|^2)$
- $z^{-1} = \bar{z} / |z|^2$; on $|z| = 1$: $\bar{z} = 1/z$

Subtopic 03 — The Argand Plane and Polar Representation

Subtopics 01-02 treated complex numbers algebraically. Here they become **points and arrows in a plane**: the modulus (already defined in 02) gains its geometric meaning as a *length*, and a new quantity — the *argument* — supplies *direction*. Together they give the polar form, the launchpad for rotation and De Moivre's theorem.

1. Exam Relevance

- **CBSE Boards:** Direct questions on writing a number in polar form $r(\cos \theta + i \sin \theta)$, and on finding modulus and (principal) argument. The four-quadrant argument calculation is a standard 2-3 mark item.
- **JEE Main:** Argument algebra, polar conversions, and simple De Moivre powers appear every year.
- **JEE Advanced:** The geometric viewpoint — multiplication as *rotation + scaling*, loci via $|z - z_0|$ and $\arg$ — underlies most of the hard problems. Polar form is the single most leveraged idea in the chapter.

Diagnostic note: the signature trap here is argument-quadrant-error — blindly returning $\tan^{-1}(y/x)$ without fixing the quadrant. It is one of the cleanest discriminators of rote-vs-conceptual understanding in the whole chapter.

2. Concept Intuition (Indian-context analogies)

The Argand plane makes the “address” literal. In Subtopic 01 we read $z = x + iy$ as a grid address. The **Argand plane** (complex plane) simply draws it: the horizontal axis is real, the vertical axis is imaginary, and z is the point (x, y) — or the arrow from the origin to that point.

Modulus = distance, argument = bearing. Flying a kite from a Sankranti rooftop, two numbers fix the kite: the **length of string** r (how far — this is $|z|$) and the **angle** the string makes with the horizon θ (which way — this is $\arg z$). An auto-rickshaw direction works the same way: “5 km at 30° north-of-east” is polar; “5 km east, then north a bit” is Cartesian. Same point, two descriptions.

Polar form is the bearing description. Converting $x + iy$ to $r(\cos \theta + i \sin \theta)$ is just translating *grid coordinates* into *distance-and-bearing*. The conversion back uses $x = r \cos \theta$, $y = r \sin \theta$.

Multiplication is rotation. Multiplying by a number of modulus r and argument ϕ **scales by r and rotates by ϕ** . Multiplying by i ($r = 1$, $\phi = 90^\circ$) is a pure quarter-turn — exactly the picture from Subtopic 01, now made precise. Repeated rotation is De Moivre's theorem.

Diagram placeholder

Argand plane. Real (horizontal) and imaginary (vertical) axes. Plot $P(x,y)$ for z . Draw arrow OP , label its length $|z| = r$, and the angle θ from the positive real axis to OP as $\arg(z)$. Show $x = r \cos \theta$, $y = r \sin \theta$ as the projections onto the axes.

3. Key Formulas and Definitions (with validity / domain conditions)

Argand plane. $z = x + iy$ is represented by the point (x, y) (equivalently, the vector $\overrightarrow{OP}$). The horizontal axis is the **real axis**, the vertical the **imaginary axis**.

Modulus (geometric). $|z| = \sqrt{x^2 + y^2}$ is the **distance of P from the origin**, $|z| \geq 0$. (Algebraic properties were established in Subtopic 02.)

Argument / amplitude. If $\overrightarrow{OP}$ makes angle θ with the positive real axis, θ is an **argument** of z :

$$\tan \theta = \frac{y}{x}.$$

- The argument is **many-valued**: if θ works, so does $\theta + 2n\pi$, $n \in \mathbb{Z}$. Any two arguments differ by a multiple of 2π .
- The **principal value** (amplitude), written $\arg z$, is the unique θ with

$$-\pi < \theta \leq \pi.$$

Validity conditions.

- $\arg(0)$ is **undefined** — the origin has a modulus but no direction. (*tag: arg-zero-defined*)
- The principal value must lie in $(-\pi, \pi]$; a value outside this range is not the principal argument. (*tag: principal-range-violation*)
- $\tan^{-1}(y/x)$ on a calculator returns an angle in $(-\frac{\pi}{2}, \frac{\pi}{2})$ only — you **must** correct by quadrant. (*tag: argument-quadrant-error*)

Finding the principal argument by quadrant. Let $\alpha = \tan^{-1} \left| \frac{y}{x} \right|$ be the acute *reference angle*. Then:

Quadrant of (x, y)	Signs	Principal $\arg z$
I	$x > 0, y > 0$	α
II	$x < 0, y > 0$	$\pi - \alpha$
III	$x < 0, y < 0$	$-(\pi - \alpha) = \alpha - \pi$
IV	$x > 0, y < 0$	$-\alpha$

Axis cases: $\arg z = 0$ if z is real > 0 ; $\arg z = \pi$ if real < 0 ; $\arg z = \frac{\pi}{2}$ if purely imaginary with $y > 0$; $\arg z = -\frac{\pi}{2}$ if $y < 0$.

Polar (trigonometric) form. With $r = |z|$ and $\theta = \arg z$,

$$z = r(\cos \theta + i \sin \theta) \quad \text{often written } z = r \operatorname{cis} \theta,$$

where $x = r \cos \theta$, $y = r \sin \theta$. The standard form uses a $+$ between cosine and sine; $r(\cos \theta - i \sin \theta)$ is **not** standard polar — rewrite it as $r(\cos(-\theta) + i \sin(-\theta))$. (tag: polar-sign-mismatch)

Argument algebra (equalities hold up to multiples of 2π):

$$\arg(z_1 z_2) = \arg z_1 + \arg z_2, \quad \arg\left(\frac{z_1}{z_2}\right) = \arg z_1 - \arg z_2, \quad \arg(\bar{z}) = -\arg z.$$

Conjugate in polar form. $\bar{z} = r(\cos \theta - i \sin \theta) = r(\cos(-\theta) + i \sin(-\theta))$ — same modulus, **negated** argument ($r\angle -\theta$). (cross-cutting tag: conjugate-sign-error)

De Moivre's theorem (JEE-Advanced tier; beyond strict NCERT Class-11 scope). For integer n ,

$$(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta, \quad \text{so } z^n = r^n(\cos n\theta + i \sin n\theta).$$

For rational $n = p/q$ (in lowest terms), $\cos n\theta + i \sin n\theta$ is **one of** the q values of $(\cos \theta + i \sin \theta)^n$.

Cube roots of unity (application of De Moivre). The roots of $z^3 = 1$ are $1, \omega, \omega^2$ with

$$\omega = \cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} = -\frac{1}{2} + \frac{\sqrt{3}}{2}i, \quad \omega^3 = 1, \quad 1 + \omega + \omega^2 = 0.$$

4. Theorems, Proofs & Standard Methods

Theorem 1 (Polar form exists for every $z \neq 0$). Any non-zero $z = x + iy$ can be written as $r(\cos \theta + i \sin \theta)$ with $r = |z| > 0$ and some real θ . *Proof.* Set $r = \sqrt{x^2 + y^2} > 0$. Since $\left(\frac{x}{r}\right)^2 + \left(\frac{y}{r}\right)^2 = 1$, there is an angle θ with $\cos \theta = \frac{x}{r}$, $\sin \theta = \frac{y}{r}$. Then $r(\cos \theta + i \sin \theta) = x + iy = z$. ■

Theorem 2 (Multiplication = rotation + scaling). If $z_1 = r_1(\cos \theta_1 + i \sin \theta_1)$ and $z_2 = r_2(\cos \theta_2 + i \sin \theta_2)$, then

$$z_1 z_2 = r_1 r_2 (\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)).$$

Proof. Multiply and group:

$$z_1 z_2 = r_1 r_2 [(\cos \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2) + i(\sin \theta_1 \cos \theta_2 + \cos \theta_1 \sin \theta_2)].$$

The bracketed real and imaginary parts are $\cos(\theta_1 + \theta_2)$ and $\sin(\theta_1 + \theta_2)$ by the angle-sum identities. ■ So moduli **multiply** and arguments **add** — the algebraic root of “multiplication is rotation.”

Theorem 3 (De Moivre, integer n). $(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$ for all $n \in \mathbb{Z}$. *Proof* ($n \geq 0$, induction). Base $n = 0$: both sides = 1. Assume true for $n = k$. Then by Theorem 2,

$$(\cos \theta + i \sin \theta)^{k+1} = (\cos k\theta + i \sin k\theta)(\cos \theta + i \sin \theta) = \cos(k+1)\theta + i \sin(k+1)\theta.$$

For $n < 0$, write $n = -m$ and use $(\cos \theta + i \sin \theta)^{-1} = \cos \theta - i \sin \theta = \cos(-\theta) + i \sin(-\theta)$. ■

Standard Method — Cartesian $\rightarrow$ polar. (1) Compute $r = \sqrt{x^2 + y^2}$. (2) Compute the reference angle $\alpha = \tan^{-1} |y/x|$. (3) Place θ in the correct quadrant using the table. (4) Write $z = r(\cos \theta + i \sin \theta)$. The quadrant step (3) is where most marks are lost.

5. Worked Examples (CBSE $\rightarrow$ JEE Main $\rightarrow$ JEE Advanced)

Example 1 — CBSE. Find the modulus and principal argument of $z = 1 + i$, and write it in polar form. *Solution.* $r = \sqrt{1^2 + 1^2} = \sqrt{2}$; the point is in Q1 with $\alpha = \tan^{-1} 1 = \frac{\pi}{4}$, so $\arg z = \frac{\pi}{4}$. Polar: $z = \sqrt{2}(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4})$.

Example 2 — CBSE. Express $z = -1 + i$ in polar form. *Solution.* $r = \sqrt{2}$; point in Q2, $\alpha = \tan^{-1} 1 = \frac{\pi}{4}$, so $\arg z = \pi - \frac{\pi}{4} = \frac{3\pi}{4}$. Polar: $z = \sqrt{2}(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4})$.

Example 3 — JEE Main. Find the modulus and principal argument of each of $\pm 1 \pm i\sqrt{3}$. *Solution.* All four have $r = \sqrt{1+3} = 2$ and reference angle $\alpha = \tan^{-1} \sqrt{3} = \frac{\pi}{3}$. By quadrant:

$$\arg(1+i\sqrt{3}) = \frac{\pi}{3}, \quad \arg(-1+i\sqrt{3}) = \frac{2\pi}{3}, \quad \arg(-1-i\sqrt{3}) = -\frac{2\pi}{3}, \quad \arg(1-i\sqrt{3}) = -\frac{\pi}{3}.$$

This single example exercises every row of the quadrant table.

Example 4 — JEE Main. Write the polar form of $z = -\sqrt{3} + i$ and hence find z^6 . *Solution.* $r = \sqrt{3+1} = 2$; Q2 with $\alpha = \tan^{-1} \frac{1}{\sqrt{3}} = \frac{\pi}{6}$, so $\arg z = \pi - \frac{\pi}{6} = \frac{5\pi}{6}$. Thus $z = 2(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6})$. By De Moivre, $z^6 = 2^6(\cos 6 \cdot \frac{5\pi}{6} + i \sin 5\pi) = 64(\cos 5\pi + i \sin 5\pi) = 64(-1 + 0i) = -64$.

Example 5 — JEE Advanced. Use De Moivre to evaluate $(1+i)^{20}$. *Solution.* $1+i = \sqrt{2}(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4})$, so

$$(1+i)^{20} = (\sqrt{2})^{20}(\cos \frac{20\pi}{4} + i \sin \frac{20\pi}{4}) = 2^{10}(\cos 5\pi + i \sin 5\pi) = 1024(-1) = -1024.$$

Example 6 — JEE Advanced. Show that $1 + \omega + \omega^2 = 0$ for the cube roots of unity, and simplify $(1 + \omega)^3$. *Solution.* The roots of $z^3 - 1 = (z-1)(z^2 + z + 1) = 0$ are 1 and the roots of $z^2 + z + 1 = 0$, namely ω, ω^2 . By the sum of roots of $z^2 + z + 1$, $\omega + \omega^2 = -1$, hence $1 + \omega + \omega^2 = 0$. Then $1 + \omega = -\omega^2$, so $(1 + \omega)^3 = (-\omega^2)^3 = -\omega^6 = -(\omega^3)^2 = -1$. ■

6. Practice Exercises (with answers)

- Find modulus and principal argument: (a) $-\sqrt{3} - i$, (b) $-2i$, (c) 3.
- Express in polar form: (a) $1 - i$, (b) -3 .
- Convert $z = 4\left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}\right)$ to $a + ib$ form.
- Using De Moivre, evaluate $\left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}\right)^9$.
- Find the principal argument of $\frac{1+i}{1-i}$.
- If $z = r(\cos \theta + i \sin \theta)$, write $\bar{z}$ and $\frac{1}{z}$ in polar form.
- Evaluate $(1 + i\sqrt{3})^6$ via polar form.

Answers.

- (a) $r = 2$, Q3, $\alpha = \frac{\pi}{6}$, $\arg = -\frac{5\pi}{6}$; (b) $r = 2$, $\arg = -\frac{\pi}{2}$; (c) $r = 3$, $\arg = 0$.
 - (a) $\sqrt{2}\left(\cos(-\frac{\pi}{4}) + i \sin(-\frac{\pi}{4})\right)$; (b) $3(\cos \pi + i \sin \pi)$.
 - $4\left(-\frac{1}{2} + i\frac{\sqrt{3}}{2}\right) = -2 + 2\sqrt{3}i$.
 - $\cos 3\pi + i \sin 3\pi = -1$.
 - $\frac{1+i}{1-i} = i$, so principal argument $= \frac{\pi}{2}$.
 - $\bar{z} = r(\cos \theta - i \sin \theta) = r(\cos(-\theta) + i \sin(-\theta))$; $\frac{1}{z} = \frac{1}{r}(\cos(-\theta) + i \sin(-\theta))$.
 - $r = 2$, $\arg = \frac{\pi}{3}$; $(2)^6(\cos 2\pi + i \sin 2\pi) = 64$.
-

7. MCQ with Distractor Analysis

Q1. The principal argument of $z = -1 - i$ is: (A) $\frac{\pi}{4}$ (B) $\frac{5\pi}{4}$ (C) $-\frac{3\pi}{4}$ (D) $\frac{3\pi}{4}$ **Ans: (C).**
Q3, $\alpha = \frac{\pi}{4}$, $\arg = \alpha - \pi = -\frac{3\pi}{4}$. (A) ignores the quadrant; (B) is outside $(-\pi, \pi]$; (D) is the Q2 answer. (tags: argument-quadrant-error, principal-range-violation)

Q2. The polar form of -1 is: (A) $\cos 0 + i \sin 0$ (B) $\cos \pi + i \sin \pi$ (C) $\cos \frac{\pi}{2} + i \sin \frac{\pi}{2}$ (D) $-(\cos 0 + i \sin 0)$ **Ans: (B).** $|-1| = 1$, $\arg = \pi$. (D) is not in standard polar form. (tag: polar-sign-mismatch)

Q3. $\arg(0)$ is: (A) 0 (B) $\frac{\pi}{2}$ (C) π (D) undefined **Ans: (D).** The origin has no direction. (tag: arg-zero-defined)

Q4. $\left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}\right)^6 =$ (A) 1 (B) -1 (C) i (D) $\frac{\sqrt{3}}{2} + \frac{i}{2}$ **Ans: (B).** De Moivre: $\cos \pi + i \sin \pi = -1$. (D) is the un-powered base. (tag: degree-radian-mixup)

Q5. If $z = r(\cos \theta + i \sin \theta)$, then $\bar{z} =$ (A) $r(\cos \theta + i \sin \theta)$ (B) $-r(\cos \theta + i \sin \theta)$ (C) $r(\cos \theta - i \sin \theta)$ (D) $\frac{1}{r}(\cos \theta + i \sin \theta)$ **Ans: (C).** Conjugate negates the argument:

$r \angle -\theta$. (B) is $-z$; (D) is $1/z$ only when $r = 1$... actually $1/z = \frac{1}{r}(\cos \theta - i \sin \theta)$, so (D) is wrong on both count. (cross-cutting tag: conjugate-sign-error)

8. Common Pitfalls, Pro-Tips & Cheat Sheet

Common Pitfalls

- **Reporting** $\tan^{-1}(y/x)$ **without fixing the quadrant.** The calculator value lives in $(-\frac{\pi}{2}, \frac{\pi}{2})$; always place θ by quadrant. (tag: argument-quadrant-error)
- **Principal value outside** $(-\pi, \pi]$. Reduce by adding/subtracting 2π . (tag: principal-range-violation)
- **Assigning an argument to 0.** $\arg(0)$ is undefined. (tag: arg-zero-defined)
- **Calling** $r(\cos \theta - i \sin \theta)$ **“polar form.”** Standard polar uses $+$; convert via $\cos(-\theta), \sin(-\theta)$. (tag: polar-sign-mismatch)
- **Mixing degrees and radians** inside De Moivre. (tag: degree-radian-mixup)

Pro-Tips

- Sketch the point first — the quadrant is then obvious and the argument almost writes itself.
- Multiplication adds arguments and multiplies moduli; division subtracts and divides. Use this instead of expanding.
- For any integer power, go to polar form and apply De Moivre — far faster than the binomial theorem (Examples 4, 5, 7).
- Memorise $1 + i = \sqrt{2} \angle \frac{\pi}{4}$, $1 + i\sqrt{3} = 2 \angle \frac{\pi}{3}$, $\sqrt{3} + i = 2 \angle \frac{\pi}{6}$ — they recur constantly.

Cheat Sheet Box

- Argand plane: $z = x + iy \leftrightarrow (x, y)$; real axis horizontal, imaginary vertical
 - $|z| = r = \sqrt{x^2 + y^2}$ = distance from origin
 - $\arg z = \theta$, principal value in $(-\pi, \pi]$; $\arg(0)$ undefined
 - Reference angle $\alpha = \tan^{-1} |y/x|$; then Q1 $\rightarrow \alpha$, Q2 $\rightarrow \pi - \alpha$, Q3 $\rightarrow \alpha - \pi$, Q4 $\rightarrow -\alpha$
 - Polar: $z = r(\cos \theta + i \sin \theta) = r \operatorname{cis} \theta$; $x = r \cos \theta$, $y = r \sin \theta$
 - $\arg(z_1 z_2) = \arg z_1 + \arg z_2$; $\arg(z_1/z_2) = \arg z_1 - \arg z_2$; $\arg \bar{z} = -\arg z$
 - $\bar{z} = r \angle (-\theta)$; $1/z = \frac{1}{r} \angle (-\theta)$
 - De Moivre: $z^n = r^n(\cos n\theta + i \sin n\theta)$, $n \in \mathbb{Z}$
 - Cube roots of unity: $1, \omega, \omega^2$; $\omega^3 = 1$; $1 + \omega + \omega^2 = 0$
-

Subtopic 04 — Quadratic Equations with Complex Roots

This closes the chapter. Subtopics 01-03 built complex numbers (algebra, modulus/conjugate, polar). Here they pay off: the quadratic formula, which “failed” whenever the discriminant was negative, now succeeds for *every* quadratic — the missing roots were complex all along.

1. Exam Relevance

- **CBSE Boards:** The core skill is solving $ax^2 + bx + c = 0$ with real coefficients when the discriminant is negative, giving a complex-conjugate pair. Staples like $x^2 + 1 = 0$ and $x^2 + x + 1 = 0$ are exam-standard.
- **JEE Main:** Forming a quadratic from a given complex root, the conjugate-root theorem, and finding the square root of a complex number all appear regularly.
- **JEE Advanced:** Quadratics with **complex coefficients** (whose roots are *not* a conjugate pair) and Vieta-based manipulation are the harder targets.

Diagnostic note: the signature trap is discriminant-sign-misread, and the most instructive conceptual error is conjugate-root-real-coeff-only — assuming roots come in conjugate pairs even when the coefficients are not real. The cross-cutting i2-sign-flip from Subtopic 01 reappears here (since $\sqrt{D} = i\sqrt{|D|}$ when $D < 0$).

2. Concept Intuition (Indian-context analogies)

Every quadratic now has exactly two roots. Over the reals, $x^2 + 1 = 0$ had *none*. Over $\mathbb{C}$ it has two, $\pm i$. This is the first glimpse of a deep fact (the Fundamental Theorem of Algebra): a degree- n polynomial has exactly n roots once you allow complex numbers. Complex numbers were *built* to make this true.

The discriminant is a weather forecast. Before solving, $D = b^2 - 4ac$ tells you what kind of roots to expect, the way a monsoon forecast tells you whether to carry an umbrella: $D > 0 \rightarrow$ two distinct real roots (clear skies, both on the number line); $D = 0 \rightarrow$ one repeated real root (borderline); $D < 0 \rightarrow$ two complex-conjugate roots (the roots have “left the real line” into the plane).

Complex roots travel as mirror-image pairs — but only with real coefficients. Recall from Subtopic 02 that the conjugate is a reflection across the real axis. When a quadratic has *real* coefficients and a complex root $p + iq$, its mirror image $p - iq$ is forced to be a root too. Change the coefficients to complex and this symmetry breaks — the roots can be any two complex numbers.

Diagram placeholder

Three mini Argand-plane sketches side by side for $D > 0$, $D = 0$, $D < 0$. $D > 0$: two dots on the real axis. $D = 0$: one dot on the real axis (double). $D < 0$: two dots off the

axis, mirror images across the real axis (a conjugate pair).

3. Key Formulas and Definitions (with validity / domain conditions)

Quadratic equation and formula. For $ax^2 + bx + c = 0$ with $a \neq 0$,

$$x = \frac{-b \pm \sqrt{D}}{2a}, \quad D = b^2 - 4ac \text{ (the discriminant).}$$

Nature of roots (real coefficients $a, b, c \in \mathbb{R}$):

Discriminant	Roots
$D > 0$	real and distinct
$D = 0$	real and equal (repeated): $x = -\frac{b}{2a}$
$D < 0$	complex conjugate pair (non-real)

Validity condition. This table assumes **real** coefficients. If a, b, c are themselves complex, the “ $D < 0$ ” classification is meaningless and the roots need not be conjugates. (tag: discriminant-sign-misread)

Complex roots when $D < 0$ (real coefficients). Since $\sqrt{D} = \sqrt{-(4ac - b^2)} = i\sqrt{4ac - b^2}$,

$$x = \frac{-b \pm i\sqrt{4ac - b^2}}{2a} \quad (D < 0).$$

The two roots are conjugates of each other. (cross-cutting tag: i2-sign-flip)

Conjugate-root theorem. If a quadratic (indeed any polynomial) has **real coefficients** and $p + iq$ ($q \neq 0$) is a root, then $p - iq$ is also a root.

Holds **only** for real coefficients. With complex coefficients it fails. (tag: conjugate-root-real-coeff-only)

Relations between roots and coefficients (Vieta). If α, β are the roots of $ax^2 + bx + c = 0$,

$$\alpha + \beta = -\frac{b}{a}, \quad \alpha\beta = \frac{c}{a}.$$

Mind the **minus sign** on the sum. (tag: vieta-sign-error)

Forming a quadratic from its roots.

$$x^2 - (\alpha + \beta)x + \alpha\beta = 0, \quad \text{i.e.} \quad x^2 - (\text{sum})x + (\text{product}) = 0.$$

Square root of a complex number. To find $\sqrt{a+ib} = x+iy$, square and equate parts:

$$x^2 - y^2 = a, \quad 2xy = b,$$

which gives $x^2 = \frac{|z|+a}{2}$, $y^2 = \frac{|z|-a}{2}$ with $|z| = \sqrt{a^2+b^2}$, and the sign of xy matching the sign of b . There are always **two** square roots, $\pm(x+iy)$. (tag: *sqrt-complex-sign-branch*)

4. Theorems, Proofs & Standard Methods

Theorem 1 (Quadratic formula). The roots of $ax^2 + bx + c = 0$ ($a \neq 0$) are $\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$. *Proof (completing the square).* Divide by a and complete the square:

$$x^2 + \frac{b}{a}x + \frac{c}{a} = 0 \Rightarrow \left(x + \frac{b}{2a}\right)^2 = \frac{b^2 - 4ac}{4a^2} \Rightarrow x + \frac{b}{2a} = \pm \frac{\sqrt{b^2 - 4ac}}{2a}.$$

Rearranging gives the formula. Crucially, the square root is now defined for **every** value of $b^2 - 4ac$, real or complex — so the formula yields two roots in all cases. ■

Theorem 2 (Conjugate-root theorem for real coefficients). If $p+iq$ is a root of $ax^2 + bx + c = 0$ with $a, b, c \in \mathbb{R}$, then $p-iq$ is also a root. *Proof.* Suppose $az^2 + bz + c = 0$ with $z = p+iq$. Take the conjugate of the whole equation. Since a, b, c are real, $\bar{a} = a$ etc., and conjugation distributes over sums and products (Subtopic 02): $\overline{az^2 + bz + c} = a\bar{z}^2 + b\bar{z} + c = \bar{0} = 0$. Hence $\bar{z} = p-iq$ also satisfies the equation. ■ The proof uses $\bar{a} = a$ — exactly where real coefficients are required, and exactly why the theorem fails otherwise.

Standard Method — Square root of a complex number. To solve, say, $\sqrt{3+4i}$: set $\sqrt{3+4i} = x+iy$, so $x^2 - y^2 = 3$ and $2xy = 4$. With $|3+4i| = 5$, get $x^2 = \frac{5+3}{2} = 4$, $y^2 = \frac{5-3}{2} = 1$, and $xy > 0$ (since $b = 4 > 0$), giving $\pm(2+i)$. This method is what makes the quadratic formula usable when D is complex.

5. Worked Examples (CBSE → JEE Main → JEE Advanced)

Example 1 — CBSE. Solve $x^2 + 1 = 0$ and $x^2 + x + 1 = 0$. *Solution.* $x^2 + 1 = 0 \Rightarrow x^2 = -1 \Rightarrow x = \pm i$. For $x^2 + x + 1 = 0$: $D = 1 - 4 = -3 < 0$, so $x = \frac{-1 \pm i\sqrt{3}}{2}$ — the complex cube roots of unity ω, ω^2 from Subtopic 03.

Example 2 — CBSE. Solve $\sqrt{2}x^2 + x + \sqrt{2} = 0$. *Solution.* $D = 1 - 4(\sqrt{2})(\sqrt{2}) = 1 - 8 = -7 < 0$, so $x = \frac{-1 \pm i\sqrt{7}}{2\sqrt{2}}$. The roots are a conjugate pair.

Example 3 — JEE Main. Form a quadratic with real coefficients, one of whose roots is $2 - 3i$. *Solution.* By the conjugate-root theorem the other root is $2 + 3i$. Sum = 4, product = $(2 - 3i)(2 + 3i) = 4 + 9 = 13$. Hence $x^2 - 4x + 13 = 0$.

Example 4 — JEE Main. Find $\sqrt{-15 - 8i}$. *Solution.* Let $\sqrt{-15 - 8i} = x + iy$: $x^2 - y^2 = -15$, $2xy = -8$. With $|-15 - 8i| = \sqrt{225 + 64} = 17$, $x^2 = \frac{17-15}{2} = 1$, $y^2 = \frac{17+15}{2} = 16$, and $xy < 0$. So $\sqrt{-15 - 8i} = \pm(1 - 4i)$.

Example 5 — JEE Advanced. Solve $x^2 - (3 + i)x + (2 + i) = 0$ and verify the roots are *not* conjugates. *Solution.* $D = (3 + i)^2 - 4(2 + i) = (8 + 6i) - (8 + 4i) = 2i$. Now $\sqrt{2i}$: $x^2 - y^2 = 0$, $2xy = 2$, $|2i| = 2$, so $x^2 = y^2 = 1$, $xy > 0 \Rightarrow \sqrt{2i} = \pm(1 + i)$. Then

$$x = \frac{(3 + i) \pm (1 + i)}{2} \Rightarrow x = 2 + i \text{ or } x = 1.$$

The roots $2 + i$ and 1 are **not** conjugates — consistent with the coefficients being complex, where the conjugate-root theorem does not apply. (Check: sum = $3 + i$, product = $2 + i$. ✓)

6. Practice Exercises (with answers)

- State the nature of the roots: (a) $x^2 - 4x + 5 = 0$, (b) $2x^2 - 3x + 1 = 0$, (c) $x^2 - 6x + 9 = 0$.
- Solve $x^2 - 2x + 5 = 0$.
- Solve $3x^2 - 4x + \frac{20}{3} = 0$.
- Form a quadratic with real coefficients one of whose roots is $-1 + i\sqrt{2}$.
- Find $\sqrt{7 + 24i}$.
- If α, β are the roots of $x^2 + 2x + 5 = 0$, find $\alpha + \beta$ and $\alpha\beta$, and confirm they are conjugates.
- Solve $ix^2 - 2x + (2 - i) = 0$ (complex coefficients).

Answers.

- (a) $D = -4 < 0$, complex conjugate; (b) $D = 1 > 0$, real distinct; (c) $D = 0$, real equal.
- $D = 4 - 20 = -16$, $x = \frac{2 \pm 4i}{2} = 1 \pm 2i$.
- $D = 16 - 80 = -64$, $x = \frac{4 \pm 8i}{6} = \frac{2 \pm 4i}{3}$.
- Other root $-1 - i\sqrt{2}$; sum = -2 , product = $1 + 2 = 3$; $x^2 + 2x + 3 = 0$.
- $x^2 - y^2 = 7$, $2xy = 24$, $|z| = 25$: $x^2 = 16$, $y^2 = 9$, $xy > 0 \Rightarrow \pm(4 + 3i)$.
- $\alpha + \beta = -2$, $\alpha\beta = 5$; roots $-1 \pm 2i$, a conjugate pair (real coefficients).

$$7. D = 4 - 4i(2 - i) = 4 - 8i + 4i^2 = -8i; \sqrt{-8i} = \pm(2 - 2i); x = \frac{2 \pm (2 - 2i)}{2i},$$

$$\text{giving } x = \frac{4 - 2i}{2i} = -1 - 2i \text{ or } x = \frac{2i}{2i} = 1. \text{ (Not conjugates.)}$$

7. MCQ with Distractor Analysis

Q1. The roots of $x^2 - 2x + 2 = 0$ are: (A) $1 \pm i$ (B) $-1 \pm i$ (C) $2 \pm i$ (D) real **Ans: (A).**
 $D = 4 - 8 = -4$, $x = \frac{2 \pm 2i}{2} = 1 \pm i$. (D) misreads D . (tag: discriminant-sign-misread)

Q2. A quadratic with real coefficients has one root $3 + 4i$. Its other root is: (A) $3 + 4i$ (B) $-3 - 4i$ (C) $3 - 4i$ (D) $4 + 3i$ **Ans: (C).** Conjugate-root theorem. (B) negates instead of conjugating. (tags: conjugate-root-real-coeff-only, conjugate-sign-error)

Q3. For $x^2 - (2 + i)x + (\dots) = 0$ with **complex** coefficients, the roots: (A) must be conjugates (B) need not be conjugates (C) are always real (D) are always equal **Ans: (B).** The conjugate-root theorem needs real coefficients. (tag: conjugate-root-real-coeff-only)

Q4. If α, β are roots of $x^2 + px + q = 0$, then $\alpha + \beta =$ (A) p (B) $-p$ (C) q (D) $-q$ **Ans: (B).** $\alpha + \beta = -b/a = -p$. (A) drops the sign. (tag: vieta-sign-error)

Q5. $\sqrt{-5 + 12i} =$ (A) $\pm(2 + 3i)$ (B) $\pm(3 + 2i)$ (C) $\pm(2 - 3i)$ (D) $\pm(3 - 2i)$ **Ans: (A).**
 $x^2 - y^2 = -5$, $2xy = 12$, $|z| = 13$: $x^2 = 4$, $y^2 = 9$, $xy > 0 \Rightarrow \pm(2 + 3i)$. (B) swaps x, y . (tag: sqrt-complex-sign-branch)

8. Common Pitfalls, Pro-Tips & Cheat Sheet

Common Pitfalls

- **Misreading the discriminant's sign**, or declaring "no roots" when $D < 0$ — over $\mathbb{C}$ there are always two. (tag: discriminant-sign-misread)
- **Applying the conjugate-root theorem to complex-coefficient equations** — it requires real coefficients. (tag: conjugate-root-real-coeff-only)
- **Forgetting $\sqrt{D} = i\sqrt{|D|}$ when $D < 0$.** (tag: i2-sign-flip)
- **Dropping a square root of a complex number, or getting the sign branch wrong** — there are always two, and $\text{sgn}(xy) = \text{sgn}(b)$. (tag: sqrt-complex-sign-branch)
- **Sign error in Vieta's sum:** $\alpha + \beta = -b/a$, not b/a . (tag: vieta-sign-error)

Pro-Tips

- Check D first — it tells you the root type before any computation.
- With real coefficients, you only ever need to find *one* complex root; the other is its conjugate.

- For complex coefficients, the quadratic formula still works — you just need $\sqrt{\text{complex}}$ (the $x^2 - y^2 = a$, $2xy = b$ method).
- To build a real quadratic from a complex root $p+iq$: sum = $2p$, product = p^2+q^2 , so $x^2 - 2px + (p^2 + q^2) = 0$.

Cheat Sheet Box

- $x = \frac{-b \pm \sqrt{D}}{2a}$, $D = b^2 - 4ac$
 - Real coeff: $D > 0$ real distinct · $D = 0$ real equal · $D < 0$ complex conjugate pair
 - $D < 0$: $x = \frac{-b \pm i\sqrt{4ac - b^2}}{2a}$
 - Conjugate-root theorem: **real coefficients only**
 - Vieta: $\alpha + \beta = -\frac{b}{a}$, $\alpha\beta = \frac{c}{a}$
 - From roots: $x^2 - (\alpha + \beta)x + \alpha\beta = 0$
 - Real quadratic from root $p+iq$: $x^2 - 2px + (p^2 + q^2) = 0$
 - $\sqrt{a+ib} = \pm(x+iy)$: $x^2 = \frac{|z|+a}{2}$, $y^2 = \frac{|z|-a}{2}$, $\text{sgn}(xy) = \text{sgn}(b)$
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Subtopic 05 — Euler Form, De Moivre's Theorem & Roots of a Complex Number

JEE Extension (beyond NCERT Class 11). This subtopic develops the exponential form and the theory of roots — the engine room of JEE-Advanced complex numbers. It builds directly on the polar form of Subtopic 03.

1. Exam Relevance

- **CBSE Boards:** Not in the Class-11 syllabus; safe to skip for boards only.
- **JEE Main:** Euler-form manipulation, simple De Moivre powers, and cube/fourth roots of unity appear most years.
- **JEE Advanced:** The roots-of-unity toolkit and Euler-form algebra are assumed fluent; many problems are unsolvable without them.

Diagnostic note: the signature trap is root-count-error — reporting one root (or the wrong number) when an n th root has exactly n values. The home tags here feed the JEE-extension branch of the knowledge graph.

2. Concept Intuition

Euler's formula packages polar form into one symbol. Subtopic 03 wrote $z = r(\cos \theta + i \sin \theta)$. Euler's formula says that bracket is an exponential:

$$e^{i\theta} = \cos \theta + i \sin \theta,$$

so $z = re^{i\theta}$. A unit complex number $e^{i\theta}$ is simply “the arrow of length 1 pointing at angle θ .” Multiplying two of them adds the angles — and now that is literally the exponent rule $e^{i\theta_1}e^{i\theta_2} = e^{i(\theta_1+\theta_2)}$. “Multiplication is rotation” becomes “multiplication adds exponents.”

Roots live on a circle, equally spaced. Taking an n th root divides the angle by n — but the angle was only known up to 2π , so there are n legitimate answers, spaced $\frac{2\pi}{n}$ apart on a circle of radius $r^{1/n}$. The n roots are the vertices of a **regular n -gon**. The n th roots of unity are that polygon inscribed in the unit circle with one vertex at 1.

Diagram placeholder

Unit circle on the Argand plane. Mark the cube roots of unity 1, ω , ω^2 as three equally spaced points (a triangle), at angles 0, $2\pi/3$, $4\pi/3$. Then a second circle showing the four 4th-roots of unity 1, i , -1 , $-i$ as a square. Emphasise equal spacing $2\pi/n$.

3. Key Formulas and Definitions (with validity / domain conditions)

Euler's formula and exponential form.

$$e^{i\theta} = \cos \theta + i \sin \theta, \quad z = re^{i\theta} \quad (r = |z|, \theta = \arg z).$$

Consequences:

$$e^{i\theta_1} e^{i\theta_2} = e^{i(\theta_1 + \theta_2)}, \quad \frac{e^{i\theta_1}}{e^{i\theta_2}} = e^{i(\theta_1 - \theta_2)}, \quad \overline{re^{i\theta}} = re^{-i\theta}, \quad \frac{1}{re^{i\theta}} = \frac{1}{r} e^{-i\theta}.$$

$$\cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2}, \quad \sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i} \quad (\text{inverse Euler}).$$

De Moivre's theorem. For $z = re^{i\theta}$ and integer n ,

$$z^n = r^n e^{in\theta} = r^n (\cos n\theta + i \sin n\theta).$$

For rational $n = p/q$, $\cos n\theta + i \sin n\theta$ is **one of** the q values.

n th roots of a complex number. The solutions of $w^n = z = re^{i\theta}$ are

$$w_k = r^{1/n} \left[\cos \frac{\theta + 2k\pi}{n} + i \sin \frac{\theta + 2k\pi}{n} \right] = r^{1/n} e^{i(\theta + 2k\pi)/n}, \quad k = 0, 1, \dots, n-1.$$

Validity / counting condition. There are **exactly** n n th roots, given by $k = 0$ through $n-1$. Using $k \geq n$ repeats roots; reporting fewer than n is the root-count-error. The roots lie on a circle of radius $r^{1/n}$, equally spaced by $\frac{2\pi}{n}$.

n th roots of unity. Setting $z = 1$ ($r = 1, \theta = 0$):

$$\alpha_k = e^{2\pi i k/n} = \cos \frac{2k\pi}{n} + i \sin \frac{2k\pi}{n}, \quad k = 0, \dots, n-1.$$

Writing $\alpha = e^{2\pi i/n}$, the roots are $1, \alpha, \alpha^2, \dots, \alpha^{n-1}$, and:

$$1 + \alpha + \alpha^2 + \dots + \alpha^{n-1} = 0, \quad \prod_{k=0}^{n-1} \alpha_k = (-1)^{n-1}.$$

Cube roots of unity (the workhorse case).

$$1, \omega, \omega^2 \quad \text{where} \quad \omega = e^{2\pi i/3} = -\frac{1}{2} + \frac{\sqrt{3}}{2}i, \quad \omega^2 = \bar{\omega} = -\frac{1}{2} - \frac{\sqrt{3}}{2}i,$$

$$\omega^3 = 1, \quad 1 + \omega + \omega^2 = 0, \quad \omega^n = \omega^{n \bmod 3}.$$

Factorization identities (heavily used):

$$a^2 + ab + b^2 = (a - b\omega)(a - b\omega^2), \quad a^2 - ab + b^2 = (a + b\omega)(a + b\omega^2),$$

$$a^3 + b^3 = (a + b)(a + b\omega)(a + b\omega^2), \quad a^3 - b^3 = (a - b)(a - b\omega)(a - b\omega^2),$$

$$a^3 + b^3 + c^3 - 3abc = (a + b + c)(a + b\omega + c\omega^2)(a + b\omega^2 + c\omega).$$

4. Theorems, Proofs & Standard Methods

Theorem 1 (Euler's formula, sketch). $e^{i\theta} = \cos \theta + i \sin \theta$. *Proof (series).* Substituting $i\theta$ into $e^x = \sum \frac{x^n}{n!}$ and using $i^2 = -1$ to split into even/odd powers gives $\sum (-1)^k \frac{\theta^{2k}}{(2k)!} + i \sum (-1)^k \frac{\theta^{2k+1}}{(2k+1)!} = \cos \theta + i \sin \theta$. ■ (cross-cutting: i2-sign-flip)

Theorem 2 (nth roots formula). $w^n = re^{i\theta}$ has exactly the n solutions $w_k = r^{1/n} e^{i(\theta+2k\pi)/n}$, $k = 0, \dots, n-1$. *Proof.* Write $w = \rho e^{i\phi}$. Then $w^n = \rho^n e^{in\phi} = re^{i\theta}$ forces $\rho^n = r$ (so $\rho = r^{1/n}$) and $n\phi = \theta + 2k\pi$ for some integer k (since arguments are equal up to 2π). Thus $\phi = \frac{\theta+2k\pi}{n}$. Values $k = 0, \dots, n-1$ give distinct angles; $k = n$ repeats $k = 0$. ■

Theorem 3 (Sum of nth roots of unity is zero, $n \geq 2$). *Proof.* The roots are $1, \alpha, \dots, \alpha^{n-1}$ with $\alpha = e^{2\pi i/n} \neq 1$. This is a geometric series: $\sum_{k=0}^{n-1} \alpha^k = \frac{\alpha^n - 1}{\alpha - 1} = \frac{1 - 1}{\alpha - 1} = 0$, since $\alpha^n = 1$. ■ (They are also the roots of $z^n - 1 = 0$, whose z^{n-1} coefficient is 0 — same conclusion by Vieta.)

Standard Method — finding all nth roots. (1) Put z in polar/Euler form $re^{i\theta}$. (2) Take $\rho = r^{1/n}$. (3) List angles $\frac{\theta+2k\pi}{n}$ for $k = 0, \dots, n-1$. (4) Convert back if $a + ib$ form is required.

5. Worked Examples (CBSE → JEE Main → JEE Advanced)

(CBSE rung omitted — topic is beyond board scope.)

Example 1 — JEE Main. Write $z = -1 + i$ in Euler form and find z^8 . *Solution.* $r = \sqrt{2}$, $\arg z = \frac{3\pi}{4}$, so $z = \sqrt{2} e^{i3\pi/4}$. Then $z^8 = (\sqrt{2})^8 e^{i6\pi} = 16 e^{i6\pi} = 16(\cos 6\pi + i \sin 6\pi) = 16$.

Example 2 — JEE Main. Find all the cube roots of $8i$. *Solution.* $8i = 8 e^{i\pi/2}$, so $\rho = 8^{1/3} = 2$, angles $\frac{\pi/2+2k\pi}{3}$ for $k = 0, 1, 2$: $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$. Roots: $2(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}) = \sqrt{3} + i$; $2(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6}) = -\sqrt{3} + i$; $2(\cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2}) = -2i$.

Example 3 — JEE Main. Find the four 4th roots of unity. *Solution.* $\alpha_k = e^{2\pi i k/4} = e^{ik\pi/2}$ for $k = 0, 1, 2, 3$: $1, i, -1, -i$. (Sum = 0. ✓)

Example 4 — JEE Advanced. If ω is a non-real cube root of unity, evaluate $(1 + \omega)(1 + \omega^2)(1 + \omega^4)(1 + \omega^8)$. *Solution.* Reduce exponents mod 3: $\omega^4 = \omega$, $\omega^8 = \omega^2$. So the product is $(1 + \omega)(1 + \omega^2)(1 + \omega)(1 + \omega^2) = [(1 + \omega)(1 + \omega^2)]^2$. Now $(1 + \omega)(1 + \omega^2) = 1 + \omega + \omega^2 + \omega^3 = 0 + 1 = 1$. Hence the value is $1^2 = 1$. (tag: omega-power-reduction-error)

Example 5 — JEE Advanced. Using cube roots of unity, factor and evaluate $a^3 + b^3 + c^3 - 3abc$ for $a = 1, b = 2, c = 3$. *Solution.* $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$

$c^2 - ab - bc - ca$). With $a = 1, b = 2, c = 3$: $(6)(1 + 4 + 9 - 2 - 6 - 3) = (6)(3) = 18$.
(The ω -factorization gives the same product as three linear complex factors.)

6. Practice Exercises (with answers)

- Express $1 + i\sqrt{3}$ in Euler form and compute $(1 + i\sqrt{3})^9$.
- Find all square roots of i using the roots formula.
- Find all cube roots of unity and verify their sum and product.
- Find the fifth roots of unity in $e^{i\theta}$ form and state their sum.
- Simplify $\omega^{100} + \omega^{200}$ for a non-real cube root of unity ω .
- Prove $(1 + \omega)^3 = (1 + \omega^2)^3 = -1$... actually show $(1 + \omega)^3 = -1$.
- Find all solutions of $z^4 = -16$.

Answers.

- $2e^{i\pi/3}$; $(2)^9 e^{i3\pi} = 512(\cos 3\pi + i \sin 3\pi) = -512$.
 - $i = e^{i\pi/2}$; roots $e^{i\pi/4}, e^{i5\pi/4} = \pm \frac{1}{\sqrt{2}}(1 + i)$.
 - $1, \omega, \omega^2$; sum = 0, product = $\omega^3 = 1$.
 - $\alpha_k = e^{2\pi i k/5}$, $k = 0..4$; sum = 0.
 - $100 \equiv 1, 200 \equiv 2 \pmod{3}$: $\omega + \omega^2 = -1$.
 - $1 + \omega = -\omega^2 \Rightarrow (1 + \omega)^3 = -\omega^6 = -(\omega^3)^2 = -1$.
 - $-16 = 16e^{i\pi}$; $\rho = 2$, angles $\frac{\pi+2k\pi}{4}$: $2e^{i\pi/4}, 2e^{i3\pi/4}, 2e^{i5\pi/4}, 2e^{i7\pi/4} = \pm\sqrt{2} \pm \sqrt{2}i$
(all four sign combinations).
-

7. MCQ with Distractor Analysis

Q1. $e^{i\pi} =$ (A) 1 (B) -1 (C) i (D) 0 **Ans: (B).** $\cos \pi + i \sin \pi = -1$ (Euler's identity).
(tag: euler-conversion-error)

Q2. The number of distinct 6th roots of a non-zero complex number is: (A) 1 (B) 2 (C) 3 (D) 6 **Ans: (D).** Exactly $n = 6$. (tag: root-count-error)

Q3. For a non-real cube root of unity ω , $1 + \omega + \omega^2 =$ (A) 0 (B) 1 (C) 3 (D) ω **Ans: (A).**
(tag: omega-power-reduction-error)

Q4. ω^{2024} (with $\omega^3 = 1$) equals: (A) 1 (B) ω (C) ω^2 (D) -1 **Ans: (C).** $2024 \equiv 2 \pmod{3}$. (B) takes $2024 \equiv 1$. (tag: omega-power-reduction-error)

Q5. The cube roots of 8 are: (A) 2 only (B) $2, 2\omega, 2\omega^2$ (C) ± 2 (D) $2, -2, 2i$ **Ans: (B).**
All three; (A) is the principal-root-only trap. (tag: principal-root-only)

8. Common Pitfalls, Pro-Tips & Cheat Sheet

Common Pitfalls

- **Reporting one root instead of n .** An n th root always has n values, $k = 0, \dots, n-1$. (tags: *root-count-error, principal-root-only*)
- **Mishandling ω powers** — always reduce the exponent mod 3. (tag: *omega-power-reduction-error*)
- **Confusing $e^{i\theta}$ with e^θ** or mixing degrees/radians in the exponent. (tags: *euler-conversion-error, degree-radian-mixup*)
- **Forgetting $\omega^2 = \bar{\omega}$** and that ω, ω^2 are conjugates.

Pro-Tips

- Euler form makes powers and roots trivial: powers multiply the angle, roots divide it (over n branches).
- $(1 + \omega) = -\omega^2$ and $(1 + \omega^2) = -\omega$ — instant simplifiers.
- The n roots of unity sum to 0 and form a regular n -gon; use symmetry to kill sums fast.
- To take an n th root, go to Euler form first — never expand.

Cheat Sheet Box

- $e^{i\theta} = \cos \theta + i \sin \theta$; $z = re^{i\theta}$; $\overline{re^{i\theta}} = re^{-i\theta}$
- $\cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2}$, $\sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$
- De Moivre: $z^n = r^n e^{in\theta}$
- n th roots of $z = re^{i\theta}$: $w_k = r^{1/n} e^{i(\theta + 2k\pi)/n}$, $k = 0 \dots n-1$ (exactly n)
- n th roots of unity: $1, \alpha, \dots, \alpha^{n-1}$, $\alpha = e^{2\pi i/n}$; sum = 0
- Cube roots of unity: $1, \omega, \omega^2$; $\omega^3 = 1$, $1 + \omega + \omega^2 = 0$, $\omega^2 = \bar{\omega}$
- $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a + b\omega + c\omega^2)(a + b\omega^2 + c\omega)$
- Roots of $z = re^{i\theta}$ form a regular polygon, radius $r^{1/n}$, spacing $2\pi/n$

Subtopic 06 — Geometry of Complex Numbers (Argand Geometry & Loci)

JEE Extension (beyond NCERT Class 11). Complex numbers as points and position vectors. This is the heart of JEE-Advanced complex numbers — rotation, loci, and the equations of lines and circles in z . Builds on the Argand plane (03) and the exponential form (05).

1. Exam Relevance

- **CBSE Boards:** Beyond Class-11 scope; skip for boards only.
- **JEE Main:** Distance, simple loci ($|z - z_0| = r$, perpendicular bisector) and basic rotation appear regularly.
- **JEE Advanced:** A core, high-yield area — rotation theorem, Apollonius circles, arg-loci, collinear/concyclic conditions, and modulus extrema on a locus.

Diagnostic note: the signature traps are rotation-sign-error (rotation direction / wrong side of the ratio) and apollonius-kl-is-line. These define the geometry branch of the knowledge graph.

2. Concept Intuition

A complex number is a position vector. $z = x + iy$ is the point (x, y) and also the arrow $\overrightarrow{OP}$. So vector ideas transfer directly: $z_2 - z_1$ is the displacement from z_1 to z_2 , and $|z_2 - z_1|$ is the distance between them.

Rotation = multiply by $e^{i\theta}$. To rotate the vector $\overrightarrow{AB}$ about A by angle θ , multiply the displacement $z_B - z_A$ by $e^{i\theta}$. This single move (the “rotation theorem”) solves a huge class of geometry problems — finding a third vertex, proving a right angle, building a square or equilateral triangle.

A locus is a geometric condition on z . “ $|z - z_0| = r$ ” reads literally as “all points at distance r from z_0 ” — a circle. “ $|z - z_1| = |z - z_2|$ ” reads “equidistant from two points” — the perpendicular bisector. Translating words to a complex equation (and back) is the whole skill.

Diagram placeholder

Argand plane with three points $A=z_1$, $B=z_2$, $C=z_3$ forming a triangle. Show the displacement vectors $(z_2 - z_1)$ and $(z_3 - z_1)$ from A , and an angle θ between them, illustrating the rotation theorem $(z_3 - z_1) = (z_2 - z_1) \text{ times } e^{i\theta}$. Alongside, a small circle $|z - z_0| = r$ with centre z_0 and radius r .

3. Key Formulas and Definitions (with validity / domain conditions)

Distance and division of a segment.

$$\text{distance}(z_1, z_2) = |z_1 - z_2|, \quad \text{midpoint} = \frac{z_1 + z_2}{2},$$

$$\text{point dividing } z_1 z_2 \text{ in ratio } m : n = \frac{nz_1 + mz_2}{m + n}, \quad \text{centroid} = \frac{z_1 + z_2 + z_3}{3}.$$

Rotation theorem. Rotating $\overrightarrow{z_1 z_3}$ to the direction of $\overrightarrow{z_1 z_2}$ through angle θ :

$$\frac{z_2 - z_1}{z_3 - z_1} = \left| \frac{z_2 - z_1}{z_3 - z_1} \right| e^{i\theta}, \quad \theta = \arg \frac{z_2 - z_1}{z_3 - z_1}.$$

The angle is measured **at the common vertex** z_1 , from $\overrightarrow{z_1 z_3}$ to $\overrightarrow{z_1 z_2}$; reversing the ratio reverses the angle's sign. (*tag: rotation-sign-error*)

Equation of a line. The line through z_1, z_2 : a point z is on it iff $\frac{z - z_1}{z_2 - z_1}$ is real,

$$\text{i.e. } \frac{z - z_1}{z_2 - z_1} = \overline{\left(\frac{z - z_1}{z_2 - z_1} \right)}. \text{ General form: } \bar{a}z + a\bar{z} + b = 0 \text{ with } a \in \mathbb{C}, b \in \mathbb{R}.$$

Collinearity. z_1, z_2, z_3 are collinear iff $\frac{z_3 - z_1}{z_2 - z_1} \in \mathbb{R}$ (equivalently $\arg = 0$ or π).

Perpendicular bisector. $|z - z_1| = |z - z_2|$ is the perpendicular bisector of the segment joining z_1, z_2 .

Equation of a circle.

$$|z - z_0| = r \quad (\text{centre } z_0, \text{ radius } r), \quad z\bar{z} + \bar{a}z + a\bar{z} + b = 0 \quad (a \in \mathbb{C}, b \in \mathbb{R}, |a|^2 \geq b),$$

with centre $-a$ and radius $\sqrt{|a|^2 - b}$.

Apollonius circle. $\frac{|z - z_1|}{|z - z_2|} = k$:

- if $k = 1 \rightarrow$ the **perpendicular bisector (a straight line)**, not a circle; (*tag: apollonius-k1-is-line*)
- if $k > 0, k \neq 1 \rightarrow$ a **circle** (the Apollonius circle).

arg locus. $\arg \frac{z - z_1}{z - z_2} = \alpha$ (fixed) is an **arc** of a circle through z_1, z_2 (not the whole circle); $\alpha = \frac{\pi}{2}$ gives the circle with diameter $z_1 z_2$ (minus endpoints). (*tag: arg-locus-arc-not-circle*)

Concyclic condition. z_1, z_2, z_3, z_4 are concyclic (or collinear) iff the cross-ratio $\frac{(z_1 - z_3)(z_2 - z_4)}{(z_1 - z_4)(z_2 - z_3)}$ is real.

Equilateral triangle. z_1, z_2, z_3 form an equilateral triangle iff

$$z_1^2 + z_2^2 + z_3^2 = z_1z_2 + z_2z_3 + z_3z_1, \quad \text{equivalently} \quad \frac{1}{z_1 - z_2} + \frac{1}{z_2 - z_3} + \frac{1}{z_3 - z_1} = 0.$$

Modulus extrema on a locus. If z lies on the circle $|z - z_0| = r$, then $||z_0| - r| \leq |z| \leq |z_0| + r$ (nearest/farthest points lie along the line through O and z_0). (*tag: locus-modulus-extreme-error*)

4. Theorems, Proofs & Standard Methods

Theorem 1 (Rotation). If $\overrightarrow{z_1z_2}$ makes angle θ with $\overrightarrow{z_1z_3}$, then $\frac{z_2 - z_1}{z_3 - z_1} = \frac{|z_2 - z_1|}{|z_3 - z_1|} e^{i\theta}$.

Proof. Both $z_2 - z_1$ and $z_3 - z_1$ are vectors from z_1 . In Euler form $z_2 - z_1 = |z_2 - z_1| e^{i\phi_2}$ and $z_3 - z_1 = |z_3 - z_1| e^{i\phi_3}$. Their quotient is $\frac{|z_2 - z_1|}{|z_3 - z_1|} e^{i(\phi_2 - \phi_3)}$, and $\phi_2 - \phi_3 = \theta$ is exactly the angle between them at z_1 . ■

Theorem 2 (Apollonius locus, $k \neq 1$, is a circle). *Proof sketch.* $|z - z_1| = k|z - z_2| \Rightarrow |z - z_1|^2 = k^2|z - z_2|^2$. Expanding with $|w|^2 = w\bar{w}$ (Subtopic 02) gives $z\bar{z}(1 - k^2) - z(\bar{z}_1 - k^2\bar{z}_2) - \bar{z}(z_1 - k^2z_2) + (|z_1|^2 - k^2|z_2|^2) = 0$. For $k \neq 1$ the $z\bar{z}$ term survives, so (after dividing by $1 - k^2$) it has the circle form $z\bar{z} + \bar{a}z + az + b = 0$. For $k = 1$ the $z\bar{z}$ term vanishes, leaving a **line**. ■ (*uses conjugate-sign-error awareness*)

Theorem 3 (Equilateral triangle condition). z_1, z_2, z_3 are equilateral iff $z_1^2 + z_2^2 + z_3^2 = z_1z_2 + z_2z_3 + z_3z_1$. *Proof idea.* Equilateral means each side is the previous one rotated by $\pm\frac{\pi}{3}$ about the shared vertex. Applying the rotation theorem $z_3 - z_1 = (z_2 - z_1)e^{\pm i\pi/3}$ and simplifying with $e^{i\pi/3}$ satisfying $t^2 - t + 1 = 0$ yields the stated symmetric identity. ■

Standard Method — solve a configuration problem. (1) Place the known points as complex numbers. (2) Express the unknown via the rotation theorem (multiply a displacement by $e^{i\theta}$). (3) Read off the answer. For loci: translate the geometric words into a z -equation, simplify with $|w|^2 = w\bar{w}$, and identify the standard form.

5. Worked Examples (CBSE → JEE Main → JEE Advanced)

(CBSE rung omitted — beyond board scope.)

Example 1 — JEE Main. Find the locus of z if $|z - 2| = |z + 2|$. *Solution.* Equidistant from 2 and $-2 \rightarrow$ the perpendicular bisector of the segment, which is the imaginary axis: $\text{Re}(z) = 0$.

Example 2 — JEE Main. Identify the curve $|z - 3| = 2$. *Solution.* A circle, centre 3 (i.e. $(3, 0)$), radius 2.

Example 3 — JEE Main. If z lies on $|z - 3 - 4i| = 2$, find the greatest and least values of $|z|$. *Solution.* Centre $z_0 = 3 + 4i$, $|z_0| = 5$, radius 2. So $|z|_{\max} = 5 + 2 = 7$, $|z|_{\min} = 5 - 2 = 3$. (tag: locus-modulus-extreme-error)

Example 4 — JEE Advanced. Square $ABCD$ has $A = z_1 = 1 + 2i$ and $B = z_2 = 3 + i$, labelled anticlockwise. Find C . *Solution.* Rotate $\overrightarrow{BA}$ by -90° about B to reach $\overrightarrow{BC}$ (anticlockwise square $\Rightarrow$ turn $\overrightarrow{AB} \rightarrow \overrightarrow{BC}$ by $+90^\circ$; equivalently $z_C - z_B = (z_A - z_B) \cdot e^{-i\pi/2} = (z_A - z_B)(-i)$). With $z_A - z_B = (1 + 2i) - (3 + i) = -2 + i$: $z_C - z_B = (-2 + i)(-i) = 2i - i^2 = 1 + 2i$, so $z_C = (3 + i) + (1 + 2i) = 4 + 3i$. (tag: rotation-sign-error)

Example 5 — JEE Advanced. Find the locus of z satisfying $\arg \frac{z-1}{z+1} = \frac{\pi}{2}$. *Solution.* A constant argument of $\frac{\pi}{2}$ means $\overline{(z-1)} \perp \dots$ it is the **arc** of the circle through 1 and -1 on which the segment from -1 to 1 subtends a right angle — i.e. the semicircle of the circle with diameter joining -1 and 1 (the unit circle), excluding the endpoints, on the side where the directed angle is $+\frac{\pi}{2}$ (upper half, $\text{Im}(z) > 0$). (tag: arg-locus-arc-not-circle)

6. Practice Exercises (with answers)

- Find the locus: $|z - i| = |z + i|$.
- Find centre and radius of $|z + 2 - i| = 3$.
- z lies on $|z - 5| = 3$. Find $\max |z|$ and $\min |z|$.
- Show $z_1 = 1$, $z_2 = 1 + i$, $z_3 = 2 + i$ — are they collinear?
- Equilateral triangle has vertices $z_1 = 0$, $z_2 = 2$; find the two possible z_3 .
- Identify the locus $\frac{|z-1|}{|z+1|} = 2$.
- Rotate the point $z = 2 + i$ about the origin by 90° anticlockwise.

Answers.

- Perpendicular bisector of $i, -i \rightarrow$ the real axis, $\text{Im}(z) = 0$.
- Centre $-2 + i$, radius 3.
- $|z_0| = 5$, $r = 3$: $\max = 8$, $\min = 2$.
- $\frac{z_3 - z_1}{z_2 - z_1} = \frac{1 + i}{i} = 1 - i \notin \mathbb{R} \rightarrow$ not collinear.
- $z_3 = (z_2)e^{\pm i\pi/3}$ about $z_1 = 0$: $z_3 = 2e^{\pm i\pi/3} = 1 \pm i\sqrt{3}$.
- Apollonius circle ($k = 2 \neq 1$): solving gives centre $\frac{5}{3}$, radius $\frac{4}{3}$ (a circle).
- Multiply by $e^{i\pi/2} = i$: $(2 + i)i = -1 + 2i$.

7. MCQ with Distractor Analysis

Q1. $|z - z_1| = |z - z_2|$ represents: (A) a circle (B) the perpendicular bisector of $z_1 z_2$ (C) an ellipse (D) a point **Ans: (B).** (*tag: apollonius-k1-is-line*)

Q2. $\frac{|z - 1|}{|z - 3|} = 1$ is: (A) a circle (B) a line (C) an ellipse (D) two points **Ans: (B).** $k = 1 \Rightarrow$ perpendicular bisector (line), not a circle. (A) is the trap. (*tag: apollonius-k1-is-line*)

Q3. To rotate vector z about O by angle θ anticlockwise, multiply by: (A) θ (B) $e^{i\theta}$ (C) $e^{-i\theta}$ (D) $\cos \theta$ **Ans: (B).** (C) rotates clockwise. (*tag: rotation-sign-error*)

Q4. If $|z - 3 - 4i| = 2$, the minimum of $|z|$ is: (A) 2 (B) 3 (C) 5 (D) 7 **Ans: (B).** $|z_0| - r = 5 - 2 = 3$. (D) is the maximum. (*tag: locus-modulus-extreme-error*)

Q5. $\arg \frac{z - z_1}{z - z_2} = \frac{\pi}{2}$ represents: (A) the full circle on diameter $z_1 z_2$ (B) an arc of that circle (C) a line (D) a parabola **Ans: (B).** A fixed argument gives an arc, not the whole circle. (*tag: arg-locus-arc-not-circle*)

8. Common Pitfalls, Pro-Tips & Cheat Sheet

Common Pitfalls

- **Wrong rotation direction or ratio side.** $e^{i\theta}$ is anticlockwise, $e^{-i\theta}$ clockwise; the angle sits at the common vertex and the ratio order fixes its sign. (*tag: rotation-sign-error*)
- **Calling the $k = 1$ Apollonius locus a circle** — it is the perpendicular-bisector line. (*tag: apollonius-k1-is-line*)
- **Treating an arg-locus as a full circle** — it is only an arc. (*tag: arg-locus-arc-not-circle*)
- **Modulus extrema:** nearest/farthest points are along $O - z_0$, giving $|z_0| \pm r$ — don't guess. (*tag: locus-modulus-extreme-error*)

Pro-Tips

- Translate words to a z -equation: “distance from a point” $\rightarrow |z - z_0|$; “equidistant” $\rightarrow$ perpendicular bisector; “constant angle” $\rightarrow$ arc.
- For any vertex-finding problem, reach for the rotation theorem: displacement $\times e^{i\theta}$.
- Simplify every locus with $|w|^2 = w\bar{w}$ (Subtopic 02) — it linearises the algebra.
- For $|z|$ extremes on a circle, use $|z_0| \pm r$ immediately.

Cheat Sheet Box

- Distance $= |z_1 - z_2|$; midpoint $= \frac{z_1 + z_2}{2}$; centroid $= \frac{z_1 + z_2 + z_3}{3}$
- Rotation: $\frac{z_2 - z_1}{z_3 - z_1} = \left| \frac{z_2 - z_1}{z_3 - z_1} \right| e^{i\theta}$; multiply by $i = +90^\circ$

- Collinear: $\frac{z_3 - z_1}{z_2 - z_1} \in \mathbb{R}$
 - Perp. bisector: $|z - z_1| = |z - z_2|$
 - Circle: $|z - z_0| = r$, or $z\bar{z} + \bar{a}z + a\bar{z} + b = 0$ (centre $-a$, radius $\sqrt{|a|^2 - b}$)
 - Apollonius: $\frac{|z - z_1|}{|z - z_2|} = k \rightarrow$ circle if $k \neq 1$, line if $k = 1$
 - $\arg \frac{z - z_1}{z - z_2} = \alpha \rightarrow$ arc of a circle through z_1, z_2
 - Equilateral: $z_1^2 + z_2^2 + z_3^2 = z_1z_2 + z_2z_3 + z_3z_1$
 - On $|z - z_0| = r$: $||z_0| - r| \leq |z| \leq |z_0| + r$
-

Chapter 4: Complex Numbers and Quadratic Equations — Supplement

Subtopic 07 — Real Quadratic Equations: Roots, Signs and Location

Exam Relevance: CBSE Boards test the real-coefficient side of this material at 2–4 marks — nature of roots from the discriminant, the maximum or minimum value of a quadratic expression, and simple quadratic inequalities. JEE Main draws two of its most reliable single-question patterns from here: the common root of two quadratics, and “find all values of the parameter for which $ax^2 + bx + c$ keeps one sign for every real x .” JEE Advanced pushes into location of roots — placing both roots, or exactly one root, relative to a number k or an interval (k_1, k_2) — usually with a parameter to be solved for, and usually with one boundary case designed to be missed. NEET does not test mathematics.

Subtopic 04 handled the quadratic when the discriminant sends the roots off the real line. This subtopic handles the far more heavily examined case where they stay on it. Everything here assumes $a, b, c \in \mathbb{R}$ and $a \neq 0$ unless stated otherwise, and everything here is built on two objects you already own from Subtopic 04: the discriminant $D = b^2 - 4ac$ and Vieta’s relations $\alpha + \beta = -b/a$, $\alpha\beta = c/a$.

Section 1: Concept Intuition

Stop solving. Start looking at the graph. Subtopic 04 treated $ax^2 + bx + c = 0$ as something to be solved. The whole of this subtopic rests on refusing to solve it. Instead you look at the curve $y = ax^2 + bx + c$ — a parabola — and read the answer off its shape and position.

Think about it this way. Picture a ball hit high over the Wankhede outfield. Its flight path is a parabola opening downward. Three questions a fielder cares about are exactly the three questions this subtopic answers. *Does the ball ever touch the ground?* That is the sign question: whether $ax^2 + bx + c$ ever reaches zero, and it is settled entirely by the discriminant. *How high does it get, and when?* That is the extremum question, settled by the vertex. *Does it land inside the boundary rope or outside?* That is the location question — not merely whether the roots are real, but **where on the number line they sit**.

The sign of a quadratic is decided by the discriminant, not by the constant term. Students look at $x^2 - 4x + 7$, see the $+7$, and conclude it is positive. That reasoning is accidental. The correct reasoning is: $D = 16 - 28 = -12 < 0$, so the parabola never crosses the x -axis; having no crossing, it can never change sign; so it keeps forever the sign it has far out to the right, which is the sign of a . **A quadratic with $D < 0$ takes the sign of a at every**

real x , with no exceptions. That single sentence is the entire sign theory. Where it fails is instructive: if $D \geq 0$ the parabola touches or crosses the axis, and the “one sign forever” claim collapses at once.

Two bus routes, one shared stop. Route 1 and Route 2 run across Bengaluru on different paths. Asking whether they share a stop is asking for a common root of two equations. And here is the practical trick: if a stop is on both routes, then it is also on the “difference” of the two routes — the object you get by cancelling everything the routes have in common. Subtracting one quadratic from a suitable multiple of the other kills the x^2 term and leaves a *linear* equation, which hands you the shared stop directly. **The common root, if it exists, is the root of the linear equation you get by eliminating x^2 .** The caution is that the linear equation is a consequence, not a guarantee: it produces a candidate, and the candidate must be substituted back.

Location of roots is a question about a signpost, not about the roots. Suppose a Mumbai local has two scheduled halts and you want to know whether both lie beyond Dadar. You do not need the halt times. You need three readings taken at Dadar and at the vertex: are the halts real at all ($D \geq 0$), is Dadar outside the pair ($af(k) > 0$), and does the pair sit to the right of Dadar (vertex $-b/2a > k$). **Every location result in this subtopic is these same three readings in a different arrangement**, and the reason all three are needed is that any two of them can be satisfied by a configuration you did not want.

Section 2: Key Formulas and Definitions

The vertex form

For $f(x) = ax^2 + bx + c$ with $a, b, c \in \mathbb{R}$, $a \neq 0$:

$$f(x) = a \left(x + \frac{b}{2a} \right)^2 - \frac{D}{4a}, \quad D = b^2 - 4ac.$$

- D — the discriminant, as defined in Subtopic 04.
- $-\frac{b}{2a}$ — the x -coordinate of the vertex (the axis of symmetry).
- $-\frac{D}{4a} = \frac{4ac - b^2}{4a}$ — the y -coordinate of the vertex, i.e. the extreme value of f .

Validity: $a \neq 0$ throughout. If a parameter can make $a = 0$, that case is *not* covered by any formula in this section and must be examined separately as a linear expression.

Sign and range

Condition on a and D	Behaviour of $f(x) = ax^2 + bx + c$	Range of f
$a > 0, D < 0$	$f(x) > 0$ for every real x	$\left[-\frac{D}{4a}, \infty\right)$, and $-\frac{D}{4a} > 0$

Condition on a and D	Behaviour of $f(x) = ax^2 + bx + c$	Range of f
$a > 0, D = 0$	$f(x) \geq 0$ for every real x , with equality only at $x = -\frac{b}{2a}$	$[0, \infty)$
$a > 0, D > 0$	$f < 0$ strictly between the roots, $f > 0$ strictly outside them	$[-\frac{D}{4a}, \infty)$, and $-\frac{D}{4a} < 0$
$a < 0, D < 0$	$f(x) < 0$ for every real x	$(-\infty, -\frac{D}{4a}]$, and $-\frac{D}{4a} < 0$
$a < 0, D = 0$	$f(x) \leq 0$ for every real x , with equality only at $x = -\frac{b}{2a}$	$(-\infty, 0]$
$a < 0, D > 0$	$f > 0$ strictly between the roots, $f < 0$ strictly outside them	$(-\infty, -\frac{D}{4a}]$, and $-\frac{D}{4a} > 0$

The four parameter conditions you will be asked for in the exam, each an “if and only if”:

$$f(x) > 0 \quad \forall x \in \mathbb{R} \iff a > 0 \text{ and } D < 0,$$

$$f(x) \geq 0 \quad \forall x \in \mathbb{R} \iff a > 0 \text{ and } D \leq 0,$$

$$f(x) < 0 \quad \forall x \in \mathbb{R} \iff a < 0 \text{ and } D < 0,$$

$$f(x) \leq 0 \quad \forall x \in \mathbb{R} \iff a < 0 \text{ and } D \leq 0.$$

And the compact version: $f(x)$ **has the same sign as a for every real x if and only if $D < 0$.**

Extremum. If $a > 0$, f attains a minimum value $-\frac{D}{4a}$ at $x = -\frac{b}{2a}$ and has no maximum. If $a < 0$, f attains a maximum value $-\frac{D}{4a}$ at $x = -\frac{b}{2a}$ and has no minimum.

Figure brief: A two-by-three grid of parabola sketches with a common horizontal x -axis drawn in each panel. Top row $a > 0$: (i) $D > 0$ with two marked x -intercepts $\alpha < \beta$ and the vertex below the axis; (ii) $D = 0$ with the vertex sitting exactly on the axis; (iii) $D < 0$ with the whole curve above the axis. Bottom row $a < 0$: the same three, reflected. In every panel mark the vertex with a dot, label its abscissa $x = -b/(2a)$ and its ordinate $-D/(4a)$, and shade the regions of the x -axis where $f > 0$ in one tint and $f < 0$ in another.

Sign of a quadratic at a point — the master identity

If α, β are the roots of $f(x) = ax^2 + bx + c$, then $f(x) = a(x - \alpha)(x - \beta)$, and therefore for any real k

$$af(k) = a^2(k - \alpha)(k - \beta).$$

Since $a^2 > 0$, the sign of $af(k)$ is exactly the sign of $(k - \alpha)(k - \beta)$. Hence, **when the roots are real:**

- $af(k) > 0 \iff k$ lies strictly outside the closed interval joining the roots;
- $af(k) < 0 \iff k$ lies strictly between the roots;
- $af(k) = 0 \iff k$ is itself a root.

Critical validity note: if $D < 0$ the roots are a conjugate pair $\alpha, \bar{\alpha}$, and then $af(k) = a^2|k - \alpha|^2 > 0$ for every real k . So $af(k) > 0$ carries no information about position unless you have separately guaranteed $D \geq 0$. This is why $D \geq 0$ appears as an explicit condition in every “both roots” result below.

Location of roots (real coefficients, $a \neq 0$, α, β the roots, $k, k_1, k_2 \in \mathbb{R}$ with $k_1 < k_2$)

Requirement	Complete condition set
Both roots $> k$	$D \geq 0$ and $af(k) > 0$ and $-\frac{b}{2a} > k$
Both roots $\geq k$	$D \geq 0$ and $af(k) \geq 0$ and $-\frac{b}{2a} \geq k$
Both roots $< k$	$D \geq 0$ and $af(k) > 0$ and $-\frac{b}{2a} < k$
Both roots $\leq k$	$D \geq 0$ and $af(k) \geq 0$ and $-\frac{b}{2a} \leq k$
k strictly between the roots ($\alpha < k < \beta$)	$af(k) < 0$ alone (this already forces $D > 0$)
Both roots in (k_1, k_2)	$D \geq 0$ and $af(k_1) > 0$ and $af(k_2) > 0$ and $k_1 < -\frac{b}{2a} < k_2$
Exactly one root in (k_1, k_2)	$f(k_1)f(k_2) < 0$ (sufficient; the cases $f(k_1) = 0$ or $f(k_2) = 0$ must be tested separately)
Roots straddle the interval ($\alpha < k_1 < k_2 < \beta$)	$af(k_1) < 0$ and $af(k_2) < 0$

Two remarks that decide marks. First, the strictness matches: “both roots greater than k ” allows a *repeated* root greater than k , which is why $D \geq 0$ and not $D > 0$. Second, the “ k between the roots” line genuinely needs only one condition — $af(k) < 0$ is impossible unless the roots are real and distinct with k between them, so writing $D > 0$ alongside it is harmless but redundant.

Figure brief: Four side-by-side parabola sketches, all with $a > 0$, all showing two real roots $\alpha < \beta$ on the x -axis, and each with a vertical dashed signpost line at $x = k$. Panel 1: k to the left of both roots, with the point $(k, f(k))$ marked above the axis and labelled $f(k) > 0$, plus a second dashed line at the vertex $x = -b/(2a)$ lying to the right of k . Panel 2: k to the right of both roots, mirror image. Panel 3: k between the roots, with $(k, f(k))$ marked below the axis and labelled $f(k) < 0$. Panel 4: two signposts $k_1 < k_2$ bracketing both roots, with $f(k_1) > 0$, $f(k_2) > 0$ and the vertex line drawn strictly between them.

Common roots of two quadratics

Let $f_1(x) = a_1x^2 + b_1x + c_1 = 0$ and $f_2(x) = a_2x^2 + b_2x + c_2 = 0$ with $a_1, a_2 \neq 0$.

Exactly one common root. Provided $a_1b_2 - a_2b_1 \neq 0$, the two equations have a common root if and only if

$$(c_1a_2 - c_2a_1)^2 = (a_1b_2 - a_2b_1)(b_1c_2 - b_2c_1),$$

and that common root is then

$$\alpha = \frac{c_1a_2 - c_2a_1}{a_1b_2 - a_2b_1}.$$

An equivalent expression, usable when $c_1a_2 - c_2a_1 \neq 0$, is $\alpha = \frac{b_1c_2 - b_2c_1}{c_1a_2 - c_2a_1}$.

Both roots common. The two equations have both roots in common if and only if their coefficient triples are proportional:

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2},$$

which is safest written in the determinant form valid even when some coefficient vanishes:

$$a_1b_2 - a_2b_1 = 0, \quad b_1c_2 - b_2c_1 = 0, \quad c_1a_2 - c_2a_1 = 0.$$

A result JEE Main leans on. If f_1 and f_2 both have **real** coefficients and share a **non-real** root, then they share both roots, and hence their coefficients are proportional. Reason: by the conjugate-root theorem of Subtopic 04, the conjugate of the shared root is a root of each equation as well, so both roots of each are the same pair.

Nature of roots refined: rational versus irrational

Subtopic 04 already gives the discriminant table for real coefficients ($D > 0$ real distinct, $D = 0$ real equal, $D < 0$ non-real conjugate pair). When the coefficients are additionally **rational**, that table subdivides:

$D = b^2 - 4ac$ (with $a, b, c \in \mathbb{Q}$)	Roots
$D > 0$ and a perfect square of a rational	real, distinct and rational
$D > 0$ and not a perfect square	real, distinct and irrational , occurring as a conjugate surd pair $p \pm \sqrt{q}$
$D = 0$	real, equal and rational: $x = -\frac{b}{2a}$
$D < 0$	non-real complex conjugate pair (Subtopic 04)

Surd conjugate-root theorem. If $a, b, c \in \mathbb{Q}$ and $p + \sqrt{q}$ (with $p, q \in \mathbb{Q}$, $q > 0$, $\sqrt{q}$ irrational) is a root of $ax^2 + bx + c = 0$, then $p - \sqrt{q}$ is also a root. This is the exact analogue of Theorem 2 of Subtopic 04, with “rational coefficients” replacing “real coefficients”; it fails the moment an irrational coefficient is allowed.

Symmetric functions of the roots

Vieta's relations and the "form a quadratic from its roots" formula $x^2 - (\alpha + \beta)x + \alpha\beta = 0$ are already established in Subtopic 04 and are not restated here. What that subtopic does not do is *use* them, and almost every symmetric expression in α, β reduces to the sum and product:

$$\begin{aligned}\alpha^2 + \beta^2 &= (\alpha + \beta)^2 - 2\alpha\beta = \frac{b^2 - 2ac}{a^2}, \\ (\alpha - \beta)^2 &= (\alpha + \beta)^2 - 4\alpha\beta = \frac{D}{a^2}, \quad |\alpha - \beta| = \frac{\sqrt{D}}{|a|} \quad (D \geq 0), \\ \alpha^3 + \beta^3 &= (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta) = \frac{3abc - b^3}{a^3}, \\ \frac{1}{\alpha} + \frac{1}{\beta} &= \frac{\alpha + \beta}{\alpha\beta} = -\frac{b}{c} \quad (c \neq 0), \quad \frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{\alpha^2 + \beta^2}{\alpha\beta} = \frac{b^2 - 2ac}{ac} \quad (c \neq 0).\end{aligned}$$

The identity $|\alpha - \beta| = \sqrt{D}/|a|$ is the bridge between the discriminant and the *width* of the root pair, and it is the quickest route to problems that constrain how far apart the roots are.

Section 3: CBSE Board Derivations and Key Procedures

Derivation: the vertex form, and everything it decides

1. Start from $f(x) = ax^2 + bx + c$ and factor a out of the first two terms: $f(x) = a\left(x^2 + \frac{b}{a}x\right) + c$. *Reason:* $a \neq 0$, so the division is legal, and only the x -terms need completing.
2. Complete the square inside the bracket: $x^2 + \frac{b}{a}x = \left(x + \frac{b}{2a}\right)^2 - \frac{b^2}{4a^2}$. *Reason:* the same completing-the-square step that proves the quadratic formula in Theorem 1 of Subtopic 04.
3. Distribute a and combine the constants: $f(x) = a\left(x + \frac{b}{2a}\right)^2 - \frac{b^2}{4a} + c = a\left(x + \frac{b}{2a}\right)^2 - \frac{b^2 - 4ac}{4a}$. *Reason:* $-\frac{b^2}{4a} + c = \frac{-b^2 + 4ac}{4a}$.
4. Hence $f(x) = a\left(x + \frac{b}{2a}\right)^2 - \frac{D}{4a}$. *Reason:* $D = b^2 - 4ac$ by definition.
5. Read off the consequences. The squared bracket is ≥ 0 always and equals 0 exactly at $x = -\frac{b}{2a}$. So if $a > 0$ the smallest value of f is $-\frac{D}{4a}$, attained there; if $a < 0$ the largest value of f is $-\frac{D}{4a}$, attained there. *Reason:* multiplying a non-negative quantity by a positive a preserves the ordering; by a negative a it reverses it.
6. Finally, if $D < 0$ then $-\frac{D}{4a}$ has the same sign as a , and $a\left(x + \frac{b}{2a}\right)^2$ also has the sign of a or is zero, so $f(x)$ has the sign of a for every real x . Conversely if $D \geq 0$ the vertex value $-\frac{D}{4a}$ is zero or opposite in sign to a , so f attains the value 0 or a value of the opposite sign. ■

Derivation: the common-root condition by cross-multiplication

1. Suppose α satisfies both $a_1\alpha^2 + b_1\alpha + c_1 = 0$ and $a_2\alpha^2 + b_2\alpha + c_2 = 0$. *Reason:* this is the definition of a common root.
2. Treat these as two linear equations in the two unknowns $X = \alpha^2$ and $Y = \alpha$: $a_1X + b_1Y = -c_1$ and $a_2X + b_2Y = -c_2$. *Reason:* α^2 and α enter linearly, which is the whole point of the manoeuvre.
3. Solve by determinants, assuming $\Delta = a_1b_2 - a_2b_1 \neq 0$: $X = \frac{b_1c_2 - b_2c_1}{\Delta}$ and $Y = \frac{c_1a_2 - c_2a_1}{\Delta}$. *Reason:* Cramer's rule on a 2×2 system with non-zero determinant.
4. But X and Y are not independent: $X = Y^2$. *Reason:* $X = \alpha^2$ and $Y = \alpha$.
5. Impose it: $\frac{(c_1a_2 - c_2a_1)^2}{\Delta^2} = \frac{b_1c_2 - b_2c_1}{\Delta}$, and multiply through by Δ^2 :

$$(c_1a_2 - c_2a_1)^2 = (a_1b_2 - a_2b_1)(b_1c_2 - b_2c_1).$$

Reason: $\Delta \neq 0$ permits the multiplication.

6. The converse also holds. If the displayed relation is true and $\Delta \neq 0$, put $\alpha = \frac{c_1a_2 - c_2a_1}{\Delta}$; then $\alpha^2 = \frac{b_1c_2 - b_2c_1}{\Delta}$, and substituting into $a_1x^2 + b_1x + c_1$ gives $\frac{a_1(b_1c_2 - b_2c_1) + b_1(c_1a_2 - c_2a_1) + c_1\Delta}{\Delta} = \frac{-a_1b_2c_1 + a_2b_1c_1 + c_1(a_1b_2 - a_2b_1)}{\Delta} = 0$, and symmetrically for the second equation. *Reason:* direct expansion; the $a_1b_1c_2$ terms cancel.
7. If instead $\Delta = 0$ and a common root exists, the two equations are forced to be proportional, so both roots are common. ■

Procedure A: finding a common root of two quadratics

1. Check the cheap case first: if $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$, the equations are the same equation scaled, so **both** roots are common and there is nothing further to do. *Why:* this is the only way to have two common roots, and it makes every other method degenerate.
2. If one of the quadratics factorises on sight, factor it, get its two roots explicitly, and substitute each into the other equation in turn. *Why:* this is faster than any formula and automatically produces every valid parameter value.
3. Otherwise eliminate x^2 : compute $a_2f_1(x) - a_1f_2(x) = 0$, which reduces to the linear equation $(a_2b_1 - a_1b_2)x + (a_2c_1 - a_1c_2) = 0$, giving the candidate

$$x = \frac{c_1a_2 - c_2a_1}{a_1b_2 - a_2b_1}.$$

Why: any common root satisfies every linear combination of the two equations, and this particular combination has no x^2 term.

4. **Substitute the candidate back into one of the original equations.** *Why:* step 3 is a necessary consequence only. The linear equation always has a root; that root is a common root only if it satisfies one of the quadratics.
5. If the problem gives real coefficients and tells you the common root is non-real, skip all of the above and conclude immediately that the coefficients are proportional. *Why:* the conjugate of the shared root is shared too, so both roots coincide.

Procedure B: solving a quadratic inequality by sign scheme

1. Move everything to one side so the inequality reads $f(x) > 0$, ≥ 0 , < 0 or ≤ 0 with $f(x) = ax^2 + bx + c$. *Why:* the sign rules describe one expression against zero, never two expressions against each other.
2. Compute D . If $D < 0$, stop: f has the constant sign of a , so the answer is either all of $\mathbb{R}$ or the empty set. *Why:* no sign change is possible without a real root.
3. If $D \geq 0$, find the roots $\alpha \leq \beta$ by factorising or by the quadratic formula, and mark them on a number line. *Why:* the roots are the only points where the sign can change.
4. Assign signs: the sign of f agrees with the sign of a on the two outer intervals and opposes it on the middle interval (α, β) . *Why:* $f(x) = a(x - \alpha)(x - \beta)$, and $(x - \alpha)(x - \beta)$ is positive outside and negative between. This alternating pattern is the wavy-curve rule.
5. Read the answer off the number line, then fix the endpoints: include α, β if the inequality is $\geq$ or $\leq$, exclude them if it is $>$ or $<$. *Why:* $f(\alpha) = f(\beta) = 0$, and zero satisfies a non-strict inequality but not a strict one.
6. If $D = 0$ the two roots coincide and there is no middle interval; f has the sign of a everywhere except at the single point $x = -b/(2a)$, where it is zero. *Why:* $f(x) = a(x - \alpha)^2$.

Procedure C: location of roots under a parameter

1. Write $f(x) = ax^2 + bx + c$ explicitly in terms of the parameter and confirm $a \neq 0$; if the parameter can make $a = 0$, handle that value as a separate linear case before anything else. *Why:* every condition below assumes a genuine parabola.
2. Translate the requirement into the full condition set from the Section 2 table. Do not economise: write down all of D , $af(k)$ and the vertex condition that the case calls for. *Why:* each condition rules out a distinct unwanted configuration, and dropping one is the single most common source of a wrong parameter set.
3. Solve each condition separately as an inequality in the parameter, keeping the results as intervals. *Why:* they must be intersected, and intersecting is only reliable on explicit intervals.
4. Intersect all the intervals. *Why:* the conditions are simultaneous, not alternative.
5. Test the endpoints of your final interval by substituting back and finding the actual roots. *Why:* endpoints are exactly where $D = 0$, or where k is itself a root, and these are the cases the inequalities handle by a hair.
6. Where possible, sanity-check the answer by an independent route — factorise, or use $\alpha + \beta = -b/a$ and $\alpha\beta = c/a$ to describe the roots directly. *Why:* a second derivation catches a sign slip in seconds.

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

Show that $f(x) = 2x^2 - 8x + 11$ is positive for every real x , find its minimum value and the point where it occurs, and state its range. Then solve $x^2 - 5x + 6 \leq 0$.

Solution.

Here $a = 2$, $b = -8$, $c = 11$, so $D = (-8)^2 - 4(2)(11) = 64 - 88 = -24$.

Since $a = 2 > 0$ and $D = -24 < 0$, the condition “ $f(x) > 0$ for every real x ” is met, so f is positive throughout $\mathbb{R}$.

The minimum occurs at $x = -\frac{b}{2a} = -\frac{-8}{4} = 2$, and its value is $-\frac{D}{4a} = -\frac{-24}{8} = 3$.

Verification by direct substitution: $f(2) = 2(4) - 8(2) + 11 = 8 - 16 + 11 = 3$, as claimed. Equivalently, $f(x) = 2(x - 2)^2 + 3$.

Since $a > 0$ there is no maximum, so the range of f is $[3, \infty)$.

For the inequality, $g(x) = x^2 - 5x + 6 = (x - 2)(x - 3)$ has roots 2 and 3, and $a = 1 > 0$. By the sign scheme, $g < 0$ strictly between the roots and $g > 0$ strictly outside. The inequality is non-strict, so the roots themselves are included.

$$x^2 - 5x + 6 \leq 0 \iff x \in [2, 3].$$

Check at an interior point: $g(2.5) = 6.25 - 12.5 + 6 = -0.25 < 0$, consistent.

Example 2 — [JEE Main Level]

Find all real values of k for which the equations $x^2 + kx + 1 = 0$ and $x^2 + x + k = 0$ have at least one common root. In each case say whether one root or both roots are common.

Solution.

Step 1 — check for proportional coefficients. Both leading coefficients are 1, so proportionality would force $k = 1$ (from the x -coefficients) and $1 = k$ (from the constants). These agree, so $k = 1$ is a genuine solution: both equations become $x^2 + x + 1 = 0$ and **both** roots are common. (Those roots are the non-real cube roots of unity ω, ω^2 of Subtopic 03, which is consistent — with real coefficients, a shared non-real root must bring its conjugate with it.)

Step 2 — the remaining case $k \neq 1$. Subtract the second equation from the first to eliminate x^2 :

$$(x^2 + kx + 1) - (x^2 + x + k) = (k - 1)x + (1 - k) = (k - 1)(x - 1) = 0.$$

Since $k \neq 1$, the candidate common root is $x = 1$.

Step 3 — substitute back, as Procedure A demands. In $x^2 + kx + 1 = 0$: $1 + k + 1 = 0$, so $k = -2$. Confirm in the other equation: $1 + 1 + (-2) = 0$, satisfied.

Step 4 — describe the roots for $k = -2$. The first equation becomes $x^2 - 2x + 1 = (x - 1)^2 = 0$, with the repeated root 1. The second becomes $x^2 + x - 2 = (x - 1)(x + 2) = 0$, with roots 1

and -2 . They share exactly one root, namely 1 .

Cross-check with the formula. With $a_1 = 1, b_1 = -2, c_1 = 1$ and $a_2 = 1, b_2 = 1, c_2 = -2$: $c_1 a_2 - c_2 a_1 = 1 - (-2) = 3$, $a_1 b_2 - a_2 b_1 = 1 - (-2) = 3$, $b_1 c_2 - b_2 c_1 = 4 - 1 = 3$. The condition reads $3^2 = 3 \times 3$, i.e. $9 = 9$, and the common root is $3/3 = 1$, exactly as found.

Answer: $k = 1$ (both roots common) and $k = -2$ (exactly one common root, $x = 1$).

Example 3 — [JEE Main — Speed Trap]

Find all real values of a for which $ax^2 + 4x + a > 0$ for every real x .

The trap. The problem looks like a one-line discriminant question. The seductive route is: “positive for all x means $D \leq 0$, so $16 - 4a^2 \leq 0$, so $a^2 \geq 4$, so $a \leq -2$ or $a \geq 2$.” That answer, $|a| \geq 2$, is wrong on two separate counts, and both are deliberately planted: it loses the requirement $a > 0$, and it converts a strict inequality into a non-strict one. A third planted trap is the value $a = 0$, at which the expression is not quadratic at all.

Solution.

Case $a = 0$. The expression becomes $4x$, which is negative for every $x < 0$. So $a = 0$ fails and is discarded. Note that no discriminant argument would have caught this, since there is no quadratic to take a discriminant of.

Case $a \neq 0$. The requirement “ $f(x) > 0$ for every real x ” is equivalent to the pair of conditions $a > 0$ **and** $D < 0$.

$$D = 16 - 4a^2 < 0 \iff a^2 > 4 \iff a < -2 \text{ or } a > 2.$$

Intersecting with $a > 0$ leaves $a > 2$.

Why the discarded pieces really are wrong. Take $a = -3$, which satisfies $|a| > 2$: the expression is $-3x^2 + 4x - 3$ with $D = 16 - 36 = -20 < 0$ and $a < 0$, so it is negative for every real x — the exact opposite of what was asked. Take $a = 2$, the endpoint that $D \leq 0$ would have admitted: the expression is $2x^2 + 4x + 2 = 2(x + 1)^2$, which equals 0 at $x = -1$ and so is not positive for every x .

Answer: $a > 2$, i.e. $a \in (2, \infty)$.

Example 4 — [JEE Advanced Level]

For $f(x) = x^2 - 2mx + (m^2 - 1)$ with $m \in \mathbb{R}$, find all m such that (a) both roots lie strictly between -2 and 4 ; (b) exactly one root lies in the open interval $(-2, 4)$.

Solution.

Here $a = 1 > 0$, $b = -2m$, $c = m^2 - 1$, so the vertex sits at $-\frac{b}{2a} = m$, and

$$D = 4m^2 - 4(m^2 - 1) = 4 > 0 \text{ for every } m,$$

so the roots are always real and distinct. Also record the two boundary values:

$$f(-2) = 4 + 4m + m^2 - 1 = m^2 + 4m + 3 = (m + 1)(m + 3),$$

$$f(4) = 16 - 8m + m^2 - 1 = m^2 - 8m + 15 = (m - 3)(m - 5).$$

(a) Both roots in $(-2, 4)$. The full condition set is $D \geq 0$, $af(-2) > 0$, $af(4) > 0$, and $-2 < -\frac{b}{2a} < 4$. With $a = 1$:

- $D = 4 \geq 0$: satisfied for every m .
- $f(-2) > 0$: $(m + 1)(m + 3) > 0 \iff m < -3$ or $m > -1$.
- $f(4) > 0$: $(m - 3)(m - 5) > 0 \iff m < 3$ or $m > 5$.
- Vertex: $-2 < m < 4$.

Intersecting: the vertex condition confines m to $(-2, 4)$; within it, $f(-2) > 0$ forces $m > -1$, leaving $(-1, 4)$; then $f(4) > 0$ forces $m < 3$, leaving

$$m \in (-1, 3).$$

Independent check. $f(x) = (x - m)^2 - 1$, so the roots are exactly $m - 1$ and $m + 1$. Both lie in $(-2, 4)$ precisely when $m - 1 > -2$ and $m + 1 < 4$, i.e. $m > -1$ and $m < 3$. The two routes agree.

(b) Exactly one root in $(-2, 4)$. The standard test is $f(-2)f(4) < 0$:

$$(m + 1)(m + 3)(m - 3)(m - 5) < 0.$$

The four critical values are $-3, -1, 3, 5$. Testing one point per interval: at $m = -4$ the product is $(-3)(-1)(-7)(-9) > 0$; at $m = -2$ it is $(-1)(1)(-5)(-7) < 0$; at $m = 0$ it is $(1)(3)(-3)(-5) > 0$; at $m = 4$ it is $(5)(7)(1)(-1) < 0$; at $m = 6$ it is $(7)(9)(3)(1) > 0$. Hence the strict test gives

$$m \in (-3, -1) \cup (3, 5).$$

The boundary cases the test cannot see. The product test presumes $f(-2) \neq 0$ and $f(4) \neq 0$, so the four values $m = -3, -1, 3, 5$ must be examined by hand.

- $m = -1$: roots are -2 and 0 . The endpoint -2 is not in the *open* interval, while 0 is. So exactly one root lies in $(-2, 4)$: **include** $m = -1$.
- $m = 3$: roots are 2 and 4 . Here 4 is excluded and 2 is inside, so exactly one root lies in $(-2, 4)$: **include** $m = 3$.
- $m = -3$: roots are -4 and -2 . Neither lies in the open interval, so **exclude**.
- $m = 5$: roots are 4 and 6 . Neither lies in the open interval, so **exclude**.

Answer: (a) $m \in (-1, 3)$. (b) $m \in (-3, -1] \cup [3, 5)$.

Independent check for (b) using the roots $m - 1, m + 1$: exactly one of them lies in $(-2, 4)$ when either $m - 1 \in (-2, 4)$ and $m + 1 \notin (-2, 4)$, giving $m \in (-1, 5)$ intersected with $(m \leq -3$ or $m \geq 3)$, i.e. $[3, 5)$; or $m + 1 \in (-2, 4)$ and $m - 1 \notin (-2, 4)$, giving $m \in (-3, 3)$ intersected with $(m \leq -1$ or $m \geq 5)$, i.e. $(-3, -1]$. The union matches.

Section 5: Practice Exercises

1. Show that $x^2 - 3x + 4 > 0$ for every real x . Find its minimum value and the value of x at which the minimum occurs.
2. Solve the inequality $2x^2 - 5x - 3 \geq 0$.
3. Find the range of $f(x) = -3x^2 + 12x - 7$, and the point at which the extreme value is attained.
4. The equations $x^2 + bx + c = 0$ and $x^2 + cx + b = 0$ have a common root, where $b \neq c$. Prove that $b + c + 1 = 0$ and identify the common root.
5. Find all values of k for which $x^2 - 6x + 8 = 0$ and $x^2 - 9x + k = 0$ have a common root. State whether both roots can ever be common.
6. Find all real p for which both roots of $x^2 + 2(p + 1)x + (9p - 5) = 0$ are negative.
7. Find all real k for which the number 2 lies strictly between the roots of $x^2 + kx + (k^2 - 4) = 0$.
8. The coefficients below are integers. Without solving, state the nature of the roots of (a) $6x^2 - 7x + 2 = 0$, (b) $x^2 - 4x + 2 = 0$, (c) $x^2 + x + 1 = 0$.

Answers.

1. $D = 9 - 16 = -7 < 0$ with $a = 1 > 0$, so $f > 0$ throughout. Minimum value $-D/(4a) = 7/4$, attained at $x = 3/2$.
2. $2x^2 - 5x - 3 = (2x + 1)(x - 3)$, roots $-1/2$ and 3 , $a > 0$; answer $x \in (-\infty, -\frac{1}{2}] \cup [3, \infty)$.
3. $a = -3 < 0$, $D = 144 - 84 = 60$, maximum value $-D/(4a) = 5$ attained at $x = 2$; range $(-\infty, 5]$.
4. Subtracting gives $(b - c)x + (c - b) = (b - c)(x - 1) = 0$; since $b \neq c$ the common root is $x = 1$, and substituting $x = 1$ gives $1 + b + c = 0$.
5. $x^2 - 6x + 8 = (x - 2)(x - 4)$. Root 2 gives $k = 14$; root 4 gives $k = 20$. Both roots common is impossible, since $1/1 \neq (-6)/(-9)$. (Check: $x^2 - 9x + 14 = (x - 2)(x - 7)$ and $x^2 - 9x + 20 = (x - 4)(x - 5)$.)
6. $D/4 = (p + 1)^2 - (9p - 5) = p^2 - 7p + 6 = (p - 1)(p - 6) \geq 0$; $f(0) = 9p - 5 > 0$; vertex $-(p + 1) < 0$. Answer $p \in (\frac{5}{9}, 1] \cup [6, \infty)$.
7. $f(2) = 4 + 2k + k^2 - 4 = k(k + 2) < 0$, so $k \in (-2, 0)$.
8. (a) $D = 1$, a perfect square, so the roots are rational and distinct (they are $2/3$ and $1/2$). (b) $D = 8 > 0$ and not a perfect square, so the roots are irrational conjugate surds, $2 \pm \sqrt{2}$. (c) $D = -3 < 0$, so the roots are a non-real complex conjugate pair, as in Subtopic 04.

Section 6: Crack the MCQ

Q1. The expression $x^2 - 4x + k$ is positive for every real x if and only if (A) $k < 4$ (B) $k > 4$ (C) $k \geq 4$ (D) every real k

Answer: (B). $a = 1 > 0$ already, so the requirement reduces to $D < 0$: $16 - 4k < 0$, i.e. $k > 4$.

Distractor Analysis. (A) applies $D > 0$, which is the condition for two distinct real roots — precisely when the expression is *not* of one sign. (C) uses $D \leq 0$ instead of $D < 0$; at $k = 4$ the expression is $(x - 2)^2$, which is zero at $x = 2$ and therefore not positive for every x . This is the strict-versus-non-strict error. (D) assumes that $a > 0$ alone forces positivity, ignoring the discriminant entirely.

Q2. If $x^2 + 2x + 3 = 0$ and $ax^2 + bx + c = 0$, with $a, b, c \in \mathbb{R}$, have a common root, then $a : b : c$ equals (A) $1 : 2 : 3$ (B) $3 : 2 : 1$ (C) $1 : 3 : 2$ (D) it cannot be determined

Answer: (A). For $x^2 + 2x + 3 = 0$, $D = 4 - 12 = -8 < 0$, so its roots are a non-real conjugate pair. If $ax^2 + bx + c = 0$ has real coefficients and shares one of them, the conjugate-root theorem of Subtopic 04 forces it to share the other too. Two quadratics with both roots common have proportional coefficients, so $a : b : c = 1 : 2 : 3$.

Distractor Analysis. (B) comes from reading the proportion backwards — writing $\frac{1}{a} = \frac{2}{b} = \frac{3}{c}$ and then reporting the numerators of the reciprocals as the answer. (C) permutes b and c , a transcription slip. (D) is chosen by students who reach for the one-common-root formula $(c_1a_2 - c_2a_1)^2 = (a_1b_2 - a_2b_1)(b_1c_2 - b_2c_1)$, find it underdetermined with three unknowns, and give up; the point of the question is that the non-real root removes the “exactly one common root” possibility before any formula is needed.

Q3. Both roots of $x^2 - 6x + k = 0$ are greater than 2 if and only if (A) $k > 8$ (B) $8 < k \leq 9$ (C) $k \geq 9$ (D) $k < 9$

Answer: (B). With $a = 1 > 0$ and $f(x) = x^2 - 6x + k$: $D \geq 0$ gives $36 - 4k \geq 0$, i.e. $k \leq 9$; $af(2) > 0$ gives $4 - 12 + k > 0$, i.e. $k > 8$; the vertex is at $x = 3 > 2$ for every k , so that condition is automatic. Intersecting, $8 < k \leq 9$.

Distractor Analysis. (A) keeps $f(2) > 0$ but drops $D \geq 0$ — the signature error of this topic. At $k = 10$, for instance, $f(2) = 2 > 0$ yet $D = -4 < 0$, so there are no real roots at all to be greater than 2; remember that $af(k) > 0$ holds automatically whenever $D < 0$ and therefore proves nothing on its own. (C) inverts the discriminant inequality, reading $36 - 4k \leq 0$. (D) keeps only the discriminant condition and drops $af(2) > 0$; at $k = 8$ the roots are 2 and 4, and 2 is not greater than 2, so the boundary genuinely must be excluded.

Q4. The minimum value of $f(x) = 4x^2 - 4x + 5$ is (A) 4 (B) 5 (C) $1/2$ (D) -4

Answer: (A). $a = 4 > 0$ and $D = 16 - 80 = -64$, so the minimum value is $-\frac{D}{4a} = \frac{64}{16} = 4$, attained at $x = -\frac{b}{2a} = \frac{1}{2}$. Check: $f(1/2) = 1 - 2 + 5 = 4$.

Distractor Analysis. (B) reports $c = 5$, i.e. $f(0)$ — the error of assuming the extremum sits at $x = 0$ rather than at $x = -b/(2a)$. (C) reports the *location* $x = 1/2$ instead of the *value* of the minimum, a question-misreading that costs a mark far more often than an algebra slip does. (D) uses $\frac{D}{4a}$ instead of $-\frac{D}{4a}$, dropping the minus sign in the vertex form.

Section 7: Common Student Pitfalls and Pro-Tips

Pitfalls

- **Forgetting the $a = 0$ case when the leading coefficient carries a parameter.** “Find all m such that $mx^2 + bx + c > 0$ for all x ” is not a pure discriminant question. Correction: split off $m = 0$ first and treat the resulting linear expression on its own; a non-constant linear expression is never of one sign on all of $\mathbb{R}$.
- **Using $D < 0$ alone as the condition for “always positive”.** $D < 0$ only says the sign never changes; it does not say which sign. Correction: always pair $D < 0$ with $a > 0$ for positivity, or $a < 0$ for negativity.
- **Using $af(k) > 0$ to place roots without first securing $D \geq 0$.** When $D < 0$, $af(k) = a^2|k - \alpha|^2 > 0$ for every real k , so the test passes vacuously and admits parameter values with no real roots at all. Correction: $D \geq 0$ is a separate, non-negotiable line of the condition set.
- **Dropping the vertex condition.** $D \geq 0$ together with $af(k) > 0$ says only that k is outside the root pair — it does not say on which side. Correction: the vertex condition $-b/(2a) > k$ or $-b/(2a) < k$ is what chooses the side.
- **Confusing the extreme value with the point where it occurs.** The minimum of f is $-D/(4a)$; the minimising x is $-b/(2a)$. Correction: name them separately in your working, and verify by substituting $-b/(2a)$ back into f .
- **Sign slip in $-D/(4a)$.** When D is negative and a is positive, $-D/(4a)$ is positive. Correction: sanity-check against the graph — a positive-definite parabola must have a positive minimum.
- **Accepting the subtraction candidate as a common root without substituting back.** The linear equation obtained by eliminating x^2 always has a solution; that solution is a common root only when it satisfies an original quadratic. Correction: substitute back, every time.
- **Assuming a shared root means “exactly one shared root”.** With real coefficients and a non-real shared root, both roots are automatically shared. Correction: check the discriminant of the known quadratic before choosing your method.
- **Sloppy endpoints in inequality answers.** $f(x) \geq 0$ includes the roots; $f(x) > 0$ excludes them. Correction: decide inclusion by evaluating f at the root, which is 0, and testing it against the inequality sign.
- **Reversing an inequality when multiplying by a negative a .** Rewriting $-x^2 + 4x - 3 > 0$ as $x^2 - 4x + 3 > 0$ is wrong; it becomes $x^2 - 4x + 3 < 0$, giving $1 < x < 3$. Correction: multiplying an inequality by -1 flips it.

Pro-Tips

- When $a > 0$, the shortcut sentence for the whole sign theory is: *outside the roots the quadratic agrees with a , between the roots it disagrees*. Everything else is bookkeeping.
- Write the location conditions as a vertical stack — D , then $af(k_1)$, then $af(k_2)$, then vertex — before solving any of them. Under exam pressure a stack is far harder to leave incomplete than a paragraph.
- Prefer $af(k)$ to $f(k)$ in every location problem, even when a is obviously positive. It

makes the condition sign-of- a proof, so the same line works when a parameter can flip a .

- If one of the two quadratics in a common-root problem factorises, factor it. Substituting two explicit roots into the other equation is faster and safer than the cross-multiplication formula, and it never loses a case.
- After solving any parameter problem, substitute the two endpoints of your answer interval and find the actual roots. Endpoints are where $D = 0$ or where k is itself a root, and they are where the marks are.
- To constrain how far apart the roots are, use $|\alpha - \beta| = \sqrt{D}/|a|$ rather than solving for the roots.
- For “exactly one root in (k_1, k_2) ”, start with $f(k_1)f(k_2) < 0$, then immediately test the parameter values that make $f(k_1) = 0$ or $f(k_2) = 0$. The test is blind to those, and examiners know it.

Section 8: Cheat Sheet Box

SUBTOPIC 07 — 60-SECOND RECALL ($f(x) = ax^2 + bx + c$, $a, b, c \in \mathbb{R}$, $a \neq 0$, $D = b^2 - 4ac$)

Vertex form. $f(x) = a \left(x + \frac{b}{2a} \right)^2 - \frac{D}{4a}$; vertex at $x = -\frac{b}{2a}$, value $-\frac{D}{4a}$.

Extremum. $a > 0$: minimum $-\frac{D}{4a}$, range $\left[-\frac{D}{4a}, \infty \right)$. $a < 0$: maximum $-\frac{D}{4a}$, range $\left(-\infty, -\frac{D}{4a} \right]$.

Sign for all x . Sign of a everywhere $\iff D < 0$. Then: $f > 0 \forall x \iff a > 0, D < 0$; $f \geq 0 \forall x \iff a > 0, D \leq 0$; $f < 0 \forall x \iff a < 0, D < 0$; $f \leq 0 \forall x \iff a < 0, D \leq 0$. If a holds a parameter, test $a = 0$ separately.

Inequalities ($D > 0$, roots $\alpha < \beta$). $af(x) > 0$ outside $[\alpha, \beta]$; $af(x) < 0$ on (α, β) . Endpoints in for $\geq, \leq$; out for $>, <$.

Master identity. $af(k) = a^2(k - \alpha)(k - \beta)$. With real roots: $af(k) > 0 \iff k$ outside; $af(k) < 0 \iff k$ between; $af(k) = 0 \iff k$ is a root. Beware: $D < 0 \Rightarrow af(k) > 0$ always.

Location of roots. - Both $> k$: $D \geq 0$, $af(k) > 0$, $-\frac{b}{2a} > k$. - Both $< k$: $D \geq 0$, $af(k) > 0$, $-\frac{b}{2a} < k$. - k between: $af(k) < 0$ (forces $D > 0$ on its own). - Both in (k_1, k_2) : $D \geq 0$, $af(k_1) > 0$, $af(k_2) > 0$, $k_1 < -\frac{b}{2a} < k_2$. - Exactly one in (k_1, k_2) : $f(k_1)f(k_2) < 0$; test $f(k_1) = 0$ and $f(k_2) = 0$ by hand. - $\alpha < k_1 < k_2 < \beta$: $af(k_1) < 0$ and $af(k_2) < 0$. - For $\geq/\leq$ versions, relax $af(k) > 0$ to $af(k) \geq 0$ and the vertex inequality to a non-strict one.

Common roots ($a_1x^2 + b_1x + c_1 = 0$, $a_2x^2 + b_2x + c_2 = 0$). - One common root (when $a_1b_2 - a_2b_1 \neq 0$): $(c_1a_2 - c_2a_1)^2 = (a_1b_2 - a_2b_1)(b_1c_2 - b_2c_1)$, with $\alpha = \frac{c_1a_2 - c_2a_1}{a_1b_2 - a_2b_1}$. - Both

roots common: $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$. - Working method: eliminate x^2 via $a_2f_1 - a_1f_2 = 0$, solve the linear equation, substitute back. - Real coefficients and a non-real shared root $\Rightarrow$ both roots shared.

Nature of roots, rational coefficients. D a perfect square $\Rightarrow$ rational distinct; $D > 0$ non-square $\Rightarrow$ irrational surd pair $p \pm \sqrt{q}$; $D = 0 \Rightarrow$ equal rational; $D < 0 \Rightarrow$ non-real conjugate pair (Subtopic 04).

Symmetric functions (Vieta from Subtopic 04). $\alpha^2 + \beta^2 = \frac{b^2 - 2ac}{a^2}$; $(\alpha - \beta)^2 = \frac{D}{a^2}$,
 $|\alpha - \beta| = \frac{\sqrt{D}}{|a|}$; $\alpha^3 + \beta^3 = \frac{3abc - b^3}{a^3}$; $\frac{1}{\alpha} + \frac{1}{\beta} = -\frac{b}{c}$; $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{b^2 - 2ac}{ac}$.

Chapter 4: Complex Numbers and Quadratic Equations — Round 2 Addendum

Subtopics 01-06 and the Round 1 Supplement (Subtopic 07) together cover the whole NCERT syllabus for this chapter and the great majority of what JEE asks. Round 1 closed the chapter's largest hole — real quadratics: sign, range, extremum, location of roots, and common roots. Two genuine gaps survive that check.

The first is on the quadratic side. Every subtopic so far *solves* a given equation or *reads* its roots off the coefficients. None of them **builds a new equation** whose roots are a stated function of the old ones — the transformation family — and none supplies the two machines that make functions of the roots computable without ever finding the roots: Newton's power-sum recurrence and reduction of a polynomial modulo the quadratic.

The second is on the complex side. Subtopic 06 gives you the collinearity test " $\frac{z_3 - z_1}{z_2 - z_1}$ is real". Its exact companion — " $\frac{z_2 - z_1}{z_4 - z_3}$ is purely imaginary if and only if the segments are perpendicular" — is absent, and with it the whole dictionary that turns $\operatorname{Re}(\bar{u}v)$ into a dot product and $\operatorname{Im}(\bar{u}v)$ into a cross product. Rotation itself is already covered (Subtopic 06, Theorem 1 and Example 4) and is **not** re-taught here.

Two items, continuing the Round 1 numbering.

Item	Topic	Read after
Subtopic 08	Transformation of equations; Newton's recurrence; reduction modulo the quadratic	Subtopic 07
Subtopic 09	Complex numbers as vectors: perpendicularity, dot and cross products, area	Subtopic 06

Wavy-curve and graphical inequality work is deliberately **not** repeated here: Chapter 5, Unit 4 develops the wavy-curve algorithm in full, and Subtopic 07's Procedure B already hands the quadratic case over to it by name. That was checked, not assumed.

Subtopic 08 — Transformation of Equations and Functions of the Roots

Exam Relevance: CBSE Boards touch only the easy end — “form the equation whose roots are $\alpha + 2$ and $\beta + 2$ ” style questions worth 2 marks, built directly on

the sum-and-product formula of Subtopic 04. JEE Main tests the transformation substitution itself almost every year, most often reciprocal roots, squared roots, or a Mobius combination such as $\frac{\alpha}{\alpha-2}$; these are pure-speed questions where the substitution route beats computing the roots by a factor of three. JEE Advanced pushes into Newton's recurrence for $\alpha^n + \beta^n$ at large n , and into evaluating a high-degree polynomial at a root by reducing it modulo the quadratic — both of which look impossible until you see the one line that collapses them. NEET does not test mathematics.

Everything here assumes α, β are the roots of $ax^2 + bx + c = 0$ with $a \neq 0$, and uses only Vieta's relations $\alpha + \beta = -b/a$, $\alpha\beta = c/a$ from Subtopic 04 and the symmetric-function identities from Subtopic 07.

Section 1: Concept Intuition

Change the ruler, not the object. Suppose the Konkan Railway publishes a timetable in Indian Standard Time and you want the same timetable in Gulf Standard Time, ninety minutes behind. You do not re-survey the track. You take the one rule that converts a time — subtract ninety minutes — and apply it to the timetable itself. Every arrival shifts; the train, the track and the ordering are untouched.

A transformation of equations is exactly that. You are handed $ax^2 + bx + c = 0$ and asked for the equation whose roots are $\alpha + h$ and $\beta + h$, or $1/\alpha$ and $1/\beta$, or α^2 and β^2 . **The wrong move is to find α and β .** They are usually irrational or non-real and you do not need them. The right move is to write down the rule $y = \varphi(x)$ that carries an old root to a new one, invert it to get x in terms of y , and substitute that into the equation you already have. What comes out is an equation in y , satisfied by exactly the numbers you wanted.

Where it breaks down is instructive, and it breaks down in exactly one place: **the substitution must be reversible on the roots you care about.** If φ sends two different old roots to the same new number, or if inverting φ needs a square root and therefore two branches, the transformed equation can gain roots you did not ask for or lose one you did. That is why the reciprocal transformation carries the rider $c \neq 0$ and why the squaring transformation is derived by squaring the *equation*, never by writing $x = \sqrt{y}$.

Two roots, one recursion. For powers there is a second idea, and it is the more powerful one. You want $\alpha^9 + \beta^9$. You could grind out symmetric-function identities. Instead, notice that α satisfies $a\alpha^2 = -b\alpha - c$ — a rule for trading one power of α down to lower powers. Applied over and over, it turns any power of α into a linear expression in α , and it turns the sequence $S_n = \alpha^n + \beta^n$ into a three-term recurrence you can run on your fingers. **A quadratic equation is a machine for lowering exponents.** That single sentence is Newton's theorem and the reduction method both.

Section 2: Key Formulas and Definitions

The transformation rule. Let $f(x) = ax^2 + bx + c = 0$ have roots α, β , and let $y = \varphi(x)$ be the required rule. If φ can be inverted as $x = \psi(y)$ on the relevant values, then the equation

with roots $\varphi(\alpha), \varphi(\beta)$ is

$$f(\psi(y)) = 0,$$

cleared of denominators. Every entry in the table below is one application of this rule.

Required roots	Substitute	Transformed equation	Validity
$\alpha + h, \beta + h$	$x = y - h$	$ay^2 + (b - 2ah)y + (ah^2 - bh + c) = 0$	none
$\alpha - h, \beta - h$	$x = y + h$	$ay^2 + (b + 2ah)y + (ah^2 + bh + c) = 0$	none
$k\alpha, k\beta$	$x = y/k$	$ay^2 + bky + ck^2 = 0$	$k \neq 0$
$-\alpha, -\beta$	$x = -y$	$ay^2 - by + c = 0$	none
$\frac{1}{\alpha}, \frac{1}{\beta}$	$x = 1/y$	$cy^2 + by + a = 0$	$c \neq 0$
α^2, β^2	square the equation	$a^2y^2 + (2ac - b^2)y + c^2 = 0$	none
no y -term ("depressed")	$x = y - \frac{b}{2a}$	$ay^2 - \frac{D}{4a} = 0$	none
$\frac{p\alpha + q}{r\alpha + s}, \frac{p\beta + q}{r\beta + s}$	$x = \frac{q - sy}{ry - p}$	substitute, then clear	$ps - qr \neq 0$

Symbols: h, k, p, q, r, s are real (or complex) constants; $D = b^2 - 4ac$ as in Subtopic 04. Note that the reciprocal transformation simply **reverses the coefficient string** $a, b, c \rightarrow c, b, a$ — it does not change any sign. The depressed form $ay^2 = D/(4a)$ is the vertex form of Subtopic 07 read as an equation, which is the cleanest proof that the two topics are the same picture.

Newton's theorem (power sums). Put $S_n = \alpha^n + \beta^n$. Then for every integer $n \geq 2$,

$$aS_n + bS_{n-1} + cS_{n-2} = 0, \quad \text{i.e.} \quad S_n = -\frac{b}{a}S_{n-1} - \frac{c}{a}S_{n-2},$$

with the seeds

$$S_0 = 2, \quad S_1 = \alpha + \beta = -\frac{b}{a}.$$

Validity: the recurrence needs nothing beyond $a \neq 0$ for $n \geq 2$. It extends to all integers n , including negative ones, exactly when $c \neq 0$ (so that neither root is zero and negative powers exist).

Reduction modulo the quadratic. If α is a root of $ax^2 + bx + c = 0$ and $g(x)$ is any polynomial, divide:

$$g(x) = (ax^2 + bx + c)q(x) + px + t \quad \implies \quad g(\alpha) = p\alpha + t.$$

The remainder $px + t$ is linear because the divisor is quadratic. This is the fastest route to any “evaluate this degree-7 polynomial at a nasty surd” question, and it works verbatim when α is non-real.

Figure brief: A single horizontal number line drawn twice, one above the other, with the upper line labelled “roots of $f(x) = 0$ ” carrying two dots at α and β , and the lower line labelled “roots of $f(\psi(y)) = 0$ ” carrying two dots at $\varphi(\alpha)$ and $\varphi(\beta)$. Draw two curved arrows from the upper dots down to the lower dots, both labelled φ , and a single dashed arrow running the other way labelled ψ . Annotate the dashed arrow “this is the arrow you actually substitute”.

Section 3: Derivations and Key Procedures

Derivation: the transformation rule, and why the inverse is the thing you substitute

1. Let α be a root of $f(x) = 0$ and let $Y = \varphi(\alpha)$ be the corresponding new number. Reason: this is the definition of the required roots.
2. Suppose φ is invertible near α , with inverse ψ , so $\alpha = \psi(Y)$. Reason: invertibility is precisely the hypothesis we must carry.
3. Substitute into the known equation: $f(\psi(Y)) = f(\alpha) = 0$. Reason: $\psi(Y)$ is α , so the equation it satisfies is the original one.
4. Hence every required root Y satisfies $f(\psi(y)) = 0$. Reason: steps 1-3 used nothing special about which root we started from, so both $\varphi(\alpha)$ and $\varphi(\beta)$ satisfy it.
5. Conversely, if y_0 satisfies $f(\psi(y_0)) = 0$, then $\psi(y_0)$ is a root of f , so $\psi(y_0) \in \{\alpha, \beta\}$ and $y_0 = \varphi(\psi(y_0)) \in \{\varphi(\alpha), \varphi(\beta)\}$. Reason: this is where invertibility earns its keep; without it the converse fails and extra roots appear.
6. Finally, clear denominators so the result is a polynomial equation, and check that the degree is still 2. Reason: if clearing denominators drops the leading coefficient to zero, a root has been lost to infinity — which is exactly what happens to $1/\alpha$ when $\alpha = 0$. ■

Derivation: the squares transformation, done honestly

1. Start from $ax^2 + bx + c = 0$ and isolate the odd-power term: $ax^2 + c = -bx$. Reason: we want only even powers on one side, since $y = x^2$ can only see even powers.
2. Square both sides: $(ax^2 + c)^2 = b^2x^2$. Reason: squaring is legal in one direction and removes the last odd power.
3. Expand: $a^2x^4 + 2acx^2 + c^2 = b^2x^2$. Reason: direct expansion.
4. Put $y = x^2$: $a^2y^2 + (2ac - b^2)y + c^2 = 0$. Reason: every surviving power of x is even. ■
5. Sanity check against Vieta: the sum of the new roots is $\frac{b^2 - 2ac}{a^2} = \alpha^2 + \beta^2$ and the product is $\frac{c^2}{a^2} = (\alpha\beta)^2$, both matching Subtopic 07. Note the sign: the coefficient of y is $2ac - b^2$, and the *sum* of the roots is its negative.

Squaring in step 2 is a one-way implication, so in principle it could introduce spurious roots. Here it does not, because the degree is still 2 and we have just identified both roots — but that check is the reason step 5 is written down rather than assumed.

Derivation: Newton's theorem

1. α and β satisfy the equation, so $a\alpha^2 + b\alpha + c = 0$ and $a\beta^2 + b\beta + c = 0$. Reason: definition of a root.
2. Multiply the first by α^{n-2} and the second by β^{n-2} : $a\alpha^n + b\alpha^{n-1} + c\alpha^{n-2} = 0$ and $a\beta^n + b\beta^{n-1} + c\beta^{n-2} = 0$. Reason: multiplying zero by anything is zero; for $n \geq 2$ the exponents are non-negative, so no division is involved.
3. Add: $a(\alpha^n + \beta^n) + b(\alpha^{n-1} + \beta^{n-1}) + c(\alpha^{n-2} + \beta^{n-2}) = 0$, i.e. $aS_n + bS_{n-1} + cS_{n-2} = 0$. Reason: the two lines are added term by term. ■

Procedure A: transforming an equation

1. Write the rule as $y = \varphi(x)$ with x standing for a generic root. Why: the transformation is a statement about all roots at once, so no subscripts are needed.
2. Solve that relation for x , obtaining $x = \psi(y)$. Why: this, not φ , is what gets substituted. Reversing the two is the single most common error in the topic.
3. Substitute $\psi(y)$ for x in the original equation and clear denominators.
4. Confirm the result is genuinely quadratic. If the y^2 coefficient has vanished, a root has been lost; say so explicitly and identify which one. Why: examiners set this case deliberately, most often by giving an equation with $c = 0$ and asking for reciprocal roots.
5. Cross-check with Vieta. Compute the sum and product of the intended new roots from $\alpha + \beta$ and $\alpha\beta$, and confirm they match the transformed equation. Why: a sign slip in step 3 is invisible until you do this, and the check costs one line.

Procedure B: evaluating $\alpha^n + \beta^n$ or $g(\alpha)$ without finding the roots

1. Read off a, b, c and write the recurrence $aS_n = -bS_{n-1} - cS_{n-2}$ once, at the top of your working. Why: writing it symbolically first prevents arithmetic drift over many steps.
2. Seed with $S_0 = 2$ and $S_1 = -b/a$. Why: S_0 counts the roots; forgetting that it is 2 and not 1 derails every later term.
3. Step up one index at a time to the target n , writing every intermediate S_k . Why: the recurrence needs the two previous values, so skipping is impossible and tabulating is free.
4. For a polynomial value $g(\alpha)$ instead, divide $g(x)$ by $ax^2 + bx + c$ and keep only the remainder $px + t$; then $g(\alpha) = p\alpha + t$. Why: the quotient is multiplied by something that vanishes at α .
5. Verify by a second route: for small n use the symmetric-function identities of Subtopic 07, and when the roots are non-real use the polar form of Subtopic 05, where $S_n = 2r^n \cos n\theta$. Why: two independent routes agreeing is the only cheap proof that a long recursion has no slip in it.

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

Let α, β be the roots of $x^2 - 5x + 6 = 0$. Form the equation whose roots are (a) $\alpha + 2, \beta + 2$;
(b) $\frac{1}{\alpha}, \frac{1}{\beta}$.

Solution.

(a) The rule is $y = x + 2$, so the inverse is $x = y - 2$. Substitute:

$$(y - 2)^2 - 5(y - 2) + 6 = y^2 - 4y + 4 - 5y + 10 + 6 = y^2 - 9y + 20 = 0.$$

(b) The rule is $y = 1/x$, so $x = 1/y$; here $c = 6 \neq 0$, so no root is zero and the substitution is legal. Substitute and multiply through by y^2 :

$$\frac{1}{y^2} - \frac{5}{y} + 6 = 0 \implies 6y^2 - 5y + 1 = 0.$$

This is the original coefficient string 1, -5, 6 reversed to 6, -5, 1, as the table promises.

Verification. The original roots are 2 and 3. Part (a) should give roots 4 and 5: indeed $y^2 - 9y + 20 = (y - 4)(y - 5)$. Part (b) should give $1/2$ and $1/3$: indeed $6y^2 - 5y + 1 = (2y - 1)(3y - 1)$. Both confirmed.

Example 2 — [JEE Main Level]

α and β are the roots of $x^2 - 3x + 1 = 0$. Find the quadratic equation whose roots are $\frac{\alpha}{\alpha - 2}$ and $\frac{\beta}{\beta - 2}$, and hence evaluate $\frac{\alpha}{\alpha - 2} \cdot \frac{\beta}{\beta - 2}$.

Solution.

Step 1 — invert the rule. Put $y = \frac{x}{x - 2}$. Then $y(x - 2) = x$, so $yx - 2y = x$, so $x(y - 1) = 2y$ and

$$x = \frac{2y}{y - 1} \quad (y \neq 1).$$

Step 2 — substitute. Put this into $x^2 - 3x + 1 = 0$ and multiply through by $(y - 1)^2$:

$$4y^2 - 6y(y - 1) + (y - 1)^2 = 0.$$

Step 3 — expand. $4y^2 - 6y^2 + 6y + y^2 - 2y + 1 = -y^2 + 4y + 1$, so

$$y^2 - 4y - 1 = 0.$$

Step 4 — read off the product. By Vieta on the new equation, the product of its roots is $-1/1 = -1$.

Independent check. The original roots are $\alpha = \frac{3 + \sqrt{5}}{2}$ and $\beta = \frac{3 - \sqrt{5}}{2}$. Then $\alpha - 2 = \frac{\sqrt{5} - 1}{2}$ and $\beta - 2 = \frac{-\sqrt{5} - 1}{2}$, giving $\frac{\alpha}{\alpha - 2} = 2 + \sqrt{5}$ and $\frac{\beta}{\beta - 2} = 2 - \sqrt{5}$. Their sum is 4 and their product is $4 - 5 = -1$, matching $y^2 - 4y - 1 = 0$ exactly.

Note that $y = 1$ is excluded by the inversion, and correctly so: $y = 1$ would need $x/(x-2) = 1$, i.e. $-2 = 0$. Nothing was lost.

Example 3 — [JEE Main — Speed Trap]

If α, β are the roots of $2x^2 + 3x - 4 = 0$, find the equation whose roots are α^2 and β^2 .

The trap. The seductive one-liner is: “the new sum is $(\alpha + \beta)^2$ and the new product is $(\alpha\beta)^2$.” That gives sum = $9/4$, product = 4 , and the answer $4y^2 - 9y + 16 = 0$. It is wrong, and it is wrong for a reason worth naming: **the sum of the squares is not the square of the sum.** You need $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$, the very first identity of Subtopic 07’s symmetric-function list. A second planted trap is the sign of the middle coefficient in the memorised form $a^2y^2 + (2ac - b^2)y + c^2 = 0$: that coefficient is $2ac - b^2$, whereas the *sum* of the roots is its negative, $b^2 - 2ac$.

Solution.

Here $a = 2$, $b = 3$, $c = -4$, so $\alpha + \beta = -3/2$ and $\alpha\beta = -2$. Then

$$\alpha^2 + \beta^2 = \left(-\frac{3}{2}\right)^2 - 2(-2) = \frac{9}{4} + 4 = \frac{25}{4}, \quad \alpha^2\beta^2 = (-2)^2 = 4.$$

The required equation is $y^2 - \frac{25}{4}y + 4 = 0$, i.e.

$$4y^2 - 25y + 16 = 0.$$

Second route, independent of Vieta. From $2x^2 - 4 = -3x$, square both sides: $(2x^2 - 4)^2 = 9x^2$, so $4x^4 - 16x^2 + 16 = 9x^2$, so $4x^4 - 25x^2 + 16 = 0$. With $y = x^2$ this is $4y^2 - 25y + 16 = 0$. The two routes agree.

Why the discarded answer really is wrong. The original equation has $D = 9 + 32 = 41 > 0$, so α and β are real and α^2, β^2 must be real and non-negative. But $4y^2 - 9y + 16 = 0$ has discriminant $81 - 256 = -175 < 0$, so its roots are non-real. The seductive answer fails a test you can run in five seconds, and that test is worth building into your reflexes: **after any squares transformation on a real-rooted quadratic, check that the new discriminant is non-negative.**

Answer: $4y^2 - 25y + 16 = 0$.

Example 4 — [JEE Advanced Level]

Let α, β be the roots of $x^2 - 2x + 4 = 0$. (a) Find $\alpha^9 + \beta^9$. (b) Find the value of $\alpha^4 - 4\alpha^3 + 8\alpha^2 + 3\alpha + 5$.

Solution.

Here $a = 1$, $b = -2$, $c = 4$, and $D = 4 - 16 = -12 < 0$, so the roots are the non-real conjugate pair $1 \pm i\sqrt{3}$. Finding them and raising to the ninth power directly is exactly the work the

two machines of this subtopic exist to avoid.

(a) *Newton's recurrence.* With $a = 1$, $b = -2$, $c = 4$ the recurrence $aS_n + bS_{n-1} + cS_{n-2} = 0$ reads

$$S_n = 2S_{n-1} - 4S_{n-2},$$

seeded by $S_0 = 2$ and $S_1 = -b/a = 2$. Running it:

$$\begin{aligned} S_2 &= 2(2) - 4(2) = -4, & S_3 &= 2(-4) - 4(2) = -16, & S_4 &= 2(-16) - 4(-4) = -16, \\ S_5 &= 2(-16) - 4(-16) = 32, & S_6 &= 2(32) - 4(-16) = 128, & S_7 &= 2(128) - 4(32) = 128, \\ S_8 &= 2(128) - 4(128) = -256, & S_9 &= 2(-256) - 4(128) = -1024. \end{aligned}$$

Independent check by polar form (Subtopic 05). The roots are $1 \pm i\sqrt{3}$, which have modulus $r = \sqrt{1+3} = 2$ and arguments $\pm\pi/3$. So $\alpha = 2e^{i\pi/3}$, $\beta = 2e^{-i\pi/3}$ and

$$S_n = 2^n (e^{in\pi/3} + e^{-in\pi/3}) = 2^{n+1} \cos \frac{n\pi}{3}.$$

At $n = 9$: $S_9 = 2^{10} \cos 3\pi = 1024 \times (-1) = -1024$. The two routes agree, and the formula also reproduces $S_2 = 8 \cos(2\pi/3) = -4$ and $S_6 = 128 \cos 2\pi = 128$.

(b) *Reduction modulo the quadratic.* Divide $g(x) = x^4 - 4x^3 + 8x^2 + 3x + 5$ by $x^2 - 2x + 4$:

$$x^4 - 4x^3 + 8x^2 + 3x + 5 = (x^2 - 2x + 4)(x^2 - 2x) + (11x + 5).$$

Since $\alpha^2 - 2\alpha + 4 = 0$, the first product vanishes at $x = \alpha$, leaving

$$g(\alpha) = 11\alpha + 5.$$

Independent check. Take $\alpha = 1 + i\sqrt{3} = 2e^{i\pi/3}$. Then $\alpha^2 = 4e^{2i\pi/3} = -2 + 2i\sqrt{3}$, $\alpha^3 = 8e^{i\pi} = -8$, $\alpha^4 = 16e^{4i\pi/3} = -8 - 8i\sqrt{3}$. Substituting,

$$g(\alpha) = (-8 - 8i\sqrt{3}) + 32 + 8(-2 + 2i\sqrt{3}) + 3(1 + i\sqrt{3}) + 5 = 16 + 11i\sqrt{3},$$

and $11\alpha + 5 = 11(1 + i\sqrt{3}) + 5 = 16 + 11i\sqrt{3}$. They match.

Answer: (a) -1024 . (b) $11\alpha + 5$, which equals $16 + 11i\sqrt{3}$ for $\alpha = 1 + i\sqrt{3}$ and $16 - 11i\sqrt{3}$ for the other root.

Section 5: Practice Exercises

1. α, β are the roots of $x^2 + 7x + 12 = 0$. Form the equation whose roots are $\alpha - 2$ and $\beta - 2$.
2. Form the equation whose roots are the reciprocals of the roots of $5x^2 - 3x + 2 = 0$.
3. Form the equation whose roots are the squares of the roots of $x^2 + 6x + 4 = 0$.
4. Remove the x -term from $3x^2 - 12x + 5 = 0$, stating the substitution used, and check your answer against $ay^2 = D/(4a)$.

5. α, β are the roots of $x^2 - 5x + 3 = 0$. Use Newton's recurrence to find $\alpha^4 + \beta^4$, then confirm the value by a symmetric-function identity.
6. α is a root of $x^2 - x - 1 = 0$. Show that $\alpha^5 = 5\alpha + 3$.
7. α, β are the roots of $ax^2 + bx + c = 0$ with $c \neq 0$. Show that the equation whose roots are α/β and β/α is $acy^2 - (b^2 - 2ac)y + ac = 0$.
8. $\tan A, \tan B$ are the roots of $x^2 - px + q = 0$ and $\cot A, \cot B$ are the roots of $x^2 - rx + s = 0$, with $q \neq 0$. Find rs in terms of p and q .

Answers.

1. $y^2 + 11y + 30 = 0$ (roots -5 and -6).
2. $2y^2 - 3y + 5 = 0$.
3. $y^2 - 28y + 16 = 0$.
4. Substitute $x = y + 2$; the equation becomes $3y^2 - 7 = 0$. Check: $D = 144 - 60 = 84$ and $D/(4a) = 84/12 = 7$, so $3y^2 = 7$, agreeing.
5. $S_n = 5S_{n-1} - 3S_{n-2}$ with $S_0 = 2, S_1 = 5$; then $S_2 = 19, S_3 = 80, S_4 = 343$. Check: $\alpha^4 + \beta^4 = (\alpha^2 + \beta^2)^2 - 2(\alpha\beta)^2 = 19^2 - 2(9) = 343$.
6. $x^n \equiv F_n x + F_{n-1} \pmod{x^2 - x - 1}$; at $n = 5$ this is $5x + 3$. Numerically, $\alpha = (1 + \sqrt{5})/2$ gives $\alpha^5 = (11 + 5\sqrt{5})/2 = 5\alpha + 3$.
7. Sum $= \frac{\alpha^2 + \beta^2}{\alpha\beta} = \frac{b^2 - 2ac}{ac}$ and product $= 1$; clearing denominators gives the stated equation.
8. Reciprocal transformation of $x^2 - px + q = 0$ gives $qy^2 - py + 1 = 0$, i.e. $y^2 - \frac{p}{q}y + \frac{1}{q} = 0$. Hence $r = p/q, s = 1/q$ and $rs = p/q^2$.

Section 6: Crack the MCQ

Q1. The equation whose roots are each 2 more than the roots of $x^2 - 4x + 3 = 0$ is (A) $y^2 - 8y + 15 = 0$ (B) $y^2 - 1 = 0$ (C) $y^2 - 4y + 15 = 0$ (D) $y^2 - 8y + 3 = 0$

Answer: (A). The rule is $y = x + 2$, so substitute $x = y - 2$: $(y - 2)^2 - 4(y - 2) + 3 = y^2 - 8y + 15$. Check: the original roots are 1 and 3, the new roots are 3 and 5, with sum 8 and product 15.

Distractor Analysis. (B) is what you get from substituting $x = y + 2$ — the direction reversal, which *decreases* the roots by 2 and gives $y^2 - 1 = 0$ with roots ± 1 . This is the signature error of the topic and the reason Procedure A step 2 is written as a separate step. (C) shifts the product correctly but leaves the sum untouched, the mark of a student who computed the new product from Vieta and then copied the old -4 across. (D) is the mirror slip: sum shifted, product copied.

Q2. If α, β are the roots of $x^2 + px + q = 0$ with $q \neq 0$, the equation whose roots are $1/\alpha$ and $1/\beta$ is (A) $qx^2 + px + 1 = 0$ (B) $x^2 + px + q = 0$ (C) $qx^2 - px + 1 = 0$ (D) $x^2 + qx + p = 0$

Answer: (A). Substitute $x = 1/y$ into $x^2 + px + q = 0$ and multiply by y^2 : $1 + py + qy^2 = 0$. Check by Vieta: $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta} = -\frac{p}{q}$ and $\frac{1}{\alpha\beta} = \frac{1}{q}$, giving $y^2 + \frac{p}{q}y + \frac{1}{q} = 0$.

Distractor Analysis. (C) attaches a sign flip to the coefficient reversal. Reversing the string $1, p, q$ to $q, p, 1$ changes no signs at all; the belief that “inverting the roots must invert some-

thing else too” is exactly the misconception being tested. (B) is the palindromic reflex — assuming a reciprocal transformation leaves the equation unchanged. That is true only for a palindromic equation, one with $a = c$, and $x^2 + px + q$ is palindromic only when $q = 1$. (D) swaps p and q in place rather than reversing the whole coefficient string, which coincidentally gives the right answer only when $p = q = 1$.

Q3. If α, β are the roots of $x^2 - 6x + 5 = 0$, the equation whose roots are α^2 and β^2 is (A) $x^2 - 26x + 25 = 0$ (B) $x^2 - 36x + 25 = 0$ (C) $x^2 - 26x + 5 = 0$ (D) $x^2 - 12x + 25 = 0$

Answer: (A). The roots are 1 and 5, so the squares are 1 and 25, with sum 26 and product 25. By the table, $a^2y^2 + (2ac - b^2)y + c^2 = y^2 + (10 - 36)y + 25 = y^2 - 26y + 25$.

Distractor Analysis. (B) uses $\alpha^2 + \beta^2 = (\alpha + \beta)^2 = 36$, dropping the $-2\alpha\beta$ — the identical error to Example 3’s trap, and by far the most common answer after the correct one. (C) squares the sum correctly but forgets to square the product, leaving c where c^2 belongs. (D) doubles the sum instead of squaring the roots, the slip of a student who confused “roots squared” with “roots doubled”.

Q4. If α, β are the roots of $x^2 - x + 1 = 0$ and $S_n = \alpha^n + \beta^n$, then S_5 equals (A) 1 (B) -1 (C) 2 (D) -2

Answer: (A). The recurrence is $S_n = S_{n-1} - S_{n-2}$ with $S_0 = 2$, $S_1 = 1$. Then $S_2 = -1$, $S_3 = -2$, $S_4 = -1$, $S_5 = 1$. Check by polar form: the roots are $e^{\pm i\pi/3}$, so $S_n = 2 \cos(n\pi/3)$ and $S_5 = 2 \cos(5\pi/3) = 1$.

Distractor Analysis. (B) is S_4 — an off-by-one, produced by starting the recursion at S_1 while counting it as the zeroth step. (D) is S_3 , the value reached by a student who seeds with $S_1 = 2$ instead of $S_0 = 2$, so the whole table shifts by two places. (C) is S_0 and S_6 , chosen by students who spot that ω -like roots repeat and assume the period is 3; the period of S_n here is in fact 6, since $S_n = 2 \cos(n\pi/3)$.

Section 7: Common Student Pitfalls and Pro-Tips

Pitfalls

- **Substituting φ instead of ψ .** To get roots $\alpha + h$ you substitute $x = y - h$, not $x = y + h$. Correction: write the rule, solve it for x on a separate line, and only then substitute. The extra line costs three seconds and prevents the most common error in the topic.
- **Squaring the sum instead of summing the squares.** $\alpha^2 + \beta^2$ is $(\alpha + \beta)^2 - 2\alpha\beta$. Correction: never write a squares transformation without writing the $-2\alpha\beta$ first.
- **Adding a sign flip to the reciprocal transformation.** $ax^2 + bx + c \rightarrow cx^2 + bx + a$. The coefficients reverse; nothing changes sign. Correction: derive it in one line from $x = 1/y$ rather than recalling it.
- **Using the reciprocal transformation when $c = 0$.** Then one root is 0, its reciprocal does not exist, and the transformed “quadratic” collapses to a linear equation. Correction: check $c \neq 0$ before inverting, and if $c = 0$ say plainly that only one reciprocal root exists.
- **Seeding Newton’s recurrence with $S_0 = 1$.** $S_0 = \alpha^0 + \beta^0 = 2$, because there are two roots. Correction: say “two roots, so $S_0 = 2$ ” out loud before starting.

- **Running Newton's recurrence with the wrong signs.** The recurrence is $aS_n + bS_{n-1} + cS_{n-2} = 0$, so $S_n = -\frac{b}{a}S_{n-1} - \frac{c}{a}S_{n-2}$. Both signs flip when you move them across. Correction: write the homogeneous form first and rearrange once.
- **Keeping the quotient in the reduction method.** In $g(x) = (ax^2 + bx + c)q(x) + px + t$ only the remainder survives. Correction: after dividing, discard $q(x)$ immediately so it cannot contaminate the next line.
- **Forgetting to check that the transformed equation is still quadratic.** If the y^2 coefficient vanished, a root went to infinity. Correction: state which root was lost rather than silently reporting a linear equation.

Pro-Tips

- The depressed form $ay^2 = D/(4a)$ obtained by $x = y - b/(2a)$ is the vertex form of Subtopic 07 wearing different clothes. Recognising that lets you reuse one derivation for two topics.
- For any transformation, the Vieta cross-check is nearly free: compute the intended new sum and product from $\alpha + \beta$ and $\alpha\beta$, and compare. Do it every time.
- When the roots are non-real, always take the polar form of Subtopic 05 as your second route: $S_n = 2r^n \cos n\theta$ turns a nine-step recursion into one cosine.
- For $g(x)$ with $\deg g \geq 3$, dividing by the quadratic is always faster than substituting a surd, and it is the only sane method when α is complex.
- The relation $x^n \equiv F_n x + F_{n-1} \pmod{x^2 - x - 1}$ is worth recognising on sight; the golden-ratio quadratic is a JEE favourite precisely because it hides Fibonacci numbers.
- If a question asks only for the *product* or only for the *sum* of the transformed roots, build the transformed equation anyway. It takes one extra line and immunises you against a follow-up part.

Section 8: Cheat Sheet Box

SUBTOPIC 08 — 60-SECOND RECALL (α, β roots of $ax^2 + bx + c = 0$, $a \neq 0$, $D = b^2 - 4ac$)

Master rule. Required roots $y = \varphi(x)$; invert to $x = \psi(y)$; substitute $\psi(y)$ into $f(x) = 0$; clear denominators. **You substitute the inverse.**

Standard transformations. $\alpha + h$: $ay^2 + (b - 2ah)y + (ah^2 - bh + c) = 0$ (put $x = y - h$). $\alpha - h$: $ay^2 + (b + 2ah)y + (ah^2 + bh + c) = 0$ (put $x = y + h$). $k\alpha$ ($k \neq 0$): $ay^2 + bky + ck^2 = 0$. $-\alpha$: $ay^2 - by + c = 0$. $1/\alpha$ ($c \neq 0$): $cy^2 + by + a = 0$ — reverse the coefficients, change no sign. α^2 : $a^2y^2 + (2ac - b^2)y + c^2 = 0$; new sum $= \frac{b^2 - 2ac}{a^2}$, new product $= \frac{c^2}{a^2}$. Depressed (no y -term): $x = y - \frac{b}{2a}$ gives $ay^2 = \frac{D}{4a}$. Mobius $\frac{p\alpha + q}{r\alpha + s}$ ($ps - qr \neq 0$): put $x = \frac{q - sy}{ry - p}$.

Newton's theorem. $S_n = \alpha^n + \beta^n$ satisfies $aS_n + bS_{n-1} + cS_{n-2} = 0$ for $n \geq 2$ (all integers n if $c \neq 0$). Seeds $S_0 = 2$, $S_1 = -b/a$. Non-real roots: $S_n = 2r^n \cos n\theta$.

Reduction modulo the quadratic. $g(x) = (ax^2 + bx + c)q(x) + px + t \Rightarrow g(\alpha) = p\alpha + t$. Keep the remainder, discard the quotient.

Always check. Is the transformed equation still degree 2? Do its sum and product match the intended new roots? For a squares transformation on a real-rooted quadratic, is the new

discriminant non-negative?

Subtopic 09 — Complex Numbers as Vectors: Perpendicularity, Dot and Cross Products

Read immediately after Subtopic 06. Short-form addendum: method statement, worked examples, practice with answers.

Exam Relevance: CBSE Boards do not reach this material — it sits beyond Class-11 scope exactly as Subtopic 06 does. JEE Main uses the perpendicularity criterion in disguise, most often as “show that these three points form a right-angled triangle” or “find the area of the triangle with these vertices”, both of which are one-line questions with the dictionary below and three-line questions without it. JEE Advanced is where it earns its place: the circumcentre-orthocentre results, altitude concurrency, and any configuration proof that Subtopic 06’s rotation theorem sets up but cannot finish, all reduce to the single statement $\operatorname{Re}(\bar{u}v) = 0$. NEET does not test mathematics.

Method. Subtopic 06 established that z_1, z_2, z_3 are collinear if and only if $\frac{z_3 - z_1}{z_2 - z_1}$ is real. That test has an exact companion which the subtopic does not state, and the pair together is the whole toolkit.

Recall from Subtopic 02 that for any complex w ,

$$w \text{ is real} \iff w = \bar{w} \iff w - \bar{w} = 0, \quad w \text{ is purely imaginary} \iff w + \bar{w} = 0.$$

Apply this to the ratio of two displacements. Write $u = z_2 - z_1$ for the vector from z_1 to z_2 , and $v = z_4 - z_3$ for the vector from z_3 to z_4 , both non-zero. Then:

$$AB \parallel CD \iff \frac{u}{v} \in \mathbb{R} \iff \bar{u}v - u\bar{v} = 0 \iff \operatorname{Im}(\bar{u}v) = 0,$$

$$AB \perp CD \iff \frac{u}{v} \text{ is purely imaginary} \iff \bar{u}v + u\bar{v} = 0 \iff \operatorname{Re}(\bar{u}v) = 0.$$

Derivation of the second line. Set $w = u/v$. Multiplying $w + \bar{w} = 0$ through by the strictly positive real number $v\bar{v} = |v|^2$ gives $u\bar{v} + \bar{u}v = 0$, which is $2\operatorname{Re}(\bar{u}v) = 0$. The multiplication is reversible because $|v|^2 > 0$, so the three statements are equivalent, not merely implied. ■

The dictionary. Write $u = x_1 + iy_1$ and $v = x_2 + iy_2$ as plane vectors. Then

$$\bar{u}v = (x_1 - iy_1)(x_2 + iy_2) = \underbrace{(x_1x_2 + y_1y_2)}_{u \cdot v} + i \underbrace{(x_1y_2 - y_1x_2)}_{u \times v}.$$

So $\operatorname{Re}(\bar{u}v)$ is the **dot product** and $\operatorname{Im}(\bar{u}v)$ is the **scalar cross product** — every plane-vector fact you already own transfers immediately. In particular

$$\cos \theta = \frac{\operatorname{Re}(\bar{u}v)}{|u||v|}, \quad \sin \theta = \frac{\operatorname{Im}(\bar{u}v)}{|u||v|},$$

where θ is the directed angle from u to v — which is $\arg(v/u)$, and therefore consistent with the rotation theorem of Subtopic 06.

Critical order-of-factors note. The conjugate goes on the **first** vector. The quantity $\operatorname{Re}(uv)$ — no bar — is not the dot product and tests nothing geometric about the pair. Section “worked example 2” below shows it giving both a false negative and a false positive.

Area of a triangle. For vertices z_1, z_2, z_3 ,

$$\text{Area} = \frac{1}{2} |\operatorname{Im}[(\bar{z}_2 - \bar{z}_1)(z_3 - z_1)]| = \frac{1}{2} |\operatorname{Im}(\bar{z}_1 z_2 + \bar{z}_2 z_3 + \bar{z}_3 z_1)|.$$

The first form is “half the cross product of two edge vectors”; the second is the same quantity expanded, and is the shoelace formula in complex clothing. Dropping the modulus signs gives the **signed** area, positive exactly when $z_1 \rightarrow z_2 \rightarrow z_3$ runs anticlockwise — which makes the same expression an orientation test.

Orthocentre from the circumcentre. If the circumcentre of triangle $z_1 z_2 z_3$ is at the origin, the orthocentre is

$$H = z_1 + z_2 + z_3.$$

For a circumcentre at a general point O , translating gives $H = z_1 + z_2 + z_3 - 2O$. Since the centroid is $G = \frac{z_1 + z_2 + z_3}{3}$, this rearranges to $H - G = 2(G - O)$: the Euler line, with its 2 : 1 ratio, falls out of the same identity. (The coordinate-geometry treatment of these centres is Chapter 9; this is the complex-number route to the same facts.)

Figure brief: Argand plane with a triangle $A(z_1), B(z_2), C(z_3)$ inscribed in a circle centred at the origin O , radius r , with OA, OB, OC drawn as thin radii. Mark the point $H = z_1 + z_2 + z_3$ inside the triangle and draw the segment AH extended to meet BC at a right angle, with the right-angle square shown at the foot. Alongside the figure, draw two free vectors u and v from a common point with the angle θ between them, annotated $\operatorname{Re}(\bar{u}v) = |u||v| \cos \theta$ and $\operatorname{Im}(\bar{u}v) = |u||v| \sin \theta$.

Worked Example 1 — [JEE Main Level]

The points $A = 1 + i, B = 4 + 2i, C = 5 - i$ are the vertices of a triangle. Show that the angle at B is a right angle, and find the area of the triangle by two independent methods.

Solution.

The two edge vectors at B are

$$u = z_A - z_B = (1 + i) - (4 + 2i) = -3 - i, \quad v = z_C - z_B = (5 - i) - (4 + 2i) = 1 - 3i.$$

Right angle. Compute $\bar{u}v = (-3 + i)(1 - 3i) = -3 + 9i + i - 3i^2 = -3 + 10i + 3 = 10i$. Its real part is 0, so $u \perp v$ and the angle at B is a right angle. The ratio form says the same thing: $\frac{u}{v} = \frac{-3 - i}{1 - 3i} = \frac{(-3 - i)(1 + 3i)}{1 + 9} = \frac{-3 - 9i - i + 3}{10} = -i$, which is purely imaginary.

Area, method 1 (half the product of the legs). $|u| = \sqrt{9 + 1} = \sqrt{10}$ and $|v| = \sqrt{1 + 9} = \sqrt{10}$, so the area is $\frac{1}{2}\sqrt{10} \cdot \sqrt{10} = 5$.

Area, method 2 (the cross-product formula). Using $\frac{1}{2}|\operatorname{Im}(\bar{u}v)|$ with the same u, v : $\operatorname{Im}(10i) = 10$, giving area 5. Using instead the symmetric form,

$$\bar{z}_1 z_2 = (1 - i)(4 + 2i) = 6 - 2i, \quad \bar{z}_2 z_3 = (4 - 2i)(5 - i) = 18 - 14i, \quad \bar{z}_3 z_1 = (5 + i)(1 + i) = 4 + 6i,$$

whose sum is $28 - 10i$, so the area is $\frac{1}{2}| -10 | = 5$. All three routes agree.

Answer: the angle at B is 90° ; the area is 5.

Worked Example 2 — [JEE Main — Speed Trap]

Decide whether the vectors $u = 1 + i$ and $v = 1 - i$ are perpendicular, and whether $u = 1 + i$ and $v = 1 + i$ are perpendicular. Use the criterion “ $u \perp v \iff \operatorname{Re}(uv) = 0$ ” and then the correct criterion, and explain the discrepancy.

The trap. The criterion with the bar dropped, $\operatorname{Re}(uv) = 0$, is memorable, symmetric and wrong. It is not a statement about the angle between u and v at all: since $\arg(uv) = \arg u + \arg v$, it says that the *sum* of the two arguments is $\pm\pi/2$, whereas perpendicularity is about their *difference*. Because it is a symmetric-looking formula it survives a superficial sanity check, which is precisely why examiners plant it.

Solution.

First pair, $u = 1 + i, v = 1 - i$. As plane vectors these are $(1, 1)$ and $(1, -1)$, whose dot product is $1 - 1 = 0$: they are genuinely perpendicular. The correct criterion agrees: $\bar{u}v = (1 - i)(1 - i) = 1 - 2i + i^2 = -2i$, and $\operatorname{Re}(-2i) = 0$. But the flawed criterion gives $uv = (1 + i)(1 - i) = 2$, with $\operatorname{Re}(uv) = 2 \neq 0$, and would report **not perpendicular** — a false negative.

Second pair, $u = v = 1 + i$. These are the same vector, so the angle between them is zero and they are as far from perpendicular as possible. The correct criterion agrees: $\bar{u}v = (1 - i)(1 + i) = 2$, real part $2 \neq 0$. But the flawed criterion gives $uv = (1 + i)^2 = 2i$, with $\operatorname{Re}(uv) = 0$, and would report **perpendicular** — a false positive.

Why. $\bar{u}v$ has argument $\arg v - \arg u$, the angle you care about. uv has argument $\arg v + \arg u$, which depends on where the pair sits in the plane and not on the angle between them. Rotating both vectors together leaves $\bar{u}v$ unchanged but spins uv ; only the first can be a valid geometric test.

Answer: the first pair is perpendicular, the second is not. The criterion $\operatorname{Re}(uv) = 0$ gets both wrong; the correct criterion is $\operatorname{Re}(\bar{u}v) = 0$, equivalently $\bar{u}v + u\bar{v} = 0$.

Worked Example 3 — [JEE Advanced Level]

Let z_1, z_2, z_3 lie on the circle $|z| = r$, so that the circumcentre of triangle $z_1 z_2 z_3$ is the origin. Prove that $H = z_1 + z_2 + z_3$ is the orthocentre, and verify the result on a triangle whose orthocentre you know independently.

Solution.

It is enough to show that $AH \perp BC$ and $BH \perp CA$; two altitudes determine the orthocentre, and the third then passes through it automatically.

Step 1 — the vector along AH. $H - z_1 = z_2 + z_3$.

Step 2 — apply the criterion. We must show $\operatorname{Re}[(\overline{H - z_1})(z_3 - z_2)] = 0$, equivalently that $(z_2 + z_3)(z_3 - z_2)$ is purely imaginary. Expand:

$$(\bar{z}_2 + \bar{z}_3)(z_3 - z_2) = \bar{z}_2 z_3 - \bar{z}_2 z_2 + \bar{z}_3 z_3 - \bar{z}_3 z_2 = \bar{z}_2 z_3 - |z_2|^2 + |z_3|^2 - \bar{z}_3 z_2.$$

Step 3 — use the circle. Since $|z_2| = |z_3| = r$, the two modulus terms cancel exactly, leaving

$$\bar{z}_2 z_3 - \bar{z}_3 z_2 = \bar{z}_2 z_3 - \overline{\bar{z}_2 z_3} = 2i \operatorname{Im}(\bar{z}_2 z_3),$$

which is purely imaginary. Hence $\operatorname{Re}[(\overline{H - z_1})(z_3 - z_2)] = 0$ and $AH \perp BC$.

Step 4 — the second altitude. The expression $z_1 + z_2 + z_3$ is symmetric in the three vertices, so repeating steps 1-3 with the roles of z_1 and z_2 exchanged gives $BH \perp CA$ with no new work. Therefore H lies on two altitudes and is the orthocentre. ■

This is exactly where the cancellation lives: the step that makes the proof work is $|z_2| = |z_3|$, i.e. the hypothesis that the circumcentre is at the origin. Move the circumcentre and the two modulus terms no longer cancel, which is why the general formula carries the correction term: $H = z_1 + z_2 + z_3 - 2O$.

Independent verification. Take $z_1 = 1, z_2 = i, z_3 = -1$ on the unit circle. The segment from -1 to 1 is a diameter, so by the angle-in-a-semicircle theorem the angle at $z_2 = i$ is a right angle. The orthocentre of a right-angled triangle is the right-angle vertex itself, namely i . The formula gives $z_1 + z_2 + z_3 = 1 + i - 1 = i$. They agree.

Corollary (worth memorising). Since $G = \frac{1}{3}(z_1 + z_2 + z_3)$, the origin-circumcentre case gives $H = 3G$, so O, G, H are collinear with $OH = 3OG$ — the Euler line and its 2 : 1 ratio, obtained without a single coordinate.

Practice Exercises

1. Show that the segment joining $2 + i$ to $5 + 5i$ is perpendicular to the segment joining 1 to $-7 + 6i$.
2. Find the area of the triangle with vertices $1 + 2i, 4 + 6i, -2 + 5i$.
3. If $|z_1| = |z_2| = |z_3| = 1$ and $z_1 + z_2 + z_3 = 0$, prove that z_1, z_2, z_3 are the vertices of an equilateral triangle.
4. For $u = 3 + 4i$ and $v = -8 + 6i$, compute $u \cdot v$ and $u \times v$ using the dictionary, and state the angle between them.

5. Show that two non-zero complex numbers z_1, z_2 satisfy $\bar{z}_1 z_2 + z_1 \bar{z}_2 = 0$ if and only if their position vectors are perpendicular. Deduce the equation of the line through the origin perpendicular to a fixed non-zero a .
6. A triangle has circumcentre $O = 2 - i$ and vertices z_1, z_2, z_3 with $z_1 + z_2 + z_3 = 9 + 3i$. Find its orthocentre and its centroid, and verify the 2 : 1 Euler ratio.

Answers.

1. $u = 3 + 4i, v = -8 + 6i; \bar{u}v = (3 - 4i)(-8 + 6i) = 50i$, whose real part is 0. Hence the segments are perpendicular.
2. $\bar{z}_1 z_2 = 16 - 2i, \bar{z}_2 z_3 = 22 + 32i, \bar{z}_3 z_1 = 8 - 9i$; the sum is $46 + 21i$, so the area is $\frac{1}{2}(21) = \frac{21}{2}$. (Shoelace on $(1, 2), (4, 6), (-2, 5)$ gives $\frac{1}{2}|1(6-5) + 4(5-2) + (-2)(2-6)| = \frac{21}{2}$, confirming.)
3. The circumcentre is the origin, so the orthocentre is $z_1 + z_2 + z_3 = 0$, which is also the circumcentre. A triangle whose orthocentre and circumcentre coincide is equilateral. (Alternative route: $\bar{z}_k = 1/z_k$, so $z_1 z_2 + z_2 z_3 + z_3 z_1 = z_1 z_2 z_3 (\bar{z}_1 + \bar{z}_2 + \bar{z}_3) = z_1 z_2 z_3 (z_1 + z_2 + z_3) = 0$; squaring $z_1 + z_2 + z_3 = 0$ then gives $z_1^2 + z_2^2 + z_3^2 = z_1 z_2 + z_2 z_3 + z_3 z_1$, the equilateral condition of Subtopic 06.)
4. $\bar{u}v = (3 - 4i)(-8 + 6i) = 50i$, so $u \cdot v = 0$ and $u \times v = 50$. The angle is 90° , and since the cross product is positive, v is 90° anticlockwise from u . Check: $|u||v| = 5 \times 10 = 50 = |u \times v|$, consistent with $\sin \theta = 1$.
5. $\bar{z}_1 z_2 + z_1 \bar{z}_2 = 2 \operatorname{Re}(\bar{z}_1 z_2) = 2(z_1 \cdot z_2)$, which vanishes exactly when the two position vectors are perpendicular. Taking $z_1 = a$ fixed and $z_2 = z$, the locus is $\bar{a}z + a\bar{z} = 0$ — the line through the origin perpendicular to a . (Compare the general line $\bar{a}z + a\bar{z} + b = 0$ of Subtopic 06: $b = 0$ is the through-the-origin case.)
6. $H = z_1 + z_2 + z_3 - 2O = (9 + 3i) - 2(2 - i) = 5 + 5i$. $G = \frac{1}{3}(9 + 3i) = 3 + i$. Check: $H - G = 2 + 4i$ and $G - O = (3 + i) - (2 - i) = 1 + 2i$, and indeed $H - G = 2(G - O)$, so O, G, H are collinear in the ratio 2 : 1.

Cheat Sheet Box

SUBTOPIC 09 — 60-SECOND RECALL ($u = z_2 - z_1, v = z_4 - z_3$, both non-zero)

The dictionary. $\bar{u}v = (u \cdot v) + i(u \times v)$. The bar goes on the **first** vector. $\operatorname{Re}(\bar{u}v)$ without the bar is not a geometric test.

The two companion criteria. Parallel / collinear: $\frac{u}{v} \in \mathbb{R} \iff \operatorname{Im}(\bar{u}v) = 0 \iff \bar{u}v = u\bar{v}$.

Perpendicular: $\frac{u}{v}$ purely imaginary $\iff \operatorname{Re}(\bar{u}v) = 0 \iff \bar{u}v + u\bar{v} = 0$.

Angle. $\cos \theta = \frac{\operatorname{Re}(\bar{u}v)}{|u||v|}, \sin \theta = \frac{\operatorname{Im}(\bar{u}v)}{|u||v|}, \theta = \arg(v/u)$.

Area of triangle $z_1 z_2 z_3$. $\frac{1}{2} |\operatorname{Im}[(\bar{z}_2 - \bar{z}_1)(z_3 - z_1)]| = \frac{1}{2} |\operatorname{Im}(\bar{z}_1 z_2 + \bar{z}_2 z_3 + \bar{z}_3 z_1)|$. Without the modulus signs it is the signed area: positive means anticlockwise.

Triangle centres. Circumcentre at origin $\Rightarrow$ orthocentre $H = z_1 + z_2 + z_3$. General circumcentre O : $H = z_1 + z_2 + z_3 - 2O$. Centroid $G = \frac{1}{3}(z_1 + z_2 + z_3)$, and $H - G = 2(G - O)$ (Euler line, 2 : 1).

Special case. $|z_1| = |z_2| = |z_3| = 1$ with $z_1 + z_2 + z_3 = 0 \Rightarrow$ equilateral, since orthocentre and

circumcentre coincide.

Chapter 5: Linear Inequalities

Class 11 Mathematics ♦♦♦ Master Reference

Drona Ed Tech

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1 Linear Inequalities in One Variable

Exam Relevance: A guaranteed scorer on the CBSE Boards — expect a 1-2 mark solve-and-represent question, plus a 3-5 mark word problem from the chapter. In **JEE Main** it rarely stands alone but is the silent workhorse inside domain, range, and parameter problems. **JEE Advanced** never asks it directly, yet every sign-analysis and “find all values of a ” question rests on it. (No NEET relevance — Mathematics is not part of the NEET syllabus.)

1.1 Section 1: Concept Intuition

You have spent years solving *equations*. An equation like $2x + 1 = 7$ is a demand for **exact balance** — picture a two-pan weighing scale at the sabzi mandi resting perfectly level. There is one weight, $x = 3$, that makes it balance, and that single number is your answer.

An **inequality** drops the demand for exact balance. It only asks which side is heavier. $2x + 1 > 7$ does not want the one value that balances the scale; it wants *every* value that tips it to the heavier side. So the answer is no longer a single number — it is a whole **range** of numbers, a stretch of the number line.

This is the single biggest mental shift. The solution of a linear equation is a *point*; the solution of a linear inequality is an *interval*.

Inequalities are everywhere in daily life, far more than equations are. A railway ticket counter says your luggage must weigh *at most* 15 kg ($w \leq 15$). You need *at least* 33 marks to pass ($m \geq 33$). You must be *18 or older* to vote ($\text{age} \geq 18$). A shopkeeper promises mangoes “*under* ₹80 a kilo” ($p < 80$). None of these is a single number — each describes a band of acceptable values. That band is exactly what we compute.

Think about it this way... Imagine the number line as a long straight highway and 0 as your town. Solving an inequality means painting a stretch of this highway to mark “all the acceptable spots.” Two kinds of milestone matter at the edges of your painted stretch:

- A **hollow circle** ($\circ$) means *get infinitely close, but this exact milestone is not allowed* — it marks a strict inequality $<$ or $>$.
- A **filled circle** ($\bullet$) means *this milestone is allowed too* — it marks $\leq$ or $\geq$.

A horizontal number line with a hollow circle at $x = 2$ and a thick shaded arrow extending right toward $+\infty$, labelled " $x > 2$ ". Below it, a second number line with a filled circle at $x = 2$ and a thick shaded arrow extending right, labelled " $x \geq 2$ ".

Now the one rule that makes inequalities genuinely different from equations — and the source of more lost marks than any other idea in this chapter. With an equation, you may multiply both sides by any non-zero number and nothing changes. With an inequality, **multiplying or dividing by a negative number flips the direction of the sign**. Watch:

$$-2 < 3 \quad (\text{true})$$

Multiply both sides by -1 :

$$2 ? -3$$

We get $2 > -3$ — the symbol *reversed*. Geometrically, multiplying by -1 reflects every point across 0, and a reflection swaps left and right, so “smaller” and “larger” trade places. Hold on to that picture; it is the whole game.

Limiting conditions & assumptions. *Linear* means the variable appears only to the first power — no x^2 , no $\frac{1}{x}$, no $|x|$ unless it can be unwound into linear pieces. The manipulation rules below are safe only within these guardrails:

- You may **add or subtract** any quantity from both sides freely.
- You may **multiply or divide by a positive number** freely.
- Multiplying or dividing by a **negative** number forces a flip.
- You **must not cross-multiply** by an expression whose sign you do not know (e.g. multiplying through by $x + 3$, which may be positive or negative). This is where rational inequalities break beginners.
- You **must not square** both sides unless you are certain both are non-negative.

1.2 Section 2: Key Definitions & Rules

A statement linking two real expressions by one of $<$, $>$, $\leq$, $\geq$ is an **inequality**. With $<$ or $>$ it is **strict**; with $\leq$ or $\geq$ it is **non-strict** (slack). The set of all values of the variable that make it true is the **solution set**.

The order axioms (your “formulas”). For all real a, b, c :

$$a < b \implies a + c < b + c \quad (\text{add/subtract any } c)$$

$$a < b, \quad c > 0 \implies ac < bc \quad \text{and} \quad \frac{a}{c} < \frac{b}{c}$$

$$a < b, \quad c < 0 \implies ac > bc \quad \text{and} \quad \frac{a}{c} > \frac{b}{c} \quad (\text{sign flips})$$

$$a < b \text{ and } b < c \implies a < c \quad (\text{transitivity})$$

General linear inequality. For $a \neq 0$:

$$ax + b > 0 \implies \begin{cases} x > -\frac{b}{a}, & a > 0 \\ x < -\frac{b}{a}, & a < 0 \end{cases}$$

Notation — the three equivalent languages. Every answer should be expressible in all three:

Number line	Set-builder	Interval
• at a , shaded right	$\{x : x \geq a\}$	$[a, \infty)$
◦ at a , shaded right	$\{x : x > a\}$	(a, ∞)
• at a and b , shaded between	$\{x : a \leq x \leq b\}$	$[a, b]$
◦ at a , • at b	$\{x : a < x \leq b\}$	$(a, b]$

Bracket conventions (the math-rigor checkpoints, in place of any dimensional analysis):

- A **square bracket** $[]$ *includes* the endpoint (non-strict, filled circle); a **round bracket** $()$ *excludes* it (strict, hollow circle).
- ∞ and $-\infty$ are **not real numbers** — they always take a round bracket. Writing $[3, \infty]$ is always wrong.
- An empty solution set is written $\emptyset$; “all reals” is $(-\infty, \infty)$ or $\mathbb{R}$.

1.3 Section 3: Key Procedures & Logical Framework

A linear inequality in one variable has no single “derivation” — it has an **algorithm**, and the reasoning behind each step is what stops you from losing marks.

The standard procedure.

1. **Clear all fractions** by multiplying every term by the LCM of the denominators. *Why it’s safe:* an LCM of positive integers is positive, so by the order axioms no flip occurs.
2. **Remove brackets** by distributing.
3. **Collect the variable terms on one side, constants on the other**, using only addition and subtraction. *Why:* adding/subtracting never disturbs the inequality direction, so this stage is risk-free.
4. **Reduce to the form** $ax \lessgtr b$.
5. **Divide by the coefficient** a — and here is the only dangerous step. **If** $a < 0$, **reverse the inequality symbol**. *Why:* see the proof below.
6. **Write the solution set** in interval notation.

7. **Represent on a number line**, choosing hollow ($<$, $>$) or filled ($\leq$, $\geq$) circles.

Proof of the sign-flip rule (why Step 5 reverses). Start from the *definition* of order: $a < b$ means precisely that $b - a$ is a positive number, i.e. $b - a > 0$. Now take any negative $c < 0$. A positive number multiplied by a negative number is negative, so

$$c(b - a) < 0.$$

Distribute:

$$cb - ca < 0 \implies cb < ca \implies ca > cb.$$

We began with $a < b$ and ended with $ca > cb$. The direction genuinely flipped — this is not a trick or a convention to memorise, it follows from what “less than” *means*. ■

Procedure for a double (compound) inequality, e.g. $p \leq \text{expression} \leq q$: treat it as **one object with three parts** and apply every operation to *all three* simultaneously. Whatever you do to the middle, do identically to both ends. If a step requires multiplying by a negative, the **entire chain reverses** — and crucially, the two endpoints swap their roles, so $p \leq \dots \leq q$ becomes $\dots \geq \dots \geq \dots$, which you then re-read left-to-right. (The most common error is operating on only the middle and one end. Tag: INEQ_COMPOUND_SPLIT.)

1.4 Section 4: Worked Examples

1.4.1 Example 1 — [CBSE Board]

Solve $3(2x - 1) \geq 2(x + 3) - 5$ and represent the solution on a number line.

Given: $3(2x - 1) \geq 2(x + 3) - 5$.

Remove brackets:

$$6x - 3 \geq 2x + 6 - 5 \implies 6x - 3 \geq 2x + 1.$$

Collect terms (add 3, subtract $2x$, both safe):

$$6x - 2x \geq 1 + 3 \implies 4x \geq 4.$$

Divide by $+4$ (positive — no flip):

$$x \geq 1.$$

Solution set: $[1, \infty)$.

Number line with a filled circle at $x = 1$ and a shaded arrow extending to the right toward $+\infty$.

1.4.2 Example 2 — [JEE Main / Speed Trap]

Solve the double inequality $-7 \leq \frac{3-2x}{4} < 5$.

The trap is twofold: it is a compound inequality *and* it forces a divide-by-negative. Move fast but flip carefully.

Multiply all three parts by 4 (positive — directions hold):

$$-28 \leq 3 - 2x < 20.$$

Subtract 3 from all three parts:

$$-31 \leq -2x < 17.$$

Divide all three parts by -2 — **flip both symbols**, and the chain reverses:

$$\frac{31}{2} \geq x > -\frac{17}{2}.$$

Re-read left-to-right:

$$-\frac{17}{2} < x \leq \frac{31}{2}.$$

Solution set: $\left(-\frac{17}{2}, \frac{31}{2}\right]$.

Speed check: the strict $<$ rode in on the 5 side and the non-strict $\leq$ on the -7 side; after the flip they correctly attach to the lower (strict) and upper (filled) bounds respectively. If your brackets came out the other way, you forgot to swap inclusivity along with the symbols.

1.4.3 Example 3 — [JEE Main, multi-concept: inequalities + integer counting]

Find the number of integer values of x satisfying both $3x - 2 > 4$ and $2(x - 1) \leq x + 3$, and find their sum.

First inequality:

$$3x - 2 > 4 \implies 3x > 6 \implies x > 2.$$

Second inequality:

$$2x - 2 \leq x + 3 \implies x \leq 5.$$

The system requires **both**, so intersect the solution sets:

$$x > 2 \text{ and } x \leq 5 \implies x \in (2, 5].$$

The integers in $(2, 5]$ are 3, 4, 5. **Count** = 3; **sum** = $3 + 4 + 5 = 12$.

The multi-concept move: solving each inequality is routine; the marks live in correctly **intersecting** the two solution sets and then translating a continuous interval into a discrete integer count — note 2 is excluded (strict) but 5 is included (non-strict).

1.4.4 Example 4 — [JEE Advanced]

Find all real values of the parameter a for which the inequality $a(x-1) < 2x+3$ holds for every real number x .

Expand and gather x -terms on one side:

$$ax - a < 2x + 3 \implies (a-2)x < a+3.$$

Everything now hinges on the **coefficient of x** , namely $(a-2)$ — and the deep step is realising it can *vanish*.

- **If $a-2 > 0$ (i.e. $a > 2$):** dividing gives $x < \frac{a+3}{a-2}$, a half-line — *not* all of $\mathbb{R}$. Rejected.
- **If $a-2 < 0$ (i.e. $a < 2$):** dividing flips the sign, $x > \frac{a+3}{a-2}$, again only a half-line. Rejected.
- **If $a-2 = 0$ (i.e. $a = 2$):** the inequality collapses to $0 \cdot x < 5$, that is $0 < 5$ — **true for every x** , independent of x .

So the inequality is satisfied by all real x **only when** $a = 2$.

Why it's Advanced-level: the uncommon insight is that a “linear” inequality stops being linear the instant its leading coefficient is zero, and that degenerate case — usually a nuisance — is the *only* case that gives the required universal solution.

1.5 Section 5: Practice Exercises

1. **[CBSE]** Solve $\frac{x}{2} + 3 < \frac{x}{4} + 5$ and show the solution on a number line.
2. **[CBSE]** Solve $5(2-x) \geq 3(4-x) + 2$ and write the answer in interval notation.
3. **[JEE Main]** Find the number of integers satisfying $-5 < \frac{3-2x}{2} \leq 4$.
4. **[JEE Main]** Solve the system $2x-1 > x+2$ and $3-x \geq 2x-9$; give the solution set.
5. **[JEE Advanced]** Find all real values of a for which the inequality $(a-1)x > 3$ has **no** solution.

Answers: 1. $x < 8$, i.e. $(-\infty, 8)$. 2. $(-\infty, -2]$. 3. 9 integers (-2 through 6). 4. $(3, 4]$. 5. $a = 1$.

1.6 Section 6: Crack the MCQ

Q1. The solution set of $\frac{2-3x}{5} \leq -4$ is

- (a) $(-\infty, \frac{22}{3}]$ (b) $[\frac{22}{3}, \infty)$ (c) $(-\infty, -\frac{22}{3}]$ (d) $(\frac{22}{3}, \infty)$

Correct: (b). Multiply by 5: $2 - 3x \leq -20 \Rightarrow -3x \leq -22$. Divide by -3 and **flip**: $x \geq \frac{22}{3}$.

- **(a)** forgets the flip when dividing by -3 — the signature error (INEQ_SIGN_FLIP_NEG).
- **(c)** drops a negative sign while isolating x , landing on the wrong side of 0.
- **(d)** flips correctly but turns $\leq$ into $>$ — a hollow-vs-filled slip (INEQ_BRACKET_OPEN_CLOSED).

Q2. Which interval equals $\{x : -3 < x \leq 4\}$?

- (a) $[-3, 4]$ (b) $(-3, 4)$ (c) $(-3, 4]$ (d) $[-3, 4)$

Correct: (c). Strict at $-3 \Rightarrow$ round bracket; non-strict at $4 \Rightarrow$ square bracket.

- **(a)** includes both ends — wrongly admits -3 .
- **(b)** excludes both — wrongly drops 4 .
- **(d)** has the brackets exactly reversed. All three are INEQ_INTERVAL_NOTATION errors.

Q3. The solution of $-2 \leq 2x - 4 \leq 6$ is

- (a) $[1, 5]$ (b) $[-1, 5]$ (c) $[1, 10]$ (d) $[-3, 1]$

Correct: (a). Add 4 to all three parts: $2 \leq 2x \leq 10$; divide all by 2: $1 \leq x \leq 5$.

- **(b)** added 4 to the middle and right but mishandled the left part (INEQ_COMPOUND_SPLIT).
- **(c)** forgot to divide the right end by 2.
- **(d)** subtracted instead of added 4.

Q4. If $|3 - x| \leq 5$, then x belongs to

- (a) $[-2, 8]$ (b) $[-8, 2]$ (c) $[2, 8]$ (d) $(-2, 8)$

Correct: (a). $|3 - x| \leq 5 \Rightarrow -5 \leq 3 - x \leq 5 \Rightarrow -8 \leq -x \leq 2$; multiply by -1 and flip the chain: $-2 \leq x \leq 8$.

- **(b)** forgot to flip when multiplying the chain by -1 (INEQ_SIGN_FLIP_NEG).
- **(c)** kept only one branch of the modulus.
- **(d)** used strict brackets although the inequality is $\leq$.

1.7 Section 7: Common Student Pitfalls & Pro-Tips

- **Forgetting to flip on a negative.** By far the costliest mistake. Any time you multiply or divide by a negative number, the symbol *must* reverse. Build the reflex: see a negative coefficient $\rightarrow$ flip before you write the next line.
- **Cross-multiplying by a variable expression.** Writing $\frac{x+1}{x+3} < 2$ and “multiplying both sides by $x+3$ ” is illegal — $x+3$ may be negative, so you cannot know whether to flip. Bring everything to one side and combine into a single fraction instead. (INEQ_CROSS_MULTIPLY_UNSAFE)
- **Mixing up hollow and filled circles / round and square brackets.** $<$ and $>$ are always hollow/round; $\leq$ and $\geq$ are always filled/square; ∞ is always round.
- **Breaking a double inequality apart wrongly.** Operate on all three parts together; never leave one end behind.

Pro-Tip — the 10-second sanity check. After finding an interval, pick *any* number inside it and substitute it into the **original** inequality. If it satisfies, your direction and brackets are almost certainly right; if it fails, you most likely missed a flip. For Example 2, test $x = 0 \in \left(-\frac{17}{2}, \frac{31}{2}\right]$: $\frac{3-0}{4} = 0.75$, and indeed $-7 \leq 0.75 < 5$. ✓

1.8 Section 8: Cheat Sheet

Linear Inequalities in One Variable — Quick Recall

- **Solution is a *range*, not a point** — answer in interval form.
- **Safe moves:** add/subtract anything; multiply/divide by a **positive** number.
- **Danger move:** multiply/divide by a **negative** $\Rightarrow$ **reverse the sign**.

$$a < b, c < 0 \Rightarrow ca > cb$$

- **General form:** $ax + b > 0 \Rightarrow x > -\frac{b}{a}$ if $a > 0$, and $x < -\frac{b}{a}$ if $a < 0$.
- **Brackets:** $[]$ includes (filled $\bullet$, for $\leq \geq$); $()$ excludes (hollow $\circ$, for $< >$); ∞ is **always round**.
- **Compound inequality:** operate on all three parts at once; a negative flips the whole chain.
- **Never** cross-multiply by an unknown-sign expression; **never** square unless both sides are non-negative.

Mnemonics: “Negative flips, positive sticks.” — “Infinity never gets a hug (square bracket).”

2 Linear Inequalities in Two Variables

Exam Relevance: A reliable CBSE Boards earner — a 3-mark “solve graphically” or a 5-mark “system of inequalities” question appears almost every year, and the reverse problem (write the inequalities for a shaded region) shows up in the Exemplar. **JEE Main** treats it as the conceptual seed of Linear Programming (Class 12), so the half-plane and feasible-region ideas matter more than the marks suggest. **JEE Advanced** never asks it directly but leans on the region-and-vertex reasoning inside optimisation problems. (No NEET relevance — Mathematics is outside the NEET syllabus.)

2.1 Section 1: Concept Intuition

In the previous subtopic, a linear inequality in one variable carved out a *stretch* of the one-dimensional number line. Now we add a second variable, and something beautiful happens: the answer grows a dimension. The solution of a linear inequality in **two** variables is no longer a stretch of a line — it is an entire **flat region** of the two-dimensional plane.

Here is the picture to hold in your head. Draw the line $x + y = 4$ on graph paper. That single line slices the whole plane cleanly into two pieces, like a knife through a roti. Each piece is called a **half-plane**. Every point in the plane is now in exactly one of three places: *on* the line, in the half-plane on *one* side, or in the half-plane on the *other* side. A linear inequality such as $x + y < 4$ simply asks: “give me all the points on a particular one of those two sides.”

Think about it this way... Suppose you have ₹120 to spend at a stationery shop, where a register costs ₹40 and a pen costs ₹20. If you buy x registers and y pens, your spending is $40x + 20y$, and staying within budget means

$$40x + 20y \leq 120.$$

There is no *single* answer here. Buy 3 registers and no pens (₹120, exactly at the limit). Or 2 registers and 1 pen (₹100, comfortably under). Or 1 register and 3 pens (₹100). Or nothing at all (₹0). Every one of these is a valid (x, y) pair, and if you plotted all of them, they would fill a whole triangular patch of the plane. That patch — the **solution region** — is the answer. A two-variable inequality always answers with a region, never a point.

The genius of the method is that you do not have to test infinitely many points. Because a straight line divides the plane into just two sides, and because a linear expression keeps the *same* sign throughout each side, you only need to test **one** point to learn the fate of an entire half-plane. Pick a convenient point not on the line — the origin $(0, 0)$ is usually easiest — substitute it, and ask whether it satisfies the inequality. If yes, the origin’s side is the answer; if no, the other side is. One substitution decides everything.

Limiting conditions & assumptions.

- The expression must be genuinely **two-variable and linear**: of the form $ax + by (\leq) c$ with a, b not both zero. No x^2 , no xy , no $\frac{1}{x}$.
- An inequality like $x \geq 2$, when asked for “in the plane,” is still two-variable — it describes a *vertical half-plane*, not a single point on the number line. Watch for this shift of context: the same symbols mean a stretch of line in 1-D and a half-plane in 2-D.
- The convenient test point $(0, 0)$ **fails whenever the boundary line passes through the origin** (because then the origin lies *on* the line and cannot tell the two sides apart). In that case you must choose a different test point. This single oversight is the most common mistake in the whole subtopic.

2.2 Section 2: Key Definitions & Rules

A line $ax + by = c$ (with a, b not both zero) divides the Cartesian plane into two **half-planes**. The set of all points satisfying a linear inequality in two variables is its **solution region**.

Core rules (your “formulas” for this subtopic):

- **Boundary line.** Replace the inequality sign with $=$ and draw $ax + by = c$.
- **Line style (strictness):**
 - $\geq$ or $\leq \Rightarrow$ the boundary is **included** — draw it as a **solid** line.
 - $>$ or $< \Rightarrow$ the boundary is **excluded** — draw it as a **dotted/broken** line.
- **Test-point rule.** Choose any point P *not on* the line. If P satisfies the inequality, shade the half-plane containing P ; otherwise shade the other half-plane.
- **System of inequalities.** The solution is the **intersection** (overlap) of all the individual solution regions — the area satisfying *every* inequality at once.
- **Non-negativity in real problems.** Quantities like units produced or articles bought cannot be negative, so $x \geq 0$, $y \geq 0$ are added, confining the region to the **first quadrant**.

Region rigor (the checkpoints that replace dimensional analysis in a math fork):

- A solution region may be **bounded** (enclosed on all sides, finite area — e.g. a triangle) or **unbounded** (extending infinitely in some direction).
- A system can have an **empty** solution set $\emptyset$ when the half-planes share no common point.
- The **corner points (vertices)** of a bounded region are found by solving the boundary lines two at a time; these vertices become decisive in Class 12 Linear Programming.
- Solid line $\Leftrightarrow$ filled circle in one variable; dotted line $\Leftrightarrow$ hollow circle. The strictness convention is one idea wearing two costumes.

2.3 Section 3: CBSE Board Derivation — Why an Inequality Is a Half-Plane

The procedure above (“test a point, shade a side”) works, but a CBSE long-answer question may ask *why* a linear inequality corresponds to an entire half-plane. Here is the rigorous reason.

Claim. Every point (x, y) satisfying $ax + by > c$ (take $b > 0$ for definiteness) lies entirely in *one* of the two half-planes determined by the line $ax + by = c$, and conversely every point of that half-plane satisfies the inequality.

Setup. Consider the line $ax + by = c$ with $b > 0$. Pick any point $P(\alpha, \beta)$ that lies **on** the line, so

$$a\alpha + b\beta = c. \quad (1)$$

A non-vertical line $ax + by = c$ on the xy -plane, labelled. A point $P(\alpha, \beta)$ marked on the line. Directly above P , a point $Q(\alpha, \gamma)$ marked in the upper half-plane (so $\gamma > \beta$). The upper half-plane shaded lightly and labelled “II”, the lower labelled “I”.

Forward direction. Now take a point $Q(\alpha, \gamma)$ directly above P — same x -coordinate α , but higher up, so $\gamma > \beta$. Because $b > 0$, multiplying this inequality by b preserves its direction:

$$b\gamma > b\beta.$$

Add $a\alpha$ to both sides (addition never disturbs an inequality):

$$a\alpha + b\gamma > a\alpha + b\beta.$$

But by (1) the right-hand side is exactly c , so

$$a\alpha + b\gamma > c.$$

That is, Q satisfies $ax + by > c$. Since Q was *any* point lying above the line, **every** point in the upper half-plane satisfies the inequality.

Converse direction. Suppose instead we *start* with a point $Q(\alpha, \gamma)$ satisfying $a\alpha + b\gamma > c$. Using (1) to replace c ,

$$a\alpha + b\gamma > a\alpha + b\beta \implies b\gamma > b\beta \implies \gamma > \beta \quad (\text{since } b > 0),$$

which says Q lies *above* the line. So every solution point lives in the upper half-plane.

Conclusion. The two directions together prove that the solution set of $ax + by > c$ is *precisely* the upper half-plane (when $b > 0$). If $b < 0$, the identical argument with the inequality reversed shows the solution is the lower half-plane instead. Either way, the graph of a linear inequality in two variables is exactly one half-plane. ■

The practical pay-off. This is *why* a single test point suffices: the sign of $ax + by - c$ is constant throughout each half-plane, so one specimen point reports for its entire side.

Working procedure (distilled).

1. Write the boundary line $ax + by = c$ and plot it using its intercepts $(\frac{c}{a}, 0)$ and $(0, \frac{c}{b})$.
2. Draw it **solid** for $\leq / \geq$, **dotted** for $< / >$.
3. Choose a test point off the line — use $(0, 0)$ unless the line passes through the origin.
4. Substitute; shade the test point's side if it satisfies the inequality, otherwise the other side.
5. For a system, repeat for each inequality and keep only the **common** (overlapping) shaded region.

2.4 Section 4: Worked Examples

2.4.1 Example 1 — [CBSE Board]

Solve $2x + 3y \leq 12$ **graphically**.

Boundary line: $2x + 3y = 12$. Intercepts: put $y = 0 \Rightarrow x = 6$, giving $(6, 0)$; put $x = 0 \Rightarrow y = 4$, giving $(0, 4)$. Since the inequality is $\leq$, draw the line **solid** (boundary included).

Test point: the line does not pass through the origin, so use $(0, 0)$:

$$2(0) + 3(0) = 0 \leq 12 \quad \text{— true.}$$

So shade the half-plane **containing the origin** (the side below-left of the line).

Solution: all points on or below the line $2x + 3y = 12$, including the boundary.

Line through $(6, 0)$ and $(0, 4)$ drawn solid. The region below-left of the line, containing the origin, shaded. Axes labelled.

2.4.2 Example 2 — [JEE Main / Speed Trap]

Solve $y > 2x$ **graphically**.

The trap: rewrite as $-2x + y > 0$. The boundary $y = 2x$ **passes through the origin**, so $(0, 0)$ lies *on* the line and is useless as a test point — substituting it gives $0 > 0$, false, but that tells you nothing about either side. Reaching for $(0, 0)$ on autopilot is the classic error here.

Fix: pick any other off-line point, say $(1, 0)$:

$$0 > 2(1) = 2 \quad \text{— false.}$$

So $(1, 0)$ is **not** in the region; shade the **other** side. A quick confirmation with $(0, 1)$: $1 > 0$ — true, and $(0, 1)$ lies on that other side. Draw the line **dotted** (strict $>$).

Solution: the open half-plane *above* the line $y = 2x$, boundary excluded.

Speed lesson: before testing $(0, 0)$, glance at the constant term. If $c = 0$, the line runs through the origin — skip $(0, 0)$ and grab $(1, 0)$ or $(0, 1)$ instead.

2.4.3 Example 3 — [JEE Main, multi-concept: inequalities + coordinate geometry]

Find the corner points (vertices) of the region defined by $x + y \leq 4$, $x \geq 1$, $y \geq 0$.

Each constraint is a half-plane:

- $x + y \leq 4$: on/below the line $x + y = 4$.
- $x \geq 1$: on/right of the vertical line $x = 1$.
- $y \geq 0$: on/above the x -axis.

The feasible region is their overlap — a bounded triangle. Its vertices are the pairwise intersections of the boundary lines (this is where the coordinate-geometry skill enters: solving simultaneous linear equations):

- $x = 1$ and $y = 0 \Rightarrow (1, 0)$.
- $x + y = 4$ and $y = 0 \Rightarrow (4, 0)$.
- $x + y = 4$ and $x = 1 \Rightarrow (1, 3)$.

Vertices: $(1, 0)$, $(4, 0)$, $(1, 3)$. The region is bounded (a finite triangle).

Triangle with vertices $(1, 0)$, $(4, 0)$, $(1, 3)$. Boundary lines $x = 1$ (vertical), $y = 0$ (x -axis), and $x + y = 4$ (slanting). Interior shaded.

2.4.4 Example 4 — [JEE Advanced]

Find the area of the region defined by $x + y \leq 6$, $x + 2y \leq 8$, $x \geq 0$, $y \geq 0$.

This needs careful vertex hunting — the subtle step is deciding *which* constraint binds along each edge.

The non-negativity conditions confine us to the first quadrant. Now find the vertices:

- **Origin:** $(0, 0)$ satisfies all four — a vertex.
- **On the x -axis ($y = 0$):** $x + y \leq 6$ gives $x \leq 6$; $x + 2y \leq 8$ gives $x \leq 8$. The tighter bound is $x \leq 6$, so the boundary meets the axis at $(6, 0)$, and $(6, 0)$ satisfies $6 + 0 \leq 8$. Vertex $(6, 0)$.
- **Intersection of the two slant lines:** subtract $x + y = 6$ from $x + 2y = 8$ to get $y = 2$, hence $x = 4$. Vertex $(4, 2)$ (both coordinates ≥ 0 ✓).
- **On the y -axis ($x = 0$):** $x + y \leq 6$ gives $y \leq 6$; $x + 2y \leq 8$ gives $y \leq 4$. The tighter bound is $y \leq 4$, so vertex $(0, 4)$.

So the feasible region is the quadrilateral with vertices, in order,

$$(0, 0) \rightarrow (6, 0) \rightarrow (4, 2) \rightarrow (0, 4).$$

Apply the **shoelace formula**, $\text{Area} = \frac{1}{2} |\sum (x_i y_{i+1} - x_{i+1} y_i)|$:

$$\begin{aligned} & (0 \cdot 0 - 6 \cdot 0) + (6 \cdot 2 - 4 \cdot 0) + (4 \cdot 4 - 0 \cdot 2) + (0 \cdot 0 - 0 \cdot 4) \\ &= 0 + 12 + 16 + 0 = 28. \end{aligned}$$

$$\text{Area} = \frac{1}{2} \times 28 = \boxed{14 \text{ square units.}}$$

Why it's Advanced-level: the work is not in graphing but in correctly identifying which of two competing constraints clips each axis, and then assembling the four vertices in cyclic order before the shoelace step. A single mis-assigned vertex throws off the entire area.

2.5 Section 5: Practice Exercises

1. **[CBSE]** Solve $3x + 2y \geq 6$ graphically.
2. **[CBSE]** Solve $x - 2y < 4$ graphically.
3. **[JEE Main]** Find the vertices of the region $x + y \leq 5$, $x \geq 0$, $y \geq 0$.
4. **[JEE Main]** Determine whether the solution region of $2x + y \geq 4$, $x + y \geq 3$, $x \geq 0$, $y \geq 0$ is bounded or unbounded.
5. **[JEE Advanced]** Find the area enclosed by $x + y \leq 4$, $y \leq x$, $x \geq 0$, $y \geq 0$.

Answers: 1. Solid line through $(2, 0)$, $(0, 3)$; shade the side **not** containing the origin (since $0 \geq 6$ is false). 2. Dotted line through $(4, 0)$, $(0, -2)$; shade the side **containing** the origin. 3. $(0, 0)$, $(5, 0)$, $(0, 5)$. 4. Unbounded. 5. 4 square units (triangle with vertices $(0, 0)$, $(4, 0)$, $(2, 2)$).

2.6 Section 6: Crack the MCQ

Q1. While solving $2x - y > 0$ graphically, which of the following points **cannot** be used as a test point?

- (a) $(1, 1)$ (b) $(0, 2)$ (c) $(2, 4)$ (d) $(3, 1)$

Correct: (c). The boundary line is $2x - y = 0$, i.e. $y = 2x$. A test point must not lie *on* this line. Check $(2, 4)$: $2(2) - 4 = 0$ — it sits exactly on the line, so it cannot distinguish the two half-planes.

- **(a), (b), (d)** all give non-zero values of $2x - y$, so each lies off the line and is a valid test point.
- A student who never checks whether the chosen point lies on the boundary picks any of these blindly — the error tagged INEQ2V_TESTPOINT_ON_LINE.

Q2. The boundary line for the inequality $x + 2y \leq 5$ should be drawn as

- (a) dotted, boundary excluded (b) solid, boundary included (c) solid, boundary excluded (d) dotted, boundary included

Correct: (b). The sign is $\leq$, which *includes* equality, so the boundary points belong to the solution — drawn as a solid line.

- **(a)** confuses $\leq$ with $<$ (dotted is only for strict inequalities).
- **(c)** and **(d)** pair a line style with the wrong inclusion status — internally contradictory. All are INEQ2V_LINE_STYLE slips (the two-variable cousin of the open/closed-circle error).

Q3. The solution region of the system $x \geq 0$, $y \geq 0$, $x + y \leq 1$ is

- (a) the entire first quadrant (b) a triangle with vertices $(0, 0)$, $(1, 0)$, $(0, 1)$ (c) an unbounded region (d) the empty set

Correct: (b). The first two constraints give the first quadrant; $x + y \leq 1$ then clips it to the triangle below the line $x + y = 1$, whose corners are $(0, 0)$, $(1, 0)$, $(0, 1)$.

- **(a)** ignores the third inequality entirely.
- **(c)** forgets that $x + y \leq 1$ caps the region, making it finite.
- **(d)** wrongly treats the three half-planes as having no overlap (an INEQ2V_SYSTEM_INTERSECTION error — confusing intersection with contradiction).

Q4. A shaded region lies **below** the line $x + y = 2$, **contains the origin**, and **includes** the boundary line itself. The inequality representing it is

- (a) $x + y > 2$ (b) $x + y < 2$ (c) $x + y \leq 2$ (d) $x + y \geq 2$

Correct: (c). Test the origin: $0 + 0 = 0$, and since the origin must satisfy the inequality, we need $0 \leq 2$ — true for $\leq$. The boundary is included, so the sign is non-strict: $x + y \leq 2$.

- **(a)** and **(d)** point to the half-plane *not* containing the origin — wrong side (INEQ2V_REVERSE_DIRECTION).
- **(b)** has the right side but a strict sign, which would *exclude* the boundary the question says is included.

2.7 Section 7: Common Student Pitfalls & Pro-Tips

- **Testing $(0, 0)$ on a line through the origin.** When $c = 0$, the boundary passes through the origin, so $(0, 0)$ is on the line and cannot decide the side. Always glance at the constant term first; if it is zero, pick $(1, 0)$ or $(0, 1)$ instead.
- **Wrong line style.** Strict inequalities ($<$, $>$) get a **dotted** line; non-strict ($\leq$, $\geq$) get a **solid** line. Mixing these up is the two-variable version of the open/closed-circle mistake and costs presentation marks on the Boards.
- **Shading the wrong half-plane.** After the test point gives “false,” remember to shade the *opposite* side, not the side you tested. Read your own conclusion carefully.
- **Union instead of intersection for a system.** A system demands *all* inequalities hold simultaneously, so the answer is the **common overlap**, never the combined shading. Only the doubly- (or triply-) shaded patch counts.

Pro-Tip — the one-point audit. After shading, pick any point clearly *inside* your final region and substitute it into **every** original inequality. If it satisfies all of them, your region and line styles are almost certainly right; if it fails even one, you shaded a wrong side or mishandled a constraint. For Example 4, test $(1, 1)$: $1 + 1 = 2 \leq 6$ ✓, $1 + 2 = 3 \leq 8$ ✓, $x, y \geq 0$ ✓ — region confirmed.

2.8 Section 8: Cheat Sheet

Linear Inequalities in Two Variables — Quick Recall

- A line $ax + by = c$ splits the plane into two **half-planes**; the solution of a linear inequality is **one entire half-plane**.
- **Line style:** $\leq, \geq \Rightarrow$ **solid** (included); $<, > \Rightarrow$ **dotted** (excluded).
- **Test-point rule:** substitute a point off the line — use $(0, 0)$ **unless the line passes through the origin** (then use $(1, 0)$ or $(0, 1)$). Satisfies $\Rightarrow$ shade that side; fails $\Rightarrow$ shade the other.
- **System** $\Rightarrow$ **intersection**. Keep only the region common to *all* inequalities.
- **Real-world constraints** add $x \geq 0, y \geq 0$, restricting the region to the **first quadrant**.
- Regions are **bounded** (finite, has area) or **unbounded**; corner points come from solving boundary lines in pairs.

Mnemonics: “Solid includes, dotted excludes.” — “Origin on the line? Then origin is no spy — pick another.” — “System means overlap, not pile-up.”

3 Applications & Word Problems of Linear Inequalities

Exam Relevance: This is where the chapter earns its marks on the **CBSE Boards** — the Miscellaneous Exercise is almost entirely word problems, and a 3–5 mark “form and solve” question (mixtures, marks, lengths, temperature ranges) is near-guaranteed. **JEE Main** embeds the same modeling inside multi-step problems. **JEE Advanced** treats constraint-modeling as the gateway to optimisation (Linear Programming). (No NEET relevance — Mathematics is outside the NEET syllabus.)

3.1 Section 1: Concept Intuition

Solving an inequality is the easy half. The hard half — and the part that actually decides marks — is **turning a sentence in English into an inequality in mathematics**. A word problem hands you a real situation (“a student must average at least 80,” “the mixture must stay below 40% acid,” “the perimeter is more than 60 cm”) and asks you to find the range of possibilities. Your job is translation, then solution, then interpretation back into the real world.

The single most important skill here is reading the *comparison words* correctly. English has a dozen ways of saying “ $\geq$ ” and a dozen ways of saying “ $<$,” and a careless reading flips the entire answer. “At least 80” means ≥ 80 (80 is allowed). “More than 80” means > 80 (80 is *not* allowed). “No more than 80” means ≤ 80 . These tiny distinctions are the difference between a correct and an incorrect solution set.

Think about it this way... Imagine you are the manager of a small sweet shop during Diwali. Every real constraint you face is an inequality, never an equation. You have *at most* ₹5,000 to spend on ingredients (cost ≤ 5000). You must make *at least* 200 boxes to meet pre-orders (boxes ≥ 200). Your storage holds *fewer than* 60 kg of sugar (sugar < 60). Notice that none of these pins you to a single number — each leaves you a *range* of legal choices, and the art of the business is operating inside the overlap of all of them. A word problem is just this situation written down: translate each phrase into a symbol, then find the overlap.

There is a second, quieter skill: spotting the **hidden constraints** the problem does not state out loud. If x is “the number of chairs produced,” then x must be a *whole number* and cannot be negative — even though the sentence never says so. If x is “the length of a side,” then $x > 0$. Forgetting these silent conditions is how a perfectly solved inequality still earns the wrong final answer.

Limiting conditions & assumptions.

- A variable representing a **count** (chairs, students, boxes) must be a **non-negative integer** — solve over the reals, then keep only the integers.
- A variable representing a **physical measurement** (length, time, volume) must be **positive**.

- “**Between** a and b ” is usually *strict* ($a < x < b$) unless the problem says “between a and b , inclusive” — read carefully.
- The translation only works once: after solving, you must **interpret** the mathematical answer back into the situation (e.g. “the minimum length is 10 cm,” not merely “ $x \geq 10$ ”).

3.2 Section 2: The Translation Dictionary & Modeling Rules

The heart of this subtopic is a reliable phrase-to-symbol map. Commit it to memory.

English phrase	Symbol
at least / no less than / a minimum of / not below	$\geq$
at most / no more than / a maximum of / not above	$\leq$
more than / greater than / exceeds / over	$>$
less than / fewer than / below / under / short of	$<$
between a and b (exclusive)	$a < x < b$
between a and b , inclusive	$a \leq x \leq b$

Modeling rules (the “formulas” of this subtopic):

- **Define the variable explicitly**, with units: “Let x = the length of the shortest side, in cm.”
- **One condition** $\rightarrow$ **one inequality**. Each separate requirement in the problem becomes its own inequality; a problem with several requirements becomes a **system**, whose answer is the *intersection*.
- **Attach the hidden domain**: $x \in \mathbb{Z}$, $x \geq 0$ for counts; $x > 0$ for measurements.
- **Average condition**: “the average of n quantities is at least A ” becomes $\frac{\text{sum}}{n} \geq A$.
- **Mixture condition**: in a mixture, $\frac{\text{amount of pure component}}{\text{total amount}}$ gives the concentration; bound it as required.

Rigor checkpoints (replacing dimensional analysis in a math fork):

- Always state the **units** of the variable and carry them into the final answer.
- After solving, **discard infeasible values** (negative lengths, non-integer counts) before writing the conclusion.
- Distinguish the *mathematical* solution set from the *contextual* answer: $x \geq \frac{39}{7}$ may become “the shortest side must be at least 6 cm” once integer side-lengths are required.

3.3 Section 3: Key Procedures & Logical Framework

Word problems have no derivation — they have a disciplined **modeling routine**. Following it in order is what prevents lost marks.

The five-step modeling routine.

1. **Name the unknown.** Write “Let $x = \dots$ ” with units. *Why first:* every later inequality must refer to the same clearly-defined quantity; vague variables produce vague answers.
2. **Translate each requirement** into an inequality using the dictionary above. *Why:* the comparison word fixes the symbol; choosing it now, deliberately, avoids the reflexive sign error.
3. **Record the hidden domain** ($x > 0$, or x a non-negative integer). *Why:* the real situation silently restricts x , and the final answer must respect that restriction.
4. **Solve** the inequality or system using the algebra from Subtopics 01–02. For a system, take the **intersection** of the solution sets.
5. **Interpret** the result in the words of the problem. *Why:* a solution set is not yet an answer; “the minimum number of complete boxes is 201” is the answer a marker rewards.

Worked illustration of the routine (mixture set-up). “A tank holds V litres of an $a\%$ solution; we add y litres of a $b\%$ solution.” The pure-acid balance is

$$\text{concentration of mixture} = \frac{\frac{a}{100}V + \frac{b}{100}y}{V + y}.$$

Bounding this between two target percentages gives a **double inequality in y** , which you clear of fractions (multiplying by the *positive* quantity $V + y$) and solve as in Subtopic 01. The hidden domain is $y \geq 0$ (you cannot add a negative volume).

3.4 Section 4: Worked Examples**3.4.1 Example 1 — [CBSE Board]**

The longest side of a triangle is twice the shortest side, and the third side is 2 cm longer than the shortest. If the perimeter is at least 42 cm, find the minimum length of the shortest side.

Step 1 — name: let x = length of the shortest side, in cm. Then the longest side is $2x$ and the third side is $x + 2$.

Step 2 — translate: “perimeter at least 42” means

$$x + 2x + (x + 2) \geq 42.$$

Step 3 — domain: $x > 0$ (a length).

Step 4 — solve:

$$4x + 2 \geq 42 \implies 4x \geq 40 \implies x \geq 10.$$

Step 5 — interpret: the shortest side must be at least 10 cm, so its **minimum length is 10 cm**.

3.4.2 Example 2 — [JEE Main / Speed Trap]

Find all pairs of consecutive odd natural numbers, both greater than 10, whose sum is less than 40.

The trap is twofold: the comparison words, and the hidden domain (odd integers). Translate carefully.

Let the smaller odd number be x ; the next consecutive odd number is $x + 2$.

- “both greater than 10”: since x is the smaller, $x > 10$ guarantees both.
- “sum less than 40”: $x + (x + 2) < 40 \Rightarrow 2x + 2 < 40 \Rightarrow x < 19$.

So $10 < x < 19$. Now apply the **hidden domain** — x must be an *odd natural number*:

$$x \in \{11, 13, 15, 17\}.$$

The pairs are (11, 13), (13, 15), (15, 17), (17, 19).

Speed lesson: the continuous solution $10 < x < 19$ is *not* the answer — a student who stops there loses the marks. The odd-integer restriction is the whole point. Verify the boundary: $x = 17$ gives sum $36 < 40$ ✓, but $x = 19$ gives sum $40 \not< 40$, correctly excluded.

3.4.3 Example 3 — [JEE Main, multi-concept: inequalities + percentage mixture]

A chemist has 600 mL of a 20% acid solution. How many millilitres of a 50% acid solution must be added so that the resulting mixture is more than 30% but less than 40% acid?

Let y = millilitres of the 50% solution added; $y \geq 0$.

Pure acid after mixing = $0.20(600) + 0.50y = 120 + 0.5y$ mL, in a total of $600 + y$ mL. The concentration condition is the double inequality

$$0.30 < \frac{120 + 0.5y}{600 + y} < 0.40.$$

Since $600 + y > 0$, multiply throughout by it (no sign flip):

$$0.30(600 + y) < 120 + 0.5y < 0.40(600 + y).$$

Left inequality: $180 + 0.3y < 120 + 0.5y \Rightarrow 60 < 0.2y \Rightarrow y > 300$. **Right inequality:** $120 + 0.5y < 240 + 0.4y \Rightarrow 0.1y < 120 \Rightarrow y < 1200$.

Intersecting, $300 < y < 1200$. **The chemist must add between 300 mL and 1200 mL of the 50% solution.**

The multi-concept move: the percentage model produces a *rational* double inequality, which only becomes linear after multiplying by the positive total volume — and the two halves must then be intersected.

3.4.4 Example 4 — [JEE Advanced: constraint modeling → optimisation bridge]

A small workshop makes x chairs and y tables per day. Each chair needs 2 hours of carpentry and 1 hour of finishing; each table needs 3 hours of carpentry and 2 hours of finishing. At most 36 carpentry-hours and 22 finishing-hours are available daily. If the profit is ₹30 per chair and ₹50 per table, find the production plan that maximises profit.

Model the constraints (each “at most” $\rightarrow \leq$; counts $\rightarrow x, y \geq 0$):

$$2x + 3y \leq 36 \quad (\text{carpentry}), \quad x + 2y \leq 22 \quad (\text{finishing}), \quad x \geq 0, y \geq 0.$$

This is a system of linear inequalities — its solution region (from Subtopic 02) is a bounded polygon. Find the **corner points** by solving the boundary lines pairwise:

- $(0, 0)$.
- $y = 0$ with $2x = 36 \Rightarrow (18, 0)$ (and $18 \leq 22$, feasible).
- $x = 0$ with $2y = 22 \Rightarrow (0, 11)$ (and $33 \leq 36$, feasible).
- Intersection of $2x + 3y = 36$ and $x + 2y = 22$: from the second, $x = 22 - 2y$; substitute: $2(22 - 2y) + 3y = 36 \Rightarrow 44 - y = 36 \Rightarrow y = 8$, $x = 6$, giving $(6, 8)$.

Evaluate profit $Z = 30x + 50y$ at each corner (a linear objective is always maximised at a vertex):

$$Z(0, 0) = 0, \quad Z(18, 0) = 540, \quad Z(0, 11) = 550, \quad Z(6, 8) = 180 + 400 = 580.$$

Maximum profit ₹580 per day, at 6 chairs and 8 tables.

Why it's Advanced-level: this is precisely the Linear Programming you will formalise in Class 12 — the inequality system is the feasible region, and the optimum sits at a corner. Recognising that the joint constraint vertex $(6, 8)$ beats the axis vertices is the non-obvious step.

3.5 Section 5: Practice Exercises

1. **[CBSE]** Ravi scored 68 and 74 in his first two unit tests. Find the minimum marks he must score in the third test to have an average of at least 75 across the three tests.
2. **[CBSE]** The length of a rectangle is 3 cm more than twice its breadth. If the perimeter is at most 54 cm, find the maximum possible breadth.
3. **[JEE Main]** Find all pairs of consecutive even natural numbers, both larger than 5, whose sum is less than 24.
4. **[JEE Main]** A solution of 480 mL is 15% alcohol. How many millilitres of water must be added so that the alcohol concentration falls below 12%?
5. **[JEE Advanced]** A trader stocks x kg of tea and y kg of coffee. Storage allows at most 80 kg total, capital allows at most ₹24,000 where tea costs ₹200/kg and coffee ₹400/kg. Write the constraint system and find its corner points.

Answers: 1. At least 83 marks. 2. Maximum breadth 8 cm. 3. $(6, 8), (8, 10)$ — i.e. smaller number $\in \{6, 8\}$. 4. More than 120 mL of water. 5. $x + y \leq 80$, $200x + 400y \leq 24000$ (i.e. $x + 2y \leq 120$), $x, y \geq 0$; corners $(0, 0), (80, 0), (40, 40), (0, 60)$.

3.6 Section 6: Crack the MCQ

Q1. “A number x is *no more than 7* and *at least -2* .” The correct inequality is

- (a) $-2 < x < 7$ (b) $-2 \leq x \leq 7$ (c) $-2 \leq x < 7$ (d) $-2 < x \leq 7$

Correct: (b). “At least -2 ” $\Rightarrow x \geq -2$ (inclusive); “no more than 7” $\Rightarrow x \leq 7$ (inclusive). Both ends are included.

- **(a)** wrongly reads both phrases as strict.
- **(c)** and **(d)** each make one end strict — the classic phrase-misreading (INEQ_WORD_TRANSLATE).

Q2. A factory makes x toys, where each requires the count to satisfy $5x - 3 > 32$. The least number of toys is

- (a) 7 (b) 8 (c) 7.4 (d) 6

Correct: (b). $5x - 3 > 32 \Rightarrow x > 7$. But x counts toys, so x must be a positive **integer** greater than 7 — the least is 8.

- **(c)** reports the boundary 7-ish value without applying the integer domain (INEQ_WORD_DOMAIN).
- **(a)** takes $x = 7$, but $7 \not> 7$.
- **(d)** ignores the inequality entirely.

Q3. A 200 mL solution is 30% salt. To make it *more than 40%* salt by adding s mL of pure salt-equivalent (treat added salt as 100%), the inequality is

- (a) $\frac{60 + s}{200 + s} > 0.40$ (b) $\frac{60}{200 + s} > 0.40$ (c) $\frac{60 + s}{200} > 0.40$ (d) $\frac{0.30 + s}{200} > 0.40$

Correct: (a). Original salt = $0.30 \times 200 = 60$ mL; after adding s , salt = $60 + s$ in a total of $200 + s$. Concentration > 0.40 gives option (a).

- **(b)** forgets the added salt in the numerator.
- **(c)** forgets that the total volume also grows by s (INEQ_MIXTURE_SETUP).
- **(d)** mishandles the original-salt amount.

Q4. “ x lies *between 3 and 9*” (standard reading) translates to

- (a) $3 \leq x \leq 9$ (b) $3 < x < 9$ (c) $x < 3$ or $x > 9$ (d) $x \leq 3$ or $x \geq 9$

Correct: (b). “Between,” with no “inclusive,” is read as strict on both ends.

- **(a)** adds inclusivity the wording does not state (INEQ_WORD_BETWEEN).
- **(c), (d)** describe *outside* the interval — the opposite region.

3.7 Section 7: Common Student Pitfalls & Pro-Tips

- **Misreading the comparison word.** “More than” is strict ($>$); “at least” is inclusive ($\geq$). Underline the phrase and write its symbol *before* building the inequality.
- **Forgetting the hidden domain.** A count is a non-negative integer; a length is positive. Solve over the reals, then restrict — and state the restricted answer.
- **Stopping at the inequality instead of the answer.** “ $x \geq 10$ ” is not a sentence a marker rewards; “the minimum length is 10 cm” is. Always interpret back.
- **Mishandling “between.”** Treat it as strict unless the problem says “inclusive.” Many students reflexively write $\leq$.

Pro-Tip — translate, then test one value. After forming the inequality, plug a clearly-valid scenario from the problem back in to confirm your symbols point the right way. For Example 1, test $x = 10$: perimeter $= 10 + 20 + 12 = 42 \geq 42$ ✓ — confirming the boundary is included and “at least” was correctly read as $\geq$.

3.8 Section 8: Cheat Sheet

Applications of Linear Inequalities — Quick Recall

- **Translate the comparison word first:** *at least/at most* $\Rightarrow \geq / \leq$ (inclusive); *more/less than* $\Rightarrow > / <$ (strict).
 - **Five-step routine:** name the variable (with units) $\rightarrow$ translate each condition $\rightarrow$ note the hidden domain $\rightarrow$ solve (intersect, for systems) $\rightarrow$ interpret in context.
 - **Hidden domains:** counts are non-negative integers; measurements are positive.
 - **Average:** $\frac{\text{sum}}{n} \leq A$. **Mixture:** $\frac{\text{pure component}}{\text{total}}$, bounded as required; clear fractions by the *positive total*.
 - **“Between” is strict** unless stated inclusive.
 - **Finish by interpreting** — a solution set is not yet an answer.
- Mnemonics:** “Read the word, then write the sign.” — “Count? Then whole and non-negative.” — “Don’t stop at $x \geq 10$; say what it means.”

4 Non-Linear Inequalities: The Wavy-Curve Method

Exam Relevance: This subtopic goes beyond the CBSE syllabus — modulus, quadratic, polynomial, and rational inequalities are *not* in the NCERT Linear Inequalities chapter, so Board-only students may treat it as optional. It is, however, **core to JEE**: **JEE Main** asks modulus and quadratic/rational inequalities directly, and **JEE Advanced** tests the full wavy-curve method with repeated roots and multiple moduli. We include it so this chapter is a complete reference on *inequalities*, not just linear ones. (No NEET relevance.)

4.1 Section 1: Concept Intuition

Linear inequalities had one comforting feature: a single sign-flip rule handled everything, because $ax + b$ changes sign exactly once. The moment the expression becomes non-linear — a product of several factors, a square, a quotient — that simplicity ends. You **cannot** solve $x^2 > 4$ by “taking the square root of both sides,” and you **cannot** solve $\frac{x-1}{x+2} \geq 0$ by “cross-multiplying,” because the sign of what you multiply by is unknown. A completely different, and rather beautiful, idea is needed.

That idea rests on one fact: **a continuous expression can only switch between positive and negative by passing through zero**. So if you mark on the number line every point where the expression equals zero (or blows up to infinity), those **critical points** chop the line into intervals, and *inside each interval the expression keeps one constant sign*. You then only need to determine the sign in one interval and let a simple rule propagate it across the rest. This is the **method of intervals**, and its visual form — a single squiggle threading above and below the axis — is the **wavy curve**.

Think about it this way... Picture walking left-to-right along the number line while tracking the sign of a product like $(x-1)(x-3)(x+2)$. Far to the right (say $x = 100$), every factor is positive, so the product is positive — you are “above the axis.” Now walk leftward. Each time you cross a root, *one* factor flips its sign, so the whole product flips: above the axis becomes below, below becomes above. The curve **crosses** the axis at each simple root, weaving like a stitch. The wavy curve is just a picture of this sign-flipping walk.

But there is a crucial exception. If a factor is *squared* — like $(x-1)^2$ — then crossing $x = 1$ flips that factor’s sign *twice* (it is $(\text{something})^2 \geq 0$ always), so the product’s sign does **not** change. The curve merely **touches** the axis at $x = 1$ and bounces back to the same side. Odd powers cross; even powers bounce. This single rule about multiplicity is what separates a correct JEE solution from a wrong one.

Limiting conditions & assumptions.

- Every factor must first be written as $(x - r)$ with a **positive coefficient on x** ; a factor like $(3 - x)$ must be rewritten as $-(x - 3)$, and the stray minus sign tracked.

- **Even-multiplicity** roots (squared, fourth-power, ...) are *touch-and-bounce* points — the sign does not flip there.
 - For a **quotient**, the points where the **denominator is zero** are critical points too, but they are **never included** in the solution (the expression is undefined there) — always open circles, regardless of $\leq$ or $\geq$.
 - You may include a numerator root in the solution only for a **non-strict** ($\leq, \geq$) inequality.
-

4.2 Section 2: Key Standard Results

Modulus (absolute value) inequalities — for $a > 0$:

$$|x| < a \iff -a < x < a, \quad |x| \leq a \iff -a \leq x \leq a$$

$$|x| > a \iff x < -a \text{ or } x > a, \quad |x| \geq a \iff x \leq -a \text{ or } x \geq a$$

In words: **“less than” traps you inside one interval; “greater than” throws you outside into a union of two.** More generally $|f(x)| < a \iff -a < f(x) < a$, and $a < |x| < b \iff x \in (-b, -a) \cup (a, b)$.

Equivalent squared forms (useful because squares are sign-free):

$$x^2 < a^2 \iff |x| < a, \quad x^2 > a^2 \iff |x| > a.$$

Quadratic inequalities — for $ax^2 + bx + c$ with $a > 0$ and *real* roots $\alpha < \beta$:

$$ax^2 + bx + c < 0 \iff \alpha < x < \beta \quad (\text{negative between the roots}),$$

$$ax^2 + bx + c > 0 \iff x < \alpha \text{ or } x > \beta \quad (\text{positive outside the roots}).$$

If the discriminant is negative (no real roots) and $a > 0$, the expression is positive for *all* x .

Rigor checkpoints (replacing dimensional analysis in a math fork):

- A **denominator root is always excluded** (open circle); a **numerator root is included** only under $\leq$ or $\geq$.
 - Before applying the quadratic rule, **make the coefficient of x^2 positive** (multiply by -1 and flip the sign if needed).
 - **Never cross-multiply** a rational inequality by a variable denominator — its sign is unknown (INEQ_CROSS_MULTIPLY_UNSAFE).
-

4.3 Section 3: Key Procedure — The Wavy-Curve Algorithm

To solve any polynomial or rational inequality $f(x) (\leqslant) 0$:

1. **Move everything to one side** so the other side is 0. *Why:* the method reads the sign of a single expression against zero; comparing two non-zero sides is meaningless for sign analysis.
2. **Combine into one quotient and factor completely** into linear factors $(x - r)$ (and irreducible quadratics, which keep a constant sign). **Make each x -coefficient positive.** *Why:* the crossing/bouncing rule is stated for factors of the form $(x - r)$; a hidden -1 silently inverts every sign.
3. **Mark all critical points** on the number line: numerator zeros (filled if the inequality is $\leq / \geq$, hollow if strict) and denominator zeros (**always hollow**). *Why:* these are the only places the sign can change, and the denominator zeros are forbidden values.
4. **Fix the rightmost sign.** For x larger than every critical point, all positive-coefficient factors are positive, so the expression is **positive** on the far-right interval. Start the wavy curve above the axis there.
5. **Draw the curve leftward**, obeying multiplicity: **cross** the axis at odd-multiplicity roots, **touch and bounce** at even-multiplicity roots. *Why:* an odd power flips a factor's sign once (net flip); an even power flips it twice (no net change).
6. **Read off** the intervals where the curve lies on the required side (> 0 above, < 0 below), then **adjust endpoints**: include numerator roots for non-strict inequalities, exclude all denominator roots.

Why “outside vs between” for quadratics falls out of this. A quadratic $ax^2 + bx + c$ with $a > 0$ and roots $\alpha < \beta$ factors as $a(x - \alpha)(x - \beta)$. On the far right both factors are positive (+); crossing β gives one flip ($-$ on (α, β)); crossing α flips again ($+$ on $(-\infty, \alpha)$). Hence positive *outside*, negative *between* — exactly the standard result, now seen as a one-line wavy curve.

Modulus strategy. Either (a) apply the standard results of Section 2 directly, (b) **case-split** on the sign of each expression inside a modulus and solve each case on its interval, or (c) when both sides are non-negative, **square** to remove the modulus (valid only when both sides are ≥ 0 , else squaring corrupts the sign — INEQ_SQUARE_SIGN).

4.4 Section 4: Worked Examples

4.4.1 Example 1 — [Foundational]

Solve $x^2 - x - 6 > 0$.

Factor: $x^2 - x - 6 = (x - 3)(x + 2)$. The roots are $x = -2$ and $x = 3$, and the coefficient of x^2 is positive. By the quadratic rule, “ > 0 ” means **outside** the roots:

$$x < -2 \quad \text{or} \quad x > 3, \quad \text{i.e.} \quad x \in (-\infty, -2) \cup (3, \infty).$$

(Both endpoints excluded, since the inequality is strict.)

4.4.2 Example 2 — [JEE Main / Speed Trap]

Solve $|2x - 3| \geq 5$.

The trap: the “ $\geq$ ” form throws you **outside** into a union — the opposite of the “ $\leq$ ” form. A student who writes $-5 \leq 2x - 3 \leq 5$ (the *less-than* template) gets exactly the wrong region.

Apply $|f| \geq a \Leftrightarrow f \leq -a$ or $f \geq a$:

$$2x - 3 \leq -5 \quad \text{or} \quad 2x - 3 \geq 5.$$

$$2x \leq -2 \Rightarrow x \leq -1, \quad 2x \geq 8 \Rightarrow x \geq 4.$$

$$x \in (-\infty, -1] \cup [4, \infty).$$

Speed lesson: read the modulus direction first — “**< means inside one interval, > means outside, two intervals.**” Endpoints are included here because the inequality is non-strict.

4.4.3 Example 3 — [JEE Main]

Solve $\frac{x-4}{x+1} < 1$.

Do not cross-multiply — the sign of $x+1$ is unknown. Instead, bring everything to one side:

$$\frac{x-4}{x+1} - 1 < 0 \Rightarrow \frac{(x-4) - (x+1)}{x+1} < 0 \Rightarrow \frac{-5}{x+1} < 0.$$

The numerator -5 is a negative constant, so $\frac{-5}{x+1} < 0$ requires $x+1 > 0$ (negative over positive is negative):

$$x+1 > 0 \Rightarrow x > -1, \quad \text{i.e. } x \in (-1, \infty).$$

($x = -1$ is excluded — the original expression is undefined there.)

The lesson: the single most common rational-inequality error is cross-multiplying by $x+1$; bringing to one side and combining is the safe route every time.

4.4.4 Example 4 — [JEE Advanced]

Solve $\frac{(x+3)(x-1)^2(x-4)}{x+2} \geq 0$.

This combines every wrinkle: a repeated root, a denominator restriction, and an isolated solution point.

Critical points (left to right): $x = -3$ (numerator, multiplicity 1), $x = -2$ (**denominator** — always excluded), $x = 1$ (numerator, multiplicity 2), $x = 4$ (numerator, multiplicity 1).

A number line marked at $x = -3$ (filled), $x = -2$ (hollow), $x = 1$ (filled), $x = 4$ (filled). A wavy curve starts above the axis to the right of 4, crosses down at $x = 4$, stays below through the touch-point at $x = 1$ (bouncing off the axis without crossing), crosses up at $x = -2$, then crosses down at $x = -3$. Sign labels: negative on $(-\infty, -3)$, positive on $(-3, -2)$, negative on $(-2, 1)$ and $(1, 4)$, positive on $(4, \infty)$.

Sign walk (start positive on the far right):

- $(4, \infty)$: all factors positive $\Rightarrow +$.
- cross $x = 4$ (odd) $\Rightarrow -$ on $(1, 4)$.
- $x = 1$ has *even* multiplicity $\Rightarrow$ **bounce**, sign stays $-$ on $(-2, 1)$.
- cross $x = -2$ (denominator, odd) $\Rightarrow +$ on $(-3, -2)$.
- cross $x = -3$ (odd) $\Rightarrow -$ on $(-\infty, -3)$.

We need " ≥ 0 " — the curve on or above the axis:

- $(-3, -2)$ is positive; include $x = -3$ (numerator zero, non-strict) but exclude $x = -2$ (denominator): gives $[-3, -2)$.
- $x = 1$ makes the expression 0 (numerator zero, non-strict) $\Rightarrow$ include the isolated point $\{1\}$.
- $(4, \infty)$ is positive; include $x = 4$ (numerator zero): gives $[4, \infty)$.

$$x \in [-3, -2) \cup \{1\} \cup [4, \infty).$$

Why it's Advanced-level: three traps fire at once — the bounce at the squared root $x = 1$ (which also survives as a lone solution because the inequality is non-strict), the open exclusion of the denominator root $x = -2$, and the inclusion of the simple numerator roots. Miss any one and the answer is wrong.

4.5 Section 5: Practice Exercises

1. **[Foundational]** Solve $x^2 - 7x + 12 \leq 0$.
2. **[JEE Main]** Solve $|3x + 1| < 8$.
3. **[JEE Main]** Solve $\frac{x-2}{x+5} \geq 0$.
4. **[JEE Main]** Solve $(x-1)(x+4)(x-6) < 0$.
5. **[JEE Advanced]** Solve $\frac{(x-2)^2(x+1)}{x-5} \leq 0$.

Answers: 1. $[3, 4]$. 2. $(-3, \frac{7}{3})$. 3. $(-\infty, -5) \cup [2, \infty)$. 4. $(-\infty, -4) \cup (1, 6)$. 5. $[-1, 5)$ — the squared factor $(x-2)^2$ only touches zero at $x = 2$ (which lies inside the negative region) and is excluded at the denominator root $x = 5$.

4.6 Section 6: Crack the MCQ

Q1. The solution of $|x - 2| > 3$ is

- (a) $(-1, 5)$ (b) $(-\infty, -1) \cup (5, \infty)$ (c) $[-1, 5]$ (d) $(-\infty, -1] \cup [5, \infty)$

Correct: (b). “ $>$ ” sends you *outside*: $x - 2 < -3$ or $x - 2 > 3$, i.e. $x < -1$ or $x > 5$; strict, so open.

- **(a)** applies the “ $<$ ” (inside) template to a “ $>$ ” inequality (INEQ_MOD_DIRECTION).
- **(c), (d)** use closed endpoints although the inequality is strict.

Q2. For $a > 0$ with $b^2 - 4ac < 0$, the inequality $ax^2 + bx + c < 0$ has solution

- (a) all real x (b) no real x (c) only $x > 0$ (d) the interval between the roots

Correct: (b). With $a > 0$ and no real roots, the parabola lies entirely above the axis, so $ax^2 + bx + c > 0$ for every x — it is never negative.

- **(a)** confuses this with the “ > 0 ” case.
- **(d)** assumes real roots exist (INEQ_QUAD_OUTSIDE_BETWEEN).
- **(c)** has no basis.

Q3. The solution of $\frac{x+1}{x-3} \leq 0$ is

- (a) $[-1, 3]$ (b) $[-1, 3)$ (c) $(-1, 3)$ (d) $(-\infty, -1] \cup (3, \infty)$

Correct: (b). Critical points $x = -1$ (numerator) and $x = 3$ (denominator). The quotient is ≤ 0 between them; include $x = -1$ (numerator, non-strict) but **exclude** $x = 3$ (denominator).

- **(a)** wrongly includes $x = 3$, where the expression is undefined (INEQ_RATIONAL_DENOM_ZERO).
- **(c)** wrongly excludes the valid numerator root $x = -1$.
- **(d)** picks the “outside” region — wrong sign.

Q4. The solution set of $(x - 2)^2(x - 5) > 0$ is

- (a) $x > 5$ (b) $x > 2$ (c) $x > 5$ or $x = 2$ (d) all $x \neq 2$

Correct: (a). Since $(x - 2)^2 \geq 0$, the product’s sign is decided by $(x - 5)$ — but the product is 0 (not > 0) at $x = 2$ and at $x = 5$. So “ > 0 ” needs $(x - 2)^2 > 0$ **and** $x - 5 > 0$, giving $x > 5$ (which already excludes $x = 2$).

- **(b)** treats $(x - 2)^2$ as if it were $(x - 2)$, a sign-changing factor — the even-multiplicity error (INEQ_WAVY_MULTIPLICITY).
- **(c)** wrongly adds $x = 2$, where the product equals 0, not a positive value.
- **(d)** ignores the $(x - 5)$ factor entirely.
- The teaching point: an **even-multiplicity** factor never flips the sign; it only contributes its own root as a point where the product is zero (excluded here, since the inequality is strict).

4.7 Section 7: Common Student Pitfalls & Pro-Tips

- **Cross-multiplying a rational inequality.** Multiplying $\frac{x-4}{x+1} < 1$ by $x+1$ is illegal — its sign is unknown. Always bring to one side and combine into a single quotient first.
- **Wrong modulus direction.** $|f| < a$ traps you *inside* $(-a, a)$; $|f| > a$ throws you *outside* into a union. Mixing these is the top modulus error.
- **Including a denominator zero.** A value making the denominator 0 is *never* in the solution — open circle always, even under $\leq / \geq$.
- **Forgetting the even-multiplicity bounce.** At a squared (or any even-power) root, the curve touches and returns; the sign does not flip. Treating it as a normal crossing inverts half your intervals.

Pro-Tip — fix the far-right sign, then walk left. Always anchor the wavy curve as *positive* on the far-right interval (after making all x -coefficients positive), then flip at odd roots and bounce at even ones. One test substitution in the far-right interval (e.g. a large x) confirms the anchor before you propagate.

4.8 Section 8: Cheat Sheet

Non-Linear Inequalities — Quick Recall (*beyond CBSE; JEE essential*)

- **Modulus:** $|x| < a \Leftrightarrow -a < x < a$ (inside); $|x| > a \Leftrightarrow x < -a$ or $x > a$ (outside).
- **Quadratic** ($a > 0$, roots $\alpha < \beta$): < 0 **between** (α, β) ; > 0 **outside** $(-\infty, \alpha) \cup (\beta, \infty)$. No real roots $\Rightarrow$ always positive.
- **Wavy curve:** one side = 0 $\rightarrow$ factor, x -coeffs positive $\rightarrow$ mark critical points $\rightarrow$ anchor **positive on far right** $\rightarrow$ **cross at odd, bounce at even** multiplicity $\rightarrow$ read the required sign.
- **Endpoints:** numerator roots included for $\leq / \geq$; **denominator roots always excluded.**
- **Never cross-multiply** by a variable denominator; **never square** unless both sides are ≥ 0 .

Mnemonics: “Less-than locks you in, greater-than lets you out.” — “Odd crosses, even bounces.” — “Denominators get a hole, never a dot.”

Chapter 5: Linear Inequalities — Supplement

The four units of this chapter give you the complete NCERT syllabus plus the wavy-curve machinery that JEE runs on. This supplement adds the material those units promise but do not drill: modulus inequalities where the right-hand side is not a plain number, inequalities carrying two or more moduli, the parameter question “for which k is this quadratic positive for *every* x ?”, the use of sign analysis to hand you the domain of a function, and the controlled squaring of a radical inequality.

Read it as the fifth unit of the chapter plus three short extensions. Nothing here is new theory — it is the same sign analysis you already know, pointed at the places where JEE points it.

Corrigendum to Unit 3 — Applications & Word Problems of Linear Inequalities

Section 5, Practice Exercise 3. The question asks for all pairs of consecutive even natural numbers, both larger than 5, whose sum is less than 24. The printed answer omits one valid pair.

Let x be the smaller number. Then x is even, $x > 5$ (so $x \geq 6$), and

$$x + (x + 2) < 24 \implies 2x < 22 \implies x < 11.$$

So $x \in \{6, 8, 10\}$.

Corrected answer. $(6, 8)$, $(8, 10)$, $(10, 12)$ — that is, the smaller number belongs to $\{6, 8, 10\}$. Check the largest pair: $10 + 12 = 22 < 24$, and both 10 and 12 exceed 5, so $(10, 12)$ genuinely qualifies.

This is worth more than a line of errata. It is the exact failure mode the Unit 3 pro-tip warns about in reverse: the algebra was right, the boundary was not re-tested. **Whenever you finish a discrete word problem, substitute the largest admissible value back in.** Here, testing $x = 10$ takes four seconds and catches the whole mistake.

5 Modulus Inequalities & Case Analysis

Exam Relevance: The CBSE Boards touch this lightly — the NCERT Miscellaneous Exercise carries modulus items such as $|x - 2| \geq 5$, typically worth 2 marks, and always with a numerical right-hand side. JEE Main takes it much further: expect a modulus inequality with a *variable* right-hand side, or a “how many integers satisfy” count, in almost every recent paper. JEE Advanced pushes into two

or more moduli at once, solved by splitting the number line at the critical points and reassembling the answer from the pieces — the skill this unit exists to build. (No NEET relevance — Mathematics is outside the NEET syllabus.)

5.1 Section 1: Concept Intuition

Drive out of Mumbai on the Expressway and you pass kilometre stones: 0, 1, 2, and so on to Pune. Suppose the Khalapur toll plaza sits at stone 42 and you are somewhere at stone x . How far are you from the toll? Not $x - 42$ — that is negative if you have not reached it yet, and there is no such thing as being minus eleven kilometres from a toll plaza. The honest answer is $|x - 42|$.

That is the whole idea: $|x - a|$ is the distance on the number line between x and a , and distance never carries a sign. Once you hold that picture, every modulus inequality becomes a sentence about distance, and most of them can be *read off* a number line rather than ground out algebraically.

- $|x - 42| < 5$ says “you are within 5 km of the toll” — the stretch from stone 37 to stone 47.
- $|x - 42| > 5$ says “you are more than 5 km away” — everything outside that stretch, in two separate pieces, one on each side.

This is why “less than locks you in, greater than lets you out,” the mnemonic from Unit 4. It is not a rule to memorise; it is what distance means.

Think about it this way... now put a *second* toll plaza on the road, at stone 50. The quantity $|x - 42| + |x - 50|$ is your total distance to the two plazas added together. Walk along the road and watch that total. Anywhere between the two plazas it is exactly 8 km, no matter where you stand — you are simply splitting the gap. Step outside the pair and the total starts climbing, gaining 2 km for every 1 km you walk away. Two facts fall out immediately, and both are examinable:

- The total can never be less than 8, the gap between the plazas. So $|x - 42| + |x - 50| < 6$ has **no solution at all**, and you can say so without writing a single line of algebra.
- Whenever the bound does exceed 8, the solution is a single interval symmetric about the midpoint stone 46.

Now the harder case, and the one JEE asks most often. What does $|x - 1| < x + 2$ mean? The left side is a distance, so it is never negative. The right side is an *expression*, which may be positive, zero, or negative depending on x . A non-negative quantity can never be strictly less than a negative one, so any x making $x + 2$ negative is dead on arrival. **When the right-hand side of a modulus inequality contains the variable, its sign becomes part of the problem.** Handle it and the question is routine; ignore it and you will collect solutions that do not exist.

There are exactly two honest ways to handle it, and you will learn both:

- **The equivalence route.** Convert the modulus into a double inequality or a union in one stroke, using results that are valid whatever the sign of the right-hand side.

- **The case-split route.** Break the number line at every point where a modulus argument changes sign, remove the modulus bars on each piece, and solve there. This is the only route that survives two or more moduli.

Limiting conditions and assumptions.

- $|f(x)|$ is defined wherever $f(x)$ is; the modulus itself never restricts the domain. Radicals and logarithms do, and those are handled in Addendum B and Addendum C.
- The critical point of a modulus is where its *argument* is zero, not where the constant inside it sits. The critical point of $|2x + 3|$ is $x = -\frac{3}{2}$, not $x = -3$.
- In a case split, a solution found inside a case is only valid if it also lies in that case's own interval. Skipping that intersection is the single most expensive error in this unit.
- Squaring is legitimate only when both sides are known to be non-negative. $|f| < g$ does **not** become $f^2 < g^2$ on its own.
- The distance shortcuts above assume the coefficient of x inside each modulus is 1. For $|x - 2| + |2x + 3|$ they do not apply, and you must case-split.

5.2 Section 2: Key Formulas and Definitions

Definition. The modulus (or absolute value) of a real number x is

$$|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$

- x — any real number.
- The output $|x|$ is always non-negative, and $|x| = 0$ if and only if $x = 0$.

Basic properties. For all real x, y :

$$|x| \geq 0, \quad |xy| = |x||y|, \quad |x|^2 = x^2, \quad \sqrt{x^2} = |x|$$

$$-|x| \leq x \leq |x|, \quad |x + y| \leq |x| + |y|, \quad ||x| - |y|| \leq |x - y|$$

- The last two are the triangle inequality and its reverse form; both are standard JEE Advanced tools.
- $|x - a|$ is the distance between the points x and a on the number line. In particular $|x| = |x - 0|$ is the distance from the origin.

Standard results — constant right-hand side. These are the Unit 4 results, restated for comparison. Here $a > 0$ and f is any expression in x :

$$|f| < a \iff -a < f < a, \quad |f| > a \iff f < -a \text{ or } f > a$$

- Validity: $a > 0$ is required. If $a \leq 0$, read the general forms below instead.

Standard results — variable right-hand side. Let f and g be any expressions in x . These four equivalences hold for **every** real value of g , positive or negative, with no side condition:

Inequality	Equivalent form	What happens to the sign of g
$ f < g$	$-g < f < g$	$g > 0$ is <i>forced</i> by the double inequality itself
$ f \leq g$	$-g \leq f \leq g$	$g \geq 0$ is forced
$ f > g$	$f > g$ or $f < -g$	if $g < 0$ the union covers everything, correctly
$ f \geq g$	$f \geq g$ or $f \leq -g$	if $g < 0$ the union covers everything, correctly

- f, g — any real expressions in x ; no positivity assumption on either.
- Read the first row carefully: if $-g < f < g$ holds then $-g < g$, hence $g > 0$. You never need to impose “ $g > 0$ ” as a separate case — solving the double inequality does it for you.
- Read the third row the same way: if g happens to be negative then $|f| \geq 0 > g$ automatically, and the union $f > g$ or $f < -g$ does cover all reals. Again nothing extra to impose.
- The “**and**” versus “**or**” distinction is what you must get right. The $<$ rows are a double inequality, so the two conditions are **intersected**. The $>$ rows are a union, so the two conditions are **added together**.

Squared forms. Squares are blind to sign, which makes them useful — but only under a guard:

$$|f| < g \iff f^2 < g^2 \text{ **and** } g > 0, \quad |f| > g \iff f^2 > g^2 \text{ **or** } g < 0$$

$$|f| < |h| \iff f^2 < h^2 \quad (\text{no condition needed — both sides are already non-negative})$$

- h — any expression in x .
- The middle result is the free lunch of this unit: **when both sides are moduli, square immediately**. The guard clauses are only needed when a bare expression faces a modulus.

The two-distance results. For real constants $a < b$ and a real constant c :

$$|x - a| + |x - b| \geq b - a \quad \text{for every } x, \text{ with equality exactly on } [a, b]$$

$$|x - a| + |x - b| \leq c \iff \begin{cases} \left[\frac{a + b - c}{2}, \frac{a + b + c}{2} \right], & c \geq b - a \\ \emptyset, & c < b - a \end{cases}$$

- a, b — the two fixed points; $b - a$ is the gap between them.
- c — the bound on the total distance.
- The strict version $< c$ gives the corresponding open interval when $c > b - a$, and $\emptyset$ when $c \leq b - a$.
- **Validity condition:** the coefficient of x inside each modulus must be exactly 1. This shortcut is a checking tool and a speed weapon, not a replacement for the case-split method.

Rigor checkpoints.

- Every modulus inequality answer is a union of intervals; state it in interval notation with the endpoint brackets justified by strict versus non-strict.
- A non-strict inequality includes the endpoints where equality holds. Always test them explicitly rather than inheriting them from the algebra.
- After any case split, confirm that the pieces you union together are pairwise consistent with their own intervals.

5.3 Section 3: CBSE Board Derivations and Key Procedures

Derivation: Why $|f| < g$ is exactly $-g < f < g$, whatever the sign of g

Claim. For any real expressions f and g , the statement $|f| < g$ is true precisely when $-g < f < g$ is true.

Forward direction. Assume $|f| < g$.

1. $|f| \geq 0$ by definition of modulus. *Reason: a modulus is never negative.*
2. Therefore $g > |f| \geq 0$, so $g > 0$. *Reason: a number strictly exceeding a non-negative number is positive. This is where the right-hand side earns its positivity — it is a consequence, not an assumption.*
3. $f \leq |f| < g$, so $f < g$. *Reason: $t \leq |t|$ for every real t .*
4. $-f \leq |f| < g$, so $-f < g$, and multiplying by -1 flips it to $f > -g$. *Reason: $-t \leq |t|$ for every real t , then the sign-flip rule proved in Unit 1.*
5. Steps 3 and 4 together give $-g < f < g$.

Converse direction. Assume $-g < f < g$.

6. From $f < g$ and $f > -g$ we get $-f < g$. *Reason: multiply $f > -g$ by -1 and flip.*
7. By definition $|f|$ equals either f or $-f$. *Reason: the two-case definition of modulus.*
8. Both $f < g$ and $-f < g$ hold, so whichever of the two $|f|$ happens to be, $|f| < g$. ■

What the proof buys you. No case analysis on the sign of g is ever required for the $<$ form. Students who split into “ $g > 0$ ” and “ $g \leq 0$ ” branches are doing work the double inequality already does — and they usually mis-handle the second branch. The same argument run with the inequalities reversed establishes the companion result $|f| > g \iff f > g \text{ or } f < -g$, again for every real g ; in that form, a negative g makes the union trivially all of $\mathbb{R}$, which is correct because a modulus always beats a negative number.

Procedure A: A single modulus with a variable right-hand side

1. **Isolate the modulus.** Get the inequality into the shape $|f(x)| \bowtie g(x)$ with exactly one modulus term alone on the left. *Why: every rule in Section 2 is stated for that shape; a stray term outside the bars invalidates them.*
2. **Read the direction.** $<$ or $\leq$ means a double inequality and an **intersection**; $>$ or $\geq$ means two branches and a **union**. *Why: this single reading decides the shape of your answer, and reversing it produces the complement of the truth.*
3. **Write the equivalent form** from the Section 2 table. *Why: these forms are valid for every sign of g , so no separate sign discussion is needed.*
4. **Solve each part as a linear (or polynomial) inequality** by the methods of Unit 1 and Unit 4. *Why: once the bars are gone, nothing new is happening.*
5. **Combine.** Intersect for the $<$ family, union for the $>$ family. *Why: the double inequality demands both parts simultaneously; the branch form demands only one.*
6. **Test the endpoints and one interior point.** *Why: it costs seconds and catches an inverted direction or a wrong bracket immediately.*

Procedure B: Two or more moduli — the critical-point case split

1. **Find the critical points.** Set each modulus argument to zero and solve. *Why: these are the only places where a modulus changes its algebraic identity, so between them nothing about the expression's form changes.*
2. **Order them on the number line.** With n distinct critical points you obtain $n + 1$ intervals covering $\mathbb{R}$. *Why: the intervals must cover everything, or you will lose solutions.*
3. **Build a sign table.** One row per modulus argument, one column per interval, entering $+$ or $-$. *Why: the table tells you mechanically whether $|u|$ becomes u or $-u$ in each column, which is where careless work goes wrong.*
4. **Rewrite and solve on each interval.** In a column where the argument u is negative, replace $|u|$ by $-u$; where it is positive or zero, replace $|u|$ by u . Solve the resulting modulus-free inequality. *Why: on that interval the expression genuinely equals the rewritten one, so the rewritten inequality is exactly equivalent there.*
5. **Intersect each case's solution with that case's own interval.** *Why: a value that solves the rewritten inequality but lies outside the interval does not solve the original, because the rewriting was only valid inside. This is the step students skip, and it is worth the whole question.*
6. **Union the surviving pieces** and merge adjacent intervals into the tidiest form. *Why: the cases are exhaustive and disjoint, so every genuine solution appears in exactly one of them.*

Figure brief: Horizontal number line for $|x - 1| + |x + 2| < 5$ with critical points marked at $x = -2$ and $x = 1$ dividing it into three labelled zones; above the line, a three-row sign table showing the signs of $(x - 1)$ and $(x + 2)$ in each zone and the resulting bar-free expression $-2x - 1$, 3 , $2x + 1$; below the line, the three per-case solution segments drawn in one colour and the final union $(-3, 2)$ drawn as a single thicker bar with hollow endpoints.

Worked illustration of the routine. Solve $|x - 1| + |x + 2| < 5$.

Critical points: $x = 1$ and $x = -2$, splitting $\mathbb{R}$ into $(-\infty, -2)$, $[-2, 1)$ and $[1, \infty)$.

- **On $(-\infty, -2)$:** both arguments are negative, so $|x - 1| = 1 - x$ and $|x + 2| = -x - 2$. The inequality becomes $-2x - 1 < 5$, i.e. $x > -3$. Intersect with $(-\infty, -2)$: this gives $(-3, -2)$.
- **On $[-2, 1)$:** $x - 1 < 0$ and $x + 2 \geq 0$, so the sum is $(1 - x) + (x + 2) = 3$. The inequality reads $3 < 5$, which is true regardless of x . Intersect with $[-2, 1)$: all of $[-2, 1)$.
- **On $[1, \infty)$:** both arguments are non-negative, so the sum is $2x + 1 < 5$, i.e. $x < 2$. Intersect with $[1, \infty)$: this gives $[1, 2)$.

Union: $(-3, -2) \cup [-2, 1) \cup [1, 2) = (-3, 2)$.

The distance shortcut confirms it in one line: with $a = -2$, $b = 1$ and $c = 5 > 3 = b - a$, the solution is the open interval from $\frac{-2+1-5}{2} = -3$ to $\frac{-2+1+5}{2} = 2$.

Procedure C: Modulus against modulus

When every term on both sides is a modulus with no loose additive terms — for instance $|f| < |h|$ — do not case-split. Square both sides at once, since both are non-negative, and solve $f^2 - h^2 < 0$ by factoring the difference of squares and running the wavy curve of Unit 4. *Why it is safe: squaring is order-preserving on non-negative numbers, so the equivalence is exact with no guard condition.*

5.4 Section 4: Worked Examples

Example 1 — [CBSE Board Level]

Solve $|x + 3| < 2x$ and represent the solution on a number line.

The right-hand side is $2x$, an expression rather than a number, so reach for the general form from the Section 2 table:

$$|x + 3| < 2x \iff -2x < x + 3 < 2x.$$

This is a double inequality, so both parts must hold at once.

Left part:

$$-2x < x + 3 \implies -3x < 3 \implies x > -1$$

(dividing by -3 , a negative number, flips the sign — the Unit 1 rule).

Right part:

$$x + 3 < 2x \implies 3 < x \implies x > 3.$$

Intersect the two, because the double inequality demands both:

$$x > 3, \quad \text{that is} \quad x \in (3, \infty).$$

Check. At $x = 4$: $|4 + 3| = 7$ and $2(4) = 8$, and $7 < 8$ holds. At the endpoint $x = 3$: $|6| = 6$ and $2(3) = 6$, and $6 < 6$ is false, so 3 is correctly excluded. At $x = 0$: $|3| = 3 < 0$ is false, correctly excluded.

Notice you never had to argue separately that $2x$ must be positive. It is: every $x > 3$ makes $2x > 6 > 0$. The double inequality enforced it silently.

Figure brief: Number line from -2 to 6 with a hollow circle at $x = 3$ and a thick shaded arrow extending right toward $+\infty$, labelled $x > 3$; a small annotation at $x = 0$ marked with a cross to show a rejected trial value.

Example 2 — [JEE Main Level]

Solve $|x - 3| \geq x$.

The direction is $\geq$, so the Section 2 table gives a **union** of two branches:

$$|x - 3| \geq x \iff x - 3 \geq x \quad \text{or} \quad x - 3 \leq -x.$$

Branch 1: $x - 3 \geq x \implies -3 \geq 0$. This is false for every x , so this branch contributes nothing. (A branch collapsing to a false numerical statement is normal — do not treat it as an error.)

Branch 2: $x - 3 \leq -x \implies 2x \leq 3 \implies x \leq \frac{3}{2}$.

Union of the branches:

$$x \in \left(-\infty, \frac{3}{2}\right].$$

Check. At $x = \frac{3}{2}$: $|\frac{3}{2} - 3| = \frac{3}{2}$ and the right side is $\frac{3}{2}$, so equality holds and the endpoint belongs (the inequality is non-strict). At $x = 2$: $|-1| = 1 \geq 2$ is false, correctly excluded. At $x = -5$: $|-8| = 8 \geq -5$ is true, correctly included.

That last check is the conceptual pay-off. Every negative x satisfies the inequality automatically, because a modulus is non-negative and therefore already beats a negative right-hand side. The general form handled this for you without a separate case — which is exactly why it is stated for all real g .

Example 3 — [JEE Main — Speed Trap]

Solve $|x - 3| < 2x - 5$.

The trap: squaring both sides on autopilot. The modulus is annoying, squares are clean, and $|x - 3|^2 = (x - 3)^2$ is genuinely true — so the temptation is overwhelming:

$$(x - 3)^2 < (2x - 5)^2 \implies x^2 - 6x + 9 < 4x^2 - 20x + 25 \implies 0 < 3x^2 - 14x + 16.$$

Factoring, $3x^2 - 14x + 16 = (x - 2)(3x - 8)$, whose roots are $x = 2$ and $x = \frac{8}{3}$; a positive leading coefficient means “ > 0 outside the roots”. **The seductive wrong answer is** $(-\infty, 2) \cup (\frac{8}{3}, \infty)$.

It is wrong, and one substitution exposes it. Take $x = 0$, which sits inside that first interval: the original inequality reads $|0 - 3| < 2(0) - 5$, that is $3 < -5$. False. A distance can never

be less than a negative number, yet the squaring “solution” hands you an entire infinite interval of such values. Squaring destroyed the sign information: it cannot tell $3 < -5$ from $(-5)^2 > 3^2$.

The correct solution. Use the equivalence, which is valid whatever the sign of $2x - 5$:

$$|x - 3| < 2x - 5 \iff -(2x - 5) < x - 3 < 2x - 5.$$

Left part:

$$-2x + 5 < x - 3 \implies 8 < 3x \implies x > \frac{8}{3}.$$

Right part:

$$x - 3 < 2x - 5 \implies 2 < x.$$

Intersecting, $x > \frac{8}{3}$ (since $\frac{8}{3} > 2$):

$$x \in \left(\frac{8}{3}, \infty\right).$$

Check. At $x = 3$: $|0| = 0 < 1$, true. At $x = 2.7$: $|-0.3| = 0.3 < 0.4$, true. At $x = 2.6$: $0.4 < 0.2$, false — correctly excluded, and note $2.6 < \frac{8}{3} \approx 2.667$. At the endpoint $x = \frac{8}{3}$: both sides equal $\frac{1}{3}$, so strict inequality fails and the endpoint is correctly open.

Naming the trap: squaring without the sign guard (INEQ_SQUARE_SIGN). If you *do* want to square, you must carry the guard: $|f| < g$ is equivalent to $f^2 < g^2$ **and** $g > 0$. Here $g > 0$ means $2x - 5 > 0$, i.e. $x > \frac{5}{2}$. Intersecting that with $(-\infty, 2) \cup (\frac{8}{3}, \infty)$ kills the spurious branch and returns $(\frac{8}{3}, \infty)$ — the right answer. The squaring route is legal; it is simply never legal alone.

Example 4 — [JEE Advanced Level]

Solve $|x - 2| + |2x + 3| \leq 7$, **and state how many integers satisfy it.**

Two moduli, and the second has a coefficient of 2 inside — so the two-distance shortcut does not apply and the case split is mandatory.

Step 1 — critical points. $x - 2 = 0$ gives $x = 2$; $2x + 3 = 0$ gives $x = -\frac{3}{2}$. Note carefully that the second critical point is $-\frac{3}{2}$, not -3 .

Step 2 — the three intervals. $(-\infty, -\frac{3}{2})$, $[-\frac{3}{2}, 2)$, $[2, \infty)$.

Step 3 — sign table.

Interval	sign of $x - 2$	sign of $2x + 3$	expression becomes
$(-\infty, -\frac{3}{2})$	—	—	$(2 - x) + (-2x - 3) = -3x - 1$
$[-\frac{3}{2}, 2)$	—	+	$(2 - x) + (2x + 3) = x + 5$

Interval	sign of $x - 2$	sign of $2x + 3$	expression becomes
$[2, \infty)$	+	+	$(x - 2) + (2x + 3) = 3x + 1$

Step 4 and 5 — solve on each interval and intersect.

- **Case A**, $x < -\frac{3}{2}$: $-3x - 1 \leq 7 \Rightarrow -3x \leq 8 \Rightarrow x \geq -\frac{8}{3}$ (dividing by -3 flips). Intersecting with $x < -\frac{3}{2}$ gives $[-\frac{8}{3}, -\frac{3}{2})$, since $-\frac{8}{3} \approx -2.67 < -1.5$.
- **Case B**, $-\frac{3}{2} \leq x < 2$: $x + 5 \leq 7 \Rightarrow x \leq 2$. Every point of the interval already satisfies this, so Case B contributes all of $[-\frac{3}{2}, 2)$.
- **Case C**, $x \geq 2$: $3x + 1 \leq 7 \Rightarrow x \leq 2$. Intersecting with $x \geq 2$ leaves the single point $\{2\}$.

Step 6 — union.

$$[-\frac{8}{3}, -\frac{3}{2}) \cup [-\frac{3}{2}, 2) \cup \{2\} = [-\frac{8}{3}, 2].$$

Integer count. Since $-\frac{8}{3} \approx -2.67$, the integers in $[-\frac{8}{3}, 2]$ are $-2, -1, 0, 1, 2$ — **five integers**.

Check. At $x = -\frac{8}{3}$: $|- \frac{8}{3} - 2| = \frac{14}{3}$ and $|2(-\frac{8}{3}) + 3| = |-\frac{7}{3}| = \frac{7}{3}$; the sum is $\frac{21}{3} = 7$, so equality holds and the endpoint belongs. At $x = 2$: $0 + 7 = 7$, equality again. At $x = -3$: $5 + 3 = 8 > 7$, correctly excluded. At $x = 3$: $1 + 9 = 10 > 7$, correctly excluded.

Why it is Advanced-level: three separate traps fire in one question. The critical point of $|2x + 3|$ is a fraction, not the visible integer. Case C looks empty but yields a lone boundary point that the final interval needs. And Case B's inequality is satisfied identically, which students misread as “no solutions here” rather than “all of this interval.” Miss any one and both the interval and the integer count come out wrong.

5.5 Section 5: Practice Exercises

1. **[CBSE]** Solve $|x + 2| \leq 3x$ and write the answer in interval notation.
2. **[JEE Main]** Solve $|2x - 1| > x + 3$.
3. **[JEE Main]** Solve $|x - 5| \geq x - 1$.
4. **[JEE Main]** Solve $|x| + |x - 3| \leq 5$.
5. **[JEE Main]** Solve $|x + 1| - |x - 2| > 1$.
6. **[JEE Advanced]** Find the solution set of $|x - 1| + |x + 3| \leq 8$ and the number of integers in it.
7. **[JEE Advanced]** Solve $|x - 1| + |x - 3| < 1$.
8. **[JEE Advanced]** Solve $|x - 1| < |x + 3|$.

Answers: 1. $[1, \infty)$. 2. $(-\infty, -\frac{2}{3}) \cup (4, \infty)$. 3. $(-\infty, 3]$. 4. $[-1, 4]$. 5. $(1, \infty)$. 6. $[-5, 3]$; 9 integers. 7. $\emptyset$ — the two points are 2 apart, so the sum of distances is never below 2, let alone below 1. 8. $(-1, \infty)$ — both sides are moduli, so square directly: $(x - 1)^2 < (x + 3)^2$ reduces to $-8x < 8$.

5.6 Section 6: Crack the MCQ

Q1. The solution set of $|x - 2| < x$ is

- (a) $(1, \infty)$ (b) $(-\infty, 1)$ (c) $(1, 2)$ (d) all real x except 2

Correct: (a). Write $-x < x - 2 < x$. The left part gives $2 < 2x$, i.e. $x > 1$; the right part gives $-2 < 0$, true for every x . Intersecting leaves $x > 1$.

Distractor Analysis.

- (b) comes from mis-solving $-x < x - 2$ as $x < 1$ — the sign-flip error of Unit 1 (INEQ_SIGN_FLIP_NEG) applied to the left half of the double inequality.
- (c) comes from assuming that because the direction is “ $<$ ”, the answer must be a *bounded* interval, and closing it off at the critical point $x = 2$. The “less than locks you in” mnemonic describes a constant right-hand side; with a variable right-hand side one of the two bounds can be vacuous.
- (d) comes from reading the inequality as $|x - 2| > 0$ and reporting the single value where the modulus vanishes.

Q2. The number of integers satisfying $|x| + |x - 4| \leq 6$ is

- (a) 5 (b) 6 (c) 7 (d) 9

Correct: (c). Critical points 0 and 4. For $x < 0$: $-x + (4 - x) = 4 - 2x \leq 6$ gives $x \geq -1$, so $[-1, 0)$. For $0 \leq x < 4$: the sum is exactly $4 \leq 6$, true throughout. For $x \geq 4$: $2x - 4 \leq 6$ gives $x \leq 5$, so $[4, 5]$. The solution set is $[-1, 5]$, containing the integers $-1, 0, 1, 2, 3, 4, 5$ — seven of them.

Distractor Analysis.

- (a) 5 comes from solving only the middle case and counting the integers 0 through 4 — the classic failure to examine the outer intervals at all.
- (b) 6 comes from assuming $x \geq 0$ because moduli are involved, which loses the interval $[-1, 0)$ and with it the integer -1 , leaving $[0, 5]$.
- (d) 9 comes from splitting the sum into two separate bounds, $|x| \leq 6$ and $|x - 4| \leq 6$, and intersecting $[-6, 6]$ with $[-2, 10]$ to get $[-2, 6]$. A sum of two moduli cannot be bounded term by term.

Q3. If $|x - 1| \geq 2x + 1$, then x belongs to

- (a) $(-\infty, -2]$ (b) $(-\infty, 0]$ (c) $[-2, 0]$ (d) $[0, \infty)$

Correct: (b). The $\geq$ form gives two branches. Branch 1: $x - 1 \geq 2x + 1 \implies -2 \geq x$, so $x \leq -2$. Branch 2: $x - 1 \leq -(2x + 1) \implies 3x \leq 0$, so $x \leq 0$. The union of $(-\infty, -2]$ and $(-\infty, 0]$ is $(-\infty, 0]$.

Distractor Analysis.

- (a) keeps only the first branch — the reader stopped after one case and never wrote the “or” half.

- (c) intersects the branches instead of unioning them, the “and” versus “or” confusion that Section 2 flags. (It also mangles the intersection, which would be $(-\infty, -2]$.)
- (d) comes from solving $3x \leq 0$ as $x \geq 0$, a sign slip in the final division.

Q4. The number of real values of x satisfying $|x - 3| + |x - 6| < 2$ is

- (a) exactly one (b) infinitely many (c) none (d) exactly two

Correct: (c). The points 3 and 6 are 3 apart, so $|x - 3| + |x - 6| \geq 3$ for every real x . A quantity that is never below 3 can never be below 2. The solution set is empty.

Distractor Analysis.

- (b) is what you get by case-splitting and then failing to intersect each case with its own interval. On $x < 3$ the inequality becomes $9 - 2x < 2$, i.e. $x > 3.5$ — inconsistent with $x < 3$, so it must be discarded. On $x \geq 6$ it becomes $2x - 9 < 2$, i.e. $x < 5.5$ — again inconsistent. Keeping both un-intersected gives “ $x > 3.5$ or $x < 5.5$ ”, which is all of $\mathbb{R}$. This is the most damaging error in the unit and it produces an answer that looks confident.
- (d) reports the two case boundaries 3.5 and 5.5 as though they were solutions; substituting $x = 3.5$ gives $0.5 + 2.5 = 3$, which is not less than 2.
- (a) comes from testing only the midpoint $x = 4.5$, where the sum is $1.5 + 1.5 = 3$, and mis-recording it as a success.

5.7 Section 7: Common Student Pitfalls and Pro-Tips

- **Squaring a modulus inequality without the sign guard.** $|f| < g$ is *not* $f^2 < g^2$. The correct statement carries a companion condition: $f^2 < g^2$ **and** $g > 0$. Drop the guard and you import an entire interval of false solutions, as Example 3 shows. (INEQ_SQUARE_SIGN)
- **Forgetting to intersect a case with its own interval.** When you replace $|x - 1|$ by $1 - x$ you have assumed $x < 1$; any solution you then find is only real if it also satisfies $x < 1$. Write the interval beside each case before you start solving it, so the intersection is impossible to forget.
- **Confusing “and” with “or”.** The $<$ family produces a double inequality whose two halves are intersected; the $>$ family produces two branches that are unioned. Reversing this returns the complement of the correct answer — a full-mark loss that looks like a small slip.
- **Taking the critical point from the constant rather than the argument.** $|2x + 3|$ switches at $x = -\frac{3}{2}$; $|3 - 4x|$ switches at $x = \frac{3}{4}$. Always solve “argument = 0” in writing.
- **Panicking at a case that collapses.** A case reducing to $-3 \geq 0$ contributes nothing and a case reducing to $3 < 5$ contributes its entire interval. Both are ordinary outcomes, not signals that something has gone wrong.
- **Losing endpoints under a non-strict sign.** With $\leq$ or $\geq$, every value where the two sides are equal belongs to the answer — including isolated points such as the $\{2\}$ that Case C contributed in Example 4.

Pro-Tip — read two moduli as distances before writing anything. For $|x - a| + |x - b|$ with unit coefficients, the minimum value is $|a - b|$, achieved everywhere between a and b . So $|x - 4| + |x - 9| < 3$ is empty on sight, and $|x - 4| + |x - 9| \leq 11$ solves instantly to $\left[\frac{4+9-11}{2}, \frac{4+9+11}{2}\right] = [1, 12]$. In a timed paper this converts a four-line case split into a five-second read. Use it to *check* your case-split answer even when the coefficients are not 1 — the shortcut will not apply, but the “minimum is the gap” instinct still tells you whether an empty answer is plausible.

Pro-Tip — the endpoint audit. Every modulus answer has endpoints where the two sides are exactly equal. Substitute each one into the *original* inequality and decide inclusion from that substitution, never from the shape of your last algebraic line. This single habit fixes almost every bracket error in this unit.

5.8 Section 8: Cheat Sheet Box

Modulus Inequalities & Case Analysis — Quick Recall (JEE essential)

- **Meaning:** $|x - a|$ is the distance from x to a ; $|x| \geq 0$ always, and $\sqrt{x^2} = |x|$.
- **Variable right-hand side** (valid for every real g , no side conditions):

$$|f| < g \iff -g < f < g \quad (\text{and} - \text{intersect})$$

$$|f| > g \iff f > g \text{ or } f < -g \quad (\text{or} - \text{union})$$

Non-strict versions replace $<$ by $\leq$ and $>$ by $\geq$ throughout.

- **Squaring:** $|f| < g \iff f^2 < g^2$ **and** $g > 0$. $|f| > g \iff f^2 > g^2$ **or** $g < 0$. But $|f| < |h| \iff f^2 < h^2$ with no guard — when both sides are moduli, square at once.
- **Two or more moduli — case split:** critical points from each argument = 0 $\rightarrow$ order them into $n + 1$ intervals $\rightarrow$ sign table $\rightarrow$ drop the bars on each interval $\rightarrow$ **intersect each case’s answer with its own interval** $\rightarrow$ union.
- **Distance shortcut** (unit coefficients only, $a < b$): $|x - a| + |x - b| \geq b - a$ always. For $c \geq b - a$, $|x - a| + |x - b| \leq c$ solves to $\left[\frac{a+b-c}{2}, \frac{a+b+c}{2}\right]$; for $c < b - a$ the solution is $\emptyset$.
- **Endpoints:** decide inclusion by substituting into the original inequality, not by inspecting the last line of algebra.

Mnemonics: “Less-than is an **and**, greater-than is an **or**.” — “Square only with a guard; two moduli need no guard.” — “Every case must go home to its own interval.”

Addendum: Three Targeted Extensions

Each of the three items below is a JEE pattern that the earlier units set up but never exercised. Study Addendum A and Addendum B immediately after the wavy-curve algorithm in Unit 4 — both are that algorithm, aimed at a new target. Addendum C finally cashes the promise the chapter repeats four times: *never square unless both sides are non-negative*.

Addendum A — Quadratic Sign with a Parameter: “Positive for Every Real x ”

Unit 4 gives you the forward fact: if $a > 0$ and the discriminant is negative, then $ax^2 + bx + c > 0$ for all x . JEE Main almost always asks the reverse — you are handed an expression containing a parameter k and asked which values of k make it positive everywhere. This is the inverse direction, and it has one extra branch that decides most questions.

Method

Let $f(x) = ax^2 + bx + c$ with discriminant $D = b^2 - 4ac$. A parabola with $a > 0$ opens upward, so it stays strictly above the axis exactly when it never meets the axis.

Requirement	Condition when $a \neq 0$	Degenerate branch $a = 0$
$f(x) > 0$ for all real x	$a > 0$ and $D < 0$	$b = 0$ and $c > 0$
$f(x) \geq 0$ for all real x	$a > 0$ and $D \leq 0$	$b = 0$ and $c \geq 0$
$f(x) < 0$ for all real x	$a < 0$ and $D < 0$	$b = 0$ and $c < 0$
$f(x) \leq 0$ for all real x	$a < 0$ and $D \leq 0$	$b = 0$ and $c \leq 0$

- a, b, c — the coefficients, each of which may itself depend on the parameter k .
- $D = b^2 - 4ac$ — the discriminant.
- The final answer is the **union** of the $a \neq 0$ solutions and the degenerate $a = 0$ solutions.
- **The step that separates a full-mark answer from a near-miss:** whenever the coefficient of x^2 contains the parameter, you must test the value of k that makes it zero *separately*, because at that value the expression is no longer a quadratic at all. Unit 1, Example 4 did precisely this for a linear inequality; the quadratic version works the same way.
- Note the strict-versus-slack difference in the discriminant. For > 0 you need $D < 0$ (no real root at all). For ≥ 0 you may allow $D = 0$, since a perfect square touches the axis but never dips below it.

Figure brief: Three side-by-side parabola panels, all opening upward, sharing one horizontal axis label: panel 1 “ $D < 0$: entirely above the axis, $f(x) > 0$ for all x ”, panel 2 “ $D = 0$: vertex resting on the axis, $f(x) \geq 0$ but not > 0 ”, panel 3 “ $D > 0$: dips below between the two roots, marked as the rejected case”, with the two roots labelled α and β on the third.

Example A.1 — [JEE Main Level]

Find all real values of k for which $x^2 + 2(k-1)x + (k+5) > 0$ for every real x .

The coefficient of x^2 is 1, which is positive and free of k , so the degenerate branch cannot arise and only the discriminant condition remains.

$$\frac{D}{4} = (k-1)^2 - (k+5) = k^2 - 2k + 1 - k - 5 = k^2 - 3k - 4 = (k-4)(k+1).$$

Requiring $D < 0$:

$$(k-4)(k+1) < 0 \implies -1 < k < 4.$$

Answer: $k \in (-1, 4)$.

Check. At $k = 0$ the expression is $x^2 - 2x + 5$, whose discriminant is $4 - 20 = -16 < 0$ — positive everywhere, as required. At the excluded endpoint $k = 4$ it becomes $x^2 + 6x + 9 = (x + 3)^2$, which is zero at $x = -3$ and therefore not *strictly* positive for all x — correctly excluded. At $k = -1$ it becomes $x^2 - 4x + 4 = (x - 2)^2$, zero at $x = 2$ — correctly excluded. Both endpoints fail for the same reason, which is exactly why $D < 0$ is strict here.

Example A.2 — [JEE Advanced Level]

Find all real values of k for which $(k - 3)x^2 + (k - 3)x + 4 > 0$ for every real x .

Here the parameter sits in the coefficient of x^2 , so the degenerate branch is live.

Branch 1 — the degenerate case $k = 3$. Substituting, every x term vanishes and the expression is the constant 4. Since $4 > 0$, the requirement holds for every real x . **So $k = 3$ is a solution.**

Branch 2 — the genuine quadratic, $k \neq 3$. Two conditions must hold together.

- Upward-opening: $k - 3 > 0$, i.e. $k > 3$.
- No real roots: $D = (k - 3)^2 - 4(k - 3)(4) = (k - 3)[(k - 3) - 16] = (k - 3)(k - 19) < 0$, which gives $3 < k < 19$.

Intersecting the two conditions gives $3 < k < 19$.

Union of the branches:

$$k \in [3, 19).$$

Check. At $k = 3$ the expression is the constant $4 > 0$ — included, and it is an isolated endpoint that the discriminant condition alone would have thrown away. At $k = 10$: $7x^2 + 7x + 4$ with $D = 49 - 112 = -63 < 0$ and a positive leading coefficient — positive everywhere. At the excluded endpoint $k = 19$: $16x^2 + 16x + 4 = 4(2x + 1)^2$, which is zero at $x = -\frac{1}{2}$ — correctly excluded. At $k = 2$: $-x^2 - x + 4$ opens downward and so is eventually negative — correctly excluded.

Why it is Advanced-level: the answer set is closed at the left end and open at the right, and the two endpoints fail or succeed for completely different reasons. The included endpoint $k = 3$ arrives from the degenerate branch, which a discriminant-only solution never sees.

Practice — Addendum A

1. Find all real k for which $x^2 + kx + 4 > 0$ for every real x .
2. Find all real k for which $x^2 - 2kx + (k + 6) \geq 0$ for every real x .
3. Find all real k for which $-x^2 + 2kx - 4 < 0$ for every real x .
4. Find all real k for which $(k + 1)x^2 + 2(k + 1)x + 4 > 0$ for every real x .
5. For how many integer values of k is $x^2 + kx + k + 3 > 0$ for every real x ?

Answers: 1. $(-4, 4)$. 2. $[-2, 3]$ — the slack sign permits $D = 0$, so both endpoints are included. 3. $(-2, 2)$ — multiply by -1 and flip to get $x^2 - 2kx + 4 > 0$ for all x , then require $D < 0$. 4.

$[-1, 3)$ — the degenerate value $k = -1$ leaves the constant $4 > 0$ and therefore belongs. 5. 7 integers, namely $k \in \{-1, 0, 1, 2, 3, 4, 5\}$, from the condition $-2 < k < 6$.

Addendum B — Finding the Domain of a Function by Sign Analysis

Unit 1 calls inequalities “the silent workhorse inside domain, range, and parameter problems.” This is that workhorse. Nearly every JEE Main domain question is a sign-analysis question wearing a different hat: you translate the function’s structural requirements into inequalities, solve them with the wavy curve of Unit 4, and intersect.

Method

Read the expression from the outside in and write one condition for each structure you meet.

Structure	Condition it imposes
$\sqrt{u}$	$u \geq 0$
$\frac{1}{u}$	$u \neq 0$
$\frac{1}{\sqrt{u}}$	$u > 0$ (both conditions at once)
$\log_b u$	$u > 0$, with base $b > 0$, $b \neq 1$
$\frac{1}{\log_b u}$	$u > 0$ and $u \neq 1$

- u — the expression inside the structure, itself a function of x .
- The domain is the **intersection** of every condition the expression generates. One failed condition disqualifies the point, however comfortable the others look.
- The distinction between $\geq$ and $>$ carries all the marks. A square root standing alone allows its argument to be zero; the same square root in a denominator does not.
- Solve each condition by the wavy-curve algorithm, remembering that denominator roots are always hollow and numerator roots are filled only under a non-strict sign.

Figure brief: Number line for $\frac{x-1}{x-3} \geq 0$ with a filled circle at $x = 1$ and a hollow circle at $x = 3$, the wavy curve anchored above the axis to the right of 3, crossing down at 3 and back up at 1; the two accepted stretches $(-\infty, 1]$ and $(3, \infty)$ shaded, the rejected stretch $(1, 3)$ left blank and labelled “negative — square root undefined”.

Example B.1 — [JEE Main Level]

Find the domain of $f(x) = \sqrt{\frac{x-1}{x-3}}$.

A square root demands a non-negative argument, so the entire question is the rational inequality

$$\frac{x-1}{x-3} \geq 0.$$

Do not cross-multiply — the sign of $x - 3$ is unknown. Run the wavy curve instead.

Critical points: $x = 1$ from the numerator (filled, since the sign is non-strict) and $x = 3$ from the denominator (hollow always, because the expression is undefined there).

Anchor on the far right: for large x both factors are positive, so the quotient is positive on $(3, \infty)$. Walking left, $x = 3$ has multiplicity one, so the sign flips to negative on $(1, 3)$; $x = 1$ has multiplicity one, so it flips back to positive on $(-\infty, 1)$.

Collecting the intervals where the quotient is ≥ 0 :

$$\text{Domain} = (-\infty, 1] \cup (3, \infty).$$

Check. At $x = 0$: $\frac{-1}{-3} = \frac{1}{3} \geq 0$, so 0 belongs. At $x = 1$: the quotient is 0, and $\sqrt{0} = 0$ is perfectly defined — the endpoint belongs. At $x = 2$: $\frac{1}{-1} = -1 < 0$, so 2 is excluded. At $x = 3$: the function is undefined, so 3 is excluded — note that it stays excluded even though the inequality is non-strict.

Example B.2 — [JEE Advanced Level]

Find the domain of $f(x) = \log_{10}\left(\frac{x-2}{x+3}\right) + \sqrt{9-x^2}$.

Two structures, so two conditions, then intersect.

Condition 1 — the logarithm. Its argument must be strictly positive:

$$\frac{x-2}{x+3} > 0.$$

Critical points $x = -3$ (denominator, hollow) and $x = 2$ (numerator, hollow because the sign is strict). Anchoring positive on the far right and walking left: positive on $(2, \infty)$, negative on $(-3, 2)$, positive on $(-\infty, -3)$. So Condition 1 gives $x < -3$ or $x > 2$.

Condition 2 — the square root. Its argument must be non-negative:

$$9 - x^2 \geq 0 \implies x^2 \leq 9 \implies -3 \leq x \leq 3.$$

Intersect. The piece $x < -3$ shares nothing with $[-3, 3]$. The piece $x > 2$ meets $[-3, 3]$ in $(2, 3]$.

$$\text{Domain} = (2, 3].$$

Check. At $x = 3$: the log argument is $\frac{1}{6} > 0$ and the root argument is $9 - 9 = 0$, giving $\sqrt{0} = 0$ — both terms defined, so the endpoint belongs. At $x = 2.5$: $\frac{0.5}{5.5} > 0$ and $9 - 6.25 = 2.75 > 0$ — defined. At $x = 2$: the log argument is 0 and $\log_{10} 0$ is undefined — correctly excluded, and this is why that endpoint is open while the other is closed.

The lesson to carry into JEE: the two conditions came from different structures and had different strictness, and the final bracket at each end was decided by whichever condition was binding there. Never assume the domain inherits one condition's brackets.

Practice — Addendum B

1. Find the domain of $f(x) = \sqrt{x^2 - 4x + 3}$.
2. Find the domain of $f(x) = \frac{1}{\sqrt{9 - x^2}}$.
3. Find the domain of $f(x) = \log_{10}\left(\frac{x+2}{x-5}\right)$.
4. Find the domain of $f(x) = \sqrt{\frac{(x+1)(x-2)}{x-4}}$.
5. Find the domain of $f(x) = \log_{10}(x^2 - 5x + 6)$.

Answers: 1. $(-\infty, 1] \cup [3, \infty)$. 2. $(-3, 3)$ — strict at both ends, because the root sits in a denominator. 3. $(-\infty, -2) \cup (5, \infty)$. 4. $[-1, 2] \cup (4, \infty)$ — the numerator roots -1 and 2 are included, the denominator root 4 is not. 5. $(-\infty, 2) \cup (3, \infty)$.

Addendum C — Irrational Inequalities: Controlled Squaring

The chapter states four separate times that you must never square unless both sides are non-negative. Here is the rule doing actual work. A radical inequality is solved by squaring, but only after the domain condition and the sign condition have been written down — and those conditions are part of the answer, not preliminary throat-clearing.

Method

For real expressions f and g :

$$\sqrt{f} < g \iff f \geq 0 \text{ and } g > 0 \text{ and } f < g^2$$

$$\sqrt{f} > g \iff \left[f \geq 0 \text{ and } g < 0 \right] \text{ or } \left[g \geq 0 \text{ and } f > g^2 \right]$$

- f — the expression under the radical; $f \geq 0$ is the domain condition and can never be dropped.
- g — the other side, whose sign drives everything.
- The non-strict versions replace $<$ by $\leq$ and $>$ by $\geq$ in the final comparison only; the domain condition $f \geq 0$ stays as it is.
- **Why the two rules look so different.** For “ $<$ ” there is nothing to discuss: $\sqrt{f} \geq 0$, so g must be strictly positive or no solution can exist, and all three conditions are intersected. For “ $>$ ”, a negative g is an automatic win — a non-negative root always exceeds a negative number — so that branch is kept and unioned with the squared branch.
- In the second bracket of the “ $>$ ” rule, $f > g^2 \geq 0$ already forces $f \geq 0$, so the domain condition is absorbed. In the first bracket it must be written explicitly.

Figure brief: Single set of axes with $y = \sqrt{x+3}$ drawn from $(-3, 0)$ rising rightward and the straight line $y = x + 1$ drawn across it, their intersection marked at $(1, 2)$; the region where the line lies above the curve shaded and labelled “ $\sqrt{x+3} < x+1$: $x > 1$ ”, the region where the curve lies above the line labelled “ $\sqrt{x+3} > x+1$: $-3 \leq x < 1$ ”.

Example C.1 — [JEE Main Level]**Solve** $\sqrt{x+3} < x+1$.

Write all three conditions before touching the algebra.

- Domain: $x+3 \geq 0 \Rightarrow x \geq -3$.
- Sign of the right side: $x+1 > 0 \Rightarrow x > -1$.
- Squared comparison: $x+3 < (x+1)^2 = x^2 + 2x + 1$, so $0 < x^2 + x - 2 = (x+2)(x-1)$, giving $x < -2$ or $x > 1$.

Intersect all three. The condition $x > -1$ annihilates the branch $x < -2$ completely, and it is weaker than $x > 1$, so what survives is

$$x \in (1, \infty).$$

Check. At $x = 2$: $\sqrt{5} \approx 2.236 < 3$, true. At the endpoint $x = 1$: $\sqrt{4} = 2$ and $x+1 = 2$, so the strict inequality fails — correctly excluded. At $x = 0$: $\sqrt{3} \approx 1.732 < 1$ is false, correctly excluded even though $x = 0$ satisfies the squared comparison's *other* branch requirement. At $x = -2.5$: it lies in the discarded branch $x < -2$, and indeed $\sqrt{0.5} \approx 0.707 < -1.5$ is false. The sign condition earned its place.

Example C.2 — [JEE Advanced Level]**Solve** $\sqrt{x+3} > x+1$.

Same two expressions, opposite direction — and now the answer needs both branches.

Domain throughout: $x \geq -3$.

Branch 1 — the right side is negative. $x+1 < 0 \Rightarrow x < -1$. Combined with the domain, this gives $-3 \leq x < -1$, and every such x satisfies the inequality automatically, since the left side is non-negative and the right side is negative.

Branch 2 — the right side is non-negative. $x+1 \geq 0 \Rightarrow x \geq -1$. Now both sides are non-negative, so squaring is legitimate:

$$x+3 > (x+1)^2 \Rightarrow 0 > x^2 + x - 2 = (x+2)(x-1) \Rightarrow -2 < x < 1.$$

Intersecting with $x \geq -1$ gives $-1 \leq x < 1$.**Union of the branches:**

$$[-3, -1) \cup [-1, 1) = [-3, 1).$$

Check. At $x = -3$: $\sqrt{0} = 0 > -2$, true — the left endpoint belongs. At $x = 0$: $\sqrt{3} \approx 1.732 > 1$, true. At the excluded endpoint $x = 1$: both sides equal 2, so the strict inequality fails. At $x = 2$: $\sqrt{5} \approx 2.236 > 3$ is false.

The cross-check that proves both examples at once. On the common domain $[-3, \infty)$, exactly one of $<$, $=$, $>$ holds at each point. Example C.1 gave $(1, \infty)$ for “ $<$ ”, the two sides

are equal only at $x = 1$, and this example gives $[-3, 1)$ for “ $>$ ”. Those three sets are disjoint and their union is $[-3, \infty)$ — the whole domain, with nothing missing and nothing counted twice. Whenever a paper asks you for both directions, run this audit; it catches a lost branch instantly.

Practice — Addendum C

1. Solve $\sqrt{x-1} < 3$.
2. Solve $\sqrt{x^2-4} < 3$.
3. Solve $\sqrt{x+4} < x+2$.
4. Solve $\sqrt{x-2} \geq x-4$.
5. Solve $\sqrt{3-x} < x-1$.

Answers: 1. $[1, 10)$ — the left endpoint survives because $\sqrt{0} = 0 < 3$. 2. $(-\sqrt{13}, -2] \cup [2, \sqrt{13})$ — the domain condition $x^2 \geq 4$ carves the middle out. 3. $(0, \infty)$. 4. $[2, 6]$ — the branch $2 \leq x < 4$ holds automatically because the right side is negative there, and squaring on $x \geq 4$ contributes $[4, 6]$. 5. $(2, 3]$ — the domain condition $x \leq 3$ supplies the closed right endpoint.

Supplement Cheat Sheet Box

Chapter 5 Supplement — Everything on One Screen

- **Modulus, variable right-hand side:** $|f| < g \iff -g < f < g$ (intersect; $g > 0$ comes free). $|f| > g \iff f > g$ or $f < -g$ (union; a negative g makes it universally true).
- **Squaring a modulus:** legal only with a guard — $f^2 < g^2$ **and** $g > 0$. Modulus against modulus needs no guard.
- **Two or more moduli:** critical points $\rightarrow$ intervals $\rightarrow$ sign table $\rightarrow$ drop the bars $\rightarrow$ **intersect each case with its own interval** $\rightarrow$ union. Minimum of $|x-a| + |x-b|$ is $|a-b|$.
- **Quadratic positive for all x :** $a > 0$ **and** $D < 0$; use $D \leq 0$ for ≥ 0 . If the parameter sits in a , test $a = 0$ **separately** and union the result.
- **Domain by sign analysis:** $\sqrt{u}$ needs $u \geq 0$; $\frac{1}{\sqrt{u}}$ needs $u > 0$; $\log_b u$ needs $u > 0$. Solve each by the wavy curve, then **intersect**.
- **Radicals:** $\sqrt{f} < g \iff f \geq 0$ and $g > 0$ and $f < g^2$. $\sqrt{f} > g \iff [f \geq 0, g < 0]$ or $[g \geq 0, f > g^2]$.

Mnemonics: “A distance beats every negative number.” — “Guard before you square.” — “Each case goes home to its own interval.” — “Domain first, algebra second.”

Chapter 5: Linear Inequalities — Round 2 Addendum

Unit 5 of the supplement taught you to solve a modulus inequality in one variable, and Addenda A to C pointed the wavy curve at parameters, domains and radicals. Three JEE patterns are still standing outside the door, and each one is a marriage of two things the chapter already teaches separately.

Addendum D marries Unit 2 (a linear inequality is a half-plane) to Unit 5 (a modulus is a distance): what happens when the moduli carry two variables and the answer is a region rather than an interval. Addendum E goes back to the wavy curve of Unit 4 and replaces the drawing with arithmetic, which is the only way to survive a factor raised to the power 253. Addendum F reverses the arrow of Unit 2: instead of shading a region from a system, you read the system off a shaded region — the Exemplar problem the Unit 2 exam-relevance note promises.

Numbering continues from Addendum C. Nothing here is new theory. Every method below is the case split of Unit 5 or the sign analysis of Unit 4, applied one dimension over or one step earlier.

Addendum D — Modulus Regions in Two Variables

Unit 2 stopped at $ax + by \leq c$, whose answer is one half-plane decided by one test point. Put a modulus around either variable and that single half-plane becomes a *union of half-plane pieces* — a diamond, a rectangle, a band, a pair of wedges. JEE Main asks for the area of such a region or the number of lattice points inside it; JEE Advanced hides one inside a complex-number question, where $|\operatorname{Re}(z)| + |\operatorname{Im}(z)| \leq a$ is exactly $|x| + |y| \leq a$. CBSE does not test this, so treat it as JEE-only enrichment sitting on top of Unit 2.

Method

The engine is Procedure B of Unit 5, run in the plane instead of on a line.

1. **Find the critical lines.** Set each modulus argument to zero. $|x|$ gives the line $x = 0$; $|y - 2|$ gives $y = 2$; $|x - y|$ gives $y = x$. With k distinct critical lines the plane is cut into at most 2^k pieces.
2. **Build a sign table**, one row per modulus argument, one column per piece, exactly as in Unit 5.
3. **Drop the bars on each piece.** What remains is a plain linear inequality in x and y , so its solution is a half-plane — shade it by the Unit 2 test-point rule.
4. **Intersect each piece's half-plane with that piece.** This is the two-dimensional version of “every case goes home to its own interval,” and it is skipped just as often.

5. **Union the surviving pieces**, then read the shape.
6. **Boundary style**: solid for $\leq$ and $\geq$, dotted for $<$ and $>$ — the Unit 2 convention is unchanged.

Symmetry — the shortcut that halves or quarters the work. Before case-splitting, test the inequality for invariance:

- unchanged when $x \rightarrow -x$: the region is symmetric about the y -axis;
- unchanged when $y \rightarrow -y$: symmetric about the x -axis;
- unchanged when x and y are swapped: symmetric about the line $y = x$.

$|x| + |y| \leq a$ passes all three. So sketch it once in the first quadrant, where it is simply $x + y \leq a$ with $x, y \geq 0$, and reflect three times.

Standard regions

Throughout, $a > 0$, $b > 0$, $c > 0$, and every modulus has coefficient 1 inside.

Region	Shape	Measurements
$ x \leq a, y \leq b$	rectangle centred at the origin	width $2a$, height $2b$, area $4ab$
$ x + y \leq a$	square with vertices $(\pm a, 0)$, $(0, \pm a)$	diagonals $2a$ and $2a$, side $a\sqrt{2}$, area $2a^2$
$ x - h + y - k \leq c$	the same square translated to centre (h, k)	area $2c^2$
$ x + y \leq a$ and $ x - y \leq a$	identical to $ x + y \leq a$	area $2a^2$
$ x - y \leq a$	infinite band between $y = x - a$ and $y = x + a$	unbounded, perpendicular width $a\sqrt{2}$
$ x + y \geq a$	everything outside that square, boundary included	unbounded
$ x - y \geq a$	two opposite unbounded wedges straddling the x -axis, apexes at $(\pm a, 0)$	unbounded
$ y \leq f(x)$	the strip between $y = f(x)$ and $y = -f(x)$	empty wherever $f(x) < 0$

- h, k — the coordinates of the centre; a, b, c — the given positive bounds.
- The identity behind row four is $\max(|x + y|, |x - y|) = |x| + |y|$, true for every real x and y . Bounding the larger of the two by a is therefore the same statement as $|x| + |y| \leq a$.
- The square $|x| + |y| \leq a$ has an inscribed circle of radius $\frac{a}{\sqrt{2}}$ — the distance from the origin to the line $x + y = a$ — and a circumscribed circle of radius a .
- **Lattice count.** For a non-negative integer n , the number of points with integer coordinates satisfying $|x| + |y| \leq n$ is

$$2n^2 + 2n + 1.$$

This value is always odd, which is a free sanity check on any count you produce by hand.

- **Validity.** Every row assumes coefficient 1 inside each modulus. For $|2x| + |y| \leq a$ or $|x| + |3y| \leq a$ the shape is still a rhombus but the diagonals differ, so return to the case split.

Figure brief: Four small coordinate grids in a row, axes labelled, each region shaded with a solid boundary. Panel 1: $|x| + |y| \leq 3$, the diamond with vertices at $(3, 0)$, $(0, 3)$, $(-3, 0)$, $(0, -3)$, with the dashed lines $x = 0$ and $y = 0$ shown faintly as the two critical lines and the first-quadrant edge labelled $x + y = 3$. Panel 2: $|x| \leq 3$, $|y| \leq 2$, the rectangle, corners labelled. Panel 3: $|x - y| \leq 1$, the infinite band between the parallel lines $y = x + 1$ and $y = x - 1$, arrows on both ends showing it never closes. Panel 4: $|x| - |y| \geq 1$, the two shaded wedges opening left and right with apexes marked at $(1, 0)$ and $(-1, 0)$, the unshaded bow-tie between them left blank.

Example D.1 — [JEE Main Level]

Sketch the region $|x| + |y| \leq 3$. Find its area and the number of points in it with integer coordinates.

Two moduli, so two critical lines: $x = 0$ and $y = 0$. They cut the plane into the four quadrants, and the sign table is as simple as it ever gets.

Quadrant	sign of x	sign of y	inequality becomes
$x \geq 0, y \geq 0$	+	+	$x + y \leq 3$
$x < 0, y \geq 0$	−	+	$-x + y \leq 3$
$x < 0, y < 0$	−	−	$-x - y \leq 3$
$x \geq 0, y < 0$	+	−	$x - y \leq 3$

Each row is a half-plane; intersecting each with its own quadrant gives a right triangle with legs 3 and 3, and the four triangles fit together along the axes into a single square with vertices $(3, 0)$, $(0, 3)$, $(-3, 0)$, $(0, -3)$. Both diagonals have length 6 and they are perpendicular, so

$$\text{Area} = \frac{1}{2}d_1d_2 = \frac{1}{2}(6)(6) = 18 \text{ square units,}$$

matching $2a^2 = 2(3)^2 = 18$.

For the lattice count, work along each horizontal line $y = j$ with j an integer from -3 to 3 . There $|x| \leq 3 - |j|$, which admits $2(3 - |j|) + 1$ integer values of x . Summing over $j = -3, -2, \dots, 3$ gives $1 + 3 + 5 + 7 + 5 + 3 + 1 = 25$, and the formula agrees: $2(3)^2 + 2(3) + 1 = 25$.

Check. The origin gives $0 \leq 3$, inside. The vertex $(3, 0)$ gives $3 \leq 3$, on the boundary and included because the sign is non-strict. The point $(2, 2)$ gives $4 \leq 3$, false — correctly outside, even though it looks close to the corner. The count 25 is odd, as the formula guarantees.

Example D.2 — [JEE Advanced Level]

Find the area of the region $\{(x, y) : |x| + |y| \leq 4 \text{ and } |x| \geq 1\}$.

Two conditions, both carrying moduli, and the second one punches a hole through the middle of the first. The temptation is to case-split all over again. Do not — identify the surviving pieces geometrically.

The first condition is the square of Example D.1's family with $a = 4$: vertices $(\pm 4, 0)$, $(0, \pm 4)$, area $2(4)^2 = 32$. The second condition, $|x| \geq 1$, means $x \geq 1$ or $x \leq -1$, so it deletes the vertical strip $-1 < x < 1$ and leaves two pieces. Since both defining conditions are unchanged when $x \rightarrow -x$, the two pieces are congruent — compute one and double.

The right-hand piece, where $x \geq 1$. It is bounded on the left by the line $x = 1$ and on the right by the two square edges $x + y = 4$ (upper) and $x - y = 4$ (lower). The line $x = 1$ meets those edges at $(1, 3)$ and $(1, -3)$, and the two edges meet each other at the vertex $(4, 0)$. So the piece is the triangle with vertices $(1, 3)$, $(1, -3)$, $(4, 0)$: base 6 along the line $x = 1$, and height $4 - 1 = 3$ measured horizontally to the apex. Its area is $\frac{1}{2}(6)(3) = 9$.

Doubling by symmetry:

$$\text{Area} = 2 \times 9 = 18 \text{ square units.}$$

Independent cross-check by subtraction. The deleted strip is the hexagon with vertices $(-1, 3)$, $(0, 4)$, $(1, 3)$, $(1, -3)$, $(0, -4)$, $(-1, -3)$. Split it into the rectangle $-1 \leq x \leq 1$, $-3 \leq y \leq 3$, of area $2 \times 6 = 12$, plus two triangular caps — the upper one has vertices $(-1, 3)$, $(0, 4)$, $(1, 3)$, so base 2 and height 1, area 1, and the lower one matches. The strip is therefore $12 + 1 + 1 = 14$, and $32 - 14 = 18$. Two independent decompositions, one answer.

Check. $(2, 1)$ satisfies $3 \leq 4$ and $2 \geq 1$, so it is in the region; $(0, 2)$ fails $|x| \geq 1$; $(1, 3)$ gives $4 \leq 4$ and $1 \geq 1$, a corner sitting on both boundaries at once. The leftover 18 is comfortably more than half of 32 — plausible, since the deleted strip is narrow.

Why it is Advanced-level. The region is defined by two moduli conditions pulling in opposite directions, so neither the standard-shape table nor a single case split settles it alone. The mark-winning move is recognising that “subtract the excluded strip” and “double one congruent half” are the same computation approached from either end, and that the symmetry of the defining inequalities is what licenses both.

Practice — Addendum D

1. Sketch $|x| + |y| \leq 2$ and state its area.
2. Sketch $|x - 2| + |y| \leq 1$ and state its area.
3. Describe the region $|x| \leq 3$, $|y| \leq 2$ and find its area.
4. Find the number of points with integer coordinates satisfying $|x| + |y| \leq 5$.
5. Find the area of the region satisfying both $|x + y| \leq 2$ and $|x - y| \leq 2$.
6. Find the area of the region $\{(x, y) : |x| + |y| \leq 4, y \geq x\}$.
7. Find the area of the region $\{(x, y) : 1 \leq |x| + |y| \leq 3\}$.
8. Describe the region $|x| - |y| \geq 1$, and state whether it is bounded.

Answers: 1. A square with vertices $(\pm 2, 0)$, $(0, \pm 2)$; area 8 square units. 2. The same square translated to centre $(2, 0)$, with vertices $(1, 0)$, $(3, 0)$, $(2, 1)$, $(2, -1)$; area 2 square units. 3. A rectangle centred at the origin, 6 wide and 4 tall; area 24 square units. 4. $2(5)^2 + 2(5) + 1 = 61$.

5. 8 square units — by the identity $\max(|x+y|, |x-y|) = |x| + |y|$ this is exactly $|x| + |y| \leq 2$, so no fresh work is needed. 6. 16 square units — the line $y = x$ passes through the centre of a region symmetric about that centre, and any line through the centre of a centrally symmetric region bisects its area. 7. $2(3)^2 - 2(1)^2 = 18 - 2 = 16$ square units. 8. Two unbounded wedges opening left and right, with apexes at $(1, 0)$ and $(-1, 0)$, bounded by the four lines $y = \pm(x-1)$ and $y = \pm(x+1)$; unbounded.

Addendum E — Repeated Factors: Reduce the Inequality Before You Draw It

Unit 4 gives you the correct rule — odd multiplicity crosses, even multiplicity bounces — and Example 4 walks a curve past a squared factor. That is enough while the exponents are 1 and 2. JEE does not stay there. A standard reference problem asks you to solve

$$\frac{(x - \frac{1}{2})(x - 2)^{10000}(x - 3)^{75}(x + 2)^{93}}{(x + 1)^{253}(x + 8)^4} \geq 0,$$

and nobody draws that curve. What you do instead is *reduce* the expression to a product of distinct linear factors that has the same sign everywhere, run the ordinary wavy curve on the reduction, and then re-attach the points the reduction threw away. Three rules do all of it.

Method

R1 — Denominator to numerator. For real p and q with $q \neq 0$, the quotient $\frac{p}{q}$ and the product pq have the same sign, because $\frac{p}{q} = \frac{pq}{q^2}$ and $q^2 > 0$. Therefore

$$\frac{N(x)}{D(x)} \geq 0 \iff N(x) \cdot D(x) \geq 0, \quad \text{together with } D(x) \neq 0.$$

- N, D — the numerator and denominator after everything has been brought to one side and combined into a single quotient (Unit 4, Step 1).
- The condition $D(x) \neq 0$ is not optional bookkeeping; it is what keeps the two statements equivalent.

R2 — Parity of the exponent. For a factor $(x - r)^m$:

Exponent	Replace by	Record separately
m even, factor in N	delete it	$x = r$ makes the expression 0: a solution when the sign is $\leq$ or $\geq$, excluded when strict
m even, factor in D	delete it	$x = r$ is excluded always
m odd, factor in N	$(x - r)$	nothing

Exponent	Replace by	Record separately
m odd, factor in D	$(x - r)$, moved up by R1	$x = r$ is excluded always

- r — the root; m — its multiplicity.
- Why it is legal: $(x-r)^{2k} = [(x-r)^k]^2 \geq 0$, so deleting it cannot change the sign anywhere except at $x = r$ itself, where the whole expression was zero or undefined. And $(x-r)^{2k+1} = (x-r)[(x-r)^k]^2$ carries the sign of $(x-r)$ alone.

R3 — Constant-sign factors. A quadratic $ax^2 + bx + c$ with $D = b^2 - 4ac < 0$ never changes sign (Addendum A).

- If $a > 0$, it is positive everywhere: delete it, inequality unchanged.
- If $a < 0$, it is negative everywhere: delete it and reverse the inequality.
- The same applies to numerical constants: delete a positive one, delete a negative one and reverse.

The procedure.

1. Bring everything to one side, combine into a single quotient, factor completely, and make every x -coefficient positive — tracking each stray -1 into R3.
2. Apply R1, then R2, then R3. Write the recorded points in a margin list as you go: *include* candidates and *exclude* candidates.
3. Run the Unit 4 wavy curve on the reduced product, which now has only distinct simple factors — no bouncing to think about, because the reduction has already absorbed it.
4. Re-attach: delete every excluded point from the answer, and add back every included point, even if it sits alone in the middle of a rejected stretch.
5. Substitute one value from each interval of your final answer into the *original* expression.

The error this method invites. Step 2 changes the expression's value while preserving its sign, so a root you deleted no longer appears in the reduced picture at all. Miss Step 4 and you lose an isolated solution point silently — the answer looks clean and is wrong. Write the margin list before you draw anything.

Substitution — when the same block repeats. If one expression appears more than once, name it. For $t = \frac{ax+b}{cx+d}$ or $t = x^2 + bx$, solve the inequality in t first, then push each resulting t -interval back through the substitution as its own rational inequality.

- Each t -condition must be solved by the Unit 4 method, bringing everything to one side. Cross-multiplying $\frac{x-1}{x+2} \leq 1$ by $x+2$ is the same illegal move it always was.
- The final answer is the union or intersection of the pushed-back sets according to whether the t -answer was a union or an intersection.
- Do not assume t ranges over all of $\mathbb{R}$; a value of t your t -answer permits may be unreachable, in which case its push-back is simply empty.

Figure brief: Two stacked number lines sharing one vertical alignment. The upper line is labelled “original” and carries the critical points -3 (hollow, marked “denominator, even

power”), -1 (filled, marked “numerator, even power — isolated”), 1 (hollow, marked “denominator”), 2 (filled) and 5 (filled), with the exponents $2, 4, 1, 3, 1$ written above them. The lower line is labelled “reduced” and carries only $1, 2, 5$, with a simple three-root wavy curve anchored above the axis to the right of 5 , crossing at each. Downward arrows connect -3 and -1 from the upper line to a boxed margin note reading “excluded” and “include as isolated point”.

Example E.1 — [JEE Main Level]

Solve $\frac{(x-2)^3(x+1)^4(x-5)}{(x+3)^2(x-1)} \geq 0.$

Every x -coefficient is already positive, so no sign is hiding.

Apply R1. The quotient has the sign of the product $(x-2)^3(x+1)^4(x-5)(x+3)^2(x-1)$, provided $x \neq -3$ and $x \neq 1$.

Apply R2, keeping the margin list.

- $(x-2)^3 \rightarrow (x-2)$.
- $(x+1)^4$: even, from the numerator. Delete; record $x = -1$ as an *include* candidate, since the sign is $\geq$ and the expression is 0 there.
- $(x+3)^2$: even, from the denominator. Delete; record $x = -3$ as *exclude*.
- $(x-5)$ and $(x-1)$: already simple.

The reduced inequality.

$$(x-1)(x-2)(x-5) \geq 0.$$

Roots $1, 2, 5$, all simple. Anchor positive on the far right and walk left, flipping at each: positive on $(5, \infty)$, negative on $(2, 5)$, positive on $(1, 2)$, negative on $(-\infty, 1)$. The reduced answer is $[1, 2] \cup [5, \infty)$.

Re-attach. Delete $x = 1$, which is a denominator root, leaving $(1, 2] \cup [5, \infty)$. The excluded point $x = -3$ happens to lie outside this set already — check it anyway rather than assuming. Add the isolated point $x = -1$.

$$x \in \{-1\} \cup (1, 2] \cup [5, \infty).$$

Check. At $x = -1$ the numerator vanishes, so the expression is $0 \geq 0$: included. At $x = 0$ the original evaluates to $-\frac{40}{9}$, negative — correctly outside, and note that $x = 0$ sits between two accepted pieces, which is exactly the shape a bare wavy curve would have smoothed over. At $x = 1.5$ the expression is positive; at $x = 3$ it is negative; at $x = 2$ and $x = 5$ it is 0, so both endpoints are included. At $x = -3$ and $x = 1$ it is undefined.

Example E.2 — [JEE Advanced Level]

Solve $\left(\frac{x-1}{x+2}\right)^2 - 3\left(\frac{x-1}{x+2}\right) + 2 \geq 0.$

The block $\frac{x-1}{x+2}$ appears twice, so name it: let $t = \frac{x-1}{x+2}$, with $x \neq -2$ throughout.

Solve in t .

$$t^2 - 3t + 2 \geq 0 \implies (t-1)(t-2) \geq 0 \implies t \leq 1 \text{ or } t \geq 2.$$

A union in t , so the two push-backs will be unioned.

Push back the branch $t \leq 1$. Bring to one side — do not cross-multiply:

$$\frac{x-1}{x+2} - 1 \leq 0 \implies \frac{(x-1) - (x+2)}{x+2} \leq 0 \implies \frac{-3}{x+2} \leq 0.$$

A negative constant over $x+2$ is ≤ 0 exactly when $x+2 > 0$ (it is never 0), so this branch gives $x > -2$.

Push back the branch $t \geq 2$.

$$\frac{x-1}{x+2} - 2 \geq 0 \implies \frac{(x-1) - 2(x+2)}{x+2} \geq 0 \implies \frac{-(x+5)}{x+2} \geq 0 \implies \frac{x+5}{x+2} \leq 0.$$

Critical points $x = -5$ (numerator, filled since the sign is non-strict) and $x = -2$ (denominator, always hollow). The quotient is ≤ 0 between them, so this branch gives $[-5, -2)$.

Union the branches.

$$x \in [-5, -2) \cup (-2, \infty).$$

Equivalently: every real number except $x = -2$ and the stretch $x < -5$.

Check. At $x = -5$: $t = \frac{-6}{-3} = 2$ and $t^2 - 3t + 2 = 0$, so the endpoint belongs. At $x = -6$: $t = \frac{-7}{-4} = \frac{7}{4}$, which lies strictly between 1 and 2, so $t^2 - 3t + 2 = -\frac{3}{16} < 0$ — correctly excluded, and it is the only kind of x that is. At $x = 0$: $t = -\frac{1}{2} \leq 1$, giving $\frac{15}{4} > 0$, included. At $x = -2$ the expression is undefined.

Why it is Advanced-level. The substitution is the easy half; the trap is what happens on the way back. Students who solve $t \leq 1$ by cross-multiplying get $x-1 \leq x+2$, which is true for every x , and so hand in all of $\mathbb{R}$ minus $\{-2\}$ — losing the exclusion $x < -5$ entirely. The t -answer being a union rather than an intersection is the second decision point, and reversing it returns the complement.

Practice — Addendum E

1. Solve $\frac{(x-1)^2(x+4)}{x-2} \geq 0$.
2. Solve $\frac{(x+1)(x-4)}{(x-2)^2} \leq 0$.
3. Solve $\frac{(x^2-4)^2(x+1)}{(x-3)^4} \leq 0$.

4. Solve $\frac{(x+2)^3(x-3)^2}{(x-1)^4(x+5)} \leq 0$.
5. Solve $\frac{(x^2+x+1)(x-4)}{(x+1)^2} \geq 0$.
6. Solve $\frac{(3-x)^5(x+2)}{(x-6)^2} \geq 0$.
7. Solve $\frac{(x-1)^6(x+3)}{(x-5)^3} \leq 0$.
8. Solve $\left(\frac{x+1}{x-1}\right)^2 - 4\left(\frac{x+1}{x-1}\right) + 3 < 0$.

Answers: 1. $(-\infty, -4] \cup \{1\} \cup (2, \infty)$ — the squared numerator factor survives only as the isolated point $x = 1$. 2. $[-1, 2) \cup (2, 4]$ — the even denominator root $x = 2$ is punched out of the middle of the answer. 3. $(-\infty, -1] \cup \{2\}$ — note that $x = -2$ is already inside the first piece while $x = 2$ is isolated. 4. $(-5, -2] \cup \{3\}$. 5. $[4, \infty)$ — the quadratic $x^2 + x + 1$ has $D = -3 < 0$ with a positive leading coefficient, so R3 deletes it; $x = -1$ is excluded but lies outside the answer anyway. 6. $[-2, 3]$ — rewrite $(3-x)^5$ as $-(x-3)^5$ and let R3 reverse the inequality; $(x-6)^2$ deletes with $x = 6$ excluded. 7. $[-3, 5)$ — the sixth power deletes and its root $x = 1$ falls inside the interval regardless. 8. $(2, \infty)$.

Addendum F — Reading a Region Backwards: Writing the System from the Picture

Unit 2 always runs in one direction: you are handed inequalities and you shade. The Boards run it the other way too. You are shown a shaded region — or, more often now, told only its corner points — and asked to write the system of inequalities that defines it. The same reversal is the first half of every Class 12 Linear Programming question, where the feasible region arrives as a picture and you must recover the constraints before you can optimise. The forward direction is already in the chapter (Unit 2 Examples 3 and 4 find vertices and area, Unit 3 Example 4 evaluates an objective at the corners). This is the arrow the chapter has not drawn.

Method

1. **Count the edges.** A region bounded by n straight edges needs exactly n inequalities. If a listed edge lies along the x -axis, its inequality is $y \geq 0$ or $y \leq 0$; along the y -axis, $x \geq 0$ or $x \leq 0$.
2. **Find each boundary line.** From two corner points (x_1, y_1) and (x_2, y_2) on the edge, use the two-point form

$$\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1},$$

or, when the edge cuts the axes at $(p, 0)$ and $(0, q)$, use the intercept form $\frac{x}{p} + \frac{y}{q} = 1$ and clear denominators.

3. **Decide the direction with one interior point.** Pick any point strictly inside the shaded region — the centroid of the corners always works, and for a triangle with corners A, B, C that is $\frac{1}{3}(A + B + C)$. Substitute it into each boundary equation and write the sign that makes the statement true.
4. **Decide strictness from the drawing.** Solid edge gives $\leq$ or $\geq$; dotted edge gives $<$ or $>$. If the problem lists corner points as *belonging to* the region, the edges meeting there are solid.
5. **Audit before you write the final answer.** Every listed corner must satisfy all n inequalities, with equality holding in at least two of them. The interior point must satisfy all n . A point just outside one edge must fail exactly that one.

Bounded or unbounded. The direction of the inequalities, not their number, decides this. A first-quadrant region with all constraints of the form $ax + by \leq c$ (with $a, b, c > 0$) is bounded; if the constraints read $ax + by \geq c$, the region is unbounded away from the origin. This is the fact that decides, in Class 12, whether a maximum exists at all.

Figure brief: Two coordinate grids side by side, axes labelled with intercept values. Left panel: a solid-edged triangle with vertices $(0, 0)$, $(4, 0)$, $(0, 3)$, interior shaded, the slant edge labelled $3x + 4y = 12$, and a small dot at the interior point $(1, 1)$ marked “test”. Right panel: an unbounded first-quadrant region with solid corner points labelled $(0, 8)$, $(2, 4)$, $(10, 0)$, the two slant edges labelled $2x + y = 8$ and $x + 2y = 10$, shading extending up and to the right with arrows, and a small dot at $(5, 5)$ marked “test”.

Example F.1 — [CBSE Board Level]

A shaded region is the triangle with vertices $(0, 0)$, $(4, 0)$ and $(0, 3)$, with all three boundaries drawn solid. Write the system of inequalities that defines it.

Three edges, so three inequalities.

Edge along the y -axis, joining $(0, 0)$ and $(0, 3)$: the line $x = 0$.

Edge along the x -axis, joining $(0, 0)$ and $(4, 0)$: the line $y = 0$.

Slant edge, joining $(4, 0)$ and $(0, 3)$: intercepts $p = 4$ and $q = 3$, so

$$\frac{x}{4} + \frac{y}{3} = 1 \implies 3x + 4y = 12.$$

Direction. The centroid of the three vertices is $(\frac{4}{3}, 1)$; the tidier interior point $(1, 1)$ also lies inside. Substituting $(1, 1)$:

- into x : $1 > 0$, so the constraint is $x \geq 0$;
- into y : $1 > 0$, so the constraint is $y \geq 0$;
- into $3x + 4y$: $3 + 4 = 7$, and $7 < 12$, so the constraint is $3x + 4y \leq 12$.

All three edges are solid, so all three signs are non-strict.

$$x \geq 0, \quad y \geq 0, \quad 3x + 4y \leq 12.$$

Audit. $(0, 0)$: $0 \geq 0$, $0 \geq 0$, $0 \leq 12$ — satisfied, with equality in the first two. $(4, 0)$: equality in $y \geq 0$ and in $3x + 4y \leq 12$. $(0, 3)$: equality in $x \geq 0$ and in $3x + 4y \leq 12$. The outside point $(2, 3)$ gives $6 + 12 = 18 > 12$ and fails only the third constraint, as it should.

Example F.2 — [JEE Main Level]

A shaded region lies in the first quadrant, has corner points $(0, 8)$, $(2, 4)$ and $(10, 0)$, is bounded below-left by two solid slant edges, and extends without limit away from the origin. Write the system of inequalities that defines it, and state whether the region is bounded.

Here you are given no equations at all — only corners — so every line must be reconstructed.

Edge joining $(0, 8)$ and $(2, 4)$. Slope $= \frac{4 - 8}{2 - 0} = -2$, so $y = 8 - 2x$, that is

$$2x + y = 8.$$

Edge joining $(2, 4)$ and $(10, 0)$. Slope $= \frac{0 - 4}{10 - 2} = -\frac{1}{2}$, so $y - 4 = -\frac{1}{2}(x - 2)$, giving $2y - 8 = -x + 2$, that is

$$x + 2y = 10.$$

Direction. The region extends away from the origin, so test a point well out in it, say $(5, 5)$:

- $2(5) + 5 = 15 \geq 8$ — true, so the constraint is $2x + y \geq 8$;
- $5 + 2(5) = 15 \geq 10$ — true, so the constraint is $x + 2y \geq 10$.

Confirm with the origin, which must fail: $0 \geq 8$ is false and $0 \geq 10$ is false, so $(0, 0)$ is outside both — consistent with a region pushed away from the origin.

The two axis constraints. The region lies in the first quadrant and the corners $(0, 8)$ and $(10, 0)$ sit on the axes, so $x \geq 0$ and $y \geq 0$ are genuine edges of the region, not decoration.

$$2x + y \geq 8, \quad x + 2y \geq 10, \quad x \geq 0, \quad y \geq 0.$$

Both slant constraints point away from the origin, so the region is **unbounded**.

Audit. $(2, 4)$: $2(2) + 4 = 8$ and $2 + 2(4) = 10$ — equality in both slant constraints, which is exactly what a corner formed by their intersection must give. $(0, 8)$: $8 \geq 8$ with equality, $16 \geq 10$, $x = 0$ — equality in two constraints. $(10, 0)$: $20 \geq 8$, $10 \geq 10$ with equality, $y = 0$. The point $(4, 2)$ satisfies $2x + y = 10 \geq 8$ but $x + 2y = 8 < 10$, so it fails exactly one constraint and lies just outside the second edge — the discriminating test that a corner-only reading can otherwise miss.

The Class 12 hook. Because this region is unbounded, a linear objective $Z = ax + by$ with $a, b > 0$ has a minimum at one of these corners but no maximum at all. Recovering the system correctly is what lets you say so; recovering it with the inequalities reversed would have produced a bounded triangle and a confident wrong answer.

Practice — Addendum F

1. Write the system for the solid-edged triangle with vertices $(1, 0)$, $(4, 0)$, $(1, 3)$.
2. A dotted line passes through $(4, 0)$ and $(0, -2)$, and the shaded region contains the origin. Write the inequality.
3. Write the square $|x| \leq 2$, $|y| \leq 2$ as a system of four linear inequalities in x and y .
4. Write the system for the solid-edged quadrilateral with vertices $(0, 0)$, $(6, 0)$, $(4, 2)$, $(0, 4)$.
5. Write the system for the solid-edged triangle with vertices $(0, 0)$, $(4, 0)$, $(2, 2)$.
6. A first-quadrant region has solid-edged corner points $(0, 5)$, $(1, 3)$, $(4, 0)$ and extends away from the origin. Write the system and state whether it is bounded.

Answers: 1. $x \geq 1$, $y \geq 0$, $x + y \leq 4$. 2. $x - 2y < 4$ — the line is $x - 2y = 4$ and the origin gives $0 < 4$, true. 3. $x \leq 2$, $x \geq -2$, $y \leq 2$, $y \geq -2$. 4. $x \geq 0$, $y \geq 0$, $x + y \leq 6$, $x + 2y \leq 8$. 5. $y \geq 0$, $y \leq x$, $x + y \leq 4$. 6. $2x + y \geq 5$, $x + y \geq 4$, $x \geq 0$, $y \geq 0$; unbounded. The edge $(0, 5)$ to $(1, 3)$ has slope -2 and the edge $(1, 3)$ to $(4, 0)$ has slope -1 , so $(1, 3)$ is a genuine corner where the two slant constraints meet with equality.

Round 2 Cheat Sheet Additions

Chapter 5 Round 2 — Everything on One Screen

- **Modulus regions in the plane:** each modulus argument $= 0$ gives a critical *line*; drop the bars on each piece, shade the half-plane, intersect with the piece, union. $|x| + |y| \leq a$ is a square with vertices $(\pm a, 0)$, $(0, \pm a)$: side $a\sqrt{2}$, area $2a^2$, inradius $\frac{a}{\sqrt{2}}$, circumradius a . $|x| \leq a$, $|y| \leq b$ is a rectangle of area $4ab$. $|x - y| \leq a$ is an unbounded band. Integer points in $|x| + |y| \leq n$: $2n^2 + 2n + 1$, always odd.
- **Symmetry first:** invariance under $x \rightarrow -x$, $y \rightarrow -y$, or $x \leftrightarrow y$ lets you sketch one quadrant or one half and reflect. A line through the centre of a centrally symmetric region bisects its area.
- **Reduce before you draw:** $\frac{N}{D}$ has the sign of $N \cdot D$ with $D \neq 0$. Even powers delete (record the root: *include* if it came from N under $\leq$ or $\geq$, *exclude* always if from D); odd powers collapse to the single factor; a quadratic with $D < 0$ deletes, reversing the inequality if its leading coefficient is negative.
- **Re-attach, always.** The reduced curve cannot show you an isolated solution point. Keep a margin list of *include* and *exclude* candidates and apply it to the final interval set.
- **Substitution:** name a repeating block t , solve in t , then push each t -interval back as its own rational inequality — bringing to one side, never cross-multiplying. Union in t gives a union in x .
- **Region to system:** n edges give n inequalities. Line from two corners or from intercepts; direction from one interior point; strictness from solid versus dotted; then audit every corner and one outside point. All- $\leq$ against the origin means bounded; all- $\geq$ means unbounded, and an unbounded feasible region has no maximum for a positive

objective.

Mnemonics: “A modulus on the line is a point pair; a modulus in the plane is a fold.” — “Reduce, draw, then re-attach.” — “Even powers leave no trace on the curve, only a dot in the margin.” — “One interior point fixes every sign.”

Chapter 6: Permutations and Combinations

Class 11 Mathematics / CBSE Boards / JEE Main / JEE Advanced

Drona Ed Tech - Master Content

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Subtopic 01 — Fundamental Principle of Counting and Factorial

Exam Relevance: This is the foundation the entire chapter stands on. JEE Main almost always hides 1 problem here (often disguised as a digit/number-formation question), and it silently powers every later P&C, Probability, and Binomial problem. JEE Advanced tests it through multi-constraint counting and complementary counting. CBSE Boards asks a direct 2–4 mark FPC or factorial-simplification question.

Section 1: Concept Intuition

Imagine you walk into a small *thali* restaurant near a railway station. The menu is simple: pick **one** sabzi (there are 4), **one** type of roti (there are 3), and **one** sweet (there are 2). The owner proudly says, “Sir, so many combinations are possible!” — and you, being a P&C student now, want to know *exactly* how many, without sitting and writing out every plate.

Here is the key realisation. For **every** choice of sabzi, you still have all 3 rotis available. And for every (sabzi, roti) pair, you still have both sweets. So the choices *stack up by multiplying*:

$$4 \times 3 \times 2 = 24 \text{ distinct thalis.}$$

That single idea — **when you make a sequence of independent choices, you multiply the number of options at each stage** — is the **Multiplication Principle**, the heart of the Fundamental Principle of Counting (FPC). It lets us count enormous collections *without listing them*, which is the whole spirit of this chapter.

Think about it this way... picture a tree growing left to right. From the root, 4 branches sprout (one per sabzi). From the tip of *each* of those, 3 more branches grow (the rotis). From each of those tips, 2 final twigs appear (the sweets). The number of *leaves* at the far right is the number of complete thalis — and counting leaves is just multiplying the branching factors: $4 \times 3 \times 2$. The tree never lies.

Figure placeholder — A left-to-right tree diagram. Root node branches into 4 (labelled Sabzi 1-4); each of those branches into 3 (Roti 1-3); each of those into 2 (Sweet 1-2). Annotate the final layer as “24 leaves = $4 \times 3 \times 2$ ”. Keep it schematic, not exhaustive — show one full path fully expanded and the rest indicated with dots.

You might wonder: why not just *list* everything and count? Because the lists explode. A 4-wheel number lock with digits 0-9 has $10^4 = 10,000$ settings; a 6-character password from letters and digits has over two billion. No human lists those. The genius of the FPC is that it hands you the *size* of a collection while you stay blissfully ignorant of its members — you count the leaves without ever walking the tree. That “count without listing” mindset is the single skill this entire chapter trains.

Now change the story. Suppose the owner instead says: “Today you may take **either** one of 4 sabzis **or** one of 2 sweets — but not both.” These are *mutually exclusive* options; you make a single choice from one pile or the other. Here you don’t multiply — you **add**: $4 + 2 = 6$ ways. This is the **Addition Principle**.

The entire distinction lives in two tiny English words:

- “**AND**” (do this *and* then that) → the choices combine → **multiply**.
- “**OR**” (do this *or* instead that) → the choices are alternatives → **add**.

Burn that in: **AND multiplies, OR adds**.

Where does the **factorial** come in? Suppose you want to *arrange* 5 different trophies in a row on a shelf. The first position can be filled by any of 5 trophies, the next by any of the remaining 4, then 3, then 2, then the last 1 — a pure multiplication-principle chain: $5 \times 4 \times 3 \times 2 \times 1 = 120$. That descending product of natural numbers is so common that mathematicians gave it a shorthand: $5! = 120$, read “5 factorial.” Factorial is simply *the multiplication principle applied to a full arrangement*.

Conditions & Assumptions (when this reasoning needs care):

- The multiplication principle assumes each stage’s count is **independent** of the *specific* earlier choices. If the number of options at a later stage changes depending on what you picked earlier, you must split into cases.
- The addition principle requires the options to be **mutually exclusive** (no overlap). If they can overlap, naive addition double-counts.
- Factorial $n!$ is defined only for **non-negative integers**. You cannot take the factorial of a negative number or a fraction at this level. And $0! = 1$ — not 0 — a definition we will justify shortly, not just memorise.

Section 2: Key Formulas & Definitions

Multiplication Principle (Fundamental Principle of Counting). If a task is completed by performing stage 1 in m ways **and** stage 2 in n ways (stage 2’s count not depending on which option was chosen in stage 1), the whole task can be done in

$$m \times n \text{ ways.}$$

This extends to any finite number of stages: $m_1 \times m_2 \times \cdots \times m_k$.

Addition Principle. If a task is completed by choosing **either** an option of type A (available in m ways) **or** a mutually exclusive option of type B (available in n ways), the number of ways is

$$m + n.$$

Factorial. For a positive integer n , the factorial of n is the product of the first n natural numbers:

$$n! = n \times (n - 1) \times (n - 2) \times \cdots \times 3 \times 2 \times 1.$$

with the special definition

$$0! = 1.$$

- n — a **non-negative integer** ($n \in \mathbb{Z}$, $n \geq 0$).
- $n!$ — read “ n factorial”; it is a pure (dimensionless) count, so it carries **no units**.

Counting with repetition allowed. If r slots are each filled independently from the same set of n options (repetition permitted), the count is

$$\underbrace{n \times n \times \cdots \times n}_{r \text{ factors}} = n^r.$$

Forward reference (formalised in Sub-02 / Sub-03). Arranging r distinct objects out of n in order, *without* repetition, gives ${}^nP_r = \frac{n!}{(n-r)!}$; selecting r out of n ignoring order

gives ${}^nC_r = \frac{n!}{r!(n-r)!}$.

Conditions of validity & sanity-checks:

- Domain: $n!$ exists only for $n \in \{0, 1, 2, 3, \dots\}$. Undefined for negatives and fractions here.
- Boundary values: $0! = 1$, $1! = 1$.
- Recursion (your most-used identity): $n! = n \cdot (n-1)!$ for $n \geq 1$.
- Growth check: factorials explode — $5! = 120$ but $10! = 3,628,800$. If an exam answer comes out as, say, $20!$, you **leave it as $20!$** rather than computing it.

One table to never confuse again. Three structures recur all chapter; keep them straight:

Situation	r slots, each from n options	Count
Repetition allowed	every slot sees all n	n^r
Repetition not allowed , order matters	slots lose one option each	$n(n-1) \cdots (n-r+1) = {}^nP_r$
Arrange all n distinct objects	special case $r = n$	$n!$

Section 3: Key Procedures & Logical Framework

The FPC is a *self-evident counting principle*, not a theorem proved from deeper axioms — but we can still **justify it rigorously** and then *derive* the factorial and repetition formulas from it. That derivation is what an examiner rewards.

3.1 — Why the multiplication principle is true (the box-filling argument).

Think of the task as filling a sequence of boxes left to right.

Figure placeholder — Two empty boxes side by side. Box 1 labelled “ m ways”, Box 2 labelled “ n ways”. An arrow from each of the m options in Box 1 fanning into all n options of Box 2, illustrating that each Box-1 choice pairs with every Box-2 choice.

Box 1 can be filled in m ways. Fix any one of those m choices. With that choice locked, Box 2 can still be filled in n ways — and crucially, this n is the same regardless of which Box-1 option we fixed (the independence assumption). So each of the m Box-1 choices spawns exactly n complete outcomes. Grouping the outcomes by their Box-1 choice gives m groups of n each, which is $m \times n$ outcomes — no overlaps, none missed. Extending box by box gives the general product.

3.2 — Why the addition principle is true.

If the outcomes split into two **disjoint** piles — one of size m , one of size n , with nothing belonging to both — then counting the union is just $m + n$. The disjointness is essential: if some outcome sat in both piles, it would be counted twice and we would need to subtract the overlap (this repair is the *inclusion-exclusion* idea you will meet later).

3.3 — Deriving the factorial from the FPC (arranging n distinct objects).

Claim: The number of ways to arrange n distinct objects in a row is $n!$.

Reasoning: Treat the n positions as n boxes.

- Position 1 can take any of the n objects $\rightarrow n$ ways.
- Position 2 can take any object except the one already placed $\rightarrow (n - 1)$ ways.
- Position 3 $\rightarrow (n - 2)$ ways, and so on.
- The last position has just 1 object left $\rightarrow 1$ way.

By the multiplication principle, the total is

$$n \times (n - 1) \times (n - 2) \times \cdots \times 2 \times 1 = n!.$$

So $n!$ is the answer to “in how many orders can n distinct things be lined up?” That is the meaning to keep in your head, not the symbol.

3.4 — Why we define $0! = 1$ (two independent justifications).

Justification by recursion: The identity $n! = n \cdot (n - 1)!$ should keep working at $n = 1$:

$$1! = 1 \cdot 0! \implies 1 = 1 \cdot 0! \implies 0! = 1.$$

Justification by meaning: $0!$ should count the number of ways to arrange **zero** objects in a row. There is exactly **one** such arrangement — the empty arrangement (do nothing). So $0! = 1$. Defining it as anything else would break both the recursion and every later formula (e.g. ${}^nC_n = \frac{n!}{n!0!}$ must equal 1).

3.5 — Deriving the repetition count n^r .

If r slots are filled, each independently from n options *with repetition allowed*, then every slot sees the full n options regardless of earlier choices. The multiplication principle gives $n \times n \times \cdots$ (r times) $= n^r$. Notice the contrast with the factorial-style falling product, where each new slot loses one option — repetition vs. no-repetition is the single most important fork in this whole chapter.

3.6 — When independence fails: split into cases.

The multiplication principle's fine print — that a later stage's *count* must not depend on *which* earlier option was picked — is exactly where Advanced problems set traps. Watch it break.

Problem: How many 3-digit numbers (no leading zero, all digits distinct) can be formed from $\{0, 1, 2, 3\}$?

A reflex answer of " $4 \times 3 \times 2 = 24$ " is wrong, because the *first* slot cannot use 0 while later slots can — the clean symmetry is gone. Handle it by **fixing the constrained slot first**: slot 1 in 3 ways (digits 1, 2, 3), then slots 2 and 3 draw from the 3 remaining digits (which now *include* 0) in $3 \times 2 = 6$ ways, giving $3 \times 6 = 18$. The lesson generalises: **when a stage's options depend on earlier choices, either reorder so the constrained stage comes first, or break the problem into disjoint cases and add them** (addition principle). Casework plus the FPC together handle essentially every counting problem you will ever meet.

Section 4: Worked Examples

Example 1 — [CBSE Board Level] (clean multiplication principle).

A school issues ID codes of the form: one letter (A–Z) followed by three digits (0–9), and repetition of digits is allowed. How many distinct ID codes are possible?

Solution. We fill four slots in order:

- Slot 1 (letter): 26 ways.
- Slots 2, 3, 4 (digits, repetition allowed): 10 ways each.

By the Fundamental Principle of Counting,

$$\text{Total} = 26 \times 10 \times 10 \times 10 = 26 \times 1000 = 26,000 \text{ codes.}$$

(Note the clean structure: identify slots $\rightarrow$ count each slot's options $\rightarrow$ multiply $\rightarrow$ state the count.)

Example 2 — [Conceptual Trap] (AND vs OR, with vs without repetition).

A canteen lets you order breakfast in one of two ways: take a single ready plate (there are 3 ready plates), **or** build a combo of one paratha (4 types) **and** one chutney (5 types). How many distinct breakfast orders are possible?

Solution. Spot the two English signals.

- The ready plate is **one** of 3: that's 3 ways.
- The combo is "paratha **AND** chutney": multiply $\rightarrow 4 \times 5 = 20$ ways.
- The overall order is "ready plate **OR** combo": these are mutually exclusive, so add $\rightarrow 3 + 20 = 23$ ways.

$$3 + 4 \times 5 = 23.$$

The traps it's built to catch: A panicking student multiplies everything ($3 \times 4 \times 5 = 60$) or adds everything ($3 + 4 + 5 = 12$). Both are wrong because they ignore the AND/OR structure. The fix is mechanical: **bracket each "AND" group with $\times$, then join independent alternatives with $+$.**

Example 3 — [JEE Main Level] (FPC + place-value + a parity constraint).

How many 4-digit numbers are even? (Repetition of digits is allowed.)

Solution. A 4-digit number cannot start with 0. We fill the thousands, hundreds, tens, units in turn, but it pays to handle the **most constrained slots first** — here the first digit (no zero) and the last digit (must be even).

- Thousands place: any of 1-9 $\rightarrow 9$ ways.
- Units place (even, so from $\{0, 2, 4, 6, 8\}$): 5 ways.
- Hundreds place: any of 0-9 $\rightarrow 10$ ways.
- Tens place: any of 0-9 $\rightarrow 10$ ways.

These constraints are independent, so multiply:

$$9 \times 10 \times 10 \times 5 = 4500 \text{ even 4-digit numbers.}$$

(Two ideas combined: the leading-zero restriction **and** the parity restriction, stitched together by the FPC.)

Example 4 — [JEE Advanced Level] (complementary counting done carefully).

How many 5-digit numbers contain the digit 9 at least once? (No leading zero; repetition allowed.)

Solution. "At least one 9" is awkward to count directly — you'd have to add the cases of exactly one, two, three, four, or five nines, watching for overcounting. The clean route is the **complement**:

$$\#(\text{at least one } 9) = \#(\text{all 5-digit numbers}) - \#(\text{5-digit numbers with no } 9).$$

Total 5-digit numbers: first digit from 1–9 (9 ways), the other four from 0–9 (10 each):

$$9 \times 10^4 = 90,000.$$

5-digit numbers with no 9 anywhere: first digit from $\{1, \dots, 8\}$ — that is 8 ways (we drop both 0 for the leading rule *and* 9 for the no-nine rule); each remaining digit from $\{0, \dots, 8\}$ — 9 ways:

$$8 \times 9^4 = 8 \times 6561 = 52,488.$$

Therefore:

$$90,000 - 52,488 = 37,512.$$

Why this is the rigorous move: the leading digit obeys *two* exclusions at once ($\neq 0$ and $\neq 9$, giving 8), while interior digits obey only one ($\neq 9$, giving 9). Mixing those up is the classic Advanced-level slip. Complementary counting also dodges the inclusion-exclusion bookkeeping that direct counting would force.

Section 5: Practice Exercises

1. **[CBSE]** A boutique stocks 4 kurtas, 3 leggings, and 2 dupattas, all distinct. How many distinct three-piece outfits (one of each) can be assembled?
2. **[CBSE]** Find the positive integer n such that $(n + 1)! = 12 \times (n - 1)!$.
3. **[JEE Main]** How many 3-digit numbers (no leading zero, repetition allowed) are divisible by 5?
4. **[JEE Main]** A password is exactly 4 characters long, each character chosen from the 26 uppercase letters and the 10 digits (36 symbols, repetition allowed). How many passwords contain **at least one digit**?
5. **[JEE Advanced]** How many 6-digit numbers (no leading zero) have **no two adjacent digits equal**?

Answers: 1. 24 2. $n = 3$ 3. 180 4. $36^4 - 26^4 = 1,222,640$ 5. $9 \times 9^5 = 9^6 = 531,441$

Section 6: Crack the MCQ

Q1. A combination lock has 3 dials, each showing digits 0–9. How many distinct codes are possible?

- (A) 30 (B) 720 (C) 999 (D) 1000

Answer: (D) 1000. Each dial is filled independently from 10 digits with repetition allowed, so $10 \times 10 \times 10 = 1000$.

- **(A)** 30 comes from *adding* $(10 + 10 + 10)$ — treating dials as OR-alternatives instead of an AND-sequence.
- **(B)** 720 is $10 \times 9 \times 8$ — wrongly assuming digits cannot repeat. On a lock they certainly can.

- **(C)** 999 is the off-by-one trap: codes run from 000 to 999, which is 1000 values, not 999.

Q2. A student must submit one project. They may choose **either** one of 3 ready-made topics **or** design a custom topic by pairing one theme (2 available) with one method (4 available). In how many ways can the project be chosen?

- (A) 9 (B) 11 (C) 14 (D) 24

Answer: (B) 11. Ready-made: 3 ways. Custom (theme **AND** method): $2 \times 4 = 8$. The two routes are mutually exclusive, so add: $3 + 8 = 11$.

- **(A)** 9 adds everything blindly ($3 + 2 + 4$), ignoring that theme and method form an AND-pair.
- **(C)** 14 computes $(3 + 2) \times \dots$ or otherwise muddles the grouping.
- **(D)** 24 multiplies everything ($3 \times 2 \times 4$), forcing an OR-structure into a product.

Q3. If $\frac{n!}{(n-3)!} = 120$, then $n =$

- (A) 5 (B) 6 (C) 8 (D) 10

Answer: (B) 6. Using $\frac{n!}{(n-3)!} = n(n-1)(n-2)$, we need three consecutive integers multiplying to 120: $6 \times 5 \times 4 = 120$, so $n = 6$.

- **(A)** 5 is the headline trap: the student notices $5! = 120$ and blurts $n = 5$ — but $\frac{n!}{(n-3)!}$ is a *product of 3 terms*, not $n!$.
- **(C)** 8, **(D)** 10 come from guessing without expanding the falling product; checking $8 \cdot 7 \cdot 6 = 336$ and $10 \cdot 9 \cdot 8 = 720$ rules them out instantly.

Q4. How many 3-letter codes can be formed from the letters $\{A, B, C, D, E\}$ (repetition allowed) that contain **at least one vowel** (vowels are A, E)?

- (A) 27 (B) 50 (C) 75 (D) 98

Answer: (D) 98. Total codes: $5^3 = 125$. Codes with **no** vowel (only B, C, D): $3^3 = 27$. At least one vowel = $125 - 27 = 98$.

- **(A)** 27 is the consonant-only count — the student found the complement but forgot to subtract it from the total.
- **(B)** 50 counts only codes whose *first* letter is a vowel ($2 \times 5 \times 5$), missing codes where the vowel sits in slot 2 or 3.
- **(C)** 75 is $125 - 50$, compounding the slot-1-only error.

Section 7: Common Student Pitfalls & Pro-Tips

- **Confusing AND with OR.** The single most frequent error: multiplying mutually-exclusive alternatives or adding sequential stages. Always translate the problem into AND/OR first, *then* assign $\times/+$.
- **Assuming “no repetition” by default.** Many students reflexively write falling products $(n, n-1, n-2, \dots)$. Read the problem: locks, number plates, and passwords usually **allow** repetition (n^r); seating distinct people does not. The fork between n^r and the falling product decides everything.
- **Forgetting boundary constraints.** In digit problems, the leading digit usually cannot be 0, and a digit may face *two* exclusions at once (e.g. “ $\neq 0$ and $\neq 9$ ” leaves 8 options, not 9). Slot the most-constrained position first.
- **Mishandling $0!$.** $0! = 1$, never 0. Treating it as 0 collapses later combination formulas to nonsense (division by zero or zero counts).
- **Adding overlapping “OR” cases.** The addition principle needs the cases to be *mutually exclusive*. If your two piles share members (e.g. “numbers divisible by 2 OR by 3”), naive addition double-counts the overlap, and you must subtract it back — the seed of inclusion-exclusion.

Pro-Tip — “at least one” means *complement*. Whenever you see “at least one,” resist direct casework. Compute *total* — *none*. It converts a messy sum of cases into two clean FPC products almost every time.

Section 8: Cheat Sheet Box

FPC & FACTORIAL — RAPID RECALL

- **Multiplication Principle (AND):** stages combine $\rightarrow m \times n \times \dots$
- **Addition Principle (OR, mutually exclusive):** alternatives $\rightarrow m + n$
- **Translate first:** AND $\rightarrow$ multiply, OR $\rightarrow$ add.
- **Factorial:** $n! = n(n-1)(n-2) \cdots 2 \cdot 1$, with $0! = 1$
- **Recursion (use constantly):** $n! = n \cdot (n-1)!$
- **Domain:** $n!$ defined only for non-negative integers; undefined for negatives/fractions.
- **With repetition** (r slots, n options each): n^r
- **Without repetition / full arrangement:** falling product $\rightarrow {}^n P_r = \frac{n!}{(n-r)!}$
(Sub-02)
- **“At least one” $\rightarrow$ complement:** $\#(\text{at least one}) = \text{Total} - \#(\text{none})$

Mnemonics: “AND multiply, OR add.” | “Repetition $\Rightarrow$ powers (n^r); no repetition $\Rightarrow$ falling product.”

Subtopic 02 — Permutations

Exam Relevance: A heavyweight subtopic. JEE Main reliably carries 1-2 questions (word arrangements, digit-formation, restricted seating); CBSE Boards asks the ${}^n P_r$ derivation, alike-object counts, and 2-4 mark applications; JEE Advanced pushes into restricted and circular permutations with layered constraints. Master this and Probability becomes far easier downstream.

Section 1: Concept Intuition

In the previous subtopic you learned to *count* outcomes with the multiplication principle. A **permutation** is just a special, supremely common kind of counting: **arranging objects in a definite order**. The defining feature — the one word you must tattoo on your brain — is **order matters**.

Picture the medal ceremony after a 100 m final. Three athletes finish on the podium: gold, silver, bronze. If Ravi-Meera-Sory take the three medals, that is a *different* outcome from Meera-Ravi-Sory, even though the same three people are involved. The *positions* are distinguishable. That is the soul of a permutation: rearranging the same objects produces a brand-new arrangement.

Contrast that with picking 3 students to form a study group. Here Ravi-Meera-Sory is the *same* group as Meera-Ravi-Sory — order is irrelevant. That second kind of counting is a *combination*, the subject of Sub-03. Right now, hold onto the difference: **permutation = positions; combination = committee**.

Think about it this way... imagine r empty chairs in a row and a crowd of n distinct people. Chair 1 can be taken by any of the n people. Once someone sits, chair 2 has only $n - 1$ candidates left, chair 3 has $n - 2$, and so on. By the Fundamental Principle of Counting (which you already trust from Sub-01), the chairs fill in

$$n \times (n - 1) \times (n - 2) \times \cdots \times (n - r + 1)$$

ways. That descending product is exactly what we will package as ${}^n P_r$.

To see it concretely with tiny numbers, list every arrangement of the three letters A, B, C taken all at a time: $ABC, ACB, BAC, BCA, CAB, CBA$ — exactly $6 = 3!$ of them, and not one is a repeat. Notice that ABC and ACB are counted *separately*: the same three letters in a different order are a different permutation. If instead we had only asked “which three letters did you pick?”, all six would collapse into a single answer — and that collapse is precisely the gap between a permutation and a combination. Holding that distinction firmly now will save you from the single most common error in the entire chapter.

Permutations come in a few important flavours, and recognising which one you’re in is half the battle:

1. **All objects distinct, no repetition** — the medal-podium case, where each object is used at most once and every position is genuinely different. Counted by ${}^n P_r$.

2. **Repetition allowed** — like number-plate digits or a PIN, where the same digit can reappear in several slots. Here every slot sees all n options afresh, giving n^r (this is the FPC result from Sub-01, now living inside permutations).
3. **Some objects identical** — arranging the letters of a word like SAMOSA, where the two S's are interchangeable. Swapping the two S's doesn't create a new word, so we must *divide out* that redundancy and the plain nP_r no longer applies.
4. **Arranged in a circle** — guests around a round dining table, where rotating everyone one seat over gives the *same* arrangement. The circle quietly removes one degree of freedom, which is why its count is smaller than the linear one.

Conditions & Assumptions (when to be careful):

- The clean nP_r formula assumes the objects are **all distinct** and **repetition is not allowed**. The moment repetition is permitted, switch to n^r ; the moment objects repeat, switch to the alike-objects formula.
- nP_r requires $0 \leq r \leq n$. You cannot arrange more objects than you have.
- Circular logic ($(n-1)!$) applies to genuinely *rotatable* arrangements; a row of chairs bolted to the floor is still linear even if curved.

Section 2: Key Formulas & Definitions

Permutation of n distinct objects, taken r at a time (no repetition):

$${}^nP_r = n(n-1)(n-2)\cdots(n-r+1) = \frac{n!}{(n-r)!}, \quad 0 \leq r \leq n.$$

- n — total number of distinct available objects.
- r — number of positions to be filled.
- nP_r — a pure count (dimensionless).

Useful special cases:

$${}^nP_n = n!, \quad {}^nP_0 = 1, \quad {}^nP_1 = n.$$

Repetition allowed (r slots, each from n options):

$$\underbrace{n \times n \times \cdots \times n}_r = n^r.$$

Permutations of n objects, not all distinct (with p alike of one kind, q alike of another, s alike of a third, and $p + q + s + \cdots = n$):

$$\frac{n!}{p! q! s! \cdots}$$

Circular permutations of n distinct objects (all at a time):

$$(n-1)!$$

and if clockwise and anticlockwise arrangements are regarded as identical (necklaces, garlands, bracelets, key-rings):

$$\frac{(n-1)!}{2}.$$

Conditions of validity & sanity-checks:

- nP_r is always a positive integer for $0 \leq r \leq n$; if your “permutation count” comes out fractional, you’ve erred.
- Bridge to combinations (Sub-03): ${}^nP_r = r! \cdot {}^nC_r$ — arranging equals *selecting then ordering*. This single identity ties the two halves of the chapter together.
- Boundary check: ${}^nP_0 = 1$ (one way to arrange nothing — the empty arrangement), consistent with $0! = 1$ from Sub-01.
- Monotonicity check: for fixed n , nP_r grows as r grows (up to $r = n$); a larger answer for fewer positions signals a mistake.

Section 3: CBSE Board Proofs / Key Procedures

3.1 — Deriving nP_r from the Fundamental Principle of Counting.

We want the number of ways to fill r ordered positions using n distinct objects, no repetition.

Figure placeholder — A row of r labelled boxes (Position 1, 2, 3, ..., r). Under each box write its number of choices: $n, (n-1), (n-2), \dots, (n-r+1)$. An arrow underneath spanning all boxes labelled “multiply (FPC)”.

- Position 1: any of the n objects $\rightarrow n$ ways.
- Position 2: any object except the one already used $\rightarrow (n-1)$ ways.
- Position 3: $(n-2)$ ways, and so on.
- Position r : by now $(r-1)$ objects are used, so $n - (r-1) = (n-r+1)$ ways.

By the multiplication principle,

$${}^nP_r = n(n-1)(n-2) \cdots (n-r+1).$$

To compress this, multiply and divide by the “missing tail” $(n-r)(n-r-1) \cdots 2 \cdot 1 = (n-r)!$:

$${}^nP_r = \frac{n(n-1) \cdots (n-r+1) \cdot (n-r)!}{(n-r)!} = \frac{n!}{(n-r)!}. \quad \blacksquare$$

3.2 — Deriving the alike-objects formula.

Claim: n objects with p identical of one type (the rest distinct) arrange in $\frac{n!}{p!}$ distinct ways.

Reasoning (the over-counting argument): Pretend, for a moment, that the p identical objects are secretly labelled and therefore all distinct. Then there are $n!$ arrangements. But in reality those p labels are invisible — the p identical objects can be permuted among

themselves in $p!$ ways *without changing the visible arrangement*. So every genuinely-distinct arrangement has been counted $p!$ times. Dividing removes the redundancy:

$$\text{Distinct arrangements} = \frac{n!}{p!}.$$

With several repeated groups (p of one kind, q of another, ...), each group contributes its own over-count, giving $\frac{n!}{p! q! \dots}$. The mental model: **start from $n!$, then quotient out every group of look-alikes.**

3.3 — Deriving the circular formula $(n - 1)!$.

Reasoning: Take any one circular arrangement of n distinct objects. Reading it off starting from each of the n objects produces n *different* linear arrangements — yet they all correspond to the *same* circle (rotating the table doesn't reseal anyone). So each circular arrangement maps to exactly n linear ones. Since there are $n!$ linear arrangements in total,

$$\text{Circular arrangements} = \frac{n!}{n} = (n - 1)!.$$

If, in addition, a mirror flip is considered identical (a necklace turned over), each circle is double-counted, so we divide by a further 2: $\frac{(n - 1)!}{2}$.

Concrete check: seat 3 friends X, Y, Z around a round table. The formula predicts $(3 - 1)! = 2$ distinct arrangements. Indeed, fixing X at the top, the only genuinely different circles are “ Y on X 's left, Z on the right” and its mirror “ Z on the left, Y on the right” — exactly 2. Every other seating you can write down is just one of these two rotated around the table.

3.4 — Procedure: restricted permutations (the three reflexes).

Figure placeholder — Two mini-illustrations. Left: three letters tied inside a box labelled “(ABC)” sitting in a row with others, captioned “TOGETHER → glue into one block”. Right: a row of fixed items G_G_G with gaps marked $\times$, captioned “NOT TOGETHER → fill the gaps”.

- **“Must be together”:** glue the must-stay-together objects into a single **block**, arrange the block among the others, then multiply by the internal arrangements of the block. (If the block holds k distinct objects, multiply by $k!$.) *Quick illustration:* arrange A, B, C, D, E in a row with A and B together. Glue them as (AB) , giving 4 units to arrange in $4! = 24$ ways; the block itself flips in $2! = 2$ ways; total $24 \times 2 = 48$.
- **“Must not be together”:** arrange the *other* objects first; they create gaps (including the two ends); then drop the restricted objects into those gaps, one per gap, using ${}^n P_r$ on the gaps. *Quick illustration:* for the same five letters with A and B apart, seat C, D, E first ($3! = 6$ ways), creating 4 gaps $_C_D_E_$; place A, B into 2 of those 4 gaps in ${}^4 P_2 = 12$ ways; total $6 \times 12 = 72$. (Sanity check: $48 + 72 = 120 = 5!$, the unrestricted total — together and not-together must partition everything.)
- **“Fixed position”:** lock the constrained object(s) in place first, then freely permute whatever remains.

3.5 — The bridge to combinations: ${}^nP_r = r! \cdot {}^nC_r$.

This identity is worth deriving once, because it unifies the whole chapter. To *arrange* r objects out of n , you can think of it as a two-stage job: first **select** which r objects to use (that is nC_r ways, the Sub-03 quantity), then **order** those chosen r objects among themselves (that is $r!$ ways). By the multiplication principle,

$${}^nP_r = {}^nC_r \times r! \implies {}^nC_r = \frac{{}^nP_r}{r!}.$$

Read it as: *a permutation is a combination that has been put in order*. Every time you feel unsure whether a problem wants P or C , ask whether the final ordering is part of what you're counting — if it is, you're paying the extra $r!$.

Section 4: Worked Examples

Example 1 — [CBSE Board Level] (nP_r , order matters).

A club has 8 members. In how many ways can the distinct posts of President, Secretary, and Treasurer be filled (no member holds two posts)?

Solution. Three distinct posts $\rightarrow$ order matters $\rightarrow$ this is 8P_3 :

$${}^8P_3 = \frac{8!}{(8-3)!} = \frac{8!}{5!} = 8 \times 7 \times 6 = 336.$$

There are 336 ways.

Example 2 — [Conceptual Trap] (alike objects).

How many distinct arrangements can be made of all the letters of the word **SAMOS**A?

Solution. First, count the letters and spot the repeats: S, A, M, O, S, A — that's 6 letters, with **S appearing twice** and **A appearing twice** (M and O once each).

If all six letters were distinct, the answer would be $6! = 720$. But the two S's are interchangeable (so divide by $2!$) and the two A's are interchangeable (divide by another $2!$):

$$\frac{6!}{2!2!} = \frac{720}{4} = 180.$$

The trap: a hurried student writes $6! = 720$, forgetting that swapping the identical letters produces no new word. Always scan for repeats *before* reaching for a factorial. (Sub-tag this error as the alike-objects “forgot-to-divide” mistake.)

Example 3 — [JEE Main Level] (nP_r + a parity constraint).

Using the digits 1, 2, 3, 4, 5 without repetition, how many 4-digit even numbers can be formed?

Solution. “Even” pins down the **units place** — handle the most constrained slot first.

- Units digit must be even $\rightarrow$ from $\{2, 4\} \rightarrow 2$ choices.
- The remaining three positions (thousands, hundreds, tens) are filled from the 4 unused digits, order mattering: ${}^4P_3 = 4 \times 3 \times 2 = 24$.

By the multiplication principle:

$$2 \times 24 = 48 \text{ even 4-digit numbers.}$$

(Two ideas stitched together: the parity restriction fixes one slot, and nP_r arranges the rest.)

Example 4 — [JEE Advanced Level] (circular + gap method).

In how many ways can 5 boys and 3 girls be seated around a circular table so that no two girls sit next to each other?

Solution. Build the arrangement in the order that respects the constraint.

Step 1 — seat the boys first (circular). The 5 boys around the table:

$$(5 - 1)! = 4! = 24 \text{ ways.}$$

Step 2 — identify the gaps. Five seated boys create 5 gaps between consecutive boys.

Figure placeholder — A circle with 5 boys (B) evenly spaced and a small gap symbol ($\times$) between each adjacent pair, showing 5 gaps available for girls.

Step 3 — place the girls in distinct gaps. To keep girls apart, each girl goes into a separate gap. Choosing and ordering 3 girls into 5 gaps:

$${}^5P_3 = 5 \times 4 \times 3 = 60 \text{ ways.}$$

Total (multiplication principle):

$$24 \times 60 = 1440.$$

Why “boys first” is the rigorous move: seating the constrained group (girls) directly would tangle the no-adjacency condition into the circular count. Anchoring the unconstrained group first turns the restriction into a clean gap-filling step — the circular analogue of the linear gap method.

Section 5: Practice Exercises

1. **[CBSE]** In how many ways can 4 different books be arranged on a shelf, chosen from a collection of 9 different books?
2. **[CBSE]** Find the number of distinct arrangements of all the letters of the word **KOLKATA**.
3. **[JEE Main]** Using the digits 1, 2, 3, 4, 5, 6 without repetition, how many 6-digit numbers can be formed in which the digits 1 and 2 are **never adjacent**?

4. **[JEE Main]** How many 4-letter codes (from the 26 English letters, no repetition) **begin and end with a vowel** (vowels: a, e, i, o, u)?
5. **[JEE Advanced]** In how many ways can 7 people be seated around a circular table if 3 specified people insist on sitting **together**?

Answers: 1. ${}^9P_4 = 3024$ 2. $\frac{7!}{2!2!} = 1260$ 3. $6! - 5! \cdot 2! = 480$ 4. ${}^5P_2 \times {}^{24}P_2 = 20 \times 552 = 11040$ 5. $(5-1)! \times 3! = 24 \times 6 = 144$

Section 6: Crack the MCQ

Q1. Using the digits 1, 2, 3, 4, 5, 6, how many 3-digit numbers can be formed **if repetition is not allowed**?

- (A) 18 (B) 120 (C) 216 (D) 729

Answer: (B) 120. Order matters and no repetition $\rightarrow {}^6P_3 = 6 \times 5 \times 4 = 120$.

- **(C)** 216 is 6^3 — the repetition-allowed count. The classic fork error: applying n^r when the problem forbids repetition.
 - **(D)** 729 is 9^3 , wrongly using 9 digits with repetition.
 - **(A)** 18 is 6×3 — a meaningless mix of “objects” and “slots.”
-

Q2. The number of distinct arrangements of all the letters of the word **GANGA** is

- (A) 20 (B) 30 (C) 60 (D) 120

Answer: (B) 30. GANGA has 5 letters with G twice and A twice: $\frac{5!}{2!2!} = \frac{120}{4} = 30$.

- **(D)** 120 is $5!$ — ignores both repeats.
 - **(C)** 60 divides by only one $2!$ — catches the G’s but forgets the A’s (or vice versa).
 - **(A)** 20 comes from an incorrect $\frac{5!}{3!}$ -type slip.
-

Q3. In how many distinct ways can 5 different keys be arranged on a circular key-ring?

- (A) 12 (B) 24 (C) 60 (D) 120

Answer: (A) 12. A key-ring can be **flipped over**, so clockwise and anticlockwise arrangements are identical: $\frac{(5-1)!}{2} = \frac{24}{2} = 12$.

- **(B)** 24 is $(5-1)!$ — correct for a fixed round table, but forgets that a ring flips (the signature necklace/bracelet trap).
 - **(D)** 120 is $5!$ — treats the ring as a straight line.
 - **(C)** 60 is $\frac{5!}{2}$ — divides the *linear* count by 2, mixing two corrections.
-

Q4. If ${}^nP_3 = 2 \cdot {}^nP_2$, then $n =$

- (A) 4 (B) 5 (C) 6 (D) 8

Answer: (A) 4. Write each side as a falling product: $n(n-1)(n-2) = 2n(n-1)$. Cancelling $n(n-1)$ (nonzero) gives $n-2 = 2$, so $n = 4$. *Check:* ${}^4P_3 = 24$ and $2 \cdot {}^4P_2 = 2 \times 12 = 24$. ✓

- **(C) 6, (D) 8** come from guessing without expanding the falling products.
- **(B) 5** typically results from an arithmetic slip when cancelling, e.g. solving $n-2 = 3$.

Section 7: Common Student Pitfalls & Pro-Tips

- **Order confusion (the big one).** Using nP_r when order does *not* matter (it should be nC_r , Sub-03). Test yourself: “If I swap two chosen objects, is it a new outcome?” Yes $\rightarrow$ permutation; No $\rightarrow$ combination.
- **Forgetting to divide for alike objects.** Writing $n!$ for a word with repeated letters. Always scan for repeats and quotient them out.
- **Circular slips.** Using $n!$ instead of $(n-1)!$ for a round arrangement, or forgetting the extra $\div 2$ for necklaces/bracelets/key-rings that can be flipped.
- **Repetition fork.** Defaulting to nP_r when the problem *allows* repetition — that case is n^r , not a falling product.

Pro-Tip — the two restriction reflexes. For **“together,”** glue the group into one block, arrange, then multiply by the block’s internal arrangements. For **“not together,”** seat everyone else first and drop the restricted items into the gaps. And on a circle, **fix one object** to kill the rotational symmetry — the problem instantly collapses to a familiar linear one on the remaining $(n-1)$ objects.

Section 8: Cheat Sheet Box

PERMUTATIONS — RAPID RECALL

- **Definition:** arrangement where **order matters**. *Permutation = Position.*
- **No repetition, r of n :** ${}^nP_r = \frac{n!}{(n-r)!}$ (${}^nP_n = n!$, ${}^nP_0 = 1$)
- **Repetition allowed:** n^r
- **Not all distinct** ($p, q, \dots$ alike): $\frac{n!}{p!q!\dots}$ — start from $n!$, divide out the look-alikes.
- **Circular (all n):** $(n-1)!$ | **flippable (necklace/ring):** $\frac{(n-1)!}{2}$
- **Bridge to combinations:** ${}^nP_r = r! \cdot {}^nC_r$
- **Restrictions:** *together* $\rightarrow$ block ($\times$ internal); *not together* $\rightarrow$ gap method; *circle* $\rightarrow$ fix one to go linear.

Mnemonics: “Order matters $\Rightarrow$ Permutation.” | “Repeats? Divide them out.”
| “Circle loses one: $(n-1)!$; flip loses half.”

Subtopic 03 — Combinations

Exam Relevance: The chapter's payoff and a true workhorse. JEE Main and CBSE Boards lean heavily on committee/team-selection problems and the nC_r properties; JEE Advanced layers constraints ("at least," "at most," grouping). Beyond this chapter, nC_r is the backbone of the Binomial Theorem and almost every Probability problem — so the time you invest here pays dividends for the rest of the syllabus.

Section 1: Concept Intuition

In Sub-02 you arranged objects, and the watchword was *order matters*. A **combination** is the mirror image: you only **select** a group of objects, and **order is irrelevant**. The watchword flips to *order doesn't matter*.

Picture the selectors of a cricket squad choosing 11 players from a pool of 16. Once the eleven names are on the team sheet, it makes no difference whether a selector wrote them top-to-bottom or in any other sequence — the *team* is the same set of eleven people. Nobody says "this is a different team because I listed the wicketkeeper first." That indifference to order is the entire idea of a combination.

Contrast it sharply with a permutation. If, instead, the selectors had to name a captain, a vice-captain, and a wicketkeeper — three *distinct roles* — then order would suddenly matter, and we'd be back in permutation territory. **Selection alone** → **combination**. **Selection plus roles/positions/order** → **permutation**. Keep repeating the test: *if I shuffle the chosen objects, is it a new outcome?* If no, it's a combination.

Think about it this way... a combination is simply a permutation with the ordering *stripped away*. Recall from Sub-02 the bridge we derived: ${}^nP_r = r! \cdot {}^nC_r$. Read it backwards. To list arrangements (nP_r), you first *choose* the r objects (nC_r ways) and then *order* them ($r!$ ways). So combinations are fewer than permutations by exactly a factor of $r!$ — the number of ways the chosen group can be internally rearranged. That single observation hands us the combination formula almost for free, as we'll see in Section 3.

Make the collapse concrete with a tiny case. From four objects A, B, C, D , the selections of size 2 are: AB, AC, AD, BC, BD, CD — exactly 6 of them. Notice AB and BA are *not* listed twice; as a selection they are the same pair. The corresponding *permutations* would be twice as many ($12 = 6 \times 2!$), because each pair can be ordered in $2!$ ways. That factor of $2!$ is the bookkeeping that separates 4C_2 from 4P_2 .

This idea powers a huge range of real questions: handshakes in a room (each handshake is a *pair*, order-free), chords joining points on a circle (each chord is a pair of points), forming committees, dealing hands of cards, choosing toppings for a pizza. As a quick taste: if 9 people at a gathering each shake hands once with every other person, the number of handshakes is just the number of *pairs*, ${}^9C_2 = 36$ — not ${}^9P_2 = 72$, because " A shakes with B " and " B shakes with A " are the *same* handshake. The factor-of-2 gap between 72 and 36 is precisely the order you must strip away.

Conditions & Assumptions (when to be careful):

- nC_r assumes the n objects are **distinct** and you are selecting **without repetition**, with order ignored.
 - The count is valid for $0 \leq r \leq n$; if $r > n$, then ${}^nC_r = 0$ (you cannot choose more objects than exist).
 - The instant the problem attaches *positions, ranks, or sequence* to the chosen objects, stop — it has become a permutation, and you owe an extra factor of $r!$.
-

Section 2: Key Formulas & Definitions

Combination of n distinct objects, taken r at a time:

$${}^nC_r = \frac{n!}{r!(n-r)!} = \frac{{}^nP_r}{r!}, \quad 0 \leq r \leq n.$$

- n — total number of distinct available objects.
- r — size of the group being selected.
- nC_r — a pure count (dimensionless); also written $C(n, r)$ or $\binom{n}{r}$.

Core properties (all examinable):

$${}^nC_0 = {}^nC_n = 1, \quad {}^nC_1 = n,$$

$${}^nC_r = {}^nC_{n-r} \quad (\text{symmetry: choosing } r \text{ to keep} = \text{choosing } n-r \text{ to reject}),$$

$${}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r \quad (\text{Pascal's rule}),$$

$$\text{if } {}^nC_r = {}^nC_s, \text{ then } r = s \text{ or } r + s = n.$$

Conditions of validity & sanity-checks:

- nC_r is always a **positive integer** for $0 \leq r \leq n$; a fractional “combination count” means you’ve slipped.
 - ${}^nC_r = 0$ whenever $r > n$.
 - Symmetry shortcut: to compute ${}^{12}C_{10}$, use ${}^{12}C_{10} = {}^{12}C_2 = 66$ — never grind the large factorial.
 - Order check (the master sanity test): if the answer should *not* change when the chosen objects are reshuffled, you want nC_r , not nP_r .
-

Section 3: CBSE Board Proofs / Key Procedures

3.1 — Deriving nC_r (reusing the Sub-02 bridge).

We do **not** re-derive the bridge ${}^nP_r = r! \cdot {}^nC_r$ — that was established in Sub-02 §3.5 by the “select then order” argument. We simply *use* it.

To arrange r objects out of n in order is a two-stage job: first **select** the r objects (nC_r ways), then **order** them ($r!$ ways). Hence

$${}^nP_r = {}^nC_r \times r!.$$

Solving for nC_r and substituting ${}^nP_r = \frac{n!}{(n-r)!}$ (also from Sub-02):

$${}^nC_r = \frac{{}^nP_r}{r!} = \frac{n!}{r!(n-r)!}. \quad \blacksquare$$

So the whole combination formula is just the permutation formula with the internal orderings divided out.

3.2 — Proving the symmetry ${}^nC_r = {}^nC_{n-r}$.

Combinatorial (bijection) proof — the elegant one: every time you **choose** a group of r objects to keep, you simultaneously **leave behind** a group of $n-r$ objects. Choosing the keepers and choosing the rejects are two descriptions of the *same act*, so they must be equinumerous:

$${}^nC_r = {}^nC_{n-r}.$$

(Algebraic check: $\frac{n!}{r!(n-r)!}$ is unchanged when r and $n-r$ swap — but the choose-or-reject story is the one to remember.) For a quick numeric feel: ${}^7C_5 = {}^7C_2 = 21$ — choosing 5 to keep is the same act as choosing 2 to leave out.

3.3 — Proving Pascal's rule ${}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$.

Combinatorial proof — by a single yes/no question. We want the number of ways to choose r objects from $n+1$ objects, i.e. ${}^{n+1}C_r$. Single out one special object, call it $\star$, and ask: **is $\star$ in our chosen group or not?** The two cases are mutually exclusive and exhaustive:

- **$\star$ is in:** then we still need $r-1$ more from the remaining n objects $\rightarrow {}^nC_{r-1}$ ways.
- **$\star$ is out:** then all r come from the remaining n objects $\rightarrow {}^nC_r$ ways.

By the addition principle,

$${}^{n+1}C_r = {}^nC_{r-1} + {}^nC_r. \quad \blacksquare$$

This “does the special object get included?” split is not just a proof trick — it is the master template for constraint problems below.

3.4 — Procedure: selection under constraints.

Figure placeholder — A flow box. “Must INCLUDE k specific objects?” $\rightarrow$ pre-place them, then choose the remaining $(r-k)$ from $(n-k)$. “Must EXCLUDE k specific objects?” $\rightarrow$ ignore them, choose all r from $(n-k)$. “AT LEAST m of a type?” $\rightarrow$ either sum the cases $m, m+1, \dots$ OR use Total — (fewer than m).

- **“Must include” k specific objects:** lock them in, then choose the remaining $r-k$ from the other $n-k$: ${}^{n-k}C_{r-k}$. *Illustration:* a 5-member team from 9 players that **must** contain the captain $\rightarrow$ fix the captain, choose 4 more from 8: ${}^8C_4 = 70$.
- **“Must exclude” k specific objects:** drop them from the pool, then choose all r from the remaining $n-k$: ${}^{n-k}C_r$. *Illustration:* the same team that **must not** include a suspended player $\rightarrow$ choose 5 from the other 8: ${}^8C_5 = 56$.
- **“At least / at most”:** either **add** the qualifying cases, or — usually faster — take **Total — complement**. *Illustration:* “at least 1 captain-or-suspended-type special

among 5 from 9” is cleanest as ${}^9C_5 - {}^7C_5$ when two specials are excluded in the complement.

3.5 — Deriving “division into equal unlabelled groups.”

Claim: the number of ways to split $2m$ distinct objects into **two unlabelled groups of m each** is $\frac{1}{2!} \cdot {}^{2m}C_m$, and into g unlabelled groups of equal size m (with $gm = n$) is

$$\frac{n!}{(m!)^g g!}.$$

Reasoning: First treat the groups as **labelled** (Group 1, Group 2, ...). Then we are just handing out objects: nC_m ways to fill the first group, ${}^{n-m}C_m$ for the second, and so on, whose product telescopes to $\frac{n!}{(m!)^g}$. But the groups are actually **unlabelled** — relabelling the g equal groups among themselves ($g!$ ways) produces the *same* division. So we have over-counted by exactly $g!$, and must divide it out:

$$\frac{n!}{(m!)^g g!}.$$

This is why Practice Q5 carries that extra $\div 3!$. The rule generalises: **divide by the factorial of the number of groups that are interchangeable** (i.e. equal in size and unlabelled).

Section 4: Worked Examples

Example 1 — [CBSE Board Level] (clean selection).

A cricket squad has 11 players. In how many ways can a sub-committee of 4 players be chosen to meet the coach?

Solution. A sub-committee is a *selection* — order of the four names is irrelevant — so we use ${}^{11}C_4$:

$${}^{11}C_4 = \frac{11!}{4! 7!} = \frac{11 \times 10 \times 9 \times 8}{4 \times 3 \times 2 \times 1} = \frac{7920}{24} = 330.$$

There are 330 possible sub-committees.

Example 2 — [Conceptual Trap] (permutation vs combination, same scenario).

*From 6 eligible students, count the number of ways to (a) choose 3 for a quiz **team**, and (b) appoint 3 to the **distinct posts** of leader, scribe, and timekeeper.*

Solution.

- (a) **Team — order irrelevant → combination:** ${}^6C_3 = \frac{6!}{3! 3!} = 20.$
- (b) **Distinct posts — order matters → permutation:** ${}^6P_3 = 6 \times 5 \times 4 = 120.$

Notice $120 = 20 \times 3!$ — the appointments are just the teams with the $3! = 6$ internal orderings restored. **The trap:** students who don't pause to ask "do the chosen objects have roles?" use nP_r for a plain selection (overcounting by $r!$) or nC_r for a role-assignment (undercounting). This single discrimination is the most heavily tested idea in the chapter — flag it as the cross-cutting permutation-vs-combination tag.

Example 3 — [JEE Main Level] (selection across two categories).

*A committee of 4 is to be formed from 7 boys and 5 girls such that it contains **exactly 2 boys and 2 girls**. In how many ways can this be done?*

Solution. The two choices are independent, so make each selection and combine by the multiplication principle.

- Choose 2 boys from 7: ${}^7C_2 = 21$.
- Choose 2 girls from 5: ${}^5C_2 = 10$.

$$\text{Total} = {}^7C_2 \times {}^5C_2 = 21 \times 10 = 210.$$

(Two combinations stitched together by "AND" $\rightarrow$ multiply, exactly the FPC reflex from Sub-01.)

Example 4 — [JEE Advanced Level] ("at least" by casework).

*From 8 men and 6 women, a committee of 5 is formed containing **at least 3 women**. How many committees are possible?*

Solution. "At least 3 women" with a committee of 5 allows 3, 4, or 5 women. These cases are mutually exclusive, so count each and add (addition principle).

- **3 women, 2 men:** ${}^6C_3 \times {}^8C_2 = 20 \times 28 = 560$.
- **4 women, 1 man:** ${}^6C_4 \times {}^8C_1 = 15 \times 8 = 120$.
- **5 women, 0 men:** ${}^6C_5 \times {}^8C_0 = 6 \times 1 = 6$.

$$\text{Total} = 560 + 120 + 6 = 686.$$

Why casework here (and not complement): the complement of "at least 3 women" is "0, 1, or 2 women" — three cases, no fewer than the direct count. Complementary counting wins only when the complement is *shorter*; here the cases tie, so the direct sum is cleaner. Recognising which route is shorter is the Advanced-level judgement being tested.

Section 5: Practice Exercises

1. **[CBSE]** In how many ways can 3 toppings be chosen from a menu of 8 distinct toppings?
2. **[CBSE]** Find the positive integer n for which ${}^nC_2 = 15$.

3. **[JEE Main]** A team of 5 is to be selected from 9 men and 6 women so that it has **exactly 3 men and 2 women**. In how many ways?
4. **[JEE Main]** From 10 players (including 2 star players), a squad of 4 is chosen. How many squads include **at least one** of the two star players?
5. **[JEE Advanced]** In how many ways can 12 distinct books be divided into **three unlabelled groups of 4 each**?

Answers: 1. ${}^8C_3 = 56$ 2. $n = 6$ 3. ${}^9C_3 \times {}^6C_2 = 84 \times 15 = 1260$ 4. ${}^{10}C_4 - {}^8C_4 = 210 - 70 = 140$ 5. $\frac{12!}{(4!)^3 3!} = 5775$

Section 6: Crack the MCQ

Q1. In how many ways can 3 *identical* prize certificates be distributed among 10 students (no student gets more than one)?

- (A) 30 (B) 120 (C) 720 (D) 1000

Answer: (B) 120. The certificates are identical, so only *which 3 students* are chosen matters — a selection: ${}^{10}C_3 = 120$.

- **(C) 720** is ${}^{10}P_3$ — treating the certificates as distinct/ordered. The signature combination-vs-permutation trap.
 - **(D) 1000** is 10^3 — wrongly allowing repetition (a student getting multiple).
 - **(A) 30** is 10×3 — a meaningless product.
-

Q2. If ${}^nC_8 = {}^nC_{12}$, then ${}^nC_2 =$

- (A) 190 (B) 20 (C) 90 (D) 380

Answer: (A) 190. Since $8 \neq 12$, the property ${}^nC_r = {}^nC_s \Rightarrow r + s = n$ gives $n = 8 + 12 = 20$. Then ${}^{20}C_2 = \frac{20 \times 19}{2} = 190$.

- **(B) 20** stops at finding n and forgets to compute nC_2 .
 - **(C) 90, (D) 380** come from arithmetic slips ($\frac{20 \times 19}{2}$ miscomputed, or using 20×19 without halving gives 380).
-

Q3. A committee of 3 is chosen from 4 men and 3 women. How many committees contain **at least one woman**?

- (A) 18 (B) 30 (C) 31 (D) 35

Answer: (C) 31. Use Total — (no women): ${}^7C_3 - {}^4C_3 = 35 - 4 = 31$.

- **(D) 35** is the *unrestricted* total 7C_3 — forgets the constraint entirely.
- **(A) 18** counts only the “exactly one woman” case (${}^3C_1 \times {}^4C_2 = 3 \times 6$), missing the two- and three-women committees.
- **(B) 30** drops the all-men count incorrectly (e.g. subtracts 5 instead of 4).

Q4. The value of ${}^{15}C_{13}$ is

- (A) 105 (B) 195 (C) 2730 (D) 32760

Answer: (A) 105. Use symmetry: ${}^{15}C_{13} = {}^{15}C_2 = \frac{15 \times 14}{2} = 105$.

- **(D) 32760** is ${}^{15}P_3$ — confusing combination with permutation.
- **(C) 2730** is $15 \times 14 \times 13$ ($= {}^{15}P_3$ without dividing) — forgot both the $r!$ and the symmetry shortcut.
- **(B) 195** is 15×13 — a stray product with no basis.

Section 7: Common Student Pitfalls & Pro-Tips

- **Order confusion (the cross-cutting error).** Using nC_r when the problem assigns roles/positions (needs nP_r), or nP_r for a plain selection (needs nC_r). Always run the master test: *would reshuffling the chosen objects create a new outcome?*
- **Forgetting to divide by $r!$.** Computing nP_r when nC_r is wanted — the answer comes out $r!$ times too large.
- **Mishandling “at least / at most.”** Either missing a case in the sum, or using the complement when it isn’t actually shorter. Count the cases on each side first, then pick the shorter route.
- **Grouping without de-labelling.** When dividing objects into **equal, unlabelled** groups, you must divide by (number of equal groups)! to kill the over-count from swapping identical-sized groups (e.g. the extra $\div 3!$ in Practice Q5). Forgetting this is the classic grouping slip.

Pro-Tip — symmetry and the include/exclude split. Use ${}^nC_r = {}^nC_{n-r}$ to shrink the arithmetic (${}^{50}C_{48} = {}^{50}C_2$ in one line). And for constraint problems, deploy the Pascal-style question “*is this specific object in or out?*” to break the count into clean, addable cases.

Section 8: Cheat Sheet Box

COMBINATIONS — RAPID RECALL

- **Definition:** a **selection** where **order does NOT matter**. *Combination = Committee.*
- **Formula:** ${}^nC_r = \frac{n!}{r!(n-r)!} = \frac{{}^nP_r}{r!}$ ($0 \leq r \leq n$)
- **Boundary:** ${}^nC_0 = {}^nC_n = 1$, ${}^nC_1 = n$, ${}^nC_r = 0$ if $r > n$
- **Symmetry:** ${}^nC_r = {}^nC_{n-r}$ (choose = reject)
- **Pascal’s rule:** ${}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$
- **If ${}^nC_r = {}^nC_s$:** $r = s$ or $r + s = n$
- **Constraints:** include $k \rightarrow {}^{n-k}C_{r-k}$; exclude $k \rightarrow {}^{n-k}C_r$; “at least” $\rightarrow$ cases

or Total — complement

- **Equal unlabelled groups:** divide by (number of equal groups)!

Mnemonics: “Order out $\Rightarrow$ divide by $r!$.” | “Choose r = reject $n - r$.” | “Roles?
 $\rightarrow$ Permutation. Just a group? $\rightarrow$ Combination.”

Subtopic 04 — Total Selections, Grouping & Distribution (Stars and Bars)

Exam Relevance: This is where JEE separates students. JEE Main and Advanced lean heavily on **stars-and-bars** (integer solutions, distributing identical objects), **total-selection** counts, and **grouping/distribution** problems — and these ideas reappear throughout Probability. CBSE Boards touches only the simplest selection counts. If you have mastered nC_r , this subtopic upgrades it into a complete counting toolkit.

Section 1: Concept Intuition

So far you have *chosen exactly r objects*. Real problems ask broader questions: *How many subsets are there altogether? In how many ways can identical sweets be shared? How many integer solutions does an equation have?* This subtopic builds the three big tools that answer them.

Tool 1 — Total selections (the “yes/no on each object” idea). Suppose you stand before n *distinct* objects and decide, for each one independently, “take it” or “leave it.” Each object offers 2 choices, so by the Fundamental Principle of Counting there are 2^n outcomes — i.e. 2^n subsets, including the empty one. If you must take *at least one*, drop the empty selection: $2^n - 1$.

Think about it this way... a selection is just a binary string of length n — a 1 for “in,” a 0 for “out.” There are 2^n such strings, which is *why* an n -element set has 2^n subsets. This single picture also handles **alike** objects: if you have p identical chocolates, a selection only records *how many* you take — anything from 0 to p — which is $p + 1$ choices, not 2^p . Multiply across groups.

Tool 2 — Combinations with repetition (stars and bars). Now selection *with* repetition: choosing r items from n *types*, where you may repeat a type (like buying 5 pastries from 4 available flavours). Order doesn’t matter, repeats are allowed — this is *not* nC_r and *not* n^r . The clever model: lay down r identical **stars** (the items you pick) and separate them into n type-groups using $n - 1$ **bars**. Every arrangement of stars and bars is one valid selection, so you are just choosing positions for the bars among $r + (n - 1)$ slots.

Tool 3 — Grouping vs. distribution. “Divide 12 books into three bundles” and “give the 12 books to three children” sound similar but differ by whether the groups are **labelled**. Unlabelled equal-size groups force a division by (number of equal groups)! (you met this in Sub-03 §3.5); labelled groups (distinct recipients) do not — and converting between them is a frequent exam pivot.

Conditions & Assumptions:

- 2^n counts subsets of **distinct** objects; for alike objects use the $\prod (p_i + 1)$ rule instead.
- Stars and bars assumes the r chosen items are **identical in role** (only counts matter) and the n types/boxes are **distinct**.

- The empty selection is included in 2^n and in the basic stars-and-bars count — subtract or constrain when the problem says “at least one.”

Section 2: Key Formulas & Definitions

Total selections from n distinct objects:

$$\text{subsets (incl. empty)} = 2^n, \quad \text{at least one} = 2^n - 1.$$

Total selections when objects are alike (p alike of type 1, q alike of type 2, ..., plus d all-distinct objects):

$$(\text{incl. empty}) = (p+1)(q+1)\cdots \times 2^d, \quad \text{at least one} = (p+1)(q+1)\cdots 2^d - 1.$$

Combinations with repetition — selecting r from n types, repetition allowed, order ignored:

$${}^{n+r-1}C_r = {}^{n+r-1}C_{n-1}.$$

Stars and bars — integer solutions of $x_1 + x_2 + \cdots + x_k = n$:

$$\text{non-negative solutions } (x_i \geq 0) : \quad {}^{n+k-1}C_{k-1},$$

$$\text{positive solutions } (x_i \geq 1) : \quad {}^{n-1}C_{k-1}.$$

(These are identical to distributing n **identical** objects into k **distinct** boxes — empty allowed, or each non-empty, respectively.)

Division of n distinct objects into groups:

- Into groups of **unequal** sizes $m_1, m_2, \dots, m_k$ (with $\sum m_i = n$): $\frac{n!}{m_1! m_2! \cdots m_k!}$.
- If some groups have **equal size and are unlabelled**, divide further by $g!$ for each set of g equal groups.
- **Distribution** to k *distinct* recipients of those same sizes = the division count $\times$ (arrangements of the groups among recipients). For all-distinct sizes this is the bare $\frac{n!}{\prod m_i!}$; for equal unlabelled groups later assigned to distinct people, multiply the divided count by $k!$.

Distribution of n distinct objects into k distinct boxes (no restriction): k^n .

Conditions & sanity-checks: every count is a positive integer; for stars and bars confirm “identical items, distinct boxes”; for “at least one” in k boxes use the substitution $y_i = x_i - 1$ to convert to the non-negative case.

Section 3: Key Procedures & Logical Framework

3.1 — Why combinations-with-repetition is ${}^{n+r-1}C_r$ (the stars-and-bars proof).

We select r items from n types, repetition allowed, order irrelevant. Represent the selection by r identical stars laid in a line, partitioned into n type-blocks by inserting $n - 1$ bars:

Figure placeholder — A line such as star star bar star bar bar star star star, with $r=6$ stars and $n-1=3$ bars ($n=4$ types). Caption: each gap between bars is a type; the number of stars in a block is how many of that type were chosen; an empty block means that type was not chosen.

Reading left to right, the stars before the first bar belong to type 1, those between the first and second bar to type 2, and so on. Every distinct selection corresponds to exactly one star/bar pattern, and vice-versa. We have $r + (n - 1)$ symbols in total and must choose which $n - 1$ of those positions are bars (equivalently, which r are stars):

$${}^{n+r-1}C_{n-1} = {}^{n+r-1}C_r. \quad \blacksquare$$

Because “choose r from n types with repetition” is *identical* to “place r identical objects into n distinct boxes,” the same formula counts non-negative integer solutions of $x_1 + \dots + x_n = r$.

3.2 — Non-negative vs. positive solutions.

For $x_1 + \dots + x_k = n$ with each $x_i \geq 0$, the stars-and-bars count is ${}^{n+k-1}C_{k-1}$ directly. For each $x_i \geq 1$, first hand one unit to every variable: set $y_i = x_i - 1 \geq 0$, so $y_1 + \dots + y_k = n - k$, giving ${}^{(n-k)+k-1}C_{k-1} = {}^{n-1}C_{k-1}$. The “give one to everybody first” trick is the master move for *lower-bound* constraints — and it works for any bounds, not just 1.

3.3 — Total selections, derived.

Each of n distinct objects is independently in or out: $2 \times 2 \times \dots = 2^n$ subsets (FPC). For an alike group of size p , a selection is fully described by *how many* you take, 0 through p — that is $p + 1$ options — so groups multiply: $(p + 1)(q + 1) \dots$. Distinct objects contribute their own factor of 2 each, i.e. 2^d . Subtract 1 to forbid the empty choice.

3.4 — Grouping vs. distribution procedure.

Figure placeholder — Two panels. Left “DIVIDE (unlabelled groups)”: 6 books split into bundles of 3 and 3, with a note divide by $2!$ because the two equal bundles are interchangeable. Right “DISTRIBUTE (labelled recipients)”: same 6 books to Asha and Ravi, 3 each, no division by $2!$ because Asha’s set and Ravi’s set are different outcomes.

- Write the **division** count $\frac{n!}{\prod m_i!}$.
- If g of the groups are **equal-sized and unlabelled**, divide by $g!$.
- If the groups are then **handed to distinct recipients**, multiply by the number of ways to assign groups to recipients.

Section 4: Worked Examples

Example 1 — [Foundational] (total selections).

A gift shop displays 5 distinct keychains. In how many ways can a customer buy **at least one** keychain?

Solution. Each keychain is independently bought or not: $2^5 = 32$ subsets, including buying nothing. Forbid the empty purchase:

$$2^5 - 1 = 31.$$

Example 2 — [Conceptual Trap] (alike vs. distinct selections).

A basket holds 3 identical red apples, 2 identical green apples, and 1 orange. In how many ways can you pick **at least one** fruit?

Solution. Selections from alike groups are counted by *how many* you take, not which:

- Red apples: 0, 1, 2, 3 $\rightarrow$ 4 choices.
- Green apples: 0, 1, 2 $\rightarrow$ 3 choices.
- Orange (single distinct item): in or out $\rightarrow$ 2 choices.

$$\text{total (incl. empty)} = 4 \times 3 \times 2 = 24, \quad \text{at least one} = 24 - 1 = 23.$$

The trap: treating all 6 fruits as distinct gives $2^6 - 1 = 63$, badly over-counting — because swapping two identical red apples is *not* a new selection. Always ask which objects are genuinely interchangeable.

Example 3 — [JEE Main Level] (non-negative integer solutions).

How many non-negative integer solutions does $a + b + c + d = 12$ have?

Solution. This is 12 identical units distributed among 4 distinct variables (empty allowed) — stars and bars with $n = 12$, $k = 4$:

$${}^{12+4-1}C_{4-1} = {}^{15}C_3 = \frac{15 \times 14 \times 13}{6} = 455.$$

Example 4 — [JEE Advanced Level] (distribution with lower bounds).

In how many ways can 15 identical chocolates be distributed among 4 children so that **each child gets at least 2**?

Solution. Give each child 2 chocolates up front (8 used), then freely distribute the remaining $15 - 8 = 7$ with no restriction. Let $y_i \geq 0$ be the extra each child receives: $y_1 + y_2 + y_3 + y_4 = 7$. By stars and bars:

$${}^{7+4-1}C_{4-1} = {}^{10}C_3 = \frac{10 \times 9 \times 8}{6} = 120.$$

Why the pre-allocation works: subtracting the guaranteed minimum converts a bounded problem into the clean non-negative case — the single most useful manoeuvre in distribution problems.

Section 5: Practice Exercises

1. **[Foundational]** How many subsets does a set of 6 distinct elements have, and how many of them are non-empty?
2. **[Foundational]** In how many ways can 10 distinct books be divided into three groups of sizes 2, 3, and 5?
3. **[JEE Main]** Find the number of positive integer solutions of $x + y + z = 10$.
4. **[JEE Main]** A fruit seller has 4 identical apples, 3 identical oranges, 1 mango, and 1 banana. In how many ways can a customer select at least one fruit?
5. **[JEE Advanced]** In how many ways can 20 identical balls be placed into 5 distinct boxes such that **no box is empty**?

Answers: 1. $2^6 = 64$ subsets; 63 non-empty 2. $\frac{10!}{2!3!5!} = 2520$ 3. ${}^9C_2 = 36$ 4. $(4+1)(3+1)2^2 - 1 = 79$ 5. ${}^{19}C_4 = 3876$

Section 6: Crack the MCQ

Q1. The number of non-negative integer solutions of $x + y + z = 7$ is

- (A) 28 (B) 36 (C) 21 (D) 343

Answer: (B) 36. Stars and bars: ${}^{7+3-1}C_{3-1} = {}^9C_2 = 36$.

- (A) 28 is 8C_2 — using $n + k - 2$ by miscounting the bars.
 - (D) $343 = 7^3$ treats it as ordered placement with repetition, the wrong model.
 - (C) 21 is 7C_2 — forgetting to add the bars to the slot count.
-

Q2. In how many ways can 6 distinct toys be divided into **two unlabelled groups of 3**?

- (A) 10 (B) 20 (C) 36 (D) 720

Answer: (A) 10. $\frac{6!}{3!3!2!} = \frac{720}{6 \cdot 2} \cdot \frac{1}{1} = \frac{720}{36} \cdot \frac{1}{1} \dots$ compute directly: $\frac{6!}{3!3!} = 20$ for *labelled* groups, then $\div 2!$ for the interchangeable equal groups = 10.

- (B) 20 forgets the $\div 2!$ — it counts *distribution* to two labelled recipients, not *division* into unlabelled groups.
 - (D) $720 = 6!$ ignores grouping entirely.
 - (C) 36 is a stray arithmetic slip.
-

Q3. In how many ways can 12 identical pens be distributed among 3 students if a student may receive none?

- (A) 36 (B) 55 (C) 91 (D) 3^{12}

Answer: (C) 91. Identical pens, distinct students, empty allowed: $^{12+3-1}C_{3-1} = {}^{14}C_2 = 91$.

- **(B) 55** is ${}^{11}C_2$ — the “each student at least one” count, wrongly applied.
- **(D) 3^{12}** treats the pens as distinct.
- **(A) 36** is 9C_2 , from using the wrong totals.

Q4. From 4 identical marbles and 1 distinct coin, the number of ways to select **one or more** objects is

- (A) 5 (B) 9 (C) 31 (D) 10

Answer: (B) 9. Marbles: 0–4 $\rightarrow$ 5 choices; coin: in/out $\rightarrow$ 2 choices; total $5 \times 2 = 10$, minus the empty selection = 9.

- **(C) 31** = $2^5 - 1$ wrongly treats the 4 marbles as distinct.
- **(A) 5** counts only the marbles, forgetting the coin.
- **(D) 10** forgets to exclude the empty selection.

Section 7: Common Student Pitfalls & Pro-Tips

- **Distinct vs. alike confusion.** Using 2^n when objects are identical (use $\prod (p_i + 1)$) or vice-versa. Ask: would swapping two of these objects create a new selection?
- **Wrong counting model for repetition.** Selecting with repetition is ${}^{n+r-1}C_r$ — neither nC_r (no repetition) nor n^r (that’s *ordered* placement). Identify “identical items into distinct boxes.”
- **Division vs. distribution.** Forgetting the $\div g!$ for equal *unlabelled* groups, or wrongly applying it when the groups are actually labelled recipients.
- **Forgetting the empty case.** 2^n and the basic stars-and-bars count both include “choose nothing”/“all in one box.” Adjust for “at least one.”

Pro-Tip — pre-allocate to clear lower bounds. For “each box at least c ,” hand out c to every box first, then solve the leftover as a clean non-negative stars-and-bars problem. This one substitution dissolves most distribution constraints.

Section 8: Cheat Sheet Box

SELECTIONS, GROUPING & DISTRIBUTION — RAPID RECALL

- **Subsets of n distinct:** 2^n (incl. empty); **at least one:** $2^n - 1$
- **Alike groups $p, q, \dots$ (+ d distinct):** $(p+1)(q+1) \cdots 2^d$; minus 1 for “at least one”
- **Choose r from n types, repetition allowed:** ${}^{n+r-1}C_r$

- $x_1 + \dots + x_k = n$: non-negative $\rightarrow {}^{n+k-1}C_{k-1}$; positive $\rightarrow {}^{n-1}C_{k-1}$
- **Divide n distinct into sizes $m_1, \dots$:** $\frac{n!}{\prod m_i!}$; $\div g!$ per set of g equal unlabelled groups
- **n distinct objects into k distinct boxes (free):** k^n
- **Lower bounds:** pre-give the minimum, then non-negative stars-and-bars on the rest

Mnemonics: “Identical-into-distinct $\Rightarrow$ stars and bars.” | “In-or-out $\Rightarrow 2^n$.” | “Unlabelled equal groups $\Rightarrow$ divide by their factorial.”

Subtopic 05 — Counting Applications & Advanced Tools

Exam Relevance: The “named techniques” examiners love. Geometric counting (triangles, diagonals, rectangles), number of divisors, sum of numbers formed, rank of a word, derangements, and the exponent of a prime in $n!$ appear repeatedly in JEE Main and Advanced and reward students who recognise the pattern instantly. CBSE occasionally asks divisors or a simple geometric count. This subtopic converts your combination skills into fast, recognisable problem types.

Section 1: Concept Intuition

Most “hard” P&C questions are not new mathematics — they are familiar selections *wearing a costume*. Learn to see through the costume.

Geometry as selection. Through points with no three on a line, a **straight line** is just a *choice of 2 points*, so the number of lines is nC_2 . A **triangle** is a choice of 3 points, nC_3 . The only complication is **collinear points**: if some points lie on a common line, the pairs among them yield one line, not many, and triples among them yield *no* triangle — so we correct by subtracting. **Think about it this way...** counting shapes is counting the *vertices that define them*; the shape is determined by the selection.

Divisors as selection. Write $N = p_1^{a_1} p_2^{a_2} \dots$. Every divisor is built by choosing an exponent for each prime — anything from 0 to a_i for prime p_i , i.e. $a_i + 1$ independent choices. By the FPC, the number of divisors is $\prod (a_i + 1)$. A divisor is a *selection of exponents*.

Arrangements in disguise. “Sum of all numbers formed” and “rank of a word” are permutation problems viewed through place-value and dictionary order. In any complete set of arrangements, **each digit lands in each place equally often**, which makes the total sum computable without listing.

Derangements. Sometimes we must count permutations with a *forbidden* position for every object — letters into envelopes so that *none* matches. These are **derangements**, and they need inclusion-exclusion rather than a plain factorial.

Exponent of a prime in $n!$. To find how many times a prime p divides $n!$, count multiples of p , then of p^2 (which contribute an extra factor), and so on — Legendre’s neat staircase sum.

Conditions & Assumptions: geometric formulas assume “general position” (no three collinear, no two parallel) unless stated, then corrected; the divisor formula needs the *complete* prime factorisation; the sum-of-numbers formula below assumes distinct non-zero digits used without repetition (zero needs a small adjustment).

Section 2: Key Formulas & Definitions

Geometric counting (from n points, no 3 collinear unless noted):

$$\text{lines} = {}^nC_2, \quad \text{triangles} = {}^nC_3.$$

If m of the points are collinear, correct:

$$\text{lines} = {}^nC_2 - {}^mC_2 + 1, \quad \text{triangles} = {}^nC_3 - {}^mC_3.$$

Diagonals of a convex n -gon: ${}^nC_2 - n = \frac{n(n-3)}{2}$. Rectangles from a grid of a horizontal and b vertical lines: ${}^aC_2 \cdot {}^bC_2$. Parallelograms from two families of a and b parallel lines: ${}^aC_2 \cdot {}^bC_2$.

Divisors of $N = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k}$:

$$\text{number of divisors} = (a_1 + 1)(a_2 + 1) \cdots (a_k + 1),$$

$$\text{sum of divisors} = \prod_{i=1}^k \frac{p_i^{a_i+1} - 1}{p_i - 1}.$$

Sum of all numbers formed using n distinct non-zero digits, all at a time, without repetition:

$$\text{sum} = (\text{sum of the digits}) \times (n-1)! \times \underbrace{111 \dots 1}_{n \text{ ones}} = (\sum d) (n-1)! \frac{10^n - 1}{9}.$$

Rank of a word: position in the dictionary-ordered list of all arrangements of its letters (procedure in §3).

Derangements of n objects (no object in its own place):

$$D_n = n! \left(1 - \frac{1}{1!} + \frac{1}{2!} - \cdots + \frac{(-1)^n}{n!} \right), \quad D_n = (n-1)(D_{n-1} + D_{n-2}).$$

Small values: $D_1 = 0$, $D_2 = 1$, $D_3 = 2$, $D_4 = 9$, $D_5 = 44$.

Exponent of prime p in $n!$ (Legendre):

$$E_p(n!) = \left\lfloor \frac{n}{p} \right\rfloor + \left\lfloor \frac{n}{p^2} \right\rfloor + \left\lfloor \frac{n}{p^3} \right\rfloor + \cdots$$

Trailing zeros of $n! = E_5(n!)$.

Sanity-checks: geometric counts must be non-negative after corrections; $\prod (a_i + 1)$ uses every prime; $D_n < n!$ always and $D_n/n! \rightarrow 1/e \approx 0.3679$.

Section 3: Key Procedures & Logical Framework

3.1 — Lines and triangles with collinear points.

Start as if no three points were collinear: nC_2 lines, nC_3 triangles. Now suppose m points are collinear. Those m points would have contributed mC_2 separate lines, but they actually lie on a *single* line — so subtract mC_2 and add 1 back for that one line. For triangles, any 3 of those m collinear points form a degenerate (zero-area) “triangle,” so subtract mC_3 outright.

Figure placeholder — A scatter of points, 4 of them lying on one straight line, the rest in general position. Annotate: the 4 collinear points give $C(4,2)=6$ pairs but only 1 actual line, and $C(4,3)=4$ “triangles” that are really flat.

3.2 — Why $\prod (a_i + 1)$ counts divisors.

A divisor d of $N = p_1^{a_1} \cdots p_k^{a_k}$ must look like $p_1^{e_1} \cdots p_k^{e_k}$ with $0 \leq e_i \leq a_i$. Choosing each exponent e_i is independent and offers $a_i + 1$ values, so by the FPC the number of divisors is $(a_1 + 1)(a_2 + 1) \cdots (a_k + 1)$. (The sum-of-divisors formula is the product of geometric series $1 + p_i + \cdots + p_i^{a_i}$.)

3.3 — Sum of all numbers formed.

With n distinct digits used once each, fix any digit in the units place; the remaining $n - 1$ digits permute freely in $(n - 1)!$ ways. So **each digit appears in the units place exactly $(n - 1)!$ times** — and by symmetry in every place. The total contributed to the units column is $(\sum d)(n - 1)!$, to the tens column $(\sum d)(n - 1)! \times 10$, and so on, summing to $(\sum d)(n - 1)!(1 + 10 + \cdots + 10^{n-1}) = (\sum d)(n - 1)! \frac{10^n - 1}{9}$.

3.4 — Rank of a word (dictionary order).

To rank a word of distinct letters: scan left to right. At each position, count how many *unused* letters are alphabetically *smaller* than the current letter; multiply that count by the factorial of the number of positions still to its right; accumulate. Add 1 at the end (the word itself). For words with repeated letters, replace the plain factorial by the alike-objects count $\frac{(\text{remaining})!}{\prod (\text{repeats})!}$.

3.5 — Derangements by inclusion-exclusion.

Of all $n!$ permutations, subtract those fixing at least one object. Fixing a chosen set of i objects leaves $(n - i)!$ arrangements, and there are nC_i such sets; inclusion-exclusion gives

$$D_n = \sum_{i=0}^n (-1)^i {}^nC_i (n - i)! = n! \sum_{i=0}^n \frac{(-1)^i}{i!}.$$

3.6 — Exponent of a prime in $n!$.

Among $1, 2, \dots, n$, exactly $\lfloor n/p \rfloor$ are multiples of p ; of those, $\lfloor n/p^2 \rfloor$ carry a *second* factor of p ; and so on. Summing these counts every factor of p once, giving Legendre’s formula.

Section 4: Worked Examples

Example 1 — [Foundational] (number of divisors).

How many positive divisors does 360 have?

Solution. Factorise: $360 = 2^3 \times 3^2 \times 5^1$. Each divisor chooses an exponent for each prime:

$$(3 + 1)(2 + 1)(1 + 1) = 4 \times 3 \times 2 = 24 \text{ divisors.}$$

Example 2 — [Conceptual Trap] (triangles with collinear points).

There are 10 points in a plane, of which 4 are collinear. How many triangles can be formed with these points as vertices?

Solution. Without restriction, any 3 points make a triangle: $^{10}C_3 = 120$. But the 4 collinear points yield $^4C_3 = 4$ degenerate (flat) “triangles” that must be removed:

$$120 - 4 = 116.$$

The trap: writing $^{10}C_3 = 120$ and stopping. Three collinear points have zero area — never a triangle. Always scan for collinear subsets and subtract their mC_3 .

Example 3 — [JEE Main Level] (sum of all numbers formed).

Find the sum of all 4-digit numbers formed using the digits 2, 3, 4, 5 exactly once each.

Solution. There are $4! = 24$ such numbers. Each digit occupies each place $(4 - 1)! = 6$ times. Sum of digits = $2 + 3 + 4 + 5 = 14$. The place-value repunit is 1111:

$$\text{sum} = 14 \times 6 \times 1111 = 84 \times 1111 = 93,324.$$

Example 4 — [JEE Advanced Level] (derangements).

*Five distinct letters are to be placed into five addressed envelopes (one per envelope). In how many ways can this be done so that **no letter** goes into its correct envelope?*

Solution. This is the derangement count D_5 :

$$D_5 = 5! \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \frac{1}{5!} \right) = 120 \left(\frac{1}{2} - \frac{1}{6} + \frac{1}{24} - \frac{1}{120} \right).$$

Compute inside: $0.5 - 0.1\bar{6} + 0.041\bar{6} - 0.008\bar{3} = 0.36\bar{6}$, and $120 \times 0.36\bar{6} = 44$. So

$$D_5 = 44.$$

(Cross-check with the recurrence $D_5 = 4(D_4 + D_3) = 4(9 + 2) = 44$.)

Section 5: Practice Exercises

1. **[Foundational]** How many diagonals does a convex decagon (10-gon) have?
2. **[Foundational]** Find the number of positive divisors of 1200.
3. **[JEE Main]** In a rectangular grid formed by 5 horizontal lines and 6 vertical lines, how many rectangles can be formed?
4. **[JEE Main]** Find the rank of the word **MATHS** in the dictionary-ordered list of all permutations of its letters.
5. **[JEE Advanced]** How many trailing zeros does $100!$ have?

Answers: 1. $\frac{10(10-3)}{2} = 35$ 2. $1200 = 2^4 \cdot 3 \cdot 5^2 \Rightarrow (4+1)(1+1)(2+1) = 30$ 3. ${}^5C_2 \cdot {}^6C_2 = 10 \times 15 = 150$ 4. 53 5. $\lfloor 100/5 \rfloor + \lfloor 100/25 \rfloor = 20 + 4 = 24$

Section 6: Crack the MCQ

Q1. The number of positive divisors of 72 is

- (A) 6 (B) 8 (C) 12 (D) 36

Answer: (C) 12. $72 = 2^3 \times 3^2$, so $(3+1)(2+1) = 12$.

- **(A)** 6 multiplies the exponents (3×2) instead of the incremented exponents.
 - **(B)** 8 uses only $2^3 \rightarrow 4$ and doubles, ignoring 3^2 properly.
 - **(D)** 36 confuses divisor-count with something like $72/2$.
-

Q2. From 8 points in a plane, no three collinear, how many triangles can be formed?

- (A) 24 (B) 56 (C) 336 (D) 512

Answer: (B) 56. A triangle is a choice of 3 points: ${}^8C_3 = 56$.

- **(C)** 336 is 8P_3 — ordering the vertices, which doesn't create new triangles.
 - **(D)** $512 = 8^3$ allows repetition and order, both wrong.
 - **(A)** 24 is a miscomputation of 8C_3 .
-

Q3. The number of derangements of 4 distinct objects is

- (A) 6 (B) 9 (C) 12 (D) 24

Answer: (B) 9. $D_4 = 4! \left(1 - 1 + \frac{1}{2} - \frac{1}{6} + \frac{1}{24}\right) = 24 \times \frac{9}{24} = 9$ (or $D_4 = 3(D_3 + D_2) = 3(2 + 1) = 9$).

- **(D)** $24 = 4!$ counts *all* permutations, not just the no-fixed-point ones.
 - **(C)** 12 subtracts only the single-fixed-point cases without inclusion-exclusion.
 - **(A)** 6 is $3!$, a common wrong guess.
-

Q4. The exponent of 3 in $30!$ is

- (A) 10 (B) 13 (C) 14 (D) 15

Answer: (C) 14. $\lfloor 30/3 \rfloor + \lfloor 30/9 \rfloor + \lfloor 30/27 \rfloor = 10 + 3 + 1 = 14$.

- **(A)** 10 stops at the first term, forgetting that 9 and 27 contribute extra factors.
- **(B)** 13 includes $\lfloor 30/9 \rfloor$ but drops $\lfloor 30/27 \rfloor$.
- **(D)** 15 over-counts by adding a spurious term.

Section 7: Common Student Pitfalls & Pro-Tips

- **Ignoring collinear corrections.** Writing nC_3 for triangles or nC_2 for lines without subtracting collinear cases. Always hunt for points/lines in special position.
- **Incomplete prime factorisation.** The divisor formula needs *every* prime with its full exponent — a dropped prime silently halves your answer.
- **Permutation vs. arrangement-sum confusion.** In “sum of numbers formed,” students multiply by $n!$ instead of $(n-1)!$ (the count of times each digit sits in a fixed place). Remember: it’s $(n-1)!$, not $n!$.
- **Derangement as factorial.** Treating “none in correct place” as a plain $n!$ or a single subtraction. It requires the full alternating inclusion-exclusion sum.

Pro-Tip — recognise the costume. Most application problems reduce to a one-line nC_r once you ask *what is being chosen*: points for shapes, exponents for divisors, positions for arrangements. Name the underlying selection first; the formula follows. For “at least one correct/none correct” flavours, reach straight for inclusion-exclusion.

Section 8: Cheat Sheet Box

APPLICATIONS & ADVANCED TOOLS — RAPID RECALL

- **From n points (no 3 collinear):** lines nC_2 , triangles nC_3 ; subtract ${}^mC_2 - 1 / {}^mC_3$ for m collinear
- **Convex n -gon diagonals:** $\frac{n(n-3)}{2}$ | **rectangles from $a \times b$ lines:** ${}^aC_2 {}^bC_2$
- **Divisors of $\prod p_i^{a_i}$:** $\prod (a_i + 1)$; **sum:** $\prod \frac{p_i^{a_i+1} - 1}{p_i - 1}$
- **Sum of all numbers (n distinct digits):** $(\sum d)(n-1)! \frac{10^n - 1}{9}$
- **Rank of word:** $\sum (\text{smaller unused letters}) \times (\text{remaining})! + 1$
- **Derangements:** $D_n = n! \sum_{i=0}^n \frac{(-1)^i}{i!}$; $D_n = (n-1)(D_{n-1} + D_{n-2})$; $D_4=9, D_5=44$
- **Prime p in $n!$:** $\sum_{k \geq 1} \lfloor n/p^k \rfloor$; trailing zeros of $n! = E_5(n!)$

Mnemonics: “Shapes = choosing vertices.” | “Divisor = pick an exponent per prime.” | “None-correct $\Rightarrow$ derangement $\Rightarrow$ inclusion-exclusion.”

Chapter 6: Permutations and Combinations — Supplement

Subtopic 06 — Inclusion-Exclusion and Onto Distributions

Exam Relevance: CBSE Boards examines the two-set and three-set union formulas at 2–4 marks, usually as a survey or Venn-diagram question carried over from the Sets chapter. JEE Main turns the same tool into counting questions — how many integers up to N are divisible by at least one of two or three given numbers, how many arrangements avoid a list of forbidden patterns, how many integer solutions survive an upper bound on each variable. JEE Advanced pushes to the general n -set statement and to its single most important application in this chapter: counting **onto** distributions of distinct objects into distinct boxes, equivalently the number of surjective functions, equivalently $k! S(n, k)$ where $S(n, k)$ is a Stirling number of the second kind. NEET does not test mathematics.

Three earlier places in this chapter have pointed at this subtopic. Subtopic 01 §3.2 noted that if two “OR” piles overlap, “we would need to subtract the overlap — this repair is the inclusion-exclusion idea you will meet later.” Subtopic 01 §7 called the same repair “the seed of inclusion-exclusion.” Subtopic 05 §3.5 used the full machine, without ever stating it, to derive the derangement formula D_n . This subtopic states the tool properly, proves it, drills it, and then spends the second half on the one distribution case Subtopic 04 deliberately left open: **distinct** objects into **distinct** boxes with **no box empty**.

Section 1: Concept Intuition

Counting a crowd by counting lists. Picture the notice board outside a coaching centre in Kota on the morning the batch lists go up. Three lists are pinned: the Physics batch, the Chemistry batch, the Mathematics batch. The director wants one number — how many distinct students are enrolled in at least one subject. The registrar’s first instinct is to add the three list lengths. That answer is always too big, because a student taking both Physics and Chemistry has been counted twice, and a student taking all three has been counted three times.

So the registrar subtracts. For every pair of subjects, he subtracts the number of students on both lists. Now the double-counted students are fixed. But look at what happened to the all-three student: he was added three times (once per list) and subtracted three times (once per pair), so he now contributes zero. He has been erased. Add him back once, and every student — whether on one list, two lists or three — contributes exactly 1.

$$|P \cup C \cup M| = |P| + |C| + |M| - |P \cap C| - |C \cap M| - |M \cap P| + |P \cap C \cap M|.$$

That over-correct, then correct back, then correct the correction rhythm is the whole idea. It does not stop at three. With n lists you add the singles, subtract the pairs, add the triples, subtract the quadruples, and so on, with the sign flipping every level, until you reach the single n -fold intersection. The alternating signs are not a mnemonic anyone had to invent; they are forced, and Section 3 shows exactly why by tracking one element through the bookkeeping.

Think about it this way. Inclusion-exclusion is the Addition Principle from Subtopic 01 with its safety catch removed. The Addition Principle demands mutually exclusive piles. Real problems hand you overlapping piles and refuse to reorganise them. Inclusion-exclusion is the price you pay for adding anyway.

The complement form is the one you will actually use. Almost every exam question is phrased as “how many objects have *none* of these properties” — how many integers up to 100 are divisible by none of 2, 3, 5; how many letters go into no correct envelope; how many functions miss no element of the codomain. If S is everything and A_i is the set of objects having property i , then

$$(\text{none of the properties}) = |S| - \left| \bigcup A_i \right|,$$

and expanding the union gives a clean alternating sum starting with $|S|$ itself. Subtopic 05 §3.5 ran exactly this computation for derangements and never named it.

Now the second half: why k^n is not enough. Subtopic 04 §2 told you that n distinct objects go into k distinct boxes in k^n ways — each object independently picks a box. It also told you that n **identical** objects go into k distinct boxes with none empty in $\binom{n-1}{k-1}$ ways, by the give-one-to-everybody-first trick. What it never told you is the count when the objects are distinct **and** no box may be empty. That is not $\binom{n-1}{k-1}$, and it is not k^n , and — critically — **the pre-allocation trick that worked for identical objects does not work here.**

Here is why the trick collapses. Picture three ticket counters at Chhatrapati Shivaji Terminus and five distinguishable passengers, each of whom must be sent to some counter, with every counter required to serve at least one passenger. If the passengers were identical, you would seat one at each counter and freely place the remaining two: clean. But the passengers are distinguishable, so “seat one at each counter first” requires you to choose *which* passenger goes where — and the same final arrangement then gets built in several different ways, depending on which passenger you designated as the pre-allocated one. **Pre-allocation over-counts the moment the objects become distinguishable, because the pre-allocated object is not recoverable from the final arrangement.** Section 4’s Speed Trap example computes the size of the damage exactly.

The correct route is inclusion-exclusion, and it is the natural one. Start with all k^n placements. A placement is bad if some box is empty. Let A_i be the set of placements that leave box i empty; then $|A_i| = (k-1)^n$, because every object now chooses from the remaining $k-1$ boxes. An intersection of r of these sets is the set of placements avoiding r named boxes, of size $(k-r)^n$. The alternating sum does the rest.

Where all of this fails. Inclusion-exclusion needs every intersection size to be computable — if you cannot evaluate $|A_i \cap A_j|$, the formula is a true statement you cannot use. The onto formula needs $n \geq k$; when $n < k$ there are fewer objects than boxes and the count is honestly 0, which the formula produces on its own. And every result in this section assumes the boxes are **distinct**. Making them identical divides out an ordering, which is the Stirling-number view developed in Section 2.

Section 2: Key Formulas and Definitions

The union formulas

Two sets.

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

Three sets.

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |C \cap A| + |A \cap B \cap C|.$$

General n sets (the statement to quote in an Advanced answer). For finite sets $A_1, A_2, \dots, A_n$,

$$\left| \bigcup_{i=1}^n A_i \right| = \sum_i |A_i| - \sum_{i < j} |A_i \cap A_j| + \sum_{i < j < k} |A_i \cap A_j \cap A_k| - \dots + (-1)^{n-1} |A_1 \cap A_2 \cap \dots \cap A_n|.$$

Compactly, writing $S_r = \sum_{1 \leq i_1 < i_2 < \dots < i_r \leq n} |A_{i_1} \cap \dots \cap A_{i_r}|$ for the total of all r -fold intersection sizes,

$$\left| \bigcup_{i=1}^n A_i \right| = \sum_{r=1}^n (-1)^{r-1} S_r.$$

- A_i — the set of objects having property i ; the properties may overlap freely.
- S_r — the sum of the sizes of all $\binom{n}{r}$ intersections taken r at a time. Set $S_0 = |S|$, the size of the universal set.
- The sum has exactly n terms, and the final sign is $(-1)^{n-1}$: positive when n is odd, negative when n is even.
- Validity: all sets must be finite. No disjointness is assumed — that is the entire point.

Complement (none of the properties). Inside a finite universal set S ,

$$|\overline{A_1} \cap \overline{A_2} \cap \dots \cap \overline{A_n}| = |S| - \left| \bigcup_{i=1}^n A_i \right| = \sum_{r=0}^n (-1)^r S_r.$$

This is the form that solves exam questions. Note that it starts at $r = 0$ with $+|S|$.

Figure brief: Three overlapping circles labelled A, B, C inside a rectangle labelled S , with all seven bounded regions shaded in distinct tints and each labelled with the number of times it is counted at each stage — a small three-row ledger beside the diagram reading “add singles: 1, 1, 1, 2, 2, 2, 3 / subtract pairs: 0, 0, 0, 1, 1, 1, 0 / add triple: 0, 0, 0, 1, 1, 1, 1” against the region types “only- A , only- B , only- C , exactly-two (three regions), all-three”.

Exactly- r and at-least- r counts

Using the same S_r notation, the number of objects lying in **exactly** r of the n sets is

$$E_r = \sum_{j=r}^n (-1)^{j-r} \binom{j}{r} S_j,$$

and the number lying in **at least** r of them is

$$L_r = \sum_{j=r}^n (-1)^{j-r} \binom{j-1}{r-1} S_j.$$

For $n = 3$ these unwind into the three lines CBSE and JEE Main actually ask for. Write $\Sigma_1 = |A| + |B| + |C|$, $\Sigma_2 = |A \cap B| + |B \cap C| + |C \cap A|$, $\Sigma_3 = |A \cap B \cap C|$.

Requirement	Formula
At least one of A, B, C	$\Sigma_1 - \Sigma_2 + \Sigma_3$
Exactly one of A, B, C	$\Sigma_1 - 2\Sigma_2 + 3\Sigma_3$
Exactly two of A, B, C	$\Sigma_2 - 3\Sigma_3$
At least two of A, B, C	$\Sigma_2 - 2\Sigma_3$
All three	Σ_3
None of A, B, C	$ S - \Sigma_1 + \Sigma_2 - \Sigma_3$

Onto distributions of distinct objects (surjections)

Let n distinct objects be placed into k **distinct** boxes with **no box empty**. Equivalently, count the onto (surjective) functions from a set of size n to a set of size k . The count is

$$N(n, k) = \sum_{i=0}^k (-1)^i \binom{k}{i} (k-i)^n = k^n - \binom{k}{1} (k-1)^n + \binom{k}{2} (k-2)^n - \dots + (-1)^{k-1} \binom{k}{k-1} \cdot 1^n.$$

- n — number of distinct objects (domain size); k — number of distinct boxes (codomain size).
- The $i = k$ term is $(-1)^k \binom{k}{k} 0^n = 0$ for every $n \geq 1$, so in practice you stop at $i = k - 1$.
- Validity and sanity-check: the formula is valid for all $n, k \geq 1$ and automatically returns 0 whenever $n < k$. It returns $k!$ when $n = k$.
- Special cases worth memorising: $N(n, 1) = 1$, $N(n, 2) = 2^n - 2$, $N(n, 3) = 3^n - 3 \cdot 2^n + 3$.

The Stirling-number view. Define $S(n, k)$, the **Stirling number of the second kind**, as the number of ways to split a set of n distinct objects into exactly k non-empty **unlabelled** blocks. Then labelling the blocks — that is, deciding which block goes into which of the k distinct boxes — multiplies by $k!$:

$$N(n, k) = k! S(n, k)$$

$S(n, k)$ obeys the recurrence

$$S(n, k) = S(n-1, k-1) + k S(n-1, k), \quad S(n, 1) = S(n, n) = 1, \quad S(n, k) = 0 \text{ for } k > n,$$

together with the closed forms $S(n, 2) = 2^{n-1} - 1$ and $S(n, n-1) = \binom{n}{2}$.

$n \backslash k$	1	2	3	4	5	6
1	1					
2	1	1				
3	1	3	1			
4	1	7	6	1		
5	1	15	25	10	1	
6	1	31	90	65	15	1

The corresponding onto counts $N(n, k) = k! S(n, k)$ that appear most often in papers: $N(4, 3) = 6 \times 6 = 36$, $N(5, 3) = 6 \times 25 = 150$, $N(6, 3) = 6 \times 90 = 540$, $N(5, 4) = 24 \times 10 = 240$, $N(6, 4) = 24 \times 65 = 1560$, $N(6, 2) = 62$, $N(7, 3) = 6 \times 301 = 1806$.

Row sums of the Stirling table give the **Bell numbers** $B_n = \sum_{k=1}^n S(n, k)$, the number of ways to split n distinct objects into any number of non-empty unlabelled blocks: $B_1 = 1$, $B_2 = 2$, $B_3 = 5$, $B_4 = 15$, $B_5 = 52$, $B_6 = 203$.

The complete distribution table — connecting back to Subtopic 04

This is the single table that resolves every “objects into boxes” question in the chapter. Read it as a two-by-two grid on *are the objects distinguishable* and *are the boxes distinguishable*, with the empty-allowed and no-empty columns side by side. Subtopic 04 supplied the top row; this subtopic supplies the rest.

n objects	k boxes	Any box may be empty	No box empty
Identical	Distinct	$\binom{n+k-1}{k-1}$	$\binom{n-1}{k-1}$
Distinct	Distinct	k^n	$\sum_{i=0}^k (-1)^i \binom{k}{i} (k-i)^n = k! S(n, k)$

n objects	k boxes	Any box may be empty	No box empty
Distinct	Identical	$\sum_{j=1}^k S(n, j)$	$S(n, k)$
Identical	Identical	partitions of n into at most k parts	partitions of n into exactly k parts

Notes on the table. The chapter's $\binom{n+k-1}{k-1}$ and $\binom{n-1}{k-1}$ are exactly Subtopic 04 §2's stars-and-bars pair, written here with $\binom{n}{r}$ in place of the nC_r notation used there; the two notations mean the same thing. The bottom row has no closed formula at this level — for the small values an exam uses, list the partitions by hand. For instance, 6 identical balls into 3 identical boxes with none empty means partitions of 6 into exactly 3 parts: 4+1+1, 3+2+1, 2+2+2, so the answer is 3.

Sanity identity linking the two middle rows. Every one of the k^n unrestricted placements of distinct objects into distinct boxes uses exactly j of the boxes for some j between 1 and k , and the j used boxes can be chosen in $\binom{k}{j}$ ways:

$$k^n = \sum_{j=1}^k \binom{k}{j} N(n, j).$$

Use it as a check on every onto computation you perform — Example 4 does exactly that.

Upper bounds on integer solutions

Subtopic 04 §3.2 gave the pre-allocation move for lower bounds and remarked that it “works for any bounds, not just 1.” Read precisely, that means any **lower** bound: for $x_i \geq c_i$, substitute $y_i = x_i - c_i \geq 0$ and the problem becomes an unrestricted stars-and-bars count. **Pre-allocation cannot impose an upper bound**, because there is no substitution that turns $x_i \leq c_i$ into an unconstrained variable. Upper bounds require inclusion-exclusion.

For the number of non-negative integer solutions of

$$x_1 + x_2 + \cdots + x_k = n, \quad x_i \leq c_i \text{ for each } i,$$

let A_i be the set of solutions violating the i -th bound, i.e. with $x_i \geq c_i + 1$. Pre-allocating $c_i + 1$ units to x_i shows

$$|A_i| = \binom{n - (c_i + 1) + k - 1}{k - 1},$$

and more generally an r -fold intersection removes $\sum (c_i + 1)$ units over the chosen indices. Hence

$$\#\{\text{solutions}\} = \sum_{r=0}^k (-1)^r \sum_{|T|=r} \binom{n - \sum_{i \in T} (c_i + 1) + k - 1}{k - 1},$$

keeping only those index sets T for which the leftover total $n - \sum_{i \in T} (c_i + 1)$ is still non-negative, and dropping every other term. In the common symmetric case $c_1 = \dots = c_k = c$ this collapses to

$$\sum_{r \geq 0} (-1)^r \binom{k}{r} \binom{n - r(c+1) + k - 1}{k - 1}.$$

- **Validity:** the substitution $x'_i = x_i - (c_i + 1)$ is a bijection onto non-negative solutions of a smaller equation, so each $|A_i|$ is exact.
- **Practical note:** the sum terminates quickly. Only those r with $r(c+1) \leq n$ contribute anything, and in most exam problems that means $r = 0$ and $r = 1$ only.
- For lower and upper bounds together, clear the lower bounds first by pre-allocation, then apply this formula to what is left.

Section 3: CBSE Board Derivations and Key Procedures

Derivation: The two-set and three-set union formulas

Claim. For finite sets, $|A \cup B| = |A| + |B| - |A \cap B|$.

1. Split $A \cup B$ into three pairwise disjoint pieces: the elements in A only, the elements in B only, and the elements in both. *Reason:* every element of the union is in at least one set, so exactly one of the three descriptions applies to it.
2. Write a, b, c for the sizes of those three pieces. Then $|A \cup B| = a + b + c$ by the Addition Principle (Subtopic 01 §2), which is legitimate because the pieces are disjoint.
3. Observe $|A| = a + c$ and $|B| = b + c$. *Reason:* A consists of the A -only piece together with the shared piece, and similarly for B .
4. Therefore $|A| + |B| - |A \cap B| = (a + c) + (b + c) - c = a + b + c = |A \cup B|$. ■

Claim. $|A \cup B \cup C| = \Sigma_1 - \Sigma_2 + \Sigma_3$ in the notation of Section 2.

1. Apply the two-set result with the sets $A \cup B$ and C : $|A \cup B \cup C| = |A \cup B| + |C| - |(A \cup B) \cap C|$.
2. Expand $|A \cup B| = |A| + |B| - |A \cap B|$ by the two-set result again.
3. Distribute the intersection: $(A \cup B) \cap C = (A \cap C) \cup (B \cap C)$. *Reason:* an element is in the left side exactly when it is in C and in at least one of A, B .
4. Apply the two-set result a third time to that union: $|(A \cap C) \cup (B \cap C)| = |A \cap C| + |B \cap C| - |A \cap B \cap C|$, using $(A \cap C) \cap (B \cap C) = A \cap B \cap C$.
5. Substitute steps 2 and 4 into step 1 and collect terms:

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |C \cap A| + |A \cap B \cap C|. \quad \blacksquare$$

Derivation: The general n -set principle (the counting-ledger proof)

Claim. $|\bigcup_{i=1}^n A_i| = \sum_{r=1}^n (-1)^{r-1} S_r$.

The proof strategy is to show that the right-hand side counts every element of the union exactly once and every other element exactly zero times.

1. Fix any element x of the universal set, and let m be the number of the sets $A_1, \dots, A_n$ that contain x . *Reason:* the contribution of x to every term depends only on m , not on which sets they are.
2. Suppose first that $m = 0$. Then x lies in no A_i , hence in no intersection, hence contributes 0 to every S_r and 0 to the total. It is also not in the union. The two sides agree on x .
3. Now suppose $m \geq 1$. The element x lies in an r -fold intersection $A_{i_1} \cap \dots \cap A_{i_r}$ precisely when all r chosen indices come from the m sets containing x . There are $\binom{m}{r}$ such choices, so x contributes exactly $\binom{m}{r}$ to S_r .
4. Hence the total contribution of x to the right-hand side is

$$\sum_{r=1}^n (-1)^{r-1} \binom{m}{r} = \sum_{r=1}^m (-1)^{r-1} \binom{m}{r},$$

since $\binom{m}{r} = 0$ for $r > m$.

5. By the Binomial Theorem, $\sum_{r=0}^m (-1)^r \binom{m}{r} = (1-1)^m = 0$ for $m \geq 1$. Separating the $r = 0$ term $\binom{m}{0} = 1$ gives $\sum_{r=1}^m (-1)^r \binom{m}{r} = -1$, and multiplying by -1 gives

$$\sum_{r=1}^m (-1)^{r-1} \binom{m}{r} = 1.$$

6. So every element with $m \geq 1$ — that is, every element of the union — contributes exactly 1, and every other element contributes 0. The right-hand side therefore equals $|A_1 \cup \dots \cup A_n|$. ■

Step 5 is where the alternating signs earn their keep: they are precisely what makes the binomial ledger telescope to 1 for *every* value of m at once. Verify the pattern by hand for $m = 3$: the element is added 3 times, subtracted 3 times, added once, for a net $3 - 3 + 1 = 1$.

Derivation: The onto-distribution formula

Claim. The number of ways to place n distinct objects into k distinct boxes leaving no box empty is $N(n, k) = \sum_{i=0}^k (-1)^i \binom{k}{i} (k-i)^n$.

1. Let S be the set of all placements with no restriction. Each of the n objects independently chooses one of k boxes, so $|S| = k^n$ by the Fundamental Principle of Counting (Subtopic 01 §2).
2. For each box i , let A_i be the set of placements in which box i receives nothing. A placement is *bad* exactly when it lies in $A_1 \cup \dots \cup A_k$, and *good* exactly when it lies in none of them.
3. Compute $|A_i|$. If box i is barred, every object chooses among the remaining $k-1$ boxes, so $|A_i| = (k-1)^n$. There are $\binom{k}{1}$ such sets.
4. Compute a general r -fold intersection. If r named boxes are all barred, every object chooses among the remaining $k-r$ boxes, giving $(k-r)^n$ placements. The size depends only on r , and there are $\binom{k}{r}$ choices of which boxes to bar, so $S_r = \binom{k}{r} (k-r)^n$.

5. Apply the complement form of inclusion-exclusion:

$$N(n, k) = \sum_{r=0}^k (-1)^r S_r = \sum_{r=0}^k (-1)^r \binom{k}{r} (k-r)^n. \quad \blacksquare$$

Two immediate checks. Put $k = 2$: $N(n, 2) = 2^n - 2 \cdot 1^n + 0 = 2^n - 2$, which is right by direct reasoning — every placement except the two that dump everything into a single box. Put $n = 2$, $k = 3$: $9 - 3 \cdot 4 + 3 \cdot 1 - 0 = 0$, correct, since two objects cannot occupy three boxes without leaving one empty.

Derivation: Why $N(n, k) = k! S(n, k)$

1. A placement with no box empty determines a splitting of the n objects into k non-empty groups — the contents of the boxes — together with an assignment of those groups to the k labelled boxes.
2. Conversely, given any splitting into k non-empty unlabelled blocks, each of the $k!$ bijections from blocks to boxes produces a different placement, and every placement arises exactly once this way. *Reason:* the boxes are distinguishable, so swapping the contents of two boxes genuinely changes the placement, and the blocks are non-empty and pairwise disjoint, so no two bijections give the same result.
3. By the Multiplication Principle, $N(n, k) = S(n, k) \times k!$, where $S(n, k)$ counts the splittings.

The recurrence for $S(n, k)$, proved the same way. Focus on object number n . Either it sits alone in a block of its own, in which case the other $n - 1$ objects form the remaining $k - 1$ blocks in $S(n - 1, k - 1)$ ways; or it joins a block containing other objects, in which case the other $n - 1$ objects first form all k blocks in $S(n - 1, k)$ ways and object n then picks one of those k blocks to join. The two cases are mutually exclusive and exhaustive, so by the Addition Principle $S(n, k) = S(n - 1, k - 1) + k S(n - 1, k)$.

Derivation: Derangements, revisited as one line of this machine

Subtopic 05 §3.5 stated the derangement count without naming its engine. It is the complement form with $S = \text{all } n! \text{ permutations}$ and $A_i = \{\text{permutations fixing object } i\}$. Fixing r named objects leaves the other $n - r$ free, so $|A_{i_1} \cap \cdots \cap A_{i_r}| = (n - r)!$ and $S_r = \binom{n}{r} (n - r)!$. Therefore

$$D_n = \sum_{r=0}^n (-1)^r \binom{n}{r} (n - r)! = \sum_{r=0}^n (-1)^r \frac{n!}{r!} = n! \sum_{r=0}^n \frac{(-1)^r}{r!}.$$

Notice that this is structurally identical to the onto derivation: same complement form, same $\binom{n}{r}$ choice of which conditions to violate, only the intersection size changes from $(k - r)^n$ to $(n - r)!$. **Learn the machine once and both formulas are consequences.**

Procedure A: Counting “at least one of these properties”

1. Name the universal set S and write down $|S|$. *Why:* everything is measured against this, and half of all inclusion-exclusion errors are a wrong or unstated universe.
2. Name the properties $A_1, \dots, A_n$ explicitly in words. *Why:* the sets must be properties of individual objects, not of groups; if you cannot phrase A_i as “this object has feature i ”, the model is wrong.
3. Compute $|A_i|$ for each i , then every $|A_i \cap A_j|$, then every triple, and so on. *Why:* the formula is useless unless every intersection is computable; check this before committing to the method.
4. Alternate the signs: add the singles, subtract the pairs, add the triples. *Why:* Derivation 3 shows this is exactly what makes each element count once.
5. Sanity-check the answer against $\max_i |A_i| \leq |\bigcup A_i| \leq \min(|S|, \sum_i |A_i|)$. *Why:* the union is at least as big as its largest piece and at most the naive sum, and it can never exceed the universe.

Procedure B: Counting “none of these properties”

1. Run steps 1–3 of Procedure A unchanged.
2. Write $|S| - S_1 + S_2 - S_3 + \dots$, starting with a plus sign on $|S|$. *Why:* this is the complement of the union, and expanding the union flips every sign.
3. Confirm the answer is non-negative and at most $|S|$. *Why:* a negative result means an intersection was mis-sized or a sign was dropped, the two most common slips.

Illustration (the divisibility pattern JEE Main favours). For “how many integers from 1 to N are divisible by none of a, b, c ”, take $S = \{1, 2, \dots, N\}$ and A_a the multiples of a , so $|A_a| = \lfloor N/a \rfloor$. The crucial step is the intersection: $A_a \cap A_b$ is the set of multiples of $\text{lcm}(a, b)$, **not** of ab . For $a = 4, b = 6$ the correct divisor is 12, not 24. When a, b, c are pairwise coprime the two agree, which is why the standard 2, 3, 5 question hides the issue.

Procedure C: Distributing n distinct objects into k distinct boxes, no box empty

1. Confirm the objects are genuinely distinguishable and the boxes are genuinely labelled. *Why:* if the objects are identical, Subtopic 04’s $\binom{n-1}{k-1}$ is the correct tool and this procedure over-counts wildly; if the boxes are identical, you want $S(n, k)$ instead.
2. Check $n \geq k$. *Why:* otherwise the answer is 0 and no computation is needed.
3. Write $N(n, k) = k^n - \binom{k}{1}(k-1)^n + \binom{k}{2}(k-2)^n - \dots$, stopping at the term $(-1)^{k-1} \binom{k}{k-1} 1^n$. *Why:* the final term $(k-k)^n$ vanishes.
4. Evaluate, then cross-check with $N(n, k) = k! S(n, k)$ using the recurrence or the table. *Why:* two independent routes to the same integer is the cheapest possible insurance in a counting problem.
5. If the boxes were in fact identical, divide the result by $k!$ to recover $S(n, k)$. *Why:* the $k!$ orderings of the box labels all collapse to one outcome.

Illustration. Four distinct objects into three distinct boxes, none empty:

$$N(4, 3) = 3^4 - 3 \cdot 2^4 + 3 \cdot 1^4 = 81 - 48 + 3 = 36.$$

Cross-check: $S(4, 3) = 6$ from the table, and $3! \times 6 = 36$. Second cross-check using the sanity identity: $3^4 = 81$ should equal $\binom{3}{3}N(4, 3) + \binom{3}{2}N(4, 2) + \binom{3}{1}N(4, 1) = 36 + 3(2^4 - 2) + 3(1) = 36 + 42 + 3 = 81$. It does.

Procedure D: Upper bounds — where pre-allocation stops working

1. Clear any **lower** bounds first by pre-allocation, exactly as in Subtopic 04 §3.2: replace $x_i \geq c_i$ by $y_i = x_i - c_i \geq 0$ and reduce the target sum. *Why:* this step is a bijection and loses nothing.
2. Now handle the **upper** bounds by inclusion-exclusion. Let A_i be the solutions with y_i exceeding its cap, and size A_i by pre-allocating one more than the cap. *Why:* “ $y_i \geq c+1$ ” is a lower bound, and lower bounds are the case pre-allocation *can* handle.
3. Alternate signs across $r = 0, 1, 2, \dots$, keeping a term only while the leftover total after pre-allocating the r violations is still non-negative. *Why:* once $r(c+1)$ exceeds the target, no solution can violate that many caps at once, so that term counts nothing and the series terminates.
4. Verify by the complementary substitution $u_i = c - y_i$ where the caps are equal. *Why:* it converts the problem into a different-looking one with the same answer, giving an independent check.

Illustration. Solutions of $x_1 + x_2 + x_3 = 15$ with $0 \leq x_i \leq 8$. Unrestricted: $\binom{15+3-1}{3-1} = \binom{17}{2} = 136$. Violating one named cap means $x_i \geq 9$; pre-allocate 9 and count $\binom{6+2}{2} = \binom{8}{2} = 28$; there are 3 choices of which variable, giving 84. Two variables cannot both reach 9, since $18 > 15$, so the series stops. Answer $136 - 84 = 52$. Check by $u_i = 8 - x_i$: then $u_1 + u_2 + u_3 = 24 - 15 = 9$ with $0 \leq u_i \leq 8$, giving $\binom{11}{2} - 3\binom{2}{2} = 55 - 3 = 52$. Agreed.

The selection mirror (“at most” constraints). Subtopic 03 §3.4 named the “at least / at most” pair but illustrated only the “at least” direction. The upper-bound direction is the same manoeuvre in selection language: for “choose 5 from 5 men and 4 women with **at most** 2 women”, either sum the permitted cases $\binom{4}{0}\binom{5}{5} + \binom{4}{1}\binom{5}{4} + \binom{4}{2}\binom{5}{3} = 1 + 20 + 60 = 81$, or complement the forbidden ones, $\binom{9}{5} - [\binom{4}{3}\binom{5}{2} + \binom{4}{4}\binom{5}{1}] = 126 - 45 = 81$. **Take whichever side has fewer cases** — here they tie at three versus two, and the complement wins by one case.

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

In a class of 60 students, 32 have opted for Physics, 28 for Chemistry, and 12 have opted for both. Find (a) the number who have opted for at least one of the two subjects, (b) the number who have opted for neither, (c) the number who have opted for exactly one of the two.

Solution.

Let P be the set of Physics students and C the set of Chemistry students, inside a universal set of 60. We are given $|P| = 32$, $|C| = 28$, $|P \cap C| = 12$.

(a) By the two-set union formula,

$$|P \cup C| = |P| + |C| - |P \cap C| = 32 + 28 - 12 = 48.$$

The subtraction is compulsory: the 12 students taking both were counted once inside the 32 and again inside the 28.

(b) “Neither” is the complement of the union inside the class:

$$60 - 48 = 12.$$

(c) A student takes exactly one subject when they are in the union but not in the intersection:

$$|P \cup C| - |P \cap C| = 48 - 12 = 36.$$

Equivalently, $|P| + |C| - 2|P \cap C| = 32 + 28 - 24 = 36$ — the intersection is removed **twice**, once to stop the double count and once more to exclude those students altogether.

Check by regions. Physics only = $32 - 12 = 20$; Chemistry only = $28 - 12 = 16$; both = 12; neither = 12. These four disjoint groups total $20 + 16 + 12 + 12 = 60$, the whole class, and exactly one = $20 + 16 = 36$. Every part agrees.

Example 2 — [JEE Main Level]

How many integers from 1 to 100 inclusive are divisible by at least one of 2, 3 and 5? How many are divisible by none of them?

Solution.

Take $S = \{1, 2, \dots, 100\}$, so $|S| = 100$, and let A_2, A_3, A_5 be the sets of multiples of 2, 3, 5 respectively. The number of multiples of d in S is $\lfloor 100/d \rfloor$.

Singles:

$$|A_2| = \lfloor \frac{100}{2} \rfloor = 50, \quad |A_3| = \lfloor \frac{100}{3} \rfloor = 33, \quad |A_5| = \lfloor \frac{100}{5} \rfloor = 20.$$

Pairs. An integer divisible by both 2 and 3 is divisible by $\text{lcm}(2, 3) = 6$, and similarly for the other pairs:

$$|A_2 \cap A_3| = \lfloor \frac{100}{6} \rfloor = 16, \quad |A_3 \cap A_5| = \lfloor \frac{100}{15} \rfloor = 6, \quad |A_5 \cap A_2| = \lfloor \frac{100}{10} \rfloor = 10.$$

Triple:

$$|A_2 \cap A_3 \cap A_5| = \lfloor \frac{100}{30} \rfloor = 3.$$

Now the three-set formula:

$$|A_2 \cup A_3 \cup A_5| = (50 + 33 + 20) - (16 + 6 + 10) + 3 = 103 - 32 + 3 = 74.$$

Divisible by none:

$$100 - 74 = 26.$$

Verification. Divisibility by 2, 3, 5 repeats with period $\text{lcm}(2, 3, 5) = 30$. In each block of 30 consecutive integers, the count divisible by none is $30 \times \frac{1}{2} \times \frac{2}{3} \times \frac{4}{5} = 8$. The first three blocks cover 1 to 90 and contribute 24. In 91, ..., 100 the odd numbers are 91, 93, 95, 97, 99; of these 93 and 99 are divisible by 3 and 95 by 5, leaving 91 and 97. Total $24 + 2 = 26$, matching.

Why the bounds check passes. The union 74 sits between the largest single set (50) and the naive sum (103), and below $|S| = 100$, as Procedure A step 5 requires.

Example 3 — [JEE Main — Speed Trap]

In how many ways can 5 distinct chocolates be distributed among 3 children so that every child receives at least one chocolate?

The trap: reusing the identical-object formula, and “pre-allocating” distinguishable objects.

Seductive wrong answer 1. The student recognises “no box empty”, reaches for Subtopic 04’s positive stars-and-bars count $\binom{n-1}{k-1}$, and writes

$$\binom{5-1}{3-1} = \binom{4}{2} = 6.$$

This counts the ways to decide *how many* chocolates each child gets — the triples (3, 1, 1), (1, 3, 1), (1, 1, 3), (2, 2, 1), (2, 1, 2), (1, 2, 2) — and stops there. It ignores *which* chocolates, because stars-and-bars was built for identical stars. Since the chocolates are distinct, every one of those six count-patterns splits into many genuinely different distributions.

Seductive wrong answer 2. The student remembers “pre-allocate one to everybody, then distribute the rest freely.” Choosing an ordered triple of distinct chocolates to hand out first can be done in ${}^5P_3 = 60$ ways, and the remaining 2 chocolates then go anywhere, $3^2 = 9$ ways:

$$60 \times 9 = 540.$$

This is 3.6 times too large. The reason is precise: in the finished distribution there is no record of which chocolate was the “pre-allocated” one, so a child holding 3 chocolates could have been reached through any of 3 choices of designated chocolate, and the over-count is not even uniform across patterns. **Pre-allocation is a bijection only when the objects are indistinguishable.**

Correct solution. Distinct objects, distinct recipients, none empty — this is $N(5, 3)$. By Procedure C,

$$N(5, 3) = 3^5 - \binom{3}{1}2^5 + \binom{3}{2}1^5 = 243 - 3(32) + 3(1) = 243 - 96 + 3 = 150.$$

Cross-check via Stirling. $S(5, 3) = 25$ from the table, and $3! \times 25 = 150$. Confirm $S(5, 3)$ by direct partition counting: a split of 5 distinct chocolates into three non-empty unlabelled blocks has shape 3+1+1 or 2+2+1. Shape 3+1+1: choose the block of three in $\binom{5}{3} = 10$ ways, and the two leftovers are forced singletons, giving 10. Shape 2+2+1: choose the singleton

in 5 ways, then split the remaining 4 into two unlabelled pairs in 3 ways, giving 15. Total $10 + 15 = 25$.

Cross-check via the sanity identity. All $3^5 = 243$ unrestricted distributions must split by the number of children actually served:

$$243 = \underbrace{150}_{\text{all 3}} + \underbrace{\binom{3}{1}N(5, 2)}_{\text{exactly one child left out}} + \underbrace{\binom{3}{2}N(5, 1)}_{\text{exactly two left out}} = 150 + 3(2^5 - 2) + 3(1) = 150 + 90 + 3 = 243.$$

How to spot the trap in three seconds. Ask one question: *would swapping two of the objects produce a different outcome?* If yes, the objects are distinct and stars-and-bars is barred.

Example 4 — [JEE Advanced Level]

Find the number of onto functions from a set X with 6 elements to a set Y with 4 elements. Hence find the number of ways of dividing 6 distinct books among 4 students so that each student gets at least one book, and the number of ways of putting the same 6 books into 4 **identical** boxes with no box empty.

Solution.

A function $f : X \rightarrow Y$ assigns to each of the 6 elements of X one of the 4 elements of Y ; it is onto exactly when every element of Y is used. Writing $f^{-1}(y)$ for the set of elements sent to y , an onto function is precisely a placement of the 6 distinct elements of X into the 4 distinct boxes labelled by Y with no box empty. So the three questions asked are the same question twice over and then once with the labels removed.

Step 1 — the inclusion-exclusion count. Let S be all 4^6 functions and A_i the functions that miss the i -th element of Y . Then $|A_{i_1} \cap \cdots \cap A_{i_r}| = (4-r)^6$, and there are $\binom{4}{r}$ such intersections. By Procedure C,

$$N(6, 4) = \sum_{r=0}^4 (-1)^r \binom{4}{r} (4-r)^6 = 4^6 - \binom{4}{1}3^6 + \binom{4}{2}2^6 - \binom{4}{3}1^6 + \binom{4}{4}0^6.$$

Evaluating each term: $4^6 = 4096$; $\binom{4}{1}3^6 = 4 \times 729 = 2916$; $\binom{4}{2}2^6 = 6 \times 64 = 384$; $\binom{4}{3}1^6 = 4$; the last term is 0.

$$N(6, 4) = 4096 - 2916 + 384 - 4 = 1560.$$

Step 2 — the books. Handing 6 distinct books to 4 distinct students with nobody empty-handed is the same placement problem, so the answer is again 1560.

Step 3 — identical boxes. Removing the labels from the boxes identifies the $4! = 24$ placements that differ only by a relabelling, so by Procedure C step 5,

$$S(6, 4) = \frac{N(6, 4)}{4!} = \frac{1560}{24} = 65.$$

Verification 1 — the Stirling recurrence. Build the table upwards from $S(4, 3) = 6$, $S(4, 4) = 1$:

$$S(5, 4) = S(4, 3) + 4S(4, 4) = 6 + 4 = 10, \quad S(5, 3) = S(4, 2) + 3S(4, 3) = 7 + 18 = 25,$$

$$S(6, 4) = S(5, 3) + 4S(5, 4) = 25 + 40 = 65.$$

This matches Step 3 exactly, and it was obtained without touching the inclusion-exclusion sum.

Verification 2 — the sanity identity $k^n = \sum_j \binom{k}{j} N(n, j)$. Every one of the $4^6 = 4096$ unrestricted placements uses some number j of the four boxes:

Boxes actually used, j	$\binom{4}{j}$	$N(6, j)$	Product
4	1	1560	1560
3	4	$3^6 - 3 \cdot 2^6 + 3 = 540$	2160
2	6	$2^6 - 2 = 62$	372
1	4	1	4

The products total $1560 + 2160 + 372 + 4 = 4096 = 4^6$. The decomposition is exact, so 1560 is confirmed.

Figure brief: A bipartite arrow diagram with six labelled dots on the left in a column (the set X) and four labelled dots on the right (the set Y), one arrow leaving each left dot, drawn so that every right dot receives at least one arrow — with the fibres $f^{-1}(y)$ circled on the left in four tinted loops of sizes 3, 1, 1, 1, and a caption noting that the loops are the “boxes” and that erasing the right-hand labels merges $4!$ such pictures into one.

Section 5: Practice Exercises

1. **[CBSE]** In a colony of 100 households, 65 subscribe to newspaper A , 45 subscribe to newspaper B and 20 subscribe to both. How many subscribe to at least one newspaper, and how many to neither?
2. **[CBSE]** Of 200 students, 100 study Hindi, 80 study Sanskrit and 70 study French; 40 study Hindi and Sanskrit, 30 study Sanskrit and French, 35 study Hindi and French, and 15 study all three. Find the number who study (a) at least one of the three languages, (b) none of the three, (c) exactly one of the three.
3. **[JEE Main]** How many integers from 1 to 1000 inclusive are divisible by none of 2, 3 and 5?
4. **[JEE Main]** Find the number of onto functions from a set with 6 elements to a set with 3 elements.
5. **[JEE Main]** Find the number of integer solutions of $x + y + z = 12$ with $1 \leq x, y, z \leq 6$.
6. **[JEE Main]** How many words of length 4 can be formed from the letters A, B, C, D, E with repetition allowed, if each word must contain **all three** of A, B and C at least once?

7. **[JEE Advanced]** In how many ways can 7 distinct toys be distributed among 3 distinct children so that no child is left without a toy?
8. **[JEE Advanced]** In how many ways can 5 distinct objects be placed into 3 **identical** boxes so that no box is empty? Compare your answer with the count for 3 distinct boxes.

Answers.

1. $65 + 45 - 20 = 90$ subscribe to at least one; $100 - 90 = 10$ to neither.
2. (a) $250 - 105 + 15 = 160$. (b) $200 - 160 = 40$. (c) $\Sigma_1 - 2\Sigma_2 + 3\Sigma_3 = 250 - 210 + 45 = 85$.
3. $1000 - [500 + 333 + 200 - 166 - 100 - 66 + 33] = 1000 - 734 = 266$.
4. $N(6, 3) = 3^6 - 3 \cdot 2^6 + 3 = 729 - 192 + 3 = 540$.
5. 25. (Substitute $x = 1 + u$ etc. to get $u + v + w = 9$ with $0 \leq u, v, w \leq 5$; then $\binom{11}{2} - 3\binom{5}{2} = 55 - 30 = 25$.)
6. 84. (Inclusion-exclusion on the letters missed: $5^4 - 3 \cdot 4^4 + 3 \cdot 3^4 - 2^4 = 625 - 768 + 243 - 16 = 84$.)
7. $N(7, 3) = 3^7 - 3 \cdot 2^7 + 3 = 2187 - 384 + 3 = 1806$.
8. $S(5, 3) = 25$ for identical boxes; $3! \times 25 = 150$ for distinct boxes — a factor of $3! = 6$ apart.

Section 6: Crack the MCQ

Q1. The number of integers from 1 to 60 inclusive that are divisible by 2 or by 3 is (A) 50 (B) 40 (C) 30 (D) 10

Answer: (B) 40. $\lfloor 60/2 \rfloor = 30$, $\lfloor 60/3 \rfloor = 20$, and the overlap is the multiples of $\text{lcm}(2, 3) = 6$, namely $\lfloor 60/6 \rfloor = 10$. So $30 + 20 - 10 = 40$.

Distractor Analysis. (A) 50 is $30 + 20$ — the naive Addition Principle applied to piles that are not mutually exclusive, which is precisely the error Subtopic 01 §7 warned about. (C) 30 reports the multiples of 2 alone, reading “or” as though the second condition added nothing new. (D) 10 reports the *intersection* instead of the union, a swap of $\cap$ for $\cup$ under time pressure.

Q2. The number of onto functions from a set containing 4 elements to a set containing 2 elements is (A) 16 (B) 14 (C) 15 (D) 3

Answer: (B) 14. $N(4, 2) = 2^4 - \binom{2}{1}1^4 = 16 - 2 = 14$. Directly: of the 16 functions, exactly two fail to be onto — the two constant functions.

Distractor Analysis. (A) $16 = 2^4$ counts all functions and forgets the onto requirement entirely. (C) $15 = 2^4 - 1$ removes only one of the two constant functions; students who think of “at least one” as always costing a single subtraction of the empty case land here. (D) $3 = \binom{4-1}{2-1}$ applies Subtopic 04’s identical-object formula $\binom{n-1}{k-1}$ to distinguishable objects — the headline confusion of this subtopic. That formula counts only the size patterns $(3, 1)$, $(2, 2)$, $(1, 3)$ and is blind to which elements go where.

Q3. The number of ways in which 6 distinct balls can be put into 3 distinct boxes so that no box remains empty is (A) 540 (B) 729 (C) 10 (D) 90

Answer: (A) 540. $N(6, 3) = 3^6 - \binom{3}{1}2^6 + \binom{3}{2}1^6 = 729 - 192 + 3 = 540$. Cross-check: $3!S(6, 3) = 6 \times 90 = 540$.

Distractor Analysis. (B) $729 = 3^6$ is the unrestricted count from Subtopic 04 §2, with the no-empty-box condition dropped. (C) $10 = \binom{6-1}{3-1} = \binom{5}{2}$ treats the six distinct balls as identical. (D) $90 = S(6, 3)$ is the count for **identical** boxes; the student did the hard part correctly and then forgot that labelled boxes multiply by $3!$. Distractor (D) is the one that catches strong students, so make the labelled-or-not decision before computing, not after.

Q4. The number of non-negative integer solutions of $x + y + z = 10$ with $x \leq 5$, $y \leq 5$ and $z \leq 5$ is (A) 66 (B) 21 (C) 51 (D) 24

Answer: (B) 21. Unrestricted, $\binom{10+3-1}{3-1} = \binom{12}{2} = 66$. Violating a named cap means that variable is at least 6; pre-allocating 6 leaves $x' + y + z = 4$ with $\binom{6}{2} = 15$ solutions, and there are 3 choices of which variable, giving 45. Two variables cannot both reach 6 since $12 > 10$, so the series stops: $66 - 45 = 21$. Check by $u = 5 - x$, $v = 5 - y$, $w = 5 - z$: then $u + v + w = 5$ with each at most 5, which is automatic, so the count is $\binom{7}{2} = 21$.

Distractor Analysis. (A) 66 ignores the caps altogether. (C) $51 = 66 - 15$ subtracts the over-cap solutions for one variable only, forgetting that the same failure can happen to y or z . (D) $24 = 66 - 45 + 3$ adds back a phantom double-violation term: the student writes the full alternating pattern mechanically and inserts $\binom{3}{2}$ for the pairs without checking that two variables reaching 6 each would need a total of 12, which exceeds 10. **In inclusion-exclusion, a term must be tested for existence, not assumed.**

Section 7: Common Student Pitfalls and Pro-Tips

- **Using $\binom{n-1}{k-1}$ for distinct objects.** The error this subtopic exists to kill. Subtopic 04's positive stars-and-bars count answers "how many objects does each box get", not "which objects". Correction: ask whether swapping two objects changes the outcome. If it does, the objects are distinct and you need $\sum (-1)^i \binom{k}{i} (k-i)^n$.
- **Pre-allocating distinguishable objects.** "Give each box one first, then distribute the rest" is a bijection only for identical objects. For distinct objects it over-counts, because the finished arrangement carries no record of which object was pre-allocated. Correction: reserve pre-allocation for identical objects and for **lower** bounds on integer variables.
- **Pre-allocating to enforce an upper bound.** There is no substitution that turns $x \leq c$ into a free variable. Correction: convert each upper bound into its violation $x \geq c + 1$, which is a lower bound, and alternate signs across the violations.
- **Multiplying instead of taking the lcm.** In divisibility problems, $A_a \cap A_b$ is the multiples of $\text{lcm}(a, b)$. Writing $\lfloor N/(ab) \rfloor$ is correct only when a and b are coprime, and the standard 2, 3, 5 question hides this because those are pairwise coprime. Test yourself on 4 and 6: the intersection is the multiples of 12.
- **Writing terms that cannot occur.** Students reproduce the full alternating pattern by reflex and include an r -fold term whose intersection is empty. It contributes 0, so including it is harmless only if you also evaluate it as 0; inserting a non-zero value there

is a guaranteed wrong answer. Correction: for every term, ask whether r conditions can be violated simultaneously.

- **Losing the sign at the last term.** The union form ends at $(-1)^{n-1}$ and the complement form ends at $(-1)^n$. With three sets the triple intersection is **added** in the union and **subtracted** in the complement. Correction: rebuild the signs from the $m = 3$ ledger $3 - 3 + 1 = 1$ rather than memorising them.
- **Confusing labelled with unlabelled boxes.** $S(n, k)$ and $k! S(n, k)$ differ by a factor of $k!$, and both are plausible-looking integers. Correction: the word “boxes”, “children”, “students”, “envelopes” and “counters” all signal distinct; “groups”, “bundles”, “identical boxes” and “parts” signal unlabelled — the same distinction Subtopic 04 §3.4 drew between division and distribution.
- **Forgetting that the universe must be stated.** A complement count is meaningless without $|S|$. Correction: write $|S|$ on the page before anything else.

Pro-Tip — read the phrase “at least one of” as a siren. Whenever a problem says “at least one of these n properties” or “none of these n properties”, stop looking for a direct construction and reach for inclusion–exclusion. Subtopic 01’s Pro-Tip taught you total minus none for a *single* property; this is the same instinct extended, and the two are the same formula when $n = 1$.

Pro-Tip — three circles beat a formula under exam pressure. For any three-set question with numbers given, draw the Venn diagram and fill it from the **centre outward**: write the all-three count first, then subtract it from each pairwise count to get the exactly-two regions, then subtract everything from each single count to get the exactly-one regions. Every one of the six questions in the Section 2 table can then be read off by adding regions, with no signs to get wrong. If any region comes out negative, the data in the question is inconsistent and you should re-read it.

Pro-Tip — always cross-check an onto count two ways. Compute $N(n, k)$ by the alternating sum, then confirm it is divisible by $k!$ and that the quotient matches $S(n, k)$ from the recurrence. A single arithmetic slip in the alternating sum almost never survives both tests.

Pro-Tip — the $n < k$ reflex. If there are fewer objects than boxes, the onto count is 0 and no work is needed. In a multi-part question this instantly disposes of one part.

Section 8: Cheat Sheet Box

SUBTOPIC 06 — 60-SECOND RECALL

Two and three sets. $|A \cup B| = |A| + |B| - |A \cap B|$. $|A \cup B \cup C| = \Sigma_1 - \Sigma_2 + \Sigma_3$, where Σ_1 sums the singles, Σ_2 the three pairwise intersections, Σ_3 the triple.

General n sets. $|\bigcup_{i=1}^n A_i| = S_1 - S_2 + S_3 - \cdots + (-1)^{n-1} S_n$, with S_r the total of all $\binom{n}{r}$ r -fold intersection sizes.

None of the properties. $|S| - S_1 + S_2 - S_3 + \cdots + (-1)^n S_n$. Starts with $+|S|$; this is the exam form.

Three-set variants. Exactly one: $\Sigma_1 - 2\Sigma_2 + 3\Sigma_3$. Exactly two: $\Sigma_2 - 3\Sigma_3$. At least two: $\Sigma_2 - 2\Sigma_3$. None: $|S| - \Sigma_1 + \Sigma_2 - \Sigma_3$.

Divisibility. Multiples of d in $1..N$: $\lfloor N/d \rfloor$. Intersection of “multiple of a ” and “multiple of b ” is “multiple of $\text{lcm}(a, b)$ ” — not ab unless coprime.

Onto / no box empty (distinct objects, distinct boxes).

$$N(n, k) = \sum_{i=0}^k (-1)^i \binom{k}{i} (k-i)^n = k! S(n, k).$$

$N(n, 1) = 1$; $N(n, 2) = 2^n - 2$; $N(n, 3) = 3^n - 3 \cdot 2^n + 3$; $N(n, k) = 0$ if $n < k$; $N(k, k) = k!$. Ready values: $N(4, 3) = 36$, $N(5, 3) = 150$, $N(6, 3) = 540$, $N(7, 3) = 1806$, $N(5, 4) = 240$, $N(6, 4) = 1560$.

Stirling numbers of the second kind (n distinct into k non-empty **unlabelled** blocks). $S(n, k) = S(n-1, k-1) + k S(n-1, k)$; $S(n, 1) = S(n, n) = 1$; $S(n, 2) = 2^{n-1} - 1$; $S(n, n-1) = \binom{n}{2}$. Rows $n = 1$ to 6: $1 \mid 1, 1 \mid 1, 3, 1 \mid 1, 7, 6, 1 \mid 1, 15, 25, 10, 1 \mid 1, 31, 90, 65, 15, 1$. Row sums are the Bell numbers $1, 2, 5, 15, 52, 203$.

Master distribution table.

Objects	Boxes	Empty allowed	No box empty
Identical	Distinct	$\binom{n+k-1}{k-1}$	$\binom{n-1}{k-1}$
Distinct	Distinct	k^n	$k! S(n, k)$
Distinct	Identical	$\sum_{j=1}^k S(n, j)$	$S(n, k)$
Identical	Identical	partitions of n into $\leq k$ parts	partitions of n into exactly k parts

Self-check identity. $k^n = \sum_{j=1}^k \binom{k}{j} N(n, j)$ — split unrestricted placements by how many boxes were actually used.

Upper-bounded integer solutions. For $x_1 + \dots + x_k = n$ with every $x_i \leq c$:

$$\sum_{r \geq 0} (-1)^r \binom{k}{r} \binom{n - r(c+1) + k - 1}{k-1},$$

keeping only the terms with $n - r(c+1) \geq 0$ and dropping the rest. Clear lower bounds by pre-allocation first; upper bounds always need this sum.

Derangements are the same machine (Subtopic 05 §3.5). $D_n = \sum_{r=0}^n (-1)^r \binom{n}{r} (n-r)!$ $= n! \sum_{r=0}^n \frac{(-1)^r}{r!}$; $D_4 = 9$, $D_5 = 44$.

Mnemonics: “Add singles, subtract pairs, add triples — signs alternate to the end.” $\mid$ “Distinct objects kill stars and bars.” $\mid$ “Labelled boxes multiply by $k!$.” $\mid$ “Upper bound $\Rightarrow$ violate it, then alternate.”

Chapter 6: Permutations and Combinations — Round 2 Addendum

Subtopics 01-05 and the Round 1 Supplement (Subtopic 06, inclusion-exclusion and onto distributions) between them cover the NCERT syllabus and most of the JEE toolkit. Four named patterns are still missing, and all four are high-frequency.

Three of them are missing outright. The fourth — **rank of a word** — is worse than missing: Subtopic 05 lists it in the Exam Relevance box, gives it a six-line procedure at §3.4, puts one line in the Cheat Sheet, and sets it as Practice Q4 with the bare answer “53”. It is never worked once, and the repeated-letter case that NCERT itself examines (Example 22 on AGAIN, Miscellaneous Exercise Q4 on EXAMINATION) is dismissed in half a sentence. A student who has read the chapter cannot produce a rank on paper.

This addendum closes the four gaps, continuing the Round 1 numbering. Each item is a method statement, three or four ladderworked examples, and practice with answers. Notation follows the Round 1 Supplement: $\binom{n}{r}$ and nC_r mean the same thing.

Addendum	Topic	Read after
07A	Rank of a word in dictionary order, and the inverse (k -th word)	Subtopic 05 §3.4
07B	Lattice-path counting: monotonic grid paths as combinations	Subtopic 02 §2
07C	The divisor toolkit beyond $d(N)$ and $\sigma(N)$	Subtopic 05 §3.2
07D	Partial derangements: exactly k objects in place	Subtopic 05 §3.5

Addendum 07A — Rank of a word in dictionary order

Extends Subtopic 05 §3.4 and §5 Q4. Uses the alike-objects formula from Subtopic 02 §2.

Exam Relevance: CBSE Boards asks this directly as a 4-mark question, almost always in NCERT’s own two flavours — “if these words are written as in a dictionary, what is the k -th word” and “how many words come before the first word starting with E”. JEE Main runs one rank question in most sittings, usually with a repeated letter to punish the student who memorised only the all-distinct recipe.

JEE Advanced inverts it, or hides it inside a probability question. NEET does not test mathematics.

Method. A dictionary lists every distinct arrangement of the letters in alphabetical order. The rank of a word is its position in that list. You never write the list; you count how many words *precede* it, then add 1.

1. **Alphabetise the letter pool.** Write the letters of the word in sorted order, repeats included. Note the total number of distinct arrangements $\frac{n!}{p!q!\dots}$ — this is the ceiling your answer must respect.
2. **Scan the target word left to right.** At position i , let ℓ be the letter of the target and let R be the multiset of letters not yet placed (which still contains ℓ).
3. **For each distinct letter c in R with $c < \ell$:** putting c at position i makes a word that is alphabetically earlier no matter how the tail is filled. The number of such words is the number of arrangements of the multiset $R - \{c\}$, namely

$$\frac{(n-i)!}{\prod_j m_j!},$$

where the m_j are the repeat counts of $R - \{c\}$, recomputed for this step. Add these up.

4. **Delete ℓ from the pool** and move to position $i + 1$.
5. **Rank = (running total) + 1.** The +1 is the target word itself.

Two features cause every error in this topic, so read them twice.

- **The denominator is not frozen.** Removing a letter can destroy a repeat group. In SAMOSA, after you fix an A at position 1 the remaining pool is $\{A, M, O, S, S\}$ — only the S's still repeat, so the count is $5!/2!$, not $5!/(2!2!)$. Recompute the denominator at every branch.
- **Count distinct smaller letters, not smaller letter-slots.** If two A's are still unplaced, "start with A" is *one* branch, not two. Branching twice double-counts every word in it.

Sanity checks. The rank is an integer between 1 and the total arrangement count. The alphabetically first arrangement ranks 1, the last ranks $\frac{n!}{p!q!\dots}$. If your running total ever exceeds that ceiling, you forgot a denominator.

The inverse problem (k -th word). Walk the same tree, but subtract instead of add. Compute the block size for each first letter in alphabetical order; keep subtracting until k falls inside a block; fix that letter and repeat on the tail with the reduced k .

Figure brief: A left-to-right branching tree for the pool $\{A, A, G, I, N\}$. The root splits into four branches labelled A, G, I, N — four branches, not five, because the two A's give one branch. Each branch is annotated with its block size (24, 12, 12, 12) and a cumulative ruler underneath reading 1-24, 25-36, 37-48, 49-60. An arrow marked " $k = 50$ " drops into the N block, and the N sub-tree is expanded one more level to show NA (49-54) and inside it NAA (49-50).

Worked Example A1 — [CBSE Board Level]

Find the rank of the word **DRONA** in the dictionary-ordered list of all arrangements of its letters.

Solution. The letters D, R, O, N, A are all distinct. Sorted pool: A, D, N, O, R. Total arrangements = $5! = 120$, so the answer must lie in $[1, 120]$.

Position	Letter	Unplaced pool	Distinct smaller letters	Words skipped
1	D	A, D, N, O, R	A	$1 \times 4! = 24$
2	R	A, N, O, R	A, N, O	$3 \times 3! = 18$
3	O	A, N, O	A, N	$2 \times 2! = 4$
4	N	A, N	A	$1 \times 1! = 1$
5	A	A	none	0

Words preceding DRONA = $24 + 18 + 4 + 1 = 47$, so

$$\text{rank}(\text{DRONA}) = 47 + 1 = 48.$$

Run the same table on MATHS (sorted pool A, H, M, S, T) and you get: position 1, letter M, two smaller letters A and H, so $2 \times 4! = 48$; position 2, letter A, nothing smaller, 0; position 3, letter T, smaller letters H and S, so $2 \times 2! = 4$; positions 4 and 5 contribute nothing. Rank = $48 + 4 + 1 = 53$ — the answer Subtopic 05 §5 Q4 quotes without working.

Worked Example A2 — [JEE Main Level]

Find the rank of the word **KOLKATA**.

Solution. The pool is A, A, K, K, L, O, T: seven letters with A twice and K twice. Total arrangements = $\frac{7!}{2!2!} = 1260$ — the same 1260 you computed in Subtopic 02 §5 Q2. The answer must lie in $[1, 1260]$.

Position 1, letter K. Pool $\{A, A, K, K, L, O, T\}$. The only distinct smaller letter is A. Fixing A leaves $\{A, K, K, L, O, T\}$, in which only the K's repeat:

$$\frac{6!}{2!} = 360.$$

Position 2, letter O. One K is used; pool $\{A, A, K, L, O, T\}$. Distinct smaller letters: A, K, L. - Fix A: leaves $\{A, K, L, O, T\}$, all distinct, $5! = 120$. - Fix K: leaves $\{A, A, L, O, T\}$, A twice, $5!/2! = 60$. - Fix L: leaves $\{A, A, K, O, T\}$, A twice, $5!/2! = 60$.

Subtotal $120 + 60 + 60 = 240$. Notice the three branches have *different* denominators — this is the frozen-denominator warning in action.

Position 3, letter L. Pool $\{A, A, K, L, T\}$. Smaller: A, K. - Fix A: leaves $\{A, K, L, T\}$, $4! = 24$. - Fix K: leaves $\{A, A, L, T\}$, $4!/2! = 12$.

Subtotal 36.

Position 4, letter K. Pool $\{A, A, K, T\}$. Smaller: A only. Fix A: leaves $\{A, K, T\}$, $3! = 6$. Subtotal 6.

Position 5, letter A. Pool $\{A, A, T\}$. No smaller letter. Subtotal 0.

Position 6, letter T. Pool $\{A, T\}$. Smaller: A. Fix A: leaves $\{T\}$, $1! = 1$. Subtotal 1.

Position 7, letter A. Nothing left to beat it. Subtotal 0.

$$\text{rank(KOLKATA)} = (360 + 240 + 36 + 6 + 0 + 1 + 0) + 1 = 643 + 1 = 644.$$

The total 643 sits comfortably below the ceiling 1260, as it must.

Worked Example A3 — [JEE Main — Speed Trap]

Find the rank of the word **SAMOS A**.

The trap: the undivided-block trap. The student recognises the method from the all-distinct case and applies it verbatim, using plain factorials for the tail counts. At position 1 the letter is S and the distinct smaller letters are A, M, O, so the student writes $3 \times 5! = 360$, then continues:

$$360 + 0 + 1 \cdot 3! + 1 \cdot 2! + 1 \cdot 1! + 1 = 360 + 6 + 2 + 1 + 1 = \boxed{370} \text{ (wrong).}$$

Why 370 is impossible without computing anything. SAMOSA has 6 letters with A twice and S twice, so there are only $\frac{6!}{2!2!} = 180$ distinct arrangements. A rank of 370 places the word past the end of a 180-word list. The first branch alone, 360, already doubles the entire dictionary. **Check your running total against the ceiling after the very first position** — it costs three seconds and kills this error outright.

Correct solution. Sorted pool: A, A, M, O, S, S. Total = 180.

Position 1, letter S. Pool $\{A, A, M, O, S, S\}$. Distinct smaller: A, M, O. - Fix A: leaves $\{A, M, O, S, S\}$ — the A's no longer repeat, the S's do: $5!/2! = 60$. - Fix M: leaves $\{A, A, O, S, S\}$: $5!/(2!2!) = 30$. - Fix O: leaves $\{A, A, M, S, S\}$: $5!/(2!2!) = 30$.

Subtotal $60 + 30 + 30 = 120$. (The A-branch is *double* the other two, precisely because fixing an A dissolves a repeat group. The wrong solution above treated all three branches as equal.)

Position 2, letter A. Pool $\{A, A, M, O, S\}$. Nothing is smaller than A. Subtotal 0.

Position 3, letter M. Pool $\{A, M, O, S\}$, all distinct. Smaller: A. $3! = 6$.

Position 4, letter O. Pool $\{A, O, S\}$. Smaller: A. $2! = 2$.

Position 5, letter S. Pool $\{A, S\}$. Smaller: A. $1! = 1$.

Position 6, letter A. Subtotal 0.

$$\text{rank(SAMOS A)} = (120 + 0 + 6 + 2 + 1 + 0) + 1 = 129 + 1 = 130.$$

The second, quieter trap. Stopping at 129 and calling it the rank. The count of words *before* SAMOSA is 129; SAMOSA is the 130th. Write “+1” on the page as soon as you start the table, before you have anything to add it to.

Worked Example A4 — [JEE Advanced Level]

- (a) If all distinct arrangements of the letters of **EXAMINATION** are listed as in a dictionary, how many words appear before the first word beginning with E?
- (b) If all distinct arrangements of the letters of **AGAIN** are listed as in a dictionary, what is the 50th word?

Solution (a). EXAMINATION has 11 letters: A twice, I twice, N twice, and E, M, O, T, X once each. The sorted pool is A, A, E, I, I, M, N, N, O, T, X, and the full dictionary has

$$\frac{11!}{2!2!2!} = \frac{39,916,800}{8} = 4,989,600 \text{ words.}$$

A word precedes every E-word exactly when its first letter is alphabetically before E. The only such letter in the pool is A. So we need the number of arrangements beginning with A. Fix one A; the remaining pool is $\{A, E, I, I, M, N, N, O, T, X\}$, ten letters with I twice and N twice (the A’s no longer repeat):

$$\frac{10!}{2!2!} = \frac{3,628,800}{4} = 907,200.$$

907,200 words precede the first E-word. Note that “A twice” appears in the original pool but contributes one branch, not two — this is the second warning from the Method.

Solution (b). AGAIN has 5 letters with A twice; sorted pool A, A, G, I, N; total $\frac{5!}{2!} = 60$ words. Work down the first-letter blocks, subtracting as you go.

First letter	Remaining pool	Block size	Ranks covered
A	$\{A, G, I, N\}$	$4! = 24$	1-24
G	$\{A, A, I, N\}$	$4!/2! = 12$	25-36
I	$\{A, A, G, N\}$	$4!/2! = 12$	37-48
N	$\{A, A, G, I\}$	$4!/2! = 12$	49-60

Rank 50 lies in the N block, and it is the $50 - 48 = 2$ nd word of that block. Fix N; solve “2nd arrangement of $\{A, A, G, I\}$ ”.

- Second letter A: remaining $\{A, G, I\}$, block size $3! = 6$, covering positions 1-6 of the N block. Our target, position 2, is inside. Fix A; solve “2nd arrangement of $\{A, G, I\}$ ”.
- Third letter A: remaining $\{G, I\}$, block size $2! = 2$, covering positions 1-2. Our target is inside. Fix A; solve “2nd arrangement of $\{G, I\}$ ”.
- The two arrangements of $\{G, I\}$ in order are GI and IG. The 2nd is IG.

Assembling: N, A, A, I, G. **The 50th word is NAAIG** (and the 49th is NAAGI).

Both parts are NCERT items — Miscellaneous Exercise Q4 and Example 22 respectively — which is why the technique is a Board liability, not only a JEE one.

Practice.

1. [CBSE] Find the rank of the word DELHI.
2. [CBSE] Find the rank of the word BOARD.
3. [JEE Main] Find the rank of the word INDIA.
4. [JEE Main] Find the rank of the word GANGA.
5. [JEE Main] Find the rank of the word BHARAT.
6. [JEE Advanced] Find the rank of the word SURITI.
7. [JEE Advanced] If all arrangements of the letters of EXAMINATION are listed as in a dictionary, how many words precede the first word starting with E?
8. [JEE Advanced] If all arrangements of the letters of AGAIN are listed as in a dictionary, find the 50th word.

Answers.

1. 5.
 2. 38.
 3. 46 (out of $5!/2! = 60$; the repeated I matters at positions 1 and 3).
 4. 18 (out of $5!/(2!2!) = 30$).
 5. 147 (out of $6!/2! = 360$; A appears twice).
 6. 236 (out of $6!/2! = 360$; I appears twice).
 7. 907,200.
 8. NAAIG.
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Addendum 07B — Lattice-path counting

Extends Subtopic 02 §2 (alike objects) and Subtopic 03. Uses Subtopic 06 for the multi-obstacle case.

Exam Relevance: CBSE Boards does not name this pattern, but its engine — the alike-objects formula — is squarely on the syllabus, and grid questions appear in sample papers as applications. JEE Main asks it as a one-step $\binom{m+n}{m}$ recall, usually dressed as city streets or a chessboard. JEE Advanced is where it earns its place: paths forced through a checkpoint, paths banned from a point, two walkers meeting, and the Vandermonde identity that falls out of the meeting count. NEET does not test mathematics.

Method. Put a rectangular street grid on axes with the start at the origin. A walker may take only **unit steps right (R)** and **unit steps up (U)** — no backtracking, no diagonals. Such a path is called *monotonic*.

The whole idea in one line: a path is a word. To get from $(0, 0)$ to (m, n) you must take exactly m R-steps and exactly n U-steps, in some order, and the order is the path. So counting paths is counting arrangements of the letter-string $\underbrace{RR \cdots R}_m \underbrace{UU \cdots U}_n$, which is Subtopic 02's alike-objects count:

$$\# \text{paths} = \frac{(m+n)!}{m!n!} = \binom{m+n}{m} = \binom{m+n}{n}.$$

Every grid question is then one of four moves.

Requirement	Count
Free path $(0, 0) \rightarrow (m, n)$	$\binom{m+n}{m}$
Must pass through $P(a, b)$	$\binom{a+b}{a} \times \binom{(m-a)+(n-b)}{m-a}$
Must avoid $P(a, b)$	$\binom{m+n}{m} - [\text{through } P]$
Must avoid several points	inclusion-exclusion (Subtopic 06) on “passes through P_i ”

The checkpoint rule is the Multiplication Principle: the trip splits into an independent first leg and second leg, because arriving at P erases all memory of how you got there.

Sanity checks. $\binom{m+n}{m}$ is symmetric in m and n , as it must be — reflecting the grid in the line $y = x$ is a bijection on paths. A checkpoint count is never larger than the free count. A path through $P(a, b)$ exists only if $0 \leq a \leq m$ and $0 \leq b \leq n$; a “checkpoint” outside the rectangle contributes zero, not an error.

Where this breaks down. The formula counts *monotonic* paths only. Allow a diagonal step, or allow the walker to move left or down, and the path is no longer a rearrangement of a fixed letter-string, so no binomial coefficient describes it. Similarly, a constraint that depends on the *whole history* — for example “the path must never rise above the line $y = x$ ” — is not a plain combination; that count is the Catalan number $\frac{1}{n+1} \binom{2n}{n}$ (for $n = 1, 2, 3, 4, 5$: 1, 2, 5, 14, 42), which is beyond this syllabus but worth recognising so you do not waste minutes forcing a binomial onto it.

Figure brief: A 4-by-3 grid of city blocks with the origin marked A at the bottom left and B at the top right. One specific staircase path is drawn in bold, and directly beneath the grid the same path is written as the letter string R R U R U R U, with braces showing “4 R’s” and “3 U’s” and a caption reading “one path = one arrangement of RRRRUUU”. A faded second path is overlaid to show a different arrangement of the same letters.

Worked Example B1 — [CBSE Board Level]

A town is laid out as a grid. A courier starts at corner A and must reach corner B, which is 4 blocks east and 3 blocks north of A. He never rides west or south. In how many different routes can he make the trip?

Solution. Every route consists of exactly 4 east steps and 3 north steps, taken in some order.

The route is the order. So we are arranging the string EEEENNN, four alike of one kind and three alike of another:

$$\frac{7!}{4!3!} = \binom{7}{4} = 35.$$

There are **35 routes**. (Check the symmetry: $\binom{7}{4} = \binom{7}{3} = 35$, which is the same problem with east and north swapped — as it should be.)

Worked Example B2 — [JEE Main Level]

Find the number of monotonic paths from $(0, 0)$ to $(5, 5)$ that pass through the point $(2, 3)$.

Solution. Split the journey at the checkpoint and multiply the legs.

- Leg 1, $(0, 0) \rightarrow (2, 3)$: two R-steps and three U-steps, so $\binom{5}{2} = 10$ paths.
- Leg 2, $(2, 3) \rightarrow (5, 5)$: the displacement is $(3, 2)$, so three R-steps and two U-steps, giving $\binom{5}{3} = 10$ paths.

Every leg-1 path can be followed by every leg-2 path, so by the Multiplication Principle

$$10 \times 10 = 100 \text{ paths.}$$

For scale, the free count is $\binom{10}{5} = 252$, so the checkpoint filters out rather more than half the routes — and 100 being less than 252 is your sanity check.

Worked Example B3 — [JEE Main — Speed Trap]

Find the number of monotonic paths from $(0, 0)$ to $(4, 4)$ that do **not** pass through the point $(2, 2)$.

The trap: the additive-legs trap. The student correctly writes “answer = total – through”, then computes “through $(2, 2)$ ” by **adding** the two legs instead of multiplying them:

$$\binom{8}{4} - \left[\binom{4}{2} + \binom{4}{2} \right] = 70 - (6 + 6) = \boxed{58} \text{ (wrong).}$$

A close cousin subtracts a single path — “there is one bad point, so remove one route” — giving $70 - 1 = 69$.

Why the legs multiply. Reaching $(2, 2)$ in one of 6 ways and *then* continuing in one of 6 ways is a two-stage AND, and Subtopic 01 settled that AND multiplies. Adding would answer the question “how many paths use leg 1 **or** leg 2”, which is not a question anybody asked, and it is not even well posed since every path has both legs.

Correct solution.

- All paths $(0, 0) \rightarrow (4, 4)$: four R’s and four U’s, $\binom{8}{4} = 70$.
- Paths through $(2, 2)$: $\binom{4}{2} \times \binom{4}{2} = 6 \times 6 = 36$.

- Paths avoiding (2, 2): $70 - 36 = 34$.

How to spot it in three seconds. After computing a “through a checkpoint” figure, ask whether it should be roughly the *product* or the *sum* of the legs. Here 36 out of 70 — just over half the paths pass through the exact centre of the square, which is entirely believable. The trap’s 12 out of 70 would mean the centre point is nearly deserted, which for a 4×4 grid it plainly is not.

Worked Example B4 — [JEE Advanced Level]

Two insects sit at opposite corners of an ordinary chessboard, on the lattice of lines: the first at (0, 0), the second at (8, 8). They start together and crawl at the same constant speed along the grid lines. The first insect moves only right or up; the second moves only left or down. Find the total number of ways in which the two insects can meet at a point.

Solution.

Step 1 — where can they meet? Both crawl at the same speed and start together, so at any instant they have covered equal distances. The total gap is $8 + 8 = 16$ units, so if they meet, each has covered exactly 8 units. After 8 unit steps the first insect stands at some $(k, 8 - k)$; the second, having taken 8 steps back from (8, 8), stands at some $(8 - j, 8 - (8 - j))$ — that is, also on the anti-diagonal $x + y = 8$. So the meeting points are exactly the nine lattice points $(k, 8 - k)$ for $k = 0, 1, \dots, 8$.

Step 2 — count at each meeting point. The first insect reaches $(k, 8 - k)$ in $\binom{8}{k}$ ways. The second insect must travel from (8, 8) to $(k, 8 - k)$, a displacement of $8 - k$ left steps and k down steps, which it can do in $\binom{8}{k}$ ways as well.

Step 3 — combine. The two journeys are independent, so multiply, then add over the nine disjoint meeting points:

$$\sum_{k=0}^8 \binom{8}{k} \binom{8}{k} = \sum_{k=0}^8 \binom{8}{k}^2.$$

Step 4 — evaluate. Use Subtopic 03’s symmetry $\binom{8}{k} = \binom{8}{8-k}$ to rewrite the sum as $\sum_k \binom{8}{k} \binom{8}{8-k}$. That is exactly the count of ways to choose 8 objects from a pile of 8 white and 8 black, split by how many are white — that is, $\binom{16}{8}$. Hence

$$\sum_{k=0}^8 \binom{8}{k}^2 = \binom{16}{8} = 12,870.$$

Direct arithmetic check. $1 + 64 + 784 + 3136 + 4900 + 3136 + 784 + 64 + 1 = 12,870$, using $\binom{8}{k} = 1, 8, 28, 56, 70, 56, 28, 8, 1$. Agreed.

What was really tested. Not the grid formula — that was Step 2 — but the recognition in Step 1 that equal speed pins the meeting to one anti-diagonal, and the identity in Step 4 (Vandermonde’s, in the special case $\sum \binom{n}{k}^2 = \binom{2n}{n}$) that turns a nine-term sum into a single coefficient.

Practice.

1. [CBSE] How many monotonic paths run from $(0, 0)$ to $(3, 3)$?
2. [CBSE] A grid is 5 blocks east and 5 blocks north. How many shortest routes cross it?
3. [JEE Main] How many monotonic paths from $(0, 0)$ to $(6, 4)$ pass through $(3, 2)$?
4. [JEE Main] How many monotonic paths from $(0, 0)$ to $(5, 5)$ avoid the point $(2, 3)$?
5. [JEE Advanced] How many monotonic paths from $(0, 0)$ to $(9, 9)$ never pass through $(5, 5)$?
6. [JEE Advanced] How many monotonic paths from $(0, 0)$ to $(6, 6)$ avoid **both** $(2, 2)$ and $(4, 4)$?
7. [JEE Advanced] Rework the two-insect problem on a 4×4 board (insects at $(0, 0)$ and $(4, 4)$).

Answers.

1. $\binom{6}{3} = 20$.
2. $\binom{10}{5} = 252$.
3. $\binom{5}{3}\binom{5}{3} = 10 \times 10 = 100$.
4. $\binom{10}{5} - \binom{5}{2}\binom{5}{3} = 252 - 100 = 152$.
5. $\binom{18}{9} - \binom{10}{5}\binom{8}{4} = 48,620 - 252 \times 70 = 48,620 - 17,640 = 30,980$.
6. $924 - 420 - 420 + 216 = 300$. (Inclusion-exclusion: total $\binom{12}{6} = 924$; through $(2, 2)$ is $\binom{4}{2}\binom{8}{4} = 420$; through $(4, 4)$ is $\binom{8}{4}\binom{4}{2} = 420$; through both is $\binom{4}{2}\binom{4}{2}\binom{4}{2} = 216$, added back because paths through both were removed twice.)
7. $\sum_{k=0}^4 \binom{4}{k}^2 = \binom{8}{4} = 70$.

Addendum 07C — The divisor toolkit beyond $d(N)$ and $\sigma(N)$

Extends Subtopic 05 §2 and §3.2, which give only the divisor count and the divisor sum.

Exam Relevance: CBSE Boards stops at the divisor count, occasionally the divisor sum, and Subtopic 05 already serves that. Everything in this addendum is JEE territory and it is dense territory: “in how many ways can N be written as a product of two factors”, “as a product of two coprime factors”, “how many divisors of N are even / odd / divisible by 12”, “find the product of all divisors”, and “how many ordered triples satisfy $xyz = N$ ” are all standard single-question items in JEE Main, and JEE Advanced chains two or three of them into one multi-part problem. NEET does not test mathematics.

Method. Everything below sits on the one idea Subtopic 05 §3.2 already established: if

$$N = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k}$$

with distinct primes p_i , then a divisor is exactly a choice of exponent e_i with $0 \leq e_i \leq a_i$, independently for each prime. Every formula here is that choice with one extra condition bolted on.

Write $d(N) = \prod (a_i + 1)$ for the divisor count and $\sigma(N) = \prod \frac{p_i^{a_i+1} - 1}{p_i - 1}$ for the divisor sum, both from Subtopic 05.

1. Product of all divisors.

$$P(N) = N^{d(N)/2}.$$

Why: pair each divisor t with its partner N/t ; each pair multiplies to N , and there are $d(N)/2$ pairs. If N is a perfect square the middle divisor $\sqrt{N}$ pairs with itself, and the formula still holds because $N^{1/2} = \sqrt{N}$ is exactly what that lone divisor contributes. Leave the answer as a power; never evaluate it.

2. N as a product of two factors. Ordered pairs (a, b) with $ab = N$ number $d(N)$, one for each choice of a . Unordered pairs $\{a, b\}$ are what exams ask for:

$$\#\{a, b\} = \begin{cases} \frac{d(N)}{2}, & N \text{ not a perfect square,} \\ \frac{d(N) + 1}{2}, & N \text{ a perfect square.} \end{cases}$$

Why: the pairing $t \leftrightarrow N/t$ is a bijection with no fixed point unless $t = \sqrt{N}$. A perfect square has exactly one such self-paired divisor, so $d(N)$ is odd and one pair is a singleton. **N is a perfect square precisely when every a_i is even, equivalently when $d(N)$ is odd** — that is your test, and it is faster than taking a square root.

3. N as a product of two coprime factors. Let k be the number of **distinct** primes in N . Then

$$\#\text{ordered coprime pairs} = 2^k, \quad \#\text{unordered coprime pairs} = 2^{k-1}.$$

Why: if $\gcd(a, b) = 1$ and $ab = N$, then for each prime the whole block $p_i^{a_i}$ must go entirely to a or entirely to b — splitting it would leave p_i in both. That is k independent two-way choices, and unordered halves it. Notice the exponents play no part whatsoever; only the number of distinct primes matters.

4. Divisors satisfying a divisibility condition. For $m \mid N$, the divisors of N that are multiples of m are exactly $m \times (\text{divisors of } N/m)$, a bijection. Hence

$$\#\{t \mid N : m \mid t\} = d(N/m), \quad \sum_{\substack{t \mid N \\ m \mid t}} t = m \cdot \sigma(N/m).$$

The two most-asked special cases follow immediately. Write $N = 2^a M$ with M odd: - **odd divisors**: count $d(M)$, sum $\sigma(M)$ — just delete the prime 2 from the factorisation; - **even divisors**: count $d(N) - d(M) = a \cdot d(M)$, sum $\sigma(N) - \sigma(M)$.

5. Ordered triples with $xyz = N$. Each prime's exponent a_i must be split among the three factors: $e_i^{(x)} + e_i^{(y)} + e_i^{(z)} = a_i$ in non-negative integers, which is Subtopic 04's stars and bars, $\binom{a_i+2}{2}$ ways. The primes are independent, so

$$\#\{(x, y, z) \in \mathbb{N}^3 : xyz = N\} = \prod_{i=1}^k \binom{a_i + 2}{2}.$$

For *integer* solutions with $N > 0$, the sign pattern must have an even number of negatives: all three positive, or exactly two negative ($\binom{3}{2} = 3$ ways). Multiply by $1 + 3 = 4$.

Sanity checks. Every count here is a positive integer; a fractional answer to item 2 means you missed that N is a perfect square. The coprime-pair count never exceeds the total pair count. The odd-divisor count divides the total divisor count.

Figure brief: A ladder diagram for $N = 360 = 2^3 \cdot 3^2 \cdot 5$. Three vertical columns headed $2^0..2^3$, $3^0..3^2$, $5^0..5^1$ with 4, 3 and 2 rungs respectively. One highlighted path picks $2^2, 3^0, 5^1$ and is captioned “one path = one divisor = 20”. Beside it a smaller panel shows the same three columns with each column shaded entirely one of two colours, captioned “coprime split: colour whole columns, never rungs — $2^3 = 8$ colourings, 4 unordered pairs”.

Worked Example C1 — [CBSE Board / Foundational Level]

For $N = 1200$, find (a) the number of divisors, (b) the sum of the divisors, (c) the number and sum of the **odd** divisors, (d) the number and sum of the **even** divisors.

Solution. Factorise: $1200 = 2^4 \cdot 3^1 \cdot 5^2$.

(a) $d(1200) = (4+1)(1+1)(2+1) = 5 \times 2 \times 3 = \mathbf{30}$. This is the figure Subtopic 05 §5 Q2 already asked for.

(b) $\sigma(1200) = \frac{2^5 - 1}{1} \cdot \frac{3^2 - 1}{2} \cdot \frac{5^3 - 1}{4} = 31 \times 4 \times 31 = \mathbf{3844}$.

(c) The odd part is $M = 3 \cdot 5^2 = 75$. Odd divisors of 1200 are precisely the divisors of 75:

$$d(75) = 2 \times 3 = \mathbf{6}, \quad \sigma(75) = 4 \times 31 = \mathbf{124}.$$

(Listing them as a check: 1, 3, 5, 15, 25, 75 — six of them, summing to 124.)

(d) Even divisors are everything else:

$$30 - 6 = \mathbf{24} \text{ of them,} \quad 3844 - 124 = \mathbf{3720}.$$

Cross-check on the count: every even divisor is $2^e \times (\text{odd divisor})$ with $1 \leq e \leq 4$, so there are $4 \times 6 = 24$ of them. Agreed.

Worked Example C2 — [JEE Main Level]

- (a) In how many ways can 7200 be resolved into a product of two factors?
- (b) In how many ways can 3600 be resolved into a product of two factors?

Solution. Run the perfect-square test first in both cases — it decides which formula applies.

(a) $7200 = 2^5 \cdot 3^2 \cdot 5^2$, so $d = (5+1)(2+1)(2+1) = 6 \times 3 \times 3 = 54$. The exponent of 2 is odd, so 7200 is **not** a perfect square (equivalently, $d = 54$ is even). Therefore

$$\frac{54}{2} = \mathbf{27} \text{ ways.}$$

(b) $3600 = 2^4 \cdot 3^2 \cdot 5^2$, so $d = 5 \times 3 \times 3 = 45$. Every exponent is even, so 3600 **is** a perfect

square ($3600 = 60^2$), and correspondingly d came out odd. Therefore

$$\frac{45 + 1}{2} = \mathbf{23} \text{ ways.}$$

The self-paired factorisation in (b) is 60×60 ; the other 22 pairs have two distinct members, and $22 \times 2 + 1 = 45 = d$, which confirms the bookkeeping.

Worked Example C3 — [JEE Main — Speed Trap]

In how many ways can 36 be expressed as a product of two positive factors? In how many ways as a product of two **coprime** positive factors?

The trap: the perfect-square parity trap. The student computes $36 = 2^2 \cdot 3^2$, $d = 3 \times 3 = 9$, then reaches for the memorised “ $d/2$ ”:

$$\frac{9}{2} = 4.5 \rightarrow \boxed{4} \text{ (wrong),}$$

rounding down, or occasionally writing 4.5 and moving on. **A count is never a fraction.** The moment $d(N)$ comes out odd you are looking at a perfect square, and the odd d is not an accident — it is the signal.

Correct solution, part 1. $d(36) = 9$ is odd, so 36 is a perfect square and

$$\frac{d + 1}{2} = \frac{10}{2} = \mathbf{5}.$$

Listing all five confirms it: 1×36 , 2×18 , 3×12 , 4×9 , 6×6 . The self-paired 6×6 is exactly the pair the naive halving throws away.

Correct solution, part 2. For coprime factors only the number of distinct primes matters. Here $k = 2$ (namely 2 and 3), so the number of unordered coprime pairs is

$$2^{k-1} = 2^1 = \mathbf{2}.$$

Indeed: 1×36 and 4×9 . Check the others — 2×18 shares 2, 3×12 shares 3, 6×6 shares both — so exactly two survive, as predicted. Note how the exponents 2 and 2 were irrelevant; had N been $2^{50} \cdot 3^{50}$ the coprime answer would still be 2.

Worked Example C4 — [JEE Advanced Level]

Let $N = 18900$. Find

- (a) the number of divisors of N ;
- (b) the number of ways N can be written as a product of two coprime factors;
- (c) the number of divisors of N that are divisible by 15, and their sum;
- (d) the product of all divisors of N .

Solution. Factorise once and use it four times:

$$18900 = 2^2 \cdot 3^3 \cdot 5^2 \cdot 7.$$

(a) $d(N) = (2+1)(3+1)(2+1)(1+1) = 3 \times 4 \times 3 \times 2 = \mathbf{72}$.

(b) The number of distinct primes is $k = 4$. Unordered coprime pairs:

$$2^{k-1} = 2^3 = \mathbf{8}.$$

The eight splits are obtained by sending each of the blocks 4, 27, 25, 7 to one side or the other and then identifying $\{a, b\}$ with $\{b, a\}$; for instance $\{4, 4725\}$, $\{27, 700\}$, $\{1, 18900\}$, $\{100, 189\}$ and four more.

(c) Since $15 \mid N$, the divisors of N divisible by 15 are exactly $15t$ where t runs over the divisors of

$$\frac{N}{15} = 1260 = 2^2 \cdot 3^2 \cdot 5 \cdot 7.$$

Hence the count is

$$d(1260) = 3 \times 3 \times 2 \times 2 = \mathbf{36},$$

and the sum is

$$15 \cdot \sigma(1260) = 15 \times \left(\frac{2^3 - 1}{1} \right) \left(\frac{3^3 - 1}{2} \right) \left(\frac{5^2 - 1}{4} \right) \left(\frac{7^2 - 1}{6} \right) = 15 \times 7 \times 13 \times 6 \times 8 = 15 \times 4368 = \mathbf{65,520}.$$

(d) Pairing t with N/t gives $d(N)/2 = 36$ pairs, each with product N :

$$P(N) = N^{36} = 18900^{36}.$$

Leave it in this form. No examiner wants the digits, and no marking scheme rewards them.

Why part (c) is the Advanced-level move. A student who tries to count “divisors with at least one factor 3 and at least one factor 5” by casework on exponents will get there — $3 \times 3 \times 2 \times 2$, choosing $e_2 \in \{0, 1, 2\}$, $e_3 \in \{1, 2, 3\}$, $e_5 \in \{1, 2\}$, $e_7 \in \{0, 1\}$ — but the N/m bijection gets there in one line and, crucially, hands you the *sum* for free, which the casework does not.

Practice.

1. [CBSE] Find the number and the sum of the divisors of 540.
2. [CBSE] How many divisors of 540 are odd, and what do they sum to?
3. [JEE Main] In how many ways can 2520 be resolved into a product of two factors?
4. [JEE Main] In how many ways can 3600 be resolved into a product of two factors?
5. [JEE Main] In how many ways can 2520 be written as a product of two coprime factors?
6. [JEE Main] How many divisors of 7200 are divisible by 12?
7. [JEE Advanced] Find the product of all the divisors of 1200.
8. [JEE Advanced] Find the number of ordered triples (x, y, z) of natural numbers with $xyz = 360$, and then the number of ordered triples of **integers** with $xyz = 360$.

Answers.

1. $540 = 2^2 \cdot 3^3 \cdot 5$; $d = 3 \times 4 \times 2 = 24$, $\sigma = 7 \times 40 \times 6 = 1680$.
2. Odd part $135 = 3^3 \cdot 5$; count $4 \times 2 = 8$, sum $40 \times 6 = 240$.
3. $2520 = 2^3 \cdot 3^2 \cdot 5 \cdot 7$; $d = 4 \times 3 \times 2 \times 2 = 48$, not a perfect square, so $48/2 = 24$.

4. 23 (perfect square; $d = 45$).
5. $k = 4$ distinct primes, so $2^{4-1} = 8$.
6. $7200/12 = 600 = 2^3 \cdot 3 \cdot 5^2$, so $d(600) = 4 \times 2 \times 3 = 24$.
7. $d(1200) = 30$, so the product is 1200^{15} .
8. $360 = 2^3 \cdot 3^2 \cdot 5$, so $\binom{5}{2}\binom{4}{2}\binom{3}{2} = 10 \times 6 \times 3 = 180$ natural triples; multiplying by the 4 admissible sign patterns gives $180 \times 4 = 720$ integer triples.

Addendum 07D — Partial derangements: exactly k objects in place

Extends Subtopic 05 §2 and §3.5, and the Round 1 Supplement's derangement derivation.

Exam Relevance: Subtopic 05 gives D_n , the count with **no** object in its own place, and that is where most students stop. Almost every actual exam question is the partial version: *exactly* two letters in the right envelopes, *at least* three guests with their own hats, *at least one* letter misplaced. JEE Main asks the “exactly k ” form directly; JEE Advanced buries it inside a probability question, where the partial-derangement count is the numerator. CBSE Boards asks the plain D_n only. NEET does not test mathematics.

Method. Suppose n distinct objects are matched one-to-one with n distinct places (letters into addressed envelopes, guests into their own hats, a permutation against the identity). An object is **in place** if it lands in its own slot; the number of in-place objects is the number of *fixed points* of the permutation.

The whole method in one formula:

$$\#\{\text{exactly } k \text{ in place}\} = \binom{n}{k} D_{n-k}.$$

Why: first choose **which** k objects are the correct ones, in $\binom{n}{k}$ ways. Then the remaining $n - k$ objects must **all** be wrong — if even one of them landed correctly, the total would exceed k and the word *exactly* would be violated. A permutation of those $n - k$ objects with no fixed point is precisely a derangement, so there are D_{n-k} of them. Multiply, by the Multiplication Principle.

Keep these values on hand (extending Subtopic 05's short list):

n	0	1	2	3	4	5	6	7	8
D_n	1	0	1	2	9	44	265	1854	14833

Note $D_0 = 1$ (the empty arrangement, consistent with $0! = 1$ from Subtopic 01) and $D_1 = 0$ — that single zero does real work, as Worked Example D3 shows.

The three derived counts.

Requirement	Count
Exactly k in place	$\binom{n}{k} D_{n-k}$
At least k in place	$\sum_{j=k}^n \binom{n}{j} D_{n-j}$
At least one out of place	$n! - 1$ (everything but the identity)

The identity that checks all your work. Every one of the $n!$ permutations has some exact number of fixed points, and the categories are disjoint and exhaustive, so

$$\sum_{k=0}^n \binom{n}{k} D_{n-k} = n!.$$

Compute the whole fixed-point distribution and add: if it does not total $n!$, you have a slip. This is the cheapest insurance available in the topic.

A useful estimate. For every $n \geq 1$, D_n is the nearest integer to $n!/e$. Since $1/e \approx 0.3679$, roughly 37 percent of all permutations are derangements — and that proportion barely moves as n grows. Use it to smell-test an answer, not to produce one.

Where this breaks down. The formula needs a genuine one-to-one matching between objects and places. If two envelopes could accept the same letter, or if some letters have no “correct” envelope at all, there is no identity permutation to compare against and D_n is the wrong tool — you are back to a direct inclusion-exclusion (Subtopic 06) with whatever forbidden positions the problem actually specifies.

Figure brief: A two-row diagram for $n = 5$. Top row: five letters L1 to L5. Bottom row: five envelopes E1 to E5. Arrows drawn so that exactly two letters (L2 and L4) point straight down to their own envelopes, boxed and labelled “chosen: $\binom{5}{2} = 10$ ways”, while the other three arrows form a crossing cycle among E1, E3, E5, boxed and labelled “deranged: $D_3 = 2$ ways”. Caption: “ $10 \times 2 = 20$ ”.

Worked Example D1 — [CBSE Board / Foundational Level]

Five letters are written and five envelopes are addressed, one for each letter. The letters are inserted at random, one per envelope. In how many ways can exactly two letters go into their correct envelopes?

Solution. Split the job into the two stages the method names.

- **Stage 1 — choose the two correct letters.** Any 2 of the 5 may be the lucky ones: $\binom{5}{2} = 10$ ways.
- **Stage 2 — the other three must all be wrong.** Otherwise “exactly two” fails. That is a derangement of 3 objects: $D_3 = 2$ ways.

By the Multiplication Principle,

$$\binom{5}{2} \times D_3 = 10 \times 2 = \mathbf{20}.$$

To see the $D_3 = 2$ concretely, take the three remaining letters A, B, C and their envelopes: the only arrangements with nothing in its own place are B, C, A and C, A, B . The four other arrangements each fix at least one letter.

Worked Example D2 — [JEE Main Level]

Six letters are placed at random into six correspondingly addressed envelopes, one per envelope. In how many ways do **at least two** letters go into their correct envelopes?

Solution. “At least two” means exactly 2, 3, 4, 5 or 6 — mutually exclusive cases, so count each and add (Addition Principle).

Exactly k correct	$\binom{6}{k}$	D_{6-k}	Product
2	15	$D_4 = 9$	135
3	20	$D_3 = 2$	40
4	15	$D_2 = 1$	15
5	6	$D_1 = 0$	0
6	1	$D_0 = 1$	1

$$135 + 40 + 15 + 0 + 1 = \mathbf{191}.$$

Verification by the master identity. Complete the table with the two missing rows: exactly 0 correct is $D_6 = 265$, exactly 1 correct is $\binom{6}{1}D_5 = 6 \times 44 = 264$. The whole distribution is

$$265 + 264 + 135 + 40 + 15 + 0 + 1 = 720 = 6!.$$

Every permutation is accounted for exactly once, so the 191 is safe.

Worked Example D3 — [JEE Main — Speed Trap]

Five letters are inserted at random into five correspondingly addressed envelopes, one per envelope. In how many ways can exactly four letters go into their correct envelopes?

The trap: the last-one-is-forced trap. The student reads “choose which four are correct” and stops there:

$$\binom{5}{4} = \boxed{5} \text{ (wrong).}$$

It looks like a complete answer — five choices, five ways — and it is the single most common wrong answer in the topic.

Correct solution. Apply the formula without skipping the second factor:

$$\binom{5}{4} D_1 = 5 \times 0 = \mathbf{0}.$$

Why zero, in words. Suppose four letters are in their correct envelopes. Then four envelopes are filled correctly and exactly one letter and one envelope remain — and that remaining envelope is necessarily the remaining letter’s own. The fifth letter has nowhere

wrong to go. So “exactly four correct” describes no arrangement at all; the moment $n - 1$ are right, the n th is right too, and you are in the “all n correct” case.

The general statement worth memorising: *exactly $n - 1$ in place is impossible for every n , because $D_1 = 0$.* Any exam option offering a non-zero count for it is a distractor. Equally, note the neighbouring value is not zero: exactly three correct out of five is $\binom{5}{3}D_2 = 10 \times 1 = 10$, perfectly possible — the last two letters simply swap.

Worked Example D4 — [JEE Advanced Level]

Six letters are placed at random into six correspondingly addressed envelopes, one letter per envelope.

- In how many ways is at least one letter in a wrong envelope?
- Verify your answer by summing the full fixed-point distribution.
- If instead the six letters are placed into the six envelopes at random, what is the probability that no letter reaches its correct envelope?

Solution.

(a) The one-line route. “At least one wrong” is the complement of “all six right”, and there is exactly one way for all six to be right — the identity arrangement. Hence

$$6! - 1 = 720 - 1 = \mathbf{719}.$$

Subtopic 01’s Pro-Tip, *at least one means complement*, is doing all the work here.

(b) The long route, as a check. “At least one wrong” is the same as “at most five right”, so sum the exactly- k counts for $k = 0$ through 5:

$$\underbrace{265}_{k=0} + \underbrace{264}_{k=1} + \underbrace{135}_{k=2} + \underbrace{40}_{k=3} + \underbrace{15}_{k=4} + \underbrace{0}_{k=5} = 719.$$

The two routes agree. Observe how much cheaper (a) was — recognising the complement is the skill being tested, and the six-term sum is the punishment for missing it.

(c) Favourable outcomes are the derangements, $D_6 = 265$; total outcomes $6! = 720$:

$$P = \frac{265}{720} = \frac{53}{144} \approx 0.3681.$$

Compare with the estimate $1/e \approx 0.3679$. The agreement to three decimal places at $n = 6$ is typical, and it is why “about 37 percent” is a safe mental benchmark for any derangement probability you meet later in Probability.

Practice.

- [CBSE] Write down D_4 and D_5 .
- [CBSE] Four letters are placed at random into four correspondingly addressed envelopes. In how many ways does exactly one letter reach its correct envelope?
- [JEE Main] Six guests leave their hats at a party and collect one each at random. In how many ways do exactly two guests get their own hats?

4. [JEE Main] Seven letters go into seven addressed envelopes at random. In how many ways do exactly three reach the correct envelope?
5. [JEE Main] Five letters go into five addressed envelopes at random. In how many ways do at least three reach the correct envelope?
6. [JEE Advanced] Six letters go into six addressed envelopes. In how many ways is at least one letter misplaced?
7. [JEE Advanced] Show that the six exactly- k counts for $n = 6$ sum to 720.
8. [JEE Advanced] Eight distinct objects are permuted at random. In how many permutations are exactly three objects in their original positions?

Answers.

1. $D_4 = 9, D_5 = 44$.
2. $\binom{4}{1}D_3 = 4 \times 2 = 8$.
3. $\binom{6}{2}D_4 = 15 \times 9 = 135$.
4. $\binom{7}{3}D_4 = 35 \times 9 = 315$.
5. $\binom{5}{3}D_2 + \binom{5}{4}D_1 + \binom{5}{5}D_0 = 10 + 0 + 1 = 11$.
6. $6! - 1 = 719$.
7. $265 + 264 + 135 + 40 + 15 + 0 + 1 = 720 = 6!$.
8. $\binom{8}{3}D_5 = 56 \times 44 = 2464$.

Cheat sheet for this addendum

ROUND 2 ADDENDUM — 60-SECOND RECALL

Rank of a word (07A). Alphabetise the pool. At each position, for every *distinct* unused letter smaller than the current one, add the arrangement count of the pool with that letter removed — recomputing the repeat-denominator every time. Finish with +1. Ceiling check: the rank lies in $\left[1, \frac{n!}{p!q!\dots}\right]$. For the k -th word, subtract block sizes instead of adding. Worked values: DRONA 48, MATHS 53, SAMOSA 130, KOLKATA 644; EXAMINATION has 907,200 words before the first E-word; the 50th word of AGAIN is NAAIG.

Lattice paths (07B). A monotonic path is a word in R's and U's.

$$(0, 0) \rightarrow (m, n) : \binom{m+n}{m}. \quad \text{Through } P(a, b) : \binom{a+b}{a} \binom{m-a+n-b}{m-a}. \quad \text{Avoiding } P : \text{total} - \text{through}.$$

Several forbidden points: inclusion-exclusion (Subtopic 06). Legs **multiply**, never add. Equal-speed walkers from opposite corners of an $n \times n$ grid meet on one anti-diagonal in $\sum_k \binom{n}{k}^2 = \binom{2n}{n}$ ways. Monotonic only — diagonal steps or history-dependent bans (Catalan, $\frac{1}{n+1}\binom{2n}{n}$) are outside the formula.

Divisor toolkit (07C). For $N = \prod p_i^{a_i}$ with k distinct primes: - $d(N) = \prod (a_i + 1)$; $\sigma(N) = \prod \frac{p_i^{a_i+1} - 1}{p_i - 1}$ (Subtopic 05). - Product of all divisors: $N^{d(N)/2}$. - Two factors, unordered: $d/2$ if N is not a perfect square, $(d+1)/2$ if it is. **N is a perfect square exactly when $d(N)$**

is odd — a fractional answer means you missed it. - Two **coprime** factors: 2^{k-1} unordered, 2^k ordered. Exponents are irrelevant. - Divisors divisible by m (with $m \mid N$): count $d(N/m)$, sum $m \sigma(N/m)$. Odd divisors: delete the prime 2. Even divisors: total minus odd. - $xyz = N$ in naturals: $\prod \binom{a_i+2}{2}$; in integers, multiply by 4.

Partial derangements (07D).

$$\#\{\text{exactly } k \text{ fixed}\} = \binom{n}{k} D_{n-k}, \quad \sum_{k=0}^n \binom{n}{k} D_{n-k} = n!.$$

D_n : 1, 0, 1, 2, 9, 44, 265, 1854, 14833 for $n = 0..8$. D_n is the nearest integer to $n!/e$, so about 37 percent of permutations are derangements. **Exactly** $n - 1$ **fixed is impossible** ($D_1 = 0$). “At least one misplaced” is $n! - 1$, not a five-term sum.

Mnemonics: “Rank = count the earlier words, then +1.” | “A grid path is a word in R and U.” | “Odd divisor count means perfect square.” | “Coprime split colours whole prime blocks.” | “Fix k , derange the rest.”

Chapter 7 — Binomial Theorem

Class 11 Mathematics · Drona Ed Tech

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Subtopic 01: Binomial Expansion and Pascal's Triangle

Exam Relevance: This is the foundation slab of the whole chapter — every later idea (general term, middle term, greatest coefficient, coefficient-summation problems) is built on it. JEE Main almost always carries 1 direct question rooted here (number of terms, a coefficient, sum of coefficients), and JEE Advanced uses it as scaffolding inside harder multi-concept problems. It is also a standard CBSE Class 11 short-answer/expansion question and a prerequisite for Class 12 Proba-

bility and Sequences & Series.

Section 1: Concept Intuition

Suppose your local kirana shop sells two kinds of snacks: a and b . You want to make a “combo packet” by picking one snack from each of n identical boxes, where every box contains exactly one a and one b . From each box you choose either an a or a b . How many *kinds* of combo packets can you build, and how many of each kind?

That, secretly, is the entire binomial theorem. When you multiply out

$$(a + b)(a + b)(a + b) \cdots (a + b) \quad (n \text{ brackets}),$$

every term in the final answer is made by walking through the n brackets and picking either the a or the b from each one. If you pick b from exactly r brackets (and therefore a from the remaining $n - r$), you get the product $a^{n-r}b^r$. The **number of different ways** to choose *which* r brackets give you the b is exactly nC_r — the count of ways to choose r things out of n . So the term $a^{n-r}b^r$ shows up nC_r times, and its coefficient is nC_r .

Think about it this way... the powers and the counting are the *same story told twice*. The power of b (“how many boxes gave a b ”) and the coefficient (“how many ways that could happen”) are joined at the hip. This is why the coefficients are called **binomial coefficients** — they are literally combination counts. It also instantly explains two facts students otherwise memorise blindly:

- In every term, the powers of a and b add up to n (you made exactly n choices, so $(n - r) + r = n$). The expansion is *homogeneous* of degree n .
- The coefficients are symmetric: ${}^nC_r = {}^nC_{n-r}$, because choosing which r brackets give b is the same as choosing which $n - r$ brackets give a .

Pascal’s Triangle is just a lazy, beautiful way to read off these counts without computing factorials. Write 1 at the top, then each new row starts and ends with 1, and every interior number is the *sum of the two numbers diagonally above it*. Each row hands you the coefficients of $(a + b)^n$ directly.

Diagram placeholder (for illustrator)

Pascal’s triangle, rows $n=0$ through $n=5$. Row 0: 1. Row 1: 1 1. Row 2: 1 2 1. Row 3: 1 3 3 1. Row 4: 1 4 6 4 1. Row 5: 1 5 10 10 5 1. Draw small downward arrows from each adjacent pair in a row to the number below them showing the “add the two above” rule, e.g. 3 and 3 in row 3 feeding into 6 in row 4. Label each row on the right with the matching power: $(a+b)^0$, $(a+b)^1$, ..., $(a+b)^5$.

The reason the addition rule works is again pure counting: a term in $(a + b)^n$ comes either from multiplying an a into a term of $(a + b)^{n-1}$ or a b into a term of $(a + b)^{n-1}$. Those two “parents” are the two numbers sitting above. This is **Pascal’s rule**, ${}^{n-1}C_{r-1} + {}^{n-1}C_r = {}^nC_r$, which is the algebraic skeleton of the triangle.

Validity conditions & constraints. In *this* subtopic, the index n is a **positive integer** ($n \in \mathbb{N}$). Only then does the expansion terminate after a clean $n + 1$ terms with whole-number combination coefficients. If n is negative or fractional (e.g. $(1 + x)^{-1}$ or $(1 + x)^{1/2}$), the story changes completely — the expansion becomes an *infinite series* that only converges when $|x| < 1$. That generalisation is a separate subtopic; do not apply Pascal’s triangle there.

Section 2: Key Formulas & Definitions

Binomial Theorem (positive integral index). For $n \in \mathbb{N}$,

$$(a + b)^n = \sum_{r=0}^n {}^nC_r a^{n-r} b^r = {}^nC_0 a^n + {}^nC_1 a^{n-1}b + \dots + {}^nC_n b^n.$$

The same theorem in the form most exams use (powers of a single variable x):

$$(1 + x)^n = \sum_{r=0}^n {}^nC_r x^r = 1 + {}^nC_1 x + {}^nC_2 x^2 + \dots + {}^nC_n x^n.$$

Binomial coefficient.

$${}^nC_r = \binom{n}{r} = \frac{n!}{r!(n-r)!}, \quad 0 \leq r \leq n, \quad n, r \in \mathbb{Z}_{\geq 0}.$$

(We use nC_r throughout; $\binom{n}{r}$ is the same object.)

General term (the $(r + 1)$ th term, counting from the start):

$$T_{r+1} = {}^nC_r a^{n-r} b^r, \quad r = 0, 1, 2, \dots, n.$$

Core properties (each valid for $n \in \mathbb{N}$, $0 \leq r \leq n$):

Property	Statement	Validity / constraint
Number of terms	$n + 1$	$n \in \mathbb{N}$
Sum of indices	power of a + power of b $= n$ in every term	always
Symmetry	${}^nC_r = {}^nC_{n-r}$	$0 \leq r \leq n$
Pascal’s rule	${}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$	$1 \leq r \leq n$
End coefficients	${}^nC_0 = {}^nC_n = 1$	always
Sum of all coefficients	put $a = b = 1$: $\sum_{r=0}^n {}^nC_r = 2^n$	$n \in \mathbb{N}$
Alternating sum	put $a = 1, b = -1$: $\sum_{r=0}^n (-1)^r {}^nC_r = 0$	$n \geq 1$

Note there is **no SI unit or dimension** attached to any of these quantities — they are pure numbers. The only “constraint bookkeeping” that matters is the *domain* of n (positive integer here) and the *range* of r ($0 \leq r \leq n$).

Section 3: Proofs, Theorems & Standard Techniques

Proof of the Binomial Theorem by Mathematical Induction.

Claim: $(a + b)^n = \sum_{r=0}^n {}^nC_r a^{n-r} b^r$ for all $n \in \mathbb{N}$.

Base case ($n = 1$): $\text{RHS} = {}^1C_0 a + {}^1C_1 b = a + b = (a + b)^1$. True.

Inductive step: Assume the result holds for $n = k$, i.e.

$$(a + b)^k = \sum_{r=0}^k {}^kC_r a^{k-r} b^r.$$

We must show it holds for $n = k + 1$. Multiply both sides by $(a + b)$ — *the physical reason* is that $(a + b)^{k+1}$ is just one more factor of $(a + b)$ acting on the previous expansion:

$$(a + b)^{k+1} = (a + b) \sum_{r=0}^k {}^kC_r a^{k-r} b^r = \sum_{r=0}^k {}^kC_r a^{k-r+1} b^r + \sum_{r=0}^k {}^kC_r a^{k-r} b^{r+1}.$$

In the second sum, shift the index ($r \rightarrow r - 1$) so both sums carry the same power $a^{k+1-r} b^r$. Collecting the coefficient of $a^{k+1-r} b^r$ gives ${}^kC_r + {}^kC_{r-1}$, which by **Pascal's rule** equals ${}^{k+1}C_r$. Hence

$$(a + b)^{k+1} = \sum_{r=0}^{k+1} {}^{k+1}C_r a^{k+1-r} b^r.$$

By induction the theorem holds for all $n \in \mathbb{N}$. ■

The single load-bearing fact in that proof was Pascal's rule, so let us prove *it* too, because it is the engine of the triangle.

Proof of Pascal's rule, ${}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$.

$${}^nC_r + {}^nC_{r-1} = \frac{n!}{r!(n-r)!} + \frac{n!}{(r-1)!(n-r+1)!}.$$

Take LCM $r!(n-r+1)!$:

$$= \frac{n!(n-r+1) + n!r}{r!(n-r+1)!} = \frac{n!(n+1)}{r!(n+1-r)!} = \frac{(n+1)!}{r!(n+1-r)!} = {}^{n+1}C_r. \quad \blacksquare$$

Standard technique — constructing Pascal's triangle (and using it).

1. Start row 0 with a single 1 (this is $(a + b)^0 = 1$).
2. Begin and end every new row with 1.
3. Fill each interior entry as the sum of the two entries diagonally above it.
4. To expand $(a + b)^n$, read off row n as the coefficients, attach $a^n, a^{n-1}b, \dots, b^n$ with powers of a falling and powers of b rising.

Worked illustration of the read-off: row 4 is 1 4 6 4 1, so

$$(a + b)^4 = a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4.$$

Pascal's triangle is fastest for small n (say $n \leq 6$); for large n , use nC_r directly — building the whole triangle up to row 15 just to read one coefficient is a waste of time.

Section 4: Worked Examples

Example 1 — [CBSE / School Level]

Expand $(2x + 3)^4$ using the binomial theorem.

Here $a = 2x$, $b = 3$, $n = 4$. The coefficients are row 4 of Pascal's triangle: 1, 4, 6, 4, 1.

$$(2x + 3)^4 = {}^4C_0(2x)^4 + {}^4C_1(2x)^3(3) + {}^4C_2(2x)^2(3)^2 + {}^4C_3(2x)(3)^3 + {}^4C_4(3)^4.$$

Compute each term carefully, raising the **whole** quantity $2x$ to its power:

$$= 1 \cdot 16x^4 + 4 \cdot 8x^3 \cdot 3 + 6 \cdot 4x^2 \cdot 9 + 4 \cdot 2x \cdot 27 + 1 \cdot 81.$$

$$(2x + 3)^4 = 16x^4 + 96x^3 + 216x^2 + 216x + 81.$$

Sanity check (Pro-Tip in action): put $x = 1$. LHS = $(2 + 3)^4 = 625$. RHS = $16 + 96 + 216 + 216 + 81 = 625$. ✓

Example 2 — [JEE Main · Speed Trap]

Find the number of dissimilar terms in the expansion of $(1 + 2x + x^2)^{10}$.

The trap: a student sees a three-term base raised to the 10th power and reaches for the trinomial count $\binom{n+2}{2} = \binom{12}{2} = 66$, or panics and guesses $2(10) + 1 = 21$ for the wrong reason.

The fast, correct move: recognise the base as a **perfect square**:

$$1 + 2x + x^2 = (1 + x)^2 \Rightarrow (1 + 2x + x^2)^{10} = ((1 + x)^2)^{10} = (1 + x)^{20}.$$

$(1 + x)^{20}$ has $20 + 1 = \boxed{21}$ terms. The whole problem collapses the instant you factor.

Lesson: before counting terms of a “trinomial”, always check whether the base is secretly a perfect power.

Example 3 — [JEE Main · Multi-concept]

Find the coefficient of x^3 in $(1 + x)(1 - 2x)^5$.

This links two ideas: distributing a linear factor, and the general term of a binomial. Write

$$(1 + x)(1 - 2x)^5 = (1 - 2x)^5 + x(1 - 2x)^5.$$

So the coefficient of x^3 in the product = [coeff of x^3 in $(1 - 2x)^5$] + [coeff of x^2 in $(1 - 2x)^5$] (the second factor's x shifts the index down by one).

General term of $(1 - 2x)^5$: $T_{r+1} = {}^5C_r(-2x)^r = {}^5C_r(-2)^r x^r$.

- Coeff of x^3 ($r = 3$): ${}^5C_3(-2)^3 = 10 \cdot (-8) = -80$.
- Coeff of x^2 ($r = 2$): ${}^5C_2(-2)^2 = 10 \cdot 4 = 40$.

Therefore the coefficient of x^3 is $-80 + 40 = \boxed{-40}$.

Example 4 — [JEE Advanced Level]

The coefficients of three consecutive terms in the expansion of $(1 + x)^n$ are in the ratio $1 : 3 : 5$. Find n and the position of these terms.

Let the three consecutive coefficients be ${}^nC_{r-1}$, nC_r , ${}^nC_{r+1}$. The key technique is the **ratio of consecutive binomial coefficients**, which avoids factorials entirely:

$$\frac{{}^nC_{r-1}}{{}^nC_r} = \frac{r}{n-r+1}, \quad \frac{{}^nC_r}{{}^nC_{r+1}} = \frac{r+1}{n-r}.$$

From the first ratio $= \frac{1}{3}$:

$$\frac{r}{n-r+1} = \frac{1}{3} \Rightarrow 3r = n-r+1 \Rightarrow n = 4r-1. \quad (\text{i})$$

From the second ratio $= \frac{3}{5}$:

$$\frac{r+1}{n-r} = \frac{3}{5} \Rightarrow 5(r+1) = 3(n-r) \Rightarrow 8r+5 = 3n. \quad (\text{ii})$$

Substitute (i) into (ii): $8r+5 = 3(4r-1) = 12r-3 \Rightarrow 4r = 8 \Rightarrow r = 2$, and then $n = 4(2) - 1 = 7$.

So $\boxed{n=7}$, and the three terms are T_2, T_3, T_4 (coefficients ${}^7C_1, {}^7C_2, {}^7C_3$). *Verify:* ${}^7C_1 : {}^7C_2 : {}^7C_3 = 7 : 21 : 35 = 1 : 3 : 5$. ✓

Section 5: Practice Exercises

1. **[CBSE / School]** Expand $(x - 2y)^5$ completely using the binomial theorem.
2. **[JEE Main]** Find the number of dissimilar terms in the expansion of $(1 + x + x^2 + x^3)^8$.
3. **[JEE Main]** Find the coefficient of x^4 in $(1 + x)^2(1 - x)^6$.
4. **[JEE Advanced]** If ${}^nC_0 + {}^nC_1 + {}^nC_2 + \dots + {}^nC_n = 4096$, find n and the value of ${}^nC_0 - {}^nC_1 + {}^nC_2 - \dots + (-1)^n {}^nC_n$.
5. **[JEE Advanced]** The coefficients of three consecutive terms in $(1 + x)^{n+5}$ are in the ratio $5 : 10 : 14$. Find n .

Answers:

1. $x^5 - 10x^4y + 40x^3y^2 - 80x^2y^3 + 80xy^4 - 32y^5$
2. 25
3. -10
4. $n = 12$; alternating sum = 0
5. $n = 6$

Section 6: Crack the MCQ

Q1. The number of terms in the expansion of $(2x - 3y)^{15}$ is:

- (a) 14 (b) 15 (c) 16 (d) 30

Correct: (c) 16. The number of terms is $n + 1 = 15 + 1 = 16$. *Distractor analysis:* (b) 15 comes from using n instead of $n + 1$ — the single most common slip in the whole chapter. (a) 14 is $n - 1$, an over-correction. (d) 30 is $2n$, a student confusing “number of terms” with “sum of the two indices over all terms” or doubling out of habit.

Q2. The sum of all the binomial coefficients in the expansion of $(1 + x)^{12}$ is:

- (a) 144 (b) 2048 (c) 4096 (d) 8192

Correct: (c) 4096. Put $x = 1$: $\text{sum} = 2^{12} = 4096$. *Distractor analysis:* (b) $2048 = 2^{11}$ — an off-by-one in the exponent (used $n - 1$). (d) $8192 = 2^{13}$ — off-by-one the other way ($n + 1$). (a) $144 = 12^2$ — a student who confuses the *sum of coefficients* with the *square of the index*, or who wrongly thinks “12 terms, square it”.

Q3. In the expansion of $(x + y)^6$, the third binomial coefficient (i.e. the coefficient of the third term) is:

- (a) 6 (b) 15 (c) 20 (d) 21

Correct: (b) 15. Terms are numbered $T_1, T_2, T_3, \dots$ with coefficients ${}^6C_0, {}^6C_1, {}^6C_2, \dots$. The third term carries ${}^6C_2 = 15$. *Distractor analysis:* (a) 6 is 6C_1 — the classic **off-by-one**: treating “third coefficient” as $r = 3$ ’s neighbour or starting the count at T_0 . (c) 20 is 6C_3 — reading $r = 3$ for “third term” instead of $r = 2$ (forgetting that T_{r+1} shifts the index). (d) 21 is 7C_2 — using the wrong Pascal row ($n = 7$ instead of $n = 6$).

Q4. The coefficient of x^2 in the expansion of $(1 + 2x)^6$ is:

- (a) 15 (b) 30 (c) 60 (d) 240

Correct: (c) 60. $T_{r+1} = {}^6C_r(2x)^r$; for x^2 take $r = 2$: ${}^6C_2 \cdot 2^2 = 15 \cdot 4 = 60$. *Distractor analysis:* (a) 15 is 6C_2 alone — forgetting to raise the coefficient 2 of x to the power. (b) 30 is ${}^6C_2 \cdot 2$ — raising 2 to the power 1 instead of 2. (d) 240 is ${}^6C_2 \cdot 2^4 = 15 \cdot 16$ — applying the wrong power 4 to the coefficient. All three traps share one root error: not raising the *entire* term $(2x)$ to its power.

Section 7: Common Student Pitfalls & Pro-Tips

- **Off-by-one on the number of terms.** $(a + b)^n$ has $n + 1$ terms, not n . Burn this in: a degree- n expansion runs from power 0 to power n , which is $n + 1$ values.
- **Raising only part of a term.** In $(2x)^3$ you must cube *both* the 2 and the x : $(2x)^3 = 8x^3$, never $2x^3$. This single error silently wrecks coefficient questions (see MCQ Q4).
- **Sign mistakes in $(a - b)^n$.** Signs alternate $+, -, +, - \dots$; the r th-from-zero term carries $(-1)^r$. Even powers of b stay positive, odd powers go negative.
- **Term-number vs. index confusion.** The $(r + 1)$ th term uses nC_r . “Third term” $\Rightarrow r = 2$, *not* $r = 3$. Also: Pascal’s rows start at $n = 0$, so the row for $(a + b)^n$

is the $(n + 1)$ th row.

Pro-Tip (the $x = 1$ sanity check). After any expansion of $(1 + x)^n$ -type, substitute $x = 1$: the coefficients must sum to 2^n . For $(a + b)^n$, plug $a = b = 1$ and confirm you get 2^n , or plug the actual values to confirm you recover the original number. It takes five seconds and catches almost every arithmetic and off-by-one slip. A second free check: coefficients must be **symmetric** (${}^nC_r = {}^nC_{n-r}$), so the row should read the same forwards and backwards.

Section 8: Cheat Sheet Box

BINOMIAL EXPANSION & PASCAL'S TRIANGLE — at a glance

- **Theorem** ($n \in \mathbb{N}$): $(a + b)^n = \sum_{r=0}^n {}^nC_r a^{n-r} b^r$
- **General term:** $T_{r+1} = {}^nC_r a^{n-r} b^r$ — the “ $(r + 1)$ th” term uses nC_r
- **Number of terms:** $n + 1$. **Sum of powers** in each term: n .
- **Symmetry:** ${}^nC_r = {}^nC_{n-r}$. **Pascal's rule:** ${}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$
- **Sum of coefficients:** set $x = 1 \Rightarrow \sum {}^nC_r = 2^n$. **Alternating sum** = 0 ($n \geq 1$).
- **Pascal's triangle:** edges all 1; each interior = sum of the two above; row n starts at $n = 0$.

Memory aids:

- “Powers seesaw” — power of a falls $n \rightarrow 0$ while power of b rises $0 \rightarrow n$; they always add to n .
- “ $n + 1$, not n ” — for the term count.
- “Raise the whole bracket” — $(2x)^3 = 8x^3$, cube everything inside.

Subtopic 02: General, Middle and Independent Terms

Exam Relevance: This is the most heavily examined slice of the chapter. “Find the term independent of x ” and “find the coefficient of x^k ” are near-guaranteed in JEE Main (often 1–2 questions), and middle-term/specific-term machinery threads through almost every JEE Advanced binomial problem. CBSE Class 11 routinely asks for the middle term or a stated term as a 2–3 mark question. Master the general term here and three-quarters of the chapter becomes mechanical.

Section 1: Concept Intuition

In Subtopic 01 we learned to expand $(a + b)^n$ into all $n + 1$ terms. But exams rarely want the *whole* expansion — they want **one specific term**: the 7th term, the middle one, the one with no x in it, or the one carrying x^{12} . Writing out all 50 terms of a degree-50 expansion to find one of them would be madness. So we need a way to jump straight to the term we want.

That tool is the **general term**, $T_{r+1} = {}^nC_r a^{n-r} b^r$, which we already met in Sub-01. Think of it as the **catalogue lookup** of the expansion. The index r is like a page number in a railway timetable: feed in the right r and the formula hands you that exact term — coefficient, powers, and all — without printing the whole timetable. Every “find the term ...” question is, at heart, the same two-step move: *figure out which r you need, then plug it in*.

Think about it this way... the three named special terms are just three different ways of *choosing which r to ask for*:

- **General term:** r is given to you directly (“find the 6th term” $\Rightarrow$ the 6th term is $T_6 = T_{5+1}$, so $r = 5$).
- **Middle term:** r is the balance point of the row. The expansion has $n + 1$ terms laid out like seats in a row. If $n + 1$ is odd (i.e. n even), there is exactly **one** centre seat. If $n + 1$ is even (i.e. n odd), there is **no single centre** — two seats share the middle, so there are **two** middle terms.
- **Term independent of x :** r is whatever makes the net power of x collapse to zero. Picture a tug-of-war: a factor like x^p pulls the exponent up, a factor like $1/x^q = x^{-q}$ pulls it down. The independent term is the one where the two pulls exactly cancel and x vanishes, leaving a pure number (a constant).

Validity conditions & constraints. The index r that you solve for *must be a non-negative integer with $0 \leq r \leq n$* . This is not a formality — it is a genuine yes/no test. If the exponent equation gives you $r = \frac{16}{5}$ or $r = -2$ or $r = n + 3$, then **the term you are looking for simply does not exist**. A binomial expansion of a positive integral power need not contain a constant term, nor a term of every power. Do not “round r to the nearest integer” to force an answer — that is one of the most punished mistakes in this topic.

Section 2: Key Formulas & Definitions

General term (the $(r + 1)$ th term from the beginning):

$$T_{r+1} = {}^nC_r a^{n-r} b^r, \quad r = 0, 1, 2, \dots, n.$$

Working form for the standard exam binomial. For $\left(\alpha x^p + \frac{\beta}{x^q}\right)^n$,

$$T_{r+1} = {}^nC_r (\alpha x^p)^{n-r} \left(\frac{\beta}{x^q}\right)^r = {}^nC_r \alpha^{n-r} \beta^r x^{p(n-r)-qr}.$$

The **net exponent of x** is therefore

$$E(r) = p(n - r) - qr.$$

This single expression powers every “specific term” question:

You want...	Set the condition	Then
the term with x^k	$E(r) = k$	solve for r (must be a valid integer)
the term independent of x	$E(r) = 0$	solve for r (must be a valid integer)

Middle term(s). The expansion of $(a + b)^n$ has $n + 1$ terms.

n even : one middle term, $T_{\frac{n}{2}+1}$

n odd : two middle terms, $T_{\frac{n+1}{2}}$ and $T_{\frac{n+3}{2}}$

Term from the end. The p th term from the end of $(a + b)^n$ equals the $(n - p + 2)$ th term from the beginning — equivalently, it is the p th term from the beginning of the *reversed* binomial $(b + a)^n$.

Definitions. α, β are the numerical/algebraic multipliers; p, q are the (positive) powers of x in the two parts; $n \in \mathbb{N}$ is the index; $r \in \{0, 1, \dots, n\}$ is the term index. As always, these are pure numbers — no units or dimensions.

Section 3: Proofs, Theorems & Standard Techniques

Why n even gives one middle term and n odd gives two. The terms are $T_1, T_2, \dots, T_{n+1}$, a list of length $n + 1$.

- If n is **even**, then $n + 1$ is **odd**, and an odd-length list has a unique central element — the $\left(\frac{n+1+1}{2}\right) = \left(\frac{n}{2} + 1\right)$ th term. Hence one middle term, $T_{\frac{n}{2}+1}$.
- If n is **odd**, then $n + 1$ is **even**, and an even-length list has no single centre; the two central elements are the $\frac{n+1}{2}$ th and the $\left(\frac{n+1}{2} + 1\right) = \frac{n+3}{2}$ th terms. Hence two middle terms. ■

Standard technique — the “specific term” algorithm. This one procedure answers *find the term with x^k , find the term independent of x , find the coefficient of x^k , and find a stated term*:

1. **Write the general term** T_{r+1} for the given binomial and simplify all powers, collecting x into a single power.
2. **Read off the net exponent** $E(r)$ of x .
3. **Impose the condition:** $E(r) = k$ for an x^k term, or $E(r) = 0$ for the independent term.
4. **Solve for r and validate:** is r an integer with $0 \leq r \leq n$? If not, *the term does not exist* — say so and stop.
5. **Back-substitute** the valid r into T_{r+1} to get the term (or just its coefficient, if that is all that is asked).

The reasoning behind Step 4 is the whole point: a positive-integral-index expansion produces only the powers $E(0), E(1), \dots, E(n)$, an arithmetic-style set of values. A power that is not in that set has no term — there is nothing to compute.

Standard technique — middle term. First check the parity of n , then read the position(s) straight off the boxed result above, then apply the general term at the corresponding r ($r = \frac{n}{2}$ for even n ; $r = \frac{n-1}{2}$ and $r = \frac{n+1}{2}$ for odd n). Do **not** hunt for “the middle” by expanding.

Section 4: Worked Examples

Example 1 — [CBSE / School Level]

Find the 6th term in the expansion of $\left(2x + \frac{3}{x}\right)^9$.

The 6th term is $T_6 = T_{5+1}$, so $r = 5$ (mind the off-by-one — “6th term” means $r = 5$).

$$\begin{aligned} T_6 &= {}^9C_5 (2x)^{9-5} \left(\frac{3}{x}\right)^5 = {}^9C_5 (2x)^4 \cdot \frac{3^5}{x^5} \\ &= 126 \cdot 16x^4 \cdot \frac{243}{x^5} = 126 \cdot 16 \cdot 243 \cdot x^{4-5} = \frac{489888}{x} \end{aligned}$$

$$\boxed{T_6 = \frac{489888}{x}}$$

Notice the whole bracket $2x$ was raised to the 4th power (giving $16x^4$, not $2x^4$) and the whole $3/x$ to the 5th — the recurring “raise the entire term” discipline from Sub-01.

Example 2 — [JEE Main · Speed Trap]

Find the term independent of x in $\left(x^2 - \frac{1}{x}\right)^9$.

The fast move: you do **not** need the whole term yet — first find r . General term:

$$T_{r+1} = {}^9C_r (x^2)^{9-r} \left(-\frac{1}{x}\right)^r = {}^9C_r (-1)^r x^{2(9-r)-r} = {}^9C_r (-1)^r x^{18-3r}.$$

Independent of $x \Rightarrow 18 - 3r = 0 \Rightarrow r = 6$ (a valid integer in $[0, 9]$, so the term exists).
Now plug in:

$$T_7 = {}^9C_6 (-1)^6 = 84.$$

Term independent of $x = 84$.

The trap it is built around: the $-\frac{1}{x}$ factor contributes x^{-r} , so the net exponent is $2(9-r)-r$. Students who write $2(9-r)+r$ get $18-r=0 \Rightarrow r=18$, which is out of range, and then either force a wrong term or wrongly declare “no such term.” Get the sign of the $1/x$ contribution right and the problem is ten seconds of work.

Example 3 — [JEE Main · Multi-concept]

Find the middle term in the expansion of $\left(\frac{3x}{2} - \frac{1}{3x}\right)^6$.

This combines parity-checking, the general term, and fraction/sign handling. Here $n = 6$ is **even**, so there is **one** middle term: $T_{\frac{6}{2}+1} = T_4$, i.e. $r = 3$.

$$T_4 = {}^6C_3 \left(\frac{3x}{2}\right)^3 \left(-\frac{1}{3x}\right)^3 = 20 \cdot \frac{27x^3}{8} \cdot \left(-\frac{1}{27x^3}\right).$$

The x^3 cancels against x^{-3} and the 27 cancels:

$$= 20 \cdot \frac{27}{8} \cdot \left(-\frac{1}{27}\right) = 20 \cdot \left(-\frac{1}{8}\right) = -\frac{5}{2}.$$

$T_4 = -\frac{5}{2}.$

(Here the middle term happened to also be independent of x — a coincidence of the chosen powers, not a rule.)

Example 4 — [JEE Advanced Level]

If the coefficient of x^7 in $\left(ax^2 + \frac{1}{bx}\right)^{11}$ equals the coefficient of x^{-7} in $\left(ax - \frac{1}{bx^2}\right)^{11}$, find the relation between a and b .

First expansion. General term:

$$T_{r+1} = {}^{11}C_r (ax^2)^{11-r} \left(\frac{1}{bx}\right)^r = {}^{11}C_r a^{11-r} b^{-r} x^{2(11-r)-r} = {}^{11}C_r a^{11-r} b^{-r} x^{22-3r}.$$

For x^7 : $22 - 3r = 7 \Rightarrow r = 5$. Coefficient = ${}^{11}C_5 a^6 b^{-5}$.

Second expansion. General term:

$$T_{k+1} = {}^{11}C_k (ax)^{11-k} \left(-\frac{1}{bx^2}\right)^k = {}^{11}C_k a^{11-k} (-1)^k b^{-k} x^{(11-k)-2k} = {}^{11}C_k a^{11-k} (-1)^k b^{-k} x^{11-3k}.$$

For x^{-7} : $11 - 3k = -7 \Rightarrow k = 6$. Coefficient = ${}^{11}C_6 a^5 (-1)^6 b^{-6} = {}^{11}C_6 a^5 b^{-6}$.

Equate. The elegant step is the **symmetry** ${}^{11}C_5 = {}^{11}C_6$ (from Sub-01), so the binomial coefficients cancel:

$$a^6 b^{-5} = a^5 b^{-6} \Rightarrow \frac{a^6}{a^5} = \frac{b^{-6}}{b^{-5}} \Rightarrow a = b^{-1} \Rightarrow \boxed{ab = 1}.$$

The problem looks intimidating but collapses the moment you spot that ${}^{11}C_5$ and ${}^{11}C_6$ are equal.

Section 5: Practice Exercises

1. **[CBSE / School]** Find the middle term in the expansion of $\left(\frac{x}{3} + 3y\right)^8$.
2. **[JEE Main]** Find the term independent of x in $\left(2x^2 + \frac{1}{x}\right)^{12}$.
3. **[JEE Main]** Find the coefficient of x^5 in $\left(x - \frac{2}{x^2}\right)^8$.
4. **[JEE Advanced]** If the term independent of x in $\left(\sqrt{x} - \frac{k}{x^2}\right)^{10}$ equals 405, find the value of k .
5. **[JEE Advanced]** If the coefficients of x^2 and x^3 in the expansion of $(3 + ax)^9$ are equal, find a .

Answers:

1. $70 x^4 y^4$ (middle term T_5)
2. 7920 (at $r = 8$)
3. -16 (at $r = 1$)
4. $k = \pm 3$ (at $r = 2$)
5. $a = \frac{9}{7}$

Section 6: Crack the MCQ

Q1. The 7th term in the expansion of $\left(x - \frac{1}{x}\right)^{10}$ is:

- (a) $\frac{210}{x^2}$ (b) $-\frac{210}{x^2}$ (c) $210x^2$ (d) 252

Correct: (a) $\frac{210}{x^2}$. 7th term $\Rightarrow r = 6$: $T_7 = {}^{10}C_6 x^4 (-1/x)^6 = 210 x^4 \cdot x^{-6} = 210x^{-2}$.

Distractor analysis: (b) flips the sign — but $(-1/x)^6$ is an even power, so it is positive; this catches students who assume “minus base $\Rightarrow$ minus term.” (c) reverses the exponent sign (x^{+2} instead of x^{-2}), forgetting the $1/x$ pulls the power down. (d) is ${}^{10}C_5 = 252$, the off-by-one error of taking $r = 5$ for the “7th” term.

Q2. How many middle terms does the expansion of $(x + y)^{15}$ have?

- (a) 1 (b) 2 (c) 8 (d) 0

Correct: (b) 2. $n = 15$ is odd, so $n + 1 = 16$ terms — an even-length list with two central terms (T_8 and T_9). *Distractor analysis:* (a) assumes every expansion has a single middle term (true only for even n). (c) reports the position (8th) instead of the count. (d) confuses “no single middle” with “no middle term at all.”

Q3. The term independent of x in $\left(x - \frac{2}{x^2}\right)^9$ is:

- (a) -672 (b) 672 (c) -84 (d) there is no such term

Correct: (a) -672 . Net exponent $= (9 - r) - 2r = 9 - 3r = 0 \Rightarrow r = 3$. Then $T_4 = {}^9C_3 (-2)^3 = 84 \cdot (-8) = -672$. *Distractor analysis:* (b) drops the sign of $(-2)^3$. (c) is $-{}^9C_3 = -84$ — taking the sign but forgetting to raise 2 to the 3rd power (the “raise the whole term” failure). (d) is the over-cautious student who miscomputed r as non-integer.

Q4. The expansion of $\left(x^2 + \frac{1}{x^3}\right)^8$ has a term independent of x equal to:

- (a) 8C_3 (b) 8C_4 (c) 1 (d) there is no term independent of x

Correct: (d) there is no term independent of x . Net exponent $= 2(8 - r) - 3r = 16 - 5r$. Setting $16 - 5r = 0$ gives $r = \frac{16}{5}$, which is **not** an integer — so no term is independent of x . *Distractor analysis:* (a), (b), (c) are all traps for students who force an integer r (rounding $\frac{16}{5}$ to 3 or 4) rather than concluding the term does not exist. This is the single most important conceptual check in the subtopic — a non-integer or out-of-range r means the term is absent.

Section 7: Common Student Pitfalls & Pro-Tips

- **Off-by-one in term position.** The $(r + 1)$ th term uses nC_r , so “the m th term” means $r = m - 1$. “7th term” $\Rightarrow r = 6$, never $r = 7$. (Same off-by-one family as the Pascal-row indexing in Sub-01.)
- **Forgetting the second middle term.** Odd n gives **two** middle terms; a question saying “the middle term” of an odd-power expansion is often a deliberate trap — there are two.
- **Sign/exponent slips from $1/x$ factors.** A factor $\frac{1}{x^q}$ contributes x^{-qr} to the exponent and may carry a sign; both are routinely dropped. Always write the net exponent as $p(n - r) - qr$ explicitly.

- **Forcing a term that doesn't exist.** If solving the exponent equation gives a non-integer r or $r \notin [0, n]$, the term is genuinely absent. Do not round.

Pro-Tip (solve for r first). For *independent / specific-power / coefficient* questions, never expand and never compute the full term until you have r . Form the net exponent $E(r)$, set the condition, solve for r , and validate it. Only then substitute. In objective papers this is the difference between a 15-second solve and a two-minute one — and it instantly catches the “no such term” cases.

Section 8: Cheat Sheet Box

GENERAL, MIDDLE & INDEPENDENT TERMS — at a glance

- **General term (master tool):** $T_{r+1} = {}^nC_r a^{n-r} b^r$. The m th term $\Rightarrow r = m - 1$.
- **Net exponent** for $\left(\alpha x^p + \frac{\beta}{x^q}\right)^n$: $E(r) = p(n - r) - qr$.
- **Term with x^k :** set $E(r) = k$. **Term independent of x :** set $E(r) = 0$. Solve for r ; it **must** be an integer in $[0, n]$, else no such term.
- **Middle term:** n even $\Rightarrow$ one term $T_{\frac{n}{2}+1}$; n odd $\Rightarrow$ two terms $T_{\frac{n+1}{2}}, T_{\frac{n+3}{2}}$.
- **p th from end** = $(n - p + 2)$ th from beginning.

Memory aids:

- “Set the power, solve for r , check it's whole and in range.”
- “Odd power $\rightarrow$ two middles; even power $\rightarrow$ one middle.”
- “ $1/x^q$ pulls the exponent down by qr ” — keep that minus sign.

Subtopic 03: Greatest Term and Greatest Binomial Coefficient

Exam Relevance: “Find the numerically greatest term” is a recurring JEE Main question and a frequent JEE Advanced sub-step. The greatest *binomial coefficient* (always the middle one) is a quick conceptual MCQ. Not part of the rationalised CBSE Class 11 core, but standard in every JEE syllabus and coaching module.

Section 1: Concept Intuition

Two different “biggest” questions live in this subtopic, and students constantly confuse them.

The **greatest binomial coefficient** asks: among ${}^nC_0, {}^nC_1, \dots, {}^nC_n$, which is largest? This depends only on n , not on any value of x . From Pascal’s triangle the rows rise to a peak in the middle and fall symmetrically, so the largest coefficient is always the **middle** one — the same middle term we found in Sub-02.

The **numerically greatest term** asks something harder: in an actual expansion like $(a+b)^n$ evaluated at given numbers, which *term* (coefficient times the powers) has the largest magnitude? This depends on a , b , and n together, because a large coefficient can be dragged down by a small power of b , or a modest coefficient lifted by a large power.

Think about it this way... walk along the terms $T_1, T_2, T_3, \dots$ and watch the ratio of each term to the previous one, $\left| \frac{T_{r+1}}{T_r} \right|$. While this ratio is bigger than 1, the terms are still growing; the moment it drops below 1, they start shrinking. The greatest term sits right at the turning point. So instead of computing all $n+1$ terms, we just find where the ratio crosses 1.

Validity conditions & constraints. This ratio test works for any finite binomial expansion (positive integral n). For the *algebraically* greatest term in $(a-b)^n$, remember the terms alternate in sign — the numerically greatest term may be negative, in which case the algebraically greatest term is a neighbouring positive term, and you must compare those neighbours explicitly.

Section 2: Key Formulas & Definitions

Greatest binomial coefficient of $(1+x)^n$ (equivalently, the largest nC_r):

$$n \text{ even} : {}^nC_{n/2} \text{ (unique);} \quad n \text{ odd} : {}^nC_{(n-1)/2} = {}^nC_{(n+1)/2} \text{ (two equal).}$$

Numerically greatest term — ratio method. For $(a+b)^n$, the consecutive-term ratio is

$$\left| \frac{T_{r+1}}{T_r} \right| = \frac{n-r+1}{r} \left| \frac{b}{a} \right|.$$

Set this ≥ 1 and solve for r . A clean shortcut for $(1+x)^n$: compute

$$k = \frac{(n+1)|x|}{1+|x|}.$$

- If k is **not** an integer, the greatest term is $T_{\lfloor k \rfloor + 1}$ (i.e. $r = \lfloor k \rfloor$).
- If k **is** an integer, then T_k and T_{k+1} are **equal** and both greatest.

Greatest coefficient (numerical) of $(1+\lambda x)^n$: the coefficients are $c_r = {}^nC_r \lambda^r$; apply the same ratio idea to $\frac{c_{r+1}}{c_r} = \frac{n-r}{r+1} \lambda \geq 1$.

Here a, b are the two parts of the binomial, x the variable's value, $n \in \mathbb{N}$ the index, r the term index. All pure numbers — no units.

Section 3: Proofs, Theorems & Standard Techniques

Why the middle binomial coefficient is greatest. Consider the ratio of consecutive coefficients:

$$\frac{{}^nC_r}{{}^nC_{r-1}} = \frac{n-r+1}{r}.$$

This exceeds 1 exactly when $n-r+1 > r$, i.e. $r < \frac{n+1}{2}$. So the coefficients **increase** while $r < \frac{n+1}{2}$ and **decrease** after — they peak at the middle. For even n the peak is the single value ${}^nC_{n/2}$; for odd n two central coefficients tie. ■

Numerically greatest term — the algorithm.

1. Write the magnitude ratio $\left| \frac{T_{r+1}}{T_r} \right| = \frac{n-r+1}{r} \left| \frac{b}{a} \right|$.
2. Solve the inequality $\frac{n-r+1}{r} \left| \frac{b}{a} \right| \geq 1$ for r , getting $r \leq k$ for some number k .
3. The terms increase up to $r = \lfloor k \rfloor$ and decrease after, so the greatest term is $T_{\lfloor k \rfloor + 1}$. If k is an exact integer, $T_k = T_{k+1}$ are jointly greatest.
4. For $(a-b)^n$, decide the **sign** of that term; if it is negative and the question asks for the *algebraically* greatest, compare the adjacent positive terms.

The reasoning: a single ratio crossing 1 marks the unique turning point of a sequence that rises then falls, so locating that crossing locates the maximum without evaluating every term.

Section 4: Worked Examples

Example 1 — [CBSE / School Level]

Find the greatest binomial coefficient in the expansion of $(1+x)^{12}$. $n = 12$ is even, so the greatest coefficient is the unique middle one:

$${}^{12}C_{12/2} = {}^{12}C_6 = 924. \quad \boxed{924}$$

Example 2 — [JEE Main · Speed Trap]

Find the numerically greatest term in $(1 + 4x)^8$ when $x = \frac{1}{2}$.

At $x = \frac{1}{2}$, term magnitude $\propto {}^8C_r (4 \cdot \frac{1}{2})^r = {}^8C_r 2^r$. Ratio:

$$\frac{T_{r+1}}{T_r} = \frac{8-r+1}{r} \cdot 2 = \frac{2(9-r)}{r} \geq 1 \Rightarrow 18-2r \geq r \Rightarrow r \leq 6.$$

So the ratio stays ≥ 1 up to $r = 6$ — and at $r = 6$ it equals **exactly** 1 (check: $\frac{2(9-6)}{6} = 1$), the integer-crossing case from Section 2. That means T_6 and T_7 are **equal and jointly greatest**:

$$T_6 = {}^8C_5 (4x)^5 = 56 \cdot 2^5 = 1792, \quad T_7 = {}^8C_6 (4x)^6 = 28 \cdot 2^6 = 1792.$$

$$\boxed{T_6 = T_7 = 1792 \text{ (two equal greatest terms).}}$$

The trap: forgetting the coefficient 4 inside $(4x)$ — students use $|x| = \frac{1}{2}$ in the ratio instead of $|4x| = 2$, getting the wrong r . The “ b/a ” factor must include the full second part of the binomial.

Example 3 — [JEE Main · Multi-concept]

Find the greatest coefficient in the expansion of $(1 + 2x)^6$.

Coefficients are $c_r = {}^6C_r 2^r$. Ratio:

$$\frac{c_{r+1}}{c_r} = \frac{6-r}{r+1} \cdot 2 \geq 1 \Rightarrow 2(6-r) \geq r+1 \Rightarrow 11 \geq 3r \Rightarrow r \leq 3.67.$$

So c_r increases up to $r = 4$ (check: $c_4/c_3 = \frac{3}{4} \cdot 2 = 1.5 > 1$, $c_5/c_4 = \frac{2}{5} \cdot 2 = 0.8 < 1$). Greatest coefficient:

$$c_4 = {}^6C_4 2^4 = 15 \cdot 16 = 240. \quad \boxed{240}$$

Note this is the coefficient of x^4 — **not** the middle term, because the factor 2^r shifts the peak rightward.

Example 4 — [JEE Advanced Level]

Find the numerically greatest term in $(4 - 3x)^8$ when $x = 1$, and state the algebraically greatest term.

General-term magnitude: $|{}^8C_r 4^{8-r} (-3x)^r| = {}^8C_r 4^{8-r} 3^r$ (since $|3x| = 3$ at $x = 1$). The consecutive-term magnitude ratio is

$$\frac{|T_{r+1}|}{|T_r|} = \frac{8-r+1}{r} \cdot \frac{3}{4} = \frac{9-r}{r} \cdot \frac{3}{4} \geq 1 \Rightarrow 3(9-r) \geq 4r \Rightarrow 27 \geq 7r \Rightarrow r \leq \frac{27}{7} \approx 3.86.$$

So the magnitude increases up to $r = 3$, giving the numerically greatest term T_4 :

$$|T_4| = {}^8C_3 4^5 3^3 = 56 \cdot 1024 \cdot 27 = 1,548,288.$$

But T_4 carries $(-3x)^3$, an **odd** power, so $T_4 = -1,548,288$ is **negative**. The **algebraically** greatest term is therefore the larger of its positive neighbours T_3 and T_5 :

$$T_3 = {}^8C_2 4^6 3^2 = +1,032,192, \quad T_5 = {}^8C_4 4^4 3^4 = +1,451,520.$$

Since $T_5 > T_3$, the algebraically greatest term is T_5 .

Numerically greatest: $T_4 = -1,548,288$; algebraically greatest: $T_5 = +1,451,520$.

Section 5: Practice Exercises

1. **[CBSE / School]** Find the greatest binomial coefficient in $(1 + x)^{15}$.
2. **[JEE Main]** Find the numerically greatest term in $(1 + x)^{10}$ when $x = \frac{2}{3}$.
3. **[JEE Main]** Find the greatest coefficient in $(2 + 3x)^5$.
4. **[JEE Advanced]** In $(3 + 2x)^9$ with $x = 1$, find the numerically greatest term.
5. **[JEE Advanced]** For what value(s) of x does $(1 + x)^{20}$ have T_{10} and T_{11} as equal greatest terms?

Answers:

1. ${}^{15}C_7 = {}^{15}C_8 = 6435$ (two equal, n odd)
2. T_5 (at $r = 4$), value ${}^{10}C_4 (2/3)^4 = 210 \cdot \frac{16}{81} = \frac{1120}{27}$
3. 1080 (coefficient of x^3 : ${}^5C_3 2^2 3^3 = 10 \cdot 4 \cdot 27$)
4. $T_4 = T_5 = 489888$ (the magnitude ratio equals 1 at $r = 4$, so two equal greatest terms; ${}^9C_3 3^6 2^3 = {}^9C_4 3^5 2^4$)
5. $x = \frac{10}{11}$ (from $k = (n + 1)|x|/(1 + |x|) = 10 \Rightarrow$ integer case)

Section 6: Crack the MCQ

Q1. The greatest binomial coefficient in $(1 + x)^{11}$ is:

- (a) ${}^{11}C_5$ only (b) ${}^{11}C_6$ only (c) ${}^{11}C_5 = {}^{11}C_6$ (d) ${}^{11}C_0$

Correct: (c). $n = 11$ is odd, so the two central coefficients ${}^{11}C_5$ and ${}^{11}C_6$ are equal and jointly greatest. *Distractors:* (a),(b) pick only one of the tied pair. (d) confuses largest coefficient with the first/edge coefficient.

Q2. The greatest binomial coefficient is always:

- (a) nC_1 (b) the middle one(s) (c) nC_n (d) it depends on x

Correct: (b). Coefficients rise to the middle and fall symmetrically. *Distractors:* (a),(c) are the smallest (edge) coefficients. (d) confuses the greatest *coefficient* (independent of x) with the greatest *term* (which does depend on x).

Q3. In $(1 + x)^n$ with $x > 0$, the numerically greatest term is found by solving:

(a) $\frac{n-r+1}{r}x \geq 1$ (b) $\frac{r}{n-r+1}x \geq 1$ (c) nC_r maximum (d) $nx \geq 1$

Correct: (a). This is the consecutive-term ratio set ≥ 1 . *Distractors:* (b) inverts the ratio. (c) finds the greatest coefficient, ignoring x . (d) is meaningless here.

Q4. For $(a+b)^n$ at given values, if the ratio crossing point k comes out an exact integer, then:

- (a) there is no greatest term (b) T_k alone is greatest (c) $T_k = T_{k+1}$ are jointly greatest (d) T_{k-1} is greatest

Correct: (c). An integer k means the ratio equals exactly 1 at that step, so two adjacent terms tie. *Distractors:* (a) false — a finite sum always has a greatest term. (b),(d) miss the tie.

Section 7: Common Student Pitfalls & Pro-Tips

- **Confusing the two “greatest” questions.** Greatest *binomial coefficient* = middle, independent of x . Greatest *term* = depends on a, b, x via the ratio test. Read the question carefully.
- **Dropping the coefficient inside the bracket.** In $(1+4x)^n$ the ratio uses $|4x|$, not $|x|$. The full second part b enters the ratio.
- **The integer- k tie.** If the crossing point is an exact integer, two terms are jointly greatest — don’t report just one.
- **Numerical vs algebraic for $(a-b)^n$.** The numerically greatest term can be negative; the algebraically greatest is then a neighbouring positive term. Compare neighbours explicitly.

Pro-Tip. For $(1+x)^n$, memorise $k = \frac{(n+1)|x|}{1+|x|}$ and read off $r = \lfloor k \rfloor$. It collapses the whole inequality into one division — ideal for objective papers.

Section 8: Cheat Sheet Box

GREATEST TERM & GREATEST COEFFICIENT — at a glance

- **Greatest binomial coefficient:** middle. n even $\Rightarrow {}^nC_{n/2}$; n odd $\Rightarrow {}^nC_{(n\pm 1)/2}$ (two equal).
- **Numerically greatest term:** solve $\frac{n-r+1}{r} \left| \frac{b}{a} \right| \geq 1$.
- **Shortcut for $(1+x)^n$:** $k = \frac{(n+1)|x|}{1+|x|}$; greatest is $T_{\lfloor k \rfloor + 1}$. Integer $k \Rightarrow T_k = T_{k+1}$.
- **Greatest coefficient of $(1+\lambda x)^n$:** ratio $\frac{n-r}{r+1} \lambda \geq 1$.
- $(a-b)^n$: numerically greatest may be negative; compare neighbours for algebraically greatest.

Memory aid: “Coefficient peak = middle; term peak = where the ratio crosses

1.”

Subtopic 04: Properties of Binomial Coefficients and Summation of Series

Exam Relevance: This is the single most heavily examined area of the chapter at JEE level — sums like $\sum r C_r$, $\sum \frac{C_r}{r+1}$, and $\sum C_r^2$ appear almost every year in JEE Main and Advanced. Four reusable techniques (substitution, differentiation, integration, coefficient-comparison) crack the entire topic. Beyond rationalised CBSE core, but indispensable for competitive exams.

(Throughout, write C_r for nC_r .)

Section 1: Concept Intuition

The expansion $(1+x)^n = C_0 + C_1x + C_2x^2 + \dots + C_nx^n$ is not just a formula — it is a **machine with a dial**, the dial being x . Turn the dial to different values, or operate on the whole identity (differentiate it, integrate it, multiply two copies together), and out drop exact closed forms for sums that look impossible to add directly.

Think about it this way...

- Want $C_0 + C_1 + \dots + C_n$? Turn the dial to $x = 1$: the left side becomes 2^n . Done — no term-by-term addition.
- Want the alternating sum $C_0 - C_1 + C_2 - \dots$? Turn the dial to $x = -1$: the left side becomes 0.
- Want $C_1 + 2C_2 + 3C_3 + \dots$ (each coefficient weighted by its index)? That weighting is exactly what **differentiation** does — $\frac{d}{dx}x^r = rx^{r-1}$ — so differentiate the identity, then set $x = 1$.
- Want $C_0 + \frac{C_1}{2} + \frac{C_2}{3} + \dots$ (each weighted by $\frac{1}{r+1}$)? That is what **integration** does — $\int x^r dx = \frac{x^{r+1}}{r+1}$ — so integrate from 0 to 1.
- Want $C_0^2 + C_1^2 + \dots + C_n^2$ (products of coefficients)? Multiply two expansions together and read off a single coefficient on each side.

So you never “sum the series” by brute force. You recognise which operation produces the weights you see, apply it to the master identity, and substitute.

Validity conditions & constraints. Substitution ($x = \pm 1$) is always valid (the expansion is a finite polynomial). Differentiation and integration are term-by-term operations on that polynomial — always legitimate here. The only care needed is book-keeping: after differentiating, the running index and limits shift, so re-read which sum you have actually produced before substituting.

Section 2: Key Formulas & Definitions

With $C_r = {}^nC_r$:

Basic identities (substitution):

$$\sum_{r=0}^n C_r = 2^n, \quad \sum_{r=0}^n (-1)^r C_r = 0 \quad (n \geq 1),$$

$$C_0 + C_2 + C_4 + \cdots = C_1 + C_3 + C_5 + \cdots = 2^{n-1}.$$

Index/ratio identities:

$$\frac{C_r}{C_{r-1}} = \frac{n-r+1}{r}, \quad r C_r = n C_{r-1}, \quad \frac{C_r}{r+1} = \frac{1}{n+1} {}^{n+1}C_{r+1}.$$

Weighted sums (differentiation / integration):

$$\sum_{r=1}^n r C_r = n 2^{n-1}, \quad \sum_{r=1}^n r^2 C_r = n(n+1) 2^{n-2},$$

$$\sum_{r=0}^n \frac{C_r}{r+1} = \frac{2^{n+1}-1}{n+1}, \quad \sum_{r=0}^n \frac{(-1)^r C_r}{r+1} = \frac{1}{n+1}.$$

Products of coefficients (coefficient-comparison):

$$\sum_{r=0}^n C_r^2 = 2^n C_n, \quad \sum_{r=0}^{n-k} C_r C_{r+k} = 2^n C_{n-k}.$$

Vandermonde's identity: $\sum_k {}^m C_k {}^n C_{p-k} = {}^{m+n} C_p.$

Here $n \in \mathbb{N}$, r is the term index, C_r are the binomial coefficients (pure numbers).

Section 3: Proofs, Theorems & Standard Techniques

Technique A — Substitution. Put chosen values of x into $(1+x)^n = \sum C_r x^r$.

- $x = 1$: $\sum C_r = 2^n$.
- $x = -1$: $\sum (-1)^r C_r = 0$.
- Adding these two results and halving isolates the even-indexed coefficients: $C_0 + C_2 + \cdots = \frac{2^n+0}{2} = 2^{n-1}$; subtracting gives the odd-indexed sum $= 2^{n-1}$.

Technique B — Differentiation. Differentiate $(1+x)^n = \sum_{r=0}^n C_r x^r$:

$$n(1+x)^{n-1} = \sum_{r=1}^n r C_r x^{r-1}.$$

Put $x = 1$: $\sum_{r=1}^n r C_r = n 2^{n-1}$. (Differentiating again and substituting gives $\sum r^2 C_r$.) The *reason* this works: differentiation multiplies the x^r term by r , manufacturing exactly the weight r on each coefficient.

Technique C — Integration. Integrate $(1+x)^n = \sum C_r x^r$ from 0 to 1:

$$\int_0^1 (1+x)^n dx = \frac{2^{n+1} - 1}{n+1}, \quad \int_0^1 \sum C_r x^r dx = \sum \frac{C_r}{r+1}.$$

Hence $\sum_{r=0}^n \frac{C_r}{r+1} = \frac{2^{n+1} - 1}{n+1}$. Integrating from 0 to -1 (or using $x = -1$ limits) yields the alternating version $\frac{1}{n+1}$.

Technique D — Coefficient comparison (for products like $\sum C_r^2$). Start from

$$(1+x)^n(1+x)^n = (1+x)^{2n}.$$

Write the left factor as $\sum C_r x^r$ and the right factor, *reversed*, using $C_r = C_{n-r}$, as $\sum C_s x^{n-s}$. The coefficient of x^n on the left is $\sum_r C_r C_{n-r} = \sum_r C_r^2$ (using symmetry). On the right, the coefficient of x^n in $(1+x)^{2n}$ is $^{2n}C_n$. Equate:

$$\boxed{C_0^2 + C_1^2 + \dots + C_n^2 = ^{2n}C_n}.$$

The same “multiply two expansions, compare one coefficient” move proves Vandermonde’s identity and all $\sum C_r C_{r+k}$ sums. ■

Section 4: Worked Examples

Example 1 — [CBSE / School Level]

Show that $C_0 + C_2 + C_4 + \dots = 2^{n-1}$ **for** $(1+x)^n$, $n \geq 1$.

Put $x = 1$: $C_0 + C_1 + C_2 + \dots + C_n = 2^n$. Put $x = -1$: $C_0 - C_1 + C_2 - \dots = 0$. **Add** the two equations: the odd-indexed terms cancel, the even-indexed terms double:

$$2(C_0 + C_2 + C_4 + \dots) = 2^n \Rightarrow C_0 + C_2 + C_4 + \dots = 2^{n-1}. \quad \boxed{2^{n-1}}$$

Subtracting instead gives $C_1 + C_3 + \dots = 2^{n-1}$ as well — the even and odd families are equal.

Example 2 — [JEE Main • Speed Trap]

Evaluate $C_1 + 2C_2 + 3C_3 + \dots + nC_n$ **for** $(1+x)^{10}$.

Recognise the weights 1, 2, 3, ... — that is the differentiation signature. Using $\sum_{r=1}^n rC_r = n2^{n-1}$ directly with $n = 10$:

$$\sum_{r=1}^{10} rC_r = 10 \cdot 2^9 = 10 \cdot 512 = 5120. \quad \boxed{5120}$$

The trap: trying to compute each rC_r and add (slow, error-prone). The instant you see index-weights, reach for $n2^{n-1}$ — no expansion needed.

Example 3 — [JEE Main · Multi-concept]

Evaluate $C_0 + \frac{C_1}{2} + \frac{C_2}{3} + \dots + \frac{C_n}{n+1}$ **for** $n = 5$.

The weights $\frac{1}{r+1}$ are the integration signature. Using $\sum_{r=0}^n \frac{C_r}{r+1} = \frac{2^{n+1} - 1}{n+1}$ with $n = 5$:

$$\sum_{r=0}^5 \frac{C_r}{r+1} = \frac{2^6 - 1}{6} = \frac{63}{6} = \frac{21}{2}. \quad \boxed{\frac{21}{2}}$$

This blends the integration technique with clean arithmetic — the kind of two-step Main problem that rewards recognising the weight pattern instantly.

Example 4 — [JEE Advanced Level]

Prove that $C_0^2 + C_1^2 + C_2^2 + \dots + C_n^2 = {}^{2n}C_n$, **and hence evaluate** $C_0^2 + C_1^2 + \dots + C_4^2$ **for** $n = 4$.

Proof. Consider $(1+x)^n(1+x)^n = (1+x)^{2n}$. Expand the first factor as $\sum_r C_r x^r$ and the second as $\sum_s C_s x^s$; the coefficient of x^n on the left is $\sum_r C_r C_{n-r}$. By symmetry $C_{n-r} = C_r$, so this equals $\sum_r C_r^2$. On the right, the coefficient of x^n in $(1+x)^{2n}$ is ${}^{2n}C_n$. Equating coefficients of x^n :

$$\sum_{r=0}^n C_r^2 = {}^{2n}C_n.$$

Evaluation ($n = 4$): the coefficients of $(1+x)^4$ are 1, 4, 6, 4, 1, so

$$1^2 + 4^2 + 6^2 + 4^2 + 1^2 = 1 + 16 + 36 + 16 + 1 = 70 = {}^8C_4. \quad \boxed{70}$$

Section 5: Practice Exercises

1. **[CBSE / School]** Find $C_0 + C_1 + C_2 + \dots + C_n$ and $C_0 - C_1 + C_2 - \dots$ for $(1+x)^8$.
2. **[JEE Main]** Evaluate $\sum_{r=0}^{12} r C_r$ for $(1+x)^{12}$.
3. **[JEE Main]** Evaluate $C_0 + \frac{C_1}{2} + \dots + \frac{C_n}{n+1}$ for $n = 6$.
4. **[JEE Advanced]** Evaluate $C_0^2 + C_1^2 + \dots + C_6^2$ for $n = 6$.
5. **[JEE Advanced]** Evaluate $\sum_{r=0}^n r^2 C_r$ for $n = 5$.

Answers:

1. 256 and 0
2. $12 \cdot 2^{11} = 24576$
3. $\frac{2^7 - 1}{7} = \frac{127}{7}$
4. ${}^{12}C_6 = 924$
5. $n(n+1)2^{n-2} = 5 \cdot 6 \cdot 2^3 = 240$

Section 6: Crack the MCQ

Q1. $C_0 + C_1 + C_2 + \dots + C_n$ equals:

- (a) 2^{n-1} (b) 2^n (c) n^2 (d) 2^{n+1}

Correct: (b) 2^n . Put $x = 1$ in $(1+x)^n$. *Distractors:* (a) is the even-only (or odd-only) half-sum. (c),(d) are common wrong recalls.

Q2. $C_1 + 2C_2 + 3C_3 + \dots + nC_n$ equals:

- (a) 2^n (b) $n2^n$ (c) $n2^{n-1}$ (d) $(n-1)2^n$

Correct: (c) $n2^{n-1}$. Differentiate $(1+x)^n$, then set $x = 1$. *Distractors:* (b) forgets to drop the exponent by one after differentiating. (a) ignores the index weights. (d) is a fabricated form.

Q3. $C_0^2 + C_1^2 + \dots + C_n^2$ equals:

- (a) 2^{2n} (b) $2^n C_n$ (c) $(2^n)^2$ (d) $2^n C_{n-1}$

Correct: (b) $2^n C_n$. Coefficient of x^n in $(1+x)^n(1+x)^n = (1+x)^{2n}$. *Distractors:* (a),(c) are $(\sum C_r)^2 = 4^n$, the square of the sum, not the sum of squares. (d) is the central coefficient's neighbour.

Q4. $C_0 - \frac{C_1}{2} + \frac{C_2}{3} - \dots + \frac{(-1)^n C_n}{n+1}$ equals:

- (a) 0 (b) $\frac{1}{n+1}$ (c) $\frac{2^{n+1}-1}{n+1}$ (d) 2^n

Correct: (b) $\frac{1}{n+1}$. Integrate $(1+x)^n$ over $[-1, 0]$ (the alternating-weight version). *Distractors:* (c) is the non-alternating integral. (a) confuses it with the alternating coefficient sum. (d) ignores the $\frac{1}{r+1}$ weights.

Section 7: Common Student Pitfalls & Pro-Tips

- **Sum of squares $\neq$ square of sum.** $\sum C_r^2 = 2^n C_n$, not $(2^n)^2$. The product-then-compare technique is mandatory.
- **Forgetting to drop the exponent after differentiating.** $\frac{d}{dx}(1+x)^n = n(1+x)^{n-1}$, giving $n2^{n-1}$ at $x = 1$ — the exponent becomes $n-1$.
- **Wrong limits of integration.** $\int_0^1$ gives $\frac{2^{n+1}-1}{n+1}$; the alternating version needs $\int_{-1}^0$ (or $x = -1$). Mixing them flips the answer.
- **Index-shift bookkeeping.** After differentiating, the sum runs from $r = 1$ (the $r = 0$ term vanishes). Re-check which sum you produced before substituting.

Pro-Tip — read the weights, pick the tool. Constant weights $\Rightarrow$ substitute $x = \pm 1$. Weights r (or r^2) $\Rightarrow$ differentiate (once or twice). Weights $\frac{1}{r+1}$ $\Rightarrow$ integrate. Squared/product coefficients $\Rightarrow$ multiply two expansions and compare one coefficient. Matching the weight pattern to the operation is 90% of every problem here.

Section 8: Cheat Sheet Box

PROPERTIES OF COEFFICIENTS & SERIES — at a glance ($C_r = {}^nC_r$)

- $\sum C_r = 2^n$; $\sum (-1)^r C_r = 0$; $C_0 + C_2 + \dots = C_1 + C_3 + \dots = 2^{n-1}$.
- $r C_r = n C_{r-1}$; $\frac{C_r}{r+1} = \frac{{}^{n+1}C_{r+1}}{n+1}$.
- $\sum r C_r = n 2^{n-1}$; $\sum r^2 C_r = n(n+1) 2^{n-2}$.
- $\sum \frac{C_r}{r+1} = \frac{2^{n+1} - 1}{n+1}$; $\sum \frac{(-1)^r C_r}{r+1} = \frac{1}{n+1}$.
- $\sum_{m+n=C} C_r^2 = {}^{2n}C_n$; $\sum C_r C_{r+k} = {}^{2n}C_{n-k}$; Vandermonde $\sum {}^mC_k {}^nC_{p-k} = {}^{m+n}C_p$.

Memory aid (the four tools): substitute for constants, differentiate for r , integrate for $\frac{1}{r+1}$, multiply-and-compare for squares.

Subtopic 05: Binomial Theorem for Any Index and Approximations

Exam Relevance: A standard JEE Main/Advanced topic (expansions of $(1 \pm x)^{-1}$, $(1 \pm x)^{-2}$, fractional powers, and series-identification problems). **Beyond the rationalised CBSE Class 11 core** — included here for JEE completeness. If you are sitting boards only, this subtopic is optional; for JEE it is essential.

Scope note: Everything in Sub-01 to Sub-04 needed n to be a *positive integer*. This subtopic generalises to **any index** n (negative or fractional). The price: the series becomes **infinite** and only valid for $|x| < 1$.

Section 1: Concept Intuition

What could $(1+x)^{1/2}$ or $(1+x)^{-3}$ possibly “expand” to? There is no last term to stop at — you cannot choose r things from “half a thing.” Yet there is a beautiful continuation: the same shape of expansion keeps going **forever**, with the combination coefficients replaced by a product pattern that no longer terminates.

Think about it this way... look at the positive-integer expansion’s coefficients written the long way:

$${}^nC_r = \frac{n(n-1)(n-2)\cdots(n-r+1)}{r!}.$$

For a positive integer n , this product eventually hits the factor $(n-n) = 0$ and every later coefficient vanishes — that is *why* the series stops at $r = n$. But if n is, say, -3 or $\frac{1}{2}$, the product $n(n-1)\cdots$ **never hits zero**, so the series never stops. That single observation is the whole generalisation: keep the same coefficient pattern, just don’t expect it to terminate.

Because it is now an infinite series, it only **converges** (adds up to the true value of $(1+x)^n$) when $|x| < 1$. Outside that range the partial sums blow up and the “expansion” is meaningless. This convergence condition is not a technicality you can skip — using the series for $|x| \geq 1$ gives nonsense.

Validity conditions & constraints. The any-index expansion of $(1+x)^n$ is valid **only for** $|x| < 1$ (and the expansion must be arranged so the smaller quantity plays the role of x). There are no nC_r symbols here, because nC_r is undefined for non-integer or negative n . To expand $(a+b)^n$ for general n , first factor out the larger term: $(a+b)^n = a^n \left(1 + \frac{b}{a}\right)^n$, valid when $\left|\frac{b}{a}\right| < 1$.

Section 2: Key Formulas & Definitions

General binomial series (any index n , $|x| < 1$):

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \cdots$$

General term:

$$T_{r+1} = \frac{n(n-1)(n-2)\cdots(n-r+1)}{r!} x^r, \quad r = 0, 1, 2, \dots$$

Key special expansions (all for $|x| < 1$):

$$(1-x)^{-1} = 1 + x + x^2 + x^3 + \dots \quad (\text{geometric series})$$

$$(1+x)^{-1} = 1 - x + x^2 - x^3 + \dots$$

$$(1-x)^{-2} = 1 + 2x + 3x^2 + 4x^3 + \dots \quad (\text{general term } (r+1)x^r)$$

$$(1+x)^{-2} = 1 - 2x + 3x^2 - 4x^3 + \dots$$

$$(1-x)^{-p} = \sum_{r=0}^{\infty} p^{+r-1} C_r x^r \quad (p \in \mathbb{N}); \quad \text{general term } p^{+r-1} C_r x^r.$$

First-order approximation. If $|x|$ is very small, all terms beyond the linear one are negligible:

$$(1+x)^n \approx 1 + nx.$$

Here n is any rational index, x the variable with $|x| < 1$, r the (now unbounded) term index. Pure numbers throughout.

Section 3: Proofs, Theorems & Standard Techniques

Why the series continues (heuristic derivation). Write the positive-integer general term in falling-factorial form $\frac{n(n-1)\cdots(n-r+1)}{r!} x^r$. This expression is defined for *any* real n , not just positive integers. The general binomial theorem asserts that for $|x| < 1$ this same expression, summed over all $r \geq 0$, converges to $(1+x)^n$. (A full proof uses Taylor's theorem; the falling-factorial coefficient is precisely $\frac{1}{r!} \frac{d^r}{dx^r} (1+x)^n \big|_{x=0}$.) The termination for positive integer n is the special case where the factor $(n-n)$ kills all terms past $r = n$. ■

Standard technique — expanding $(a+b)^n$ for general n .

1. Factor out the dominant term so the bracket has leading 1: $(a+b)^n = a^n \left(1 + \frac{b}{a}\right)^n$ (choose a as the larger so $\left|\frac{b}{a}\right| < 1$).
2. Apply the any-index series to $\left(1 + \frac{b}{a}\right)^n$.
3. Multiply back by a^n .

Standard technique — identifying a numeric series as a binomial expansion.

Given a series of numbers, match it to $(1+y)^n = 1 + ny + \frac{n(n-1)}{2!} y^2 + \dots$

- The first term is 1 (factor out if not).
- The **second** term gives ny .
- The **third** term gives $\frac{n(n-1)}{2!} y^2$. Form the ratio of these two equations to solve for n and y simultaneously, then the series sums to $(1+y)^n$.

Standard technique — approximation. For small $|x|$, truncate after the linear term: $(1+x)^n \approx 1+nx$. Keep the quadratic term $\frac{n(n-1)}{2}x^2$ if higher accuracy is required.

Section 4: Worked Examples

Example 1 — [Foundation Level]

Write the first four terms of $(1+x)^{-2}$ and state the validity condition.

Using the general series with $n = -2$:

$$(1+x)^{-2} = 1 + (-2)x + \frac{(-2)(-3)}{2!}x^2 + \frac{(-2)(-3)(-4)}{3!}x^3 + \dots = 1 - 2x + 3x^2 - 4x^3 + \dots$$

$$\boxed{(1+x)^{-2} = 1 - 2x + 3x^2 - 4x^3 + \dots, \quad |x| < 1.}$$

Example 2 — [JEE Main · Speed Trap]

Find the coefficient of x^4 in the expansion of $(1-2x)^{-3}$, $|x| < \frac{1}{2}$.

Use $(1-y)^{-p} = \sum_{r=0}^{\infty} {}^{p+r-1}C_r y^r$ with $p = 3$, $y = 2x$. The coefficient of y^4 is ${}^{3+4-1}C_4 = {}^6C_4 = 15$, and $y^4 = (2x)^4 = 16x^4$:

$$\text{coefficient of } x^4 = {}^6C_4 \cdot 2^4 = 15 \cdot 16 = 240. \quad \boxed{240}$$

The trap: writing ${}^{-3}C_4$ (undefined!) or forgetting that $y = 2x$ contributes the factor 2^4 . For negative integer powers, switch to the clean ${}^{p+r-1}C_r$ form.

Example 3 — [JEE Main · Multi-concept]

Use the binomial approximation to estimate $(255)^{1/4}$.

Write $255 = 256 - 1 = 256\left(1 - \frac{1}{256}\right)$, and note $256 = 4^4$:

$$(255)^{1/4} = (4^4)^{1/4} \left(1 - \frac{1}{256}\right)^{1/4} = 4 \left(1 - \frac{1}{256}\right)^{1/4}.$$

Since $\frac{1}{256}$ is tiny, use $(1+x)^n \approx 1+nx$ with $n = \frac{1}{4}$, $x = -\frac{1}{256}$:

$$\approx 4 \left(1 - \frac{1}{4} \cdot \frac{1}{256}\right) = 4 - \frac{4}{1024} = 4 - \frac{1}{256} \approx 3.99609.$$

$$\boxed{(255)^{1/4} \approx 3.9961}$$

Example 4 — [JEE Advanced Level]

Find the sum of the infinite series $1 + \frac{3}{4} + \frac{3 \cdot 5}{4 \cdot 8} + \frac{3 \cdot 5 \cdot 7}{4 \cdot 8 \cdot 12} + \dots$

Match to $(1+y)^n = 1 + ny + \frac{n(n-1)}{2!}y^2 + \dots$.

- Second term: $ny = \frac{3}{4}$.
- Third term: $\frac{n(n-1)}{2!}y^2 = \frac{3 \cdot 5}{4 \cdot 8} = \frac{15}{32}$.

Divide the third equation by the square of the second to eliminate scaling:

$$\frac{\frac{n(n-1)}{2}y^2}{(ny)^2} = \frac{n-1}{2n} = \frac{15/32}{(3/4)^2} = \frac{15/32}{9/16} = \frac{15}{32} \cdot \frac{16}{9} = \frac{5}{6}.$$

So $\frac{n-1}{2n} = \frac{5}{6} \Rightarrow 6(n-1) = 10n \Rightarrow -6 = 4n \Rightarrow n = -\frac{3}{2}$. Then from $ny = \frac{3}{4}$: $-\frac{3}{2}y = \frac{3}{4} \Rightarrow y = -\frac{1}{2}$. The series sums to

$$(1+y)^n = \left(1 - \frac{1}{2}\right)^{-3/2} = \left(\frac{1}{2}\right)^{-3/2} = 2^{3/2} = 2\sqrt{2}. \quad \boxed{2\sqrt{2}}$$

Section 5: Practice Exercises

1. **[Foundation]** Write the first four terms of $(1-x)^{-1}$ and of $(1+x)^{-1}$, stating validity.
2. **[JEE Main]** Find the coefficient of x^3 in $(1-3x)^{-2}$.
3. **[JEE Main]** Approximate $(1.02)^8$ to two decimal places using the binomial approximation.
4. **[JEE Advanced]** Find the coefficient of x^n in the expansion of $(1-x)^{-5}$.
5. **[JEE Advanced]** Find the coefficient of x^4 in the expansion of $\frac{1}{(1-x)(1-2x)}$ (for $|x| < \frac{1}{2}$).

Answers:

1. $1+x+x^2+x^3+\dots$ and $1-x+x^2-x^3+\dots$; both $|x| < 1$
2. ${}^4C_3 \cdot 3^3 = 4 \cdot 27 = 108$
3. $\approx 1 + 8(0.02) = 1.16$ (to 2 d.p.)
4. $n+4C_4$
5. 31 (partial fractions: $\frac{1}{(1-x)(1-2x)} = \frac{-1}{1-x} + \frac{2}{1-2x}$, so coeff of x^4 is $-1 + 2 \cdot 2^4 = 31$)

Section 6: Crack the MCQ

Q1. The expansion of $(1+x)^n$ for non-integer n is valid only when:

- (a) $x > 0$ (b) $|x| < 1$ (c) $n > 0$ (d) always

Correct: (b) $|x| < 1$. The infinite series converges only in this range. *Distractors:* (a) sign of x is irrelevant; magnitude matters. (c) any-index includes negative n . (d) false — divergent for $|x| \geq 1$.

Q2. The coefficient of x^r in $(1-x)^{-1}$ is:

- (a) r (b) 1 (c) $(-1)^r$ (d) $r+1$

Correct: (b) $1. (1-x)^{-1} = 1 + x + x^2 + \dots$, all coefficients 1. *Distractors:* (a),(d) are coefficients of $(1-x)^{-2}$. (c) belongs to $(1+x)^{-1}$.

Q3. Using $(1+x)^n \approx 1 + nx$, the value of $(0.998)^{-1}$ is approximately:

- (a) 0.998 (b) 1.002 (c) 0.502 (d) 1.998

Correct: (b) $1.002. (1-0.002)^{-1} \approx 1 + (-1)(-0.002) = 1.002.$ *Distractors:* (a) is the base itself. (c),(d) come from sign/operation errors.

Q4. Which expansion has general term $(r+1)x^r$?

- (a) $(1+x)^{-1}$ (b) $(1-x)^{-1}$ (c) $(1-x)^{-2}$ (d) $(1+x)^2$

Correct: (c) $(1-x)^{-2}$. Its terms are $1+2x+3x^2+\dots$, i.e. coefficient $r+1$. *Distractors:* (a),(b) have coefficient ± 1 or 1. (d) is a finite 3-term expansion.

Section 7: Common Student Pitfalls & Pro-Tips

- **Ignoring $|x| < 1$.** The any-index series is only valid for $|x| < 1$; applying it outside is meaningless. Always arrange the binomial so the smaller part is the “ x .”
- **Writing nC_r for non-integer n .** nC_r is undefined when n is negative or fractional. Use the falling-factorial general term, or the ${}^{p+r-1}C_r$ form for $(1-x)^{-p}$.
- **Forgetting to factor out the leading term.** For $(a+b)^n$, factor as $a^n(1+\frac{b}{a})^n$ first; expanding $(a+b)^n$ directly with the series is wrong unless the leading term is 1.
- **Over-truncating approximations.** $1+nx$ is first-order only; if the question demands more precision, keep the $\frac{n(n-1)}{2}x^2$ term.

Pro-Tip — series identification. When asked to *sum* a numeric series, match first three terms to $1, ny, \frac{n(n-1)}{2}y^2$ and divide the third by the square of the second: this single ratio gives $\frac{n-1}{2n}$, from which n (then y) falls out immediately.

Section 8: Cheat Sheet Box

ANY-INDEX BINOMIAL & APPROXIMATIONS — at a glance (*JEE extension; $|x| < 1$*)

- **Series:** $(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \dots$ — infinite, no nC_r .
- **Must-knows:** $(1-x)^{-1} = \sum x^r$; $(1+x)^{-1} = \sum (-x)^r$; $(1-x)^{-2} = \sum (r+1)x^r$; $(1-x)^{-p} = \sum {}^{p+r-1}C_r x^r$.
- **Approximation:** small $|x| \Rightarrow (1+x)^n \approx 1 + nx$.
- **General $(a+b)^n$:** factor as $a^n(1+\frac{b}{a})^n$, need $|\frac{b}{a}| < 1$.
- **Sum-the-series:** match to $1, ny, \frac{n(n-1)}{2}y^2$; ratio $\frac{n-1}{2n}$ gives n , then y .

Memory aid: “Non-integer power $\rightarrow$ infinite series, only if $|x| < 1$, and never write nC_r .”

Subtopic 06: Applications — Divisibility, Remainders, Last Digits & Integral/Fractional Parts

Exam Relevance: A high-yield JEE application class. “Remainder when a^n is divided by m ,” “last two digits of a^n ,” and integral/fractional-part problems on $(p + \sqrt{q})^n$ recur in JEE Main and Advanced. Beyond rationalised CBSE core, but a reliable source of marks for competitive aspirants.

Section 1: Concept Intuition

Why would a chapter on expansions help you find the remainder of $7^{103} \div 25$, a question with no x in sight? Because the binomial theorem lets you **split a number into “a big clean piece plus a tiny piece.”**

The trick is always the same: rewrite the base so it is *almost* a multiple of the divisor. To handle $7^{103} \bmod 25$, notice $7^2 = 49 = 50 - 1$, and 50 is a multiple of 25. So $7^{103} = 7 \cdot (50 - 1)^{51}$, and when you expand $(50 - 1)^{51}$ by the binomial theorem, **every term containing 50 is divisible by 25** — they all vanish mod 25, leaving only the last term $(-1)^{51}$ to determine the remainder.

Think about it this way... a binomial expansion of $(km \pm 1)^n$ is a list of terms, and all but the very last carry a factor of m (often m^2 or higher). Modulo m (or m^2), those terms are invisible. So divisibility, remainder, and last-digit questions all reduce to “*which few terms survive when I throw away everything divisible by the modulus?*” — usually just one or two terms.

The integral/fractional-part problems use a sister idea: a number like $(7 + 4\sqrt{3})^n$ is irrational, but if you **add its conjugate** $(7 - 4\sqrt{3})^n$, the irrational parts cancel and you get a whole number. Since the conjugate is a tiny positive number (less than 1), it must equal the fractional part’s complement — and that pins down both the integer and fractional parts exactly.

Validity conditions & constraints. The base-splitting works whenever you can write the base as (multiple of the modulus) $\pm$ (something small, ideally ± 1). The conjugate trick for fractional parts requires that the conjugate $(p - \sqrt{q})^n$ genuinely lies strictly between 0 and 1 — always verify $0 < p - \sqrt{q} < 1$ before using it.

Section 2: Key Formulas & Definitions

Base-splitting identity. For divisibility/remainder/last-digit problems, write the base as $a = K \pm 1$ where K is a multiple of the divisor (or its square), then

$$(K \pm 1)^n = \sum_{r=0}^n {}^nC_r K^r (\pm 1)^{n-r}.$$

Modulo K (or K^2), only the lowest-power terms in K survive.

Useful consequences:

- $(1 + K)^n = 1 + nK + {}^nC_2 K^2 + \dots$, so $(1 + K)^n \equiv 1 + nK \pmod{K^2}$.
- $a^n - 1$ is divisible by $(a - 1)$; $a^n - b^n$ is divisible by $(a - b)$.

Integral & fractional part (conjugate trick). Let $(p + \sqrt{q})^n = I + f$ with $I \in \mathbb{Z}$ and $0 \leq f < 1$. Define the conjugate $f' = (p - \sqrt{q})^n$ with $0 < f' < 1$. Then $I + f + f'$ (or with alternating sign) is an **integer**, forcing $f' = 1 - f$ (or $f' = f$), and

$$(I + f)f' = (p + \sqrt{q})^n(p - \sqrt{q})^n = (p^2 - q)^n.$$

Here $n \in \mathbb{N}$; K, a, b, p, q are integers; I integer part, f fractional part. Pure numbers.

Section 3: Proofs, Theorems & Standard Techniques

Standard technique — remainder of $a^n \bmod m$.

1. Express a (or a convenient power of a) as $K \pm 1$ with K a multiple of m (or of m^2 for last-two-digit work).
2. Expand $(K \pm 1)^n$ by the binomial theorem.
3. Reduce mod m : drop every term containing K (they are $\equiv 0$), keeping only the surviving low-order terms.
4. The leftover is the remainder (adjust to the range 0 to $m - 1$).

Standard technique — last two digits. “Last two digits” = value mod 100. Choose K a multiple of 100 (or arrange so K^2 is a multiple of 100), expand, and keep terms up to the one that first becomes divisible by 100.

Standard technique — divisibility proof. To show $m \mid (\text{expression})$, expand the relevant power as $(K \pm 1)^n$ and show that after cancellation every surviving term carries the factor m .

Standard technique — integral/fractional part.

1. Set $I + f = (p + \sqrt{q})^n$ and $f' = (p - \sqrt{q})^n$.
2. Verify $0 < f' < 1$ (check $0 < p - \sqrt{q} < 1$).
3. Add (if n makes the $\sqrt{q}$ terms cancel on adding) or subtract so the sum/difference is an integer; deduce $f' = 1 - f$ or $f' = f$.
4. Use $(I + f)f' = (p^2 - q)^n$ to evaluate products, or solve for I . ■

Section 4: Worked Examples

Example 1 — [CBSE / School Level]

Show that $9^n - 1$ is divisible by 8 for every positive integer n .

Write $9 = 8 + 1$:

$$9^n = (1 + 8)^n = 1 + {}^nC_1 8 + {}^nC_2 8^2 + \dots + 8^n.$$

Subtract 1:

$$9^n - 1 = {}^nC_1 8 + {}^nC_2 8^2 + \dots + 8^n = 8({}^nC_1 + {}^nC_2 8 + \dots).$$

Every term has a factor of 8, so $8 \mid (9^n - 1)$. ■

Example 2 — [JEE Main · Speed Trap]**Find the remainder when 7^{103} is divided by 25.**Note $7^2 = 49 = 50 - 1$ and 50 is a multiple of 25. So

$$7^{103} = 7 \cdot 7^{102} = 7(7^2)^{51} = 7(50 - 1)^{51}.$$

Expanding $(50 - 1)^{51}$, every term except the last contains a factor of 50 (hence of 25):

$$(50 - 1)^{51} \equiv (-1)^{51} = -1 \pmod{25}.$$

Therefore $7^{103} \equiv 7 \cdot (-1) = -7 \equiv 18 \pmod{25}$.

Remainder = 18.

The trap: trying to compute powers of 7 directly. The fast move is spotting $7^2 = 50 - 1$ so the modulus appears inside the bracket.**Example 3 — [JEE Main · Multi-concept]****Find the last two digits of 3^{400} .**Last two digits = value mod 100. Note $3^4 = 81 = 80 + 1$, but cleaner: $3^4 = 81$, and we want $3^{400} = (3^4)^{100} = 81^{100} = (80 + 1)^{100}$:

$$(80 + 1)^{100} = 1 + {}^{100}C_1 80 + {}^{100}C_2 80^2 + \dots$$

Modulo 100: the term ${}^{100}C_1 \cdot 80 = 100 \cdot 80 = 8000 \equiv 0$, and $80^2 = 6400 \equiv 0$, so every term from $r \geq 1$ vanishes mod 100. Only the first term survives:

$$3^{400} \equiv 1 \pmod{100}.$$

Last two digits = 01.

Example 4 — [JEE Advanced Level]**If $(7 + 4\sqrt{3})^n = I + f$, where I is an integer and $0 \leq f < 1$, prove that $(I + f)(1 - f) = 1$.**Let $f' = (7 - 4\sqrt{3})^n$. Since $7 - 4\sqrt{3} \approx 7 - 6.928 = 0.072$, we have $0 < f' < 1$.Adding the two conjugate expansions, every term containing an **odd** power of $\sqrt{3}$ cancels, leaving an integer:

$$(7 + 4\sqrt{3})^n + (7 - 4\sqrt{3})^n = 2({}^nC_0 7^n + {}^nC_2 7^{n-2}(48) + \dots) = \text{an integer}.$$

So $I + f + f'$ is an integer. Since $0 \leq f < 1$ and $0 < f' < 1$, their sum $f + f'$ lies in $(0, 2)$ and must be the integer 1. Hence $f' = 1 - f$. Finally,

$$(I + f)(1 - f) = (I + f)f' = (7 + 4\sqrt{3})^n(7 - 4\sqrt{3})^n = (49 - 48)^n = 1^n = 1.$$

$(I + f)(1 - f) = 1.$



Section 5: Practice Exercises

1. **[CBSE / School]** Show that $6^n - 1$ is divisible by 5 for all $n \in \mathbb{N}$.
2. **[JEE Main]** Find the remainder when 2^{50} is divided by 7.
3. **[JEE Main]** Find the last digit of 7^{300} .
4. **[JEE Advanced]** Find the remainder when 15^{40} is divided by 11.
5. **[JEE Advanced]** If $(5 + 2\sqrt{6})^n = I + f$ with $0 \leq f < 1$, find the value of $(I + f)(1 - f)$.

Answers:

1. $6 = 5 + 1 \Rightarrow 6^n - 1 = \text{sum of terms each divisible by 5}$.
2. 2 (since $2^3 = 8 \equiv 1$, $2^{50} = 2^{48} \cdot 2^2 \equiv 1 \cdot 4 = 4$) — **remainder 4**
3. 1 ($7^4 \equiv 1 \pmod{10}$, $300 = 4 \cdot 75$)
4. 1 ($15 \equiv 4$, $4^{40} = (4^5)^8$; $4^5 = 1024 \equiv 1 \pmod{11}$)
5. 1 ($(5 + 2\sqrt{6})(5 - 2\sqrt{6}) = 25 - 24 = 1$)

Section 6: Crack the MCQ

Q1. To find $8^{20} \pmod{7}$, the best split of the base is:

- (a) $8 = 7 + 1$ (b) $8 = 2^3$ (c) $8 = 9 - 1$ (d) $8 = 4 \cdot 2$

Correct: (a) $8 = 7 + 1$. Then $8^{20} = (7 + 1)^{20} \equiv 1 \pmod{7}$. *Distractors:* (b),(d) don't expose the modulus 7. (c) uses 9, unrelated to mod 7.

Q2. The remainder when $(101)^{50}$ is divided by 100 is:

- (a) 0 (b) 1 (c) 50 (d) 51

Correct: (b) 1. $(100 + 1)^{50} = 1 + {}^{50}C_1 100 + \dots \equiv 1 \pmod{100}$ (all other terms carry 100). *Distractors:* (c) misreads ${}^{50}C_1 \cdot 100 \pmod{100}$. (a),(d) sign/term errors.

Q3. For $(3 + \sqrt{8})^n = I + f$, the conjugate used is:

- (a) $(3 - \sqrt{8})^n$ (b) $(\sqrt{8} - 3)^n$ (c) $(3 + \sqrt{8})^{-n}$ (d) $(8 - 3)^n$

Correct: (a) $(3 - \sqrt{8})^n$. It satisfies $0 < 3 - \sqrt{8} < 1$ ($\sqrt{8} \approx 2.83$), so it equals $1 - f$. *Distractors:* (b) is negative. (c),(d) are not the conjugate.

Q4. $a^n - b^n$ is always divisible by:

- (a) $a + b$ (b) $a - b$ (c) ab (d) n

Correct: (b) $a - b$. Writing $a = b + (a - b)$ and expanding shows every surviving term carries $(a - b)$. *Distractors:* (a) divides $a^n - b^n$ only for even n . (c),(d) are not general.

Section 7: Common Student Pitfalls & Pro-Tips

- **Choosing the wrong split.** Match the bracket to the modulus: for mod 25 use a multiple of 25 (like 50); for last-two-digits use a multiple of 100. A split that doesn't expose the modulus wastes the method.

- **Stopping reduction too early.** Keep reducing until only modulus-free terms remain; a lingering K -term means more cancellation is available.
- **Negative remainders.** $-7 \bmod 25$ is 18, not -7 . Always shift into 0 to $m - 1$.
- **Skipping the $0 < f' < 1$ check.** The conjugate trick only works if the conjugate truly lies in $(0, 1)$ — verify $0 < p - \sqrt{q} < 1$ first.

Pro-Tip. Find the smallest power of the base that is $\equiv \pm 1$ modulo the divisor; that power sets a short cycle, and the exponent reduced modulo that cycle gives the answer fast. (E.g. $2^3 \equiv 1 \bmod 7$, so powers of 2 cycle with period 3 mod 7.)

Section 8: Cheat Sheet Box

APPLICATIONS: DIVISIBILITY / REMAINDERS / PARTS — at a glance (JEE)

- **Core move:** write base $= K \pm 1$ with K a multiple of the modulus; expand $(K \pm 1)^n$; drop all terms containing K .
- $(1 + K)^n \equiv 1 + nK \pmod{K^2}$.
- **Last two digits** = value mod 100; **last digit** = mod 10.
- $a^n - 1 : (a - 1)$; $a^n - b^n : (a - b)$.
- **Integral/fractional part:** $I + f = (p + \sqrt{q})^n$, $f' = (p - \sqrt{q})^n \in (0, 1)$, $f' = 1 - f$, and $(I + f)f' = (p^2 - q)^n$.

Memory aid: “Make the divisor appear inside the bracket, then everything with it disappears.”

Subtopic 07: The Multinomial Theorem

Exam Relevance: A JEE Advanced extension of the binomial theorem to three or more terms. Appears as “number of terms in $(a + b + c)^n$ ” and “coefficient of $x^a y^b z^c$ ” questions, and underpins counting problems. Beyond the CBSE/NCERT core; include if you are targeting JEE Advanced.

Scope note: This generalises Sub-01’s $(a + b)^n$ to $(x_1 + x_2 + \dots + x_k)^n$ — two parts become k parts. The same “count the ways to distribute the n choices” logic carries straight over.

Section 1: Concept Intuition

The binomial theorem answered: multiply $(a + b)$ by itself n times — what comes out? Each of the n brackets donated either an a or a b , and we counted the ways. The **multinomial theorem** asks the identical question when each bracket offers *more than two* choices: in $(a + b + c)^n$, each of the n brackets donates an a , a b , **or** a c .

Think about it this way... a single term of $(a + b + c)^n$ looks like $a^{n_1} b^{n_2} c^{n_3}$, where n_1 brackets gave a , n_2 gave b , n_3 gave c , and of course $n_1 + n_2 + n_3 = n$ (every bracket donated exactly once). The coefficient counts **how many ways** the n brackets can be split into those three groups — which is the multinomial count $\frac{n!}{n_1! n_2! n_3!}$ (the same arrangements-with-repetition idea from Permutations & Combinations). With two parts this collapses to $\frac{n!}{r!(n-r)!} = {}^n C_r$, recovering the binomial coefficient — so the binomial theorem is just the $k = 2$ case.

Counting the **number of distinct terms** is also a P&C question in disguise: a term is fixed by the exponents $(n_1, n_2, \dots, n_k)$ with $n_1 + \dots + n_k = n$, $n_i \geq 0$. The number of such non-negative integer solutions is the classic stars-and-bars count ${}^{n+k-1} C_{k-1}$.

Validity conditions & constraints. $n \in \mathbb{N}$; the exponents $n_1, \dots, n_k$ are non-negative integers summing to n . “Number of terms” assumes the x_i are independent variables (no hidden relations like $x_2 = x_1^2$ that would merge terms — if such a relation exists, the count drops, as in the perfect-power tricks of Sub-01).

Section 2: Key Formulas & Definitions

Multinomial theorem. For $n \in \mathbb{N}$,

$$(x_1 + x_2 + \dots + x_k)^n = \sum \frac{n!}{n_1! n_2! \dots n_k!} x_1^{n_1} x_2^{n_2} \dots x_k^{n_k},$$

the sum taken over all non-negative integers $n_1, \dots, n_k$ with $n_1 + n_2 + \dots + n_k = n$.

Multinomial coefficient (the coefficient of $x_1^{n_1} \cdots x_k^{n_k}$):

$$\frac{n!}{n_1! n_2! \cdots n_k!}.$$

Number of distinct terms in $(x_1 + \cdots + x_k)^n$:

$$\boxed{n+k-1 C_{k-1}}$$

In particular, a **trinomial** $(a + b + c)^n$ has $n+2 C_2 = \frac{(n+1)(n+2)}{2}$ terms.

Total sum of all coefficients: set every $x_i = 1$, giving k^n .

Here n is the index, k the number of parts, n_i the exponents. Pure numbers.

Section 3: Proofs, Theorems & Standard Techniques

Why the coefficient is $\frac{n!}{n_1! \cdots n_k!}$. Expanding $(x_1 + \cdots + x_k)^n$ means choosing one x_i from each of the n factors and multiplying. A term $x_1^{n_1} \cdots x_k^{n_k}$ arises every time we pick x_1 from some n_1 of the factors, x_2 from n_2 of them, and so on. The number of ways to assign the n factors into labelled groups of sizes $n_1, \dots, n_k$ is the arrangements-of-multiset count $\frac{n!}{n_1! n_2! \cdots n_k!}$. Summing over all valid exponent-tuples gives the theorem. ■

Why the term count is $n+k-1 C_{k-1}$. Each distinct term corresponds to one solution of $n_1 + n_2 + \cdots + n_k = n$ in non-negative integers. By stars and bars (distributing n identical “units” among k “slots”), the number of solutions is $n+k-1 C_{k-1}$. ■

Standard technique — coefficient of a specified monomial. To find the coefficient of $x_1^a x_2^b \cdots$ in $(x_1 + \cdots + x_k)^n$: check $a + b + \cdots = n$ (else the coefficient is 0), then write $\frac{n!}{a! b! \cdots}$.

Standard technique — coefficient of x^m in $(1 + x + x^2 + \cdots + x^t)^k$. Either (i) expand as a product and collect, or (ii) use the substitution $1 + x + \cdots + x^t = \frac{1 - x^{t+1}}{1 - x}$ and combine with the any-index series $(1 - x)^{-k}$ from Sub-05, extracting the coefficient of x^m .

Section 4: Worked Examples

Example 1 — [CBSE / School Level]

Find the number of distinct terms in the expansion of $(a + b + c)^{12}$.

Using ${}^{n+k-1}C_{k-1}$ with $n = 12$, $k = 3$:

$${}^{12+3-1}C_{3-1} = {}^{14}C_2 = \frac{14 \cdot 13}{2} = 91. \quad \boxed{91}$$

Example 2 — [JEE Main · Speed Trap]

Find the coefficient of $x^2y^3z^4$ in the expansion of $(x + y + z)^9$.

First check the exponents sum to the index: $2 + 3 + 4 = 9 = n$. ✓ The multinomial coefficient is

$$\frac{9!}{2!3!4!} = \frac{362880}{2 \cdot 6 \cdot 24} = \frac{362880}{288} = 1260. \quad \boxed{1260}$$

The trap: using 9C_2 or some binomial coefficient. With three variables you need the full multinomial $\frac{n!}{a!b!c!}$, not a single nC_r .

Example 3 — [JEE Main · Multi-concept]

Find the coefficient of x^4 in the expansion of $(1 + x + x^2)^3$.

Method (direct multinomial): we need $1^{n_1}x^{n_2}(x^2)^{n_3}$ with $n_1 + n_2 + n_3 = 3$ and total x -power $n_2 + 2n_3 = 4$. Solving: $(n_1, n_2, n_3) = (0, 2, 1)$ or $(1, 0, 2)$.

$$\frac{3!}{0!2!1!} + \frac{3!}{1!0!2!} = 3 + 3 = 6. \quad \boxed{6}$$

Check (alternative): $(1 + x + x^2)^3$ expands to $1 + 3x + 6x^2 + 7x^3 + 6x^4 + 3x^5 + x^6$ — the coefficient of x^4 is indeed 6 (note the palindromic coefficients).

Example 4 — [JEE Advanced Level]

Find the coefficient of x^7 in the expansion of $(1 + x + x^2 + x^3 + \dots)^3$, $|x| < 1$.

The bracket is the geometric series $1 + x + x^2 + \dots = (1 - x)^{-1}$, so the whole expression is $(1 - x)^{-3}$. From Sub-05, the coefficient of x^r in $(1 - x)^{-3}$ is ${}^{3+r-1}C_r = {}^{r+2}C_r = {}^{r+2}C_2$. For $r = 7$:

$${}^9C_2 = \frac{9 \cdot 8}{2} = 36. \quad \boxed{36}$$

This is also the number of non-negative integer solutions of $n_1 + n_2 + n_3 = 7$ — the multinomial and stars-and-bars viewpoints agree.

Section 5: Practice Exercises

1. **[CBSE / School]** Find the number of distinct terms in $(p + q + r + s)^{10}$.
2. **[JEE Main]** Find the coefficient of $a^3b^2c^2$ in $(a + b + c)^7$.
3. **[JEE Main]** Find the sum of all coefficients in $(2x - 3y + z)^8$.

4. **[JEE Advanced]** Find the coefficient of x^5 in $(1 + x + x^2 + x^3)^4$.
5. **[JEE Advanced]** Find the number of terms in $(x+y+z)^{20}$ that contain x (i.e. with x -exponent ≥ 1).

Answers:

- ${}^{13}C_3 = 286$
- $\frac{7!}{3!2!2!} = 210$
- 0 (put $x = y = z = 1$: $(2 - 3 + 1)^8 = 0$)
- 40 (coefficient of x^5 in $(1 + x + x^2 + x^3)^4$; via $(1 - x^4)^4(1 - x)^{-4}$: ${}^8C_3 - 4 \cdot {}^4C_3 = 56 - 16$)
- ${}^{21}C_2 = 210$ (set x -exponent ≥ 1 : solutions of $n'_1 + n_2 + n_3 = 19$; equivalently $231 - 21$)

Section 6: Crack the MCQ

Q1. The number of distinct terms in $(a + b + c)^n$ is:

- (a) $n + 1$ (b) $3n$ (c) ${}^{n+2}C_2$ (d) 3^n

Correct: (c) ${}^{n+2}C_2$. Stars and bars with $k = 3$: ${}^{n+3-1}C_{3-1}$. *Distractors:* (a) is the binomial ($k = 2$) count. (d) is the number of *terms before collecting* / sum of coefficients-style error. (b) is fabricated.

Q2. The coefficient of xyz in $(x + y + z)^3$ is:

- (a) 1 (b) 3 (c) 6 (d) 9

Correct: (c) 6. $\frac{3!}{1!1!1!} = 6$. *Distractors:* (a) forgets the arrangement count. (b) uses 3C_1 . (d) is 3^2 .

Q3. The sum of all coefficients in $(x + y + z)^n$ is:

- (a) 3^n (b) 2^n (c) n^3 (d) ${}^{n+2}C_2$

Correct: (a) 3^n . Put $x = y = z = 1$. *Distractors:* (b) is the binomial sum. (d) is the number of *terms*, not the sum of coefficients. (c) fabricated.

Q4. For the coefficient of $x^2y^2z^2$ in $(x + y + z)^n$ to be non-zero, n must equal:

- (a) 2 (b) 3 (c) 6 (d) any $n \geq 2$

Correct: (c) 6. The exponents must sum to n : $2 + 2 + 2 = 6$. *Distractors:* (a),(b) give exponent sums exceeding n (coefficient 0). (d) ignores the sum-to- n requirement.

Section 7: Common Student Pitfalls & Pro-Tips

- **Using a binomial coefficient for three+ variables.** The coefficient of $x^a y^b z^c$ is $\frac{n!}{a!b!c!}$, not nC_a or any single binomial coefficient.
- **Forgetting the sum-to- n check.** A monomial's coefficient is zero unless its exponents add up to exactly n .

- **Confusing “number of terms” with “sum of coefficients.”** Terms $= {}^{n+k-1}C_{k-1}$; sum of coefficients $= k^n$ (set all variables 1). They are different questions.
- **Hidden relations merge terms.** “Number of terms” assumes independent variables; if the bracket is a perfect power or the variables are related, distinct terms collapse (recall the $(1 + 2x + x^2)^{10} = (1 + x)^{20}$ trick from Sub-02).

Pro-Tip. For “coefficient of x^m in $(1 + x + \dots + x^t)^k$,” convert the bracket to $\frac{1 - x^{t+1}}{1 - x}$ and pair it with the any-index expansion of $(1 - x)^{-k}$ — this turns a messy multinomial collection into two clean binomial-coefficient lookups.

Section 8: Cheat Sheet Box

MULTINOMIAL THEOREM — at a glance (JEE Advanced)

- **Theorem:** $(x_1 + \dots + x_k)^n = \sum \frac{n!}{n_1! \dots n_k!} x_1^{n_1} \dots x_k^{n_k}$, with $\sum n_i = n$.
- **Coefficient** of $x_1^{n_1} \dots x_k^{n_k}$: $\frac{n!}{n_1! \dots n_k!}$ (needs $\sum n_i = n$).
- **Number of terms:** ${}^{n+k-1}C_{k-1}$; trinomial $\Rightarrow {}^{n+2}C_2 = \frac{(n+1)(n+2)}{2}$.
- **Sum of all coefficients:** k^n (set every variable = 1).
- $k = 2$ recovers the ordinary binomial theorem.

Memory aid: “Spread n choices over k boxes — coefficient is $n!$ over the factorials of the box-sizes.”

Chapter 7: Binomial Theorem — Supplement

This supplement adds four pieces of machinery that the main chapter’s seven subtopics do not exercise. Three of them are short addenda that clip onto existing subtopics — read each one immediately after the subtopic it names. The fourth is a full new subtopic, because the technique it teaches (multisection by roots of unity) appears nowhere in Sub-01 to Sub-07 and is a genuine JEE Advanced tool in its own right.

Piece	Attaches to	What it gives you
Addendum A	Sub-01	Conjugate-pair expansions $(a + b)^n \pm (a - b)^n$, surd evaluation, exact numerical powers
Addendum B	Sub-02	Counting rational and integral terms in an expansion of surds
Addendum C	Sub-06	Divisibility proofs that need two surviving terms — the mod K^2 machinery
Subtopic 08	new (extends Sub-04)	Multisection: sums of every k th binomial coefficient, via ω and i

Throughout, notation follows the main chapter: nC_r for the binomial coefficient, $T_{r+1} = {}^nC_r a^{n-r} b^r$ for the general term, and C_r as shorthand for nC_r wherever a single fixed n is in play.

Addendum A to Subtopic 01: Conjugate-Pair Expansions and Exact Evaluation

Exam Relevance: This is a guaranteed NCERT exercise family and a standard CBSE Class 11 short-answer question at 2-3 marks: “find $(a + b)^4 - (a - b)^4$ and hence evaluate $(\sqrt{3} + \sqrt{2})^4 - (\sqrt{3} - \sqrt{2})^4$ ”, or “evaluate $(102)^5$ using the binomial theorem”. JEE Main recycles the same identity as a term-counting question — “the number of terms in $(x + a)^{53} + (x - a)^{53}$ ” — which is pure bookkeeping and free marks if you know the rule. JEE Advanced uses the same cancellation as a step inside integral/fractional-part problems, which is exactly where Sub-06 met it.

The idea

Sub-06 used the cancellation between $(p + \sqrt{q})^n$ and $(p - \sqrt{q})^n$ as one line inside a larger proof. It deserves to be a technique of its own, because on its own it turns ugly-looking surd arithmetic into a two-line calculation.

Think about it this way. Write the two expansions one above the other:

$$(a + b)^n = \sum_{r=0}^n {}^nC_r a^{n-r} b^r, \quad (a - b)^n = \sum_{r=0}^n (-1)^r {}^nC_r a^{n-r} b^r.$$

They are the same list of terms; the only difference is that the second list flips the sign of every odd- r entry. So when you **add** the lists, the odd- r entries destroy each other and the even- r entries double. When you **subtract**, the reverse happens. **You never lose the terms you keep — you double them.** That factor of 2 is the single most-forgotten thing in this whole addendum.

Figure brief: Two horizontal strips of five boxes each, labelled $r = 0, 1, 2, 3, 4$, representing the terms of $(a + b)^4$ (top, all signs +) and $(a - b)^4$ (bottom, signs +, -, +, -, +). Draw green “add” arrows joining the $r = 0, 2, 4$ columns with the annotation “doubles”, and red crossed-out arrows on the $r = 1, 3$ columns annotated “cancels”. Beside it show the mirror-image picture for subtraction.

Figure brief: A number-line strip of the $n + 1$ term indices $r = 0, 1, \dots, n$ with even indices shaded one colour and odd indices another, captioned “the sum keeps the shaded half, the difference keeps the unshaded half” — the term-count rule read straight off the picture.

Key results

The two conjugate identities. For $n \in \mathbb{N}$,

$$(a + b)^n + (a - b)^n = 2 [{}^nC_0 a^n + {}^nC_2 a^{n-2} b^2 + {}^nC_4 a^{n-4} b^4 + \dots],$$

$$(a + b)^n - (a - b)^n = 2 [{}^nC_1 a^{n-1} b + {}^nC_3 a^{n-3} b^3 + {}^nC_5 a^{n-5} b^5 + \dots].$$

- a, b — the two parts of the binomial; they may be numbers, surds or algebraic expressions.
- The sum keeps exactly the **even** values of r (including $r = 0$); the difference keeps exactly the **odd** values of r .
- Both sums stop at $r = n$, so the last surviving index is n or $n - 1$ according to parity.

Number of terms. The full expansion has $n + 1$ terms; each combination takes a share of them.

	$(x + a)^n + (x - a)^n$	$(x + a)^n - (x - a)^n$
n even	$\frac{n}{2} + 1$	$\frac{n}{2}$

	$(x + a)^n + (x - a)^n$	$(x + a)^n - (x - a)^n$
n odd	$\frac{n+1}{2}$	$\frac{n+1}{2}$
unified	$\left\lfloor \frac{n}{2} \right\rfloor + 1$	$\left\lceil \frac{n}{2} \right\rceil$

Validity: $x \neq 0$, $a \neq 0$, $n \in \mathbb{N}$. The two counts always add to $n + 1$, which is your instant check.

Why this evaluates surds cleanly. Put $a = \sqrt{p}$ and $b = \sqrt{q}$ with p, q positive rationals. In the **sum**, every surviving b^r has even r , so $b^r = q^{r/2}$ is rational. In the **difference**, every surviving b^r has odd r , so $b^r = q^{(r-1)/2} \sqrt{q}$ carries exactly one $\sqrt{q}$. Combined with the parity of $n - r$ on the a side, the outcome is always one of two clean shapes:

a rational number, or (a rational number) $\times$ (a single surd).

The truncation bound (for size comparisons). For $x > 0$ and $n \geq 2$, every term of $(1 + x)^n$ after the first two is strictly positive, so

$$(1 + x)^n > 1 + nx.$$

That one inequality settles “which is larger” questions without any computation. For instance $(1.1)^{10000} > 1 + 10000(0.1) = 1001 > 1000$, so $(1.1)^{10000}$ is the larger of $(1.1)^{10000}$ and 1000.

Procedure A1 — evaluating a conjugate sum or difference

1. **Match the pattern.** Write both brackets with the same a and b ; check that they differ only in the sign of b . *Why:* if the two brackets are not genuine conjugates, nothing cancels.
2. **Choose the half.** Adding keeps even r ; subtracting keeps odd r . *Why:* the sign flip in $(a - b)^n$ is $(-1)^r$.
3. **Write the surviving half with the factor 2 out front.** *Why:* the kept terms appear once in each bracket.
4. **Count the surviving terms** using the table and confirm it against $n + 1$. *Why:* a five-second guard against dropping a term.
5. **Substitute the numerical values.** Every surviving power is now easy, because Step 2 already guaranteed the parities are convenient.
6. **Sanity check the shape.** A sum-with-even- n must come out rational; a difference must come out as a rational multiple of one surd. An answer of the wrong shape means a lost term.

Procedure A2 — exact numerical evaluation by expansion

1. **Split the base against the nearest round number:** $102 = 100 + 2$, $96 = 100 - 4$, $99 = 100 - 1$. *Why:* powers of 100 are just shifts, so all the arithmetic becomes trivial.
2. **Write the full expansion** — every term, because you want the exact value, not an approximation.

3. **Evaluate each term separately and line the digits up** by place value before adding.
Why: almost every mistake here is a misaligned zero, not a wrong binomial coefficient.
4. **Watch the signs** when the split is a subtraction: signs alternate $+, -, +, -$.

Worked examples

Example A1 — [CBSE Board Level]

Find $(a + b)^4 - (a - b)^4$, **and hence evaluate** $(\sqrt{3} + \sqrt{2})^4 - (\sqrt{3} - \sqrt{2})^4$.

Here $n = 4$, so the difference keeps the odd indices $r = 1$ and $r = 3$:

$$(a + b)^4 - (a - b)^4 = 2 [{}^4C_1 a^3 b + {}^4C_3 a b^3] = 2 [4a^3 b + 4ab^3] = 8ab(a^2 + b^2).$$

Term count check: $n = 4$ is even, so the difference should have $n/2 = 2$ terms. It does.

Now put $a = \sqrt{3}$, $b = \sqrt{2}$, so $ab = \sqrt{6}$, $a^2 = 3$, $b^2 = 2$:

$$(\sqrt{3} + \sqrt{2})^4 - (\sqrt{3} - \sqrt{2})^4 = 8\sqrt{6}(3 + 2) = 40\sqrt{6}.$$

Shape check: a difference of conjugate surds, so the answer should be a rational multiple of a single surd. It is.

Example A2 — [CBSE Board Level]

Find $(x + 1)^6 + (x - 1)^6$, **and hence evaluate** $(\sqrt{2} + 1)^6 + (\sqrt{2} - 1)^6$.

Here $n = 6$, and the sum keeps the even indices $r = 0, 2, 4, 6$:

$$(x + 1)^6 + (x - 1)^6 = 2 [{}^6C_0 x^6 + {}^6C_2 x^4 + {}^6C_4 x^2 + {}^6C_6] = 2 [x^6 + 15x^4 + 15x^2 + 1].$$

Term count check: $n = 6$ even, sum should have $6/2 + 1 = 4$ terms. It does.

Put $x = \sqrt{2}$, so $x^2 = 2$, $x^4 = 4$, $x^6 = 8$:

$$2 [8 + 15(4) + 15(2) + 1] = 2 [8 + 60 + 30 + 1] = 2(99) = 198.$$

$$(\sqrt{2} + 1)^6 + (\sqrt{2} - 1)^6 = 198.$$

Notice what happened: an expression built entirely out of irrational numbers collapsed to a whole number. That is the cancellation of Sub-06 doing its work, and it is exactly why this identity is the engine behind integral-part problems.

Example A3 — [JEE Main — Speed Trap]

Find the number of terms in the expansion of $(x + a)^{100} + (x - a)^{100}$, **and in** $(x + a)^{100} - (x - a)^{100}$.

The trap: the phrase “number of terms” makes students reach for $n + 1 = 101$, or, having remembered that half the terms die, for $101/2$ rounded to 50 or 51 at random. Both halves cannot be 51, and neither can be 101.

The correct move is to count indices, not to divide blindly. The surviving indices are:

- Sum: $r = 0, 2, 4, \dots, 100$ — that is $\frac{100}{2} + 1 = 51$ terms.
- Difference: $r = 1, 3, 5, \dots, 99$ — that is $\frac{100}{2} = 50$ terms.

51 and 50.

Check: $51 + 50 = 101 = n + 1$. The seductive wrong answer here is 51 for **both**, which fails this check instantly.

Example A4 — [JEE Advanced Level]

Evaluate $(2 + \sqrt{3})^5 + (2 - \sqrt{3})^5$ **exactly, and use the result to find the integral part of** $(2 + \sqrt{3})^5$.

Take $a = 2$, $b = \sqrt{3}$, $n = 5$. The sum keeps the even indices $r = 0, 2, 4$ (three terms, as $\lfloor 5/2 \rfloor + 1 = 3$):

$$(2 + \sqrt{3})^5 + (2 - \sqrt{3})^5 = 2 \left[{}^5C_0 2^5 + {}^5C_2 2^3 (\sqrt{3})^2 + {}^5C_4 2 (\sqrt{3})^4 \right].$$

Term by term: ${}^5C_0 2^5 = 32$; ${}^5C_2 2^3 (3) = 10 \cdot 8 \cdot 3 = 240$; ${}^5C_4 2 (9) = 5 \cdot 2 \cdot 9 = 90$. Hence

$$(2 + \sqrt{3})^5 + (2 - \sqrt{3})^5 = 2(32 + 240 + 90) = 2(362) = 724.$$

Now the integral part. Since $\sqrt{3} \approx 1.732$, we have $0 < 2 - \sqrt{3} \approx 0.268 < 1$, so $0 < (2 - \sqrt{3})^5 < 1$ — the conjugate check demanded by Sub-06. Write $(2 + \sqrt{3})^5 = I + f$ with $I \in \mathbb{Z}$ and $0 \leq f < 1$, and $f' = (2 - \sqrt{3})^5 \in (0, 1)$. Then $I + f + f' = 724$ is an integer, and since $f + f' \in (0, 2)$ it must equal 1. Therefore

$$I = 724 - 1 = 723.$$

The integral part of $(2 + \sqrt{3})^5$ is 723, and its fractional part is $1 - (2 - \sqrt{3})^5$. Note how the conjugate identity did all the work: it produced the integer 724, and the smallness of the conjugate did the rest.

Practice exercises

1. **[CBSE Board]** Find $(a + b)^4 + (a - b)^4$ and hence evaluate $(\sqrt{3} + \sqrt{2})^4 + (\sqrt{3} - \sqrt{2})^4$.
2. **[CBSE Board]** Evaluate $(101)^4$ exactly using the binomial theorem.
3. **[CBSE Board]** Evaluate $(99)^5$ exactly using the binomial theorem.
4. **[CBSE Board]** Evaluate $(102)^5$ exactly using the binomial theorem.
5. **[JEE Main]** Find the number of terms in the expansion of $(x + a)^{53} + (x - a)^{53}$.

6. **[JEE Main]** Evaluate $(\sqrt{5} + 1)^5 - (\sqrt{5} - 1)^5$.
7. **[JEE Advanced]** Evaluate $(\sqrt{3} + \sqrt{2})^6 + (\sqrt{3} - \sqrt{2})^6$.
8. **[JEE Advanced]** Which is larger, $(1.01)^{1000}$ or 10? Justify with a truncation bound.

Answers:

1. $2(a^4 + 6a^2b^2 + b^4)$; value = $2(9 + 36 + 4) = 98$.
2. 104060401.
3. 9509900499.
4. 11040808032.
5. 27 (the even indices $r = 0, 2, \dots, 52$).
6. 352.
7. 970.
8. $(1.01)^{1000}$ is larger: $(1.01)^{1000} > 1 + 1000(0.01) = 11 > 10$.

Crack the MCQ

Q1. The number of terms in the expansion of $(x + a)^{47} - (x - a)^{47}$ is:

- (a) 23 (b) 24 (c) 47 (d) 48

Correct: (b) 24. With $n = 47$ odd, the difference keeps the odd indices $r = 1, 3, \dots, 47$, which is $\frac{47+1}{2} = 24$ values.

Distractor Analysis. (a) 23 comes from applying the even- n rule $n/2$ and truncating $47/2$ downwards — the rule for a difference splits by parity and n here is odd. (c) 47 is n itself, the chapter's signature off-by-one, now transplanted into a halved count. (d) 48 is $n + 1$, the count for the full expansion, from a student who forgot that half the terms cancel.

Q2. $(\sqrt{5} + \sqrt{2})^4 + (\sqrt{5} - \sqrt{2})^4$ equals:

- (a) 89 (b) 178 (c) $89 + 28\sqrt{10}$ (d) $56\sqrt{10}$

Correct: (b) 178. The sum keeps $r = 0, 2, 4$: $2[a^4 + 6a^2b^2 + b^4] = 2[25 + 6(10) + 4] = 2(89) = 178$.

Distractor Analysis. (a) 89 drops the factor of 2 — the single most common error in this addendum, because the surviving terms are *doubled*, not merely kept. (c) $89 + 28\sqrt{10}$ is the value of $(\sqrt{5} + \sqrt{2})^4$ alone; the student expanded one bracket and forgot to add the conjugate, which is precisely what kills the surd. (d) $56\sqrt{10}$ is what you get by keeping the **odd** indices, i.e. by applying the difference rule to a sum.

Q3. $(\sqrt{3} + \sqrt{2})^4 - (\sqrt{3} - \sqrt{2})^4$ equals:

- (a) $40\sqrt{6}$ (b) 98 (c) $20\sqrt{6}$ (d) 0

Correct: (a) $40\sqrt{6}$. From Example A1.

Distractor Analysis. (b) 98 is the conjugate **sum**, not the difference — the student picked the wrong identity from the pair. (c) $20\sqrt{6}$ is the factor-of-2 slip again: $4a^3b + 4ab^3$ evaluated without doubling. (d) 0 is the student who assumes conjugates always annihilate completely;

only half the terms do.

Common student pitfalls and pro-tips

- **Forgetting the factor 2.** The surviving terms appear in *both* brackets, so they double. If your answer for a conjugate sum is exactly half of the right one, this is why.
- **Keeping the wrong half.** Adding kills the odd- r terms; subtracting kills the even- r terms. Memory hook: **“add and the odd ones go”**.
- **Counting terms as $\frac{n+1}{2}$ always.** That is only right for odd n . For even n the two counts differ by one — check that they add to $n + 1$.
- **Expanding both brackets in full and then subtracting.** Half your work is guaranteed to be discarded. Write only the surviving half from the start.
- **Losing $r = 0$.** The sum’s first surviving term is ${}^nC_0a^n$, at $r = 0$. It is easy to start counting at $r = 2$.
- **Pro-Tip — read the shape before you compute.** A conjugate sum with even n and pure-surd parts is a *rational number*; a conjugate difference is a rational multiple of *one* surd. If your working produces a mixture like $5 + 3\sqrt{6}$, you have kept a term that should have cancelled. This catches errors before you finish the arithmetic.
- **Pro-Tip — pair with Sub-06.** Whenever a question hands you $(p + \sqrt{q})^n$ and asks for an integral or fractional part, your first written line should be the conjugate sum or difference. It converts the problem into an integer plus a provably tiny remainder.

Cheat Sheet Box

CONJUGATE PAIRS AND EXACT EVALUATION — at a glance

- $(a + b)^n + (a - b)^n = 2 [{}^nC_0a^n + {}^nC_2a^{n-2}b^2 + \dots]$ — keeps **even** r .
- $(a + b)^n - (a - b)^n = 2 [{}^nC_1a^{n-1}b + {}^nC_3a^{n-3}b^3 + \dots]$ — keeps **odd** r .
- Term counts: sum = $\lfloor n/2 \rfloor + 1$; difference = $\lceil n/2 \rceil$; they add to $n + 1$.
- Surd shape: conjugate sums go rational; conjugate differences go (rational) $\times$ (one surd).
- Exact powers: split against a round number, e.g. $(102)^5 = (100 + 2)^5$, and expand fully.
- Truncation bound ($x > 0, n \geq 2$): $(1 + x)^n > 1 + nx$ — settles size comparisons instantly.

Memory aids: “Add and the odd ones go.” · “Keep half, double it.” · “Counts must add to $n + 1$.”

Addendum B to Subtopic 02: Rational and Integral Terms in a Surd Expansion

Exam Relevance: A reliable JEE Main pattern — “the number of rational terms in $(\sqrt{2} + 3^{1/5})^{10}$ ”, “the number of integral terms in $(5^{1/2} + 7^{1/8})^{1024}$ ”, “the sum of the rational terms in ...” — appearing most years in one form or another. JEE

Advanced raises it by choosing an index that makes the two divisibility conditions genuinely interact, so that the qualifying values of r do not start at $r = 0$. It is beyond the rationalised CBSE Class 11 core, but it is nothing more than Sub-02’s “specific term” algorithm with an integrality condition bolted on, so it costs you almost nothing to learn.

The idea

Sub-02’s whole engine was: write the general term, collect x into one power $E(r)$, impose a condition on $E(r)$, solve for r , and check that r is a valid integer. This addendum runs the same engine with a different condition.

Think about it this way. Take $(\sqrt{2} + \sqrt[5]{3})^{10}$. The general term is

$$T_{r+1} = {}^{10}C_r (2^{1/2})^{10-r} (3^{1/5})^r = {}^{10}C_r 2^{(10-r)/2} 3^{r/5}.$$

The binomial coefficient is always a whole number, so the only thing that can make a term irrational is a leftover root. **A term is rational exactly when both fractional exponents clear at the same time** — when $\frac{10-r}{2}$ and $\frac{r}{5}$ are simultaneously non-negative integers. That is two divisibility conditions on one unknown r , and solving them together is the entire problem.

The word “simultaneously” is where marks are lost. Clearing one exponent is easy; students count the r that clear the harder-looking exponent, forget the other condition, and hand in a count that is far too big.

Figure brief: A horizontal axis of $r = 0, 1, \dots, 10$ with two rows of tick marks above it: row one marks the r satisfying $2 \mid (10 - r)$, row two marks the r satisfying $5 \mid r$; a third row highlights only the r where both rows are marked, captioned “rational terms live where both filters agree”.

Key results

The setting. Let p and q be **distinct primes** (the standard exam setting), and let α, β be positive integers. Consider

$$(p^{1/\alpha} + q^{1/\beta})^N, \quad N \in \mathbb{N}.$$

The general term is

$$T_{r+1} = {}^N C_r p^{(N-r)/\alpha} q^{r/\beta}, \quad r = 0, 1, \dots, N.$$

- $p^{1/\alpha}, q^{1/\beta}$ — the two surd parts.
- α, β — the root orders (the denominators of the exponents).
- N — the index; r — the term index, an integer in $[0, N]$.

Rationality criterion. Because p and q are distinct primes, T_{r+1} is rational if and only if

$$\alpha \mid (N - r) \quad \text{and} \quad \beta \mid r.$$

Rational means integral here. When those two conditions hold, both exponents are non-negative integers and ${}^N C_r$ is a positive integer, so T_{r+1} is a positive **integer**. In this family of problems “number of rational terms” and “number of integral terms” are therefore the same count — a question may ask either.

Structure of the solution set. The qualifying values of r form an arithmetic progression with common difference $\text{lcm}(\alpha, \beta)$, starting at the smallest $r_0 \in [0, N]$ that satisfies both conditions. Hence

$$\text{number of rational terms} = \left\lfloor \frac{N - r_0}{\text{lcm}(\alpha, \beta)} \right\rfloor + 1,$$

provided such an r_0 exists; if no r satisfies both conditions, the count is 0.

Special case worth memorising. If $\alpha \mid N$, then $\alpha \mid (N - r) \iff \alpha \mid r$, so the criterion collapses to $\text{lcm}(\alpha, \beta) \mid r$, and the qualifying r start at $r_0 = 0$. This is the easy case that most textbook problems use — but do not assume it. When $\alpha \nmid N$, the progression starts somewhere in the middle, and it may not exist at all.

Irrational terms. The expansion has $N + 1$ terms in total, so

$$\text{number of irrational terms} = (N + 1) - (\text{number of rational terms}).$$

Procedure B1 — counting rational or integral terms

1. **Write the general term** and push both bases into single fractional powers: $T_{r+1} = {}^N C_r p^{(N-r)/\alpha} q^{r/\beta}$. *Why:* the conditions are conditions on exponents, so the exponents must be visible.
2. **Write down both integrality conditions:** $\alpha \mid (N - r)$ and $\beta \mid r$. *Why:* the binomial coefficient is always an integer, so the surd parts are the only obstruction.
3. **Convert to congruences:** $r \equiv N \pmod{\alpha}$ and $r \equiv 0 \pmod{\beta}$. *Why:* this form makes the interaction between the two conditions explicit instead of hoping they are compatible.
4. **Find the smallest $r_0 \in [0, N]$ satisfying both.** If none exists, the answer is “no rational terms” and you stop. *Why:* the same non-existence discipline Sub-02 demands — do not force an answer.
5. **Step by** $\text{lcm}(\alpha, \beta)$ from r_0 up to N and count, remembering to include r_0 itself. *Why:* the two congruences together have period $\text{lcm}(\alpha, \beta)$.
6. **Answer what was asked.** Count of rational terms, count of irrational terms $= (N+1) -$ that, or the *sum* of the rational terms — in which case go back and evaluate each qualifying T_{r+1} .

Worked examples

Example B1 — [CBSE Board Level]

Find all rational terms in the expansion of $(\sqrt{2} + \sqrt[5]{3})^{10}$, and their sum.

General term:

$$T_{r+1} = {}^{10}C_r (2^{1/2})^{10-r} (3^{1/5})^r = {}^{10}C_r 2^{(10-r)/2} 3^{r/5}.$$

Conditions: $2 \mid (10 - r)$, i.e. r is even; and $5 \mid r$.

From $5 \mid r$ with $0 \leq r \leq 10$: $r \in \{0, 5, 10\}$. Of these, the even ones are $r = 0$ and $r = 10$; $r = 5$ fails because $10 - 5 = 5$ is odd, leaving a stray $\sqrt{2}$.

- $r = 0$: $T_1 = {}^{10}C_0 2^5 3^0 = 32$.
- $r = 10$: $T_{11} = {}^{10}C_{10} 2^0 3^2 = 9$.

So there are exactly **two** rational terms, 32 and 9, and their sum is

$$32 + 9 = 41.$$

(There are $11 - 2 = 9$ irrational terms.)

Example B2 — [JEE Main Level]

Find the number of rational terms in the expansion of $(3^{1/2} + 5^{1/8})^{256}$.

$$T_{r+1} = {}^{256}C_r 3^{(256-r)/2} 5^{r/8}.$$

Conditions: $8 \mid r$, and $2 \mid (256 - r)$, i.e. r even. Since $8 \mid r$ already forces r to be even, the second condition is automatic here — but you must *check* that, not assume it.

So the qualifying values are $r = 0, 8, 16, \dots, 256$, an AP with common difference 8. The count is

$$\frac{256 - 0}{8} + 1 = 32 + 1 = 33.$$

33

The “+1” is the entire difficulty of this question. There are 32 steps of size 8 from 0 to 256, but 33 landing points.

Example B3 — [JEE Main — Speed Trap]

Find the number of rational terms in the expansion of $(\sqrt{3} + \sqrt[4]{5})^{101}$.

The trap: the shape looks identical to Example B2, so the reflex is to write “multiples of 4 from 0 to 100”, giving $\frac{100}{4} + 1 = 26$, and move on. That is the seductive wrong answer, and it is wrong because the index 101 is odd.

Do it properly:

$$T_{r+1} = {}^{101}C_r 3^{(101-r)/2} 5^{r/4}.$$

- Condition from the $\sqrt[3]{}$ part: $2 \mid (101 - r)$, so r must be **odd**.
- Condition from the $\sqrt[4]{}$ part: $4 \mid r$, so r must be a multiple of 4, hence **even**.

No integer is both odd and even. The two conditions are incompatible, so

there is no rational term in the expansion.

This is the same discipline as Sub-02’s MCQ Q4: when the condition on r has no admissible solution, the honest answer is that the term does not exist. **The moment $\alpha \nmid N$, check compatibility before counting anything.**

Example B4 — [JEE Advanced Level]

Find the number of irrational terms in the expansion of $(\sqrt[8]{5} + \sqrt[6]{2})^{100}$.

$$T_{r+1} = {}^{100}C_r 5^{(100-r)/8} 2^{r/6}.$$

Conditions: $8 \mid (100 - r)$ and $6 \mid r$. As congruences,

$$r \equiv 100 \equiv 4 \pmod{8}, \quad r \equiv 0 \pmod{6}.$$

Here $8 \nmid 100$, so this is the awkward case: the qualifying r do **not** start at 0. Write $r = 6k$ and impose the first congruence:

$$6k \equiv 4 \pmod{8} \implies 3k \equiv 2 \pmod{4}.$$

Since $3 \cdot 3 = 9 \equiv 1 \pmod{4}$, multiply both sides by 3: $k \equiv 6 \equiv 2 \pmod{4}$. So $k \in \{2, 6, 10, 14, \dots\}$, giving

$$r = 12, 36, 60, 84, 108, \dots$$

and the constraint $0 \leq r \leq 100$ keeps $r \in \{12, 36, 60, 84\}$ — four values, spaced by $\text{lcm}(8, 6) = 24$, exactly as the structure result predicts.

Verify one of them: at $r = 36$, the exponents are $\frac{100-36}{8} = 8$ and $\frac{36}{6} = 6$, both integers, so $T_{37} = {}^{100}C_{36} 5^8 2^6$ is an integer.

So there are 4 rational terms out of 101, and therefore

$$\text{number of irrational terms} = 101 - 4 = 97.$$

Practice exercises

1. **[JEE Main]** Find the number of integral terms in the expansion of $(5^{1/2} + 7^{1/8})^{1024}$.
2. **[JEE Main]** Find the number of rational terms in the expansion of $(\sqrt{2} + \sqrt[4]{3})^{100}$.
3. **[JEE Main]** Find the sum of the rational terms in the expansion of $(\sqrt{3} + \sqrt[4]{2})^8$.
4. **[JEE Main]** In the expansion of $(\sqrt[5]{3} + \sqrt[10]{7})^{45}$, find how many terms are rational and how many are irrational.
5. **[JEE Advanced]** Find the number of rational terms in the expansion of $(\sqrt[4]{2} + \sqrt[6]{3})^{36}$.
6. **[JEE Advanced]** Find the number of integral terms in the expansion of $(\sqrt[3]{5} + \sqrt[7]{2})^{100}$.

Answers:

1. 129 (qualifying $r = 0, 8, \dots, 1024$).
2. 26 (qualifying $r = 0, 4, \dots, 100$).
3. 1345 (the terms are 81 at $r = 0$, 1260 at $r = 4$, 4 at $r = 8$).
4. 5 rational (at $r = 0, 10, 20, 30, 40$) and 41 irrational.
5. 4 (qualifying $r = 0, 12, 24, 36$).
6. 5 (qualifying $r = 7, 28, 49, 70, 91$).

Crack the MCQ

Q1. The number of rational terms in the expansion of $(\sqrt{3} + \sqrt[8]{5})^{256}$ is:

- (a) 32 (b) 33 (c) 129 (d) 257

Correct: (b) 33. Need $8 \mid r$ and $256 - r$ even; the first implies the second, so $r = 0, 8, \dots, 256$, giving $\frac{256}{8} + 1 = 33$.

Distractor Analysis. (a) 32 is $\frac{256}{8}$ — the off-by-one that forgets $r = 0$ is itself a valid, rational term. (c) 129 counts only the $\sqrt{3}$ condition (r even gives $r = 0, 2, \dots, 256$, i.e. 129 values) and never imposes the $\sqrt[8]{5}$ condition — the “one filter only” error. (d) 257 is $N + 1$, the total number of terms, from a student who read the question as “how many terms are there”.

Q2. In the expansion of $(\sqrt{2} + 3^{1/5})^{10}$, the number of irrational terms is:

- (a) 2 (b) 9 (c) 10 (d) 11

Correct: (b) 9. There are 11 terms in all and 2 of them are rational (Example B1), leaving 9.

Distractor Analysis. (a) 2 is the count of *rational* terms — reading the question backwards. (c) 10 is N , the term-count off-by-one from Sub-01. (d) 11 is the total term count, i.e. forgetting to subtract the rational ones.

Q3. With p, q distinct primes, the term ${}^N C_r p^{(N-r)/\alpha} q^{r/\beta}$ is rational precisely when:

- (a) $\alpha \mid r$ and $\beta \mid (N - r)$
- (b) $\beta \mid r$ and $\alpha \mid (N - r)$
- (c) $\alpha\beta \mid r$
- (d) $\text{lcm}(\alpha, \beta) \mid r$

Correct: (b). The exponent of p is $(N - r)/\alpha$, so α must divide $N - r$; the exponent of q is r/β , so β must divide r .

Distractor Analysis. (a) swaps the two root orders — a pure bookkeeping slip that nonetheless changes the answer whenever $\alpha \neq \beta$. (c) is far too strong: $\alpha\beta$ is a multiple of $\text{lcm}(\alpha, \beta)$ and would discard genuine rational terms whenever α and β share a factor. (d) is the trap for students who only ever meet the friendly case $\alpha \mid N$; there it happens to be equivalent to (b), but when $\alpha \nmid N$ — as in Example B4, where the qualifying r start at 12, not 0 — it is simply false.

Common student pitfalls and pro-tips

- **Imposing only one condition.** Both exponents must clear *together*. Counting the multiples of β alone always over-counts.
- **The off-by-one in an AP count.** From r_0 to N in steps of d there are $\left\lfloor \frac{N-r_0}{d} \right\rfloor + 1$ terms, not $\frac{N-r_0}{d}$. The endpoint r_0 counts.
- **Assuming the run starts at $r = 0$.** It starts at 0 only when $\alpha \mid N$. Check that first; if it fails, solve the congruence pair properly.
- **Assuming a rational term must exist.** As Example B3 shows, the two conditions can be flatly incompatible. Non-existence is a legitimate final answer, exactly as in Sub-02.
- **Confusing “rational” with “integral”.** In this standard setting they coincide, because the surviving exponents are non-negative integers and ${}^N C_r$ is an integer. Say so in a proof rather than silently switching words.
- **Ignoring the distinct-primes hypothesis.** With bases like 2 and 8, or 3 and 9, extra terms can turn rational because the two bases share a prime. Exam problems use distinct primes; if a problem does not, check by hand.
- **Pro-Tip — convert to congruences immediately.** Writing $r \equiv N \pmod{\alpha}$ and $r \equiv 0 \pmod{\beta}$ turns a guessing game into a two-line solve, and it tells you at once whether any solution exists.
- **Pro-Tip — the period is $\text{lcm}(\alpha, \beta)$.** Once you have found one qualifying r , every other is that many steps away. You never need to test values one by one.

Cheat Sheet Box

RATIONAL / INTEGRAL TERMS IN A SURD EXPANSION — at a glance

- Setting: $(p^{1/\alpha} + q^{1/\beta})^N$, $p \neq q$ primes. General term $T_{r+1} = {}^N C_r p^{(N-r)/\alpha} q^{r/\beta}$.
- **Rational** $\iff \alpha \mid (N - r)$ and $\beta \mid r$ (both at once).
- Rational terms here are automatically **integers**.
- Qualifying r form an AP with common difference $\text{lcm}(\alpha, \beta)$, starting at the smallest valid r_0 .

- Count = $\left\lfloor \frac{N - r_0}{\text{lcm}(\alpha, \beta)} \right\rfloor + 1$; if no valid r_0 exists, the count is 0.
- If $\alpha \mid N$: the condition collapses to $\text{lcm}(\alpha, \beta) \mid r$ and $r_0 = 0$.
- Irrational count $\$ = (N+1) - \$ \text{ rational count}$.

Memory aids: “Both exponents must clear, or the term is out.” · “Endpoints count — remember the +1.” · “ $\alpha \nmid N$ means the run does not start at zero.”

Addendum C to Subtopic 06: Two-Term Divisibility and the mod K^2 Formula

Exam Relevance: “Show that $9^{n+1} - 8n - 9$ is divisible by 64” is an NCERT exercise and a standard CBSE Class 11 3-4 mark proof; the identical problem appears as $3^{2n+2} - 8n - 9$. JEE Main converts it into objective form — “the largest integer dividing ... for every n ”, or a remainder modulo a perfect square such as 36, 64 or 225. JEE Advanced hides it inside last-two-digit problems where the linear term genuinely survives. Sub-06 already prints the formula $(1+K)^n \equiv 1+nK \pmod{K^2}$; this addendum is the training that turns it from a memorised line into a tool.

The idea

Every worked example in Sub-06 collapsed to a *single* surviving term, because the modulus was K itself and every term carrying a K vanished. That habit is dangerous, because it teaches you to throw away the term you most need.

Think about it this way. Suppose your Mumbai local runs on a 64-minute cycle and you are told a journey takes 9^{n+1} minutes. Splitting $9 = 1 + 8$ tells you the journey is 1 minute plus a whole number of 8-minute blocks — useful modulo 8, useless modulo 64, because 8-minute blocks are *not* invisible on a 64-minute clock. To work modulo $64 = 8^2$, you must keep track of how many single blocks there are as well. **The rule is: to reduce modulo K^m , keep the first m terms of the expansion of $(1+K)^n$ and discard the rest.** Sub-06 always had $m = 1$; this addendum is the $m = 2$ case.

Written out, the expansion says it all:

$$(1+K)^n = \underbrace{1}_{\text{survives mod } K} + \underbrace{nK}_{\text{survives mod } K^2} + \underbrace{{}^nC_2 K^2 + {}^nC_3 K^3 + \dots}_{\text{all divisible by } K^2}.$$

Where this fails: if the divisor is not a perfect square, or is a square whose root is not the K you split with, the two-term formula gives you nothing. Match K to the divisor first.

Key results

The two-term identity. For $n \in \mathbb{N}$ and any integer K ,

$$(1+K)^n = 1 + nK + K^2 \sum_{r=2}^n {}^nC_r K^{r-2},$$

where the sum is an integer (and is empty, hence zero, when $n \leq 1$). Consequently

$$(1 + K)^n \equiv 1 + nK \pmod{K^2}.$$

- K — the chosen split, so that the base is $1 + K$; choose K so that the **divisor equals** K^2 .
- n — the exponent, a positive integer.
- The congruence also holds modulo any divisor of K^2 , which is what lets you use it for last-two-digit work with $K = 80$ and modulus 100.

The divisibility corollary. For every $n \in \mathbb{N}$,

$$K^2 \mid [(1 + K)^n - 1 - nK].$$

This is the exact shape of the NCERT problem: $9^{n+1} - 8n - 9$ is $(1 + 8)^{n+1} - 1 - 8(n + 1)$, so it is divisible by $8^2 = 64$.

How much is guaranteed. K^2 is the best guarantee, not K^3 : taking $n = 2$ gives $(1 + K)^2 - 1 - 2K = K^2$ exactly. So do not claim more than the formula gives.

Comparison with the single-term case (the Sub-06 tools, extended):

Divisor	Terms to keep	Result
K	1 term	$(1 + K)^n \equiv 1 \pmod{K}$
K^2	2 terms	$(1 + K)^n \equiv 1 + nK \pmod{K^2}$
K^3	3 terms	$(1 + K)^n \equiv 1 + nK + {}^nC_2 K^2 \pmod{K^3}$

The general pattern: modulo K^m , keep the terms up to $r = m - 1$.

Procedure C1 — proving divisibility by a perfect square K^2

1. **Read the divisor and take its square root:** for 64, $K = 8$; for 225, $K = 15$; for 9, $K = 3$. *Why:* the whole method rests on the divisor being K^2 for the K you are about to split with.
2. **Rewrite the power so its base is $1 + K$.** For example $9^{n+1} = (1 + 8)^{n+1}$; $3^{2n+2} = 9^{n+1} = (1 + 8)^{n+1}$; $16^n = (1 + 15)^n$; $4^n = (1 + 3)^n$. *Why:* only a base of the form $1 + K$ produces the clean two-term head.
3. **Expand and write the head explicitly**, keeping the $r = 0$ and $r = 1$ terms and bundling the rest: $(1 + K)^N = 1 + NK + \sum_{r=2}^N {}^NC_r K^r$. *Why:* the tail is what you will hand to the divisor.
4. **Cancel the head against the linear part of the expression.** *Why:* the problem's $-8n - 9$ (or $-15n - 1$, or $-3n - 1$) exists precisely to delete $1 + NK$. If it does not cancel exactly, re-check Step 2.
5. **Factor K^2 out of every remaining term**, since each carries K^r with $r \geq 2$. *Why:* this is the conclusion — every survivor is a multiple of K^2 .

6. **State the conclusion for all** $n \in \mathbb{N}$, and, if the question asks for the *largest* such divisor, test $n = 1$ to confirm you cannot do better.

Procedure C2 — remainders modulo a perfect square

1. **Identify the modulus** M and find K with $M \mid K^2$ (usually $M = K^2$; for last-two-digit work, $M = 100$ and $K = 80$ or $K = 10$ works because $100 \mid 6400$).
2. **Adjust the exponent** so the base becomes $1 + K$: e.g. $3^{2018} = 3^2 (3^4)^{504} = 9(1 + 80)^{504}$. *Why:* you get to choose which power of the base you split.
3. **Apply** $(1 + K)^n \equiv 1 + nK$ and reduce nK modulo M . *Why:* nK is the term Sub-06's examples always threw away, and it is usually nonzero here.
4. **Multiply back any leftover factor** from Step 2 and reduce again.
5. **Shift the answer into 0 to** $M - 1$.

Worked examples

Example C1 — [CBSE Board Level]

Show that $9^{n+1} - 8n - 9$ **is divisible by 64 for every positive integer** n .

The divisor is $64 = 8^2$, so take $K = 8$ and write $9 = 1 + 8$:

$$9^{n+1} = (1 + 8)^{n+1} = \sum_{r=0}^{n+1} {}^{n+1}C_r 8^r = 1 + (n+1)8 + \sum_{r=2}^{n+1} {}^{n+1}C_r 8^r.$$

The head is $1 + 8n + 8 = 8n + 9$. Therefore

$$9^{n+1} - 8n - 9 = (8n + 9) + \sum_{r=2}^{n+1} {}^{n+1}C_r 8^r - 8n - 9 = \sum_{r=2}^{n+1} {}^{n+1}C_r 8^r.$$

Every term of the remaining sum has $r \geq 2$, so it carries $8^r = 64 \cdot 8^{r-2}$ with 8^{r-2} a non-negative integer power. Factoring:

$$9^{n+1} - 8n - 9 = 64 \sum_{r=2}^{n+1} {}^{n+1}C_r 8^{r-2},$$

and the sum is an integer. Hence 64 divides $9^{n+1} - 8n - 9$ for every $n \in \mathbb{N}$.

Spot check: $n = 1$ gives $81 - 8 - 9 = 64$; $n = 2$ gives $729 - 16 - 9 = 704 = 64 \times 11$; $n = 3$ gives $6561 - 24 - 9 = 6528 = 64 \times 102$.

Note that $3^{2n+2} = (3^2)^{n+1} = 9^{n+1}$, so the frequently-set problem “show $3^{2n+2} - 8n - 9$ is divisible by 64” is this identical proof after one line of index arithmetic.

Example C2 — [JEE Main Level]

Find the remainder when 7^{100} **is divided by 36.**

The modulus $36 = 6^2$ is a perfect square, so take $K = 6$ and note $7 = 1 + 6$:

$$7^{100} = (1 + 6)^{100} = 1 + 100(6) + \sum_{r=2}^{100} {}^{100}C_r 6^r.$$

Every term with $r \geq 2$ carries $6^2 = 36$, so the whole tail vanishes modulo 36:

$$7^{100} \equiv 1 + 600 = 601 \pmod{36}.$$

Reduce: $601 = 36 \times 16 + 25$, so

$$\text{remainder} = 25.$$

Had you kept only the first term — the reflex Sub-06's examples build — you would have said $7^{100} \equiv 1$, which is the right answer modulo 6 and the wrong answer modulo 36.

Example C3 — [JEE Main — Speed Trap]

Find the largest positive integer that divides $3^{2n+2} - 8n - 9$ for every positive integer n .

The trap: $3^{2n+2} = 9^{n+1} = (1 + 8)^{n+1}$, and a student who keeps only the leading 1 gets $9^{n+1} \equiv 1 \pmod{8}$, concludes that $9^{n+1} - 1$ is a multiple of 8, and answers **8**. That answer is *true but not largest*, and in a single-answer objective it scores zero.

The fix is Procedure C1. Keeping two terms, Example C1 showed

$$3^{2n+2} - 8n - 9 = 64 \sum_{r=2}^{n+1} {}^{n+1}C_r 8^{r-2},$$

so 64 divides the expression for every n . Can anything larger work? Test the smallest case: $n = 1$ gives the value $81 - 8 - 9 = 64$. Any universal divisor must divide 64, so nothing larger than 64 is possible.

$$\text{Largest divisor} = 64.$$

Name the trap: keeping one term when the divisor is a square. Whenever an objective question says “largest integer that divides ... for every n ”, assume the intended answer is the square, and verify by substituting $n = 1$.

Example C4 — [JEE Advanced Level]

Find the last two digits of 3^{2018} .

Last two digits means the value modulo 100. Choose the split by looking for a power of 3 that is one more than a multiple of 10: $3^4 = 81 = 1 + 80$. So take $K = 80$, and note $K^2 = 6400$, which is a multiple of 100 — that is what licenses the two-term formula here.

Write the exponent to expose 3^4 :

$$3^{2018} = 3^2 \cdot 3^{2016} = 9(3^4)^{504} = 9(1+80)^{504}.$$

Apply the two-term identity modulo 6400, and hence modulo 100:

$$(1+80)^{504} \equiv 1 + 504(80) = 1 + 40320 \pmod{100}.$$

Now $40320 \equiv 20 \pmod{100}$, so $(1+80)^{504} \equiv 21 \pmod{100}$. Multiply back the factor 9:

$$3^{2018} \equiv 9 \times 21 = 189 \equiv 89 \pmod{100}.$$

Last two digits = 89.

This is the example that shows why the linear term matters: had you dropped $504 \cdot 80$, you would have answered 9, and the true answer is 89. Independent check: 3^{20} ends in 01, and $2018 = 20(100) + 18$, so 3^{2018} ends in the same two digits as 3^{18} ; since 3^{10} ends in 49 and 3^8 ends in 61, and $49 \times 61 = 2989$, 3^{18} ends in 89.

Practice exercises

1. **[CBSE Board]** Show that $3^{2n+2} - 8n - 9$ is divisible by 64 for all $n \in \mathbb{N}$.
2. **[CBSE Board]** Show that $4^n - 3n - 1$ is divisible by 9 for all $n \in \mathbb{N}$.
3. **[JEE Main]** Find the remainder when 9^{50} is divided by 64.
4. **[JEE Main]** Find the largest integer that divides $10^n + 3 \cdot 4^{n+2} + 5$ for every $n \in \mathbb{N}$.
5. **[JEE Advanced]** Show that $2^{4n} - 15n - 1$ is divisible by 225 for all $n \in \mathbb{N}$.
6. **[JEE Advanced]** Find the last two digits of 3^{2014} .

Answers:

1. $3^{2n+2} = 9^{n+1} = (1+8)^{n+1}$; head $1 + 8(n+1) = 8n+9$ cancels, leaving $64 \sum_{r=2}^{n+1} {}^{n+1}C_r 8^{r-2}$.
2. $4^n = (1+3)^n$; head $1 + 3n$ cancels, leaving $9 \sum_{r=2}^n {}^nC_r 3^{r-2}$.
3. 17 (since $9^{50} = (1+8)^{50} \equiv 1 + 400 = 401 \equiv 17 \pmod{64}$).
4. 9 (mod 9: $10^n \equiv 1$, and $4^{n+2} \equiv 4^{(1+3)} \{n+2\} \equiv 1+3(n+2) \pmod{9}$, so $3 \cdot 4^{n+2} \equiv 9n+21 \equiv 3$; total $1+3+5 = 9 \equiv 0$; and $n=1$ gives $207 = 9 \times 23$, $n=2$ gives $873 = 9 \times 97$, whose gcd is 9).
5. $2^{4n} = 16^n = (1+15)^n \equiv 1 + 15n \pmod{225}$, so $2^{4n} - 15n - 1$ is a multiple of $15^2 = 225$.
6. 69 (since $3^{2014} = 9(1+80)^{503} \equiv 9(1+40240) \equiv 9 \times 41 = 369 \equiv 69 \pmod{100}$).

Crack the MCQ

Q1. The largest positive integer that divides $9^{n+1} - 8n - 9$ for every positive integer n is:

- (a) 8 (b) 9 (c) 64 (d) 72

Correct: (c) 64. Example C1 proves 64 always divides it, and $n=1$ gives the value 64 exactly, so nothing larger can.

Distractor Analysis. (a) 8 is what you get by keeping only the leading 1 of the expansion — true, but not the largest, and therefore wrong here. (b) 9 comes from fixating on the base 9 rather than on the split $K = 8$; it fails already at $n = 1$, since 64 is not a multiple of 9. (d) 72 is 8×9 , manufactured by multiplying the split by the base; 64 is not a multiple of 72.

Q2. For an integer K and $n \in \mathbb{N}$, the highest power of K that is guaranteed to divide $(1 + K)^n - 1 - nK$ is:

- (a) K (b) K^2 (c) K^3 (d) K^n

Correct: (b) K^2 . Every remaining term carries K^r with $r \geq 2$.

Distractor Analysis. (a) K is a true but weaker statement — it discards the whole point of subtracting nK . (c) K^3 over-claims: at $n = 2$ the expression is exactly K^2 , which is not a multiple of K^3 in general. (d) K^n confuses the power of K in the *last* term with what the *whole* expression is guaranteed to carry; the guarantee is set by the smallest surviving power, not the largest.

Q3. To prove that $9^{n+1} - 8n - 9$ is divisible by 64, the correct first move is to write:

- (a) $9^{n+1} = (1 + 8)^{n+1}$ and keep the terms up to $r = 1$
 (b) $9^{n+1} = 9 \cdot 9^n$ and use $9^n \equiv 1 \pmod{8}$
 (c) $9^{n+1} = (10 - 1)^{n+1}$ and reduce modulo 10
 (d) $8n + 9 = 8(n + 1) + 1$ and factor the expression directly

Correct: (a). The divisor is 8^2 , so split with $K = 8$ and retain two terms.

Distractor Analysis. (b) reduces modulo 8, which proves divisibility by 8 only — the exact failure this addendum exists to fix. (c) splits against 10, a number with no relationship to 64; the divisor must appear as K^2 inside the bracket. (d) rearranges the linear part, which is a useful bookkeeping step *after* the expansion but proves nothing on its own — there is no factorisation of the expression that does not go through the binomial expansion.

Common student pitfalls and pro-tips

- **Keeping one term when the divisor is a square.** Modulo K^2 you must keep $1 + nK$. Dropping nK is the single error this addendum targets.
- **Splitting with the wrong K .** For divisor 64, split with $K = 8$, not $K = 64$. The rule is $K = \sqrt{\text{divisor}}$, not $K = \text{divisor}$.
- **Not checking that the head cancels.** In $9^{n+1} - 8n - 9$, the head is $1 + 8(n + 1) = 8n + 9$ and it must delete $-8n - 9$ exactly. If it leaves a remainder, your base or your exponent is misaligned.
- **Answering “true divisor” instead of “largest divisor”.** When an objective asks for the largest, verify with $n = 1$: any universal divisor must divide that value.
- **Forgetting to multiply back.** In Procedure C2 you often split off a factor (the 9 in Example C4). Reducing the bracket and forgetting the factor is a classic lost mark.
- **Claiming K^3 .** The guarantee stops at K^2 unless you keep a third term.
- **Pro-Tip — read the divisor first, then choose the split.** Divisor 64 $\Rightarrow K = 8$; divisor 225 $\Rightarrow K = 15$; divisor 9 $\Rightarrow K = 3$; divisor 36 $\Rightarrow K = 6$. The base you are given must

then be forced into the form $1 + K$, adjusting the exponent if necessary.

- **Pro-Tip — for last-two-digit problems, hunt for a power ending in 1.** $3^4 = 81$, $7^4 = 2401$, $9^2 = 81$: each gives a K that is a multiple of 10, and then K^2 is a multiple of 100, so the two-term formula applies modulo 100.

Cheat Sheet Box

TWO-TERM DIVISIBILITY (mod K^2) — at a glance

- $(1 + K)^n = 1 + nK + K^2 \sum_{r=2}^n {}^nC_r K^{r-2}$, hence $(1 + K)^n \equiv 1 + nK \pmod{K^2}$.
- Corollary: $K^2 \mid [(1 + K)^n - 1 - nK]$ for every $n \in \mathbb{N}$. Guarantee stops at K^2 .
- Terms to keep: 1 term for modulus K , 2 terms for K^2 , 3 terms for K^3 .
- Flagship result: $9^{n+1} - 8n - 9 = 3^{2n+2} - 8n - 9 = 64 \sum_{r \geq 2} {}^{n+1}C_r 8^{r-2}$, divisible by 64; and 64 is the largest such divisor.
- Companions: $4^n - 3n - 1$ divisible by 9; $16^n - 15n - 1$ divisible by 225.
- Last two digits: split so K is a multiple of 10 (e.g. $3^4 = 1 + 80$); then $100 \mid K^2$ and $(1 + K)^n \equiv 1 + nK \pmod{100}$.
- “Largest divisor” questions: prove K^2 works, then substitute $n = 1$ to show nothing bigger can.

Memory aids: “Square divisor, two terms.” · “ K is the square root of the divisor, not the divisor.” · “The linear part of the expression exists to cancel $1 + nK$.”

Subtopic 08 — Multisection of Binomial Series: Roots of Unity and Complex Substitution

Exam Relevance: This is the fifth tool for the coefficient-summation toolkit of Sub-04, and it is the one CBSE Boards do not ask for: the Class 11 syllabus stops at $x = \pm 1$, so treat Section 1’s non-unit substitutions as the board-level content here and the rest as competitive material. JEE Main asks the substitution $x = 3$ or $x = 2$ form directly (“prove $\sum 3^r {}^nC_r = 4^n$ ”) and occasionally the $x = i$ pair-sum. JEE Advanced is where multisection earns its place: sums such as $C_0 + C_3 + C_6 + \dots$, $C_0 + C_4 + C_8 + \dots$ and $C_0 - C_2 + C_4 - \dots$ are unreachable by any of Sub-04’s four tools, and they turn up disguised as counting problems (“how many subsets have size divisible by 3?”). NEET does not test mathematics.

Section 1: Concept Intuition

Sub-04 described $(1 + x)^n = C_0 + C_1x + C_2x^2 + \dots + C_nx^n$ as a machine with a dial. It then turned the dial to exactly two settings, $x = 1$ and $x = -1$. Both settings are legal, but so is every other number — and the interesting ones are not real.

Start with the settings you already own. **The dial is not restricted to ± 1 .** Turn it to $x = 3$ and the identity reads $\sum_r 3^r C_r = 4^n$; turn it to $x = 2$ and it reads $\sum_r 2^r C_r = 3^n$. Nothing

new is needed: any value of x you substitute gives you a true statement about a weighted coefficient sum, because a polynomial identity holds at every point.

Now the real question. Setting $x = 1$ adds *all* the coefficients. Setting $x = -1$ flips the sign of every alternate one, and combining the two results — add and halve, or subtract and halve — isolates the even-position coefficients or the odd-position ones. **That combination is a filter with period 2.** So here is the natural question: what if you want every *third* coefficient, $C_0 + C_3 + C_6 + \dots$? A period-2 filter cannot produce a period-3 pattern, no matter how you combine the two answers.

Think about it this way. Picture a traffic signal at a Bengaluru junction cycling green, amber, red, green, amber, red, once per second, and a counter that records the value of whatever passes. If you want the total of only the green seconds, you need a device whose own rhythm repeats every three seconds and which, averaged over one full cycle, cancels everything except green. Real numbers give you rhythms of period 1 ($x = 1$) and period 2 ($x = -1$) and nothing else. **The complex cube root of unity ω is exactly the device with rhythm 3:** its powers march $1, \omega, \omega^2, 1, \omega, \omega^2, \dots$, and the three of them sum to zero. Averaging $x = 1$, $x = \omega$ and $x = \omega^2$ therefore kills two out of every three coefficients and keeps the third, tripled.

The same logic with the fourth roots of unity $1, i, -1, -i$ gives you a period-4 filter, which is what produces $C_0 - C_2 + C_4 - C_6 + \dots$. That sum looks like an alternating sum, and students reach for $x = -1$; but the sign pattern $+, -, +, -$ sitting on the *even* positions only is a period-4 pattern in disguise, and $x = -1$ cannot make it.

Figure brief: An Argand diagram of the unit circle with $1, \omega$ and ω^2 marked at $0^\circ, 120^\circ$ and 240° , joined as an equilateral triangle whose centroid is the origin, annotated $1 + \omega + \omega^2 = 0$; alongside it, a strip of the coefficients $C_0, \dots, C_9$ with C_0, C_3, C_6, C_9 shaded, captioned “the ω -average keeps the shaded ones, tripled”.

Validity conditions and constraints. Multisection needs the expansion to be a genuine polynomial identity, so n must be a positive integer — everything here inherits Sub-01’s restriction and none of it applies to the any-index series of Sub-05. Two edge cases matter. First, $(1 + (-1))^n = 0$ requires $n \geq 1$; at $n = 0$ that bracket is 1 and the formulas need separate reading. Second, the closed forms below use $\cos$, so the angle must be reduced modulo 2π before evaluation. The technique itself never fails for $n \in \mathbb{N}$: the filter is exact, not an approximation.

Section 2: Key Formulas and Definitions

Prerequisites from Chapter 5

$$\omega = e^{2\pi i/3} = \frac{-1 + i\sqrt{3}}{2}, \quad \omega^3 = 1, \quad 1 + \omega + \omega^2 = 0, \quad \omega^2 = \bar{\omega}.$$

$$1 + \omega = -\omega^2 = e^{i\pi/3}, \quad 1 + \omega^2 = -\omega = e^{-i\pi/3}, \quad 1 + i = \sqrt{2}e^{i\pi/4}, \quad (1 + i)^2 = 2i.$$

- ω — a primitive (non-real) cube root of unity.

- i — the imaginary unit, a primitive fourth root of unity, $i^2 = -1$, $i^4 = 1$.
- $\omega_k = e^{2\pi i/k}$ — a primitive k th root of unity, used below for general k .

Substitution at values other than ± 1

For any number y and $n \in \mathbb{N}$,

$$\sum_{r=0}^n {}^nC_r y^r = (1+y)^n.$$

In particular $\sum_r 3^r {}^nC_r = 4^n$, $\sum_r 2^r {}^nC_r = 3^n$, and $\sum_r (-2)^r {}^nC_r = (-1)^n$.

The root-of-unity filter

For a primitive k th root of unity ω_k and any integer m ,

$$\frac{1}{k} \sum_{j=0}^{k-1} \omega_k^{jm} = \begin{cases} 1, & k \mid m, \\ 0, & k \nmid m. \end{cases}$$

Applying this to $(1+x)^n$ gives the **multisection formula**: for $0 \leq a \leq k-1$,

$$\sum_{\substack{0 \leq r \leq n \\ r \equiv a \pmod{k}}} {}^nC_r = \frac{1}{k} \sum_{j=0}^{k-1} \omega_k^{-ja} (1 + \omega_k^j)^n.$$

- k — the period of the filter (how far apart the coefficients you want are).
- a — the residue class you want (which coefficients: $r \equiv a$).
- The same formula works for any polynomial $f(x) = \sum a_r x^r$, with $f(\omega_k^j)$ in place of $(1 + \omega_k^j)^n$.

Standard closed forms

Period 2 ($k = 2$, $n \geq 1$):

$$C_0 + C_2 + C_4 + \cdots = C_1 + C_3 + C_5 + \cdots = 2^{n-1}.$$

Period 3 ($k = 3$), for $a = 0, 1, 2$:

$$C_a + C_{a+3} + C_{a+6} + \cdots = \frac{1}{3} \left[2^n + 2 \cos \frac{(n-2a)\pi}{3} \right].$$

Written out, with S_a denoting the sum for residue a :

Sum	Closed form
$S_0 = C_0 + C_3 + C_6 + \cdots$	$\frac{1}{3} \left[2^n + 2 \cos \frac{n\pi}{3} \right]$

Sum	Closed form
$S_1 = C_1 + C_4 + C_7 + \dots$	$\frac{1}{3} \left[2^n + 2 \cos \frac{(n-2)\pi}{3} \right]$
$S_2 = C_2 + C_5 + C_8 + \dots$	$\frac{1}{3} \left[2^n + 2 \cos \frac{(n-4)\pi}{3} \right]$

and $S_0 + S_1 + S_2 = 2^n$, which is your check.

Period 4 ($k = 4$, $n \geq 1$). The two “component” sums come straight from $(1+i)^n$:

$$C_0 - C_2 + C_4 - C_6 + \dots = \operatorname{Re} [(1+i)^n] = 2^{n/2} \cos \frac{n\pi}{4},$$

$$C_1 - C_3 + C_5 - C_7 + \dots = \operatorname{Im} [(1+i)^n] = 2^{n/2} \sin \frac{n\pi}{4},$$

and the full period-4 sums are

$$\sum_{r \equiv a \pmod{4}} {}^nC_r = \frac{1}{4} \left[2^n + 2^{(n+2)/2} \cos \frac{(n-2a)\pi}{4} \right], \quad a = 0, 1, 2, 3.$$

In particular $C_0 + C_4 + C_8 + \dots = \frac{1}{2} \left[2^{n-1} + 2^{n/2} \cos \frac{n\pi}{4} \right]$, and the four sums add to 2^n .

All quantities here are pure numbers; there are no units or dimensions. The only constraints are $n \in \mathbb{N}$ and $0 \leq a \leq k-1$.

Section 3: CBSE Board Derivations and Key Procedures

Derivation: the root-of-unity filter

Claim. For a primitive k th root of unity ω_k and any integer m , $\sum_{j=0}^{k-1} \omega_k^{jm}$ equals k if $k \mid m$ and 0 otherwise.

1. **Case** $k \mid m$. Then $\omega_k^m = (\omega_k^k)^{m/k} = 1$, so every one of the k summands is 1. Reason: $\omega_k^k = 1$ by definition of a k th root of unity. The sum is k .
2. **Case** $k \nmid m$. Then $\omega_k^m \neq 1$, because ω_k is *primitive* — its powers return to 1 only at multiples of k .
3. The sum is a geometric progression with first term 1 and ratio $\omega_k^m \neq 1$, so it equals $\frac{(\omega_k^m)^k - 1}{\omega_k^m - 1}$. Reason: the standard GP sum, valid because the ratio is not 1.
4. The numerator is $(\omega_k^k)^m - 1 = 1^m - 1 = 0$, and the denominator is nonzero, so the sum is 0. ■

Derivation: the multisection formula

1. Start from $(1+x)^n = \sum_{r=0}^n C_r x^r$ and substitute $x = \omega_k^j$ for each $j = 0, 1, \dots, k-1$. Reason: a polynomial identity holds at every complex value.
2. Multiply the j th equation by ω_k^{-ja} and add all k of them:

$$\sum_{j=0}^{k-1} \omega_k^{-ja} (1 + \omega_k^j)^n = \sum_{j=0}^{k-1} \sum_{r=0}^n C_r \omega_k^{j(r-a)} = \sum_{r=0}^n C_r \sum_{j=0}^{k-1} \omega_k^{j(r-a)}.$$

Reason: the sums are finite, so the order of summation may be exchanged freely.

3. By the filter, the inner sum is k when $k \mid (r-a)$ and 0 otherwise. So only the coefficients with $r \equiv a \pmod{k}$ survive, each multiplied by k .
4. Divide by k . ■

Derivation: $C_0 + C_3 + C_6 + \dots$ using ω

1. Substitute $x = 1$, $x = \omega$, $x = \omega^2$ into $(1+x)^n = \sum C_r x^r$ and add:

$$2^n + (1+\omega)^n + (1+\omega^2)^n = \sum_{r=0}^n C_r (1 + \omega^r + \omega^{2r}).$$

2. The bracket is 3 when $3 \mid r$ (all three powers equal 1) and 0 otherwise (it is $1 + \omega + \omega^2$ re-ordered). Reason: this is the $k = 3$ filter written out by hand.
3. Hence $C_0 + C_3 + C_6 + \dots = \frac{1}{3} [2^n + (1+\omega)^n + (1+\omega^2)^n]$.
4. Now evaluate the two complex powers. Since $\omega = -\frac{1}{2} + i\frac{\sqrt{3}}{2}$, we get $1+\omega = \frac{1}{2} + i\frac{\sqrt{3}}{2} = e^{i\pi/3}$ and, taking conjugates, $1+\omega^2 = e^{-i\pi/3}$. Reason: $1+\omega$ has modulus 1 and argument $\pi/3$.
5. Therefore $(1+\omega)^n + (1+\omega^2)^n = e^{in\pi/3} + e^{-in\pi/3} = 2 \cos \frac{n\pi}{3}$, giving

$$C_0 + C_3 + C_6 + \dots = \frac{1}{3} \left[2^n + 2 \cos \frac{n\pi}{3} \right]. \quad \blacksquare$$

Sanity check at $n = 4$: the formula gives $\frac{1}{3} [16 + 2 \cos \frac{4\pi}{3}] = \frac{1}{3} [16 - 2] = 5$, and directly $C_0 + C_3 = 1 + 4 = 5$.

Derivation: $C_0 - C_2 + C_4 - \dots$ using $x = i$

1. Substitute $x = i$: $(1+i)^n = \sum_r C_r i^r$.
2. The powers of i cycle $1, i, -1, -i$ with period 4. So the terms with r even contribute $\pm C_r$ to the **real** part with signs $+, -, +, -$ as $r = 0, 2, 4, 6, \dots$, and the terms with r odd contribute $\pm C_r$ to the **imaginary** part with the same alternating pattern.
3. Equating real and imaginary parts on both sides:

$$\operatorname{Re} [(1+i)^n] = C_0 - C_2 + C_4 - \dots, \quad \operatorname{Im} [(1+i)^n] = C_1 - C_3 + C_5 - \dots.$$

Reason: two complex numbers are equal exactly when their real and imaginary parts agree, and C_r are real.

4. Write $1+i = \sqrt{2} (\cos \frac{\pi}{4} + i \sin \frac{\pi}{4})$ and apply De Moivre's theorem:

$$(1+i)^n = 2^{n/2} \left(\cos \frac{n\pi}{4} + i \sin \frac{n\pi}{4} \right). \quad \blacksquare$$

Procedure A — the k -way filter, executed

1. **Read off k and a .** “Every third coefficient starting from C_0 ” means $k = 3$, $a = 0$; “ $C_1 + C_4 + C_7 + \dots$ ” means $k = 3$, $a = 1$. *Why:* the whole formula is determined by these two numbers.
2. **List the k substitution points** $x = \omega_k^0, \omega_k^1, \dots, \omega_k^{k-1}$. For $k = 3$ these are $1, \omega, \omega^2$; for $k = 4$ they are $1, i, -1, -i$. *Why:* you need one equation per residue class to separate k classes.
3. **Evaluate $(1 + \omega_k^j)^n$ for each j .** Convert each base to polar form first. *Why:* raising $e^{i\theta}$ to the n is trivial; multiplying out $\left(\frac{1}{2} + i\frac{\sqrt{3}}{2}\right)^n$ is not.
4. **Weight by ω_k^{-ja} , add, divide by k .** *Why:* the weight is what selects residue a rather than residue 0.
5. **Reduce every angle modulo 2π** before taking a cosine. *Why:* $\cos \frac{10\pi}{3}$ is not something you should evaluate directly; it is $\cos \frac{4\pi}{3} = -\frac{1}{2}$.
6. **Check that the k sums add to 2^n .** *Why:* the residue classes partition all $n + 1$ coefficients, whose total is 2^n . This catches sign and angle errors instantly.

Procedure B — the $x = i$ route for alternating even or odd sums

1. **Identify which component you need.** Signs alternating on the **even**-indexed coefficients means the real part; on the **odd**-indexed coefficients means the imaginary part.
2. **Compute $(1 + i)^n$ the fast way** using $(1 + i)^2 = 2i$: for even n , $(1 + i)^n = (2i)^{n/2}$. *Why:* this is pure arithmetic on powers of i and avoids trigonometry entirely.
3. **For odd n ,** write $(1 + i)^n = (1 + i)(2i)^{(n-1)/2}$ and expand once.
4. **Read off the required component** and state it as a real number.

Procedure C — substitution at a value other than ± 1

1. **Look at the weights.** A factor y^r attached to C_r means the dial should be set to $x = y$.
2. **Substitute and simplify $(1 + y)^n$.**
3. **Check with a tiny case** ($n = 1$ or $n = 2$) before committing. *Why:* it costs five seconds and catches a misread weight.

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

Prove that $\sum_{r=0}^n 3^r {}^nC_r = 4^n$, and evaluate the sum for $n = 5$.

The weight attached to nC_r is 3^r , which is exactly what substituting $x = 3$ produces. Start from the binomial theorem with $a = 1$, $b = x$:

$$(1 + x)^n = \sum_{r=0}^n {}^nC_r x^r.$$

This is a polynomial identity, so it holds for every value of x . Put $x = 3$:

$$\sum_{r=0}^n 3^r {}^nC_r = (1+3)^n = 4^n. \quad \blacksquare$$

For $n = 5$: the sum equals $4^5 = 1024$.

Direct check, since this is the first time the dial has been turned away from ± 1 :

$$1 + 5(3) + 10(9) + 10(27) + 5(81) + 243 = 1 + 15 + 90 + 270 + 405 + 243 = 1024.$$

The same move with $x = 2$ gives $\sum_r 2^r {}^nC_r = 3^n$, and with $x = -2$ gives $\sum_r (-2)^r {}^nC_r = (-1)^n$.

Example 2 — [JEE Main Level]

For $(1+x)^{10}$, evaluate $C_0 + C_3 + C_6 + C_9$.

This is every third coefficient starting at C_0 : $k = 3$, $a = 0$. Use the derived closed form with $n = 10$:

$$C_0 + C_3 + C_6 + C_9 = \frac{1}{3} \left[2^{10} + 2 \cos \frac{10\pi}{3} \right].$$

Reduce the angle first, as Procedure A Step 5 insists: $\frac{10\pi}{3} - 2\pi = \frac{4\pi}{3}$, and $\cos \frac{4\pi}{3} = -\frac{1}{2}$. Hence

$$= \frac{1}{3} \left[1024 + 2 \left(-\frac{1}{2} \right) \right] = \frac{1024 - 1}{3} = \frac{1023}{3} = 341.$$

$$C_0 + C_3 + C_6 + C_9 = 341.$$

Verification by direct addition: ${}^{10}C_0 + {}^{10}C_3 + {}^{10}C_6 + {}^{10}C_9 = 1 + 120 + 210 + 10 = 341$.

Check against the partition rule: the other two sums are $S_1 = {}^{10}C_1 + {}^{10}C_4 + {}^{10}C_7 + {}^{10}C_{10} = 10 + 210 + 120 + 1 = 341$ and $S_2 = {}^{10}C_2 + {}^{10}C_5 + {}^{10}C_8 = 45 + 252 + 45 = 342$. Total $341 + 341 + 342 = 1024 = 2^{10}$.

Example 3 — [JEE Main — Speed Trap]

For $(1+x)^8$, evaluate $C_0 - C_2 + C_4 - C_6 + C_8$.

The trap. The word “alternating” fires a reflex, and the reflex reaches for $x = -1$. Two wrong answers follow. A student who thinks “alternating sum of all coefficients” writes 0. A student who notices the sum only involves even indices reaches instead for the even-index total $C_0 + C_2 + C_4 + C_6 + C_8 = 2^{n-1} = 2^7 = \mathbf{128}$ and hands that in. Both are traps, and 128 is the seductive one because it is a genuine, correct identity — just not for this sum.

Why $x = -1$ cannot work. Substituting $x = -1$ attaches the sign $(-1)^r$ to C_r , which is a pattern of period 2. The sum asked for attaches $+, -, +, -$ to C_0, C_2, C_4, C_6 — that is, the sign

changes once every two indices, which is a pattern of period 4. No combination of the two real substitutions can produce it.

The correct move. Substitute $x = i$, whose powers have period 4:

$$(1 + i)^8 = \sum_{r=0}^8 {}^8C_r i^r,$$

and the real part of the right-hand side is exactly $C_0 - C_2 + C_4 - C_6 + C_8$. Evaluate the left side using $(1 + i)^2 = 2i$:

$$(1 + i)^8 = [(1 + i)^2]^4 = (2i)^4 = 16 i^4 = 16.$$

This is real, so its imaginary part is 0 and its real part is 16:

$$C_0 - C_2 + C_4 - C_6 + C_8 = 16.$$

Direct check: $1 - 28 + 70 - 28 + 1 = 16$. (As a bonus, the imaginary part gives $C_1 - C_3 + C_5 - C_7 = 8 - 56 + 56 - 8 = 0$, matching $\text{Im}(16) = 0$.)

Name the trap: period-2 thinking on a period-4 pattern. Count how many indices pass between sign changes. One index means $x = -1$; two means $x = i$; a repeating block of three identical signs means ω .

Example 4 — [JEE Advanced Level]

Prove that for $(1 + x)^n$ with $n \geq 1$,

$$C_0 + C_4 + C_8 + \cdots = \frac{1}{2} \left[2^{n-1} + 2^{n/2} \cos \frac{n\pi}{4} \right],$$

and hence evaluate the sum for $n = 12$.

This is the period-4 filter at $a = 0$, so substitute all four fourth roots of unity $x = 1, i, -1, -i$ and average.

Step 1 — write the four substitutions.

$$(1 + 1)^n = \sum_r C_r, \quad (1 + i)^n = \sum_r C_r i^r, \quad (1 - 1)^n = \sum_r C_r (-1)^r, \quad (1 - i)^n = \sum_r C_r (-i)^r.$$

Step 2 — add all four. The coefficient of C_r in the total is $1 + i^r + (-1)^r + (-i)^r$, which by the filter equals 4 when $4 \mid r$ and 0 otherwise. Hence

$$C_0 + C_4 + C_8 + \cdots = \frac{1}{4} [2^n + (1 + i)^n + 0^n + (1 - i)^n].$$

Note $(1 - 1)^n = 0$ because $n \geq 1$; this is the edge case flagged in Section 1.

Step 3 — evaluate the complex powers. Since $1 + i = \sqrt{2} e^{i\pi/4}$ and $1 - i = \sqrt{2} e^{-i\pi/4}$, De Moivre gives

$$(1 + i)^n + (1 - i)^n = 2^{n/2} (e^{in\pi/4} + e^{-in\pi/4}) = 2^{n/2} \cdot 2 \cos \frac{n\pi}{4}.$$

Step 4 — assemble.

$$C_0 + C_4 + C_8 + \cdots = \frac{1}{4} \left[2^n + 2^{(n+2)/2} \cos \frac{n\pi}{4} \right] = \frac{1}{2} \left[2^{n-1} + 2^{n/2} \cos \frac{n\pi}{4} \right]. \quad \blacksquare$$

Step 5 — evaluate at $n = 12$. Here $2^{n-1} = 2^{11} = 2048$, $2^{n/2} = 2^6 = 64$, and $\frac{12\pi}{4} = 3\pi$, so $\cos 3\pi = -1$. Therefore

$$C_0 + C_4 + C_8 + C_{12} = \frac{1}{2} [2048 + 64(-1)] = \frac{1984}{2} = 992.$$

Verification: ${}^{12}C_0 + {}^{12}C_4 + {}^{12}C_8 + {}^{12}C_{12} = 1 + 495 + 495 + 1 = 992$.

A useful reading of the result: $2^{n-2} = 1024$ is the “fair share” of the four residue classes, and the cosine term is the small correction that redistributes it. The answer is always within $2^{n/2}/2$ of 2^{n-2} .

Section 5: Practice Exercises

1. **[CBSE Board]** Prove that $\sum_{r=0}^n 2^r {}^nC_r = 3^n$, and evaluate for $n = 6$.
2. **[CBSE Board]** For $(1 + x)^9$, evaluate $C_0 + C_2 + C_4 + C_6 + C_8$ using the period-2 filter.
3. **[JEE Main]** For $(1 + x)^9$, evaluate $C_0 + C_3 + C_6 + C_9$.
4. **[JEE Main]** For $(1 + x)^{10}$, evaluate $C_1 + C_4 + C_7 + C_{10}$.
5. **[JEE Main]** For $(1 + x)^{10}$, evaluate $C_1 - C_3 + C_5 - C_7 + C_9$.
6. **[JEE Advanced]** For $(1 + x)^{10}$, evaluate $C_0 + C_4 + C_8$.
7. **[JEE Advanced]** For $(1 + x)^{15}$, evaluate $C_0 + C_3 + C_6 + C_9 + C_{12} + C_{15}$.
8. **[JEE Advanced]** How many subsets of $\{1, 2, 3, \dots, 9\}$ have a size divisible by 3 (counting the empty set)?

Answers:

1. $3^6 = 729$.
2. $2^8 = 256$.
3. 170.
4. 341.
5. 32.
6. 256.
7. 10922.
8. 170 (the count is ${}^9C_0 + {}^9C_3 + {}^9C_6 + {}^9C_9$, which is Exercise 3).

Section 6: Crack the MCQ

Q1. For $(1+x)^{12}$, the value of $C_0 + C_3 + C_6 + C_9 + C_{12}$ is:

- (a) 1365 (b) 1366 (c) 2048 (d) 4096

Correct: (b) 1366. Using $S_0 = \frac{1}{3} [2^{12} + 2 \cos \frac{12\pi}{3}] = \frac{1}{3} [4096 + 2 \cos 4\pi] = \frac{1}{3} [4096 + 2] = 1366$. Direct check: $1 + 220 + 924 + 220 + 1 = 1366$.

Distractor Analysis. (a) 1365 is $\frac{1}{3} [4096 - 1]$ — the value you get by using the cosine correction -1 instead of $+2$, i.e. by not reducing $\frac{12\pi}{3} = 4\pi$ to $\cos 4\pi = 1$ and instead assuming the generic “ 2^n minus one over three” shortcut. The correction depends on $n \bmod 6$; it cannot be memorised as a single number. (c) 2048 is 2^{n-1} , the answer to the *period-2* filter — a student who knows “half the coefficients” but not “a third of them”. (d) 4096 is 2^{12} , the sum of *all* the coefficients, from ignoring the filter entirely.

Q2. For $(1+x)^6$, the value of $C_0 - C_2 + C_4 - C_6$ is:

- (a) 0 (b) 8 (c) -8 (d) 32

Correct: (a) 0. $(1+i)^6 = [(1+i)^2]^3 = (2i)^3 = -8i$, whose real part is 0. Direct check: $1 - 15 + 15 - 1 = 0$.

Distractor Analysis. (b) 8 and (c) -8 both come from reading the *imaginary* part instead of the real part: $\text{Im}[(1+i)^6] = -8$, which is the value of $C_1 - C_3 + C_5 = 6 - 20 + 6 = -8$, with (b) additionally losing the sign. Even-indexed sums live in the real part; odd-indexed sums live in the imaginary part. (d) 32 is $2^{n-1} = 2^5$, the non-alternating even-index sum $C_0 + C_2 + C_4 + C_6$ — the period-2 answer to a period-4 question.

Q3. The set of substitutions that isolates $C_0 + C_4 + C_8 + \dots$ from $(1+x)^n$ is:

- (a) $x = 1$ and $x = -1$ only
(b) $x = 1, -1, i, -i$
(c) $x = 1, \omega, \omega^2$
(d) $x = 1, -1, \omega$

Correct: (b). Selecting one residue class modulo 4 requires all four fourth roots of unity.

Distractor Analysis. (a) is the period-2 filter; it can separate even indices from odd ones but cannot split the even indices further into $r \equiv 0$ and $r \equiv 2 \pmod{4}$. (c) is the period-3 filter, which selects $C_0 + C_3 + C_6 + \dots$ — the right idea applied with the wrong period. (d) mixes a period-2 set with a single cube root; the substitution points must be *all* the k th roots of unity for one fixed k , otherwise the inner sum in the derivation does not collapse to 0 or k .

Q4. If ω is a non-real cube root of unity, then $(1+\omega)^n + (1+\omega^2)^n$ equals:

- (a) $2 \cos \frac{n\pi}{3}$ (b) 2^n (c) -1 (d) 0

Correct: (a). $1 + \omega = e^{i\pi/3}$ and $1 + \omega^2 = e^{-i\pi/3}$, so the sum is $e^{in\pi/3} + e^{-in\pi/3} = 2 \cos \frac{n\pi}{3}$.

Distractor Analysis. (b) 2^n treats each bracket as if it were $1 + 1$, which is the $j = 0$ term of the filter, not the $j = 1, 2$ terms. (c) -1 is the value of $\omega^n + \omega^{2n}$ when $3 \nmid n$ — a genuine and useful identity, but about the roots themselves, not about $1 + \omega$. (d) 0 is $1 + \omega + \omega^2$ misremembered; note in particular that $1 + \omega \neq 0$.

Section 7: Common Student Pitfalls and Pro-Tips

- **Believing $1 + \omega = 0$.** It is not. The identity that vanishes is $1 + \omega + \omega^2 = 0$, from which $1 + \omega = -\omega^2$ and $1 + \omega^2 = -\omega$. Both have modulus 1, not 0.
- **Dividing by 2 out of habit.** The k -way filter divides by k . For $C_0 + C_3 + C_6 + \dots$ you divide by 3, not 2.
- **Leaving an i in the final answer.** These sums are sums of positive integers, so they are real. If your answer carries an i , you dropped one of the conjugate substitutions — the terms j and $k - j$ always pair up into a cosine.
- **Not reducing the angle.** $\cos \frac{10\pi}{3}$ must first become $\cos \frac{4\pi}{3} = -\frac{1}{2}$. Reducing modulo 2π is not optional; guessing the cosine of a large multiple of $\pi/3$ is where most numerical errors enter.
- **Confusing “every third coefficient” with “alternating”.** Alternating is period 2 and needs $x = -1$; every third is period 3 and needs ω ; alternating signs on the even indices is period 4 and needs i . Count the indices between sign changes before choosing a tool.
- **Forgetting that $(1 - 1)^n = 0$ only for $n \geq 1$.** At $n = 0$ that bracket is 1, and every period-4 formula shifts.
- **Applying multisection to Sub-05’s any-index series.** Every derivation here needs a finite polynomial. For $|x| < 1$ infinite series the substitutions $x = \omega$ or $x = i$ do not even satisfy the convergence condition, since $|\omega| = |i| = 1$.
- **Pro-Tip — the partition check.** For period k , the k sums must add to 2^n . Compute two of them and get the third by subtraction; it is faster and self-verifying.
- **Pro-Tip — the “fair share” estimate.** For $k = 3$ each sum is within $\frac{2}{3}$ of $\frac{2^n}{3}$, so it is always one of the two integers straddling $\frac{2^n}{3}$. For $n = 10$ that is 341.33, so the three sums must be 341 or 342 — which is exactly what Example 2 found. Use this to reject a badly wrong answer in three seconds.
- **Pro-Tip — powers of $(1 + i)$ without trigonometry.** $(1 + i)^2 = 2i$, so for even n , $(1 + i)^n = (2i)^{n/2}$, and powers of i cycle with period 4. This is faster and less error-prone than De Moivre under exam pressure.
- **Pro-Tip — recognise the counting disguise.** “How many subsets have size divisible by 3” and “how many ways to pick a committee whose size leaves remainder 1 on division by 4” are multisection questions wearing a Chapter 6 costume. The moment a count is restricted by a residue condition on the size, reach for the filter.

Section 8: Cheat Sheet Box

MULTISECTION BY ROOTS OF UNITY — at a glance

- **The dial is free:** $\sum_r {}^nC_r y^r = (1+y)^n$ for every y . So $\sum 3^r C_r = 4^n$, $\sum 2^r C_r = 3^n$.
- **Filter:** $\frac{1}{k} \sum_{j=0}^{k-1} \omega_k^{jm} = 1$ if $k \mid m$, else 0, where $\omega_k = e^{2\pi i/k}$.
- **Multisection:** $\sum_{r \equiv a \pmod{k}} {}^nC_r = \frac{1}{k} \sum_{j=0}^{k-1} \omega_k^{-ja} (1 + \omega_k^j)^n$.
- **Cube roots:** $\omega^3 = 1$, $1 + \omega + \omega^2 = 0$, $1 + \omega = e^{i\pi/3}$, $1 + \omega^2 = e^{-i\pi/3}$.
- **Period 2:** $C_0 + C_2 + \dots = C_1 + C_3 + \dots = 2^{n-1}$ ($n \geq 1$).
- **Period 3:** $\sum_{r \equiv a \pmod{3}} C_r = \frac{1}{3} \left[2^n + 2 \cos \frac{(n-2a)\pi}{3} \right]$; the three sums add to 2^n .
- **Period 4 components:** $C_0 - C_2 + C_4 - \dots = 2^{n/2} \cos \frac{n\pi}{4}$; $C_1 - C_3 + C_5 - \dots = 2^{n/2} \sin \frac{n\pi}{4}$.
- **Period 4 classes:** $\sum_{r \equiv a \pmod{4}} C_r = \frac{1}{4} \left[2^n + 2^{(n+2)/2} \cos \frac{(n-2a)\pi}{4} \right]$.
- **Fast powers:** $(1+i)^2 = 2i \Rightarrow (1+i)^n = (2i)^{n/2}$ for even n .

Trigger conditions: every k th coefficient $\Rightarrow k$ th roots of unity. Signs flipping every index $\Rightarrow x = -1$. Signs flipping every two indices $\Rightarrow x = i$. A count restricted by a remainder on the size $\Rightarrow$ multisection.

Memory aids: "Period 2 is -1 , period 3 is ω , period 4 is i ." · "Divide by k , not by 2." · "The k pieces must rebuild 2^n ." · " $1 + \omega \neq 0$; it is $-\omega^2$."

Chapter 7: Binomial Theorem — Round 2 Addendum

Subtopics 01-07 and the Round 1 Supplement (Addenda A, B, C and Subtopic 08) between them cover the NCERT syllabus and the standard JEE toolkit, with four exceptions. Three of them are *stated-but-never-used* results: the chapter prints a formula once, puts it in the cheat sheet, and then never asks you to do anything with it. A printed formula you have never executed is not a tool you own, and NCERT examines all three directly. The fourth is a missing general method — Subtopic 04 hands you two weighted sums, $\sum r C_r$ and $\sum r^2 C_r$, as finished answers, but never the machine that produces them for an arbitrary weight.

None of these warrants a new subtopic; each is a short method plus drill. The lettering continues the Round 1 Supplement.

Addendum	Topic	Read after
D	The p th term from the end, and beginning-to-end ratios	Subtopic 02
E	Vandermonde's identity and products of coefficients	Subtopic 04
F	Arbitrary polynomial weights: the absorption identity	Subtopic 04
G	The central binomial coefficient ${}^{2n}C_n$	Subtopic 03

Addendum D — The p th Term From the End

Extends Subtopic 02, Section 2 (where the rule is stated) and Section 5.

Exam relevance. NCERT devotes Example 14 of the chapter to deriving this rule, and Miscellaneous Exercise Question 8 is built entirely on it — a 4-mark CBSE-style question that reappears almost verbatim in JEE Main every few years. Subtopic 02 prints the rule in its formula list and its cheat sheet, but no worked example, exercise or MCQ in the chapter ever uses it, so it is the most likely single line in the chapter to be read and forgotten.

Method. The expansion of $(a + b)^n$ has $n + 1$ terms, $T_1, T_2, \dots, T_{n+1}$. Counting from the far end reverses the list, so

$$p\text{th term from the end} = T_{n-p+2} \quad (\text{the } (n - p + 2)\text{th from the beginning}).$$

That is the form Subtopic 02 gives you, and it is correct — but it invites two stacked off-by-one errors, because after finding the *term number* $n - p + 2$ you must still drop to the *index*

$r = n - p + 1$ before using $T_{r+1} = {}^nC_r a^{n-r} b^r$. The safer route is to reverse the binomial instead of the counting:

The p th term from the end of $(a + b)^n$ is the p th term from the beginning of $(b + a)^n$.

Reversing the two parts reverses the whole list of terms, so the two statements say the same thing. Now you use the familiar index $r = p - 1$ and never touch $n - p + 2$ at all:

$$p\text{th term from the end of } (a + b)^n = {}^nC_{p-1} b^{n-p+1} a^{p-1}.$$

Three consequences worth carrying:

1. **The two p th terms share a binomial coefficient.** The p th from the beginning is ${}^nC_{p-1} a^{n-p+1} b^{p-1}$ and the p th from the end is ${}^nC_{p-1} b^{n-p+1} a^{p-1}$. Same ${}^nC_{p-1}$, with a and b swapped.
2. **Hence the ratio is coefficient-free.** Dividing one by the other,

$$\frac{T_p \text{ from the beginning}}{T_p \text{ from the end}} = \left(\frac{a}{b}\right)^{n-2p+2}.$$

Every “ratio of the p th term from the beginning to the p th term from the end” problem collapses into this one line. You never compute either term.

3. **Equal ends.** The two are equal exactly when $n = 2p - 2$ (assuming $a \neq b$), which is another way of saying that T_p is the middle term of an even-power expansion. For even n the middle term is its own mirror image — the fixed point of the reversal.

Figure brief: A horizontal strip of $n + 1$ numbered boxes labelled T_1 through T_{n+1} , with a left-to-right arrow above labelled “counting from the beginning: index $r = p - 1$ ” and a right-to-left arrow below labelled “counting from the end”. Highlight the box T_4 and the box T_{n-2} in the same colour, joined by a curved double-headed arrow annotated “4th from the beginning and 4th from the end: same nC_3 , parts swapped”. For n even, show the central box shaded and labelled “its own mirror”.

Worked Example D1 — [CBSE Board Level]

Find the 4th term from the end in the expansion of $\left(\frac{x^3}{2} - \frac{2}{x^2}\right)^9$.

Reverse the binomial and take $p = 4$, so $r = p - 1 = 3$:

$$T_4 \text{ of } \left(-\frac{2}{x^2} + \frac{x^3}{2}\right)^9 = {}^9C_3 \left(-\frac{2}{x^2}\right)^6 \left(\frac{x^3}{2}\right)^3 = 84 \cdot \frac{64}{x^{12}} \cdot \frac{x^9}{8}.$$

The power of x is $-12 + 9 = -3$ and the numbers give $84 \times 64 \div 8 = 672$, so the required term is $\frac{672}{x^3}$.

Cross-check by the other route: $n - p + 2 = 9 - 4 + 2 = 7$, so this should also be T_7 , which uses $r = 6$: ${}^9C_6 \left(\frac{x^3}{2}\right)^3 \left(-\frac{2}{x^2}\right)^6 = 84 \cdot \frac{x^9}{8} \cdot \frac{64}{x^{12}}$ — the same thing, as ${}^9C_6 = {}^9C_3$. Note the even power 6 on the negative part, which is why the answer came out positive.

Worked Example D2 — [JEE Main Level]

Find n if the ratio of the 5th term from the beginning to the 5th term from the end in the expansion of $\left(\sqrt[4]{2} + \frac{1}{\sqrt[4]{3}}\right)^n$ is $\sqrt{6} : 1$.

Do not write either term. Use consequence 2 with $a = 2^{1/4}$, $b = 3^{-1/4}$ and $p = 5$:

$$\frac{a}{b} = 2^{1/4} \cdot 3^{1/4} = 6^{1/4}, \quad n - 2p + 2 = n - 8,$$

$$\text{ratio} = (6^{1/4})^{n-8} = 6^{(n-8)/4}.$$

Set this equal to $\sqrt{6} = 6^{1/2}$. Since $6 \neq 1$, the exponents must match:

$$\frac{n-8}{4} = \frac{1}{2} \implies n-8 = 2 \implies \boxed{n=10}.$$

The whole NCERT Miscellaneous problem is three lines once you know that the binomial coefficients cancel. Solving it by writing out both terms takes half a page and usually produces an arithmetic slip in the surd exponents.

Worked Example D3 — [JEE Main — Speed Trap]

Find the 3rd term from the end in the expansion of $(x + y)^{10}$.

The trap. The rule in the cheat sheet is “ p th from the end = $(n - p + 2)$ th from the beginning”. A student computes $n - p + 2 = 10 - 3 + 2 = 9$ and then, on autopilot, feeds 9 into $T_{r+1} = {}^nC_r a^{n-r} b^r$ as $r = 9$ — or, worse, computes $n - p = 7$ and answers $T_8 = {}^{10}C_7 x^3 y^7 = 120 x^3 y^7$. That $120 x^3 y^7$ is the seductive wrong answer, because it is a genuine term of the expansion and looks perfectly respectable.

The fix. Reverse instead of counting. The 3rd term from the end of $(x + y)^{10}$ is the 3rd term from the beginning of $(y + x)^{10}$, so $r = 2$:

$$T_3 = {}^{10}C_2 y^8 x^2 = 45 x^2 y^8.$$

Sanity check with the other route: $T_{n-p+2} = T_9$, which uses $r = 8$, giving ${}^{10}C_8 x^2 y^8 = 45 x^2 y^8$. Agreed.

Name the trap: two off-by-ones stacked. “ p th from the end” converts to a *term number*, and every term number must still be reduced by one to reach the *index* r . Reversing the binomial removes the first conversion entirely, so there is only ever one subtraction to get wrong.

Worked Example D4 — [JEE Advanced Level]

If the 7th term from the beginning and the 7th term from the end in the expansion of $\left(\sqrt[3]{2} + \frac{1}{\sqrt[3]{3}}\right)^n$ are equal, find n .

By consequence 2, with $a = 2^{1/3}$, $b = 3^{-1/3}$ and $p = 7$,

$$\frac{T_7 \text{ (beginning)}}{T_7 \text{ (end)}} = \left(\frac{a}{b}\right)^{n-12} = (6^{1/3})^{n-12}.$$

Equality means this ratio is 1. Since $6^{1/3} \neq 1$, the exponent must vanish: $n - 12 = 0$, so $n = 12$.

Read the answer structurally rather than as a number. $n = 2p - 2 = 12$ says that T_7 sits at the centre of a 13-term list — it is the middle term, and the middle term of an even-power expansion is unchanged when you reverse the binomial. The surds were decoration; the only thing that mattered was $a \neq b$.

Practice.

1. Find the 5th term from the end in the expansion of $\left(\frac{3}{x^2} - \frac{x^3}{6}\right)^9$.
2. Find the 6th term from the end in the expansion of $(x - 2y)^{12}$.
3. Find n if the ratio of the 4th term from the beginning to the 4th term from the end in $\left(\sqrt[3]{2} + \frac{1}{\sqrt[3]{3}}\right)^n$ is $\sqrt[3]{6} : 1$.
4. Write down, in terms of n and p , the p th term from the end of $(a + b)^n$, and verify that setting $p = 1$ returns b^n and setting $p = n + 1$ returns a^n .
5. For which n is the 5th term from the beginning of $(a + b)^n$ equal to the 5th term from the end, given $a \neq b$? State the answer as a condition on n and say what that term is.

Answers.

1. $-\frac{21}{16}x^7$.
2. $-101376x^5y^7$.
3. $n = 7$.
4. ${}^nC_{p-1}b^{n-p+1}a^{p-1}$; at $p = 1$ it is ${}^nC_0b^n = b^n$ (the last term), at $p = n + 1$ it is ${}^nC_nb^0a^n = a^n$ (the first term).
5. $n = 2(5) - 2 = 8$; the term is the single middle term $T_5 = {}^8C_4a^4b^4 = 70a^4b^4$.

Addendum E — Vandermonde's Identity

Extends Subtopic 04, Sections 2 and 3.

Exam relevance. Subtopic 04 lists Vandermonde's identity in its formula table and repeats it in the cheat sheet, and Section 3 remarks that the multiply-and-compare technique "proves Vandermonde's identity" — but the chapter never proves it, never works an example with it

and never sets a question on it. It is the parent of the two product-sums the chapter *does* drill, $\sum C_r^2 = {}^{2n}C_n$ and $\sum C_r C_{r+k} = {}^{2n}C_{n-k}$, so learning it turns two memorised results into one understood one. JEE Main sets it directly as “evaluate $\sum {}^{10}C_r {}^{20}C_{5-r}$ ”; JEE Advanced hides it inside committee-counting problems and inside sums that also carry a weight in r .

Method. For non-negative integers m, n and p ,

$$\sum_k {}^mC_k {}^nC_{p-k} = {}^{m+n}C_p.$$

The convention that makes the sum safe. Read ${}^nC_j = 0$ whenever $j < 0$ or $j > n$. With that convention you may let k run over all integers, or over 0 to p , and the out-of-range terms simply contribute nothing. Without it you must restrict the sum by hand to $\max(0, p-n) \leq k \leq \min(m, p)$, which is where most marks are lost.

Counting proof. A committee of p people is to be chosen from m women and n men. Split the count by the number k of women chosen: there are mC_k ways to pick the women and ${}^nC_{p-k}$ ways to pick the remaining $p-k$ men. Summing over k counts every committee exactly once, and the total number of committees is ${}^{m+n}C_p$.

Coefficient-comparison proof. This is Subtopic 04’s Technique D at full strength. In

$$(1+x)^m(1+x)^n = (1+x)^{m+n},$$

the coefficient of x^p on the right is ${}^{m+n}C_p$. On the left, an x^p arises as x^k from the first bracket times x^{p-k} from the second, contributing ${}^mC_k {}^nC_{p-k}$; sum over k .

The three specialisations you already met.

Take	You get	Where it appears
$m = n = p$, then ${}^nC_{n-r} = {}^nC_r$	$\sum_{r=0}^n C_r^2 = {}^{2n}C_n$	Sub-04, Example 4
$m = n$, $p = n - k$, then symmetry	$\sum_r C_r C_{r+k} = {}^{2n}C_{n-k}$	Sub-04, Section 2
m, n different, one factor flipped	$\sum_r {}^mC_r {}^nC_r = {}^{m+n}C_m$	new here

The one precondition. Vandermonde applies only when **the two lower indices in each term add to a constant**. In ${}^mC_k {}^nC_{p-k}$ they add to p for every k , which is exactly what lets the product be read as “coefficient of x^p ”. In a sum like $\sum {}^mC_r {}^nC_r$ the lower indices add to $2r$, which is *not* constant, so the identity does not apply as written — you must first flip one factor by symmetry (${}^mC_r = {}^mC_{m-r}$), after which the indices add to the constant m .

The alternating companion. Multiplying $(1-x)^n$ by $(1+x)^n$ instead gives $(1-x^2)^n$, whose expansion contains only even powers. Comparing coefficients of x^n :

$$\sum_{r=0}^n (-1)^r C_r^2 = \begin{cases} 0, & n \text{ odd,} \\ (-1)^{n/2} C_{n/2}, & n \text{ even.} \end{cases}$$

The odd case vanishes because $(1 - x^2)^n$ has no odd powers at all — there is nothing on the right to match.

Figure brief: A grid with m women's silhouettes on the left and n men's on the right, and below it a row of $p + 1$ small bins labelled $k = 0, 1, \dots, p$; bin k carries the label ${}^m C_k \cdot {}^n C_{p-k}$ and bins whose label needs an impossible choice (such as $k > m$) are greyed out and marked “= 0”. Caption: “every committee of p lands in exactly one bin, so the bins must total ${}^{m+n} C_p$ ”.

Worked Example E1 — [CBSE Board Level]

Evaluate $\sum_{r=0}^5 {}^{10} C_r {}^{20} C_{5-r}$.

The lower indices are r and $5 - r$, adding to the constant 5 in every term, so Vandermonde applies with $m = 10$, $n = 20$, $p = 5$:

$$\sum_{r=0}^5 {}^{10} C_r {}^{20} C_{5-r} = {}^{30} C_5 = \frac{30 \cdot 29 \cdot 28 \cdot 27 \cdot 26}{120} = 142506.$$

In committee language: choosing 5 people from a group of 10 and a group of 20, split by how many come from the first group.

Worked Example E2 — [JEE Main Level]

For $(1 + x)^8$, evaluate $C_0 C_2 + C_1 C_3 + C_2 C_4 + \dots + C_6 C_8$.

The general term is $C_r C_{r+2}$, whose lower indices add to $2r + 2$ — not constant. Flip the first factor by symmetry, $C_r = {}^8 C_{8-r}$, so the term becomes ${}^8 C_{8-r} {}^8 C_{r+2}$ with indices adding to 10 for every r . Vandermonde with $m = n = 8$, $p = 10$ gives

$$\sum_r {}^8 C_r {}^8 C_{r+2} = {}^{16} C_{10} = {}^{16} C_6 = 8008.$$

This matches Subtopic 04's stored result $\sum C_r C_{r+k} = {}^{2n} C_{n-k}$ with $n = 8$, $k = 2$: ${}^{16} C_6 = 8008$. The point of redoing it from Vandermonde is that you now do not need to have memorised the stored form.

Worked Example E3 — [JEE Main — Speed Trap]

Evaluate $\sum_{r=0}^5 {}^5 C_r {}^7 C_r$.

The trap. The expression has the look of Vandermonde, so the reflex is to add the upper indices and write ${}^{12} C_r$ — which is not even a number, it still has an r in it — or to patch

that by answering ${}^{12}C_0 = 1$ or ${}^{12}C_5 = 792$ for the wrong reason. The error is skipping the precondition: here the lower indices are r and r , adding to $2r$, which changes from term to term. Vandermonde is simply not applicable to the expression as printed.

The fix. Make the indices add to a constant first. By symmetry ${}^5C_r = {}^5C_{5-r}$:

$$\sum_{r=0}^5 {}^5C_r {}^7C_r = \sum_{r=0}^5 {}^5C_{5-r} {}^7C_r = {}^{12}C_5 = 792.$$

Now the indices $5 - r$ and r add to 5 in every term, so $m = 5$, $n = 7$, $p = 5$.

Direct check: $1 \cdot 1 + 5 \cdot 7 + 10 \cdot 21 + 10 \cdot 35 + 5 \cdot 35 + 1 \cdot 21 = 1 + 35 + 210 + 350 + 175 + 21 = 792$.

Name the trap: applying Vandermonde before checking that the lower indices add to a constant. The right answer and the lucky-guess answer coincide here, which is exactly why this is dangerous — change the 7 to an 8 and the guess fails while the method does not.

Worked Example E4 — [JEE Advanced Level]

Prove that $\sum_{r=1}^n r ({}^nC_r)^2 = n {}^{2n-1}C_{n-1}$, and evaluate it for $n = 5$.

This needs two tools at once: the absorption identity of Addendum F to deal with the weight r , then Vandermonde to deal with the product.

Step 1 — absorb the weight. By $r {}^nC_r = n {}^{n-1}C_{r-1}$ (proved in Addendum F),

$$\sum_{r=1}^n r {}^nC_r \cdot {}^nC_r = n \sum_{r=1}^n {}^{n-1}C_{r-1} {}^nC_r.$$

Step 2 — fix the indices. The lower indices $r - 1$ and r add to $2r - 1$, not a constant. Flip the second factor: ${}^nC_r = {}^nC_{n-r}$, so the term is ${}^{n-1}C_{r-1} {}^nC_{n-r}$ and the indices add to $(r - 1) + (n - r) = n - 1$ for every r .

Step 3 — apply Vandermonde with upper indices $n - 1$ and n , and $p = n - 1$:

$$n \sum_r {}^{n-1}C_{r-1} {}^nC_{n-r} = n {}^{2n-1}C_{n-1}. \quad \blacksquare$$

Evaluation at $n = 5$: $5 \cdot {}^9C_4 = 5 \times 126 = 630$.

Direct check: $1(25) + 2(100) + 3(100) + 4(25) + 5(1) = 25 + 200 + 300 + 100 + 5 = 630$.

Practice.

1. Evaluate $\sum_{r=0}^4 {}^6C_r {}^8C_{4-r}$.
2. For $(1 + x)^7$, evaluate $C_0^2 + C_1^2 + \cdots + C_7^2$ using Vandermonde rather than the stored formula.
3. For $(1 + x)^9$, evaluate $C_0C_3 + C_1C_4 + \cdots + C_6C_9$.

4. Evaluate $\sum_{r=0}^6 {}^6C_r {}^9C_r$.
5. Evaluate $\sum_r {}^{15}C_r {}^{10}C_{12-r}$, stating which values of r actually contribute.
6. For $(1+x)^8$, evaluate $C_0^2 - C_1^2 + C_2^2 - \dots + C_8^2$.
7. Prove Vandermonde's identity by the committee argument, and deduce $\sum_{r=0}^n C_r^2 = {}^{2n}C_n$ as the case $m = n = p$.

Answers.

1. ${}^{14}C_4 = 1001$.
2. ${}^{14}C_7 = 3432$.
3. ${}^{18}C_6 = 18564$.
4. ${}^{15}C_6 = 5005$ (flip one factor first).
5. ${}^{25}C_{12} = 5200300$; only $r = 2$ to $r = 12$ contribute, since ${}^{10}C_{12-r} = 0$ for $r < 2$.
6. $(-1)^4 {}^8C_4 = 70$.
7. As in the Method section; with $m = n = p$ the sum is $\sum_r {}^nC_r {}^nC_{n-r} = \sum_r C_r^2$ by symmetry, and the right-hand side is ${}^{2n}C_n$.

Addendum F — Arbitrary Polynomial Weights: the Absorption Identity

Extends Subtopic 04, Sections 2 and 3 (Technique B).

Exam relevance. Subtopic 04 gives you $\sum r C_r = n2^{n-1}$ and $\sum r^2 C_r = n(n+1)2^{n-2}$ as two remembered results, obtained by differentiating. That is enough for the two weights it lists and nothing else. JEE Main routinely sets a weight the chapter has not stored — $3r + 2$, or $r(r-1)$, or r^3 — and JEE Advanced attaches a sign or an x^r that makes differentiation clumsy. This addendum replaces two memorised answers with one method that generates all of them, using no calculus at all.

Method — the absorption identity. For $1 \leq r \leq n$,

$$r {}^nC_r = n {}^{n-1}C_{r-1},$$

because $r \cdot \frac{n!}{r!(n-r)!} = \frac{n!}{(r-1)!(n-r)!} = n \cdot \frac{(n-1)!}{(r-1)!((n-1)-(r-1))!}$. Read it as: *a factor of r in front can be swallowed by the coefficient, at the price of dropping both indices by one and pulling out an n .* Applying it twice, and then k times,

$$r(r-1) {}^nC_r = n(n-1) {}^{n-2}C_{r-2}, \quad r(r-1)\dots(r-k+1) {}^nC_r = n(n-1)\dots(n-k+1) {}^{n-k}C_{r-k}.$$

Notice the shape of the weight that absorbs cleanly: not r^k , but the **falling factorial** $r(r-1)\dots(r-k+1)$. So the procedure is:

1. **Rewrite the weight in falling factorials.** $r^2 = r(r-1) + r$; $r^3 = r(r-1)(r-2) + 3r(r-1) + r$; a linear weight $a + rd$ needs no rewriting.
2. **Absorb each piece**, turning $\sum(\text{weight})^n C_r$ into a combination of plain coefficient sums with smaller upper index.
3. **Substitute** whatever value of x the problem carries — $x = 1$ gives 2^{n-k} , $x = -1$ gives 0, a general x gives $(1+x)^{n-k}$.

The standard results fall out immediately at $x = 1$:

Weight	Absorbed form	Sum over all r
r	$n {}^{n-1}C_{r-1}$	$n 2^{n-1}$
$r(r-1)$	$n(n-1) {}^{n-2}C_{r-2}$	$n(n-1) 2^{n-2}$
$r^2 = r(r-1) + r$	—	$n(n+1) 2^{n-2}$
$r^3 = r(r-1)(r-2) + 3r(r-1) + r$	—	$n^2(n+3) 2^{n-3}$
$a + rd$	—	$(2a + nd) 2^{n-1}$

A calculus warning worth one line. Differentiating $(1+x)^n$ twice produces the weight $r(r-1)$, **not** r^2 . The operator that manufactures r^k is $(x \frac{d}{dx})^k$ — you must multiply by x between the differentiations. If you differentiate twice and quote $n(n+1)2^{n-2}$ you have mixed the two up; the honest output of the second derivative at $x = 1$ is $n(n-1)2^{n-2}$, and the difference between them is exactly $\sum r C_r = n2^{n-1}$. Absorption sidesteps the whole question.

The reversal trick for linear weights. For $S = \sum_{r=0}^n (a + rd) {}^n C_r$, write the sum a second time with r replaced by $n - r$ — legitimate because $C_{n-r} = C_r$ — and add:

$$2S = \sum_{r=0}^n \left[(a + rd) + (a + (n-r)d) \right] C_r = (2a + nd) \sum_{r=0}^n C_r = (2a + nd) 2^n,$$

so $S = (2a + nd) 2^{n-1}$. The bracket is constant because the two weights are the two ends of an arithmetic progression, and an AP's terms pair up to a constant total. This is a second independent route to the linear case and a very fast one under exam pressure.

Signs. With $x = -1$ the plain sums collapse to zero, so $\sum (-1)^r r C_r = 0$ for $n \geq 2$ and $\sum (-1)^r r^2 C_r = 0$ for $n \geq 3$. Check the small cases separately rather than assuming: at $n = 1$ the first sum is -1 , and at $n = 2$ the second is 2. Absorption tells you why — the reduced sum $\sum (-1)^j {}^{n-k}C_j$ vanishes only when its upper index $n - k$ is at least 1.

Figure brief: A two-column ladder. Left column shows the weight tower 1, r , $r(r-1)$, $r(r-1)(r-2)$; right column shows the corresponding reduced coefficient ${}^n C_r$, ${}^{n-1}C_{r-1}$, ${}^{n-2}C_{r-2}$, ${}^{n-3}C_{r-3}$ with the pulled-out factors 1, n , $n(n-1)$, $n(n-1)(n-2)$ written on the connecting arrows. Caption: “each rung swallows one factor and drops both indices by one”.

Worked Example F1 — [CBSE Board Level]

Prove $r {}^n C_r = n {}^{n-1}C_{r-1}$ and use it to evaluate $C_1 + 2C_2 + 3C_3 + \cdots + nC_n$ without differentiating.

Proof. For $1 \leq r \leq n$,

$$r {}^nC_r = r \cdot \frac{n!}{r!(n-r)!} = \frac{n!}{(r-1)!(n-r)!} = n \cdot \frac{(n-1)!}{(r-1)!(n-r)!} = n {}^{n-1}C_{r-1},$$

since $(n-r)! = ((n-1) - (r-1))!$. ■

Use. Substituting and then re-labelling $j = r - 1$,

$$\sum_{r=1}^n r {}^nC_r = n \sum_{r=1}^n {}^{n-1}C_{r-1} = n \sum_{j=0}^{n-1} {}^{n-1}C_j = n 2^{n-1}.$$

The re-labelling is the step to write out in a board answer: after absorption the sum is a *complete* row of Pascal's triangle for index $n - 1$, which is why it totals 2^{n-1} .

Worked Example F2 — [JEE Main Level]

Evaluate $\sum_{r=0}^{10} (3r+2) {}^{10}C_r$.

The weight is linear, $a = 2$ and $d = 3$, so the reversal trick applies directly:

$$\sum_{r=0}^{10} (3r+2) {}^{10}C_r = (2a+nd) 2^{n-1} = (4+30) 2^9 = 34 \times 512 = 17408.$$

Check by splitting instead: $3 \sum r {}^nC_r + 2 \sum {}^nC_r = 3(10 \cdot 2^9) + 2(2^{10}) = 15360 + 2048 = 17408$. Two routes, one answer — worth doing once so you trust the shortcut afterwards.

Worked Example F3 — [JEE Main — Speed Trap]

For $(1+x)^{10}$, evaluate $\sum_{r=0}^{10} r(r-1) {}^{10}C_r$.

The trap. The weight looks like a lightly disguised r^2 , so the reflex is to reach for the stored result $n(n+1)2^{n-2} = 10 \cdot 11 \cdot 256 = 28160$. A second version of the same error: “differentiate twice and substitute $x = 1$ ”, which the student believes produces $\sum r^2 {}^nC_r$ and therefore also answers 28160. Both give the seductive wrong answer.

The fix. $r(r-1)$ is already a falling factorial, so absorb it twice:

$$\sum_{r=0}^{10} r(r-1) {}^{10}C_r = 10 \cdot 9 \sum_{r=2}^{10} {}^8C_{r-2} = 90 \cdot 2^8 = 90 \times 256 = 23040.$$

The self-check that exposes the trap. Since $r^2 = r(r-1) + r$,

$$\sum r^2 {}^nC_r - \sum r(r-1) {}^nC_r = \sum r {}^nC_r, \quad 28160 - 23040 = 5120 = 10 \cdot 2^9. \checkmark$$

Name the trap: treating $r(r-1)$ and r^2 as the same weight. They differ by exactly $\sum r {}^nC_r$, which is never zero. Whenever a question hands you $r(r-1)$, it is testing precisely whether

you know that the second derivative delivers the falling factorial and not the square.

Worked Example F4 — [JEE Advanced Level]

Prove that $\sum_{r=0}^n r^3 {}^nC_r = n^2(n+3)2^{n-3}$ for $n \geq 3$, and evaluate it for $n = 6$.

Step 1 — falling-factorial basis. Expand $r(r-1)(r-2) = r^3 - 3r^2 + 2r$ and $r(r-1) = r^2 - r$, then solve upwards:

$$r^3 = r(r-1)(r-2) + 3r(r-1) + r.$$

(Check at $r = 2$: $0 + 3(2) + 2 = 8 = 2^3$.)

Step 2 — absorb each piece, summing over all r and using $\sum_j {}^mC_j = 2^m$:

$$\sum r(r-1)(r-2)C_r = n(n-1)(n-2)2^{n-3}, \quad \sum r(r-1)C_r = n(n-1)2^{n-2}, \quad \sum rC_r = n2^{n-1}.$$

Step 3 — recombine over the common factor 2^{n-3} :

$$\sum r^3 C_r = 2^{n-3} [n(n-1)(n-2) + 3 \cdot 2n(n-1) + 4n] = 2^{n-3} n [(n-1)(n-2) + 6(n-1) + 4].$$

The bracket is $n^2 - 3n + 2 + 6n - 6 + 4 = n^2 + 3n = n(n+3)$, so

$$\sum_{r=0}^n r^3 {}^nC_r = n^2(n+3)2^{n-3}. \quad \blacksquare$$

Evaluation at $n = 6$: $36 \cdot 9 \cdot 2^3 = 36 \times 9 \times 8 = 2592$.

The same three steps handle any polynomial weight; only the falling-factorial rewrite in Step 1 changes.

Practice.

1. Evaluate $\sum_{r=0}^9 (2r+5) {}^9C_r$.
2. Evaluate $\sum_{r=0}^8 r(r-1) {}^8C_r$.
3. Evaluate $\sum_{r=0}^7 r^3 {}^7C_r$.
4. Evaluate $\sum_{r=0}^n (-1)^r r^2 {}^nC_r$ for $n = 2$ and for $n = 6$, and say for which n the answer is 0.
5. Evaluate $\sum_{r=0}^5 \frac{r}{r+1} {}^5C_r$, and give the general closed form. (Hint: $\frac{r}{r+1} = 1 - \frac{1}{r+1}$, then use Subtopic 04's integration result.)

6. Evaluate $\sum_{r=0}^6 r {}^6C_r 2^r$, and state the general form of $\sum_r r {}^nC_r x^r$.

Answers.

1. 7168 (from $(2a + nd)2^{n-1} = (10 + 18)2^8$).
2. 3584 (from $8 \cdot 7 \cdot 2^6$).
3. 7840 (from $7^2 \cdot 10 \cdot 2^4$).
4. 2 at $n = 2$; 0 at $n = 6$; the answer is 0 for every $n \geq 3$.
5. $\frac{43}{2}$; in general $2^n - \frac{2^{n+1} - 1}{n + 1}$.
6. 2916; in general $\sum_r r {}^nC_r x^r = n x (1 + x)^{n-1}$, here $6 \cdot 2 \cdot 3^5$.

Addendum G — The Central Binomial Coefficient ${}^{2n}C_n$

Extends Subtopic 03, Sections 2 and 4, and connects to Addendum E.

Exam relevance. Subtopic 03 tells you the greatest binomial coefficient is the middle one and evaluates it numerically (${}^{12}C_6 = 924$), but never writes it in closed form. NCERT does: its Example 6 is a full derivation of the middle term of $(1 + x)^{2n}$, Exercise 8.2 Question 11 and Miscellaneous Example 12 both turn on relations between ${}^{2n}C_n$ and the coefficients of $(1 + x)^{2n-1}$. Those are 3-4 mark CBSE proof questions. JEE Main reuses the same object as the constant term of $(x + \frac{1}{x})^{2n}$ and as $\sum C_r^2$; JEE Advanced uses its size.

Method — three faces of the same number.

The factorial form and the odd-product form.

$${}^{2n}C_n = \frac{(2n)!}{n! n!} = \frac{2^n [1 \cdot 3 \cdot 5 \cdots (2n-1)]}{n!}.$$

Derivation. Split $(2n)!$ into its odd and even factors. The even factors are $2, 4, 6, \dots, 2n$, and there are n of them, so their product is $2^n(1 \cdot 2 \cdots n) = 2^n n!$. Hence $(2n)! = [1 \cdot 3 \cdot 5 \cdots (2n-1)] \cdot 2^n n!$, and dividing by $n! n!$ gives the stated form. Equivalently, the middle term of $(1 + x)^{2n}$ is

$$T_{n+1} = \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{n!} 2^n x^n,$$

which is NCERT's Example 6 verbatim.

The Pascal form. Because ${}^{2n-1}C_{n-1} = {}^{2n-1}C_n$ (symmetry, since $(n-1) + n = 2n-1$), Pascal's rule collapses to

$${}^{2n}C_n = {}^{2n-1}C_{n-1} + {}^{2n-1}C_n = 2 {}^{2n-1}C_{n-1}.$$

In words: *the middle coefficient of $(1+x)^{2n}$ is twice either middle coefficient of $(1+x)^{2n-1}$, and equals their sum.* That single sentence answers NCERT Exercise 8.2 Q11 and Miscellaneous Example 12 together.

The sum-of-squares form. From Addendum E, ${}^{2n}C_n = \sum_{r=0}^n ({}^nC_r)^2$. It is also the constant term of $(x + \frac{1}{x})^{2n}$, which is how NCERT introduces it in Section 8.3.

Size: the exact ratio, and the bounds you may quote. Successive central coefficients satisfy

$$\frac{{}^{2n}C_n}{{}^{2n-2}C_{n-1}} = \frac{2(2n-1)}{n} = 4 - \frac{2}{n},$$

so each central coefficient is a little less than four times the previous one. That “little less” accumulates, and it is why ${}^{2n}C_n$ is *not* 4^n . The two bounds worth carrying are

$$\frac{4^n}{2n+1} \leq {}^{2n}C_n < 4^n \quad (n \geq 1),$$

both proved in one line from $\sum_{k=0}^{2n} {}^{2n}C_k = 2^{2n} = 4^n$: the sum has $2n+1$ terms of which ${}^{2n}C_n$ is the largest, giving the lower bound; and all the other terms are strictly positive, giving the strict upper bound.

Scope note. Sharper asymptotics — the Stirling estimate ${}^{2n}C_n \sim 4^n / \sqrt{\pi n}$ — are not part of either the CBSE syllabus or the JEE syllabus, and no JEE question requires them. Do not quote them in a paper. The elementary bounds above are enough for every “estimate the size” or “which is larger” question you will be set, and they are provable from material already in this chapter.

Worked Example G1 — [CBSE Board Level]

Show that the middle term in the expansion of $(1+x)^{2n}$ is $\frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{n!} 2^n x^n$, and verify for $n=3$.

Since $2n$ is even, there is exactly one middle term, the $(n+1)$ th, with $r=n$:

$$T_{n+1} = {}^{2n}C_n x^n = \frac{(2n)!}{n! n!} x^n.$$

Now separate the even factors of $(2n)!$:

$$(2n)! = \underbrace{[1 \cdot 3 \cdot 5 \cdots (2n-1)]}_{n \text{ odd factors}} \cdot \underbrace{[2 \cdot 4 \cdot 6 \cdots 2n]}_{n \text{ even factors}} = [1 \cdot 3 \cdots (2n-1)] \cdot 2^n n!,$$

because pulling a 2 out of each of the n even factors leaves $1 \cdot 2 \cdots n = n!$. Substituting and cancelling one $n!$,

$$T_{n+1} = \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{n!} 2^n x^n. \quad \blacksquare$$

Verification at $n=3$: the formula gives $\frac{1 \cdot 3 \cdot 5}{3!} \cdot 2^3 = \frac{15}{6} \times 8 = 20$, and directly ${}^6C_3 = 20$.

Worked Example G2 — [JEE Main Level]

Prove that the coefficient of x^n in $(1+x)^{2n}$ is twice the coefficient of x^n in $(1+x)^{2n-1}$.

The two coefficients are ${}^{2n}C_n$ and ${}^{2n-1}C_n$. By Pascal's rule,

$${}^{2n}C_n = {}^{2n-1}C_{n-1} + {}^{2n-1}C_n.$$

Now ${}^{2n-1}C_{n-1} = {}^{2n-1}C_{(2n-1)-(n-1)} = {}^{2n-1}C_n$, so the two terms on the right are equal and

$${}^{2n}C_n = 2 {}^{2n-1}C_n. \quad \blacksquare$$

The load-bearing observation is that $(n-1) + n = 2n-1$, which is exactly the symmetry condition — the two middle coefficients of an *odd*-index row are always equal, which is what Subtopic 03 means when it says an odd index gives two tied greatest coefficients.

Worked Example G3 — [JEE Main — Speed Trap]

Find the value of $\frac{{}^{10}C_5}{{}^8C_4}$, and say whether the ratio $\frac{{}^{2n}C_n}{{}^{2n-2}C_{n-1}}$ can ever equal 4.

The trap. “Central binomial coefficients grow like 4^n ” is a slogan students collect, and it makes 4 look like the obvious answer — going from 8C_4 to ${}^{10}C_5$ raises n by one, so surely the ratio is 4. That gives $70 \times 4 = 280$, and the seductive part is that the true value 252 is close enough to make the error survive a rough sanity check.

The fix. Compute the ratio exactly:

$$\frac{{}^{2n}C_n}{{}^{2n-2}C_{n-1}} = \frac{(2n)!}{n!n!} \cdot \frac{(n-1)!(n-1)!}{(2n-2)!} = \frac{2n(2n-1)}{n \cdot n} = \frac{2(2n-1)}{n} = 4 - \frac{2}{n}.$$

At $n = 5$ this is $4 - \frac{2}{5} = \frac{18}{5} = 3.6$, so ${}^{10}C_5 = 3.6 \times 70 = 252$. And since $\frac{2}{n} > 0$ for every n , the ratio is **strictly less than 4 for all n** — it approaches 4 but never reaches it.

Name the trap: reading an asymptotic slogan as an exact identity. The gap is what makes ${}^{2n}C_n$ strictly smaller than 4^n ; if the ratio really were 4 every time, the two would be equal.

Worked Example G4 — [JEE Advanced Level]

Prove that $\frac{4^n}{2n+1} \leq {}^{2n}C_n < 4^n$ for every $n \geq 1$, and use it to bracket ${}^{40}C_{20}$.

Setup. Put $x = 1$ in $(1+x)^{2n}$:

$$\sum_{k=0}^{2n} {}^{2n}C_k = 2^{2n} = 4^n.$$

This is a sum of exactly $2n+1$ positive terms.

Upper bound. ${}^{2n}C_n$ is one of those terms and the remaining $2n$ terms are all strictly positive (each is at least 1, since ${}^{2n}C_0 = 1$), so ${}^{2n}C_n < 4^n$ whenever $n \geq 1$.

Lower bound. By Subtopic 03, ${}^{2n}C_n$ is the *largest* of the $2n + 1$ terms. A sum of $2n + 1$ terms cannot exceed $2n + 1$ copies of its largest term, so $4^n \leq (2n + 1) {}^{2n}C_n$, that is, ${}^{2n}C_n \geq \frac{4^n}{2n + 1}$. ■

Application with $n = 20$. Here $4^{20} = 1099511627776$, so the bounds read

$$\frac{1099511627776}{41} \approx 2.68 \times 10^{10} \leq {}^{40}C_{20} < 1.10 \times 10^{12}.$$

The true value is ${}^{40}C_{20} = 137846528820 \approx 1.38 \times 10^{11}$, comfortably inside. The bracket spans a factor of 41, which is coarse but entirely sufficient for the order-of-magnitude and “which is larger” questions these bounds are set for — and, unlike Stirling’s formula, every step is inside your syllabus.

Practice.

- Write ${}^{10}C_5$ in the form $\frac{2^n [1 \cdot 3 \cdots (2n - 1)]}{n!}$ and confirm the value.
- Evaluate $\frac{{}^{12}C_6}{{}^{10}C_5}$ using the ratio formula, and check against the actual values.
- Find the term independent of x in $\left(x + \frac{1}{x}\right)^{14}$.
- Prove that ${}^{2n}C_n = 2 {}^{2n-1}C_{n-1}$, and state which NCERT result this is.
- Use the bounds of Example G4 to bracket ${}^{30}C_{15}$, and compare with the true value 155117520.
- Using Addendum E, express ${}^{2n}C_n$ as a sum of squares and verify the identity for $n = 4$.

Answers.

- $\frac{2^5 (1 \cdot 3 \cdot 5 \cdot 7 \cdot 9)}{5!} = \frac{32 \times 945}{120} = 252$.
- $4 - \frac{2}{6} = \frac{11}{3}$; check $924/252 = 11/3$.
- ${}^{14}C_7 = 3432$ (the central binomial coefficient with $n = 7$).
- Pascal’s rule plus ${}^{2n-1}C_{n-1} = {}^{2n-1}C_n$; this is NCERT Exercise 8.2 Q11 and Miscellaneous Example 12.
- $4^{15} = 1073741824$, so $34636833 \leq {}^{30}C_{15} < 1073741824$; the true value 155117520 lies inside.
- ${}^{2n}C_n = \sum_{r=0}^n ({}^nC_r)^2$; at $n = 4$, $1 + 16 + 36 + 16 + 1 = 70 = {}^8C_4$.

Sequences and Series

Class 11 Mathematics — Chapter 8 — Complete Master Content

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June 2026

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Unit 1 — Foundations: Sequences and Series

1. Exam Relevance

This is the vocabulary and notation that the entire chapter is built on. It is rarely tested *directly* in JEE, but every later question silently assumes it.

- **CBSE Boards:** Direct questions on writing the first few terms from a general term a_n , or finding a_n from a pattern, are common 1-2 mark items.
- **JEE Main / Advanced:** Almost never asked in isolation, but the language (“general term,” “ $\sum$,” “finite/infinite series”) and the habit of writing a *general term before summing* are prerequisites for AP, GP, AGP, and special-series problems. A student weak here makes silent errors everywhere downstream.

Diagnostic value: if a student cannot move fluently between “the pattern,” “the general term a_n ,” and “the series $\sum a_n$,” every later subtopic will leak errors. This unit is the foundation that makes the rest learnable.

2. Concept Intuition

A **sequence** is just an ordered list of numbers that follows a rule — like house numbers along one side of a street, or the dates of every Sunday this year. Order matters: the 1st term, 2nd term, 3rd term each have a fixed position.

Formally, a sequence is a **function whose inputs are the counting numbers** 1, 2, 3, ... and whose output is the term at that position. Feed in n , get out a_n — the n th term. That is the single most important habit in this chapter: *think of the term as a function of its position*.

A **series** is what you get when you stop listing the terms and start **adding them up**. The sequence 1, 2, 3, 4 becomes the series $1 + 2 + 3 + 4$. Sequences are about *the terms*; series are about *the running total*.

Indian-context anchors:

- **Cricket over-by-over score:** the runs in each over form a sequence; the cumulative score after n overs is the series.
- **A staircase’s step numbers** 1, 2, 3, ... — a sequence; the total number of steps climbed — a series.
- **The Fibonacci pattern** 1, 1, 2, 3, 5, 8, ... (each term the sum of the previous two) appears in flower petals and the spirals of a sunflower — a *recursively defined* sequence, studied in India by Pingala and Hemachandra centuries before Fibonacci.

Figure (placeholder): A function-machine illustration — input box “ $n = 1, 2, 3, 4, \dots$ ” feeding an arrow into a box labelled “rule a_n ”, outputting “ $a_1, a_2, a_3, a_4, \dots$ ”. Below, the same terms shown being summed under a $\sum$ sign to form the series.

3. Key Formulas and Definitions

Sequence. A function $a : \mathbb{N} \rightarrow \mathbb{R}$ (or a subset of $\mathbb{N}$). Its values $a_1, a_2, a_3, \dots$ are the **terms**; a_n is the **general term** (or n th term).

- **Finite sequence:** finitely many terms, $a_1, a_2, \dots, a_m$.
- **Infinite sequence:** unending, $a_1, a_2, a_3, \dots$
- **Two ways to define a sequence:**
 1. **Explicit:** a formula for a_n directly (e.g. $a_n = 2n - 1$ gives 1, 3, 5, 7, ...).
 2. **Recursive:** each term from earlier ones plus seed value(s) (e.g. Fibonacci $a_1 = a_2 = 1, a_n = a_{n-1} + a_{n-2}$).

Series. The sum of the terms of a sequence:

$$a_1 + a_2 + \dots + a_n = \sum_{k=1}^n a_k$$

Finite series (sum of finitely many terms) or infinite series (sum continued without end).

Sigma (summation) notation and its properties:

$$\sum_{k=1}^n (a_k + b_k) = \sum_{k=1}^n a_k + \sum_{k=1}^n b_k, \quad \sum_{k=1}^n c a_k = c \sum_{k=1}^n a_k, \quad \sum_{k=1}^n c = cn.$$

Validity / scope conditions:

1. **The index is a positive integer.** $\sum_{k=1}^n$ runs over $k = 1, 2, \dots, n$; the lower limit need not be 1 (e.g. $\sum_{k=3}^n$), and a partial sum is $\sum_1^n - \sum_1^{m-1}$.
2. **A sequence is not its set of values.** Order and repetition matter: 1, 1, 2 is a valid sequence; as a set it would collapse to $\{1, 2\}$.
3. **“General term” must hold for all n ,** not just the first few. A pattern fitting a_1, a_2, a_3 is a conjecture until the rule is stated.

4. Key Procedures

Procedure 1 — from general term to terms. Substitute $n = 1, 2, 3, \dots$ into a_n .

Example: $a_n = \frac{n}{n+1}$ gives $\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \dots$

Procedure 2 — from a pattern to the general term. Inspect how each term relates to its position n . Linear pattern $\rightarrow a_n = An + B$; differences constant $\rightarrow$ AP-type; ratios constant $\rightarrow$ GP-type; differences-of-differences constant $\rightarrow$ quadratic a_n .

Procedure 3 — expanding sigma notation. $\sum_{k=1}^4 (2k - 1) = 1 + 3 + 5 + 7 = 16$. Read the index range, substitute, add.

Procedure 4 — recursive to explicit (when possible). For simple recursions, unfold a few terms to spot the closed form; for AP/GP recursions, this is the n th-term formula derived in later units.

5. Worked Examples

Ladder: CBSE Board $\rightarrow$ JEE Main $\rightarrow$ JEE Advanced. (Standard curriculum problems.)

Example 1 — CBSE Board. Write the first four terms of the sequence $a_n = \frac{2n - 3}{4}$.

Solution. $a_1 = \frac{-1}{4}$, $a_2 = \frac{1}{4}$, $a_3 = \frac{3}{4}$, $a_4 = \frac{5}{4}$. (Note the terms themselves form an AP with $d = \frac{1}{2}$ — a preview of Unit 2.)

Example 2 — CBSE Board. Find the general term of 3, 8, 15, 24, 35, ...

Solution. First differences 5, 7, 9, 11 form an AP, so a_n is quadratic. Observe $a_n = n^2 + 2n = n(n + 2)$ (check: $1 \cdot 3 = 3$, $2 \cdot 4 = 8$, $3 \cdot 5 = 15$ ✓). So $a_n = n(n + 2)$.

Example 3 — JEE Main. Write the first five terms of the recursive sequence $a_1 = 2$, $a_n = 3a_{n-1} + 1$.

Solution. $a_1 = 2$, $a_2 = 3(2) + 1 = 7$, $a_3 = 3(7) + 1 = 22$, $a_4 = 67$, $a_5 = 202$. Recursive sequences need not be AP or GP — here neither differences nor ratios are constant.

Example 4 — JEE Advanced. Evaluate $\sum_{k=1}^n (3k^2 - 2k + 1)$ using sigma linearity (formulas from Unit 7).

Solution. By linearity, $3 \sum k^2 - 2 \sum k + \sum 1 = 3 \cdot \frac{n(n+1)(2n+1)}{6} - 2 \cdot \frac{n(n+1)}{2} + n$.

Simplifying: $\frac{n(n+1)(2n+1)}{2} - n(n+1) + n = \boxed{n^3 + \frac{1}{2}n^2 + \frac{1}{2}n}$ (i.e. $\frac{n(2n^2+n+1)}{2}$). This is the foundational habit — *split the sum, apply known sums* — formalised in Unit 7.

6. Practice Exercises

1. **(Board)** Write the first three terms of $a_n = (-1)^{n-1} 5^{n+1}$.
2. **(Board)** Find the general term of 1, 4, 9, 16, 25, ...
3. **(Main)** A sequence is defined by $a_1 = 3$, $a_{n+1} = 2a_n - 1$. Find a_4 .
4. **(Main)** Expand and evaluate $\sum_{k=2}^5 (k^2 - 1)$.

5. **(Adv)** If $a_n = \frac{n^2 - 1}{n^2 + 1}$, find the value approached by a_n as n grows large, and explain in terms of the sequence-as-function idea.

Answers: 1. 25, -125, 625 | 2. $a_n = n^2$ | 3. $a_4 = 17$ | 4. $3 + 8 + 15 + 24 = 50$ | 5. approaches 1 (numerator and denominator both grow like n^2).

7. MCQ with Distractor Analysis

Q1. The general term of 2, 5, 10, 17, 26, ... is: (A) $n^2 + 1$ (B) $2n + 1$ (C) $n^2 + n$ (D) $3n - 1$

Correct: (A). Terms are $1 + 1, 4 + 1, 9 + 1, \dots = n^2 + 1$. **(B),(D)** are linear (would give constant differences; here differences 3, 5, 7, 9 grow). **(C)** gives 2, 6, 12, ... — fails at $n = 2$. *Tests differences-of-differences reasoning.*

Q2. Which is the correct expansion of $\sum_{k=1}^3 k(k+1)$? (A) $2 + 6 + 12 = 20$ (B) $1 + 2 + 3 = 6$ (C) $2 + 6 + 12 + 20 = 40$ (D) $1 + 4 + 9 = 14$

Correct: (A). $1 \cdot 2 + 2 \cdot 3 + 3 \cdot 4 = 20$. **(B)** sums only k . **(C)** runs to $k = 4$. **(D)** sums k^2 . *Tests reading the index range and the summand exactly.*

Q3. A sequence defined by $a_n = a_{n-1} + a_{n-2}$ with $a_1 = a_2 = 1$ is: (A) an AP (B) a GP (C) the Fibonacci sequence (D) constant

Correct: (C). **(A),(B)** fail — neither differences nor ratios are constant. *Recognises a recursively-defined sequence.*

8. Common Pitfalls, Pro-Tips & Cheat Sheet

Common Pitfalls (candidate mistake-tags):

- **seq-term-vs-series:** Confusing the n th term a_n with the sum S_n — the root error behind countless AP/GP slips later.
- **seq-pattern-overfit:** Declaring a general term from 3 terms without verifying a 4th (a pattern is a conjecture until the rule holds for all n).
- **seq-index-range:** Misreading the limits of $\sum$ (starting at 1 when it starts at 2, etc.).
- **seq-set-vs-sequence:** Treating a sequence as an unordered set, losing order/repetition.

Schema note: seq-term-vs-series is the *foundational* tag — many downstream AP/GP/special-series errors trace back to it. Worth flagging as a high-level parent in the knowledge-tracing graph.

Pro-Tips:

- Always **write the general term a_n first**, then decide what to do with it (list, sum, test for AP/GP).

- To find a_n from a pattern: check **differences** (constant $\Rightarrow$ linear), **second differences** (constant $\Rightarrow$ quadratic), **ratios** (constant $\Rightarrow$ exponential/GP).
- Sigma is just shorthand for addition — expand it whenever you're unsure.

Cheat Sheet Box — Foundations

Idea	Statement
Sequence	function $a : \mathbb{N} \rightarrow \mathbb{R}$; terms $a_1, a_2, \dots$; general term a_n
Finite vs infinite	finitely many terms vs unending
Explicit definition	formula for a_n (e.g. $a_n = 2n - 1$)
Recursive definition	each term from earlier ones + seeds (e.g. Fibonacci)
Series	$\sum_{k=1}^n a_k = a_1 + \dots + a_n$
Sigma linearity	$\sum (a_k + b_k) = \sum a_k + \sum b_k$; $\sum c a_k = c \sum a_k$; $\sum c = cn$
Partial range	$\sum_m^n = \sum_1^n - \sum_1^{m-1}$

Golden habit: position $\rightarrow$ general term $a_n \rightarrow$ then list or sum.

Unit 2 — Arithmetic Progression (AP)

1. Exam Relevance

Arithmetic progressions are the **foundational unit of the entire Sequences and Series chapter** — every later idea (GP, special series $\sum n$, $\sum n^2$, $\sum n^3$, AGP, telescoping) leans on the counting and summation reflexes you build here.

- **CBSE Boards:** AP is almost guaranteed to appear. Expect one short-answer question (find a term, find n , find the sum) worth 2–3 marks and frequently one long-answer “word problem” (savings, seating, production) worth 5 marks. The derivation of S_n is a named board derivation and is directly askable.
- **JEE Main:** Typically **1–2 questions per paper**, blending AP with GP/HP (e.g. “if a, b, c are in AP and a^2, b^2, c^2 are in HP”), ratio-of-sums setups, and the $S_n = An^2 + Bn$ characterization.
- **JEE Advanced:** AP rarely stands alone — it appears fused with logarithms, trigonometry, coordinate geometry, or as the scaffolding of a multi-concept problem (e.g. *JEE Adv 2019*: intersection of three APs as sets). Mastery is assumed, not tested in isolation.

Why it’s a high-leverage diagnostic topic: AP errors are almost always *procedural slips* on a tiny formula set, not conceptual gaps. That makes it the cleanest place in the chapter to separate a student who “knows the formula” from one who can *choose the right form of the formula* under exam pressure.

2. Concept Intuition

An arithmetic progression is the mathematics of **adding the same step, over and over**.

Picture a **staircase where every step is exactly the same height**. Start at floor level a . Each step lifts you by the same amount d . After 1 step you’re at $a + d$; after 2 steps, $a + 2d$; after $(n - 1)$ steps, $a + (n - 1)d$. That last expression is the n th-term formula — you’ve already derived it just by climbing.

The single defining property: **the gap between consecutive terms never changes**. That constant gap is the *common difference* d .

$$d = a_2 - a_1 = a_3 - a_2 = \cdots = a_k - a_{k-1}$$

A few everyday Indian-context anchors that *are* APs:

- **EMI-style fixed savings:** You save ₹500 in month 1, then ₹50 more each month — ₹550, ₹600, ₹650, ... The monthly amounts form an AP with $a = 500$, $d = 50$. (This is literally an old AIEEE problem — see Worked Example 7.)

- **Cinema / stadium seating:** Row 1 has 20 seats, each row behind has 2 more. Rows form an AP; total seating is the *sum* of the AP.
- **Auto-rickshaw fare:** ₹25 for the first kilometre, then a fixed ₹15 per additional km. Fare-after- n -km is an AP.
- **Stacked logs / water-tank steps:** The classic “balls arranged in rows” Olympiad-flavoured problem (chunk 42).

The deep idea worth installing early: **an AP is “linear.”** Plot a_n against n and you get points on a straight line of slope d . This single mental image explains almost every property below — equidistant terms averaging out, the sum being a quadratic, three-term symmetry — because all of them are just facts about straight lines in disguise.

Figure (placeholder): A staircase with equal-height steps; floor labelled a , each riser labelled d , step tops labelled $a, a + d, a + 2d, a + 3d$. Beside it, a scatter plot of a_n vs n showing collinear points on a line of slope d .

3. Key Formulas and Definitions

Notation is **locked chapter-wide** (carried into Unit 3 onward):

Symbol	Meaning
a	first term (a_1)
d	common difference
a_n	n th term (general term); JEE texts also write t_n or T_n
S_n	sum of the first n terms
l	last term
$A, A_1, A_2, \dots, A_m$	arithmetic mean(s) inserted between two numbers

Definition. A sequence $a_1, a_2, a_3, \dots$ is an AP if $a_k - a_{k-1} = d$ (a constant) for all $k \geq 2$.

Core formulas:

$$a_n = a + (n - 1)d \quad (\text{nth term from the beginning})$$

$$a'_n = l - (n - 1)d \quad (\text{nth term from the end})$$

$$S_n = \frac{n}{2} [2a + (n - 1)d] = \frac{n}{2} (a + l)$$

Arithmetic Mean (AM). If a, b, c are in AP, then b is the AM of a and c :

$$b = \frac{a + c}{2}$$

Inserting m AMs between a and b . If $A_1, \dots, A_m$ are m AMs between a and b , the common difference of the extended AP is $d = \frac{b - a}{m + 1}$, and

$$A_k = a + k \cdot \frac{b - a}{m + 1}, \quad \sum_{k=1}^m A_k = m \left(\frac{a + b}{2} \right)$$

(the sum of m AMs equals m times the single AM of the endpoints).

Characterization (powerful, JEE-favourite):

$$S_n = An^2 + Bn \iff \text{the sequence is an AP}$$

with first term $A + B$ and common difference $2A$. Equivalently: a sequence is an AP **iff** a_n is linear in n (i.e. $a_n = An + B$, where the coefficient A is the common difference).

Validity / scope conditions (the AP analogue of dimensional checks — *always state these in solutions*):

1. $n \in \mathbb{N}$. Every formula above assumes a positive-integer term count. A non-integer or negative n from algebra signals an error or a “no such term” answer.
2. d **may be any real (or even complex/imaginary) constant** — positive (increasing AP), zero (constant AP), or negative (decreasing AP). Do not assume $d > 0$.
3. **The “ $S_n = An^2 + Bn$ ” test requires the constant term to be zero.** If $S_n = An^2 + Bn + c$ with $c \neq 0$, the sequence is *not* an AP (only the terms from the 2nd onward are). This is the single most common trap on this formula.
4. **Three-term symmetry** (a_1, a_2, a_3 in AP $\Leftrightarrow 2a_2 = a_1 + a_3$) holds for *any* three consecutive terms — a clean test that needs no knowledge of a or d .

4. Theorem Proofs and Key Derivations

These are the board-derivable results. Reproduce them line-by-line in exams.

Derivation 1 — n th term, $a_n = a + (n - 1)d$. By definition each term exceeds its predecessor by d :

$$a_2 = a_1 + d, \quad a_3 = a_2 + d, \quad \dots, \quad a_n = a_{n-1} + d.$$

Adding these $(n - 1)$ equations, all intermediate terms cancel telescopically:

$$a_n = a_1 + (n - 1)d = a + (n - 1)d. \quad \blacksquare$$

Derivation 2 — sum of n terms, $S_n = \frac{n}{2}[2a + (n-1)d]$ (the “Gauss reversal”). This is the trick the 10-year-old Gauss reputedly used to sum 1 to 100 in seconds. Write the sum forwards and backwards:

$$S_n = a + (a + d) + \cdots + (a + (n-1)d)$$

$$S_n = (a + (n-1)d) + \cdots + (a + d) + a$$

Add column by column. **Each of the n columns sums to the same value $2a + (n-1)d$** (the first-plus-last identity):

$$2S_n = n[2a + (n-1)d] \implies S_n = \frac{n}{2}[2a + (n-1)d].$$

Since $a + (n-1)d = l$, this is also $S_n = \frac{n}{2}(a + l)$. ■

Derivation 3 — the characterization $S_n = An^2 + Bn \Leftrightarrow \text{AP}$. ($\Rightarrow$) From $S_n = \frac{n}{2}[2a + (n-1)d] = \frac{d}{2}n^2 + (a - \frac{d}{2})n$, which has the form $An^2 + Bn$ with $A = \frac{d}{2}$, $B = a - \frac{d}{2}$, and zero constant term. ($\Rightarrow$) Given $S_n = An^2 + Bn$, recover the n th term via $a_n = S_n - S_{n-1}$:

$$a_n = (An^2 + Bn) - (A(n-1)^2 + B(n-1)) = A(2n-1) + B.$$

Then $a_n - a_{n-1} = 2A$, a constant — so it's an AP with common difference $2A$ and first term $a_1 = A + B$. ■

Key Procedure — symmetric selection of terms. When a problem gives the *sum* of terms in an AP, choose terms symmetrically about the middle so the d 's cancel:

- 3 terms: $a - d, a, a + d$ (sum = $3a$)
- 4 terms: $a - 3d, a - d, a + d, a + 3d$ (sum = $4a$, common difference $2d$)
- 5 terms: $a - 2d, a - d, a, a + d, a + 2d$ (sum = $5a$)

This converts a two-unknown system into a one-unknown system instantly. (Note the *even* case uses common difference $2d$ — a frequent slip.)

5. Worked Examples

Difficulty ladder: CBSE Board $\rightarrow$ JEE Main $\rightarrow$ JEE Advanced.

Example 1 — CBSE Board. Find the 28th term and the first negative term of the AP $20, 19\frac{1}{4}, 18\frac{1}{2}, 17\frac{3}{4}, \dots$

Solution. Here $a = 20$, $d = 19\frac{1}{4} - 20 = -\frac{3}{4}$.

$$a_{28} = 20 + 27\left(-\frac{3}{4}\right) = 20 - \frac{81}{4} = -\frac{1}{4}.$$

First negative term needs $a_n < 0$:

$$20 + (n - 1) \left(-\frac{3}{4}\right) < 0 \Rightarrow 80 - 3(n - 1) < 0 \Rightarrow n > 27\frac{2}{3}.$$

Smallest integer: $n = 28$. So the **28th term is the first negative term**, equal to $-\frac{1}{4}$. (Note $n \in \mathbb{N}$ forced us to round up.)

Example 2 — CBSE Board. The sum of the first 24 terms of an AP is needed, given $a_1 + a_5 + a_{10} + a_{15} + a_{20} + a_{24} = 225$.

Solution. Use the equidistant property $a_1 + a_{24} = a_5 + a_{20} = a_{10} + a_{15}$. The six given terms pair into three equal sums:

$$3(a_1 + a_{24}) = 225 \Rightarrow a_1 + a_{24} = 75.$$

Then $S_{24} = \frac{24}{2}(a_1 + a_{24}) = 12 \times 75 = \boxed{900}$. No need for a or d individually — the linearity of an AP does the work.

Example 3 — JEE Main. The sums of n terms of two APs are in the ratio $(7n + 1) : (4n + 17)$. Find the ratio of their 24th terms.

Solution. For ratio-of-sums problems, the standard device: the ratio of the **m th terms** equals the ratio of S_n evaluated at $n = 2m - 1$.

$$\frac{a_m^{(1)}}{a_m^{(2)}} = \frac{S_{2m-1}^{(1)}}{S_{2m-1}^{(2)}}.$$

For the 24th terms, $m = 24 \Rightarrow n = 2(24) - 1 = 47$:

$$\frac{a_{24}^{(1)}}{a_{24}^{(2)}} = \frac{7(47) + 1}{4(47) + 17} = \frac{330}{205} = \boxed{\frac{66}{41}}.$$

Why $n = 2m - 1$? Because $\frac{a_m}{a'_m} = \frac{a + (m - 1)d}{a' + (m - 1)d'}$, and $\frac{S_{2m-1}}{S'_{2m-1}} = \frac{\frac{2m-1}{2}[2a + (2m - 2)d]}{\frac{2m-1}{2}[2a' + (2m - 2)d']} = \frac{a + (m - 1)d}{a' + (m - 1)d'}$ — identical.

Example 4 — JEE Main. Insert 11 AMs between 28 and 10, and find the three middle inserted means.

Solution. With $m = 11$ means, the extended AP 28, $A_1, \dots, A_{11}$, 10 has 12 gaps:

$$d = \frac{10 - 28}{11 + 1} = \frac{-18}{12} = -\frac{3}{2}.$$

Eleven means is odd, so the three middle ones are A_5, A_6, A_7 :

$$A_5 = 28 + 5d = 28 - \frac{15}{2} = \frac{41}{2}, \quad A_6 = 28 + 6d = 19, \quad A_7 = 28 + 7d = \frac{35}{2}.$$

Answer: $\boxed{\frac{41}{2}, 19, \frac{35}{2}}$ (and note $A_6 = 19$ is itself the AM of 28 and 10, as it must be — it's the centre of a symmetric AP).

Example 5 — JEE Main (AIEEE-style word problem). A man saves ₹200 in each of the first three months of service. In each subsequent month his saving increases by ₹40 over the previous month. After how many months will his total savings be ₹11040?

Solution. Savings from month 3 onward form an AP. Total = first three months + AP sum. Take the savings as ₹200, ₹200, ₹200, ₹240, ₹280, ... The cleanest model: from month 1 treat it as ₹200 for months 1-3, then arithmetic. Summing all n months and solving $S_n = 11040$ yields a quadratic in n ; the positive integer root is

$$\boxed{n = 21 \text{ months}}.$$

(Pipeline note: this is the ₹-savings archetype — perfect for the “translate-a-word-problem-into-a, d” mistake-tag.)

Example 6 — JEE Advanced (AP + logarithms). If $\log_{10} 2, \log_{10}(2^x - 1), \log_{10}(2^x + 3)$ are in AP, find x .

Solution. Three terms in AP $\Rightarrow$ middle is the AM:

$$\begin{aligned} 2\log_{10}(2^x - 1) &= \log_{10} 2 + \log_{10}(2^x + 3) \\ \Rightarrow (2^x - 1)^2 &= 2(2^x + 3). \end{aligned}$$

Let $u = 2^x$: $u^2 - 2u + 1 = 2u + 6 \Rightarrow u^2 - 4u - 5 = 0 \Rightarrow (u - 5)(u + 1) = 0$. Since $u = 2^x > 0$, reject $u = -1$. So $2^x = 5 \Rightarrow \boxed{x = \log_2 5}$. The validity condition ($2^x > 0$) is what kills the extraneous root — never skip it.

Example 7 — JEE Advanced. Suppose all terms of an AP are natural numbers. The ratio of the sum of the first seven terms to the sum of the first eleven terms is 6 : 11, and the seventh term lies between 130 and 140. Find the common difference. (JEE Main 2015)

Solution.

$$\frac{S_7}{S_{11}} = \frac{\frac{7}{2}(2a + 6d)}{\frac{11}{2}(2a + 10d)} = \frac{7(a + 3d)}{11(a + 5d)} = \frac{6}{11}.$$

The 11's cancel against the 11 in the ratio: $7(a + 3d) = 6(a + 5d) \Rightarrow a = 9d$. Seventh term $a_7 = a + 6d = 9d + 6d = 15d$. Given $130 < 15d < 140$:

$$8.67 < d < 9.33 \Rightarrow d = 9 \quad (\text{integer, since terms are natural}).$$

Answer: $d = 9$. The natural-number constraint is doing essential work — without it d wouldn't be pinned down.

Example 8 — JEE Advanced (sets / multi-AP). Let $AP(a; d)$ denote the set of all terms of the infinite AP with first term a and common difference $d > 0$. If

$$AP(1; 3) \cap AP(2; 5) \cap AP(3; 7) = AP(a; d),$$

find $a + d$. (JEE Adv 2019)

Solution. A number lies in all three sets iff it satisfies three congruences:

$$x \equiv 1 \pmod{3}, \quad x \equiv 2 \pmod{5}, \quad x \equiv 3 \pmod{7}.$$

By CRT the common solutions themselves form an AP with common difference $d = \text{lcm}(3, 5, 7) = 105$. The smallest positive solution: checking $x = 52$ gives $52 = 3(17) + 1 \checkmark$, $52 = 5(10) + 2 \checkmark$, $52 = 7(7) + 3 \checkmark$. So $a = 52$, $d = 105$, and $a + d = 157$. This is the AP-as-residue-class viewpoint that Advanced loves.

6. Practice Exercises

(Answers below; full worked solutions available on request.)

1. **(Board)** Find the sum of all three-digit natural numbers divisible by 7.
2. **(Board)** The 17th term of an AP exceeds its 10th term by 7. Find the common difference.
3. **(Board)** How many terms of the AP 9, 17, 25, ... must be taken to give a sum of 636?
4. **(Main)** If the p th, q th, r th terms of an AP are a, b, c respectively, find the value of $a(q - r) + b(r - p) + c(p - q)$.
5. **(Main)** In an AP, 5 times the 5th term equals 8 times the 8th term. Find its 13th term.
6. **(Main)** If $S_n = cn^2$ is the sum of the first n terms of an AP, find the sum of the squares of these n terms (in terms of c, n).
7. **(Adv)** If a, b, c are in AP, prove that $a^2(b+c) + b^2(c+a) + c^2(a+b) = \frac{2}{9}(a+b+c)^3$.
8. **(Adv)** The first term of an AP is unity and the common difference is $d \neq 0$. For what value of d is $a_1a_3 + a_2a_3$ minimum?

Answers:

1. 70336 | 2. $d = 1$ | 3. $n = 12$ | 4. 0 | 5. $a_{13} = 0$ | 6. $\frac{n(4n^2 - 1)c^2}{3}$ | 7. (identity — verify via $b = \frac{a+c}{2}$) | 8. $d = -\frac{2}{3}$ region; minimise the quadratic in d .
-

7. MCQ with Distractor Analysis

Q1. If $S_n = 3n^2 + 5n$ is the sum of n terms of an AP, its common difference is: (A) 3 (B) 5 (C) 6 (D) 8

Correct: (C). $d = 2A = 2(3) = 6$. **(A)** picks the coefficient of n^2 raw — forgetting the factor of 2. **(B)** picks the coefficient of n (that gives B , related to the first term, not d). **(D)** computes $a_1 = A + B = 8$ and mislabels it as d . *This single question discriminates students who memorised “ $S_n = An^2 + Bn$ ” from those who know what each constant means.*

Q2. If the 9th term of an AP is zero, the ratio of its 29th term to its 19th term is: (A) 1 : 2 (B) 2 : 1 (C) 1 : 3 (D) 3 : 1

Correct: (B). $a_9 = a + 8d = 0 \Rightarrow a = -8d$. Then $a_{29} = a + 28d = 20d$, $a_{19} = a + 18d = 10d$, ratio = 2 : 1. **(A)** inverts the ratio (a sign/order slip). **(C), (D)** arise from arithmetic errors in computing a from $a_9 = 0$ (e.g. using $a + 9d = 0$).

Q3. Three numbers in AP have sum -3 and product 8. The sum of their squares is: (A) 9 (B) 10 (C) 12 (D) 21

Correct: (D). Take terms $a - d, a, a + d$. Sum = $3a = -3 \Rightarrow a = -1$. Product $(a^2 - d^2)a = 8 \Rightarrow (1 - d^2)(-1) = 8 \Rightarrow d^2 = 9$. Sum of squares = $(a - d)^2 + a^2 + (a + d)^2 = 3a^2 + 2d^2 = 3 + 18 = 21$. **(A)** is $3a^2 + 2d^2$ computed with d^2 omitted (= 3, then mis-added). **(C)** uses the wrong symmetric set (four-term spacing $2d$). **(B)** is a near-miss arithmetic trap. *The signature trap here is forgetting the -1 from the product equation's sign.*

Q4. If 100 times the 100th term of an AP (with $d \neq 0$) equals 50 times its 50th term, then its 150th term is: (A) 150 (B) -150 (C) 0 (D) $150 \times a_{50}$

Correct: (C). $100(a + 99d) = 50(a + 49d) \Rightarrow 2a + 198d = a + 49d \Rightarrow a = -149d$. Then $a_{150} = a + 149d = 0$. **(A), (B)** are sign/value lures for students who solve a correctly but mis-substitute. **(D)** tempts pattern-matchers who recall “term equals multiple of another term” results from similar problems. (AIEEE 2012.)

8. Common Pitfalls, Pro-Tips & Cheat Sheet

Common Pitfalls (candidate mistake-tags):

- **ap-nth-term-off-by-one:** Using $a + nd$ instead of $a + (n - 1)d$. The single most frequent AP error. *Always say “n minus one.”*

- **ap-sum-const-term:** Treating $S_n = An^2 + Bn + c$ (with $c \neq 0$) as an AP. It isn't — only terms from the 2nd onward are.
- **ap-even-symmetric-spacing:** Using common difference d (not $2d$) when picking an *even* count of symmetric terms ($a - 3d, a - d, a + d, a + 3d$).
- **ap-sign-of-d:** Assuming $d > 0$ and dropping the decreasing-AP / negative-term cases (kills “first negative term” and inequality problems).
- **ap-n-not-natural:** Accepting a non-integer or negative n as a valid term count instead of reading it as “no such term.”
- **ap-extraneous-root:** In AP-with-logs/quadratics problems, forgetting the domain condition (e.g. $2^x > 0$) that rejects an extraneous root.
- **ap-ratio-of-sums-index:** Forgetting the $n = 2m - 1$ substitution and instead trying to recover a, d separately.

Cross-subtopic tag note (for knowledge-tracing graph): ap-sign-of-d and the GP analogue gp-sign-of-r should share a parent tag (e.g. progression-overlooked-sign-case) — it's the *same physics-of-the-mistake* surfacing in two subtopics, exactly the cross-cutting pattern the schema wants unified.

Pro-Tips:

- When a problem gives a **sum**, switch to symmetric terms ($a - d, a, a + d$) — it halves the unknowns.
- When it gives **specific terms**, stay with $a + (n - 1)d$.
- The **equidistant identity** ($a_1 + a_n = a_2 + a_{n-1} = \dots$) collapses many “sum of scattered terms” problems in one line — reach for it before grinding algebra.
- To *test* whether a sequence is an AP fast: check if a_n is **linear in n** , or S_n is **quadratic with zero constant**.

Cheat Sheet Box — Arithmetic Progression

Quantity	Formula	Condition
nth term	$a_n = a + (n - 1)d$	$n \in \mathbb{N}$
nth term from end	$a'_n = l - (n - 1)d$	$n \in \mathbb{N}$
Sum of n terms	$S_n = \frac{n}{2}[2a + (n - 1)d] = \frac{n}{2}(a + l)$	—
AM of a, c	$b = \frac{a+c}{2}$	a, b, c in AP
m AMs between a, b	$A_k = a + k \frac{b-a}{m+1};$ $\sum A_k = m \frac{a+b}{2}$	—
AP test (term)	$a_n = An + B$ linear $\Rightarrow d = A$	—
AP test (sum)	$S_n = An^2 + Bn \Rightarrow$ $d = 2A, a_1 = A + B$	zero constant term
Symmetric pick (odd)	$a - d, a, a + d$	sum = $3a$

Quantity	Formula	Condition
Symmetric pick (even)	$a - 3d, a - d, a + d, a + 3d$	$CD = 2d$
Equidistant identity	$a_1 + a_n = a_2 + a_{n-1} = \dots$	finite AP

Three-term AP test: $2a_2 = a_1 + a_3$. **Ratio of mth terms** = S_{2m-1} ratio.

Unit 3 — Geometric Progression (GP)

1. Exam Relevance

A geometric progression is the multiplicative twin of the AP: where an AP *adds* a fixed step, a GP *multiplies* by a fixed factor. It is one of the two most heavily tested ideas in the chapter.

- **CBSE Boards:** Reliably one short-answer question (nth term, sum, GM) and frequently a 5-mark application — compound-growth, repaying-loan, or bacterial-doubling word problems. The S_n derivation and the sum-to-infinity formula are both named, directly askable results.
- **JEE Main:** Typically **1-2 questions**, often fusing GP with AP/HP, recurring decimals, or the $b^2 = ac$ GM condition. Sum-to-infinity and the AM-GM inequality are perennial.
- **JEE Advanced:** GP underpins a large fraction of “three-numbers-in-progression” problems, AM-GM optimisation, infinite-geometric-figure problems (areas of nested squares/triangles), and recurring-decimal-to-fraction conversions. As with AP, it rarely stands alone — it’s load-bearing scaffolding.

Why it’s high-leverage diagnostically: GP errors cluster around three precise decision points — the **sign of r** , the **convergence condition** $|r| < 1$ for S_∞ , and the **$\pm$ ambiguity of the GM**. Each is a clean, single-decision discriminator between rote recall and genuine understanding.

2. Concept Intuition

A geometric progression is the mathematics of **repeated multiplication** — of things that *scale* rather than *step*.

The classic Indian anchor is the **chessboard and the grains of wheat**. Legend says the inventor of chess, Sissa Ben Dhair, asked King Shirham for one grain on the first square, two on the second, four on the third — doubling each time. The king laughed at so modest a request. He shouldn’t have: the squares hold $1, 2, 4, 8, \dots, 2^{63}$ grains — a GP with $a = 1, r = 2$ — and the total $(2^{64} - 1)$ exceeds all the wheat that has ever existed. **Geometric growth feels small at the start and detonates at the end.** That single intuition explains compound interest, viral spread, and Moore’s law.

The defining property: **the ratio between consecutive terms never changes.**

$$r = \frac{a_2}{a_1} = \frac{a_3}{a_2} = \dots = \frac{a_k}{a_{k-1}}$$

Everyday Indian-context GPs:

- **Compound interest / SIP growth:** ₹10,000 growing at 10% per year $\rightarrow$ ₹11,000, ₹12,100, ₹13,310, ... a GP with $r = 1.1$.

- **Depreciation:** A bike worth ₹80,000 losing 20% of its *current* value yearly $\rightarrow$ ₹64,000, ₹51,200, ... a GP with $r = 0.8$ (a *shrinking* GP, $0 < r < 1$).
- **Nested squares:** Join the midpoints of a square's sides to get a new square of half the area; repeat forever. The areas form a GP with $r = \frac{1}{2}$ — and their *infinite* sum is finite (Worked Example 5).

The deep idea, mirroring AP's "linearity": **a GP is "exponential."** Plot a_n against n and you get points on an exponential curve $a \cdot r^{n-1}$; plot $\log a_n$ and you get a *straight line* — which is exactly why **logs convert a GP into an AP** (a property we'll use repeatedly). Two contrasts to install early:

- $r > 1$: explosive growth. $0 < r < 1$: decay toward zero. $r < 0$: terms **alternate in sign** — neither increasing nor decreasing.
- An infinite AP always diverges; an infinite GP **converges** precisely when $|r| < 1$. This is the single most important structural difference between the two.

Figure (placeholder): Left — a chessboard with the first few squares labelled 1, 2, 4, 8, 16 grains, growing visibly. Right — two overlaid plots vs n : an exponential curve ar^{n-1} for $r > 1$, and a decaying curve for $0 < r < 1$ approaching a horizontal asymptote; inset shows $\log a_n$ as a straight line.

3. Key Formulas and Definitions

Notation (chained from Unit 2; new GP symbols added):

Symbol	Meaning
a	first term ($a \neq 0$)
r	common ratio ($r \neq 0$)
a_n	n th term; JEE texts also write t_n, T_n
S_n	sum of first n terms
S_∞	sum to infinity (only when $ r < 1$)
l	last term
$G, G_1, \dots, G_n$	geometric mean(s) inserted between two numbers

Definition. A sequence $a_1, a_2, \dots$ of **non-zero** terms is a GP if $\frac{a_k}{a_{k-1}} = r$ (a constant) for all $k \geq 2$.

Core formulas:

$$a_n = a r^{n-1}$$

(n th term)

$$a'_n = \frac{l}{r^{n-1}}$$

(n th term from the end)

$$S_n = \frac{a(r^n - 1)}{r - 1} = \frac{a(1 - r^n)}{1 - r} \quad (r \neq 1), \quad S_n = na \quad (r = 1)$$

$$S_\infty = \frac{a}{1 - r} \quad \text{valid only when } |r| < 1$$

Geometric Mean (GM). If a, b, c are in GP, then b is the GM of a and c :

$$b^2 = ac \iff b = \pm\sqrt{ac}$$

For two positive numbers, the single GM is $G = \sqrt{ab}$.

Inserting n GMs between a and b . If $G_1, \dots, G_n$ are n GMs between a and b , the common ratio of the extended GP is $r = \left(\frac{b}{a}\right)^{1/(n+1)}$, and

$$G_k = a \left(\frac{b}{a}\right)^{k/(n+1)}, \quad \prod_{k=1}^n G_k = (\sqrt{ab})^n = (\text{single GM})^n.$$

AM-GM inequality (carried into every optimisation problem). For positive reals $a_1, \dots, a_n$,

$$\frac{a_1 + \dots + a_n}{n} \geq (a_1 a_2 \dots a_n)^{1/n}, \quad \text{equality iff all } a_i \text{ are equal.}$$

For two positive numbers: $\frac{a+b}{2} \geq \sqrt{ab}$, i.e. $\text{AM} \geq \text{GM}$, with the further chain $\text{AM} \geq \text{GM} \geq \text{HM}$ and the identity $\text{AM} \times \text{HM} = \text{GM}^2$.

Validity / scope conditions (the GP analogue of dimensional checks — *state these explicitly*):

1. **No term is ever zero**, and $r \neq 0$. A GP cannot contain 0; if algebra forces a term to 0, the assumption was wrong.
2. **Sign of r matters.** $r > 0 \rightarrow$ all terms share the first term's sign (monotone). $r < 0 \rightarrow$ terms **alternate** in sign (neither increasing nor decreasing). Never silently assume $r > 0$.
3. S_∞ **exists iff** $|r| < 1$. For $|r| \geq 1$ the infinite sum diverges (or oscillates at $r = -1$). Writing $\frac{a}{1-r}$ without checking $|r| < 1$ is the chapter's most punished error.
4. **GM sign.** $b = \pm\sqrt{ac}$ — choose the sign that keeps the three terms a genuine GP. For real GMs you also need $ac > 0$ (same-sign endpoints).

4. Theorem Proofs and Key Derivations

Derivation 1 — nth term, $a_n = ar^{n-1}$. Each term is the previous one times r : $a_2 = ar$, $a_3 = a_2r = ar^2$, ... Telescoping the multiplications, $a_n = ar^{n-1}$. ■

Derivation 2 — sum of n terms (the “multiply-by- r and subtract” trick).

$$S_n = a + ar + ar^2 + \dots + ar^{n-1} \quad (i)$$

Multiply through by r :

$$rS_n = ar + ar^2 + \dots + ar^{n-1} + ar^n \quad (ii)$$

Subtract (ii) from (i). **Every interior term cancels**, leaving only the first term of (i) and the last of (ii):

$$S_n - rS_n = a - ar^n \implies S_n(1 - r) = a(1 - r^n) \implies S_n = \frac{a(1 - r^n)}{1 - r} \quad (r \neq 1). \quad \blacksquare$$

Derivation 3 — sum to infinity, $S_\infty = \frac{a}{1 - r}$ for $|r| < 1$. Start from $S_n = \frac{a}{1 - r} - \frac{ar^n}{1 - r}$. When $|r| < 1$, the term $r^n \rightarrow 0$ as $n \rightarrow \infty$ (each multiplication shrinks it). Therefore

$$S_\infty = \lim_{n \rightarrow \infty} S_n = \frac{a}{1 - r}. \quad \blacksquare$$

(If $|r| \geq 1$, r^n does not vanish, so the limit fails to exist — this is the proof of the convergence condition, not a side note.)

Derivation 4 — product of n GMs equals (single GM) ^{n} . With $G_k = ar^k$ and $r = (b/a)^{1/(n+1)}$,

$$\prod_{k=1}^n G_k = a^n r^{1+2+\dots+n} = a^n r^{n(n+1)/2} = a^n \left(\frac{b}{a}\right)^{\frac{1}{n+1} \cdot \frac{n(n+1)}{2}} = a^n \left(\frac{b}{a}\right)^{n/2} = (ab)^{n/2} = (\sqrt{ab})^n. \quad \blacksquare$$

Derivation 5 — AM $\geq$ GM for two positives. For $a, b > 0$, $(\sqrt{a} - \sqrt{b})^2 \geq 0 \implies a + b - 2\sqrt{ab} \geq 0 \implies \frac{a+b}{2} \geq \sqrt{ab}$, with equality iff $\sqrt{a} = \sqrt{b}$, i.e. $a = b$. ■

Key Procedure — symmetric selection. When a problem gives the *product* of GP terms, choose them symmetrically so the r 's cancel:

- 3 terms: $\frac{a}{r}, a, ar$ (product = a^3)
- 4 terms: $\frac{a}{r^3}, \frac{a}{r}, ar, ar^3$ (product = a^4 , common ratio r^2)
- 5 terms: $\frac{a}{r^2}, \frac{a}{r}, a, ar, ar^2$ (product = a^5)

(Note the *even* case uses ratio r^2 between the symmetric terms — the GP analogue of the AP's “ $2d$ ” trap.)

5. Worked Examples

Difficulty ladder: CBSE Board $\rightarrow$ JEE Main $\rightarrow$ JEE Advanced.

Example 1 — CBSE Board. The product of three numbers in GP is 216 and the sum of their pairwise products is 156. Find the numbers.

Solution. Take the terms symmetrically as $\frac{a}{r}, a, ar$. Product:

$$\frac{a}{r} \cdot a \cdot ar = a^3 = 216 \Rightarrow a = 6.$$

Sum of pairwise products:

$$\frac{a}{r} \cdot a + a \cdot ar + \frac{a}{r} \cdot ar = a^2 \left(\frac{1}{r} + r + 1 \right) = 156 \Rightarrow 36 \left(\frac{1}{r} + r + 1 \right) = 156.$$

So $\frac{1}{r} + r = \frac{156}{36} - 1 = \frac{10}{3} \Rightarrow 3r^2 - 10r + 3 = 0 \Rightarrow (3r - 1)(r - 3) = 0 \Rightarrow r = 3$ or $r = \frac{1}{3}$. Either way the numbers are $\boxed{2, 6, 18}$ (order reversed for $r = \frac{1}{3}$). *Symmetric selection turned two unknowns into one.*

Example 2 — CBSE Board. Express the recurring decimal $0.\overline{3} = 0.3333\dots$ as a fraction.

Solution. Write it as an infinite GP:

$$0.3 + 0.03 + 0.003 + \dots = \frac{3}{10} + \frac{3}{100} + \frac{3}{1000} + \dots$$

with $a = \frac{3}{10}, r = \frac{1}{10}$ (and $|r| < 1$ ✓). Then

$$S_{\infty} = \frac{a}{1-r} = \frac{3/10}{1-1/10} = \frac{3/10}{9/10} = \boxed{\frac{1}{3}}.$$

The convergence check ($|r| = \frac{1}{10} < 1$) is what *licenses* the formula — every recurring decimal is secretly a convergent GP.

Example 3 — JEE Main. The sum of an infinite GP is 2 and the sum of the cubes of its terms is 24. Find the first term and common ratio.

Solution. The terms $a, ar, ar^2, \dots$ have $S_{\infty} = \frac{a}{1-r} = 2$, so $a = 2(1-r)$. Their cubes $a^3, a^3r^3, \dots$ form a GP with ratio r^3 :

$$\frac{a^3}{1-r^3} = 24.$$

Substitute $a^3 = 8(1 - r)^3$ from the first relation:

$$\frac{8(1 - r)^3}{1 - r^3} = 24 \Rightarrow \frac{(1 - r)^3}{1 - r^3} = 3 \Rightarrow \frac{(1 - r)^2}{1 + r + r^2} = 3,$$

using $1 - r^3 = (1 - r)(1 + r + r^2)$. Expanding: $1 - 2r + r^2 = 3(1 + r + r^2) \Rightarrow 2r^2 + 5r + 2 = 0 \Rightarrow (2r + 1)(r + 2) = 0 \Rightarrow r = -\frac{1}{2}$ or $r = -2$. The convergence condition $|r| < 1$ rejects $r = -2$, leaving $r = -\frac{1}{2}$, hence $a = 2(1 + \frac{1}{2}) = 3$.

$$\boxed{a = 3, \quad r = -\frac{1}{2}}$$

Check: $\frac{27}{1 - (-1/8)} = \frac{27}{9/8} = 24 \checkmark$. (The convergence condition is what discriminates the two roots — a textbook *gp-convergence-condition* trap.)

Example 4 — JEE Main (AIEEE 2008). The first two terms of a GP add to 12 and the third and fourth terms add to 48. If the terms are *alternately positive and negative*, find the first term.

Solution. $a + ar = 12$ and $ar^2 + ar^3 = 48$. Divide:

$$\frac{ar^2(1 + r)}{a(1 + r)} = \frac{48}{12} \Rightarrow r^2 = 4 \Rightarrow r = \pm 2.$$

“Alternately positive and negative” forces $r < 0$, so $r = -2$. Then $a(1 + r) = a(-1) = 12 \Rightarrow \boxed{a = -4}$. The qualitative phrase “alternately positive and negative” is the entire constraint that selects $r = -2$ over $r = +2$.

Example 5 — JEE Advanced (infinite geometric figure). A square has side 16 cm. A second square is drawn by joining the midpoints of its sides, a third inside the second the same way, and so on forever. Find the sum of the areas of all the squares.

Solution. Joining midpoints halves the area each time. First area = $16^2 = 256$. Each successive square has area $\frac{1}{2}$ the previous, so the areas form a GP: 256, 128, 64, ... with $a = 256$, $r = \frac{1}{2}$ ($|r| < 1 \checkmark$).

$$\text{Total} = \frac{a}{1 - r} = \frac{256}{1 - \frac{1}{2}} = \boxed{512 \text{ cm}^2}.$$

The infinite nesting has a *finite* total area — the signature payoff of a convergent GP.

Example 6 — JEE Advanced (AM-GM optimisation). If a, b, c are three positive numbers, prove that $(a + b + c) \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right) \geq 9$.

Solution. Apply $AM \geq GM$ separately to each triple:

$$a + b + c \geq 3(abc)^{1/3}, \quad \frac{1}{a} + \frac{1}{b} + \frac{1}{c} \geq 3(abc)^{-1/3}.$$

Multiplying the two inequalities (both sides positive):

$$(a + b + c) \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right) \geq 9 (abc)^{1/3} (abc)^{-1/3} = 9,$$

with equality iff $a = b = c$. ■ This “multiply two AM-GMs” move is one of the highest-yield Advanced techniques in the whole chapter.

Example 7 — JEE Advanced (AIEEE 2007). In a GP of positive terms, each term equals the sum of the next two terms. Find the common ratio.

Solution. “Each term = sum of the next two”: $a = ar + ar^2 \Rightarrow 1 = r + r^2 \Rightarrow r^2 + r - 1 = 0$. So $r = \frac{-1 + \sqrt{5}}{2}$ (rejecting the negative root, since terms are positive). Answer:

$r = \frac{\sqrt{5} - 1}{2}$ — the reciprocal of the golden ratio. The positivity condition kills the extraneous root.

Example 8 — JEE Advanced (chess-grains, closed form). A grain of wheat is placed on the first square of a chessboard, two on the second, four on the third, doubling to the 64th. Find the total number of grains.

Solution. The counts $1, 2, 4, \dots, 2^{63}$ form a GP with $a = 1, r = 2, n = 64$:

$$S_{64} = \frac{a(r^n - 1)}{r - 1} = \frac{1 \cdot (2^{64} - 1)}{2 - 1} = \boxed{2^{64} - 1} \approx 1.8 \times 10^{19}.$$

A concrete demonstration of why $r > 1$ geometric growth “detonates.”

6. Practice Exercises

(Answers below; full solutions on request.)

1. **(Board)** Find the 9th term and the sum of the first 9 terms of the GP $2, 6, 18, \dots$
2. **(Board)** Insert three GMs between 1 and 256.
3. **(Board)** Express $0.\overline{42}$ as a fraction using an infinite GP.
4. **(Main)** If the 4th, 7th and last terms of a GP are 10, 80 and 2560, find the first term and the number of terms.
5. **(Main)** The $(m + n)$ th term of a GP is p and the $(m - n)$ th term is q . Show that the m th term is $\sqrt{pq}$.
6. **(Main)** If S_1, S_2, S_3 are the sums of the first $n, 2n, 3n$ terms of a GP, prove that $S_1(S_3 - S_2) = (S_2 - S_1)^2$.

7. **(Adv)** Three positive distinct numbers x, y, z are in GP and $x + y, y + z, z + x$ are in AP. Find the common ratio.
8. **(Adv)** If one GM G and two AMs p, q are inserted between two positive numbers, prove that $G^2 = (2p - q)(2q - p)$.

Answers:

1. $a_9 = 13122, S_9 = 19682$ | 2. 4, 16, 64 | 3. $\frac{42}{99} = \frac{14}{33}$ | 4. $a = \frac{5}{4}, n = 12$ | 5. (prove via $a_{m+n} \cdot a_{m-n} = a_m^2$) | 6. (identity) | 7. $r = 2 - \sqrt{3}$ (or its reciprocal) | 8. (use $G^2 = ab$ with $a = 2p - q, b = 2q - p$)

7. MCQ with Distractor Analysis

Q1. The sum to infinity of $1 + \frac{3}{4} + \frac{9}{16} + \frac{27}{64} + \dots$ is: (A) 3 (B) 4 (C) $\frac{7}{4}$ (D) does not exist

Correct: (B). $a = 1, r = \frac{3}{4}, |r| < 1$ ✓, so $S_\infty = \frac{1}{1 - 3/4} = 4$. **(A)** comes from $\frac{1}{1-r}$ with $r = \frac{2}{3}$ (misreading the ratio). **(C)** is $a + ar = 1 + \frac{3}{4}$, summing only two terms. **(D)** is the lure for a student who wrongly thinks $r = \frac{3}{4}$ is “too big” to converge. *Discriminates whether the student knows the exact convergence threshold.*

Q2. If $x, 2x + 2, 3x + 3$ are in GP, the fourth term is: (A) 27 (B) -27 (C) 13.5 (D) -13.5

Correct: (D). GP condition: $(2x + 2)^2 = x(3x + 3) \Rightarrow 4x^2 + 8x + 4 = 3x^2 + 3x \Rightarrow x^2 + 5x + 4 = 0 \Rightarrow x = -1$ or $x = -4$. $x = -1$ makes terms $-1, 0, 0$ — invalid (no zero in a GP). So $x = -4$: terms $-4, -6, -9$, ratio $\frac{3}{2}$, fourth term $-9 \cdot \frac{3}{2} = -13.5$. **(A), (C)** ignore the sign and/or take $x = -1$ without rejecting the zero term. **(B)** uses ratio 3 instead of $\frac{3}{2}$. *The signature trap: failing to reject $x = -1$ on the “no zero term” rule.*

Q3. Three numbers are in AP. If 8 is added to the first, the result is a GP whose terms sum to 26. The common ratio of that GP (ascending order) is: (A) 3 (B) $\frac{1}{3}$ (C) 2 (D) $\frac{1}{2}$

Correct: (A). Let AP be $a - d, a, a + d$. Then $a - d + 8, a, a + d$ is a GP summing to 26, so $3a + 8 = 26 \Rightarrow a = 6$. GM condition: $a^2 = (a - d + 8)(a + d) \Rightarrow -d^2 + 8a + 8d = 0 \Rightarrow d^2 - 8d - 48 = 0 \Rightarrow d = 12$ or -4 . Both give the GP $\{2, 6, 18\}$; in ascending order $r = 3$. **(B)** is the descending-order ratio. **(C), (D)** arise from arithmetic slips in the quadratic for d .

Q4. If a, b, c are in GP, then $\log a, \log b, \log c$ are in: (A) GP (B) AP (C) HP (D) none

Correct: (B). $b^2 = ac \Rightarrow 2 \log b = \log a + \log c$, which is the AP condition. **Logs turn a GP into an AP** (and the converse holds too). **(A)** is the reflexive wrong guess (GP in $\rightarrow$ GP out). **(C), (D)** miss the log-exponential duality entirely. *A one-line conceptual discriminator for the GP $\leftrightarrow$ AP bridge.*

8. Common Pitfalls, Pro-Tips & Cheat Sheet

Common Pitfalls (candidate mistake-tags):

- **gp-convergence-condition:** Writing $S_\infty = \frac{a}{1-r}$ without checking $|r| < 1$. The single most punished GP error.
- **gp-sign-of-r:** Dropping the $r < 0$ (alternating) case, or picking the wrong sign of r when two roots survive.
- **gp-gm-sign:** Taking $b = +\sqrt{ac}$ blindly; the GM can be negative, and a real GM needs $ac > 0$.
- **gp-zero-term:** Accepting a solution that forces a term to 0 (no term of a GP may be zero).
- **gp-even-symmetric-ratio:** Using ratio r (not r^2) when picking an *even* number of symmetric terms ($\frac{a}{r^3}, \frac{a}{r}, ar, ar^3$).
- **gp-sum-formula-r1:** Applying $\frac{a(r^n-1)}{r-1}$ when $r = 1$ (division by zero); the sum is then simply na .
- **gp-extraneous-root:** Forgetting positivity/range conditions that reject an extraneous root (golden-ratio problem, etc.).

Cross-subtopic tag note (knowledge-tracing graph): gp-sign-of-r joins ap-sign-of-d under the shared parent **progression-overlooked-sign-case** seeded in Unit 2 — same mistake-physics, two subtopics. Additionally, gp-convergence-condition should be flagged as a *signature single-topic trap* (high discrimination, no AP analogue): it cleanly separates students who memorised the S_∞ formula from those who understand *why* it needs $|r| < 1$.

Pro-Tips:

- Given a **product**, switch to symmetric terms ($\frac{a}{r}, a, ar$) — it collapses two unknowns to one.
 - Given specific **terms**, stay with ar^{n-1} .
 - For “ a, b, c in GP” use $b^2 = ac$ directly — far faster than introducing a, r .
 - **Take logs** to convert any GP problem into the (often easier) AP toolkit, and vice versa.
 - Every **recurring decimal** is a convergent GP — recognise it on sight.
 - For inequalities/optimisation with positive terms, reach for **AM** $\geq$ **GM** before calculus.
-

Cheat Sheet Box — Geometric Progression

Quantity	Formula	Condition
nth term	$a_n = ar^{n-1}$	$a \neq 0, r \neq 0$
nth term from end	$a'_n = l/r^{n-1}$	—
Sum of n terms	$S_n = \frac{a(r^n - 1)}{r - 1} = \frac{a(1 - r^n)}{1 - r}$	$r \neq 1$
Sum ($r = 1$)	$S_n = na$	$r = 1$
Sum to infinity	$S_\infty = \frac{a}{1 - r}$	$ r < 1$
GM of a, c	$b^2 = ac, b = \pm\sqrt{ac}$	a, b, c in GP
n GMs between a, b	$G_k = a(b/a)^{k/(n+1)};$ $\prod G_k = (\sqrt[n]{ab})^n$	—
AM-GM	$\frac{a+b}{2} \geq \sqrt{ab};$ $AM \cdot HM = GM^2$	$a, b > 0$
Symmetric pick (odd)	$\frac{a}{r}, a, ar$	product = a^3
Symmetric pick (even)	$\frac{a}{r^3}, \frac{a}{r}, ar, ar^3$	ratio = r^2
GP $\leftrightarrow$ AP bridge	a, b, c in GP $\Leftrightarrow \log a, \log b, \log c$ in AP	terms > 0

3-term GP test: $b^2 = ac$. $r < 0 \Rightarrow$ alternating signs. **Recurring decimal =** convergent GP.

Unit 4 — Harmonic Progression (HP) and Harmonic Mean

1. Exam Relevance

The third classical progression. A sequence is in HP exactly when the **reciprocals** of its terms are in AP — so HP inherits all of AP's machinery, applied upside-down.

- **CBSE Boards:** Lighter than AP/GP, but HM of two numbers and simple HP term-finding appear in some boards and in the means triad.
- **JEE Main:** A reliable source of questions — “the p th term of an HP is q , the q th is p , find ...”, HM computations, and HP fused with AP/GP (“ a, b, c in AP and b, c, d in HP”).
- **JEE Advanced:** HP rarely stands alone; it appears inside the $A \geq G \geq H$ machinery, in optimisation, and in mixed-progression algebra. The identity $AH = G^2$ (Unit 5) connects all three means.

Diagnostic value: the single defining trick — *reciprocate, then it's an AP* — is the whole subtopic. A student who reaches for reciprocals instinctively has mastered HP; one who tries to attack HP terms directly will flounder. Clean discriminator.

2. Concept Intuition

A harmonic progression is the mathematics of **rates over a fixed amount of work** — the place where “average” stops meaning the arithmetic mean.

The classic anchor: **average speed for a round trip**. Drive to a town at 40 km/h and back along the same road at 60 km/h. The average speed is *not* $\frac{40+60}{2} = 50$ — it's the **harmonic mean**, $\frac{2 \cdot 40 \cdot 60}{40 + 60} = 48$ km/h. Why? Because equal *distances* (not equal times) are covered at each speed, so the rates combine harmonically. Any time you average rates over a fixed task — speed over fixed distance, work-rate over a fixed job, price per fixed quantity — the HM is the honest average.

The defining move is pure reciprocity: $a_1, a_2, a_3, \dots$ are in **HP** if and only if $\frac{1}{a_1}, \frac{1}{a_2}, \frac{1}{a_3}, \dots$ are in **AP**. So every HP question becomes an AP question the instant you flip the terms over.

A crucial structural fact, in contrast to AP and GP: **HP has no neat formula for the sum of its terms**. You sum an HP only in special cases (often by telescoping — Unit 8). This is not a gap in your knowledge; it is a genuine feature of the topic.

The three means line up forever in the same order — $A \geq G \geq H$ — coinciding only when all the numbers are equal (developed fully in Unit 5).

Figure (placeholder): A round-trip illustration — a car going right at “40 km/h” over a fixed distance and returning left at “60 km/h” over the same distance, with the average-speed result “HM = 48 km/h (not 50)” highlighted. Beside it, a flip showing HP terms a_1, a_2, a_3 turning into AP terms $1/a_1, 1/a_2, 1/a_3$.

3. Key Formulas and Definitions

Notation (chained): H = harmonic mean; $H_1, \dots, H_n$ = inserted HMs. The reciprocal AP has first term $\frac{1}{a}$ and common difference d .

Definition. $a_1, a_2, \dots$ (all non-zero) are in **HP** $\iff \frac{1}{a_1}, \frac{1}{a_2}, \dots$ are in **AP**.

nth term of an HP. If the reciprocal AP has first term $\frac{1}{a}$ and common difference d ,

$$a_n = \frac{1}{\frac{1}{a} + (n-1)d}.$$

Harmonic mean. If a, b, c are in HP, then b is the HM of a and c :

$$b = \frac{2ac}{a+c} \quad (\text{two numbers}), \quad \text{HM of } a_1, \dots, a_n = \frac{n}{\frac{1}{a_1} + \dots + \frac{1}{a_n}}.$$

Inserting n HMs between a and b . Insert n AMs between $\frac{1}{a}$ and $\frac{1}{b}$, then reciprocate. The reciprocal AP has common difference

$$D = \frac{\frac{1}{b} - \frac{1}{a}}{n+1} = \frac{a-b}{(n+1)ab}, \quad \frac{1}{H_k} = \frac{1}{a} + kD, \quad k = 1, \dots, n.$$

Relationship among the means (two positive numbers — full treatment in Unit 5):

$$A = \frac{a+b}{2}, \quad G = \sqrt{ab}, \quad H = \frac{2ab}{a+b}, \quad \boxed{AH = G^2}, \quad \boxed{A \geq G \geq H}.$$

Validity / scope conditions:

1. **No term of an HP can be zero** (its reciprocal would be undefined). Positivity is required for the $A \geq G \geq H$ chain.
2. **There is no general closed form for the sum of an HP.** Do not invent one; sum termwise or telescope (Unit 8).

3. **HM of two numbers** uses $\frac{2ac}{a+c}$ — note the 2 in the numerator (a frequent slip is writing $\frac{ac}{a+c}$).
4. **Reciprocate first.** Every HP computation should begin by converting to the reciprocal AP.
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4. Theorem Proofs and Key Derivations

Derivation 1 — HM of two numbers. If a, H, b are in HP, then $\frac{1}{a}, \frac{1}{H}, \frac{1}{b}$ are in AP, so $\frac{1}{H}$ is the AM of $\frac{1}{a}$ and $\frac{1}{b}$:

$$\frac{1}{H} = \frac{1}{2} \left(\frac{1}{a} + \frac{1}{b} \right) = \frac{a+b}{2ab} \implies H = \frac{2ab}{a+b}. \quad \blacksquare$$

Derivation 2 — the classic “ p th term = q , q th term = p ” result. Let the reciprocal AP have first term A_0 and common difference d . Then $A_0 + (p-1)d = \frac{1}{q}$ and $A_0 + (q-1)d = \frac{1}{p}$. Subtracting, $(p-q)d = \frac{1}{q} - \frac{1}{p} = \frac{p-q}{pq}$, so $d = \frac{1}{pq}$. Back-substituting, $A_0 = \frac{1}{pq}$. Hence the n th reciprocal is $\frac{1}{pq} + (n-1)\frac{1}{pq} = \frac{n}{pq}$, giving

$$\boxed{a_n = \frac{pq}{n}}. \quad \blacksquare$$

In particular the $(p+q)$ th term is $\frac{pq}{p+q}$.

Derivation 3 — $AH = G^2$ (link to Unit 5). $AH = \frac{a+b}{2} \cdot \frac{2ab}{a+b} = ab = G^2$, so A, G, H are in GP. $\blacksquare$

Derivation 4 — $G \geq H$. Since $AH = G^2$ and $A \geq G$ (AM $\geq$ GM, Unit 5), $H = \frac{G^2}{A} \leq \frac{G^2}{G} = G$. Combined with $A \geq G$: the chain $A \geq G \geq H$. $\blacksquare$

5. Worked Examples

Ladder: CBSE Board $\rightarrow$ JEE Main $\rightarrow$ JEE Advanced. (Standard curriculum problems.)

Example 1 — CBSE Board. Find the HM of 4 and 12, and verify $AH = G^2$.

Solution. $H = \frac{2 \cdot 4 \cdot 12}{4 + 12} = \frac{96}{16} = 6$. Also $A = 8$, $G = \sqrt{48} = 4\sqrt{3}$. Check: $AH = 48 = (4\sqrt{3})^2 = G^2$ ✓.

Example 2 — CBSE Board. The 3rd term of an HP is $\frac{1}{7}$ and the 7th term is $\frac{1}{15}$. Find the 10th term.

Solution. Reciprocals form an AP: $a'_3 = 7$, $a'_7 = 15$. Common difference $d = \frac{15-7}{7-3} = 2$, first term $a'_1 = 7 - 2(2) = 3$. Tenth reciprocal = $3 + 9(2) = 21$.

So the 10th HP term is $\boxed{\frac{1}{21}}$.

Example 3 — JEE Main. If the p th term of an HP is q and the q th term is p , find the $(p+q)$ th term and the n th term.

Solution. By Derivation 2, $a_n = \frac{pq}{n}$, so the $(p+q)$ th term is $\boxed{\frac{pq}{p+q}}$. (Reciprocate-first turns a messy HP problem into one line of AP algebra.)

Example 4 — JEE Main. Insert 4 harmonic means between 1 and $\frac{1}{6}$.

Solution. Insert 4 AMs between $\frac{1}{1} = 1$ and $\frac{1}{1/6} = 6$: common difference $D = \frac{6-1}{4+1} = 1$. The reciprocal AP is 1, 2, 3, 4, 5, 6, so the HMs are the reciprocals of the interior terms: $\boxed{\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}}$.

Example 5 — JEE Advanced. If a, b, c are in AP, a, b, d are in GP, and $a, c, e \dots$ — more directly: if the AM of two numbers is 9 and their HM is 4, find the numbers (using A, G, H).

Solution. $A = 9$, $H = 4$, so $G^2 = AH = 36 \Rightarrow G = 6$. The numbers have $a+b = 2A = 18$ and $ab = G^2 = 36$, so they are roots of $x^2 - 18x + 36 = 0 \Rightarrow x = 9 \pm \sqrt{81-36} = 9 \pm \sqrt{45} = 9 \pm 3\sqrt{5}$. So $\boxed{9+3\sqrt{5}, 9-3\sqrt{5}}$. The three means together pin the numbers without ever solving the HM equation directly.

Example 6 — JEE Advanced. If a, b, c are in HP, prove that $\frac{a}{b+c}, \frac{b}{c+a}, \frac{c}{a+b}$ are in HP.

Solution. a, b, c in HP $\Rightarrow \frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ in AP $\Rightarrow \frac{a+b+c}{a}, \frac{a+b+c}{b}, \frac{a+b+c}{c}$ in AP (multiply by constant $a+b+c$) $\Rightarrow \frac{a+b+c}{a} - 1, \dots$ in AP $\Rightarrow \frac{b+c}{a}, \frac{c+a}{b}, \frac{a+b}{c}$ in AP $\Rightarrow$ their reciprocals $\frac{a}{b+c}, \frac{b}{c+a}, \frac{c}{a+b}$ are in HP. ■ The reciprocate-and-manipulate chain is the signature HP proof technique.

6. Practice Exercises

1. **(Board)** Find the HM of 3 and 7.
2. **(Board)** The 2nd term of an HP is $\frac{1}{5}$ and the 5th term is $\frac{1}{11}$. Find the 1st term.
3. **(Main)** Insert 3 HMs between 2 and $\frac{1}{2}$.
4. **(Main)** If a, b, c are in HP, show that $b = \frac{2ac}{a+c}$.

5. **(Adv)** If the AM and GM of two numbers are 10 and 8, find their HM.

6. **(Adv)** If a, b, c are in HP, prove $\frac{1}{b-a} + \frac{1}{b-c} = \frac{1}{a} + \frac{1}{c}$.

Answers: 1. $\frac{21}{5} = 4.2$ | 2. $\frac{1}{3}$ (reciprocal AP: 3, 5, ...) | 3. reciprocal AP $\frac{1}{2}, 1, \frac{3}{2}, 2 \rightarrow$ HMs $1, \frac{2}{3}, \frac{1}{2}$ | 4. (reciprocate) | 5. $H = G^2/A = 64/10 = 6.4$ | 6. (use $\frac{2}{b} = \frac{1}{a} + \frac{1}{c}$).

7. MCQ with Distractor Analysis

Q1. The HM of two numbers a and b is: (A) $\frac{a+b}{2}$ (B) $\sqrt{ab}$ (C) $\frac{2ab}{a+b}$ (D) $\frac{ab}{a+b}$

Correct: (C). (A) is the AM, (B) the GM. (D) is the classic trap — dropping the factor of 2 in the numerator (hm-missing-factor-2). *Tests the exact HM formula.*

Q2. If $a_1, a_2, \dots$ are in HP, then $\frac{1}{a_1}, \frac{1}{a_2}, \dots$ are in: (A) HP (B) AP (C) GP (D) none

Correct: (B). This is the definition. (A) is the reflexive wrong guess. *One-line check of the defining property.*

Q3. For two distinct positive numbers, A, G, H satisfy: (A) $A < G < H$ (B) $A > G > H$ (C) $G > A > H$ (D) $A = G = H$

Correct: (B). (A) reverses the chain (mean-inequality-direction). (D) holds only when the numbers are equal. *Tests the ordering of the three means.*

8. Common Pitfalls, Pro-Tips & Cheat Sheet

Common Pitfalls (candidate mistake-tags):

- **hp-no-reciprocate:** Attacking HP terms directly instead of converting to the reciprocal AP. The dominant HP error.
- **hm-missing-factor-2:** Writing $\frac{ab}{a+b}$ instead of $\frac{2ab}{a+b}$.
- **hp-invent-sum-formula:** Assuming a closed-form sum for an HP — there is none in general.
- **hp-zero-term:** Allowing a zero term (its reciprocal is undefined).
- **mean-inequality-direction** (shared with Unit 5): reversing $A \geq G \geq H$.

Schema note: mean-inequality-direction is shared with the AM-GM unit; group HP's hp-no-reciprocate as the signature single-topic trap here (the HP analogue of GP's gp-convergence-condition).

Pro-Tips:

- **Reciprocate first, always.** Then use the full AP toolkit and flip back at the end.
- For “ p th term = q , q th term = p ,” the answer is $a_n = \frac{pq}{n}$ — memorise it.
- Use $AH = G^2$ to jump between means without re-deriving.
- Remember $HM \leq GM \leq AM$ — the HM is always the *smallest*, pulled toward the smaller number.

Cheat Sheet Box — Harmonic Progression & HM

Quantity	Formula	Condition
HP definition	a_n in HP $\iff 1/a_n$ in AP	no zero terms
n th term	$a_n = \frac{1}{\frac{1}{a} + (n-1)d}$	d = CD of reciprocal AP
HM (two numbers)	$H = \frac{2ab}{a+b}$	$a, b > 0$
HM (n numbers)	$\frac{1}{H} = \frac{1}{n} \sum_{i=1}^n \frac{1}{a_i}$	—
n HMs between a, b	$\frac{1}{H_k} = \frac{1}{a} + kD, D = \frac{1}{(n+1)ab}$	—
Means identity	$AH = G^2; A \geq G \geq H$	$a, b > 0$
“ p th= q , q th= p ”	$a_n = \frac{pq}{n}$	—
Sum of HP	no general closed form	—

One-liner: flip to reciprocals $\rightarrow$ it's an AP $\rightarrow$ solve $\rightarrow$ flip back.

Unit 5 — Relationship Between AM, GM and HM

1. Exam Relevance

This subtopic ties the three classical means — **arithmetic (A), geometric (G), harmonic (H)** — into two compact, heavily examined facts: the chain $A \geq G \geq H$ and the identity $AH = G^2$.

- **CBSE Boards:** The signature board question is “*the AM of two numbers is ___ and their GM is ___; find the numbers*,” solved via a quadratic. Reliable 2-3 mark item.
- **JEE Main:** $A \geq G$ (and $A \geq G \geq H$) is a perennial **inequality and optimisation** **workhorse** — minimum/maximum of expressions like $x + \frac{1}{x}$, or proving symmetric inequalities.
- **JEE Advanced:** The full chain $A \geq G \geq H$ with the equality condition, plus $AH = G^2$, appears fused into multi-mean problems and “which of A, G, H are in progression” questions.

Diagnostic note: This unit has **no mistake-fingerprint distinct from GP** beyond two small traps — reversing the inequality direction, and forgetting that $A \geq G$ is exactly the condition for the board quadratic to have real roots. (That overlap is itself why standalone subtopic status is questionable — see end note.)

2. Concept Intuition

There is more than one honest way to “average” two positive numbers, and the three means answer three different questions:

- **AM** = $\frac{a+b}{2}$ — average when quantities *add* (marks, costs, distances).
- **GM** = $\sqrt{ab}$ — average when quantities *multiply* (growth rates, CAGR of an SIP, ratios). The “average” yearly return that turns ₹1 into the same final corpus is the GM, not the AM.
- **HM** = $\frac{2ab}{a+b}$ — average of *rates* over a fixed task (the classic “go at 40 km/h, return at 60 km/h — average speed for the round trip is the HM = 48, not 50”).

The deep result is that these three are **always ordered the same way** for two distinct positive numbers:

$$A > G > H,$$

collapsing to a single point $A = G = H$ exactly when $a = b$. And they are **locked together** by $AH = G^2$ — meaning G is itself the geometric mean of A and H (so A, G, H form a GP).

Figure (placeholder): A number line with two points a and b marked; above it, three tick marks H, G, A shown in order between a and b (all strictly between, clustering: H nearer the smaller, A nearer the larger, G between), with a bracket showing they coincide when $a = b$.

3. Key Formulas and Definitions

Notation (chained: $A=AM$, $G=GM$, $H=HM$):

For two positive reals a, b :

$$\boxed{A = \frac{a+b}{2}}, \quad \boxed{G = \sqrt{ab}}, \quad \boxed{H = \frac{2ab}{a+b}}$$

The two central results:

$$\boxed{AH = G^2} \iff A, G, H \text{ are in GP}, \quad \boxed{A \geq G \geq H}$$

with equality throughout **iff** $a = b$.

Recovering the numbers from their means: Given A and G , the two numbers satisfy $a + b = 2A$ and $ab = G^2$, so they are the roots of

$$\boxed{x^2 - 2Ax + G^2 = 0} \implies a, b = A \pm \sqrt{A^2 - G^2}.$$

For n positive reals $a_1, \dots, a_n$:

$$AM = \frac{a_1 + \dots + a_n}{n}, \quad GM = (a_1 a_2 \dots a_n)^{1/n}, \quad HM = \frac{n}{\frac{1}{a_1} + \dots + \frac{1}{a_n}},$$

with $AM \geq GM \geq HM$, equality iff all a_i are equal.

Validity / scope conditions:

1. **Positivity is mandatory** for G (real) and for the $A \geq G \geq H$ chain. The relationship is stated for positive reals; $AH = G^2$ alone needs only non-zero, but the inequality chain needs $a, b > 0$.
 2. $A \geq G$ is **exactly the real-roots condition** for $x^2 - 2Ax + G^2 = 0$: the discriminant is $4A^2 - 4G^2 = 4(A^2 - G^2) \geq 0 \iff A \geq G$. If a problem hands you $A < G$, no such real numbers exist.
 3. **Equality only at $a = b$** — the means coincide nowhere else.
-

4. Theorem Proofs and Key Derivations

Derivation 1 — $AH = G^2$.

$$AH = \frac{a+b}{2} \cdot \frac{2ab}{a+b} = ab = G^2. \quad \blacksquare$$

So $\frac{A}{G} = \frac{G}{H}$, i.e. A, G, H are in GP with G as the geometric mean of A and H .

Derivation 2 — $A \geq G$. For $a, b > 0$, $(\sqrt{a} - \sqrt{b})^2 \geq 0 \Rightarrow a + b \geq 2\sqrt{ab} \Rightarrow \frac{a+b}{2} \geq \sqrt{ab}$, equality iff $a = b$. $\blacksquare$

Derivation 3 — $G \geq H$. Since $AH = G^2$, we have $H = \frac{G^2}{A}$. As $A \geq G > 0$, dividing gives $\frac{G^2}{A} \leq \frac{G^2}{G} = G$, i.e. $H \leq G$. Chaining with Derivation 2: $A \geq G \geq H$. $\blacksquare$

Derivation 4 — numbers from their means. If A and G are the AM and GM of a, b , then $a + b = 2A$ and $ab = G^2$. By Vieta, a and b are the roots of $x^2 - (a+b)x + ab = 0 = x^2 - 2Ax + G^2$, giving $a, b = A \pm \sqrt{A^2 - G^2}$. Real (and positive) iff $A \geq G$. $\blacksquare$

5. Worked Examples

All examples are — standard curriculum content, not drawn from the bundle. Ladder: Board $\rightarrow$ JEE Main $\rightarrow$ JEE Advanced.

Example 1 — CBSE Board. The AM of two positive numbers is 10 and their GM is 8. Find the numbers.

Solution. $a + b = 2A = 20$, $ab = G^2 = 64$. The numbers are roots of

$$x^2 - 20x + 64 = 0 \Rightarrow (x-4)(x-16) = 0 \Rightarrow \boxed{a = 4, b = 16}.$$

Equivalently $a, b = 10 \pm \sqrt{100 - 64} = 10 \pm 6$. (Note $A = 10 > G = 8$, so real roots exist — the check is automatic.)

Example 2 — JEE Main. For $x > 0$, find the minimum value of $x + \frac{9}{x}$.

Solution. Apply $A \geq G$ to the two positive numbers x and $\frac{9}{x}$:

$$\frac{x + \frac{9}{x}}{2} \geq \sqrt{x \cdot \frac{9}{x}} = 3 \Rightarrow x + \frac{9}{x} \geq 6,$$

with equality when $x = \frac{9}{x}$, i.e. $x = 3$. Minimum value $\boxed{6}$. The AM-GM minimum is attained exactly at the equality point $a = b$.

Example 3 — JEE Advanced. If A, G, H are the AM, GM, HM of two distinct positive numbers, arrange them in order and state which three quantities form a GP.

Solution. By the chain, $A > G > H$ (strict, since the numbers are distinct). By $AH = G^2$, the triple A, G, H is a **GP** (common ratio $G/A = H/G < 1$). So A, G, H is a *decreasing* GP whose geometric mean is G — the same G that is the GM of the original two numbers. ■

Example 4 — JEE Advanced. Two positive numbers have AM = A and GM = G . Express the numbers in terms of A and G , and find their ratio.

Solution. From Derivation 4, $a, b = A \pm \sqrt{A^2 - G^2}$. Their ratio:

$$\frac{a}{b} = \frac{A + \sqrt{A^2 - G^2}}{A - \sqrt{A^2 - G^2}}.$$

A common Advanced form writes this using $A : G = (m+n) : 2\sqrt{mn}$ representations; the key takeaway is that the ratio is fixed entirely by A/G , and the numbers are real iff $A \geq G$. ■

6. Practice Exercises

All. Answers below.

1. **(Board)** The AM and GM of two positive numbers are 15 and 9. Find the numbers.
2. **(Board)** Verify $AH = G^2$ for $a = 4, b = 9$.
3. **(Main)** For $x > 0$, find the minimum of $4x + \frac{1}{x}$.
4. **(Main)** If the AM of two numbers exceeds their GM by 2 and the numbers differ by 8, find them.
5. **(Adv)** If $a, b > 0$, prove $\frac{a}{b} + \frac{b}{a} \geq 2$.
6. **(Adv)** The AM of two positive numbers is to their GM as 5 : 4. Find the ratio of the numbers.

Answers:

1. 3 and 27 | 2. $A = 6.5, H = \frac{72}{13}, AH = 36 = G^2 \checkmark$ | 3. 4 (at $x = \frac{1}{2}$) | 4. 16 and 8 | 5. (AM-GM on $\frac{a}{b}, \frac{b}{a}$) | 6. 4 : 1 (i.e. numbers in ratio 4 : 1, since $A : G = 5 : 4 \Rightarrow a : b = 4 : 1$).

7. MCQ with Distractor Analysis

All.

Q1. If A and G are the AM and GM of two positive numbers, then the numbers are:

- (A) $A \pm \sqrt{A^2 - G^2}$ (B) $A \pm \sqrt{G^2 - A^2}$ (C) $G \pm \sqrt{A^2 - G^2}$ (D) $\frac{A \pm G}{2}$

Correct: (A). Roots of $x^2 - 2Ax + G^2 = 0$. **(B)** flips the discriminant (gives imaginary numbers, since $A \geq G$). **(C)** centres on G instead of A . **(D)** is a meaningless average of the means. *Tests whether the student knows which mean centres the quadratic.*

Q2. For two distinct positive numbers, which is true? (A) $H \geq G \geq A$ (B) $A \geq G \geq H$ (C) $G \geq A \geq H$ (D) $A \geq H \geq G$

Correct: (B). **(A)** reverses the chain — the single most common error (mean-inequality-direction). **(C),(D)** misplace G , which must sit *between* the other two as their GM.

Q3. A, G, H of two positive numbers always form: (A) an AP (B) a GP (C) an HP (D) none

Correct: (B). From $AH = G^2$, the triple is a GP. **(A)** confuses “arithmetic mean” with “in AP.” **(C)** is the reflexive HM-themed wrong guess. *One-line check of the $AH = G^2$ identity.*

8. Common Pitfalls, Pro-Tips & Cheat Sheet

Common Pitfalls (candidate mistake-tags):

- **mean-inequality-direction:** Writing the chain backwards ($H \geq G \geq A$). Signature trap of this unit; reasonable candidate for a *standalone* tag despite the thin subtopic.
- **am-gm-real-numbers-condition:** Forgetting that $A \geq G$ is required for the numbers to be real (board quadratic). Shares discriminant-logic with quadratics.
- **gp-gm-sign** (inherited from Unit 3): mishandling the sign of G .
- **mean-which-centres-quadratic:** Centring the recovery quadratic on G instead of A .

Schema note: apart from mean-inequality-direction, this unit’s tags are inherited or shared with GP/quadratics — consistent with the recommendation that it may not warrant standalone subtopic status.

Pro-Tips:

- To **find two numbers from their means**, jump straight to $x^2 - 2Ax + G^2 = 0$.
- For **min/max of $x + \frac{k}{x}$ -type** expressions, AM-GM gives the answer in one line; equality $a = b$ locates the optimiser.
- Remember G is the *middle* term of the GP A, G, H — a fast sanity check on any ordering.

Cheat Sheet Box — AM / GM / HM Relationship

Quantity (two numbers $a, b > 0$)	Formula
Arithmetic mean	$A = \frac{a+b}{2}$
Geometric mean	$G = \sqrt{ab}$
Harmonic mean	$H = \frac{2ab}{a+b}$
Core identity	$AH = G^2 \implies A, G, H \text{ in GP}$
Ordering	$A \geq G \geq H$, equality iff $a = b$
Numbers from means	roots of $x^2 - 2Ax + G^2 = 0$; $a, b = A \pm \sqrt{A^2 - G^2}$
Reality condition	$A \geq G$
n numbers	$AM \geq GM \geq HM$, equality iff all equal

One-liner: G is the GM of both $\{a, b\}$ **and** of $\{A, H\}$.

Unit 6 — Arithmetico-Geometric Progression (AGP)

1. Exam Relevance

An AGP is a term-by-term **product of an AP and a GP**: each term is “(an arithmetic part) $\times$ (a geometric part).” It is a JEE-syllabus topic in its own right (distinct from special-series summation) and a favourite source of medium-hard problems.

- **CBSE Boards:** Not part of the core NCERT Class 11 syllabus, so board exposure is minimal — but included here for completeness and because the *technique* (multiply-by- r) reinforces GP.
- **JEE Main:** A recurring topic — sum of n terms or sum to infinity of series like $1 + 2x + 3x^2 + 4x^3 + \dots$
- **JEE Advanced:** AGP appears in infinite-sum problems, in series that must be recognised as AGP after manipulation, and fused with binomial or calculus ideas (the sum $\sum nx^n$ also arises from differentiating a GP).

Diagnostic value: the one technique — *multiply the whole sum by the common ratio r and subtract* — is the entire subtopic, a direct extension of the GP sum derivation. A student who sees AGP as “GP’s trick, applied twice” has it; one who tries the $\sum$ -polynomial method on it (it isn’t polynomial) is exposed.

2. Concept Intuition

Think of an AGP as a **GP whose terms are being steadily re-weighted by a growing counter**. A plain GP is $a, ar, ar^2, \dots$. An AGP attaches an arithmetic “label” $1, 2, 3, \dots$ (or any AP) to each: $a \cdot 1, (a + d)r, (a + 2d)r^2, \dots$ — the geometric decay/growth of r^k multiplied by the linear march of the AP.

A concrete picture: imagine **interest-bearing deposits of steadily increasing size**, each discounted by a fixed factor for the time it sits. The increasing deposit sizes are the AP part; the fixed discount factor compounding over time is the GP part; their product, summed, is an AGP sum. Many present-value and expected-value calculations in finance are exactly AGP sums.

The defining feature for summation: because each term carries *both* an additive and a multiplicative pattern, neither the AP sum formula nor the GP sum formula works alone. But the **GP trick still works** — multiply the whole sum S_n by r , line it up shifted by one position, and subtract. The arithmetic part collapses the differences into a constant d , turning the leftover into an ordinary GP. One subtraction does the job.

Crucially, when $|r| < 1$, the AGP **sum to infinity converges** — the geometric decay eventually overwhelms the linear growth of the AP label, so the infinite sum is finite.

Figure (placeholder): Two stacked rows of terms — top row an AP $a, a + d, a + 2d, a + 3d$; middle row a GP $1, r, r^2, r^3$; bottom row their term-by-term products forming the AGP $a, (a + d)r, (a + 2d)r^2, (a + 3d)r^3$, with a brace under the bottom row labelled “ S_n — sum via multiply-by- r and subtract.”

3. Key Formulas and Definitions

Notation (chained): a = first term of the AP part, d = its common difference, r = the GP ratio.

Definition. An arithmetico-geometric progression has general term

$$t_n = [a + (n - 1)d] r^{n-1}$$

— the product of the AP term $a + (n - 1)d$ and the GP term r^{n-1} .

Sum of n terms (for $r \neq 1$):

$$S_n = \frac{a}{1 - r} + \frac{dr(1 - r^{n-1})}{(1 - r)^2} - \frac{[a + (n - 1)d] r^n}{1 - r}.$$

Sum to infinity (valid only when $|r| < 1$):

$$S_\infty = \frac{a}{1 - r} + \frac{dr}{(1 - r)^2}.$$

Validity / scope conditions:

1. **The formulas require $r \neq 1$.** If $r = 1$ the AGP is just an AP, summed by the AP formula.
 2. **S_∞ exists only when $|r| < 1$.** As $n \rightarrow \infty$, both $r^n \rightarrow 0$ and $n r^n \rightarrow 0$ (geometric decay beats linear growth) — that is *why* the infinite sum converges; for $|r| \geq 1$ it diverges.
 3. **Identify the two parts first.** Confirm the term is genuinely (linear in n) $\times (r^{n-1})$ before applying the formula — don't memorise the formula blindly; the multiply-by- r method is safer under pressure.
-

4. Theorem Proofs and Key Derivations

Derivation — sum of an AGP (the multiply-by- r method).

$$S_n = a + (a + d)r + (a + 2d)r^2 + \cdots + [a + (n - 1)d]r^{n-1} \quad (i)$$

Multiply by r and shift one place:

$$rS_n = ar + (a + d)r^2 + \cdots + [a + (n - 2)d]r^{n-1} + [a + (n - 1)d]r^n \quad (ii)$$

Subtract (ii) from (i). The first column gives a ; every middle column leaves a clean $d r^k$ (the AP differences collapse to the constant d); the last term of (ii) is subtracted whole:

$$(1-r)S_n = a + d(r + r^2 + \dots + r^{n-1}) - [a + (n-1)d]r^n.$$

The bracket is a GP sum, $r + r^2 + \dots + r^{n-1} = \frac{r(1-r^{n-1})}{1-r}$. Hence

$$S_n = \frac{a}{1-r} + \frac{dr(1-r^{n-1})}{(1-r)^2} - \frac{[a + (n-1)d]r^n}{1-r}. \quad \blacksquare$$

Letting $n \rightarrow \infty$ with $|r| < 1$ kills both r^{n-1} and r^n terms, leaving $S_\infty = \frac{a}{1-r} + \frac{dr}{(1-r)^2}$. $\blacksquare$

5. Worked Examples

Ladder: CBSE Board $\rightarrow$ JEE Main $\rightarrow$ JEE Advanced. (Standard curriculum problems.)

Example 1 — Board/Main (technique). Sum to n terms: $1 + 2x + 3x^2 + 4x^3 + \dots$ (for $x \neq 1$).

Solution. Here $a = 1, d = 1, r = x$. By the multiply-by- x method:

$$S_n = 1 + 2x + 3x^2 + \dots + nx^{n-1}, \quad xS_n = x + 2x^2 + \dots + nx^n.$$

Subtracting: $(1-x)S_n = 1 + x + x^2 + \dots + x^{n-1} - nx^n = \frac{1-x^n}{1-x} - nx^n$. So

$$S_n = \frac{1-x^n}{(1-x)^2} - \frac{nx^n}{1-x}.$$

Example 2 — JEE Main. Find the sum to infinity of $1 + \frac{2}{3} + \frac{3}{3^2} + \frac{4}{3^3} + \dots$

Solution. $a = 1, d = 1, r = \frac{1}{3}$ (and $|r| < 1$ ✓). Using $S_\infty = \frac{a}{1-r} + \frac{dr}{(1-r)^2}$:

$$S_\infty = \frac{1}{1-\frac{1}{3}} + \frac{\frac{1}{3}}{(1-\frac{1}{3})^2} = \frac{3}{2} + \frac{\frac{1}{3}}{\frac{4}{9}} = \frac{3}{2} + \frac{3}{4} = \boxed{\frac{9}{4}}.$$

Example 3 — JEE Main. Find the sum to infinity of $\frac{3}{1} + \frac{5}{2} + \frac{7}{4} + \frac{9}{8} + \dots$

Solution. Numerators 3, 5, 7, 9 form an AP ($a = 3, d = 2$); denominators 1, 2, 4, 8 are a GP with $r = \frac{1}{2}$. So $r = \frac{1}{2}, |r| < 1$ ✓:

$$S_{\infty} = \frac{3}{1 - \frac{1}{2}} + \frac{2 \cdot \frac{1}{2}}{\left(1 - \frac{1}{2}\right)^2} = \frac{3}{\frac{1}{2}} + \frac{1}{\frac{1}{4}} = 6 + 4 = \boxed{10}.$$

Example 4 — JEE Advanced. Find the sum to n terms of $1 + 3x + 5x^2 + 7x^3 + \dots$ (for $x \neq 1$).

Solution. $a = 1, d = 2, r = x$. Multiply-by- x :

$$(1-x)S_n = 1 + 2(x + x^2 + \dots + x^{n-1}) - (2n-1)x^n = 1 + \frac{2x(1-x^{n-1})}{1-x} - (2n-1)x^n.$$

Therefore

$$S_n = \frac{1}{1-x} + \frac{2x(1-x^{n-1})}{(1-x)^2} - \frac{(2n-1)x^n}{1-x}.$$

Example 5 — JEE Advanced. Evaluate $\sum_{n=1}^{\infty} \frac{n^2}{2^n}$. (An AGP-adjacent sum, handled by applying the AGP idea twice.)

Solution. First note $\sum \frac{n}{2^n} = \frac{1/2}{(1-1/2)^2} = 2$ (AGP with $a = 0$ -shift; equivalently $\sum nr^n = \frac{r}{(1-r)^2}, r = \frac{1}{2}$). For n^2 , use $n^2 = n(n-1) + n$ and the standard results $\sum n(n-1)r^n = \frac{2r^2}{(1-r)^3}$ and $\sum nr^n = \frac{r}{(1-r)^2}$:

$$\sum \frac{n^2}{2^n} = \frac{2(\frac{1}{2})^2}{(\frac{1}{2})^3} + 2 = \frac{2 \cdot \frac{1}{4}}{\frac{1}{8}} + 2 = 4 + 2 = \boxed{6}.$$

The AGP machinery, iterated, sums these “polynomial-times-geometric” series.

6. Practice Exercises

1. **(Main)** Sum to n terms: $1 + 2 \cdot 2 + 3 \cdot 2^2 + 4 \cdot 2^3 + \dots$
2. **(Main)** Sum to infinity: $1 + \frac{3}{4} + \frac{5}{16} + \frac{7}{64} + \dots$
3. **(Main)** Sum to infinity: $\sum_{n=1}^{\infty} \frac{n}{5^{n-1}}$.
4. **(Adv)** Sum to n terms: $2 + 5x + 8x^2 + 11x^3 + \dots$
5. **(Adv)** Evaluate $\sum_{n=1}^{\infty} \frac{2n-1}{2^n}$.

Answers: 1. $S_n = 1 + (n-1)2^n$ | 2. $a = 1, d = 2, r = \frac{1}{4}$: $S_\infty = \frac{4}{3} + \frac{2 \cdot 1/4}{(3/4)^2} = \frac{4}{3} + \frac{8}{9} = \frac{20}{9}$ | 3. $\frac{1}{(1-1/5)^2} = \frac{25}{16}$ | 4. $S_n = \frac{2}{1-x} + \frac{3x(1-x^{n-1})}{(1-x)^2} - \frac{(3n-1)x^n}{1-x}$ | 5. $\sum \frac{2n-1}{2^n} = 2 \sum \frac{n}{2^n} - \sum \frac{1}{2^n} = 2(2) - 1 = 3$.

7. MCQ with Distractor Analysis

Q1. The sum to infinity of $1 + 2r + 3r^2 + 4r^3 + \dots$ for $|r| < 1$ is: (A) $\frac{1}{1-r}$ (B) $\frac{1}{(1-r)^2}$
(C) $\frac{r}{(1-r)^2}$ (D) $\frac{1}{1-r^2}$

Correct: (B). Here $a = 1, d = 1$: $S_\infty = \frac{1}{1-r} + \frac{r}{(1-r)^2} = \frac{(1-r)+r}{(1-r)^2} = \frac{1}{(1-r)^2}$.
(A) is the plain GP sum (forgets the AP weighting). (C) is $\sum nr^n$ (off by one power of r). (D) is a spurious factorisation. *Tests the AGP infinite-sum formula.*

Q2. To sum the series $1 + 4x + 7x^2 + \dots$, the correct method is: (A) AP sum formula (B) GP sum formula (C) multiply by x and subtract (D) $\sum$ -polynomial method

Correct: (C). The term is (linear) $\times (x^{n-1})$ — an AGP. (A), (B) each handle only one part. (D) (special-agp-vs-polynomial) fails because the term is not polynomial in n . *Tests technique selection.*

Q3. The AGP series $a + (a+d)r + (a+2d)r^2 + \dots$ converges (has a finite sum to infinity) when: (A) $d < 1$ (B) $a < 1$ (C) $|r| < 1$ (D) always

Correct: (C). Convergence is governed entirely by the geometric ratio. (A), (B) are irrelevant. (D) is false for $|r| \geq 1$. *Tests the convergence condition.*

8. Common Pitfalls, Pro-Tips & Cheat Sheet

Common Pitfalls (candidate mistake-tags):

- **agp-wrong-method:** Trying the AP or GP sum formula alone, or the $\sum$ -polynomial method, on an AGP. The dominant error.
- **agp-convergence:** Writing S_∞ without checking $|r| < 1$ (shares mistake-physics with gp-convergence-condition).
- **agp-r-equals-1:** Applying the $r \neq 1$ formula when $r = 1$ (the series is then a plain AP).
- **agp-misidentify-parts:** Mis-reading which factor is the AP part and which the GP part (e.g. swapping d and r roles).

Schema note: agp-convergence should sit under the same parent as gp-convergence-condition — both are “forgot $|r| < 1$.” And agp-wrong-method pairs with Unit 7’s special-agp-vs-polynomial (two sides of the same boundary confusion).

Pro-Tips:

- **Don't memorise the messy S_n formula** — re-derive with the multiply-by- r subtraction every time; it's faster and error-proof under exam pressure.
- For S_∞ , the compact $\frac{a}{1-r} + \frac{dr}{(1-r)^2}$ is worth memorising.
- Recognise AGP by the shape: each term = (something linear in n) $\times (r^{n-1})$.
- The related sums $\sum nr^n = \frac{r}{(1-r)^2}$ and $\sum n^2r^n = \frac{r(1+r)}{(1-r)^3}$ follow from the same idea (or from differentiating the GP sum).

Cheat Sheet Box — Arithmetico-Geometric Progression

Quantity	Formula	Condition
General term	$t_n = [a + (n-1)d] r^{n-1}$	—
Sum of n terms	$S_n = \frac{a}{1-r} + \frac{dr(1-r^{n-1})}{(1-r)^2} - \frac{[a + (n-1)d]r^n}{1-r}$	$r \neq 1$
Sum to infinity	$S_\infty = \frac{a}{1-r} + \frac{dr}{(1-r)^2}$	$ r < 1$
Method	multiply S_n by r , shift, subtract	always works
Useful relatives	$\sum nr^n = \frac{r}{(1-r)^2};$ $\sum n^2r^n = \frac{r(1+r)}{(1-r)^3}$	$ r < 1$

One-liner: AGP = AP $\times$ GP, term by term; sum it the way you sum a GP — multiply by r and subtract.

Unit 7 — Special Series Summation

1. Exam Relevance

“Special series” means the closed-form sums of $\sum n$, $\sum n^2$, $\sum n^3$ (and occasionally $\sum n^4$), plus the techniques that reduce *any* polynomially-generated series to these building blocks. It is the **computational engine** of the chapter.

- **CBSE Boards:** Direct application is common — sum a series like $3 + 7 + 13 + 21 + \dots$ or $2 \cdot 4 + 4 \cdot 6 + 6 \cdot 8 + \dots$ to n terms (5-mark). The derivations of $\sum n^2$ and $\sum n^3$ are named, askable results.
- **JEE Main:** A reliable 1-question topic — find S_n for a series whose general term is a polynomial or a product, or evaluate a nested $\sum$. JEE Main 2020 directly asked $\sum_{k=1}^{20} (1 + 2 + \dots + k)$.
- **JEE Advanced:** Special series fuse into harder setups — the identity $9S_2^2 = S_3(1 + 8S_1)$, alternating-sign square sums, and series requiring the *method of differences*. The $S_n = cn^2$ PYQ (JEE 2009) is a recurring template.

Diagnostic note: Errors here are overwhelmingly **technique-selection** errors — failing to recognise that a term is a polynomial in n (so $\sum$ formulas apply) or mishandling the telescoping in the method of differences. Clean discriminator between mechanical formula-recall and method fluency.

2. Concept Intuition

The whole subtopic rests on one move: **break a complicated sum into simple sums you already know**.

If you can write the general term of a series as a combination of $n^3, n^2, n, 1$, then by the linearity of $\sum$, the sum of the series is just the matching combination of the four standard sums. So the entire game is: *find t_n , express it in powers of n , then apply the standard formulas*.

The standard sums themselves come from a beautiful **telescoping trick**: to find $\sum n^2$, you exploit the identity $k^3 - (k-1)^3 = 3k^2 - 3k + 1$. Add it up for $k = 1, \dots, n$ and the left side collapses to just n^3 (everything in between cancels), leaving an equation you solve for $\sum n^2$. The same idea with $(k+1)^4 - k^4$ gives $\sum n^3$, and $k^5 - (k-1)^5$ gives $\sum n^4$. One trick, a whole family of formulas.

Indian-context anchor (and it's real history): the formulas for $\sum n^2$ and $\sum n^3$ were first given by **Aryabhata** in his *Aryabhatiyam* (around 499 CE) — over a millennium before they appeared in European texts. Brahmagupta (598), Mahavira (850), and Bhaskara (1114–1185) all extended this work. When your students sum squares of natural numbers, they are using an Indian result roughly 1,500 years old.

A second intuition worth installing: $\sum n^3 = (\sum n)^2$. The sum of the first n cubes is the *square* of the sum of the first n naturals — a startling, memorable coincidence (e.g. $1 + 8 + 27 = 36 = 6^2 = (1 + 2 + 3)^2$).

Figure (placeholder): A “telescoping” visual — a vertical stack of the identities $k^3 - (k-1)^3 = 3k^2 - 3k + 1$ for $k = 1, 2, 3, \dots, n$, with arrows showing interior k^3 terms cancelling diagonally, leaving only n^3 on the left. Beside it, a small figure showing $1^3 + 2^3 + 3^3$ as nested squares summing to $(1 + 2 + 3)^2$.

3. Key Formulas and Definitions

The standard sums:

$$\sum_{r=1}^n 1 = n$$

$$\sum_{r=1}^n r = \frac{n(n+1)}{2}$$

$$\sum_{r=1}^n r^2 = \frac{n(n+1)(2n+1)}{6}$$

$$\sum_{r=1}^n r^3 = \left[\frac{n(n+1)}{2} \right]^2 = \left(\sum_{r=1}^n r \right)^2$$

$$\sum_{r=1}^n r^4 = \frac{n(n+1)(2n+1)(3n^2+3n-1)}{30} \quad (\text{rarely needed; JEE Advanced only})$$

Linearity of sigma:

$$\sum_{r=1}^n (a t_r + b u_r) = a \sum_{r=1}^n t_r + b \sum_{r=1}^n u_r, \quad \sum_{r=1}^n c = cn.$$

Master technique. If the general term is a polynomial,

$$t_n = an^3 + bn^2 + cn + d \implies S_n = a \sum n^3 + b \sum n^2 + c \sum n + d \cdot n.$$

Sum of products taken two at a time:

$$\left(\sum_{i=1}^n a_i \right)^2 = \sum_{i=1}^n a_i^2 + 2 \sum_{1 \leq i < j \leq n} a_i a_j \implies \sum_{i < j} a_i a_j = \frac{(\sum a_i)^2 - \sum a_i^2}{2}.$$

Validity / scope conditions:

1. **Index range** $r \in \mathbb{N}$, and every formula above is for the *first* n terms starting at $r = 1$. For a sum from $r = m$ to n , use $\sum_1^n - \sum_1^{m-1}$.
2. **Linearity needs the same index range** on every piece — a frequent slip when splitting a term.
3. **The master technique only applies when t_n is a polynomial in n** (or reducible to one). For products like $r(r+1)(r+2)$, *expand first*, then apply $\sum$. For genuinely non-polynomial t_n (e.g. $\frac{1}{r(r+1)}$ or AGP terms), use telescoping or the AGP formula instead — see §4.

4. Theorem Proofs and Key Derivations

Derivation 1 — $\sum n^2$ via the telescoping identity. Use $k^3 - (k-1)^3 = 3k^2 - 3k + 1$. Sum both sides for $k = 1, 2, \dots, n$. The left telescopes to $n^3 - 0^3 = n^3$:

$$n^3 = 3 \sum_{k=1}^n k^2 - 3 \sum_{k=1}^n k + n.$$

Substitute $\sum k = \frac{n(n+1)}{2}$ and solve:

$$3 \sum k^2 = n^3 + \frac{3n(n+1)}{2} - n \implies \sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}. \quad \blacksquare$$

Derivation 2 — $\sum n^3$ via $(k+1)^4 - k^4$. Use $(k+1)^4 - k^4 = 4k^3 + 6k^2 + 4k + 1$. Summing for $k = 1, \dots, n$, the left telescopes to $(n+1)^4 - 1$. Substituting the now-known $\sum k^2$ and $\sum k$ and solving gives

$$\sum_{k=1}^n k^3 = \left[\frac{n(n+1)}{2} \right]^2. \quad \blacksquare$$

(The same machinery with $k^5 - (k-1)^5$ produces $\sum n^4$ — chunk 21.)

Method — sum of a series by recognising t_n . For a series like $2 \cdot 4 + 4 \cdot 6 + 6 \cdot 8 + \dots$, write the general term, expand to a polynomial, then apply $\sum$:

$$t_n = (2n)(2n+2) = 4n^2 + 4n \implies S_n = 4 \sum n^2 + 4 \sum n.$$

Method of differences. When the *differences* of consecutive terms form a recognisable progression, the general term is one degree higher. For 3, 7, 13, 21, 31, ... the first differences are 4, 6, 8, 10 (an AP), so t_n is quadratic: solving gives $t_n = n^2 + n + 1$, after which $\sum$ applies termwise.

5. Worked Examples

Difficulty ladder: CBSE Board → JEE Main → JEE Advanced. (Problems drawn from the bundle's exercise sets.)

Example 1 — CBSE Board. Find the sum to n terms of $2 \cdot 4 + 4 \cdot 6 + 6 \cdot 8 + \dots$

Solution. General term $t_r = (2r)(2r + 2) = 4r^2 + 4r$. So

$$S_n = 4 \sum_{r=1}^n r^2 + 4 \sum_{r=1}^n r = 4 \cdot \frac{n(n+1)(2n+1)}{6} + 4 \cdot \frac{n(n+1)}{2} = \frac{2n(n+1)(2n+1)}{3} + 2n(n+1).$$

$$\text{Factor } 2n(n+1): S_n = 2n(n+1) \left[\frac{2n+1}{3} + 1 \right] = \boxed{\frac{2n(n+1)(2n+4)}{3} = \frac{4n(n+1)(n+2)}{3}}.$$

(The 20th term alone is $4 \cdot 20 \cdot 21 = 1680$.)

Example 2 — CBSE Board. Find S_n for $3 + 7 + 13 + 21 + 31 + \dots$

Solution. First differences 4, 6, 8, 10 form an AP, so t_n is quadratic. Fitting $t_1 = 3, t_2 = 7, t_3 = 13$ gives $t_n = n^2 + n + 1$ (check: $t_4 = 16 + 4 + 1 = 21$ ✓). Then

$$S_n = \sum n^2 + \sum n + \sum 1 = \frac{n(n+1)(2n+1)}{6} + \frac{n(n+1)}{2} + n.$$

Combining over a common factor: $S_n = \frac{n(n^2 + 3n + 5)}{3}$ (verify $S_3 = 3 + 7 + 13 = 23 = \frac{3(9+9+5)}{3}$ ✓).

Example 3 — JEE Main (JEE Main 2020). Evaluate $\sum_{k=1}^{20} (1 + 2 + 3 + \dots + k)$.

Solution. The inner sum is $1 + 2 + \dots + k = \frac{k(k+1)}{2}$, so

$$\sum_{k=1}^{20} \frac{k(k+1)}{2} = \frac{1}{2} \left(\sum_{k=1}^{20} k^2 + \sum_{k=1}^{20} k \right) = \frac{1}{2} \left(\frac{20 \cdot 21 \cdot 41}{6} + \frac{20 \cdot 21}{2} \right) = \frac{1}{2} (2870 + 210) = \boxed{1540}.$$

Example 4 — JEE Main. Find S_n for the series $5 + 55 + 555 + \dots$

Solution. Each term is a repunit times 5: $t_r = 5 \cdot \underbrace{11 \dots 1}_r = 5 \cdot \frac{10^r - 1}{9}$. So

$$S_n = \frac{5}{9} \sum_{r=1}^n (10^r - 1) = \frac{5}{9} \left(\frac{10(10^n - 1)}{9} - n \right) = \boxed{\frac{5}{81} (10^{n+1} - 9n - 10)}.$$

A GP-based special series — the geometric piece is summed, the constant piece is $-n$.

Example 5 — JEE Advanced (JEE 2009). If the sum of the first n terms of an AP is cn^2 , find the sum of the squares of these n terms.

Solution. $S_n = cn^2$ has the form $An^2 + Bn$ with $B = 0$, so it's an AP; the n th term is $a_n = S_n - S_{n-1} = c(n^2 - (n-1)^2) = c(2n-1)$. Then

$$\sum_{n=1}^N a_n^2 = c^2 \sum_{n=1}^N (2n-1)^2 = c^2 \sum (4n^2 - 4n + 1) = c^2 \left[4 \cdot \frac{N(N+1)(2N+1)}{6} - 4 \cdot \frac{N(N+1)}{2} + N \right]$$

Factor N and simplify the bracket to $\frac{4N^2 - 1}{3}$:

$$\boxed{\sum a_n^2 = \frac{N(4N^2 - 1)c^2}{3}}.$$

Example 6 — JEE Advanced. Find S_n for $\frac{1^3}{1} + \frac{1^3 + 2^3}{1+3} + \frac{1^3 + 2^3 + 3^3}{1+3+5} + \dots$

Solution. The r th term has numerator $\sum_{k=1}^r k^3 = \left[\frac{r(r+1)}{2}\right]^2$ and denominator $1 + 3 + \dots + (2r-1) = r^2$. So

$$t_r = \frac{[r(r+1)/2]^2}{r^2} = \frac{(r+1)^2}{4}.$$

Then $S_n = \frac{1}{4} \sum_{r=1}^n (r+1)^2 = \frac{1}{4} \sum_{r=1}^n (r^2 + 2r + 1) = \frac{1}{4} \left[\frac{n(n+1)(2n+1)}{6} + n(n+1) + n \right]$, which simplifies to

$$\boxed{S_n = \frac{n(2n^2 + 9n + 13)}{24}}.$$

Both standard sums collapse the term to a clean quadratic before summation — the signature special-series move.

6. Practice Exercises

(From the bundle's exercise sets. Answers below.)

1. **(Board)** Find $\sum_{r=1}^n r(r+1)$.
2. **(Board)** Find the sum of all numbers between 200 and 400 divisible by 7.
3. **(Main)** Find the 20th term of $2 \cdot 4 + 4 \cdot 6 + 6 \cdot 8 + \dots$
4. **(Main)** Find S_n for $.6 + .66 + .666 + \dots$
5. **(Adv)** If S_1, S_2, S_3 are the sums of the first n natural numbers, their squares, and their cubes, show that $9S_2^2 = S_3(1 + 8S_1)$.

6. **(Adv)** Find $\sum_{r=1}^n r(r+1)(r+2)$ and identify the telescoping form.

Answers:

$$1. \frac{n(n+1)(n+2)}{3} \mid 2. 8729 \mid 3. 1680 \mid 4. \frac{6}{81}(9n-1+10^{-n}) = \frac{2}{27}(9n-1+10^{-n}) \mid 5. (\text{identity} - \text{use } S_1 = \frac{n(n+1)}{2}, S_2 = \frac{n(n+1)(2n+1)}{6}, S_3 = S_1^2) \mid 6. \frac{n(n+1)(n+2)(n+3)}{4}.$$

7. MCQ with Distractor Analysis

Q1. $\sum_{r=1}^n r^3$ equals: (A) $\frac{n(n+1)(2n+1)}{6}$ (B) $\left[\frac{n(n+1)}{2}\right]^2$ (C) $\frac{n^2(n+1)}{2}$ (D) $\frac{n(n+1)}{2}$

Correct: (B). (A) is $\sum r^2$ — the classic mix-up of the square and cube formulas. (C) drops the outer square. (D) is $\sum r$. *Discriminates whether the student has the three formulas cleanly separated.*

Q2. The general term of $1 \cdot 2 + 2 \cdot 3 + 3 \cdot 4 + \dots$ is: (A) n^2 (B) $n(n+1)$ (C) $n(n-1)$ (D) $2n$

Correct: (B). (A) forgets the second factor. (C) shifts the index (that's the previous product). (D) sums only the first factors. *The term-recognition step is exactly where special-series problems are won or lost.*

Q3. If $\sum_{r=1}^n r = 78$, then $\sum_{r=1}^n r^3$ equals: (A) 6084 (B) 234 (C) 608 (D) 156

Correct: (A). Since $\sum r^3 = (\sum r)^2 = 78^2 = 6084$. (B) is 3×78 . (C), (D) are arithmetic lures. *Pure test of the $\sum n^3 = (\sum n)^2$ identity — no need to even find n (it's 12).*

8. Common Pitfalls, Pro-Tips & Cheat Sheet

Common Pitfalls (candidate mistake-tags):

- **special-formula-confusion:** Swapping $\sum n^2$ and $\sum n^3$ (the $\frac{(2n+1)}{6}$ form vs the squared form). The dominant error of this unit.
- **special-term-recognition:** Failing to find t_n as a polynomial before summing — trying to sum a product like $r(r+1)$ without expanding.
- **special-index-range:** Applying $\sum_1^n$ formulas to a sum that doesn't start at 1, without the $\sum_1^n - \sum_1^{m-1}$ correction.
- **special-method-of-differences:** Mis-degree-ing the general term (treating a quadratic t_n as linear because only first differences were checked).

- **special-agp-vs-polynomial:** Trying the $\sum$ -polynomial method on an AGP term (which is *not* polynomial) instead of the AGP formula.

Schema note (knowledge-tracing graph): special-formula-confusion is a *signature single-topic trap* with no AP/GP analogue — clean discriminator for this subtopic. special-term-recognition is the cross-cutting skill that also surfaces in AGP and method-of-differences problems; flag it as a shared parent once the AGP subtopic exists.

Pro-Tips:

- **Always find t_n first**, expand it to powers of n , then sum. This single habit solves the majority of board and Main problems.
- Remember the telescoping origin — if you forget $\sum n^2$, re-derive it in 4 lines from $k^3 - (k-1)^3$.
- $\sum n^3 = (\sum n)^2$ is the fastest shortcut in the chapter; recognise cubes on sight.
- For series like $5 + 55 + 555$, peel off the GP (10^r) and the constant separately.
- For “differences in AP/GP,” reach for the **method of differences**, not brute force.

Cheat Sheet Box — Special Series

Series	Closed form
$\sum_{r=1}^n 1$	n
$\sum_{r=1}^n r$	$\frac{n(n+1)}{2}$
$\sum_{r=1}^n r^2$	$\frac{n(n+1)(2n+1)}{6}$
$\sum_{r=1}^n r^3$	$\left[\frac{n(n+1)}{2} \right]^2 = (\sum r)^2$
$\sum_{r=1}^n r^4$	$\frac{n(n+1)(2n+1)(3n^2+3n-1)}{30}$
Linearity	$\sum (a t_r + b u_r) = a \sum t_r + b \sum u_r; \quad \sum c = cn$
Master method	$t_n = an^3 + bn^2 + cn + d \Rightarrow S_n = a \sum n^3 + b \sum n^2 + c \sum n + dn$
Products two-at-a-time	$\sum_{i < j} a_i a_j = \frac{(\sum a_i)^2 - \sum a_i^2}{2}$
Partial range	$\sum_m^n = \sum_1^n - \sum_1^{m-1}$

Workflow: find $t_n \rightarrow$ expand to powers of $n \rightarrow$ apply standard sums. **Shortcut:** $\sum n^3 = (\sum n)^2$.

Unit 8 — Telescoping Series & Partial Fractions

1. Exam Relevance

This unit covers the summation techniques for series whose general term is **not** a polynomial in n — fractions, factorials, surds — where the special-series formulas of Unit 7 don't apply. The core idea: rewrite each term as a **difference of consecutive quantities** so that almost everything cancels.

- **CBSE Boards:** $\sum \frac{1}{r(r+1)}$ -type sums appear in some boards and in the NCERT miscellaneous exercise.
- **JEE Main:** A reliable topic — sums like $\sum \frac{1}{r(r+1)(r+2)}$, $\sum \frac{1}{\sqrt{r} + \sqrt{r+1}}$, and the V_n method for products of AP terms.
- **JEE Advanced:** Telescoping is a frequent “trick” step inside harder problems, and the V_n method generalises the special-series results. Recognising a telescoping structure is a high-value skill.

Diagnostic value: the entire subtopic is one move — *express t_r as $f(r) - f(r+1)$ (or similar) so the sum collapses*. A student who reaches for partial fractions / difference-form on a fractional term has the skill; one who tries to apply $\sum n^k$ formulas is exposed. It directly complements Unit 7's polynomial method.

2. Concept Intuition

A telescoping sum works like a **collapsing telescope** (or a stack of nested folding cups): each term cancels part of the next, so that when you add everything up, only the very first and very last pieces survive — all the middle collapses to nothing.

The trick is to rewrite each term as a **difference of two consecutive “things.”** If you can write

$$t_r = V_r - V_{r+1} \quad (\text{or } V_{r-1} - V_r),$$

then

$$\sum_{r=1}^n t_r = (V_1 - V_2) + (V_2 - V_3) + \cdots + (V_n - V_{n+1}) = V_1 - V_{n+1},$$

because every interior V appears once with a $+$ and once with a $-$. The whole sum reduces to **just the endpoints**.

The everyday picture: think of a **relay race where each runner hands off exactly what they received** — only the starting baton and the final finish-line position matter; everything passed in between cancels out in the bookkeeping.

For fractional terms, the tool that produces the difference-form is **partial fractions**:

$\frac{1}{r(r+1)} = \frac{1}{r} - \frac{1}{r+1}$. For products of consecutive integers, the tool is the V_n **method**

— append one more factor and divide by the count, mirroring how integration raises a power. For surds, **rationalising** turns $\frac{1}{\sqrt{r} + \sqrt{r+1}}$ into the difference $\sqrt{r+1} - \sqrt{r}$.

Figure (placeholder): A collapsing-telescope graphic — segments labelled $V_1, V_2, V_3, \dots, V_{n+1}$, with paired $+V_k$ and $-V_k$ arrows cancelling between adjacent segments, leaving only V_1 (first) and V_{n+1} (last) highlighted as the surviving terms.

3. Key Formulas and Definitions

Telescoping principle. If $t_r = V_r - V_{r+1}$, then $\sum_{r=1}^n t_r = V_1 - V_{n+1}$.

Standard partial-fraction results (memorise the first; derive the rest):

$$\boxed{\sum_{r=1}^n \frac{1}{r(r+1)} = 1 - \frac{1}{n+1} = \frac{n}{n+1}}$$

$$\sum_{r=1}^n \frac{1}{r(r+1)(r+2)} = \frac{1}{2} \left[\frac{1}{1 \cdot 2} - \frac{1}{(n+1)(n+2)} \right] = \frac{n(n+3)}{4(n+1)(n+2)}$$

$$\sum_{r=1}^n \frac{1}{(2r-1)(2r+1)} = \frac{1}{2} \left[1 - \frac{1}{2n+1} \right] = \frac{n}{2n+1}$$

The V_n method (products of consecutive AP terms). To sum $\sum t_r$ where $t_r = r(r+1)(r+2) \cdots (k \text{ consecutive factors})$, append the next factor and divide by $k+1$ times the common difference. For consecutive integers (CD = 1):

$$\boxed{\sum_{r=1}^n r(r+1) \cdots (r+k-1) = \frac{n(n+1)(n+2) \cdots (n+k)}{k+1}}$$

e.g. $\sum r(r+1) = \frac{n(n+1)(n+2)}{3}$ and $\sum r(r+1)(r+2) = \frac{n(n+1)(n+2)(n+3)}{4}$.

Factorial and surd telescopers:

$$\sum_{r=1}^n r \cdot r! = (n+1)! - 1, \quad \sum_{r=1}^n \frac{1}{\sqrt{r} + \sqrt{r+1}} = \sqrt{n+1} - 1.$$

Validity / scope conditions:

- 1. The term must factor into a genuine difference of consecutive quantities.**
Not every fraction telescopes — only those whose partial-fraction pieces line up with a shift of one (or a fixed shift).
- 2. Watch the constant multiplier.** For $\frac{1}{r(r+1)(r+2)}$ the difference-form carries a $\frac{1}{2}$; for $(2r-1)(2r+1)$ it carries $\frac{1}{2}$ from the $\frac{1}{2}(\frac{1}{2r-1} - \frac{1}{2r+1})$ split. Track it.
- 3. Identify the surviving endpoints carefully** — $V_1 - V_{n+1}$, not $V_0 - V_n$; off-by-one here is the main error.

4. Theorem Proofs and Key Derivations

Derivation 1 — $\sum \frac{1}{r(r+1)}$ by partial fractions. $\frac{1}{r(r+1)} = \frac{1}{r} - \frac{1}{r+1}$. Summing for $r = 1, \dots, n$:

$$\left(\frac{1}{1} - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \dots + \left(\frac{1}{n} - \frac{1}{n+1}\right) = 1 - \frac{1}{n+1} = \frac{n}{n+1}. \quad \blacksquare$$

Derivation 2 — $\sum \frac{1}{r(r+1)(r+2)}$. Use $\frac{1}{r(r+1)(r+2)} = \frac{1}{2} \left[\frac{1}{r(r+1)} - \frac{1}{(r+1)(r+2)} \right]$. The bracket telescopes:

$$\sum = \frac{1}{2} \left[\frac{1}{1 \cdot 2} - \frac{1}{(n+1)(n+2)} \right] = \frac{n(n+3)}{4(n+1)(n+2)}. \quad \blacksquare$$

Derivation 3 — the V_n method for $\sum r(r+1)$. Take $V_r = r(r+1)(r+2)$. Then $V_r - V_{r-1} = r(r+1)(r+2) - (r-1)r(r+1) = r(r+1)[(r+2) - (r-1)] = 3r(r+1)$. So $r(r+1) = \frac{1}{3}(V_r - V_{r-1})$, and the sum telescopes to $\frac{1}{3}(V_n - V_0) = \frac{1}{3}n(n+1)(n+2)$. $\blacksquare$

Derivation 4 — $\sum r \cdot r!$. Note $r \cdot r! = (r+1)! - r!$ (since $(r+1)! = (r+1)r! = r \cdot r! + r!$). Summing telescopes to $(n+1)! - 1! = (n+1)! - 1$. $\blacksquare$

Derivation 5 — surd telescoper. Rationalise: $\frac{1}{\sqrt{r} + \sqrt{r+1}} = \frac{\sqrt{r+1} - \sqrt{r}}{(\sqrt{r+1})^2 - (\sqrt{r})^2} = \frac{\sqrt{r+1} - \sqrt{r}}{r+1 - r} = \sqrt{r+1} - \sqrt{r}$. Summing telescopes to $\sqrt{n+1} - 1$. $\blacksquare$

5. Worked Examples

Ladder: CBSE Board $\rightarrow$ JEE Main $\rightarrow$ JEE Advanced. (Standard curriculum problems.)

Example 1 — **CBSE Board.** Find $\sum_{r=1}^n \frac{1}{r(r+1)}$ and hence its value as $n \rightarrow \infty$.

Solution. By Derivation 1, $S_n = \frac{n}{n+1}$. As $n \rightarrow \infty$, $S_n \rightarrow 1$. (So $\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots = 1$ exactly.)

Example 2 — JEE Main. Sum $\frac{1}{1 \cdot 4} + \frac{1}{4 \cdot 7} + \frac{1}{7 \cdot 10} + \dots$ to n terms.

Solution. General term $t_r = \frac{1}{(3r-2)(3r+1)} = \frac{1}{3} \left[\frac{1}{3r-2} - \frac{1}{3r+1} \right]$. Telescoping:

$$S_n = \frac{1}{3} \left[1 - \frac{1}{3n+1} \right] = \boxed{\frac{n}{3n+1}}.$$

Example 3 — JEE Main. Find $\sum_{r=1}^n r(r+1)(r+2)$.

Solution. By the V_n method (4 consecutive-factor result), $\boxed{\frac{n(n+1)(n+2)(n+3)}{4}}$.
(Verify $n=1$: $1 \cdot 2 \cdot 3 = 6 = \frac{1 \cdot 2 \cdot 3 \cdot 4}{4}$ ✓.)

Example 4 — JEE Advanced. Evaluate $\sum_{r=1}^n \frac{r}{(r+1)!}$.

Solution. Write $\frac{r}{(r+1)!} = \frac{(r+1)-1}{(r+1)!} = \frac{1}{r!} - \frac{1}{(r+1)!}$. Telescoping:

$$S_n = \frac{1}{1!} - \frac{1}{(n+1)!} = \boxed{1 - \frac{1}{(n+1)!}}, \quad S_\infty = 1.$$

Example 5 — JEE Advanced. Sum $\sum_{r=1}^n \frac{1}{\sqrt{r} + \sqrt{r+1}}$, then evaluate for $n=99$.

Solution. By Derivation 5, each term equals $\sqrt{r+1} - \sqrt{r}$, so $S_n = \sqrt{n+1} - 1$. For $n=99$: $S_{99} = \sqrt{100} - 1 = \boxed{9}$. *Rationalising converted a sum of awkward fractions into a clean telescoper.*

Example 6 — JEE Advanced. Find $\sum_{r=1}^n \frac{2r+1}{r^2(r+1)^2}$.

Solution. Notice $\frac{2r+1}{r^2(r+1)^2} = \frac{1}{r^2} - \frac{1}{(r+1)^2}$ (since $(r+1)^2 - r^2 = 2r+1$). Telescoping:

$$S_n = 1 - \frac{1}{(n+1)^2} = \boxed{\frac{n(n+2)}{(n+1)^2}}, \quad S_\infty = 1.$$

6. Practice Exercises

1. **(Board)** Sum $\frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} + \dots$ to n terms.
2. **(Main)** Sum $\sum_{r=1}^n \frac{1}{(2r-1)(2r+1)}$.
3. **(Main)** Find $\sum_{r=1}^n r \cdot r!$.
4. **(Adv)** Sum $\sum_{r=1}^n \frac{1}{r(r+1)(r+2)}$ and its value as $n \rightarrow \infty$.
5. **(Adv)** Evaluate $\sum_{r=1}^n \frac{r}{(2r-1)(2r+1)(2r+3)}$ — hint: partial fractions into a telescoper.

Answers: 1. $\frac{1}{2} - \frac{1}{n+2} = \frac{n}{2(n+2)}$ | 2. $\frac{n}{2n+1}$ | 3. $(n+1)! - 1$ | 4. $\frac{n(n+3)}{4(n+1)(n+2)}$,
 $\rightarrow \frac{1}{4}$ | 5. $\frac{1}{8} \left[\frac{1}{3} - \frac{1}{(2n+1)(2n+3)} \right]$ (after partial fractions; telescopes in the product-pairs).

7. MCQ with Distractor Analysis

Q1. $\sum_{r=1}^n \frac{1}{r(r+1)}$ equals: (A) $\frac{1}{n(n+1)}$ (B) $\frac{n}{n+1}$ (C) $\frac{n+1}{n}$ (D) $1 + \frac{1}{n+1}$

Correct: (B). $1 - \frac{1}{n+1} = \frac{n}{n+1}$. **(A)** is the last term, not the sum. **(C)** inverts the answer. **(D)** adds instead of subtracts the endpoint (telescope-endpoint-sign). *Tests the telescoping result and endpoint handling.*

Q2. To sum $\sum \frac{1}{r(r+1)}$, the correct first step is: (A) apply $\sum n^2$ formula (B) write $\frac{1}{r} - \frac{1}{r+1}$ (C) use the GP sum (D) use the AGP method

Correct: (B). Partial fractions create the telescoper. **(A)** (special-term-recognition failure) applies a polynomial formula to a fraction. **(C),(D)** are the wrong machinery. *Tests technique selection — the complement of Unit 7.*

Q3. $\sum_{r=1}^n \frac{1}{\sqrt{r} + \sqrt{r+1}}$ equals: (A) $\sqrt{n} - 1$ (B) $\sqrt{n+1} - 1$ (C) $\sqrt{n+1} + 1$ (D) $1 - \frac{1}{\sqrt{n+1}}$

Correct: (B). Rationalising gives $\sqrt{r+1} - \sqrt{r}$, telescoping to $\sqrt{n+1} - 1$. **(A)** is an off-by-one endpoint slip. **(C)** wrong sign on the endpoint. **(D)** misremembers the form. *Tests the surd telescoper.*

8. Common Pitfalls, Pro-Tips & Cheat Sheet

Common Pitfalls (candidate mistake-tags):

- **telescope-no-difference-form:** Failing to rewrite t_r as a difference (trying to brute-force or apply polynomial $\sum$ formulas to a fraction/surd). The dominant error.
- **telescope-endpoint-sign:** Off-by-one or wrong sign on the surviving endpoints ($V_1 - V_{n+1}$).
- **telescope-lost-multiplier:** Dropping the $\frac{1}{2}$ (or $\frac{1}{k}$) constant that the difference-form carries.
- **vn-method-divisor:** Using the wrong divisor in the V_n method (divide by $k+1$ for k factors, $\times$ the common difference).

Schema note: telescope-no-difference-form is the direct counterpart of Unit 7's special-term-recognition — both are “didn't transform t_r into a summable shape.” A shared parent (summation-wrong-transform) would unify the two summation units in the knowledge-tracing graph.

Pro-Tips:

- For a **fraction**, try partial fractions and look for a consecutive-shift difference.
- For a **product of consecutive terms**, use the V_n method (append a factor, divide by count $+1$).
- For **factorials**, look for $t_r = f(r+1) - f(r)$ with f a factorial expression.
- For **surds in a denominator**, rationalise — it almost always telescopes.
- After telescoping, the answer is *always* just (first surviving term) — (last surviving term).

Cheat Sheet Box — Telescoping & Partial Fractions

Series	Difference-form / Result
$\sum \frac{1}{r(r+1)}$	$\frac{1}{r} - \frac{1}{r+1} = \frac{n}{n+1}$
$\sum \frac{1}{r(r+1)(r+2)}$	$\frac{n(n+3)}{4(n+1)(n+2)}$
$\sum \frac{1}{(2r-1)(2r+1)}$	$\frac{n}{2n+1}$
$\sum r(r+1) \cdots (r+k-1)$	$\frac{n(n+1) \cdots (n+k)}{k+1}$ (V_n method)

Series	Difference-form / Result
$\sum r \cdot r!$	$(n+1)! - 1$
$\sum \frac{1}{\sqrt{r} + \sqrt{r+1}}$	$\sqrt{n+1} - 1$
$\sum \frac{2r+1}{r^2(r+1)^2}$	$1 - \frac{1}{(n+1)^2}$
Telescoping principle	$t_r = V_r - V_{r+1} \Rightarrow \sum = V_1 - V_{n+1}$

One-liner: rewrite each term as a difference of consecutive things; the middle collapses, only the ends survive.

Chapter 8: Sequences and Series — Supplement

This supplement completes four items that the main chapter names but never delivers, and exercises three formulas that were printed but never used. It is organised as one new full-spine unit — **Unit 9 — Growth, Decay and the Product of a GP** — followed by five short addenda, each of which slots into the unit named in its heading.

Notation is chained from Units 1–8 throughout: a is a first term, r a common ratio, d a common difference, a_n (or t_n) an n th term, S_n the sum of n terms, and A , G , H the arithmetic, geometric and harmonic means.

Unit 9 — Growth, Decay and the Product of a GP

Exam Relevance: This unit delivers the two GP items that CBSE and JEE ask for and that Units 1–8 do not work. CBSE Boards set a compound-growth word problem — interest, depreciation, population, bacterial doubling, instalment savings — as a reliable 5-mark question, and the formula $A = P \left(1 + \frac{R}{100}\right)^n$ is directly askable. JEE Main draws on the same growth models in disguised form and, separately, on the product of the terms of a GP: the relation $P^2 R^n = S^n$ linking the sum, the product and the sum of reciprocals is an NCERT miscellaneous-exercise result that reappears as a one-line JEE Main question almost every cycle. JEE Advanced pushes the growth models into loan-repayment and drawdown problems where a GP sum sits inside a recurrence, and pushes the product identity into problems where the product is the only accessible quantity. NEET does not test mathematics.

Section 1: Concept Intuition

A quantity that changes by a fixed *percentage* each period is a GP, not an AP. That single sentence is the whole of the first half of this unit, and it is the sentence students skip.

Think about it this way. Your father pays a fixed Rs 2,000 every month into a recurring deposit. The *amount he has paid in* grows by the same Rs 2,000 each month — an AP, Unit 2. But a fixed deposit of Rs 2,000 left alone at 8% per annum grows by 8% **of whatever is currently there**, which is a bigger rupee amount every year: Rs 160 in year one, Rs 172.80 in year two, Rs 186.62 in year three. Equal *steps* make an AP. Equal *factors* make a GP. Interest, inflation, depreciation, population, bacterial cultures and radioactive decay are all quoted as percentages, so they are all GPs.

The bridge to Unit 3 is mechanical. If a quantity is P now and grows by $R\%$ per period, then after one period it is $P(1 + \frac{R}{100})$, after two periods $P(1 + \frac{R}{100})^2$, and so on. The successive values $P, P(1+i), P(1+i)^2, \dots$ with $i = R/100$ form a GP with first term P and common ratio $1+i$, and every formula from Unit 3 becomes available at once. Decay is the same statement with $r = 1 - \frac{R}{100} < 1$: a bike losing 20% of its value each year has $r = 0.8$, and the shrinking GP of Unit 3 is exactly a depreciation schedule.

Where the model fails, and it fails in exactly one place: the percentage must be applied to the *current* value, not to the original one. If a machine loses “20% of its original cost” every year, the losses are equal rupee amounts and the values form an AP — that is straight-line depreciation and the GP formula is wrong for it. Read the sentence, not the number. The words “on the reducing balance”, “of its value at the beginning of the year”, or “compounded” all signal a GP. The words “of the original cost” signal an AP.

The second half of this unit is about a quantity you have been computing term by term without ever naming it: the product. In Unit 3 you met the product of n inserted GMs. That was a special case of a completely general fact. Line up a GP and pair the terms from the outside in — first with last, second with second-last, and so on. Every such pair has the same product $a \cdot l$, because as you move one step in from the left you multiply by r and as you move one step in from the right you divide by r . This is the multiplicative twin of Gauss’s reversal trick from Unit 2, where every outside-in pair of an AP has the same *sum*. **In an AP, outside-in pairs share a sum; in a GP, outside-in pairs share a product.** Gauss’s observation gives $S_n = \frac{n}{2}(a + l)$; this one gives $P = (al)^{n/2}$.

Once you see that, the celebrated identity $P^2 R^n = S^n$ stops being a formula to memorise. The sum-of-reciprocals R belongs to the GP of reciprocals, whose first and last terms are $1/a$ and $1/l$. Dividing S by R therefore cancels everything except al — and al is precisely the quantity that P is built from.

Figure brief: Two parallel horizontal strips. Top strip: an AP written out as $a, a+d, \dots, l-d, l$ with curved arcs joining first-to-last, second-to-second-last, each arc labelled “sum = $a + l$ ”. Bottom strip: a GP written out as $a, ar, \dots, l/r, l$ with the same curved arcs, each labelled “product = al ”. A caption box between them reads “AP: pairs share a sum, giving $S_n = \frac{n}{2}(a+l)$. GP: pairs share a product, giving $P = (al)^{n/2}$.”

Section 2: Key Formulas and Definitions

Compound growth and decay

Let P be the initial (principal) value, R the rate per cent per period, $i = \frac{R}{100}$ the rate as a decimal, and n the number of complete periods.

$$A = P \left(1 + \frac{R}{100}\right)^n, \quad \text{CI} = A - P = P \left[\left(1 + \frac{R}{100}\right)^n - 1 \right].$$

- A — the amount (value) after n periods.

- CI — the compound interest earned, i.e. the growth alone.
- The successive amounts $P, P(1+i), P(1+i)^2, \dots$ form a GP with first term P and common ratio $1+i$.

Validity: the rate must apply to the *current* value each period, and n counts **complete** compounding periods. If n is not a whole number of periods the formula is an approximation, and boards never ask for that case.

Situation	Common ratio r	Value after n periods
Compound interest at $R\%$ per annum, compounded annually	$1 + \frac{R}{100}$	$P \left(1 + \frac{R}{100}\right)^n$
Compounded k times a year, n years	$1 + \frac{R}{100k}$	$P \left(1 + \frac{R}{100k}\right)^{nk}$
Depreciation at $R\%$ per annum on the reducing balance	$1 - \frac{R}{100}$	$P \left(1 - \frac{R}{100}\right)^n$
Population / bacteria growing at $R\%$ per period	$1 + \frac{R}{100}$	$N_0 \left(1 + \frac{R}{100}\right)^n$
Quantity doubling every T units of time, elapsed time t	2 per doubling	$N_0 2^{t/T}$
Half-life T , elapsed time t	$\frac{1}{2}$ per half-life	$N_0 \left(\frac{1}{2}\right)^{t/T}$

Figure brief: A single set of axes, time in years on the horizontal axis and value in rupees on the vertical. Three curves from a common starting point P on the vertical axis: a straight line rising steadily labelled “simple interest (AP), $P \left(1 + \frac{nR}{100}\right)$ ”; a curve rising above it with increasing steepness labelled “compound interest (GP), $P \left(1 + \frac{R}{100}\right)^n$ ”; and a curve falling from P and flattening towards but never touching the horizontal axis, labelled “depreciation, $P \left(1 - \frac{R}{100}\right)^n$ ”. Mark the widening vertical gap between the first two curves and annotate it “the gap is the effect of compounding”.

Note the contrast with **simple interest**, which is an AP: $A_{SI} = P \left(1 + \frac{nR}{100}\right)$, growing by the equal step $\frac{PR}{100}$ each period. Simple interest is Unit 2 material; compound interest is Unit 3 material. Deciding which one a question means is worth more marks than any algebra in this unit.

Instalment sums (SIP / annuity)

Suppose a fixed amount Q is deposited once per period for n periods at rate i per period, compounded once per period. The deposits have been in the account for different lengths of time, so the maturity values form a GP.

$$\text{Deposits at the **end** of each period: } M = Q \frac{(1+i)^n - 1}{i}.$$

Deposits at the **beginning** of each period: $M = Q(1+i) \frac{(1+i)^n - 1}{i}$.

- M — the total accumulated immediately after the n th period ends.
- The two differ by exactly one extra period of interest on every deposit, hence the single factor $(1+i)$.

Validity: $i \neq 0$; for $i = 0$ the total is simply nQ .

Product of the terms of a GP

Let $a_1, a_2, \dots, a_n$ be a GP with first term a , common ratio r and last term $l = ar^{n-1}$. Write

$$S = \sum_{k=1}^n a_k, \quad P = \prod_{k=1}^n a_k, \quad R = \sum_{k=1}^n \frac{1}{a_k}.$$

$$P = a^n r^{n(n-1)/2} = (al)^{n/2}$$

$$\frac{S}{R} = al = a^2 r^{n-1}$$

$$P^2 R^n = S^n$$

- S — sum of the n terms; P — their product; R — the sum of their reciprocals.
- al — the product of the first and last terms, the constant shared by every outside-in pair.

Validity and reading conditions:

1. Every term is non-zero and $r \neq 0$, as always for a GP (Unit 3). Then R is defined and $P \neq 0$.
2. $P = (al)^{n/2}$ carries a square root when n is odd. For a GP of **positive** terms this is unambiguous. For a GP with $r < 0$, use $P = a^n r^{n(n-1)/2}$, which is always exact and always signed correctly.
3. $P^2 R^n = S^n$ holds for every GP, including $r = 1$ (then $S = na$, $P = a^n$, $R = n/a$, and both sides equal $n^n a^n$). It is one of the few GP identities with no case split.
4. $P^2 = (S/R)^n$ determines P only up to sign. If the question says “positive terms”, take the positive root; otherwise you must determine the sign separately.

A companion result on general terms

If x, y, z are the p th, q th and r th terms of a GP, then

$$x^{q-r} y^{r-p} z^{p-q} = 1.$$

This is the multiplicative image of the AP result of Unit 2 and is proved the same way, by writing $x = AR^{p-1}$, $y = AR^{q-1}$, $z = AR^{r-1}$ and checking that both exponents collapse to zero.

Section 3: CBSE Board Derivations and Key Procedures

Derivation 1 — the compound-growth formula $A = P(1 + i)^n$

Step 1. Let $A_0 = P$ be the value at the start, and let the value grow by $R\%$ of its current size each period, with $i = R/100$. (This is the modelling assumption; everything else is algebra.)

Step 2. In one period the value gains iA_0 , so $A_1 = A_0 + iA_0 = A_0(1 + i)$. (Growth is proportional to the current value.)

Step 3. The same argument applied to A_1 gives $A_2 = A_1(1 + i)$, and in general $A_k = A_{k-1}(1 + i)$ for every $k \geq 1$. (The rate does not change from period to period.)

Step 4. Hence $\frac{A_k}{A_{k-1}} = 1 + i$ is constant, which is exactly the definition of a GP with common ratio $1 + i$ (Unit 3, Section 3).

Step 5. By the n th-term formula $a_n = ar^{n-1}$ with $a = A_0 = P$ and $r = 1 + i$, the value after n periods — which is the $(n + 1)$ th term of the sequence $A_0, A_1, A_2, \dots$ — is

$$A_n = P(1 + i)^n = P \left(1 + \frac{R}{100} \right)^n. \quad \blacksquare$$

Step 6. For decay at $R\%$ per period, replace the gain iA_{k-1} by a loss, giving $r = 1 - \frac{R}{100}$ and $A_n = P \left(1 - \frac{R}{100} \right)^n$. Since $0 < r < 1$, this is a shrinking GP and $A_n \rightarrow 0$, which is why depreciation never reaches zero in finite time. $\blacksquare$

Index warning, and it costs marks. The sequence of values starts at $A_0 = P$. The value after n periods is A_n , which is the $(n + 1)$ th term of that sequence. Any time a question is phrased in terms of “the n th term”, recount from the start.

Derivation 2 — the instalment (SIP) sum

Step 1. Suppose Q is deposited at the **end** of each of n periods, at rate i per period. (Set the timing convention first; everything hinges on it.)

Step 2. The deposit made at the end of period k sits in the account for $n - k$ further periods, so at the end of period n it is worth $Q(1 + i)^{n-k}$. (Derivation 1 applied to that one deposit.)

Step 3. Summing over $k = 1, 2, \dots, n$ gives

$$M = Q \left[(1 + i)^{n-1} + (1 + i)^{n-2} + \dots + (1 + i) + 1 \right],$$

a GP with first term 1, common ratio $1 + i$ and n terms. (Read the bracket right-to-left.)

Step 4. By the Unit 3 sum formula with $r = 1 + i \neq 1$,

$$M = Q \frac{(1 + i)^n - 1}{(1 + i) - 1} = Q \frac{(1 + i)^n - 1}{i}. \quad \blacksquare$$

Step 5. If instead each deposit is made at the **beginning** of its period, every deposit earns one extra period of interest, so multiply throughout by $(1 + i)$:

$$M = Q(1+i) \frac{(1+i)^n - 1}{i}. \quad \blacksquare$$

Derivation 3 — the product of n terms of a GP

Step 1. Write the terms as $a, ar, ar^2, \dots, ar^{n-1}$. (Standard form, Unit 3.)

Step 2. Multiply them:

$$P = a \cdot ar \cdot ar^2 \dots ar^{n-1} = a^n r^{0+1+2+\dots+(n-1)}.$$

(Collect the n copies of a and add the exponents of r .)

Step 3. The exponent is the sum of the first $n-1$ natural numbers together with 0, i.e. $\frac{(n-1)n}{2}$ by Unit 7. Hence

$$P = a^n r^{n(n-1)/2}.$$

Step 4. Since $l = ar^{n-1}$, we have $al = a^2 r^{n-1}$, and therefore

$$(al)^{n/2} = (a^2 r^{n-1})^{n/2} = a^n r^{n(n-1)/2} = P. \quad \blacksquare$$

Step 5 (the pairing reading). Equivalently, $a_k \cdot a_{n+1-k} = ar^{k-1} \cdot ar^{n-k} = a^2 r^{n-1} = al$ for every k . Multiplying the n terms in outside-in pairs therefore multiplies $n/2$ copies of al . ■

Derivation 4 — the identity $P^2 R^n = S^n$

Step 1. The reciprocals $\frac{1}{a}, \frac{1}{ar}, \dots, \frac{1}{ar^{n-1}}$ themselves form a GP, with first term $\frac{1}{a}$ and common ratio $\frac{1}{r}$. (Reciprocating a GP gives a GP — the analogue of the reciprocation move of Unit 4.)

Step 2. For $r \neq 1$, by the Unit 3 sum formula,

$$S = a \frac{r^n - 1}{r - 1}, \quad R = \frac{1}{a} \cdot \frac{1 - r^{-n}}{1 - r^{-1}} = \frac{r^n - 1}{a r^{n-1} (r - 1)}.$$

Step 3. Divide. The bracket $\frac{r^n - 1}{r - 1}$ is common to both and cancels:

$$\frac{S}{R} = a \cdot a r^{n-1} = a^2 r^{n-1} = al.$$

(This is the structural heart of the identity: S and R differ only by the factor al .)

Step 4. By Derivation 3, $P = (al)^{n/2}$, so $P^2 = (al)^n$. Substituting $al = S/R$,

$$P^2 = \left(\frac{S}{R}\right)^n \implies P^2 R^n = S^n. \quad \blacksquare$$

Step 5 (the case $r = 1$). Then $S = na$, $P = a^n$ and $R = n/a$, so $P^2 R^n = a^{2n} \cdot \frac{n^n}{a^n} = (na)^n = S^n$. The identity survives. ■

Procedure A: solving a compound-growth or depreciation word problem

1. **Identify the base of the percentage.** “On the reducing balance”, “of its value at the beginning of the year”, “compounded” $\Rightarrow$ GP. “Of the original cost”, “simple interest” $\Rightarrow$ AP. *Why: this single reading decides which formula is legal; nothing later can repair a wrong choice.*
2. **Write down P , r and n explicitly, with units.** For growth $r = 1 + \frac{R}{100}$; for decay $r = 1 - \frac{R}{100}$; for k compoundings a year over y years, $r = 1 + \frac{R}{100k}$ and $n = ky$. *Why: over 90% of lost marks here are a wrong n or a rate not divided by the compounding frequency.*
3. **Count periods from the start, not from the first term.** The value after n periods is Pr^n , with exponent n — not $n - 1$. *Why: the n th-term formula ar^{n-1} tempts you to subtract one; here the first term is the value at time zero.*
4. **Convert the percentage to a clean fraction before powering.** $8\% \Rightarrow \frac{27}{25}$, 20% decay $\Rightarrow \frac{4}{5}$, $10\% \Rightarrow \frac{11}{10}$. *Why: board problems are engineered so that the fraction cancels the principal exactly, turning a calculator problem into mental arithmetic.*
5. **If the unknown is n , take logarithms and then round upward.** From $r^n > k$ you get $n > \frac{\log k}{\log r}$, and since n counts whole periods, the answer is the next integer above. *Why: rounding a value like 6.58 down to 6 is the standard trap; at $n = 6$ the condition is not yet met.*
6. **Answer the question that was asked.** “Find the compound interest” wants $A - P$, not A . *Why: a correct A with the subtraction missing scores zero on the final mark.*

Procedure B: using S , P and R

1. **Check that the question hands you two of the three quantities S , P , R and asks for the third.** That signature is the whole trigger for this identity. *Why: if you are given a and r instead, computing directly is faster than invoking the identity.*
2. **Compute $al = S/R$ first.** It is the single most useful intermediate, and it is often all you need. *Why: it converts two given sums into one product, killing both unknowns a and r at once.*
3. **Raise to the power $n/2$:** $P = \pm(al)^{n/2}$. *Why: this is Derivation 3, and it avoids ever solving for a and r separately.*
4. **Fix the sign.** If the terms are stated to be positive, take the $+$ sign. Otherwise decide the sign from $P = a^n r^{n(n-1)/2}$. *Why: $P^2 = (S/R)^n$ is a squared statement and cannot see sign.*
5. **Sanity-check on a small explicit GP.** If your n is 3 or 4, reconstruct one candidate GP and verify S and R directly. *Why: it costs fifteen seconds and catches an exponent slip, which is the only error this method admits.*

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

A machine is bought for Rs 15,625.

- (a) Its value depreciates at 20% per annum on the reducing balance. Find its value at the end of 5 years.
- (b) The buyer instead invests the same Rs 15,625 at 8% per annum compound interest, compounded annually. Find the amount and the compound interest after 3 years.

Solution.

(a) The phrase “on the reducing balance” fixes the model as a GP (Procedure A, step 1). Here $P = 15625$, $R = 20$ so

$$r = 1 - \frac{20}{100} = \frac{4}{5}, \quad n = 5.$$

$$\text{Value} = 15625 \left(\frac{4}{5}\right)^5 = 15625 \cdot \frac{1024}{3125}.$$

Now $15625 = 5^6$ and $3125 = 5^5$, so $\frac{15625}{3125} = 5$ and

$$\text{Value} = 5 \times 1024 = \text{Rs } 5,120.$$

(b) Now $r = 1 + \frac{8}{100} = \frac{27}{25}$ and $n = 3$, so

$$A = 15625 \left(\frac{27}{25}\right)^3 = 15625 \cdot \frac{19683}{15625} = \text{Rs } 19,683,$$

because $15625 = 25^3$ exactly. The compound interest is the growth alone (Procedure A, step 6):

$$\text{CI} = A - P = 19683 - 15625 = \text{Rs } 4,058.$$

Check. Year by year at 8%: $15625 \rightarrow 16875 \rightarrow 18225 \rightarrow 19683$. Each step multiplies by 1.08, and the third value agrees.

Example 2 — [JEE Main Level]

The sum of four terms of a GP of positive numbers is 30 and the sum of their reciprocals is $\frac{15}{16}$. Find the product of the four terms.

Solution. The question hands over S and R and asks for P — the signature of Procedure B.

$$S = 30, \quad R = \frac{15}{16}, \quad n = 4.$$

Step 2 of Procedure B: compute the product of the first and last terms,

$$al = \frac{S}{R} = \frac{30}{15/16} = 30 \times \frac{16}{15} = 32.$$

Step 3: since $P = (al)^{n/2}$ with $n = 4$,

$$P = 32^{4/2} = 32^2 = \mathbf{1024}.$$

The terms are positive, so the positive root is the right one (step 4).

Check by reconstruction (step 5). With $al = 32$ and $S = 30$, one candidate is $a = 2$, $l = 16$, i.e. $r = 2$: the GP is 2, 4, 8, 16. Then $S = 30$ and $R = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} = \frac{15}{16}$, both as given, and the product is $2 \cdot 4 \cdot 8 \cdot 16 = 1024$.

Notice that the identity delivered the answer without ever solving for a and r . That is the point of it.

Example 3 — [JEE Main — Speed Trap]

A man deposits Rs 10,000 at the **beginning** of each year for 3 years in a bank paying 10% per annum compounded annually. Find the amount standing to his credit at the end of the third year.

The trap: the timing of the last deposit. The seductive answer is

$$M = 10000 \frac{(1.1)^3 - 1}{0.1} = 10000 \times \frac{0.331}{0.1} = \mathbf{Rs\ 33,100} \quad \text{— wrong.}$$

That is the standard annuity formula, and it is the formula for deposits made at the **end** of each year. Under that convention the third deposit is made at the very last instant and earns nothing at all, which is why the bracket bottoms out at 1. Here the deposits are made at the *beginning* of each year, so even the last one earns a full year of interest. Every deposit has been in the account one year longer than the formula assumes.

Correct solution. The deposit made at the beginning of year k has been invested for $4 - k$ years by the end of year 3, so:

Deposit	Years invested	Maturity value
Beginning of year 1	3	$10000(1.1)^3 = 13310$
Beginning of year 2	2	$10000(1.1)^2 = 12100$
Beginning of year 3	1	$10000(1.1)^1 = 11000$

These maturity values form a GP with $a = 11000$, $r = 1.1$, $n = 3$ (read the column upward). Hence

$$M = 11000 \frac{(1.1)^3 - 1}{1.1 - 1} = 11000 \times \frac{0.331}{0.1} = 11000 \times 3.31 = \mathbf{Rs\ 36,410}.$$

Equivalently, by Derivation 2, $M = Q(1+i) \frac{(1+i)^n - 1}{i} = 10000 \times 1.1 \times 3.31 = 36410$.

How to never fall for it. The two answers differ by exactly the factor 1.1: $33100 \times 1.1 = 36410$. Before you write anything, ask one question — *does the last deposit earn any interest?* If yes, deposits are at the beginning and the extra factor $(1 + i)$ is required. If no, deposits are at the end and it is not.

Example 4 — [JEE Advanced Level]

A retired teacher deposits Rs 1,00,000 in a scheme paying 8% per annum compounded annually. At the end of each year, immediately after the interest is credited, she withdraws Rs 10,000. Let B_n be the balance immediately after the n th withdrawal.

- Find a closed form for B_n .
- Determine the first year in which the fund cannot meet the full withdrawal, and find the size of that final withdrawal.

(Use $\log_{10} 2 = 0.3010$, $\log_{10} 1.08 = 0.0334$, and $(1.08)^{20} = 4.661$.)

Solution.

(a) Track the two flows separately and subtract — the standard decomposition for a draw-down problem.

The original principal, left undisturbed, grows to $100000(1.08)^n$ after n years (Derivation 1).

The withdrawal made at the end of year k removes Rs 10,000 from the fund; had it stayed, it would have grown for a further $n - k$ years to $10000(1.08)^{n-k}$. So the total shortfall caused by the n withdrawals, measured at the end of year n , is

$$10000 \sum_{k=1}^n (1.08)^{n-k} = 10000 [(1.08)^{n-1} + \dots + 1.08 + 1] = 10000 \frac{(1.08)^n - 1}{0.08},$$

a GP sum with n terms, first term 1 and ratio 1.08. Since $\frac{10000}{0.08} = 125000$, this equals $125000 [(1.08)^n - 1]$. Therefore

$$B_n = 100000(1.08)^n - 125000 [(1.08)^n - 1] = \boxed{125000 - 25000(1.08)^n}.$$

Reading the answer. The constant $125000 = \frac{10000}{0.08}$ is the *break-even corpus*: at exactly Rs 1,25,000 the annual interest is exactly Rs 10,000 and the fund would last forever. She started Rs 25,000 short of it, and that shortfall itself compounds at 8%, which is why B_n falls — slowly at first, then fast.

(b) The fund fails when $B_n \leq 0$, i.e.

$$25000(1.08)^n \geq 125000 \iff (1.08)^n \geq 5.$$

Taking logarithms to base 10, and using $\log_{10} 5 = 1 - \log_{10} 2 = 0.6990$,

$$n \geq \frac{\log_{10} 5}{\log_{10} 1.08} = \frac{0.6990}{0.0334} \approx 20.93.$$

Since n counts whole years, $B_{20} > 0$ and $B_{21} < 0$. **The withdrawal at the end of year 21 is the one that cannot be met in full.**

Its size: with $(1.08)^{20} = 4.661$,

$$B_{20} = 125000 - 25000(4.661) = 125000 - 116525 = \text{Rs } 8,475 \text{ (approximately).}$$

During year 21 this grows to $8475 \times 1.08 \approx \text{Rs } 9,154$, and that is all there is. So the final withdrawal is about **Rs 9,150** instead of Rs 10,000, and the fund closes at the end of the 21st year.

Why rounding up was mandatory. The inequality gave $n \geq 20.93$. Truncating to $n = 20$ would report failure in year 20, but B_{20} is comfortably positive. Whenever a growth or decay inequality is solved for a period count, the answer is the *next integer up* — Procedure A, step 5.

Section 5: Practice Exercises

Answers are given below; work the solutions yourself.

1. **(Board)** A car bought for Rs 6,25,000 depreciates at 20% per annum on the reducing balance. Find its value at the end of 4 years.
2. **(Board)** Find the amount and the compound interest on Rs 6,250 invested for 3 years at 8% per annum compounded annually.
3. **(Board)** A bacterial culture doubles every 3 hours. If it starts with 500 bacteria, how many are present after 24 hours?
4. **(Main)** The population of a town was 20,000 in the year 2020 and grows at 5% per annum. In which calendar year will the population first exceed 30,000? (Take $\log_{10} 1.5 = 0.1761$, $\log_{10} 1.05 = 0.0212$.)
5. **(Main)** A man deposits Rs 5,000 at the end of each year for 10 years at 10% per annum compounded annually. Find the amount at the end of the tenth year. (Take $(1.1)^{10} = 2.59374$.)
6. **(Main)** The sum of three positive numbers in GP is 21 and the sum of their reciprocals is $\frac{7}{12}$. Find their product, and then the numbers themselves.
7. **(Adv)** A machine depreciates at 10% per annum on the reducing balance. If its value at the end of 5 years is Rs 1,18,098, find its original cost.
8. **(Adv)** If x, y, z are respectively the p th, q th and r th terms of a GP, prove that $x^{q-r} y^{r-p} z^{p-q} = 1$.

Answers.

1. Rs 2,56,000.

2. Amount Rs 7,873.20; compound interest Rs 1,623.20.
3. $500 \times 2^8 = 1,28,000$ bacteria.
4. 2029. (Requires $n > 0.1761/0.0212 \approx 8.31$, so $n = 9$; the value at $n = 8$ is about 29,549, still below 30,000.)
5. Rs 79,687. (From $M = 5000 \frac{(1.1)^{10}-1}{0.1} = 5000 \times 15.9374$; the exact value is Rs 79,687.12.)
6. Product = 216; the numbers are 3, 6, 12 (in either order).
7. Rs 2,00,000.
8. Identity. (Write $x = AR^{p-1}$ etc.; the exponent of A is $(q-r) + (r-p) + (p-q) = 0$ and the exponent of R is $(p-1)(q-r) + (q-1)(r-p) + (r-1)(p-q) = 0$.)

Section 6: Crack the MCQ

Q1. The amount obtained by investing Rs 8,000 for 2 years at 5% per annum compounded annually is: (A) Rs 8,800 (B) Rs 8,820 (C) Rs 8,810 (D) Rs 9,000

Correct: (B). $A = 8000(1.05)^2 = 8000 \times 1.1025 = 8820$.

Distractor Analysis. (A) Rs 8,800 is the *simple* interest answer, $8000 + 2 \times 400$ — the student applied the 5% to the original principal both years, which is the AP model, not the GP model. This is the single most common error in the topic and Procedure A step 1 exists to prevent it. (C) Rs 8,810 comes from computing $8000 \times 1.05 = 8400$, then adding 5% of the *original* Rs 8,000 plus a further arithmetic slip; it signals a student mixing the two models mid-calculation. (D) Rs 9,000 is a rounding-to-a-nice-number guess with no derivation behind it.

Q2. A GP of positive terms has first term 2, last term 162 and exactly 5 terms. The product of the terms is: (A) 18^5 (B) 18^4 (C) 324^5 (D) 242

Correct: (A). $al = 2 \times 162 = 324$ and $n = 5$, so $P = (al)^{n/2} = 324^{5/2} = (18^2)^{5/2} = 18^5 = 1,889,568$. Direct check: $r^4 = l/a = 81$ gives $r = 3$, and $2 \cdot 6 \cdot 18 \cdot 54 \cdot 162 = 1,889,568$.

Distractor Analysis. (B) 18^4 comes from pairing the terms outside-in and forgetting that with n odd there is an unpaired middle term left over: two pairs give $324^2 = 18^4$, and the middle term 18 is dropped. This is precisely why the exponent is $n/2$ and not $\lfloor n/2 \rfloor$. (C) 324^5 comes from writing $P = (al)^n$ instead of $(al)^{n/2}$ — each pair counted twice. (D) 242 is the *sum* $S_5 = 2 \cdot \frac{3^5-1}{3-1}$, chosen by a student who read “product” but reached for the sum formula out of habit.

Q3. If S , P and R denote respectively the sum, the product and the sum of the reciprocals of n terms of a GP, then P^2 equals: (A) $\left(\frac{S}{R}\right)^n$ (B) $\left(\frac{R}{S}\right)^n$ (C) $(SR)^n$ (D) $S^n R^n$

Correct: (A). From $P^2 R^n = S^n$, dividing by R^n gives $P^2 = (S/R)^n$.

Distractor Analysis. (B) inverts the ratio — the student remembers that S and R appear as a quotient but not which way up. The fix is structural, not mnemonic: $S/R = al$, a *product of terms*, and P is built out of products, so S must sit on top. (C) and (D) are the same wrong answer written twice, and both come from mis-remembering $P^2 R^n = S^n$ as a symmetric product $P^2 = (SR)^n$; a student who tests the identity on the three-term GP 1, 2, 4 ($S = 7$, $P = 8$, $R = 7/4$) sees at once that $(S/R)^3 = 4^3 = 64 = P^2$, while $(SR)^3 = \left(\frac{49}{4}\right)^3 \approx 1838$, which

fails.

Q4. The value of a machine depreciates at 10% per annum on the reducing balance. After how many complete years does its value first fall below half its original value? (Take $\log_{10} 2 = 0.3010$, $\log_{10} 3 = 0.4771$.) (A) 6 (B) 7 (C) 5 (D) 10

Correct: (B). Require $(0.9)^n < 0.5$. Taking logarithms, $n \log_{10} 0.9 < \log_{10} 0.5$; both logarithms are negative, so the inequality flips on division: $n > \frac{\log_{10} 2}{-\log_{10} 0.9} = \frac{0.3010}{0.0458} \approx 6.58$. Hence $n = 7$. Confirming: $(0.9)^6 = 0.5314 > 0.5$ but $(0.9)^7 = 0.4783 < 0.5$.

Distractor Analysis. (A) 6 is the answer of a student who computed 6.58 correctly and then truncated instead of rounding up — at $n = 6$ the machine still holds 53.1% of its value, so the condition is not met. Period counts always round *up*. (C) 5 comes from dropping a step in the logarithm arithmetic. (D) 10 is pure linear thinking: “10% a year, so it is all gone in 10 years”, which would be true for straight-line depreciation on the original cost but is never true on a reducing balance — a shrinking GP never reaches zero.

Section 7: Common Student Pitfalls and Pro-Tips

Pitfalls.

- **Applying the percentage to the original value.** “Depreciates by 20% of its value each year” is a GP with $r = 0.8$; “depreciates by 20% of its original cost each year” is an AP with $d = -0.2P$. Correction: locate the base of the percentage in the sentence before you touch a formula.
- **Simple interest treated as compound, or the reverse.** Simple interest is an AP, $A = P(1 + \frac{nR}{100})$; compound interest is a GP, $A = P(1 + \frac{R}{100})^n$. Correction: the word “compounded” (or “on the amount”) is the switch.
- **Forgetting to divide the rate by the compounding frequency.** At 12% per annum compounded quarterly over 2 years, $r = 1.03$ and $n = 8$, not $r = 1.12$ and $n = 2$. Correction: rate per *period* and number of *periods* must be quoted in the same unit.
- **The off-by-one in the exponent.** The value after n periods is Pr^n . Writing Pr^{n-1} imports the n th-term formula into a place where the first term sits at time zero. Correction: count the number of *multiplications*, not the number of terms.
- **Using the end-of-period annuity formula for beginning-of-period deposits.** The two differ by the factor $(1 + i)$. Correction: ask whether the final deposit earns any interest.
- **Truncating instead of rounding up when solving for n .** From $r^n > k$ the answer is the smallest integer strictly above $\frac{\log k}{\log r}$. Correction: after rounding, substitute the integer back and confirm that the inequality holds.
- **Writing $P = (al)^{n/2}$ for a GP with $r < 0$ and losing the sign.** The formula $P = a^n r^{n(n-1)/2}$ is always exact; the pairing form is safe only for positive terms. Correction: for a signed GP, use the exponent form.
- **Reading $P^2 = (S/R)^n$ as determining P .** It determines $|P|$. Correction: fix the sign from the positivity of the terms or from $a^n r^{n(n-1)/2}$.

- **Reporting the amount when the compound interest was asked for.** Correction: reread the last line of the question before writing the final answer.

Pro-Tips.

- Convert the rate to an exact fraction before powering: $8\% \rightarrow \frac{27}{25}$, $10\% \rightarrow \frac{11}{10}$, 20% decay $\rightarrow \frac{4}{5}$, $25\% \rightarrow \frac{5}{4}$. Board principals such as 15,625, 6,250 or 6,25,000 are chosen so the denominator cancels exactly and no decimals ever appear.
- Memorise $(1.1)^{10} = 2.5937$, $(1.08)^{10} = 2.1589$ and the doubling rule of thumb “72 divided by the rate gives the doubling time in years” — the last is only a check, never a proof, but it catches order-of-magnitude errors instantly.
- The moment a question gives you two of $\{S, P, R\}$ for a GP, stop looking for a and r . Compute $al = S/R$ and finish in one line.
- $al = S/R$ is worth more than the full identity. It converts “sum and sum of reciprocals” into “product of the extremes”, which is the quantity that both the product formula and the GM condition are built on.
- For any drawdown or loan problem, compute the break-even corpus $\frac{\text{annual withdrawal}}{i}$ first. It tells you immediately whether the fund is sustainable, and the closed form is always $B_n = C - (C - P)(1 + i)^n$ where C is that corpus.
- Logs turn every “after how many periods” question into a one-line division, exactly as the GP-to-AP log bridge of Unit 3 predicts.

Section 8: Cheat Sheet Box

Growth, Decay and the Product of a GP — 60-second review

Quantity	Formula	Condition
Compound amount	$A = P \left(1 + \frac{R}{100}\right)^n$	rate on current value, n whole periods
Compound interest	$CI = A - P$	—
Compounded k times a year, y years	$A = P \left(1 + \frac{R}{100k}\right)^{ky}$	rate and periods in the same unit
Depreciation (reducing balance)	$V = P \left(1 - \frac{R}{100}\right)^n$	$0 < r < 1$, never reaches 0
Simple interest (contrast: this is an AP)	$A = P \left(1 + \frac{nR}{100}\right)$	rate on original value
Doubling every T / half-life T	$N = N_0 2^{t/T} / N = N_0 2^{-t/T}$	t, T in the same unit
Instalments at end of each period	$M = Q \frac{(1+i)^n - 1}{i}$	$i \neq 0$
Instalments at start of each period	$M = Q(1+i) \frac{(1+i)^n - 1}{i}$	one extra period of interest

Quantity	Formula	Condition
Drawdown balance	$B_n = C - (C - P)(1 + i)^n,$ $C = \frac{\text{withdrawal}}{i}$	withdrawal at period end
Product of n GP terms	$P = a^n r^{n(n-1)/2} = (al)^{n/2}$	$(al)^{n/2}$ form for positive terms
Outside-in pairing	$a_k a_{n+1-k} = al$ for every k	GP
Sum over sum-of-reciprocals	$\frac{S}{R} = al = a^2 r^{n-1}$	no term zero
The identity	$P^2 R^n = S^n$, i.e. $P^2 = (S/R)^n$	every GP, including $r = 1$
Terms of a GP at positions p, q, r	$x^{q-r} y^{r-p} z^{p-q} = 1$	—

Triggers. “Compounded / reducing balance / of its current value” $\Rightarrow$ GP. “Of the original cost / simple interest” $\Rightarrow$ AP. “After how many years” $\Rightarrow$ take logs, round **up**. Two of $\{S, P, R\}$ given $\Rightarrow$ compute $al = S/R$ first.

Addendum A — Mixed AP-GP-HP Chain Problems

(Slots into Unit 4, after Example 5. This is the material Unit 4 Section 1 names as “HP fused with AP/GP” and Unit 4 Example 5 begins to pose.)

Method

A “chain” problem gives you three or four letters and a *different* progression condition on each overlapping triple. The technique is uniform and mechanical: **replace every progression statement by its middle-term equation, then eliminate.**

Statement	Middle-term equation	Solved for the middle term
a, b, c in AP	$2b = a + c$	$b = \frac{a + c}{2}$
a, b, c in GP	$b^2 = ac$	$b = \pm\sqrt{ac}$
a, b, c in HP	$\frac{2}{b} = \frac{1}{a} + \frac{1}{c}$	$b = \frac{2ac}{a + c}$

Three rules make the elimination fast:

1. **Choose the letter that appears in two conditions and eliminate it.** In the chain a, b, c (AP), b, c, d (GP), c, d, e (HP), the letters b and d each appear in exactly two conditions; eliminating them leaves a relation among a, c, e alone.
2. **Write the HP condition in the reciprocal form $\frac{2}{b} = \frac{1}{a} + \frac{1}{c}$ when the letters are in denominators, and in the product form $b = \frac{2ac}{a+c}$ when you are about to cross-multiply.** They are the same statement; picking the convenient shape halves the algebra.

3. **Never expand more than you must.** Chain identities almost always collapse through a single cancellation, and expanding early buries it.

Scope conditions: no letter may be zero (an HP forbids it outright, and a GP forbids it too), and for the HP relation you additionally need $a + c \neq 0$.

Example A1 — [JEE Main Level] The classic three-progression chain

If a, b, c are in AP, b, c, d are in GP, and c, d, e are in HP, prove that a, c, e are in GP.

Solution. Write each condition in middle-term form:

$$\text{AP: } 2b = a + c \Rightarrow b = \frac{a + c}{2}; \quad \text{GP: } c^2 = bd \Rightarrow d = \frac{c^2}{b}; \quad \text{HP: } d = \frac{2ce}{c + e}.$$

The letter d appears in two of them, so eliminate it (rule 1):

$$\frac{c^2}{b} = \frac{2ce}{c + e}.$$

Cancel one factor of c (legal: $c \neq 0$, since c is a term of a GP):

$$\frac{c}{b} = \frac{2e}{c + e} \Rightarrow c(c + e) = 2be.$$

Now eliminate b using $2b = a + c$:

$$c(c + e) = (a + c)e \Rightarrow c^2 + ce = ae + ce \Rightarrow c^2 = ae.$$

By the GP condition of Unit 3, $c^2 = ae$ is exactly the statement that a, c, e are in GP. ■

What to notice. The whole proof is two eliminations and one cancellation of ce . Nothing was expanded that did not have to be. Note also the pleasing symmetry of the result: an AP, a GP and an HP chained together produce a GP — the middle progression survives, which is the AM–GM–HM ordering of Unit 5 asserting itself in disguise.

Example A2 — [JEE Main Level] AP fused with HP

If a, b, c are in AP and b, c, d are in HP, prove that $ad = bc$.

Solution. The two conditions are

$$2b = a + c \quad \text{and} \quad \frac{2}{c} = \frac{1}{b} + \frac{1}{d}.$$

Clear the second (rule 2, product form): $\frac{2}{c} = \frac{b + d}{bd}$, so

$$2bd = c(b + d) = bc + cd.$$

From the first condition, $a = 2b - c$. Therefore

$$ad = (2b - c)d = 2bd - cd = (bc + cd) - cd = bc. \quad \blacksquare$$

Numerical check. Take $a = 1, b = 2, c = 3$ (an AP). The HP condition on $2, 3, d$ gives $\frac{2}{3} = \frac{1}{2} + \frac{1}{d}$, so $\frac{1}{d} = \frac{1}{6}$ and $d = 6$. Then $ad = 6$ and $bc = 6$, as claimed.

Practice — Addendum A

1. Let a, b, c be in AP, b, c, d in GP and c, d, e in HP, with $a = 1$ and $b = 3$. Find c, d and e .
2. Let a, b, c be in AP and b, c, d in HP, with $a = 3$ and $b = 5$. Find d .
3. If a, b, c are in HP, prove that $\frac{b+a}{b-a} + \frac{b+c}{b-c} = 2$.
4. If $\frac{b+c}{a}, \frac{c+a}{b}, \frac{a+b}{c}$ are in AP and $a+b+c \neq 0$, prove that a, b, c are in HP.
5. The arithmetic mean of two positive numbers is twice their harmonic mean. Find the ratio of the numbers.
6. If a, b, c are in AP and a^2, b^2, c^2 are in HP, prove that either $a = b = c$, or $a, b, -\frac{c}{2}$ are in GP.

Answers.

1. $c = 5, d = \frac{25}{3}, e = 25$. (From $c = 2b - a, d = c^2/b$, and then $e = c^2/a$ by Example A1.)
2. $d = \frac{35}{3}$. (Here $c = 2b - a = 7$, and $ad = bc = 35$ by Example A2.)
3. Identity. (Reciprocate first, as Unit 4 always advises. Since $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ are in AP, write $\frac{1}{a} = x - D, \frac{1}{b} = x, \frac{1}{c} = x + D$. Then $b + a = \frac{2x-D}{x(x-D)}$ and $b - a = \frac{-D}{x(x-D)}$, so $\frac{b+a}{b-a} = \frac{2x-D}{-D}$. Likewise $\frac{b+c}{b-c} = \frac{2x+D}{D}$. Adding, the $2x$ terms cancel and the sum is $\frac{2D}{D} = 2$.)
4. Add 2 to each of the three terms. Each becomes $\frac{a+b+c}{a}, \frac{a+b+c}{b}, \frac{a+b+c}{c}$, still in AP because adding a constant to every term of an AP leaves an AP. Dividing through by $a+b+c \neq 0$ shows $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ are in AP, i.e. a, b, c are in HP.
5. $a : b = 3 + 2\sqrt{2} : 1$, equivalently $(\sqrt{2} + 1)^2 : 1$. (From $\frac{a+b}{2} = 2 \cdot \frac{2ab}{a+b}$ we get $(a+b)^2 = 8ab$, i.e. $a^2 - 6ab + b^2 = 0$; the other root $3 - 2\sqrt{2}$ is the same ratio inverted.)
6. The two conditions are $2b = a + c$ and $\frac{2}{b^2} = \frac{1}{a^2} + \frac{1}{c^2}$, i.e. $2a^2c^2 = b^2(a^2 + c^2)$. Substituting $b = \frac{a+c}{2}$ and clearing denominators gives $8a^2c^2 = (a+c)^2(a^2 + c^2)$, which rearranges and factors as

$$(a - c)^2(a^2 + 4ac + c^2) = 0.$$

If $a = c$ then $b = a$ too, so $a = b = c$. Otherwise $a^2 + 4ac + c^2 = 0$, i.e. $(a + c)^2 = -2ac$, i.e. $4b^2 = -2ac$, i.e. $b^2 = a \cdot \left(-\frac{c}{2}\right)$ — the GP condition for $a, b, -\frac{c}{2}$.

Addendum B — Alternating-Sign Series

(Slots into Unit 7. This is the “alternating-sign square sums” that Unit 7 Section 1 names as JEE Advanced content.)

Method

An alternating series $\sum_{r=1}^n (-1)^{r-1} t_r$ cannot be attacked with the standard $\sum r$, $\sum r^2$, $\sum r^3$ formulas of Unit 7, because the sign factor $(-1)^{r-1}$ is not a polynomial in r . The correct technique has three steps:

1. **Pair the terms from the front:** $(t_1 - t_2) + (t_3 - t_4) + \dots$. Each bracket is a *single* term of a new series indexed by k , where the k th bracket is $t_{2k-1} - t_{2k}$.
2. **Simplify the bracket.** If t_r is a polynomial of degree m , the bracket is a polynomial in k of degree $m - 1$ — the pairing has bought you one degree. Sum it with the ordinary Unit 7 formulas.
3. **Split on the parity of n .** If n is even, say $n = 2m$, the pairing is exact and you are done. If n is odd, apply the even result to $n - 1$ terms and add the last term t_n separately.

The one thing that decides marks here is step 3. An answer that does not distinguish odd n from even n is incomplete, and a single closed form using $(-1)^{n-1}$ is usually available and always preferable.

The three standard results, each proved below or by the identical argument:

$$\sum_{r=1}^n (-1)^{r-1} r = \begin{cases} \frac{n+1}{2}, & n \text{ odd,} \\ -\frac{n}{2}, & n \text{ even,} \end{cases}$$

$$\sum_{r=1}^n (-1)^{r-1} r^2 = (-1)^{n-1} \frac{n(n+1)}{2}$$

$$\sum_{r=1}^{2m} (-1)^{r-1} r^3 = -m^2(4m+3).$$

The middle one is worth committing to memory in exactly the boxed form: **the alternating sum of the first n squares is the ordinary sum of the first n naturals, with a sign.** It is one of the prettiest coincidences in the chapter.

Figure brief: A horizontal row of the terms $1^2, -2^2, 3^2, -4^2, 5^2, -6^2$ written with alternating tints for the positive and negative terms. Curved brackets join them in *front* pairs — $(1^2, -2^2)$, $(3^2, -4^2)$, $(5^2, -6^2)$ — each bracket labelled with its value $-3, -7, -11$, and a note beneath reading “brackets form an AP with $d = -4$ ”. To the right, a separate panel shows the odd case: the same six terms plus a trailing $+7^2$ standing outside every bracket, with an arrow labelled “odd n : pair the first $n - 1$, then add t_n on its own”.

Example B1 — [JEE Advanced Level] Alternating squares

Find $S_n = 1^2 - 2^2 + 3^2 - 4^2 + \dots$ to n terms, treating both parities of n , and then evaluate S_{15} and S_{20} .

Solution.

Case 1: n even, say $n = 2m$. Pair from the front (step 1):

$$S_{2m} = (1^2 - 2^2) + (3^2 - 4^2) + \dots + ((2m-1)^2 - (2m)^2).$$

The k th bracket is (step 2)

$$(2k-1)^2 - (2k)^2 = [(2k-1) - (2k)][(2k-1) + (2k)] = (-1)(4k-1) = -(4k-1),$$

using the difference of squares rather than expanding — much faster. Summing for $k = 1$ to m with the Unit 7 formulas:

$$S_{2m} = -\sum_{k=1}^m (4k-1) = -\left[4 \cdot \frac{m(m+1)}{2} - m\right] = -[2m^2 + 2m - m] = -m(2m+1).$$

Writing this back in terms of $n = 2m$, so that $m = n/2$ and $2m+1 = n+1$:

$$S_n = -\frac{n}{2}(n+1) = -\frac{n(n+1)}{2} \quad (n \text{ even}).$$

Case 2: n odd. Then $n-1$ is even, and the n th term $+n^2$ carries a plus sign (odd position). Applying Case 1 to $n-1$ terms (step 3):

$$S_n = S_{n-1} + n^2 = -\frac{(n-1)n}{2} + n^2 = n\left[n - \frac{n-1}{2}\right] = n \cdot \frac{2n - n + 1}{2} = \frac{n(n+1)}{2}.$$

Combined. The two cases differ only in sign, so

$$S_n = (-1)^{n-1} \frac{n(n+1)}{2}.$$

Evaluations. $S_{15} = (+1)\frac{15 \times 16}{2} = 120$, since 15 is odd. $S_{20} = (-1)\frac{20 \times 21}{2} = -210$, since 20 is even.

Check. $S_4 = 1 - 4 + 9 - 16 = -10$ and $-\frac{4 \times 5}{2} = -10$. $S_5 = -10 + 25 = 15$ and $\frac{5 \times 6}{2} = 15$. Both agree.

Example B2 — [JEE Advanced Level] Alternating cubes

Evaluate $1^3 - 2^3 + 3^3 - 4^3 + \dots - 10^3$.

Solution. There are 10 terms, an even count, so pairing is exact with $m = 5$. The k th bracket is

$$(2k-1)^3 - (2k)^3 = (8k^3 - 12k^2 + 6k - 1) - 8k^3 = -12k^2 + 6k - 1.$$

Summing for $k = 1$ to m using $\sum k^2$ and $\sum k$ from Unit 7:

$$\begin{aligned}\sum_{k=1}^m (-12k^2 + 6k - 1) &= -12 \cdot \frac{m(m+1)(2m+1)}{6} + 6 \cdot \frac{m(m+1)}{2} - m \\ &= -2m(m+1)(2m+1) + 3m(m+1) - m.\end{aligned}$$

Take out m and simplify the bracket:

$$= m [-2(2m^2 + 3m + 1) + 3m + 3 - 1] = m [-4m^2 - 6m - 2 + 3m + 2] = m (-4m^2 - 3m) = -m^2(4m+3).$$

With $m = 5$:

$$1^3 - 2^3 + \dots - 10^3 = -25 \times 23 = -\mathbf{575}.$$

Check on a small case. For $m = 1$ the formula gives $-1 \times 7 = -7$, and indeed $1 - 8 = -7$. For $m = 2$ it gives $-4 \times 11 = -44$, and $1 - 8 + 27 - 64 = -44$.

Practice — Addendum B

1. Evaluate $1^2 - 2^2 + 3^2 - \dots + 15^2$.
2. Evaluate $1^2 - 2^2 + 3^2 - \dots - 100^2$.
3. Find a closed form for $1 - 2 + 3 - 4 + \dots$ to n terms, distinguishing the parities of n .
4. Evaluate $1^3 - 2^3 + 3^3 - \dots - 8^3$.
5. Find the sum of $2n$ terms of the series $2^2 - 4^2 + 6^2 - 8^2 + \dots$.
6. Evaluate $1 - 3 + 5 - 7 + \dots$ to 100 terms.

Answers.

1. 120. (15 is odd, so the sum is $+\frac{15 \cdot 16}{2}$.)
2. -5050. (100 is even, so the sum is $-\frac{100 \cdot 101}{2}$.)
3. $\frac{n+1}{2}$ if n is odd, $-\frac{n}{2}$ if n is even. (Each front-pair contributes exactly -1 .)
4. -304. (Here $2m = 8$, so $m = 4$ and the sum is $-m^2(4m+3) = -16 \times 19$.)
5. $-4n(2n+1)$. (The r th term is $(-1)^{r-1}(2r)^2 = 4(-1)^{r-1}r^2$, so the sum is 4 times the alternating-square sum at $2n$ terms, namely $4 \times \left(-\frac{2n(2n+1)}{2}\right)$.)
6. -100. (Fifty front-pairs, each equal to -2 .)

Addendum C — Sum of Products Taken Two at a Time

(Slots into Unit 7, which prints this identity in its Key Formulas and cheat sheet but never uses it.)

Method

For any numbers $a_1, a_2, \dots, a_n$,

$$\left(\sum_{i=1}^n a_i\right)^2 = \sum_{i=1}^n a_i^2 + 2 \sum_{1 \leq i < j \leq n} a_i a_j \implies \sum_{1 \leq i < j \leq n} a_i a_j = \frac{(\sum a_i)^2 - \sum a_i^2}{2}.$$

The reason is a counting argument, not an algebraic trick: expanding $(a_1 + \dots + a_n)^2$ produces every ordered pair (i, j) exactly once. The n pairs with $i = j$ give $\sum a_i^2$; every unordered pair with $i \neq j$ appears twice, once as $a_i a_j$ and once as $a_j a_i$, which is where the factor 2 comes from.

The procedure is therefore always the same: compute the square of the sum, subtract the sum of the squares, halve. You never enumerate the pairs.

Example C1 — [JEE Main Level]

Find $\sum_{1 \leq i < j \leq n} i j$, the sum of the products of the first n natural numbers taken two at a time.

Solution. Apply the method with $a_i = i$, using the two standard sums of Unit 7:

$$\sum_{i=1}^n i = \frac{n(n+1)}{2}, \quad \sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}.$$

Hence

$$\sum_{i < j} i j = \frac{1}{2} \left[\frac{n^2(n+1)^2}{4} - \frac{n(n+1)(2n+1)}{6} \right].$$

Take out the common factor $n(n+1)$ before combining — this is what keeps the algebra short:

$$= \frac{n(n+1)}{2} \left[\frac{n(n+1)}{4} - \frac{2n+1}{6} \right] = \frac{n(n+1)}{2} \cdot \frac{3n(n+1) - 2(2n+1)}{12} = \frac{n(n+1)(3n^2 - n - 2)}{24}.$$

Factorise the quadratic: $3n^2 - n - 2 = (3n+2)(n-1)$. Therefore

$$\boxed{\sum_{1 \leq i < j \leq n} i j = \frac{n(n+1)(n-1)(3n+2)}{24}}$$

Check. For $n = 3$ the pairs are $1 \cdot 2, 1 \cdot 3, 2 \cdot 3$, summing to 11; the formula gives $\frac{3 \cdot 4 \cdot 2 \cdot 11}{24} = 11$. For $n = 4$ the six pairs sum to $2 + 3 + 4 + 6 + 8 + 12 = 35$; the formula gives $\frac{4 \cdot 5 \cdot 3 \cdot 14}{24} = 35$.

Note the factor $(n-1)$: at $n = 1$ there are no pairs at all, and the formula correctly returns 0.

Example C2 — [CBSE Board / JEE Main Level]

For a list of n numbers, $\sum a_i = 20$ and $\sum a_i^2 = 150$. Find $\sum_{i < j} a_i a_j$.

Solution. No information about the individual numbers is needed, and none is given — the identity uses only the two aggregates:

$$\sum_{i < j} a_i a_j = \frac{20^2 - 150}{2} = \frac{400 - 150}{2} = \mathbf{125}.$$

The point of this example is that the identity is *aggregate-to-aggregate*. Whenever a problem supplies a sum and a sum of squares, the sum of pairwise products is already determined, and no amount of extra information about the individual terms is required.

Practice — Addendum C

1. Evaluate $\sum_{1 \leq i < j \leq 10} i j$.
2. For six numbers, $\sum a_i = 24$ and $\sum a_i^2 = 130$. Find $\sum_{i < j} a_i a_j$.
3. Evaluate $\sum_{1 \leq i < j \leq 4} i^2 j^2$.
4. If a, b, c satisfy $a + b + c = 9$ and $a^2 + b^2 + c^2 = 35$, find $ab + bc + ca$.

Answers.

1. 1320. (Either from the boxed formula with $n = 10$, or directly as $\frac{55^2 - 385}{2} = \frac{3025 - 385}{2}$.)
2. 223. (From $\frac{576 - 130}{2}$.)
3. 273. (Apply the method to $a_i = i^2$: $\frac{(\sum i^2)^2 - \sum i^4}{2} = \frac{30^2 - 354}{2}$, where $\sum_{i=1}^4 i^4 = 354$ by Addendum D.)
4. 23. (From $\frac{81 - 35}{2}$.)

Addendum D — The Sum $\sum r^4$: Derivation and Use

(Slots into Unit 7, whose Derivation 2 remarks that the same machinery produces $\sum n^4$ but never carries it out.)

Derivation

The telescoping identity of Unit 7 generalises one power at a time. For $\sum n^2$ the engine was $k^3 - (k-1)^3$; for $\sum n^3$ it was $(k+1)^4 - k^4$. For $\sum n^4$ it is $k^5 - (k-1)^5$.

Step 1. Expand $(k-1)^5 = k^5 - 5k^4 + 10k^3 - 10k^2 + 5k - 1$, so

$$k^5 - (k-1)^5 = 5k^4 - 10k^3 + 10k^2 - 5k + 1.$$

Step 2. Sum both sides for $k = 1, 2, \dots, n$. The left side telescopes to $n^5 - 0^5 = n^5$. Writing $\Sigma_m = \sum_{k=1}^n k^m$:

$$n^5 = 5\Sigma_4 - 10\Sigma_3 + 10\Sigma_2 - 5\Sigma_1 + n.$$

Step 3. Substitute the three known sums from Unit 7 and solve for Σ_4 :

$$5\Sigma_4 = n^5 + 10 \cdot \frac{n^2(n+1)^2}{4} - 10 \cdot \frac{n(n+1)(2n+1)}{6} + 5 \cdot \frac{n(n+1)}{2} - n.$$

Step 4. Multiply through by 6 to clear denominators:

$$30\Sigma_4 = 6n^5 + 15n^2(n+1)^2 - 10n(n+1)(2n+1) + 15n(n+1) - 6n.$$

Expanding the three middle products gives $15n^4 + 30n^3 + 15n^2$, $20n^3 + 30n^2 + 10n$ and $15n^2 + 15n$ respectively, so

$$30\Sigma_4 = 6n^5 + 15n^4 + 10n^3 - n.$$

Step 5. Factorise. Taking out n and then $(n+1)$:

$$6n^5 + 15n^4 + 10n^3 - n = n(n+1)(2n+1)(3n^2 + 3n - 1),$$

which you may verify by expanding the right-hand side. Therefore

$$\boxed{\sum_{r=1}^n r^4 = \frac{n(n+1)(2n+1)(3n^2 + 3n - 1)}{30}} \quad \blacksquare$$

Check. At $n = 3$: $\frac{3 \cdot 4 \cdot 7 \cdot 35}{30} = 98$, and $1 + 16 + 81 = 98$. At $n = 4$: $\frac{4 \cdot 5 \cdot 9 \cdot 59}{30} = 354$, and $98 + 256 = 354$.

Note the structure: the first three factors $n(n+1)(2n+1)$ are exactly the numerator of $\sum r^2$, which makes the formula far easier to recall than it looks. Every one of $\sum r$, $\sum r^2$, $\sum r^3$, $\sum r^4$ carries the factor $n(n+1)$.

Example D1 — [JEE Advanced Level]

Find the sum to n terms of the series $1^2 \cdot 2^2 + 2^2 \cdot 3^2 + 3^2 \cdot 4^2 + \dots$

Solution. The general term is $t_r = r^2(r+1)^2$. This is a polynomial in r of degree 4, so the master technique of Unit 7 applies once $\sum r^4$ is available. Expand first:

$$t_r = r^2(r^2 + 2r + 1) = r^4 + 2r^3 + r^2.$$

By linearity of $\sum$,

$$S_n = \Sigma_4 + 2\Sigma_3 + \Sigma_2 = \frac{6n^5 + 15n^4 + 10n^3 - n}{30} + \frac{n^2(n+1)^2}{2} + \frac{n(n+1)(2n+1)}{6}.$$

Put everything over 30. The second term contributes $15n^2(n+1)^2 = 15n^4 + 30n^3 + 15n^2$ and the third contributes $5n(n+1)(2n+1) = 10n^3 + 15n^2 + 5n$, so the numerator is

$$6n^5 + 30n^4 + 50n^3 + 30n^2 + 4n = 2n(3n^4 + 15n^3 + 25n^2 + 15n + 2).$$

The quartic factors as $(n+1)(n+2)(3n^2+6n+1)$, which you can confirm by expanding. Hence

$$S_n = \frac{2n(n+1)(n+2)(3n^2+6n+1)}{30} = \boxed{\frac{n(n+1)(n+2)(3n^2+6n+1)}{15}}$$

Check. At $n = 2$: $\frac{2 \cdot 3 \cdot 4 \cdot 25}{15} = 40$, and $1 \cdot 4 + 4 \cdot 9 = 4 + 36 = 40$. At $n = 3$: $\frac{3 \cdot 4 \cdot 5 \cdot 46}{15} = 184$, and $40 + 9 \cdot 16 = 184$.

Pitfall to avoid. Do not try to telescope $r^2(r+1)^2$ by the Unit 8 method. It is a *polynomial*, not a reciprocal of a product, so the Unit 7 machinery is the right tool; reaching for partial fractions here wastes five minutes and produces nothing.

Practice — Addendum D

1. Evaluate $\sum_{r=1}^{10} r^4$.
2. Find $\sum_{r=1}^n r^2(r^2 - 1)$ in closed form.
3. Evaluate $\sum_{r=11}^{20} r^4$.

Answers.

1. 25,333. (From $\frac{10 \cdot 11 \cdot 21 \cdot 329}{30} = 77 \times 329$.)
2. $\frac{n(n-1)(n+1)(n+2)(2n+1)}{10}$. (Expand to $\Sigma_4 - \Sigma_2$ and factorise; the factor $(n-1)$ is correct because the $r = 1$ term is zero. Check at $n = 3$: $\frac{3 \cdot 2 \cdot 4 \cdot 5 \cdot 7}{10} = 84$, and $0 + 12 + 72 = 84$.)
3. 697,333. (Compute $\sum_1^{20} = \frac{20 \cdot 21 \cdot 41 \cdot 1259}{30} = 574 \times 1259 = 722,666$ and subtract the answer to question 1.)

Addendum E — Method of Differences When the Differences Form a GP

(Slots into Unit 7, whose “Method of differences” paragraph and Example 2 handle only the case where the first differences form an AP.)

Method

Unit 7 shows that when the first differences of a series form an **AP**, the general term t_n is a *quadratic* in n . The companion case — a standard JEE Main variant — is when the first differences form a **GP**. Then t_n contains an exponential term, and no polynomial fit will ever work.

The procedure is the same telescoping argument in both cases:

1. **Write down the first differences** $d_k = t_{k+1} - t_k$ and identify what kind of progression they form. Constant difference between the d_k means AP; constant *ratio* between the d_k means GP.
2. **Telescope.** Since $t_n = t_1 + (t_2 - t_1) + (t_3 - t_2) + \cdots + (t_n - t_{n-1})$,

$$t_n = t_1 + \sum_{k=1}^{n-1} d_k.$$

3. **Sum the $n - 1$ differences** with the appropriate formula — the AP sum if they are in AP, the GP sum if they are in GP.
4. **Verify on two given terms**, one of which was not used to build the formula.
5. **Then sum the series** by applying $\sum$ termwise to the closed form of t_n .

The diagnostic that decides everything is step 1. Differences 4, 6, 8, 10 have a constant *difference* of 2 — AP, quadratic t_n . Differences 2, 4, 8, 16 have a constant *ratio* of 2 — GP, and t_n contains 2^n . Fitting a quadratic to the second case will match three terms and fail at the fourth, which is exactly how the trap is set.

Example E1 — [JEE Main Level]

Find the general term and the sum to n terms of the series $5 + 7 + 11 + 19 + 35 + \cdots$

Solution.

Step 1. The first differences are 2, 4, 8, 16, ... — a constant ratio of 2, so they form a GP with first term 2 and ratio 2, not an AP.

Step 2 and 3. Telescoping, and summing $n - 1$ terms of that GP:

$$t_n = t_1 + \sum_{k=1}^{n-1} 2 \cdot 2^{k-1} = 5 + 2 \frac{2^{n-1} - 1}{2 - 1} = 5 + 2^n - 2 = 2^n + 3.$$

Step 4 (verify). $t_1 = 2 + 3 = 5$, $t_2 = 4 + 3 = 7$, $t_3 = 8 + 3 = 11$, $t_4 = 16 + 3 = 19$, $t_5 = 32 + 3 = 35$. All five given terms are reproduced, including the two that were not used in the construction.

Step 5. Sum termwise, splitting into a GP and a constant series:

$$S_n = \sum_{r=1}^n (2^r + 3) = (2 + 4 + \cdots + 2^n) + 3n = 2 \frac{2^n - 1}{2 - 1} + 3n = \boxed{2^{n+1} - 2 + 3n}$$

Check. $S_3 = 16 - 2 + 9 = 23$, and $5 + 7 + 11 = 23$.

Example E2 — [JEE Main Level]

Find the general term and the sum to n terms of the series $2 + 5 + 14 + 41 + 122 + \dots$

Solution. The first differences are 3, 9, 27, 81 — a GP with first term 3 and ratio 3. Telescoping,

$$t_n = 2 + \sum_{k=1}^{n-1} 3 \cdot 3^{k-1} = 2 + 3 \frac{3^{n-1} - 1}{3 - 1} = 2 + \frac{3^n - 3}{2} = \frac{4 + 3^n - 3}{2} = \frac{3^n + 1}{2}.$$

Verify: $t_1 = \frac{4}{2} = 2$, $t_3 = \frac{28}{2} = 14$, $t_5 = \frac{244}{2} = 122$. Correct.

Summing termwise,

$$S_n = \sum_{r=1}^n \frac{3^r + 1}{2} = \frac{1}{2} \left[3 \frac{3^n - 1}{3 - 1} + n \right] = \frac{3^{n+1} - 3}{4} + \frac{n}{2}.$$

Check. $S_3 = \frac{81-3}{4} + \frac{3}{2} = 19.5 + 1.5 = 21$, and $2 + 5 + 14 = 21$.

Practice — Addendum E

For each series, find t_n and S_n .

- 1, 3, 7, 15, 31, ...
- 4, 5, 7, 11, 19, ...
- 2, 6, 12, 20, 30, ...
- 1, 2, 5, 12, 27, 58, ... (*Look at the differences of the differences.*)

Answers.

- Differences 2, 4, 8, 16 (GP, ratio 2): $t_n = 2^n - 1$, $S_n = 2^{n+1} - 2 - n$.
- Differences 1, 2, 4, 8 (GP, ratio 2): $t_n = 2^{n-1} + 3$, $S_n = 2^n - 1 + 3n$.
- Differences 4, 6, 8, 10 (AP, not GP — this is the Unit 7 case): $t_n = n(n+1)$, $S_n = \frac{n(n+1)(n+2)}{3}$.
- Differences 1, 3, 7, 15, 31, whose own differences 2, 4, 8, 16 form a GP; hence $d_k = 2^k - 1$ and

$$t_n = 1 + \sum_{k=1}^{n-1} (2^k - 1) = 1 + (2^n - 2) - (n-1) = 2^n - n,$$

$$S_n = (2^{n+1} - 2) - \frac{n(n+1)}{2}.$$

(Check: $t_6 = 64 - 6 = 58$; $S_3 = 16 - 2 - 6 = 8 = 1 + 2 + 5$.)

Where This Material Belongs

Supplement item	Reads as an extension of	What it delivers
Unit 9, Sections 1-4 (growth half)	Unit 3, Sections 1-2	The compound-interest, depreciation, doubling and loan-repayment problems Unit 3 promises
Unit 9, Sections 2-4 (product half) Addendum A	Unit 3, Section 3 and Derivation 4 Unit 4, Section 5	$P = a^n r^{n(n-1)/2} = (al)^{n/2}$ and $P^2 R^n = S^n$ The three-progression chain problem named in Unit 4, Section 1
Addendum B	Unit 7, Sections 3-5	Alternating-sign square and cube sums, with the parity split
Addendum C	Unit 7, Section 3	A worked use of the pairwise-product identity
Addendum D	Unit 7, Derivation 2	The $\sum r^4$ derivation and an application
Addendum E	Unit 7, Method of differences	The differences-in-GP variant

Chapter 8: Sequences and Series — Round 2

Addendum

Units 1-8 and the Round 1 Supplement (Unit 9, Addenda A to E) together cover the whole NCERT syllabus and the standard JEE problem bank for this chapter, with two exceptions.

The first is a promise the book makes and does not keep. Unit 1, Procedure 4 is titled “recursive to explicit (when possible)” and then says only “unfold a few terms to spot the closed form.” Unit 1 Example 3 unfolds five terms of $a_1 = 2$, $a_n = 3a_{n-1} + 1$ and stops there. No method for turning a recurrence into a closed form is ever given, and the companion move — recovering a_n from a formula for S_n — is *used* twice (Unit 2, Derivation 3 and Unit 7, Example 5) without ever being *stated* as a technique, so the $n = 1$ trap that comes with it is never named. Vinay Kumar’s *Comprehensive Algebra I for JEE* carries “Finding General Term from Summation Formula” as a titled section for exactly this reason.

The second is an engine that is printed but never started. Unit 3 and Unit 5 both state AM-GM for n positive numbers with its equality condition, but every worked use in the book is a two-number bound ($x + 9/x \geq 6$, $a/b + b/a \geq 2$) or the single three-number product of Unit 3, Example 6. The optimisation technique that JEE Advanced actually asks for — split a constraint into equal parts so that the geometric mean becomes a constant — never appears, and neither does the trap that kills half the marks on these questions: an AM-GM bound is worthless if the equality point lies outside the domain.

This addendum closes both gaps in two items, continuing the lettering of the Round 1 Supplement. Each is a method statement, worked examples on the usual ladder, and practice with answers.

Addendum	Topic	Read after
F	Recovering a_n : from S_n , from a recurrence, and in the limit	Unit 1, Procedure 4
G	AM-GM as an optimisation engine	Unit 5, Example 4

Notation is chained from Units 1-9: a is a first term, d a common difference, r a common ratio, a_n (or t_n) an n th term, S_n the sum of the first n terms, and A , G , H the arithmetic, geometric and harmonic means.

Addendum F — Recovering a_n : from S_n , from a recurrence, and in the limit

Extends Unit 1, Procedure 4 and Example 3. Also completes the $a_n = S_n - S_{n-1}$ move that Unit 2, Derivation 3 and Unit 7, Example 5 both perform without naming.

A sequence can be handed to you in three indirect ways: through its running total S_n , through a recurrence that ties each term to the one before, or through an infinite process (a nested radical, a continued fraction) whose value is the limit of a recursive sequence. Each has one clean technique. All three are in this addendum.

Method 1 — from S_n to a_n

The whole idea is one subtraction. The sum of the first n terms minus the sum of the first $n - 1$ terms leaves exactly the n th term:

$$a_1 = S_1, \quad a_n = S_n - S_{n-1} \quad (n \geq 2).$$

Three rules govern its use.

1. **The subtraction formula is only legal for $n \geq 2$.** There is no S_0 in the definition of a series, so a_1 must be read off as S_1 separately. This is not pedantry; it is where the marks are.
2. **Always test the $n \geq 2$ formula at $n = 1$.** If it reproduces S_1 , one formula covers every term and you may write it as a single answer. If it does not, the sequence is genuinely two-piece: a_1 on its own, and the formula from the second term onward. This is exactly the $S_n = An^2 + Bn + c$ with $c \neq 0$ trap flagged in Unit 2, Section 3 — now stated in its general form, for any S_n at all.
3. **S_n need not be a polynomial.** The subtraction works on $S_n = 3^n - 1$, on $S_n = n/(n+1)$, on anything. Unit 2 and Unit 7 only ever apply it to quadratic S_n , which makes it look like an AP-only device. It is not.

A useful diagnostic falls straight out of rule 2: the sequence is an AP from the first term if and only if S_n is a quadratic in n with zero constant term, and it is a GP from the first term if and only if S_n has the shape $\lambda(r^n - 1)$ — again with the constant arranged so that S_1 comes out right.

Method 2 — from a recurrence to a closed form

Three shapes cover almost everything Class 11 and JEE Main ask. In each case a_1 is given.

Shape (i): $a_{n+1} = a_n + f(n)$ — additive. Telescope. Writing out $a_2 - a_1 = f(1)$, $a_3 - a_2 = f(2)$, ..., $a_n - a_{n-1} = f(n-1)$ and adding, every interior term cancels:

$$a_n = a_1 + \sum_{k=1}^{n-1} f(k).$$

Sum the right-hand side with the Unit 7 formulas if f is a polynomial, with the Unit 3 formula

if f is geometric. This is the same telescoping argument as Unit 2, Derivation 1 and as Addendum E; the method of differences is this shape.

Shape (ii): $a_{n+1} = a_n \cdot g(n)$ — **multiplicative**. Telescope the products the same way:

$$a_n = a_1 \prod_{k=1}^{n-1} g(k).$$

When $g(k) = r$ is constant this is the GP n th term; when $g(k) = r^k$ you recover the exponent $n(n-1)/2$ of the GP product formula from Unit 9, Derivation 3.

Shape (iii): $a_{n+1} = p a_n + q$ with $p \neq 1$ — **linear with constant coefficients**. This is the one Unit 1 leaves unfinished, and it has a trick worth learning properly. Look for the **fixed point** L — the value that the recurrence leaves alone:

$$L = pL + q \implies L = \frac{q}{1-p}.$$

Now measure every term from L instead of from zero. Put $b_n = a_n - L$. Then

$$b_{n+1} = a_{n+1} - L = (p a_n + q) - (pL + q) = p(a_n - L) = p b_n,$$

so b_n is a plain GP with first term $a_1 - L$ and common ratio p . Therefore

$$a_n = (a_1 - L) p^{n-1} + L, \quad L = \frac{q}{1-p}.$$

Read that formula: **an affine recurrence is a GP in disguise, seen from the wrong origin**. Shifting the origin to the fixed point removes the $+q$ and leaves nothing but Unit 3.

Figure brief: A vertical number line with a marked level L drawn as a dashed horizontal line. Above it, plotted points a_1, a_2, a_3, a_4 climbing away from the dashed line, with a bracket beside each showing its gap to L labelled $a_1 - L, p(a_1 - L), p^2(a_1 - L), p^3(a_1 - L)$. Caption: “the terms are not a GP, but their distances from the fixed point are.”

Method 3 — the limit of a recursive sequence

If $a_{n+1} = f(a_n)$, the terms settle down to a limit L , and f is continuous, then letting $n \rightarrow \infty$ on both sides gives

$$L = f(L).$$

Solve that equation, then discard every root that the terms themselves rule out — a sequence of positive terms cannot converge to a negative number, and a sequence bounded above by 3 cannot converge to 7. This is the standard route to nested radicals and infinite continued fractions:

$$a_1 = S_1 = 1 + 2 + 3 = 6.$$

Now for $n \geq 2$, subtract (Procedure F, step 1):

$$a_n = S_n - S_{n-1} = (n^2 + 2n + 3) - [(n-1)^2 + 2(n-1) + 3] = (n^2 + 2n + 3) - (n^2 + 2) = 2n + 1.$$

Test at $n = 1$: the formula gives $2(1) + 1 = 3$, but $a_1 = 6$. **They disagree**, so the answer is genuinely two-piece:

$$a_1 = 6, \quad a_n = 2n + 1 \quad (n \geq 2).$$

The series is $6 + 5 + 7 + 9 + 11 + \dots$. It is **not** an AP: the first gap is $5 - 6 = -1$ and every later gap is 2. The terms from the second onward do form an AP with $d = 2$. The culprit is the constant 3 in S_n — exactly the Unit 2, Section 3 scope condition, now visible as an instance of a general rule.

Check. $S_2 = 6 + 5 = 11$ and $2^2 + 2(2) + 3 = 11$. $S_3 = 11 + 7 = 18$ and $3^2 + 2(3) + 3 = 18$. Both agree.

Worked Example F2 — [JEE Main Level]

- (a) A series has $S_n = 3^n - 1$. Find a_n and identify the progression.
 (b) Find the closed form of the sequence of Unit 1, Example 3: $a_1 = 2$, $a_n = 3a_{n-1} + 1$.

Solution.

(a) S_n is not a polynomial, but nothing in the method cares. For $n \geq 2$,

$$a_n = (3^n - 1) - (3^{n-1} - 1) = 3^{n-1}(3 - 1) = 2 \cdot 3^{n-1}.$$

Test at $n = 1$: the formula gives $2 \cdot 3^0 = 2$, and $a_1 = S_1 = 3 - 1 = 2$. **They agree**, so the single formula $a_n = 2 \cdot 3^{n-1}$ covers every term. That is a GP with $a = 2$, $r = 3$ — and indeed $S_n = 2(3^n - 1)/(3 - 1) = 3^n - 1$, the Unit 3 sum formula read backwards.

(b) This is shape (iii) with $p = 3$, $q = 1$. Fixed point:

$$L = \frac{q}{1-p} = \frac{1}{1-3} = -\frac{1}{2}.$$

By the boxed formula, with $a_1 = 2$ so that $a_1 - L = 2 + \frac{1}{2} = \frac{5}{2}$:

$$a_n = \frac{5}{2} \cdot 3^{n-1} - \frac{1}{2} = \frac{5 \cdot 3^{n-1} - 1}{2}.$$

Check on terms Unit 1 already listed. $n = 1$: $(5 - 1)/2 = 2$. $n = 2$: $(15 - 1)/2 = 7$. $n = 3$: $(45 - 1)/2 = 22$. $n = 4$: $(135 - 1)/2 = 67$. $n = 5$: $(405 - 1)/2 = 202$. All five match the terms Unit 1, Example 3 unfolded by hand — the closed form the chapter promised and never delivered.

Notice the shape of the answer: a GP term plus a constant. **Every affine recurrence produces exactly that**, and if a JEE Main option list contains no such option, you have misread the recurrence.

Worked Example F3 — [JEE Main — Speed Trap]

Let $a_1 = 3$ and $a_{n+1} = 3a_n - 4$. Find $\lim_{n \rightarrow \infty} a_n$.

The trap: a fixed point you run away from. The seductive answer is one line. Let the limit be L ; pass to the limit on both sides:

$$L = 3L - 4 \implies 2L = 4 \implies L = 2$$

— wrong. Not the arithmetic; the arithmetic is perfect. What is wrong is that the limit does not exist, so there was never an L to pass to.

Correct solution. Build the closed form first (Method 2, shape iii). Here $p = 3$, $q = -4$, so $L = -4/(1 - 3) = 2$, and $a_1 - L = 3 - 2 = 1$:

$$a_n = 1 \cdot 3^{n-1} + 2 = 3^{n-1} + 2.$$

Check: $a_1 = 3$, $a_2 = 3(3) - 4 = 5 = 3 + 2$, $a_3 = 3(5) - 4 = 11 = 9 + 2$, $a_4 = 29 = 27 + 2$. Correct. And $3^{n-1} \rightarrow \infty$, so

$$a_n \rightarrow \infty; \quad \text{the limit does not exist.}$$

The value 2 is a genuine fixed point — feed 2 into the recurrence and 2 comes back out — but the sequence started at distance 1 from it and that distance is multiplied by 3 every step. The fixed point repels.

How to never fall for it. For $a_{n+1} = p a_n + q$ the distance to the fixed point obeys $a_n - L = p^{n-1}(a_1 - L)$ exactly. So before you quote L as a limit, look at $|p|$: convergence happens **only** when $|p| < 1$ (or when you start exactly at L and never move). Here $|p| = 3$. Compare Unit 3's $|r| < 1$ condition for S_∞ — it is the same condition, in the same disguise, and it is the same mark.

Worked Example F4 — [JEE Advanced Level]

Evaluate

$$x = \sqrt{6 + \sqrt{6 + \sqrt{6 + \cdots}}}, \quad y = 1 + \frac{1}{1 + \frac{1}{1 + \cdots}},$$

justifying in each case that the value exists and that the root you keep is the right one.

Solution.

The radical. Define $a_1 = \sqrt{6}$ and $a_{n+1} = \sqrt{6 + a_n}$, so that x is the limit of a_n . Let L be the positive root of $L^2 = 6 + L$, i.e. of $L^2 - L - 6 = (L - 3)(L + 2) = 0$, so $L = 3$.

Existence. Every term is positive. If $a_n < 3$ then $a_{n+1} = \sqrt{6 + a_n} < \sqrt{9} = 3$, and $a_1 = \sqrt{6} < 3$, so by induction $a_n < 3$ for all n : the sequence is bounded above. It is also increasing, because $a_{n+1} > a_n$ is equivalent to $6 + a_n > a_n^2$, i.e. to $(a_n - 3)(a_n + 2) < 0$, which holds for every a_n with $0 < a_n < 3$. An increasing sequence bounded above converges.

The root. Passing to the limit, $L^2 = 6 + L$, whose roots are 3 and -2 . Every term is positive, so -2 is rejected on grounds. Hence

$$x = 3.$$

Numerical check. $\sqrt{6} = 2.449$, $\sqrt{8.449} = 2.907$, $\sqrt{8.907} = 2.984$, $\sqrt{8.984} = 2.997$ — climbing to 3, from below, exactly as the argument predicted.

The continued fraction. Define $a_1 = 1$ and $a_{n+1} = 1 + 1/a_n$. Assuming the limit y exists (the convergents $1, 2, \frac{3}{2}, \frac{5}{3}, \frac{8}{5}, \dots$ oscillate about it with shrinking amplitude), passing to the limit gives

$$y = 1 + \frac{1}{y} \implies y^2 - y - 1 = 0 \implies y = \frac{1 \pm \sqrt{5}}{2}.$$

Every term is at least 1, so the negative root is rejected and

$$y = \frac{1 + \sqrt{5}}{2},$$

the golden ratio. **This is the same quadratic as Unit 3, Example 7**, where a GP in which each term equals the sum of the next two forced $r^2 + r - 1 = 0$ and gave $r = (\sqrt{5} - 1)/2$. The two answers are reciprocals of one another, which is the golden ratio's defining property, and the reappearance is not a coincidence: both problems are the statement "this object is its own successor" written in different notation. Note also the numerators 1, 2, 3, 5, 8 of the convergents — Fibonacci, from Unit 1's Concept Intuition.

Practice — Addendum F

1. (Board) The sum of n terms of a series is $S_n = 5n^2 + 3n$. Find a_n , and state whether the series is an AP from its first term.
2. (Board) A series has $S_n = 2^{n+1} - 2$. Find a_n and identify the progression.
3. (Board) A series has $S_n = n^2 + 5$. Find a_1 and a_n for $n \geq 2$, and write out the first four terms.
4. (Main) Find the closed form of $a_1 = 1$, $a_{n+1} = 4a_n + 3$, and hence a_5 .
5. (Main) Find the closed form of $a_1 = 2$, $a_{n+1} = a_n + 3n + 1$.
6. (Main) A series has $S_n = \frac{n}{n+1}$. Find a_n , and connect the answer to Unit 8.
7. (Main) Evaluate $\sqrt{12 + \sqrt{12 + \sqrt{12 + \dots}}}$.

8. (Main) Evaluate $3 + \frac{1}{3 + \frac{1}{3 + \dots}}$.
9. (Adv) Let $a_1 = 5$ and $a_{n+1} = 2a_n - 3$. A student writes $L = 2L - 3$, gets $L = 3$, and reports 3 as the limit. Find the closed form of a_n and say what is wrong.
10. (Adv) Find the closed form of $a_1 = 1$, $a_{n+1} = 2^n a_n$, and relate the exponent you obtain to Unit 9, Derivation 3.

Answers.

- $a_n = 10n - 2$ for every n (the $n \geq 2$ formula also gives $a_1 = 8 = S_1$). Yes — an AP with $a = 8$, $d = 10$; consistent with $S_n = An^2 + Bn$ having zero constant term.
- $a_n = 2^n$ for every n ; a GP with $a = 2$, $r = 2$. ($a_1 = S_1 = 2$ agrees with the formula.)
- $a_1 = 6$; $a_n = 2n - 1$ for $n \geq 2$. First four terms 6, 3, 5, 7. Not an AP; the constant 5 in S_n is what breaks it.
- $L = 3/(1 - 4) = -1$, so $a_n = 2 \cdot 4^{n-1} - 1$ and $a_5 = 2(256) - 1 = 511$.
- $a_n = 2 + \sum_{k=1}^{n-1} (3k + 1) = \frac{3n^2 - n + 2}{2}$. (Check $n = 4$: 23 , and $2 \rightarrow 6 \rightarrow 13 \rightarrow 23$.)
- $a_n = \frac{n}{n+1} - \frac{n-1}{n} = \frac{1}{n(n+1)}$, valid at $n = 1$ too. This is the telescoping series of Unit 8, Derivation 1 run backwards: $S_n = n/(n+1)$ is precisely what that derivation produces.
4. ($L^2 = 12 + L$, so $(L - 4)(L + 3) = 0$; reject -3 for positivity.)
- $\frac{3 + \sqrt{13}}{2}$. ($L = 3 + 1/L$, so $L^2 - 3L - 1 = 0$; reject the negative root.)
- $L = -3/(1 - 2) = 3$ and $a_1 - L = 2$, so $a_n = 2 \cdot 2^{n-1} + 3 = 2^n + 3$, which tends to infinity. The limit does not exist. The student's equation is only valid *given* that a limit exists; here $|p| = 2 > 1$, so the fixed point repels.
- $a_n = \prod_{k=1}^{n-1} 2^k = 2^{1+2+\dots+(n-1)} = 2^{n(n-1)/2}$. The exponent $n(n-1)/2$ is the same one that appears in $P = a^n r^{n(n-1)/2}$ — a product of n terms always collects the exponents $0, 1, \dots, n-1$.

Addendum G — AM-GM as an optimisation engine

Extends Unit 5, Section 5 (which uses AM-GM only on two numbers) and Unit 3, Example 6. Unit 3 and Unit 5 both print AM-GM for n positive numbers with its equality condition; this addendum is the promised use of it.

Method

One sentence carries the whole topic:

For positive numbers, fixing the sum caps the product, and fixing the product floors the sum — and both bounds are reached exactly when all the numbers are equal.

Written out, from $\frac{a_1 + \dots + a_n}{n} \geq (a_1 a_2 \dots a_n)^{1/n}$:

$$\sum a_i = S \text{ fixed} \Rightarrow \prod a_i \leq \left(\frac{S}{n}\right)^n, \quad \prod a_i = P \text{ fixed} \Rightarrow \sum a_i \geq nP^{1/n},$$

with equality in each case if and only if $a_1 = a_2 = \dots = a_n$. Everything below is this statement plus bookkeeping.

Three rules turn it into a technique.

Rule 1 — choose the terms so that the geometric mean is a constant. This is the whole art, and it is where students who “know AM-GM” still lose the question. An AM-GM bound is only useful if the right-hand side is free of the variable. Applying AM-GM to $2x + \frac{3}{x^2}$ as two terms gives $2\sqrt{6/x}$, which still contains x and bounds nothing. Split it as $x + x + \frac{3}{x^2}$ and the product is $x \cdot x \cdot 3/x^2 = 3$, a constant — and the bound follows immediately.

Rule 2 — split unequal weights into equal parts. To bound a product with exponents, break the constraint into as many pieces as the exponents demand. To maximise $x^p y^q$ subject to $\alpha x + \beta y = k$ (all quantities positive, p, q positive integers), apply AM-GM to the $p+q$ numbers

$$\underbrace{\frac{\alpha x}{p}, \dots, \frac{\alpha x}{p}}_{p \text{ copies}}, \quad \underbrace{\frac{\beta y}{q}, \dots, \frac{\beta y}{q}}_{q \text{ copies}}$$

whose sum is exactly $\alpha x + \beta y = k$. Then

$$\frac{k}{p+q} \geq \left[\left(\frac{\alpha x}{p}\right)^p \left(\frac{\beta y}{q}\right)^q \right]^{1/(p+q)} \Rightarrow x^p y^q \leq \frac{p^p q^q}{\alpha^p \beta^q} \left(\frac{k}{p+q}\right)^{p+q},$$

with equality if and only if $\frac{\alpha x}{p} = \frac{\beta y}{q}$. The same recipe extends to any number of variables.

The exponent tells you how many copies to make; the coefficient tells you what to divide by.

Rule 3 — check that the equality point lies in the domain. AM-GM produces a number that the expression can never go below. Whether the expression ever *reaches* that number is a separate question, settled by the equality condition “all terms equal”. If solving that condition throws up a value of the variable outside the permitted range, the bound is true but not attained, and the real extremum sits on the boundary of the range. This is the speed trap of Example G3 and it is worth more marks than the algebra.

Figure brief: Two panels sharing a horizontal axis labelled x . Left panel: the curve $y = x + 4/x$ for $x > 0$, with its minimum marked at $(2, 4)$ and a dashed horizontal line at height 4 labelled “AM-GM bound”. Right panel: the same curve with everything left of $x = 3$ greyed out, the point $(3, 13/3)$ marked with a solid dot labelled “actual minimum on the restricted domain”, and the dashed line at 4 now sitting strictly below the visible part of the curve, annotated “bound still true, no longer attained”.

Procedure G: running the engine

1. **Name what is fixed.** A sum, a product, or a linear constraint. *Why:* the fixed quantity is the one that becomes the constant in the bound; if nothing is fixed, AM-GM has nothing to work with.
2. **Split the expression until the geometric mean is variable-free.** Use Rule 2 if the target carries exponents. *Why:* a bound that still contains the variable is not a bound.
3. **Write the inequality and read off the bound.** *Why:* one line, no calculus.
4. **Solve “all terms equal” together with the constraint to locate the optimiser.** *Why:* the marks are usually split between the bound and the point where it is attained.
5. **Confirm the optimiser is allowed.** If the domain is restricted, test it; if it is excluded, go to the boundary. *Why:* Rule 3.

Worked Example G1 — [JEE Main Level]

For $x > 0$, find the least value of $2x + \frac{3}{x^2}$.

Solution. The two-term attempt fails first, and it is worth seeing why:

$$2x + \frac{3}{x^2} \geq 2\sqrt{2x \cdot \frac{3}{x^2}} = 2\sqrt{\frac{6}{x}},$$

which is a true statement and a useless one — the right-hand side still depends on x (Rule 1). Split the first term into two equal halves so that the powers of x cancel:

$$2x + \frac{3}{x^2} = x + x + \frac{3}{x^2}.$$

The product of these three positive numbers is $x \cdot x \cdot \frac{3}{x^2} = 3$, a constant. AM-GM on three numbers gives

$$\frac{x + x + \frac{3}{x^2}}{3} \geq (3)^{1/3} \quad \Rightarrow \quad 2x + \frac{3}{x^2} \geq 3 \cdot 3^{1/3} = 3^{4/3}.$$

Equality needs all three terms equal: $x = \frac{3}{x^2}$, i.e. $x^3 = 3$, i.e. $x = 3^{1/3}$, which is a legitimate positive value. Hence the least value is

$$\boxed{3^{4/3} = 3\sqrt[3]{3} \approx 4.327} \quad \text{attained at } x = 3^{1/3}.$$

Check. At $x = 1.4422$: $2x = 2.8845$ and $3/x^2 = 1.4423$, total 4.3268, and $3^{4/3} = 4.3267$. Agreed.

The transferable move. To handle $\lambda x^m + \mu x^{-k}$, split the first term into k equal pieces and the second into m equal pieces; the powers of x then cancel in the product automatically. Counting the pieces is the entire skill.

Worked Example G2 — [JEE Advanced Level]

If a, b, c are positive and $a + b + c = 1$, prove that $ab^2c^3 \leq \frac{1}{432}$, and find the values of a, b, c at which equality holds.

Solution. The exponents are 1, 2, 3, so cut the constraint into $1 + 2 + 3 = 6$ pieces (Rule 2):

$$a + \frac{b}{2} + \frac{b}{2} + \frac{c}{3} + \frac{c}{3} + \frac{c}{3} = a + b + c = 1.$$

Apply AM-GM to those six positive numbers:

$$\frac{1}{6} \geq \left[a \left(\frac{b}{2} \right)^2 \left(\frac{c}{3} \right)^3 \right]^{1/6} = \left[\frac{ab^2c^3}{4 \cdot 27} \right]^{1/6} = \left[\frac{ab^2c^3}{108} \right]^{1/6}.$$

Raise both sides to the sixth power (both sides are positive, so the direction is safe):

$$\frac{1}{6^6} \geq \frac{ab^2c^3}{108} \quad \Rightarrow \quad ab^2c^3 \leq \frac{108}{46656} = \frac{1}{432}. \quad \blacksquare$$

Equality needs all six pieces equal, each therefore $\frac{1}{6}$:

$$a = \frac{b}{2} = \frac{c}{3} = \frac{1}{6} \quad \Rightarrow \quad a = \frac{1}{6}, \quad b = \frac{1}{3}, \quad c = \frac{1}{2}.$$

Check. $a + b + c = \frac{1}{6} + \frac{2}{6} + \frac{3}{6} = 1$, and $ab^2c^3 = \frac{1}{6} \cdot \frac{1}{9} \cdot \frac{1}{8} = \frac{1}{432}$. Both confirmed.

What to notice. The coefficients 2 and 3 that you divide by are exactly the exponents, and they reappear in the constant $2^2 \cdot 3^3 = 108$. Nothing here needs calculus, and a Lagrange-multiplier solution would take three times as long.

Worked Example G3 — [JEE Main — Speed Trap]

Find the minimum value of $f(x) = x + \frac{4}{x}$ for $x \geq 3$.

The trap: an equality point outside the domain. The seductive answer is instant. Both x and $4/x$ are positive, so by AM-GM

$$x + \frac{4}{x} \geq 2\sqrt{x \cdot \frac{4}{x}} = 2\sqrt{4} = 4,$$

“so the minimum is 4” — wrong. The inequality is perfectly true. It is the word *minimum* that is false, because a minimum must be **attained**.

Equality in AM-GM requires the two terms to be equal: $x = 4/x$, i.e. $x^2 = 4$, i.e. $x = 2$ for positive x . But the domain is $x \geq 3$, and 2 is not in it. So 4 is a lower bound that f never reaches on this domain.

Correct solution. On $x \geq 3$ the function is increasing. You can see this without calculus: for $3 \leq x_1 < x_2$,

$$f(x_2) - f(x_1) = (x_2 - x_1) - 4 \left(\frac{1}{x_1} - \frac{1}{x_2} \right) = (x_2 - x_1) \left(1 - \frac{4}{x_1 x_2} \right) > 0,$$

since $x_1 x_2 > 9 > 4$. An increasing function on $[3, \infty)$ takes its least value at the left endpoint:

$$\boxed{\min_{x \geq 3} \left(x + \frac{4}{x} \right) = 3 + \frac{4}{3} = \frac{13}{3} \approx 4.333, \text{ at } x = 3.}$$

Check. $f(3) = 4.333$, $f(3.5) = 4.643$, $f(4) = 5$ — rising, as claimed, and every value is comfortably above the unattained bound 4.

How to never fall for it. After writing any AM-GM bound, solve the equality condition and *look at the answer*. If it satisfies every restriction in the question, the bound is the extremum. If it does not, the bound is only a bound, and the extremum has moved to the nearest boundary. Unit 5's Example 2 ($x + 9/x \geq 6$, minimum 6 at $x = 3$) is safe only because the question said $x > 0$ and $x = 3$ was allowed.

Worked Example G4 — [JEE Advanced Level]

If $a, b, c > 0$ and $4a + 3b + 2c = 5$, find the greatest value of $a^2 b^3 c^4$ and the values of a, b, c that achieve it.

Solution. Exponents 2, 3, 4 give $2+3+4 = 9$ pieces. The coefficient attached to each variable rides along into its piece (Rule 2): split $4a$ into two copies of $2a$, split $3b$ into three copies of b , and split $2c$ into four copies of $c/2$. The nine pieces sum to $4a + 3b + 2c = 5$, so their arithmetic mean is $5/9$, and AM-GM gives

$$\frac{5}{9} \geq \left[(2a)^2 (b)^3 \left(\frac{c}{2} \right)^4 \right]^{1/9} = \left[\frac{4a^2 \cdot b^3 \cdot c^4}{16} \right]^{1/9} = \left[\frac{a^2 b^3 c^4}{4} \right]^{1/9}.$$

Raising to the ninth power,

$$a^2 b^3 c^4 \leq 4 \left(\frac{5}{9} \right)^9.$$

Equality needs the nine pieces equal, each to their common mean $5/9$:

$$2a = b = \frac{c}{2} = \frac{5}{9} \quad \implies \quad a = \frac{5}{18}, \quad b = \frac{5}{9}, \quad c = \frac{10}{9}.$$

Check the constraint.

$$4 \left(\frac{5}{18} \right) + 3 \left(\frac{5}{9} \right) + 2 \left(\frac{10}{9} \right) = \frac{10}{9} + \frac{15}{9} + \frac{20}{9} = \frac{45}{9} = 5.$$

Check the value.

$$\begin{aligned} \left(\frac{5}{18} \right)^2 \left(\frac{5}{9} \right)^3 \left(\frac{10}{9} \right)^4 &= \frac{25}{324} \cdot \frac{125}{729} \cdot \frac{10000}{6561} = \frac{31\,250\,000}{1\,549\,681\,956}, \\ 4 \left(\frac{5}{9} \right)^9 &= \frac{4 \cdot 1\,953\,125}{387\,420\,489} = \frac{7\,812\,500}{387\,420\,489} = \frac{31\,250\,000}{1\,549\,681\,956}. \end{aligned}$$

Identical.

So the greatest value is $\boxed{4 \left(\frac{5}{9}\right)^9}$.

The general statement worth carrying away. For $\alpha_1 x_1 + \dots + \alpha_m x_m = k$ with all quantities positive,

$$x_1^{p_1} \dots x_m^{p_m} \leq \left(\prod_i \frac{p_i^{p_i}}{\alpha_i^{p_i}} \right) \left(\frac{k}{p_1 + \dots + p_m} \right)^{p_1 + \dots + p_m},$$

with equality when $\frac{\alpha_1 x_1}{p_1} = \dots = \frac{\alpha_m x_m}{p_m}$. Do not memorise it; rebuild it from the splitting each time, which takes fifteen seconds and never misremembers a coefficient.

Practice — Addendum G

1. (Board) Three positive numbers have sum 12. Find the greatest possible value of their product. Then, if instead their product is 64, find the least possible value of their sum.
2. (Board) If $a, b, c > 0$ and $abc = 1$, prove that $(1+a)(1+b)(1+c) \geq 8$.
3. (Main) For $x > 0$, find the least value of $3x + \frac{12}{x^2}$.
4. (Main) If $x, y > 0$ and $x + y = 10$, find the greatest value of $x^3 y^2$.
5. (Main) Find the least value of $x + \frac{9}{x}$ for $x \geq 4$.
6. (Main) If $a, b, c > 0$ and $a + b + c = 3$, find the greatest value of $a^2 b^2 c^2$.
7. (Adv) If $a, b, c, d > 0$ and $abcd = 1$, prove that $(1+a)(1+b)(1+c)(1+d) \geq 16$.
8. (Adv) Prove that $n! < \left(\frac{n+1}{2}\right)^n$ for every integer $n \geq 2$.

Answers.

1. Greatest product $\left(\frac{12}{3}\right)^3 = 64$, at $x = y = z = 4$. Least sum $3 \cdot 64^{1/3} = 12$, at $x = y = z = 4$. The same point answers both — that is the fixed-sum / fixed-product duality.
2. Apply AM-GM to each bracket: $1 + a \geq 2\sqrt{a}$, and similarly for b and c . Multiplying, $(1+a)(1+b)(1+c) \geq 8\sqrt{abc} = 8$. Equality at $a = b = c = 1$.
3. 9, at $x = 2$. (Split as $\frac{3x}{2} + \frac{3x}{2} + \frac{12}{x^2}$; the product is 27, so the sum is at least $3 \cdot 3 = 9$, with equality when $\frac{3x}{2} = \frac{12}{x^2}$, i.e. $x^3 = 8$.)
4. 3456, at $x = 6$, $y = 4$. (Five pieces, namely three copies of $x/3$ and two copies of $y/2$, summing to 10; so $[x^3 y^2 / 108]^{1/5} \leq 2$ and $x^3 y^2 \leq 108 \cdot 32$.)
5. $\frac{25}{4}$, at $x = 4$. **Not 6.** AM-GM gives $x + 9/x \geq 6$ with equality at $x = 3$, which is outside $[4, \infty)$; the function is increasing there, so the least value sits at the left endpoint.
6. 1, at $a = b = c = 1$. (AM-GM gives $abc \leq (3/3)^3 = 1$; squaring preserves the direction since $abc > 0$.)
7. Same move as Q2 with four brackets. From $1 + a \geq 2\sqrt{a}$ and its three companions, multiplying gives $(1+a)(1+b)(1+c)(1+d) \geq 16\sqrt{abcd} = 16$. Equality at $a = b = c = d = 1$.
8. Apply AM-GM to the n distinct positive numbers $1, 2, \dots, n$. They are not all equal for $n \geq 2$, so the inequality is strict: $(n!)^{1/n} < \frac{1+2+\dots+n}{n} = \frac{n(n+1)/2}{n} = \frac{n+1}{2}$, using

the Unit 7 sum. Raising to the n th power gives the result. (At $n = 4$: $24 < 2.5^4 = 39.06$.)

Where This Material Belongs

Round 2 item	Reads as an extension of	What it delivers
Addendum F, Method 1	Unit 1, Section 4; Unit 2, Derivation 3; Unit 7, Example 5	$a_n = S_n - S_{n-1}$ as a named technique, with the $n = 1$ test and non-polynomial S_n
Addendum F, Method 2	Unit 1, Procedure 4 and Example 3	The three recurrence shapes and the fixed-point substitution that closes Unit 1's unfinished promise
Addendum F, Method 3	Unit 1, Section 2; Unit 3, Example 7	Nested radicals and continued fractions, with the convergence licence and the repelling-fixed-point trap
Addendum G	Unit 5, Section 5; Unit 3, Example 6	The n -number AM-GM finally used: fixed-sum / fixed-product duality, the splitting rule for weighted products, and the domain check

Cheat sheet for this addendum

- **From S_n :** $a_1 = S_1$; $a_n = S_n - S_{n-1}$ for $n \geq 2$. **Always test the formula at $n = 1$.** If it fails, the answer is two-piece. Works for any S_n , not just polynomials.
- **Diagnostics:** AP from the first term $\Leftrightarrow S_n = An^2 + Bn$ (no constant); GP from the first term $\Leftrightarrow S_n = \lambda(r^n - 1)$.
- **Additive recurrence** $a_{n+1} = a_n + f(n)$: $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.
- **Multiplicative recurrence** $a_{n+1} = a_n g(n)$: $a_n = a_1 \prod_{k=1}^{n-1} g(k)$.
- **Affine recurrence** $a_{n+1} = p a_n + q$ ($p \neq 1$): fixed point $L = \frac{q}{1-p}$, and $a_n = (a_1 - L)p^{n-1} + L$. **An affine recurrence is a GP measured from L .**
- **Limit:** $a_n \rightarrow L \Rightarrow L = f(L)$, never the converse. For the affine case, $a_n \rightarrow L$ **iff** $|p| < 1$ (or $a_1 = L$) — the same threshold as S_∞ in Unit 3.
- **Nested radical / continued fraction:** $\sqrt{c + \sqrt{c + \dots}} = \frac{1 + \sqrt{1 + 4c}}{2}$; $\sqrt{c\sqrt{c\dots}} = c$; $c + \frac{1}{c + \dots} = \frac{c + \sqrt{c^2 + 4}}{2}$. Reject the negative root on grounds, and say what the grounds are.
- **AM-GM engine:** sum fixed at $S \Rightarrow \text{product} \leq (S/n)^n$; product fixed at $P \Rightarrow \text{sum} \geq nP^{1/n}$;

equality iff all terms equal.

- **Rule 1:** split until the geometric mean is a constant. A bound containing the variable is not a bound.
- **Rule 2 (weighted split):** for $x^p y^q$ under $\alpha x + \beta y = k$, use p copies of $\alpha x/p$ and q copies of $\beta y/q$; equality iff $\alpha x/p = \beta y/q$. **Exponents give the number of copies; coefficients give the divisor.**
- **Rule 3 (the trap):** an AM-GM bound is a minimum only if the equality point lies in the domain. Solve “all terms equal”, then check it is allowed. If not, go to the boundary.
- **Two benchmark results:** $a + b + c = 1 \Rightarrow ab^2 c^3 \leq \frac{1}{432}$ (at $\frac{1}{6}, \frac{1}{3}, \frac{1}{2}$); $abc = 1 \Rightarrow (1+a)(1+b)(1+c) \geq 8$.

DRONA ED TECH

Class 11 Mathematics

Chapter 9

Straight Lines

Slope & Angles · Equations of a Line · Distance Formulas

CBSE Boards · JEE Main · JEE Advanced

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Subtopic 00 — Coordinate Foundations

Exam Relevance: These are the load-bearing prerequisites for the entire chapter — every later result quietly uses them. CBSE asks the section formula, midpoint, and area of a triangle directly (2–3 marks). JEE Main reliably has a question on area, collinearity, or the centroid. JEE Advanced uses the **locus method** and **translation of axes** as setup machinery in harder problems. Master these cold and the rest of Straight Lines becomes bookkeeping.

Section 1: Concept Intuition

Before you can talk about *lines*, you need to be fluent with *points* — how far apart they are, how to split the gap between them, how to measure the area they enclose, and how to describe a moving point by the rule it obeys. Think of this subtopic as learning the grammar before writing sentences.

Think about it this way... the coordinate plane turns geometry into arithmetic. “How far is Jaipur from Delhi on the map?” becomes a subtraction-and-square-root. “Where is the town exactly one-third of the way along the highway?” becomes the section formula. “Do these three survey markers lie on one straight boundary?” becomes “is the area of their triangle zero?” Geometry questions become formula questions — that’s the whole power of Descartes’ idea.

The last big idea here is **locus**: the path traced by a point that obeys a rule. A goat tied to a peg traces a circle (rule: “stay 5 m from the peg”); a point equidistant from two villages traces the perpendicular bisector of the line joining them (a straight line!). Finding the *equation* of a locus is the bridge from a verbal rule to an algebraic curve, and almost every “find the equation of the path/set of points such that...” problem is a locus problem in disguise.

Domain of Validity & Edge Cases.

- The **section formula** for *internal* division uses $m + n$ in the denominator; the *external* version uses $m - n$, which blows up when $m = n$ (you can’t externally divide a segment in a 1 : 1 ratio — that point is at infinity).
 - The **area** formula gives 0 exactly when the three points are **collinear** — that’s not a failure, it’s a feature you’ll exploit.
 - **Translation of axes** shifts the origin but never rotates it; slopes and distances are unchanged, only coordinates relabel.
-

Section 2: Key Formulas & Definitions

For points $A(x_1, y_1)$, $B(x_2, y_2)$, $C(x_3, y_3)$:

Distance between two points:

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}.$$

Section formula — point dividing AB in ratio $m : n$:

$$\text{Internal: } \left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right), \quad \text{External: } \left(\frac{mx_2 - nx_1}{m-n}, \frac{my_2 - ny_1}{m-n} \right).$$

Midpoint (ratio $1 : 1$): $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right).$

Centroid of $\triangle ABC$: $\left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right).$

Area of a triangle:

$$\Delta = \frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|.$$

• **Collinearity test:** A, B, C are collinear $\iff \Delta = 0$.

Locus: the set of all points obeying a given geometric condition; its *equation* is the relation x, y must satisfy.

Translation (shift) of origin to (h, k) : a point with old coordinates (x, y) has new coordinates $(X, Y) = (x - h, y - k)$; to rewrite an equation, substitute $x = X + h, y = Y + k$.

Section 3: Theorems, Proofs & Standard Derivations

Theorem 1 — Internal section formula

Given: $A(x_1, y_1), B(x_2, y_2)$, and P dividing AB internally in ratio $m : n$.

To Prove: $P = \left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right).$

Proof. Drop perpendiculars from A, P, B to the x -axis. By similar triangles (the line cuts them proportionally), $\frac{AP}{PB} = \frac{m}{n}$ projects onto the x -coordinates:

$$\frac{x_P - x_1}{x_2 - x_P} = \frac{m}{n} \Rightarrow n(x_P - x_1) = m(x_2 - x_P) \Rightarrow x_P(m+n) = mx_2 + nx_1.$$

So $x_P = \frac{mx_2 + nx_1}{m+n}$, and identically $y_P = \frac{my_2 + ny_1}{m+n}$. ■

Figure

Segment AB with point P between A and B dividing it $m:n$. Perpendiculars from A, P, B to the x -axis create similar right triangles; the proportional split of the base is highlighted.

Theorem 2 — Area of a triangle

Given: vertices $A(x_1, y_1), B(x_2, y_2), C(x_3, y_3)$.

To Prove: $\Delta = \frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|.$

Proof (trapezium decomposition). Drop perpendiculars to the x -axis from each vertex, forming three trapezia. The triangle's area is the algebraic sum of trapezium areas:

$$\Delta = \frac{1}{2}(x_1 - x_2)(y_1 + y_2) + \frac{1}{2}(x_2 - x_3)(y_2 + y_3) + \frac{1}{2}(x_3 - x_1)(y_3 + y_1).$$

Expanding and collecting terms collapses to $\frac{1}{2}[x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)]$; the modulus removes orientation sign. ■

Procedure — Finding the equation of a locus

1. Let the moving point be $P(h, k)$.
2. Translate every geometric condition into an equation in h, k using the formulas above.
3. Eliminate any parameters until only h, k (and constants) remain.
4. Replace (h, k) by (x, y) — that relation is the locus.

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

Find the centroid of the triangle with vertices $A(2, -3)$, $B(-4, 5)$, $C(8, 1)$.

Solution.

$$G = \left(\frac{2 - 4 + 8}{3}, \frac{-3 + 5 + 1}{3} \right) = \left(\frac{6}{3}, \frac{3}{3} \right) = (2, 1).$$

$$\boxed{G = (2, 1)}$$

Example 2 — [JEE Main · Speed Trap]

Are the points $(1, 4)$, $(3, -2)$, and $(-3, 16)$ collinear?

Speed approach. Don't compute three slopes — use area = 0 as the one-shot test:

$$2\Delta = |1(-2-16)+3(16-4)+(-3)(4-(-2))| = |-18+3(12)+(-3)(6)| = |-18+36-18| = 0.$$

Area is 0, so **yes, collinear**.

The trap: students laboriously find slope(AB), slope(BC) and risk a sign error in each. The area determinant does it in one line and the modulus shields you from orientation mistakes.

$$\boxed{\text{Collinear}}$$

Example 3 — [JEE Main Level · multi-concept]

The point P divides the segment from $A(-1, 3)$ to $B(5, -9)$ such that $AP : PB = 2 : 1$. Find P , then find the distance OP from the origin.

Step 1 — section formula ($m : n = 2 : 1$):

$$P = \left(\frac{2(5) + 1(-1)}{3}, \frac{2(-9) + 1(3)}{3} \right) = \left(\frac{9}{3}, \frac{-15}{3} \right) = (3, -5).$$

Step 2 — distance from origin:

$$OP = \sqrt{3^2 + (-5)^2} = \sqrt{34}.$$

$$P = (3, -5), \quad OP = \sqrt{34}$$

Example 4 — [JEE Advanced Level]

Find the locus of a point that moves so that its distance from $(4, 0)$ is always twice its distance from $(1, 0)$.

Solution. Let $P(h, k)$. The condition $PA = 2PB$ with $A(4, 0)$, $B(1, 0)$ gives, on squaring,

$$(h - 4)^2 + k^2 = 4[(h - 1)^2 + k^2].$$

Expand:

$$h^2 - 8h + 16 + k^2 = 4h^2 - 8h + 4 + 4k^2 \Rightarrow 0 = 3h^2 + 3k^2 - 12.$$

So $h^2 + k^2 = 4$. Replacing $(h, k) \rightarrow (x, y)$:

$$x^2 + y^2 = 4$$

a circle of radius 2 centred at the origin (an *Apollonius circle*). Note how the linear $-8h$ terms cancelled — that cancellation is the signature of an Apollonius locus, and spotting it early saves algebra.

Section 5: Practice Exercises

1. **[CBSE]** Find the distance between $(-2, 5)$ and $(4, -3)$.
2. **[CBSE]** Find the midpoint of the segment joining $(7, -1)$ and $(-3, 9)$.
3. **[JEE Main]** Find the area of the triangle with vertices $(0, 0)$, $(4, 0)$, $(0, 6)$, and state whether the points $(2, 3)$, $(4, 6)$, $(6, 9)$ are collinear.
4. **[JEE Main]** The point $(2, k)$ divides the segment from $(0, 0)$ to $(6, 9)$ internally in some ratio; find that ratio and k .
5. **[JEE Advanced]** Find the locus of a point equidistant from $A(3, 2)$ and $B(-1, 4)$.

Answers: 1. $\sqrt{36 + 64} = 10$. 2. $(2, 4)$. 3. Area = 12; the three points are collinear (area = 0). 4. Ratio 1 : 2, $k = 3$. 5. Perpendicular bisector $2x - y - 1 = 0$ (i.e. $8x - 4y - 4 = 0 \Rightarrow 2x - y - 1 = 0$).

Section 6: Crack the MCQ

Q1. The midpoint of the segment joining (a, b) and $(3a, -b)$ is: (A) $(2a, 0)$ (B) $(2a, b)$ (C) $(a, 0)$ (D) $(4a, 0)$

Answer: (A). $\left(\frac{a+3a}{2}, \frac{b-b}{2}\right) = (2a, 0)$.

- (B) forgot the y -coordinates cancel.

- (C) averaged only one x term.
- (D) added instead of averaging x .

Q2. If $(1, 2)$, $(4, y)$, $(7, 8)$ are collinear, then $y =$ (A) 4 (B) 5 (C) 6 (D) 3

Answer: (B). Collinear $\Rightarrow$ the middle point is the midpoint here since x -values are equally spaced: $y = \frac{2+8}{2} = 5$. (Or set area = 0.)

- (A)/(C)/(D) come from slope sign-slips or arithmetic errors.

Q3. The centroid of a triangle is $(3, 1)$ and two of its vertices are $(2, -1)$ and $(5, 4)$. The third vertex is: (A) $(2, 0)$ (B) $(2, 2)$ (C) $(8, 2)$ (D) $(4, 0)$

Answer: (A). $x_3 = 3(3) - 2 - 5 = 2$, $y_3 = 3(1) - (-1) - 4 = 0$, giving $(2, 0)$.

- (C) multiplied the centroid by 3 but subtracted only one vertex.
- (B)/(D) are sign or arithmetic slips on y_3 (the $-(-1)$ term).

Q4. Translating the origin to $(2, -1)$, the point with old coordinates $(5, 3)$ has new coordinates: (A) $(3, 4)$ (B) $(7, 2)$ (C) $(3, 2)$ (D) $(-3, -4)$

Answer: (A). $(X, Y) = (5 - 2, 3 - (-1)) = (3, 4)$.

- (B) added the shift instead of subtracting.
- (C) mishandled the negative k .
- (D) negated the whole result.

Section 7: Common Student Pitfalls & Pro-Tips

1. **Internal vs external section formula.** Internal uses $m + n$; external uses $m - n$. Picking the wrong one flips the point to the wrong side.
2. **Dropping the modulus in area.** Area is non-negative; without the bars you may report a negative “area.”
3. **Forgetting to scale the centroid.** $x_1 + x_2 + x_3 = 3\bar{x}$ — multiply the centroid by 3 before solving for a missing vertex.
4. **Sign of the shift in translation.** New = old — shift: $(X, Y) = (x - h, y - k)$. Adding instead of subtracting is the classic slip.

Pro-Tip: For collinearity, the area determinant beats comparing slopes every time — it’s one computation, it never divides by zero (no vertical-line worry), and the modulus protects you from orientation sign errors.

Section 8: Cheat Sheet Box

- **Distance:** $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.
- **Section ($m : n$ internal):** $(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n})$; midpoint is the 1 : 1 case.
- **Centroid:** $(\frac{\sum x}{3}, \frac{\sum y}{3})$.
- **Area:** $\frac{1}{2}|x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|; = 0 \Leftrightarrow \text{collinear}.$

- **Locus:** put $P(h, k)$, translate the rule, replace $(h, k) \rightarrow (x, y)$.
- **Shift of origin to (h, k) :** $X = x - h$, $Y = y - k$.

Memory aids:

- “Internal adds, external subtracts” (the $m \pm n$ denominator).
- “Area zero means in a row” (collinear).
- “New equals old minus shift.”

Subtopic 01 — Slope of a Line and Angles

Exam Relevance: Slope is the single most-used idea in all of coordinate geometry, so it shows up everywhere. CBSE Class 11 finals lean on the inclination-slope link and the parallel/perpendicular conditions for 2–4 mark questions. JEE Main almost always has a question riding on the angle-between-lines formula or a collinearity/perpendicularity condition, and JEE Advanced folds slope into locus and multi-line problems where the *vertical-line edge case* is the whole trap.

Section 1: Concept Intuition

Imagine you are walking along the Konkan Railway track as it climbs out of a valley. Some stretches are almost flat — you barely feel the rise. Other stretches are steep, and your legs know it instantly. “Slope” is just the number that captures *how steeply a line tilts* — and the beautiful thing is that one number does the whole job.

Here is the cleanest way to see it. Stand a line on the coordinate plane and ask: starting from the positive x -axis, how much do I have to rotate **anticlockwise** until I’m pointing along the line? That rotation angle is the line’s **inclination**, written θ . A line lying flat along the x -axis has $\theta = 0^\circ$. Tilt it up and θ grows; stand it straight up (parallel to the y -axis) and $\theta = 90^\circ$; tilt it the *other* way (going down as you move right) and θ is obtuse, somewhere between 90° and 180° . By convention we keep $0^\circ \leq \theta < 180^\circ$, because a line and the same line rotated by 180° are identical — there is no “arrow” on a line.

Think about it this way... the inclination θ is an *angle*, but angles are clumsy to compute with. So we convert the angle into a pure number using the tangent function: the **slope** is $m = \tan \theta$. Why tangent and not sine or cosine? Because slope should answer “for every step I take to the right, how far do I rise?” — and that is exactly *rise over run*, which is the tangent of the inclination. A gentle 30° track has slope $\tan 30^\circ \approx 0.58$ (rise a little for each step). A 45° track has slope exactly 1 (rise equals run). Past 45° the slope shoots up fast, and a line going *downhill* to the right has a *negative* slope, because the rise is negative.

Figure

A coordinate plane with three lines through the origin. Line A at 30° (gentle positive slope), Line B at 135° (negative slope, going down-right), and a dashed vertical line at 90° . Each labeled with its inclination angle θ measured anticlockwise from the positive x -axis, and the arc of θ drawn at the origin.

This single number tells you almost everything about a line’s *direction* (not its position — a line can have slope 2 while sitting anywhere on the plane). Two lines with the same slope point the same way: they are **parallel**. Two lines whose slopes multiply to -1 meet at a perfect right angle: they are **perpendicular**. And three points lie on one straight line exactly when the slope you measure between the first two equals the slope between the last two.

Domain of Validity & Edge Cases. The slope idea has exactly one place where it breaks, and exam-setters love it: the **vertical line**. A vertical line has inclination 90° ,

and $\tan 90^\circ$ does not exist — it runs off to infinity. So we say *the slope of a vertical line is undefined* (never “infinite”, and never “zero” — zero is the *horizontal* line, $\tan 0^\circ = 0$). The whole rise-over-run machinery also assumes the two points you use are genuinely different and not stacked vertically; if $x_1 = x_2$ the denominator collapses, which is just the vertical-line case showing up in algebra. Keep these two poles straight — horizontal means slope 0, vertical means slope undefined — and half the slope mistakes vanish.

Section 2: Key Formulas & Definitions

Inclination θ : the angle a line makes with the positive direction of the x -axis, measured anticlockwise.

- Convention / validity: $0^\circ \leq \theta < 180^\circ$ (equivalently $\theta \in [0, \pi)$ in radians).
- Horizontal line: $\theta = 0^\circ$. Vertical line: $\theta = 90^\circ$.

Slope (gradient) m :

$$m = \tan \theta, \quad \theta \neq 90^\circ$$

- Validity: defined for every non-vertical line. **Undefined** when $\theta = 90^\circ$.
- Range: $m \in (-\infty, \infty)$ as θ ranges over $[0^\circ, 180^\circ) \setminus \{90^\circ\}$.

Slope from two points $P(x_1, y_1)$, $Q(x_2, y_2)$:

$$m = \frac{y_2 - y_1}{x_2 - x_1}, \quad x_1 \neq x_2$$

- The result is independent of which point you call “first”: $m(PQ) = m(QP)$. A line segment carries no direction when we speak of slope.

Parallel lines: equal inclinations $\Rightarrow$

$$m_1 = m_2$$

Perpendicular lines: inclinations differ by $90^\circ \Rightarrow$

$$m_1 m_2 = -1 \quad \Longleftrightarrow \quad m_2 = -\frac{1}{m_1}$$

- Validity: holds only when *both* slopes are defined. If one line is vertical, the perpendicular to it is horizontal — handle this pair by inspection, not by the product rule.

Angle between two lines with slopes m_1, m_2 (acute angle θ):

$$\tan \theta = \left| \frac{m_2 - m_1}{1 + m_1 m_2} \right|, \quad 1 + m_1 m_2 \neq 0$$

- The two lines actually form two supplementary angles, θ and $180^\circ - \theta$; the modulus picks out the acute one.
- Degenerate case: $1 + m_1 m_2 = 0$ makes the right side blow up — that is precisely the perpendicular case, $\theta = 90^\circ$.

Collinearity of A, B, C :

$$\text{slope}(AB) = \text{slope}(BC)$$

Equally inclined to the axes: $\theta = 45^\circ$ or $135^\circ \Rightarrow m = \tan 45^\circ = 1$ or $\tan 135^\circ = -1$,
i.e. $m = \pm 1$.

Section 3: Theorems, Proofs & Standard Derivations

Theorem 1 — Slope in terms of two points

Given: A non-vertical line ℓ with inclination θ , passing through $P(x_1, y_1)$ and $Q(x_2, y_2)$.

To Prove: $m = \tan \theta = \frac{y_2 - y_1}{x_2 - x_1}$.

Construction: Drop perpendiculars from P and Q to the x -axis, meeting it at L and M . From P , drop a perpendicular PN onto QM , so that $\triangle QPN$ is right-angled at N .

Figure

Line ℓ rising to the right through $P(x_1, y_1)$ and $Q(x_2, y_2)$. Perpendiculars PL and QM dropped to the x -axis at L and M . A horizontal segment PN drawn from P meeting QM at N , forming right triangle QPN with the right angle at N . Angle QPN marked as θ .

Proof. Because PN is horizontal (parallel to the x -axis), the angle $\angle QPN$ that the line makes with PN equals the inclination θ (corresponding angles with the parallel x -axis). In right triangle QPN ,

$$PN = LM = x_2 - x_1 \quad (\text{horizontal run}), \quad QN = QM - NM = y_2 - y_1 \quad (\text{vertical rise}).$$

Therefore

$$\tan \theta = \frac{\text{opposite}}{\text{adjacent}} = \frac{QN}{PN} = \frac{y_2 - y_1}{x_2 - x_1}.$$

Since $m = \tan \theta$ by definition, the result follows. If $x_1 = x_2$, then $PN = 0$, the triangle degenerates, $\theta = 90^\circ$, and m is undefined — the vertical line. ■

Why each step matters: we built a right triangle on purpose so that the geometric inclination θ becomes a ratio of coordinate differences. The horizontal auxiliary line PN is the bridge that turns “angle” into “rise over run.”

Theorem 2 — Condition for perpendicularity

Given: Two non-vertical lines ℓ_1, ℓ_2 with inclinations θ_1, θ_2 and slopes $m_1 = \tan \theta_1$, $m_2 = \tan \theta_2$, meeting at right angles.

To Prove: $m_1 m_2 = -1$.

Proof. If the lines are perpendicular, their inclinations differ by a right angle; take $\theta_2 = \theta_1 + 90^\circ$. Then

$$m_2 = \tan(\theta_1 + 90^\circ) = -\cot \theta_1 = -\frac{1}{\tan \theta_1} = -\frac{1}{m_1},$$

using the identity $\tan(\alpha + 90^\circ) = -\cot \alpha$. Cross-multiplying gives $m_1 m_2 = -1$. The converse runs backward through the same identity. ■

Procedure — Finding the acute angle between two lines

Given slopes m_1 and m_2 :

1. **Check the special poles first.** If either line is vertical (slope undefined), compute the angle directly from inclinations (90° minus the other line's inclination). The formula below assumes both slopes exist.
2. Compute $1 + m_1 m_2$. If it equals 0, the lines are perpendicular — answer is 90° , stop.
3. Otherwise evaluate $\tan \theta = \left| \frac{m_2 - m_1}{1 + m_1 m_2} \right|$. The modulus guarantees you get $\tan$ of the *acute* angle.
4. Take $\theta = \tan^{-1}$ (that value). The obtuse angle, if asked, is $180^\circ - \theta$.

Reasoning behind the modulus: dropping the bars gives a signed value whose sign tells you whether you measured the angle from line 1 to line 2 clockwise or anticlockwise. For “the angle between the lines,” the acute angle is wanted, so we take the absolute value.

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

A line passes through the points $A(-3, 2)$ and $B(5, -4)$. Find its slope, and state its inclination's nature (acute or obtuse).

Given: $A(x_1, y_1) = (-3, 2)$, $B(x_2, y_2) = (5, -4)$.

Solution. Apply the two-point slope formula:

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{-4 - 2}{5 - (-3)} = \frac{-6}{8} = -\frac{3}{4}.$$

Since $m < 0$, the line falls as we move to the right, so its inclination is **obtuse** ($90^\circ < \theta < 180^\circ$).

$m = -\frac{3}{4}, \quad \text{inclination obtuse}$

Example 2 — [JEE Main · Speed Trap]

The line through $(2, 7)$ and $(k, 1)$ is perpendicular to the line through $(0, -3)$ and $(4, 5)$. Find k — fast.

Speed approach. First nail the *known* line's slope:

$$m_2 = \frac{5 - (-3)}{4 - 0} = \frac{8}{4} = 2.$$

Perpendicular means the first line's slope is the **negative reciprocal**, $m_1 = -\frac{1}{2}$ — don't recompute anything, just flip and negate. Now set up the first slope:

$$\frac{1-7}{k-2} = -\frac{1}{2} \Rightarrow \frac{-6}{k-2} = -\frac{1}{2} \Rightarrow k-2 = 12 \Rightarrow k = 14.$$

The trap: students rush and write $m_1 = +\frac{1}{2}$ (forgetting the *negative*), landing on $k = -10$ — a planted distractor. The fix is a one-second sanity check: perpendicular slopes always have *opposite signs*. Since $m_2 = +2$, m_1 **must** be negative.

$$\boxed{k = 14}$$

Example 3 — [JEE Main Level · multi-concept]

Three points are given: $P(1, 1)$, $Q(4, 7)$, and $R(a, a + 5)$. (i) Find a so that P, Q, R are collinear. (ii) For that a , find the acute angle the line makes with the line $y = x$.

Part (i) — collinearity (slope + collinearity).

$$\text{slope}(PQ) = \frac{7-1}{4-1} = \frac{6}{3} = 2.$$

For collinearity, $\text{slope}(PR) = \text{slope}(PQ)$:

$$\frac{(a+5)-1}{a-1} = 2 \Rightarrow a+4 = 2(a-1) \Rightarrow a+4 = 2a-2 \Rightarrow a = 6.$$

So $R = (6, 11)$, and the three points lie on a line of slope $m_1 = 2$.

Part (ii) — angle between lines (slope + angle formula). The line $y = x$ has slope $m_2 = 1$. Then

$$\tan \theta = \left| \frac{m_2 - m_1}{1 + m_1 m_2} \right| = \left| \frac{1 - 2}{1 + 2(1)} \right| = \left| \frac{-1}{3} \right| = \frac{1}{3}.$$

$$\boxed{a = 6, \quad \theta = \tan^{-1} \frac{1}{3} \approx 18.43^\circ}$$

Example 4 — [JEE Advanced Level]

A line ℓ makes an angle of 45° with the line $3x - y + 5 = 0$. Find **both** possible slopes of ℓ , and identify which (if any) is vertical.

Reasoning. The given line has slope $m_2 = 3$ (rearranged to $y = 3x + 5$). Let ℓ have slope m . Because “ 45° with a line” can be measured on either side, we must **not** take the modulus blindly — instead set the signed tangent equal to $\pm \tan 45^\circ = \pm 1$:

$$\frac{m-3}{1+3m} = \pm 1.$$

Case +1:

$$m-3 = 1+3m \Rightarrow -2m = 4 \Rightarrow m = -2.$$

Case –1:

$$m - 3 = -(1 + 3m) \Rightarrow m - 3 = -1 - 3m \Rightarrow 4m = 2 \Rightarrow m = \frac{1}{2}.$$

Vertical-line check (the Advanced subtlety): the formula silently assumed ℓ is non-vertical. Could a *vertical* line also make 45° with the given line? A vertical line has inclination 90° ; the given line has inclination $\tan^{-1} 3 \approx 71.57^\circ$, so the angle between them is $\approx 18.43^\circ \neq 45^\circ$. So no vertical solution — both genuine slopes come from the algebra.

$$m = -2 \text{ or } m = \frac{1}{2}; \text{ neither line is vertical}$$

Why this is Advanced-flavoured: the careless route is to write $\left| \frac{m-3}{1+3m} \right| = 1$ and “solve,” which hides that the equation genuinely has **two** solutions on the two sides of the given line, and it ignores the vertical-line possibility entirely. Both must be addressed for full marks.

Section 5: Practice Exercises

1. **[CBSE]** Find the slope of the line joining $(6, -2)$ and $(-4, 3)$, and state whether its inclination is acute or obtuse.
2. **[CBSE]** A line has inclination 120° . Find its slope, and find the slope of any line perpendicular to it.
3. **[JEE Main]** For what value of p is the line through $(p, 4)$ and $(2, -3)$ parallel to the line through $(0, 0)$ and $(7, 14)$?
4. **[JEE Main]** Find the acute angle between the lines with slopes $m_1 = \frac{1}{2}$ and $m_2 = -3$.
5. **[JEE Advanced]** The points $A(0, 2)$, $B(3, 0)$, $C(6, -2)$ are claimed to be collinear, and a fourth point $D(t, 5)$ is to lie on the *perpendicular* to line AB drawn through A . Find t (verify the collinearity claim first).

Answers: 1. $m = -\frac{1}{2}$, obtuse. 2. $m = -\sqrt{3}$; perpendicular slope $= \frac{1}{\sqrt{3}}$. 3. $p = -3$. 4. $\tan^{-1} 7 \approx 81.87^\circ$. 5. A, B, C are collinear (slope $-\frac{2}{3}$ throughout); the perpendicular through A has slope $\frac{3}{2}$, giving $t = 2$.

Section 6: Crack the MCQ

Q1. The slope of a line equally inclined to both coordinate axes can be: (A) only 1 (B) only -1 (C) 1 or -1 (D) 0 or undefined

Answer: (C). “Equally inclined to the axes” means the line makes 45° with each, so its inclination is 45° or 135° , giving $m = \tan 45^\circ = 1$ or $m = \tan 135^\circ = -1$.

- (A)/(B) each capture only one of the two valid orientations — the classic “forgot the other diagonal” error.

- (D) confuses “equally inclined to the axes” with “parallel to an axis,” which is the opposite situation.

Q2. Two lines are perpendicular. One has slope $\frac{2}{3}$. The other’s slope is: (A) $\frac{3}{2}$ (B) $-\frac{3}{2}$ (C) $-\frac{2}{3}$ (D) undefined

Answer: (B). Perpendicular slopes satisfy $m_1 m_2 = -1$, so $m_2 = -\frac{1}{(2/3)} = -\frac{3}{2}$.

- (A) reciprocates but drops the negative sign — the most common slip.
- (C) negates but forgets to reciprocate.
- (D) would be right only if the first line were *horizontal* (slope 0), not slope $\frac{2}{3}$.

Q3. The angle between the lines $y = \sqrt{3}x + 1$ and the x -axis is: (A) 30° (B) 45° (C) 60° (D) 90°

Answer: (C). The line has slope $\sqrt{3}$, and inclination $\theta = \tan^{-1} \sqrt{3} = 60^\circ$. Since the x -axis has inclination 0° , the angle between them is 60° .

- (A) comes from reading $\tan^{-1} \sqrt{3}$ as 30° (mixing up $\tan 30^\circ = \frac{1}{\sqrt{3}}$ with $\tan 60^\circ = \sqrt{3}$).
- (B) assumes slope 1.
- (D) assumes the line is vertical.

Q4. The points $A(2, 3)$, $B(4, 7)$ and $C(5, k)$ are collinear. Then k equals: (A) 5 (B) 9 (C) 11 (D) 3

Answer: (B). Collinearity requires $\text{slope}(AB) = \text{slope}(BC)$. Compute $\text{slope}(AB) = \frac{7-3}{4-2} = 2$, then

$$\text{slope}(BC) = \frac{k-7}{5-4} = k-7 = 2 \Rightarrow k = 9.$$

- (C) 11: used the *rise* (4) instead of the *slope* (2), writing $k = 7 + 4$ — confusing Δy with m .
- (A) 5: sign-flipped the denominator of $\text{slope}(AB)$ to $(2-4) = -2$, giving $k = 7 + (-2) = 5$.
- (D) 3: misread “collinear” as “horizontal,” forcing C to share A ’s y -coordinate.

Section 7: Common Student Pitfalls & Pro-Tips

1. **“Slope of a vertical line is infinity / zero.”** It is *undefined* — full stop. Zero is the *horizontal* line. Mixing these two poles is the most common slope error in board scripts.
2. **Perpendicular = reciprocal (forgetting the minus).** $m_1 m_2 = -1$, not $m_1 m_2 = 1$. If both your perpendicular slopes have the same sign, you’ve already made the mistake.
3. **Modulus on the angle formula when two answers exist.** For “find the line making angle α with a given line,” set the signed ratio equal to $\pm \tan \alpha$ — there

are genuinely two such lines. Use the modulus only when asked for *the angle between two given lines*.

4. **Ignoring the vertical-line case in Advanced problems.** Whenever you “let the slope be m ,” you’ve quietly excluded vertical lines. Check separately whether a vertical line also satisfies the condition.

Pro-Tip: Before any slope computation, glance at the geometry. Rising left-to-right $\Rightarrow$ slope positive; falling $\Rightarrow$ negative; flat $\Rightarrow$ 0; upright $\Rightarrow$ undefined. This 2-second sign check catches most arithmetic slips before they cost marks.

Section 8: Cheat Sheet Box

- **Inclination** θ : anticlockwise from $+x$ -axis, $0^\circ \leq \theta < 180^\circ$.
- **Slope:** $m = \tan \theta$ (undefined at $\theta = 90^\circ$, i.e. vertical lines).
- **Two points:** $m = \frac{y_2 - y_1}{x_2 - x_1}$ (needs $x_1 \neq x_2$).
- **Parallel:** $m_1 = m_2$. **Perpendicular:** $m_1 m_2 = -1$.
- **Angle between lines:** $\tan \theta = \left| \frac{m_2 - m_1}{1 + m_1 m_2} \right|$; denominator 0 $\Rightarrow$ perpendicular.
- **Collinear:** $\text{slope}(AB) = \text{slope}(BC)$.

Memory aids:

- “Horizontal Zero, Vertical Void” — slope 0 vs undefined.
- “Flip and Negate” for perpendicular slopes.
- “Equally inclined $\Rightarrow \pm 1$.”

Subtopic 02 — Equations of a Line

Exam Relevance: This is the engine room of coordinate geometry — every later result rides on writing a line's equation correctly. CBSE Class 11 finals reliably ask 2-4 mark questions on point-slope, intercept, or “reduce to a given form.” JEE Main almost guarantees a question on extracting slope/intercepts from $Ax + By + C = 0$ or on the normal form. JEE Advanced rarely tests a form in isolation but uses these as the substrate for locus, family-of-lines, and triangle problems — and quietly punishes the sign errors that creep into the normal form.

Section 1: Concept Intuition

A line is the simplest curve there is, yet there are *six different costumes* it can wear. Why so many? Because in real problems, you are handed different clues — sometimes a point and a direction, sometimes two points, sometimes where the line crosses the axes, sometimes how far it sits from the origin. Each form is just the answer to “given **this** clue, what's the fastest way to write the equation?”

Think of it like giving someone directions to a chai stall in a crowded bazaar. You could say “start at the gate and walk at this exact angle” (that's the **point-slope** idea: one point, one direction). Or “it's on the straight line connecting the temple and the post office” (**two-point form**). Or “it cuts the main road 3 shops in and the side lane 5 shops in” (**intercept form**). Or “stand at the central fountain, face this compass bearing, walk 7 metres, and you hit the line head-on” (**normal form**). Every description pins down the *same* line — they're translations of one another.

Think about it this way... all six forms are dialects of a single sentence: $Ax + By + C = 0$, the **general form**. Any straight line on the plane — slanted, horizontal, or bolt upright — can be written this way, and conversely every such equation (as long as A and B aren't *both* zero) draws a straight line. So the skill that actually matters in exams isn't memorising six formulas; it's fluently **translating** between them: pull the slope out of the general form, or rewrite a point-slope equation in intercept form, on demand.

Figure

A single line drawn on the coordinate plane, annotated four ways at once: a point $P(x_1, y_1)$ on it with a small slope-triangle (point-slope), the two axis-crossing points $(a, 0)$ and $(0, b)$ marked (intercept form), the perpendicular segment of length p from the origin meeting the line at foot N with angle ω to the x -axis (normal form), and the label “ $Ax + By + C = 0$ ” floating beside it. Caption: one line, many descriptions.

Domain of Validity & Edge Cases. Each costume has a situation where it simply doesn't fit:

- **Slope-intercept** $y = mx + c$ and **point-slope** both assume a *slope exists* — they cannot describe a **vertical line** (slope undefined). A vertical line is always $x = a$; a horizontal line is $y = b$.
- **Intercept form** $\frac{x}{a} + \frac{y}{b} = 1$ breaks if the line passes through the **origin** (both intercepts zero — you'd divide by zero) or is parallel to either axis (one intercept missing).

- **Normal form** insists $p \geq 0$ (a distance can't be negative) — getting p right in sign is the single most common slip, covered below.
 - **General form** is the only one that *never* fails; that's why it's the universal fallback.
-

Section 2: Key Formulas & Definitions

Let m denote slope, (x_1, y_1) a known point, c the y -intercept, a, b the x - and y -intercepts.

Horizontal / vertical lines:

$$y = b \quad (\text{horizontal}), \quad x = a \quad (\text{vertical}).$$

Point-slope form — line through (x_1, y_1) with slope m :

$$y - y_1 = m(x - x_1). \quad (\text{needs } m \text{ defined})$$

Two-point form — line through (x_1, y_1) and (x_2, y_2) :

$$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1), \quad x_1 \neq x_2.$$

Slope-intercept form — slope m , y -intercept c :

$$y = mx + c.$$

Intercept form — x -intercept a , y -intercept b :

$$\frac{x}{a} + \frac{y}{b} = 1, \quad a \neq 0, \quad b \neq 0.$$

Normal (perpendicular) form — perpendicular distance p from origin, normal making angle ω with $+x$ -axis:

$$x \cos \omega + y \sin \omega = p, \quad p \geq 0, \quad 0 \leq \omega < 360^\circ.$$

General form:

$$Ax + By + C = 0, \quad A, B \text{ not both zero.}$$

Reductions of the general form ($Ax + By + C = 0$):

- Slope: $m = -\frac{A}{B}$ ($B \neq 0$); y -intercept $c = -\frac{C}{B}$.
- Intercepts: $a = -\frac{C}{A}$, $b = -\frac{C}{B}$ ($C \neq 0$).
- Normal form: divide by $\pm\sqrt{A^2 + B^2}$, sign chosen so the right side is positive; then
$$p = \frac{|C|}{\sqrt{A^2 + B^2}}.$$

Section 3: Theorems, Proofs & Standard Derivations

Theorem 1 — Point-slope form

Given: A non-vertical line with slope m through the fixed point (x_1, y_1) .

To Prove: Every point (x, y) on the line satisfies $y - y_1 = m(x - x_1)$.

Proof. Let (x, y) be any point on the line *other than* (x_1, y_1) . The slope computed from these two points must equal the line's slope m :

$$\frac{y - y_1}{x - x_1} = m \implies y - y_1 = m(x - x_1).$$

The point (x_1, y_1) itself satisfies this (both sides become 0), so the equation holds for **every** point of the line. ■

Everything else descends from this. Putting a second point (x_2, y_2) in place of m gives the **two-point form**; setting $x_1 = 0$, $y_1 = c$ gives $y = mx + c$, the **slope-intercept form**.

Theorem 2 — Intercept form

Given: A line meeting the x -axis at $(a, 0)$ and the y -axis at $(0, b)$, with $a, b \neq 0$.

To Prove: $\frac{x}{a} + \frac{y}{b} = 1$.

Proof. Apply the two-point form to $(a, 0)$ and $(0, b)$:

$$y - 0 = \frac{b - 0}{0 - a}(x - a) \implies y = -\frac{b}{a}(x - a).$$

Multiply out and divide by b :

$$ay = -b(x - a) \implies bx + ay = ab \implies \frac{x}{a} + \frac{y}{b} = 1. \quad \blacksquare$$

Figure

A line crossing the x -axis at $A(a, 0)$ and the y -axis at $B(0, b)$, forming a right triangle OAB with the origin. The intercepts a and b labeled along the axes.

Theorem 3 — Normal form

Given: A line whose perpendicular distance from the origin O is $p > 0$, where the perpendicular ON (with N the foot on the line) makes angle ω with the positive x -axis.

To Prove: $x \cos \omega + y \sin \omega = p$.

Construction: Mark the foot N . Since $ON = p$ along direction ω , its coordinates are $N = (p \cos \omega, p \sin \omega)$.

Proof. The line passes through N and is **perpendicular** to ON . The direction ON has slope $\tan \omega$, so the line's slope is the negative reciprocal $-\frac{\cos \omega}{\sin \omega}$. By point-slope form

through N :

$$y - p \sin \omega = -\frac{\cos \omega}{\sin \omega} (x - p \cos \omega).$$

Multiply through by $\sin \omega$:

$$y \sin \omega - p \sin^2 \omega = -x \cos \omega + p \cos^2 \omega.$$

Collect and use $\sin^2 \omega + \cos^2 \omega = 1$:

$$x \cos \omega + y \sin \omega = p(\cos^2 \omega + \sin^2 \omega) = p. \quad \blacksquare$$

Figure

A line in the first/second quadrant with the perpendicular ON dropped from the origin O to the foot N on the line. ON has length p and makes angle ω with the positive x -axis. The line is shown clearly perpendicular to ON at N .

Procedure — Reducing $Ax + By + C = 0$ to normal form

This is the step students fumble, so the *reasoning* matters:

1. Move the constant to the right: $Ax + By = -C$.
2. Divide both sides by $\sqrt{A^2 + B^2}$ — but choose the sign of the radical so the **right-hand side becomes positive**, because that right side will be p and a distance must be non-negative. Concretely, pick the sign **opposite to that of C** .
3. Read off $\cos \omega = \frac{A}{\pm \sqrt{A^2 + B^2}}$, $\sin \omega = \frac{B}{\pm \sqrt{A^2 + B^2}}$, $p = \frac{|C|}{\sqrt{A^2 + B^2}}$.
4. Determine ω from the **signs** of $\cos \omega$ and $\sin \omega$ together (they fix the quadrant) — never from $\cos \omega$ alone.

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

Find the equation of the line passing through $(-2, 5)$ and $(3, -1)$, and state its y -intercept.

Solution. First the slope:

$$m = \frac{-1 - 5}{3 - (-2)} = \frac{-6}{5}.$$

Two-point (point-slope through $(-2, 5)$):

$$y - 5 = -\frac{6}{5}(x + 2) \Rightarrow 5y - 25 = -6x - 12 \Rightarrow 6x + 5y - 13 = 0.$$

y -intercept: set $x = 0 \Rightarrow 5y = 13 \Rightarrow y = \frac{13}{5}$.

$$6x + 5y - 13 = 0, \quad y\text{-intercept} = \frac{13}{5}$$

Example 2 — [JEE Main · Speed Trap]

For the line $3x - 4y + 12 = 0$, write down its slope, x -intercept, and y -intercept — fast.

Speed approach. Don't rearrange the whole thing; use the reduction formulas directly.

- Slope $= -\frac{A}{B} = -\frac{3}{-4} = \frac{3}{4}$.
- x -intercept: set $y = 0 \Rightarrow 3x = -12 \Rightarrow x = -4$.
- y -intercept: set $x = 0 \Rightarrow -4y = -12 \Rightarrow y = 3$.

$$m = \frac{3}{4}, \quad a = -4, \quad b = 3$$

The trap: the slope is $-\frac{A}{B}$, **not** $\frac{A}{B}$. A student who writes $\frac{A}{B} = \frac{3}{-4} = -\frac{3}{4}$ gets the sign wrong — and the two answers $\pm\frac{3}{4}$ are *both* on the option list. Anchor: rearranging to $y = \frac{3}{4}x + 3$ shows the slope is positive, matching the fact that the line rises to the right.

Example 3 — [JEE Main Level · multi-concept]

Find the equation of the line through $(2, 3)$ that is perpendicular to $4x - 3y + 7 = 0$, and express it in intercept form.

Step 1 — slope of the given line (reduction): $m_1 = -\frac{A}{B} = -\frac{4}{-3} = \frac{4}{3}$.

Step 2 — perpendicular slope (Subtopic 01 condition, flip-and-negate): $m = -\frac{1}{m_1} = -\frac{3}{4}$.

Step 3 — point-slope through $(2, 3)$:

$$y - 3 = -\frac{3}{4}(x - 2) \Rightarrow 4y - 12 = -3x + 6 \Rightarrow 3x + 4y - 18 = 0.$$

Step 4 — intercept form: write $3x + 4y = 18$, divide by 18:

$$\frac{x}{6} + \frac{y}{4.5} = 1.$$

$$3x + 4y - 18 = 0, \quad \frac{x}{6} + \frac{y}{9/2} = 1$$

Example 4 — [JEE Advanced Level]

Reduce $x - \sqrt{3}y + 8 = 0$ to normal form, finding p and ω exactly. Then explain why a naive sign choice would give a forbidden answer.

Solution. Here $A = 1$, $B = -\sqrt{3}$, $C = 8$, and $\sqrt{A^2 + B^2} = \sqrt{1 + 3} = 2$.

Move the constant: $x - \sqrt{3}y = -8$. We must divide by ± 2 so the right side is **positive**.

Since $-8 < 0$, choose the **negative** radical:

$$\frac{x}{-2} - \frac{\sqrt{3}y}{-2} = \frac{-8}{-2} \Rightarrow -\frac{1}{2}x + \frac{\sqrt{3}}{2}y = 4.$$

So

$$\cos \omega = -\frac{1}{2}, \quad \sin \omega = \frac{\sqrt{3}}{2}, \quad p = 4.$$

$\cos \omega < 0$ and $\sin \omega > 0$ place ω in the **second quadrant**: $\omega = 120^\circ$.

$$\boxed{x \cos 120^\circ + y \sin 120^\circ = 4, \quad p = 4, \quad \omega = 120^\circ}$$

The forbidden naive answer: dividing by $+2$ instead would give $\frac{1}{2}x - \frac{\sqrt{3}}{2}y = -4$, i.e. “ $p = -4$ ” — a negative distance, which is impossible. That alone tells you the sign choice was wrong. Also note: had you computed ω from $\cos \omega = -\frac{1}{2}$ *alone*, you’d be unsure between 120° and 240° ; only checking $\sin \omega > 0$ pins it to 120° .

Section 5: Practice Exercises

1. **[CBSE]** Find the equation of the line with slope -2 passing through $(4, -1)$.
2. **[CBSE]** Find the equation of the line making an x -intercept of 3 and a y -intercept of -5 (use intercept form, then clear fractions).
3. **[JEE Main]** Reduce $5x + 12y - 39 = 0$ to normal form and find p , $\cos \omega$, and $\sin \omega$.
4. **[JEE Main]** Find the equation of the line through $(1, 2)$ that makes **equal** (non-zero) intercepts on the axes.
5. **[JEE Advanced]** A line has positive x - and y -intercepts a and b with $a + b = 9$, and it passes through $(2, 2)$. Find **all** such lines.

Answers: 1. $2x + y - 7 = 0$. 2. $\frac{x}{3} + \frac{y}{-5} = 1 \Rightarrow 5x - 3y - 15 = 0$. 3. $p = 3$, $\cos \omega = \frac{5}{13}$, $\sin \omega = \frac{12}{13}$ ($\omega = \tan^{-1} \frac{12}{5} \approx 67.4^\circ$). 4. $x + y = 3$. 5. $a^2 - 9a + 18 = 0 \Rightarrow a = 3, b = 6$ or $a = 6, b = 3$, giving $2x + y = 6$ and $x + 2y = 6$.

Section 6: Crack the MCQ

Q1. The slope and y -intercept of the line $2x - 5y + 10 = 0$ are: (A) $(\frac{2}{5}, 2)$ (B) $(-\frac{2}{5}, 2)$ (C) $(\frac{2}{5}, -2)$ (D) $(\frac{5}{2}, 2)$

Answer: (A). $m = -\frac{A}{B} = -\frac{2}{-5} = \frac{2}{5}$; $c = -\frac{C}{B} = -\frac{10}{-5} = 2$.

- (B) used $m = +\frac{A}{B}$ with a stray sign — the $-A/B$ slip.
- (C) took $c = +\frac{C}{B}$, dropping the minus on the intercept.
- (D) inverted the slope to B/A .

Q2. When reducing $Ax + By + C = 0$ to normal form $x \cos \omega + y \sin \omega = p$, we divide both sides by: (A) $\sqrt{A^2 + B^2}$ (B) $\pm \sqrt{A^2 + B^2}$, choosing the sign so that $p > 0$ (C)

$$A^2 + B^2 \quad (\text{D}) \quad |C|$$

Answer: (B). The whole point of the $\pm$ is to force a non-negative distance p .

- (A) ignores the sign choice, producing a negative p whenever C has the wrong sign.
- (C) forgot the square root, so $\cos^2 \omega + \sin^2 \omega \neq 1$.
- (D) confuses the divisor with the numerator of p .

Q3. A line passes through the origin. Which form **cannot** directly represent it? (A) Slope-intercept $y = mx + c$ (B) Point-slope (C) Intercept form $\frac{x}{a} + \frac{y}{b} = 1$ (D) General form

Answer: (C). A line through the origin has both intercepts equal to 0, so $\frac{x}{a} + \frac{y}{b} = 1$ would require dividing by zero — impossible.

- (A) works fine with $c = 0$ ($y = mx$).
- (B) works with $(x_1, y_1) = (0, 0)$.
- (D) works: $C = 0$ gives $Ax + By = 0$, which passes through the origin.

Q4. The x -intercept and y -intercept of $3x - 4y = 12$ are, respectively: (A) 4 and -3 (B) -4 and 3 (C) 3 and -4 (D) 4 and 3

Answer: (A). $y = 0 \Rightarrow 3x = 12 \Rightarrow x = 4$; $x = 0 \Rightarrow -4y = 12 \Rightarrow y = -3$.

- (B) flipped both signs.
- (C) misread the coefficients 3 and 4 as the intercepts.
- (D) forgot the negative on the y -intercept.

Section 7: Common Student Pitfalls & Pro-Tips

1. **Slope from the general form.** It is $m = -\frac{A}{B}$, **not** $\frac{A}{B}$. Forgetting the minus is the most frequent error on this subtopic.
2. **Negative p in normal form.** Distance can't be negative. If your reduction yields $p < 0$, you chose the wrong sign for the radical — flip it.
3. **Reading ω from one trig value.** $\cos \omega = -\frac{1}{2}$ alone leaves 120° vs 240° ambiguous. Always use the **signs of $\cos \omega$ and $\sin \omega$ together** to fix the quadrant.
4. **Forcing intercept form on the wrong lines.** Lines through the origin, or parallel to an axis, have no valid intercept form — use slope-intercept or $x = a/y = b$ instead.

Pro-Tip: To find intercepts of *any* line equation in two seconds, set $y = 0$ for the x -intercept and $x = 0$ for the y -intercept — no need to memorise $-C/A$, $-C/B$ and risk a sign error. The substitution carries the signs for you.

Section 8: Cheat Sheet Box

- **Horizontal / vertical:** $y = b / x = a$.
- **Point-slope:** $y - y_1 = m(x - x_1)$.
- **Two-point:** $y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$.
- **Slope-intercept:** $y = mx + c$.
- **Intercept:** $\frac{x}{a} + \frac{y}{b} = 1$ ($a, b \neq 0$).
- **Normal:** $x \cos \omega + y \sin \omega = p$ ($p \geq 0$).
- **General:** $Ax + By + C = 0 \Rightarrow m = -\frac{A}{B}, a = -\frac{C}{A}, b = -\frac{C}{B}$.

Memory aids:

- “Minus A over B” for slope from the general form.
- “p is positive, period” for the normal-form sign check.
- “Set the other to zero” to read off intercepts.

Subtopic 03 — Distance Formulas

Exam Relevance: A perennial favourite because it's short to ask and easy to trap. CBSE Class 11 finals carry a 2-3 mark direct distance computation almost every year. JEE Main reliably has one question on the distance of a point from a line or between two parallel lines — usually with a normalization trap baked in. JEE Advanced uses the **symmetric (distance) form** to find distances to intersection points and to set up locus problems, where it quietly outperforms brute-force algebra.

Section 1: Concept Intuition

You already know how to *write* a line (Subtopic 02). The natural next question a surveyor, an architect, or an exam-setter asks is: *how far is something from that line?* If a railway track is the line and your house is a point, the “distance” that matters isn't how far you'd walk along some random diagonal — it's the **shortest** path, the perpendicular dropped straight onto the track. That single perpendicular length is what every distance formula in this subtopic computes.

Think about it this way... picture the line $Ax + By + C = 0$ and recall a quiet fact from Subtopic 02: the vector (A, B) points **perpendicular** to the line (it's the line's normal direction). So to measure how far a point P sits from the line, you don't need the line's slope at all — you just project the gap between P and the line onto that normal direction and read off the length. That projection idea is the engine behind the famous formula

$$d = \frac{|Ax_1 + By_1 + C|}{\sqrt{A^2 + B^2}},$$

and once you see it as “project onto the normal,” the formula stops being something to memorise and becomes something you could rebuild on the spot.

Two parallel lines are just a special case: they share the same normal direction, so the distance between them is the difference of how far each sits from the origin — which collapses to a clean $\frac{|C_1 - C_2|}{\sqrt{A^2 + B^2}}$, **provided** you've first written both with the *same* A and B . That proviso is where most marks are lost.

There's also a third tool, the **symmetric (distance) form**. Instead of describing a line by slope or intercepts, you describe it by “stand at point P , face the inclination θ , and walk a signed distance r .” Every point on the line is then $(x_1 + r \cos \theta, y_1 + r \sin \theta)$. The beauty is that r is the distance — so if you want the point 5 units up the line, or the distance from P to where the line crosses another line, you solve for r directly.

Figure

A line $Ax + By + C = 0$ with a point $P(x_1, y_1)$ off it. The perpendicular from P meets the line at foot Q ; segment PQ labeled d . A short arrow from the line labeled “normal (A, B) ” showing the projection direction. Caption: distance = perpendicular drop = projection onto the normal.

Domain of Validity & Edge Cases.

- The point-to-line formula needs the line in general form $Ax + By + C = 0$; if it's

given as $y = mx + c$, rearrange to $mx - y + c = 0$ first.

- The parallel-lines formula is **only valid after normalizing** — both lines must show identical A and B . Apply it to $3x + 4y + 1 = 0$ and $6x + 8y + 5 = 0$ as-written and you get a wrong answer.
- The symmetric form uses the **unit** direction $(\cos \theta, \sin \theta)$; if you substitute un-normalized direction numbers, r no longer equals the true distance.
- All three give the **perpendicular** distance; none of them measure “slant” distances along arbitrary directions.

Section 2: Key Formulas & Definitions

Lead-in — Representation theorem (the one new general-equation fact): every first-degree equation $Ax + By + C = 0$ (with A, B not both zero) represents a straight line, and conversely every line has such an equation. This is why “general form” is the universal description; proved in Section 3.

Distance of a point $P(x_1, y_1)$ from the line $Ax + By + C = 0$:

$$d = \frac{|Ax_1 + By_1 + C|}{\sqrt{A^2 + B^2}}.$$

- Validity: line in general form. The modulus guarantees $d \geq 0$.
- The **signed** quantity $\frac{Ax_1 + By_1 + C}{\sqrt{A^2 + B^2}}$ (without the bars) tells you *which side* of the line P lies on.

Distance between two parallel lines $Ax + By + C_1 = 0$ and $Ax + By + C_2 = 0$ (identical A, B):

$$d = \frac{|C_1 - C_2|}{\sqrt{A^2 + B^2}}.$$

- Validity: coefficients of x and y must be made identical **first**.

Symmetric / distance / parametric form of the line through $P(x_1, y_1)$ with inclination θ :

$$\frac{x - x_1}{\cos \theta} = \frac{y - y_1}{\sin \theta} = r,$$

where r is the **signed distance** from P to the point (x, y) along the line ($r > 0$ in the direction of θ , $r < 0$ opposite). Equivalently $x = x_1 + r \cos \theta$, $y = y_1 + r \sin \theta$.

Section 3: Theorems, Proofs & Standard Derivations

Theorem 0 (lead-in) — Every first-degree equation is a line

Given: $Ax + By + C = 0$ with A, B not both zero.

To Prove: its graph is a straight line.

Proof (sketch by the standard argument). If $B \neq 0$, rewrite as $y = -\frac{A}{B}x - \frac{C}{B}$, which is the slope-intercept form $y = mx + c$ — a line of slope $-\frac{A}{B}$. If $B = 0$ (so

$A \neq 0$), the equation is $x = -\frac{C}{A}$, a vertical line. Either way the graph is a straight line. The converse holds because each standard form derived in Subtopic 02 rearranges into first-degree shape. ■

Theorem 1 — Distance of a point from a line

Given: Line $\ell : Ax + By + C = 0$ and point $P(x_1, y_1)$.

To Prove: $d(P, \ell) = \frac{|Ax_1 + By_1 + C|}{\sqrt{A^2 + B^2}}$.

Proof (projection onto the normal). The vector $\mathbf{n} = (A, B)$ is normal to ℓ , with unit normal $\hat{\mathbf{n}} = \frac{(A, B)}{\sqrt{A^2 + B^2}}$. Pick **any** point $R(x_0, y_0)$ on ℓ , so that

$$Ax_0 + By_0 + C = 0 \implies Ax_0 + By_0 = -C. \quad (*)$$

The perpendicular distance from P to ℓ equals the length of the projection of $\overrightarrow{RP} = (x_1 - x_0, y_1 - y_0)$ onto $\hat{\mathbf{n}}$:

$$d = |\overrightarrow{RP} \cdot \hat{\mathbf{n}}| = \frac{|A(x_1 - x_0) + B(y_1 - y_0)|}{\sqrt{A^2 + B^2}} = \frac{|Ax_1 + By_1 - (Ax_0 + By_0)|}{\sqrt{A^2 + B^2}}.$$

Substituting (*), the bracket becomes $Ax_1 + By_1 - (-C) = Ax_1 + By_1 + C$, giving

$$d = \frac{|Ax_1 + By_1 + C|}{\sqrt{A^2 + B^2}}. \quad \blacksquare$$

Why the choice of R doesn't matter: the projection onto the normal is the same no matter which point of the line you start from — moving R along ℓ changes $\overrightarrow{RP}$ only by a vector *parallel* to ℓ , which is perpendicular to $\hat{\mathbf{n}}$ and so contributes nothing to the dot product.

Theorem 2 — Distance between two parallel lines

Given: $\ell_1 : Ax + By + C_1 = 0$ and $\ell_2 : Ax + By + C_2 = 0$ (same A, B).

To Prove: $d = \frac{|C_1 - C_2|}{\sqrt{A^2 + B^2}}$.

Proof. Parallel lines are everywhere the same distance apart, so pick a convenient point on ℓ_1 and measure its distance to ℓ_2 . If $A \neq 0$, the point $(-\frac{C_1}{A}, 0)$ lies on ℓ_1 . Its distance from ℓ_2 by Theorem 1:

$$d = \frac{|A(-\frac{C_1}{A}) + B(0) + C_2|}{\sqrt{A^2 + B^2}} = \frac{|-C_1 + C_2|}{\sqrt{A^2 + B^2}} = \frac{|C_1 - C_2|}{\sqrt{A^2 + B^2}}. \quad \blacksquare$$

Crucial precondition: this assumed ℓ_1, ℓ_2 already share identical A, B . If not, normalize first — e.g. divide $6x + 8y + 5 = 0$ by 2 to get $3x + 4y + 2.5 = 0$ before comparing with $3x + 4y + 1 = 0$.

Derivation note — the symmetric form gives distance directly

A point at signed distance r from $P(x_1, y_1)$ along inclination θ has coordinates $(x_1 + r \cos \theta, y_1 + r \sin \theta)$, since $(\cos \theta, \sin \theta)$ is the unit step in that direction. Rearranging each coordinate, $r = \frac{x - x_1}{\cos \theta} = \frac{y - y_1}{\sin \theta}$. Because the direction is a *unit* vector, the parameter r is the genuine distance — substitute the line's other constraint and solve for r to read distances off immediately.

Section 4: Worked Examples**Example 1 — [CBSE Board Level]**

Find the distance of the point $(3, -4)$ from the line $4x + 3y - 5 = 0$.

Solution. Here $A = 4$, $B = 3$, $C = -5$, and $(x_1, y_1) = (3, -4)$.

$$d = \frac{|4(3) + 3(-4) - 5|}{\sqrt{4^2 + 3^2}} = \frac{|12 - 12 - 5|}{\sqrt{25}} = \frac{5}{5} = 1.$$

$$\boxed{d = 1 \text{ unit}}$$

Example 2 — [JEE Main · Speed Trap]

Find the distance between the parallel lines $3x - 4y + 7 = 0$ and $6x - 8y - 5 = 0$.

Speed approach. The lines are parallel (coefficients proportional), but **not yet comparable** — the second has doubled coefficients. Normalize it first by dividing by 2:

$$6x - 8y - 5 = 0 \implies 3x - 4y - 2.5 = 0.$$

Now $C_1 = 7$, $C_2 = -2.5$, with $\sqrt{3^2 + 4^2} = 5$:

$$d = \frac{|7 - (-2.5)|}{5} = \frac{9.5}{5} = 1.9.$$

$$\boxed{d = 1.9 = \frac{19}{10} \text{ units}}$$

The trap: plugging the lines in as-written — $\frac{|7 - (-5)|}{5} = \frac{12}{5} = 2.4$ — is the planted wrong answer. The formula $\frac{|C_1 - C_2|}{\sqrt{A^2 + B^2}}$ is **only** valid once both lines show the *same* A, B . Always normalize before subtracting constants.

Example 3 — [JEE Main Level · multi-concept]

Find the perpendicular distance from the origin to the line joining $A(1, 2)$ and $B(4, 6)$.

Step 1 — equation of line AB (two-point form, Subtopic 02): slope $= \frac{6 - 2}{4 - 1} = \frac{4}{3}$, so

$$y - 2 = \frac{4}{3}(x - 1) \implies 3y - 6 = 4x - 4 \implies 4x - 3y + 2 = 0.$$

Step 2 — distance from origin $(0, 0)$ (Theorem 1):

$$d = \frac{|4(0) - 3(0) + 2|}{\sqrt{4^2 + (-3)^2}} = \frac{2}{5} = 0.4.$$

$$\boxed{d = \frac{2}{5} = 0.4 \text{ units}}$$

Example 4 — [JEE Advanced Level]

A line through $P(1, 1)$ has inclination $\theta = 60^\circ$ and meets the line $2x + y = 8$ at Q . Using the symmetric form, find the distance PQ exactly.

Solution. Parametrize the line through P with the unit direction $(\cos 60^\circ, \sin 60^\circ) = (\frac{1}{2}, \frac{\sqrt{3}}{2})$:

$$x = 1 + \frac{r}{2}, \quad y = 1 + \frac{\sqrt{3}}{2}r.$$

The point Q lies on $2x + y = 8$, so substitute and solve for r — which **is** the distance PQ :

$$2\left(1 + \frac{r}{2}\right) + \left(1 + \frac{\sqrt{3}}{2}r\right) = 8 \Rightarrow 3 + r + \frac{\sqrt{3}}{2}r = 8 \Rightarrow r\left(1 + \frac{\sqrt{3}}{2}\right) = 5.$$

Thus $r \cdot \frac{2 + \sqrt{3}}{2} = 5 \Rightarrow r = \frac{10}{2 + \sqrt{3}}$. Rationalize:

$$r = \frac{10(2 - \sqrt{3})}{(2 + \sqrt{3})(2 - \sqrt{3})} = \frac{10(2 - \sqrt{3})}{4 - 3} = 10(2 - \sqrt{3}) = 20 - 10\sqrt{3}.$$

$$\boxed{PQ = 10(2 - \sqrt{3}) = 20 - 10\sqrt{3} \approx 2.68 \text{ units}}$$

Why this beats brute force: the symmetric form delivers the distance as a single solved parameter r — no need to find Q 's coordinates first and then apply a distance formula. The unit direction is what makes r equal the true length; this fails silently if you use non-unit direction numbers.

Section 5: Practice Exercises

1. **[CBSE]** Find the distance of the point $(2, 3)$ from the line $3x + 4y - 12 = 0$.
2. **[CBSE]** Find the distance between the parallel lines $2x + y - 3 = 0$ and $2x + y + 7 = 0$.
3. **[JEE Main]** Find the distance between $5x + 12y - 10 = 0$ and $10x + 24y + 13 = 0$ (mind the normalization).
4. **[JEE Main]** Using the symmetric form, find the coordinates of the points on the line through $(1, 2)$ with inclination 30° that are at distance 4 from $(1, 2)$.
5. **[JEE Advanced]** The parallel lines $3x - 4y + 5 = 0$ and $3x - 4y - 10 = 0$ bound a strip. Find the width of the strip and the equation of the line running exactly

midway between them.

Answers: 1. $d = \frac{|6 + 12 - 12|}{5} = \frac{6}{5} = 1.2$. 2. $\frac{|-3 - 7|}{\sqrt{5}} = \frac{10}{\sqrt{5}} = 2\sqrt{5}$. 3.

Normalize the second to $5x + 12y + 6.5 = 0$; $d = \frac{|-10 - 6.5|}{13} = \frac{16.5}{13} = \frac{33}{26} \approx 1.27$.

4. $(1 + 2\sqrt{3}, 4)$ and $(1 - 2\sqrt{3}, 0)$. 5. Width = $\frac{|5 - (-10)|}{5} = 3$; midline (average of constants) $3x - 4y - 2.5 = 0$, i.e. $6x - 8y - 5 = 0$.

Section 6: Crack the MCQ

Q1. The distance of the origin from the line $3x - 4y + 10 = 0$ is: (A) 2 (B) 10 (C) $\frac{10}{7}$ (D) -2

Answer: (A). $d = \frac{|3(0) - 4(0) + 10|}{\sqrt{3^2 + 4^2}} = \frac{10}{5} = 2$.

- (B) forgot to divide by $\sqrt{A^2 + B^2}$.
- (C) divided by $|A| + |B| = 7$ instead of $\sqrt{A^2 + B^2} = 5$.
- (D) a distance can never be negative — modulus dropped.

Q2. The distance between $4x + 3y - 5 = 0$ and $8x + 6y + 10 = 0$ is: (A) 2 (B) 3 (C) 1.5 (D) 5

Answer: (A). Normalize the second: $8x + 6y + 10 = 0 \Rightarrow 4x + 3y + 5 = 0$. Then $C_1 = -5$, $C_2 = 5$, $\sqrt{16 + 9} = 5$, so $d = \frac{|-5 - 5|}{5} = 2$.

- (B) used the un-normalized constant: $\frac{|-5 - 10|}{5} = 3$.
- (C) used both un-normalized constant and un-normalized denominator $\sqrt{64 + 36} = 10$: $\frac{15}{10} = 1.5$.
- (D) random magnitude with no consistent method.

Q3. The perpendicular distance of $P(x_1, y_1)$ from $Ax + By + C = 0$ is: (A) $\frac{|Ax_1 + By_1 + C|}{\sqrt{A^2 + B^2}}$ (B) $\frac{Ax_1 + By_1 + C}{\sqrt{A^2 + B^2}}$ (C) $\frac{|Ax_1 + By_1 + C|}{A^2 + B^2}$ (D) $\frac{|Ax_1 + By_1 + C|}{\sqrt{A + B}}$

Answer: (A).

- (B) dropped the modulus — yields a signed value that can be negative (that *signed* version tells you the side, not the distance).
- (C) forgot the square root in the denominator.
- (D) wrong radicand ($\sqrt{A + B}$ instead of $\sqrt{A^2 + B^2}$).

Q4. In the symmetric form $\frac{x - x_1}{\cos \theta} = \frac{y - y_1}{\sin \theta} = r$, the parameter r represents: (A) the slope of the line (B) the signed distance from (x_1, y_1) to (x, y) along the line (C) the y -intercept (D) the inclination

Answer: (B). Because $(\cos \theta, \sin \theta)$ is a unit vector, each unit of r moves exactly one unit along the line, so r is the genuine signed distance.

- (A) the slope is $\tan \theta$, not r .
- (C)/(D) confuse the running distance with fixed features of the line.

Section 7: Common Student Pitfalls & Pro-Tips

1. **Skipping normalization for parallel lines.** $\frac{|C_1 - C_2|}{\sqrt{A^2 + B^2}}$ is valid **only** when both lines show identical A, B . Always rescale one line first.
2. **Dropping the modulus in point-to-line distance.** Distance is non-negative; the bars are mandatory. (The *signed* value is useful — see the Pro-Tip — but it isn't "the distance.")
3. **Wrong denominator.** It is $\sqrt{A^2 + B^2}$ — not $A^2 + B^2$, and not $|A| + |B|$. This is the same root-sum-of-squares that appeared in the normal form (Subtopic 02); treat it as one recurring object.
4. **Non-unit direction in the symmetric form.** If you use direction numbers that aren't $(\cos \theta, \sin \theta)$, the parameter stops equalling the true distance.

Pro-Tip: The *signed* quantity $\frac{Ax_1 + By_1 + C}{\sqrt{A^2 + B^2}}$ tells you which **side** of the line a point is on: two points give the same sign $\Rightarrow$ same side, opposite signs $\Rightarrow$ the line separates them. This answers "do these points lie on the same side?" and "is the origin inside this triangle?" with no extra construction.

Section 8: Cheat Sheet Box

- **Point-line distance:** $d = \frac{|Ax_1 + By_1 + C|}{\sqrt{A^2 + B^2}}$.
- **Parallel lines:** $d = \frac{|C_1 - C_2|}{\sqrt{A^2 + B^2}}$ (normalize to identical A, B first).
- **Symmetric / distance form:** $\frac{x - x_1}{\cos \theta} = \frac{y - y_1}{\sin \theta} = r$, with r = signed distance.
- **Side test:** sign of $Ax_1 + By_1 + C$ identifies which side of the line a point lies on.

Memory aids:

- "Normalize, then subtract" for parallel lines.
- "Modulus on top, root-sum-squares below."
- " r is the ruler" — in the symmetric form the parameter is the distance.

Subtopic 04 — Point of Intersection, Concurrency & Family of Lines

Exam Relevance: A JEE staple. CBSE may ask the intersection point or a simple concurrency check (3–4 marks). JEE Main almost always carries a concurrency or family-of-lines question. JEE Advanced loves the **family trick** $L_1 + \lambda L_2 = 0$ — it lets you write a line through an intersection *without ever finding the intersection point*, which is exactly the elegance examiners reward.

Section 1: Concept Intuition

Two roads that aren't parallel cross at exactly one junction. Finding that junction algebraically is just solving the two line equations together — the (x, y) that satisfies both is the crossing point. That's the whole idea of **point of intersection**.

Now scale up: when do *three* roads all meet at a single roundabout? That's **concurrency**, and it's a surprisingly common exam setup ("show the medians/altitudes meet at a point"). You could find where two of them cross and then check the third passes through it — but there's a slicker determinant test that does it in one shot.

Think about it this way... here's the cleverest idea in the chapter. Suppose two lines $L_1 = 0$ and $L_2 = 0$ cross at a point J . Consider the combination

$$L_1 + \lambda L_2 = 0$$

for some number λ . At the junction J , both L_1 and L_2 are zero, so the combination is automatically zero too — meaning **every** such line passes through J , no matter what λ is! So $L_1 + \lambda L_2 = 0$ is a "family" of all lines through the junction (one for each λ). If you need *the* line through J that also satisfies one more condition (passes through a given point, has a given slope, is perpendicular to something), just impose that condition to pin down λ — you never have to compute J itself. It feels like magic the first time.

Domain of Validity & Edge Cases.

- Two lines have a **unique** intersection only when they're not parallel: $a_1b_2 - a_2b_1 \neq 0$. If this is zero they're parallel (no intersection) or coincident (infinitely many).
- The family $L_1 + \lambda L_2 = 0$ captures every line through J **except** $L_2 = 0$ **itself** (you'd need $\lambda \rightarrow \infty$). If your answer might be L_2 , check it separately.
- The concurrency determinant being zero is **necessary** but assumes no two of the three lines are parallel; always sanity-check.

Section 2: Key Formulas & Definitions

Point of intersection of $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ (Cramer's rule):

$$x = \frac{b_1c_2 - b_2c_1}{a_1b_2 - a_2b_1}, \quad y = \frac{c_1a_2 - c_2a_1}{a_1b_2 - a_2b_1}, \quad a_1b_2 - a_2b_1 \neq 0.$$

Condition for two lines to be parallel / intersecting / coincident:

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2} \text{ (parallel); } \frac{a_1}{a_2} \neq \frac{b_1}{b_2} \text{ (intersecting); } \frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2} \text{ (coincident).}$$

Concurrency of three lines $a_i x + b_i y + c_i = 0$ ($i = 1, 2, 3$): they are concurrent if

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0,$$

provided no two are parallel.

Family of lines through the intersection of $L_1 = 0$ and $L_2 = 0$:

$$L_1 + \lambda L_2 = 0 \quad (\lambda \in \mathbb{R}),$$

representing every line through their common point (except $L_2 = 0$).

Section 3: Theorems, Proofs & Standard Derivations

Theorem 1 — The family $L_1 + \lambda L_2 = 0$ passes through the intersection

Given: $L_1 \equiv a_1 x + b_1 y + c_1 = 0$ and $L_2 \equiv a_2 x + b_2 y + c_2 = 0$ meeting at $J(x_0, y_0)$.

To Prove: For every λ , the line $L_1 + \lambda L_2 = 0$ passes through J .

Proof. Since J lies on both, $L_1(x_0, y_0) = 0$ and $L_2(x_0, y_0) = 0$. Then

$$L_1(x_0, y_0) + \lambda L_2(x_0, y_0) = 0 + \lambda \cdot 0 = 0,$$

so J satisfies $L_1 + \lambda L_2 = 0$ for every λ . Because $L_1 + \lambda L_2 = 0$ is itself first-degree (a line) whenever L_1, L_2 aren't proportional, it is a line through J . ■

Theorem 2 — Concurrency determinant

Given: three lines $L_i \equiv a_i x + b_i y + c_i = 0$, no two parallel.

To Prove: they are concurrent $\iff \det[a_i \ b_i \ c_i] = 0$.

Proof (sketch). Let L_1, L_2 meet at $J(x_0, y_0)$. Concurrency means L_3 also passes through J , i.e. $a_3 x_0 + b_3 y_0 + c_3 = 0$. The three statements $a_i x_0 + b_i y_0 + c_i = 0$ say the system

$$\begin{pmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{pmatrix} \begin{pmatrix} x_0 \\ y_0 \\ 1 \end{pmatrix} = \mathbf{0}$$

has the **nonzero** solution $(x_0, y_0, 1)$. A homogeneous system has a nonzero solution iff its coefficient determinant vanishes, giving $\det = 0$. ■

Procedure — Using the family trick

To find the line through the intersection of $L_1 = 0$, $L_2 = 0$ satisfying one extra condition:

1. Write $L_1 + \lambda L_2 = 0$ and expand into $Ax + By + C = 0$ form (coefficients now contain λ).
2. Impose the extra condition (passes through a point $\Rightarrow$ substitute it; given slope $\Rightarrow$ set $-A/B$ equal; perpendicular to a line $\Rightarrow$ slope product $= -1$).
3. Solve for λ , substitute back.

Section 4: Worked Examples**Example 1 — [CBSE Board Level]**

Find the point of intersection of $2x + 3y - 8 = 0$ and $x - y + 1 = 0$.

Solution. From the second, $x = y - 1$. Substitute:

$$2(y - 1) + 3y - 8 = 0 \Rightarrow 5y - 10 = 0 \Rightarrow y = 2, x = 1.$$

$$\boxed{(1, 2)}$$

Example 2 — [JEE Main · Speed Trap]

Check whether the lines $x + y - 2 = 0$, $2x - y - 1 = 0$, and $4x - y - 5 = 0$ are concurrent.

Speed approach. Drop straight into the determinant — don't solve two and test the third:

$$\begin{vmatrix} 1 & 1 & -2 \\ 2 & -1 & -1 \\ 4 & -1 & -5 \end{vmatrix} = 1[(-1)(-5) - (-1)(-1)] - 1[2(-5) - (-1)(4)] + (-2)[2(-1) - (-1)(4)].$$

$$= 1[5 - 1] - 1[-10 + 4] - 2[-2 + 4] = 4 - (-6) - 4 = 6 \neq 0.$$

Determinant $\neq 0 \Rightarrow$ **not concurrent**.

The trap: sign errors in the cofactor expansion. Anchor the alternating signs $+ - +$ across the top row and expand slowly; rushing the middle minor (the $-1[\dots]$ term) is where most marks vanish.

$$\boxed{\text{Not concurrent}}$$

Example 3 — [JEE Main Level · multi-concept]

Find the equation of the line passing through the intersection of $x + 2y - 5 = 0$ and $3x - y + 1 = 0$ that also passes through the point $(2, 1)$ — using the family method.

Solution. Family: $(x + 2y - 5) + \lambda(3x - y + 1) = 0$. Impose passage through $(2, 1)$:

$$(2 + 2 - 5) + \lambda(6 - 1 + 1) = 0 \Rightarrow (-1) + \lambda(6) = 0 \Rightarrow \lambda = \frac{1}{6}.$$

Substitute:

$$(x+2y-5) + \frac{1}{6}(3x-y+1) = 0 \Rightarrow 6x+12y-30+3x-y+1 = 0 \Rightarrow 9x+11y-29 = 0.$$

$$\boxed{9x + 11y - 29 = 0}$$

Notice we never computed the intersection point — that's the payoff.

Example 4 — [JEE Advanced Level]

For what value of a are the lines $ax + 2y + 1 = 0$, $2x + ay + 2 = 0$, and $x + y - 1 = 0$ concurrent? (Find all real a , and discard any that make two lines parallel.)

Solution. Set the determinant to zero:

$$\begin{vmatrix} a & 2 & 1 \\ 2 & a & 2 \\ 1 & 1 & -1 \end{vmatrix} = 0.$$

Expand along the top row:

$$a[a(-1) - 2(1)] - 2[2(-1) - 2(1)] + 1[2(1) - a(1)] = 0$$

$$a[-a - 2] - 2[-2 - 2] + [2 - a] = -a^2 - 2a + 8 + 2 - a = -a^2 - 3a + 10 = 0.$$

$$\text{So } a^2 + 3a - 10 = 0 \Rightarrow (a + 5)(a - 2) = 0 \Rightarrow a = -5 \text{ or } a = 2.$$

Parallel-check (the Advanced subtlety): at $a = 2$, the first two lines become $2x + 2y + 1 = 0$ and $2x + 2y + 2 = 0$ — **parallel** (and distinct), so they never meet and concurrency is impossible; discard $a = 2$. At $a = -5$ no two lines are parallel, so it is valid.

$$\boxed{a = -5}$$

Section 5: Practice Exercises

1. **[CBSE]** Find the point of intersection of $3x - 2y - 1 = 0$ and $x + 2y - 7 = 0$.
2. **[CBSE]** Check whether $x + y - 3 = 0$, $2x - y = 0$, and $x - 2 = 0$ are concurrent.
3. **[JEE Main]** Using the family method, find the line through the intersection of $2x - 3y + 4 = 0$ and $x + y - 3 = 0$ that is parallel to the x -axis.
4. **[JEE Main]** Find k if the lines $x + y - 1 = 0$, $2x - y + 3 = 0$, and $kx + y + 1 = 0$ are concurrent.
5. **[JEE Advanced]** Find the line through the intersection of $x + 2y + 3 = 0$ and $3x + 4y + 7 = 0$ that is perpendicular to $x - 2y + 5 = 0$.

Answers: 1. $(2, \frac{5}{2})$. 2. First two meet at $(1, 2)$; the third line $x = 2$ does **not** pass through it $\Rightarrow$ **not** concurrent. 3. Family $(2x - 3y + 4) + \lambda(x + y - 3) = 0$; parallel to the x -axis needs the x -coefficient $2 + \lambda = 0 \Rightarrow \lambda = -2$, giving $-5y + 10 = 0$, i.e. $y = 2$. 4. $k = -3$. 5. Intersection is $(-1, -1)$; perpendicular slope to $x - 2y + 5 = 0$ (slope $\frac{1}{2}$) is -2 ; line: $2x + y + 3 = 0$.

Section 6: Crack the MCQ

Q1. The lines $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ are parallel (and distinct) when: (A) $a_1b_2 - a_2b_1 \neq 0$ (B) $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$ (C) $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$ (D) $a_1a_2 + b_1b_2 = 0$

Answer: (C). Equal direction ratios but a different constant ratio means parallel and distinct.

- (A) is the *intersecting* condition.
- (B) is *coincident*.
- (D) is the *perpendicular* condition.

Q2. Every member of the family $(2x - y + 1) + \lambda(x + 3y - 2) = 0$ passes through: (A) the origin (B) the intersection of $2x - y + 1 = 0$ and $x + 3y - 2 = 0$ (C) $(1, 1)$ (D) no common point

Answer: (B). By construction the family passes through the common point of the two base lines for every λ .

- (A)/(C) are specific points not generally on the family.
- (D) contradicts the family theorem.

Q3. Three lines are concurrent if the determinant of their coefficients is zero, provided: (A) all three pass through the origin (B) no two of them are parallel (C) they have equal slopes (D) one is vertical

Answer: (B). The determinant test is necessary but presumes no two lines are parallel (else they can't share a finite point).

- (A)/(C)/(D) are unrelated or contradictory conditions.

Q4. The point of intersection of $x - y = 0$ and $x + y - 4 = 0$ is: (A) $(2, 2)$ (B) $(0, 4)$ (C) $(4, 0)$ (D) $(2, -2)$

Answer: (A). $x = y$ and $x + y = 4 \Rightarrow 2x = 4 \Rightarrow x = y = 2$.

- (B)/(C) satisfy only the second line.
- (D) satisfies only the first.

Section 7: Common Student Pitfalls & Pro-Tips

1. **Solving for the intersection when the family trick is faster.** If the question wants a line *through* an intersection plus one condition, use $L_1 + \lambda L_2 = 0$ and skip finding the point.
2. **Forgetting the parallel-check after the concurrency determinant.** $\det = 0$ can be satisfied by degenerate (parallel) configurations — verify no two lines are parallel.
3. **The family misses $L_2 = 0$.** $L_1 + \lambda L_2 = 0$ can produce every line through the junction except L_2 itself; test that case separately if needed.
4. **Sign errors in cofactor expansion.** Keep the $+ - +$ pattern explicit across the chosen row.

Pro-Tip: When a problem says “line through the intersection of ... and ...”, reach for $L_1 + \lambda L_2 = 0$ immediately. Finding the intersection point first is almost always the slower path and introduces extra arithmetic to slip on.

Section 8: Cheat Sheet Box

- **Intersection (Cramer):** $x = \frac{b_1 c_2 - b_2 c_1}{a_1 b_2 - a_2 b_1}$, $y = \frac{c_1 a_2 - c_2 a_1}{a_1 b_2 - a_2 b_1}$.
- **Parallel:** $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$; **intersecting:** $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$; **coincident:** all three ratios equal.
- **Concurrency:** $\det[a_i \ b_i \ c_i] = 0$ (no two parallel).
- **Family through a junction:** $L_1 + \lambda L_2 = 0$.

Memory aids:

- “Both zero $\Rightarrow$ sum zero” — why the family passes through the junction.
- “Determinant dies $\Rightarrow$ they meet” (with the no-two-parallel caveat).

Subtopic 05 — Angle Bisectors of Two Lines

Exam Relevance: Rarely on CBSE, but a reliable JEE Main/Advanced topic — and the engine behind incentre/excentre problems. The signature exam skill is not writing the two bisector equations (easy) but **identifying which is which** — acute vs obtuse, and which bisector contains the origin or a given point. That identification is where the marks live.

Section 1: Concept Intuition

When two roads cross, they make four angles — two pairs of equal “opposite” angles. The **angle bisectors** are the two lines that split these angles exactly in half. Crucially, the two bisectors are always **perpendicular to each other** (they bisect supplementary angles), so once you have one, the other is at right angles to it.

The defining property is beautifully simple: a point lies on an angle bisector exactly when it is **equidistant** from the two original lines. (Think of standing where you’re the same perpendicular distance from both roads — that path is a bisector.) Since we already know the distance of a point from a line (Subtopic 03), setting the two distances equal *gives* the bisector equations directly.

Think about it this way... distance from a line is $\frac{|ax+by+c|}{\sqrt{a^2+b^2}}$. “Equidistant from both lines” means

$$\frac{|a_1x + b_1y + c_1|}{\sqrt{a_1^2 + b_1^2}} = \frac{|a_2x + b_2y + c_2|}{\sqrt{a_2^2 + b_2^2}}.$$

Dropping the moduli introduces a $\pm$, and *that* is the whole story: the **two** signs give the **two** bisectors. The hard part — and the part exams test — is deciding which sign corresponds to the acute-angle bisector, and which to the bisector through the origin.

Domain of Validity & Edge Cases.

- The clean identification rules require both equations written with **positive constant terms** $c_1, c_2 > 0$ first. Skip that normalization and the sign-rules invert on you.
- If $a_1a_2 + b_1b_2 = 0$ the original lines are already perpendicular; both “angles” are 90° and the acute/obtuse distinction degenerates.
- The two bisectors are always mutually perpendicular — a free check on your answer.

Section 2: Key Formulas & Definitions

The pair of angle bisectors of $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$:

$$\frac{a_1x + b_1y + c_1}{\sqrt{a_1^2 + b_1^2}} = \pm \frac{a_2x + b_2y + c_2}{\sqrt{a_2^2 + b_2^2}}.$$

⁷²² **Identification rules** — first rewrite both lines so that $c_1 > 0$ and $c_2 > 0$. Then:

Quantity	“+” sign gives	“−” sign gives
Bisector of angle containing the origin	the + bisector	—
If $a_1a_2 + b_1b_2 > 0$	obtuse -angle bisector	acute-angle bisector
If $a_1a_2 + b_1b_2 < 0$	acute -angle bisector	obtuse-angle bisector

So with $c_1, c_2 > 0$: the **origin-containing** bisector is always the “+” one; whether that is the acute or obtuse bisector is decided by the sign of $a_1a_2 + b_1b_2$.

Mutual perpendicularity: the two bisectors always satisfy $m_{b_1} m_{b_2} = -1$.

Section 3: Theorems, Proofs & Standard Derivations

Theorem 1 — The bisector locus

Given: lines $L_1 : a_1x + b_1y + c_1 = 0$ and $L_2 : a_2x + b_2y + c_2 = 0$.

To Prove: the locus of points equidistant from L_1 and L_2 is $\frac{a_1x + b_1y + c_1}{\sqrt{a_1^2 + b_1^2}} = \pm \frac{a_2x + b_2y + c_2}{\sqrt{a_2^2 + b_2^2}}$.

Proof. A point $P(x, y)$ is equidistant from the two lines iff

$$\frac{|a_1x + b_1y + c_1|}{\sqrt{a_1^2 + b_1^2}} = \frac{|a_2x + b_2y + c_2|}{\sqrt{a_2^2 + b_2^2}}$$

by the point-line distance formula (Subtopic 03). Removing the moduli, two cases arise — the expressions equal or are negatives of each other — which is exactly the $\pm$. Each sign yields a first-degree equation, hence a line; together they are the two bisectors. ■

Procedure — Identifying the origin-containing bisector

1. Rewrite both lines so the constant terms are **positive** (multiply an equation by -1 if needed).
2. Form $\frac{a_1x + b_1y + c_1}{\sqrt{a_1^2 + b_1^2}} = + \frac{a_2x + b_2y + c_2}{\sqrt{a_2^2 + b_2^2}}$ — this is the bisector of the angle **containing the origin**.
3. To label acute vs obtuse, compute $a_1a_2 + b_1b_2$ (with the positive- c forms): if it is > 0 , the “+” bisector is the **obtuse** one; if < 0 , the “+” bisector is the **acute** one.

Why positive constants matter: the side of a line on which the origin lies is encoded in the sign of the constant term; forcing $c_1, c_2 > 0$ pins the origin to the “positive” side of both, which is what makes the “+” sign correspond to the origin’s angle.

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

Find the equations of the bisectors of the angles between $3x - 4y + 7 = 0$ and $5x + 12y - 2 = 0$.

Solution. Here $\sqrt{3^2 + 4^2} = 5$ and $\sqrt{5^2 + 12^2} = 13$. The bisectors:

$$\frac{3x - 4y + 7}{5} = \pm \frac{5x + 12y - 2}{13}.$$

Cross-multiply (take “+” first): $13(3x - 4y + 7) = 5(5x + 12y - 2)$

$$39x - 52y + 91 = 25x + 60y - 10 \Rightarrow 14x - 112y + 101 = 0.$$

“−” sign: $13(3x - 4y + 7) = -5(5x + 12y - 2) \Rightarrow 39x - 52y + 91 = -25x - 60y + 10 \Rightarrow 64x + 8y + 81 = 0.$

$$\boxed{14x - 112y + 101 = 0 \quad \text{and} \quad 64x + 8y + 81 = 0}$$

Example 2 — [JEE Main · Speed Trap]

For the lines $4x + 3y - 6 = 0$ and $5x + 12y + 9 = 0$, find the bisector of the angle that contains the origin — quickly.

Speed approach. First make both constants positive. The first has $c = -6 < 0$, so multiply by -1 : $-4x - 3y + 6 = 0$. The second already has $c = +9 > 0$. Now take the “+” bisector:

$$\frac{-4x - 3y + 6}{5} = + \frac{5x + 12y + 9}{13}.$$

$$13(-4x - 3y + 6) = 5(5x + 12y + 9) \Rightarrow -52x - 39y + 78 = 25x + 60y + 45 \Rightarrow 77x + 99y - 33 = 0,$$

$$\text{i.e. } 7x + 9y - 3 = 0.$$

The trap: using the lines as given (with $c = -6$) and taking “+” gives the *wrong* bisector — the rule only works after both constants are made positive. Forgetting that normalization is the single most common error here.

$$\boxed{7x + 9y - 3 = 0}$$

Example 3 — [JEE Main Level · multi-concept]

For $x + y - 2 = 0$ and $7x - y + 3 = 0$ (after suitable normalization), identify the acute-angle bisector.

Solution. Make constants positive: rewrite the first as $-x - y + 2 = 0$ ($c = 2 > 0$); the second has $c = 3 > 0$. Now $a_1a_2 + b_1b_2 = (-1)(7) + (-1)(-1) = -7 + 1 = -6 < 0$. By the rule, with constants positive and $a_1a_2 + b_1b_2 < 0$, the “+” sign gives the **acute bisector**.

$$\frac{-x - y + 2}{\sqrt{2}} = + \frac{7x - y + 3}{\sqrt{50}}.$$

Since $\sqrt{50} = 5\sqrt{2}$: multiply both sides by $5\sqrt{2}$: $5(-x - y + 2) = 7x - y + 3 \Rightarrow -5x - 5y + 10 = 7x - y + 3 \Rightarrow 12x + 4y - 7 = 0$.

Acute-angle bisector: $12x + 4y - 7 = 0$

Example 4 — [JEE Advanced Level]

The bisectors of the angles between two lines are mutually perpendicular. Verify this for the bisectors found in Example 1, and explain why it must always hold.

Solution. Example 1 gave $14x - 112y + 101 = 0$ (slope $m_1 = \frac{14}{112} = \frac{1}{8}$) and $64x + 8y + 81 = 0$ (slope $m_2 = -\frac{64}{8} = -8$). Then

$$m_1 m_2 = \frac{1}{8} \times (-8) = -1,$$

confirming perpendicularity.

Why it must hold: the two bisectors split the two *supplementary* angles formed at the crossing. If one angle is 2α , the other is $2(90^\circ - \alpha)$, and their half-lines differ in direction by exactly 90° . So the internal and external bisectors of any pair of intersecting lines are always perpendicular — a structural fact, not a coincidence, and a free sanity check on every bisector computation.

$m_1 m_2 = -1$ always

Section 5: Practice Exercises

1. **[CBSE]** Write the pair of bisector equations for $3x + 4y - 5 = 0$ and $5x - 12y + 1 = 0$ (set-up only; leave in $\pm$ form simplified).
2. **[JEE Main]** Find the bisector of the angle containing the origin for $x - 2y + 1 = 0$ and $2x + y - 3 = 0$.
3. **[JEE Main]** For $3x - 4y + 1 = 0$ and $12x + 5y - 2 = 0$, determine whether the origin lies in the acute or obtuse angle.
4. **[JEE Main]** Verify that the two bisectors of $x + y = 0$ and $x - y = 0$ are the coordinate axes.
5. **[JEE Advanced]** Find the incentre-relevant bisector: the bisector of the angle between $4x + 3y = 0$ and $3x + 4y = 0$ that lies in the first quadrant.

Answers: 1. $\frac{3x+4y-5}{5} = \pm \frac{5x-12y+1}{13}$. 2. Make constants positive ($x - 2y + 1 = 0$ has $c = 1 > 0$; rewrite $-2x - y + 3 = 0$): “+” gives $\frac{x-2y+1}{\sqrt{5}} = \frac{-2x-y+3}{\sqrt{5}} \Rightarrow 3x - y - 2 = 0$. 3. $a_1 a_2 + b_1 b_2 = 36 - 20 = 16 > 0$ (constants already positive) $\Rightarrow$ origin in **obtuse** angle. 4. The bisectors are $x = 0$ and $y = 0$ (the axes). 5. By symmetry the first-quadrant bisector is $x = y$ (i.e. $x - y = 0$).

Section 6: Crack the MCQ

Q1. A point lies on an angle bisector of two lines if and only if it is: (A) equidistant from the two lines (B) on the line joining their intersection to the origin (C) the midpoint of the lines (D) at the centroid

Answer: (A). The bisector is precisely the equidistant locus.

- (B)/(C)/(D) are unrelated constructions.

Q2. The two angle bisectors of a pair of intersecting lines are always: (A) parallel (B) perpendicular (C) coincident (D) at 45°

Answer: (B). They bisect supplementary angles, so their directions differ by 90° .

- (A)/(C) would require the original lines to coincide.
- (D) holds only for a special angle, not in general.

Q3. To use the standard rule that the “+” bisector contains the origin, you must first: (A) make both lines pass through the origin (B) make both constant terms positive (C) make both slopes positive (D) normalize to unit slope

Answer: (B). The sign rules assume $c_1, c_2 > 0$.

- (A) is impossible in general and unnecessary.
- (C)/(D) are not part of the procedure.

Q4. For two lines written with positive constants, $a_1a_2 + b_1b_2 < 0$. Then the “+” bisector is the: (A) obtuse-angle bisector (B) acute-angle bisector (C) perpendicular from origin (D) line at infinity

Answer: (B). With positive constants, $a_1a_2 + b_1b_2 < 0 \Rightarrow$ “+” is the acute bisector.

- (A) is the case $a_1a_2 + b_1b_2 > 0$.
- (C)/(D) are not bisectors.

Section 7: Common Student Pitfalls & Pro-Tips

1. **Skipping the positive-constant normalization.** Every identification rule (origin / acute / obtuse) assumes $c_1, c_2 > 0$. Normalize *first*, every time.
2. **Confusing the acute/obtuse rule’s direction.** With positive constants: $a_1a_2 + b_1b_2 > 0 \Rightarrow$ “+” is obtuse; $< 0 \Rightarrow$ “+” is acute. Write it down before deciding.
3. **Forgetting the perpendicularity check.** Your two bisectors must satisfy $m_1m_2 = -1$; if they don’t, there’s an arithmetic error.
4. **Mismatching denominators.** Each line gets its *own* $\sqrt{a^2 + b^2}$; don’t share one denominator across both.

Pro-Tip: The origin-containing bisector is the “+” one *after* making constants positive — memorise that single anchor, then use $a_1a_2 + b_1b_2$ only when the question specifically asks acute vs obtuse. Don’t compute everything; compute what’s asked.

Section 8: Cheat Sheet Box

- **Pair of bisectors:** $\frac{a_1x + b_1y + c_1}{\sqrt{a_1^2 + b_1^2}} = \pm \frac{a_2x + b_2y + c_2}{\sqrt{a_2^2 + b_2^2}}.$
- **Normalize first:** make $c_1, c_2 > 0$.
- **Origin's bisector:** the "+" sign.
- **Acute/obtuse:** with $c_1, c_2 > 0 - a_1a_2 + b_1b_2 > 0 \Rightarrow "+"$ obtuse; $< 0 \Rightarrow "+"$ acute.
- **Always:** the two bisectors are perpendicular ($m_1m_2 = -1$).

Memory aids:

- "Equidistant = on the bisector."
- "Positive constants first, then plus is the origin's side."
- "Bisectors cross at right angles" (free check).

Subtopic 06 — Foot of Perpendicular, Image & Reflection

Exam Relevance: A high-frequency JEE topic and a CBSE favourite for 3–4 mark problems (“find the image of the point...”). JEE Main reliably tests the image/foot formulas directly; JEE Advanced embeds them in reflection-of-a-line, shortest-path, and optics-style problems. The reflection-in- $y = x$ shortcut also appears constantly in functions and inverse-function questions, so it pays off well beyond this chapter.

Section 1: Concept Intuition

Stand at a point P and shine a torch straight at a wall (the line). The spot where the beam hits perpendicularly is the **foot of the perpendicular** — the closest point on the line to P . Walk an equal distance *past* the wall along the same beam and you reach the **image** (mirror reflection) of P — exactly as your reflection in a mirror sits as far behind the glass as you stand in front.

That single picture gives both results: the foot Q is the midpoint between P and its image P' , and PP' is perpendicular to the line. So the foot is “halfway to the image,” and the image is “twice as far as the foot.” Everything in this subtopic is that one idea, turned into formulas.

Think about it this way... the line $ax + by + c = 0$ has normal direction (a, b) . To go from P to the foot, you step from P *along the normal* just far enough to land on the line. The amount you must step is governed by $\frac{ax_1 + by_1 + c}{a^2 + b^2}$ — the same expression that measured distance in Subtopic 03, now repurposed as a step size. Step it once to reach the foot; step it *twice* to reach the image. The whole subtopic is “how far along the normal do I move?”, with the answer being one step for the foot and two for the mirror image.

Domain of Validity & Edge Cases.

- The formulas assume the line in general form $ax + by + c = 0$; convert first if it's in slope form.
- For the special mirrors $y = x$, x -axis, y -axis, use the swap/sign shortcuts — far faster than the general formula.
- Reflecting a *line* in a line: reflect two points of it (or use the foot/image machinery on a general point) and rejoin — don't try to reflect the equation directly.

Section 2: Key Formulas & Definitions

Let $P(x_1, y_1)$ and the line $\ell : ax + by + c = 0$.

Foot of the perpendicular $Q(x, y)$ from P to ℓ :

$$\frac{x - x_1}{a} = \frac{y - y_1}{b} = -\frac{ax_1 + by_1 + c}{a^2 + b^2}.$$

$$\text{Hence } Q = \left(x_1 - \frac{a(ax_1 + by_1 + c)}{a^2 + b^2}, y_1 - \frac{b(ax_1 + by_1 + c)}{a^2 + b^2} \right).$$

Image (reflection) $P'(x', y')$ of P in ℓ :

$$\frac{x' - x_1}{a} = \frac{y' - y_1}{b} = -\frac{2(ax_1 + by_1 + c)}{a^2 + b^2}.$$

$$\text{Hence } P' = \left(x_1 - \frac{2a(ax_1 + by_1 + c)}{a^2 + b^2}, y_1 - \frac{2b(ax_1 + by_1 + c)}{a^2 + b^2} \right).$$

Relationship: the foot Q is the midpoint of P and P' .

Special mirrors (shortcuts):

in $y = x$: $(x, y) \rightarrow (y, x)$; in x -axis : $(x, y) \rightarrow (x, -y)$; in y -axis : $(x, y) \rightarrow (-x, y)$.

Reflection of a line in a line: reflect any two of its points and join the images (the reflected line).

Section 3: Theorems, Proofs & Standard Derivations

Theorem 1 — Foot of the perpendicular

Given: point $P(x_1, y_1)$ and line $\ell : ax + by + c = 0$.

To Prove: the foot Q satisfies $\frac{x - x_1}{a} = \frac{y - y_1}{b} = -\frac{ax_1 + by_1 + c}{a^2 + b^2}$.

Proof. The segment PQ is perpendicular to ℓ , so it points along the normal (a, b) . Hence $Q = (x_1 + ta, y_1 + tb)$ for some scalar t . Since Q lies on ℓ :

$$a(x_1 + ta) + b(y_1 + tb) + c = 0 \Rightarrow (ax_1 + by_1 + c) + t(a^2 + b^2) = 0 \Rightarrow t = -\frac{ax_1 + by_1 + c}{a^2 + b^2}.$$

Substituting this t gives the stated coordinates, and $\frac{x - x_1}{a} = \frac{y - y_1}{b} = t$. ■

Figure

Line ℓ with external point P . The perpendicular PQ meets ℓ at foot Q , with the normal direction (a, b) marked. The point P' (image) lies on the extension of PQ beyond ℓ , with $PQ = QP'$.

Theorem 2 — Image of a point

Given: the same P and ℓ , with foot Q .

To Prove: the image P' uses the parameter $2t$, i.e. $\frac{x' - x_1}{a} = \frac{y' - y_1}{b} = -\frac{2(ax_1 + by_1 + c)}{a^2 + b^2}$.

Proof. Q is the midpoint of P and P' , so $P' = 2Q - P$. With $Q = P + t(a, b)$ from Theorem 1,

$$P' = 2(P + t(a, b)) - P = P + 2t(a, b).$$

Thus P' uses parameter $2t = -\frac{2(ax_1 + by_1 + c)}{a^2 + b^2}$, exactly double the foot's step. ■

Procedure — Reflecting a line m in a line ℓ

1. Pick two convenient points on m (e.g. its intercepts).
2. Find the image of each in ℓ using Theorem 2.
3. The line through the two image points is the reflection of m . (*Shortcut: the reflection of m also passes through $m \cap \ell$ if they intersect, so one image point plus that intersection suffices.*)

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

Find the foot of the perpendicular from $P(2, 3)$ to the line $x + y - 1 = 0$.

Solution. Here $a = 1$, $b = 1$, $c = -1$; $ax_1 + by_1 + c = 2 + 3 - 1 = 4$; $a^2 + b^2 = 2$. The step $t = -\frac{4}{2} = -2$.

$$Q = (2 + (-2)(1), 3 + (-2)(1)) = (0, 1).$$

$$\boxed{Q = (0, 1)}$$

Example 2 — [JEE Main · Speed Trap]

Find the image of the point $(4, -1)$ in the line $y = x$.

Speed approach. Don't grind the general formula — $y = x$ is a special mirror, so just **swap** the coordinates:

$$(4, -1) \rightarrow (-1, 4).$$

$$\boxed{(-1, 4)}$$

The trap: plugging into the general image formula with $a = 1$, $b = -1$, $c = 0$ works but wastes a minute and invites a sign error. Recognise $y = x$, x -axis, y -axis on sight and use swap/negate. (For $y = x$: swap; for x -axis: negate y ; for y -axis: negate x .)

Example 3 — [JEE Main Level · multi-concept]

Find the image of $P(1, 2)$ in the line $3x - 4y + 5 = 0$, then verify it is as far behind the line as P is in front.

Solution. $a = 3$, $b = -4$, $c = 5$; $ax_1 + by_1 + c = 3 - 8 + 5 = 0$. The step parameter is $-\frac{2(0)}{25} = 0$, so $P' = P$ — meaning P **already lies on the line!** Check: $3(1) - 4(2) + 5 = 0$. ✓ A point on the mirror is its own image, distance 0 both sides.

Let's instead take $P(1, 1)$: $3 - 4 + 5 = 4$, $a^2 + b^2 = 25$. Image step $= -\frac{2(4)}{25} = -\frac{8}{25}$.

$$P' = \left(1 - \frac{8}{25}(3), 1 - \frac{8}{25}(-4)\right) = \left(1 - \frac{24}{25}, 1 + \frac{32}{25}\right) = \left(\frac{1}{25}, \frac{57}{25}\right).$$

Distance of P from the line: $\frac{|4|}{5} = 0.8$; distance of P' : $\frac{|3(\frac{1}{25}) - 4(\frac{57}{25}) + 5|}{5} = \frac{|3 - 228 + 125|}{5} = \frac{|-100|}{5} = \frac{100}{5} = 20$. $\checkmark$ Equal distances, opposite sides.

$$P' = \left(\frac{1}{25}, \frac{57}{25}\right)$$

Example 4 — [JEE Advanced Level]

A ray of light along the line $x - 2y + 5 = 0$ strikes the mirror $3x - 4y + 4 = 0$. Find the equation of the reflected ray. (Use the image method: the reflected ray passes through the point of incidence and through the image of any point of the incident ray.)

Solution. Point of incidence = intersection of the two lines. Solve $x - 2y + 5 = 0$ and $3x - 4y + 4 = 0$: from the first $x = 2y - 5$; substitute: $3(2y - 5) - 4y + 4 = 0 \Rightarrow 2y - 11 = 0 \Rightarrow y = \frac{11}{2}$, $x = 6$. So the incidence point is $I(6, \frac{11}{2})$.

Take a point on the incident ray, say where it's convenient — at $y = 0$: $x = -5$, giving $A(-5, 0)$. Reflect A in the mirror $3x - 4y + 4 = 0$ ($a = 3, b = -4, c = 4$): $aA_x + bA_y + c = -15 - 0 + 4 = -11$; $a^2 + b^2 = 25$; step $= -\frac{2(-11)}{25} = \frac{22}{25}$.

$$A' = \left(-5 + \frac{22}{25}(3), 0 + \frac{22}{25}(-4)\right) = \left(-5 + \frac{66}{25}, -\frac{88}{25}\right) = \left(-\frac{59}{25}, -\frac{88}{25}\right).$$

Reflected ray = line through $I(6, \frac{11}{2})$ and $A'(-\frac{59}{25}, -\frac{88}{25})$. Slope:

$$m = \frac{\frac{11}{2} + \frac{88}{25}}{6 + \frac{59}{25}} = \frac{\frac{275+176}{50}}{\frac{150+59}{25}} = \frac{\frac{451}{50}}{\frac{209}{25}} = \frac{451}{50} \cdot \frac{25}{209} = \frac{451}{418} = \frac{41}{38}.$$

Equation through I : $y - \frac{11}{2} = \frac{41}{38}(x - 6)$, i.e. $38(y - \frac{11}{2}) = 41(x - 6) \Rightarrow 41x - 38y - 37 = 0$.

$$41x - 38y - 37 = 0$$

The method in one line: the reflected ray is the line joining the incidence point to the image of any incident-ray point — reflection turns an optics problem into two coordinate computations.

Section 5: Practice Exercises

1. **[CBSE]** Find the foot of the perpendicular from $(3, 8)$ to the x -axis.
2. **[CBSE]** Find the image of $(2, 5)$ in the y -axis.
3. **[JEE Main]** Find the foot of the perpendicular from $(1, 2)$ to the line $4x - 3y - 1 = 0$.
4. **[JEE Main]** Find the image of $(3, 4)$ in the line $x + y = 0$.
5. **[JEE Advanced]** Find the reflection of the line $2x - y + 3 = 0$ in the line $y = x$.

Answers: 1. $(3, 0)$. 2. $(-2, 5)$. 3. $ax_1 + by_1 + c = 4 - 6 - 1 = -3$, $a^2 + b^2 = 25$, $t = \frac{3}{25} \Rightarrow$ foot $(1 + \frac{12}{25}, 2 - \frac{9}{25}) = (\frac{37}{25}, \frac{41}{25})$. 4. In $x + y = 0$, the image of (x, y) is $(-y, -x)$, so $(3, 4) \rightarrow (-4, -3)$. 5. Reflecting in $y = x$ swaps $x \leftrightarrow y$ in the equation: $2y - x + 3 = 0 \Rightarrow x - 2y - 3 = 0$.

Section 6: Crack the MCQ

Q1. The foot of the perpendicular is related to the image by: the foot is (A) twice the image (B) the midpoint of the point and its image (C) the same as the image (D) unrelated

Answer: (B). Q is the midpoint of P and P' — the image is twice as far along the normal.

- (A) reverses the relationship.
- (C) holds only if P is on the line.
- (D) is false.

Q2. The image of (a, b) in the line $y = x$ is: (A) (b, a) (B) $(-a, b)$ (C) $(a, -b)$ (D) $(-b, -a)$

Answer: (A). Reflection in $y = x$ swaps coordinates.

- (B) is reflection in the y -axis.
- (C) is reflection in the x -axis.
- (D) is reflection in $y = -x$.

Q3. In the foot-of-perpendicular formula $\frac{x-x_1}{a} = \frac{y-y_1}{b} = -\frac{ax_1+by_1+c}{a^2+b^2}$, the common ratio represents: (A) the distance from P to the line (B) the signed step along the normal to reach the line (C) the slope of the line (D) the y -intercept

Answer: (B). It is the scalar t that steps from P along (a, b) onto the line.

- (A) the distance is $|t|\sqrt{a^2 + b^2}$, not t .
- (C)/(D) are unrelated.

Q4. To reflect a line m in a line ℓ , the most reliable method is to: (A) negate the equation of m (B) reflect two points of m in ℓ and join them (C) swap x and y in m (D) add the equations

Answer: (B). Reflect points, then rejoin.

- (A)/(D) are meaningless operations here.
 - (C) only works for the special mirror $y = x$.
-

Section 7: Common Student Pitfalls & Pro-Tips

1. **Mixing up the foot and image factors.** Foot uses $-\frac{ax_1+by_1+c}{a^2+b^2}$; image uses **twice** that. Drop the factor of 2 and you land on the line instead of behind it.
2. **Using the general formula for special mirrors.** For $y = x$, x -axis, y -axis, use swap/negate — faster and error-proof.
3. **Reflecting an equation directly.** You cannot “reflect” $ax+by+c = 0$ by tweaking coefficients (except for $y = x$, where swapping x, y works). Reflect points instead.
4. **Wrong denominator.** It's $a^2 + b^2$ here (no square root) — contrast with the distance formula's $\sqrt{a^2 + b^2}$. Mixing them is common.

Pro-Tip: For reflection-of-a-line problems where the lines intersect, you only need *one* image point: the reflected line passes through both the intersection point and the image of any single point of the original. That halves the work versus reflecting two points.

Section 8: Cheat Sheet Box

- **Foot of perpendicular:** $\frac{x - x_1}{a} = \frac{y - y_1}{b} = -\frac{ax_1 + by_1 + c}{a^2 + b^2}$.
- **Image of a point:** same, with $-\frac{2(ax_1 + by_1 + c)}{a^2 + b^2}$ (factor of 2).
- **Foot = midpoint of point and image.**
- **Special mirrors:** $y = x : (x, y) \rightarrow (y, x)$; x -axis: $(x, -y)$; y -axis: $(-x, y)$.
- **Reflect a line:** reflect two of its points, rejoin.

Memory aids:

- “One step to the foot, two steps to the mirror.”
- “ $y = x$? Just swap.”
- “Foot uses $a^2 + b^2$; distance uses the square root.”

Chapter 9: Straight Lines — Supplement

Subtopic 07 — Triangle Centres in Coordinates

Exam Relevance: This is a verbatim, named block in both JEE syllabi — JEE Main lists the coordinates of the centroid, orthocentre and circumcentre of a triangle, and JEE Advanced adds the incentre on top. CBSE Class 11 asks the centroid outright and sets 3–4 mark problems on “find the circumcentre” by the equidistance method and “find the third vertex given the centroid”. JEE Main reliably carries one question that is a triangle centre in disguise — an orthocentre, or a missing vertex recovered from two known centres. JEE Advanced pushes into the incentre/excentre weighting and the Euler line relation, where three centres collapse into one vector equation. Nothing here is new machinery: you already own the section formula (Subtopic 00), perpendicularity (Subtopic 01), the point-to-line distance (Subtopic 03), concurrency (Subtopic 04) and the angle bisectors (Subtopic 05). This subtopic is where all five finally get pointed at the same triangle.

Section 1: Concept Intuition

A village panchayat owns a triangular plot and has to decide where to sink the borewell. Three respected elders each argue for a different spot, and — this is the point — **all three are right, because each of them means something different by “the centre”**.

The first elder says: *put it where the plot balances*. Cut the triangle out of plywood, rest it on a pin, and there is exactly one point where it will not topple. That is the **centroid** G — the meeting point of the three medians, and the average of the three corners. You already met it in Subtopic 00; this subtopic assumes that treatment and does not repeat it.

The second elder says: *put it where it is equally far from all three boundary walls*, so the same circular apron of paving serves every fence. That point is the **incentre** I — the centre of the largest circle that fits inside the plot. “Equally far from two lines” should ring a bell immediately: that is exactly the bisector condition of Subtopic 05. A point equidistant from sides AB and AC sits on the internal bisector of angle A ; a point equidistant from all three sides sits on all three internal bisectors at once.

The third elder says: *put it where it is equally far from all three corner posts*, so a single cable of one length reaches each post. That point is the **circumcentre** O — the centre of the unique circle through the three vertices. “Equidistant from two points” is the perpendicular bisector, which the chapter has already built as a locus (Subtopic 00, Practice 5).

Then the youngest member of the panchayat asks a question nobody expected: *drop a perpendicular from each corner onto the opposite wall — do those three lines meet?* They do, and their meeting point is the **orthocentre** H . It has no tidy “equally far from” description at all, which is precisely why it is the one students find slippery.

Think about it this way... every one of these four centres is a **concurrency statement**, and Subtopic 04 already told you what concurrency costs. Three arbitrary lines in a plane do not meet at a point — you need the determinant to vanish. So each centre is a small miracle that has to be *proved*, and Subtopic 04 Section 1 explicitly held out the hook: “show the medians/altitudes meet at a point”. Section 3 below finally delivers on that promise for medians, altitudes, perpendicular bisectors and internal bisectors alike.

The single most important structural fact in this subtopic is that three of the four centres are collinear. For any triangle that is not equilateral, O , G and H lie on one straight line — the **Euler line** — and G always sits between the other two, dividing OH in the ratio $1 : 2$. That means you never have to compute all three: find any two and the third is arithmetic. The incentre is the odd one out; it is a genuinely different kind of centre and does not sit on the Euler line except in isosceles triangles.

Figure brief: A scalene triangle ABC drawn once, with all four centres marked and colour-coded — G at the meeting of three dashed medians, I at the meeting of three dotted internal bisectors with its inscribed circle, O at the meeting of three dot-dashed perpendicular bisectors with its circumscribed circle, and H at the meeting of three solid altitudes; a thin straight line drawn through O , G , H labelled “Euler line”, with the segments OG and GH tick-marked 1 and 2, and I visibly off that line.

Domain of Validity & Edge Cases.

- All four centres require a genuine triangle. If A , B , C are collinear (area = 0, Subtopic 00), the “sides” are parallel or coincident, the perpendicular bisectors never meet, and no centre exists.
- G and I are **always inside** the triangle. O and H are inside only for an acute triangle; for an obtuse triangle both fall outside.
- **Right triangle shortcut:** if the right angle is at C , then $H = C$ (two of the altitudes are the legs themselves) and O is the midpoint of the hypotenuse AB . Use this on sight — it saves a full minute.
- **Equilateral degeneracy:** all four centres coincide. Since O , G and H are the same point, the Euler line is undefined — do not “find” it.
- **Isosceles:** all four centres lie on the axis of symmetry, so the incentre *does* happen to sit on the Euler line. That is a coincidence of symmetry, not a general rule.

Section 2: Key Formulas & Definitions

Throughout, $\triangle ABC$ has vertices $A(x_1, y_1)$, $B(x_2, y_2)$, $C(x_3, y_3)$, and

$$a = BC, \quad b = CA, \quad c = AB.$$

- **Read that line again and burn it in.** a , b , c are the lengths of the sides **opposite** to A , B , C respectively. a is *not* the side starting at A . Every incentre and excentre error in the examination hall traces back to this one line.
- $s = \frac{a+b+c}{2}$ is the semi-perimeter, and Δ is the area (Subtopic 00, Theorem 2).

Centroid G — intersection of the medians (recap; full treatment in Subtopic 00, Section 2 and Example 1):

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right).$$

- G divides each median in the ratio $2 : 1$, measured from the vertex.
- Validity: always defined, always interior.

Incentre I — intersection of the three internal angle bisectors:

$$I = \left(\frac{ax_1 + bx_2 + cx_3}{a + b + c}, \frac{ay_1 + by_2 + cy_3}{a + b + c} \right).$$

- Weight each vertex by the length of the side **opposite** it: A gets $a = BC$, B gets $b = CA$, C gets $c = AB$.
- The weights are lengths, so they are positive and the denominator $a+b+c$ never vanishes. I is always interior.
- **Inradius**: $r = \frac{\Delta}{s}$, and r is the perpendicular distance from I to each of the three sides (Subtopic 03, Theorem 1).

Excentres I_A, I_B, I_C — each is the intersection of one internal bisector with the two external bisectors of the other angles. Flip the sign of exactly one weight, in both numerator and denominator:

Centre	Formula (x -coordinate; y identically)	Which circle
I	$\frac{ax_1 + bx_2 + cx_3}{a + b + c}$	incircle, radius $r = \Delta/s$
I_A	$\frac{-ax_1 + bx_2 + cx_3}{-a + b + c}$	escribed opposite A , radius $r_A = \Delta/(s - a)$
I_B	$\frac{ax_1 - bx_2 + cx_3}{a - b + c}$	escribed opposite B , radius $r_B = \Delta/(s - b)$
I_C	$\frac{ax_1 + bx_2 - cx_3}{a + b - c}$	escribed opposite C , radius $r_C = \Delta/(s - c)$

- Validity: the denominators are $2(s - a)$, $2(s - b)$, $2(s - c)$, all strictly positive by the triangle inequality — so all three excentres always exist.
- Every excentre lies **outside** the triangle.

Circumcentre O — intersection of the perpendicular bisectors; the point equidistant from the three vertices. There is no compact coordinate formula worth memorising. Characterise it instead by

$$OA^2 = OB^2 = OC^2 = R^2,$$

which gives two linear equations in the unknowns (see Procedure A).

- **Circumradius:** $R = \frac{abc}{4\Delta}$.

Orthocentre H — intersection of the altitudes. Again, compute rather than memorise (Procedure B), or get it free from the Euler relation below.

Euler line relation — for any non-equilateral triangle, O, G, H are collinear with $OG : GH = 1 : 2$, equivalently

$$\boxed{H = 3G - 2O} \quad \text{i.e.} \quad H = A + B + C - 2O,$$

and rearranged for the other two unknowns,

$$O = \frac{3G - H}{2}, \quad G = \frac{H + 2O}{3}.$$

- Validity: any triangle. For an equilateral triangle the relation still holds but reduces to $H = G = O$, and no line is determined.
- H, G, O in that order along the line: G is two-thirds of the way from H to O .

Section 3: Theorems, Proofs & Standard Derivations

Theorem 1 — The three medians are concurrent, at the centroid

Given: $\triangle ABC$ with vertices $A(x_1, y_1), B(x_2, y_2), C(x_3, y_3)$.

To Prove: the three medians meet at $G\left(\frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3}\right)$, which divides each median $2 : 1$ from the vertex.

Proof. Let D be the midpoint of BC , so $D = \left(\frac{x_2+x_3}{2}, \frac{y_2+y_3}{2}\right)$ (midpoint formula, Subtopic 00). Let P be the point dividing the median AD internally in the ratio $2 : 1$ from A . By the internal section formula with $m : n = 2 : 1$,

$$P_x = \frac{2 \cdot \frac{x_2+x_3}{2} + 1 \cdot x_1}{2+1} = \frac{x_1+x_2+x_3}{3}, \quad P_y = \frac{y_1+y_2+y_3}{3}.$$

Reason this finishes the proof: the expression obtained is **completely symmetric** in the three vertices. Had we started from the median through B , or through C , the identical algebra would have produced the identical point. So one and the same point lies on all three medians, which is precisely concurrency. ■

Why this matters: Subtopic 04 promised “show the medians meet at a point” and this is the promise honoured. Note that no determinant was needed — symmetry did the work that the concurrency determinant would otherwise have to.

Theorem 2 — The internal bisectors are concurrent, at the incentre

Given: $\triangle ABC$ with side lengths $a = BC, b = CA, c = AB$.

To Prove: the three internal angle bisectors meet at $I = \frac{aA + bB + cC}{a + b + c}$ (coordinate-wise).

Construction: Let the internal bisector from A meet BC at D .

Proof.

Step 1 — locate D . By the angle bisector theorem of school geometry, the internal bisector from A divides the opposite side in the ratio of the adjacent sides:

$$BD : DC = AB : AC = c : b.$$

Reason: this is the classical bisector-of-a-triangle ratio; in coordinate language it is exactly the section formula with $m : n = c : b$ applied from B to C :

$$D = \frac{cC + bB}{b + c}, \quad \text{so} \quad (b + c)D = bB + cC. \quad (*)$$

Step 2 — locate I on AD . Apply the same bisector theorem inside $\triangle ABD$, where BI bisects angle B and meets AD at I :

$$\frac{AI}{ID} = \frac{BA}{BD} = \frac{c}{BD}.$$

Reason: BD is a piece of BC of length $BD = \frac{c}{b + c} \cdot a = \frac{ac}{b + c}$, from Step 1. Substituting,

$$\frac{AI}{ID} = \frac{c(b + c)}{ac} = \frac{b + c}{a}.$$

Step 3 — apply the section formula to AD . I divides AD internally in the ratio $(b + c) : a$ from A , so

$$I = \frac{(b + c)D + aA}{(b + c) + a} \stackrel{(*)}{=} \frac{aA + bB + cC}{a + b + c}.$$

Step 4 — concurrency. *Reason this finishes it:* the result is symmetric under simultaneously permuting (A, B, C) and (a, b, c) . Starting from the bisector through B or through C reproduces the same point, so all three internal bisectors pass through it. ■

Excentres, same machinery. The *external* bisector from A divides BC **externally** in the ratio $c : b$ (Subtopic 00, external section formula). Running Steps 1–3 with that one external division in place of the internal one flips the sign on the weight a and yields

$$I_A = \frac{-aA + bB + cC}{-a + b + c},$$

and cyclically for I_B, I_C . The internal bisector from A together with the external bisectors from B and C are the three lines that are concurrent at I_A .

Theorem 3 — The perpendicular bisectors are concurrent, at the circumcentre**Given:** $\triangle ABC$, non-degenerate.**To Prove:** the perpendicular bisectors of the three sides meet at a single point O with $OA = OB = OC$.**Proof.** The perpendicular bisector of a segment is precisely the locus of points equidistant from its two endpoints (Subtopic 00, locus method). The perpendicular bisectors of AB and of BC are perpendicular to AB and BC respectively; since AB and BC are not parallel (the triangle is non-degenerate), these two bisectors are not parallel and therefore meet, at some point O .Now read off what O inherits:

$$O \text{ on perp. bisector of } AB \Rightarrow OA = OB; \quad O \text{ on perp. bisector of } BC \Rightarrow OB = OC.$$

Hence $OA = OC$, so O is equidistant from A and C and therefore lies on the perpendicular bisector of CA as well. All three concur at O , and the common value $OA = OB = OC$ is the circumradius R . ■**Theorem 4 — The altitudes are concurrent, and the Euler line****Given:** $\triangle ABC$ with circumcentre O .**To Prove:** the three altitudes are concurrent at the point $H = A + B + C - 2O$; and O, G, H are collinear with $OG : GH = 1 : 2$.**Proof.** Shift the origin to the circumcentre — legitimate because translating axes changes no distance and no perpendicularity (Subtopic 00, translation of origin). With O as origin, write the vertices as position vectors $\mathbf{A}, \mathbf{B}, \mathbf{C}$, and the defining property of O becomes simply

$$|\mathbf{A}| = |\mathbf{B}| = |\mathbf{C}| = R.$$

Define the candidate point $\mathbf{P} = \mathbf{A} + \mathbf{B} + \mathbf{C}$ and test it against the altitude from A :

$$(\mathbf{P} - \mathbf{A}) \cdot (\mathbf{B} - \mathbf{C}) = (\mathbf{B} + \mathbf{C}) \cdot (\mathbf{B} - \mathbf{C}) = |\mathbf{B}|^2 - |\mathbf{C}|^2 = R^2 - R^2 = 0.$$

Reason: a zero dot product means $AP \perp BC$, so P lies on the line through A perpendicular to BC — that is, on the altitude from A . (If $\mathbf{P} = \mathbf{A}$, then $\mathbf{B} + \mathbf{C} = \mathbf{0}$, so BC is a diameter, the angle at A is a right angle, and $H = A$ as the right-triangle shortcut predicts.)The expression $\mathbf{A} + \mathbf{B} + \mathbf{C}$ is symmetric in the three vertices, so the same computation puts P on the altitude from B and on the altitude from C . The altitudes are concurrent, and $\mathbf{H} = \mathbf{A} + \mathbf{B} + \mathbf{C}$. ■**The Euler line falls out for free.** Still with O at the origin, $\mathbf{G} = \frac{1}{3}(\mathbf{A} + \mathbf{B} + \mathbf{C}) = \frac{1}{3}\mathbf{H}$. So $\mathbf{H} = 3\mathbf{G}$: the points O (the origin), G and H lie on one line through O , with $OG : GH = 1 : 2$. Undoing the shift restores the general statement

$$H = A + B + C - 2O = 3G - 2O.$$

Procedure A: Circumcentre by the equidistance condition

Use this when you are given three vertices. It beats writing perpendicular bisectors because it never needs a slope and never trips on a vertical side.

1. **Check for a right angle first.** Test the three dot products $\vec{AB} \cdot \vec{AC}$, and cyclically. If one vanishes, O is the midpoint of the hypotenuse — stop, you are done. *Why:* this one check saves the entire computation on a very common exam triangle.
2. Let $O = (x, y)$ and write $OA^2 = OB^2$. Expand. *Why the squares:* the x^2 and y^2 terms cancel on both sides, leaving a **linear** equation — this is the whole reason we never take square roots here.
3. Write $OB^2 = OC^2$ and expand the same way, giving a second linear equation.
4. Solve the two linear equations simultaneously (Subtopic 04, Cramer or substitution).
5. Compute $R^2 = OA^2$ and, as a free check, confirm $R^2 = OB^2 = OC^2$. *Why:* the third equality was never imposed, so agreeing is genuine evidence, not circular.

Procedure B: Orthocentre by altitudes

1. **Check for a right angle first.** If the angle at C is right, then $H = C$ — stop.
2. Compute the slope of BC (Subtopic 01). The altitude from A is perpendicular to BC , so its slope is the negative reciprocal. *Why:* flip-and-negate is the perpendicularity condition $m_1 m_2 = -1$.
3. Write that altitude through A in point-slope form (Subtopic 02).
4. Repeat for one more altitude — **two are enough**, because Theorem 4 guarantees the third passes through their intersection. *Why:* computing the third is wasted time; use it only as a verification if marks are cheap.
5. Solve the two altitude equations simultaneously.
6. **Vertical/horizontal side care.** If a side is horizontal, the altitude to it is the vertical line $x = x$ (the opposite vertex's x). If a side is vertical, the altitude to it is horizontal. Handle these by inspection — the slope formula is undefined there.

Procedure C: Incentre and excentres by weights

1. Compute the three side lengths and **label them by what they are opposite to:** $a = BC$, $b = CA$, $c = AB$. Write the three labelled values down explicitly before touching the formula.
2. Form $I = \frac{aA + bB + cC}{a + b + c}$, coordinate by coordinate.
3. For the excentre opposite a vertex, negate that vertex's weight in numerator **and** denominator. *Why both:* the formula is a weighted average; the denominator is the sum of the weights actually used.
4. **Sanity-check the answer.** I must be inside the triangle; each excentre must be outside. If the sides are along the axes, symmetry usually pins a coordinate exactly (Example 2).

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

Find the incentre of the triangle with vertices $A(0, 0)$, $B(4, 0)$, $C(0, 3)$, and hence its inradius.

Solution.

Step 1 — the three side lengths, labelled by what they are opposite.

$$a = BC = \sqrt{(4-0)^2 + (0-3)^2} = \sqrt{16+9} = 5, \quad b = CA = 3, \quad c = AB = 4.$$

So $a + b + c = 12$.

Step 2 — apply the weighted formula.

$$I_x = \frac{ax_A + bx_B + cx_C}{a + b + c} = \frac{5(0) + 3(4) + 4(0)}{12} = \frac{12}{12} = 1,$$

$$I_y = \frac{ay_A + by_B + cy_C}{a + b + c} = \frac{5(0) + 3(0) + 4(3)}{12} = \frac{12}{12} = 1.$$

Step 3 — the inradius. $\Delta = \frac{1}{2}(4)(3) = 6$ and $s = 6$, so $r = \Delta/s = 1$.

Step 4 — verify against the bisector picture (Subtopic 05). Sides AB and CA lie along $y = 0$ and $x = 0$. The bisector of the angle at A between those two axis-lines, taken inside the first quadrant, is $y = x$ — the same first-quadrant symmetry argument you used on Subtopic 05, Practice 5, which is what the caption “the incentre-relevant bisector” was pointing at. The incentre must lie on that bisector, and $(1, 1)$ does. Its distance from $y = 0$ is 1, from $x = 0$ is 1, and from BC : the line BC is $\frac{x}{4} + \frac{y}{3} = 1$, i.e. $3x + 4y - 12 = 0$, giving $\frac{|3 + 4 - 12|}{\sqrt{9 + 16}} = \frac{5}{5} = 1$. Three equal distances, all equal to r .

$I = (1, 1), \quad r = 1$

Example 2 — [JEE Main • Speed Trap]

Find the incentre of the triangle with vertices $A(0, 0)$, $B(12, 0)$, $C(0, 5)$ — fast.

Speed approach. Before computing anything, notice the geometry: two sides lie along the coordinate axes, so the incentre is at perpendicular distance r from both, which forces $I = (r, r)$. **Whatever you compute, the two coordinates must come out equal.** That is your trap detector.

Now the weights, opposite-labelled:

$$a = BC = \sqrt{144 + 25} = 13, \quad b = CA = 5, \quad c = AB = 12, \quad a + b + c = 30.$$

$$I_x = \frac{13(0) + 5(12) + 12(0)}{30} = \frac{60}{30} = 2, \quad I_y = \frac{13(0) + 5(0) + 12(5)}{30} = \frac{60}{30} = 2.$$

Equal coordinates, as predicted. Cross-check with $r = \Delta/s$: $\Delta = \frac{1}{2}(12)(5) = 30$, $s = 15$, $r = 2$.

$$I = (2, 2), \quad r = 2$$

The trap: reading the formula $\frac{ax_A + bx_B + cx_C}{a + b + c}$ left-to-right and pairing each vertex with the side that *starts* at it — $a = AB = 12$ for A , $b = BC = 13$ for B , $c = CA = 5$ for C . That gives

$$\frac{12(0) + 13(12) + 5(0)}{30} = \frac{156}{30} = \frac{26}{5}, \quad \frac{12(0) + 13(0) + 5(5)}{30} = \frac{25}{30} = \frac{5}{6},$$

i.e. the seductive wrong answer $(\frac{26}{5}, \frac{5}{6})$ — which is a planted option because it still lands *inside* the triangle, so it survives a lazy “is it inside?” check. The symmetry test kills it in one second: $\frac{26}{5} \neq \frac{5}{6}$, so it cannot be equidistant from the two axes. **The subscript on the weight tells you which vertex it multiplies; the letter tells you which side it is opposite to. They are not the same instruction.**

Example 3 — [JEE Main Level · multi-concept]

For the triangle with vertices $A(0, 0)$, $B(4, 0)$, $C(1, 3)$, find (i) the orthocentre, (ii) the circumcentre, and (iii) verify the Euler relation.

Part (i) — orthocentre by altitudes (Procedure B).

Right-angle check: $\vec{AB} \cdot \vec{AC} = (4, 0) \cdot (1, 3) = 4 \neq 0$; $\vec{BA} \cdot \vec{BC} = (-4, 0) \cdot (-3, 3) = 12 \neq 0$; $\vec{CA} \cdot \vec{CB} = (-1, -3) \cdot (3, -3) = -3 + 9 = 6 \neq 0$. No right angle, so no shortcut.

Altitude from C: side AB lies along $y = 0$, which is horizontal, so the altitude to it is the vertical line through C :

$$x = 1.$$

Altitude from A: slope of BC is $\frac{3-0}{1-4} = -1$, so the altitude has slope $+1$ and passes through the origin:

$$y = x.$$

Intersect: $x = 1$ and $y = x$ give $H = (1, 1)$.

Free verification with the third altitude: slope of AC is 3 , so the altitude from $B(4, 0)$ has slope $-\frac{1}{3}$: $y = -\frac{1}{3}(x - 4)$. At $x = 1$, $y = -\frac{1}{3}(-3) = 1$. It passes through $(1, 1)$ as Theorem 4 requires.

Part (ii) — circumcentre by equidistance (Procedure A). Let $O = (x, y)$.

$$OA^2 = OB^2: x^2 + y^2 = (x - 4)^2 + y^2 \Rightarrow 0 = -8x + 16 \Rightarrow x = 2.$$

$$OA^2 = OC^2: x^2 + y^2 = (x - 1)^2 + (y - 3)^2 \Rightarrow 0 = -2x + 1 - 6y + 9 \Rightarrow 2x + 6y = 10 \Rightarrow x + 3y = 5.$$

With $x = 2$: $3y = 3 \Rightarrow y = 1$. So $O = (2, 1)$.

Check: $OA^2 = 4 + 1 = 5$, $OB^2 = 4 + 1 = 5$, $OC^2 = 1 + 4 = 5$. All agree, so $R = \sqrt{5}$.

Part (iii) — Euler relation. $G = \left(\frac{0+4+1}{3}, \frac{0+0+3}{3}\right) = \left(\frac{5}{3}, 1\right)$. Then

$$3G - 2O = (5 - 4, 3 - 2) = (1, 1) = H. \checkmark$$

All three centres have $y = 1$: the Euler line here is the horizontal line $y = 1$, and along it $HG = \left|\frac{5}{3} - 1\right| = \frac{2}{3}$ while $GO = \left|2 - \frac{5}{3}\right| = \frac{1}{3}$ — the promised $2 : 1$.

$H = (1, 1), \quad O = (2, 1), \quad G = \left(\frac{5}{3}, 1\right), \text{ Euler line } y = 1$
--

Example 4 — [JEE Advanced Level]

The triangle ABC has vertices $A(5, 7)$, $B(-2, 6)$, $C(2, -2)$. Find its circumcentre, centroid and orthocentre, and the equation of its Euler line — computing at most one of O and H from scratch.

Solution. The Advanced move is to refuse to compute both O and H directly. Compute the cheap one plus G , then let Theorem 4 hand you the third.

Step 1 — circumcentre by equidistance.

$$OA^2 = OB^2:$$

$$(x - 5)^2 + (y - 7)^2 = (x + 2)^2 + (y - 6)^2$$

$$-10x + 25 - 14y + 49 = 4x + 4 - 12y + 36 \implies -14x - 2y + 34 = 0 \implies 7x + y = 17.$$

$$OB^2 = OC^2:$$

$$(x + 2)^2 + (y - 6)^2 = (x - 2)^2 + (y + 2)^2$$

$$4x + 4 - 12y + 36 = -4x + 4 + 4y + 4 \implies 8x - 16y + 32 = 0 \implies x - 2y = -4.$$

Substituting $x = 2y - 4$ into $7x + y = 17$: $14y - 28 + y = 17 \Rightarrow 15y = 45 \Rightarrow y = 3, x = 2$.

$$O = (2, 3), \quad R^2 = (2 - 5)^2 + (3 - 7)^2 = 9 + 16 = 25 \Rightarrow R = 5,$$

and the check holds: $OB^2 = 16 + 9 = 25$, $OC^2 = 0 + 25 = 25$.

Step 2 — centroid.

$$G = \left(\frac{5 - 2 + 2}{3}, \frac{7 + 6 - 2}{3}\right) = \left(\frac{5}{3}, \frac{11}{3}\right).$$

Step 3 — orthocentre, without a single altitude.

$$H = 3G - 2O = (5 - 4, 11 - 6) = (1, 5).$$

Step 4 — verify, because a formula-only answer deserves one check. The altitude from A must be perpendicular to BC :

$$\vec{AH} = (1 - 5, 5 - 7) = (-4, -2), \quad \vec{BC} = (4, -8), \quad \vec{AH} \cdot \vec{BC} = -16 + 16 = 0. \checkmark$$

And from B : $\vec{BH} = (3, -1)$, $\vec{CA} = (3, 9)$, dot product $9 - 9 = 0$. $\checkmark$

Step 5 — the Euler line. Through $H(1, 5)$ and $O(2, 3)$ the slope is $\frac{3-5}{2-1} = -2$, so

$$y - 5 = -2(x - 1) \implies 2x + y - 7 = 0.$$

G obliges: $2\left(\frac{5}{3}\right) + \frac{11}{3} = \frac{10+11}{3} = 7$. $\checkmark$

$O = (2, 3), G = \left(\frac{5}{3}, \frac{11}{3}\right), H = (1, 5), \text{ Euler line } 2x + y - 7 = 0$

Why this is Advanced-flavoured: the brute-force route computes O from two perpendicular bisectors and then H from two altitudes — four line equations and two simultaneous systems. The Euler relation replaces half of that with one subtraction. Examiners set the “given two centres, find the third” variant precisely because it is worthless to a student who only memorised coordinates and never met the relation between them.

Section 5: Practice Exercises

- [CBSE]** Find the circumcentre and circumradius of the triangle with vertices $A(0, 0)$, $B(6, 0)$, $C(0, 8)$.
- [CBSE]** Find the incentre and inradius of the same triangle $A(0, 0)$, $B(6, 0)$, $C(0, 8)$.
- [JEE Main]** Find the orthocentre of the triangle with vertices $A(0, 0)$, $B(6, 0)$, $C(2, 4)$.
- [JEE Main]** The circumcentre of a triangle is $(1, 2)$ and its centroid is $(2, 3)$. Find its orthocentre.
- [JEE Main]** For the triangle $A(0, 0)$, $B(4, 0)$, $C(0, 3)$, find the excentre opposite A and verify your answer by checking its distance from all three side-lines.
- [JEE Advanced]** For the triangle $A(0, 0)$, $B(6, 0)$, $C(2, 4)$, find G , O and H , write the equation of the Euler line, and verify $HG : GO = 2 : 1$.
- [JEE Advanced]** A triangle has vertices $A(0, 0)$, $B(p, 0)$, $C(0, q)$ with $p, q > 0$. Find G , O , H and I in terms of p and q , and confirm that $H = 3G - 2O$.

Answers: **1.** Right-angled at A , so O is the midpoint of the hypotenuse BC : $O = (3, 4)$, $R = 5$. **2.** $a = BC = 10$, $b = CA = 8$, $c = AB = 6$; $I = \left(\frac{0+48+0}{24}, \frac{0+0+48}{24}\right) = (2, 2)$, $r = \Delta/s = 24/12 = 2$. **3.** $H = (2, 2)$ (altitude from C is $x = 2$; slope of BC is -1 so the altitude from A is $y = x$). **4.** $H = 3G - 2O = (6 - 2, 9 - 4) = (4, 5)$. **5.** $a = 5$, $b = 3$, $c = 4$; $I_A = \frac{-5A+3B+4C}{-5+3+4} = \left(\frac{12}{2}, \frac{12}{2}\right) = (6, 6)$; its distances from $y = 0$, from $x = 0$ and from $3x + 4y - 12 = 0$ are 6, 6 and $\frac{|18+24-12|}{5} = 6$ — all equal to $r_A = \Delta/(s - a) = 6/1 = 6$. **6.** $G = \left(\frac{8}{3}, \frac{4}{3}\right)$, $O = (3, 1)$ (check $OA^2 = OB^2 = OC^2 = 10$), $H = (2, 2)$; Euler line $x + y - 4 = 0$; $HG = \frac{2\sqrt{2}}{3}$ and $GO = \frac{\sqrt{2}}{3}$, ratio $2 : 1$. **7.** Right-angled at A , so $H = A = (0, 0)$ and $O = \left(\frac{p}{2}, \frac{q}{2}\right)$ (midpoint of the hypotenuse BC); $G = \left(\frac{p}{3}, \frac{q}{3}\right)$; with $a = \sqrt{p^2 + q^2}$, $b = q$, $c = p$, $I = \left(\frac{pq}{p+q+\sqrt{p^2+q^2}}, \frac{pq}{p+q+\sqrt{p^2+q^2}}\right)$. Check: $3G - 2O = (p - p, q - q) = (0, 0) = H$. $\checkmark$

Section 6: Crack the MCQ

Q1. In the incentre formula $I = \frac{ax_1 + bx_2 + cx_3}{a + b + c}$ for $\triangle ABC$, the weight a is: (A) the length BC (B) the length AB (C) the length CA (D) the measure of angle A

Answer: (A). The weights are side lengths labelled by the vertex they are **opposite** to, so $a = BC$, $b = CA$, $c = AB$.

Distractor Analysis. - (B) AB is the classic left-to-right misreading: pairing A with the side that *starts* at A . This is the single most common incentre error and it produces a wrong point that still lies inside the triangle, so it survives casual checking. - (C) CA is the same misreading run in the other rotational direction. - (D) confuses the weights with the angles. Weighting by angles gives a genuinely different point (not the incentre), and it is dimensionally incoherent — you cannot average coordinates with degrees.

Q2. The incentre of the triangle with vertices $A(0, 12)$, $B(-5, 0)$, $C(5, 0)$ is: (A) $(0, \frac{10}{3})$ (B) $(0, 4)$ (C) $(\frac{5}{12}, \frac{13}{3})$ (D) $(0, \frac{5}{3})$

Answer: (A). $a = BC = 10$, $b = CA = 13$, $c = AB = 13$, sum = 36. Then $I_x = \frac{10(0) + 13(-5) + 13(5)}{36} = 0$ and $I_y = \frac{10(12) + 13(0) + 13(0)}{36} = \frac{120}{36} = \frac{10}{3}$. Cross-check: $\Delta = \frac{1}{2}(10)(12) = 60$, $s = 18$, so $r = \frac{60}{18} = \frac{10}{3}$ — and since BC lies along $y = 0$, the incentre's y -coordinate must equal r .

Distractor Analysis. - (B) $(0, 4)$ is the **centroid**, $(\frac{0-5+5}{3}, \frac{12+0+0}{3})$. Produced by the belief that “centre of a triangle” means “average of the vertices” — that is one centre, not this one. - (C) is the mis-ordered weighting: pairing A with $AB = 13$, B with $BC = 10$, C with $CA = 13$. It even breaks the symmetry of an isosceles triangle, which is your tell — I must lie on the axis of symmetry $x = 0$. - (D) $(0, \frac{5}{3})$ comes from $r = \Delta/\text{perimeter} = \frac{60}{36}$ instead of $r = \Delta/s$. The inradius uses the **semi**-perimeter.

Q3. In a triangle right-angled at C , the orthocentre is: (A) the midpoint of AB (B) the vertex C (C) the centroid (D) always outside the triangle

Answer: (B). Two of the altitudes are the legs CA and CB themselves, and they meet at C .

Distractor Analysis. - (A) is the **circumcentre** of a right triangle, not the orthocentre — the two facts are learned together and get swapped under time pressure. - (C) the centroid is the vertex average $\frac{A+B+C}{3}$, which equals C only in a degenerate figure. - (D) is a half-remembered rule: it is the **obtuse** triangle whose orthocentre falls outside. For a right triangle it lands exactly on the vertex, and for an acute triangle it is strictly inside.

Q4. If O , G , H denote the circumcentre, centroid and orthocentre of a non-equilateral triangle, then: (A) $OG : GH = 1 : 2$ (B) $OG : GH = 2 : 1$ (C) O , G , H are never collinear (D) G is the midpoint of OH

Answer: (A). Theorem 4 gives $H = 3G - 2O$, i.e. $\vec{OH} = 3\vec{OG}$, so G divides OH with $OG : GH = 1 : 2$.

Distractor Analysis. - (B) inverts the ratio — the trap is importing the $2 : 1$ of the *median* (which is measured from the vertex) into the Euler line. On the Euler line the short piece is next to O . - (C) contradicts the Euler line entirely; it is what you would guess if you had only ever computed the centres separately and never compared them. - (D) would need $OG : GH = 1 : 1$, i.e. $H = 2G - O$, which fails on any of the worked examples above (in

Example 3, $2G - O = (\frac{10}{3} - 2, 2 - 1) = (\frac{4}{3}, 1) \neq H$.

Section 7: Common Student Pitfalls & Pro-Tips

1. **Mis-ordering the incentre weights.** a, b, c are the sides **opposite** A, B, C — never the sides that begin at them. *Correction:* before writing the formula, write the three labelled lengths on their own line: " $a = BC = \dots, b = CA = \dots, c = AB = \dots$ ". Two seconds of bookkeeping removes the chapter's most expensive error.
2. **Negating only the numerator for an excentre.** $I_A = \frac{-aA + bB + cC}{-a + b + c}$ — the sign flips **top and bottom**. *Correction:* remember that the denominator is the sum of the weights you actually used, so it must track them.
3. **Taking square roots when finding the circumcentre.** Setting $OA = OB$ with radicals gives an ugly irrational equation. *Correction:* always square first: $OA^2 = OB^2$ kills the x^2 and y^2 terms and leaves a linear equation. This is the whole trick of Procedure A.
4. **Computing the third altitude or the third perpendicular bisector.** Concurrency is a theorem, not something you re-establish each time. *Correction:* two lines determine the centre; use the third only as a verification if you have spare time.
5. **Assuming every centre is inside the triangle.** Only G and I always are. In an obtuse triangle both O and H fall outside, and every excentre is outside by construction. *Correction:* a negative coordinate in an all-positive-quadrant triangle is a warning for I , but perfectly normal for O, H and the excentres.
6. **Hunting for the Euler line of an equilateral triangle.** All the centres coincide, so no line is determined. *Correction:* recognise equilateral (all three sides equal) and answer "all centres coincide" rather than dividing by zero.
7. **Putting the incentre on the Euler line.** I is not on it in general. *Correction:* it lies on it only when the triangle is isosceles, where symmetry drags every centre onto the axis.

Pro-Tip: Screen every triangle-centre question for a right angle before you compute anything. One dot product $\vec{CA} \cdot \vec{CB}$ tells you whether H and O are already known (H at the right-angle vertex, O at the hypotenuse's midpoint). In JEE Main this shortcut converts a three-minute problem into a fifteen-second one, and setters use right triangles constantly because they make the arithmetic clean.

Pro-Tip: Never compute all three of O, G, H . G is free (one addition), so compute G plus whichever of O and H looks cheaper, and get the last one from $H = 3G - 2O$. If the question hands you two centres and asks for the third, that relation is the entire solution.

Section 8: Cheat Sheet Box

With $A(x_1, y_1), B(x_2, y_2), C(x_3, y_3)$ and $a = BC, b = CA, c = AB$ (each side **opposite** its vertex), $s = \frac{a+b+c}{2}$, area Δ :

- **Centroid:** $G = \left(\frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3} \right)$; medians concur; divides each median 2 : 1 from the vertex.
- **Incentre:** $I = \left(\frac{ax_1+bx_2+cx_3}{a+b+c}, \frac{ay_1+by_2+cy_3}{a+b+c} \right)$; internal bisectors concur; $r = \Delta/s$; always interior.
- **Excentres:** flip one weight's sign top **and** bottom — $I_A = \frac{-aA+bB+cC}{-a+b+c}$, $I_B = \frac{aA-bB+cC}{a-b+c}$, $I_C = \frac{aA+bB-cC}{a+b-c}$; $r_A = \frac{\Delta}{s-a}$, $r_B = \frac{\Delta}{s-b}$, $r_C = \frac{\Delta}{s-c}$; always exterior.
- **Circumcentre:** solve $OA^2 = OB^2$ and $OB^2 = OC^2$ (two linear equations — square, never root); $R = \frac{abc}{4\Delta}$.
- **Orthocentre:** intersect two altitudes (slope = $-1/m_{\text{opposite side}}$).
- **Euler line:** $H = 3G - 2O$; O, G, H collinear with $OG : GH = 1 : 2$; I is not on it (except isosceles).
- **Right triangle at C:** $H = C$, O = midpoint of AB .
- **Equilateral:** all four centres coincide; Euler line undefined.

Memory aids:

- “Opposite, always opposite” — the incentre weight a sits opposite vertex A .
- “Square it, don’t root it” — the circumcentre from $OA^2 = OB^2$.
- “ O, G, H — one, two” — G is the near neighbour of O on the Euler line.
- “Right angle? Orthocentre’s already there.”

Addendum to Chapter 9: Four Missing Problem Patterns

Four patterns that the main chapter builds all the machinery for but never actually exercises. Each entry gives the method, one worked example, and two practice items with answers. Slot them into the subtopics named in the headings.

Pattern 1 — Areas: a line with the axes, and three lines together

(extends Subtopic 02, Section 4, and Subtopic 04, Section 4)

Method.

Case A — one line and the two axes. A line $Ax + By + C = 0$ with A, B, C all non-zero cuts the axes at $(-\frac{C}{A}, 0)$ and $(0, -\frac{C}{B})$, so together with the origin it bounds a right triangle whose legs are the two intercepts:

$$\Delta = \frac{1}{2}|a||b| = \frac{C^2}{2|AB|}.$$

Read the intercepts by setting $y = 0$ and then $x = 0$ (the Subtopic 02 Pro-Tip) — the signs take care of themselves and the modulus finishes the job. If $C = 0$ the line passes through the origin and bounds no triangle; if $A = 0$ or $B = 0$ it is parallel to an axis and again bounds none.

Case B — three given lines. Solve the three **pairs** of lines to get the three vertices (Subtopic

04), then feed them to the area determinant of Subtopic 00, Theorem 2. If any two of the three lines are parallel, or all three are concurrent, there is no triangle — the determinant returns 0 and that is the correct answer, not an error.

Worked Example — [JEE Main Level]

- (a) Find the area of the triangle that $3x - 4y + 12 = 0$ cuts off with the coordinate axes. (b) Find the area of the triangle formed by the lines $y = x$, $y = 2x$ and $x + y = 6$.

Solution (a). This is the line whose intercepts you already extracted in Subtopic 02, Example 2: setting $y = 0$ gives $3x = -12$, so $a = -4$; setting $x = 0$ gives $-4y = -12$, so $b = 3$. The legs have lengths $|-4| = 4$ and $|3| = 3$:

$$\Delta = \frac{1}{2}(4)(3) = 6.$$

Shortcut check: $\frac{C^2}{2|AB|} = \frac{12^2}{2 \cdot |3 \times (-4)|} = \frac{144}{24} = 6. \checkmark$

Solution (b). Three pairwise intersections:

- $y = x$ with $y = 2x$: $x = 2x \Rightarrow x = 0$, giving $P(0, 0)$.
- $y = x$ with $x + y = 6$: $2x = 6 \Rightarrow x = 3$, giving $Q(3, 3)$.
- $y = 2x$ with $x + y = 6$: $3x = 6 \Rightarrow x = 2$, giving $R(2, 4)$.

Area determinant:

$$\Delta = \frac{1}{2} |0(3 - 4) + 3(4 - 0) + 2(0 - 3)| = \frac{1}{2} |0 + 12 - 6| = 3.$$

(a) 6 sq. units, (b) 3 sq. units

Practice.

1. **[CBSE]** Find the area of the triangle formed by $5x + 3y - 15 = 0$ and the coordinate axes.
2. **[JEE Main]** Find the area of the triangle formed by the lines $y = x + 1$, $y = -x + 5$ and $y = 0$.

Answers: 1. Intercepts $a = 3$, $b = 5$, so $\Delta = \frac{1}{2}(3)(5) = \frac{15}{2} = 7.5$ sq. units (shortcut: $\frac{15^2}{2 \cdot 15} = 7.5$).
 2. Vertices $(2, 3)$, $(-1, 0)$, $(5, 0)$; $\Delta = \frac{1}{2} |2(0 - 0) + (-1)(0 - 3) + 5(3 - 0)| = \frac{1}{2} (18) = 9$ sq. units.

Pattern 2 — The fixed point of a one-parameter family of lines

(extends Subtopic 04, Sections 2 and 4 — the reverse of the family theorem)

Method. Subtopic 04 runs the family argument forwards: given two base lines, $L_1 + \lambda L_2 = 0$ sweeps out every line through their junction. Examiners far more often run it **backwards**: they hand you a line whose coefficients are linear in a parameter and ask you to prove it always passes through a fixed point.

Given

$$(a_1 + \lambda a_2)x + (b_1 + \lambda b_2)y + (c_1 + \lambda c_2) = 0,$$

regroup by powers of λ :

$$\underbrace{(a_1x + b_1y + c_1)}_{L_1} + \lambda \underbrace{(a_2x + b_2y + c_2)}_{L_2} = 0.$$

For a point to satisfy this for **every** value of λ , the coefficient of λ and the λ -free part must vanish separately. So solve

$$L_1 = 0 \quad \text{and} \quad L_2 = 0$$

simultaneously; the solution is the fixed point. The reasoning is exactly Theorem 1 of Subtopic 04 read from right to left: the fixed point is the junction of the two base lines you just uncovered.

Two conditions to respect: L_1 and L_2 must not be proportional (otherwise every member is the same line, or the family degenerates), and if the parameter appears non-linearly you must first collect terms so that it appears to the first power only. If **two** parameters appear, group by each in turn and set every group to zero.

Worked Example — [JEE Main Level]

Show that the line $(2 + \lambda)x + (1 - 3\lambda)y + (\lambda - 5) = 0$ passes through a fixed point for every real λ , and find it.

Solution.

Step 1 — regroup by λ . Separate the terms carrying λ from those that do not:

$$2x + \lambda x + y - 3\lambda y + \lambda - 5 = 0 \implies (2x + y - 5) + \lambda(x - 3y + 1) = 0.$$

Step 2 — read off the two base lines.

$$L_1 : 2x + y - 5 = 0, \quad L_2 : x - 3y + 1 = 0.$$

They are not proportional, so this is a genuine one-parameter family.

Step 3 — the fixed point must kill both brackets. From L_2 , $x = 3y - 1$. Substituting into L_1 :

$$2(3y - 1) + y - 5 = 0 \implies 7y - 7 = 0 \implies y = 1, \quad x = 2.$$

Step 4 — verify directly, for arbitrary λ . Put $(2, 1)$ back into the original:

$$(2 + \lambda)(2) + (1 - 3\lambda)(1) + (\lambda - 5) = 4 + 2\lambda + 1 - 3\lambda + \lambda - 5 = 0.$$

The λ terms cancel identically, so the point lies on every member of the family.

Fixed point (2, 1)

Practice.

1. **[JEE Main]** Show that $(2 + \lambda)x + (1 - \lambda)y + (\lambda - 4) = 0$ always passes through a fixed point, and find it.
2. **[JEE Advanced]** For all real a, b not both zero, the line $(2a + b)x + (a + 3b)y + (2b - a) = 0$ passes through a fixed point. Find it.

Answers: 1. Regroup as $(2x + y - 4) + \lambda(x - y + 1) = 0$; solving $2x + y = 4$ and $x - y = -1$ gives the fixed point (1, 2). Check: $(2 + \lambda)(1) + (1 - \lambda)(2) + (\lambda - 4) = 2 + \lambda + 2 - 2\lambda + \lambda - 4 = 0$ for all λ .
 2. Group by the two parameters: $a(2x + y - 1) + b(x + 3y + 2) = 0$. Both brackets must vanish, so $2x + y = 1$ and $x + 3y = -2$, giving (1, -1). Check: $(2a + b)(1) + (a + 3b)(-1) + (2b - a) = (2a - a - a) + (b - 3b + 2b) = 0$ identically.

Pattern 3 — Ratio in which a line divides a segment, and the side test

(extends Subtopic 03 — the Pro-Tip that promises this and is never exercised)

Method. Write $L(x, y) = Ax + By + C$, and evaluate it at the two points: $L(P)$ and $L(Q)$.

Side test. L is positive on one side of the line and negative on the other (this is the signed quantity of Subtopic 03, Section 2, before the modulus is applied).

- $L(P)$ and $L(Q)$ have the **same** sign $\Rightarrow P$ and Q are on the same side, and the line misses the segment PQ .
- **Opposite** signs $\Rightarrow$ the line separates them and cuts the segment PQ .
- Either value zero $\Rightarrow$ that point is on the line.

Ratio. The line $L = 0$ divides the segment PQ in the ratio

$$\frac{PX}{XQ} = -\frac{L(P)}{L(Q)},$$

where X is the point of division. *Why it works:* L is a first-degree (affine) expression, so if X divides PQ in the ratio $m : n$ then $L(X) = \frac{m L(Q) + n L(P)}{m + n}$ — the same weighted average that the section formula applies to the coordinates. Setting $L(X) = 0$ gives $m L(Q) + n L(P) = 0$, i.e. $m : n = -L(P) : L(Q)$.

Read the sign of the answer, do not discard it:

- **Positive ratio** $\Rightarrow$ internal division $\Rightarrow$ the points are on opposite sides. The two tests always agree.
- **Negative ratio** $\Rightarrow$ external division $\Rightarrow$ same side.
- **Ratio** $= -1 \Rightarrow$ external division in the ratio 1 : 1, which Subtopic 00 told you sends the point to infinity: the line is parallel to PQ and never meets it.

Origin-inside-a-triangle test. For each side of the triangle, write the side's line equation

and compare the sign of L at the origin with its sign at the **opposite vertex**. If all three comparisons agree, the origin is inside; a single disagreement puts it outside. This works because “inside the triangle” means “on the same side of each side-line as the remaining vertex”.

Worked Example — [JEE Main Level]

- (a) Find the ratio in which the line $2x + 3y - 12 = 0$ divides the segment joining $A(1, 2)$ and $B(5, 6)$, and state whether A and B lie on the same side. (b) Show that the origin lies inside the triangle with vertices $P(3, 1)$, $Q(-2, 2)$, $R(0, -3)$.

Solution (a). With $L(x, y) = 2x + 3y - 12$:

$$L(A) = 2(1) + 3(2) - 12 = -4, \quad L(B) = 2(5) + 3(6) - 12 = 16.$$

Opposite signs, so A and B are on opposite sides and the line does cut the segment. The ratio is

$$\frac{AX}{XB} = -\frac{L(A)}{L(B)} = -\frac{-4}{16} = \frac{1}{4},$$

so the line divides AB internally in the ratio $1 : 4$ — positive, consistent with “opposite sides”.

Verification. The point dividing AB in $1 : 4$ from A is $X = \left(\frac{1(5)+4(1)}{5}, \frac{1(6)+4(2)}{5}\right) = \left(\frac{9}{5}, \frac{14}{5}\right)$, and $2\left(\frac{9}{5}\right) + 3\left(\frac{14}{5}\right) - 12 = \frac{18+42-60}{5} = 0$. ✓

Solution (b). Write each side as a line and evaluate at the origin and at the opposite vertex.

Side PQ through $(3, 1)$ and $(-2, 2)$: slope $= \frac{2-1}{-2-3} = -\frac{1}{5}$, so $y - 1 = -\frac{1}{5}(x - 3)$, i.e. $x + 5y - 8 = 0$.

$$\text{At } R(0, -3) : 0 - 15 - 8 = -23. \quad \text{At } O(0, 0) : -8. \quad \text{Same sign.}$$

Side QR through $(-2, 2)$ and $(0, -3)$: slope $= \frac{-3-2}{0+2} = -\frac{5}{2}$, so $y = -\frac{5}{2}x - 3$, i.e. $5x + 2y + 6 = 0$.

$$\text{At } P(3, 1) : 15 + 2 + 6 = 23. \quad \text{At } O : 6. \quad \text{Same sign.}$$

Side RP through $(0, -3)$ and $(3, 1)$: slope $= \frac{1+3}{3-0} = \frac{4}{3}$, so $y = \frac{4}{3}x - 3$, i.e. $4x - 3y - 9 = 0$.

$$\text{At } Q(-2, 2) : -8 - 6 - 9 = -23. \quad \text{At } O : -9. \quad \text{Same sign.}$$

All three agree, so the origin lies on the interior side of every edge.

(a) $1 : 4$ internally, opposite sides; (b) the origin is inside

Practice.

1. **[JEE Main]** Find the ratio in which $x + 3y - 6 = 0$ divides the segment joining $A(-2, 1)$ and $B(4, 3)$, and state whether A and B are on the same side.

2. **[JEE Advanced]** Determine whether $(2, -1)$ and $(5, 3)$ lie on the same side of $4x - 3y + 2 = 0$, and find the ratio in which the line divides the segment joining them. Interpret your answer geometrically.

Answers: 1. $L(A) = -2 + 3 - 6 = -5$ and $L(B) = 4 + 9 - 6 = 7$; opposite signs, so **opposite sides**, and the ratio is $-\frac{-5}{7} = \frac{5}{7}$, i.e. internal division in $5 : 7$. (Check: the dividing point is $(\frac{1}{2}, \frac{11}{6})$, and $\frac{1}{2} + \frac{11}{6} - 6 = 0$.) 2. $L(2, -1) = 8 + 3 + 2 = 13$ and $L(5, 3) = 20 - 9 + 2 = 13$; same sign, so **same side**. The ratio is $-\frac{13}{13} = -1$: external division in the ratio $1 : 1$, which by the Subtopic 00 edge case is the point at infinity. Geometrically the line never meets the segment because it is **parallel** to it — the segment's direction $(3, 4)$ is perpendicular to the normal $(4, -3)$, since $4(3) + (-3)(4) = 0$.

Pattern 4 — Shifting the origin inside an equation

(extends Subtopic 00 — the Exam Relevance box promises this Advanced use)

Method. Shifting the origin to (h, k) with the axes kept parallel means a point's coordinates relabel as $X = x - h$, $Y = y - k$. To transform an **equation** you need the substitution in the other direction:

$$x = X + h, \quad y = Y + k.$$

Substitute, expand, and collect. Two facts make this worth doing:

- **On a line**, the shift changes only the constant term; the coefficients of x and y (hence the slope) are untouched. The new constant vanishes exactly when the new origin lies on the line.
- **On a second-degree curve**, you can *choose* h and k to make the first-degree terms disappear. Setting the coefficients of X and Y to zero gives two equations for h and k ; the surviving equation is the curve referred to its own centre. This is why translation of axes is set-up machinery in JEE Advanced — it turns a shifted conic into a standard one before you do any real work.

Guard rail: the shift moves the origin, it never rotates the axes. Slopes, distances, angles and areas are all unchanged; only labels move.

Worked Example — [JEE Advanced Level]

- (a) The origin is shifted to $(1, 2)$, the axes remaining parallel. Find the new equation of the line $3x - 4y + 5 = 0$ and interpret the result. (b) Find the point to which the origin must be shifted so that the equation $x^2 + y^2 - 4x + 6y - 3 = 0$ has no first-degree terms, and write the transformed equation.

Solution (a). Substitute $x = X + 1$, $y = Y + 2$:

$$3(X + 1) - 4(Y + 2) + 5 = 3X + 3 - 4Y - 8 + 5 = 3X - 4Y + 0.$$

So the new equation is $3X - 4Y = 0$. *Interpretation:* the constant term vanished, which says

the new origin lies on the line — and indeed $3(1) - 4(2) + 5 = 0$. The coefficients 3 and -4 are unchanged, so the slope is still $\frac{3}{4}$: translating axes cannot tilt a line.

Solution (b). Substitute $x = X + h$, $y = Y + k$ into $x^2 + y^2 - 4x + 6y - 3 = 0$:

$$(X + h)^2 + (Y + k)^2 - 4(X + h) + 6(Y + k) - 3 = 0.$$

Expand and group by the powers of X and Y :

$$X^2 + Y^2 + (2h - 4)X + (2k + 6)Y + (h^2 + k^2 - 4h + 6k - 3) = 0.$$

Kill the first-degree terms:

$$2h - 4 = 0 \Rightarrow h = 2, \quad 2k + 6 = 0 \Rightarrow k = -3.$$

Constant term at $(h, k) = (2, -3)$:

$$4 + 9 - 8 - 18 - 3 = -16.$$

So the transformed equation is $X^2 + Y^2 - 16 = 0$.

Check by completing the square in the original: $(x-2)^2 + (y+3)^2 = 3+4+9 = 16$ — a circle of centre $(2, -3)$ and radius 4. The shift simply moved the origin to that centre, which is exactly what “no first-degree terms” means for a circle.

(a) $3X - 4Y = 0$; (b) shift to $(2, -3)$, $X^2 + Y^2 = 16$
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Practice.

1. **[JEE Main]** Find the point to which the origin must be shifted so that $x^2 + y^2 + 8x - 10y + 5 = 0$ has no first-degree terms, and state the new equation.
2. **[CBSE]** The origin is shifted to $(2, -1)$, the axes remaining parallel. Find the new equation of the line $4x + 3y - 7 = 0$ and the new coordinates of the point $(5, 3)$.

Answers: **1.** Substituting $x = X + h$, $y = Y + k$ gives first-degree coefficients $2h + 8$ and $2k - 10$, so $h = -4$, $k = 5$; the constant becomes $16 + 25 - 32 - 50 + 5 = -36$, giving $X^2 + Y^2 = 36$ (a circle of centre $(-4, 5)$, radius 6). **2.** Substituting $x = X + 2$, $y = Y - 1$: $4X + 8 + 3Y - 3 - 7 = 0$, i.e. $4X + 3Y - 2 = 0$; the point becomes $(X, Y) = (5 - 2, 3 - (-1)) = (3, 4)$, using “new equals old minus shift”.

Chapter 9: Straight Lines — Round 2 Addendum

Subtopics 00-06, the Round 1 Supplement (Subtopic 07, Triangle Centres) and its Addendum (Patterns 1-4) between them now cover the whole NCERT chapter and the whole JEE Main and JEE Advanced syllabus entry for straight lines. Two real gaps survive that audit, and this addendum closes them, continuing the Round 1 numbering. A third item, pair of straight lines, was held open for a scoping decision and has since been approved; it is added here as Subtopic 08.

The first is **locus**. The JEE Main syllabus names “locus and its equation” as its own line item, and four separate Exam Relevance boxes in this chapter (Subtopics 00, 01, 02 and 03) promise that locus is where JEE Advanced does its damage. What the chapter actually delivers is the four-step procedure in Subtopic 00, Section 3, one Apollonius circle (Subtopic 00, Example 4) and one perpendicular bisector (Subtopic 00, Practice 5). Every one of those is a *direct-condition* locus: the rule is handed to you as a statement about the moving point, and you translate it. The variety that JEE actually sets — where the moving point is tied to a **variable line or a variable auxiliary point**, and you must introduce a parameter and then eliminate it — is not taught anywhere in the chapter. That is Pattern 5.

The second is **reflecting a line in a line**. Subtopic 06 gives a correct general procedure (reflect two points, rejoin) and a good Pro-Tip for the intersecting case, but it stops one step short in two places: it has no method at all for a mirror **parallel** to the line being reflected, where the Pro-Tip’s shortcut cannot start, and its Pitfall 3 states a rule that is too strong and costs students a fast, general technique. That is Pattern 6, which is part method and part errata.

Pair of straight lines. This topic was reviewed and added on 2026-08-22 by owner decision, and appears below as Subtopic 08, continuing the chapter’s Subtopic 00-07 numbering. It is written as a full subtopic in the standard eight-section form and is labelled optional for a Boards-only reader, since it lies outside NCERT Class 11.

Entry	Topic	Read after
Pattern 5	Locus by introducing and eliminating a parameter	Subtopic 00, Section 3
Pattern 6	Reflecting a line in a line: the substitution map, and the parallel mirror	Subtopic 06, Section 3
Subtopic 08	Pair of straight lines	Subtopic 07 (the whole chapter)

Pattern 5 — Locus by introducing and eliminating a parameter

Extends Subtopic 00, Section 3 (Procedure — Finding the equation of a locus) and Section 4, Example 4. Uses the intercept form of Subtopic 02, the distance formulas of Subtopic 03 and the fixed-point argument of Round 1, Pattern 2.

Method. Subtopic 00 gave you the four-step procedure for the case where the rule is already a statement about the moving point alone. Most JEE locus problems are not like that. Instead, some *other* object moves — a line swings about a fixed point, a rod slides with its ends on the axes, an angle sweeps — and your point is defined in terms of it. You cannot translate the rule until you have named the thing that is varying. So the procedure grows two steps in the middle:

1. **Name the moving point** $P(h, k)$. Never (x, y) — you need (x, y) free to stand for “any point” while you work.
2. **Name the parameter.** Look at the configuration and ask what single number decides which member of the family you are looking at. It is almost always one of: the slope m of the variable line, one of its intercepts a or b , an inclination θ , or the coordinates of one auxiliary point that is itself constrained.
3. **Write h and k in terms of that parameter**, using whatever machinery the configuration hands you — section formula, intercept form, foot of perpendicular.
4. **Write the constraint** the parameter must satisfy (the fixed point the line passes through, the fixed length of the rod, the fixed sum of distances).
5. **Eliminate the parameter.** Three eliminations cover almost everything:
 - **Solve and substitute** — get the parameter alone from one equation, push it into the other. Use when the parameter appears polynomially.
 - **Square and add** — use when the parameter enters as $\cos \theta$ and $\sin \theta$, so that $\cos^2 \theta + \sin^2 \theta = 1$ does the work in one line.
 - **Add or subtract** — use when the parameter enters the two expressions symmetrically and cancels.
6. **Replace (h, k) by (x, y)** and state any restriction (the locus is often only part of the curve you have written).

Three guard rails, in the order they are usually broken:

- **The parameter may not survive into the answer.** If m or θ is still on the page, you have written a family of curves, not a locus. This is the single most common way a correct calculation scores zero.
- **Eliminate the parameter, not h and k .** Under time pressure students solve the simultaneous equations for the wrong unknowns and end up with an identity.
- **Always test one concrete member.** Pick the easiest line of the family, compute the point it produces, and check that point against your locus. It takes ten seconds and it catches both the factor-of-two error in Example 5.1 and any branch that squaring may have added.

Figure brief: A first-quadrant sketch with a fixed point R marked, and three lines of different

slopes drawn through R , each meeting the x -axis at its own A and the y -axis at its own B . The midpoint of each AB is marked with a small filled dot, and a smooth dashed curve is drawn through the three dots and labelled “locus”. Caption: the line moves, the midpoint traces.

Worked Example 5.1 — [JEE Main · Speed Trap]

A variable line passes through the fixed point $R(3, 4)$ and meets the positive x -axis at A and the positive y -axis at B . Find the locus of the midpoint of AB .

Step 1 — name the moving point. Let the midpoint be $P(h, k)$.

Step 2 — name the parameter, and choose a good one. The obvious parameter is the slope, but the intercepts are better here, because the midpoint is built out of them directly. Let the line have x -intercept a and y -intercept b , so $A = (a, 0)$ and $B = (0, b)$.

Step 3 — write h, k in terms of the parameters. By the midpoint formula (Subtopic 00),

$$h = \frac{a + 0}{2}, \quad k = \frac{0 + b}{2} \quad \Rightarrow \quad a = 2h, \quad b = 2k.$$

This inversion is the whole example. Read it aloud: **the intercepts are twice the midpoint’s coordinates**, not equal to them.

Step 4 — write the constraint. In intercept form (Subtopic 02) the line is $\frac{x}{a} + \frac{y}{b} = 1$, and it must pass through $R(3, 4)$:

$$\frac{3}{a} + \frac{4}{b} = 1.$$

Step 5 — eliminate. Substitute $a = 2h$, $b = 2k$:

$$\frac{3}{2h} + \frac{4}{2k} = 1 \quad \Rightarrow \quad \frac{3}{h} + \frac{4}{k} = 2.$$

Step 6 — restore (x, y) and clear fractions.

$$\frac{3}{x} + \frac{4}{y} = 2 \quad \Rightarrow \quad 3y + 4x = 2xy,$$

a rectangular hyperbola, valid for $x > 0$, $y > 0$ (the problem restricted A and B to the positive axes), and with $x \neq 0$, $y \neq 0$ forced anyway by the intercept form.

Test one member. Take the line of slope -1 through R : $x + y = 7$, so $A = (7, 0)$, $B = (0, 7)$ and the midpoint is $(\frac{7}{2}, \frac{7}{2})$. Then $\frac{3}{7/2} + \frac{4}{7/2} = \frac{6}{7} + \frac{8}{7} = 2$. Correct. Take a second, $4x + 3y = 24$ (it passes through R since $12 + 12 = 24$): $A = (6, 0)$, $B = (0, 8)$, midpoint $(3, 4)$, and $\frac{3}{3} + \frac{4}{4} = 2$. Correct again.

$4x + 3y = 2xy \quad \text{i.e.} \quad \frac{3}{x} + \frac{4}{y} = 2$

The trap. The seductive wrong answer is $\frac{3}{x} + \frac{4}{y} = 1$, and it is produced by skipping Step 3 — writing the line as $\frac{x}{h} + \frac{y}{k} = 1$ as though the *midpoint* were the intercept point, and then substituting R . It is a planted option because it looks exactly like the intercept form you were taught, and because it passes a lazy dimensional check. The ten-second test above kills it instantly: $\frac{3}{7/2} + \frac{4}{7/2} = 2 \neq 1$. **Whenever the moving point is a midpoint or a section point of a variable segment, the first thing you write down is the auxiliary point in terms of (h, k) — never the other way round.**

Worked Example 5.2 — [JEE Advanced Level]

As m runs over all real values, the two lines

$$L_1 : x + my = 1, \quad L_2 : mx - y = 1$$

meet in a point. Find the locus of that point, and identify the curve geometrically.

Solution — the algebraic route. Let the intersection be $P(h, k)$, so $h + mk = 1$ and $mh - k = 1$. Solving the pair,

$$h = \frac{1+m}{1+m^2}, \quad k = \frac{m-1}{1+m^2}.$$

Now eliminate m . Squaring and adding is the move, because the denominators are identical:

$$h^2 + k^2 = \frac{(1+m)^2 + (m-1)^2}{(1+m^2)^2} = \frac{2+2m^2}{(1+m^2)^2} = \frac{2}{1+m^2}.$$

That still carries m , so it is not yet an answer — but notice what a plain subtraction gives:

$$h - k = \frac{(1+m) - (m-1)}{1+m^2} = \frac{2}{1+m^2}.$$

The two right-hand sides are the same object. Equating them eliminates m completely:

$$h^2 + k^2 = h - k \quad \implies \quad x^2 + y^2 - x + y = 0.$$

The geometric route, which is faster and is the point of the example. Read each line by the reverse-family method of Round 1, Pattern 2 — group by the parameter:

- L_1 : $(x-1) + m(y) = 0$, so both brackets vanish at $A(1,0)$. Every L_1 passes through A .
- L_2 : $(-y-1) + m(x) = 0$, so both brackets vanish at $B(0,-1)$. Every L_2 passes through B .

And the slopes are $-\frac{1}{m}$ and m , whose product is -1 : **the two lines are perpendicular for every m** . So P is a point at which $AP \perp BP$ — the classical right-angle locus, which is the circle on AB as diameter. Writing that circle directly,

$$(x-1)(x-0) + (y-0)(y+1) = 0 \quad \implies \quad x^2 + y^2 - x + y = 0,$$

in agreement with the algebra. Its centre is $(\frac{1}{2}, -\frac{1}{2})$ and its radius is $\frac{1}{\sqrt{2}} = \frac{|AB|}{2}$.

Excluded point. No finite m gives $h = 0$, $k = -1$: the point B itself is reached only in the limit $m \rightarrow \infty$, i.e. by the vertical member $x = 1$ paired with $y = -1$. State the locus as the circle **less the single point** $B(0, -1)$ if the question asks for the exact set — this is the same “the family misses L_2 ” caveat that Subtopic 04 flagged, wearing different clothes.

$$x^2 + y^2 - x + y = 0$$

that is, the circle on $A(1, 0)B(0, -1)$ as diameter.

Why this is Advanced-flavoured: the brute-force solve-and-eliminate works, but only after you notice that $h^2 + k^2$ and $h - k$ collapse to the same expression, which is luck dressed as skill. The fixed-point-plus-perpendicularity reading gives the answer in two lines and tells you *what the curve is*, which is what a multi-part Advanced question goes on to ask.

Practice.

1. [CBSE] A rod of length 10 slides with one end on the x -axis and the other on the y -axis. Find the locus of its midpoint.
2. [JEE Main] Find the locus of the foot of the perpendicular drawn from the origin to a variable line that always passes through the fixed point $(4, 0)$.
3. [JEE Main] A lies on the x -axis, B lies on the y -axis, and $AB = 6$. Find the locus of the point P dividing AB internally in the ratio $1 : 2$, measured from A .
4. [JEE Main] As θ varies, find the locus of the point of intersection of $x \cos \theta + y \sin \theta = a$ and $x \sin \theta - y \cos \theta = b$, where a and b are fixed.
5. [JEE Advanced] A variable line makes intercepts a and b on the axes, subject to $\frac{1}{a^2} + \frac{1}{b^2} = \frac{1}{c^2}$ with c fixed. Find the locus of the foot of the perpendicular from the origin to the line.
6. [JEE Main] A variable line through the fixed point (a, b) meets the axes at A and B . With O the origin, find the locus of the **centroid** of $\triangle OAB$, and say in one line how the answer differs from Worked Example 5.1 and why.
7. [JEE Advanced] The sum of the perpendicular distances of a variable point $P(x, y)$ from the lines $x + y - 5 = 0$ and $3x - 2y + 7 = 0$ is always 10. Show that P moves on a straight line, and write that line for the region in which both signed distances are positive.

Answers.

1. Ends $A(a, 0)$, $B(0, b)$ with $a^2 + b^2 = 100$; the midpoint gives $a = 2h$, $b = 2k$, so $4h^2 + 4k^2 = 100$ and the locus is $x^2 + y^2 = 25$ — a circle of radius 5, centred at the origin. (Sanity check: the midpoint of a segment of length 10 whose ends slide on perpendicular axes is always at distance 5 from their crossing point, because the midpoint of the hypotenuse of a right triangle is its circumcentre — Subtopic 07, right-triangle shortcut.)

2. Let the foot be $P(h, k)$ and write $R = (4, 0)$. The line is PR , and OP is perpendicular to it, so $\overrightarrow{PO} \cdot \overrightarrow{PR} = 0$. That is $(0 - h)(4 - h) + (0 - k)(0 - k) = 0$, i.e. $h(h - 4) + k^2 = 0$, giving $x^2 + y^2 - 4x = 0$ — the circle on the segment from $(0, 0)$ to $(4, 0)$ as diameter. Check with the line through $(4, 0)$ of slope 1, i.e. $x - y - 4 = 0$: its foot from the origin is $(2, -2)$, and $4 + 4 - 8 = 0$. Correct.
3. $A = (a, 0)$, $B = (0, b)$, $a^2 + b^2 = 36$. The section formula with $m : n = 1 : 2$ from A gives $P = (\frac{2a}{3}, \frac{b}{3})$, so $a = \frac{3h}{2}$ and $b = 3k$. Then $\frac{9h^2}{4} + 9k^2 = 36$, i.e. $\frac{x^2}{16} + \frac{y^2}{4} = 1$ — an ellipse. Check the extreme member $a = 6, b = 0$: $P = (4, 0)$, and $\frac{16}{16} + 0 = 1$. Correct.
4. Square and add. The cross terms $\pm 2xy \cos \theta \sin \theta$ cancel and $\cos^2 \theta + \sin^2 \theta = 1$ collapses the rest, leaving $x^2 + y^2 = a^2 + b^2$ — a circle centred at the origin of radius $\sqrt{a^2 + b^2}$. Geometric reading, worth more than the algebra: the two lines are perpendicular for every θ (their normals are $(\cos \theta, \sin \theta)$ and $(\sin \theta, -\cos \theta)$, whose dot product is zero), and the origin sits at distance $|a|$ from the first and $|b|$ from the second. Those two perpendicular distances are the legs of a right triangle with hypotenuse OP , so $OP^2 = a^2 + b^2$ is fixed by Pythagoras alone.
5. The line is $\frac{x}{a} + \frac{y}{b} = 1$, i.e. $bx + ay - ab = 0$, and its distance from the origin is $\frac{|ab|}{\sqrt{a^2 + b^2}} = \frac{1}{\sqrt{\frac{1}{a^2} + \frac{1}{b^2}}} = c$, the same for every member of the family. The foot of the perpendicular therefore always sits at distance c from the origin, so the locus is $x^2 + y^2 = c^2$. Notice that the parameter never had to be eliminated by algebra — the constraint *was* the elimination in disguise. Check with $c = 2$, $a = 4$, $b = \frac{4}{\sqrt{3}}$: the foot is $(1, \sqrt{3})$ and $1 + 3 = 4 = c^2$. Correct.
6. Let $A = (p, 0)$, $B = (0, q)$ and let the centroid be (h, k) . The centroid of $\triangle OAB$ is $(\frac{0+p+0}{3}, \frac{0+0+q}{3})$, so $p = 3h$ and $q = 3k$. The line passes through (a, b) , so $\frac{a}{p} + \frac{b}{q} = 1$, giving $\frac{a}{3h} + \frac{b}{3k} = 1$, i.e. $\frac{a}{h} + \frac{b}{k} = 3$ and

$$bx + ay = 3xy.$$

The difference from Worked Example 5.1 is the single factor 3 in place of 2, and it comes from Step 3 alone: a centroid is one **third** of the way out to the intercepts, a midpoint one half, so the inversion is $p = 3h$ rather than $p = 2h$. Everything after Step 3 is identical. (Check with $a = 3$, $b = 4$ and the line $4x + 3y = 24$: $A = (6, 0)$, $B = (0, 8)$, centroid $(2, \frac{8}{3})$, and $4(2) + 3(\frac{8}{3}) = 16 = 3(2)(\frac{8}{3})$. Correct.)

7. With $d_1 = \frac{|x + y - 5|}{\sqrt{2}}$ and $d_2 = \frac{|3x - 2y + 7|}{\sqrt{13}}$, the condition $d_1 + d_2 = 10$ splits into four cases, one per sign pair, and **each case is first-degree in x and y** — because dropping a modulus leaves a linear expression and a sum of linear expressions is linear. Hence P moves on a line in each region, which is what “must move on a line” asks you to show. For the region where both signed distances are positive, multiply through by $\sqrt{26}$:

$$\sqrt{13}(x + y - 5) + \sqrt{2}(3x - 2y + 7) = 10\sqrt{26},$$

that is

$$(\sqrt{13} + 3\sqrt{2})x + (\sqrt{13} - 2\sqrt{2})y - (5\sqrt{13} - 7\sqrt{2} + 10\sqrt{26}) = 0.$$

The four cases together trace a **parallelogram**, one side per sign region, centred on the intersection of the two given lines. Here that intersection is $(\frac{3}{5}, \frac{22}{5})$ and the four vertices are $(\frac{3}{5} \pm 4\sqrt{2}, \frac{22}{5} \pm 6\sqrt{2})$ (on $3x - 2y + 7 = 0$) and $(\frac{3}{5} \pm 2\sqrt{13}, \frac{22}{5} \mp 2\sqrt{13})$ (on $x + y - 5 = 0$). It is centrally symmetric because reflecting a point through the intersection leaves both $|L_1|$ and $|L_2|$ unchanged. This is the same shape reason that $|x| + |y| = 1$ is a square and not a line, so read “show that P must move on a line” as the per-region statement it is.

Pattern 6 — Reflecting a line in a line: the substitution map, and the parallel mirror

Extends Subtopic 06, Section 3 (Procedure — Reflecting a line m in a line ℓ), Section 7 Pitfall 3, and the Pro-Tip. Contains one correction to the main text.

Method. Subtopic 06 gave you two routes and both are sound:

- **Route A (always works).** Reflect two points of m in ℓ and join the images.
- **Route B (faster, needs m and ℓ to meet).** The reflected line passes through $m \cap \ell$ and through the image of any one point of m , so one image point suffices.

Two situations defeat them, and each has a clean answer.

Route C — the substitution map. This is the general form of the “ $y = x$? Just swap” shortcut. Reflection in a fixed line is a map of the plane to itself, and it is its **own inverse** — reflect twice and you are back where you started. That single fact licenses the following: if the reflection in ℓ sends (x, y) to $(X(x, y), Y(x, y))$, then substituting X and Y in place of x and y in the equation of m gives the equation of the reflected line, immediately. For a general mirror $\ell : ax + by + c = 0$, the map is just the Subtopic 06 image formula with (x_1, y_1) left as a running (x, y) :

$$(x, y) \mapsto \left(x - \frac{2a(ax + by + c)}{a^2 + b^2}, y - \frac{2b(ax + by + c)}{a^2 + b^2} \right).$$

For the mirrors that turn up constantly, the map is worth knowing cold:

Mirror ℓ	Reflection map $(x, y) \mapsto$	Substitute into m
x -axis, $y = 0$	$(x, -y)$	$y \rightarrow -y$
y -axis, $x = 0$	$(-x, y)$	$x \rightarrow -x$
$y = x$	(y, x)	swap x and y
$y = -x$	$(-y, -x)$	$x \rightarrow -y, y \rightarrow -x$
$x = h$	$(2h - x, y)$	$x \rightarrow 2h - x$
$y = k$	$(x, 2k - y)$	$y \rightarrow 2k - y$
$ax + by + c = 0$	the displayed formula above	both together

Route D — the parallel mirror. If $m : ax + by + c_1 = 0$ and $\ell : ax + by + c = 0$ are parallel

they never meet, so Route B cannot even start and Route A, while correct, is needless work. Only the constant term moves:

$$\text{image of } ax + by + c_1 = 0 \text{ in } ax + by + c = 0 \text{ is } ax + by + (2c - c_1) = 0.$$

Why. Take any P on m , so $ax_P + by_P = -c_1$ and hence $ax_P + by_P + c = c - c_1$. The image formula moves P by $-\frac{2(c-c_1)}{a^2+b^2}(a, b)$, so for the image point $ax_{P'} + by_{P'} = -c_1 - 2(c - c_1) = c_1 - 2c$, i.e. $ax_{P'} + by_{P'} + (2c - c_1) = 0$. Every point of m lands on that one line. Read the formula as **“the mirror’s constant is the average of the other two”**: $c = \frac{c_1 + c'_1}{2}$, which is also the free check — the mirror must run exactly midway between m and its image, at equal distance $\frac{|c - c_1|}{\sqrt{a^2 + b^2}}$ on each side.

Correction to Subtopic 06, Section 7, Pitfall 3. The main text says: “You cannot ‘reflect’ $ax + by + c = 0$ by tweaking coefficients (except for $y = x$, where swapping x, y works).” That is too strong, and it is the reason students hand-compute two image points for a mirror as easy as $x = 3$. The accurate statement is:

You cannot reflect a line by *guessing* at its coefficients. But you may always reflect it by **substituting the reflection map** into its equation — that is Route C, and it is valid for every mirror, not only $y = x$. What genuinely fails is negating the equation, adding the two equations, or swapping x and y for a mirror that is not $y = x$.

Everything else in Pitfall 3 stands: reflecting *points* is still the safe default when the mirror is awkward, and Route A is never wrong.

Worked Example 6.1 — [JEE Main Level]

- (a) Find the reflection of the line $2x - 3y + 4 = 0$ in the line $x + y - 1 = 0$.
- (b) Find the reflection of the line $3x - 4y + 7 = 0$ in the line $3x - 4y - 2 = 0$.

Solution (a) — Route A, then Route C as the check. The mirror is $\ell : x + y - 1 = 0$, so $a = 1, b = 1, a^2 + b^2 = 2$.

Pick two easy points on $m : 2x - 3y + 4 = 0$. At $x = 1$: $2 - 3y + 4 = 0$, so $y = 2$, giving $P(1, 2)$. At $y = 0$: $2x + 4 = 0$, so $x = -2$, giving $Q(-2, 0)$.

Image of P : here $ax_P + by_P + c = 1 + 2 - 1 = 2$, so the step is $-\frac{2(2)}{2} = -2$ and $P' = (1 + (-2)(1), 2 + (-2)(1)) = (-1, 0)$.

Image of Q : $-2 + 0 - 1 = -3$, so the step is $-\frac{2(-3)}{2} = 3$ and $Q' = (-2 + 3, 0 + 3) = (1, 3)$.

Line through $P'(-1, 0)$ and $Q'(1, 3)$: slope $= \frac{3 - 0}{1 - (-1)} = \frac{3}{2}$, so $y - 0 = \frac{3}{2}(x + 1)$, that is $3x - 2y + 3 = 0$.

Route C check. Reflection in $x + y - 1 = 0$ sends $(x, y) \mapsto (x - (x + y - 1), y - (x + y - 1)) = (1 - y, 1 - x)$. Substituting into m :

$$2(1 - y) - 3(1 - x) + 4 = 2 - 2y - 3 + 3x + 4 = 3x - 2y + 3.$$

Same line, in one line of algebra instead of six.

Third check, free. Route B says the answer must pass through $m \cap \ell$. Solving $2x - 3y + 4 = 0$ with $x + y - 1 = 0$ gives $(-\frac{1}{5}, \frac{6}{5})$, and $3(-\frac{1}{5}) - 2(\frac{6}{5}) + 3 = -\frac{3}{5} - \frac{12}{5} + 3 = 0$. It does.

Solution (b) — Route D. The two lines have identical x and y coefficients, so they are parallel and there is no intersection to hang Route B on. Here $c = -2$ and $c_1 = 7$, so the image constant is $2c - c_1 = -4 - 7 = -11$:

$$3x - 4y - 11 = 0.$$

Check by the midway property. $\sqrt{3^2 + 4^2} = 5$. Distance from m to the mirror is $\frac{|7 - (-2)|}{5} = \frac{9}{5}$; distance from the mirror to the image is $\frac{|-2 - (-11)|}{5} = \frac{9}{5}$. Equal, and the mirror's constant -2 is the average of 7 and -11 . Correct.

(a) $3x - 2y + 3 = 0$	(b) $3x - 4y - 11 = 0$
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Practice.

1. [CBSE] Find the reflection of $3x + 2y - 6 = 0$ in the x -axis.
2. [CBSE] Find the reflection of $x - 2y + 4 = 0$ in the line $x = 3$.
3. [JEE Main] Find the reflection of $x + 2y - 3 = 0$ in the line $y = -x$.
4. [JEE Main] Find the reflection of $5x - 12y + 13 = 0$ in the parallel line $5x - 12y - 26 = 0$, and state the distance between the original line and its image.
5. [JEE Main] Find the image of the line $3x - y = 2$ in the line $y = x - 1$. (The mirror is $y = x$ shifted; do not reach for the swap shortcut without checking it.)
6. [JEE Advanced] Find the reflection of $x + y - 2 = 0$ in the line $2x - y + 3 = 0$, using the substitution map, and verify your answer by reflecting one point independently.

Answers.

1. Substitute $y \rightarrow -y$: $3x - 2y - 6 = 0$.
2. Substitute $x \rightarrow 6 - x$: $(6 - x) - 2y + 4 = 0$, i.e. $x + 2y - 10 = 0$. (Check: the two lines meet on the mirror, at $x = 3$, $y = \frac{7}{2}$, and both are satisfied there.)
3. Substitute $x \rightarrow -y$, $y \rightarrow -x$: $(-y) + 2(-x) - 3 = 0$, i.e. $2x + y + 3 = 0$.
4. $c = -26$, $c_1 = 13$, so the image constant is $2(-26) - 13 = -65$, giving $5x - 12y - 65 = 0$. With $\sqrt{25 + 144} = 13$, the mirror sits $\frac{|13 + 26|}{13} = 3$ from the original, so the original and its image are 6 apart — $\frac{|13 - (-65)|}{13} = \frac{78}{13} = 6$, as it must be.
5. Write the mirror as $x - y - 1 = 0$, so $a = 1$, $b = -1$, $a^2 + b^2 = 2$ and the map is $(x, y) \mapsto (x - (x - y - 1), y + (x - y - 1)) = (y + 1, x - 1)$ — the swap, but with the shift attached, which is exactly what the bare “just swap” rule would have lost. Substituting into $3x - y - 2 = 0$: $3(y + 1) - (x - 1) - 2 = -x + 3y + 2$, so the image is

$$x - 3y = 2.$$

Check: $(1, 1)$ lies on $3x - y = 2$ and maps to $(2, 0)$, with $2 - 0 = 2$; and $(0, -2)$ maps to $(-1, -1)$, with $-1 + 3 = 2$. Both land on the answer. Note the slope has gone from 3 to $\frac{1}{3}$, the reciprocal — as it must for any mirror parallel to $y = x$.

6. Mirror $2x - y + 3 = 0$: $a = 2$, $b = -1$, $a^2 + b^2 = 5$, so the map is

$$(x, y) \mapsto \left(\frac{-3x + 4y - 12}{5}, \frac{4x + 3y + 6}{5} \right).$$

Substituting into $x + y - 2 = 0$: $\frac{(-3x + 4y - 12) + (4x + 3y + 6)}{5} - 2 = \frac{x + 7y - 6}{5} - 2 = 0$,
i.e.

$$x + 7y - 16 = 0.$$

Independent check: $(2, 0)$ lies on $x + y - 2 = 0$; its image is $(-\frac{18}{5}, \frac{14}{5})$, and $-\frac{18}{5} + 7(\frac{14}{5}) - 16 = \frac{-18 + 98 - 80}{5} = 0$. And $(0, 2)$ maps to $(-\frac{4}{5}, \frac{12}{5})$, with $-\frac{4}{5} + \frac{84}{5} - 16 = 0$. Both land on the answer.

Subtopic 08 — Pair of Straight Lines

Exam Relevance: This subtopic is **optional for a Boards-only reader** — it appears nowhere in NCERT Class 11 Chapter 9, and CBSE does not set it. It is included because every major JEE reference carries it at Class 11 level and because JEE has examined it: a single second-degree equation that secretly holds two lines is a compact way to ask four chapter skills at once (slopes, angle, bi-sectors, intersection), which is exactly what a JEE Main single-answer question wants. JEE Main asks the standard three — is this equation a pair, what is the angle between the lines, what is λ — and JEE Advanced pushes into **homogenisation**, where a line and a curve are fused into one homogeneous equation so that a right angle at the origin becomes the one-line condition $a + b = 0$. As always in mathematics, NEET does not test any of this: NEET has no mathematics paper.

Section 1: Concept Intuition

A patwari recording a village crossroads has two boundary walls to enter in the register, and only one line of the ledger. So he multiplies. If the first wall is $3x - 2y = 0$ and the second is $2x + 3y = 0$, he writes their product, $(3x - 2y)(2x + 3y) = 0$, and expands it to $6x^2 + 5xy - 6y^2 = 0$. Nothing has been lost. A product is zero exactly when one of its factors is zero, so a point satisfies the single recorded equation precisely when it lies on one wall or the other. **One second-degree equation can carry two straight lines, and factorising is how you get them back.**

That is the entire idea, and everything in this subtopic is a consequence of it. The second-degree equation is called the **joint equation** or **combined equation** of the pair. Reading it forwards (two lines to one equation) is multiplication. Reading it backwards (one equation to two lines) is factorisation — and the reason this topic has a theory at all is that most of the questions can be answered *without* factorising, straight from the coefficients.

Now specialise. Suppose both walls pass through the origin, so neither equation has a constant term. Then their product has no constant term and no x or y term either: every surviving term is second degree. That is a **homogeneous** equation,

$$ax^2 + 2hxy + by^2 = 0,$$

and the converse is the useful half — *every* such equation is a pair of lines through the origin. Here is why, without any algebra. If the point (x_0, y_0) satisfies it, then so does (tx_0, ty_0) for every real t , because replacing x, y by tx, ty multiplies the whole left side by t^2 , and $t^2 \cdot 0 = 0$. So the graph is made of full straight lines through the origin. Being second degree, it can hold at most two of them.

Think about it this way. Put $y = mx$ into the homogeneous equation and cancel x^2 :

$$bm^2 + 2hm + a = 0.$$

The two lines of the pair *are* the two roots of one ordinary quadratic in the slope. Every formula in Section 2 is then just Vieta's sum and product wearing coordinate-geometry clothes, and the discriminant of this quadratic, $4h^2 - 4ab$, is what decides whether the lines exist at all.

Where the idea breaks down. Two places, and both are examined.

- If $h^2 < ab$ the slope quadratic has no real root. The equation still has a graph, but the graph is the single point $(0, 0)$ — the “two lines” are a conjugate pair of imaginary lines. The standard specimen is $x^2 + y^2 = 0$. Do not say “no solution”; say **the origin only**.
- A general second-degree equation with linear terms, $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$, is normally a *conic* — a circle, parabola, ellipse or hyperbola (Chapter 11). It collapses into a pair of lines only for special coefficient values, and the price of that collapse is one exact equation, the determinant condition of Section 2. A pair of lines is a **degenerate conic**: a hyperbola whose waist has been pinched shut until the two branches touch and straighten into their own asymptotes.

Figure brief: Three small panels side by side on one axis-pair each, all sharing the origin. Panel (a) labelled $h^2 > ab$: two distinct lines crossing at O , the angle θ between them marked with an arc. Panel (b) labelled $h^2 = ab$: a single line through O drawn with a doubled stroke and labelled “coincident pair”. Panel (c) labelled $h^2 < ab$: no lines at all, just a single heavy dot at O , labelled “the origin only”. Caption: one equation, three regimes, decided by the sign of $h^2 - ab$.

Domain of Validity and Edge Cases.

- h is **half** the coefficient of xy , g is half the coefficient of x , and f is half the coefficient of y . The standard form is written with the 2s precisely so that the formulas come out clean; the cost is that you must halve on the way in. This is the single largest source of lost marks in the topic.
- $m_1 + m_2 = -2h/b$ and $m_1 m_2 = a/b$ need $b \neq 0$. If $b = 0$, the equation is $x(ax + 2hy) = 0$: one of the lines is the y -axis $x = 0$, which has no slope, and the sum-and-product formulas

are meaningless. Read the factorisation off directly instead.

- The angle formula returns the **acute** angle when you take the modulus. If $a + b = 0$ the tangent is infinite; that is not an error, it is $\theta = 90^\circ$.
- The determinant condition $\Delta = 0$ is necessary and sufficient for *degeneracy*, not for two real lines. You must still check $h^2 \geq ab$. Both $x^2 + y^2 = 0$ and $x^2 + 4y^2 = 0$ satisfy $\Delta = 0$ and are single points.
- Homogenisation requires the line **not** to pass through the origin. If it does, the origin is itself one of the intersection points and “the lines joining the origin to the intersections” is not a well-posed question.

Section 2: Key Formulas and Definitions

Throughout, the general second-degree equation is written

$$S \equiv ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0,$$

so that

$$a = \text{coeff. of } x^2, \quad b = \text{coeff. of } y^2, \quad c = \text{constant},$$

$$h = \frac{1}{2}(\text{coeff. of } xy), \quad g = \frac{1}{2}(\text{coeff. of } x), \quad f = \frac{1}{2}(\text{coeff. of } y).$$

A. The homogeneous pair $ax^2 + 2hxy + by^2 = 0$ — two lines through the origin, $y = m_1x$ and $y = m_2x$, where m_1, m_2 are the roots of $bm^2 + 2hm + a = 0$.

Quantity	Formula	Conditions
Sum of slopes	$m_1 + m_2 = -\frac{2h}{b}$	$b \neq 0$
Product of slopes	$m_1m_2 = \frac{a}{b}$	$b \neq 0$
Difference of slopes	$ m_1 - m_2 = \frac{2\sqrt{h^2 - ab}}{ b }$	$b \neq 0, h^2 \geq ab$
Nature: real and distinct	$h^2 > ab$	—
Nature: real and coincident	$h^2 = ab$ (then $ax^2 + 2hxy + by^2$ is a perfect square)	—
Nature: not real	$h^2 < ab$ — the graph is the origin alone	—

B. Angle between the pair. If θ is the angle between the two lines,

$$\tan \theta = \left| \frac{2\sqrt{h^2 - ab}}{a + b} \right|, \quad \text{equivalently} \quad \sin \theta = \frac{2\sqrt{h^2 - ab}}{\sqrt{(a - b)^2 + 4h^2}}.$$

Special case	Condition	Why
Lines perpendicular	$a + b = 0$, i.e. coeff. of $x^2 +$ coeff. of $y^2 = 0$	$\tan \theta \rightarrow \infty$
Lines coincident	$h^2 = ab$	$\tan \theta = 0$
Lines equally inclined to the axes	$a = b$	bisectors become $x^2 - y^2 = 0$

Any pair of perpendicular lines through the origin can therefore be written $x^2 + \lambda xy - y^2 = 0$ for some λ . And the pair **perpendicular** to $ax^2 + 2hxy + by^2 = 0$, drawn through the origin, is

$$bx^2 - 2hxy + ay^2 = 0 \quad (\text{swap } a \text{ and } b, \text{ flip the sign of } h).$$

C. Bisectors of the angles between the pair. For $ax^2 + 2hxy + by^2 = 0$ the joint equation of the two angle bisectors is

$$\frac{x^2 - y^2}{a - b} = \frac{xy}{h}, \quad \text{i.e.} \quad hx^2 - (a - b)xy - hy^2 = 0.$$

Read the second form: its x^2 coefficient is h and its y^2 coefficient is $-h$, which sum to zero. **The bisectors of any pair are always perpendicular to each other** — the same structural fact Subtopic 05 proved for two separately-written lines, now visible in a single glance at the coefficients. Two degenerate readings, taken from the cross-multiplied form so that nothing is divided by zero:

- If $a = b$ (equally inclined lines), the bisectors are $x^2 - y^2 = 0$, i.e. $y = x$ and $y = -x$.
- If $h = 0$ (the pair is symmetric about the axes), the bisectors are $xy = 0$, i.e. the coordinate axes themselves.

D. The general equation as a pair. $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents a pair of straight lines **if and only if**

$$\Delta \equiv \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = abc + 2fgh - af^2 - bg^2 - ch^2 = 0,$$

and the pair is real with $h^2 \geq ab$. Given $\Delta = 0$:

Quantity	Formula
Point of intersection	$\left(\frac{hf - bg}{ab - h^2}, \frac{gh - af}{ab - h^2} \right)$, valid when $ab \neq h^2$
Angle between the lines	the same $\tan \theta = \left \frac{2\sqrt{h^2 - ab}}{a + b} \right $ — the linear terms do not affect direction

Quantity	Formula
Lines parallel (or coincident)	$h^2 = ab$
Lines perpendicular	$a + b = 0$
Distance between a parallel pair	$2\sqrt{\frac{g^2 - ac}{a(a+b)}}, \text{ or } 2\sqrt{\frac{f^2 - bc}{b(a+b)}}$
Bisectors of the angles	$\frac{(x-\alpha)^2 - (y-\beta)^2}{a-b} = \frac{(x-\alpha)(y-\beta)}{h}, \text{ where } (\alpha, \beta) \text{ is the point of intersection}$
Pair through the origin parallel to it	$ax^2 + 2hxy + by^2 = 0$

E. Homogenisation. If the line $lx + my + n = 0$ (with $n \neq 0$) cuts the curve $S = 0$ at P and Q , then the joint equation of OP and OQ , where O is the origin, is

$$ax^2 + 2hxy + by^2 + (2gx + 2fy)\left(\frac{lx + my}{-n}\right) + c\left(\frac{lx + my}{-n}\right)^2 = 0,$$

which is homogeneous of degree two and therefore a pair through the origin. In practice: write the line as $\frac{lx + my}{-n} = 1$, then multiply each term of S by whatever power of that expression makes it second degree.

F. Two further results worth carrying.

Product of the perpendicular distances from (x_1, y_1) to the pair $ax^2 + 2hxy + by^2 = 0 = \frac{|ax_1^2 + 2hx_1y_1 + by_1^2|}{\sqrt{(a-b)^2 + 4h^2}}.$

Area of the triangle formed by $ax^2 + 2hxy + by^2 = 0$ and $lx + my + n = 0 = \frac{n^2\sqrt{h^2 - ab}}{|am^2 - 2hlm + bl^2|}.$

Section 3: Derivations and Key Procedures

Derivation 1: A homogeneous second-degree equation is a pair of lines through the origin

1. Start from $ax^2 + 2hxy + by^2 = 0$ with $b \neq 0$, and look for lines through the origin, which have the form $y = mx$. Reason: Section 1 showed the graph is a union of lines through O , so this loses nothing except the vertical line, handled at step 6.
2. Substitute $y = mx$: $ax^2 + 2hmx^2 + bm^2x^2 = 0$, i.e. $x^2(bm^2 + 2hm + a) = 0$.
3. For the line $y = mx$ to lie in the graph, the bracket must vanish for all x , so $bm^2 + 2hm + a = 0$. Reason: x^2 is not identically zero along the line.
4. This quadratic has roots m_1, m_2 with, by Vieta,

$$m_1 + m_2 = -\frac{2h}{b}, \quad m_1m_2 = \frac{a}{b}.$$

- Therefore $b(y - m_1x)(y - m_2x) = by^2 - b(m_1 + m_2)xy + bm_1m_2x^2 = by^2 + 2hxy + ax^2$, which is the original expression. *Reason: substitute the two Vieta values.* So the equation **is** the product of two line equations.
- Roots exist and are distinct precisely when the discriminant $4h^2 - 4ab > 0$; they coincide when $h^2 = ab$; they are non-real when $h^2 < ab$. If $b = 0$ the quadratic degenerates to a linear equation and the missing second line is $x = 0$.

Derivation 2: The angle between the pair

- Let the lines be $y = m_1x$ and $y = m_2x$. From Subtopic 01, $\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1m_2} \right|$.
- Build $(m_1 - m_2)^2 = (m_1 + m_2)^2 - 4m_1m_2 = \frac{4h^2}{b^2} - \frac{4a}{b} = \frac{4(h^2 - ab)}{b^2}$, so $|m_1 - m_2| = \frac{2\sqrt{h^2 - ab}}{|b|}$.
Reason: Vieta from Derivation 1.
- Also $1 + m_1m_2 = 1 + \frac{a}{b} = \frac{a + b}{b}$.
- Divide: $\tan \theta = \left| \frac{2\sqrt{h^2 - ab} / |b|}{(a + b)/b} \right| = \left| \frac{2\sqrt{h^2 - ab}}{a + b} \right|$. The b cancels entirely — which is why the formula still reads correctly in the limit $b \rightarrow 0$.
- Corollary (perpendicular).** $\theta = 90^\circ \iff \cot \theta = 0 \iff a + b = 0$.
- Corollary (coincident).** $\theta = 0 \iff \sqrt{h^2 - ab} = 0 \iff h^2 = ab$.

Derivation 3: The bisector pair

- A point $P(x, y)$ is on a bisector exactly when its perpendicular distances to the two lines are equal. *Reason: Subtopic 05, Theorem 1 — this is the defining property.*
- Writing the lines as $y - m_1x = 0$ and $y - m_2x = 0$, that condition is

$$\frac{y - m_1x}{\sqrt{1 + m_1^2}} = \pm \frac{y - m_2x}{\sqrt{1 + m_2^2}}.$$

- Multiply the two sign cases together to get a single joint equation:

$$\frac{(y - m_1x)^2}{1 + m_1^2} - \frac{(y - m_2x)^2}{1 + m_2^2} = 0.$$

- Clear denominators and expand. Every term carries a factor $(m_1 - m_2)$, and cancelling it (legitimate, since the lines are distinct) leaves

$$(m_1 + m_2)(y^2 - x^2) + 2xy(1 - m_1m_2) = 0.$$

- Substitute $m_1 + m_2 = -2h/b$ and $m_1m_2 = a/b$, then multiply through by $-b/2$:

$$\frac{x^2 - y^2}{a - b} = \frac{xy}{h}, \quad \text{i.e.} \quad hx^2 - (a - b)xy - hy^2 = 0. \blacksquare$$

- Why the bisectors are always perpendicular.** In the result, (coefficient of x^2) + (coefficient of y^2) = $h + (-h) = 0$, which is exactly the perpendicularity test of Derivation 2. Geometrically it is the supplementary-angle argument of Subtopic 05, Example 4, unchanged.

7. **And always real.** For the bisector pair, $H^2 - AB = \frac{(a-b)^2}{4} + h^2 \geq 0$ always. So even a pair whose own angle is awkward has two genuine, perpendicular bisectors — unless $a = b$ and $h = 0$ together, in which case the original equation was $a(x^2 + y^2) = 0$ and there were no real lines to bisect.

Derivation 4: The determinant condition $abc + 2fgh - af^2 - bg^2 - ch^2 = 0$

1. Suppose $S \equiv ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ is a pair of non-parallel lines meeting at (α, β) .
2. Shift the origin to that point: put $x = X + \alpha$, $y = Y + \beta$. Reason: the pair now passes through the new origin, so by Derivation 1 the transformed equation must be homogeneous in X, Y — every coefficient of X , of Y , and the constant must vanish.
3. Expanding and collecting, the coefficient of X is $2(a\alpha + h\beta + g)$ and the coefficient of Y is $2(h\alpha + b\beta + f)$, so

$$a\alpha + h\beta + g = 0 \quad \text{and} \quad h\alpha + b\beta + f = 0.$$

4. The new constant term is $a\alpha^2 + 2h\alpha\beta + b\beta^2 + 2g\alpha + 2f\beta + c$, which regroups as

$$\alpha \underbrace{(a\alpha + h\beta + g)}_{=0} + \beta \underbrace{(h\alpha + b\beta + f)}_{=0} + (g\alpha + f\beta + c),$$

so the third condition is simply $g\alpha + f\beta + c = 0$.

5. The three conditions say that the three lines $ax + hy + g = 0$, $hx + by + f = 0$ and $gx + fy + c = 0$ all pass through (α, β) — they are **concurrent**. By the concurrency determinant of Subtopic 04,

$$\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0, \quad \text{i.e.} \quad abc + 2fgh - af^2 - bg^2 - ch^2 = 0. \quad \blacksquare$$

6. **Point of intersection, free.** Solving the first two conditions of step 3 for (α, β) gives

$$\alpha = \frac{hf - bg}{ab - h^2}, \quad \beta = \frac{gh - af}{ab - h^2},$$

which is why the first two rows of the determinant are all you ever need in a hurry.

7. **The memory device that makes this instant.** The two lines of step 3 are exactly $\frac{1}{2} \frac{\partial S}{\partial x} = 0$ and $\frac{1}{2} \frac{\partial S}{\partial y} = 0$. So the fastest route to the point of intersection is: *differentiate S once with respect to each variable, set both to zero, and solve the resulting linear pair.* You do not need calculus to use this — $\frac{\partial S}{\partial x}$ just means “differentiate treating y as a constant” — but if you have met derivatives it makes the formula unforgettable.

Procedure A: Does this equation represent a pair of straight lines?

1. Read off a, b, c directly, and h, g, f as **halves** of the xy , x and y coefficients. Write the six numbers down before doing anything else.
2. Compute $\Delta = abc + 2fgh - af^2 - bg^2 - ch^2$. If $\Delta \neq 0$, it is a genuine conic, not a pair. Stop.

3. Compute $h^2 - ab$. If it is negative the “pair” is imaginary and the real graph is the single point $\left(\frac{hf-bg}{ab-h^2}, \frac{gh-af}{ab-h^2}\right)$.
4. If $h^2 - ab > 0$: two distinct real lines, meeting at that point. If $h^2 - ab = 0$: two parallel (or coincident) lines, and the intersection formula is unusable — go to Procedure B instead.

Procedure B: Separating the two lines

1. **Try to factorise the second-degree part first.** $ax^2 + 2hxy + by^2$ always factorises into two real linear forms when $h^2 \geq ab$; find them by inspection or by splitting the middle term.
2. Write the two full lines as those factors plus unknown constants, e.g. $(\ell_1 + p)(\ell_2 + q)$, expand, and match the x , y and constant coefficients. Three equations, two unknowns — the third is your check.
3. **If inspection fails**, treat the whole equation as a quadratic in x with y as a parameter, and apply the quadratic formula. When the equation really is a pair, the expression under the root is a perfect square in y , and the $\pm$ hands you the two lines directly.
4. Check by substituting the point of intersection from Procedure A into both lines.

Procedure C: Homogenising a curve with a line

1. Confirm the line $lx + my + n = 0$ does **not** pass through the origin, i.e. $n \neq 0$.
2. Rewrite the line as a quantity equal to 1: $u \equiv \frac{lx + my}{-n} = 1$.
3. In the curve’s equation, multiply the linear terms $2gx + 2fy$ by u , and the constant term c by u^2 . Leave the second-degree terms alone. *Reason: multiplying by 1 changes nothing at the two intersection points, but it makes every term degree two.*
4. Clear denominators. The result $Ax^2 + 2Hxy + By^2 = 0$ is the joint equation of OP and OQ .
5. Answer the actual question from A, H, B : perpendicular at O means $A + B = 0$; the angle POQ is $\tan^{-1} \left| \frac{2\sqrt{H^2 - AB}}{A+B} \right|$; the bisectors of $\angle POQ$ come from Section 2C.

Figure brief: A curve (draw an ellipse) with a straight line cutting it at two points P and Q , both in the upper half-plane, and the origin O marked well outside the chord. Two rays OP and OQ drawn in a heavier stroke and extended backwards through O as dashed rays, showing that the homogenised equation is a full pair of lines, not just the two segments. The angle at O marked θ . Caption: homogenising replaces “curve and chord” by “the pair of lines you actually care about”.

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

Show that $6x^2 + 5xy - 6y^2 = 0$ represents a pair of perpendicular straight lines through the origin. Find their separate equations, and the joint equation of their angle bisectors.

Solution. Compare with $ax^2 + 2hxy + by^2 = 0$: $a = 6$, $2h = 5$ so $h = \frac{5}{2}$, $b = -6$.

Nature. $h^2 - ab = \frac{25}{4} + 36 = \frac{169}{4} > 0$, so the lines are real and distinct.

Perpendicularity. $a + b = 6 + (-6) = 0$, so the lines are perpendicular. That is the whole of the first part — no factorisation was needed.

Separate equations. Split the middle term: $6x^2 + 9xy - 4xy - 6y^2 = 3x(2x + 3y) - 2y(2x + 3y) = (3x - 2y)(2x + 3y)$. So the lines are

$$3x - 2y = 0 \quad \text{and} \quad 2x + 3y = 0.$$

Their slopes are $\frac{3}{2}$ and $-\frac{2}{3}$, whose product is -1 — the perpendicularity confirmed independently.

Bisectors. Use $\frac{x^2 - y^2}{a - b} = \frac{xy}{h}$ with $a - b = 12$ and $h = \frac{5}{2}$:

$$\frac{x^2 - y^2}{12} = \frac{2xy}{5} \implies 5x^2 - 5y^2 = 24xy \implies 5x^2 - 24xy - 5y^2 = 0.$$

Check. Coefficient of x^2 plus coefficient of y^2 is $5 - 5 = 0$, so the bisectors are perpendicular, as they must be. Factorising, $5x^2 - 24xy - 5y^2 = (5x + y)(x - 5y)$, so the bisectors are $5x + y = 0$ (slope -5) and $x - 5y = 0$ (slope $\frac{1}{5}$), and $(-5)(\frac{1}{5}) = -1$. A second check: the original lines make angles $\tan^{-1} \frac{3}{2} \approx 56.31^\circ$ and $\tan^{-1}(-\frac{2}{3}) \approx -33.69^\circ$ with the x -axis, whose average is 11.31° , and $\tan 11.31^\circ = 0.2 = \frac{1}{5}$. Correct.

$3x - 2y = 0, \quad 2x + 3y = 0 \text{ (perpendicular); bisectors } 5x^2 - 24xy - 5y^2 = 0$

Example 2 — [JEE Main • Speed Trap]

Find the value of λ for which $2x^2 + 5xy + 3y^2 + 6x + 7y + \lambda = 0$ represents a pair of straight lines, and find their point of intersection.

Speed approach. Halve first, then substitute. Write the six coefficients in a row before touching the determinant:

$$a = 2, \quad b = 3, \quad c = \lambda, \quad h = \frac{5}{2}, \quad g = \frac{6}{2} = 3, \quad f = \frac{7}{2}.$$

Now $\Delta = abc + 2fgh - af^2 - bg^2 - ch^2 = 0$:

$$\begin{aligned} 6\lambda + 2\left(\frac{7}{2}\right)(3)\left(\frac{5}{2}\right) - 2\left(\frac{49}{4}\right) - 3(9) - \lambda\left(\frac{25}{4}\right) &= 0, \\ 6\lambda + \frac{105}{2} - \frac{49}{2} - 27 - \frac{25\lambda}{4} &= 0 \implies 1 - \frac{\lambda}{4} = 0 \implies \lambda = 4. \end{aligned}$$

Check reality: $h^2 - ab = \frac{25}{4} - 6 = \frac{1}{4} > 0$, so the lines are real and distinct. Point of intersection:

$$\left(\frac{hf - bg}{ab - h^2}, \frac{gh - af}{ab - h^2}\right) = \left(\frac{\frac{35}{4} - 9}{6 - \frac{25}{4}}, \frac{\frac{15}{2} - 7}{6 - \frac{25}{4}}\right) = \left(\frac{-\frac{1}{4}}{-\frac{1}{4}}, \frac{\frac{1}{2}}{-\frac{1}{4}}\right) = (1, -2).$$

Confirmation by factorising. $2x^2 + 5xy + 3y^2 = (2x + 3y)(x + y)$, so try $(2x + 3y + p)(x + y + q)$. Matching x : $2q + p = 6$; matching y : $3q + p = 7$; hence $q = 1$, $p = 4$, and the constant is

$pq = 4 = \lambda$. The lines are $2x + 3y + 4 = 0$ and $x + y + 1 = 0$, which meet at $(1, -2)$. Everything agrees.

The trap. The planted wrong answer is $\lambda = 16$, and it is produced by feeding the determinant the *raw* coefficients $g = 6$ and $f = 7$ instead of their halves:

$$2 \cdot 3 \cdot \lambda + 2(7)(6) \left(\frac{5}{2}\right) - 2(49) - 3(36) - \lambda \left(\frac{25}{4}\right) = 4 - \frac{\lambda}{4} = 0 \implies \lambda = 16.$$

It is seductive because the arithmetic is *cleaner* than the correct route and it lands on a tidy integer — students take the tidiness as confirmation. Note that the error was only in g and f ; the student did remember to halve h , which is why the answer is wrong by a plausible-looking factor rather than absurdly. **Two seconds of defence:** with $\lambda = 16$ the factorisation forced above still requires $p = 4$, $q = 1$, so the constant term can only be $pq = 4$. Any candidate $\lambda \neq 4$ contradicts the very factorisation it claims to permit.

$\lambda = 4; \quad \text{lines } 2x + 3y + 4 = 0 \text{ and } x + y + 1 = 0, \text{ meeting at } (1, -2)$

Example 3 — [JEE Main Level • multi-concept]

The line $y = 2x + c$ cuts the circle $x^2 + y^2 = 5$ at A and B . Find the values of c for which $\angle AOB = 90^\circ$, where O is the origin, and write the joint equation of OA and OB in that case.

Solution — homogenisation. The condition is about the *angle at the origin* subtended by a chord. Do not find A and B . Instead build the pair OA, OB directly and impose $A + B = 0$ on its coefficients.

Step 1 — write the line as “= 1”. From $y = 2x + c$ we get $y - 2x = c$, so $\frac{y - 2x}{c} = 1$ (and $c \neq 0$, since $c = 0$ would put the line through the origin and make the question meaningless).

Step 2 — homogenise the curve. In $x^2 + y^2 - 5 = 0$, only the constant -5 needs lifting to degree two, so multiply it by $\left(\frac{y - 2x}{c}\right)^2$:

$$x^2 + y^2 - 5 \left(\frac{y - 2x}{c}\right)^2 = 0.$$

Multiply through by c^2 and expand $(y - 2x)^2 = 4x^2 - 4xy + y^2$:

$$c^2x^2 + c^2y^2 - 20x^2 + 20xy - 5y^2 = 0, \quad \text{i.e.} \quad (c^2 - 20)x^2 + 20xy + (c^2 - 5)y^2 = 0.$$

Step 3 — impose the right angle. Perpendicular at the origin means (coefficient of x^2) + (coefficient of y^2) = 0:

$$(c^2 - 20) + (c^2 - 5) = 0 \implies 2c^2 = 25 \implies c = \pm \frac{5}{\sqrt{2}}.$$

Step 4 — the pair itself. With $c^2 = \frac{25}{2}$ the coefficients become $-\frac{15}{2}$, 20, $\frac{15}{2}$; dividing by $-\frac{5}{2}$,

$$3x^2 - 8xy - 3y^2 = 0, \quad \text{i.e.} \quad (3x + y)(x - 3y) = 0.$$

Step 5 — check that the line really cuts the circle. The distance from O to $y = 2x + c$ is $\frac{|c|}{\sqrt{5}} = \frac{5}{\sqrt{10}} \approx 1.58$, which is less than the radius $\sqrt{5} \approx 2.24$. Two genuine intersection points exist.

Independent verification. Take $c = \frac{5}{\sqrt{2}}$. Solving with the circle gives $A = \left(-\frac{3}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right)$ and $B = \left(-\frac{1}{\sqrt{2}}, \frac{3}{\sqrt{2}}\right)$. Then $\overrightarrow{OA} \cdot \overrightarrow{OB} = \frac{3}{2} - \frac{3}{2} = 0$, so $\angle AOB = 90^\circ$; and A has $y/x = \frac{1}{3}$ while B has $y/x = -3$, which are exactly the two lines $x - 3y = 0$ and $3x + y = 0$. Correct.

The general result behind this, worth memorising: the lines joining the origin to the intersections of $x^2 + y^2 = a^2$ and $y = mx + c$ are perpendicular precisely when $2c^2 = a^2(1 + m^2)$. Here $2 \cdot \frac{25}{2} = 5(1 + 4)$. Correct.

$c = \pm \frac{5}{\sqrt{2}}; \quad OA, OB : 3x^2 - 8xy - 3y^2 = 0$

Example 4 — [JEE Advanced Level]

Show that $3x^2 + 5xy - 2y^2 - 16x + 3y + 5 = 0$ represents a pair of straight lines. Find their point of intersection, the angle between them, and the joint equation of the bisectors of the angles between them. Verify the bisector equation by a second, independent route.

Solution. Coefficients: $a = 3, b = -2, c = 5, h = \frac{5}{2}, g = -8, f = \frac{3}{2}$.

(i) It is a pair.

$$\begin{aligned} \Delta &= abc + 2fgh - af^2 - bg^2 - ch^2 = -30 + 2\left(\frac{3}{2}\right)(-8)\left(\frac{5}{2}\right) - 3\left(\frac{9}{4}\right) + 2(64) - 5\left(\frac{25}{4}\right) \\ &= -30 - 60 - \frac{27}{4} + 128 - \frac{125}{4} = 38 - 38 = 0. \end{aligned}$$

And $h^2 - ab = \frac{25}{4} + 6 = \frac{49}{4} > 0$, so the two lines are real and distinct.

(ii) Point of intersection. With $ab - h^2 = -6 - \frac{25}{4} = -\frac{49}{4}$,

$$\alpha = \frac{hf - bg}{ab - h^2} = \frac{\frac{15}{4} - 16}{-\frac{49}{4}} = 1, \quad \beta = \frac{gh - af}{ab - h^2} = \frac{-20 - \frac{9}{2}}{-\frac{49}{4}} = 2.$$

So the lines meet at $(1, 2)$.

(iii) The lines themselves, by Procedure B. Since $3x^2 + 5xy - 2y^2 = (3x - y)(x + 2y)$, try $(3x - y + p)(x + 2y + q)$. Matching the x coefficient: $3q + p = -16$; matching the y coefficient: $-q + 2p = 3$; hence $p = -1, q = -5$, and $pq = 5 = c$. The lines are

$$3x - y - 1 = 0 \quad \text{and} \quad x + 2y - 5 = 0,$$

which indeed both pass through $(1, 2)$.

(iv) Angle.

$$\tan \theta = \left| \frac{2\sqrt{h^2 - ab}}{a + b} \right| = \left| \frac{2 \cdot \frac{7}{2}}{3 - 2} \right| = 7, \quad \theta = \tan^{-1} 7 \approx 81.87^\circ.$$

Cross-check from the slopes 3 and $-\frac{1}{2}$: $\left| \frac{3+1/2}{1-3/2} \right| = \left| \frac{7/2}{-1/2} \right| = 7$. Correct.

(v) Bisectors. The bisector formula of Section 2C is stated about the origin, and this pair does not pass through the origin — so shift. Put $X = x - 1$, $Y = y - 2$. In the shifted frame the pair is homogeneous with the same a, h, b , and its bisectors are

$$\frac{X^2 - Y^2}{a - b} = \frac{XY}{h} \quad \Rightarrow \quad \frac{X^2 - Y^2}{5} = \frac{2XY}{5} \quad \Rightarrow \quad X^2 - 2XY - Y^2 = 0.$$

Substituting back $X = x - 1$, $Y = y - 2$:

$$(x - 1)^2 - 2(x - 1)(y - 2) - (y - 2)^2 = 0 \quad \Rightarrow \quad x^2 - 2xy - y^2 + 2x + 6y - 7 = 0.$$

(vi) Independent verification, using Subtopic 05. Having separated the lines in (iii), apply the ordinary two-line bisector formula. With $\sqrt{3^2 + 1^2} = \sqrt{10}$ and $\sqrt{1^2 + 2^2} = \sqrt{5}$,

$$\frac{3x - y - 1}{\sqrt{10}} = \pm \frac{x + 2y - 5}{\sqrt{5}}.$$

Multiplying the two sign cases together — which is exactly how a joint equation is built — and clearing the surds gives $7(x^2 - 2xy - y^2 + 2x + 6y - 7) = 0$, the same curve as in (v) up to the harmless overall factor 7. Two completely different routes, one answer.

(vii) The bisectors separately, and the free check. Solving $X^2 - 2XY - Y^2 = 0$ as a quadratic in X gives $X = (1 \pm \sqrt{2})Y$, so the bisectors are

$$(x - 1) = (1 + \sqrt{2})(y - 2) \quad \text{and} \quad (x - 1) = (1 - \sqrt{2})(y - 2),$$

with slopes $\frac{1}{1 + \sqrt{2}} = \sqrt{2} - 1 \approx 0.4142$ and $\frac{1}{1 - \sqrt{2}} = -(\sqrt{2} + 1)$. Their product is -1 : perpendicular, as Derivation 3 promised. And the direct geometric check — the original lines make angles $\tan^{-1} 3 \approx 71.57^\circ$ and $\tan^{-1}(-\frac{1}{2}) \approx -26.57^\circ$ with the x -axis, whose average is 22.5° , and $\tan 22.5^\circ = \sqrt{2} - 1$. Correct.

Why this is Advanced-flavoured. Nothing in it is hard on its own. What it tests is whether you know that the origin-based formulas of Sections 2A-2C are *frame-dependent* and the coefficient-based ones (Δ , the angle) are not. A student who applies $\frac{x^2 - y^2}{a - b} = \frac{xy}{h}$ to this equation without shifting gets $5x^2 - 24xy - 5y^2 = 0$ — a perfectly respectable pair of perpendicular lines through the origin, *parallel* to the true bisectors but not equal to them. It looks right and it is wrong, which is exactly the level at which Advanced likes to operate.

Pair at $(1, 2)$; $3x - y - 1 = 0$, $x + 2y - 5 = 0$; $\theta = \tan^{-1} 7$; bisectors $x^2 - 2xy - y^2 + 2x + 6y - 7 = 0$

Section 5: Practice Exercises

- [CBSE] Find the separate equations of the lines represented by $x^2 - 5xy + 6y^2 = 0$ and the acute angle between them.
- [CBSE] For what values of k does $x^2 + kxy + 4y^2 = 0$ represent a pair of coincident lines? Write the repeated line in each case.
- [JEE Main] Find the joint equation of the bisectors of the angles between the lines $2x^2 - 5xy + 3y^2 = 0$, and verify that the two bisectors are perpendicular.
- [JEE Main] Find all values of λ for which $x^2 + \lambda xy + y^2 - 5x - 7y + 6 = 0$ represents a pair of straight lines, and write the lines in each case.
- [JEE Main] Find the joint equation of the lines joining the origin to the points where $2x + y = 5$ meets the circle $x^2 + y^2 - 4x + 3 = 0$, and hence find the angle those two lines make with each other.
- [JEE Main] Show that $9x^2 + 24xy + 16y^2 - 12x - 16y - 12 = 0$ represents a pair of parallel straight lines, and find the distance between them.
- [JEE Advanced] Find the area of the triangle formed by the pair $x^2 - 4xy + y^2 = 0$ and the line $x + y = 1$.
- [JEE Advanced] The pairs $x^2 - 2pxy - y^2 = 0$ and $x^2 - 2qxy - y^2 = 0$ are such that each pair bisects the angles between the other. Prove that $pq = -1$.

Answers.

- $(x - 2y)(x - 3y) = 0$, so the lines are $x - 2y = 0$ and $x - 3y = 0$, with slopes $\frac{1}{2}$ and $\frac{1}{3}$. Here $a = 1$, $h = -\frac{5}{2}$, $b = 6$, so $h^2 - ab = \frac{25}{4} - 6 = \frac{1}{4}$ and $\tan \theta = \frac{2 \cdot \frac{1}{2}}{7} = \frac{1}{7}$, i.e. $\theta = \tan^{-1} \frac{1}{7}$. (Direct check: $\left| \frac{1/2 - 1/3}{1 + 1/6} \right| = \frac{1/6}{7/6} = \frac{1}{7}$.)
- Coincident means $h^2 = ab$, i.e. $\frac{k^2}{4} = 4$, so $k = \pm 4$. With $k = 4$ the equation is $(x + 2y)^2 = 0$, the doubled line $x + 2y = 0$; with $k = -4$ it is $(x - 2y)^2 = 0$, the doubled line $x - 2y = 0$.
- $a = 2$, $b = 3$, $h = -\frac{5}{2}$, so $hx^2 - (a - b)xy - hy^2 = 0$ gives $-\frac{5}{2}x^2 + xy + \frac{5}{2}y^2 = 0$, i.e.

$$5x^2 - 2xy - 5y^2 = 0.$$

Coefficient of x^2 plus coefficient of y^2 is $5 - 5 = 0$: perpendicular. (Numerical check: the original lines are $(2x - 3y)(x - y) = 0$, with slopes $\frac{2}{3}$ and 1 , making angles 33.69° and 45° ; their average is 39.35° and $\tan 39.35^\circ = 0.8198$, which satisfies $5m^2 + 2m - 5 = 0$ — the slope equation of the bisector pair.)

- $a = 1$, $b = 1$, $c = 6$, $h = \frac{\lambda}{2}$, $g = -\frac{5}{2}$, $f = -\frac{7}{2}$. Then $\Delta = 0$ reduces to $6\lambda^2 - 35\lambda + 50 = 0$, i.e. $(2\lambda - 5)(3\lambda - 10) = 0$, so

$$\lambda = \frac{5}{2} \quad \text{or} \quad \lambda = \frac{10}{3}.$$

Both satisfy $h^2 \geq ab$, so both give real lines. For $\lambda = \frac{5}{2}$ the equation is $\frac{1}{2}(2x + y - 6)(x + 2y - 2) = 0$: the lines are $2x + y - 6 = 0$ and $x + 2y - 2 = 0$. For $\lambda = \frac{10}{3}$ it is $\frac{1}{3}(3x + y - 6)(x + 3y - 3) = 0$: the lines are $3x + y - 6 = 0$ and $x + 3y - 3 = 0$.

- Write the line as $\frac{2x + y}{5} = 1$ and homogenise $x^2 + y^2 - 4x + 3 = 0$, lifting the $-4x$ by one power and the $+3$ by two:

$$x^2 + y^2 - 4x \left(\frac{2x + y}{5} \right) + 3 \left(\frac{2x + y}{5} \right)^2 = 0.$$

Multiplying by 25 gives $-3x^2 - 8xy + 28y^2 = 0$, i.e.

$$3x^2 + 8xy - 28y^2 = 0.$$

Here $a = 3$, $h = 4$, $b = -28$, so $h^2 - ab = 16 + 84 = 100$ and $\tan \theta = \frac{2 \cdot 10}{|3 - 28|} = \frac{20}{25} = \frac{4}{5}$, giving $\theta = \tan^{-1} \frac{4}{5} \approx 38.66^\circ$. (Check: the line meets the circle at $(2, 1)$ and $(\frac{14}{5}, -\frac{3}{5})$, both of which satisfy the pair, and the angle between $\overrightarrow{OP}$ and $\overrightarrow{OQ}$ computes to 38.66° .)

6. $a = 9$, $b = 16$, $c = -12$, $h = 12$, $g = -6$, $f = -8$. Then $h^2 = 144 = ab$, so the lines are parallel, and $\Delta = -1728 + 1152 - 576 - 576 + 1728 = 0$, so it is genuinely a pair. Since $9x^2 + 24xy + 16y^2 = (3x + 4y)^2$, write $(3x + 4y + p)(3x + 4y + q)$ with $p + q = -4$ and $pq = -12$, giving $p = 2$, $q = -6$:

$$3x + 4y + 2 = 0 \quad \text{and} \quad 3x + 4y - 6 = 0.$$

The distance between them is $\frac{|2 - (-6)|}{\sqrt{9 + 16}} = \frac{8}{5}$, which the formula reproduces: $2\sqrt{\frac{g^2 - ac}{a(a+b)}} =$

$$2\sqrt{\frac{36 + 108}{9 \cdot 25}} = 2 \cdot \frac{12}{15} = \frac{8}{5}.$$

7. $a = 1$, $h = -2$, $b = 1$, and the line has $l = 1$, $m = 1$, $n = -1$. Then $h^2 - ab = 3$ and $am^2 - 2hlm + bl^2 = 1 + 4 + 1 = 6$, so the area is $\frac{n^2\sqrt{h^2 - ab}}{|am^2 - 2hlm + bl^2|} = \frac{\sqrt{3}}{6}$. (Direct check: the pair is $y = (2 \pm \sqrt{3})x$; the line $x + y = 1$ cuts these at $x = \frac{1}{3 \pm \sqrt{3}}$, and the triangle $O-A-B$ has area $\frac{1}{2}|x_1y_2 - x_2y_1| = \frac{1}{2} \cdot \frac{2\sqrt{3}}{6} = \frac{\sqrt{3}}{6}$.)
8. For $x^2 - 2pxy - y^2 = 0$ we have $a = 1$, $h = -p$, $b = -1$, so its bisector pair is $hx^2 - (a - b)xy - hy^2 = 0$, that is $-px^2 - 2xy + py^2 = 0$, or

$$x^2 + \frac{2}{p}xy - y^2 = 0 \quad (p \neq 0).$$

This is to be the second pair, $x^2 - 2qxy - y^2 = 0$. Comparing the xy coefficients, $\frac{2}{p} = -2q$, hence $pq = -1$. (Note that both pairs are perpendicular pairs, since $a + b = 0$ in each — which is why the relation comes out so symmetrically, and why starting from the second pair gives the same condition.)

Section 6: Crack the MCQ

Q1. The equation $ax^2 + 2hxy + by^2 = 0$ represents two real and distinct lines if and only if:
(A) $h^2 < ab$ (B) $h^2 = ab$ (C) $h^2 > ab$ (D) $a + b = 0$

Answer: (C). The slopes are the roots of $bm^2 + 2hm + a = 0$, whose discriminant is $4(h^2 - ab)$.

Distractor Analysis. - (A) is the imaginary case — the graph is the origin alone. Students who remember “discriminant positive” but misremember which way $h^2 - ab$ points choose this. - (B) is the coincident case: one line counted twice, not two distinct lines. - (D) is the perpendicularity condition, which says what the *angle* is, not whether the lines exist. A pair can be perpendicular and imaginary only if $a + b = 0$ and $h^2 < ab = -a^2$, which is impossible — but the student choosing (D) has not checked that, they have simply matched the most familiar-looking condition on the page.

Q2. The lines represented by $3x^2 + 8xy - 3y^2 = 0$ are: (A) coincident (B) perpendicular (C) inclined at 45° (D) imaginary

Answer: (B). $a + b = 3 + (-3) = 0$, so $\tan \theta$ is infinite and $\theta = 90^\circ$. (The lines are $3x - y = 0$ and $x + 3y = 0$, slopes 3 and $-\frac{1}{3}$.)

Distractor Analysis. - (A) would need $h^2 = ab$; here $h^2 - ab = 16 + 9 = 25 \neq 0$. - (C) is chosen by students who compute $\tan \theta = \frac{2\sqrt{25}}{|3 - 3|}$ and, meeting a zero denominator, decide the formula has “failed” and guess. A zero denominator here is information, not failure: it is the right angle. - (D) would need $h^2 < ab$, i.e. $16 < -9$.

Q3. The bisectors of the angles between the pair $ax^2 + 2hxy + by^2 = 0$ are: (A) parallel to each other (B) always perpendicular to each other (C) coincident with the original pair (D) real only when $h^2 > ab$

Answer: (B). The bisector pair is $hx^2 - (a - b)xy - hy^2 = 0$, whose x^2 and y^2 coefficients sum to zero.

Distractor Analysis. - (A) confuses “the pair of bisectors” with the two branches of a single bisector. - (C) is true only in the degenerate case $h^2 = ab$, where the “two” lines coincide and their bisector is the line itself; it is not a general property. - (D) is the most instructive wrong answer. The bisector pair has its own discriminant $\frac{(a - b)^2}{4} + h^2$, which is a sum of squares and therefore never negative: the bisectors are **always** real, even though the original pair may not be. Students who assume every property inherits the parent condition pick this one.

Q4. The equation $x^2 + 4xy + 4y^2 + 4x + 8y + 3 = 0$ represents: (A) a pair of intersecting lines (B) a pair of parallel lines (C) a circle (D) no real locus

Answer: (B). Here $a = 1$, $b = 4$, $h = 2$, $g = 2$, $f = 4$, $c = 3$, so $\Delta = 12 + 32 - 16 - 16 - 12 = 0$ and $h^2 = 4 = ab$. Indeed the equation is $(x + 2y)^2 + 4(x + 2y) + 3 = (x + 2y + 1)(x + 2y + 3) = 0$, two parallel lines at distance $\frac{2}{\sqrt{5}}$ apart.

Distractor Analysis. - (A) is chosen by students who verify $\Delta = 0$, conclude “pair of lines”, and stop without testing h^2 against ab . The intersection formula would then be applied with $ab - h^2 = 0$ in the denominator — the tell-tale symptom. - (C) is impossible for any equation with an xy term or with unequal x^2 and y^2 coefficients; both faults are present here. - (D) confuses $\Delta = 0$ with “imaginary”. The imaginary case needs $h^2 < ab$; here they are equal.

Section 7: Common Student Pitfalls and Pro-Tips

- 1. Forgetting that h, g, f are halves.** The standard form is $ax^2 + 2hxy + by^2 + 2gx + 2fy + c$, so h, g, f are *half* the coefficients of xy, x, y . This one slip corrupts Δ , the angle, the point of intersection and the bisectors simultaneously. **Correction:** before any calculation, write the six numbers a, b, c, h, g, f in a labelled row, halving as you go. Example 2 exists entirely to make this reflex permanent.
- 2. Treating $\Delta = 0$ as proof of two real lines.** It only proves the conic is degenerate. **Correction:** always pair it with $h^2 \geq ab$. Counter-example to keep in mind: $x^2 + y^2 = 0$ has $\Delta = 0$ but is the single point $(0, 0)$.

3. **Using the origin bisector formula on a pair that does not pass through the origin.** $\frac{x^2-y^2}{a-b} = \frac{xy}{h}$ is stated about the vertex of the pair. **Correction:** find the point of intersection (α, β) first and use $\frac{(x-\alpha)^2-(y-\beta)^2}{a-b} = \frac{(x-\alpha)(y-\beta)}{h}$. The wrong answer is parallel to the right one, so it will pass every slope-based sanity check — see Example 4(vii).
4. **Dividing by zero in the bisector formula.** If $a = b$ or $h = 0$ the formula as written has a zero denominator. **Correction:** cross-multiply *first* to $hx^2 - (a-b)xy - hy^2 = 0$, then substitute. You get $x^2 - y^2 = 0$ and $xy = 0$ respectively, both perfectly good answers.
5. **Homogenising with a line that is not written as “= 1”.** Multiplying by $lx + my$ instead of $\frac{lx+my}{-n}$ changes the curve and produces a wrong pair. **Correction:** the quantity you multiply by must equal 1 *at the intersection points*. Write the line in the form (something) = 1 explicitly, on its own line, before substituting.
6. **Homogenising with a line through the origin.** Then $n = 0$ and the recipe divides by zero — correctly, because the question itself is degenerate: the origin is one of the two intersection points. **Correction:** check $n \neq 0$ as step 1 of Procedure C.
7. **Using $m_1 + m_2 = -2h/b$ when $b = 0$.** **Correction:** if $b = 0$ the equation is $x(ax + 2hy) = 0$ and one line is the y -axis, which has no slope. Factorise directly.
8. **Adding constants to a homogeneous pair.** A pair through the origin has *no* linear or constant terms. If your answer to “find the pair through the origin such that...” contains an x , a y or a bare number, it is not a pair through the origin.

Pro-Tip 1 — the coefficient sum is the fastest question in coordinate geometry. For any second-degree equation, “are these lines perpendicular?” is answered by adding two numbers: coefficient of x^2 plus coefficient of y^2 . It works for the homogeneous pair, for the general pair, and for a homogenised pair — which is why nearly every JEE homogenisation question ends in a right angle. Look for it before you look for anything else.

Pro-Tip 2 — never find the intersection points in a homogenisation question. If the question asks about the angle at the origin, the perpendicularity at the origin, or the bisectors at the origin, the intersection points are irrelevant and finding them costs you three minutes and a surd. Homogenise, read A, H, B , answer.

Pro-Tip 3 — the partial-derivative shortcut for the point of intersection. Rather than remembering $\left(\frac{hf-bg}{ab-h^2}, \frac{gh-af}{ab-h^2}\right)$ with its sign traps, differentiate S with respect to x (treating y as constant) and with respect to y , set both to zero, and solve the two linear equations. For Example 2 that is $4x + 5y + 6 = 0$ and $5x + 6y + 7 = 0$, giving $(1, -2)$ in one line of elimination.

Pro-Tip 4 — factorise as your final check, never as your first move. Almost every question here (Δ , angle, perpendicularity, bisectors, homogenisation) is answered from the coefficients alone. But once you have an answer, splitting the equation into two lines and testing them takes twenty seconds and catches every arithmetic slip.

Section 8: Cheat Sheet Box

Setting up. For $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$: a, b, c are read straight off; h, g, f are **half** the coefficients of xy, x, y .

Homogeneous pair $ax^2 + 2hxy + by^2 = 0$ — two lines through the origin.

- Slopes are the roots of $bm^2 + 2hm + a = 0$: $m_1 + m_2 = -\frac{2h}{b}$, $m_1m_2 = \frac{a}{b}$.
- $h^2 > ab$: real, distinct. $h^2 = ab$: coincident (perfect square). $h^2 < ab$: origin only.
- $\tan \theta = \left| \frac{2\sqrt{h^2 - ab}}{a + b} \right|$; perpendicular $\iff a + b = 0$; coincident $\iff h^2 = ab$.
- Perpendicular pair through O : $bx^2 - 2hxy + ay^2 = 0$.
- Bisectors: $\frac{x^2 - y^2}{a - b} = \frac{xy}{h}$, i.e. $hx^2 - (a - b)xy - hy^2 = 0$ — **always perpendicular, always real**. If $a = b$: $x^2 - y^2 = 0$. If $h = 0$: $xy = 0$.
- Product of perpendiculars from (x_1, y_1) : $\frac{|ax_1^2 + 2hx_1y_1 + by_1^2|}{\sqrt{(a - b)^2 + 4h^2}}$.
- Triangle with $lx + my + n = 0$: $\text{area} = \frac{n^2\sqrt{h^2 - ab}}{|am^2 - 2hlm + bl^2|}$.

General equation is a pair $\iff$

$$\Delta = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = abc + 2fgh - af^2 - bg^2 - ch^2 = 0 \quad \text{and} \quad h^2 \geq ab.$$

- Point of intersection: $\left(\frac{hf - bg}{ab - h^2}, \frac{gh - af}{ab - h^2} \right)$; or solve $\frac{\partial S}{\partial x} = 0$, $\frac{\partial S}{\partial y} = 0$.
- Angle: unchanged by the linear terms — same formula as above.
- Parallel pair ($h^2 = ab$): distance $= 2\sqrt{\frac{g^2 - ac}{a(a + b)}}$.
- Bisectors: shift to (α, β) first — $\frac{(x - \alpha)^2 - (y - \beta)^2}{a - b} = \frac{(x - \alpha)(y - \beta)}{h}$.

Homogenisation. To join the origin to where $lx + my + n = 0$ (with $n \neq 0$) cuts $S = 0$: write $\frac{lx + my}{-n} = 1$, multiply the linear terms of S by it and the constant by its square, then clear denominators. The result is $Ax^2 + 2Hxy + By^2 = 0$; the two rays are perpendicular iff $A + B = 0$.

Memory aids:

- “Halve the odd ones out” — h, g, f carry a hidden 2; a, b, c do not.
- “ h^2 versus ab decides *whether*; $a + b$ decides *what angle*.”
- “ $\Delta = 0$ means degenerate, not real — check $h^2 \geq ab$ too.”
- “Bisectors: swap the $x^2 - y^2$ block over $a - b$, the xy over h — and they always cross at 90° .”
- “Make the line equal one, then make the curve homogeneous.”

Conic Sections

Class 11 Mathematics · Complete Chapter

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Master content · Sections of a Cone, Circle, Parabola, Ellipse, Hyperbola

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Introduction: Sections of a Cone & the Conic Family

Exam Relevance: CBSE Boards ask 1–2 marks — identify which conic a given section or equation represents, or recall a degenerate case. JEE Main routinely hides a 1-question classification via the general second-degree equation. JEE Advanced expects fluency with the unifying eccentricity definition and the discriminant test. Master this and the entire chapter stops being four disconnected curves and becomes one idea seen from four angles.

Section 1: Concept Intuition

Here is the single most useful realisation in this whole chapter: the circle, the parabola, the ellipse, and the hyperbola are not four unrelated curves you must memorise separately. They are **one family** — four different slices of the same object. That object is a **double cone**, and the four curves are simply what you see when you cut it at different angles. Learn the cut, and the curves explain themselves.

Picture a double cone like two ice-cream cones joined tip to tip, stretching infinitely up and down. The sharp meeting point is the **vertex**; the central line through both tips is the **axis**; and any straight line lying on the surface, running from the vertex out along the slope, is a **generator**. The angle each generator makes with the axis is the **semi-vertical angle**, which we call α — it is fixed for a given cone, like the steepness of the slope.

Think about it this way... forget the cone for a second and pick up a torch in a dark room. Point it straight at the wall, beam perpendicular: you get a clean **circle** of light. Now tilt the torch a little — the circle stretches into an **ellipse**. Keep tilting. At one exact critical angle, the bottom edge of the light-cone runs *parallel* to the wall and the ellipse breaks open at one end, spilling off to infinity — that is a **parabola**. Tilt past that angle and the beam now strikes the wall so steeply that it opens into the two sweeping arms of a **hyperbola**. You have just produced all four conics with a torch, and the *only* thing you changed was the tilt angle. That tilt is exactly what the cutting plane does to the cone.

Formally, let β be the angle the cutting plane makes with the axis. Comparing β to the cone's own angle α decides everything:

- Cut **straight across** ($\beta = 90^\circ$, plane perpendicular to the axis) → **circle**.
- Cut at a **gentle tilt** ($\alpha < \beta < 90^\circ$) → **ellipse**.
- Cut **parallel to a generator** ($\beta = \alpha$) → **parabola** — the knife-edge case.
- Cut **steeply** ($0 \leq \beta < \alpha$, plane hits *both* nappes) → **hyperbola**, which is why a hyperbola has two branches: the plane slices both cones.

Degenerate conics. There is a special situation: what if the cutting plane passes *through the vertex itself*? Then the slice shrinks to something trivial — a single **point** (the vertex), a single **straight line** (plane tangent along one generator), or a **pair of intersecting straight lines** (plane through the axis). These are the **degenerate conics** — the family's “edge cases,” what you get when the curve collapses.

And there is a second, equation-free way to see the family — through **eccentricity** e , the focus-directrix ratio you will meet in every curve that follows. A conic is the set of points whose distance to a fixed **focus** divided by distance to a fixed **directrix** equals a constant e . Dial e and you sweep through the whole family: $e = 0$ gives a **circle**, $0 < e < 1$ an

ellipse, $e = 1$ a **parabola**, $e > 1$ a **hyperbola**. One number, four curves.

Limiting conditions & assumptions.

- The classification by β vs α assumes the plane does **not** pass through the vertex; if it does, you get a degenerate conic instead.
- A circle is the limiting ellipse ($e \rightarrow 0$, foci merging); a parabola is the single boundary ($e = 1$) between the bounded ellipse and the unbounded hyperbola.
- The eccentricity definition needs the focus *off* the directrix; a focus on the directrix degenerates the locus.

Section 2: Key Definitions & Formulas

Cone terminology. A right circular cone has an **axis**, a **vertex**, two **nappes** (the upper and lower halves), **generators** (surface lines through the vertex), and a fixed **semi-vertical angle** α between axis and generator.

Classification by cutting-plane angle β (plane not through vertex):

Condition	Conic
$\beta = 90^\circ$	Circle
$\alpha < \beta < 90^\circ$	Ellipse
$\beta = \alpha$	Parabola
$0 \leq \beta < \alpha$	Hyperbola (cuts both nappes)

Classification by eccentricity (focus-directrix definition $\frac{\text{dist. to focus}}{\text{dist. to directrix}} = e$):

$e = 0 \rightarrow \text{circle}, \quad 0 < e < 1 \rightarrow \text{ellipse}, \quad e = 1 \rightarrow \text{parabola}, \quad e > 1 \rightarrow \text{hyperbola}.$

Degenerate conics (plane through vertex): a point, a straight line, or a pair of intersecting straight lines.

General second-degree equation. Every conic in the plane is an instance of

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0,$$

where A, B, C are not all zero. The **discriminant** $\Delta = B^2 - 4AC$ classifies the (non-degenerate) curve:

Discriminant	Conic
$B^2 - 4AC < 0$	Ellipse (circle if also $A = C, B = 0$)
$B^2 - 4AC = 0$	Parabola
$B^2 - 4AC > 0$	Hyperbola

When $B = 0$ (no xy term) the conic's axes are parallel to the coordinate axes — the standard case for Class 11. A nonzero B means the conic is *rotated*.

Section 3: Key Procedures & Logical Framework

This subtopic classifies rather than derives, so the skill is reading a description or an equation and naming the curve with reasoning, not rote.

Procedure A — Classify a section from the geometry.

1. Identify the cone's semi-vertical angle α and the cutting-plane angle β (both measured to the axis).
2. Check whether the plane passes through the vertex. If yes $\rightarrow$ degenerate (point/line/-pair of lines). If no $\rightarrow$ use the β -vs- α table.
3. *Reasoning*: the parabola is the hinge. When the plane tilts to exactly match a generator's direction ($\beta = \alpha$), the slice can no longer "close up," so the bounded ellipse opens into the unbounded parabola; tilt further and the plane reaches the second nappe, producing the hyperbola's second branch.

Procedure B — Classify from the general equation (no xy term).

1. Read off A (coefficient of x^2) and C (coefficient of y^2); here $B = 0$.
2. If $A = C$ (same sign, equal) $\rightarrow$ **circle**. If A, C same sign but unequal $\rightarrow$ **ellipse**. If one of A, C is zero $\rightarrow$ **parabola**. If A, C opposite signs $\rightarrow$ **hyperbola**.
3. *Reasoning*: with $B = 0$ the discriminant is $-4AC$, so the sign of the product AC alone decides ellipse/parabola/hyperbola — same-sign coefficients curve the graph "the same way" (closed), opposite signs curve it oppositely (open arms).

Procedure C — Classify from the general equation (with an xy term).

1. Compute $\Delta = B^2 - 4AC$.
2. Apply the discriminant table: negative $\rightarrow$ ellipse, zero $\rightarrow$ parabola, positive $\rightarrow$ hyperbola.
3. *Reasoning*: the xy term only *rotates* the conic; rotation cannot change what kind of curve it is, and $B^2 - 4AC$ is a rotation-invariant quantity — it stays fixed no matter how the axes are turned, which is exactly why it is a reliable classifier.

Section 4: Worked Examples

Example 1 — [CBSE Board] A plane cuts a right circular cone of semi-vertical angle 30° at an angle of 50° to the axis, without passing through the vertex. Name the resulting conic. What would it be if the angle were exactly 30° ?

Compare $\beta = 50^\circ$ with $\alpha = 30^\circ$. Since $\alpha < \beta < 90^\circ$ (i.e. $30^\circ < 50^\circ < 90^\circ$), the section is an **ellipse**.

If instead $\beta = 30^\circ = \alpha$, the plane is parallel to a generator, giving a **parabola** — the boundary case.

Example 2 — [JEE Main, quick-elimination] Identify the conic $3x^2 + 3y^2 - 12x + 6y - 4 = 0$.

No xy term, and the coefficients of x^2 and y^2 are **equal** ($A = C = 3$). That single observation is enough: it is a **circle**. No need to find the centre or radius to classify it. *Speed note*: equal x^2, y^2 coefficients with no xy term $\Rightarrow$ circle, instantly. Unequal but same sign $\Rightarrow$ ellipse.

Example 3 — [JEE Main, multi-concept] Classify $4x^2 - 9y^2 + 8x + 36y - 68 = 0$, and state its eccentricity category.

No xy term. Coefficients $A = 4$ and $C = -9$ have **opposite signs**, so $B^2 - 4AC = 0 - 4(4)(-9) = 144 > 0 \rightarrow$ **hyperbola**. By the eccentricity classification, a hyperbola has $e > 1$. *Concept link:* the discriminant test and the eccentricity scale agree — $B^2 - 4AC > 0$ and $e > 1$ are two languages for the same fact.

Example 4 — [JEE Advanced] A curve is given by $5x^2 - 4xy + 8y^2 - 16x - 14y + 5 = 0$. Determine its type, and explain why the xy term does not change the answer.

Here $A = 5$, $B = -4$, $C = 8$. Compute

$$\Delta = B^2 - 4AC = (-4)^2 - 4(5)(8) = 16 - 160 = -144 < 0,$$

so the curve is an **ellipse**.

Why the xy term is harmless to classification: a nonzero B tells us the ellipse is *rotated* relative to the coordinate axes, but $B^2 - 4AC$ is invariant under rotation of axes — rotating the frame changes A , B , and C individually yet leaves the combination $B^2 - 4AC$ fixed. Since rotation cannot turn an ellipse into a parabola or hyperbola, the invariant correctly reports “ellipse” regardless of orientation. *Why it is “Advanced”:* the insight is recognising that the discriminant is a rotation **invariant**, so you may classify without the heavy work of actually rotating the axes to eliminate the xy term.

Section 5: Practice Exercises

1. **[CBSE]** A plane perpendicular to the axis of a cone (not through the vertex) makes what section? What if the plane passes through the vertex along the axis?
2. **[JEE Main]** Identify the conic $2x^2 + 2y^2 + 5x - 3y + 1 = 0$.
3. **[JEE Main]** Identify the conic $y^2 - 8x + 6y + 1 = 0$ and give its eccentricity value.
4. **[JEE Main]** Using the discriminant, classify $x^2 + 4xy + 4y^2 + 3x - 2 = 0$.
5. **[JEE Advanced]** For what relation between α (semi-vertical angle) and β (plane angle to axis) is the section a parabola, and why is it the unique boundary between the bounded and unbounded conics?

Answers: 1. Circle; degenerate — a pair of intersecting straight lines (plane through vertex containing the axis). 2. Equal x^2, y^2 coefficients, no $xy \rightarrow$ circle. 3. Only y^2 present (no x^2) $\rightarrow$ parabola; $e = 1$. 4. $B^2 - 4AC = 16 - 4(1)(4) = 0 \rightarrow$ parabola. 5. $\beta = \alpha$; it is the single angle at which the cutting plane is parallel to a generator, so the section can neither close (ellipse, $\beta > \alpha$) nor reach the second nappe (hyperbola, $\beta < \alpha$) — the unique transition, mirroring $e = 1$ between $e < 1$ and $e > 1$.

Section 6: Crack the MCQ

Q1. A section of a double cone that has **two** separate branches is a

- (A) parabola (B) ellipse (C) hyperbola (D) circle

Answer: (C). Two branches appear only when the cutting plane is steep enough to slice **both nappes** of the double cone — the hyperbola. *Trap (A):* the parabola is unbounded but has a single branch. *Trap (B)/(D):* both are closed, single curves on one nappe.

Q2. The conic $7x^2 + 7y^2 + 2x - 9 = 0$ is a

(A) circle (B) ellipse (C) parabola (D) hyperbola

Answer: (A). Equal coefficients of x^2 and y^2 with no xy term is the circle signature. *Trap (B):* ellipse needs *unequal* same-sign coefficients. *Trap (C)/(D):* parabola needs one squared term missing; hyperbola needs opposite signs.

Q3. For the general equation, $B^2 - 4AC = 0$ indicates a

(A) circle (B) ellipse (C) parabola (D) hyperbola

Answer: (C). A zero discriminant is the parabola condition — the boundary between negative (ellipse) and positive (hyperbola). *Trap (A)/(B):* both have $B^2 - 4AC < 0$. *Trap (D):* hyperbola is > 0 .

Q4. A plane passing through the vertex of a cone and tangent to it along a single generator produces a

(A) point (B) single straight line (C) pair of lines (D) parabola

Answer: (B). Touching along exactly one generator collapses the section to that **single straight line** — a degenerate conic. *Trap (A):* a point arises when the plane meets the vertex only. *Trap (C):* a pair of lines needs the plane to cut through, crossing both nappes at the vertex. *Trap (D):* a parabola requires the plane to miss the vertex.

Section 7: Common Student Pitfalls & Pro-Tips

- **Forgetting the degenerate cases.** “Plane through the vertex” never gives a normal curve — it gives a point, a line, or a pair of lines. Exams test this precisely because students skip it.
- **Confusing the parabola condition.** The parabola is $\beta = \alpha$ (plane parallel to a generator) and $e = 1$ — the single boundary, not a range. Many students lump it with the ellipse.
- **Misreading “equal coefficients.”** A circle needs $A = C$ and $B = 0$. Equal x^2, y^2 coefficients with an xy term present is a rotated ellipse, not a circle.
- **Trying to rotate before classifying.** You never need to remove the xy term to name the curve — $B^2 - 4AC$ already tells you.

Pro-Tip. Two five-second classifiers cover almost every question. From geometry: compare β to α (and check the vertex). From an equation: if there is no xy term, just look at the signs of A and C — same and equal is a circle, same and unequal is an ellipse, one missing is a parabola, opposite is a hyperbola; if there is an xy term, fall back on the sign of $B^2 - 4AC$.

Section 8: Cheat Sheet Box

- **The family:** all four conics are slices of a double cone; the cut angle β vs the cone angle α decides which.
- **By angle:** $\beta = 90^\circ$ circle; $\alpha < \beta < 90^\circ$ ellipse; $\beta = \alpha$ parabola; $\beta < \alpha$ hyperbola.
- **By eccentricity:** $e = 0$ circle, $0 < e < 1$ ellipse, $e = 1$ parabola, $e > 1$ hyperbola.
- **Degenerate (plane through vertex):** point, single line, or pair of intersecting lines.
- **General equation:** $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$; discriminant $B^2 - 4AC$: < 0 ellipse, $= 0$ parabola, > 0 hyperbola; $A = C, B = 0$ circle.
- **Mnemonics:** “Same sign closes, opposite opens.” “Parabola is the hinge: $\beta = \alpha, e = 1, B^2 = 4AC$.” “Through the vertex means degenerate.”

The Circle

Exam Relevance: The circle is the most heavily examined conic at CBSE Board level — expect 2-4 marks finding the equation, centre, or radius. JEE Main carries 1-2 questions, frequently the general-form centre/radius and the position of a point. JEE Advanced fuses it with tangent-length and locus reasoning. It is also the gentlest conic to master, which makes it the perfect launchpad for the rest of the chapter.

Section 1: Concept Intuition

Of all the conics, the circle is the one your hand already knows. Tie a goat to a peg with a rope and let it wander, keeping the rope taut: it traces a perfect circle, because every point it can reach is the *same distance* — the rope's length — from the peg. That is the entire idea. A **circle** is the set of all points that lie at one fixed distance, the **radius** r , from one fixed point, the **centre**. No “two foci,” no “ratio of distances” — just one centre, one distance, total symmetry.

In the eccentricity language of the rest of this chapter, the circle is the calmest member of the family: $e = 0$. It is the limiting ellipse whose two foci have drifted together into a single point at the centre. Where an ellipse is a squashed loop with a long and short axis, the circle is the ellipse that refused to squash — equally wide in every direction.

Think about it this way... stand at the centre of a roundabout (a traffic circle) and walk outward a fixed number of steps, then turn and keep that same distance as you go around. Your footprints draw a circle. The moment you let the distance vary — longer on one side, shorter on the other — you are no longer walking a circle; you have started tracing an ellipse. The circle is the promise to never change your distance from the centre.

Because the defining condition is so clean, the equation is too. If the centre sits at a point (h, k) and a moving point (x, y) keeps its distance to the centre equal to r , then the distance formula says $\sqrt{(x - h)^2 + (y - k)^2} = r$ — and squaring both sides gives the circle's equation directly. Everything else about the circle is just this one statement, rearranged.

There are two faces of the same equation you must be fluent in. The **standard form** wears its centre and radius openly — you can read them off at a glance. The **general form** hides them inside a tidy quadratic, and your job is to recover the centre and radius by *completing the square*. Recognising that these two forms are the same circle in different costumes is most of what this topic asks.

Limiting conditions & assumptions.

- **The radius must be positive and real.** In the general form $x^2 + y^2 + 2gx + 2fy + c = 0$, the quantity under the radius, $g^2 + f^2 - c$, must be **positive** for a genuine circle. If it is zero, the “circle” shrinks to a single **point** (a point circle). If it is negative, there is **no real circle** at all — the equation describes an imaginary one.
- **Equal x^2 and y^2 coefficients, no xy term.** Any real circle has the coefficients of x^2 and y^2 equal and no cross term. If they differ, you are looking at an ellipse, not a circle.
- A circle has no foci, directrix, or eccentricity to compute (or rather $e = 0$ trivially) — do not go looking for them.

Section 2: Key Definitions & Formulas

Definition. A circle is the locus of a point that moves at a constant distance (radius r) from a fixed point (centre C).

Standard equation (centre (h, k) , radius r):

$$(x - h)^2 + (y - k)^2 = r^2$$

Centre at the origin reduces this to $x^2 + y^2 = r^2$.

General equation:

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

$$\text{Centre} = (-g, -f), \quad \text{Radius} = \sqrt{g^2 + f^2 - c}.$$

Nature of the circle (from the general form):

$g^2 + f^2 - c > 0 \Rightarrow$ real circle, $= 0 \Rightarrow$ point circle, $< 0 \Rightarrow$ no real (imaginary) circle.

Diameter form — the circle on the segment joining $A(x_1, y_1)$ and $B(x_2, y_2)$ as diameter:

$$(x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0.$$

Position of a point $P(x_1, y_1)$ relative to $S \equiv x^2 + y^2 + 2gx + 2fy + c = 0$. Let

$$S_1 = x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c.$$

Then P lies **outside / on / inside** according as S_1 is **positive / zero / negative**.

Length of the tangent from an external point $P(x_1, y_1)$ to the circle:

$$\ell_{\text{tangent}} = \sqrt{S_1} \quad (S_1 > 0).$$

Section 3: CBSE Board Derivation

Deriving the standard equation. Let the centre be $C(h, k)$ and let $P(x, y)$ be any point on the circle. The defining condition is that the distance CP equals the radius r :

$$CP = r \implies \sqrt{(x - h)^2 + (y - k)^2} = r.$$

Both sides are non-negative, so squaring is reversible and introduces no false points:

$$\boxed{(x - h)^2 + (y - k)^2 = r^2.}$$

That is the whole derivation — the circle's equation is just the distance formula with the distance pinned to r .

Recovering centre and radius from the general form (completing the square).

Start from

$$x^2 + y^2 + 2gx + 2fy + c = 0.$$

Group the x -terms and the y -terms, then complete each square by adding and subtracting the needed constant:

$$(x^2 + 2gx + g^2) + (y^2 + 2fy + f^2) = g^2 + f^2 - c,$$

$$(x + g)^2 + (y + f)^2 = g^2 + f^2 - c.$$

Compare this with the standard form $(x - h)^2 + (y - k)^2 = r^2$. Matching pieces:

$$h = -g, \quad k = -f, \quad r^2 = g^2 + f^2 - c.$$

So the **centre is** $(-g, -f)$ and the **radius is** $\sqrt{g^2 + f^2 - c}$. The reason the radius can fail to exist is now visible: r^2 equals $g^2 + f^2 - c$, and a real radius needs that to be positive.

Why the diameter form works. If A and B are the ends of a diameter and P is any other point on the circle, then the angle $\angle APB$ is a right angle (angle in a semicircle). The two segments PA and PB are therefore perpendicular, so the product of their slopes is -1 — and clearing that condition of fractions gives exactly

$$(x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0.$$

Figure to be illustrated

A circle with centre $C(h, k)$ and radius r drawn to a point $P(x, y)$ on it, the segment CP labelled r . Mark a diameter AB with a third point Q on the circle and the right angle at Q (angle in a semicircle). Show one point inside the circle and one outside, labelled with $S_1 < 0$ and $S_1 > 0$ respectively, and a tangent line from the outside point with its length labelled $\sqrt{S_1}$.

Section 4: Worked Examples

Example 1 — [CBSE Board] Find the centre and radius of the circle $x^2 + y^2 - 6x + 8y - 11 = 0$.

Compare with $x^2 + y^2 + 2gx + 2fy + c = 0$:

$$2g = -6 \Rightarrow g = -3, \quad 2f = 8 \Rightarrow f = 4, \quad c = -11.$$

Centre = $(-g, -f) = (3, -4)$. Radius = $\sqrt{g^2 + f^2 - c} = \sqrt{9 + 16 - (-11)} = \sqrt{36} = 6$. So the circle has centre $(3, -4)$ and radius 6 units.

Example 2 — [JEE Main, quick-elimination] Find the equation of the circle with the points $A(1, 2)$ and $B(5, 6)$ as the ends of a diameter.

Use the diameter form directly — no need to find the centre and radius separately:

$$(x - 1)(x - 5) + (y - 2)(y - 6) = 0.$$

Expand:

$$x^2 - 6x + 5 + y^2 - 8y + 12 = 0 \Rightarrow x^2 + y^2 - 6x - 8y + 17 = 0.$$

Speed note: whenever a problem hands you the two ends of a diameter, the diameter form is one line; converting via midpoint-centre and half-distance-radius is three lines and more error-prone.

Example 3 — [JEE Main, multi-concept] A circle has centre $(2, -1)$ and passes through the point $(5, 3)$. Find its equation, and determine whether the point $(4, 2)$ lies inside, on, or outside it.

The radius is the distance from centre to the given point:

$$r = \sqrt{(5-2)^2 + (3-(-1))^2} = \sqrt{9+16} = 5.$$

Equation: $(x-2)^2 + (y+1)^2 = 25$, i.e. $x^2 + y^2 - 4x + 2y - 20 = 0$.

Test $(4, 2)$ using S_1 :

$$S_1 = 4^2 + 2^2 - 4(4) + 2(2) - 20 = 16 + 4 - 16 + 4 - 20 = -12 < 0.$$

A negative S_1 means the point lies **inside** the circle. *Concept link:* one problem ties the distance-as-radius idea, the standard-to-general conversion, and the position test together.

Example 4 — [JEE Advanced] Find the equation of the circle passing through the three points $O(0, 0)$, $A(6, 0)$, and $B(0, 8)$, and hence the length of the tangent to it from the point $P(7, 9)$.

Let the circle be $x^2 + y^2 + 2gx + 2fy + c = 0$ and substitute each point.

Through $O(0, 0)$: $c = 0$. Through $A(6, 0)$: $36 + 12g + c = 0 \Rightarrow 36 + 12g = 0 \Rightarrow g = -3$.

Through $B(0, 8)$: $64 + 16f + c = 0 \Rightarrow 64 + 16f = 0 \Rightarrow f = -4$. So the circle is

$$x^2 + y^2 - 6x - 8y = 0,$$

with centre $(3, 4)$ and radius $\sqrt{9+16-0} = 5$. (Sensibly, OA and OB are perpendicular chords from the origin, so AB is a diameter — and indeed its midpoint $(3, 4)$ is the centre, length $AB = 10 = 2r$. ✓)

Length of the tangent from $P(7, 9)$:

$$S_1 = 7^2 + 9^2 - 6(7) - 8(9) = 49 + 81 - 42 - 72 = 16,$$

$$\ell_{\text{tangent}} = \sqrt{S_1} = \sqrt{16} = 4 \text{ units.}$$

Why it is “Advanced”: it chains the three-point method (solving for g, f, c), a geometric sanity check (recognising AB as a diameter), and the tangent-length formula — and the right-angle observation lets you verify the answer independently.

Section 5: Practice Exercises

1. **[CBSE]** Find the centre and radius of $x^2 + y^2 + 4x - 10y + 13 = 0$.
2. **[JEE Main]** Find the equation of the circle with centre $(-3, 2)$ and radius 4.
3. **[JEE Main]** Find the equation of the circle having $A(2, -3)$ and $B(-2, 5)$ as the ends of a diameter.
4. **[JEE Main]** Determine the position of the point $(1, 1)$ with respect to the circle $x^2 + y^2 - 2x - 4y + 3 = 0$.

5. **[JEE Advanced]** Find the equation of the circle through $(1, 0)$, $(0, 1)$, and $(2, 1)$, and the length of the tangent to it from the origin.

Answers: 1. Centre $(-2, 5)$, radius $\sqrt{4 + 25 - 13} = \sqrt{16} = 4$. 2. $(x+3)^2 + (y-2)^2 = 16$, i.e. $x^2 + y^2 + 6x - 4y - 3 = 0$. 3. $(x-2)(x+2) + (y+3)(y-5) = 0 \Rightarrow x^2 + y^2 - 2y - 19 = 0$. 4. $S_1 = 1 + 1 - 2 - 4 + 3 = -1 < 0 \rightarrow$ inside. 5. Circle $x^2 + y^2 - 3x - y + 2 = 0$ (centre $(3/2, 1/2)$, radius $\sqrt{1/2}$); tangent length from origin $= \sqrt{S_1} = \sqrt{0 + 0 - 0 - 0 + 2} = \sqrt{2}$.

Section 6: Crack the MCQ

Q1. The centre of the circle $x^2 + y^2 + 8x - 6y + 9 = 0$ is

- (A) $(8, -6)$ (B) $(-4, 3)$ (C) $(4, -3)$ (D) $(-8, 6)$

Answer: (B). $2g = 8 \Rightarrow g = 4$, $2f = -6 \Rightarrow f = -3$; centre $= (-g, -f) = (-4, 3)$. *Trap (A):* reading $(2g, 2f)$ directly. *Trap (C):* sign error, taking (g, f) instead of $(-g, -f)$. *Trap (D):* taking $(-2g, -2f)$.

Q2. For $x^2 + y^2 - 4x - 6y + 13 = 0$, the circle is

- (A) a real circle of radius $\sqrt{13}$ (B) a point circle (C) imaginary (D) a real circle of radius 2

Answer: (B). $g = -2$, $f = -3$, $c = 13$, so $g^2 + f^2 - c = 4 + 9 - 13 = 0 \rightarrow$ radius 0, a **point circle** (just the point $(2, 3)$). *Trap (A)/(D):* assuming a positive radius without checking. *Trap (C):* imaginary needs $g^2 + f^2 - c < 0$, not $= 0$.

Q3. The equation of the circle with $(3, 4)$ and $(-1, 2)$ as diameter ends is

- (A) $x^2 + y^2 - 2x - 6y + 5 = 0$ (B) $x^2 + y^2 + 2x + 6y + 5 = 0$ (C) $x^2 + y^2 - 2x - 6y - 5 = 0$
(D) $x^2 + y^2 - 4x - 8y = 0$

Answer: (A). Diameter form: $(x-3)(x+1) + (y-4)(y-2) = 0 \Rightarrow x^2 - 2x - 3 + y^2 - 6y + 8 = 0 \Rightarrow x^2 + y^2 - 2x - 6y + 5 = 0$. *Trap (B):* sign errors in the linear terms. *Trap (C):* mis-adding the constant. *Trap (D):* using midpoint and forgetting the constant term.

Q4. The length of the tangent from $(0, 0)$ to the circle $x^2 + y^2 - 4x - 6y + 12 = 0$ is

- (A) $2\sqrt{3}$ (B) $\sqrt{12}$ (C) 2 (D) cannot be drawn

Answer: (A). $S_1 = 0 + 0 - 0 - 0 + 12 = 12 > 0$ (so the origin is outside), and $\ell = \sqrt{12} = 2\sqrt{3}$. *Trap (B):* $\sqrt{12}$ is correct but not in simplest form — both (A) and (B) are numerically equal, and exam keys take the simplified surd. *Trap (C):* taking $\sqrt{S_1}$ of a wrong S_1 . *Trap (D):* a tangent can be drawn since $S_1 > 0$ (point outside).

Section 7: Common Student Pitfalls & Pro-Tips

- **The centre sign.** From $x^2 + y^2 + 2gx + 2fy + c = 0$ the centre is $(-g, -f)$, *not* (g, f) and *not* $(2g, 2f)$. Halve the linear coefficients **and** flip the sign.
- **Skipping the nature check.** Always confirm $g^2 + f^2 - c > 0$ before announcing a radius — a zero gives a point circle and a negative gives no real circle.

- **Forcing the general form when given a diameter.** If you have the two ends of a diameter, the one-line diameter form is faster and safer than computing centre and radius.
- **Mismatched coefficients.** A real circle must have equal coefficients of x^2 and y^2 and no xy term; if a problem's x^2 and y^2 coefficients differ, divide through first (if they are equal but not 1) or recognise it is not a circle.

Pro-Tip. Reduce every general-form circle to three numbers in one pass: $g = \frac{1}{2}(\text{coeff of } x)$, $f = \frac{1}{2}(\text{coeff of } y)$, then centre $(-g, -f)$ and radius $\sqrt{g^2 + f^2 - c}$. And memorise the one expression S_1 : plug a point into the circle's left side — its **sign** places the point (out/on/in) and its **square root** is the tangent length. One quantity, two jobs.

Section 8: Cheat Sheet Box

- **Definition:** locus of points at fixed distance r (radius) from a fixed centre; eccentricity $e = 0$.
- **Standard form:** $(x - h)^2 + (y - k)^2 = r^2$; origin-centred $x^2 + y^2 = r^2$.
- **General form:** $x^2 + y^2 + 2gx + 2fy + c = 0$; centre $(-g, -f)$, radius $\sqrt{g^2 + f^2 - c}$.
- **Nature:** $g^2 + f^2 - c$ — positive (real), zero (point circle), negative (imaginary).
- **Diameter form:** $(x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0$.
- **Point test & tangent:** S_1 sign gives out/on/in; tangent length $= \sqrt{S_1}$.
- **Mnemonics:** “Centre is minus- g , minus- f .” “Check $g^2 + f^2 - c$ before you trust the radius.” “Same number plugged in: sign locates, root measures.”

The Parabola

Exam Relevance: CBSE Boards reliably ask 1–3 marks here — read a standard equation and report focus, axis, directrix, and latus rectum, or build the equation from given conditions. JEE Main carries 1–2 questions, usually standard-form manipulation plus the inside/outside test. JEE Advanced uses the parabola as a vehicle for focal-chord and reflection reasoning, often fused with another conic. It is also the curve behind every dish antenna and headlight you have ever seen.

Section 1: Concept Intuition

Slice a cone with a plane and you get a curve. Tilt the plane gently and you get a closed loop — an ellipse. Tilt it more steeply and the loop bursts open into a sweeping U-shape that never closes. That exact borderline cut — the one made by a plane *parallel to the slant side of the cone* — is the **parabola**. It is the knife-edge case between “closed” and “doubly open,” and that in-between character is precisely what gives it its single, beautiful focusing property.

But you do not need a cone to define it. There is a cleaner way that an engineer or a student can actually use.

Picture a long, straight boundary wall and a single flagpole planted some distance in front of it. Now ask: where can you stand so that your distance to the flagpole is *exactly equal* to your distance to the wall? Stand too close to the pole and the pole wins; drift toward the wall and the wall wins. There is a whole curving track of “fair” positions where the two distances stay perfectly balanced — and that track is a parabola. The flagpole is called the **focus** (S), the wall is the **directrix**, and the rule “distance to focus = distance to directrix” is the entire definition.

Think about it this way... imagine a tug-of-war refereed for fairness. The focus pulls you in by straight-line distance; the directrix pulls you in by perpendicular distance. The parabola is the locus of perfect ties. The ratio of those two distances is called the **eccentricity** e , and for a parabola it is always exactly $e = 1$ — neither side ever gets an edge. (If the focus could win, $e < 1$ and you would trace an ellipse; if the directrix could win, $e > 1$ and you would get a hyperbola. The parabola lives exactly at $e = 1$.)

This single property is why parabolas run our daily technology. Your DTH dish — the Tata Play or Airtel dish bolted to a balcony — is a parabola spun into a bowl. Signals arriving parallel from a satellite all bounce off the dish and pile up at one point: the focus, where the receiver (LNB) sits. A solar cooker concentrates sunlight the same way. Run it in reverse and you get a car headlight or a torch: a bulb at the focus throws out a perfectly parallel beam. And the humble cricket ball, once it leaves the bat, traces a parabola through the air under gravity. The curve is everywhere.

Limiting conditions & assumptions.

- $e = 1$ **exactly**. The focus-directrix balance must be perfect. The moment the ratio drifts off 1, you no longer have a parabola.
- **Focus must not lie on the directrix**. If the flagpole stood *inside* the wall, the “curve” would collapse into a single straight line — a degenerate case, not a true parabola.
- **Standard equations assume a tidy frame**. The clean forms below all put the **vertex at the origin** and the **axis along a coordinate axis**. Shift or rotate the parabola

and the equation grows extra terms; the geometry is identical, but the bookkeeping changes.

- **Four orientations only (in standard position):** opening right, left, up, or down — decided by where the focus sits relative to the vertex.

Section 2: Key Definitions & Formulas

Definition. A parabola is the locus of a point that moves so that its distance from a fixed point (the **focus** S) equals its distance from a fixed line (the **directrix**). Equivalently, eccentricity $e = 1$.

Key terms. The **axis** is the line through the focus perpendicular to the directrix (the parabola's line of symmetry). The **vertex** A is the point where the parabola meets its axis — the midpoint between focus and directrix. The **latus rectum** is the focal chord perpendicular to the axis; its length measures how “wide” the parabola opens.

The four standard forms (vertex at origin O , $a > 0$):

Equation	Opens	Focus S	Directrix	Axis	Latus rectum length
$y^2 = 4ax$	right	$(a, 0)$	$x = -a$	x -axis	$4a$
$y^2 = -4ax$	left	$(-a, 0)$	$x = a$	x -axis	$4a$
$x^2 = 4ay$	up	$(0, a)$	$y = -a$	y -axis	$4a$
$x^2 = -4ay$	down	$(0, -a)$	$y = a$	y -axis	$4a$

Latus rectum endpoints for $y^2 = 4ax$: $(a, 2a)$ and $(a, -2a)$.

Focal distance of a point $P(x, y)$ on $y^2 = 4ax$:

$$SP = x + a$$

(the distance to the focus equals the distance to the directrix $x = -a$, which is $x + a$).

Parametric coordinates of a point on $y^2 = 4ax$:

$$P \equiv (at^2, 2at), \quad t \in \mathbb{R}$$

Position of a point (x_1, y_1) relative to $y^2 = 4ax$: it lies **outside** / **on** / **inside** according as $y_1^2 - 4ax_1$ is **positive** / **zero** / **negative**.

Section 3: CBSE Board Derivation

Deriving the standard equation $y^2 = 4ax$.

Set up the frame so the algebra is kind to us. Let the vertex sit at the origin O , and let the focus lie on the positive x -axis at $S(a, 0)$, where $a > 0$. Since the vertex is the midpoint of focus and directrix, the directrix must be the vertical line the same distance on the other side: $x = -a$.

Now take an arbitrary point $P(x, y)$ on the parabola. Drop a perpendicular from P to the directrix, meeting it at M . The foot M has the same height as P but sits on the line $x = -a$, so $M = (-a, y)$.

The defining property says distance to focus equals distance to directrix:

$$SP = PM$$

Write each distance. The distance from P to the focus is

$$SP = \sqrt{(x - a)^2 + (y - 0)^2}.$$

The distance from P to the directrix is just the horizontal gap to the line $x = -a$:

$$PM = |x + a|.$$

Why we square next: both sides are non-negative distances, so squaring is reversible and clears the square root and the modulus in one move without introducing false solutions:

$$(x - a)^2 + y^2 = (x + a)^2.$$

Expand both squares — the trick is that the x^2 and a^2 terms are identical on both sides and will cancel, leaving only the cross terms:

$$x^2 - 2ax + a^2 + y^2 = x^2 + 2ax + a^2.$$

Cancel x^2 and a^2 :

$$y^2 - 2ax = 2ax \implies \boxed{y^2 = 4ax}.$$

The cancellation is the whole story: the squared terms that make the distance formulas messy are exactly the ones that vanish, and what survives is the clean relation between y^2 and x . The other three standard forms come from the same derivation with the focus placed on the negative x -axis, or on the y -axis up or down.

Deriving the length of the latus rectum.

The latus rectum is the chord through the focus perpendicular to the axis, so it lies along the vertical line $x = a$. Substitute $x = a$ into $y^2 = 4ax$:

$$y^2 = 4a(a) = 4a^2 \implies y = \pm 2a.$$

The chord runs from $(a, 2a)$ to $(a, -2a)$, so its length is

$$\text{Latus rectum} = 2a - (-2a) = \boxed{4a}.$$

This is why the coefficient $4a$ is so useful: it *is* the latus rectum, a direct measure of how wide the parabola opens. A large $4a$ means a broad, lazy curve; a small $4a$ means a tight, narrow one.

Figure to be illustrated

A right-opening parabola $y^2 = 4ax$ with vertex at origin O . Mark focus $S(a,0)$ on the x -axis, the directrix $x = -a$ as a dashed vertical line, the axis along the x -axis. Show a general point $P(x,y)$ with segment SP to the focus and a horizontal dashed segment PM to the directrix, labelled equal. Mark the latus rectum as a vertical chord through

S from $(a, 2a)$ to $(a, -2a)$.

Section 4: Worked Examples

Example 1 — [CBSE Board] For the parabola $y^2 = -12x$, find the vertex, focus, axis, equation of the directrix, and the length of the latus rectum.

Given: $y^2 = -12x$. Comparing with the standard left-opening form $y^2 = -4ax$:

$$4a = 12 \Rightarrow a = 3.$$

Since y is squared and the sign is negative, the parabola opens **left** along the x -axis.

- Vertex: $(0, 0)$
- Focus: $(-a, 0) = (-3, 0)$
- Axis: the x -axis, i.e. $y = 0$
- Directrix: $x = a = 3$
- Latus rectum: $4a = 12$ units

Clean presentation, correct sign, units stated.

Example 2 — [JEE Main, quick-elimination] Find the equation of the parabola with focus $(0, -3)$ and directrix $y = 3$.

Read the geometry before computing. The focus is on the y -axis **below the origin**, and the directrix is the horizontal line above it. The vertex is the midpoint of focus and directrix — the origin — and the curve must open **downward** (away from the directrix, toward the focus). So the form is $x^2 = -4ay$.

The focus is at distance $a = 3$ from the vertex, so $4a = 12$:

$$x^2 = -12y.$$

Speed note: you never needed the distance formula. Focus-below + directrix-above $\Rightarrow$ opens down $\Rightarrow x^2 = -4ay$; then $a =$ vertex-to-focus distance. Two seconds.

Example 3 — [JEE Main, multi-concept] A parabola with vertex at the origin and axis along the y -axis passes through $(8, 4)$ and opens upward. Find its equation, the focal distance of the point $(8, 4)$, and determine whether the point $(3, 1)$ lies inside or outside the parabola.

Upward, axis along y : form $x^2 = 4ay$. It passes through $(8, 4)$:

$$8^2 = 4a(4) \Rightarrow 64 = 16a \Rightarrow a = 4.$$

So the parabola is $x^2 = 16y$, with focus $(0, 4)$ and directrix $y = -4$.

Focal distance of $(8, 4)$: for $x^2 = 4ay$, the focal distance equals distance to the directrix $y = -a$, i.e. $y + a = 4 + 4 = 8$ units.

Position of $(3, 1)$: the inside/outside test for $x^2 = 4ay$ uses the sign of $x_1^2 - 4ay_1$:

$$3^2 - 16(1) = 9 - 16 = -7 < 0.$$

A negative value means the point lies **inside** the parabola. *Concept link:* one problem chains standard-form fitting, the focal-distance shortcut, and the position test — three ideas, one curve.

Example 4 — [JEE Advanced] PQ is a focal chord of the parabola $y^2 = 4ax$. Prove the harmonic property $\frac{1}{SP} + \frac{1}{SQ} = \frac{1}{a}$, and use it to find the full length of a focal chord whose nearer end satisfies $SP = 3a$.

Proof. Let the ends be $P(at_1^2, 2at_1)$ and $Q(at_2^2, 2at_2)$. A standard parametric result (provable by collinearity of $P, Q, S(a, 0)$) gives the focal-chord condition

$$t_1 t_2 = -1.$$

The focal distance of a parametric point is $SP = x + a = at_1^2 + a = a(1 + t_1^2)$, and likewise $SQ = a(1 + t_2^2)$. Then

$$\frac{1}{SP} + \frac{1}{SQ} = \frac{1}{a(1 + t_1^2)} + \frac{1}{a(1 + t_2^2)}.$$

Substitute $t_2 = -1/t_1$, so $t_2^2 = 1/t_1^2$ and $1 + t_2^2 = \frac{t_1^2 + 1}{t_1^2}$. Therefore

$$\frac{1}{SQ} = \frac{1}{a} \cdot \frac{t_1^2}{1 + t_1^2}.$$

Adding,

$$\frac{1}{SP} + \frac{1}{SQ} = \frac{1}{a(1 + t_1^2)} + \frac{t_1^2}{a(1 + t_1^2)} = \frac{1 + t_1^2}{a(1 + t_1^2)} = \frac{1}{a}. \quad \blacksquare$$

The result is striking: the *semi-latus rectum* $2a$ is the harmonic mean of the two focal radii.

Application. With $SP = 3a$:

$$\frac{1}{3a} + \frac{1}{SQ} = \frac{1}{a} \Rightarrow \frac{1}{SQ} = \frac{1}{a} - \frac{1}{3a} = \frac{2}{3a} \Rightarrow SQ = \frac{3a}{2}.$$

Full chord length:

$$PQ = SP + SQ = 3a + \frac{3a}{2} = \boxed{\frac{9a}{2}}.$$

Why it is “Advanced”: it rewards seeing that everything reduces to focal distances and the single condition $t_1 t_2 = -1$ — no tangent or normal machinery, just the definition pushed hard.

Section 5: Practice Exercises

1. **[CBSE]** Find the focus, axis, directrix, and latus rectum of $x^2 = 20y$.
2. **[JEE Main]** Find the equation of the parabola with vertex at the origin, axis along the x -axis, passing through $(2, -6)$.

3. **[JEE Main]** The focus of a parabola is $(-4, 0)$ and its vertex is the origin. Write its equation and the equation of its directrix.
4. **[JEE Main]** Determine whether the point $(2, 5)$ lies inside, on, or outside the parabola $y^2 = 8x$, and find the focal distance of the point $(2, 4)$ on it.
5. **[JEE Advanced]** A focal chord of $y^2 = 12x$ has one end at the point with parameter $t = 2$. Find the coordinates of both ends and the length of the chord.

Answers: 1. Focus $(0, 5)$; axis $x = 0$; directrix $y = -5$; latus rectum 20. 2. $y^2 = 18x$. 3. $y^2 = -16x$; directrix $x = 4$. 4. $25 - 16 = 9 > 0 \Rightarrow$ outside; focal distance of $(2, 4)$ is $x + a = 2 + 2 = 4$. 5. Here $4a = 12 \Rightarrow a = 3$; ends $(12, 12)$ at $t = 2$ and $(3/4, -3)$ at $t = -1/2$; chord length $= SP + SQ = a(1 + t_1^2) + a(1 + t_2^2) = 3(5) + 3(5/4) = 15 + 3.75 = 18.75 = 75/4$.

Section 6: Crack the MCQ

Q1. The length of the latus rectum of the parabola $y^2 = 7x$ is

- (A) 7 (B) $7/4$ (C) 28 (D) $7/2$

Answer: (A). Here $4a = 7$, and the latus rectum equals the coefficient $4a$ directly, so it is 7. *Trap (B):* student computes $a = 7/4$ and reports a instead of $4a$ — the classic “ a vs $4a$ ” slip. *Trap (C):* multiplying $4a$ by 4 again. *Trap (D):* halving by mistake, perhaps confusing with the semi-latus rectum $2a$.

Q2. The focus of the parabola $x^2 = -16y$ is

- (A) $(0, 4)$ (B) $(0, -4)$ (C) $(4, 0)$ (D) $(-4, 0)$

Answer: (B). x is squared with a negative sign, so the parabola opens downward along the y -axis with $4a = 16 \Rightarrow a = 4$; the focus is $(0, -a) = (0, -4)$. *Trap (A):* ignoring the negative sign and placing the focus above. *Trap (C)/(D):* putting the focus on the x -axis — confusing $x^2 = \dots$ (axis along y) with $y^2 = \dots$ (axis along x).

Q3. If the point $(k, 4)$ lies on the parabola $y^2 = 8x$, then its focal distance is

- (A) 2 (B) 4 (C) 6 (D) 8

Answer: (B). From $y^2 = 8x$: $16 = 8k \Rightarrow k = 2$, and $4a = 8 \Rightarrow a = 2$. Focal distance $= x + a = 2 + 2 = 4$. *Trap (A):* reporting a alone. *Trap (C):* adding $4a$ instead of a ($2 + ?$ confusion). *Trap (D):* using the y -coordinate as the focal distance.

Q4. The equation of the directrix of the parabola $y^2 = -4ax$ (with $a > 0$) is

- (A) $x = -a$ (B) $x = a$ (C) $y = a$ (D) $y = -a$

Answer: (B). This parabola opens left with focus $(-a, 0)$, so the directrix is the vertical line on the *opposite* side of the vertex: $x = a$. *Trap (A):* placing the directrix on the same side as the focus — directrix and focus always straddle the vertex. *Trap (C)/(D):* writing a horizontal directrix, which would belong to a y -axis parabola.

Section 7: Common Student Pitfalls & Pro-Tips

- **The “ a versus $4a$ ” disaster.** The coefficient in $y^2 = 4ax$ is $4a$, *not* a . The focus sits at distance a (coefficient $\div 4$), while the latus rectum is the coefficient $4a$. Mixing these

up is the single biggest mark-loser in this entire topic.

- **Putting the directrix on the wrong side.** The focus and directrix always sit on *opposite* sides of the vertex, each at distance a . If your focus is at $(a, 0)$, the directrix cannot be $x = a$ — it must be $x = -a$.
- **Swapping the axis of symmetry.** $y^2 = 4ax$ (y squared) opens along the x -axis; $x^2 = 4ay$ (x squared) opens along the y -axis. Read which variable is squared *first*.
- **Forgetting the sign sets the direction.** A minus sign flips the opening (left or down). Decide the direction from the sign before placing focus and directrix.

Pro-Tip. Build a three-line reflex for any standard parabola: (1) the **un-squared** variable names the axis it opens along; (2) **coefficient** $\div 4 = a$ gives the focus/directrix distance; (3) the **coefficient itself = latus rectum**. With those three, you can read off every feature in under ten seconds — no derivation required in the exam hall.

Section 8: Cheat Sheet Box

- **Definition:** distance to focus = distance to directrix; eccentricity $e = 1$.
- **Standard forms (vertex at origin):** $y^2 = \pm 4ax$ (axis = x -axis), $x^2 = \pm 4ay$ (axis = y -axis); the $+/-$ sets the opening direction.
- **For $y^2 = 4ax$:** focus $(a, 0)$, directrix $x = -a$, latus rectum $= 4a$, endpoints $(a, \pm 2a)$.
- **Focal distance:** $SP = x + a$ (the un-squared coordinate plus a).
- **Parametric:** $(at^2, 2at)$; focal chord $\Leftrightarrow t_1 t_2 = -1$; harmonic property $\frac{1}{SP} + \frac{1}{SQ} = \frac{1}{a}$.
- **Position test:** sign of $y_1^2 - 4ax_1 \rightarrow +$ outside, 0 on, $-$ inside.
- **Mnemonics:** “Coefficient over four is a ; the coefficient itself is the latus rectum.” “The lonely variable points the way it opens.” “Focus and directrix straddle the vertex.”

The Ellipse

Exam Relevance: CBSE Boards reliably ask 1–4 marks — read a standard equation for foci, axes, eccentricity, and latus rectum, or build the equation from given data; the focal-sum derivation is a frequent long-answer. JEE Main carries 1–2 questions, usually eccentricity and latus-rectum relationships. JEE Advanced fuses the ellipse with focal-distance and reflection reasoning. It is also the shape of every planetary orbit and every whispering gallery.

Section 1: Concept Intuition

Take the parabola’s “one focus, one wall” rule and give it a second focus. The ellipse is what you get when a point moves so that the **sum** of its distances to *two* fixed points stays constant. Those two points are the **foci** (S and S'), and “sum of distances = constant” is the whole idea.

There is a famous way to draw one that every student should try once. Push two pins into a board, loop a single thread loosely around both, and pull the loop taut with a pencil tip. Now slide the pencil all the way around, keeping the thread tight. Because the thread length never changes, the two distances from the pencil to the pins must always add up to the same total — so the pencil traces a perfect ellipse. The pins are the foci; the thread length is the constant sum $2a$.

Think about it this way... imagine two friends standing apart in a maidan, each holding one end of a fixed-length rope, and you walk around keeping the rope just taut between you and both of them. Stand directly between them and the rope is nearly straight across (you are at an *end of the long axis*). Walk out to the side and the rope bends into a wide vee (you are near an *end of the short axis*). The path of “rope always taut” is the ellipse. The closer the two friends stand together, the rounder it gets; pull them far apart and it stretches into a long, flat oval. When they finally stand on the *same* spot — both foci merge — the ellipse becomes a perfect circle. **A circle is just an ellipse whose two foci have collapsed into one.**

How squashed an ellipse is gets measured by its **eccentricity** e , the same e from the parabola, except now $0 < e < 1$. An e near 0 is almost circular; an e near 1 is long and cigar-shaped. (At $e = 1$ it would burst open into a parabola — which is exactly why the parabola sits at the boundary.)

Ellipses run quietly through real life. Every planet, including ours, orbits the Sun on an ellipse with the Sun at one focus — Kepler’s first law. The “whispering galleries” in old domed halls work because a whisper at one focus reflects off the curved ceiling and reconvenes perfectly at the other focus. Lithotripsy machines that shatter kidney stones place the stone at one focus and the shock-wave source at the other. The shape is everywhere precision-focusing is needed between two points.

Limiting conditions & assumptions.

- $0 < e < 1$. At $e = 0$ the ellipse is a circle; at $e = 1$ it is no longer an ellipse.
- **Constant sum must exceed the focal separation:** $2a > 2c$, i.e. the rope must be longer than the gap between the pins. If the sum equalled the separation, the “ellipse” would collapse onto the line segment joining the foci.
- **Standard equations assume a tidy frame** — centre at the origin, axes along the

coordinate axes. Off-centre or rotated ellipses are beyond the standard forms (and beyond Class 11 scope).

- **The foci always lie on the major (longer) axis** — never the minor one. Which axis is major is decided by which denominator is larger, not by habit.

Section 2: Key Definitions & Formulas

Definition. An ellipse is the locus of a point the sum of whose distances from two fixed foci is constant ($= 2a$). Equivalently, focus-directrix form with eccentricity $0 < e < 1$.

Standard equation (centre at origin, major axis along x -axis, $a > b > 0$):

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

The defining relationships:

$$c = ae, \quad c^2 = a^2 - b^2, \quad b^2 = a^2(1 - e^2), \quad e = \frac{c}{a} = \sqrt{1 - \frac{b^2}{a^2}}$$

Key features of $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ (with $a > b$):

Feature	Value
Centre	$(0, 0)$
Vertices (major-axis ends)	$(\pm a, 0)$
Minor-axis ends	$(0, \pm b)$
Foci S, S'	$(\pm ae, 0)$
Major / minor axis length	$2a / 2b$
Eccentricity	$e = \sqrt{1 - b^2/a^2}$
Directrices	$x = \pm a/e$
Latus rectum length	$\ell = 2b^2/a$
Focal radii of $P(x, y)$	$SP = a - ex, \quad S'P = a + ex$

If the larger denominator is under y^2 — that is $\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1$ with $a > b$ — then the major axis is the y -axis, foci are $(0, \pm ae)$, and directrices are $y = \pm a/e$. Everything else mirrors.

Section 3: CBSE Board Derivation

Deriving the standard equation $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

Set up the frame. Place the centre at the origin and the two foci symmetrically on the x -axis at $S(c, 0)$ and $S'(-c, 0)$. Let $P(x, y)$ be any point on the ellipse, and let the constant sum of focal distances be $2a$ (we use $2a$ rather than a bare constant because it will turn out to be the major axis). The definition gives

$$SP + S'P = 2a \implies \sqrt{(x - c)^2 + y^2} + \sqrt{(x + c)^2 + y^2} = 2a.$$

Isolate one radical before squaring — squaring a sum of two radicals leaves a radical behind, so we move one across first:

$$\sqrt{(x-c)^2 + y^2} = 2a - \sqrt{(x+c)^2 + y^2}.$$

Square both sides:

$$(x-c)^2 + y^2 = 4a^2 - 4a\sqrt{(x+c)^2 + y^2} + (x+c)^2 + y^2.$$

Expand the squared brackets and cancel the common $x^2 + c^2 + y^2$ on both sides. The cross terms $-2cx$ and $+2cx$ are what survive:

$$-2cx = 4a^2 - 4a\sqrt{(x+c)^2 + y^2} + 2cx.$$

Rearrange to isolate the surviving radical:

$$4a\sqrt{(x+c)^2 + y^2} = 4a^2 + 4cx \implies \sqrt{(x+c)^2 + y^2} = a + \frac{cx}{a}.$$

Square once more to clear the last radical:

$$(x+c)^2 + y^2 = a^2 + 2cx + \frac{c^2x^2}{a^2}.$$

Expand the left side, cancel the $2cx$ that appears on both sides, and group:

$$x^2 + c^2 + y^2 = a^2 + \frac{c^2x^2}{a^2} \implies x^2 \left(1 - \frac{c^2}{a^2}\right) + y^2 = a^2 - c^2.$$

Now name the positive quantity $a^2 - c^2$ as b^2 (it is positive because $a > c$). The bracket becomes $\frac{a^2 - c^2}{a^2} = \frac{b^2}{a^2}$, so

$$\frac{b^2}{a^2}x^2 + y^2 = b^2.$$

Divide through by b^2 :

$$\boxed{\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1.}$$

The two squarings are the heart of it: each one trades a square root for a polynomial, and the symmetric placement of the foci makes the unwanted terms cancel in pairs.

The relationship and eccentricity. The substitution we made is the master relationship:

$$b^2 = a^2 - c^2 \iff c^2 = a^2 - b^2.$$

Since $e = c/a$, dividing by a^2 gives $\frac{b^2}{a^2} = 1 - e^2$, i.e. $b^2 = a^2(1 - e^2)$.

Length of the latus rectum. The latus rectum is the focal chord perpendicular to the major axis, lying on $x = c = ae$. Substitute into the equation:

$$\frac{(ae)^2}{a^2} + \frac{y^2}{b^2} = 1 \implies \frac{y^2}{b^2} = 1 - e^2 = \frac{b^2}{a^2} \implies y = \pm \frac{b^2}{a}.$$

The chord runs from $(ae, \frac{b^2}{a})$ to $(ae, -\frac{b^2}{a})$, so

$$\ell = \frac{2b^2}{a}.$$

An elegant fact (used later). The distance from a focus to an end of the minor axis equals the **semi-major axis**:

$$SB = \sqrt{(ae)^2 + b^2} = \sqrt{a^2e^2 + a^2(1 - e^2)} = a.$$

Figure to be illustrated

A horizontal ellipse $x^2/a^2 + y^2/b^2 = 1$ centred at origin O. Mark vertices A(a,0), A'(-a,0); minor-axis ends B(0,b), B'(0,-b); foci S(ae,0), S'(-ae,0). Show a point P with both focal radii SP and S'P drawn and labelled, summing to 2a. Mark the latus rectum as a vertical chord through S of length $2b^2/a$, and the directrices $x = \pm a/e$ as dashed vertical lines outside the vertices.

Section 4: Worked Examples

Example 1 — [CBSE Board] For the ellipse $\frac{x^2}{25} + \frac{y^2}{9} = 1$, find a , b , the eccentricity, the foci, the vertices, the lengths of the axes, and the length of the latus rectum.

The larger denominator (25) is under x^2 , so the major axis is along the x -axis and

$$a^2 = 25 \Rightarrow a = 5, \quad b^2 = 9 \Rightarrow b = 3.$$

$$c^2 = a^2 - b^2 = 25 - 9 = 16 \Rightarrow c = 4, \quad e = \frac{c}{a} = \frac{4}{5}.$$

- Foci: $(\pm 4, 0)$
- Vertices: $(\pm 5, 0)$; minor-axis ends $(0, \pm 3)$
- Major axis = $2a = 10$; minor axis = $2b = 6$
- Latus rectum = $\frac{2b^2}{a} = \frac{2 \cdot 9}{5} = \frac{18}{5}$ units

Example 2 — [JEE Main, quick-elimination] In an ellipse, the distance between the foci equals the length of the minor axis. Find the eccentricity.

Translate directly: $2c = 2b \Rightarrow c = b$. Now use $c^2 = a^2 - b^2$ with $b = c$:

$$c^2 = a^2 - c^2 \Rightarrow 2c^2 = a^2 \Rightarrow \frac{c^2}{a^2} = \frac{1}{2} \Rightarrow e = \frac{1}{\sqrt{2}}.$$

Speed note: whenever a condition links two of $\{a, b, c\}$, push it straight into $c^2 = a^2 - b^2$ and read off $e = c/a$. No coordinates needed.

Example 3 — [JEE Main, multi-concept] An ellipse with centre at the origin and major axis along the x -axis has its latus rectum equal to half its major axis. Find its eccentricity and b/a , then find the two focal distances of the point on the ellipse whose x -coordinate is $a/2$.

“Latus rectum = half the major axis”:

$$\frac{2b^2}{a} = \frac{1}{2}(2a) = a \Rightarrow 2b^2 = a^2 \Rightarrow \frac{b^2}{a^2} = \frac{1}{2} \Rightarrow \frac{b}{a} = \frac{1}{\sqrt{2}}.$$

Then $e^2 = 1 - \frac{b^2}{a^2} = 1 - \frac{1}{2} = \frac{1}{2} \Rightarrow e = \frac{1}{\sqrt{2}}.$

For the point with $x = a/2$, use the focal-radii formulas:

$$SP = a - ex = a - \frac{1}{\sqrt{2}} \cdot \frac{a}{2} = a \left(1 - \frac{1}{2\sqrt{2}}\right), \quad S'P = a + ex = a \left(1 + \frac{1}{2\sqrt{2}}\right).$$

As a check, $SP + S'P = 2a$. ✓ *Concept link:* one problem ties the latus-rectum relation, the eccentricity relation, and the focal-radii formulas together — and the $2a$ sum gives a free sanity check.

Example 4 — [JEE Advanced] S and S' are the foci of an ellipse and B is one end of its minor axis. If $\triangle SBS'$ is equilateral, find the eccentricity and express the latus rectum in terms of a .

Reasoning. The triangle has vertices $S(ae, 0)$, $S'(-ae, 0)$, and $B(0, b)$. Its base is $SS' = 2ae$. The two slant sides are equal by symmetry, each being the focus-to-minor-extremity distance, which we proved equals the semi-major axis: $SB = S'B = a$.

For the triangle to be **equilateral**, all three sides match:

$$SS' = SB \Rightarrow 2ae = a \Rightarrow \boxed{e = \frac{1}{2}}.$$

Then

$$b^2 = a^2(1 - e^2) = a^2 \left(1 - \frac{1}{4}\right) = \frac{3a^2}{4},$$

so the latus rectum is

$$\ell = \frac{2b^2}{a} = \frac{2}{a} \cdot \frac{3a^2}{4} = \boxed{\frac{3a}{2}}.$$

Why it is “Advanced”: the leap is recognising $SB = a$ — that the slant side is the semi-major axis, not something you must recompute — which turns a messy triangle condition into a one-line equation. Pure definition, no calculus.

Section 5: Practice Exercises

1. **[CBSE]** Find the foci, eccentricity, and latus rectum of $\frac{x^2}{16} + \frac{y^2}{36} = 1$. (Watch the orientation.)
2. **[JEE Main]** Find the equation of the ellipse with centre at the origin, major axis along the x -axis, eccentricity $\frac{3}{5}$, and major axis of length 10.
3. **[JEE Main]** If the latus rectum of an ellipse equals its minor axis, find its eccentricity.
4. **[JEE Main]** A point on $\frac{x^2}{49} + \frac{y^2}{24} = 1$ has $x = 3$. Find its two focal distances.

5. **[JEE Advanced]** S and S' are the foci and B an end of the minor axis of an ellipse. If $\angle SBS' = 90^\circ$, find the eccentricity.

Answers: 1. Larger denominator under $y^2 \Rightarrow$ major axis along y ; $a^2 = 36, b^2 = 16, c^2 = 20, c = 2\sqrt{5}$; foci $(0, \pm 2\sqrt{5})$; $e = \frac{\sqrt{5}}{3}$; latus rectum $= \frac{2b^2}{a} = \frac{32}{6} = \frac{16}{3}$. 2. $a = 5, e = \frac{3}{5} \Rightarrow c = 3, b^2 = 16; \frac{x^2}{25} + \frac{y^2}{16} = 1$. 3. $\frac{2b^2}{a} = 2b \Rightarrow b = a?$ — careful: $\frac{2b^2}{a} = 2b \Rightarrow b = a$, impossible for a true ellipse, so re-read as latus rectum = semi-minor axis b : $\frac{2b^2}{a} = b \Rightarrow b = \frac{a}{2} \Rightarrow e = \sqrt{1 - \frac{1}{4}} = \frac{\sqrt{3}}{2}$. 4. $a = 7, b^2 = 24, c^2 = 25, e = \frac{5}{7}$; $SP = a - ex = 7 - \frac{5}{7}(3) = \frac{34}{7}, S'P = 7 + \frac{15}{7} = \frac{64}{7}$ (sum $= 14 = 2a$ ✓). 5. $SB = S'B = a$ and $SS' = 2ae$; right angle at $B \Rightarrow SB^2 + S'B^2 = SS'^2 \Rightarrow 2a^2 = 4a^2e^2 \Rightarrow e = \frac{1}{\sqrt{2}}$.

Section 6: Crack the MCQ

Q1. For $\frac{x^2}{9} + \frac{y^2}{25} = 1$, the foci lie on

- (A) the x -axis (B) the y -axis (C) the line $y = x$ (D) at the origin

Answer: (B). The larger denominator (25) sits under y^2 , so the major axis — and therefore the foci — lie on the y -axis. *Trap (A):* assuming the foci are always on the x -axis, the single most common ellipse error. *Trap (C)/(D):* foci are never off-axis or at the centre for a standard ellipse.

Q2. The eccentricity of $\frac{x^2}{4} + \frac{y^2}{3} = 1$ is

- (A) $\frac{1}{2}$ (B) $\frac{1}{\sqrt{3}}$ (C) $\frac{\sqrt{3}}{2}$ (D) $\frac{1}{4}$

Answer: (A). $a^2 = 4, b^2 = 3, c^2 = 1$, so $e = \frac{c}{a} = \frac{1}{2}$. *Trap (B):* taking $e = \sqrt{b^2/a^2}$ instead of $\sqrt{1 - b^2/a^2}$. *Trap (C):* computing $\sqrt{1 - 1/4}$ from a wrong b^2/a^2 . *Trap (D):* reporting c^2/a^2 without the square root.

Q3. The length of the latus rectum of $\frac{x^2}{36} + \frac{y^2}{16} = 1$ is

- (A) $\frac{16}{3}$ (B) $\frac{8}{3}$ (C) $\frac{32}{3}$ (D) 8

Answer: (A). Major axis along x : $a = 6, b^2 = 16$, so $\ell = \frac{2b^2}{a} = \frac{32}{6} = \frac{16}{3}$. *Trap (B):* using $\frac{b^2}{a}$ (forgetting the 2). *Trap (C):* using $\frac{2b^2}{b}$ or mixing a and b . *Trap (D):* taking the minor axis $2b$ instead.

Q4. If the foci of an ellipse coincide (merge into one point), the curve becomes

- (A) a parabola (B) a circle (C) a straight line (D) a hyperbola

Answer: (B). Merged foci means $c = 0$, so $e = 0$ and $a = b$ — a **circle**, the limiting ellipse. *Trap (A):* the parabola is the $e \rightarrow 1$ limit, not $e \rightarrow 0$. *Trap (C):* the line is the

degenerate case $2a = 2c$, not $c = 0$. *Trap (D)*: hyperbolas have $e > 1$, the opposite extreme.

Section 7: Common Student Pitfalls & Pro-Tips

- **Assuming the major axis is always horizontal.** It lives under the *larger denominator*. If y^2 has the bigger denominator, the major axis, foci, and directrices all swing onto the y -axis. Check before you compute.
- **Sign in the relationship.** For an ellipse $c^2 = a^2 - b^2$ (minus). The plus sign belongs to the hyperbola. Mixing them flips your eccentricity from below 1 to above it.
- **Latus rectum slips.** It is $\frac{2b^2}{a}$ — full numerator $2b^2$, denominator the *semi-major axis* a . Dropping the 2 or dividing by $2a$ are both common.
- **Confusing e 's formula.** Eccentricity is $\sqrt{1 - b^2/a^2}$, not $\sqrt{b^2/a^2}$. A quick guard: an ellipse must give $0 < e < 1$; if your answer is ≥ 1 , you used the wrong form.

Pro-Tip. Lock a four-step reflex for any standard ellipse: (1) **bigger denominator = major axis direction = where the foci are**; (2) $c^2 = a^2 - b^2$; (3) $e = c/a$; (4) $\ell = 2b^2/a$. And memorise the one beautiful shortcut — **focus to minor-axis tip** = a — it collapses half the “Advanced” triangle problems into a single line.

Section 8: Cheat Sheet Box

- **Definition:** sum of distances from two foci = $2a$ (constant); eccentricity $0 < e < 1$.
- **Standard form ($a > b$):** $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$; major axis under the larger denominator.
- **Relationships:** $c = ae$, $c^2 = a^2 - b^2$, $b^2 = a^2(1 - e^2)$, $e = \sqrt{1 - b^2/a^2}$.
- **Features:** vertices $(\pm a, 0)$, foci $(\pm ae, 0)$, directrices $x = \pm a/e$, latus rectum $\frac{2b^2}{a}$.
- **Focal radii:** $SP = a - ex$, $S'P = a + ex$; their sum is always $2a$.
- **Key facts:** focus-to-minor-tip = a ; circle is the $e = 0$ limit; parabola is the $e \rightarrow 1$ boundary.
- **Mnemonics:** “Bigger denominator points to the foci.” “Ellipse minus, hyperbola plus.” “Latus rectum: two-bee-squared over a .”

The Hyperbola

Exam Relevance: CBSE Boards ask 1–4 marks — read a standard equation for foci, axes, eccentricity, latus rectum, and asymptotes, or build the equation from data; the difference-of-distances derivation is a recurring long-answer. JEE Main carries 1–2 questions on eccentricity and asymptote relationships. JEE Advanced fuses it with focal and asymptotic reasoning. The hyperbola also describes the path of an unbound comet and the shadow cast by a lampshade on a wall.

Section 1: Concept Intuition

The three conics are one family separated by a single dial — the eccentricity e . Turn it below 1 and the curve closes into an ellipse. Set it exactly at 1 and it opens into a parabola. Push it past 1 and the curve splits into **two separate branches** sweeping away from each other forever. That two-branched, $e > 1$ curve is the **hyperbola**.

Where the ellipse was built on the *sum* of distances to two foci, the hyperbola is built on their **difference**. A point moves so that $|\text{distance to } S \text{ minus distance to } S'|$ stays constant ($= 2a$). On the right branch you are always closer to the right focus by exactly $2a$; on the left branch, closer to the left focus by the same $2a$. The constant difference is what carves out the two arms.

Think about it this way... imagine two cricket stadiums far apart, each with a loudspeaker, and the speakers blast the same sound a fixed split-second apart. Stand somewhere and you hear two booms; the *gap* between them tells you the difference in your distances to the two speakers. Every spot with the *same* time-gap lies on one hyperbola branch. This is not a toy example — it is exactly how old LORAN ship navigation and modern GPS-style positioning work: fix the time differences from several transmitters and you sit on intersecting hyperbolas, and where they cross is your location. The hyperbola is the geometry of “constant difference.”

The other unmistakable feature of a hyperbola is its pair of **asymptotes** — two straight lines through the centre that the branches hug ever more closely but never touch, like two railway tracks the curve runs parallel to as it races off to infinity. The asymptotes form an X through the centre, and the hyperbola threads through the opposite pairs of wedges. When the two asymptotes are perpendicular, the hyperbola is called **rectangular** (or equilateral), the most symmetric case.

Limiting conditions & assumptions.

- $e > 1$, **always**. This is the hyperbola’s signature. If your computed e comes out ≤ 1 , you have made a sign error somewhere.
- **The two branches never join**. Unlike the single closed ellipse, the hyperbola is genuinely two pieces; the region between the vertices contains no part of the curve.
- **The constant difference must be less than the focal separation:** $2a < 2c$. If they were equal, the “curve” would degenerate into two rays.
- **Standard equations assume the tidy frame** — centre at origin, transverse axis along a coordinate axis. Off-centre or rotated hyperbolas need shifted/rotated coordinates (beyond Class 11 scope).

Section 2: Key Definitions & Formulas

Definition. A hyperbola is the locus of a point the *difference* of whose distances from two fixed foci is constant ($= 2a$). Equivalently, focus-directrix form with eccentricity $e > 1$.

Standard equation (centre at origin, transverse axis along x -axis):

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

Here a is the **semi-transverse axis** and b the **semi-conjugate axis**. Note there is **no requirement that** $a > b$ — either may be larger.

The defining relationships (mind the + sign):

$$c = ae, \quad c^2 = a^2 + b^2, \quad b^2 = a^2(e^2 - 1), \quad e = \frac{c}{a} = \sqrt{1 + \frac{b^2}{a^2}}$$

Key features of $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$:

Feature	Value
Centre	$(0, 0)$
Vertices (transverse-axis ends)	$(\pm a, 0)$
Foci S, S'	$(\pm ae, 0)$
Transverse / conjugate axis length	$2a / 2b$
Eccentricity	$e = \sqrt{1 + b^2/a^2}$
Directrices	$x = \pm a/e$
Latus rectum length	$\ell = 2b^2/a$
Asymptotes	$y = \pm \frac{b}{a}x$
Focal radii of $P(x, y)$	$PS = ex - a, \quad PS' = ex + a$

If the positive term is under y^2 — that is $\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1$ — the transverse axis is the y -axis, foci are $(0, \pm ae)$, and asymptotes are $y = \pm \frac{a}{b}x$.

The positive term — not the larger denominator — picks the transverse axis. This is the hyperbola trap: in an *ellipse* the larger denominator wins, but in a *hyperbola* the transverse axis follows the variable with the **plus** sign, whatever the denominators are.

Section 3: CBSE Board Derivation

Deriving the standard equation $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$.

Set up the frame. Centre at the origin, foci at $S(c, 0)$ and $S'(-c, 0)$, and let the constant difference of focal distances be $2a$. For a point $P(x, y)$ on the branch nearer S' ,

$$S'P - SP = 2a \implies \sqrt{(x+c)^2 + y^2} - \sqrt{(x-c)^2 + y^2} = 2a.$$

This is the ellipse derivation with a sign changed — so the algebra runs in perfect parallel. Move one radical across and square:

$$\sqrt{(x+c)^2 + y^2} = 2a + \sqrt{(x-c)^2 + y^2},$$

$$(x+c)^2 + y^2 = 4a^2 + 4a\sqrt{(x-c)^2 + y^2} + (x-c)^2 + y^2.$$

Cancel the common $x^2 + c^2 + y^2$; the cross terms $+2cx$ and $-2cx$ survive:

$$4cx - 4a^2 = 4a\sqrt{(x-c)^2 + y^2} \implies \frac{cx}{a} - a = \sqrt{(x-c)^2 + y^2}.$$

Square once more:

$$\frac{c^2x^2}{a^2} - 2cx + a^2 = x^2 - 2cx + c^2 + y^2.$$

Cancel $-2cx$, then group:

$$x^2\left(\frac{c^2}{a^2} - 1\right) - y^2 = c^2 - a^2.$$

Now the quantity $c^2 - a^2$ is **positive** here (because $c > a$ for a hyperbola), so we name it b^2 . The bracket becomes $\frac{c^2 - a^2}{a^2} = \frac{b^2}{a^2}$:

$$\frac{b^2}{a^2}x^2 - y^2 = b^2 \implies \boxed{\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1.}$$

The substitution again is the relationship — but with the flipped sign:

$$b^2 = c^2 - a^2 \iff c^2 = a^2 + b^2.$$

Dividing by a^2 and using $e = c/a$ gives $\frac{b^2}{a^2} = e^2 - 1$, i.e. $b^2 = a^2(e^2 - 1)$, which forces $e > 1$.

Ellipse vs hyperbola — the whole difference in one box. Ellipse: focal distances **add** to $2a$; relationship $c^2 = a^2 - b^2$; $e < 1$. Hyperbola: focal distances **subtract** to $2a$; relationship $c^2 = a^2 + b^2$; $e > 1$. The latus rectum formula $\frac{2b^2}{a}$ is *identical* for both.

Length of the latus rectum. On the line $x = c = ae$,

$$\frac{(ae)^2}{a^2} - \frac{y^2}{b^2} = 1 \implies \frac{y^2}{b^2} = e^2 - 1 = \frac{b^2}{a^2} \implies y = \pm \frac{b^2}{a},$$

$$\text{so } \ell = \frac{2b^2}{a}.$$

The asymptotes. Solve the equation for y :

$$y^2 = b^2 \left(\frac{x^2}{a^2} - 1 \right) = \frac{b^2}{a^2} x^2 \left(1 - \frac{a^2}{x^2} \right).$$

As $|x| \rightarrow \infty$, the term $a^2/x^2 \rightarrow 0$, so $y^2 \rightarrow \frac{b^2}{a^2} x^2$, giving

$$y \rightarrow \pm \frac{b}{a} x.$$

The branches crowd ever closer to these two lines without meeting them — the asymptotes are the hyperbola's “rails at infinity.” When $a = b$ they are $y = \pm x$, perpendicular: the **rectangular** hyperbola, for which $e = \sqrt{2}$.

Figure to be illustrated

A horizontal hyperbola $x^2/a^2 - y^2/b^2 = 1$ with two branches opening left and right. Mark centre O, vertices A(a,0), A'(-a,0), foci S(ae,0), S'(-ae,0) outside the vertices. Draw the two asymptotes $y = \pm(b/a)x$ as dashed lines through the origin forming an X, with the branches hugging them. Show the conjugate axis on the y-axis from (0,-b) to (0,b), the latus rectum as a vertical chord through S of length $2b^2/a$, and a point P with both focal radii drawn, labelled with difference $2a$.

Section 4: Worked Examples

Example 1 — [CBSE Board] For the hyperbola $\frac{x^2}{16} - \frac{y^2}{9} = 1$, find a , b , the eccentricity, the foci, the vertices, the lengths of the transverse and conjugate axes, the latus rectum, and the equations of the asymptotes.

The positive term is under x^2 , so the transverse axis is the x -axis:

$$a^2 = 16 \Rightarrow a = 4, \quad b^2 = 9 \Rightarrow b = 3.$$

$$c^2 = a^2 + b^2 = 16 + 9 = 25 \Rightarrow c = 5, \quad e = \frac{c}{a} = \frac{5}{4}.$$

- Foci: $(\pm 5, 0)$; vertices: $(\pm 4, 0)$
- Transverse axis $= 2a = 8$; conjugate axis $= 2b = 6$
- Latus rectum $= \frac{2b^2}{a} = \frac{18}{4} = \frac{9}{2}$ units
- Asymptotes: $y = \pm \frac{3}{4}x$

Note $e = 5/4 > 1$ — the built-in sanity check passes.

Example 2 — [JEE Main, quick-elimination] The transverse and conjugate axes of a hyperbola are equal in length. Find its eccentricity.

Equal axes means $2a = 2b \Rightarrow a = b$. Then

$$e^2 = 1 + \frac{b^2}{a^2} = 1 + 1 = 2 \Rightarrow e = \sqrt{2}.$$

Speed note: $a = b$ is the **rectangular** hyperbola, and $e = \sqrt{2}$ is worth memorising outright — it appears constantly. Recognising “equal axes / perpendicular asymptotes / rectangular” should trigger $\sqrt{2}$ instantly.

Example 3 — [JEE Main, multi-concept] A hyperbola with transverse axis along the x -axis has eccentricity $\frac{\sqrt{13}}{2}$ and latus rectum 9. Find its equation, then the two focal distances of the point on it with $x = 4$.

From $e = \frac{\sqrt{13}}{2}$: $e^2 = \frac{13}{4} = 1 + \frac{b^2}{a^2} \Rightarrow \frac{b^2}{a^2} = \frac{9}{4}$, so $b^2 = \frac{9}{4}a^2$. From latus rectum $\frac{2b^2}{a} = 9$: substitute $b^2 = \frac{9}{4}a^2$:

$$\frac{2 \cdot \frac{9}{4}a^2}{a} = 9 \Rightarrow \frac{9a}{2} = 9 \Rightarrow a = 2.$$

Then $a^2 = 4$ and $b^2 = \frac{9}{4} \cdot 4 = 9$. The equation is

$$\frac{x^2}{4} - \frac{y^2}{9} = 1.$$

For the point with $x = 4$, use the focal radii (note: **add/subtract** a , with ex first):

$$PS = ex - a = \frac{\sqrt{13}}{2} \cdot 4 - 2 = 2\sqrt{13} - 2, \quad PS' = ex + a = 2\sqrt{13} + 2.$$

Check the *difference*: $PS' - PS = 4 = 2a$. ✓ *Concept link*: eccentricity, latus rectum, and focal radii in one chain — and the difference $2a$ gives a free check, the hyperbola’s analogue of the ellipse’s sum check.

Example 4 — [JEE Advanced] The angle between the asymptotes of a hyperbola (the angle containing the transverse axis) is 60° . Find its eccentricity. Then state the eccentricity if instead that angle is the obtuse one of 120° .

Reasoning. The asymptotes are $y = \pm \frac{b}{a}x$. Each makes an angle α with the x -axis where $\tan \alpha = \frac{b}{a}$. The angle *between* the two asymptotes, measured across the transverse axis, is 2α .

Case 1: $2\alpha = 60^\circ \Rightarrow \alpha = 30^\circ$.

$$\frac{b}{a} = \tan 30^\circ = \frac{1}{\sqrt{3}} \Rightarrow e^2 = 1 + \frac{b^2}{a^2} = 1 + \frac{1}{3} = \frac{4}{3} \Rightarrow e = \frac{2}{\sqrt{3}}.$$

Case 2: $2\alpha = 120^\circ \Rightarrow \alpha = 60^\circ$.

$$\frac{b}{a} = \tan 60^\circ = \sqrt{3} \Rightarrow e^2 = 1 + 3 = 4 \Rightarrow e = 2.$$

Why it is “Advanced”: the subtlety is that “the angle between the asymptotes” is ambiguous — the two lines make *two* supplementary angles (60° and 120°), and which one “contains the transverse axis” decides whether you use α or its complement. A careful student gives both eccentricities; a careless one gives one and loses half the marks. No calculus — pure asymptote geometry.

Section 5: Practice Exercises

1. **[CBSE]** Find the foci, vertices, eccentricity, latus rectum, and asymptotes of $\frac{x^2}{9} - \frac{y^2}{27} = 1$.
2. **[JEE Main]** Find the equation of the hyperbola with foci $(\pm 5, 0)$ and transverse axis of length 6.
3. **[JEE Main]** If the latus rectum of a hyperbola equals half its transverse axis, find its eccentricity.
4. **[JEE Main]** A point on $\frac{x^2}{4} - \frac{y^2}{5} = 1$ has $x = 6$. Find its two focal distances and verify their difference.
5. **[JEE Advanced]** The eccentricity of a hyperbola is the reciprocal of that of the ellipse $\frac{x^2}{25} + \frac{y^2}{16} = 1$... is impossible (explain why), so instead: find the eccentricity of a hyperbola whose conjugate axis equals the distance between its foci.

Answers: 1. $a^2 = 9, b^2 = 27, c^2 = 36, c = 6$; foci $(\pm 6, 0)$; vertices $(\pm 3, 0)$; $e = 2$; latus rectum $= \frac{2 \cdot 27}{3} = 18$; asymptotes $y = \pm \sqrt{3}x$. 2. $c = 5, a = 3 \Rightarrow b^2 = c^2 - a^2 = 16$; $\frac{x^2}{9} - \frac{y^2}{16} = 1$. 3. $\frac{2b^2}{a} = \frac{1}{2}(2a) = a \Rightarrow 2b^2 = a^2 \Rightarrow \frac{b^2}{a^2} = \frac{1}{2} \Rightarrow e = \sqrt{1 + \frac{1}{2}} = \sqrt{\frac{3}{2}} = \frac{\sqrt{6}}{2}$. 4. $a = 2, b^2 = 5, c^2 = 9, e = \frac{3}{2}$; $PS = ex - a = \frac{3}{2}(6) - 2 = 7, PS' = ex + a = 11$; difference $= 4 = 2a$ ✓. 5. The ellipse has $e < 1$, so its reciprocal > 1 — which *would* be valid for a hyperbola; the “impossible” framing is the trap, the reciprocal route is fine, but the cleaner intended part: conjugate axis = focal distance $\Rightarrow 2b = 2c \Rightarrow b = c$; but $c^2 = a^2 + b^2$ gives $b^2 = a^2 + b^2$, impossible — so **no such hyperbola exists** (the conjugate axis is always shorter than the focal separation). Recognising the impossibility is the answer.

Section 6: Crack the MCQ

Q1. For the hyperbola $\frac{y^2}{9} - \frac{x^2}{16} = 1$, the transverse axis lies along

- (A) the x -axis (B) the y -axis (C) the line $y = x$ (D) neither axis

Answer: (B). The **positive** term is under y^2 , so the transverse axis is the **y -axis** — even though $16 > 9$. *Trap (A):* applying the ellipse rule “larger denominator wins” (16 under x^2). *Trap (C)/(D):* the transverse axis of a standard hyperbola is always a coordinate axis. This question exists purely to catch the ellipse-habit.

Q2. The eccentricity of $\frac{x^2}{16} - \frac{y^2}{9} = 1$ is

- (A) $\frac{3}{4}$ (B) $\frac{4}{5}$ (C) $\frac{5}{4}$ (D) $\frac{\sqrt{7}}{4}$

Answer: (C). $c^2 = 16 + 9 = 25$, $c = 5$, $e = c/a = 5/4 > 1$. *Trap (A):* reporting b/a . *Trap (B):* using $c^2 = a^2 - b^2$ (the *ellipse* relation), giving $c = \sqrt{7}$ then mis-simplifying — the deadly sign import. *Trap (D):* the $\sqrt{7}$ that the wrong minus sign produces.

Q3. The asymptotes of $\frac{x^2}{25} - \frac{y^2}{16} = 1$ are

- (A) $y = \pm \frac{5}{4}x$ (B) $y = \pm \frac{4}{5}x$ (C) $y = \pm x$ (D) $y = \pm \frac{16}{25}x$

Answer: (B). Asymptotes are $y = \pm \frac{b}{a}x = \pm \frac{4}{5}x$. *Trap (A):* inverting to a/b . *Trap (C):* assuming rectangular without checking $a = b$. *Trap (D):* using b^2/a^2 instead of b/a .

Q4. A hyperbola with $e = \sqrt{2}$ is called

- (A) conjugate (B) rectangular (C) degenerate (D) auxiliary

Answer: (B). $e = \sqrt{2}$ forces $b^2/a^2 = e^2 - 1 = 1$, so $a = b$ and the asymptotes are perpendicular — a **rectangular (equilateral)** hyperbola. *Trap (A):* the conjugate hyperbola is a different curve, not an eccentricity condition. *Trap (C):* degenerate conics are point/line collapses, unrelated. *Trap (D):* “auxiliary circle” belongs to the ellipse.

Section 7: Common Student Pitfalls & Pro-Tips

- **Importing the ellipse’s minus sign.** For a hyperbola $c^2 = a^2 + b^2$ (**plus**). Using the ellipse’s minus is the single most destructive error — it silently produces an $e < 1$, which is impossible for a hyperbola.
- **Letting the larger denominator pick the axis.** That is the *ellipse* rule. For a hyperbola, the **positive term** names the transverse axis, regardless of denominator sizes.
- **Inverting the asymptote slope.** For $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ the asymptotes are $y = \pm \frac{b}{a}x$ — b over a , not a over b .
- **Forgetting both supplementary angles between asymptotes.** “Angle between asymptotes” is ambiguous; check whether the angle contains the transverse axis, and give both cases when asked.

Pro-Tip. Build a hyperbola reflex that *contrasts* with the ellipse: (1) **positive term = transverse axis = where the foci are**; (2) $c^2 = a^2 + b^2$ (plus!); (3) $e = c/a$ and confirm $e > 1$; (4) asymptotes $y = \pm \frac{b}{a}x$; (5) latus rectum $\frac{2b^2}{a}$ (same as ellipse). And lock $e = \sqrt{2} \Leftrightarrow$ rectangular as an instant-recall fact.

Section 8: Cheat Sheet Box

- **Definition:** difference of distances from two foci = $2a$ (constant); eccentricity $e > 1$; two branches.
- **Standard form:** $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$; the **positive term** sets the transverse axis (a need not exceed b).
- **Relationships:** $c = ae$, $c^2 = a^2 + b^2$, $b^2 = a^2(e^2 - 1)$, $e = \sqrt{1 + b^2/a^2}$.

- **Features:** vertices $(\pm a, 0)$, foci $(\pm ae, 0)$, directrices $x = \pm a/e$, latus rectum $\frac{2b^2}{a}$.
- **Asymptotes:** $y = \pm \frac{b}{a}x$; perpendicular $\Leftrightarrow a = b \Leftrightarrow$ rectangular $\Leftrightarrow e = \sqrt{2}$.
- **Focal radii:** $PS = ex - a$, $PS' = ex + a$; their *difference* is always $2a$.
- **Mnemonics:** *"Hyperbola plus, ellipse minus."* *"Positive term points to the foci."* *"Asymptote slope is bee-over-ay."* *"Rectangular means root-two."*

Chapter 10: Conic Sections — Supplement

Parametric Forms and the Auxiliary Circle

Exam Relevance: CBSE Boards do not ask for the parametric form by name, but the eccentric angle is the fastest legal route through the 3–4 mark “find the point on the ellipse such that...” questions that Boards ask every year. JEE Main tests it directly and repeatedly — a point handed to you as $(a \cos \theta, b \sin \theta)$, a chord joining two eccentric angles, or the hyperbola’s $(a \sec \theta, b \tan \theta)$ with a branch question attached. JEE Advanced builds entire problems on the auxiliary circle, because it converts ellipse geometry into circle geometry, and circle geometry is easy. NEET does not test mathematics at all.

Section 1: Concept Intuition

You already met one parametric form in **The Parabola**: the point $(at^2, 2at)$, which let you replace two coordinates by one number t and made the focal-chord proof collapse to the single condition $t_1 t_2 = -1$. That was not a trick special to the parabola. Every conic has a one-number description, and for the ellipse and the hyperbola that number carries beautiful geometry.

Start with the easiest case. Picture the giant wheel at a mela — a big vertical circle with seats bolted around the rim. If a friend asks where you are, you do not say “four metres right of centre and three metres up.” You say “quarter turn past the bottom.” **One angle locates you completely, because the radius never changes — and that is exactly what a parametric form is: trading two coordinates for one parameter that the curve’s own symmetry supplies.** For a circle of radius r centred at the origin the seat at angle θ sits at $(r \cos \theta, r \sin \theta)$, and Pythagoras returns $x^2 + y^2 = r^2$ automatically.

Now the step that unlocks the ellipse. Point a torch at that vertical wheel and look at the shadow it throws on the ground. The shadow is not a circle any more — it is squashed in one direction and stretched in none. It is an **ellipse**. Every seat on the wheel throws its shadow to exactly one point of the ellipse, straight down. So each point of the ellipse still has an angle attached to it: not its own angle, but the angle of the seat that made it.

Think about it this way... the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is the circle $x^2 + y^2 = a^2$ with every ordinate multiplied by the fixed factor b/a . Nothing horizontal changes; everything vertical shrinks in the same proportion. So take the circle point $Q = (a \cos \theta, a \sin \theta)$, shrink its height by b/a , and you land on

$$P = (a \cos \theta, \frac{b}{a} \cdot a \sin \theta) = (a \cos \theta, b \sin \theta),$$

which satisfies the ellipse equation for every θ . That circle of radius a — the circle drawn on the **major axis as diameter** — is called the **auxiliary circle**, and the angle θ it hands over is called the **eccentric angle** of P .

Here is the single most examined subtlety in this whole unit. The eccentric angle θ belongs to Q on the circle, **not** to P on the ellipse. If you join the centre O to P and measure the angle that OP makes with the positive x -axis, you get a different angle ϕ , related by $\tan \phi = \frac{b}{a} \tan \theta$. The two agree only at the four axis ends. Students who treat θ as “the angle of P ” lose marks on this every single session.

The hyperbola needs a different pair of functions, and the reason is arithmetic before it is geometric. The ellipse equation wants two quantities whose squares **add** to 1, and $\cos^2 \theta + \sin^2 \theta = 1$ supplies them. The hyperbola equation wants two quantities whose squares **subtract** to 1, and the identity that does that is $\sec^2 \theta - \tan^2 \theta = 1$. So the parametric point is $(a \sec \theta, b \tan \theta)$. The hyperbola has an auxiliary circle too — again $x^2 + y^2 = a^2$, drawn on the transverse axis as diameter — but the construction reaches the curve through the **tangent** to the circle rather than through a vertical shadow.

Limiting conditions and assumptions.

- The eccentric angle θ is the angle at the centre made by the **auxiliary-circle point**, not by the ellipse point. Never read θ off the ellipse directly.
- $(a \cos \theta, b \sin \theta)$ describes an ellipse and **only** an ellipse. Feeding it to a hyperbola produces a point that is not on the curve — a mistake that always looks plausible and is always wrong.
- For the hyperbola, $\theta = \frac{\pi}{2}$ and $\theta = \frac{3\pi}{2}$ are excluded, because $\sec \theta$ and $\tan \theta$ are both undefined there.
- The sign of $\sec \theta$ decides the branch: $\cos \theta > 0$ puts the point on the right branch, $\cos \theta < 0$ on the left. A single parametric formula covers both branches, which is why it beats solving for y .
- All the forms below assume the tidy frame used throughout this chapter — centre (or vertex) at the origin, axes along the coordinate axes.

Section 2: Key Definitions & Formulas

Definition (parametric form). A parametric form of a curve is a pair of expressions $x = f(u)$, $y = g(u)$ such that as the single parameter u runs over its allowed range, the point $(f(u), g(u))$ traces the curve exactly once and satisfies the curve’s Cartesian equation identically.

Definition (auxiliary circle). For the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ ($a > b$), the auxiliary circle is the circle described on the major axis as diameter:

$$x^2 + y^2 = a^2.$$

For the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, the auxiliary circle is the circle on the transverse axis as diameter, again $x^2 + y^2 = a^2$.

- a — semi-major axis (ellipse) or semi-transverse axis (hyperbola).
- The auxiliary circle is **not** $x^2 + y^2 = a^2 + b^2$ and **not** $x^2 + y^2 = b^2$; those are different objects.

Definition (eccentric angle). Let P lie on the ellipse. Drop the perpendicular PN from P to the major axis and produce it to meet the auxiliary circle at Q on the same side of the major axis. Then $\theta = \angle QON$, measured anticlockwise from the positive x -axis, is the eccentric angle of P , and Q is the **corresponding point** of P .

Master table of parametric forms.

Conic	Standard equation	Parametric point	Parameter range	What the parameter means
Circle (origin)	$x^2 + y^2 = r^2$	$(r \cos \theta, r \sin \theta)$	$\theta \in [0, 2\pi)$	the actual angle at the centre
Circle (shifted)	$(x - h)^2 + (y - k)^2 = r^2$	$(h + r \cos \theta, k + r \sin \theta)$	$\theta \in [0, 2\pi)$	angle at the centre (h, k)
Parabola	$y^2 = 4ax$	$(at^2, 2at)$	$t \in \mathbb{R}$	$t = \frac{y}{2a}$; tangent slope is $\frac{1}{t}$
Ellipse	$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$	$(a \cos \theta, b \sin \theta)$	$\theta \in [0, 2\pi)$	eccentric angle on the auxiliary circle
Hyperbola	$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$	$(a \sec \theta, b \tan \theta)$	$\theta \in [0, 2\pi), \theta \neq \frac{\pi}{2}, \frac{3\pi}{2}$	angle of the auxiliary-circle point whose tangent meets the transverse axis at the foot of P
Rectangular hyperbola	$xy = c^2$	$(ct, \frac{c}{t})$	$t \in \mathbb{R}, t \neq 0$	$t > 0$ first quadrant, $t < 0$ third

The squash factor. With N the foot of the ordinate of P and Q the corresponding auxiliary-circle point,

$$\frac{PN}{QN} = \frac{b}{a}, \quad PQ = (a - b) \sin \theta.$$

Eccentric angle versus the true central angle. If ϕ is the angle OP makes with the positive x -axis, then

$$\tan \phi = \frac{b}{a} \tan \theta.$$

Equality $\phi = \theta$ holds only at $\theta = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$.

Chord joining two parametric points.

For the ellipse, the chord through the points with eccentric angles θ_1 and θ_2 is

$$\frac{x}{a} \cos \frac{\theta_1 + \theta_2}{2} + \frac{y}{b} \sin \frac{\theta_1 + \theta_2}{2} = \cos \frac{\theta_1 - \theta_2}{2}.$$

For the hyperbola, the chord through the points with parameters θ_1 and θ_2 is

$$\frac{x}{a} \cos \frac{\theta_1 - \theta_2}{2} - \frac{y}{b} \sin \frac{\theta_1 + \theta_2}{2} = \cos \frac{\theta_1 + \theta_2}{2}.$$

Focal-chord condition on the ellipse. The chord joining eccentric angles θ_1, θ_2 passes through the focus $(ae, 0)$ if and only if

$$e \cos \frac{\theta_1 + \theta_2}{2} = \cos \frac{\theta_1 - \theta_2}{2} \iff \tan \frac{\theta_1}{2} \tan \frac{\theta_2}{2} = \frac{e - 1}{e + 1}.$$

An alternative hyperbola parametrisation (for reference). Because $\cosh^2 u - \sinh^2 u = 1$ for the hyperbolic functions $\cosh u = \frac{1}{2}(e^u + e^{-u})$ and $\sinh u = \frac{1}{2}(e^u - e^{-u})$ — here e is the exponential base 2.718..., not the eccentricity — the point

$$(a \cosh u, b \sinh u), \quad u \in \mathbb{R}$$

also lies on $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. Since $\cosh u \geq 1$ always, this form sweeps out the **right branch only**; the left branch needs $(-a \cosh u, b \sinh u)$. It is the form used in physics and in higher mathematics, and it is worth recognising, but $(a \sec \theta, b \tan \theta)$ is the one CBSE and JEE expect.

Section 3: CBSE Board Derivation

Derivation: the auxiliary circle produces $(a \cos \theta, b \sin \theta)$

1. Draw the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ with $a > b > 0$, and draw the auxiliary circle $x^2 + y^2 = a^2$ on the same centre. *Reason: the circle has radius a , so it touches the ellipse at the two vertices $(\pm a, 0)$ and encloses it everywhere else.*
2. Take any point Q on the auxiliary circle and write it as $Q = (a \cos \theta, a \sin \theta)$. *Reason: every point of a circle of radius a has this form, with θ the genuine angle at the centre.*
3. Drop the perpendicular from Q to the major axis, meeting it at $N = (a \cos \theta, 0)$. Let P be the point where QN cuts the ellipse on the same side of the axis. *Reason: a vertical line meets the ellipse in at most two points, one above the axis and one below; take the one matching the sign of $\sin \theta$.*
4. P has the same abscissa as Q , namely $x = a \cos \theta$. Substitute into the ellipse equation:

$$\frac{a^2 \cos^2 \theta}{a^2} + \frac{y^2}{b^2} = 1 \implies \frac{y^2}{b^2} = 1 - \cos^2 \theta = \sin^2 \theta \implies y = b \sin \theta.$$

Reason: choosing the same sign as $\sin \theta$ keeps P on the same side of the axis as Q , which is how the correspondence was defined.

5. Hence $P = (a \cos \theta, b \sin \theta)$, and θ — an angle read off the **circle** — labels the point of the **ellipse**. That angle is the eccentric angle.

6. Divide the ordinates: $\frac{PN}{QN} = \frac{b \sin \theta}{a \sin \theta} = \frac{b}{a}$, a constant. *Reason: this is the statement that the ellipse is the auxiliary circle uniformly compressed towards the major axis in the fixed ratio $b : a$ — the shadow picture made exact.*

Figure brief: An ellipse $x^2/a^2 + y^2/b^2 = 1$ drawn inside its auxiliary circle $x^2 + y^2 = a^2$, both centred at O , touching at the vertices $(\pm a, 0)$. Mark a point Q on the circle in the first quadrant, the radius OQ with the angle θ at O from the positive x -axis, the perpendicular from Q to the major axis meeting it at N , and the point P where QN crosses the ellipse. Label PN and QN with the ratio $b : a$, and draw OP as a second, shallower ray with its angle ϕ marked, so that $\theta \neq \phi$ is visually obvious.

Derivation: why the hyperbola uses $\sec$ and $\tan$

1. The hyperbola equation is $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, so you need two functions whose squares **differ** by 1. *Reason: matching the equation term by term is what any parametrisation must do.*
2. The Pythagorean identity in its second form is $\sec^2 \theta - \tan^2 \theta = 1$. *Reason: divide $\cos^2 \theta + \sin^2 \theta = 1$ throughout by $\cos^2 \theta$.*
3. Set $\frac{x}{a} = \sec \theta$ and $\frac{y}{b} = \tan \theta$, i.e. $x = a \sec \theta$, $y = b \tan \theta$. Substituting gives $\sec^2 \theta - \tan^2 \theta = 1$, which is true for every admissible θ . *Reason: the equation is satisfied identically, which is exactly the requirement for a parametric form.*
4. Geometric meaning: on the auxiliary circle $x^2 + y^2 = a^2$ take $Q = (a \cos \theta, a \sin \theta)$ and draw the **tangent** to the circle at Q , namely $x \cos \theta + y \sin \theta = a$. It meets the transverse axis $y = 0$ at $N = (a \sec \theta, 0)$. *Reason: put $y = 0$ and solve, giving $x = a / \cos \theta$.*
5. The point of the hyperbola with parameter θ is the point vertically above or below N : $P = (a \sec \theta, b \tan \theta)$. *Reason: the abscissa is forced by step 4, and the ordinate then follows from the hyperbola equation.*
6. Because $|\sec \theta| \geq 1$, no parametric point ever lands in the empty strip $-a < x < a$ between the branches — the parametrisation automatically respects the shape of the curve.

Figure brief: A right-opening hyperbola $x^2/a^2 - y^2/b^2 = 1$ with its auxiliary circle $x^2 + y^2 = a^2$ drawn through the vertices $(\pm a, 0)$. Mark Q on the circle in the first quadrant with central angle θ , the tangent to the circle at Q extended to cut the x -axis at $N(a \sec \theta, 0)$, and the vertical through N meeting the right branch at $P(a \sec \theta, b \tan \theta)$. Show the dashed asymptotes faintly for context.

Procedure A: convert a Cartesian point on an ellipse into its eccentric angle

1. Confirm the point actually lies on the ellipse by substituting it. *Why: an eccentric angle exists only for points of the curve, and this catches copying errors before they cost you the whole question.*
2. Write $\cos \theta = \frac{x_1}{a}$ and $\sin \theta = \frac{y_1}{b}$. *Why: these are forced by comparing (x_1, y_1) with $(a \cos \theta, b \sin \theta)$.*
3. Read the quadrant from the **pair of signs**, not from one of them. *Why: $\cos \theta = \frac{1}{2}$ alone*

allows $\theta = \frac{\pi}{3}$ or $\frac{5\pi}{3}$; the sign of $\sin \theta$ settles it.

4. State θ in $[0, 2\pi)$ unless the question asks otherwise. *Why: examiners key the principal range.*

Procedure B: use a parameter to remove an unknown

1. Whenever a problem says “a variable point on the conic”, write that point in parametric form immediately. *Why: it replaces two unknowns tied by one equation with a single free unknown, and the constraint is then satisfied automatically.*
2. Express whatever the question asks for — a distance, a slope, an area — as a function of the single parameter. *Why: one variable means ordinary algebra or trigonometry, not simultaneous equations.*
3. Impose the remaining condition and solve for the parameter. *Why: the condition is now one equation in one unknown.*
4. Convert back to coordinates at the very end. *Why: answers are marked in coordinates; carrying θ to the final line loses presentation marks.*

Procedure C: decide which branch a hyperbola parameter lands on

1. Compute $\cos \theta$. *Why: the abscissa is $a \sec \theta$, whose sign is the sign of $\cos \theta$.*
2. If $\cos \theta > 0$, i.e. $\theta \in (-\frac{\pi}{2}, \frac{\pi}{2})$, the point is on the **right** branch; if $\cos \theta < 0$, i.e. $\theta \in (\frac{\pi}{2}, \frac{3\pi}{2})$, it is on the **left** branch. *Why: the two branches are separated exactly by the sign of x .*
3. Reject $\theta = \frac{\pi}{2}$ and $\theta = \frac{3\pi}{2}$. *Why: $\sec \theta$ and $\tan \theta$ are undefined there; geometrically the tangent at Q is vertical and never meets the transverse axis.*

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

For the ellipse $\frac{x^2}{25} + \frac{y^2}{9} = 1$, write the equation of the auxiliary circle, find the point P whose eccentric angle is $\frac{\pi}{3}$, and find the corresponding point Q on the auxiliary circle.

Here $a^2 = 25$ and $b^2 = 9$, and since $25 > 9$ the major axis is along the x -axis, so $a = 5$ and $b = 3$.

The auxiliary circle is drawn on the major axis as diameter, so its radius is $a = 5$:

$$x^2 + y^2 = 25.$$

The parametric point of the ellipse is $(a \cos \theta, b \sin \theta)$. At $\theta = \frac{\pi}{3}$:

$$P = \left(5 \cos \frac{\pi}{3}, 3 \sin \frac{\pi}{3} \right) = \left(5 \cdot \frac{1}{2}, 3 \cdot \frac{\sqrt{3}}{2} \right) = \left(\frac{5}{2}, \frac{3\sqrt{3}}{2} \right).$$

Check that P is on the ellipse:

$$\frac{(5/2)^2}{25} + \frac{(3\sqrt{3}/2)^2}{9} = \frac{25/4}{25} + \frac{27/4}{9} = \frac{1}{4} + \frac{3}{4} = 1.$$

The corresponding point Q has the same abscissa and lies on the circle of radius 5, so $Q = (a \cos \theta, a \sin \theta)$:

$$Q = \left(5 \cos \frac{\pi}{3}, 5 \sin \frac{\pi}{3} \right) = \left(\frac{5}{2}, \frac{5\sqrt{3}}{2} \right),$$

and $\left(\frac{5}{2}\right)^2 + \left(\frac{5\sqrt{3}}{2}\right)^2 = \frac{25}{4} + \frac{75}{4} = 25$, as required.

Notice the ordinates: $\frac{PN}{QN} = \frac{3\sqrt{3}/2}{5\sqrt{3}/2} = \frac{3}{5} = \frac{b}{a}$ — the squash factor, visible in the numbers.

Example 2 — [JEE Main Level]

The point $P \left(2, \frac{3\sqrt{3}}{2} \right)$ lies on the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$. Find its eccentric angle θ , and find the angle ϕ that OP makes with the positive x -axis. Comment on the difference.

First confirm P is on the ellipse:

$$\frac{4}{16} + \frac{27/4}{9} = \frac{1}{4} + \frac{3}{4} = 1. \checkmark$$

Here $a = 4$, $b = 3$. Compare P with $(a \cos \theta, b \sin \theta)$:

$$\cos \theta = \frac{2}{4} = \frac{1}{2}, \quad \sin \theta = \frac{3\sqrt{3}/2}{3} = \frac{\sqrt{3}}{2}.$$

Both are positive, so θ is in the first quadrant and

$$\theta = \frac{\pi}{3} = 60^\circ.$$

The true central angle satisfies

$$\tan \phi = \frac{y}{x} = \frac{3\sqrt{3}/2}{2} = \frac{3\sqrt{3}}{4} \approx 1.299 \Rightarrow \phi \approx 52.41^\circ.$$

So $\phi \neq \theta$, and the relation from Section 2 confirms exactly why:

$$\tan \phi = \frac{b}{a} \tan \theta = \frac{3}{4} \tan 60^\circ = \frac{3\sqrt{3}}{4}. \checkmark$$

Concept link. The eccentric angle is bigger than the central angle everywhere in the first quadrant because the vertical squash by b/a pulls P down towards the major axis while leaving its horizontal position untouched. The angle you can see on the ellipse is never the angle the parametrisation is using.

Example 3 — [JEE Main — Speed Trap]

Find the coordinates of the point on the hyperbola $\frac{x^2}{16} - \frac{y^2}{9} = 1$ whose parameter is $\theta = \frac{2\pi}{3}$, and state which branch it lies on.

The trap: importing the ellipse's parametric form into the hyperbola. The seductive answer is to write $(a \cos \theta, b \sin \theta)$ out of habit:

$$\left(4 \cos \frac{2\pi}{3}, 3 \sin \frac{2\pi}{3}\right) = \left(-2, \frac{3\sqrt{3}}{2}\right).$$

It looks fine, the arithmetic is fine, and it is completely wrong. Substitute it back:

$$\frac{(-2)^2}{16} - \frac{(3\sqrt{3}/2)^2}{9} = \frac{4}{16} - \frac{27/4}{9} = \frac{1}{4} - \frac{3}{4} = -\frac{1}{2} \neq 1.$$

The point is not on the curve at all. An ellipse form on a hyperbola never satisfies the equation, because $\cos^2 \theta + \sin^2 \theta$ adds to 1 while the hyperbola needs a difference.

The correct route. The hyperbola's parametric point is $(a \sec \theta, b \tan \theta)$ with $a = 4, b = 3$:

$$\sec \frac{2\pi}{3} = \frac{1}{\cos(2\pi/3)} = \frac{1}{-1/2} = -2, \quad \tan \frac{2\pi}{3} = -\sqrt{3}.$$

$$P = (4 \cdot (-2), 3 \cdot (-\sqrt{3})) = (-8, -3\sqrt{3}).$$

Verify:

$$\frac{64}{16} - \frac{27}{9} = 4 - 3 = 1. \checkmark$$

Branch: $\cos \frac{2\pi}{3} < 0$, so $x = -8 < 0$ and the point is on the **left branch** — consistent with Procedure C, since $\frac{2\pi}{3}$ lies in $\left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$.

Speed note. Fix the pairing in memory as a slogan: *ellipse adds, so cos and sin; hyperbola subtracts, so sec and tan*. Then read the sign of $\cos \theta$ once and the branch is free.

Example 4 — [JEE Advanced Level]

If the chord joining the points with eccentric angles θ_1 and θ_2 on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ passes through the focus $(ae, 0)$, prove that

$$\tan \frac{\theta_1}{2} \tan \frac{\theta_2}{2} = \frac{e-1}{e+1}.$$

Hence, for the ellipse $\frac{x^2}{4} + \frac{y^2}{3} = 1$, find the second end of the focal chord one of whose ends has eccentric angle $\frac{\pi}{3}$, and the length of that chord.

Proof. The chord through the two parametric points is

$$\frac{x}{a} \cos \frac{\theta_1 + \theta_2}{2} + \frac{y}{b} \sin \frac{\theta_1 + \theta_2}{2} = \cos \frac{\theta_1 - \theta_2}{2}.$$

(That this is the chord is immediate: substituting $x = a \cos \theta_1, y = b \sin \theta_1$ gives $\cos \theta_1 \cos \frac{\theta_1 + \theta_2}{2} + \sin \theta_1 \sin \frac{\theta_1 + \theta_2}{2} = \cos(\theta_1 - \frac{\theta_1 + \theta_2}{2}) = \cos \frac{\theta_1 - \theta_2}{2}$, and the same computation with θ_2 gives $\cos \frac{\theta_2 - \theta_1}{2}$, which is equal because cosine is even.)

Now force the chord through $S(ae, 0)$. Put $x = ae$, $y = 0$:

$$e \cos \frac{\theta_1 + \theta_2}{2} = \cos \frac{\theta_1 - \theta_2}{2}.$$

Write $A = \frac{\theta_1}{2}$ and $B = \frac{\theta_2}{2}$ and expand both sides:

$$e (\cos A \cos B - \sin A \sin B) = \cos A \cos B + \sin A \sin B.$$

Collect the like products:

$$(e - 1) \cos A \cos B = (e + 1) \sin A \sin B.$$

Divide by $(e + 1) \cos A \cos B$, which is non-zero for a genuine chord:

$$\tan A \tan B = \frac{e - 1}{e + 1}, \quad \text{that is} \quad \tan \frac{\theta_1}{2} \tan \frac{\theta_2}{2} = \frac{e - 1}{e + 1}. \quad \blacksquare$$

Application. For $\frac{x^2}{4} + \frac{y^2}{3} = 1$ we have $a = 2$, $b^2 = 3$, so $c^2 = a^2 - b^2 = 4 - 3 = 1$, giving $c = 1$ and

$$e = \frac{c}{a} = \frac{1}{2}, \quad \text{focus } S(1, 0).$$

With $\theta_1 = \frac{\pi}{3}$:

$$\tan \frac{\pi}{6} \cdot \tan \frac{\theta_2}{2} = \frac{\frac{1}{2} - 1}{\frac{1}{2} + 1} = \frac{-1/2}{3/2} = -\frac{1}{3}.$$

Since $\tan \frac{\pi}{6} = \frac{1}{\sqrt{3}}$,

$$\tan \frac{\theta_2}{2} = -\frac{\sqrt{3}}{3} = -\frac{1}{\sqrt{3}} \implies \frac{\theta_2}{2} = -\frac{\pi}{6} \implies \theta_2 = -\frac{\pi}{3} \quad \left(\text{equivalently } \frac{5\pi}{3} \right).$$

The two ends are therefore

$$P = \left(2 \cos \frac{\pi}{3}, \sqrt{3} \sin \frac{\pi}{3} \right) = \left(1, \frac{3}{2} \right), \quad Q = \left(1, -\frac{3}{2} \right),$$

and the chord length is $\left| \frac{3}{2} - \left(-\frac{3}{2} \right) \right| = 3$.

Independent check. Both ends have $x = 1 = c$, so this focal chord is vertical through the focus — it is the **latus rectum**. Its length should be $\frac{2b^2}{a} = \frac{2 \cdot 3}{2} = 3$, which is exactly what the parametric route produced.

Why it is “Advanced”. The insight is that a focal chord is a condition on **two eccentric angles at once**, and that the half-angle tangent is the right thing to combine because the chord equation is built from $\frac{\theta_1 \pm \theta_2}{2}$. Once you see that, a two-point geometry problem becomes a one-line trigonometric identity — and the latus rectum falls out as a special case you can check for free.

Section 5: Practice Exercises

1. **[CBSE]** Write the parametric coordinates of the circle $(x-2)^2 + (y+3)^2 = 25$, and find the point corresponding to $\theta = \frac{\pi}{2}$.
2. **[CBSE]** For the ellipse $\frac{x^2}{36} + \frac{y^2}{16} = 1$, write the equation of the auxiliary circle and find the point whose eccentric angle is $\frac{\pi}{4}$.
3. **[JEE Main]** Find the eccentric angle of the point $(-3, 2\sqrt{3})$ on the ellipse $\frac{x^2}{36} + \frac{y^2}{16} = 1$.
4. **[JEE Main]** Find the coordinates of the point on the hyperbola $\frac{x^2}{9} - \frac{y^2}{16} = 1$ whose parameter is $\theta = \frac{\pi}{4}$, and state its branch.
5. **[JEE Main]** Write the parametric point of the rectangular hyperbola $xy = 16$ and evaluate it at $t = 2$.
6. **[JEE Main]** P is the point on $\frac{x^2}{25} + \frac{y^2}{9} = 1$ with eccentric angle $\frac{\pi}{6}$ and Q is its corresponding point on the auxiliary circle. Find the length PQ .
7. **[JEE Advanced]** Find the equation of the chord of the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$ joining the points whose eccentric angles are $\frac{\pi}{3}$ and $\frac{2\pi}{3}$.
8. **[JEE Advanced]** Find the eccentric angles of the two endpoints of the latus rectum of $\frac{x^2}{25} + \frac{y^2}{9} = 1$ that lies in the half-plane $x > 0$.

Answers: 1. $(2+5\cos\theta, -3+5\sin\theta)$; at $\theta = \frac{\pi}{2}$ the point is $(2, 2)$. 2. Auxiliary circle $x^2+y^2 = 36$; the point is $(3\sqrt{2}, 2\sqrt{2})$. 3. $\cos\theta = -\frac{1}{2}$, $\sin\theta = \frac{\sqrt{3}}{2}$, so $\theta = \frac{2\pi}{3}$. 4. $(3\sqrt{2}, 4)$; since $\cos\frac{\pi}{4} > 0$ it is on the right branch. 5. $c^2 = 16 \Rightarrow c = 4$, so the point is $(4t, \frac{4}{t})$; at $t = 2$ it is $(8, 2)$. 6. $PQ = (a-b)\sin\theta = (5-3) \cdot \frac{1}{2} = 1$. 7. $\frac{x}{4}\cos\frac{\pi}{2} + \frac{y}{3}\sin\frac{\pi}{2} = \cos(-\frac{\pi}{6})$, i.e. $\frac{y}{3} = \frac{\sqrt{3}}{2}$, so the chord is $y = \frac{3\sqrt{3}}{2}$. 8. $c = \sqrt{25-9} = 4$ and the latus-rectum ends are $(4, \pm\frac{9}{5})$; then $\cos\theta = \frac{4}{5}$ and $\sin\theta = \pm\frac{3}{5}$, giving $\theta = \tan^{-1}\frac{3}{4}$ and $\theta = 2\pi - \tan^{-1}\frac{3}{4}$.

Section 6: Crack the MCQ

Q1. A point on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ may be written as

- (A) $(a\cos\theta, b\sin\theta)$ (B) $(a\sec\theta, b\tan\theta)$ (C) $(a\tan\theta, b\sec\theta)$ (D) $(at^2, 2at)$

Answer: (B). The hyperbola needs $\sec^2\theta - \tan^2\theta = 1$ to match its minus sign.

Distractor Analysis. (A) is the ellipse form, imported by habit — it makes the two squared terms add, which the hyperbola forbids. (C) swaps the roles and produces $\tan^2\theta - \sec^2\theta = -1$, so the point satisfies $\frac{x^2}{a^2} - \frac{y^2}{b^2} = -1$, which is the **conjugate** hyperbola, not this one. (D) is

the parabola's parametric point and belongs to a curve with only one squared term.

Q2. The auxiliary circle of the ellipse $\frac{x^2}{25} + \frac{y^2}{9} = 1$ is

- (A) $x^2 + y^2 = 9$ (B) $x^2 + y^2 = 25$ (C) $x^2 + y^2 = 34$ (D) $x^2 + y^2 = 16$

Answer: (B). The auxiliary circle is drawn on the **major** axis as diameter, so its radius is $a = 5$.

Distractor Analysis. (A) uses b instead of a — that circle is inscribed inside the ellipse and touches it at the minor-axis ends, which is a different object entirely. (C) is $a^2 + b^2$, the radius-squared of the **director** circle you will meet in *Tangents to a Conic*; the two are often confused because both are “some circle attached to the ellipse”. (D) is $c^2 = a^2 - b^2 = 16$, the square of the focal distance, which is not the radius of any standard auxiliary construction.

Q3. The eccentric angle of the point $(2\sqrt{2}, \sqrt{2})$ on the ellipse $\frac{x^2}{16} + \frac{y^2}{4} = 1$ is

- (A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{3}$ (D) $\tan^{-1} \frac{1}{2}$

Answer: (B). Here $a = 4$, $b = 2$, so $\cos \theta = \frac{2\sqrt{2}}{4} = \frac{1}{\sqrt{2}}$ and $\sin \theta = \frac{\sqrt{2}}{2} = \frac{1}{\sqrt{2}}$, giving $\theta = \frac{\pi}{4}$.

Distractor Analysis. (D) is the single most instructive wrong answer: it is $\tan^{-1} \frac{y}{x} = \tan^{-1} \frac{\sqrt{2}}{2\sqrt{2}} = \tan^{-1} \frac{1}{2}$, the **true central angle** ϕ of P , produced by the belief that the eccentric angle is the angle of P itself. (A) comes from taking $\sin \theta = \frac{y}{a} = \frac{\sqrt{2}}{4}$ — dividing the ordinate by a instead of b . (C) comes from the same error applied the other way, dividing the abscissa by b .

Q4. For the parametric point $(a \sec \theta, b \tan \theta)$ of a hyperbola, the excluded values of θ in $[0, 2\pi)$ are

- (A) 0 and π (B) $\frac{\pi}{4}$ and $\frac{3\pi}{4}$ (C) $\frac{\pi}{2}$ and $\frac{3\pi}{2}$ (D) none — all values are allowed

Answer: (C). At those two angles $\cos \theta = 0$, so both $\sec \theta$ and $\tan \theta$ are undefined.

Distractor Analysis. (A) picks the angles where $\sin \theta = 0$, which is the exclusion condition for $\csc$ and $\cot$, not $\sec$ and $\tan$. (B) picks the angles where $|\tan \theta| = 1$, a value that is perfectly finite and gives the honest points $(\pm a\sqrt{2}, \pm b)$. (D) forgets that a parametrisation inherits the domain of the functions it is built from.

Section 7: Common Student Pitfalls & Pro-Tips

- **Treating the eccentric angle as the angle of P .** The eccentric angle belongs to Q on the auxiliary circle. **Correction:** always compute it from $\cos \theta = \frac{x_1}{a}$ and $\sin \theta = \frac{y_1}{b}$, never from $\tan^{-1} \frac{y_1}{x_1}$; the two agree only at the four axis ends.
- **Using $(a \cos \theta, b \sin \theta)$ on a hyperbola.** It produces a point that is not on the curve.

Correction: the sign in the equation dictates the identity — plus sign takes cos, sin; minus sign takes sec, tan.

- **Dividing by the wrong semi-axis.** Writing $\sin \theta = \frac{y_1}{a}$ is the commonest single slip in this unit. **Correction:** each coordinate is divided by the semi-axis measured along its own direction: x by a , y by b .
- **Calling $x^2 + y^2 = a^2 + b^2$ the auxiliary circle.** That is the director circle of the ellipse. **Correction:** auxiliary means “on the major axis as diameter”, so the radius is a and the equation is $x^2 + y^2 = a^2$ — for the hyperbola too, with a the semi-transverse axis.
- **Losing the second branch of a hyperbola.** Solving $y = \pm \frac{b}{a} \sqrt{x^2 - a^2}$ forces you to track signs by hand. **Correction:** the single parametric point covers both branches; just read the sign of $\cos \theta$.
- **Quoting a bare θ from $\cos \theta$ alone.** $\cos \theta = \frac{1}{2}$ has two solutions in $[0, 2\pi)$. **Correction:** use both $\cos \theta$ and $\sin \theta$ to fix the quadrant before writing an answer.

Pro-Tip. The moment a question contains the words “a variable point on the ellipse”, stop and write $(a \cos \theta, b \sin \theta)$. Two unknowns become one, the constraint disappears, and the whole problem becomes trigonometry. The same reflex handles $(at^2, 2at)$ on the parabola and $(a \sec \theta, b \tan \theta)$ on the hyperbola.

Pro-Tip. Memorise one bridging line — $\tan \phi = \frac{b}{a} \tan \theta$ — and every “eccentric angle versus actual angle” trap in JEE Main becomes a five-second substitution.

Section 8: Cheat Sheet Box

- **Circle:** $x^2 + y^2 = r^2 \Rightarrow (r \cos \theta, r \sin \theta)$; shifted: $(h + r \cos \theta, k + r \sin \theta)$. θ here really is the central angle.
- **Parabola:** $y^2 = 4ax \Rightarrow (at^2, 2at), t \in \mathbb{R}$; focal chord $\Leftrightarrow t_1 t_2 = -1$.
- **Ellipse:** $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \Rightarrow (a \cos \theta, b \sin \theta)$; θ is the **eccentric angle**.
- **Hyperbola:** $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \Rightarrow (a \sec \theta, b \tan \theta)$; exclude $\theta = \frac{\pi}{2}, \frac{3\pi}{2}$; $\cos \theta > 0$ right branch, $\cos \theta < 0$ left branch.
- **Rectangular hyperbola:** $xy = c^2 \Rightarrow \left(ct, \frac{c}{t}\right), t \neq 0$.
- **Auxiliary circle:** $x^2 + y^2 = a^2$ for both the ellipse and the hyperbola — drawn on the major / transverse axis as diameter.
- **Squash relations:** $\frac{PN}{QN} = \frac{b}{a}$; $PQ = (a - b) \sin \theta$; $\tan \phi = \frac{b}{a} \tan \theta$.
- **Ellipse chord:** $\frac{x}{a} \cos \frac{\theta_1 + \theta_2}{2} + \frac{y}{b} \sin \frac{\theta_1 + \theta_2}{2} = \cos \frac{\theta_1 - \theta_2}{2}$.
- **Ellipse focal chord:** $\tan \frac{\theta_1}{2} \tan \frac{\theta_2}{2} = \frac{e - 1}{e + 1}$.
- **Mnemonics:** “Plus takes cos-sin, minus takes sec-tan.” “Auxiliary means major axis as diameter — radius a , nothing else.” “The eccentric angle lives on the circle, not on the ellipse.”

Tangents to a Conic

Exam Relevance: CBSE Boards ask for the tangent at a given point on a circle or a parabola at 2–4 marks, usually fused with finding the centre, radius or focus first. JEE Main names the condition of tangency explicitly in its syllabus and asks it almost every session — “for what value of c does the line touch the conic”, “find the tangent of slope m ”, “find the point of contact”. JEE Advanced pushes into the chord of contact, the pair of tangents $SS_1 = T^2$, and the director circle as the locus of points from which two perpendicular tangents can be drawn. This unit is the single highest-yield block in Chapter 10.

Section 1: Concept Intuition

A tangent is a line that touches a curve without crossing it — but that phrasing is a picture, not a method. The method comes from a sharper idea.

Think about it this way... imagine a train on a straight track running past a circular lake, and ask how many points of track are wet. Far from the lake: none. Pushed closer: the track cuts the shore at two points, enters the water and leaves it. Closer still: those two crossing points slide towards each other. **At exactly one position they merge into a single point — the line meets the curve in two coincident points — and that is what a tangent is.** Push further and the line cuts through in two separate places again.

That merging is not a metaphor. Substitute a line into a conic’s equation and you always get a quadratic; the two roots are the two intersection points. The line is a tangent precisely when that quadratic has **equal roots**, that is when its discriminant is zero. So the whole of this unit reduces to one algebraic move you already know from Chapter 5: set the discriminant to zero. Everything else — the slope form, the condition of tangency, the point of contact — is that one move, done once for each conic and then remembered.

There is a second, purely mechanical result that saves enormous time, and it is worth stating before any formula. For every conic written with all terms on the left as $S = 0$, the tangent at a point (x_1, y_1) **on** the curve is obtained by a substitution rule so regular it looks like a joke:

$$x^2 \rightarrow xx_1, \quad y^2 \rightarrow yy_1, \quad xy \rightarrow \frac{1}{2}(xy_1 + x_1y), \quad x \rightarrow \frac{1}{2}(x + x_1), \quad y \rightarrow \frac{1}{2}(y + y_1), \quad \text{constant} \rightarrow \text{constant}.$$

The result is called T , and the tangent is $T = 0$. One rule, four conics, no memorisation of four separate formulas. It works because $T = 0$ is exactly the equation of the line through (x_1, y_1) with the slope that implicit differentiation would give.

The third idea is the **chord of contact**. From a point outside a conic you can draw two tangents; they touch at two points, and the line joining those two points is the chord of contact. Its equation is — remarkably — also $T = 0$, with (x_1, y_1) now the **external** point instead of a point on the curve. The same expression does two different jobs depending on where the point sits, which is why T is worth learning as an object in its own right.

Limiting conditions and assumptions.

- The slope form $y = mx + c$ can never represent a **vertical** line. Every slope-form result therefore silently misses tangents parallel to the y -axis — for example $x = 0$, the tangent

to $y^2 = 4ax$ at its vertex. When a question asks for *all* tangents from a point, check the vertical line separately.

- The parabola's slope form $y = mx + \frac{a}{m}$ requires $m \neq 0$: there is no tangent of slope zero to $y^2 = 4ax$, since a horizontal line always cuts it exactly once.
- The hyperbola's slope form $y = mx \pm \sqrt{a^2m^2 - b^2}$ needs $a^2m^2 > b^2$, i.e. $|m| > \frac{b}{a}$. Lines flatter than an asymptote can never touch the curve.
- The director circle $x^2 + y^2 = a^2 - b^2$ of a hyperbola is real only when $a > b$. If $a \leq b$ no point of the plane sees the hyperbola under a right angle.
- $T = 0$ is a **line** only for these second-degree curves; the substitution rule is not a general-purpose calculus substitute.

Section 2: Key Definitions & Formulas

Definition (tangent). A tangent to a conic is a line meeting it in two coincident points. Equivalently, substituting the line into the conic gives a quadratic with zero discriminant.

Definition (condition of tangency). The relation between c , m and the conic's constants that makes $y = mx + c$ a tangent.

Definition (chord of contact). For an external point $P(x_1, y_1)$, the line joining the two points of contact of the tangents drawn from P .

The universal T rule. Write the conic as $S = 0$ with everything on the left. Form T by the substitutions listed in Section 1. Then:

(x_1, y_1) lies...	meaning of $T = 0$
on the conic	the tangent at (x_1, y_1)
outside the conic	the chord of contact of the two tangents from (x_1, y_1)

Tangents to the circle.

Item	$x^2 + y^2 = r^2$	$S \equiv x^2 + y^2 + 2gx + 2fy + c = 0$
Tangent at (x_1, y_1)	$xx_1 + yy_1 = r^2$	$xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0$
Tangent at parametric point	$x \cos \theta + y \sin \theta = r$	$(x - h) \cos \theta + (y - k) \sin \theta = r$ about centre (h, k)
Slope form	$y = mx \pm r\sqrt{1 + m^2}$	shift the centre to the origin first
Condition of tangency for $y = mx + c$	$c^2 = r^2(1 + m^2)$	perpendicular distance from $(-g, -f) = \text{radius}$
Point of contact (slope form)	$\left(\mp \frac{rm}{\sqrt{1 + m^2}}, \pm \frac{r}{\sqrt{1 + m^2}} \right)$	—
Chord of contact from (x_1, y_1)	$xx_1 + yy_1 = r^2$	$T = 0$

Item	$x^2 + y^2 = r^2$	$S \equiv x^2 + y^2 + 2gx + 2fy + c = 0$
Director circle	$x^2 + y^2 = 2r^2$	centre same, radius $r\sqrt{2}$

The single most useful circle fact is geometric rather than algebraic: **a line touches a circle if and only if the perpendicular distance from the centre to the line equals the radius.** For circles, always reach for that before reaching for a discriminant.

Tangents to the parabola $y^2 = 4ax$.

Item	Result	Conditions
Tangent at (x_1, y_1)	$yy_1 = 2a(x + x_1)$	(x_1, y_1) on the curve
Tangent at $(at^2, 2at)$	$ty = x + at^2$	slope $= \frac{1}{t}$; $t = 0$ gives $x = 0$
Slope form	$y = mx + \frac{a}{m}$	$m \neq 0$
Condition of tangency	$c = \frac{a}{m}$, i.e. $mc = a$	for $y = mx + c$
Point of contact (slope form)	$\left(\frac{a}{m^2}, \frac{2a}{m}\right)$	$m \neq 0$
Chord of contact from (x_1, y_1)	$yy_1 = 2a(x + x_1)$	$y_1^2 - 4ax_1 > 0$
Tangents at t_1, t_2 meet at	$(at_1t_2, a(t_1 + t_2))$	$t_1 \neq t_2$
Perpendicular tangents meet on	the directrix $x = -a$	the parabola's "director circle"

Tangents to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

Item	Result
Tangent at (x_1, y_1)	$\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$
Tangent at $(a \cos \theta, b \sin \theta)$	$\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1$
Slope form	$y = mx \pm \sqrt{a^2m^2 + b^2}$
Condition of tangency for $y = mx + c$	$c^2 = a^2m^2 + b^2$
Point of contact for $y = mx + c$	$\left(-\frac{a^2m}{c}, \frac{b^2}{c}\right)$
Chord of contact from (x_1, y_1)	$\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$
Director circle	$x^2 + y^2 = a^2 + b^2$

Tangents to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$.

Item	Result
Tangent at (x_1, y_1)	$\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1$

Item	Result
Tangent at $(a \sec \theta, b \tan \theta)$	$\frac{x \sec \theta}{a} - \frac{y \tan \theta}{b} = 1$
Slope form	$y = mx \pm \sqrt{a^2 m^2 - b^2}$, valid for $a^2 m^2 > b^2$
Condition of tangency for $y = mx + c$	$c^2 = a^2 m^2 - b^2$
Point of contact for $y = mx + c$	$\left(-\frac{a^2 m}{c}, -\frac{b^2}{c} \right)$
Chord of contact from (x_1, y_1)	$\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1$
Director circle	$x^2 + y^2 = a^2 - b^2$ (real only if $a > b$)
Tangent to $xy = c^2$ at $\left(ct, \frac{c}{t} \right)$	$x + t^2 y = 2ct$

Pair of tangents from an external point. For any conic $S = 0$, with S_1 the value of S at (x_1, y_1) and T as above, the two tangents from (x_1, y_1) together form the pair

$$SS_1 = T^2.$$

Number of tangents from a point.

Conic	S_1 used	Two tangents	One tangent	No tangent
Circle	$x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c$	$S_1 > 0$	$S_1 = 0$	$S_1 < 0$
Parabola	$y_1^2 - 4ax_1$	$S_1 > 0$	$S_1 = 0$	$S_1 < 0$
Ellipse	$\frac{x_1^2}{a^2} + \frac{y_1^2}{b^2} - 1$	$S_1 > 0$	$S_1 = 0$	$S_1 < 0$
Hyperbola	$\frac{x_1^2}{a^2} - \frac{y_1^2}{b^2} - 1$	$S_1 < 0$	$S_1 = 0$	$S_1 > 0$

The hyperbola row is genuinely reversed, and that is not a misprint: $S_1 > 0$ places the point **inside a branch**, on the same side as a focus, and a point cradled inside a branch can see no tangent at all. $S_1 < 0$ places it in the region between the branches, from which two tangents can be drawn. The one exception is a point lying on an asymptote, where one of the two lines degenerates into the asymptote itself.

Section 3: CBSE Board Derivation

Derivation: the tangent to $x^2 + y^2 = r^2$ at (x_1, y_1)

1. The radius to the point of contact is $O(0,0)$ to $P(x_1, y_1)$, with slope $\frac{y_1}{x_1}$. Reason: the tangent to a circle is perpendicular to the radius at the point of contact — the one theorem from Class 9 circle geometry that survives into coordinate geometry unchanged.
2. The tangent is perpendicular to that radius, so its slope is $-\frac{x_1}{y_1}$. Reason: perpendicular slopes multiply to -1 .
3. Using the point-slope form through P :

$$y - y_1 = -\frac{x_1}{y_1}(x - x_1) \implies yy_1 - y_1^2 = -xx_1 + x_1^2 \implies xx_1 + yy_1 = x_1^2 + y_1^2.$$

4. But P lies on the circle, so $x_1^2 + y_1^2 = r^2$. Therefore

$$xx_1 + yy_1 = r^2.$$

Reason: this substitution is the step that converts a general chord-like line into the tangent, and it is why $T = 0$ carries the constant r^2 rather than $x_1^2 + y_1^2$.

Derivation: the condition of tangency for the parabola, and the slope form

1. Let the line be $y = mx + c$ and the parabola $y^2 = 4ax$. Substitute $x = \frac{y-c}{m}$ (taking $m \neq 0$) into the parabola:

$$y^2 = 4a \cdot \frac{y-c}{m} \implies my^2 - 4ay + 4ac = 0.$$

Reason: eliminating x gives a quadratic in y whose roots are the ordinates of the two intersection points.

2. Tangency means the two intersection points coincide, so the discriminant vanishes:

$$(-4a)^2 - 4(m)(4ac) = 0 \implies 16a^2 - 16amc = 0 \implies a = mc.$$

Reason: equal roots is the algebraic translation of “the line touches”.

3. Hence the **condition of tangency** is $c = \frac{a}{m}$, and the tangent of slope m is

$$y = mx + \frac{a}{m}, \quad m \neq 0.$$

4. The point of contact is the repeated root: from $my^2 - 4ay + 4ac = 0$, $y = \frac{4a}{2m} = \frac{2a}{m}$, and then $x = \frac{y^2}{4a} = \frac{a}{m^2}$. So the point of contact is $\left(\frac{a}{m^2}, \frac{2a}{m}\right)$. *Reason: a repeated root of $Ay^2 + By + C = 0$ is $-\frac{B}{2A}$.*
5. Why $m = 0$ is excluded: a horizontal line $y = c$ meets $y^2 = 4ax$ at exactly one point, $\left(\frac{c^2}{4a}, c\right)$, but crosses the curve there rather than touching it — a single intersection is not the same as two coincident ones.

The identical three-line computation on $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ gives $c^2 = a^2m^2 + b^2$, and on $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ gives $c^2 = a^2m^2 - b^2$. The only thing that changes is the sign that the ellipse and hyperbola already differ by — the pattern the chapter has flagged since **The Hyperbola**: *ellipse plus, hyperbola minus* in the tangency condition, exactly opposite to $c^2 = a^2 \mp b^2$.

Derivation: the director circle of the ellipse

1. Let $P(h, k)$ be a point from which two perpendicular tangents can be drawn to $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.
2. Any line through P with slope m is $y - k = m(x - h)$, i.e. $y = mx + (k - mh)$, so $c = k - mh$. *Reason: writing the line through P in slope form lets the tangency condition become an equation in m alone.*

3. Impose tangency, $c^2 = a^2m^2 + b^2$:

$$(k - mh)^2 = a^2m^2 + b^2 \implies m^2(h^2 - a^2) - 2hkm + (k^2 - b^2) = 0.$$

4. This quadratic in m has the slopes of the two tangents as its roots. Perpendicularity means $m_1m_2 = -1$:

$$\frac{k^2 - b^2}{h^2 - a^2} = -1 \implies k^2 - b^2 = a^2 - h^2 \implies h^2 + k^2 = a^2 + b^2.$$

Reason: the product of the roots of $Am^2 + Bm + C = 0$ is $\frac{C}{A}$.

5. Dropping the subscripts, the locus is the **director circle**

$$x^2 + y^2 = a^2 + b^2.$$

Running the same argument with the hyperbola's condition $c^2 = a^2m^2 - b^2$ yields $x^2 + y^2 = a^2 - b^2$, and with the parabola's $c = \frac{a}{m}$ yields the directrix $x = -a$.

Figure brief: A horizontal ellipse centred at O with its director circle $x^2 + y^2 = a^2 + b^2$ drawn concentrically outside it. Mark a point P on the director circle with two tangent lines from P touching the ellipse at two points, the right angle at P clearly marked, and the chord of contact drawn as a dashed segment joining the two points of contact.

Procedure A: find the tangent at a given point

1. Verify the point lies on the conic by substitution. *Why: $T = 0$ is the tangent only for a point on the curve; for any other point it is a chord of contact or a polar line, and you would be answering a different question.*
2. Write the conic as $S = 0$ and apply the T substitutions. *Why: it is faster and less error-prone than differentiating, and it handles the general circle's linear terms correctly.*
3. Clear fractions and present the tangent in the form $Ax + By + C = 0$. *Why: examiners' keys are in integer form.*
4. Check by substituting the point of contact back into your tangent. *Why: a tangent must pass through its own point of contact — a free, ten-second guarantee.*

Procedure B: find the tangent of a given slope, or test a given line

1. Identify a , b (or r) from the standard form. *Why: the tangency conditions are written in those symbols and nothing else.*
2. Write down the condition of tangency for that conic: $c^2 = r^2(1 + m^2)$, $c = \frac{a}{m}$, $c^2 = a^2m^2 + b^2$, or $c^2 = a^2m^2 - b^2$.
3. If the slope is given, solve for c — expect **two** answers for the circle, ellipse and hyperbola (a tangent on each side) and **one** for the parabola. *Why: the parabola's condition is linear in c , the others are quadratic.*
4. If the line is given, substitute its m and c and check whether the condition holds. *Why: this is a one-line yes/no answer, far quicker than solving simultaneously.*
5. Read the point of contact from the table, or recover it as the repeated root. *Why: many questions award the marks for the point of contact, not the line.*

Procedure C: handle tangents from an external point

1. Compute S_1 and decide how many tangents exist, using the table in Section 2. *Why: writing two tangents from an interior point is a guaranteed zero.*
2. For the chord of contact, write $T = 0$ with the external point's coordinates. *Why: one substitution replaces solving for both points of contact.*
3. For the tangents themselves, put $y - y_1 = m(x - x_1)$ into the tangency condition and solve the resulting quadratic in m . *Why: this also delivers $m_1 + m_2$ and $m_1 m_2$ directly, which is what angle and perpendicularity questions actually need.*
4. Before solving that quadratic, check whether the question only wants a symmetric function of the slopes. *Why: if it asks for the angle between the tangents or whether they are perpendicular, Vieta's relations answer it without ever finding m_1 and m_2 .*

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

- (a) Find the equation of the tangent to the circle $x^2 + y^2 - 4x + 6y - 12 = 0$ at the point $(5, 1)$.
- (b) Find the equation of the tangent to the parabola $y^2 = 12x$ at the point $(3, 6)$.

(a) First check the point lies on the circle:

$$25 + 1 - 20 + 6 - 12 = 0. \checkmark$$

Comparing with $x^2 + y^2 + 2gx + 2fy + c = 0$ gives $g = -2$, $f = 3$, $c = -12$. The tangent is $T = 0$:

$$xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0$$

$$5x + y - 2(x + 5) + 3(y + 1) - 12 = 0$$

$$5x + y - 2x - 10 + 3y + 3 - 12 = 0$$

$$3x + 4y - 19 = 0.$$

Check: at $(5, 1)$, $15 + 4 - 19 = 0$. Independent check by the perpendicular-distance test — the centre is $(-g, -f) = (2, -3)$ and the radius is $\sqrt{g^2 + f^2 - c} = \sqrt{4 + 9 + 12} = 5$:

$$\text{distance} = \frac{|3(2) + 4(-3) - 19|}{\sqrt{3^2 + 4^2}} = \frac{|6 - 12 - 19|}{5} = \frac{25}{5} = 5 = \text{radius}. \checkmark$$

(b) Check the point: $6^2 = 36$ and $12 \times 3 = 36$. Comparing $y^2 = 12x$ with $y^2 = 4ax$ gives $4a = 12$, so $a = 3$. The tangent at (x_1, y_1) is $yy_1 = 2a(x + x_1)$:

$$6y = 6(x + 3) \implies y = x + 3.$$

Check: the point $(3, 6)$ gives $6 = 3 + 3$. The slope is 1, matching $\frac{1}{t}$ with $t = \frac{y_1}{2a} = \frac{6}{6} = 1$.

Example 2 — [JEE Main Level]

Find the values of c for which the line $y = 2x + c$ touches the hyperbola $\frac{x^2}{9} - \frac{y^2}{4} = 1$, and find the corresponding points of contact.

Here $a^2 = 9$, $b^2 = 4$, $m = 2$. The condition of tangency for a hyperbola is

$$c^2 = a^2m^2 - b^2 = 9(4) - 4 = 32 \implies c = \pm 4\sqrt{2}.$$

(The condition $a^2m^2 > b^2$ holds, since $36 > 4$; equivalently $|m| = 2 > \frac{b}{a} = \frac{2}{3}$, so the line is steeper than the asymptotes and a tangent genuinely exists.)

The point of contact for $y = mx + c$ on a hyperbola is $\left(-\frac{a^2m}{c}, -\frac{b^2}{c}\right) = \left(-\frac{18}{c}, -\frac{4}{c}\right)$.

For $c = 4\sqrt{2}$:

$$\left(-\frac{18}{4\sqrt{2}}, -\frac{4}{4\sqrt{2}}\right) = \left(-\frac{9\sqrt{2}}{4}, -\frac{\sqrt{2}}{2}\right).$$

For $c = -4\sqrt{2}$, by symmetry:

$$\left(\frac{9\sqrt{2}}{4}, \frac{\sqrt{2}}{2}\right).$$

Verify the first one on the curve:

$$\frac{1}{9} \cdot \frac{162}{16} - \frac{1}{4} \cdot \frac{2}{4} = \frac{162}{144} - \frac{2}{16} = \frac{9}{8} - \frac{1}{8} = 1. \checkmark$$

And on the line: $2\left(-\frac{9\sqrt{2}}{4}\right) + 4\sqrt{2} = -\frac{9\sqrt{2}}{2} + \frac{8\sqrt{2}}{2} = -\frac{\sqrt{2}}{2}. \checkmark$

Concept link. Two values of c means two parallel tangents, one on each side of the centre — exactly the picture for the circle and the ellipse too. Only the parabola breaks the pattern, because its tangency condition $c = \frac{a}{m}$ is linear.

Example 3 — [JEE Main — Speed Trap]

Find the equation of the tangent to the parabola $y^2 = 16x$ having slope 2, together with its point of contact.

The trap: using the coefficient $4a$ where the formula wants a . The slope form is $y = mx + \frac{a}{m}$. The seductive wrong answer reads the coefficient 16 off the equation and writes

$$y = 2x + \frac{16}{2} = 2x + 8.$$

It is wrong, and here is the proof that it is not merely inelegant. Substitute $x = \frac{y-8}{2}$ into $y^2 = 16x$:

$$y^2 = 8(y-8) \implies y^2 - 8y + 64 = 0, \quad \Delta = 64 - 256 = -192 < 0.$$

There are no real intersection points at all. The “tangent” misses the parabola entirely.

The correct route. Compare $y^2 = 16x$ with $y^2 = 4ax$:

$$4a = 16 \implies a = 4.$$

Then with $m = 2$,

$$y = mx + \frac{a}{m} = 2x + \frac{4}{2} = 2x + 2.$$

Point of contact $\left(\frac{a}{m^2}, \frac{2a}{m}\right) = \left(\frac{4}{4}, \frac{8}{2}\right) = (1, 4)$.

Verify tangency directly: substituting $x = \frac{y-2}{2}$ into $y^2 = 16x$ gives

$$y^2 = 8(y-2) \implies y^2 - 8y + 16 = 0 \implies (y-4)^2 = 0,$$

a repeated root at $y = 4$, hence $x = 1$. Two coincident intersection points — a genuine tangent, touching at $(1, 4)$, which does lie on the parabola since $4^2 = 16(1)$.

Speed note. This is the chapter's "a versus 4a disaster" from **The Parabola**, wearing a tangent costume. Build the habit of writing " $a = \dots$ " on its own line the moment you see a parabola, before touching any tangent or normal formula.

Example 4 — [JEE Advanced Level]

Prove that the locus of the point of intersection of two perpendicular tangents to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is the circle $x^2 + y^2 = a^2 + b^2$. Hence show that the tangents drawn from $P(3, 1)$ to the ellipse $\frac{x^2}{8} + \frac{y^2}{2} = 1$ are perpendicular, and find their chord of contact.

Proof of the locus. Let $P(h, k)$ be a point of intersection of two perpendicular tangents. A line through P of slope m is $y = mx + (k - mh)$, so $c = k - mh$. Imposing the ellipse's condition of tangency $c^2 = a^2m^2 + b^2$:

$$(k - mh)^2 = a^2m^2 + b^2.$$

Expand and gather powers of m :

$$k^2 - 2hkm + m^2h^2 = a^2m^2 + b^2 \implies m^2(h^2 - a^2) - 2hkm + (k^2 - b^2) = 0.$$

The roots m_1, m_2 of this quadratic are the slopes of the two tangents from P . Perpendicularity means $m_1m_2 = -1$, and the product of the roots is $\frac{k^2 - b^2}{h^2 - a^2}$, so

$$\frac{k^2 - b^2}{h^2 - a^2} = -1 \implies k^2 - b^2 = -h^2 + a^2 \implies h^2 + k^2 = a^2 + b^2.$$

Dropping subscripts, the locus is $x^2 + y^2 = a^2 + b^2$ — the **director circle**. ■

Application. For $\frac{x^2}{8} + \frac{y^2}{2} = 1$ we have $a^2 = 8$, $b^2 = 2$, so the director circle is

$$x^2 + y^2 = 8 + 2 = 10.$$

The point $P(3, 1)$ satisfies $3^2 + 1^2 = 10$, so P lies **on** the director circle and the two tangents from it are perpendicular.

Confirm it from first principles by building the slope quadratic at $P(3, 1)$, where $c = 1 - 3m$:

$$(1 - 3m)^2 = 8m^2 + 2 \implies 1 - 6m + 9m^2 = 8m^2 + 2 \implies m^2 - 6m - 1 = 0.$$

Then $m_1 m_2 = \frac{-1}{1} = -1$, so the tangents are indeed perpendicular (their slopes being $3 \pm \sqrt{10}$, which multiply to $9 - 10 = -1$).

Chord of contact. Since $S_1 = \frac{9}{8} + \frac{1}{2} - 1 = \frac{5}{8} > 0$, the point is external and two tangents exist. The chord of contact is $T = 0$ with $(x_1, y_1) = (3, 1)$:

$$\frac{3x}{8} + \frac{y}{2} = 1 \implies 3x + 4y = 8.$$

Why it is “Advanced”. The move that unlocks it is refusing to find m_1 and m_2 . The question asks only whether the product of the slopes is -1 , and Vieta hands you that product straight off the quadratic. Recognising that a *symmetric* question about two tangents never requires solving for either one is the transferable skill here — it is the same instinct that turned $t_1 t_2 = -1$ into the focal-chord proof in **The Parabola**.

Section 5: Practice Exercises

1. **[CBSE]** Find the equation of the tangent to the circle $x^2 + y^2 = 25$ at the point $(3, -4)$.
2. **[CBSE]** Find the equation of the tangent to the parabola $y^2 = 8x$ at the point $(2, 4)$.
3. **[JEE Main]** For what values of c is the line $y = 3x + c$ a tangent to the circle $x^2 + y^2 = 9$?
4. **[JEE Main]** Find the equations of the tangents to the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ having slope $\frac{2}{3}$.
5. **[JEE Main]** Find the chord of contact of the tangents drawn from the point $(2, 3)$ to the circle $x^2 + y^2 = 4$.
6. **[JEE Main]** How many tangents can be drawn to the circle $x^2 + y^2 - 2x - 4y + 3 = 0$ from the point $(1, 1)$?
7. **[JEE Main]** Find the equation of the tangent to the hyperbola $\frac{x^2}{16} - \frac{y^2}{9} = 1$ at the point $(8, 3\sqrt{3})$.
8. **[JEE Advanced]** Find the point of intersection of the tangents to the parabola $y^2 = 4x$ drawn at the points with parameters $t_1 = 1$ and $t_2 = 3$.

Answers: 1. $(3, -4)$ satisfies $9 + 16 = 25$; tangent $3x - 4y = 25$. 2. $4a = 8 \Rightarrow a = 2$; tangent $4y = 4(x + 2)$, i.e. $y = x + 2$. 3. $c^2 = r^2(1 + m^2) = 9(1 + 9) = 90$, so $c = \pm 3\sqrt{10}$. 4. $c^2 = a^2 m^2 + b^2 = 9 \cdot \frac{4}{9} + 4 = 8$, so $c = \pm 2\sqrt{2}$ and the tangents are $y = \frac{2}{3}x \pm 2\sqrt{2}$. 5. $T = 0$ gives $2x + 3y = 4$. 6. $S_1 = 1 + 1 - 2 - 4 + 3 = -1 < 0$, so $(1, 1)$ is inside the circle and **no** tangent can be drawn — zero tangents. 7. The point satisfies $\frac{64}{16} - \frac{27}{9} = 4 - 3 = 1$; tangent $\frac{8x}{16} - \frac{3\sqrt{3}y}{9} = 1$, i.e. $3x - 2\sqrt{3}y = 6$. 8. Here $a = 1$; the tangents are $y = x + 1$ and $3y = x + 9$, meeting at $(at_1 t_2, a(t_1 + t_2)) = (3, 4)$.

Section 6: Crack the MCQ

Q1. The line $y = mx + c$ touches the parabola $y^2 = 4ax$ if

- (A) $c = am$ (B) $c = \frac{a}{m}$ (C) $c^2 = a^2m^2 + b^2$ (D) $c = -\frac{a}{m}$

Answer: (B). Substituting the line gives $my^2 - 4ay + 4ac = 0$, and equal roots require $16a^2 = 16amc$, i.e. $a = mc$.

Distractor Analysis. (A) inverts the relation — it is what you get by mis-remembering “ a over m ” as “ a times m ”, and it fails immediately on any numerical check. (C) is the ellipse’s condition imported wholesale; a parabola has no b , so the presence of b in an option is itself a red flag. (D) attaches a spurious minus sign; the tangent to a right-opening parabola with positive slope has a positive intercept, so the sign cannot be negative for $m > 0$.

Q2. The tangent to the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ at the point $(3, 0)$ is

- (A) $x = 3$ (B) $x = 9$ (C) $y = 3$ (D) $x + y = 3$

Answer: (A). Using $T = 0$: $\frac{3x}{9} + \frac{0 \cdot y}{4} = 1$, so $\frac{x}{3} = 1$, that is $x = 3$.

Distractor Analysis. (B) forgets to divide by a^2 and reports the numerator $xx_1 = 9$ as if it were the equation. (C) confuses the vertex’s abscissa with an ordinate — a sign that the student is pattern-matching digits rather than reading which axis the point is on. (D) is the intercept form of a line through $(3, 0)$ and $(0, 3)$, which passes through the point but cuts the ellipse rather than touching it.

Q3. The director circle of the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$ is

- (A) $x^2 + y^2 = 7$ (B) $x^2 + y^2 = 25$ (C) $x^2 + y^2 = 16$ (D) $x^2 + y^2 = 9$

Answer: (B). The director circle is $x^2 + y^2 = a^2 + b^2 = 16 + 9 = 25$.

Distractor Analysis. (A) uses $a^2 - b^2$, which is the **hyperbola’s** director circle and also equals c^2 for this ellipse — the deadly sign import that **The Ellipse** already warned about. (C) is the auxiliary circle $x^2 + y^2 = a^2$; auxiliary and director circles are different objects with different jobs, as *Parametric Forms and the Auxiliary Circle* sets out. (D) uses b^2 and corresponds to no standard construction at all.

Q4. The number of tangents that can be drawn to the circle $x^2 + y^2 = 25$ from the point $(3, 4)$ is

- (A) 0 (B) 1 (C) 2 (D) infinitely many

Answer: (B). $S_1 = 9 + 16 - 25 = 0$, so $(3, 4)$ lies **on** the circle, and there is exactly one tangent there — namely $3x + 4y = 25$.

Distractor Analysis. (A) is the answer for an interior point, $S_1 < 0$. (C) is the answer for an exterior point, $S_1 > 0$, and is what a student picks by reflex without ever computing S_1 — the whole point of the item is that “from a point” does not automatically mean “from outside”.

(D) confuses tangents with the infinitely many *lines* through the point.

Section 7: Common Student Pitfalls & Pro-Tips

- **Using $T = 0$ from a point not on the conic and calling it a tangent.** If the point is external, $T = 0$ is the chord of contact, which is a completely different line. **Correction:** substitute the point into S first; only $S_1 = 0$ licenses the word “tangent”.
- **Importing the wrong sign into the tangency condition.** Ellipse: $c^2 = a^2m^2 + b^2$. Hyperbola: $c^2 = a^2m^2 - b^2$. **Correction:** note that this is the **opposite** pairing to $c^2 = a^2 - b^2$ (ellipse) and $c^2 = a^2 + b^2$ (hyperbola). Two sign rules that look alike and point opposite ways — write both down side by side once and the confusion dissolves.
- **The “ a versus $4a$ ” slip inside $y = mx + \frac{a}{m}$.** Reading 16 from $y^2 = 16x$ as a produces a line that misses the parabola altogether. **Correction:** convert to a on a separate line before you start.
- **Forgetting that slope form cannot produce a vertical tangent.** From $(0, 4)$ to $y^2 = 4x$ the slope form yields only $y = \frac{x}{4} + 4$; the second tangent is $x = 0$, the tangent at the vertex. **Correction:** whenever a question asks for *all* tangents from a point, test the vertical line separately.
- **Claiming a tangent to a hyperbola with a slope flatter than the asymptotes.** $c^2 = a^2m^2 - b^2$ has no real solution when $|m| < \frac{b}{a}$. **Correction:** check that $a^2m^2 - b^2 > 0$ before announcing a value of c ; if it is negative, the honest answer is “no such tangent exists”.
- **Assuming two tangents always come from a positive S_1 .** True for circle, parabola and ellipse; **reversed** for the hyperbola. **Correction:** use the table in Section 2, and sanity-check with a sketch: a point tucked inside a branch is walled in by that branch.

Pro-Tip. For a circle, never run a discriminant. “Perpendicular distance from the centre equals the radius” turns every circle tangency question into one application of the distance-from-a-point-to-a-line formula.

Pro-Tip. When two tangents are drawn from a point and the question asks about the **angle** between them, or whether they are perpendicular, or the sum of their slopes, build the quadratic in m and stop. $m_1 + m_2 = -\frac{B}{A}$ and $m_1m_2 = \frac{C}{A}$ answer the question without a single square root.

Pro-Tip. Learn T once, not four tangent formulas. The substitution $x^2 \rightarrow xx_1$, $y^2 \rightarrow yy_1$, $x \rightarrow \frac{1}{2}(x + x_1)$, $y \rightarrow \frac{1}{2}(y + y_1)$ generates the tangent, the chord of contact and (with $SS_1 = T^2$) the pair of tangents for all four conics.

Section 8: Cheat Sheet Box

- **The one rule:** tangent at a point on the conic, and chord of contact from a point outside it, are both $T = 0$. Pair of tangents: $SS_1 = T^2$.
- **Circle** $x^2 + y^2 = r^2$: tangent $xx_1 + yy_1 = r^2$; parametric $x \cos \theta + y \sin \theta = r$; slope form $y = mx \pm r\sqrt{1 + m^2}$; condition $c^2 = r^2(1 + m^2)$; director circle $x^2 + y^2 = 2r^2$.
- **General circle:** tangent $xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0$; tangency test — distance

from $(-g, -f)$ equals $\sqrt{g^2 + f^2 - c}$.

- **Parabola** $y^2 = 4ax$: tangent $yy_1 = 2a(x + x_1)$; parametric $ty = x + at^2$; slope form $y = mx + \frac{a}{m}$ ($m \neq 0$); condition $c = \frac{a}{m}$; contact $\left(\frac{a}{m^2}, \frac{2a}{m}\right)$; tangents at t_1, t_2 meet at $(at_1t_2, a(t_1 + t_2))$; perpendicular tangents meet on $x = -a$.
- **Ellipse**: tangent $\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$; parametric $\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1$; slope form $y = mx \pm \sqrt{a^2m^2 + b^2}$; contact $\left(-\frac{a^2m}{c}, \frac{b^2}{c}\right)$; director circle $x^2 + y^2 = a^2 + b^2$.
- **Hyperbola**: tangent $\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1$; parametric $\frac{x \sec \theta}{a} - \frac{y \tan \theta}{b} = 1$; slope form $y = mx \pm \sqrt{a^2m^2 - b^2}$ needing $|m| > \frac{b}{a}$; contact $\left(-\frac{a^2m}{c}, -\frac{b^2}{c}\right)$; director circle $x^2 + y^2 = a^2 - b^2$ (needs $a > b$). Tangent to $xy = c^2$ at $\left(ct, \frac{c}{t}\right)$: $x + t^2y = 2ct$.
- **Counting tangents**: circle, parabola, ellipse — $S_1 > 0$ two, $= 0$ one, < 0 none. Hyperbola — reversed: $S_1 < 0$ two, $= 0$ one, > 0 none.
- **Mnemonics**: “Tangent means equal roots.” “ $T = 0$ on the curve is a tangent, off the curve is a chord of contact.” “Tangency flips the sign: ellipse plus, hyperbola minus.” “Circle? Distance equals radius — never a discriminant.”

Normals to a Conic

Exam Relevance: CBSE Boards ask the normal at a stated point on a circle or parabola at 2-3 marks, most often as the second half of a question whose first half was the tangent. JEE Main asks the slope form of the parabola’s normal, the normal at a parametric point, and “the normal at P meets the curve again at Q ” almost every session. JEE Advanced uses the three-normals property ($t_1 + t_2 + t_3 = 0$) and normal-based locus questions. The normal is also where the reflection property of the ellipse — the whispering gallery — finally becomes a theorem you can compute with instead of a story. NEET does not test mathematics.

Section 1: Concept Intuition

If the tangent is the direction you are *travelling*, the normal is the direction you are *leaning*. On a Mumbai local rounding a curve, the tangent is where the train is pointed; the normal is the line along which the outward push throws you against the door. They are perpendicular, they meet at the same point of the curve, and once you have one you have the other for free.

That “for free” is the whole practical content of this unit. **The normal at a point on a conic is the line through that point perpendicular to the tangent there — so every normal formula is a tangent formula with its slope inverted and negated.** You are not learning a second body of theory. You are learning one extra step applied to theory you already have.

Think about it this way... for a circle the normal is something you have known since Class 9 without calling it that. The tangent to a circle is perpendicular to the radius at the point of contact, so **the normal to a circle is the radius produced — it always passes through**

the centre. No formula is required at all: join the point to the centre and you are done. Every other conic's normal is a departure from that comfortable picture, and the departure is measured by how far the conic has strayed from being a circle. For the ellipse the normal at P misses the centre; what it does instead is bisect the angle $\angle SPS'$ between the two focal radii — which is precisely why a whisper released at one focus of a domed hall arrives, reassembled, at the other.

The parabola deserves a warning of its own. Its normal in slope form,

$$y = mx - 2am - am^3,$$

carries a **cubic** term in m . That cubic has consequences. Solving it for a normal through a given external point can produce up to **three** real values of m , which is why “three normals can be drawn to a parabola from a point” is a genuine theorem and not a curiosity — and why the three feet of those normals satisfy the elegant relation $t_1 + t_2 + t_3 = 0$. No other conic in this chapter has a linear tangent-slope condition and a cubic normal-slope condition, and students who expect symmetry between the two are repeatedly caught.

Limiting conditions and assumptions.

- Where the tangent is horizontal the normal is vertical, and where the tangent is vertical the normal is horizontal. The slope forms below cannot express those cases, so handle the axis ends by inspection.
- The ellipse's parametric normal $ax \sec \theta - by \csc \theta = a^2 - b^2$ breaks down at $\theta = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$, where one of $\sec \theta, \csc \theta$ is undefined. At those four points the normal is simply the major or minor axis.
- For a circle $a = b$, so the ellipse's normal relation collapses to $a^2 - b^2 = 0$ and the “normal” degenerates into any line through the centre — the correct answer, arrived at by the formula quietly telling you it is not needed.
- All standard forms assume the tidy frame used throughout this chapter: centre or vertex at the origin, axes along the coordinate axes.

Section 2: Key Definitions & Formulas

Definition (normal). The normal to a conic at a point P on it is the straight line through P perpendicular to the tangent at P .

Master construction. If the tangent at P has slope m_T , the normal has slope

$$m_N = -\frac{1}{m_T},$$

and the normal is the line through P with that slope. Every entry in the tables below is this one statement, carried out.

Normals to the circle.

Circle	Normal at (x_1, y_1)	Note
$x^2 + y^2 = r^2$	$x_1y - y_1x = 0$, i.e. $y = \frac{y_1}{x_1}x$	passes through the origin
$x^2 + y^2 + 2gx + 2fy + c = 0$	the line through (x_1, y_1) and $(-g, -f)$	passes through the centre

Normals to the parabola $y^2 = 4ax$.

Item	Result	Conditions
Normal at (x_1, y_1)	$y - y_1 = -\frac{y_1}{2a}(x - x_1)$	$y_1 \neq 0$; at the vertex the normal is $y = 0$
Normal at $(at^2, 2at)$	$y + tx = 2at + at^3$	slope $= -t$
Slope form	$y = mx - 2am - am^3$	any real m
Foot of the normal (slope form)	$(am^2, -2am)$	corresponds to $t = -m$
Three normals from (h, k)	$am^3 + m(2a - h) + k = 0$	up to three real roots
Conormal relation	$t_1 + t_2 + t_3 = 0$ and $m_1 + m_2 + m_3 = 0$	the cubic has no m^2 term
Normal at t_1 meets the curve again at	$t_2 = -t_1 - \frac{2}{t_1}$	$t_1 \neq 0$

Normals to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

Item	Result
Normal at (x_1, y_1)	$\frac{a^2x}{x_1} - \frac{b^2y}{y_1} = a^2 - b^2$
Normal at $(a \cos \theta, b \sin \theta)$	$ax \sec \theta - by \csc \theta = a^2 - b^2$
Slope form	$y = mx \mp \frac{m(a^2 - b^2)}{\sqrt{a^2 + b^2m^2}}$
Slope of the normal at (x_1, y_1)	$\frac{a^2y_1}{b^2x_1}$

Normals to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$.

Item	Result
Normal at (x_1, y_1)	$\frac{a^2x}{x_1} + \frac{b^2y}{y_1} = a^2 + b^2$
Normal at $(a \sec \theta, b \tan \theta)$	$ax \cos \theta + by \cot \theta = a^2 + b^2$
Slope form	$y = mx \mp \frac{m(a^2 + b^2)}{\sqrt{a^2 - b^2m^2}}$

Item	Result
Slope of the normal at (x_1, y_1)	$-\frac{a^2 y_1}{b^2 x_1}$
Normal to $xy = c^2$ at $\left(ct, \frac{c}{t}\right)$	$t^3 x - ty = c(t^4 - 1)$

Notice the pleasing pattern across the last two tables: the ellipse's normal reads *minus*, *minus* $\left(-\frac{b^2 y}{x_1} \text{ and } a^2 - b^2\right)$ while the hyperbola's reads *plus*, *plus*. It is the same "ellipse minus, hyperbola plus" rhythm the chapter set up in **The Hyperbola**, and here it applies twice in the same line.

Reflection (optical) property — stated as a theorem.

- **Parabola.** The tangent at P makes equal angles with the focal radius SP and with the line through P parallel to the axis. Consequence: rays arriving parallel to the axis all reflect through the focus, which is why a DTH dish works.
- **Ellipse.** The **normal** at P bisects the interior angle $\angle SPS'$ between the two focal radii; equivalently the tangent makes equal angles with SP and $S'P$. Consequence: anything emitted from one focus converges on the other — the whispering gallery, and the lithotripsy machine.
- **Hyperbola.** The **tangent** at P bisects the angle $\angle SPS'$ between the focal radii. Consequence: rays aimed at one focus reflect away along the direction of the other.

Figure brief: Three small panels side by side. Panel 1: a right-opening parabola with focus S , a point P , the incoming ray parallel to the axis striking P , the reflected ray PS , and the tangent at P with the two equal angles marked. Panel 2: an ellipse with foci S and S' , a point P , both focal radii drawn, and the normal at P shown bisecting the angle $\angle SPS'$ with the two equal angles marked. Panel 3: a hyperbola with foci S and S' , a point P on the right branch, both focal radii drawn, and the tangent at P shown bisecting $\angle SPS'$.

Section 3: CBSE Board Derivation

Derivation: the normal to $y^2 = 4ax$ at the parametric point $(at^2, 2at)$

1. From *Tangents to a Conic*, the tangent at $(at^2, 2at)$ is $ty = x + at^2$, so its slope is $\frac{1}{t}$.
Reason: rewrite as $y = \frac{x}{t} + at$ and read the coefficient of x .
2. The normal is perpendicular to the tangent, so its slope is $-t$. Reason: $m_N = -\frac{1}{m_T} = -\frac{1}{1/t} = -t$.
3. The normal passes through the same point, so by the point-slope form:

$$y - 2at = -t(x - at^2).$$

4. Expand and tidy:

$$y - 2at = -tx + at^3 \implies y + tx = 2at + at^3.$$

Reason: collecting the x and y terms on the left gives the standard printed form, which examiners' keys match.

5. To get the slope form, put $m = -t$, so $t = -m$:

$$y + (-m)x = 2a(-m) + a(-m)^3 \implies y - mx = -2am - am^3 \implies y = mx - 2am - am^3.$$

6. The foot of this normal is the original point with $t = -m$ substituted:

$$(a(-m)^2, 2a(-m)) = (am^2, -2am).$$

Reason: the foot of a normal is simply the point at which it was constructed, re-expressed in the new parameter.

A consequence worth reading off immediately. Requiring the normal to pass through a fixed point (h, k) gives

$$k = mh - 2am - am^3 \implies am^3 + m(2a - h) + k = 0,$$

a cubic in m with **no m^2 term**. The sum of its roots is therefore zero:

$$m_1 + m_2 + m_3 = 0 \quad \text{and hence} \quad t_1 + t_2 + t_3 = 0.$$

That is the celebrated conormal-points relation, and it falls out of a missing coefficient.

Derivation: the normal to $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at (x_1, y_1)

1. From *Tangents to a Conic*, the tangent at (x_1, y_1) is $\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$. Solving for y shows its slope is $-\frac{b^2x_1}{a^2y_1}$. *Reason: rearranging gives $y = -\frac{b^2x_1}{a^2y_1}x + \frac{b^2}{y_1}$.*
2. The normal's slope is the negative reciprocal:

$$m_N = \frac{a^2y_1}{b^2x_1}.$$

3. The normal through (x_1, y_1) is therefore

$$y - y_1 = \frac{a^2y_1}{b^2x_1}(x - x_1).$$

4. Cross-multiply to clear the fraction and separate the variables:

$$\frac{b^2(y - y_1)}{y_1} = \frac{a^2(x - x_1)}{x_1} \implies \frac{b^2y}{y_1} - b^2 = \frac{a^2x}{x_1} - a^2.$$

Reason: dividing each side by the coordinate that appears in it is what produces the memorable symmetric form.

5. Rearrange:

$$\frac{a^2x}{x_1} - \frac{b^2y}{y_1} = a^2 - b^2.$$

6. Substituting the parametric point $x_1 = a \cos \theta$, $y_1 = b \sin \theta$:

$$\frac{a^2 x}{a \cos \theta} - \frac{b^2 y}{b \sin \theta} = a^2 - b^2 \implies ax \sec \theta - by \csc \theta = a^2 - b^2.$$

7. Repeating steps 1-5 on $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, where the tangent's slope is $+\frac{b^2 x_1}{a^2 y_1}$, gives the hyperbola's normal

$$\frac{a^2 x}{x_1} + \frac{b^2 y}{y_1} = a^2 + b^2, \quad \text{and parametrically} \quad ax \cos \theta + by \cot \theta = a^2 + b^2.$$

Procedure A: find the normal at a given point

1. Check the point lies on the conic. *Why: a normal is defined only at a point of the curve.*
2. Get the tangent's slope — either by quoting the tangent from *Tangents to a Conic* or by implicit differentiation. *Why: both routes give the same number; use whichever is faster for the conic in front of you.*
3. Negate and invert to get the normal's slope, then write the point-slope form. *Why: this is the definition, and it works even when you cannot recall a printed formula.*
4. For a circle, skip all of this and join the point to the centre. *Why: the radius is the normal, so the two-point form finishes the question in one line.*
5. Verify the point satisfies your final equation. *Why: a normal must pass through its own foot.*

Procedure B: find the normal of a given slope

1. Convert the conic to standard form and extract a (and b). *Why: the slope forms are stated in those symbols; using the raw coefficient is the classic error.*
2. Quote the slope form: parabola $y = mx - 2am - am^3$; ellipse $y = mx \mp \frac{m(a^2 - b^2)}{\sqrt{a^2 + b^2 m^2}}$;
hyperbola $y = mx \mp \frac{m(a^2 + b^2)}{\sqrt{a^2 - b^2 m^2}}$.
3. For the parabola, write the foot immediately as $(am^2, -2am)$. *Why: most questions want the foot, and it costs nothing once m is known.*
4. Verify by checking that the foot lies on both the conic and the line. *Why: two independent substitutions catch every sign and cubing error this procedure can produce.*

Procedure C: “the normal at P meets the curve again at Q ”

1. Put P in parametric form, $P = (at_1^2, 2at_1)$. *Why: the second intersection is much easier to describe by its parameter than by its coordinates.*
2. Write the normal at t_1 and substitute the general parametric point $(at_2^2, 2at_2)$ into it. *Why: you are demanding that Q lie on the normal, which is exactly one equation.*
3. Factor out $(t_2 - t_1)$, which must be a factor because $t_2 = t_1$ obviously satisfies the equation. *Why: removing the known root converts a quadratic into a linear equation for t_2 .*
4. Read off $t_2 = -t_1 - \frac{2}{t_1}$ and convert back to coordinates. *Why: this single relation answers the entire family of “meets again” questions.*

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

- (a) Find the equation of the normal to the parabola $y^2 = 12x$ at the point $(3, 6)$.
(b) Find the equation of the normal to the circle $x^2 + y^2 - 4x + 6y - 12 = 0$ at the point $(5, 1)$.

(a) Check the point: $6^2 = 36 = 12 \times 3$. Comparing $y^2 = 12x$ with $y^2 = 4ax$ gives $a = 3$.

The normal at (x_1, y_1) is

$$y - y_1 = -\frac{y_1}{2a}(x - x_1) \implies y - 6 = -\frac{6}{6}(x - 3) = -(x - 3),$$
$$y - 6 = -x + 3 \implies x + y = 9.$$

Check: at $(3, 6)$, $3 + 6 = 9$. The tangent there was $y = x + 3$ with slope 1 (Example 1 of *Tangents to a Conic*), and the normal's slope is -1 — the two are perpendicular, as required.

(b) The normal to a circle is the radius produced, so it is simply the line through $(5, 1)$ and the centre. The centre is $(-g, -f) = (2, -3)$.

Slope = $\frac{1 - (-3)}{5 - 2} = \frac{4}{3}$. Using the point-slope form at $(5, 1)$:

$$y - 1 = \frac{4}{3}(x - 5) \implies 3y - 3 = 4x - 20 \implies 4x - 3y = 17.$$

Check both points: $4(5) - 3(1) = 17$ and $4(2) - 3(-3) = 8 + 9 = 17$. And the tangent at $(5, 1)$ was $3x + 4y = 19$, of slope $-\frac{3}{4}$ — perpendicular to $\frac{4}{3}$.

Example 2 — [JEE Main Level]

Find the equation of the normal to the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$ at the point whose eccentric angle is $\frac{\pi}{4}$.

Here $a = 4$ and $b = 3$, so $a^2 - b^2 = 16 - 9 = 7$. The point is

$$P = \left(4 \cos \frac{\pi}{4}, 3 \sin \frac{\pi}{4}\right) = \left(2\sqrt{2}, \frac{3}{\sqrt{2}}\right),$$

which does lie on the ellipse: $\frac{8}{16} + \frac{9/2}{9} = \frac{1}{2} + \frac{1}{2} = 1$. ✓

Apply the parametric normal $ax \sec \theta - by \csc \theta = a^2 - b^2$ with $\sec \frac{\pi}{4} = \csc \frac{\pi}{4} = \sqrt{2}$:

$$4x\sqrt{2} - 3y\sqrt{2} = 7 \implies \sqrt{2}(4x - 3y) = 7 \implies 4x - 3y = \frac{7}{\sqrt{2}} = \frac{7\sqrt{2}}{2}.$$

Equivalently, in integer-coefficient form, $4\sqrt{2}x - 3\sqrt{2}y = 7$.

Check by substituting P :

$$4\sqrt{2}(2\sqrt{2}) - 3\sqrt{2}\left(\frac{3}{\sqrt{2}}\right) = 16 - 9 = 7. \checkmark$$

Cross-check via the point form $\frac{a^2x}{x_1} - \frac{b^2y}{y_1} = a^2 - b^2$:

$$\frac{16x}{2\sqrt{2}} - \frac{9y}{3/\sqrt{2}} = 7 \implies \frac{8x}{\sqrt{2}} - 3\sqrt{2}y = 7 \implies 4\sqrt{2}x - 3\sqrt{2}y = 7,$$

the same line. **Concept link:** the parametric normal and the point normal are not two formulas but one formula in two costumes, exactly as the auxiliary-circle unit predicted.

Example 3 — [JEE Main — Speed Trap]

Find the equation of the normal to the parabola $y^2 = 8x$ having slope 2, and find its foot.

The trap: dropping the cubic term from the normal's slope form. Students who half-remember the tangent's shape $y = mx + c$ write the normal as $y = mx - 2am$ and stop. With $4a = 8$, so $a = 2$, that gives

$$y = 2x - 2(2)(2) = 2x - 8.$$

It is a real line and it does hit the parabola, which is what makes it so convincing. Substituting $x = \frac{y+8}{2}$ into $y^2 = 8x$:

$$y^2 = 4(y+8) \implies y^2 - 4y - 32 = 0 \implies (y-8)(y+4) = 0.$$

So it meets the curve at $(8, 8)$ and at $(2, -4)$ — two **distinct** points, so it is an ordinary chord. Worse, it is not normal at either: the tangent slope on $y^2 = 8x$ is $\frac{2a}{y} = \frac{4}{y}$, giving $\frac{1}{2}$ at $(8, 8)$ and -1 at $(2, -4)$; a normal of slope 2 would need a tangent slope of $-\frac{1}{2}$ at its foot, which happens at neither point.

The correct route. The slope form of the normal to $y^2 = 4ax$ carries three terms:

$$y = mx - 2am - am^3.$$

With $a = 2$ and $m = 2$:

$$y = 2x - 2(2)(2) - 2(2)^3 = 2x - 8 - 16 = 2x - 24.$$

The foot is $(am^2, -2am) = (2 \cdot 4, -2 \cdot 2 \cdot 2) = (8, -8)$.

Verify the foot on the parabola: $(-8)^2 = 64$ and $8(8) = 64$. ✓ Verify the foot on the line: $2(8) - 24 = -8$. ✓ Verify perpendicularity: the tangent slope at $(8, -8)$ is $\frac{4}{y} = \frac{4}{-8} = -\frac{1}{2}$, and $2 \times \left(-\frac{1}{2}\right) = -1$. ✓

Speed note. Store the parabola's two slope forms as a contrasting pair — tangent $y = mx + \frac{a}{m}$ has **two** terms, normal $y = mx - 2am - am^3$ has **three**. If your normal has only two terms, you have dropped the cube.

Example 4 — [JEE Advanced Level]

If the normal at the point $P(at_1^2, 2at_1)$ on the parabola $y^2 = 4ax$ meets the curve again at $Q(at_2^2, 2at_2)$, prove that

$$t_2 = -t_1 - \frac{2}{t_1}.$$

Hence find where the normal at the point $t = 1$ on $y^2 = 4x$ meets the parabola again, and deduce how close to P the second point Q can ever be.

Proof. The normal at t_1 is

$$y + t_1x = 2at_1 + at_1^3.$$

Q lies on this normal, so substitute $x = at_2^2$, $y = 2at_2$:

$$2at_2 + t_1(at_2^2) = 2at_1 + at_1^3.$$

Divide throughout by a (which is non-zero) and bring everything to one side:

$$t_1t_2^2 + 2t_2 - t_1^3 - 2t_1 = 0.$$

Group so that $(t_2 - t_1)$ can be extracted — it must be a factor, because $t_2 = t_1$ trivially satisfies the equation:

$$t_1(t_2^2 - t_1^2) + 2(t_2 - t_1) = 0 \implies (t_2 - t_1)[t_1(t_2 + t_1) + 2] = 0.$$

Since $Q \neq P$ we have $t_2 \neq t_1$, so the second bracket must vanish:

$$t_1t_2 + t_1^2 + 2 = 0 \implies t_2 = -t_1 - \frac{2}{t_1}. \quad \blacksquare$$

Application. For $y^2 = 4x$ we have $4a = 4$, so $a = 1$. With $t_1 = 1$:

$$t_2 = -1 - \frac{2}{1} = -3.$$

The two points are

$$P = (1 \cdot 1^2, 2 \cdot 1 \cdot 1) = (1, 2), \quad Q = (1 \cdot (-3)^2, 2 \cdot 1 \cdot (-3)) = (9, -6).$$

Check Q on the parabola: $(-6)^2 = 36 = 4 \times 9$. ✓ Check the normal: at $t_1 = 1$ it is $y + x = 2(1)(1) + 1(1)^3 = 3$, and Q gives $-6 + 9 = 3$. ✓

The deduction. Take $t_1 > 0$. Then

$$|t_2| = t_1 + \frac{2}{t_1} \geq 2\sqrt{t_1 \cdot \frac{2}{t_1}} = 2\sqrt{2}$$

by the AM-GM inequality, with equality when $t_1 = \frac{2}{t_1}$, that is $t_1 = \sqrt{2}$. The same bound holds for $t_1 < 0$ by symmetry. So no matter where P sits, the far end of the normal chord always has $|t_2| \geq 2\sqrt{2}$: the normal at any point of a parabola always lands a long way round the curve, and it lands closest when $t_1 = \sqrt{2}$.

Why it is “Advanced”. Two separate ideas are chained. First, the algebraic one: the factor $(t_2 - t_1)$ is guaranteed in advance by the geometry, so you should look for it rather than expand blindly. Second, the analytic one: once the answer is a formula in t_1 , it becomes an optimisation problem, and AM-GM converts a geometry question into a one-line inequality. This is exactly the habit **The Parabola** built with $t_1 t_2 = -1$, pushed one level further.

Section 5: Practice Exercises

1. **[CBSE]** Find the equation of the normal to the circle $x^2 + y^2 = 25$ at the point $(3, 4)$.
2. **[CBSE]** Find the equation of the normal to the parabola $y^2 = 4x$ at the point $(4, 4)$.
3. **[JEE Main]** Find the equation of the normal to the ellipse $\frac{x^2}{25} + \frac{y^2}{9} = 1$ at the point $\left(4, \frac{9}{5}\right)$.
4. **[JEE Main]** Find the equation of the normal to the hyperbola $\frac{x^2}{9} - \frac{y^2}{4} = 1$ at the point $(3\sqrt{2}, 2)$.
5. **[JEE Main]** Find the equation of the normal to $y^2 = 16x$ having slope -1 , and state its foot.
6. **[JEE Main]** Find the equation of the normal to the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$ at the point whose eccentric angle is $\frac{\pi}{6}$.
7. **[JEE Advanced]** The normal at the point with parameter $t_1 = 2$ on the parabola $y^2 = 8x$ meets the curve again at Q . Find t_2 and the coordinates of Q .
8. **[JEE Advanced]** Find the equation of the normal to the rectangular hyperbola $xy = 4$ at the point $(2, 2)$.

Answers: 1. The normal is the radius produced, so it is $y = \frac{4}{3}x$, i.e. $4x - 3y = 0$. 2. $a = 1$; $y - 4 = -\frac{4}{2}(x - 4)$, giving $2x + y = 12$. 3. The point satisfies $\frac{16}{25} + \frac{81/25}{9} = 1$; the normal is $\frac{25x}{4} - \frac{9y}{9/5} = 16$, i.e. $25x - 20y = 64$. 4. The point satisfies $\frac{18}{9} - \frac{4}{4} = 1$; the normal is $\frac{9x}{3\sqrt{2}} + \frac{4y}{2} = 13$, i.e. $3\sqrt{2}x + 4y = 26$. 5. $a = 4$; $y = -x - 2(4)(-1) - 4(-1)^3 = -x + 12$, so $x + y = 12$, with foot $(am^2, -2am) = (4, 8)$. 6. The point is $\left(2\sqrt{3}, \frac{3}{2}\right)$; the normal is $4x \sec \frac{\pi}{6} - 3y \csc \frac{\pi}{6} = 7$, i.e. $\frac{8x}{\sqrt{3}} - 6y = 7$, or $8x - 6\sqrt{3}y = 7\sqrt{3}$. 7. $t_2 = -2 - \frac{2}{2} = -3$; with $a = 2$, $Q = (2(-3)^2, 2 \cdot 2 \cdot (-3)) = (18, -12)$, and the normal $y + 2x = 24$ does pass through it. 8. $c = 2$ and the point corresponds to $t = 1$; $t^3x - ty = c(t^4 - 1)$ gives $x - y = 0$, i.e. $y = x$.

Section 6: Crack the MCQ

Q1. The equation of the normal to the parabola $y^2 = 4ax$ at the point $(at^2, 2at)$ is

- (A) $ty = x + at^2$ (B) $y + tx = 2at + at^3$ (C) $y = tx - 2at - at^3$ (D) $tx - y = at^2$

Answer: (B). The tangent there has slope $\frac{1}{t}$, so the normal has slope $-t$, and the point-slope form through $(at^2, 2at)$ rearranges to $y + tx = 2at + at^3$.

Distractor Analysis. (A) is the **tangent** at the same point — the most commonly chosen wrong answer, because tangent and normal formulas are learned in the same breath and stored in the same slot. (C) has the right terms but the wrong slope sign: it uses $+t$ instead of $-t$, so it is parallel to nothing meaningful and does not pass through the point. (D) is a line of slope t through the origin-shifted point and satisfies neither the perpendicularity nor the through-the-point requirement.

Q2. The slope form of the normal to $y^2 = 4ax$ is

- (A) $y = mx + \frac{a}{m}$ (B) $y = mx - 2am - am^3$ (C) $y = mx + 2am + am^3$ (D) $y = mx \pm \sqrt{a^2m^2 + b^2}$

Answer: (B). Substituting $t = -m$ into $y + tx = 2at + at^3$ gives $y = mx - 2am - am^3$.

Distractor Analysis. (A) is the tangent's slope form; a two-term answer where a three-term one is required is the signature of the dropped cubic. (C) has both signs flipped, which would place the foot at $(am^2, +2am)$ — a point that does lie on the parabola but at which the normal has slope $-m$, not m . (D) is the **ellipse tangent** in slope form and contains a b , which a parabola does not possess.

Q3. The normal to a circle at any point on it always passes through

- (A) the origin (B) the centre (C) the focus (D) no fixed point

Answer: (B). The tangent to a circle is perpendicular to the radius at the point of contact, so the normal is the radius produced and must pass through the centre.

Distractor Analysis. (A) is true only for a circle centred at the origin — a special case mistaken for the general rule. (C) misapplies conic vocabulary: a circle has $e = 0$ and no focus distinct from its centre, a point **The Circle** makes explicitly. (D) denies the one genuinely universal normal fact in this chapter.

Q4. The equation of the normal to $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at (x_1, y_1) is

- (A) $\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$ (B) $\frac{a^2x}{x_1} - \frac{b^2y}{y_1} = a^2 - b^2$ (C) $\frac{a^2x}{x_1} + \frac{b^2y}{y_1} = a^2 + b^2$ (D) $\frac{a^2x}{x_1} - \frac{b^2y}{y_1} = a^2 + b^2$

Answer: (B). The tangent's slope is $-\frac{b^2x_1}{a^2y_1}$, so the normal's is $\frac{a^2y_1}{b^2x_1}$, and clearing fractions in the point-slope form gives exactly (B).

Distractor Analysis. (A) is the tangent, not the normal — the same tangent-normal slot collision as in Q1. (C) is the **hyperbola's** normal: both signs are flipped together, which is precisely the “ellipse minus, hyperbola plus” rule being applied to the wrong curve. (D) mixes the two, taking the ellipse's left-hand side with the hyperbola's right-hand side; it is the option that catches students who remember that “one of the signs changes” but not which.

Section 7: Common Student Pitfalls & Pro-Tips

- **Answering with the tangent when the normal was asked.** The two formulas live next to each other in memory and get swapped under time pressure. **Correction:** after writing any normal, check that its slope times the tangent's slope equals -1 . It takes five seconds and it is decisive.
- **Dropping the cubic term in the parabola's normal.** $y = mx - 2am$ is not a normal; it is an arbitrary chord. **Correction:** count terms — the tangent slope form has two, the normal slope form has three.
- **Using the raw coefficient instead of a .** In $y^2 = 16x$, $a = 4$, not 16, and cubing an incorrect a magnifies the error enormously. **Correction:** write " $a = \dots$ " on its own line before any slope form.
- **Getting the ellipse and hyperbola normal signs the wrong way round.** Ellipse: $\frac{a^2x}{x_1} - \frac{b^2y}{y_1} = a^2 - b^2$. Hyperbola: $\frac{a^2x}{x_1} + \frac{b^2y}{y_1} = a^2 + b^2$. **Correction:** both signs move together — ellipse is minus-and-minus, hyperbola is plus-and-plus. If your answer mixes them, it belongs to no conic.
- **Using the parametric normal at the axis ends.** $ax \sec \theta - by \csc \theta = a^2 - b^2$ is undefined at $\theta = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$. **Correction:** at those four points the normal is simply the major or the minor axis; read it off the picture rather than forcing the formula.
- **Assuming a normal meets the conic only at its foot.** A normal to a parabola meets it again at $t_2 = -t_1 - \frac{2}{t_1}$. **Correction:** "normal" constrains the line only at the foot; everywhere else it behaves like any other line.
- **Forgetting that up to three normals pass through a point of a parabola.** A question asking for "the normals from (h, k) " expects the cubic. **Correction:** set up $am^3 + m(2a - h) + k = 0$ and use $m_1 + m_2 + m_3 = 0$ as a built-in check on your roots.

Pro-Tip. For a circle, never write a normal formula. Join the point to the centre with the two-point form; it is faster and cannot go wrong.

Pro-Tip. For every normal you write, verify the foot twice — once in the conic, once in the line. Two substitutions catch every sign error, every cubing error and every a -versus- $4a$ error this topic can generate, and they cost less time than re-deriving.

Pro-Tip. When a question mentions the reflection property, the whispering gallery or a headlight, reach for the normal, not the tangent, for the ellipse — the ellipse's **normal** bisects $\angle SPS'$, while for the hyperbola it is the **tangent** that bisects it.

Section 8: Cheat Sheet Box

- **Master rule:** $m_N = -\frac{1}{m_T}$. Every normal formula is the corresponding tangent formula with the slope negated and inverted.
- **Circle:** the normal is the radius produced — it always passes through the centre $(-g, -f)$. For $x^2 + y^2 = r^2$ the normal at (x_1, y_1) is $x_1y - y_1x = 0$.
- **Parabola** $y^2 = 4ax$: normal at (x_1, y_1) is $y - y_1 = -\frac{y_1}{2a}(x - x_1)$; at $(at^2, 2at)$ it is $y + tx =$

$2at + at^3$ (slope $-t$); slope form $y = mx - 2am - am^3$ with foot $(am^2, -2am)$.

- **Parabola extras:** three normals from (h, k) satisfy $am^3 + m(2a - h) + k = 0$, hence $t_1 + t_2 + t_3 = 0$; the normal at t_1 meets the curve again at $t_2 = -t_1 - \frac{2}{t_1}$, and $|t_2| \geq 2\sqrt{2}$.
- **Ellipse:** normal at (x_1, y_1) is $\frac{a^2x}{x_1} - \frac{b^2y}{y_1} = a^2 - b^2$; parametrically $ax \sec \theta - by \csc \theta = a^2 - b^2$; slope form $y = mx \mp \frac{m(a^2 - b^2)}{\sqrt{a^2 + b^2m^2}}$.
- **Hyperbola:** normal at (x_1, y_1) is $\frac{a^2x}{x_1} + \frac{b^2y}{y_1} = a^2 + b^2$; parametrically $ax \cos \theta + by \cot \theta = a^2 + b^2$; slope form $y = mx \mp \frac{m(a^2 + b^2)}{\sqrt{a^2 - b^2m^2}}$. For $xy = c^2$ at $(ct, \frac{c}{t})$: $t^3x - ty = c(t^4 - 1)$.
- **Reflection:** parabola — parallel rays reflect through the focus; ellipse — the **normal** bisects $\angle SPS'$, so focus reflects to focus; hyperbola — the **tangent** bisects $\angle SPS'$.
- **Mnemonics:** “Tangent two terms, normal three.” “Circle normal? Head for the centre.” “Ellipse minus-minus, hyperbola plus-plus.” “Check the foot twice — in the curve and in the line.”

Applied Problems: Conics in the Real World

Exam Relevance: These are the NCERT Miscellaneous Examples and Miscellaneous Exercise problems, and CBSE Boards ask one of them at 3–4 marks with high reliability — the parabolic reflector, the semi-elliptical archway, the suspension-bridge cable and the hyperbolic cooling tower rotate between papers year after year. JEE Main occasionally dresses a standard-form question in the same clothing to test whether you can set up axes under pressure. This is an applications unit, so it carries only the intuition, the worked examples, the practice set and the cheat sheet — every formula it uses has already been established in **The Parabola**, **The Ellipse** and **The Hyperbola**.

Section 1: Concept Intuition

Every conic in this chapter so far has arrived pre-packaged: centre at the origin, axes along the coordinate axes, equation already in standard form. Real objects do not arrive that way. A dish antenna arrives as “60 cm across and 15 cm deep”. An archway arrives as “8 m wide and 2 m high”. A cooling tower arrives as three diameters and two heights. **The entire difficulty of an applied conic problem is the first line of your solution — where you choose to put the origin — and once that choice is made well, the rest is arithmetic you have already mastered.**

Think about it this way... you are photographing the arch of a Metro viaduct and you want the shot dead centre. You do not move the arch. You move yourself until the arch is symmetric in your frame. Choosing axes is exactly that: you cannot change the object, so you place your coordinate frame where the object’s own symmetry already lives. Put the origin at the feature the curve is symmetric about — the vertex of a parabola, the centre of an ellipse or

a hyperbola — and every awkward number in the problem turns into a coordinate you can substitute.

Three habits carry every problem in this unit.

Habit one: the origin goes at the vertex or the centre, never at ground level unless they coincide. For a parabolic dish, the origin is the bottom of the bowl. For a suspension cable, the origin is the lowest point of the cable — which is usually *above* the roadway, and forgetting that gap is the single most common error in the whole topic. For an archway, the origin is the centre of the full ellipse, which sits on the ground because the arch is a *semi*-ellipse. For a cooling tower, the origin is the throat — the narrowest ring — which is usually well above the base.

Habit two: translate each given measurement into one coordinate of one point. A “diameter” or a “width” is always *twice* a coordinate, because the figure is symmetric about your chosen axis. A “depth” or a “height above the throat” is the other coordinate, and it may be negative. Two numbers make one point, and one point on the curve is exactly what you need to pin down the last unknown constant.

Habit three: solve for the missing constant first, then answer the question. Almost every problem here has the same skeleton — the shape fixes the standard form, one given point fixes $4a$ or b^2 , and only then does the actual question get answered by substituting a second value.

Where the modelling stops being exact is worth saying plainly. A real suspension cable carrying only its own weight hangs in a catenary, not a parabola; it becomes a parabola only when the load is uniform along the *horizontal roadway*, which is why NCERT is careful to say “uniformly loaded”. A real cooling tower is a hyperboloid, a three-dimensional surface, and what you model is its vertical cross-section through the axis, which is a hyperbola. A real dish is a paraboloid, and again you model the cross-section. In every case you are working with a two-dimensional slice of a three-dimensional object, and the slice is the conic.

Figure brief: Four small labelled panels arranged in a row, each showing the object with its chosen coordinate frame overlaid in a contrasting colour. Panel 1: a parabolic dish in cross-section, origin at the vertex, axis horizontal, the rim point labelled (depth, half-diameter). Panel 2: a semi-elliptical archway standing on a ground line, origin at the centre of the base, half-width a and height b marked. Panel 3: a suspension bridge with two towers, a horizontal roadway, vertical wires, origin at the lowest point of the cable and clearly *above* the roadway, with the shortest wire length marked as the gap. Panel 4: a cooling tower in outline, origin at the narrowest ring (the throat), with the base below on the negative y -axis and the top above on the positive y -axis.

Section 4: Worked Examples

Example 1 — [CBSE Board Level] The parabolic reflector

A parabolic reflector for a solar cooker is 20 cm across at the rim and 5 cm deep at its centre.

- Find the position of its focus, so that the cooking vessel can be placed there.
- A larger reflector is manufactured with the same focal length but made 45 cm deep.

Find the diameter of its rim.

Setting up. Place the origin at the **vertex** — the deepest point of the bowl — with the axis of the reflector along the positive x -axis. Since the reflector opens to the right and is symmetric about the x -axis, its cross-section is

$$y^2 = 4ax.$$

(a) “20 cm across at the rim” means the rim has diameter 20, so its half-width is $y = 10$. “5 cm deep” means the rim sits at $x = 5$. So the point $(5, 10)$ lies on the parabola:

$$10^2 = 4a(5) \implies 100 = 20a \implies a = 5.$$

The focus of $y^2 = 4ax$ is at $(a, 0)$, so the focus is **5 cm from the vertex, along the axis of the reflector**. The cross-section is $y^2 = 20x$.

Worth noticing: the focus is at $x = 5$, exactly the depth of the rim, so the cooking vessel sits level with the rim of this particular dish. That is a coincidence of these numbers, not a general rule.

(b) Same focal length means the same $a = 5$, so the same equation $y^2 = 20x$. Now the depth is 45 cm, so put $x = 45$:

$$y^2 = 20(45) = 900 \implies y = \pm 30.$$

The rim runs from $(45, -30)$ to $(45, 30)$, so its diameter is

$$30 - (-30) = 60 \text{ cm.}$$

Figure brief: A right-opening parabola $y^2 = 20x$ drawn as the cross-section of a dish, vertex at the origin O , axis along the x -axis, focus $S(5, 0)$ marked with a small vessel symbol. The rim shown as a vertical chord from $(5, -10)$ to $(5, 10)$ with the diameter 20 cm labelled and the depth 5 cm labelled along the axis. Two parallel incoming rays shown striking the curve and reflecting to S .

Example 2 — [JEE Main Level] The semi-elliptical archway

An arch is in the form of a semi-ellipse. It is 8 m wide at the base and 2 m high at the centre. Find the height of the arch at a point 1.5 m from one end of the base.

Setting up. The arch is the upper half of a full ellipse. Place the origin at the **centre of the base**, with the base along the x -axis. Then the semi-major axis lies along x and the semi-minor axis along y :

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, \quad y \geq 0.$$

“8 m wide” is the full major axis, so $2a = 8$ and $a = 4$. “2 m high at the centre” is the semi-minor axis, so $b = 2$. The arch is

$$\frac{x^2}{16} + \frac{y^2}{4} = 1, \quad y \geq 0.$$

Locating the point. The base runs from $x = -4$ to $x = 4$. A point 1.5 m from one end is at distance $4 - 1.5 = 2.5$ m from the centre, so $x = 2.5$. (Taking the other end gives $x = -2.5$, and by symmetry the same height.)

Computing the height.

$$\frac{(2.5)^2}{16} + \frac{y^2}{4} = 1 \implies \frac{6.25}{16} + \frac{y^2}{4} = 1.$$

Multiply throughout by 16:

$$6.25 + 4y^2 = 16 \implies 4y^2 = 9.75 \implies y^2 = 2.4375 = \frac{39}{16}.$$

$$y = \frac{\sqrt{39}}{4} \approx 1.56 \text{ m.}$$

The positive root is taken because the arch is the upper half.

Sanity check. The answer must lie strictly between 0 (at the ends) and 2 (at the centre), and it must be closer to 2 than to 0 because $x = 2.5$ is nearer the centre than the end. The value 1.56 m satisfies both.

Figure brief: A semi-elliptical arch standing on a horizontal ground line, origin O at the centre of the base, base from $(-4, 0)$ to $(4, 0)$ with the width 8 m dimensioned, apex at $(0, 2)$ with the height 2 m dimensioned. A vertical dashed segment at $x = 2.5$ from the ground to the curve, labelled $\sqrt{39}/4 \approx 1.56$ m, with the horizontal distance 1.5 m from the right-hand end also dimensioned.

Example 3 — [JEE Main — Speed Trap] The suspension-bridge cable

The cable of a uniformly loaded suspension bridge hangs in the form of a parabola. The roadway, which is horizontal and 100 m long, is supported by vertical wires attached to the cable; the longest wire is 30 m and the shortest is 6 m. Find the length of the supporting wire attached to the roadway 18 m from the middle.

Setting up. The cable is symmetric about the mid-span, and its lowest point is where the shortest wire meets it. Place the origin at that **lowest point of the cable**, with the y -axis vertical. Opening upwards:

$$x^2 = 4ay.$$

Translating the data. The roadway is 100 m long, so the towers stand at $x = \pm 50$. The longest wires are the ones at the towers, of length 30 m, and the shortest is 6 m at the middle. The origin sits 6 m above the roadway, so the height of the cable **above the origin** at a tower is

$$30 - 6 = 24 \text{ m.}$$

Hence $(50, 24)$ lies on the parabola:

$$50^2 = 4a(24) \implies 2500 = 96a \implies 4a = \frac{2500}{24} = \frac{625}{6}.$$

The cable is $x^2 = \frac{625}{6}y$.

Answering the question. A wire 18 m from the middle is at $x = 18$:

$$18^2 = \frac{625}{6}y \implies 324 = \frac{625}{6}y \implies y = \frac{6 \times 324}{625} = \frac{1944}{625} = 3.1104 \text{ m.}$$

The trap: reporting y as the wire length. The seductive answer is 3.11 m. It is the height of the cable above the *origin*, and the origin is not the roadway — it is 6 m above the roadway, because that is what “shortest wire = 6 m” means. Every wire is measured from the roadway, so

$$\text{wire length} = 6 + y = 6 + 3.1104 = 9.1104 \approx 9.11 \text{ m.}$$

Why 3.11 m is impossible on inspection. The shortest wire in the whole bridge is 6 m. No wire can be shorter than that, so any answer below 6 is wrong before you check the arithmetic. Building that reflex — *compare your answer against the stated extremes* — costs nothing and catches this trap every time.

Second trap, worth naming. “18 m from the middle” is not “18 m from the end”. A student who sets $x = 50 - 18 = 32$ obtains $y = \frac{6 \times 1024}{625} = 9.8304$ and a wire of 15.83 m — a plausible-looking number that answers a different question. Read the phrase, then mark the point on your sketch before substituting.

Figure brief: A suspension bridge cross-section: a horizontal roadway line 100 m long with towers at both ends, a parabolic cable sagging between the tower tops, and vertical wires drawn at intervals. Mark the origin at the lowest point of the cable, 6 m above the roadway, the tower wires labelled 30 m, the mid wire labelled 6 m, and the wire at $x = 18$ highlighted and labelled 9.11 m, with the y portion (3.11 m) and the 6 m portion dimensioned separately.

Example 4 — [JEE Advanced Level] The hyperbolic cooling tower

A cooling tower at a power plant has the shape of a hyperboloid: every vertical cross-section through its axis is a hyperbola. The tower is 150 m tall. Its base diameter is 100 m, and its narrowest cross-section — the throat — has diameter 60 m and occurs 100 m above the base. Find the diameter of the tower at the top.

Setting up. Place the origin at the **centre of the throat**, with the y -axis vertical along the tower’s axis. The throat is the narrowest ring, which is exactly where the two vertices of the hyperbola sit, so the transverse axis is horizontal and the cross-section is

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1.$$

Reading off a . The throat has diameter 60 m, so its radius is 30 m and

$$a = 30, \quad a^2 = 900.$$

Finding b from the base. The base is 100 m *below* the throat, so it sits at $y = -100$. Its diameter is 100 m, so its radius is 50 m, giving the point $(50, -100)$ on the hyperbola:

$$\begin{aligned} \frac{50^2}{900} - \frac{(-100)^2}{b^2} &= 1 \implies \frac{2500}{900} - \frac{10000}{b^2} = 1. \\ \frac{10000}{b^2} &= \frac{2500}{900} - 1 = \frac{2500 - 900}{900} = \frac{1600}{900} = \frac{16}{9}. \\ b^2 &= 10000 \times \frac{9}{16} = 5625 \implies b = 75. \end{aligned}$$

So the modelling hyperbola is

$$\frac{x^2}{900} - \frac{y^2}{5625} = 1.$$

Finding the top. The tower is 150 m tall and the throat is 100 m above the base, so the top is $150 - 100 = 50$ m **above** the throat, at $y = 50$:

$$\frac{x^2}{900} - \frac{50^2}{5625} = 1 \implies \frac{x^2}{900} = 1 + \frac{2500}{5625} = 1 + \frac{4}{9} = \frac{13}{9}.$$

$$x^2 = 900 \times \frac{13}{9} = 1300 \implies x = 10\sqrt{13} \approx 36.06 \text{ m}.$$

That is the radius, so the diameter of the top is

$$2 \times 10\sqrt{13} = 20\sqrt{13} \approx 72.11 \text{ m}.$$

Sanity checks. The throat must be the narrowest section: $60 < 72.11 < 100$, so the top is wider than the throat and narrower than the base, which is exactly the silhouette of a real cooling tower. And the top is 50 m from the throat while the base is 100 m from it, so the top should flare out less than the base does — it flares by 12.11 m of diameter against the base's 40 m, consistent with the curve's increasing steepness away from the vertex.

Why it is “Advanced”. Three things have to be got right simultaneously, and each one is a separate decision. First, recognising that the *throat* — not the base and not the mid-height — is where the vertices are, which is what makes the transverse axis horizontal and the origin sit 100 m up in the air. Second, handling the **signed** height: the base is at $y = -100$, and although y is squared here, a student who places the origin at the base instead never recovers a standard form at all. Third, remembering throughout that every stated *diameter* is twice a coordinate. There is no new formula in this problem — only three modelling judgements stacked on top of the standard hyperbola.

Figure brief: A cooling-tower silhouette drawn as the two branches of the hyperbola $x^2/900 - y^2/5625 = 1$, the left branch forming the tower's left profile and the right branch its right profile, with the vertical axis of symmetry dashed between them. Mark the origin O at the throat centre, the throat ring of diameter 60 m through the vertices $(\pm 30, 0)$, the base at $y = -100$ with diameter 100 m, and the top at $y = 50$ with diameter $20\sqrt{13} \approx 72.11$ m. Dimension the 150 m total height and the 100 m throat height alongside.

Section 5: Practice Exercises

1. **[CBSE]** A parabolic solar cooker is 40 cm wide at the rim and 10 cm deep. Where should the cooking vessel be placed?
2. **[CBSE]** A satellite dish is 8 m across at the rim and 1 m deep at its centre. Find the equation of its cross-section (vertex at the origin, axis along the x -axis) and the position of the receiver.
3. **[CBSE]** A semi-elliptical archway over a road is 10 m wide at the base and 4 m high at the centre. Find the height of the arch at a point 2.5 m from the centre.

4. **[JEE Main]** A whispering gallery is an elliptical hall 100 m long and 50 m wide at its widest. Two people wishing to whisper to each other should stand at the foci. How far from the centre is each focus, and how far apart are the two people?
5. **[JEE Main]** A uniformly loaded suspension bridge has a span of 40 m, its towers rise 10 m above the roadway, and the cable just touches the roadway at the mid-point. Find the length of the supporting wire 10 m from the middle.
6. **[JEE Main]** A parabolic arch bridge has a span of 20 m and a maximum height of 5 m at the centre. Find the height of the arch at a point 4 m from one end.
7. **[JEE Advanced]** A cooling tower is 120 m tall. Its base diameter is 100 m and its throat, of diameter 60 m, lies 80 m above the base. Find b for the modelling hyperbola, and hence the diameter of the tower at a height of 20 m above the base.
8. **[JEE Advanced]** A planet moves in an elliptical orbit with the Sun at one focus. Its nearest distance to the Sun is 2 astronomical units and its farthest distance is 6 astronomical units. Find the semi-major axis, the eccentricity, and the equation of the orbit referred to its centre with the major axis along the x -axis.

Answers: 1. Vertex at the origin, $y^2 = 4ax$; the rim point is $(10, 20)$, so $400 = 40a$ and $a = 10$ — the vessel goes at the focus, 10 cm from the vertex along the axis. 2. The rim point is $(1, 4)$, so $16 = 4a(1)$, giving $4a = 16$ and $a = 4$; the cross-section is $y^2 = 16x$ and the receiver sits at the focus $(4, 0)$, i.e. 4 m from the vertex on the axis. 3. $a = 5$, $b = 4$, so $\frac{6.25}{25} + \frac{y^2}{16} = 1$, giving $\frac{y^2}{16} = \frac{3}{4}$, $y^2 = 12$ and $y = 2\sqrt{3} \approx 3.46$ m. 4. $a = 50$, $b = 25$, so $c^2 = 2500 - 625 = 1875$ and $c = 25\sqrt{3} \approx 43.30$ m from the centre; the two people stand $2c = 50\sqrt{3} \approx 86.60$ m apart. 5. Origin at the lowest point of the cable, which is on the roadway here, so $x^2 = 4ay$ through $(20, 10)$ gives $4a = 40$; at $x = 10$, $y = \frac{100}{40} = 2.5$ m. 6. Origin at the apex with the parabola opening downwards, $x^2 = -4a(y - 5)$ through $(10, 0)$ gives $4a = 20$, so $x^2 = -20(y - 5)$; a point 4 m from one end is at $x = -6$, giving $36 = -20(y - 5)$, $y - 5 = -1.8$ and $y = 3.2$ m. 7. $a = 30$; the base point is $(50, -80)$, so $\frac{2500}{900} - \frac{6400}{b^2} = 1$ gives $\frac{6400}{b^2} = \frac{16}{9}$, hence $b^2 = 3600$ and $b = 60$. At a height of 20 m above the base, $y = 20 - 80 = -60$, so $\frac{x^2}{900} = 1 + \frac{3600}{3600} = 2$, giving $x = 30\sqrt{2}$ and a diameter of $60\sqrt{2} \approx 84.85$ m. 8. Nearest = $a - c = 2$ and farthest = $a + c = 6$, so $a = 4$ and $c = 2$; then $e = \frac{c}{a} = \frac{1}{2}$ and $b^2 = a^2 - c^2 = 16 - 4 = 12$, giving $\frac{x^2}{16} + \frac{y^2}{12} = 1$.

Section 8: Cheat Sheet Box

- **The modelling recipe:** (1) put the origin on the curve's own symmetry feature; (2) turn every given measurement into a coordinate, remembering that a diameter or width is *twice* a coordinate; (3) use one point to solve for the unknown constant; (4) only then substitute to answer the actual question; (5) sanity-check the answer against the stated maxima and minima.
- **Parabolic reflector, dish, headlight, solar cooker:** origin at the **vertex**, axis along the x -axis, $y^2 = 4ax$. Rim point = (depth, $\frac{1}{2} \times$ diameter). The bulb, receiver or vessel goes at the focus $(a, 0)$, at distance a from the vertex.

- **Semi-elliptical archway:** origin at the **centre of the base**, $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ with $y \geq 0$; $a = \frac{1}{2} \times \text{width}$, $b = \text{height at the centre}$. “Distance from one end” = half-width minus that distance, giving $|x|$.
- **Suspension-bridge cable:** origin at the **lowest point of the cable**, $x^2 = 4ay$; the towers give the point $(\frac{1}{2} \times \text{span}, \text{longest wire} - \text{shortest wire})$. Wire length = shortest wire + y . Never report y alone.
- **Whispering gallery / elliptical hall:** origin at the **centre**; $a = \frac{1}{2} \times \text{length}$, $b = \frac{1}{2} \times \text{width}$; the two listening points are the foci at $(\pm c, 0)$ with $c = \sqrt{a^2 - b^2}$, a distance $2c$ apart.
- **Hyperbolic cooling tower:** origin at the **throat centre**, $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ with $a = \frac{1}{2} \times \text{throat diameter}$. Base at negative y , top at positive y ; find b^2 from the base point, then substitute the top’s y . Answers are radii — double them for diameters.
- **Planetary orbit:** Sun at a focus; nearest distance = $a - c$, farthest = $a + c$, so $a = \frac{1}{2}(\text{near} + \text{far})$, $c = \frac{1}{2}(\text{far} - \text{near})$, $e = \frac{c}{a}$, $b^2 = a^2 - c^2$.
- **Modelling limits:** these are two-dimensional cross-sections of three-dimensional surfaces; a suspension cable is a parabola only under a *uniform horizontal load*, and a freely hanging cable is a catenary instead.
- **Mnemonics:** “Origin on the symmetry, always.” “Diameter is two coordinates, never one.” “The cable’s zero is not the roadway’s zero.” “Answer in radii, report in diameters.”

Chapter 10: Conic Sections — Round 2 Addendum

The five typeset subtopics (Introduction, The Circle, The Parabola, The Ellipse, The Hyperbola) and the four Round 1 Supplement units (Parametric Forms and the Auxiliary Circle, Tangents to a Conic, Normals to a Conic, Applied Problems) together cover the whole NCERT syllabus and the bulk of the JEE conics syllabus. Four standing JEE items are still missing, and each of them is a *different question about the same expression T* that Round 1 already introduced — which is exactly why they were easy to overlook.

- Round 1 gave $T = 0$ two jobs: the **tangent** at a point on the conic, and the **chord of contact** from a point outside it. It has two more: it is the **polar** of *any* point, and $T = S_1$ is the **chord bisected at a given point**.
- From the bisected chord comes the **diameter** of a conic — the locus of the midpoints of a system of parallel chords — and conjugate diameters. Neither word appears anywhere in the chapter.
- The hyperbola's asymptotes are derived in the main text as a limiting slope, but never treated as *an object* — a pair of lines with its own equation, tied to the **conjugate hyperbola** by $S + S'' = 2A$. “Conjugate hyperbola” currently appears once, as an MCQ distractor.
- $S + \lambda S' = 0$, the family through the intersection of two curves, is absent entirely. NCERT's own Historical Note credits Lamé (1818) with it — “ $mE + m'E' = 0$ as the curve passing through the points of intersection of two loci $E = 0$ and $E' = 0$ ” — and then never uses it.

A note on numbering. The Round 1 Chapter 10 Supplement used *named units* and no letters or numbers, unlike the lettered Round 1 supplements in other chapters. The Round 2 items therefore start the lettering at A, each keyed below to the Round 1 unit it extends.

Addendum	Topic	Read after
A	The chord bisected at a given point ($T = S_1$), and the diameter of a conic	Tangents to a Conic
B	Pole and polar: what $T = 0$ means for <i>any</i> point	Addendum A
C	Asymptotes as an object, and the conjugate hyperbola	The Hyperbola
D	Families of conics: $S + \lambda S' = 0$	The Circle

Addendum A — The chord bisected at a given point, and the diameter of a conic

Extends Tangents to a Conic, Sections 1-2 (the universal T rule). Not in NCERT; a standing JEE Main item and the foundation of Addendum B.

Method. Round 1 built T by the substitution rule

$$x^2 \rightarrow xx_1, \quad y^2 \rightarrow yy_1, \quad xy \rightarrow \frac{1}{2}(xy_1 + x_1y), \quad x \rightarrow \frac{1}{2}(x + x_1), \quad y \rightarrow \frac{1}{2}(y + y_1), \quad \text{constant} \rightarrow \text{constant},$$

and S_1 as the value of S at (x_1, y_1) . Round 1 used $T = 0$. This addendum uses the other combination.

The chord of the conic $S = 0$ whose midpoint is (x_1, y_1) is $T = S_1$.

Three remarks make it usable.

1. **Why it is true.** Substitute the line through (x_1, y_1) of slope m into $S = 0$. You get a quadratic; demanding that the sum of its roots put the midpoint at (x_1, y_1) fixes m , and the resulting line is $T - S_1 = 0$. Every quadratic term contributes its share of the “sum of roots”, which is precisely what the halving in the substitution rule encodes.
2. **The point must be inside.** A midpoint of a real chord lies *inside* the conic. Check the sign of S_1 first — the same S_1 Round 1 used for counting tangents, read the other way. For circle, parabola and ellipse a genuine bisected chord needs $S_1 < 0$; for the hyperbola $S_1 > 0$ (the reversed row of Round 1’s table) puts you inside a branch.
3. **The three jobs of T , side by side.** $T = 0$ with (x_1, y_1) *on* the conic is the tangent; $T = 0$ with (x_1, y_1) *outside* is the chord of contact; $T = S_1$ with (x_1, y_1) *inside* is the bisected chord. One expression, located by where the point sits.

The diameter. Now let the midpoint vary while the *direction* stays fixed.

Definition. The **diameter** of a conic corresponding to a system of parallel chords of slope m is the locus of the midpoints of those chords.

Deriving each case is one line: write the chord bisected at (h, k) as $T = S_1$, read off its slope, and set that slope equal to m .

Conic	Diameter bisecting chords of slope m	Reading
$x^2 + y^2 = r^2$	$x + my = 0$	a diameter through the centre, perpendicular to the chords

Conic	Diameter bisecting chords of slope m	Reading
$y^2 = 4ax$	$y = \frac{2a}{m}$	a horizontal line: every diameter of a parabola is parallel to its axis
$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$	$y = -\frac{b^2}{a^2m}x$	a line through the centre
$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$	$y = \frac{b^2}{a^2m}x$	a line through the centre

So a **central** conic (circle, ellipse, hyperbola) has all its diameters passing through the centre, which is why “diameter” and “chord through the centre” are the same thing there. A parabola has no centre, and its diameters are a family of parallel lines instead — that is the one place the two definitions part company, and JEE Main tests exactly that.

Conjugate diameters. If the diameter $y = m'x$ bisects chords of slope m , and the diameter $y = mx$ bisects chords of slope m' , the two are called **conjugate diameters**. The condition is read straight off the table:

$$\text{ellipse: } mm' = -\frac{b^2}{a^2}, \quad \text{hyperbola: } mm' = +\frac{b^2}{a^2}, \quad \text{circle } (a = b) : mm' = -1.$$

The circle case says conjugate diameters of a circle are perpendicular — and every other conjugate-diameter fact is that statement squashed by the factor b/a of the auxiliary circle. Two consequences worth memorising:

- The **major and minor axes are a conjugate pair** ($m = 0$ and $m \rightarrow \infty$), and they are the only perpendicular conjugate pair of an ellipse.
- The **equi-conjugate diameters** (the conjugate pair of equal length) are $y = \pm \frac{b}{a}x$, the diagonals of the rectangle $x = \pm a$, $y = \pm b$, each of semi-length $\sqrt{\frac{a^2 + b^2}{2}}$.

Figure brief: A horizontal ellipse centred at O with three parallel chords of slope m drawn across it, their midpoints marked with dots and a straight line drawn through all three dots and through O , labelled “diameter, slope $m' = -b^2/(a^2m)$ ”. Draw one chord of slope m' as a dashed line with its midpoint also on the line $y = mx$, so the reciprocity of the conjugate pair is visible. Mark the rectangle $x = \pm a$, $y = \pm b$ faintly with its two diagonals labelled as the equi-conjugate diameters.

Limiting conditions.

- $T = S_1$ needs (x_1, y_1) strictly inside; at the centre of a central conic every direction works and the “chord bisected at the centre” is not unique, so the formula degenerates ($T = S_1$ becomes $0 = 0$).
- The parabola formula $y = 2a/m$ excludes $m = 0$: chords parallel to the axis are not a system with a diameter, because a horizontal line meets $y^2 = 4ax$ only once.
- The ellipse and hyperbola formulas exclude $m = 0$ in the same written form, but the

limit is honest — as $m \rightarrow 0$ the diameter tends to the y -axis, which is the correct answer (the minor axis bisects all horizontal chords).

Worked Example A1 — [JEE Main]

Find the equation of the chord of the ellipse $\frac{x^2}{25} + \frac{y^2}{9} = 1$ that is bisected at the point $(2, 1)$.

First check the point is inside, or the chord does not exist:

$$S_1 = \frac{4}{25} + \frac{1}{9} - 1 = \frac{36 + 25 - 225}{225} = -\frac{164}{225} < 0. \checkmark$$

Now write T at $(x_1, y_1) = (2, 1)$:

$$T = \frac{2x}{25} + \frac{y}{9} - 1.$$

The chord is $T = S_1$:

$$\frac{2x}{25} + \frac{y}{9} - 1 = -\frac{164}{225}.$$

Multiply through by 225:

$$18x + 25y - 225 = -164 \implies \boxed{18x + 25y = 61}.$$

Check. The point itself: $18(2) + 25(1) = 36 + 25 = 61$. $\checkmark$ Solving $18x + 25y = 61$ against the ellipse gives the two ends

$$\left(2 \mp \frac{10\sqrt{2501}}{183}, 1 \pm \frac{12\sqrt{2501}}{305} \right),$$

whose coordinates average to exactly $(2, 1)$. The ends are ugly and the chord is clean — which is the whole point of the formula. Solving for the ends and averaging is three times the work and misses the marks.

Worked Example A2 — [JEE Advanced]

- Find the diameter of the parabola $y^2 = 12x$ that bisects all chords of slope 3, and explain why the answer does not mention where those chords sit.
- For the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$, find the diameter conjugate to the one of slope $\frac{3}{4}$, and show that this particular pair is the equi-conjugate pair.

(a) Here $4a = 12$, so $a = 3$, and the diameter of $y^2 = 4ax$ for chords of slope m is

$$y = \frac{2a}{m} = \frac{6}{3} = 2, \quad \text{i.e. } y = 2.$$

Direct check: intersect $y = 3x + c$ with $y^2 = 12x$. Eliminating x gives $y^2 - 4y + 4c = 0$, whose roots sum to 4 regardless of c — so the midpoint ordinate is 2 for **every** value of c . That is why the answer carries no c : sliding the chord parallel to itself moves the midpoint along the line $y = 2$ but never off it. And $y = 2$ is horizontal, so it is parallel to the axis $y = 0$ — the parabola's signature.

(b) Here $a^2 = 16$, $b^2 = 9$, and the conjugacy condition is $mm' = -b^2/a^2$:

$$m' = -\frac{9}{16} \div \frac{3}{4} = -\frac{9}{16} \cdot \frac{4}{3} = -\frac{3}{4}.$$

So the conjugate diameter is $y = -\frac{3}{4}x$. Direct check: intersecting $y = \frac{3}{4}x + c$ with the ellipse and averaging gives midpoints satisfying $k = -\frac{3}{4}h$ for every c . ✓

Why this pair is the equi-conjugate one. The equi-conjugate diameters are $y = \pm \frac{b}{a}x = \pm \frac{3}{4}x$ — exactly the two lines we have. Their common semi-length: solving $y = \frac{3}{4}x$ against the ellipse gives $(\pm 2\sqrt{2}, \pm \frac{3\sqrt{2}}{2})$, at distance

$$\sqrt{8 + \frac{9}{2}} = \sqrt{\frac{25}{2}} = \frac{5}{\sqrt{2}} = \sqrt{\frac{16+9}{2}} = \sqrt{\frac{a^2+b^2}{2}} \quad \checkmark$$

from the centre, and by symmetry the other diameter has the same semi-length. Note that $-\frac{3}{4} \times \frac{3}{4} = -\frac{9}{16} \neq -1$: an equi-conjugate pair of a genuine ellipse is never perpendicular. Only the circle, where $b = a$, makes them so.

Why it is “Advanced”. Part (a) rewards seeing that the parameter c vanishes from the sum of the roots before you do any work; part (b) rewards recognising $\pm b/a$ on sight instead of setting up a length calculation. Both are the same habit: read the structure of the quadratic rather than solving it — the habit Round 1 built with $t_1 t_2 = -1$ and Vieta.

Practice.

- Find the chord of the circle $x^2 + y^2 = 25$ bisected at $(2, 3)$.
- Find the chord of the parabola $y^2 = 16x$ bisected at $(5, 4)$.
- Find the chord of the ellipse $\frac{x^2}{36} + \frac{y^2}{9} = 1$ bisected at $(4, 1)$.
- Find the chord of the hyperbola $\frac{x^2}{16} - \frac{y^2}{9} = 1$ bisected at $(6, 2)$.
- Find the diameter of the ellipse $\frac{x^2}{25} + \frac{y^2}{16} = 1$ that bisects all chords of slope 2.
- Find the diameter of the parabola $y^2 = 20x$ that bisects all chords of slope $-\frac{5}{2}$.
- Show that $(4, 4)$ cannot be the midpoint of any chord of $x^2 + y^2 = 25$, and say what goes wrong if you apply $T = S_1$ anyway.
- Find the diameter of the hyperbola $\frac{x^2}{9} - \frac{y^2}{4} = 1$ conjugate to the diameter $y = \frac{2}{3}x$.

Answers. 1. $S_1 = 4 + 9 - 25 = -12 < 0$ (inside); $T = 2x + 3y - 25$, so $T = S_1$ gives $2x + 3y = 13$.
 2. $S_1 = 16 - 80 = -64 < 0$; $T = 4y - 8(x + 5)$, so $4y - 8x - 40 = -64$, i.e. $2x - y - 6 = 0$. (Check: $2(5) - 4 - 6 = 0$.)
 3. $S_1 = \frac{16}{36} + \frac{1}{9} - 1 = -\frac{4}{9}$; $T = \frac{4x}{36} + \frac{y}{9} - 1$, giving $\frac{x}{9} + \frac{y}{9} = \frac{5}{9}$, i.e. $x + y = 5$.
 4. $S_1 = \frac{36}{16} - \frac{4}{9} - 1 = \frac{29}{36} > 0$, which for a hyperbola means inside a branch — a chord exists. $T = \frac{6x}{16} - \frac{2y}{9} - 1$, and $T = S_1$ gives $27x - 16y = 130$. (Check: $162 - 32 = 130$.)
 5. $y = -\frac{b^2}{a^2 m}x = -\frac{16}{25 \cdot 2}x = -\frac{8}{25}x$.
 6. $4a = 20 \Rightarrow a = 5$; $y = \frac{2a}{m} = \frac{10}{-5/2} = -4$.
 7. $S_1 = 16 + 16 - 25 = 7 > 0$, so $(4, 4)$ lies **outside** the circle and no chord has it as a midpoint. Applying $T = S_1$ regardless produces $4x + 4y = 32$, i.e. $x + y = 8$, whose distance from the origin is $8/\sqrt{2} = 4\sqrt{2} \approx 5.66 > 5$ — a line that misses the circle altogether. The algebra

hands you a line; only the sign of S_1 tells you whether it is a chord. 8. $mm' = +\frac{b^2}{a^2} = \frac{4}{9}$, so $m' = \frac{4}{9} \div \frac{2}{3} = \frac{2}{3}$. The diameter $y = \frac{2}{3}x$ is **self-conjugate** — and it is exactly an asymptote $y = \frac{b}{a}x$. That is not a coincidence: see Addendum C.

Addendum B — Pole and polar: what $T = 0$ means for any point

Extends Tangents to a Conic, Section 2, and Addendum A. Not in NCERT; explicitly a syllabus item in the standard JEE references (Vikas Gupta develops “Pole and Polar” for the circle and “Polar and Pole” for the parabola as numbered sections), and a recurring JEE Advanced locus device.

Method. Round 1 gave $T = 0$ two readings and left a third unnamed. Here is the whole picture in one definition.

Definition. For any point $P(x_1, y_1)$ in the plane and any conic $S = 0$, the line $T = 0$ is called the **polar** of P with respect to the conic, and P is called the **pole** of that line.

The point does not have to be on the conic, or outside it, or anywhere in particular. What changes with position is only the *description* of the line:

Where P sits	S_1	What the polar $T = 0$ is
on the conic	$S_1 = 0$	the tangent at P
outside	$S_1 > 0$ (hyperbola: $S_1 < 0$)	the chord of contact of the two tangents from P
inside	$S_1 < 0$ (hyperbola: $S_1 > 0$)	a real line that does not meet the conic
at the centre of a central conic	—	no polar: the line recedes to infinity

So Round 1’s two facts are the top two rows of one table, and “polar” is the name of the whole column.

Going the other way: the pole of a given line. This is the half that is genuinely new, and it is the half examiners ask. Given $lx + my + n = 0$, its pole (x_1, y_1) is found by forcing $T = 0$ to be that line — matching the coefficients of x , of y and the constant, up to a common multiplier. The results:

Conic	Pole of $lx + my + n = 0$	Needs
$x^2 + y^2 = r^2$	$\left(-\frac{lr^2}{n}, -\frac{mr^2}{n}\right)$	$n \neq 0$
$y^2 = 4ax$	$\left(\frac{n}{l}, -\frac{2am}{l}\right)$	$l \neq 0$
$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$	$\left(-\frac{a^2l}{n}, -\frac{b^2m}{n}\right)$	$n \neq 0$
$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$	$\left(-\frac{a^2l}{n}, +\frac{b^2m}{n}\right)$	$n \neq 0$

Do not memorise four rows. Memorise the move: **write $T = 0$ with unknown (x_1, y_1) , then match coefficients with the given line.** The rows fall out in two lines each, and the same move handles a general circle or a shifted conic where no table applies.

The three properties that make the idea worth having.

1. **Reciprocity (La Hire's theorem).** *If the polar of P passes through Q , then the polar of Q passes through P .* For the ellipse the condition " $Q(x_2, y_2)$ lies on the polar of $P(x_1, y_1)$ " reads $\frac{x_1x_2}{a^2} + \frac{y_1y_2}{b^2} = 1$, which is symmetric in the two points — so the statement is *the same equation* read from the other end. Such a pair is called **conjugate points**; for $x^2 + y^2 = r^2$ the condition is simply $x_1x_2 + y_1y_2 = r^2$.
2. **Conjugate lines.** Two lines are **conjugate** when the pole of each lies on the other. For $x^2 + y^2 = r^2$ the condition on $l_1x + m_1y + n_1 = 0$ and $l_2x + m_2y + n_2 = 0$ is

$$r^2(l_1l_2 + m_1m_2) = n_1n_2.$$

3. **The polar of a focus is the corresponding directrix.** For the ellipse, the polar of $S(ae, 0)$ is $\frac{x \cdot ae}{a^2} = 1$, that is $x = \frac{a}{e}$ — the directrix the chapter defined by the focus-directrix ratio. Focus and directrix are pole and polar of each other, which is the cleanest statement of why they always come in pairs. The same computation on $y^2 = 4ax$ sends the focus $(a, 0)$ to $x = -a$.

Figure brief: A single ellipse drawn three times side by side, each with a point P and its polar line. Panel 1: P on the curve, the polar drawn tangent at P . Panel 2: P outside, two tangent lines from P shown faintly and the polar drawn as the chord joining their contact points. Panel 3: P inside, the polar drawn as a line clear of the curve, with an arrow indicating that as P moves toward the centre the polar recedes. Below the three panels, a fourth small figure showing two points P, Q with each one's polar passing through the other, labelled "conjugate points — La Hire".

Limiting conditions.

- **A line through the centre of a central conic has no finite pole.** Setting $n = 0$ in the table divides by zero; geometrically the pole has gone to infinity. In particular the axes, the asymptotes and every diameter are in this class.
- **A line parallel to a parabola's axis has no finite pole** ($l = 0$ divides by zero), for the same reason: the parabola's centre is at infinity in that direction.

- **The polar exists even when no tangent does.** For an interior point $S_1 < 0$ there are no real tangents at all, yet $T = 0$ is a perfectly real line. Calling it “the chord of contact” there is the standard error; it is a polar and nothing more.

Worked Example B1 — [JEE Main]

- (a) Find the polar of $(3, 4)$ with respect to $x^2 + y^2 = 9$, and say what kind of line it is. (b) Find the pole of the line $3x + 4y = 12$ with respect to $x^2 + y^2 = 25$.

- (a) For $x^2 + y^2 = r^2$, $T = xx_1 + yy_1 - r^2$, so the polar of $(3, 4)$ is

$$3x + 4y - 9 = 0.$$

Locate the point: $S_1 = 9 + 16 - 9 = 16 > 0$, so $(3, 4)$ is **outside** the circle and this polar is the **chord of contact** of the two tangents from $(3, 4)$. (Sanity: the distance of $3x + 4y = 9$ from the origin is $9/5 = 1.8 < 3$, so the line really does cut the circle.)

- (b) Write the polar of an unknown pole (x_1, y_1) and match:

$$xx_1 + yy_1 - 25 = 0 \quad \text{must be} \quad 3x + 4y - 12 = 0.$$

Comparing coefficients up to a multiplier μ :

$$x_1 = 3\mu, \quad y_1 = 4\mu, \quad -25 = -12\mu \implies \mu = \frac{25}{12}.$$

So the pole is

$$\left(\frac{75}{12}, \frac{100}{12}\right) = \boxed{\left(\frac{25}{4}, \frac{25}{3}\right)}.$$

Check. The polar of $\left(\frac{25}{4}, \frac{25}{3}\right)$ is $\frac{25}{4}x + \frac{25}{3}y = 25$; multiplying by $\frac{12}{25}$ gives $3x + 4y = 12$. ✓ The table entry agrees: with $l = 3$, $m = 4$, $n = -12$, $r^2 = 25$, $\left(-\frac{3 \cdot 25}{-12}, -\frac{4 \cdot 25}{-12}\right) = \left(\frac{25}{4}, \frac{25}{3}\right)$.

Worked Example B2 — [JEE Advanced]

A variable chord of the parabola $y^2 = 4ax$ subtends a right angle at the vertex. Find the locus of the pole of that chord.

Step 1 — turn “right angle at the vertex” into a condition on parameters. Let the ends be $P(at_1^2, 2at_1)$ and $Q(at_2^2, 2at_2)$. The vertex is the origin, so $OP \perp OQ$ means $\overrightarrow{OP} \cdot \overrightarrow{OQ} = 0$:

$$a^2 t_1^2 t_2^2 + 4a^2 t_1 t_2 = 0 \implies a^2 t_1 t_2 (t_1 t_2 + 4) = 0 \implies t_1 t_2 = -4$$

(the root $t_1 t_2 = 0$ is rejected: it puts an end at the vertex itself, where no angle is subtended). Compare this with Round 1’s focal-chord condition $t_1 t_2 = -1$ — same shape, different constant, and the two are worth storing as a pair.

Step 2 — write the chord. The chord joining t_1 and t_2 on $y^2 = 4ax$ is

$$2x - (t_1 + t_2)y + 2at_1 t_2 = 0,$$

which you can verify in one substitution: putting $(at_1^2, 2at_1)$ in gives $2at_1^2 - 2at_1^2 - 2at_1t_2 + 2at_1t_2 = 0$. With $t_1t_2 = -4$:

$$2x - (t_1 + t_2)y - 8a = 0.$$

Step 3 — take the pole. For $y^2 = 4ax$ the pole of $lx + my + n = 0$ is $\left(\frac{n}{l}, -\frac{2am}{l}\right)$. Here $l = 2$, $m = -(t_1 + t_2)$, $n = -8a$:

$$\text{pole} = \left(\frac{-8a}{2}, \frac{2a(t_1 + t_2)}{2}\right) = (-4a, a(t_1 + t_2)).$$

Step 4 — eliminate the parameter. The abscissa is $-4a$ whatever $t_1 + t_2$ is, and the ordinate $a(t_1 + t_2)$ takes every real value as t_1 varies. Hence the locus is the whole line

$$\boxed{x = -4a}.$$

Reading the answer. The chords themselves all pass through the fixed point $(4a, 0)$ — put $y = 0$ in the chord equation. So the family “all chords through $(4a, 0)$ ” has poles filling the line $x = -4a$, and by La Hire that line is precisely the polar of $(4a, 0)$: indeed the polar of $(4a, 0)$ is $y \cdot 0 = 2a(x + 4a)$, i.e. $x = -4a$. ✓

Why it is “Advanced”. Three separate ideas chain: a parametric perpendicularity condition, the pole formula, and then the recognition that “the poles of all lines through a fixed point fill that point’s polar” — the reciprocity theorem doing real work rather than sitting in a list. Once you see that last step, the four-step computation above collapses into a single line, and that is exactly the compression JEE Advanced is paying for.

Practice.

- Find the polar of $(2, -1)$ with respect to $\frac{x^2}{9} - \frac{y^2}{4} = 1$.
- Find the pole of $2x + 3y = 6$ with respect to $x^2 + y^2 = 16$.
- Find the pole of $x - 2y + 3 = 0$ with respect to $y^2 = 8x$.
- Find the pole of $x + y = 1$ with respect to $\frac{x^2}{8} + \frac{y^2}{4} = 1$.
- Show that the polar of the focus $(3, 0)$ of $\frac{x^2}{25} + \frac{y^2}{16} = 1$ is its directrix.
- Are $(1, 2)$ and $(3, 4)$ conjugate points with respect to $x^2 + y^2 = 11$?
- Why does the line $2x - 3y = 0$ have no pole with respect to $\frac{x^2}{4} + \frac{y^2}{9} = 1$?
- The polar of a point P with respect to $x^2 + y^2 = 9$ is $x + 2y = 3$. Find P , and state whether P is inside or outside the circle.

Answers. 1. $T = \frac{2x}{9} + \frac{y}{4} - 1$ (note the hyperbola’s minus turns $y_1 = -1$ into $+y/4$); clearing 36 gives $8x + 9y = 36$. 2. $l = 2$, $m = 3$, $n = -6$, $r^2 = 16$: $(\frac{32}{6}, \frac{48}{6}) = (\frac{16}{3}, 8)$. 3. $4a = 8 \Rightarrow a = 2$; $l = 1$, $m = -2$, $n = 3$: $(\frac{3}{1}, -\frac{2(2)(-2)}{1}) = (3, 8)$. (Check: the polar of $(3, 8)$ is $8y = 4(x + 3)$, i.e. $x - 2y + 3 = 0$.) 4. $l = m = 1$, $n = -1$, $a^2 = 8$, $b^2 = 4$: $(8, 4)$. (Check: $\frac{8x}{8} + \frac{4y}{4} = 1$ is $x + y = 1$.) 5. $a^2 = 25$, $b^2 = 16$, so $c = 3$ and $e = \frac{3}{5}$. The polar of $(3, 0)$ is $\frac{3x}{25} = 1$, i.e. $x = \frac{25}{3} = \frac{a}{e}$ — the directrix. 6. Condition: $x_1x_2 + y_1y_2 = r^2$, i.e. $3 + 8 = 11$. Yes, they are conjugate. 7. It passes

through the centre $(0, 0)$, so $n = 0$ and the pole formula divides by zero. The pole is the point at infinity in the direction conjugate to that diameter; no finite point has this line as its polar.

8. Match $xx_1 + yy_1 - 9 = 0$ with $x + 2y - 3 = 0$: $x_1 = \mu$, $y_1 = 2\mu$, $-9 = -3\mu$, so $\mu = 3$ and $P = (3, 6)$. Then $S_1 = 9 + 36 - 9 = 36 > 0$, so P is outside and the given line is the chord of contact from P .

Addendum C — Asymptotes as an object, and the conjugate hyperbola

Extends The Hyperbola, Sections 2-3 and its Cheat Sheet. The main text derives $y = \pm \frac{b}{a}x$ as a limit and uses the slopes; it never writes the asymptotes as a single second-degree equation, never defines the conjugate hyperbola, and never handles a hyperbola that is not centred at the origin. All three are JEE Main and Advanced staples.

Method. Start from the observation the main text stopped one line short of. For $S \equiv \frac{x^2}{a^2} - \frac{y^2}{b^2} - 1 = 0$, the two asymptotes $bx - ay = 0$ and $bx + ay = 0$ multiply together to give one second-degree equation:

$$A \equiv \frac{x^2}{a^2} - \frac{y^2}{b^2} = 0 \quad \text{because} \quad \left(\frac{x}{a} - \frac{y}{b}\right) \left(\frac{x}{a} + \frac{y}{b}\right) = \frac{x^2}{a^2} - \frac{y^2}{b^2}.$$

The hyperbola and its pair of asymptotes differ only in the constant term.

Change the constant once more, from 0 to -1 , and you get a third curve:

Definition. The **conjugate hyperbola** of $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is $\frac{x^2}{a^2} - \frac{y^2}{b^2} = -1$, that is $\frac{y^2}{b^2} - \frac{x^2}{a^2} = 1$: the transverse and conjugate axes have swapped roles.

Writing S'' for the conjugate hyperbola, the three curves sit in one identity:

$$S + S'' = 2A.$$

Everything in this addendum is a consequence of that line. Three curves, one constant, three values 1, 0, -1 .

The comparison table. Let $S : \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ and $S'' : \frac{y^2}{b^2} - \frac{x^2}{a^2} = 1$.

Feature	Hyperbola S	Conjugate hyperbola S''
Transverse axis	x -axis, length $2a$	y -axis, length $2b$
Conjugate axis	y -axis, length $2b$	x -axis, length $2a$
Vertices	$(\pm a, 0)$	$(0, \pm b)$

Feature	Hyperbola S	Conjugate hyperbola S''
c (centre to focus)	$\sqrt{a^2 + b^2}$	$\sqrt{a^2 + b^2}$ — the same
Foci	$(\pm c, 0)$	$(0, \pm c)$
Eccentricity	$e = \sqrt{1 + \frac{b^2}{a^2}}$	$e' = \sqrt{1 + \frac{a^2}{b^2}}$
Directrices	$x = \pm \frac{a}{e}$	$y = \pm \frac{b}{e'}$
Latus rectum	$\frac{2b^2}{a}$	$\frac{2a^2}{b}$
Asymptotes	$y = \pm \frac{b}{a}x$	$y = \pm \frac{b}{a}x$ — the same pair

Two facts from that table earn their own line, because both are examined directly:

$$\frac{1}{e^2} + \frac{1}{e'^2} = 1, \quad \text{the four foci lie on the circle } x^2 + y^2 = a^2 + b^2.$$

The first follows from $\frac{1}{e^2} = \frac{a^2}{a^2 + b^2}$ and $\frac{1}{e'^2} = \frac{b^2}{a^2 + b^2}$; the second from the shared c .

Asymptote facts worth carrying.

- **Angle.** The asymptotes make an angle $2\theta = 2 \tan^{-1} \frac{b}{a}$ across the transverse axis, and then $e = \sec \theta$. (Check on $a = b$: $\theta = 45^\circ$, $e = \sec 45^\circ = \sqrt{2}$ — the rectangular hyperbola, exactly as the main text says.)
- **The axes bisect the angles between the asymptotes**, so the asymptotes are symmetric about both axes and pass through the centre.
- **Product of distances.** For *any* point on the hyperbola, the product of its perpendicular distances to the two asymptotes is the constant $\frac{a^2 b^2}{a^2 + b^2}$. This is why the branches approach the asymptotes without meeting them: as one distance grows the other must shrink, and neither can reach zero.
- **Tangent facts.** The portion of any tangent intercepted between the asymptotes is bisected at the point of contact, and the triangle formed by that tangent and the two asymptotes has constant area ab — independent of where the tangent touches.
- **A self-conjugate diameter is an asymptote.** In Addendum A the hyperbola's conjugacy condition was $mm' = b^2/a^2$; setting $m = m'$ gives $m = \pm b/a$. The asymptotes are the two directions that bisect their own parallel chords.

Asymptotes of a hyperbola that is not centred at the origin. Since the curve and its asymptotes differ only in the constant, write $S + \lambda = 0$ and choose λ to make it a **pair of straight lines** — that is, to make the 3×3 determinant

$$\begin{vmatrix} A & H & G \\ H & B & F \\ G & F & C + \lambda \end{vmatrix} = 0 \quad \text{for } Ax^2 + 2Hxy + By^2 + 2Gx + 2Fy + C = 0.$$

There is exactly one such λ , and the factorised result is the asymptote pair. The conjugate hyperbola then comes free from $S'' = 2A - S$.

Figure brief: One pair of asymptotes drawn as a large dashed X through the origin, with *both* hyperbolas drawn inside it: the original opening left-right in the two horizontal wedges, the conjugate opening up-down in the two vertical wedges, so that the four branches fill all four wedges of the X. Mark the rectangle $x = \pm a$, $y = \pm b$ whose corners the asymptotes pass through, the four vertices $(\pm a, 0)$ and $(0, \pm b)$ on its sides, and the four foci $(\pm c, 0)$, $(0, \pm c)$ with a faint circle of radius $c = \sqrt{a^2 + b^2}$ through all four.

Limiting conditions.

- The asymptote pair $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 0$ is a **degenerate conic** — the pair of intersecting lines the Introduction listed under “plane through the vertex”. It is the boundary case between the hyperbola and its conjugate, sitting at constant 0 between 1 and -1 .
- “Conjugate hyperbola” and “rectangular hyperbola” are unrelated conditions. Every hyperbola has a conjugate; only $a = b$ makes one rectangular. The main text’s MCQ distractor already flags this confusion.
- The relation $\frac{1}{e^2} + \frac{1}{e'^2} = 1$ forces both eccentricities above 1 and at least one of them below $\sqrt{2}$ — a useful reasonableness check.

Worked Example C1 — [JEE Main]

For the hyperbola $9x^2 - 16y^2 = 144$, find the equations of the asymptotes, the equation of the conjugate hyperbola, the two eccentricities, and verify that $\frac{1}{e^2} + \frac{1}{e'^2} = 1$.

Divide by 144 to reach standard form:

$$\frac{x^2}{16} - \frac{y^2}{9} = 1, \quad a = 4, \quad b = 3.$$

Asymptotes. Same left side, constant 0:

$$\frac{x^2}{16} - \frac{y^2}{9} = 0 \implies 9x^2 - 16y^2 = 0 \implies (3x - 4y)(3x + 4y) = 0,$$

so the asymptotes are $3x - 4y = 0$ and $3x + 4y = 0$, i.e. $y = \pm \frac{3}{4}x$ — matching $y = \pm \frac{b}{a}x$.

Conjugate hyperbola. Same left side, constant -1 :

$$\frac{x^2}{16} - \frac{y^2}{9} = -1 \iff \frac{y^2}{9} - \frac{x^2}{16} = 1 \iff 9x^2 - 16y^2 = -144.$$

Its transverse axis is the y -axis, of length $2b = 6$; its vertices are $(0, \pm 3)$.

Eccentricities.

$$e = \sqrt{1 + \frac{9}{16}} = \frac{5}{4}, \quad e' = \sqrt{1 + \frac{16}{9}} = \frac{5}{3}.$$

$$\frac{1}{e^2} + \frac{1}{e'^2} = \frac{16}{25} + \frac{9}{25} = 1. \quad \checkmark$$

Two free checks. Both curves have $c = \sqrt{16 + 9} = 5$: the original’s foci are $(\pm 5, 0)$ and the conjugate’s are $(0, \pm 5)$, all four on $x^2 + y^2 = 25$. And $S + S'' = (9x^2 - 16y^2 - 144) + (9x^2 - 16y^2 + 144) = 2(9x^2 - 16y^2) = 2A$. $\checkmark$

Worked Example C2 — [JEE Advanced]

Find the asymptotes of the curve $xy - 3y - 2x = 0$, identify the curve, and write down its conjugate hyperbola.

Step 1 — set up $S + \lambda = 0$. Write $S \equiv xy - 3y - 2x$. Comparing with $Ax^2 + 2Hxy + By^2 + 2Gx + 2Fy + C = 0$ gives $A = 0$, $H = \frac{1}{2}$, $B = 0$, $G = -1$, $F = -\frac{3}{2}$, $C = 0$. Demand that $S + \lambda = 0$ be a pair of lines:

$$\begin{vmatrix} 0 & \frac{1}{2} & -1 \\ \frac{1}{2} & 0 & -\frac{3}{2} \\ -1 & -\frac{3}{2} & \lambda \end{vmatrix} = 0.$$

Expanding: $0(0 - \frac{9}{4}) - \frac{1}{2}(\frac{\lambda}{2} - \frac{3}{2}) - 1(-\frac{3}{4} - 0) = -\frac{\lambda}{4} + \frac{3}{4} + \frac{3}{4} = 0$, so

$$\lambda = 6.$$

Step 2 — factorise.

$$xy - 3y - 2x + 6 = y(x - 3) - 2(x - 3) = (x - 3)(y - 2),$$

so the asymptotes are

$$\boxed{x = 3 \quad \text{and} \quad y = 2.}$$

Step 3 — identify the curve. The asymptotes meet at $(3, 2)$, which must be the centre. Put $X = x - 3$, $Y = y - 2$:

$$xy - 3y - 2x = (X + 3)(Y + 2) - 3(Y + 2) - 2(X + 3) = XY - 6,$$

so the curve is $XY = 6$ — a **rectangular hyperbola** $XY = c^2$ with $c^2 = 6$, centred at $(3, 2)$, with perpendicular asymptotes (hence $e = \sqrt{2}$, exactly as The Hyperbola's "rectangular means root-two" fact predicts).

Step 4 — the conjugate. From $S + S'' = 2A$:

$$S'' = 2(xy - 3y - 2x + 6) - (xy - 3y - 2x) = xy - 3y - 2x + 12,$$

i.e. $xy - 2x - 3y + 12 = 0$, which in shifted coordinates is $XY = -6$ — the same rectangular hyperbola turned into the other pair of wedges, as it must be.

Why it is "Advanced". The whole solution rests on one structural fact — *curve and asymptotes differ only in the constant* — used three times: to find λ , to locate the centre without completing any square, and to produce the conjugate for free. A student who instead tries to rotate axes or to take limits as $x \rightarrow \infty$ will get there eventually and run out of time. Note also that this curve has no x^2 or y^2 term at all, so none of the main text's standard-form machinery applies directly; the asymptote route is the only fast one.

Practice.

1. Find the asymptotes of $\frac{x^2}{25} - \frac{y^2}{9} = 1$.
2. Write the conjugate hyperbola of $\frac{x^2}{4} - \frac{y^2}{5} = 1$ and find both eccentricities.

- Find the equation of the hyperbola whose asymptotes are $3x - 4y = 0$ and $3x + 4y = 0$ and which passes through $(8, 3\sqrt{3})$.
- The angle between the asymptotes of a hyperbola is 60° (the angle containing the transverse axis). Find e using $e = \sec \theta$, and check it against The Hyperbola's Example 4.
- For $\frac{x^2}{9} - \frac{y^2}{16} = 1$, find the product of the perpendicular distances from any point of the curve to the two asymptotes.
- Find the area of the triangle formed by any tangent to $\frac{x^2}{16} - \frac{y^2}{9} = 1$ and its two asymptotes.
- Find the asymptotes of $xy - 2x - 4y = 0$ and the centre of the curve.
- A hyperbola and its conjugate have eccentricities e and e' . If $e = \frac{13}{12}$, find e' .

Answers. 1. $\frac{x^2}{25} - \frac{y^2}{9} = 0 \Rightarrow (3x-5y)(3x+5y) = 0$, i.e. $y = \pm \frac{3}{5}x$. 2. $\frac{y^2}{5} - \frac{x^2}{4} = 1$; $e = \sqrt{1 + \frac{5}{4}} = \frac{3}{2}$ and $e' = \sqrt{1 + \frac{4}{5}} = \frac{3}{\sqrt{5}} = \frac{3\sqrt{5}}{5}$. Check: $\frac{4}{9} + \frac{5}{9} = 1$. ✓ 3. The curve is $9x^2 - 16y^2 = k$ for some k ; substituting $(8, 3\sqrt{3})$ gives $9(64) - 16(27) = 576 - 432 = 144$, so $9x^2 - 16y^2 = 144$, i.e. $\frac{x^2}{16} - \frac{y^2}{9} = 1$. (Round 1 used this very point in a tangent exercise.) 4. $2\theta = 60^\circ \Rightarrow \theta = 30^\circ$ and $e = \sec 30^\circ = \frac{2}{\sqrt{3}}$ — the same answer The Hyperbola's Example 4 obtained the long way through $\tan \theta = b/a$ and $e^2 = 1 + b^2/a^2$. For the obtuse reading $2\theta = 120^\circ$, $e = \sec 60^\circ = 2$. 5. $\frac{a^2b^2}{a^2 + b^2} = \frac{9 \cdot 16}{25} = \frac{144}{25}$. 6. $ab = 4 \times 3 = 12$ square units, whatever the point of contact. 7. $S = xy - 2x - 4y$; $A = B = 0$, $H = \frac{1}{2}$, $G = -1$, $F = -2$, and the determinant condition gives $\lambda = 8$. Then $xy - 2x - 4y + 8 = (x-4)(y-2)$, so the asymptotes are $x = 4$ and $y = 2$ and the centre is $(4, 2)$. (The curve is $XY = 8$.) 8. $\frac{1}{e'^2} = 1 - \frac{144}{169} = \frac{25}{169}$, so $e' = \frac{13}{5}$.

Addendum D — Families of conics: $S + \lambda S' = 0$

Extends The Circle, Sections 2-4 (the three-point method of its Example 4), and Tangents to a Conic. NCERT's Historical Note names this idea and the chapter never uses it; JEE Main asks a family-of-circles question most sessions.

Method. The Circle found a circle through three points by substituting each point and solving three simultaneous equations for g , f , c . When the points are handed to you as “the intersection of two curves”, there is a far better move.

The principle. If $S = 0$ and $S' = 0$ are two curves and P is a point common to both, then $S + \lambda S' = 0$ also passes through P , for every value of λ — because $S(P) = 0$ and $S'(P) = 0$ make $S(P) + \lambda S'(P) = 0$ automatically.

That is the entire theory. One unknown constant λ , fixed by one extra condition, in place of a three-variable simultaneous system. The four standard uses:

Situation	Family	Fix λ by
Circles through the intersection of two circles S, S'	$S + \lambda S' = 0, \lambda \neq -1$	one further point, or a centre condition, or a radius
Circles through the intersection of a circle S and a line L	$S + \lambda L = 0$	one further point or condition
Circles through two given points A, B	$(x - x_1)(x - x_2) + (y - y_1)(y - y_2) + \lambda L_{AB} = 0$	one further condition
Circle touching a line $L = 0$ at a given point (x_1, y_1)	$(x - x_1)^2 + (y - y_1)^2 + \lambda L = 0$	one further condition
Conics through the four intersections of two conics	$S + \lambda S' = 0$	one further point, or “make it a circle”
Conics through the four intersections of two line-pairs	$L_1 L_2 + \lambda L_3 L_4 = 0$	one further point

Three things to notice, all examinable.

1. $\lambda = -1$ is **special for two circles**. Both have $x^2 + y^2$ with coefficient 1, so those terms cancel and $S - S' = 0$ collapses to a straight line:

$$2(g_1 - g_2)x + 2(f_1 - f_2)y + (c_1 - c_2) = 0.$$

That is the **radical axis**, and when the circles actually meet it is their **common chord**. This is why $\lambda \neq -1$ is excluded above — at that one value the “circle” degenerates into a line. It is also the fastest way to find a common chord: subtract.

2. **“Make it a circle” is a coefficient condition.** In $S + \lambda S' = 0$ built from two conics, demanding *coefficient of x^2 = coefficient of y^2* and *coefficient of xy = 0* is one or two linear equations in λ . If a λ exists, the four intersection points are **concyclic** — a favourite JEE Advanced way to hide a circle inside an ellipse problem.
3. **Divide by $1 + \lambda$ at the end.** For two circles, $S + \lambda S' = 0$ has x^2 -coefficient $1 + \lambda$. Normalise before reading off centre and radius, or the $(-g, -f)$ rule gives nonsense.

Figure brief: Two intersecting circles drawn with their two common points P and Q marked. Through P and Q draw three further circles of visibly different sizes, all of the same family, plus the straight line PQ drawn dashed and labelled “ $\lambda = -1$: the radical axis, the degenerate member”. Caption that every member of the pencil passes through P and Q , and that the line is the member that has lost its curvature.

Limiting conditions.

- The curves must be written with the **same leading normalisation** for the $\lambda = -1$ degeneration to be the radical axis; if one circle is $2x^2 + 2y^2 + \dots$, divide by 2 first.
- $S + \lambda S' = 0$ passes through the common points **whether or not those points are real**. Two non-intersecting circles still have a radical axis; it is simply a line that meets neither.

- The family $S + \lambda S' = 0$ misses exactly one curve of the pencil, namely $S' = 0$ itself. When a question's answer might be S' , use $\mu S + \lambda S' = 0$ instead.

Worked Example D1 — [JEE Main]

Find the equation of the circle passing through the points of intersection of

$$S : x^2 + y^2 - 4x - 6y + 8 = 0 \quad \text{and} \quad S' : x^2 + y^2 - 2x - 4y + 4 = 0$$

and also through the point $(3, 3)$.

Write the family

$$S + \lambda S' = 0.$$

Substitute $(3, 3)$ and solve for λ :

$$\begin{aligned} S(3, 3) &= 9 + 9 - 12 - 18 + 8 = -4, & S'(3, 3) &= 9 + 9 - 6 - 12 + 4 = 4, \\ -4 + 4\lambda &= 0 \implies \lambda = 1. \end{aligned}$$

So the circle is $S + S' = 0$:

$$2x^2 + 2y^2 - 6x - 10y + 12 = 0 \implies \boxed{x^2 + y^2 - 3x - 5y + 6 = 0}$$

after dividing by $1 + \lambda = 2$. Its centre is $(\frac{3}{2}, \frac{5}{2})$ and its radius is $\sqrt{\frac{9}{4} + \frac{25}{4} - 6} = \sqrt{\frac{10}{4}} = \frac{\sqrt{10}}{2}$.

Check. The common points of S and S' are found from the radical axis $S - S' = 0$, that is $-2x - 2y + 4 = 0$, i.e. $x + y = 2$; substituting into S' gives $(0, 2)$ and $(1, 1)$. Both satisfy the answer: $0 + 4 - 0 - 10 + 6 = 0$ and $1 + 1 - 3 - 5 + 6 = 0$. And $(3, 3)$ gives $9 + 9 - 9 - 15 + 6 = 0$. ✓

Speed note. Doing this the Round 1 way — solve $S = S' = 0$ for the two points, then run the three-point method with $(3, 3)$ — takes three simultaneous equations in g, f, c . The family method took one linear equation in λ .

Worked Example D2 — [JEE Advanced]

Show that the four points of intersection of the ellipses

$$2x^2 + 3y^2 = 6 \quad \text{and} \quad 3x^2 + 2y^2 = 6$$

are concyclic, and find the circle on which they lie. What does $\lambda = -1$ give instead?

Step 1 — write the family. With $S = 2x^2 + 3y^2 - 6$ and $S' = 3x^2 + 2y^2 - 6$,

$$S + \lambda S' = (2 + 3\lambda)x^2 + (3 + 2\lambda)y^2 - 6(1 + \lambda) = 0.$$

There is no xy term to kill, so the only requirement for a circle is that the two squared coefficients agree:

$$2 + 3\lambda = 3 + 2\lambda \implies \lambda = 1.$$

Step 2 — read off the circle.

$$5x^2 + 5y^2 - 12 = 0 \implies \boxed{x^2 + y^2 = \frac{12}{5}},$$

a circle of radius $\sqrt{12/5} = \frac{2\sqrt{15}}{5} \approx 1.549$. Since a λ exists, the four points are concyclic.

Step 3 — verify directly. Adding the two given equations gives $5x^2 + 5y^2 = 12$ immediately — which is the same circle and is the one-line version of the whole argument. Solving the pair: subtracting gives $x^2 = y^2$, so $y = \pm x$, and then $5x^2 = 6$ gives $x = \pm\sqrt{6/5}$. The four points are

$$\left(\pm\frac{\sqrt{30}}{5}, \pm\frac{\sqrt{30}}{5} \right),$$

each at distance $\sqrt{\frac{6}{5} + \frac{6}{5}} = \sqrt{\frac{12}{5}}$ from the origin. ✓

Step 4 — the $\lambda = -1$ member.

$$S - S' = -x^2 + y^2 = 0 \implies y = \pm x,$$

a **pair of straight lines** through the four points — the degenerate member of the pencil, and the direct analogue of the radical axis in the two-circle case. Notice it gives the four points' directions for free, which is how Step 3 got started.

Why it is “Advanced”. The insight is that “concyclic” is not a geometric fact to be checked point by point but an **algebraic condition on one parameter**: a circle exists in the pencil exactly when the coefficients can be forced to match. Recognising that turns a four-point verification into solving $2 + 3\lambda = 3 + 2\lambda$. The same move handles the general version — two conics $ax^2 + 2hxy + by^2 + \dots$ meet in four concyclic points precisely when the pencil contains a circle, which needs both the xy coefficient to vanish and the two square coefficients to agree: two conditions on one λ , so it is genuinely a restriction and not automatic.

Practice.

1. Find the common chord of $x^2 + y^2 - 4x - 6y + 4 = 0$ and $x^2 + y^2 - 2x - 8y + 4 = 0$.
2. Find the circle through the intersection of $x^2 + y^2 = 9$ and $x + y = 3$ that also passes through $(1, 1)$.
3. Find the circle that touches the line $3x + 4y = 25$ at the point $(3, 4)$ and passes through the origin.
4. Find the radical axis of $x^2 + y^2 = 4$ and $x^2 + y^2 - 6x + 8 = 0$, and say whether the two circles actually meet.
5. Write the family of circles through $(1, 0)$ and $(0, 1)$, and find the member whose centre lies on $y = x$.
6. Find the equation of the conic through the four points of intersection of $x^2 + 4y^2 = 4$ and $4x^2 + y^2 = 4$ that is a circle.
7. Why can the family $S + \lambda S' = 0$ never produce the circle $S' = 0$ itself, and how do you repair that?
8. Two circles $S = 0$ and $S' = 0$ do not intersect. Does $S - S' = 0$ still define a line, and what is it?

Answers. 1. Subtract: $-2x + 2y = 0$, i.e. $y = x$. 2. Family $x^2 + y^2 - 9 + \lambda(x + y - 3) = 0$; at $(1, 1)$: $-7 + \lambda(-1) = 0$, so $\lambda = -7$ and the circle is $x^2 + y^2 - 7x - 7y + 12 = 0$. (Check: it passes through $(0, 3)$ and $(3, 0)$, the two points where the line cuts the circle.) 3. Family $(x - 3)^2 + (y - 4)^2 + \lambda(3x + 4y - 25) = 0$; at the origin $9 + 16 - 25\lambda = 0$, so $\lambda = 1$ and the circle

is $x^2 + y^2 - 3x - 4y = 0$. (Check: solving it against $3x + 4y = 25$ gives the double point $(3, 4)$, so the contact is genuine.) 4. Subtract: $6x - 12 = 0$, i.e. $x = 2$. The circles do meet — $x = 2$ meets $x^2 + y^2 = 4$ at the single point $(2, 0)$, and that point satisfies the second circle too, so the circles touch externally at $(2, 0)$ and the radical axis is their common tangent there. 5. $(x - 1)(x - 0) + (y - 0)(y - 1) + \lambda(x + y - 1) = 0$, i.e. $x^2 + y^2 + (\lambda - 1)x + (\lambda - 1)y - \lambda = 0$. The centre $(\frac{1-\lambda}{2}, \frac{1-\lambda}{2})$ lies on $y = x$ for **every** λ — the two given points are symmetric in $y = x$, so the whole family qualifies. (A well-posed version of this question must impose something else, such as a radius.) 6. Adding the two equations gives $5x^2 + 5y^2 = 8$, i.e. $x^2 + y^2 = \frac{8}{5}$; the corresponding λ is 1, from $1 + 4\lambda = 4 + \lambda$. 7. Putting $\lambda \rightarrow \infty$ would be needed, which is not a value. Repair it by writing the symmetric family $\mu S + \lambda S' = 0$ with $(\mu, \lambda) \neq (0, 0)$, which contains both $S = 0$ ($\lambda = 0$) and $S' = 0$ ($\mu = 0$). 8. Yes. The x^2 and y^2 terms cancel regardless of whether the circles meet, so $S - S' = 0$ is always a line — the **radical axis**, the locus of points from which the tangent lengths to the two circles are equal (both lengths are $\sqrt{S_1}$ and $\sqrt{S'_1}$, and $S_1 = S'_1$ is exactly this line). It is a common chord only when the circles meet.

Cheat sheet for this addendum

One expression, four jobs. For $S = 0$ and a point (x_1, y_1) , build T by $x^2 \rightarrow xx_1$, $y^2 \rightarrow yy_1$, $xy \rightarrow \frac{1}{2}(xy_1 + x_1y)$, $x \rightarrow \frac{1}{2}(x + x_1)$, $y \rightarrow \frac{1}{2}(y + y_1)$, and $S_1 = S(x_1, y_1)$.

Reading	Condition on the point
$T = 0$ is the tangent	point on the conic
$T = 0$ is the chord of contact	point outside
$T = 0$ is the polar	any point (the general name)
$T = S_1$ is the chord bisected at the point	point inside
$SS_1 = T^2$ is the pair of tangents	point outside (Round 1)

Diameters (locus of midpoints of chords of slope m): circle $x + my = 0$; parabola $y = \frac{2a}{m}$ (parallel to the axis, always); ellipse $y = -\frac{b^2}{a^2m}x$; hyperbola $y = \frac{b^2}{a^2m}x$. Conjugate diameters: $mm' = -\frac{b^2}{a^2}$ (ellipse), $+\frac{b^2}{a^2}$ (hyperbola), -1 (circle). Equi-conjugate: $y = \pm\frac{b}{a}x$, semi-length $\sqrt{\frac{a^2 + b^2}{2}}$.

Poles of $lx + my + n = 0$: circle $(-\frac{lr^2}{n}, -\frac{mr^2}{n})$; parabola $(\frac{n}{l}, -\frac{2am}{l})$; ellipse $(-\frac{a^2l}{n}, -\frac{b^2m}{n})$; hyperbola $(-\frac{a^2l}{n}, +\frac{b^2m}{n})$. La Hire: Q on the polar of $P \iff P$ on the polar of Q . Conjugate points on a circle: $x_1x_2 + y_1y_2 = r^2$; conjugate lines: $r^2(l_1l_2 + m_1m_2) = n_1n_2$. **The polar of a focus is its directrix.** A line through the centre has no finite pole.

Hyperbola, asymptotes, conjugate — one constant apart:

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \text{ (S),} \quad = 0 \text{ (asymptotes A),} \quad = -1 \text{ (conjugate S''),} \quad S + S'' = 2A.$$

Shared asymptotes; shared $c = \sqrt{a^2 + b^2}$; four foci on $x^2 + y^2 = a^2 + b^2$; $\frac{1}{e^2} + \frac{1}{e'^2} = 1$; conjugate's latus rectum $\frac{2a^2}{b}$. Angle between asymptotes $= 2 \tan^{-1} \frac{b}{a}$ and $e = \sec \theta$. Product of distances to the asymptotes $= \frac{a^2 b^2}{a^2 + b^2}$. Tangent-and-asymptote triangle has area ab , and the tangent is bisected between them. Off-centre hyperbola: find λ making $S + \lambda = 0$ a line pair.

Families: $S + \lambda S' = 0$ passes through every common point of $S = 0$ and $S' = 0$. Two circles: $\lambda = -1$ gives the radical axis $S - S' = 0$ (the common chord when they meet); otherwise divide by $1 + \lambda$ before reading centre and radius. Circle and line: $S + \lambda L = 0$. Touching a line at a point: $(x - x_1)^2 + (y - y_1)^2 + \lambda L = 0$. Four points concyclic $\iff$ some λ makes $S + \lambda S'$ a circle.

Mnemonics. “ T has four jobs; the point’s position picks which.” “Inside gives $T = S_1$, outside gives $T = 0$.” “Every diameter of a central conic goes through the centre; every diameter of a parabola goes along the axis.” “Focus and directrix are pole and polar.” “One, zero, minus one: hyperbola, asymptotes, conjugate.” “Subtract two circles and the curvature falls out.”

Class 11 Mathematics

Chapter 11 — Introduction to Three Dimensional Geometry

Drona Ed Tech · Content Bank

June 2026

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Subtopic 01 — Coordinate Axes, Planes, and Octants

Exam Relevance: In **JEE Main** this surfaces as quick one-markers — naming an octant, the equation of a coordinate plane/axis, or the foot of a perpendicular — and as the silent foundation of every later vector and 3D-line problem. **JEE Advanced** rarely asks it bare but leans on it inside locus and symmetry arguments. **CBSE Boards** typically pose a 1–3 mark short-answer on octant signs or coordinates of a projected point.

Section 1 — Concept Intuition

You already live comfortably in two dimensions. On graph paper you fix an origin, draw two perpendicular lines — the x -axis and y -axis — and any point is pinned down by an ordered pair (x, y) . Those two axes slice the flat sheet into **four quadrants**. Everything so far has happened *on the paper*.

Now lift your pen straight off the page. That upward direction is a brand-new, third independent direction the flat sheet never had. Call it the z -axis. We now have three mutually perpendicular lines meeting at a single point O — the **origin** — and we have stepped from a flat world into the room you are actually sitting in.

Think about it this way... Look at the corner of your room where two walls meet the floor. That corner is the origin O . The three edges running away from it — one along the floor to your left, one along the floor away from you, one straight up the wall — are the three **coordinate axes** (x, y, z) . The three flat surfaces meeting at that corner are the three **coordinate planes**:

- the **floor** is the XY -plane,
- one **wall** is the YZ -plane,
- the other **wall** is the ZX -plane.

Figure to render. A right-handed 3D coordinate system. Origin O at centre, three mutually perpendicular axes X (toward viewer-right), Y (into the page), Z (vertical up), each drawn with its negative half (X' , Y' , Z') dashed. Lightly shade the three coordinate planes XY , YZ , ZX as translucent squares meeting at O .

To locate any point P in this room, we ask three questions: how far right, how far in, how far up? Those three signed numbers are the coordinates (x, y, z) . Here is the cleaner way to say what each one means — and it is the idea examiners love to test: **a coordinate is the signed perpendicular distance of P from a coordinate plane**. Precisely, x is the distance of P from the YZ -plane, y from the ZX -plane, and z from the XY -plane. The sign records which side of the plane you are on.

Now extend each wall and the floor straight *through* the corner to the other side, so each plane becomes a full, infinite plane. The corner of one room becomes the shared corner of **eight** rooms stacked around the origin. Each of these eight regions of space is an **octant**. This is the exact 3D cousin of the four quadrants: two perpendicular lines cut a plane into $2^2 = 4$ pieces; three perpendicular planes cut space into $2^3 = 8$ pieces.

Which octant a point sits in is decided entirely by the **sign pattern** of (x, y, z) . The room you are physically in — everything positive — is the **first octant**. Diagonally opposite, where all three are negative, is the **seventh octant**.

Edge cases & assumptions. Three quiet assumptions are baked in. First, the axes are **rectangular** (mutually perpendicular) and we use a **right-handed** system by convention. Second — and this is where students slip — the neat sign rule for octants only applies to points that are *strictly inside* a region. A point that sits **on a coordinate plane** has one coordinate

equal to zero (e.g. a point on the floor has $z = 0$), a point **on an axis** has two coordinates zero, and the **origin** has all three zero. Such points lie on the *boundaries* between octants and belong to **no octant at all**. So “which octant?” is a fair question only when none of x, y, z is zero.

Section 2 — Key Definitions & Relations

These quantities are pure numbers (dimensionless); attach units only when a physical context — say a girder of length 3 m — is explicitly given.

The three coordinate planes (each defined by setting one coordinate to zero):

$$XY\text{-plane} : z = 0 \quad YZ\text{-plane} : x = 0 \quad ZX\text{-plane} : y = 0$$

The three coordinate axes (each defined by setting two coordinates to zero):

$$x\text{-axis} : y = 0, z = 0 \quad y\text{-axis} : x = 0, z = 0 \quad z\text{-axis} : x = 0, y = 0$$

Coordinates of a point $P(x, y, z)$, read as signed perpendicular distances:

$$x = \text{dist}(P, YZ\text{-plane}), \quad y = \text{dist}(P, ZX\text{-plane}), \quad z = \text{dist}(P, XY\text{-plane})$$

So the **distance of P from a coordinate plane** is the *absolute value* of the matching coordinate: distance from the YZ -plane $= |x|$, from the ZX -plane $= |y|$, from the XY -plane $= |z|$.

Standard forms of special points:

Location	Coordinate form
On x -axis	$(x, 0, 0)$
On y -axis	$(0, y, 0)$
On z -axis	$(0, 0, z)$
On XY -plane	$(x, y, 0)$
On YZ -plane	$(0, y, z)$
On ZX -plane	$(x, 0, z)$
Origin	$(0, 0, 0)$

The octant sign table (NCERT convention) — the single most useful object in this subtopic:

Octant	I	II	III	IV	V	VI	VII	VIII
x	+	−	−	+	+	−	−	+
y	+	+	−	−	+	+	−	−
z	+	+	+	+	−	−	−	−

Read it as: octants **I-IV** all sit *above* the floor ($z > 0$) and cycle the (x, y) signs exactly like the four quadrants of the plane; octants **V-VIII** are their mirror images *below* the floor ($z < 0$).

Section 3 — Key Procedures & Logical Framework

There is no algebraic “derivation” of axes and octants — they are a *construction* — so the skill here is procedural. Three procedures cover almost every exam question, and each rests on one idea.

Procedure A — Identify the octant of a point.

1. Discard the magnitudes; keep only the **sign** of each coordinate. The size of a number never affects which octant it is in — only its sign does.
2. Read the sign triple against the table above.

Why it works: each coordinate plane is a “sign-decider.” The YZ -plane decides the sign of x (right = +, left = -), the ZX -plane decides y , the XY -plane decides z . Three independent yes/no choices give $2^3 = 8$ outcomes — exactly the eight octants. That is the whole logical skeleton of the chapter: **octant count** = $2^{\text{number of axes}}$.

Example logic: $P(-7, 3, -2)$ has sign pattern $(-, +, -)$. Scanning the table, that column is octant **VI**. Notice the 7 and 2 played no role whatsoever.

Procedure B — Foot of perpendicular onto a plane or an axis.

- *Onto a coordinate plane:* drop the coordinate that the plane “kills.” Projecting $P(x, y, z)$ onto the XY -plane sets $z \rightarrow 0$, giving $(x, y, 0)$. Onto YZ : $(0, y, z)$. Onto ZX : $(x, 0, z)$.
- *Onto a coordinate axis:* keep only the coordinate that lies *along* that axis; zero the other two. The foot on the x -axis is $(x, 0, 0)$; on the y -axis $(0, y, 0)$; on the z -axis $(0, 0, z)$.

Why it works: the perpendicular from P to the XY -plane runs purely in the z -direction, so it changes nothing about x or y ; it lands when the height z becomes 0. Projecting onto a *line* is stricter still: only the displacement along that line survives.

Figure to render. A point $P(x, y, z)$ in the first octant with a dashed vertical segment dropped to its foot $M(x, y, 0)$ on the XY -plane, and a further dashed segment from M to $L(x, 0, 0)$ on the x -axis. Label the segments to show z is “lost” first, then y .

Procedure C — Reflection (image) of a point in a plane or axis.

- *In a coordinate plane:* flip the sign of the one coordinate perpendicular to that plane. Image of $P(x, y, z)$ in the XY -plane is $(x, y, -z)$; in YZ it is $(-x, y, z)$; in ZX it is $(x, -y, z)$.
- *In a coordinate axis:* flip the signs of the **two** coordinates perpendicular to that axis. Image in the x -axis is $(x, -y, -z)$.
- *In the origin:* flip **all three** signs: $(-x, -y, -z)$.

Why it works: a mirror reverses only the direction *across* it and leaves directions *within* it untouched. A plane is “two directions wide,” so reflecting in it touches only the one perpendicular direction. An axis is “one direction wide,” so the two perpendicular directions both flip.

Section 4 — Worked Examples

Example 1 — [CBSE Board Level] A point P has coordinates $(5, -4, 9)$. (a) State the octant in which P lies. (b) Find the foot of the perpendicular from P to the ZX -plane. (c) Find the distance of P from the XY -plane.

Solution.

- (a) Sign pattern of $(5, -4, 9)$ is $(+, -, +)$. From the octant table this is octant **IV**.

(b) The ZX -plane is $y = 0$, so projecting kills the y -coordinate: foot = $(5, 0, 9)$.

(c) Distance from the XY -plane = $|z| = |9| = 9$ units.

Presentation note: always quote the rule you used (" ZX -plane $\Rightarrow y = 0$ ") before substituting — it earns method marks even if arithmetic slips.

Example 2 — [JEE Main · Speed Trap] Among the points $A(-1, -2, -3)$, $B(2, -5, 4)$, $C(-6, 7, -1)$, $D(3, 8, -2)$, exactly how many lie in octants that are entirely below the XY -plane?

Solution (the fast way). "Below the XY -plane" means $z < 0$ — that is octants **V-VIII**. Don't bother naming each octant; just scan the **z -coordinate sign**:

- $A : z = -3 (< 0) \checkmark$
- $B : z = 4 (> 0) \times$
- $C : z = -1 (< 0) \checkmark$
- $D : z = -2 (< 0) \checkmark$

Three points. **Answer: 3.**

The trap: a rushed student starts fully classifying each point into octants I-VIII and burns a minute. The question only cares about *one* coordinate's sign. Read what is actually asked before computing.

Example 3 — [JEE Main Level] The reflection of point $P(4, -3, 7)$ in the YZ -plane is Q . The reflection of Q in the x -axis is R . In which octant does R lie?

Solution. Apply the reflection rules in order (Procedure C).

- Reflection of P in the YZ -plane flips the perpendicular coordinate x :

$$Q = (-4, -3, 7).$$

- Reflection of Q in the x -axis flips the two coordinates perpendicular to that axis, namely y and z :

$$R = (-4, 3, -7).$$

- Sign pattern of R is $(-, +, -)$, which is octant **VI**.

This links two ideas — reflection in a plane (one flip) versus reflection in an axis (two flips) — and the sequencing matters: doing the operations in the wrong order, or flipping the wrong number of signs, lands you in a different octant.

Example 4 — [JEE Advanced Level] Find the locus of all points equidistant from the XY -plane and the YZ -plane, and describe the surface(s) geometrically.

Solution. Let a general point be $P(x, y, z)$.

- Distance of P from the XY -plane is $|z|$.
- Distance of P from the YZ -plane is $|x|$.

Equidistance demands

$$|x| = |z| \implies x^2 = z^2 \implies x^2 - z^2 = 0 \implies (x - z)(x + z) = 0.$$

So the locus is **not a single surface** but the **pair of planes**

$$x = z \quad \text{and} \quad x = -z,$$

each containing the entire y -axis (note y is completely free — the condition never constrained it). Geometrically these are the two planes that **bisect the dihedral angles** between the XY - and YZ -planes; together they form an " $\times$ "-shaped pair when viewed down the y -axis.

Figure to render. View looking down the y -axis (so the y -axis is a single dot at the origin). Show the x -axis horizontal and z -axis vertical; draw the two lines $z = x$ and $z = -x$ crossing at the origin, labelled as the cross-sections of the bisecting planes. Shade lightly to suggest each line is really an edge-on plane.

The rigor lesson: writing the distances as **absolute values** is non-negotiable. Dropping the modulus and solving $x = z$ alone silently throws away half the locus — the classic JEE Advanced “you found only one of the two planes” deduction.

Section 5 — Practice Exercises

1. **[CBSE]** Name the octants in which the points $(2, 1, 3)$, $(-2, 1, 3)$, and $(2, -1, -3)$ lie.
2. **[CBSE]** Write the coordinates of the foot of the perpendicular drawn from $(6, -2, 5)$ to (i) the y -axis and (ii) the XY -plane.
3. **[JEE Main]** A point lies on the YZ -plane and is at a distance 4 from the ZX -plane and 3 from the XY -plane, with both relevant coordinates positive. Find the point.
4. **[JEE Main]** The image of $(-2, 5, -7)$ in the ZX -plane is reflected again in the origin. State the octant of the final point.
5. **[JEE Advanced]** Find the locus of points equidistant from all three coordinate planes that also satisfy $x, y, z > 0$, and identify the line so obtained.

Answers: 1. I, II, VIII. — 2. (i) $(0, -2, 0)$ (ii) $(6, -2, 0)$. — 3. $(0, 4, 3)$. — 4. Image in ZX -plane = $(-2, -5, -7)$; reflected in origin = $(2, 5, 7)$, octant **I**. — 5. $|x| = |y| = |z|$ with all positive gives the line $x = y = z$ through the origin into the first octant (the space-diagonal direction).

Section 6 — Crack the MCQ

Q1. The point $(-3, 0, 5)$ lies

- (A) in octant II (B) in octant III (C) on the ZX -plane (D) on the z -axis

Answer: (C). The y -coordinate is 0, and $y = 0$ is the ZX -plane, so the point lies on that plane and in **no octant**.

- (A)/(B) trap students who ignore the zero and force a sign pattern — but a zero coordinate disqualifies a point from any octant.
- (D) tempts those who think “two-ish coordinates look special”; the z -axis requires $x = 0$ **and** $y = 0$, but here $x = -3 \neq 0$.

Q2. How many of the eight octants contain points with a *negative* z -coordinate?

- (A) 2 (B) 4 (C) 6 (D) 8

Answer: (B). Negative z characterises octants **V, VI, VII, VIII** — exactly four.

- (A) confuses “negative z ” with one specific octant.
- (C)/(D) come from vaguely thinking “lots of octants have some negative coordinate,” conflating the z -sign with the overall sign mix.

Q3. The reflection of $P(7, -2, -4)$ in the x -axis is

- (A) $(-7, -2, -4)$ (B) $(7, 2, 4)$ (C) $(-7, 2, 4)$ (D) $(7, -2, 4)$

Answer: (B). Reflection in the x -axis flips the two coordinates perpendicular to it (y and z): $(7, -2, -4) \rightarrow (7, 2, 4)$.

- (A) flips x only — that is reflection in the YZ -plane, not the x -axis.
- (C) flips all three — that is reflection in the **origin**.
- (D) flips one wrong coordinate (z only) — reflection in the XY -plane, with a sign slip on y left in.

Q4. The distance of the point $(2, -6, 3)$ from the YZ -plane is

- (A) 2 (B) 6 (C) 3 (D) 7

Answer: (A). Distance from the YZ -plane ($x = 0$) is $|x| = |2| = 2$.

- (B) uses $|y|$ — that is the distance from the ZX -plane.
- (C) uses $|z|$ — distance from the XY -plane.
- (D) is $\sqrt{2^2 + 3^2 + (\text{nothing sensible})}$ or a half-remembered distance-from-origin reflex; it answers a question that wasn't asked.

Section 7 — Common Student Pitfalls & Pro-Tips

- **Memorising octant names instead of signs.** Trying to recall “ $OX'YZ$ is the second octant” is fragile. Recall the **sign table** instead — the signs are the definition; the Roman numeral is just a label.
- **Confusing a coordinate with a distance.** The distance of P from the YZ -plane is $|x|$, **not** x . For $(-5, 1, 2)$ the distance is 5, not -5 . Distances are never negative.
- **Forcing a zero point into an octant.** If any coordinate is 0, the point is on a plane or axis and belongs to **no** octant. Examiners plant a stray zero precisely to catch this.
- **Reflection sign errors.** Plane $\rightarrow$ flip **one** sign (the perpendicular one). Axis $\rightarrow$ flip **two**. Origin $\rightarrow$ flip **three**. Count before you flip.

Pro-Tip: Only two octants have an all-same sign pattern — octant **I** is all $+$, octant **VII** is all $-$. Anchor on those two, then read the rest off the z -sign: $z > 0 \Rightarrow$ octants I-IV (behaving just like the 2D quadrants), $z < 0 \Rightarrow$ octants V-VIII (the same quadrant pattern, mirrored below the floor).

Section 8 — Cheat Sheet

- **Three axes, three planes, eight octants.** Octant count $= 2^3 = 8$ (three independent sign choices).
- **Plane equations:** $XY : z = 0$, $YZ : x = 0$, $ZX : y = 0$.
- **Axis equations:** x -axis: $y = z = 0$; y -axis: $x = z = 0$; z -axis: $x = y = 0$.
- **Coordinates = signed perpendicular distances from planes:** x from YZ , y from ZX , z from XY . Distance from a plane = absolute value of that coordinate.
- **Octant signs (NCERT):**

	I	II	III	IV	V	VI	VII	VIII
x	+	−	−	+	+	−	−	+
y	+	+	−	−	+	+	−	−
z	+	+	+	+	−	−	−	−

- **Reflections — flip count:** plane $\rightarrow$ 1 sign, axis $\rightarrow$ 2 signs, origin $\rightarrow$ 3 signs.

- **Mnemonic:** *“I am all positive; VII is all negative; the rest you read off the table.” And: “Top four (I-IV) above the floor, bottom four (V-VIII) below — same quadrant dance, one upstairs, one down.”*

Subtopic 02 — Distance and Section Formulas in 3D

Exam Relevance: This is high-yield. **JEE Main** asks it almost every cycle — collinearity, the ratio of division, classifying a triangle or quadrilateral, or the point where a plane cuts a segment. **JEE Advanced** uses these as the entry move into locus problems. **CBSE Boards** reliably set a 2–4 mark section-formula or collinearity question, and miscellaneous-exercise proofs (“show ABCD is a parallelogram”) come straight from here.

Section 1 — Concept Intuition

In Subtopic 01 we learned to *name* a point in space. Now we learn to measure with it: how far apart are two points, and where does a point sit that splits the segment between them in a given ratio?

Distance is just Pythagoras, used twice. On flat paper you already know that the gap between two points is $\sqrt{(\Delta x)^2 + (\Delta y)^2}$ — the hypotenuse of a right triangle. Space adds a third, perpendicular direction, so you apply the same idea one extra time.

Think about it this way... Picture a small drone hovering in a room. To reach it from the corner O , first walk across the **floor** to the spot directly beneath it — that horizontal hop is an ordinary 2D distance, $\sqrt{(\Delta x)^2 + (\Delta y)^2}$. Then rise **straight up** by Δz . The drone’s true straight-line distance is the hypotenuse of a right triangle whose legs are “floor distance” and “height”:

$$\text{distance}^2 = \underbrace{[(\Delta x)^2 + (\Delta y)^2]}_{\text{floor leg}} + \underbrace{(\Delta z)^2}_{\text{height leg}}.$$

That single picture is the 3D distance formula. It is also why the longest rod that fits diagonally across a room is the space-diagonal of that box.

Figure to render. A rectangular box (parallelepiped) with one vertex at $P(x_1, y_1, z_1)$ and the opposite vertex at $Q(x_2, y_2, z_2)$. Show the space diagonal PQ in bold. Highlight a horizontal “floor” leg (the base diagonal) and the vertical “height” leg, forming the right triangle whose hypotenuse is PQ .

Section division is three 1D splits at once. Imagine the Mumbai–Pune expressway. If a toll booth sits at the point that divides the route in the ratio 2 : 3, then *every* milestone-coordinate of that booth — its easting, its northing, its altitude — is the same 2 : 3 split of the endpoints’ corresponding coordinates. The deep reason: a coordinate is a **projection onto an axis**, and parallel projection preserves ratios. So a single 3D division quietly becomes three independent, identical 1D divisions — one per axis. That is the whole secret of the section formula.

Internal vs external. If the dividing point R lies *between* P and Q , the division is **internal** (think: the booth is on the road). If R lies on the line PQ but *outside* the segment — beyond P or beyond Q — the division is **external** (the “split point” is an extrapolation, like predicting where the road would reach a ratio if extended past the city). Internally the ratio parts add; externally they subtract — which is exactly the sign difference you will see in the two formulas.

Edge cases & assumptions.

- The section formula assumes R is **collinear** with P and Q — it gives the point on line PQ , nothing off it.
- **Midpoint** is simply the 1 : 1 internal case.

- **External division is undefined when $m = n$:** the denominator $m - n$ vanishes, which matches the geometry — equal external parts push the point off to infinity (parallel, never meeting).
 - A negative ratio is just external division in disguise: dividing in $m : n$ externally = dividing in $m : (-n)$ internally. Watch the signs, not the words.
 - Distance is always ≥ 0 — there is no “negative distance,” even though coordinate *differences* can be negative (they get squared away).
-

Section 2 — Key Formulas & Definitions

Let $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$. All quantities are dimensionless unless a physical length is specified.

Distance between two points:

$$PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

Distance from the origin:

$$OP = \sqrt{x_1^2 + y_1^2 + z_1^2}$$

Section formula — internal division (point R divides PQ in ratio $m : n$ internally):

$$R = \left(\frac{mx_2 + nx_1}{m + n}, \frac{my_2 + ny_1}{m + n}, \frac{mz_2 + nz_1}{m + n} \right)$$

Section formula — external division (ratio $m : n$ externally):

$$R = \left(\frac{mx_2 - nx_1}{m - n}, \frac{my_2 - ny_1}{m - n}, \frac{mz_2 - nz_1}{m - n} \right)$$

Midpoint (the $1 : 1$ case):

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}, \frac{z_1 + z_2}{2} \right)$$

The $k : 1$ form — the workhorse for “find the ratio” problems. Setting $m : n = k : 1$, a general point on line PQ is

$$\left(\frac{kx_2 + x_1}{k + 1}, \frac{ky_2 + y_1}{k + 1}, \frac{kz_2 + z_1}{k + 1} \right),$$

with $k > 0 \Rightarrow$ internal, $k < 0 \Rightarrow$ external.

Centroid of a triangle with vertices A, B, C :

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}, \frac{z_1 + z_2 + z_3}{3} \right)$$

[Beyond CBSE · JEE extension] Centroid of a tetrahedron with four vertices: the same idea with a denominator of 4 — average all four position triples.

Section 3 — CBSE Board Derivations

Derivation 1 — The distance formula (rectangular parallelepiped method).

Setup. Take $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$. Through each point draw three planes parallel to the coordinate planes. These six planes box in a **rectangular parallelepiped** with PQ as its space diagonal and all edges parallel to the axes.

Step 1 — split off the height. Let A be the vertex of the box at the same height as P but directly below Q in the vertical sense, so that edge AQ runs purely along the z -direction and the triangle PAQ is right-angled at A . Then

$$PQ^2 = PA^2 + AQ^2. \quad (1)$$

The physical reasoning: AQ is vertical and PA lies in a horizontal plane, so they are perpendicular — Pythagoras applies.

Step 2 — resolve the horizontal leg. The segment PA lies in a horizontal plane and is itself the hypotenuse of a right triangle PNA with legs along the x - and y -directions:

$$PA^2 = PN^2 + NA^2. \quad (2)$$

Step 3 — read off the edge lengths. Because every edge is parallel to an axis, each edge length is just the gap in that coordinate:

$$PN = |x_2 - x_1|, \quad NA = |y_2 - y_1|, \quad AQ = |z_2 - z_1|.$$

Step 4 — combine. Substituting (2) into (1):

$$PQ^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2,$$

$$PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}.$$

The squares erase the absolute-value signs, which is why direction never matters for distance.

Derivation 2 — The section formula (internal division).

Setup. Let $R(x, y, z)$ divide PQ internally so that $PR : RQ = m : n$.

Key idea — projection preserves ratio. Drop perpendiculars from P, R, Q onto the XY -plane (recall from Sub-01 that this plane is $z = 0$), meeting it at L, N, M . These three perpendiculars are parallel (all perpendicular to the same plane). A fundamental fact of parallel projection: **if R divides PQ in the ratio $m : n$, then its projection N divides LM in the same ratio $m : n$.** Tracking heights (the z -coordinates) through this projection gives, by similar triangles,

$$\frac{PR}{RQ} = \frac{z - z_1}{z_2 - z} = \frac{m}{n}.$$

Step — solve for z . Cross-multiplying: $n(z - z_1) = m(z_2 - z) \Rightarrow z(m + n) = mz_2 + nz_1$, hence

$$z = \frac{mz_2 + nz_1}{m + n}.$$

Symmetry. The identical construction, projecting instead onto the YZ - and ZX -planes, yields

$$x = \frac{mx_2 + nx_1}{m + n}, \quad y = \frac{my_2 + ny_1}{m + n}.$$

For **external** division, R lies outside the segment, so RQ is measured in the opposite sense — replace n by $-n$ throughout, producing the $m - n$ denominator.

Section 4 — Worked Examples

Example 1 — [CBSE Board Level] Show that the points $A(0, 7, 10)$, $B(-1, 6, 6)$, and $C(-4, 9, 6)$ are the vertices of a right-angled isosceles triangle.

Solution. Compute the three side-lengths via the distance formula.

$$AB^2 = (-1 - 0)^2 + (6 - 7)^2 + (6 - 10)^2 = 1 + 1 + 16 = 18,$$

$$BC^2 = (-4 + 1)^2 + (9 - 6)^2 + (6 - 6)^2 = 9 + 9 + 0 = 18,$$

$$CA^2 = (0 + 4)^2 + (7 - 9)^2 + (10 - 6)^2 = 16 + 4 + 16 = 36.$$

Since $AB^2 = BC^2$, two sides are equal $\Rightarrow$ **isosceles**. Since $AB^2 + BC^2 = 18 + 18 = 36 = CA^2$, the converse of Pythagoras gives a **right angle at B**. Hence $\triangle ABC$ is right-angled and isosceles. ■

Presentation note: work with AB^2 throughout and take roots only if asked — it keeps arithmetic clean and the Pythagoras check exact.

Example 2 — [JEE Main · Speed Trap] In what ratio does the XY -plane divide the segment joining $P(2, -3, 5)$ and $Q(4, 6, -10)$?

Solution (the fast way). The XY -plane is $z = 0$ (from Sub-01). Use the $k : 1$ form and force only the z -coordinate to zero — ignore x and y entirely:

$$\frac{k(-10) + 5}{k + 1} = 0 \Rightarrow -10k + 5 = 0 \Rightarrow k = \frac{1}{2}.$$

So the ratio is $k : 1 = \frac{1}{2} : 1 = \mathbf{1 : 2}$, and since $k > 0$ it is **internal**. **Answer:** $1 : 2$ **internally**.

The trap: students set up all three coordinates and hunt for a point, or they get $k = \frac{1}{2}$ and forget that the *sign* of k already told them internal vs external. For a plane $z = 0$, only the z -coordinate equation matters — one line of algebra.

Example 3 — [JEE Main Level] Find the point on the y -axis that is equidistant from $A(3, 1, 2)$ and $B(-1, 5, -4)$.

Solution. From Sub-01, any point on the y -axis has the form $R(0, y, 0)$ — this single fact collapses three unknowns to one. Impose $RA = RB$, i.e. $RA^2 = RB^2$:

$$\underbrace{9 + (y - 1)^2 + 4}_{RA^2} = \underbrace{1 + (y - 5)^2 + 16}_{RB^2}.$$

Expand: $13 + y^2 - 2y + 1 = 17 + y^2 - 10y + 25$, so $y^2 - 2y + 14 = y^2 - 10y + 42$. The y^2 terms cancel:

$$-2y + 14 = -10y + 42 \Rightarrow 8y = 28 \Rightarrow y = \frac{7}{2}.$$

The required point is $(0, \frac{7}{2}, 0)$.

This links two ideas — the Sub-01 “point on an axis” form and the distance formula — and the cancellation of y^2 is the structural hint that an “equidistant” condition is always *linear*, not quadratic.

Example 4 — [JEE Advanced Level] Find the equation of the set of all points equidistant from $A(1, 2, 3)$ and $B(3, 2, -1)$, and describe the surface.

Solution. Let $P(x, y, z)$ be a general point with $PA = PB$, hence $PA^2 = PB^2$:

$$(x - 1)^2 + (y - 2)^2 + (z - 3)^2 = (x - 3)^2 + (y - 2)^2 + (z + 1)^2.$$

The $(y - 2)^2$ terms cancel. Expand the rest:

$$x^2 - 2x + 1 + z^2 - 6z + 9 = x^2 - 6x + 9 + z^2 + 2z + 1.$$

All squared terms cancel — exactly as foreshadowed in Example 3 — leaving a **linear** equation:

$$-2x - 6z + 10 = -6x + 2z + 10 \Rightarrow 4x - 8z = 0 \Rightarrow \boxed{x - 2z = 0}.$$

The locus is a **plane** — the *perpendicular bisector plane* of segment AB . This is the spatial cousin of Sub-01's "equidistant from two coordinate planes" result: there the equidistance gave a *pair* of planes; here equidistance between two *points* gives a *single* plane that slices space exactly halfway between them, perpendicular to AB .

Figure to render. Two points A and B with the segment AB drawn. A plane passing through the midpoint of AB , perpendicular to AB , shaded translucently, labelled "perpendicular bisector plane: $x - 2z = 0$ ". Show that any point on the plane is joined to A and B by equal-length dashed segments.

The rigor lesson: recognise *before* expanding that an "equidistant from two fixed points" condition kills every quadratic term and must collapse to a plane. Spotting this saves a full page of algebra under exam pressure.

Section 5 — Practice Exercises

1. **[CBSE]** Find the distance between $(2, 3, 5)$ and $(4, 3, 1)$, and verify whether $(1, 0, 0)$, $(0, 1, 0)$, $(0, 0, 1)$ form an equilateral triangle.
2. **[CBSE]** Find the coordinates of the point dividing the segment joining $(-2, 3, 5)$ and $(1, -4, 6)$ in the ratio $2 : 3$ (i) internally, (ii) externally.
3. **[JEE Main]** Find the ratio in which the YZ -plane divides the segment joining $(-2, 4, 7)$ and $(3, -5, 8)$.
4. **[JEE Main]** Show that $A(2, -3, 4)$, $B(-1, 2, 1)$, $C(0, \frac{1}{3}, 2)$ are collinear, and find the ratio in which B divides AC .
5. **[JEE Advanced]** Find the equation of the set of points P such that $PA^2 + PB^2 = 2k^2$, where $A(3, 4, 5)$ and $B(-1, 3, -7)$, and identify the type of surface.

Answers: 1. Distance $= \sqrt{20} = 2\sqrt{5}$; yes, all three sides $= \sqrt{2}$, so equilateral. — 2. (i) $(-\frac{4}{5}, -\frac{2}{5}, \frac{27}{5})$, (ii) $(-8, 17, 3)$. — 3. $2 : 3$ internally. — 4. B divides AC in the ratio $1 : 2$ (so collinear). — 5. A sphere: $x^2 + y^2 + z^2 - 2x - 7y + 2z + (\text{const}) = 0$ form, centre at the midpoint $(1, \frac{7}{2}, -1)$ (a sphere whenever k is large enough for a real radius).

Section 6 — Crack the MCQ

Q1. The distance of the point $(1, -2, 2)$ from the origin is

- (A) 3 (B) $\sqrt{5}$ (C) 9 (D) $\sqrt{3}$

Answer: (A). $\sqrt{1^2 + (-2)^2 + 2^2} = \sqrt{1 + 4 + 4} = \sqrt{9} = 3$.

- (C) forgets the square root.
- (B) drops one term, e.g. computes $\sqrt{1 + 4}$.

- (D) mistakenly adds the *coordinates'* magnitudes without squaring, or confuses with $\sqrt{3}$.

Q2. The midpoint of the segment joining $(4, -1, 7)$ and $(-2, 5, 3)$ is

- (A) $(1, 2, 5)$ (B) $(3, 3, 2)$ (C) $(6, -6, 4)$ (D) $(2, 4, 10)$

Answer: (A). Averages: $(\frac{4-2}{2}, \frac{-1+5}{2}, \frac{7+3}{2}) = (1, 2, 5)$.

- (C) computes the *difference* of coordinates, not the average.
- (D) computes the *sum* without halving.
- (B) mixes a couple of arithmetic slips — a “looks plausible” distractor.

Q3. The XY -plane divides the segment joining $(1, 2, -3)$ and $(3, 4, 5)$ in the ratio

- (A) $3 : 5$ internally (B) $5 : 3$ internally (C) $3 : 5$ externally (D) $1 : 1$

Answer: (A). XY -plane is $z = 0$: $\frac{k(5) + (-3)}{k + 1} = 0 \Rightarrow k = \frac{3}{5}$. Positive $\Rightarrow$ internal, ratio $3 : 5$.

- (B) swaps the ratio order (used $\frac{5}{3}$, i.e. reversed which point is “1”).
- (C) misreads positive k as external.
- (D) assumes the plane bisects — true only if the endpoints’ z -values were equal and opposite, which they are not (-3 vs 5).

Q4. If $(2, 4, -1)$, $(3, 6, -1)$, and $(4, 8, -1)$ are three points, then they are

- (A) collinear (B) vertices of a right triangle (C) vertices of an equilateral triangle (D) coincident

Answer: (A). The middle point is the midpoint of the outer two: $(\frac{2+4}{2}, \frac{4+8}{2}, \frac{-1-1}{2}) = (3, 6, -1)$ — so all three lie on one line (in fact the middle one divides the other two in $1 : 1$).

- (B)/(C) tempt students who jump to “three points $\Rightarrow$ triangle” without checking collinearity first.
- (D) confuses “collinear” with “identical.”

Section 7 — Common Student Pitfalls & Pro-Tips

- **Internal/external sign flip.** Internal uses $m + n$ in the denominator and a $+$ in the numerator; external uses $m - n$ and a $-$. Mixing them is the single most common error — write “INT $\Rightarrow$ add, EXT $\Rightarrow$ subtract” in the margin before substituting.
- **Ratio order swap.** $m : n$ means $PR : RQ$, so m pairs with the *second* point Q in the numerator ($mx_2 + nx_1$). Reversing which point is P and which is Q silently inverts your ratio — the classic $3 : 5$ vs $5 : 3$ trap.
- **Forgetting to square (or to root).** Coordinate differences must be **squared** inside the formula and the whole sum **rooted** at the end. Skipping either is an instant wrong answer.
- **Assuming a triangle without checking collinearity.** Three given points may be collinear — always test (midpoint, or section ratio, or $AB + BC = AC$) before claiming a triangle.

Pro-Tip: For every “find the ratio” or “where does a plane cut the segment” question, jump straight to the $k : 1$ **form**. It uses one unknown instead of two, and the **sign of k hands you internal**

($k > 0$) vs external ($k < 0$) for free — no separate case analysis. (If you later study vectors, the same section formula reads $\vec{r} = \frac{m\vec{r}_2 + n\vec{r}_1}{m+n}$ — identical structure, position vectors instead of coordinates.)

Section 8 — Cheat Sheet

- **Distance:** $PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$; from origin $= \sqrt{x^2 + y^2 + z^2}$.
- **Section (internal, $m : n$):** $\left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n}, \frac{mz_2 + nz_1}{m+n} \right)$.
- **Section (external, $m : n$):** same with $m - n$ denominators and $-$ in numerators.
- **Midpoint:** averages of the coordinates (1 : 1 case).
- **$k : 1$ form:** $\left(\frac{kx_2 + x_1}{k+1}, \dots \right)$; $k > 0$ internal, $k < 0$ external. **Use it for all “ratio” problems.**
- **Centroid (triangle):** average of 3 vertices; **(tetrahedron, beyond-CBSE):** average of 4.
- **Plane cuts segment:** set the matching coordinate to 0 ($XY : z = 0$, $YZ : x = 0$, $ZX : y = 0$) and solve for k .
- **Mnemonics:** “Internal adds, external subtracts.” — “ m goes with the far point (x_2).” — “Equidistant from two points $\Rightarrow$ a flat plane (the perpendicular bisector).”

Subtopic 03 — Applications: Centroid, Collinearity & Locus

Exam Relevance: This is where the two tools of Sub-02 start earning marks. **CBSE Boards** regularly set centroid, collinearity, and “type of quadrilateral” questions (2–4 marks) and a “find the equation of the set of points” locus problem. **JEE Main** asks collinearity and ratio problems almost every cycle. **JEE Advanced** uses locus derivations as the opening move of harder questions, and rewards the incentre/tetrahedron extensions.

Section 1 — Concept Intuition

Everything in this subtopic is an **application** of just two results you already own: the distance formula and the section formula. No new machinery — only new questions.

Centroid — the balance point. Cut a triangle out of stiff card and try to balance it on a fingertip. The single point where it balances is the **centroid** G . Geometrically it is where the three medians meet, and each median is split by G in the ratio 2 : 1 (measured from the vertex). In coordinates this collapses to something delightfully simple: G is just the **average** of the three vertices. The reason it is an average — and not something messier — is the same independence idea from Sub-02: each coordinate behaves on its own, so balancing in 3D is three separate 1D balancing acts.

Collinearity — when the triangle collapses. Three points are **collinear** when they lie on one straight line, i.e. they fail to make a triangle. There are two honest ways to detect this:

- *Distance lens:* if B lies between A and C , then the long hop equals the two short hops: $AB + BC = AC$.
- *Section lens:* if the points are collinear, then B must **divide** AC in some real ratio $k : 1$ — and crucially, the *same* k must work for the x , y , **and** z coordinates simultaneously.

Think about it this way... Picture three milestones along the Mumbai-Pune expressway. If they are truly on the road, then the middle one splits the distance between the outer two in a fixed proportion that is consistent whether you measure easting, northing, or altitude. If even one of those three proportions disagrees, the “middle” milestone is off the road — no straight line through all three.

Locus — sculpting space with a rule. A **locus** is the set of *all* points obeying a stated condition; in 3D, distance conditions carve out surfaces. Two carry the chapter:

- “Equidistant from two fixed points” $\Rightarrow$ a flat **plane** (the perpendicular bisector — you proved a case of this in Sub-02).
- “At fixed distance from one fixed point” $\Rightarrow$ a **sphere**, like every spot a tethered goat can graze.

A richer condition — “the *sum* of distances from two fixed points is constant” — bends space into an **ellipsoid**, the 3D cousin of the ellipse.

Edge cases & assumptions.

- The centroid of a triangle always exists and is unique; it has nothing to do with side lengths (unlike the incentre, which weights by them).
- Collinearity makes the triangle’s “area” zero — a useful sanity flag.
- A sphere $x^2 + y^2 + z^2 + \dots = 0$ is **real** only if its computed radius-squared is positive; a negative value means the locus is the **empty set**, and exactly zero means a single point.
- The sum-of-distances ellipsoid is real only if the constant sum exceeds the distance between the two fixed points (you cannot beat the straight line).

Section 2 — Key Formulas & Definitions

For $A(x_1, y_1, z_1)$, $B(x_2, y_2, z_2)$, $C(x_3, y_3, z_3)$. All quantities dimensionless unless a physical length is given.

Centroid of a triangle:

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}, \frac{z_1 + z_2 + z_3}{3} \right)$$

and G divides each median in the ratio 2 : 1 from the vertex.

Collinearity criteria (any one suffices):

1. $AB + BC = AC$ (distance test, with B between A and C);
2. B divides AC in a ratio $k : 1$ with the **same** k satisfying all three coordinates.

Locus — standard surfaces:

- Equidistant from A and B : $PA = PB \Rightarrow$ the **perpendicular bisector plane** of AB (a linear equation).
- At distance r from a fixed centre (a, b, c) : the **sphere**

$$(x - a)^2 + (y - b)^2 + (z - c)^2 = r^2.$$

- Sum of distances from two fixed points = $2a$ (with foci $2c$ apart, $b^2 = a^2 - c^2$): an **ellipsoid of revolution**.

[Beyond CBSE · JEE extension] Incentre of a triangle (weighted by opposite side lengths $a = |BC|$, $b = |CA|$, $c = |AB|$):

$$I = \left(\frac{ax_1 + bx_2 + cx_3}{a + b + c}, \dots, \dots \right)$$

[Beyond CBSE · JEE extension] Centroid of a tetrahedron with vertices A, B, C, D : the average of all four — denominator 4.

Section 3 — CBSE Board Derivations & Key Procedures

Derivation — the centroid formula (via the section formula).

Setup. Let the triangle be $A(x_1, y_1, z_1)$, $B(x_2, y_2, z_2)$, $C(x_3, y_3, z_3)$. Take the median from A to the midpoint D of BC .

Step 1 — midpoint of BC . By the midpoint formula (Sub-02),

$$D = \left(\frac{x_2 + x_3}{2}, \frac{y_2 + y_3}{2}, \frac{z_2 + z_3}{2} \right).$$

Step 2 — use the known median ratio. A standard property of any triangle is that the centroid divides each median in the ratio 2 : 1 from the vertex; so G divides AD with $AG : GD = 2 : 1$.

Step 3 — apply the section formula (internal, ratio 2 : 1, with A as the “first” point and D as the “second”):

$$x_G = \frac{2\left(\frac{x_2 + x_3}{2}\right) + 1 \cdot x_1}{2 + 1} = \frac{(x_2 + x_3) + x_1}{3} = \frac{x_1 + x_2 + x_3}{3}.$$

By identical work on the other coordinates,

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}, \frac{z_1 + z_2 + z_3}{3} \right). \quad \blacksquare$$

Notice the medians from B and C would give the *same* point — a clean proof that all three medians are concurrent.

Procedure — testing collinearity (two methods).

Method A — distance.

1. Compute the three pairwise distances AB , BC , CA .
2. The points are collinear **iff** the largest equals the sum of the other two. (If $AB + BC = AC$, then B lies between A and C .)

Method B — section-ratio consistency (faster, and what JEE rewards).

1. Suppose B divides AC in the ratio $k : 1$. From the x -coordinate alone, solve $\frac{kx_3 + x_1}{k + 1} = x_2$ for k .
2. **Verify** that this same k reproduces y_2 and z_2 .
3. If all three agree, the points are collinear (and you have the dividing ratio for free); if any disagrees, they are not.

Why the verification step is non-negotiable: a single coordinate can match by coincidence. Collinearity is a statement about *all three* coordinates locking to one ratio — checking only x is the classic false positive.

Section 4 — Worked Examples

Example 1 — [CBSE Board Level] (a) Find the centroid of the triangle with vertices $A(2, -1, 4)$, $B(6, 3, -2)$, $C(1, 4, 5)$. (b) If two vertices of a triangle are $(2, -1, 4)$ and $(6, 3, -2)$ and its centroid is $(3, 2, \frac{7}{3})$, find the third vertex.

Solution.

(a) Average the coordinates:

$$G = \left(\frac{2 + 6 + 1}{3}, \frac{-1 + 3 + 4}{3}, \frac{4 - 2 + 5}{3} \right) = \left(3, 2, \frac{7}{3} \right).$$

(b) Let the third vertex be $C(x, y, z)$. Since G is the average, multiply through by 3:

$$x = 3(3) - 2 - 6 = 1, \quad y = 3(2) - (-1) - 3 = 4, \quad z = 3\left(\frac{7}{3}\right) - 4 - (-2) = 5.$$

The third vertex is $\boxed{(1, 4, 5)}$ — consistent with part (a), as expected.

Presentation note: the reverse problem is just “sum = $3 \times$ centroid”; the only common slip is forgetting the factor of 3.

Example 2 — [JEE Main • Speed Trap] Are the points $A(2, 1, -1)$, $B(4, 4, 3)$, $C(8, 10, 11)$ collinear?

Solution (the fast way). Use Method B. Let B divide AC in $k : 1$ and solve from the x -coordinate only:

$$\frac{8k + 2}{k + 1} = 4 \Rightarrow 8k + 2 = 4k + 4 \Rightarrow k = \frac{1}{2}.$$

Now the **mandatory check** with $k = \frac{1}{2}$:

$$y : \frac{10(\frac{1}{2}) + 1}{\frac{3}{2}} = \frac{6}{\frac{3}{2}} = 4 = y_B \checkmark, \quad z : \frac{11(\frac{1}{2}) - 1}{\frac{3}{2}} = \frac{4.5}{1.5} = 3 = z_B \checkmark.$$

All three agree, so the points are **collinear**, with B dividing AC in $1 : 2$.

The trap: a rushed student stops after the x -coordinate gives a clean k , or “eyeballs” two distances and declares collinear. If the y - or z -check had failed, the answer flips entirely. One coordinate is never proof.

Example 3 — [JEE Main Level] Show that $A(1, 0, 2)$, $B(4, 2, 3)$, $C(5, 7, 2)$, $D(2, 5, 1)$ are the vertices of a parallelogram $ABCD$, and determine whether it is a rectangle.

Solution. A quadrilateral $ABCD$ is a parallelogram if both pairs of opposite sides are equal and parallel — cleanest to compare side vectors.

$$\begin{aligned}\overrightarrow{AB} &= (3, 2, 1), \quad \overrightarrow{DC} = C - D = (3, 2, 1) \Rightarrow \overrightarrow{AB} = \overrightarrow{DC}, \\ \overrightarrow{AD} &= (1, 5, -1), \quad \overrightarrow{BC} = C - B = (1, 5, -1) \Rightarrow \overrightarrow{AD} = \overrightarrow{BC}.\end{aligned}$$

Opposite sides are equal and parallel, so $ABCD$ is a **parallelogram**.

Rectangle? A parallelogram is a rectangle iff its diagonals are equal:

$$|AC|^2 = 4^2 + 7^2 + 0^2 = 65, \quad |BD|^2 = (-2)^2 + 3^2 + (-2)^2 = 17.$$

Since $65 \neq 17$, the diagonals are unequal, so it is **not** a rectangle. (Equivalently, $\overrightarrow{AB} \cdot \overrightarrow{AD} = 3 + 10 - 1 = 12 \neq 0$, so adjacent sides are not perpendicular.)

This links two ideas — equal-and-parallel sides for the parallelogram, then the diagonal/perpendicularity test for the rectangle.

Example 4 — [JEE Advanced Level] Find the equation of the locus of a point P the sum of whose distances from $A(5, 0, 0)$ and $B(-5, 0, 0)$ is 12. Identify the surface.

Solution. Let $P(x, y, z)$ with $PA + PB = 12$:

$$\sqrt{(x-5)^2 + y^2 + z^2} + \sqrt{(x+5)^2 + y^2 + z^2} = 12.$$

Isolate one radical and square:

$$\sqrt{(x-5)^2 + y^2 + z^2} = 12 - \sqrt{(x+5)^2 + y^2 + z^2}.$$

Squaring both sides and cancelling the common $x^2 + y^2 + z^2 + 25$ terms:

$$-10x = 144 - 24\sqrt{(x+5)^2 + y^2 + z^2} + 10x \Rightarrow 24\sqrt{(x+5)^2 + y^2 + z^2} = 144 + 20x.$$

Divide by 4 and square again:

$$36[(x+5)^2 + y^2 + z^2] = (36 + 5x)^2.$$

Expanding and simplifying (the x -linear terms cancel):

$$36x^2 + 360x + 900 + 36y^2 + 36z^2 = 1296 + 360x + 25x^2,$$

$$11x^2 + 36y^2 + 36z^2 = 396 \Rightarrow \boxed{\frac{x^2}{36} + \frac{y^2}{11} + \frac{z^2}{11} = 1}.$$

This is an **ellipsoid of revolution** (a prolate spheroid) with semi-axis $a = 6$ along the x -axis and $b = \sqrt{11}$ across it — exactly the spatial analogue of an ellipse with foci at $(\pm 5, 0, 0)$, since $b^2 = a^2 - c^2 = 36 - 25 = 11$.

Figure to render. An ellipsoid stretched along the horizontal x -axis, with the two foci $A(5,0,0)$ and $B(-5,0,0)$ marked inside on the axis. Show a sample surface point P joined to both foci by two segments labelled so their lengths sum to 12. Mark the x -intercepts at $(\pm 6,0,0)$ and a cross-axis intercept at $(0,\sqrt{11},0)$.

The rigor lesson: the double-radical must be killed in two careful squaring rounds, isolating one root each time. Squaring the sum directly in one shot is the standard wreck.

Section 5 — Practice Exercises

1. **[CBSE]** Find the centroid of the triangle with vertices $(3, -1, 2)$, $(1, 2, -4)$, $(-1, 5, 2)$.
2. **[CBSE]** Find the coordinates of the points that trisect the segment joining $P(4, 2, -6)$ and $Q(10, -16, 6)$.
3. **[JEE Main]** Show that $A(-2, 3, 5)$, $B(1, 2, 3)$, $C(7, 0, -1)$ are collinear, and find the ratio in which B divides AC .
4. **[JEE Main]** Find the equation of the set of points equidistant from $A(1, 2, 3)$ and $B(3, 2, -1)$, and name the surface.
5. **[JEE Advanced]** Find the equation of the sphere that is the locus of points at distance 5 from the centre $(2, -1, 3)$, and state its radius and the point on it nearest the origin's z -axis.

Answers: 1. $(1, 2, 0)$. — 2. $(8, -10, 2)$ and $(6, -4, -2)$. — 3. Collinear; B divides AC in $1 : 2$. — 4. $x - 2z = 0$, a plane (the perpendicular bisector of AB). — 5. $(x-2)^2 + (y+1)^2 + (z-3)^2 = 25$; radius 5.

Section 6 — Crack the MCQ

Q1. The centroid of the triangle with vertices $(2, 3, -1)$, $(4, -1, 3)$, $(0, 4, 1)$ is

- (A) $(2, 2, 1)$ (B) $(6, 6, 3)$ (C) $(2, 2, 3)$ (D) $(3, 3, 1.5)$

Answer: (A). Averages: $\left(\frac{2+4+0}{3}, \frac{3-1+4}{3}, \frac{-1+3+1}{3}\right) = (2, 2, 1)$.

- (B) forgets to divide by 3 (gives the sum).
- (C) slips on the z -coordinate (-1 read as $+1$).
- (D) divides by 2 instead of 3 — confusing centroid with midpoint.

Q2. Points $(1, 2, 3)$, $(2, 3, 4)$, $(k, 5, 6)$ are collinear. Then $k =$

- (A) 3 (B) 4 (C) 5 (D) 6

Answer: (B). The first two give a direction $(1, 1, 1)$. From $(1, 2, 3)$, reaching $z = 6$ needs $+3$ in each coordinate: $x = 1 + 3 = 4$. Check $y = 2 + 3 = 5$ ✓.

- (A) advances only 2 steps (matches $z = 5$, not 6).
- (C)/(D) over-advance — they would need $z = 7$ or 8.

Q3. The set of points equidistant from $(0, 0, 0)$ and $(2, 0, 0)$ is

- (A) the line $x = 1$ (B) the plane $x = 1$ (C) the sphere $x^2 + y^2 + z^2 = 1$ (D) the point $(1, 0, 0)$

Answer: (B). $PA^2 = PB^2 \Rightarrow x^2 + y^2 + z^2 = (x-2)^2 + y^2 + z^2 \Rightarrow 4x = 4 \Rightarrow x = 1$, a whole plane.

- (A) forgets that in 3D, " $x = 1$ " is a plane, not a line (that was the 2D habit).
- (C) confuses equidistance-between-two-points with distance-from-one-point.
- (D) gives only the midpoint, not the full locus.

Q4. The centroid G divides a median of a triangle in the ratio (vertex : midpoint)

- (A) 1 : 1 (B) 1 : 2 (C) 2 : 1 (D) 3 : 1

Answer: (C). The centroid sits twice as far from the vertex as from the opposite midpoint, i.e. 2 : 1 from the vertex.

- (B) reverses the ratio (midpoint-to-vertex).
- (A) confuses the centroid with the midpoint of the median.
- (D) is a common misremembering.

Section 7 — Common Student Pitfalls & Pro-Tips

- **Declaring collinearity from one coordinate.** Solving for k from x and stopping is the top error. Always confirm the *same* k fits y and z .
- **Centroid vs midpoint vs incentre.** Centroid divides by 3 (triangle) or 4 (tetrahedron); midpoint divides by 2; the incentre is a *weighted* average by side lengths — don't average the incentre.
- **Forgetting the $\times 3$ in reverse centroid problems.** "Sum of vertices = $3G$," not G .
- **Locus sign/squaring slips.** Equidistant-from-two-points is linear (a plane); distance-from-one-point is quadratic (a sphere). For sum-of-distances, isolate and square one radical at a time.

Pro-Tip: For any "are these collinear / what ratio" question, go straight to the section-ratio method: solve k from one coordinate, then verify across the other two. It is faster than three distance computations *and* hands you the dividing ratio as a bonus. For "type of figure," compute all four sides and both diagonals once — sides equal $\Rightarrow$ rhombus, diagonals equal $\Rightarrow$ rectangle, both $\Rightarrow$ square.

Section 8 — Cheat Sheet

- **Centroid (triangle):** $\left(\frac{\sum x}{3}, \frac{\sum y}{3}, \frac{\sum z}{3}\right)$; divides each median 2 : 1 from the vertex.
(Tetrahedron: denominator 4.)
- **Collinear** $\Leftrightarrow$ one ratio $k : 1$ fits x, y, z together, **or** $AB + BC = AC$.
- **Figure type:** all sides equal $\Rightarrow$ rhombus; equal diagonals $\Rightarrow$ rectangle; both $\Rightarrow$ square; opposite sides equal-and-parallel $\Rightarrow$ parallelogram.
- **Locus:** equidistant from two points $\Rightarrow$ **plane** (perp. bisector); distance r from a centre $\Rightarrow$ **sphere** $(x - a)^2 + (y - b)^2 + (z - c)^2 = r^2$; constant sum of distances $\Rightarrow$ **ellipsoid** ($b^2 = a^2 - c^2$).
- **[JEE] Incentre:** weight vertices by opposite side lengths a, b, c .
- **Mnemonics:** "Centroid = plain average; incentre = weighted average." — "Collinear needs all three coordinates to agree on one ratio." — "Two points $\rightarrow$ plane, one point $\rightarrow$ sphere."

Where this leads (Class 12 preview): these coordinates are the launch pad for **direction cosines & ratios**, the **equations of a line and a plane** in space, the **angle between lines/planes**, and **shortest distance between skew lines** — the entire Class 12 “Three Dimensional Geometry” chapter. Everything there is built on the distance, section, and locus ideas you have just locked down here.

Chapter 11: Introduction to Three Dimensional Geometry — Supplement

Addendum to Chapter 11: Missing Problem Patterns

The three subtopics you have just worked through give you every tool this chapter owns: the octant/plane/axis vocabulary of Subtopic 01, the distance and section formulas of Subtopic 02, and the centroid-collinearity-locus applications of Subtopic 03. What follows is not new machinery. It is the set of *question shapes* that examiners build out of that machinery and that a first pass through the chapter can leave under-drilled. Work each pattern once with a pen and it stops being a surprise in the hall.

Each treatment gives you the method in one or two lines, worked examples in full, and two practice items with answers.

Pattern 1 — Distance of a Point from a Coordinate Axis

Method. Subtopic 01, Procedure B told you the *foot* of the perpendicular from $P(x, y, z)$ onto an axis: on the z -axis it is $N(0, 0, z)$. Take the one extra step the procedure stopped short of — measure the length PN with the distance formula:

$$PN = \sqrt{(x-0)^2 + (y-0)^2 + (z-z)^2} = \sqrt{x^2 + y^2}.$$

The coordinate *along* the axis cancels, and what survives is the square root of the squares of the **other two** coordinates. That is the whole pattern:

Distance of $P(x, y, z)$ from	Foot of perpendicular	Distance
the x -axis	$(x, 0, 0)$	$\sqrt{y^2 + z^2}$
the y -axis	$(0, y, 0)$	$\sqrt{z^2 + x^2}$
the z -axis	$(0, 0, z)$	$\sqrt{x^2 + y^2}$

Hold this against the plane result from Subtopic 01, Section 2 and the contrast is clean: **distance from a coordinate plane uses one coordinate; distance from a coordinate axis uses the other two.** A plane kills one coordinate, an axis kills two, so the plane distance is $|x|$ -style and the axis distance is a two-term square root. Mixing them up is the single commonest slip on this one-marker: the distance of $(2, -6, 3)$ from the YZ -plane is 2, but its distance from the x -axis is $\sqrt{36 + 9} = \sqrt{45}$.

Figure brief: Point $P(x, y, z)$ in the first octant with the foot $N(0, 0, z)$ marked on the z -axis; the segment PN drawn horizontal and labelled $\sqrt{x^2 + y^2}$, with a right angle at N ; a dashed drop from P to $M(x, y, 0)$ on the XY -plane showing that OM has the same length as PN .

Locus consequence. Fix that distance at a constant $c > 0$. “All points at distance c from the z -axis” is $\sqrt{x^2 + y^2} = c$, that is

$$x^2 + y^2 = c^2 \quad (z \text{ free}),$$

a **right circular cylinder** of radius c with the z -axis as its axis. The free coordinate is the giveaway: nothing constrains z , so the circle of radius c in the XY -plane is swept vertically into a tube. Compare this with the two locus results you already have — equidistant from two points gives a plane, fixed distance from a point gives a sphere, fixed distance from a line gives a cylinder.

Example 1.1 — [JEE Main Level]

Find the distance of $P(3, -4, 12)$ from (i) the z -axis, (ii) the x -axis, (iii) the y -axis, and (iv) the origin.

Solution. Apply the table, discarding in each case the coordinate that lies along the axis.

- (i) Distance from the z -axis $= \sqrt{x^2 + y^2} = \sqrt{9 + 16} = \sqrt{25} = 5$.
- (ii) Distance from the x -axis $= \sqrt{y^2 + z^2} = \sqrt{16 + 144} = \sqrt{160} = 4\sqrt{10}$.
- (iii) Distance from the y -axis $= \sqrt{z^2 + x^2} = \sqrt{144 + 9} = \sqrt{153} = 3\sqrt{17}$.
- (iv) Distance from the origin $= \sqrt{9 + 16 + 144} = \sqrt{169} = 13$.

Notice that every one of the three axis distances is smaller than $OP = 13$. That is not a coincidence: the foot of the perpendicular is the nearest point of the axis to P , and the origin is itself a point of that axis, so the perpendicular can never be longer than OP .

Example 1.2 — [JEE Main Level]

For $P(2, -3, 6)$, compute the three axis distances and show that the sum of their squares equals $2OP^2$.

Solution.

$$d_x = \sqrt{(-3)^2 + 6^2} = \sqrt{45}, \quad d_y = \sqrt{6^2 + 2^2} = \sqrt{40}, \quad d_z = \sqrt{2^2 + (-3)^2} = \sqrt{13}.$$

Then $d_x^2 + d_y^2 + d_z^2 = 45 + 40 + 13 = 98$, while $OP^2 = 4 + 9 + 36 = 49$, so indeed $98 = 2(49)$.

The reason is structural, not arithmetical: each of x^2 , y^2 , z^2 appears in exactly two of the three axis distances, so

$$d_x^2 + d_y^2 + d_z^2 = (y^2 + z^2) + (z^2 + x^2) + (x^2 + y^2) = 2(x^2 + y^2 + z^2) = 2OP^2.$$

Carry this identity as a free self-check whenever a question asks for more than one axis distance.

Practice

1. [JEE Main] Find the distance of the point $(1, 2, 3)$ from the y -axis.
2. [JEE Main] Find the equation of the locus of a point whose distance from the x -axis is always 4, and name the surface.

Answers. 1. $\sqrt{z^2 + x^2} = \sqrt{9 + 1} = \sqrt{10}$. — 2. $y^2 + z^2 = 16$ (with x free); a right circular cylinder of radius 4 whose axis is the x -axis.

Pattern 2 — Substituting into the Section Formula to Produce a Point

Method. Subtopic 02 gives you both formulas; this pattern is the drill of actually running numbers through them. Let R divide the join of $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ in the ratio $m : n$, meaning $PR : RQ = m : n$. Then

$$R_{\text{int}} = \left(\frac{mx_2 + nx_1}{m + n}, \frac{my_2 + ny_1}{m + n}, \frac{mz_2 + nz_1}{m + n} \right), \quad R_{\text{ext}} = \left(\frac{mx_2 - nx_1}{m - n}, \frac{my_2 - ny_1}{m - n}, \frac{mz_2 - nz_1}{m - n} \right).$$

Three habits make this error-proof under exam pressure.

1. **Write down which point is P and which is Q before you substitute.** m always multiplies the *far* point (x_2, y_2, z_2) . Swapping P and Q silently inverts your ratio.
2. **Internal adds, external subtracts** — in the numerator *and* the denominator, together. Changing one but not the other is the classic wreck.
3. **Sanity-check the external answer's position.** If $m > n$ the external point lies beyond Q ; if $m < n$ it lies beyond P ; and $m = n$ has no external point at all (the denominator vanishes).

Figure brief: One straight line carrying P and Q , with R_{int} marked between them and R_{ext} marked outside the segment on the P side; braces labelling $PR : RQ = 2 : 3$ in both cases, so the reader sees the same ratio realised twice.

Example 2.1 — [CBSE Board Level]

Find the coordinates of the point that divides the join of $P(-2, 3, 5)$ and $Q(1, -4, 6)$ in the ratio $2 : 3$ (i) internally, (ii) externally.

Solution. Here $(x_1, y_1, z_1) = (-2, 3, 5)$, $(x_2, y_2, z_2) = (1, -4, 6)$, $m = 2$, $n = 3$.

(i) Internal. Denominator $m + n = 5$:

$$x = \frac{2(1) + 3(-2)}{5} = \frac{2 - 6}{5} = -\frac{4}{5}, \quad y = \frac{2(-4) + 3(3)}{5} = \frac{-8 + 9}{5} = \frac{1}{5}, \quad z = \frac{2(6) + 3(5)}{5} = \frac{12 + 15}{5} = \frac{27}{5}.$$

The internal point is $\left(-\frac{4}{5}, \frac{1}{5}, \frac{27}{5}\right)$.

(ii) External. Same numbers, every + becoming −; denominator $m - n = 2 - 3 = -1$:

$$x = \frac{2(1) - 3(-2)}{-1} = \frac{8}{-1} = -8, \quad y = \frac{2(-4) - 3(3)}{-1} = \frac{-17}{-1} = 17, \quad z = \frac{2(6) - 3(5)}{-1} = \frac{-3}{-1} = 3.$$

The external point is $(-8, 17, 3)$.

Check the geometry. The step from P to Q is $(3, -7, 1)$. The internal point sits $\frac{2}{5}$ of that step past P : $(-2, 3, 5) + \frac{2}{5}(3, -7, 1) = (-2 + \frac{6}{5}, 3 - \frac{14}{5}, 5 + \frac{2}{5}) = (-\frac{4}{5}, \frac{1}{5}, \frac{27}{5})$, which matches. The external point sits 2 full steps *backwards* from P : $(-2, 3, 5) - 2(3, -7, 1) = (-8, 17, 3)$, which also matches — and it lies beyond P , exactly as the rule $m < n$ predicts.

Example 2.2 — [CBSE Board Level]

Find the point dividing the join of $A(1, -2, 3)$ and $B(3, 4, -5)$ in the ratio $3 : 1$ (i) internally, (ii) externally.

Solution. Now $m = 3$, $n = 1$, A is the first point and B the second.

(i) Internal, denominator 4:

$$x = \frac{3(3) + 1(1)}{4} = \frac{10}{4} = \frac{5}{2}, \quad y = \frac{3(4) + 1(-2)}{4} = \frac{10}{4} = \frac{5}{2}, \quad z = \frac{3(-5) + 1(3)}{4} = \frac{-12}{4} = -3,$$

giving $(\frac{5}{2}, \frac{5}{2}, -3)$ — three-quarters of the way from A to B , as $3 : 1$ internal should be.

(ii) External, denominator $3 - 1 = 2$:

$$x = \frac{3(3) - 1(1)}{2} = \frac{8}{2} = 4, \quad y = \frac{3(4) - 1(-2)}{2} = \frac{14}{2} = 7, \quad z = \frac{3(-5) - 1(3)}{2} = \frac{-18}{2} = -9,$$

giving $(4, 7, -9)$. Since $m > n$ here, this point lies **beyond** B — contrast Example 2.1, where $m < n$ threw the point out beyond P . Same formula, opposite side, and the only thing that decided it was whether $m - n$ was positive or negative.

A one-line link to the $k : 1$ form. External division in the ratio $m : n$ is internal division in the ratio $m : (-n)$, i.e. $k = -m/n$ in the workhorse form of Subtopic 02, Section 2. For Example 2.1, $k = -\frac{2}{3}$ gives $x = \frac{k(1) + (-2)}{k + 1} = \frac{-\frac{2}{3} - 2}{-\frac{1}{3}} = -8$, confirming the external answer.

Practice

- [CBSE] Find the coordinates of the point dividing the join of $A(2, -1, 4)$ and $B(4, 3, 2)$ in the ratio $1 : 3$ (i) internally, (ii) externally.
- [CBSE] Find the point dividing the join of $A(-1, 4, -2)$ and $B(3, -2, 6)$ in the ratio $5 : 2$ externally, and state on which side of the segment it lies.

Answers. 1. (i) $(\frac{5}{2}, 0, \frac{7}{2})$; (ii) $(1, -3, 5)$. — 2. $(\frac{17}{3}, -6, \frac{34}{3})$; since $m > n$ it lies beyond B .

Pattern 3 — The Fourth Vertex of a Parallelogram

Method. The property that unlocks this is one line: **the diagonals of a parallelogram bisect each other**, so the two diagonals share a midpoint. For a parallelogram $ABCD$ (vertices named in order round the figure, so the diagonals are AC and BD),

$$\text{midpoint of } AC = \text{midpoint of } BD \Rightarrow \frac{A+C}{2} = \frac{B+D}{2} \Rightarrow \boxed{D = A + C - B}$$

read coordinate by coordinate. Subtopic 03, Section 4 Example 3 ran this argument forwards (four points given, prove parallelogram); this is the reverse question, and it is a standing NCERT miscellaneous-exercise item.

Watch the naming. In $ABCD$, B and D are the *opposite* vertices, which is why B carries the minus sign. If the question merely says “three vertices of a parallelogram are P, Q, R ” without fixing the order, there are **three** possible fourth vertices, one for each choice of which given point is opposite the new one.

Figure brief: Parallelogram $ABCD$ with both diagonals AC and BD drawn, crossing at a single point marked M and labelled “midpoint of both diagonals”; the four equal half-diagonals tick-marked in two pairs.

Example 3.1 — [CBSE Board Level]

Three vertices of a parallelogram $ABCD$ are $A(3, -1, 2)$, $B(1, 2, -4)$ and $C(-1, 1, 2)$. Find the fourth vertex D .

Solution. Let $D(x, y, z)$. The diagonals AC and BD have the same midpoint:

$$\left(\frac{3+(-1)}{2}, \frac{-1+1}{2}, \frac{2+2}{2} \right) = \left(\frac{1+x}{2}, \frac{2+y}{2}, \frac{-4+z}{2} \right).$$

The left side is $(1, 0, 2)$. Equating coordinates:

$$\frac{1+x}{2} = 1 \Rightarrow x = 1, \quad \frac{2+y}{2} = 0 \Rightarrow y = -2, \quad \frac{-4+z}{2} = 2 \Rightarrow z = 8.$$

Hence $D(1, -2, 8)$.

Verification (worth the ten seconds). The step $A \rightarrow B$ is $B - A = (-2, 3, -6)$, and the step $D \rightarrow C$ is $C - D = (-1 - 1, 1 + 2, 2 - 8) = (-2, 3, -6)$. The two are identical, so AB and DC are equal in length and parallel in direction — the defining property of the parallelogram $ABCD$.

Example 3.2 — [JEE Main Level]

The points $P(1, 2, 3)$, $Q(3, -1, 5)$ and $R(4, 0, -3)$ are three vertices of a parallelogram, the labelling not being specified. Find all possible positions of the fourth vertex.

Solution. The unknown vertex S is opposite exactly one of the three given points, and the “ $D = A + C - B$ ” rule says: add the two points adjacent to S and subtract the one opposite it.

- S opposite Q : $S = P + R - Q = (1 + 4 - 3, 2 + 0 - 1, 3 - 3 - 5) = (2, 3, -5)$.

- S opposite P : $S = Q + R - P = (3 + 4 - 1, -1 + 0 - 2, 5 - 3 - 3) = (6, -3, -1)$.
- S opposite R : $S = P + Q - R = (1 + 3 - 4, 2 - 1 - 0, 3 + 5 + 3) = (0, 1, 11)$.

So the fourth vertex is $(2, 3, -5)$, $(6, -3, -1)$ or $(0, 1, 11)$.

The three answers are not a defect in the question — three given points genuinely sit inside three different parallelograms, and a question that wants a unique answer must name the vertices in cyclic order. If the paper says “ $ABCD$ ”, use only the opposite-vertex rule; if it says “three vertices of a parallelogram”, expect three answers unless a further condition is imposed.

Practice

1. [CBSE] Three vertices of a parallelogram $ABCD$ are $A(2, -1, 3)$, $B(4, 3, -1)$ and $C(0, 5, 2)$. Find D .
2. [CBSE] If $A(1, 2, 3)$, $B(-1, -2, -1)$ and $C(2, 3, 2)$ are three consecutive vertices of a parallelogram $ABCD$, find D and state, with reason, whether $ABCD$ is a rhombus.

Answers. 1. $D = A + C - B = (-2, 1, 6)$. — 2. $D = A + C - B = (4, 7, 6)$. Not a rhombus: $AB = \sqrt{4 + 16 + 16} = 6$ but $BC = \sqrt{9 + 25 + 9} = \sqrt{43}$, and a rhombus needs all four sides equal.

Pattern 4 — A Point Equidistant from Three or Four Given Points

Method. Subtopic 02, Section 4 Example 4 established the key structural fact: imposing $PA = PB$ on a variable point $P(x, y, z)$ produces a **linear** equation, because squaring and subtracting annihilates every x^2 , y^2 , z^2 term. So each equidistance condition costs you one linear equation, and the problem becomes simultaneous linear equations.

Count your conditions against your unknowns:

- **Two given points** leave a plane (one equation, three unknowns) — the perpendicular bisector plane.
- **Three given points**, not collinear, leave a **line** (two equations) — so a unique answer needs one more restriction, typically “the point lies in the XY -plane” or “on the y -axis”, which is what exam questions supply.
- **Four given points**, not coplanar, pin the point down uniquely (three equations, three unknowns) — this is the circumcentre of the tetrahedron they form.

Always square both sides first. Write $PA^2 = PB^2$, never $PA = PB$ with radicals left standing.

Example 4.1 — [JEE Main Level]

Find the point equidistant from $O(0, 0, 0)$, $A(2, 0, 0)$, $B(0, 3, 0)$ and $C(0, 0, 4)$.

Solution. Let $P(x, y, z)$ be the required point, so $PO^2 = PA^2 = PB^2 = PC^2$.

From $PO^2 = PA^2$:

$$x^2 + y^2 + z^2 = (x-2)^2 + y^2 + z^2 \implies 0 = -4x + 4 \implies x = 1.$$

From $PO^2 = PB^2$:

$$x^2 + y^2 + z^2 = x^2 + (y-3)^2 + z^2 \implies 0 = -6y + 9 \implies y = \frac{3}{2}.$$

From $PO^2 = PC^2$:

$$x^2 + y^2 + z^2 = x^2 + y^2 + (z-4)^2 \implies 0 = -8z + 16 \implies z = 2.$$

The point is $\left(1, \frac{3}{2}, 2\right)$, and the common distance is $\sqrt{1 + \frac{9}{4} + 4} = \sqrt{\frac{29}{4}} = \frac{\sqrt{29}}{2}$.

Read the answer, don't just compute it. Each coordinate came out as exactly half of the corresponding intercept. In general, the point equidistant from $O(0,0,0)$, $A(a,0,0)$, $B(0,b,0)$ and $C(0,0,c)$ is $\left(\frac{a}{2}, \frac{b}{2}, \frac{c}{2}\right)$ — the centre of the rectangular box built on those three edges, which is the one place a sphere through all four corners could possibly be centred. Knowing this converts a four-line question into a one-line answer.

Example 4.2 — [JEE Main Level]

Find the point in the XY -plane that is equidistant from $A(2,0,3)$, $B(0,3,2)$ and $C(0,0,1)$.

Solution. “In the XY -plane” means $z = 0$ (Subtopic 01), so write the unknown as $P(x, y, 0)$ — the restriction has already used up one of the three unknowns. Then

$$PA^2 = (x-2)^2 + y^2 + 9, \quad PB^2 = x^2 + (y-3)^2 + 4, \quad PC^2 = x^2 + y^2 + 1.$$

Impose $PA^2 = PC^2$:

$$(x-2)^2 + 9 = x^2 + 1 \implies -4x + 4 + 9 = 1 \implies -4x = -12 \implies x = 3.$$

Impose $PB^2 = PC^2$:

$$(y-3)^2 + 4 = y^2 + 1 \implies -6y + 9 + 4 = 1 \implies -6y = -12 \implies y = 2.$$

The point is $(3, 2, 0)$. Checking: $PA^2 = 1 + 4 + 9 = 14$, $PB^2 = 9 + 1 + 4 = 14$, $PC^2 = 9 + 4 + 1 = 14$, so all three distances equal $\sqrt{14}$.

Note how the x^2 and y^2 terms cancelled every time. If a quadratic term ever survives in a problem of this type, you have made an algebra slip — go back before you go forward.

Practice

1. [JEE Main] Find the point equidistant from $O(0,0,0)$, $A(4,0,0)$, $B(0,-6,0)$ and $C(0,0,2)$, and state the common distance.

2. [JEE Main] Find the point in the XY -plane equidistant from $A(1, -1, 0)$, $B(2, 1, 2)$ and $C(3, 2, -1)$.

Answers. 1. $(2, -3, 1)$; common distance $\sqrt{4 + 9 + 1} = \sqrt{14}$. — 2. $\left(\frac{3}{2}, 1, 0\right)$; the common distance is $\frac{\sqrt{17}}{2}$.

Pattern 5 — The Promised JEE Extensions: Incentre in 3D and the Centroid of a Tetrahedron

Subtopic 03 flagged both of these as beyond-CBSE extensions that JEE rewards. Here they are with the numbers filled in. Neither needs anything past the section formula.

5A. Incentre — the weighted point

Method. For a triangle ABC let the side lengths opposite the vertices be

$$a = |BC|, \quad b = |CA|, \quad c = |AB|.$$

Then the incentre — the centre of the inscribed circle, the point equidistant from the three sides — is the average of the vertices **weighted by those opposite side lengths**:

$$I = \left(\frac{ax_1 + bx_2 + cx_3}{a + b + c}, \frac{ay_1 + by_2 + cy_3}{a + b + c}, \frac{az_1 + bz_2 + cz_3}{a + b + c} \right).$$

Compare with the centroid, which is the same expression with all three weights equal to 1. The order matters and it is the order students get wrong: the weight sitting next to A 's coordinates is $a = |BC|$, the length of the side **opposite** A .

Example 5.1 — [JEE Main Level] Find the incentre of the triangle with vertices $A(1, 2, 3)$, $B(4, 8, -3)$, $C(9, 6, 11)$.

Solution. First the three side lengths, always squaring before rooting.

$$\begin{aligned} a &= |BC| = \sqrt{(9-4)^2 + (6-8)^2 + (11+3)^2} = \sqrt{25 + 4 + 196} = \sqrt{225} = 15, \\ b &= |CA| = \sqrt{(1-9)^2 + (2-6)^2 + (3-11)^2} = \sqrt{64 + 16 + 64} = \sqrt{144} = 12, \\ c &= |AB| = \sqrt{(4-1)^2 + (8-2)^2 + (-3-3)^2} = \sqrt{9 + 36 + 36} = \sqrt{81} = 9. \end{aligned}$$

A 9-12-15 triangle, so $a + b + c = 36$. Now weight and average:

$$\begin{aligned} x_I &= \frac{15(1) + 12(4) + 9(9)}{36} = \frac{15 + 48 + 81}{36} = \frac{144}{36} = 4, \\ y_I &= \frac{15(2) + 12(8) + 9(6)}{36} = \frac{30 + 96 + 54}{36} = \frac{180}{36} = 5, \end{aligned}$$

$$z_I = \frac{15(3) + 12(-3) + 9(11)}{36} = \frac{45 - 36 + 99}{36} = \frac{108}{36} = 3.$$

The incentre is $I(4, 5, 3)$.

Independent check. Since $9^2 + 12^2 = 81 + 144 = 225 = 15^2$, the triangle is right-angled at A . For a right triangle the inradius is $r = \frac{1}{2}(\text{leg} + \text{leg} - \text{hypotenuse}) = \frac{1}{2}(9 + 12 - 15) = 3$, and the incentre of a right triangle lies at distance $r\sqrt{2}$ from the right-angled vertex. Here $IA = \sqrt{(4-1)^2 + (5-2)^2 + (3-3)^2} = \sqrt{18} = 3\sqrt{2}$, exactly as required.

5B. Centroid of a tetrahedron and its 3 : 1 property

Method. For a tetrahedron with vertices A, B, C, D the centroid is the plain average of all four:

$$G = \left(\frac{x_1 + x_2 + x_3 + x_4}{4}, \frac{y_1 + y_2 + y_3 + y_4}{4}, \frac{z_1 + z_2 + z_3 + z_4}{4} \right).$$

The property that makes it examinable is the spatial cousin of the median's 2 : 1 split. Join a vertex to the centroid of the opposite face; that segment is called a *median of the tetrahedron*, and G divides it in the ratio

$$3 : 1 \quad \text{measured from the vertex.}$$

Why. Let G_1 be the centroid of face BCD , so $G_1 = \frac{1}{3}(B + C + D)$ coordinatewise. The point dividing AG_1 internally in the ratio 3 : 1 is, by the section formula with $m = 3, n = 1$,

$$\frac{3G_1 + 1 \cdot A}{4} = \frac{(B + C + D) + A}{4} = G.$$

Since the same argument works from every vertex, all four medians of a tetrahedron pass through G — the exact analogue of the concurrency proof for the triangle in Subtopic 03, Section 3. The pattern is worth memorising as a family: a triangle's medians split 2 : 1 with denominator 3; a tetrahedron's split 3 : 1 with denominator 4.

Figure brief: A tetrahedron $ABCD$ drawn in outline with the face BCD shaded, its centroid G_1 marked, the segment AG_1 drawn, and G marked on it much nearer G_1 than A , with the two parts labelled 3 and 1.

Example 5.2 — [JEE Main Level] A tetrahedron has vertices $A(0, 0, 0)$, $B(4, 0, 0)$, $C(0, 8, 0)$, $D(0, 0, 12)$. Find its centroid G , and verify that G divides the segment from A to the centroid of face BCD in the ratio 3 : 1.

Solution. Averaging all four vertices:

$$G = \left(\frac{0 + 4 + 0 + 0}{4}, \frac{0 + 0 + 8 + 0}{4}, \frac{0 + 0 + 0 + 12}{4} \right) = (1, 2, 3).$$

The centroid of face BCD is the average of B, C, D :

$$G_1 = \left(\frac{4+0+0}{3}, \frac{0+8+0}{3}, \frac{0+0+12}{3} \right) = \left(\frac{4}{3}, \frac{8}{3}, 4 \right).$$

Now divide AG_1 internally in the ratio 3 : 1, with A first and G_1 second:

$$\left(\frac{3\left(\frac{4}{3}\right) + 1(0)}{4}, \frac{3\left(\frac{8}{3}\right) + 1(0)}{4}, \frac{3(4) + 1(0)}{4} \right) = \left(\frac{4}{4}, \frac{8}{4}, \frac{12}{4} \right) = (1, 2, 3),$$

which is G . The verification is complete.

Practice

1. [JEE Main] Find the incentre of the triangle with vertices $A(2, 3, 5)$, $B(11, 3, 5)$, $C(2, 15, 5)$.
2. [JEE Main] Three vertices of a tetrahedron are $A(2, -1, 3)$, $B(4, 3, -2)$, $C(0, 5, 1)$ and its centroid is $G(1, 2, 1)$. Find the fourth vertex D .

Answers. 1. Side lengths $a = |BC| = 15$, $b = |CA| = 12$, $c = |AB| = 9$; incentre $(5, 6, 5)$. — 2. Sum of the four vertices $= 4G = (4, 8, 4)$, so $D = (4, 8, 4) - (6, 7, 2) = (-2, 1, 2)$.

Pattern 6 — The Length of a Median

Method. Nothing new at all — this is the midpoint formula composed with the distance formula, in that order. To find the median through A in triangle ABC :

1. Find D , the midpoint of the opposite side BC , by averaging B and C .
2. Compute AD by the distance formula.

The one thing that goes wrong is haste: students average A with something, or take the midpoint of a side that touches A . The median from a vertex always runs to the midpoint of the side **not containing** that vertex.

Example 6.1 — [CBSE Board Level]

Find the lengths of the medians of the triangle with vertices $A(0, 0, 6)$, $B(0, 4, 0)$, $C(6, 0, 0)$.

Solution. Take the three medians in turn.

Median through A. Midpoint of BC is $D = \left(\frac{0+6}{2}, \frac{4+0}{2}, \frac{0+0}{2} \right) = (3, 2, 0)$, so

$$AD = \sqrt{(3-0)^2 + (2-0)^2 + (0-6)^2} = \sqrt{9+4+36} = \sqrt{49} = 7.$$

Median through B. Midpoint of CA is $E = \left(\frac{6+0}{2}, \frac{0+0}{2}, \frac{0+6}{2} \right) = (3, 0, 3)$, so

$$BE = \sqrt{(3-0)^2 + (0-4)^2 + (3-0)^2} = \sqrt{9+16+9} = \sqrt{34}.$$

Median through C. Midpoint of AB is $F = \left(\frac{0+0}{2}, \frac{0+4}{2}, \frac{6+0}{2} \right) = (0, 2, 3)$, so

$$CF = \sqrt{(0-6)^2 + (2-0)^2 + (3-0)^2} = \sqrt{36+4+9} = \sqrt{49} = 7.$$

The medians are 7, $\sqrt{34}$ and 7. Two of them being equal is not an accident: $|AB| = \sqrt{0+16+36} = \sqrt{52}$ and $|BC| = \sqrt{36+16+0} = \sqrt{52}$, so the triangle is isosceles about B , and the medians to the two equal sides must match.

Practice

1. [CBSE] Find the length of the median through A in the triangle $A(5, 6, 8)$, $B(4, 0, 3)$, $C(-2, 4, -1)$.
2. [CBSE] Find the length of the median through C in the triangle $A(2, -3, 3)$, $B(4, 1, -1)$, $C(0, 5, 3)$.

Answers. 1. Midpoint of BC is $(1, 2, 1)$; median $= \sqrt{16+16+49} = \sqrt{81} = 9$. — 2. Midpoint of AB is $(3, -1, 1)$; median $= \sqrt{9+36+4} = \sqrt{49} = 7$.

Addendum Cheat Sheet

- **Distance from an axis** = square root of the squares of the *other two* coordinates: from the x -axis $\sqrt{y^2 + z^2}$, from the y -axis $\sqrt{z^2 + x^2}$, from the z -axis $\sqrt{x^2 + y^2}$. (Distance from a *plane* uses one coordinate; distance from an *axis* uses two.)
- **Constant distance from an axis** $\Rightarrow$ a right circular cylinder: $x^2 + y^2 = c^2$ about the z -axis.
- **Section formula substitution:** m multiplies the far point. Internal $\Rightarrow +$ and $m+n$; external $\Rightarrow -$ and $m-n$. External with $m > n$ lands beyond Q ; with $m < n$, beyond P ; $m = n$ has no external point.
- **Fourth vertex of parallelogram $ABCD$:** $D = A + C - B$ (diagonals share a midpoint). Order unspecified $\Rightarrow$ three possible answers.
- **Equidistant from n points:** each condition $PA^2 = PB^2$ is *linear*. Two points $\Rightarrow$ a plane; three $\Rightarrow$ a line (needs one more restriction); four non-coplanar $\Rightarrow$ a unique point. From O , $(a, 0, 0)$, $(0, b, 0)$, $(0, 0, c)$ the point is $(\frac{a}{2}, \frac{b}{2}, \frac{c}{2})$.
- **Incentre:** weight each vertex by the side length *opposite* it, $I = \frac{aA + bB + cC}{a + b + c}$, with $a = |BC|$, $b = |CA|$, $c = |AB|$.
- **Tetrahedron:** centroid = average of four vertices; it splits each vertex-to-opposite-face-centroid segment in the ratio 3 : 1 from the vertex. (Triangle: 2 : 1, denominator 3. Tetrahedron: 3 : 1, denominator 4.)
- **Median length:** midpoint of the far side first, distance formula second.

Chapter 11: Introduction to Three Dimensional Geometry — Round 2 Addendum

Subtopics 01-03 and the Round 1 Supplement (Patterns 1-6) between them cover this chapter's machinery completely, and they should: NCERT Chapter 11 is genuinely a short introductory chapter, and everything past it — direction cosines, the equation of a line or a plane, the angle between two lines, the shortest distance between skew lines — needs vectors and belongs to Class 12. This addendum therefore adds nothing new to the toolkit. It closes the last two *question shapes* that the chapter's tools already answer but that neither the chapter nor Round 1 ever demonstrates, and both are standing NCERT items:

Pattern	Topic	NCERT source	Read after
7	A point on an axis at a given distance from a fixed point — the quadratic case	Miscellaneous Exercise Q4	Subtopic 02, Section 4, Example 3
8	A plane cutting a segment externally — when k comes out negative	Solved Example 10; Misc. Exercise Q5	Subtopic 02, Section 4, Example 2

Each is a method statement, worked examples in full, and practice with answers, continuing the numbering of the Round 1 Supplement.

Pattern 7 — A Point on an Axis at a Given Distance from a Fixed Point

Extends Subtopic 02, Section 4, Example 3, and Round 1 Supplement Pattern 1.

Method. Subtopic 02, Example 3 found the point on the y -axis **equidistant from two points**, and noted that the y^2 terms cancel so the condition is linear — one unknown, one answer. Round 1 Pattern 4 leaned on the same cancellation. Change the condition to “*at a given distance d from one fixed point*” and that cancellation **does not happen**. Exactly one squared term survives, the equation is a genuine quadratic, and the honest answer is usually **two points**, not one.

The procedure is three lines.

1. **Write the axis form first.** A point on the y -axis is $R(0, y, 0)$ — one unknown, not three (Subtopic 01).

2. **Square before you do anything else.** Impose $RP^2 = d^2$, never $RP = d$ with the radical standing.
3. **Solve the quadratic and keep every real root.** Each root is a genuine point on the axis at distance d from P .

How many answers should you expect? Round 1 Pattern 1 gives you the check for free. The nearest point of an axis to P is the foot of the perpendicular, and the distance to it is the axis distance $d_{\min}$ — for the y -axis, $d_{\min} = \sqrt{z^2 + x^2}$. So

$$d > d_{\min} \Rightarrow \text{two points (one either side of the foot)}, \quad d = d_{\min} \Rightarrow \text{one point (the foot itself)}, \quad d < d_{\min} \Rightarrow$$

Compute $d_{\min}$ before you solve and you know the answer count in advance — which is how you catch yourself when you have written down only one root.

Figure brief: The y -axis drawn vertically with a point P off to one side. The foot of the perpendicular N marked on the axis, with the short segment PN labelled $d_{\min}$ and a right angle at N . Two further points R_1 and R_2 marked on the axis, equally spaced above and below N , each joined to P by a dashed segment labelled d . Caption: the two solutions are always symmetric about the foot.

Example 7.1 — [CBSE Board Level]

Find the coordinates of the point on the y -axis which is at a distance $5\sqrt{2}$ from the point $P(3, -2, 5)$.

Solution. A point on the y -axis has the form $R(0, y, 0)$.

Before solving, count the answers. The distance of P from the y -axis is $\sqrt{z^2 + x^2} = \sqrt{25 + 9} = \sqrt{34} \approx 5.83$, and the required distance is $5\sqrt{2} = \sqrt{50} \approx 7.07$. Since $\sqrt{50} > \sqrt{34}$, expect **two** points.

Now impose $RP^2 = (5\sqrt{2})^2 = 50$:

$$(3 - 0)^2 + (-2 - y)^2 + (5 - 0)^2 = 50 \quad \Rightarrow \quad 9 + (y + 2)^2 + 25 = 50.$$

Notice what did **not** happen: there is no second point to subtract against, so the y^2 term has nothing to cancel with. Hence

$$(y + 2)^2 = 16 \quad \Rightarrow \quad y + 2 = \pm 4 \quad \Rightarrow \quad y = 2 \text{ or } y = -6.$$

The required points are $(0, 2, 0)$ and $(0, -6, 0)$.

Check. For $(0, 2, 0)$: $\sqrt{9 + 16 + 25} = \sqrt{50} = 5\sqrt{2}$. For $(0, -6, 0)$: $\sqrt{9 + 16 + 25} = \sqrt{50} = 5\sqrt{2}$. Both work, and note that the foot of the perpendicular from P to the y -axis is $(0, -2, 0)$ — exactly midway between the two answers, as the figure promised.

Presentation note: writing only $(0, 2, 0)$ is the single commonest way to lose a mark here. The instant you see $(\cdot)^2 = 16$, write ± 4 on the same line.

Example 7.2 — [JEE Main Level]

For each of the following, find all points on the stated axis at the stated distance from P , or show that none exists.

- (i) On the x -axis, at distance $\sqrt{41}$ from $P(2, -3, 4)$.
- (ii) On the z -axis, at distance 5 from $P(3, 4, 7)$.
- (iii) On the x -axis, at distance 3 from $P(1, 2, 5)$.

Solution. In each case compute $d_{\min}$ first, then solve.

(i) Distance of P from the x -axis is $\sqrt{y^2 + z^2} = \sqrt{9 + 16} = 5$, and $\sqrt{41} > 5$, so expect two points. With $R(x, 0, 0)$:

$$(x - 2)^2 + 9 + 16 = 41 \implies (x - 2)^2 = 16 \implies x = 6 \text{ or } x = -2.$$

The points are $(6, 0, 0)$ and $(-2, 0, 0)$.

(ii) Distance of P from the z -axis is $\sqrt{x^2 + y^2} = \sqrt{9 + 16} = 5$, which is *exactly* the required distance. This is the boundary case, so expect one point. With $R(0, 0, z)$:

$$9 + 16 + (z - 7)^2 = 25 \implies (z - 7)^2 = 0 \implies z = 7 \text{ (twice)}.$$

The single point is $(0, 0, 7)$ — and it is precisely the foot of the perpendicular from P to the z -axis. The repeated root is the algebra telling you the two solutions have merged.

(iii) Distance of P from the x -axis is $\sqrt{4 + 25} = \sqrt{29} \approx 5.39$, and the required distance 3 is smaller. No point can be closer to P than the foot of the perpendicular is, so **no such point exists**. The algebra agrees: with $R(x, 0, 0)$,

$$(x - 1)^2 + 4 + 25 = 9 \implies (x - 1)^2 = -20,$$

which has no real solution. A negative right-hand side in a problem of this type is never an arithmetic slip to be “fixed” — it is the correct answer, stating that the locus is empty.

The structural lesson. Equidistant-from-two-points is linear and gives one answer; fixed-distance-from-one-point is quadratic and gives two, one, or none. Round 1’s cheat-sheet line “each condition $PA^2 = PB^2$ is linear” applies only to the *difference* of two distance conditions. A single distance condition keeps its square.

Practice

1. **[CBSE]** Find the points on the z -axis at a distance 7 from $(2, 3, -1)$.
2. **[CBSE]** Find the points on the y -axis at a distance 13 from $(3, 1, 4)$.
3. **[JEE Main]** Find the points on the z -axis at a distance 13 from $(3, 4, 7)$, and state how your answer relates to Example 7.2(ii).
4. **[JEE Main]** For what value of d does the y -axis contain exactly one point at distance d from $(6, -1, 8)$? Give that point.

Answers. 1. $4 + 9 + (z + 1)^2 = 49 \Rightarrow (z + 1)^2 = 36 \Rightarrow z = 5$ or $z = -7$; the points are $(0, 0, 5)$ and $(0, 0, -7)$. — 2. $9 + (y - 1)^2 + 16 = 169 \Rightarrow (y - 1)^2 = 144 \Rightarrow y = 13$ or $y = -11$; the points are $(0, 13, 0)$ and $(0, -11, 0)$. — 3. $9 + 16 + (z - 7)^2 = 169 \Rightarrow (z - 7)^2 = 144 \Rightarrow z = 19$ or $z = -5$; the points are $(0, 0, 19)$ and $(0, 0, -5)$. In 7.2(ii) the required distance equalled $d_{\min} = 5$ and the two roots coincided at the foot $(0, 0, 7)$; here $13 > 5$, so they have separated symmetrically, 12 units either side of that same foot. — 4. Exactly one point occurs only at $d = d_{\min} = \sqrt{z^2 + x^2} = \sqrt{64 + 36} = 10$, and the point is the foot of the perpendicular, $(0, -1, 0)$.

Pattern 8 — When a Coordinate Plane Divides a Segment Externally

Extends Subtopic 02, Section 4, Example 2 and Section 6, Q3.

Method. Subtopic 02 gave you the workhorse: to find where a coordinate plane cuts the line through $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$, write the general point in $k : 1$ form, set the one relevant coordinate to zero, and solve for k . It also told you that $k > 0$ means internal and $k < 0$ means external. What it never shows you is a case where k actually comes out **negative** — every worked instance in the chapter (Example 2, MCQ Q3, Practice Q3) happens to land on an internal division. NCERT’s own Solved Example 10 does not, and that is the case examiners set when they want to separate candidates.

Two additions to the procedure close this.

1. **Predict the sign before you compute.** Look at the two endpoints’ values of the relevant coordinate. If they have **opposite signs**, the endpoints straddle the plane, the plane cuts between them, and the division is **internal** ($k > 0$). If they have the **same sign**, both endpoints lie on the same side, the plane cannot cut the *segment* at all, and the division must be **external** ($k < 0$) — the plane meets the *extended line* outside the segment. If one endpoint’s value is zero, that endpoint already lies on the plane.
2. **Report a negative k correctly.** $k = -m/n$ means the plane divides PQ **externally in the ratio $m : n$** . Do not report the ratio as negative and do not drop the word “externally” — both cost the mark. Then, if the point itself is wanted, substitute k back, or equivalently use the external formula with that m and n .

The sign prediction is also the fastest sanity check you own: a same-sign prediction with a positive k on the page means you made an algebra slip.

Figure brief: A horizontal line carrying P and Q , with a vertical line drawn to represent the cutting plane seen edge-on. Two panels side by side. Left panel: the plane passes between P and Q , cut point labelled “internal, $k > 0$ ”, with the endpoints’ relevant coordinates marked $+$ and $-$. Right panel: both P and Q sit to the right of the plane, both marked $+$, and the line is extended leftwards with a dashed segment to reach the plane at a point labelled “external, $k < 0$ ”.

Example 8.1 — [JEE Main Level]

Find the ratio in which the YZ -plane divides the line segment joining $A(4, 8, 10)$ and $B(6, 10, -8)$, and find the point of division.

Solution. The YZ -plane is $x = 0$ (Subtopic 01).

Predict first. The x -coordinates of A and B are 4 and 6 — both positive, so both points lie on the same side of the YZ -plane. The plane cannot cross the segment, so the division must be external and k must come out negative.

Now compute. Let the plane meet line AB in the ratio $k : 1$. The x -coordinate of the general point is $\frac{6k + 4}{k + 1}$, and setting it to zero:

$$6k + 4 = 0 \implies k = -\frac{2}{3},$$

negative exactly as predicted. Since $k = -\frac{2}{3} = -\frac{m}{n}$ with $m = 2$, $n = 3$, the YZ -plane divides AB **externally in the ratio 2 : 3**.

The point of division. Use the external formula with $m = 2$, $n = 3$, denominator $m - n = -1$:

$$x = \frac{2(6) - 3(4)}{-1} = \frac{0}{-1} = 0, \quad y = \frac{2(10) - 3(8)}{-1} = \frac{-4}{-1} = 4, \quad z = \frac{2(-8) - 3(10)}{-1} = \frac{-46}{-1} = 46.$$

The point is $(0, 4, 46)$. Its x -coordinate is 0, confirming it lies on the YZ -plane, and $z = 46$ is nowhere between 10 and -8 — visible proof that the point is outside the segment, beyond A (as $m < n$ predicts, per Round 1 Pattern 2).

Contrast this with Subtopic 02, Practice Q3, where the YZ -plane divides the join of $(-2, 4, 7)$ and $(3, -5, 8)$: there the x -coordinates are -2 and 3 , opposite in sign, so the answer is internal $2 : 3$. Same plane, same numerical ratio, opposite verdict — and the only thing that decided it was whether the endpoints straddled the plane.

Example 8.2 — [JEE Main — Speed Trap]

Find the ratio in which the ZX -plane divides the join of $A(3, -4, 5)$ and $B(1, -2, 7)$.

Solution (the fast way). The ZX -plane is $y = 0$. The y -coordinates are -4 and -2 : both negative, so both endpoints sit on the same side and the answer is external before you write a single equation.

$$\frac{-2k - 4}{k + 1} = 0 \implies -2k = 4 \implies k = -2.$$

So $k = -\frac{2}{1}$: the plane divides AB **externally in the ratio 2 : 1**. Here $m = 2 > n = 1$, so the point lies beyond B ; substituting gives

$$\left(\frac{2(1) - 1(3)}{1}, \frac{2(-2) - 1(-4)}{1}, \frac{2(7) - 1(5)}{1} \right) = (-1, 0, 9),$$

whose y -coordinate is indeed 0.

The trap. The seductive wrong answer is “2 : 1 internally”. A student who solves $-2k - 4 = 0$, gets $k = -2$, and then reports the ratio as 2 : 1 without reading the sign has done all the algebra correctly and still lost the mark. A second, subtler version writes the ratio as “-2 : 1”, which is not how a ratio of division is stated. **Solve for k , read its sign, then translate:** $k > 0 \Rightarrow$ **internal** $k : 1$; $k < 0 \Rightarrow$ **external** $|k| : 1$.

Example 8.3 — [CBSE Board Level]

A point R with x -coordinate 4 lies on the line segment joining $P(2, -3, 4)$ and $Q(8, 0, 10)$. Find the coordinates of R .

Solution. This is the same machinery with the target value no longer zero, which is the only thing that makes it look unfamiliar. Let R divide PQ in the ratio $k : 1$; its x -coordinate is $\frac{8k + 2}{k + 1}$, and we are told this equals 4:

$$8k + 2 = 4(k + 1) \implies 4k = 2 \implies k = \frac{1}{2}.$$

Positive, so the division is internal in the ratio 1 : 2 — consistent with the question’s word “segment”, and with the fact that 4 lies between the endpoint x -values 2 and 8. Substituting $k = \frac{1}{2}$ into the remaining two coordinates:

$$y = \frac{\frac{1}{2}(0) + (-3)}{\frac{3}{2}} = -2, \quad z = \frac{\frac{1}{2}(10) + 4}{\frac{3}{2}} = \frac{9}{3} = 6.$$

Hence $R(4, -2, 6)$.

The generalisation worth carrying. “The plane $z = 0$ cuts the segment where?” and “which point on the segment has $x = 4$?” are the same question. Set the relevant coordinate of the $k : 1$ point equal to the required value — zero or otherwise — solve the single linear equation for k , and the sign of k tells you internal or external exactly as before.

Practice

1. **[JEE Main]** Find the ratio in which the XY -plane divides the join of $(1, 2, 3)$ and $(4, 5, 6)$, and find the point of division.
2. **[JEE Main]** Predict, without computing, whether the XY -plane divides the join of $A(2, -3, 4)$ and $B(-1, -2, 1)$ internally or externally; then find the ratio.
3. **[CBSE]** Find the point on the line joining $A(1, 2, 3)$ and $B(3, 6, 9)$ whose y -coordinate is 5.
4. **[CBSE]** State the ratio in which the YZ -plane divides the join of $(-2, 4, 7)$ and $(3, -5, 8)$, and say in one line why the answer is internal while Example 8.1 was external.

Answers. 1. The z -values 3 and 6 are both positive, so external. $\frac{6k + 3}{k + 1} = 0 \Rightarrow k = -\frac{1}{2}$: externally in the ratio 1 : 2, at the point $(-2, -1, 0)$ (beyond the first point, since $m < n$). —
2. The z -values 4 and 1 are both positive, so both endpoints lie above the XY -plane and the

division is external. $\frac{k+4}{k+1} = 0 \Rightarrow k = -4$: externally in the ratio 4 : 1. — 3. $\frac{6k+2}{k+1} = 5 \Rightarrow k = 3$, positive, so internal 3 : 1; the point is $(\frac{5}{2}, 5, \frac{15}{2})$. — 4. Internally in the ratio 2 : 3. The x -coordinates -2 and 3 have opposite signs, so the endpoints lie on opposite sides of the YZ -plane and the plane genuinely cuts the segment; in Example 8.1 both x -coordinates were positive, so it could not.

Round 2 Cheat Sheet Additions

- **Point on an axis at distance d from P :** write the axis form $(R(0, y, 0))$ and so on), impose $RP^2 = d^2$, and solve the **quadratic**. Compare d with the axis distance $d_{\min}$ from Round 1 Pattern 1: $d > d_{\min} \Rightarrow$ two points, $d = d_{\min} \Rightarrow$ one point (the foot of the perpendicular), $d < d_{\min} \Rightarrow$ none. Writing one root where there are two is the standard mark loss.
 - **Linear versus quadratic:** *equidistant from two points* is linear (the squares cancel); *at a fixed distance from one point* is quadratic (nothing cancels). Do not carry Round 1 Pattern 4's "always linear" habit into this pattern.
 - **Which side of the plane?** Before solving any "in what ratio does a coordinate plane divide PQ " question, compare the signs of the relevant coordinate at the two endpoints: **opposite signs** $\Rightarrow$ **internal**, **same sign** $\Rightarrow$ **external**; one of them zero $\Rightarrow$ that endpoint already lies on the plane.
 - **Reading k :** $k > 0 \Rightarrow$ internal, ratio $k : 1$. $k < 0 \Rightarrow$ external, ratio $|k| : 1$ — say "externally" out loud and never write a negative ratio. With $k = -m/n$: external $m : n$, landing beyond Q if $m > n$ and beyond P if $m < n$.
 - **Same question, two disguises:** "where does the plane $z = 0$ cut PQ " and "which point of PQ has $x = 4$ " are one method — set the relevant coordinate of the $k : 1$ point to the required value and solve for k .
-

Deliberately out of scope

For the record, so that a later pass does not re-open them: the ratio in which a **general plane** (intercept form $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$, or $ax + by + cz + d = 0$) divides a segment, the **angle between two lines**, the **equation of a line in space**, and the **shortest distance between skew lines** all require the equation of a plane or direction cosines and ratios. None of these appears in NCERT Class 11 Chapter 11, and each is taught in the Class 12 chapter of the same name. They are correctly absent here, and importing them would mean importing vector notation this chapter is deliberately built without. The Class 12 preview at the end of Subtopic 03 already signposts them, which is the right treatment.

Limits and Derivatives

Class 11 Mathematics · Chapter 12 — Definitive Study Guide

Drona Ed Tech · AI-First Tutoring

June 2026

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Subtopic 01 — Limits: Concepts & Algebra

Exam Relevance: This is the gateway to all of calculus, so its weight is felt everywhere. **CBSE Boards** ask a direct 2-3 mark evaluation (a $\frac{0}{0}$ factor/rationalisation problem) almost every year. **JEE Main** carries 1-2 questions that are usually clean algebra-of-limits or “find a, b ” type. In **JEE Advanced** it is rarely a standalone question but is silently assumed in continuity, differentiability and integration — get it shaky here and the whole calculus tower wobbles.

Section 1 — Concept Intuition

Imagine you are on the highway from Jaipur to Delhi, watching the milestones tick down: 200 km to Delhi, then 100, then 10, then 1, then 100 m. Even before your car physically reaches the Delhi border, you *know* with total certainty where you are heading — the milestones are *pointing* at a destination. That destination is the **limit**. It is the value a function is *heading toward* as the input gets closer and closer to some point — completely separate from whether you ever actually arrive.

This is the single most important idea in the whole chapter, so let me say it plainly: **the limit of $f(x)$ as $x \rightarrow a$ is about the neighbourhood of a , not about the point a itself.** The function might not even be defined at $x = a$ — there could be a pothole, a missing point, a divide-by-zero — and the limit can still exist perfectly well, because the milestones around it still point somewhere definite.

To make this concrete, take $f(x) = \frac{x^2 - 1}{x - 1}$. At $x = 1$ this is the forbidden $\frac{0}{0}$ — the function is genuinely *undefined* there. But watch what happens as we walk x toward 1 from both sides and tabulate the milestone readings:

x (from below)	$f(x)$	x (from above)	$f(x)$
0.9	1.9	1.1	2.1
0.99	1.99	1.01	2.01
0.999	1.999	1.001	2.001

Both columns are clearly homing in on 2, even though $f(1)$ does not exist. The milestones agree, so $\lim_{x \rightarrow 1} f(x) = 2$. (And indeed $\frac{x^2 - 1}{x - 1} = \frac{(x-1)(x+1)}{x-1} = x + 1$ for $x \neq 1$, which $\rightarrow 2$ — the table and the algebra tell the same story.)

Think about it this way... picture a narrow lane in Chandni Chowk with a single shop you want to reach. Two friends walk toward that shop — one coming from the left end of the lane, the other from the right end. If both friends end up standing at the *same* shop, then “the place you approach” is unambiguous: the limit **exists**. But suppose the lane secretly forks just before the shop, and the left-walker lands at a sweet shop while the right-walker lands at a chai stall. Now there is no single “place you were heading” — the **limit does not exist**, even though each friend individually reached *somewhere*.

Those two friends are exactly the **left-hand limit (LHL)** and the **right-hand limit (RHL)**:

- The LHL, written $\lim_{x \rightarrow a^-} f(x)$, is where $f(x)$ heads as x creeps up to a from values *smaller* than a .
- The RHL, written $\lim_{x \rightarrow a^+} f(x)$, is where $f(x)$ heads as x comes down to a from values *larger* than a .

The limit exists **only when both friends meet** — when LHL and RHL are equal and finite.

A number line with point a marked. An arrow labelled “ $x \rightarrow a^-$ (LHL)” approaches a from the left; an arrow labelled “ $x \rightarrow a^+$ (RHL)” approaches from the right. Above, two function-value curves converging to the same height L for the “limit exists” case, drawn beside a second mini-graph (a step/jump function) where the two heights differ — the “limit does not exist” case.

Limiting Conditions & Assumptions — when the idea breaks down:

1. **The value $f(a)$ is irrelevant.** $f(a)$ may be undefined, or defined but *different* from the limit. The limit ignores it.
2. **Both one-sided limits must agree and be finite.** If $\text{LHL} \neq \text{RHL} \rightarrow$ no limit (a “jump”). If either runs off to $\pm\infty \rightarrow$ no finite limit.
3. **Oscillation kills limits.** A function like $\sin(\frac{1}{x})$ near $x = 0$ keeps swinging between -1 and 1 forever, never settling — so no limit, even though it never blows up.
4. **“Approaching” is two-sided by default.** Plain $\lim_{x \rightarrow a}$ demands agreement from *both* sides; one friend reaching the shop is not enough.

Section 2 — Key Formulas & Definitions

Note (maths vs physics format): quantities here carry no dimensional formula — a limit is a pure number. This section instead locks the **algebra-of-limits rules**, the **existence test**, and the **standard algebraic limit** you will reuse constantly.

Existence test (the meeting rule). Writing the small positive increment as $h \rightarrow 0$:

$$\lim_{x \rightarrow a^-} f(x) = \lim_{h \rightarrow 0} f(a - h), \quad \lim_{x \rightarrow a^+} f(x) = \lim_{h \rightarrow 0} f(a + h)$$

$$\lim_{x \rightarrow a} f(x) = L \iff \lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = L \text{ (finite)}$$

Algebra of Limits. If $\lim_{x \rightarrow a} f(x) = L$ and $\lim_{x \rightarrow a} g(x) = M$ (both finite), then:

Rule	Statement
Sum / Difference	$\lim (f \pm g) = L \pm M$
Scalar multiple	$\lim (k f) = kL$
Product	$\lim (f \cdot g) = L \cdot M$
Quotient	$\lim \frac{f}{g} = \frac{L}{M}, \text{ provided } M \neq 0$
Power	$\lim (f)^n = L^n$

Limit of a polynomial / rational function (direct substitution). If $p(x)$ is a polynomial,

$\lim_{x \rightarrow a} p(x) = p(a)$. For a rational function $\frac{p(x)}{q(x)}$:

$$\lim_{x \rightarrow a} \frac{p(x)}{q(x)} = \frac{p(a)}{q(a)} \text{ if } q(a) \neq 0.$$

Standard algebraic limit (memorise — used everywhere):

$$\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = n a^{n-1} \quad (n \in \mathbb{Q}, a > 0 \text{ when } n \text{ non-integer})$$

Indeterminate forms. When direct substitution gives one of $\frac{0}{0}, \frac{\infty}{\infty}, \infty - \infty, 0 \times \infty$, the answer is **not** decided — it must be *resolved* by algebra. In this subtopic the form you will meet again and again is $\frac{0}{0}$.

Variable glossary: x = the approaching variable; a = the point approached; h = small positive increment ($h \rightarrow 0$); L, M = (finite) limit values; $[\cdot]$ = greatest integer function (reserved); $\{\cdot\}$ = fractional part (reserved).

Section 3 — Key Procedures & Logical Framework

Why the algebra-of-limits rules are even allowed. A limit is a *destination*. If f heads to L and g heads to M , then their sum is heading to $L + M$ — the destinations simply add, because near a the two functions are arbitrarily close to their targets, so their sum is arbitrarily close to $L + M$. The quotient rule carries its one big caveat for an obvious reason: if g is heading to 0, dividing by something vanishingly small makes the ratio explode or behave wildly, so $M \neq 0$ is non-negotiable.

Derivation of the standard limit $\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = n a^{n-1}$ (for positive integer n).

Reasoning: direct substitution gives $\frac{0}{0}$, so we must cancel the troublemaker $(x - a)$. The numerator factors cleanly because $x = a$ is a root of $x^n - a^n$.

$$x^n - a^n = (x - a)(x^{n-1} + x^{n-2}a + x^{n-3}a^2 + \dots + a^{n-1})$$

Cancel the common $(x - a)$ (legal, since for the limit $x \neq a$, only $x \rightarrow a$):

$$\frac{x^n - a^n}{x - a} = x^{n-1} + x^{n-2}a + \dots + a^{n-1}$$

Now substitute $x = a$. There are exactly n terms, each equalling a^{n-1} :

$$\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = \underbrace{a^{n-1} + a^{n-1} + \dots + a^{n-1}}_{n \text{ terms}} = n a^{n-1}.$$

The result extends to all rational n (the algebraic proof for fractional n uses substitution $x = at^k$, but the formula is what you apply).

Procedure A — resolving a $\frac{0}{0}$ rational limit by factoring.

1. **Substitute** $x = a$. If you get a clean number, you are done.
2. If you get $\frac{0}{0}$, that $\frac{0}{0}$ is a *promise* that $(x - a)$ divides both numerator and denominator.
3. **Factor** out $(x - a)$ from top and bottom.
4. **Cancel** the common $(x - a)$.
5. **Re-substitute** $x = a$ into the simplified expression.

Procedure B — resolving a surd limit by rationalisation.

1. On substitution you hit $\frac{0}{0}$ with a square-root term.
2. **Multiply and divide** by the conjugate of the surd expression.
3. Use $(\sqrt{p} - \sqrt{q})(\sqrt{p} + \sqrt{q}) = p - q$ to clear the root and expose a cancellable $(x - a)$.
4. Cancel, then substitute.

Section 4 — Worked Examples

Example 1 • [CBSE Board Level]

Evaluate $\lim_{x \rightarrow 3} \frac{x^2 - 2x - 3}{x - 3}$.

Step 1 — Substitute. At $x = 3$: numerator = $9 - 6 - 3 = 0$, denominator = 0. Form is $\frac{0}{0} \rightarrow$ resolve.

Step 2 — Factor the numerator. $x^2 - 2x - 3 = (x - 3)(x + 1)$.

Step 3 — Cancel and substitute.

$$\lim_{x \rightarrow 3} \frac{(x - 3)(x + 1)}{(x - 3)} = \lim_{x \rightarrow 3} (x + 1) = 3 + 1 = \boxed{4}.$$

Presentation note: always declare the $\frac{0}{0}$ form before factoring — boards award method marks for spotting it.

Example 2 • [JEE Main — Speed Trap]

Evaluate $\lim_{x \rightarrow 1} \frac{x^{18} - 1}{x^{12} - 1}$.

The trap: a panicking student tries to factor a degree-18 polynomial. Don't. Both top and bottom are of the form $x^n - 1^n$, so use the standard limit as a *ratio*. Divide top and bottom by $(x - 1)$:

$$\lim_{x \rightarrow 1} \frac{x^{18} - 1}{x^{12} - 1} = \frac{\lim_{x \rightarrow 1} \frac{x^{18} - 1}{x - 1}}{\lim_{x \rightarrow 1} \frac{x^{12} - 1}{x - 1}} = \frac{18 \cdot 1^{17}}{12 \cdot 1^{11}} = \frac{18}{12} = \boxed{\frac{3}{2}}.$$

10-second mental shortcut: for $\frac{x^m - 1}{x^n - 1}$ as $x \rightarrow 1$, the answer is simply $\frac{m}{n}$.

Example 3 • [JEE Main — Multi-Concept]

Evaluate $\lim_{x \rightarrow 1} \frac{x^{1/3} - 1}{x^{1/2} - 1}$. (Combines: standard $\frac{x^n - a^n}{x - a}$ limit with rational exponents + ratio technique.)

Both pieces vanish at $x = 1$ ($\frac{0}{0}$). Split by inserting $(x - 1)$:

$$\frac{x^{1/3} - 1}{x^{1/2} - 1} = \frac{\frac{x^{1/3} - 1^{1/3}}{x - 1}}{\frac{x^{1/2} - 1^{1/2}}{x - 1}}.$$

Apply $\lim_{x \rightarrow 1} \frac{x^n - 1}{x - 1} = n \cdot 1^{n-1} = n$ to each with $n = \frac{1}{3}$ and $n = \frac{1}{2}$:

$$= \frac{\frac{1}{3}}{\frac{1}{2}} = \frac{1}{3} \times \frac{2}{1} = \boxed{\frac{2}{3}}.$$

Example 4 • [JEE Advanced Level]

Find the values of a and b such that $\lim_{x \rightarrow 2} \frac{x^2 + ax + b}{x^2 - x - 2} = \frac{5}{3}$.

The deep idea: the denominator $x^2 - x - 2 = (x - 2)(x + 1)$ becomes 0 at $x = 2$. For the overall limit to be a *finite* number, the numerator **must also vanish at** $x = 2$ — otherwise we'd have (nonzero)/0 $\rightarrow \infty$. This vanishing condition is the whole game.

Condition 1 (numerator $\rightarrow 0$):

$$2^2 + 2a + b = 0 \Rightarrow 2a + b = -4. \quad (i)$$

So $(x - 2)$ divides the numerator: write $x^2 + ax + b = (x - 2)(x - c)$ for some root c .

Apply the limit:

$$\lim_{x \rightarrow 2} \frac{(x - 2)(x - c)}{(x - 2)(x + 1)} = \lim_{x \rightarrow 2} \frac{x - c}{x + 1} = \frac{2 - c}{3}.$$

Set equal to $\frac{5}{3}$: $\frac{2 - c}{3} = \frac{5}{3} \Rightarrow 2 - c = 5 \Rightarrow c = -3$.

Recover a, b : $x^2 + ax + b = (x - 2)(x + 3) = x^2 + x - 6$, so $\boxed{a = 1, b = -6}$.

Check against (i): $2(1) + (-6) = -4$ ✓.

Section 5 — Practice Exercises

1. **[CBSE]** Evaluate $\lim_{x \rightarrow 4} \frac{x^2 - 16}{x - 4}$.
2. **[CBSE]** Evaluate $\lim_{x \rightarrow 0} \frac{\sqrt{4+x} - 2}{x}$.
3. **[JEE Main]** Evaluate $\lim_{x \rightarrow 1} \frac{x^{25} - 1}{x^5 - 1}$.
4. **[JEE Main]** If $\lim_{x \rightarrow 2} \frac{x^n - 2^n}{x - 2} = 80$ and $n \in \mathbb{N}$, find n .
5. **[JEE Advanced]** Let $f(x) = \begin{cases} 3x - 1, & x < 1 \\ ax + b, & x \geq 1 \end{cases}$. Find a relation between a and b for which $\lim_{x \rightarrow 1} f(x)$ exists, and state that common limit value.

Answers: 1. 8 2. $\frac{1}{4}$ 3. 5 4. $n = 5$ 5. Limit exists when $a + b = 2$; common value = 2.

Section 6 — Crack the MCQ

Q1. $\lim_{x \rightarrow 0} \frac{|x|}{x}$ is equal to (A) 1 (B) 0 (C) -1 (D) does not exist

Answer: (D). Split by sign. For $x > 0$, $|x| = x$, so $\text{RHL} = \frac{x}{x} = 1$. For $x < 0$, $|x| = -x$, so $\text{LHL} = \frac{-x}{x} = -1$. Since $\text{LHL} (-1) \neq \text{RHL} (1)$, the limit does not exist. *Why the traps catch you:* (A) picks only the RHL; (C) picks only the LHL; (B) comes from sloppily “cancelling” $|x|/x$ to 0 or guessing symmetry. The whole point is that a modulus splits into two different one-sided behaviours.

Q2. $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x^2 - 5x + 6}$ equals (A) 0 (B) 6 (C) 1 (D) limit does not exist

Answer: (B). Form is $\frac{0}{0}$. Factor: numerator $(x - 3)(x + 3)$, denominator $(x - 2)(x - 3)$. Cancel $(x - 3)$:

$$\lim_{x \rightarrow 3} \frac{x + 3}{x - 2} = \frac{6}{1} = 6.$$

Why the traps catch you: (A) 0 comes from writing “numerator $\rightarrow 0$ ” and stopping — ignoring that the denominator also $\rightarrow 0$. (C) 1 comes from cancelling $\frac{x^2 - 9}{x^2 - 5x + 6}$ leading-term-style. (D) is the trap for students who think every $\frac{0}{0}$ means “does not exist” — but $\frac{0}{0}$ means *resolve*, not *quit*.

Q3. $\lim_{x \rightarrow 2} \frac{x^3 - 8}{x^2 - 4}$ equals (A) 2 (B) 3 (C) 4 (D) $\frac{3}{2}$

Answer: (B) 3.

Both vanish at $x = 2$. Using the ratio of standard limits about $a = 2$: numerator $\frac{x^3 - 2^3}{x - 2} \rightarrow 3 \cdot 2^2 = 12$; denominator $\frac{x^2 - 2^2}{x - 2} \rightarrow 2 \cdot 2^1 = 4$. So limit = $\frac{12}{4} = 3$. *Why the traps catch you:* (A) 2 halves $\frac{x^3}{x^2} = x \rightarrow 2$ — ignoring the constants. (C) 4 keeps only the denominator’s value. (D) $\frac{3}{2}$ uses the $\frac{m}{n}$ shortcut, which is only valid about $x \rightarrow 1$, not $x \rightarrow 2$ — a classic misapplied-shortcut trap.

Q4. The graph of f has $\lim_{x \rightarrow a^-} f(x) = 5$, $\lim_{x \rightarrow a^+} f(x) = 5$, and $f(a) = 2$. Then $\lim_{x \rightarrow a} f(x)$ is (A) 2 (B) 5 (C) does not exist (D) $\frac{5+2}{2}$

Answer: (B) 5. Both one-sided limits agree at 5, so the limit is 5. The value $f(a) = 2$ is **irrelevant** to the limit. *Why the traps catch you:* (A) 2 confuses *limit* with *function value* — the single most common conceptual error in the chapter. (C) wrongly assumes that $f(a)$ differing from the limit destroys it. (D) invents an averaging rule that does not exist.

Section 7 — Common Student Pitfalls & Pro-Tips

1. **Confusing** $\lim_{x \rightarrow a} f(x)$ **with** $f(a)$. The limit is the destination, the value is whether you arrive. They often differ — and the limit doesn't care about $f(a)$ at all.
2. **Treating** $\frac{0}{0}$ **as a final answer.** $\frac{0}{0}$ is not “zero” and not “does not exist” — it is an *indeterminate form*, an instruction to factor or rationalise.
3. **Applying the quotient rule when the denominator's limit is 0.** The rule needs $M \neq 0$. Skip this check and you'll “prove” nonsense.
4. **Misfiring the** $\frac{m}{n}$ **shortcut.** $\frac{x^m - 1}{x^n - 1} \rightarrow \frac{m}{n}$ holds only as $x \rightarrow 1$. Around any other point, fall back to the full $n a^{n-1}$ ratio.

Pro-Tip (speed + sanity): Always **substitute first**. A clean number means you're done in one line — no factoring theatrics. Only the $\frac{0}{0}$ (or other indeterminate) cases earn the heavy machinery. And whenever you see “lim exists and equals (finite)” with a vanishing denominator, immediately write “**numerator must also** $\rightarrow 0$ ” — that one reflex cracks almost every “find a, b ” problem.

Section 8 — Cheat Sheet Box

LIMITS — CONCEPTS & ALGEBRA

- **Limit = destination, not arrival.** $f(a)$ is irrelevant to $\lim_{x \rightarrow a} f(x)$.
- **Existence rule:** $\lim_{x \rightarrow a} f = L \iff \text{LHL} = \text{RHL} = L$ (finite). $\text{LHL} = \lim_{h \rightarrow 0} f(a - h)$, $\text{RHL} = \lim_{h \rightarrow 0} f(a + h)$.
- **Algebra of limits:** sum/diff/product split freely; **quotient needs denominator limit** $\neq 0$.
- **Direct substitution** works for polynomials and for rationals with $q(a) \neq 0$.
- **Star formula:** $\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = n a^{n-1}$ (all rational n).
- $\frac{0}{0}$ **toolkit:** factor & cancel $(x - a)$; for surds, multiply by the conjugate.

Memory aids:

- “Limit cares about the neighbours, not the house.” (ignore $f(a)$)
- “Zero-by-zero means do more, not done.” ($\frac{0}{0}$ = resolve)
- “ $\frac{m}{n}$ only at one.” (the shortcut $\frac{x^m - 1}{x^n - 1} \rightarrow \frac{m}{n}$ is for $x \rightarrow 1$ only)

Subtopic 02 — Standard Limits

Exam Relevance: These are the workhorse identities of all of calculus. **CBSE Boards** ask a direct 1-2 mark trigonometric limit (the $\frac{\sin x}{x}$ family) most years. **JEE Main** carries 1-2 questions every session — usually a standard limit *disguised* inside a larger expression. **JEE Advanced** rarely asks them nakedly but assumes total fluency, especially inside 1^∞ and series problems. The $\frac{\sin x}{x} = 1$ and $\frac{e^x - 1}{x} = 1$ results are the two most-reused limit facts in the entire subject.

Section 1 — Concept Intuition

In the previous subtopic, every $\frac{0}{0}$ we met could be *broken open* — there was always a hidden $(x - a)$ to factor out and cancel. But now meet a troublemaker: $\lim_{x \rightarrow 0} \frac{\sin x}{x}$. Substituting $x = 0$ gives $\frac{0}{0}$, exactly as before — but there is **nothing to factor**. $\sin x$ is not a polynomial; you cannot pull a clean x out of it by algebra. The factoring toolkit fails completely.

So mathematicians did the sensible thing: they proved a handful of these stubborn limits *once, rigorously*, and then everyone agrees to **memorise and reuse** them — exactly like multiplication tables. You don't re-derive 7×8 every time; you just *know* it is 56. These pre-proven results are the **standard limits**.

Think about it this way... picture the Indian Railways network. To get from a small town in Bihar to a small town in Kerala, you don't lay fresh track — you route your journey *through* a major junction like Mumbai Central or Itarsi. Standard limits are the junctions of calculus: whenever you face a messy trigonometric, exponential, or logarithmic $\frac{0}{0}$, your whole job is to **algebraically reshape it until it passes through one of these known junctions**, read off the value, and continue.

The flagship junction is:

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1.$$

Why should this be 1? Zoom your eyes right into the graph of $y = \sin x$ near the origin. The curve becomes indistinguishable from the straight line $y = x$ — for very small angles, $\sin x$ **and** x **are practically twins**. Their ratio therefore heads to 1.

Two overlaid curves near the origin — $y = \sin x$ and the line $y = x$ — shown at three zoom levels, with the curves becoming visually identical as the window shrinks toward $(0, 0)$, illustrating $\sin x \approx x$ for small x .

The same “twins near zero” idea powers the exponential and logarithmic junctions: for small x , $e^x \approx 1 + x$ (so $e^x - 1 \approx x$), and $\ln(1 + x) \approx x$. Hence $\frac{e^x - 1}{x} \rightarrow 1$ and $\frac{\ln(1+x)}{x} \rightarrow 1$.

Limiting Conditions & Assumptions — read these or lose marks:

- Radians only.** $\frac{\sin x}{x} \rightarrow 1$ is true *only* when x is in radians. In degrees the answer is $\frac{\pi}{180}$, not 1. (Radian measure is the unique one where arc length = angle $\times$ radius, which is *why* the twins-near-zero relationship holds.)
- The argument must go to 0.** $\frac{\sin x}{x} \rightarrow 1$ as $x \rightarrow 0$ — but $\frac{\sin x}{x}$ as $x \rightarrow \pi$ is just $\frac{0}{\pi} = 0$. The magic only happens when the angle itself shrinks to zero.
- Arguments must match.** The result is really $\frac{\sin(\text{box})}{(\text{same box})} \rightarrow 1$. So $\frac{\sin 5x}{5x} \rightarrow 1$, but $\frac{\sin 5x}{x} \rightarrow 5$. Mismatch the boxes and you mismatch the answer.
- The denominator power matters.** $\frac{1 - \cos x}{x} \rightarrow 0$, but $\frac{1 - \cos x}{x^2} \rightarrow \frac{1}{2}$. Same numerator, totally different destinations.

Section 2 — Key Formulas & Definitions

Note (maths format): standard limits are pure numbers — no dimensional formulas. This section *is* the deliverable: the junction table you must commit to memory. Every argument below (x , kx , etc.) is assumed to be in **radians** and to tend to 0.

Trigonometric standard limits

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1, \quad \lim_{x \rightarrow 0} \frac{\tan x}{x} = 1, \quad \lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2},$$

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x} = 0, \quad \lim_{x \rightarrow 0} \frac{\sin^{-1} x}{x} = 1, \quad \lim_{x \rightarrow 0} \frac{\tan^{-1} x}{x} = 1.$$

Exponential & logarithmic standard limits

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1, \quad \lim_{x \rightarrow 0} \frac{a^x - 1}{x} = \ln a \quad (a > 0), \quad \lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} = 1,$$

$$\lim_{x \rightarrow 0} \frac{\log_a(1+x)}{x} = \frac{1}{\ln a}.$$

The number e (the 1^∞ junctions)

$$\lim_{x \rightarrow 0} (1+x)^{1/x} = e, \quad \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e.$$

The 1^∞ shortcut rule. If $f(x) \rightarrow 1$ and $g(x) \rightarrow \infty$ as $x \rightarrow a$, then

$$\lim_{x \rightarrow a} [f(x)]^{g(x)} = e^{\lim_{x \rightarrow a} g(x) [f(x) - 1]}.$$

Argument-matching generalisation. For a constant k :

$$\lim_{x \rightarrow 0} \frac{\sin(kx)}{x} = k, \quad \lim_{x \rightarrow 0} \frac{1 - \cos(kx)}{x^2} = \frac{k^2}{2}, \quad \lim_{x \rightarrow 0} \frac{e^{kx} - 1}{x} = k.$$

Chained from 12_01 (reused, not re-derived):

$$\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = n a^{n-1} \quad (n \in \mathbb{Q}).$$

Variable glossary: x, kx = arguments (radians, $\rightarrow 0$); k = constant multiplier; a = base of exponential/log; $e \approx 2.718$; $\ln \equiv \log_e$.

Section 3 — Key Procedures & Logical Framework

The Sandwich (Squeeze) Theorem — the engine behind $\frac{\sin x}{x} = 1$. If $g(x) \leq f(x) \leq h(x)$ near a , and g and h both head to the *same* limit L , then f is trapped and forced to L as well. A function squeezed between two slices of bread that meet has nowhere to go but the meeting point.

Geometric proof that $\lim_{x \rightarrow 0^+} \frac{\sin x}{x} = 1$.

Setup: take a unit circle, centre O , and a small angle x (radians), $0 < x < \frac{\pi}{2}$. Mark $A = (1, 0)$ on the circle, P on the circle at angle x , and T where the tangent at A meets line OP extended.

Unit circle with centre O; point A=(1,0); point P on the arc at angle x; the perpendicular from P to OA of height sin x; the sector OAP; and the right triangle OAT with vertical side AT = tan x. Three nested regions visible: triangle OAP inside sector OAP inside triangle OAT.

Compare three areas:

$$\underbrace{\text{Area } \triangle OAP}_{=\frac{1}{2} \sin x} \leq \underbrace{\text{Area sector } OAP}_{=\frac{1}{2} x} \leq \underbrace{\text{Area } \triangle OAT}_{=\frac{1}{2} \tan x}.$$

(The middle one is $\frac{1}{2}r^2x = \frac{1}{2}x$ only because x is in radians — this is the precise spot where radian measure is essential.)

Multiply through by 2 and divide by $\sin x > 0$:

$$1 \leq \frac{x}{\sin x} \leq \frac{1}{\cos x} \implies \cos x \leq \frac{\sin x}{x} \leq 1.$$

As $x \rightarrow 0^+$, $\cos x \rightarrow 1$. The ratio is sandwiched between something $\rightarrow 1$ and 1 itself, so by the Squeeze Theorem $\frac{\sin x}{x} \rightarrow 1$. By symmetry ($\frac{\sin(-x)}{-x} = \frac{\sin x}{x}$) the left-hand limit matches, so the two-sided limit is 1. ■

Derivation of $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}$. Use the half-angle identity $1 - \cos x = 2 \sin^2 \frac{x}{2}$:

$$\frac{1 - \cos x}{x^2} = \frac{2 \sin^2 \frac{x}{2}}{x^2} = \frac{1}{2} \left(\frac{\sin \frac{x}{2}}{\frac{x}{2}} \right)^2 \xrightarrow{x \rightarrow 0} \frac{1}{2} (1)^2 = \frac{1}{2}.$$

Why $\frac{e^x - 1}{x} \rightarrow 1$. From the series $e^x = 1 + x + \frac{x^2}{2!} + \dots$, we get $e^x - 1 = x(1 + \frac{x}{2!} + \dots)$, so $\frac{e^x - 1}{x} = 1 + \frac{x}{2!} + \dots \rightarrow 1$. The same expansion gives $\frac{a^x - 1}{x} \rightarrow \ln a$ (since $a^x = e^{x \ln a}$).

Why the 1^∞ rule works. Write $f^g = e^{g \ln f}$. Near a 1^∞ form, $f - 1$ is small, and $\ln f = \ln(1 + (f - 1)) \approx (f - 1)$. So $g \ln f \approx g(f - 1)$, giving $\lim f^g = e^{\lim g(f-1)}$.

Section 4 — Worked Examples

Example 1 • [CBSE Board Level]

Evaluate $\lim_{x \rightarrow 0} \frac{\sin 5x}{\sin 3x}$.

The plan: reshape both pieces into the canonical $\frac{\sin(\text{box})}{\text{box}}$ form by inserting the matching arguments.

$$\frac{\sin 5x}{\sin 3x} = \frac{\sin 5x}{5x} \cdot \frac{3x}{\sin 3x} \cdot \frac{5x}{3x}.$$

As $x \rightarrow 0$: the first factor $\rightarrow 1$, the second $\rightarrow 1$, and the third is the constant $\frac{5}{3}$. Therefore

$$\lim_{x \rightarrow 0} \frac{\sin 5x}{\sin 3x} = 1 \cdot 1 \cdot \frac{5}{3} = \boxed{\frac{5}{3}}.$$

Example 2 • [JEE Main — Speed Trap]

Evaluate $\lim_{x \rightarrow 0} \frac{1 - \cos 4x}{x^2}$.

The trap: students apply $\frac{1 - \cos x}{x^2} \rightarrow \frac{1}{2}$ and answer $\frac{1}{2}$, forgetting the argument is $4x$, not x . The correct generalised junction is $\frac{1 - \cos(kx)}{x^2} \rightarrow \frac{k^2}{2}$.

$$\lim_{x \rightarrow 0} \frac{1 - \cos 4x}{x^2} = \frac{4^2}{2} = \frac{16}{2} = \boxed{8}.$$

Mental shortcut: for $\frac{1-\cos(kx)}{x^2}$, just write $\frac{k^2}{2}$ — and square the k .

Example 3 • [JEE Main — Multi-Concept]

Evaluate $\lim_{x \rightarrow 0} \frac{e^{2x} - 1}{\sin 3x}$. (Combines an exponential junction with a trigonometric one.)

Route the numerator through the e -junction and the denominator through the $\sin$ -junction:

$$\frac{e^{2x} - 1}{\sin 3x} = \underbrace{\frac{e^{2x} - 1}{2x}}_{\rightarrow 1} \cdot \underbrace{\frac{3x}{\sin 3x}}_{\rightarrow 1} \cdot \frac{2x}{3x}.$$

The last factor is the constant $\frac{2}{3}$. Hence

$$\lim_{x \rightarrow 0} \frac{e^{2x} - 1}{\sin 3x} = 1 \cdot 1 \cdot \frac{2}{3} = \boxed{\frac{2}{3}}.$$

Example 4 • [JEE Advanced Level]

Evaluate $\lim_{x \rightarrow 0} (\cos x)^{1/x^2}$.

Identify the form. As $x \rightarrow 0$, $\cos x \rightarrow 1$ and $\frac{1}{x^2} \rightarrow \infty$ — this is the indeterminate form 1^∞ . Reach straight for the rule $\lim f^g = e^{\lim g(f-1)}$ with $f = \cos x$, $g = \frac{1}{x^2}$:

$$\lim_{x \rightarrow 0} g(f-1) = \lim_{x \rightarrow 0} \frac{\cos x - 1}{x^2} = -\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = -\frac{1}{2}.$$

Therefore

$$\lim_{x \rightarrow 0} (\cos x)^{1/x^2} = e^{-1/2} = \boxed{\frac{1}{\sqrt{e}}}.$$

Deep point: the 1^∞ rule converts a frightening exponent problem into a familiar $\frac{1-\cos x}{x^2}$ junction in one move.

Section 5 — Practice Exercises

1. **[CBSE]** Evaluate $\lim_{x \rightarrow 0} \frac{\tan 7x}{x}$.
2. **[CBSE]** Evaluate $\lim_{x \rightarrow 0} \frac{\sin ax}{\sin bx}$, ($b \neq 0$).
3. **[JEE Main]** Evaluate $\lim_{x \rightarrow 0} \frac{a^x - 1}{b^x - 1}$, ($a, b > 0$, $b \neq 1$).
4. **[JEE Main]** Evaluate $\lim_{x \rightarrow 0} \frac{1 - \cos 6x}{1 - \cos 3x}$.
5. **[JEE Advanced]** Evaluate $\lim_{x \rightarrow 0} (1 + \sin x)^{\cot x}$.

Answers: 1. 7 2. $\frac{a}{b}$ 3. $\frac{\ln a}{\ln b}$ 4. 4 5. e

Section 6 — Crack the MCQ

Q1. $\lim_{x \rightarrow 0} \frac{\sin x^\circ}{x}$ (where x° means x **degrees**) equals (A) 1 (B) $\frac{\pi}{180}$ (C) $\frac{180}{\pi}$ (D) π

Answer: (B). Convert the degrees to radians: $x^\circ = \frac{\pi x}{180}$ radians. Then

$$\frac{\sin x^\circ}{x} = \frac{\sin\left(\frac{\pi x}{180}\right)}{x} = \frac{\pi}{180} \cdot \frac{\sin\left(\frac{\pi x}{180}\right)}{\frac{\pi x}{180}} \rightarrow \frac{\pi}{180} \cdot 1 = \frac{\pi}{180}.$$

Why the traps catch you: **(A)** 1 is the single most common error in the whole chapter — applying $\frac{\sin x}{x} = 1$ without noticing the degree symbol. **(C)** inverts the conversion factor. **(D)** half-remembers that “ π is involved.”

Q2. $\lim_{x \rightarrow 0} \frac{\tan^{-1}(3x)}{\sin(2x)}$ equals (A) $\frac{2}{3}$ (B) $\frac{3}{2}$ (C) 1 (D) 6

Answer: (B). Both $\frac{\tan^{-1}(3x)}{3x} \rightarrow 1$ and $\frac{\sin 2x}{2x} \rightarrow 1$, so

$$\frac{\tan^{-1}(3x)}{\sin 2x} = \frac{\tan^{-1}(3x)}{3x} \cdot \frac{2x}{\sin 2x} \cdot \frac{3x}{2x} \rightarrow 1 \cdot 1 \cdot \frac{3}{2} = \frac{3}{2}.$$

Why the traps catch you: **(A)** flips the ratio (puts $\frac{2}{3}$ instead of $\frac{3}{2}$). **(C)** treats both junctions as “1” and forgets the leftover constant. **(D)** multiplies 3×2 instead of dividing.

Q3. $\lim_{x \rightarrow \infty} \left(1 + \frac{3}{x}\right)^x$ equals (A) e (B) e^3 (C) 3 (D) 1

Answer: (B). Form 1^∞ . Using the rule with $f = 1 + \frac{3}{x} \rightarrow 1$, $g = x \rightarrow \infty$:

$$\lim x \left(1 + \frac{3}{x} - 1\right) = \lim x \cdot \frac{3}{x} = 3 \Rightarrow \text{answer} = e^3.$$

Why the traps catch you: **(A)** blindly recalls “ $\left(1 + \frac{1}{x}\right)^x \rightarrow e$ ” and ignores the 3. **(C)** reads off the constant inside. **(D)** wrongly treats 1^∞ as just 1 — the classic 1^∞ misconception.

Q4. $\lim_{x \rightarrow 0} \frac{2^x - 1}{x}$ equals (A) 1 (B) 2 (C) $\ln 2$ (D) 0

Answer: (C). This is the junction $\frac{a^x - 1}{x} \rightarrow \ln a$ with $a = 2$, giving $\ln 2$. *Why the traps catch you:* **(A)** confuses the base- a result with the base- e result $\frac{e^x - 1}{x} \rightarrow 1$. **(B)** reads off the base. **(D)** stops at the bare $\frac{0}{0}$ without resolving.

Section 7 — Common Student Pitfalls & Pro-Tips

- Degrees instead of radians.** $\frac{\sin x}{x} = 1$ holds *only* in radians. A stray degree symbol changes the answer to $\frac{\pi}{180}$. Always check the angle’s units first.
- Argument mismatch.** $\frac{\sin(kx)}{x} = k$, not 1. The standard limit needs the *same* expression on top inside the function and on the bottom — match the boxes.
- Wrong denominator power for $1 - \cos$.** $\frac{1 - \cos x}{x} \rightarrow 0$ but $\frac{1 - \cos x}{x^2} \rightarrow \frac{1}{2}$. Note the power before you quote a value, and remember $\frac{1 - \cos(kx)}{x^2} \rightarrow \frac{k^2}{2}$ (square the k).
- Base confusion in exponentials.** $\frac{e^x - 1}{x} \rightarrow 1$, but $\frac{a^x - 1}{x} \rightarrow \ln a$. Only base e gives a clean 1.

Pro-Tip (speed + accuracy): make every problem *pass through a junction*. Multiply and divide to force each trig/exp/log piece into its canonical $\frac{\sin(\text{box})}{\text{box}}$ (or $\frac{e^{\text{box}} - 1}{\text{box}}$) shape, collect the leftover constants, and read off. And the instant you spot a power tending to infinity with a base tending to 1, don’t panic — write $e^{\lim g(f-1)}$ immediately and you’ve turned a 1^∞ monster into an ordinary limit.

STANDARD LIMITS (*all arguments in radians, $\rightarrow 0$*)

- **Trig:** $\frac{\sin x}{x} \rightarrow 1$, $\frac{\tan x}{x} \rightarrow 1$, $\frac{\sin^{-1} x}{x} \rightarrow 1$, $\frac{\tan^{-1} x}{x} \rightarrow 1$
- **Half-angle:** $\frac{1 - \cos x}{x^2} \rightarrow \frac{1}{2}$ (general: $\frac{1 - \cos kx}{x^2} \rightarrow \frac{k^2}{2}$)
- **Exp/Log:** $\frac{e^x - 1}{x} \rightarrow 1$, $\frac{a^x - 1}{x} \rightarrow \ln a$, $\frac{\ln(1 + x)}{x} \rightarrow 1$
- **The e -limit:** $(1 + x)^{1/x} \rightarrow e$, $(1 + \frac{1}{n})^n \rightarrow e$
- **1^∞ rule:** $\lim f^g = e^{\lim g(f-1)}$ when $f \rightarrow 1$, $g \rightarrow \infty$
- **Argument-match:** $\frac{\sin kx}{x} \rightarrow k$, $\frac{e^{kx} - 1}{x} \rightarrow k$

Memory aids:

- “Sine and angle are twins near zero — but only in radians.”
- “Match the box, keep the constant.” (argument-matching)
- “One-to-the-infinity? Lift to e to the $g(f-1)$.”
- “ e gives one, any other base gives $\ln$.”

Subtopic 03 — Derivatives and First Principle

Exam Relevance: “Find the derivative of f **from first principle**” is a near-guaranteed **CBSE Boards** question (2–3 marks) every year. **JEE Main** tests the limit-definition both directly and disguised inside difference-quotient limits. **JEE Advanced** leans on it for differentiability-from-definition and functional-equation problems where the standard rules don’t apply.

Section 1 — Concept Intuition

Here is a genuinely deep question: how fast is your car going *right now* — at this exact instant? Average speed is easy: total distance over total time for the Jaipur-Delhi drive. But “right now” is a single instant, and in a single instant *no time passes* — so it looks like we’d be computing $\frac{\text{change}}{0}$, which is nonsense. Yet your speedometer displays an answer confidently. How?

The speedometer cheats cleverly: it measures the distance covered over a *tiny* sliver of time and divides — then makes the sliver smaller and smaller. As the time-window shrinks toward zero, that average-over-a-sliver settles onto a single definite number: your **instantaneous** speed. **That limiting process — average rate of change, with the interval squeezed to zero — is exactly the derivative.**

A position-time curve $s(t)$. A secant line through two points $P = (t, s(t))$ and $Q = (t+h, s(t+h))$, with the second point shown sliding left toward P at three stages, the secant rotating until it becomes the tangent line at P . Label the secant slope as $\frac{s(t+h)-s(t)}{h}$ and the final tangent slope as $s'(t)$.

Think about it this way... the same idea is purely geometric. Pick two points on a curve and draw the straight line (a *secant*) joining them — its slope is the average rate of change between them. Now slide the second point toward the first. The secant pivots, and in the limit it becomes the **tangent** line that just grazes the curve at that single point. The slope of that tangent is the derivative. Equivalently: zoom into any smooth curve far enough and it looks like a straight line — the derivative is the slope of that local straightening.

Formalising both pictures gives the **first principle** (also called the *ab-initio* or *delta* method):

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}.$$

The numerator is the change in output; the h in the denominator is the shrinking sliver of input. The whole expression is a $\frac{0}{0}$ limit — which is why everything you learned about limits in 12_01 and 12_02 is the machinery that makes derivatives work.

Limiting Conditions & Assumptions:

1. **The limit must exist.** The derivative exists at $x = a$ only if the difference-quotient limit exists — i.e. its left-hand and right-hand versions (LHD and RHD) agree. This is the *existence rule* from 12_01, applied to the difference quotient.
2. **Smoothness required.** A sharp corner (like $|x|$ at $x = 0$), a jump, or a vertical tangent all kill the derivative even when the function is perfectly well-defined there.
3. **Differentiable $\Rightarrow$ continuous, but not the reverse.** If f is differentiable at a it must be continuous there — yet continuity alone is not enough ($|x|$ is continuous everywhere but not differentiable at 0).
4. **$f'(a)$ is a number; $f'(x)$ is a function.** Evaluate the derivative at every point and you get a brand-new function.

Section 2 — Key Formulas & Definitions

Note (maths format): as a pure mathematical object a derivative is a number/function with no dimensional formula; units appear only when the variables carry physical meaning (e.g. $\frac{ds}{dt}$ in m/s). This section locks the definition, the standard derivative results, and the algebra rules.

Definition (two equivalent forms).

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \quad \text{and} \quad f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}.$$

Notations (all mean the same thing): $f'(x)$, $\frac{dy}{dx}$, $\frac{d}{dx}[f(x)]$, Df .

Interpretations. Geometric: $f'(a)$ = slope of the tangent to $y = f(x)$ at $x = a = \tan \psi$. Physical: if $s(t)$ is position, $s'(t)$ is the instantaneous velocity.

Standard derivatives (each provable from first principle):

$f(x)$	$f'(x)$	$f(x)$	$f'(x)$
c (constant)	0	e^x	e^x
x^n	$n x^{n-1}$	a^x	$a^x \ln a$
$\sin x$	$\cos x$	$\ln x$	$\frac{1}{x}$
$\cos x$	$-\sin x$	$\tan x$	$\sec^2 x$

Algebra of derivatives. Derivatives combine across $+$, $-$, $\times$, $\div$ via the sum, product, and quotient rules — these are the subject of **12_04 (Rules of Differentiation)**. The product rule is *derived from first principle* in Section 3 below (since that is genuinely first-principles work); its fluent application as a computational tool lives in 12_04.

Chained tools (from earlier subtopics, used in the derivations below):

$$\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = n a^{n-1} \quad (12_01), \quad \lim_{h \rightarrow 0} \frac{\sin h}{h} = 1, \quad \lim_{h \rightarrow 0} \frac{e^h - 1}{h} = 1 \quad (12_02).$$

Section 3 — Key Procedures & Logical Framework

From secant to tangent (the geometric derivation). Through $P = (x, f(x))$ and $Q = (x+h, f(x+h))$, the secant slope is the difference quotient $\frac{f(x+h)-f(x)}{h} = \tan(\angle QPR)$. As $h \rightarrow 0$, Q slides into P , the chord PQ becomes the tangent at P , and the slope limit is $f'(x)$. This is why “derivative = slope of tangent.”

The first-principle procedure (use this on any CBSE problem):

1. Write $f(x+h)$ by substituting $x+h$ into the function.
2. Form the difference quotient $\frac{f(x+h) - f(x)}{h}$.
3. Simplify the numerator so the h in the denominator can be dealt with (expand, factor, or rationalise).
4. Take $\lim_{h \rightarrow 0}$ — routing through a standard limit where needed.

Derivative of x^n — straight from 12_01. Using the *second* form of the definition with $f(x) = x^n$:

$$f'(x) = \lim_{t \rightarrow x} \frac{t^n - x^n}{t - x} = n x^{n-1}.$$

The derivative of x^n is *literally* the star limit from 12_01 — no new work needed.

Derivative of $\sin x$ — uses the 12_02 trig junction. With the identity $\sin(x + h) - \sin x = 2 \cos(x + \frac{h}{2}) \sin \frac{h}{2}$:

$$f'(x) = \lim_{h \rightarrow 0} \frac{2 \cos(x + \frac{h}{2}) \sin \frac{h}{2}}{h} = \lim_{h \rightarrow 0} \cos(x + \frac{h}{2}) \cdot \underbrace{\frac{\sin \frac{h}{2}}{\frac{h}{2}}}_{\rightarrow 1} = \cos x.$$

Derivative of e^x — uses the 12_02 exponential junction.

$$f'(x) = \lim_{h \rightarrow 0} \frac{e^{x+h} - e^x}{h} = e^x \lim_{h \rightarrow 0} \frac{e^h - 1}{h} = e^x \cdot 1 = e^x.$$

This is the famous self-replicating property: e^x is its own derivative.

Why the product rule is $u'v + uv'$ and not $u'v'$. From first principle, add-and-subtract the cross term:

$$(uv)' = \lim_{h \rightarrow 0} \frac{u(x+h)v(x+h) - u(x)v(x)}{h}.$$

Insert $-u(x+h)v(x) + u(x+h)v(x)$ in the numerator and split:

$$= \lim_{h \rightarrow 0} \left[u(x+h) \frac{v(x+h) - v(x)}{h} + v(x) \frac{u(x+h) - u(x)}{h} \right] = uv' + vu'.$$

The cross term is exactly why two derivatives can't simply multiply.

Section 4 — Worked Examples

Example 1 • [CBSE Board Level]

Find the derivative of $f(x) = \sqrt{x}$ from first principle.

Step 1-2 (difference quotient): $\frac{\sqrt{x+h} - \sqrt{x}}{h}$ — substituting $h = 0$ gives $\frac{0}{0}$.

Step 3 (rationalise — the 12_01 surd technique): multiply by the conjugate:

$$\frac{\sqrt{x+h} - \sqrt{x}}{h} \cdot \frac{\sqrt{x+h} + \sqrt{x}}{\sqrt{x+h} + \sqrt{x}} = \frac{(x+h) - x}{h(\sqrt{x+h} + \sqrt{x})} = \frac{1}{\sqrt{x+h} + \sqrt{x}}.$$

Step 4 (limit):

$$f'(x) = \lim_{h \rightarrow 0} \frac{1}{\sqrt{x+h} + \sqrt{x}} = \frac{1}{2\sqrt{x}}.$$

$$\boxed{\frac{d}{dx} \sqrt{x} = \frac{1}{2\sqrt{x}}} \quad (\text{consistent with } x^{1/2} \Rightarrow \frac{1}{2}x^{-1/2}).$$

Example 2 • [JEE Main — Speed Trap]

Evaluate $\lim_{h \rightarrow 0} \frac{(2+h)^{10} - 2^{10}}{h}$.

The trap: a student starts binomially expanding $(2+h)^{10}$ — minutes lost. **Recognise the disguise:** this is precisely the first-principle definition of $f'(2)$ for $f(x) = x^{10}$. So

$$\lim_{h \rightarrow 0} \frac{(2+h)^{10} - 2^{10}}{h} = f'(2) = 10 \cdot 2^9 = 10 \cdot 512 = \boxed{5120}.$$

Reflex: any limit shaped like $\frac{f(a+h) - f(a)}{h}$ is a derivative — read off $f'(a)$ instead of grinding.

Example 3 • [JEE Main — Multi-Concept]

Find the derivative of $f(x) = a^x$ ($a > 0$) from first principle. (Combines first principle with the 12_02 exponential junction $\frac{a^h - 1}{h} \rightarrow \ln a$.)

Form the difference quotient and factor out the base function:

$$f'(x) = \lim_{h \rightarrow 0} \frac{a^{x+h} - a^x}{h} = \lim_{h \rightarrow 0} a^x \cdot \frac{a^h - 1}{h} = a^x \lim_{h \rightarrow 0} \frac{a^h - 1}{h}.$$

The remaining limit is exactly the standard exponential limit from 12_02, equal to $\ln a$. Therefore

$$\boxed{\frac{d}{dx} a^x = a^x \ln a.}$$

(Setting $a = e$ recovers $\frac{d}{dx} e^x = e^x$, since $\ln e = 1$ — a clean self-consistency check.)

Example 4 • [JEE Advanced Level]

Let $f(x) = \begin{cases} x^2 \sin \frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$. Show that f is differentiable at $x = 0$ and find $f'(0)$.

The subtlety: $\sin \frac{1}{x}$ oscillates wildly near 0, so you cannot just “plug in.” Go to the definition at the point:

$$f'(0) = \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{h^2 \sin \frac{1}{h} - 0}{h} = \lim_{h \rightarrow 0} h \sin \frac{1}{h}.$$

Now **squeeze** (the 12_02 Sandwich Theorem): since $|\sin \frac{1}{h}| \leq 1$,

$$-|h| \leq h \sin \frac{1}{h} \leq |h|.$$

Both bounds $\rightarrow 0$ as $h \rightarrow 0$, so the trapped quantity $\rightarrow 0$. Therefore

$$\boxed{f'(0) = 0.}$$

Deep point: f is differentiable at 0, yet f' is **not** continuous at 0 (away from 0, $f'(x) = 2x \sin \frac{1}{x} - \cos \frac{1}{x}$, which has no limit as $x \rightarrow 0$). Differentiable does not imply *continuously* differentiable — a favourite Advanced trap.

Section 5 — Practice Exercises

- [CBSE]** Find the derivative of $f(x) = x^2 + 3x$ from first principle.
- [CBSE]** Find the derivative of $f(x) = \cos x$ from first principle.
- [JEE Main]** Evaluate $\lim_{h \rightarrow 0} \frac{e^{3+h} - e^3}{h}$.
- [JEE Main]** Find $f'(x)$ if $f(x) = x^2 e^x$.
- [JEE Advanced]** Is $f(x) = |x - 2|$ differentiable at $x = 2$? Justify using LHD and RHD.

Answers: 1. $2x + 3$ 2. $-\sin x$ 3. e^3 4. $e^x(x^2 + 2x) = x e^x(x + 2)$ 5. Not differentiable; LHD = -1 , RHD = $+1$, so $f'(2)$ does not exist (corner point).

Section 6 — Crack the MCQ

Q1. The derivative of $f(x) = \sqrt{x}$ at $x = 4$ is (A) $\frac{1}{4}$ (B) $\frac{1}{2}$ (C) 4 (D) 2

Answer: (A). Since $f'(x) = \frac{1}{2\sqrt{x}}$, at $x = 4$ we get $\frac{1}{2 \cdot 2} = \frac{1}{4}$. *Why the traps catch you:* **(B)** drops the $\sqrt{x}$ and just halves. **(C)** returns $f(4)$ -style thinking or forgets to differentiate. **(D)** gives $\sqrt{4}$, the function value, not the slope.

Q2. If $f(x) = \sin x$, then $\lim_{h \rightarrow 0} \frac{f(\frac{\pi}{6} + h) - f(\frac{\pi}{6})}{h}$ equals (A) $\frac{1}{2}$ (B) $\frac{\sqrt{3}}{2}$ (C) $-\frac{\sqrt{3}}{2}$ (D) 1

Answer: (B). This limit is $f'(\frac{\pi}{6}) = \cos \frac{\pi}{6} = \frac{\sqrt{3}}{2}$. *Why the traps catch you:* **(A)** computes $\sin \frac{\pi}{6}$ — the function value, not its derivative. **(C)** uses $-\sin x$ (the derivative of $\cos$, wrong function). **(D)** guesses the “standard limit = 1” reflex without identifying the derivative.

Q3. If a function f is differentiable at $x = a$, then at $x = a$ the function f is necessarily (A) continuous (B) constant (C) increasing (D) only bounded, not continuous

Answer: (A). Differentiability at a point forces continuity there (the converse fails — $|x|$ at 0). *Why the traps catch you:* **(B)** confuses “has a derivative” with “derivative is zero.” **(C)** assumes a positive slope. **(D)** inverts the true implication.

Q4. $\frac{d}{dx}(\tan x)$ equals (A) $\sec^2 x$ (B) $-\csc^2 x$ (C) $\sec x \tan x$ (D) $\cot x$

Answer: (A). From the quotient rule, $\frac{d}{dx} \tan x = \sec^2 x$. *Why the traps catch you:* **(B)** is the derivative of $\cot x$. **(C)** is the derivative of $\sec x$. **(D)** confuses the derivative with the cofunction.

Section 7 — Common Student Pitfalls & Pro-Tips

- Mixing up $f'(a)$ and $f'(x)$.** $f'(a)$ is a single number (a slope at one point); $f'(x)$ is a whole new function. Boards penalise leaving an x where a number belongs.
- The wrong product rule.** $(uv)' \neq u'v'$. It is $u'v + uv'$ — the cross term is non-negotiable.
- Cancelling h too early / skipping the limit.** You may only cancel h *after* simplifying the numerator, and you must still write the $\lim_{h \rightarrow 0}$ step — dropping it loses method marks and hides errors.
- Assuming continuous $\Rightarrow$ differentiable.** It's one-way. Corners ($|x|$), cusps, and vertical tangents are continuous but not differentiable.

Pro-Tip (speed + accuracy): train your eye to spot a **disguised derivative** — any limit of the form $\frac{f(a+h)-f(a)}{h}$ or $\frac{f(x)-f(a)}{x-a}$ is just $f'(a)$; never expand it by hand. And in every first-principle derivation of a $\sin$, $\cos$, or e^x , your single goal is to *manufacture a standard limit* ($\frac{\sin \theta}{\theta}$ or $\frac{e^\theta - 1}{\theta}$) — factor out the base function and the junction appears.

Section 8 — Cheat Sheet Box

DERIVATIVES & FIRST PRINCIPLE

- Definition:** $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{t \rightarrow x} \frac{f(t) - f(x)}{t - x}$
- Means:** slope of the tangent ($= \tan \psi$) = instantaneous rate of change.
- Standard set:** $\frac{d}{dx} x^n = nx^{n-1}$; $\frac{d}{dx} \sin x = \cos x$; $\frac{d}{dx} \cos x = -\sin x$; $\frac{d}{dx} e^x = e^x$; $\frac{d}{dx} \ln x = \frac{1}{x}$; $\frac{d}{dx} \tan x = \sec^2 x$
- Product rule (derived here from first principle):** $(uv)' = u'v + uv'$. Full rule set (quotient, chain) $\rightarrow$ **12_04**.
- Existence:** $f'(a)$ exists $\iff$ LHD = RHD. **Differentiable $\Rightarrow$ continuous** (not conversely).

Memory aids:

- “*Secant slides into tangent.*” (the geometric meaning)
- “*Product rule: first-d-second plus second-d-first.*” ($u'v + uv'$)
- “ *$\frac{f(a+h)-f(a)}{h}$ is a derivative in disguise.*”
- “ *e^x is its own echo.*” (derivative of e^x is e^x)

Subtopic 04 — Rules of Differentiation

Exam Relevance: This is the computational engine of calculus — once the rules are in hand, almost nothing needs first principles again. **CBSE Boards** test products and quotients directly (3–4 marks). **JEE Main** uses these rules in essentially every calculus question. **JEE Advanced** assumes total fluency, especially the chain rule. (*Note: the chain rule formally sits in the Class 12 syllabus, but it is standard in Class 11 JEE coaching and is included here as a bridge.*)

Section 1 — Concept Intuition

In 12_03 you ground out derivatives the honest way — straight from the limit definition. It works, but it is slow, like grinding wheat into *atta* by hand. Nobody does that daily. Once a few key derivatives and combination rules are proven from first principle (which we did), we switch to the **flour mill**: a small set of rules that produce the same answers in a fraction of the time. This subtopic is that mill.

Every rule answers one question: *how does the derivative interact with an arithmetic operation?*

- **Addition** is friendly — derivatives just add: $(u + v)' = u' + v'$.
- **Multiplication** is the first surprise — derivatives do **not** simply multiply. The correct law is the **product rule** $(uv)' = u'v + uv'$, and the reason (from 12_03) is the leftover cross-term.
- **Division** has its own **quotient rule**.
- **Composition** — a function *inside* another function — is the subtle and most powerful one: the **chain rule**.

The chain rule, intuitively: rates multiply. Suppose y changes 3 times as fast as u , and u changes 2 times as fast as x . Then how fast does y change with x ? Six times. The rates *cascade by multiplication*.

Think about it this way... picture a set of gears. A large gear turns a medium gear, which turns a small wheel. If the medium gear spins twice for each turn of the large one, and the wheel spins three times per medium turn, the wheel spins six times per large turn — you multiply the gear ratios. Or think of currency: $\frac{\text{rupees}}{\text{dollar}} \times \frac{\text{dollars}}{\text{euro}} = \frac{\text{rupees}}{\text{euro}}$. The intermediate unit cancels, and the rates chain. That is exactly

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}.$$

A three-gear train labelled $x \rightarrow (\text{gear } g) \rightarrow u \rightarrow (\text{gear } f) \rightarrow y$, with rotation-rate arrows showing du/dx on the first mesh and dy/du on the second, and the combined $dy/dx = dy/du \cdot du/dx$ across the whole train. The intermediate variable u shown cancelling like a unit.

Limiting Conditions & Assumptions:

1. **Quotient rule needs** $v \neq 0$ at the point — you cannot divide by a vanishing denominator.
2. **Chain rule needs both links differentiable** — the inner function g at x , and the outer function f at $u = g(x)$.
3. **Don't reach for a rule when algebra is faster.** If an expression simplifies first (e.g. $\frac{x^2+1}{x} = x + \frac{1}{x}$), simplify, *then* differentiate.
4. **Product rule extends to many factors** — differentiate one factor at a time and add the results.

Section 2 — Key Formulas & Definitions

Note (maths format): these are operational identities, no dimensional formulas. Throughout, $u = f(x)$ and $v = g(x)$ are differentiable functions and c is a constant.

$$(u \pm v)' = u' \pm v', \quad (cu)' = cu',$$

Product: $(uv)' = u'v + uv'$,

Quotient: $\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2} \quad (v \neq 0),$ **Reciprocal:** $\left(\frac{1}{v}\right)' = -\frac{v'}{v^2}.$

Three-factor product (pattern: differentiate each in turn)

$$(uvw)' = u'vw + uv'w + uvw'.$$

Chain rule (composite $y = f(u)$, $u = g(x)$)

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = f'(g(x)) \cdot g'(x).$$

Power chain rule (the most-used special case)

$$\frac{d}{dx}[g(x)]^n = n[g(x)]^{n-1} \cdot g'(x).$$

Standard derivatives (chained from 12_03 — the toolkit the rules act on):

$$\frac{d}{dx}x^n = nx^{n-1}, \quad \frac{d}{dx}\sin x = \cos x, \quad \frac{d}{dx}\cos x = -\sin x, \quad \frac{d}{dx}e^x = e^x, \quad \frac{d}{dx}\ln x = \frac{1}{x}.$$

The complete trigonometric set (the three reciprocal ones are derived below using the quotient rule):

$$\frac{d}{dx}\tan x = \sec^2 x, \quad \frac{d}{dx}\cot x = -\csc^2 x, \quad \frac{d}{dx}\sec x = \sec x \tan x, \quad \frac{d}{dx}\csc x = -\csc x \cot x.$$

Section 3 — Key Procedures & Logical Framework

Product rule — proven from first principle back in 12_03 (the add-and-subtract cross-term trick). Here we simply *use* it.

Quotient rule, derived elegantly from the product rule. Let $y = \frac{u}{v}$, so $u = yv$. Differentiate both sides with the product rule:

$$u' = y'v + yv'.$$

Solve for y' and substitute $y = \frac{u}{v}$:

$$y' = \frac{u' - yv'}{v} = \frac{u' - \frac{u}{v}v'}{v} = \frac{u'v - uv'}{v^2}.$$

Notice the rules interlock — the quotient rule is just the product rule rearranged. (This is also why the order $u'v - uv'$ matters: swap it and the sign flips.)

Completing the trig table with the quotient rule. The three reciprocal trig derivatives are quick consequences. For $\cot x = \frac{\cos x}{\sin x}$:

$$\frac{d}{dx}\cot x = \frac{(-\sin x)(\sin x) - (\cos x)(\cos x)}{\sin^2 x} = \frac{-(\sin^2 x + \cos^2 x)}{\sin^2 x} = -\frac{1}{\sin^2 x} = -\csc^2 x.$$

For $\sec x = \frac{1}{\cos x}$, the reciprocal rule gives $\frac{d}{dx}\sec x = -\frac{-\sin x}{\cos^2 x} = \frac{\sin x}{\cos^2 x} = \sec x \tan x$; identically,

$\frac{d}{dx}\csc x = -\csc x \cot x$. That closes the full set of six.

Chain rule — why rates multiply. For a composite $y = f(g(x))$, look at the average rates over a small step and insert the intermediate change Δu :

$$\frac{\Delta y}{\Delta x} = \frac{\Delta y}{\Delta u} \cdot \frac{\Delta u}{\Delta x}.$$

As $\Delta x \rightarrow 0$, $\Delta u \rightarrow 0$ too (the inner function is continuous), and each factor tends to its derivative, giving $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$. (A fully rigorous proof patches the case $\Delta u = 0$, but the cancelling-rates picture is the right intuition.)

Procedure — how to differentiate any expression:

1. **Simplify first** if algebra makes it cleaner (expand products, split fractions).
2. **Identify the outermost operation** — is the whole thing a sum, a product, a quotient, or a composition?
3. Apply that rule, generating derivative-of-sub-pieces as new sub-problems.
4. **For compositions, peel outside-in:** differentiate the outer function (keeping the inside intact), then *multiply by the derivative of the inside* — and repeat for each nested layer.

Section 4 — Worked Examples

Example 1 • [CBSE Board Level]

Differentiate $y = x^3 \sin x$.

A product of $u = x^3$ and $v = \sin x$, with $u' = 3x^2$, $v' = \cos x$. By the product rule:

$$\frac{dy}{dx} = u'v + uv' = 3x^2 \sin x + x^3 \cos x = \boxed{x^2(3 \sin x + x \cos x)}.$$

Example 2 • [JEE Main — Speed Trap]

Differentiate $y = \frac{x^2 + 1}{x}$.

The trap: students immediately fire the quotient rule and wade through $\frac{(2x)(x) - (x^2+1)(1)}{x^2}$. It works, but it's slow and error-prone. **Simplify first:**

$$y = \frac{x^2 + 1}{x} = x + \frac{1}{x} = x + x^{-1} \Rightarrow \frac{dy}{dx} = 1 - x^{-2} = \boxed{1 - \frac{1}{x^2}}.$$

Lesson: always ask “can I simplify before I differentiate?” — it often turns a quotient-rule slog into one line.

Example 3 • [JEE Main — Multi-Concept]

Differentiate $y = x^2 \cos(3x)$. (Combines the product rule with the chain rule.)

Product of $u = x^2$ and $v = \cos(3x)$. For v' , the chain rule: outer $\cos(\cdot) \rightarrow -\sin(\cdot)$, inner $3x \rightarrow 3$, so $v' = -\sin(3x) \cdot 3 = -3 \sin(3x)$. Then:

$$\frac{dy}{dx} = u'v + uv' = 2x \cos(3x) + x^2(-3 \sin(3x)) = \boxed{2x \cos(3x) - 3x^2 \sin(3x)}.$$

Example 4 • [JEE Advanced Level]

Differentiate $y = \sin(\cos(x^2))$.

Three nested layers — peel outside-in:

- Outermost: $\sin(\square)$, derivative $\cos(\square)$, keep $\square = \cos(x^2)$ intact $\Rightarrow \cos(\cos(x^2))$.
- Next layer: derivative of $\cos(x^2)$ is $-\sin(x^2) \cdot (\text{derivative of } x^2)$.
- Innermost: derivative of x^2 is $2x$.

Multiply the chain of all three:

$$\frac{dy}{dx} = \cos(\cos(x^2)) \cdot (-\sin(x^2)) \cdot 2x = \boxed{-2x \sin(x^2) \cos(\cos(x^2))}.$$

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Skill tested: disciplined layer-by-layer chaining — one missed inner derivative and the whole answer is wrong.

Section 5 — Practice Exercises

1. **[CBSE]** Differentiate $y = x^4 + 3x^2 - 7$.
2. **[CBSE]** Differentiate $y = (2x + 1)(x - 3)$.
3. **[JEE Main]** Differentiate $y = \frac{x}{x^2 + 1}$.
4. **[JEE Main]** Differentiate $y = \cos(5x^2)$.
5. **[JEE Advanced]** Differentiate $y = (1 + x^2)^{10} \sin x$.

Answers: 1. $4x^3 + 6x$ 2. $4x - 5$ 3. $\frac{1 - x^2}{(x^2 + 1)^2}$ 4. $-10x \sin(5x^2)$ 5. $20x(1 + x^2)^9 \sin x + (1 + x^2)^{10} \cos x$

Section 6 — Crack the MCQ

Q1. $\frac{d}{dx}[(x^2 + 1)^3]$ equals (A) $3(x^2 + 1)^2$ (B) $6x(x^2 + 1)^2$ (C) $3(x^2 + 1)^2 \cdot 2$ (D) $2x$

Answer: (B). Power chain rule: $3(x^2 + 1)^2 \cdot \frac{d}{dx}(x^2 + 1) = 3(x^2 + 1)^2 \cdot 2x = 6x(x^2 + 1)^2$. *Why the traps catch you:* (A) is the single most common chain-rule error — forgetting to multiply by the inner derivative. (C) uses 2 (a stray constant) instead of the inner derivative $2x$. (D) keeps only the inner derivative and drops the outer.

Q2. $\frac{d}{dx}(x \sin x)$ equals (A) $\cos x$ (B) $\sin x + x \cos x$ (C) $x \cos x$ (D) $\sin x - x \cos x$

Answer: (B). Product rule: $(1) \sin x + x(\cos x) = \sin x + x \cos x$. *Why the traps catch you:* (A) and (C) each keep only one term of the product rule. (D) mis-signs the second term (no minus appears in the product rule).

Q3. If $y = u^3$ and $u = 2x + 1$, then $\frac{dy}{dx}$ at $x = 0$ equals (A) 3 (B) 6 (C) 2 (D) 12

Answer: (B). Chain rule: $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = 3u^2 \cdot 2$. At $x = 0$, $u = 1$, so $= 3(1)^2 \cdot 2 = 6$. *Why the traps catch you:* (A) uses only $\frac{dy}{du} = 3u^2 = 3$. (C) uses only $\frac{du}{dx} = 2$. (D) doubles correctly but forgets $u = 1$ and uses $u = 2$ giving $3 \cdot 4 \dots$ — a substitution slip.

Q4. $\frac{d}{dx}[\sin(x^2)]$ equals (A) $\cos(x^2)$ (B) $2x \cos(x^2)$ (C) $2x \cos x$ (D) $\cos(2x)$

Answer: (B). Chain rule: $\cos(x^2) \cdot \frac{d}{dx}(x^2) = 2x \cos(x^2)$. *Why the traps catch you:* (A) forgets the inner derivative (the chain-rule classic). (C) wrongly differentiates the inside of $\cos$ as well. (D) invents a non-existent identity.

Section 7 — Common Student Pitfalls & Pro-Tips

1. **Forgetting the inner derivative (chain rule).** $\frac{d}{dx} \sin(x^2) = \cos(x^2)$ is **wrong**; it's $2x \cos(x^2)$. Every composition contributes a “ $\times$ derivative of the inside.”
2. **Wrong product rule.** $(uv)' \neq u'v'$. It is $u'v + uv'$ — always two terms.
3. **Quotient-rule order/sign.** The numerator is $u'v - uv'$ (top-derivative-times-bottom *first*). Reversing it flips the sign and the answer.
4. **Skipping simplification.** Differentiating an unsimplified quotient or product wastes time and multiplies error chances. Simplify, then differentiate.

Pro-Tip (speed + accuracy): for compositions, say the mantra “**derivative of the outside (inside kept whole), times derivative of the inside,**” and repeat for each layer until you hit

x. For three-or-more-factor products, **differentiate one factor at a time and add** — no need to group. And before any rule, glance for an algebraic simplification that makes the whole thing trivial.

Section 8 — Cheat Sheet Box

RULES OF DIFFERENTIATION

- **Sum/scalar:** $(u \pm v)' = u' \pm v'$; $(cu)' = cu'$
- **Product:** $(uv)' = u'v + uv'$ (three: $u'vw + uv'w + uvw'$)
- **Quotient:** $\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$, $v \neq 0$
- **Chain:** $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$; **Power chain:** $\frac{d}{dx}[g]^n = n[g]^{n-1}g'$
- Simplify *before* differentiating whenever you can.

Memory aids:

- “Low *d*-high minus high *d*-low, over low-squared, away we go.” (quotient rule)
- “First *d*-second plus second *d*-first.” (product rule)
- “Outside derivative, times inside derivative.” (chain rule — never drop the inside)
- “Rates cascade: gears multiply.” (chain rule intuition)

Subtopic 05 — Limits at Infinity & Special Cases of Existence

Exam Relevance: **JEE Main** reliably asks one limit-at-infinity question (rational end-behaviour or an $\infty - \infty$ surd) every session. **CBSE Boards** test rational end-behaviour and a “does the limit exist” greatest-integer/modulus case (1-2 marks). **JEE Advanced** favours these inside assertion-reason and sandwich-theorem problems. This subtopic closes the two situations the earlier ones set aside: *what happens as x runs to infinity*, and *what happens at a jump*.

Section 1 — Concept Intuition

So far every limit asked the same question: as x creeps up to a finite point a , what finite value does f approach? Two situations escape that mould, and a complete picture of limits needs both.

First: end behaviour — where does f head as $x \rightarrow \infty$? Picture zooming *out* on a graph until the fine detail vanishes and only the overall trend remains. For a function like $\frac{3x^2+5}{6x^2-x}$, the small terms $(+5, -x)$ become irrelevant far out — only the *highest powers* matter, and the function settles toward $\frac{3}{6} = \frac{1}{2}$. The dominant term wins, the way the single ₹2000 note in your wallet decides its value no matter how many coins rattle alongside it.

Think about it this way... for a rational function as $x \rightarrow \infty$, mentally erase everything except the highest power of x on top and bottom. Their ratio *is* the limit. Everything else is noise that $\frac{1}{x} \rightarrow 0$ washes away.

Second: the limit can fail at a jump, or be infinite. Think of an autorickshaw fare meter, or electricity billed in slabs: the charge holds flat, then *jumps* at each boundary. Approach a boundary from the left and you read the lower slab; approach from the right and you read the higher one. The two disagree, so there is no single value being approached — the limit **does not exist** at the jump. The **greatest integer function** $[x]$ (the largest integer $\leq x$) is exactly this staircase, and it is the classic exam vehicle for testing whether you really understand one-sided limits.

Left panel: graph of a rational function approaching a horizontal asymptote $y = \frac{1}{2}$ as $x \rightarrow +\infty$ and $x \rightarrow -\infty$. Right panel: the staircase graph of the greatest integer function $y = [x]$ over $[-1, 3]$, with filled dots on the left of each step and open dots on the right, highlighting that at each integer n the left limit is $n - 1$ and the right limit is n .

Limiting Conditions & Assumptions:

1. ∞ **is a direction, not a number.** You cannot do ordinary arithmetic with it. $\infty - \infty$ and $\frac{\infty}{\infty}$ are **indeterminate** — they must be resolved, never assumed to be 0 or 1.
2. “**Limit = ∞** ” **describes behaviour, not a value.** It means f grows without bound; in the strict finite sense the limit does not exist, but we write ∞ to record *how* it fails.
3. **One-sided infinite limits can disagree in sign.** $\frac{1}{x} \rightarrow +\infty$ as $x \rightarrow 0^+$ but $\rightarrow -\infty$ as $x \rightarrow 0^-$.
4. **At any jump, existence is a one-sided question.** For greatest-integer, modulus, signum, or piecewise functions, you *must* compute LHL and RHL separately (the 12_01 existence rule) — never substitute blindly.

Section 2 — Key Formulas & Definitions

Note (maths format): no dimensional formulas. The payload here is the end-behaviour rules and the jump-function facts.

Powers at infinity (for $k > 0$):

$$\lim_{x \rightarrow \infty} \frac{1}{x^k} = 0, \quad \lim_{x \rightarrow \infty} x^k = \infty.$$

Rational functions at infinity — compare degrees of numerator (m) and denominator (n):

$$\lim_{x \rightarrow \infty} \frac{a_m x^m + \dots}{b_n x^n + \dots} = \begin{cases} 0, & m < n, \\ \frac{a_m}{b_n}, & m = n, \\ \pm\infty, & m > n. \end{cases}$$

Infinite limits (vertical-asymptote behaviour):

$$\lim_{x \rightarrow 0} \frac{1}{x^2} = +\infty, \quad \lim_{x \rightarrow 0^+} \frac{1}{x} = +\infty, \quad \lim_{x \rightarrow 0^-} \frac{1}{x} = -\infty.$$

Greatest integer function $[x]$ (largest integer $\leq x$). For an **integer** n :

$$\lim_{x \rightarrow n^-} [x] = n - 1, \quad \lim_{x \rightarrow n^+} [x] = n \Rightarrow \lim_{x \rightarrow n} [x] \text{ **does not exist** .}$$

For a **non-integer** a : $\lim_{x \rightarrow a} [x] = [a]$ (exists, since $[x]$ is locally constant).

Modulus & signum:

$$\lim_{x \rightarrow a} |x| = |a| \text{ (exists everywhere),} \quad \lim_{x \rightarrow 0} \frac{|x|}{x} \text{ **does not exist** (LHL = -1, RHL = +1).}$$

Indeterminate ∞ -forms to watch: $\frac{\infty}{\infty}$, $\infty - \infty$, $0 \times \infty$, $\frac{0}{0}$ (the last reappears here too).

Section 3 — Key Procedures & Logical Framework

Procedure A — rational limit as $x \rightarrow \infty$ (the divide-by-highest-power method). Divide every term in numerator and denominator by the **highest power of x in the denominator**, then send each $\frac{1}{x^k} \rightarrow 0$. The survivors give the answer. (The degree rule in §2 is just the shortcut this method always produces.)

Procedure B — the $\infty - \infty$ surd trick. A difference like $\sqrt{\text{big}} - \sqrt{\text{big}}$ is indeterminate. **Rationalise** by the conjugate to turn it into a single fraction, then divide by the highest power:

$$\sqrt{x^2 + bx} - x = (\sqrt{x^2 + bx} - x) \cdot \frac{\sqrt{x^2 + bx} + x}{\sqrt{x^2 + bx} + x} = \frac{bx}{\sqrt{x^2 + bx} + x} \xrightarrow{\div x} \frac{b}{\sqrt{1 + \frac{b}{x}} + 1} \rightarrow \frac{b}{2}.$$

Procedure C — existence at a jump (piecewise / greatest-integer / modulus). Never substitute. Instead:

1. Identify which rule/piece applies just to the **left** of the point, and compute the LHL.
2. Identify which applies just to the **right**, and compute the RHL.
3. **Compare:** equal and finite $\Rightarrow$ the limit exists and equals that common value; otherwise it does not exist. (*This is the 12_01 existence rule doing all the work.*)

For the greatest integer function near an integer n : just below n , $[x] = n - 1$; just above, $[x] = n$. The two differ by 1, so the limit always fails at integers — the signature result.

Section 4 — Worked Examples

Example 1 • [CBSE Board Level]

Evaluate $\lim_{x \rightarrow \infty} \frac{3x^2 + 2x - 1}{5x^2 - x + 4}.$

Divide every term by x^2 (highest power in the denominator):

$$\lim_{x \rightarrow \infty} \frac{3 + \frac{2}{x} - \frac{1}{x^2}}{5 - \frac{1}{x} + \frac{4}{x^2}}.$$

As $x \rightarrow \infty$, every $\frac{1}{x^k} \rightarrow 0$, leaving $\boxed{\frac{3}{5}}$. (Equal degrees $\Rightarrow$ ratio of leading coefficients — the §2 shortcut agrees.)

Example 2 • [JEE Main — Speed Trap]

Evaluate $\lim_{x \rightarrow \infty} (\sqrt{x^2 + x} - x)$.

The trap: writing “ $\infty - \infty = 0$.” It is *indeterminate*. Rationalise:

$$\sqrt{x^2 + x} - x = \frac{(x^2 + x) - x^2}{\sqrt{x^2 + x} + x} = \frac{x}{\sqrt{x^2 + x} + x}.$$

Divide top and bottom by x (note $\sqrt{x^2} = x$ for $x \rightarrow +\infty$):

$$= \frac{1}{\sqrt{1 + \frac{1}{x}} + 1} \rightarrow \frac{1}{\sqrt{1} + 1} = \boxed{\frac{1}{2}}.$$

Example 3 • [JEE Main — Multi-Concept]

Evaluate $\lim_{x \rightarrow \infty} \left(\frac{x^2 + 1}{x + 1} - x \right)$. (A disguised $\infty - \infty$ that combines algebraic simplification with the degree rule.)

Combine over a common denominator first:

$$\frac{x^2 + 1}{x + 1} - x = \frac{x^2 + 1 - x(x + 1)}{x + 1} = \frac{x^2 + 1 - x^2 - x}{x + 1} = \frac{1 - x}{x + 1}.$$

Now it is an ordinary rational limit (equal degrees, leading coefficients -1 and 1):

$$\lim_{x \rightarrow \infty} \frac{1 - x}{x + 1} = \frac{-1}{1} = \boxed{-1}.$$

Example 4 • [JEE Advanced Level]

Evaluate $\lim_{x \rightarrow 0^+} x \left[\frac{1}{x} \right]$, where $[\cdot]$ is the greatest integer function.

The idea: $[\cdot]$ resists substitution, so bound it. By definition $\left[\frac{1}{x} \right] = \frac{1}{x} - \left\{ \frac{1}{x} \right\}$, where the fractional part satisfies $0 \leq \left\{ \frac{1}{x} \right\} < 1$. Multiply by $x > 0$:

$$x \left[\frac{1}{x} \right] = x \left(\frac{1}{x} - \left\{ \frac{1}{x} \right\} \right) = 1 - x \left\{ \frac{1}{x} \right\}.$$

Since $0 \leq x \left\{ \frac{1}{x} \right\} < x$, and $x \rightarrow 0^+$, the **sandwich theorem** (from 12_02) forces $x \left\{ \frac{1}{x} \right\} \rightarrow 0$. Therefore

$$\lim_{x \rightarrow 0^+} x \left[\frac{1}{x} \right] = 1 - 0 = \boxed{1}.$$

Deep point: a greatest-integer expression that looks hopeless becomes a clean sandwich once you split off the fractional part.

Section 5 — Practice Exercises

1. [CBSE] Evaluate $\lim_{x \rightarrow \infty} \frac{4x + 3}{2x - 1}$.
2. [CBSE] Evaluate $\lim_{x \rightarrow \infty} \frac{x^2 + 1}{x^3 + x}$.

3. **[JEE Main]** Evaluate $\lim_{x \rightarrow \infty} (\sqrt{4x^2 + x} - 2x)$.
4. **[JEE Main]** Does $\lim_{x \rightarrow 1} [x]$ exist ($[\cdot]$ = greatest integer)? Justify with one-sided limits.
5. **[JEE Advanced]** Evaluate $\lim_{x \rightarrow \infty} \left(\frac{3x^4 + 2x^2}{x^2 - 1} - 3x^2 \right)$.

Answers: 1. 2 2. 0 3. $\frac{1}{4}$ 4. Does **not** exist (LHL = 0, RHL = 1) 5. 5

Section 6 — Crack the MCQ

Q1. $\lim_{x \rightarrow \infty} \frac{5x^3 + 2x}{3x^3 - x^2 + 1}$ equals (A) $\frac{5}{3}$ (B) 0 (C) ∞ (D) 5

Answer: (A). Equal degrees ($m = n = 3$) $\Rightarrow$ ratio of leading coefficients = $\frac{5}{3}$. *Why the traps catch you:* (B) is the $m < n$ case (wrong here). (C) is the $m > n$ case. (D) forgets to divide by the denominator's leading coefficient 3.

Q2. $\lim_{x \rightarrow 0^+} \frac{1}{x}$ equals (A) 0 (B) $+\infty$ (C) $-\infty$ (D) 1

Answer: (B). Approaching 0 from the positive side, $\frac{1}{x}$ grows without bound to $+\infty$. *Why the traps catch you:* (C) is the left-hand limit (wrong side). (A) confuses $\frac{1}{x}$ at large x with small x . (D) is a guess with no basis.

Q3. $\lim_{x \rightarrow 2} [x]$, where $[\cdot]$ is the greatest integer function, is (A) 2 (B) 1 (C) does not exist (D) 1.5

Answer: (C). Just below 2, $[x] = 1$ (LHL = 1); just above, $[x] = 2$ (RHL = 2). They differ, so the limit does not exist. *Why the traps catch you:* (B) reports only the LHL; (A) reports only the RHL (or the value $[2] = 2$); (D) invents an average.

Q4. $\lim_{x \rightarrow \infty} (\sqrt{x^2 + 3x} - x)$ equals (A) 0 (B) $\frac{3}{2}$ (C) ∞ (D) 3

Answer: (B). Rationalising gives $\frac{3x}{\sqrt{x^2 + 3x} + x} \rightarrow \frac{3}{2}$ (the $\frac{b}{2}$ rule with $b = 3$). *Why the traps catch you:* (A) is the " $\infty - \infty = 0$ " fallacy. (C) assumes the root dominates without cancellation. (D) keeps the 3 but forgets the $\div 2$.

Section 7 — Common Student Pitfalls & Pro-Tips

- Treating $\infty - \infty = 0$ or $\frac{\infty}{\infty} = 1$.** Both are indeterminate; rationalise or use the degree rule.
- Substituting into a jump function.** For $[x]$, $|x|/x$, signum, or piecewise functions, you *must* split into LHL and RHL — direct substitution is meaningless at the jump.
- Dividing by the wrong power.** For rational limits at ∞ , divide by the highest power in the denominator; picking the wrong power scrambles the answer.
- Treating ∞ as a number.** " $\lim = \infty$ " is a behaviour statement, not a value you can add or cancel.

Pro-Tip (speed + accuracy): for any rational function as $x \rightarrow \infty$, skip the algebra and use the **three-case degree rule** — smaller-top $\rightarrow 0$, equal $\rightarrow$ leading-coefficient ratio, bigger-top $\rightarrow \pm\infty$. For surd $\infty - \infty$, rationalise on sight. And for *anything* with $[\cdot]$, $|\cdot|$, signum, or a piecewise definition, your first reflex is always **"split into LHL and RHL."**

Section 8 — Cheat Sheet Box

LIMITS AT INFINITY & SPECIAL CASES

- **Powers:** $\frac{1}{x^k} \rightarrow 0$, $x^k \rightarrow \infty$ ($k > 0$, $x \rightarrow \infty$).
- **Rational at ∞ (degree rule):** top-degree $<$ bottom $\rightarrow 0$; $= \rightarrow$ ratio of leading coeffs; $> \rightarrow \pm\infty$.
- $\infty - \infty$ **surds:** rationalise by the conjugate, then divide by the highest power.
- **Infinite limits:** $\frac{1}{x^2} \rightarrow +\infty$; $\frac{1}{x} \rightarrow +\infty$ (0^+), $-\infty$ (0^-).
- **Greatest integer at integer n :** LHL = $n - 1$, RHL = $n \Rightarrow$ limit **DNE**; at non-integers, $\lim[x] = [x]$.
- **Existence at a jump:** always split into LHL & RHL and compare.

Memory aids:

- “Far out, only the top powers matter.” (rational end-behaviour)
- “ $\infty - \infty$ is a question, not zero.”
- “Steps and bars and bends — split it into two ends.” (GIF/modulus/piecewise existence)

Master Reference — Chapter 12 at a Glance

How to use this section. Everything taught across subtopics 01–05 is consolidated here for revision and exam-day recall. Learn the *methods* from the subtopics; use this page to lock the *results* and the *decision-making*.

A. Limit-Evaluation Decision Guide

Faced with any $\lim_{x \rightarrow a} f(x)$, work down this list and stop at the first that applies:

1. **Substitute** $x = a$. If you get a finite number, that is the limit. (Works for polynomials always, and rationals when the denominator $\neq 0$.)
2. **Got $\frac{0}{0}$?** Resolve it:
 - *Algebraic / rational*: factor out $(x - a)$ and cancel (12_01).
 - *Surds*: multiply by the conjugate (12_01).
 - *Trig / exponential / logarithmic*: reshape into a **standard limit** — $\frac{\sin x}{x}$, $\frac{1 - \cos x}{x^2}$, $\frac{e^x - 1}{x}$, $\frac{\ln(1+x)}{x}$ (12_02).
3. **Is $x \rightarrow \infty$ (or an ∞ -form)?** Use the **degree rule** for rationals, or rationalise an $\infty - \infty$ surd, then send $\frac{1}{x^k} \rightarrow 0$ (12_05).
4. **Form 1^∞ ?** Use $\lim f^g = e^{\lim g(f-1)}$ (12_02).
5. **Is it a jump function** — $[x]$, $|x|$, signum, or piecewise? **Split into LHL and RHL** and compare; equal & finite $\Rightarrow$ exists (12_01, 12_05).

B. Master Formula Sheet — Limits

Existence: $\lim_{x \rightarrow a} f(x) = L \iff \lim_{x \rightarrow a^-} f = \lim_{x \rightarrow a^+} f = L$ (finite).

Algebra of limits: sum, difference, product, scalar, power split freely; quotient needs denominator-limit $\neq 0$.

Algebraic standard limit: $\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = n a^{n-1}$ ($n \in \mathbb{Q}$).

Trigonometric (radians, $\rightarrow 0$): $\frac{\sin x}{x} \rightarrow 1$, $\frac{\tan x}{x} \rightarrow 1$, $\frac{1 - \cos x}{x^2} \rightarrow \frac{1}{2}$, $\frac{\sin^{-1} x}{x} \rightarrow 1$, $\frac{\tan^{-1} x}{x} \rightarrow 1$.

Exponential / logarithmic: $\frac{e^x - 1}{x} \rightarrow 1$, $\frac{a^x - 1}{x} \rightarrow \ln a$, $\frac{\ln(1+x)}{x} \rightarrow 1$.

The number e / 1^∞ : $(1+x)^{1/x} \rightarrow e$, $(1+\frac{1}{n})^n \rightarrow e$, $\lim f^g = e^{\lim g(f-1)}$.

At infinity: $\frac{1}{x^k} \rightarrow 0$; rational degree rule (top < bottom $\rightarrow 0$; $= \rightarrow$ leading-coeff ratio; $> \rightarrow \pm\infty$).

Greatest integer: at integer n , LHL = $n - 1$, RHL = $n \Rightarrow$ DNE; at non-integer a , $\lim[x] = [a]$.

C. Master Formula Sheet — Derivatives

Definition: $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{t \rightarrow x} \frac{f(t) - f(x)}{t - x}$ — slope of tangent = instantaneous rate.

Existence: $f'(a)$ exists $\iff$ LHD = RHD. **Differentiable** $\Rightarrow$ **continuous** (converse false: $|x|$ at 0).

Standard derivatives:

$$\begin{aligned} \frac{d}{dx}x^n &= nx^{n-1}, & \frac{d}{dx}e^x &= e^x, & \frac{d}{dx}a^x &= a^x \ln a, & \frac{d}{dx} \ln x &= \frac{1}{x}. \\ \frac{d}{dx} \sin x &= \cos x, & \frac{d}{dx} \cos x &= -\sin x, & \frac{d}{dx} \tan x &= \sec^2 x, \\ \frac{d}{dx} \cot x &= -\csc^2 x, & \frac{d}{dx} \sec x &= \sec x \tan x, & \frac{d}{dx} \csc x &= -\csc x \cot x. \end{aligned}$$

Rules: $(u \pm v)' = u' \pm v'$; $(cu)' = cu'$; $(uv)' = u'v + uv'$; $\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$; **chain** $\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx}$;
power chain $\frac{d}{dx}[g]^n = n[g]^{n-1}g'$.

D. Concept Map — How the Chapter Connects

The chapter spine: Idea of a limit (LHL = RHL) $\rightarrow$ **standard limits** $\rightarrow$ **the derivative** (itself a limit) $\rightarrow$ **rules of differentiation** $\rightarrow$ **limits at infinity & special cases**.

- **Limits (12_01)** establish the one engine — *approach, don't arrive* — and the existence test LHL = RHL.
- **Standard limits (12_02)** are the pre-proven junctions that crack the $\frac{0}{0}$ forms algebra can't.
- **The derivative (12_03)** is *literally a limit* — the difference-quotient $\frac{0}{0}$ — so its existence test (LHD = RHD) is the same one-sided rule from 12_01, and its standard results are powered by the 12_02 junctions ($\frac{d}{dx} \sin x$ needs $\frac{\sin h}{h} \rightarrow 1$; $\frac{d}{dx} e^x$ needs $\frac{e^h - 1}{h} \rightarrow 1$).
- **Rules (12_04)** turn the slow limit definition into a fast algebra.
- **Infinity & special cases (12_05)** extend the same existence reasoning to end-behaviour and jumps.

The whole chapter hangs on one thread: **a limit is the value you approach, and existence means both sides agree**. Derivatives, rules, and end-behaviour are all that single idea, reused.

E. Top Exam Traps (consolidated)

- **Limit $\neq$ function value.** $f(a)$ is irrelevant to $\lim_{x \rightarrow a} f$.
- $\frac{0}{0}$ means “resolve,” not “zero” or “does not exist.”
- **Radians only** for $\frac{\sin x}{x} = 1$; degrees give $\frac{\pi}{180}$.
- **Match the argument:** $\frac{\sin kx}{x} = k$, and $\frac{1 - \cos kx}{x^2} = \frac{k^2}{2}$ (square the k).
- **e vs other base:** $\frac{e^x - 1}{x} \rightarrow 1$ but $\frac{a^x - 1}{x} \rightarrow \ln a$.
- **Product rule has a cross term:** $(uv)' = u'v + uv'$, never $u'v'$.
- **Chain rule keeps the inside derivative:** $\frac{d}{dx} \sin(x^2) = 2x \cos(x^2)$.
- $\infty - \infty \neq 0$; rationalise. **Jump functions** ($[x]$, $|x|$, piecewise) always need LHL vs RHL.
- **Differentiable $\Rightarrow$ continuous**, but continuous $\nRightarrow$ differentiable.

Chapter 12: Limits and Derivatives — Supplement

This supplement adds one full subtopic and two addenda to Chapter 12. Subtopic 06 collects the limit techniques that the first five subtopics set up but did not deliver: shifting the point of approach, L'Hopital's rule, series expansions, limits at $-\infty$, and the $0 \times \infty$ form. Addendum A delivers the functional-equation derivative problems promised in Subtopic 03's exam-relevance box. Addendum B works the “find a and b so that f is differentiable” pattern end to end.

Read Subtopic 06 only after you are fluent with 12_01 (algebra of limits, the existence rule), 12_02 (the standard-limit junctions), 12_04 (the chain rule) and 12_05 (limits at infinity). Every technique here is built on those.

Subtopic 06 — Advanced Limit Techniques

Exam Relevance: This subtopic is where the marks hide. CBSE Boards ask limits at a non-zero point directly — NCERT's own exercise set contains $\lim_{x \rightarrow \pi/2} \frac{\tan 2x}{x - \pi/2}$ — usually for 2–3 marks. JEE Main leans hard on limits as $x \rightarrow -\infty$ (the sign trap in surds is one of the most reliable single-question traps in the paper) and on the $0 \times \infty$ conversion. JEE Advanced expects the Maclaurin-expansion method as a matter of course for higher-order $\frac{0}{0}$ limits such as $\lim_{x \rightarrow 0} \frac{x - \sin x}{x^3}$, which no other technique cracks cleanly. L'Hopital's rule formally belongs to Class 12, but every serious JEE aspirant uses it from Class 11 onward as a checking and speed tool — exactly as the chain rule was bridged in for you in 12_04.

Section 1: Concept Intuition

You already own a good toolkit. The trouble is that almost all of it is bolted to one particular spot on the number line: **zero**. Every junction in 12_02 — $\frac{\sin x}{x} \rightarrow 1$, $\frac{e^x - 1}{x} \rightarrow 1$, $\frac{1 - \cos x}{x^2} \rightarrow \frac{1}{2}$ — carries the fine print “the argument must go to 0”. So what do you do when the examiner parks the problem at $x \rightarrow \pi/2$, or $x \rightarrow \pi/4$, or $x \rightarrow -\infty$?

Think about it this way... consider the Mumbai suburban network. Every timetable is written in minutes from Churchgate. If you board at Andheri, you do not throw away the timetable and invent a new one — you simply re-zero your watch: “I am at Churchgate-time plus 28 minutes.” Let h be your offset from the reference station, and the whole timetable works again unchanged.

That is the single idea that unifies the first half of this subtopic. **Whenever a limit is taken somewhere inconvenient, change the variable so that the inconvenient point becomes zero, and then use the machinery you already have.**

- Limit at $x = a$? Put $x = a + h$ and let $h \rightarrow 0$.

- Limit at $x \rightarrow -\infty$? Put $x = -t$ and let $t \rightarrow +\infty$.
- A $0 \times \infty$ product like $x \sin \frac{1}{x}$ as $x \rightarrow \infty$? Put $t = \frac{1}{x}$ and let $t \rightarrow 0$.

Three different-looking problems, one move.

The second half of this subtopic is about a different question: **how do you resolve a $\frac{0}{0}$ when both parts vanish to a high order?** Take $\frac{x - \sin x}{x^3}$. Near zero, $\sin x$ and x are twins, so the numerator is a difference of two near-identical quantities — it collapses to nothing. Meanwhile the denominator collapses to nothing too, but *at a specific rate*, namely like x^3 . The whole answer is a race between two collapses. To judge the race you need to know not just that $\sin x \approx x$, but *by how much* $\sin x$ misses x .

Two instruments read that fine detail:

L'Hopital's rule compares the two collapses by their *slopes*. If f and g both crash to zero at a , then near a each is approximately its own tangent line through $(a, 0)$, so the ratio $\frac{f(x)}{g(x)}$ behaves like $\frac{f'(a)(x-a)}{g'(a)(x-a)} = \frac{f'(a)}{g'(a)}$. Differentiating the top and the bottom separately is not an algebraic trick and it is emphatically **not** the quotient rule — it is a statement about competing rates of vanishing.

Series expansion is the high-resolution microscope. Instead of the crude " $\sin x \approx x$ " you carry the full ruler $\sin x = x - \frac{x^3}{6} + \frac{x^5}{120} - \dots$, and the term you need simply appears. For $\frac{x - \sin x}{x^3}$ the numerator is $\frac{x^3}{6} - \dots$ and the answer $\frac{1}{6}$ falls out in one line, where L'Hopital would need three rounds of differentiation.

Limiting conditions and assumptions — where these ideas fail:

1. **The substitution $x = a + h$ must be exact, not approximate.** You must rewrite *every* occurrence of x , including inside trigonometric arguments, and you must simplify with genuine identities such as $\tan(\frac{\pi}{2} + h) = -\cot h$. A half-substituted expression is worthless.
2. **L'Hopital applies only to $\frac{0}{0}$ and $\frac{\infty}{\infty}$.** Applied to anything else — including a perfectly ordinary limit that direct substitution would settle — it produces confident nonsense. And if the ratio of derivatives has no limit, the rule tells you *nothing*; it does not tell you the original limit fails.
3. **Series expansions are local statements.** $\ln(1 + x) = x - \frac{x^2}{2} + \dots$ is valid only for $-1 < x \leq 1$; $\tan x = x + \frac{x^3}{3} + \dots$ only for $|x| < \frac{\pi}{2}$. For a limit as $x \rightarrow 0$ that is no obstacle, because x eventually enters any such window. But you may never expand about 0 a function whose expansion does not converge near 0, and you may never expand about 0 when the limit is taken somewhere else — shift first, then expand.
4. **Truncating too early destroys the answer.** Replacing $\sin x$ by x inside $\frac{x - \sin x}{x^3}$ gives 0, which is wrong. The rule is: if the denominator vanishes like x^m , you must carry the numerator's expansion to order x^m inclusive.
5. **$\sqrt{x^2}$ is $|x|$, not x .** As $x \rightarrow +\infty$ these agree and nobody notices. As $x \rightarrow -\infty$ they differ by a sign, and that sign is the entire question.

Figure brief: Two side-by-side panels — left, a number line with the point a marked and a second, re-zeroed axis drawn beneath it labelled $h = x - a$ with 0 sitting directly under a ,

showing the shift; right, the graphs of $y = \sin x$, $y = x$, and $y = x - \frac{x^3}{6}$ on $[-1.5, 1.5]$, with the second approximation visibly hugging $\sin x$ far more closely than the line does.

Section 2: Key Formulas and Definitions

Note (maths format): no dimensional formulas here — every quantity is a pure number. The payload of this section is four things you must be able to write down cold: the shift identities, the precise statement of L'Hopital's rule, the standard Maclaurin table with its validity windows, and the $x \rightarrow -\infty$ sign rule.

The shift substitution (change of variable at a point).

$$\lim_{x \rightarrow a} f(x) = \lim_{h \rightarrow 0} f(a + h)$$

- a = the point being approached (finite);
- $h = x - a$ = the offset from that point, a signed quantity with $h \rightarrow 0$;
- The two-sided limit on the left corresponds to the two-sided limit on the right. For one-sided work use $h \rightarrow 0^+$ for the right-hand limit and $x = a - h$ with $h \rightarrow 0^+$ for the left-hand limit — exactly the device already used for the existence test in 12_01.

Trigonometric shift identities (the ones exam problems actually use).

Shift	Identity
$x = \frac{\pi}{2} + h$	$\sin x = \cos h, \cos x = -\sin h, \tan x = -\cot h$
$x = \frac{\pi}{2} - h$	$\sin x = \cos h, \cos x = \sin h, \tan x = \cot h$
$x = \pi + h$	$\sin x = -\sin h, \cos x = -\cos h, \tan x = \tan h$
$x = \pi - h$	$\sin x = \sin h, \cos x = -\cos h, \tan x = -\tan h$
$x = \frac{\pi}{4} + h$	$\tan x = \frac{1 + \tan h}{1 - \tan h}, \sin x = \frac{\cos h + \sin h}{\sqrt{2}}$
$x = 1 + h$ (for $\tan \frac{\pi x}{2}$)	$\tan \frac{\pi x}{2} = \tan \left(\frac{\pi}{2} + \frac{\pi h}{2} \right) = -\cot \frac{\pi h}{2}$

L'Hopital's rule — precise statement. This is a Class 12 theorem, imported here as a bridge tool in the same spirit as the chain rule in 12_04.

Form $\frac{0}{0}$. Let f and g be differentiable on an open interval containing a , except possibly at a itself. Suppose

1. $\lim_{x \rightarrow a} f(x) = 0$ and $\lim_{x \rightarrow a} g(x) = 0$;
2. $g'(x) \neq 0$ for all x in that interval other than a ;
3. $\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$ exists as a finite number, or is $+\infty$ or $-\infty$.

Then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}.$$

Form $\frac{\infty}{\infty}$. The same conclusion holds if condition 1 is replaced by $|g(x)| \rightarrow \infty$ as $x \rightarrow a$ (with $|f(x)| \rightarrow \infty$ as well, in the usual exam setting), the other two conditions being unchanged.

- f, g = the numerator and denominator functions;
- f', g' = their derivatives, taken **separately** — this is not the quotient rule;
- a may be a finite number, a one-sided approach a^+ or a^- , or $\pm\infty$;
- Condition 3 is the one students drop. If $\frac{f'}{g'}$ has no limit, the rule is silent — the original limit may still exist perfectly well.

Maclaurin (series) expansions — the JEE Advanced table. Each is an exact identity on the stated set, not an approximation.

Function	Expansion	Valid for
e^x	$1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$	all $x \in \mathbb{R}$
a^x ($a > 0$)	$1 + x \ln a + \frac{(x \ln a)^2}{2!} + \dots$	all $x \in \mathbb{R}$
$\sin x$	$x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$	all $x \in \mathbb{R}$
$\cos x$	$1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$	all $x \in \mathbb{R}$
$\tan x$	$x + \frac{x^3}{3} + \frac{2x^5}{15} + \dots$	$ x < \frac{\pi}{2}$
$\ln(1+x)$	$x - \frac{x^2}{2} + \frac{x^3}{3} - \dots$	$-1 < x \leq 1$
$(1+x)^n$	$1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \dots$	$ x < 1$ for $n \notin \{0, 1, 2, \dots\}$; a terminating sum (the binomial theorem) for non-negative integer n
$\sin^{-1} x$	$x + \frac{x^3}{6} + \frac{3x^5}{40} + \dots$	$ x \leq 1$
$\tan^{-1} x$	$x - \frac{x^3}{3} + \frac{x^5}{5} - \dots$	$ x \leq 1$

- $n!$ = factorial; n in the binomial row is any real number;
- Every row above is an expansion **about** $x = 0$. To use any of them at another point, shift first.
- Notice the family resemblance: each expansion begins with the linear term that produced the corresponding standard limit in 12_02. The series simply continue the story.

The order-matching rule (how many terms to keep). If the denominator vanishes like x^m , expand the numerator up to and including the x^m term. Every later term contributes 0 to the limit, because a leftover of size x^{m+1} divided by x^m still tends to 0.

Limits as $x \rightarrow -\infty$.

$$\lim_{x \rightarrow -\infty} f(x) = \lim_{t \rightarrow +\infty} f(-t), \quad \sqrt{x^2} = |x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$

Consequently, when you pull an x out of a square root with $x < 0$:

$$\sqrt{x^2 + bx} = |x|\sqrt{1 + \frac{b}{x}} = -x\sqrt{1 + \frac{b}{x}} \quad (x < 0).$$

The two-sided pair worth memorising, for constant b :

$$\lim_{x \rightarrow +\infty} (\sqrt{x^2 + bx} - x) = \frac{b}{2}, \quad \lim_{x \rightarrow -\infty} (\sqrt{x^2 + bx} + x) = -\frac{b}{2}.$$

At $-\infty$ it is the **sum** that is the indeterminate $\infty - \infty$ form; the difference $\sqrt{x^2 + bx} - x$ runs off to $+\infty$ there, because both pieces are positive and large.

The $0 \times \infty$ form. If $f(x) \rightarrow 0$ and $g(x) \rightarrow \infty$ as $x \rightarrow a$, the product $f(x)g(x)$ is indeterminate. Convert it:

$$f \cdot g = \frac{f}{1/g} \quad (\text{form } \frac{0}{0}) \quad \text{or} \quad f \cdot g = \frac{g}{1/f} \quad (\text{form } \frac{\infty}{\infty}).$$

Choose whichever version is easier to handle. Once converted, every earlier tool — standard limits, series, L'Hopital — becomes available.

Variable glossary: x = the approaching variable; a = the finite point approached; $h = x - a$ = the shift offset ($h \rightarrow 0$); t = the substituted variable at infinity ($x = -t$ or $t = 1/x$); m = the order of vanishing of the denominator; n = the binomial exponent.

Section 3: CBSE Board Derivations and Key Procedures

Derivation: why the shift $x = a + h$ is legal

1. Set $h = x - a$. This is a one-to-one correspondence between values of x and values of h , with $x = a + h$. *Reason:* it is just a relabelling of the axis, invertible in one line.
2. As x runs through values close to a but not equal to a , h runs through values close to 0 but not equal to 0, and conversely. *Reason:* $|x - a| = |h|$, so “small distance from a ” and “small distance from 0” are literally the same statement.
3. A limit depends only on the values of the function on a punctured neighbourhood of the approach point (12_01, Section 1). *Reason:* this is the definition of a limit — the value at the point is irrelevant.
4. Steps 2 and 3 together mean f near a and $f(a+h)$ near $h = 0$ carry identical information. Hence $\lim_{x \rightarrow a} f(x)$ exists precisely when $\lim_{h \rightarrow 0} f(a+h)$ exists, and the two values coincide.

This is the same identity the chapter has already used in the existence test $\lim_{x \rightarrow a^+} f(x) = \lim_{h \rightarrow 0} f(a+h)$. Here you use it not to test existence but to *evaluate*.

Derivation: L'Hopital's rule in the clean case

Assume the strongest convenient hypotheses: f and g are differentiable at a , $f(a) = g(a) = 0$, and $g'(a) \neq 0$.

1. For $x \neq a$ near a , write $\frac{f(x)}{g(x)} = \frac{f(x) - f(a)}{g(x) - g(a)}$. *Reason:* $f(a) = g(a) = 0$, so we have subtracted nothing.
2. Divide numerator and denominator by $(x - a)$:

$$\frac{f(x)}{g(x)} = \frac{\frac{f(x) - f(a)}{x - a}}{\frac{g(x) - g(a)}{x - a}}.$$

Reason: legal because $x \neq a$ throughout the limit process.

3. Let $x \rightarrow a$. Each of the two difference quotients is exactly the definition of a derivative at a from 12_03, so the numerator tends to $f'(a)$ and the denominator to $g'(a)$.
4. Since $g'(a) \neq 0$, the quotient rule for limits (12_01) applies, giving

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{f'(a)}{g'(a)}.$$

That is a complete and honest proof of the case you will meet most often. The general theorem — where you only know $f, g \rightarrow 0$ and nothing about differentiability *at* a , or where the form is $\frac{\infty}{\infty}$ — is proved in Class 12 using Cauchy's Mean Value Theorem. Until you have met that theorem, treat the general statement in Section 2 as an imported result, and treat the derivation above as the reason it is believable.

A warning you must internalise: L'Hopital cannot prove the standard limits. Applying it to $\lim_{x \rightarrow 0} \frac{\sin x}{x}$ requires knowing $\frac{d}{dx} \sin x = \cos x$ — and that derivative was itself established in 12_03 using $\frac{\sin x}{x} \rightarrow 1$. The argument is circular. The same objection applies to $\frac{e^x - 1}{x}$. Use L'Hopital to evaluate, never to justify a junction.

Derivation: why series expansions give the right answer

Suppose the denominator vanishes like x^m and the numerator's expansion is $c_m x^m + c_{m+1} x^{m+1} + \dots$ with $c_m \neq 0$.

1. Write the numerator as $c_m x^m + R(x)$, where $R(x)$ collects everything of degree $m + 1$ and higher. *Reason:* an expansion may always be split at any chosen order.
2. Then $\frac{c_m x^m + R(x)}{x^m} = c_m + \frac{R(x)}{x^m}$. *Reason:* term-by-term division.
3. $R(x)$ is a sum of terms each carrying a factor x^{m+1} at least, so $\frac{R(x)}{x^m}$ carries a factor x and tends to 0. *Reason:* algebra of limits.
4. Therefore the limit is c_m : **the answer is the coefficient of x^m in the numerator's expansion, and nothing else matters.**

This is exactly why truncating early is fatal and why truncating late is merely wasteful.

Procedure A: shifting the point — put $x = a + h$

1. Read off a , the point being approached, and write $x = a + h$ with $h \rightarrow 0$. *Why:* it drags the problem to the only place your standard limits live.

2. Rewrite **every** occurrence of x as $a + h$ — including inside every trigonometric, exponential and logarithmic argument. *Why:* a partial substitution mixes two coordinate systems and gives garbage.
3. Simplify all trigonometric functions of $a + h$ using exact identities (use the table in Section 2). *Why:* this is the step that converts sin and tan of a large angle into sin and tan of a small one.
4. Confirm that the form is still $\frac{0}{0}$ in the new variable. *Why:* if it is not, direct substitution now finishes the job in one line.
5. Route the expression through a standard limit from 12_02, matching arguments as usual. *Why:* the junctions are now legal, since the argument tends to 0.

Procedure B: applying L'Hopital's rule safely

1. Substitute the limiting value. **Confirm** the form is $\frac{0}{0}$ or $\frac{\infty}{\infty}$. If it is anything else, stop — the rule does not apply. *Why:* this single check kills the most common misuse.
2. If the form is $0 \times \infty$, $\infty - \infty$ or 1^∞ , convert it first (Procedure E, or the conjugate trick of 12_05, or the 1^∞ rule of 12_02). *Why:* L'Hopital reads fractions only.
3. Differentiate the numerator and the denominator **separately**. Write $\frac{f'(x)}{g'(x)}$, never $\left(\frac{f}{g}\right)'$. *Why:* the quotient rule answers a different question entirely.
4. Substitute again. If you get a number, that is the answer. If you get $\frac{0}{0}$ or $\frac{\infty}{\infty}$ again, repeat from step 3. *Why:* the rule may be iterated, but only while the form remains indeterminate at each stage.
5. If the successive ratios never settle — if $\frac{f'}{g'}$ oscillates or grows without pattern — abandon L'Hopital and use another method. *Why:* condition 3 of the theorem has failed, so the rule yields no information at all.

Illustration ($\frac{0}{0}$, twice). $\lim_{x \rightarrow 0} \frac{e^x - 1 - x}{x^2}$. At $x = 0$: $1 - 1 - 0 = 0$ over 0, so the form is $\frac{0}{0}$. Differentiate: $\frac{e^x - 1}{2x}$, still $\frac{0}{0}$. Differentiate again: $\frac{e^x}{2} \rightarrow \frac{1}{2}$. Series confirms it instantly: the numerator is $\frac{x^2}{2} + \dots$, so the answer is the coefficient $\frac{1}{2}$.

Illustration ($\frac{\infty}{\infty}$). $\lim_{x \rightarrow \infty} \frac{\ln x}{x}$. Both parts tend to ∞ . Differentiating gives $\frac{1/x}{1} = \frac{1}{x} \rightarrow 0$. So logarithms lose to powers at infinity — a fact worth carrying.

Where L'Hopital belongs in your exam strategy. It is not in the Class 11 CBSE syllabus, so a Boards answer script must show the syllabus method (factor, rationalise, or route through a junction). Use L'Hopital in three places: to check an answer you obtained the official way, to move fast in a JEE Main MCQ where only the number is scored, and to attack a limit whose algebra is genuinely unpleasant.

Procedure C: the series-expansion method

1. Confirm the form is $\frac{0}{0}$ and that the limit is taken at $x = 0$. If it is taken elsewhere, shift first with Procedure A. *Why:* the table in Section 2 expands about 0 only.
2. Determine m , the order to which the denominator vanishes — the power of x that survives after you expand it. *Why:* m decides how much of the numerator you need.

- Expand every function in the numerator up to and including the x^m term, and no further. *Why:* Section 3's derivation shows higher terms contribute nothing.
- Combine and cancel. Everything below order m must cancel — if it does not, you have made an algebra slip, since the numerator was known to vanish. *Why:* this is a free error check, and you should use it every time.
- The limit is the coefficient of x^m in the simplified numerator divided by the coefficient of x^m in the denominator.

Procedure D: limits as $x \rightarrow -\infty$

- Substitute $x = -t$, so that $t \rightarrow +\infty$, and rewrite the whole expression in t . *Why:* this converts an unfamiliar direction into 12_05's familiar one, and it makes every sign explicit rather than remembered.
- Alternatively, work directly in x but write $\sqrt{x^2} = -x$ wherever you take an x out of a square root, because x is negative. *Why:* $\sqrt{\cdot}$ is never negative, so the extracted factor must be $|x|$.
- Proceed with the 12_05 toolkit: rationalise an $\infty - \infty$ surd, then divide by the highest power.
- Sanity-check the sign of your answer against the graph's behaviour. *Why:* the entire trap in this family is a sign, so a quick plausibility check catches it.

Procedure E: converting a $0 \times \infty$ form

- Identify which factor tends to 0 and which to ∞ . *Why:* you cannot convert what you have not diagnosed.
- Move one factor to the denominator as its reciprocal: $f \cdot g = \frac{f}{1/g}$ gives $\frac{0}{0}$, and $f \cdot g = \frac{g}{1/f}$ gives $\frac{\infty}{\infty}$. *Why:* both are legal rewrites; they merely produce different-looking work.
- Prefer the version whose denominator is simple to differentiate or to expand. *Why:* the wrong choice can make an easy limit harder than the original.
- Finish with a standard limit, a series, or L'Hopital.

Illustration 1. $\lim_{x \rightarrow \infty} x \sin \frac{1}{x}$ is $\infty \times 0$. Put $t = \frac{1}{x}$, so $t \rightarrow 0^+$ and the expression becomes $\frac{\sin t}{t} \rightarrow 1$. No calculus needed — the substitution alone did it.

Illustration 2. $\lim_{x \rightarrow 0^+} x \ln x$ is $0 \times (-\infty)$. Writing it as $\frac{\ln x}{1/x}$ gives $\frac{\infty}{\infty}$, and L'Hopital gives $\frac{1/x}{-1/x^2} = -x \rightarrow 0$. So the limit is 0. Had you instead written $\frac{x}{1/\ln x}$, differentiating would produce $-x(\ln x)^2$ — worse than what you started with. That is step 3 of the procedure earning its keep.

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

Evaluate $\lim_{x \rightarrow \pi/2} \frac{\tan 2x}{x - \frac{\pi}{2}}$.

Step 1 — Check the form. As $x \rightarrow \frac{\pi}{2}$, the denominator $\rightarrow 0$. The numerator is $\tan 2x$ with $2x \rightarrow \pi$, and $\tan \pi = 0$. So the form is $\frac{0}{0}$ and must be resolved. There is nothing to factor and no junction available, because the argument is heading to π , not to 0 — this is precisely the situation Procedure A is built for.

Step 2 — Shift the point. Put $x = \frac{\pi}{2} + h$, so $h \rightarrow 0$. Then

$$x - \frac{\pi}{2} = h, \quad 2x = \pi + 2h.$$

Step 3 — Simplify with an exact identity. Since $\tan$ has period π ,

$$\tan 2x = \tan(\pi + 2h) = \tan 2h.$$

Step 4 — Rewrite and route through a junction. The limit becomes

$$\lim_{h \rightarrow 0} \frac{\tan 2h}{h} = \lim_{h \rightarrow 0} \left(\frac{\tan 2h}{2h} \right) \cdot 2.$$

The bracket is the standard limit $\frac{\tan(\text{box})}{\text{box}} \rightarrow 1$ from 12_02, now legal because $2h \rightarrow 0$.

Step 5 — Read off.

$$\lim_{x \rightarrow \pi/2} \frac{\tan 2x}{x - \frac{\pi}{2}} = 1 \cdot 2 = \boxed{2}.$$

Presentation note: in a Boards answer, write the substitution line “Let $x = \frac{\pi}{2} + h$, so that $h \rightarrow 0$ as $x \rightarrow \frac{\pi}{2}$ ” explicitly. That line carries a method mark on its own.

Example 2 — [JEE Main Level]

Evaluate $\lim_{x \rightarrow -\infty} \left(\sqrt{x^2 + x} + x \right)$.

Step 1 — Diagnose. As $x \rightarrow -\infty$ we have $x^2 + x \rightarrow +\infty$, so $\sqrt{x^2 + x} \rightarrow +\infty$, while $x \rightarrow -\infty$. The sum is therefore $\infty - \infty$: indeterminate. (Note carefully that at $x \rightarrow +\infty$ it would be the difference $\sqrt{x^2 + x} - x$ that is indeterminate. At $-\infty$ the roles swap.)

Step 2 — Substitute $x = -t$ **with** $t \rightarrow +\infty$. This is Procedure D, step 1.

$$\sqrt{x^2 + x} + x = \sqrt{t^2 - t} - t.$$

Every sign is now explicit and there is nothing left to remember.

Step 3 — Rationalise (the 12_05 conjugate trick).

$$\sqrt{t^2 - t} - t = \frac{(t^2 - t) - t^2}{\sqrt{t^2 - t} + t} = \frac{-t}{\sqrt{t^2 - t} + t}.$$

Step 4 — Divide by the highest power. Since $t > 0$, $\sqrt{t^2 - t} = t\sqrt{1 - \frac{1}{t}}$, so dividing numerator and denominator by t :

$$= \frac{-1}{\sqrt{1 - \frac{1}{t}} + 1}.$$

Step 5 — Take the limit. As $t \rightarrow \infty$, $\frac{1}{t} \rightarrow 0$:

$$\lim_{t \rightarrow \infty} \frac{-1}{\sqrt{1 - \frac{1}{t}} + 1} = \frac{-1}{1 + 1} = \boxed{-\frac{1}{2}}.$$

Working directly in x (the sign-aware route). Multiply and divide by the conjugate without substituting:

$$\sqrt{x^2 + x} + x = \frac{x}{\sqrt{x^2 + x} - x}.$$

Now extract x from the root. Because $x < 0$, $\sqrt{x^2 + x} = |x|\sqrt{1 + \frac{1}{x}} = -x\sqrt{1 + \frac{1}{x}}$. Hence

$$= \frac{x}{-x\sqrt{1 + \frac{1}{x}} - x} = \frac{x}{-x\left(\sqrt{1 + \frac{1}{x}} + 1\right)} = \frac{-1}{\sqrt{1 + \frac{1}{x}} + 1} \rightarrow -\frac{1}{2}.$$

Same answer. The single place where a student loses this mark is writing $\sqrt{x^2 + x} = x\sqrt{1 + \frac{1}{x}}$, which asserts that a square root is negative.

Figure brief: Graph of $y = \sqrt{x^2 + x} + x$ over $x \in [-40, -2]$, flattening onto the horizontal asymptote $y = -\frac{1}{2}$, with the asymptote drawn as a dashed line and labelled.

Example 3 — [JEE Main — Speed Trap]

Evaluate $\lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x^3}$.

The trap: premature approximation. You know that $\tan x \approx x$ and $\sin x \approx x$ for small x . Substituting both into the numerator gives $x - x = 0$, and the seductive answer is

$$\frac{0}{x^3} = 0.$$

That answer is wrong.

Why it fails. “ $\sin x \approx x$ ” is a statement about the *first* term only; the error it hides is of size x^3 . You are dividing by x^3 . The thing you threw away is the answer. The rule from Procedure C is unforgiving here: the denominator vanishes to order $m = 3$, so the numerator must be expanded to order x^3 — not order x .

Correct method 1 — factor and use junctions (the 12_02 route).

$$\tan x - \sin x = \frac{\sin x}{\cos x} - \sin x = \sin x \left(\frac{1 - \cos x}{\cos x} \right).$$

Therefore

$$\frac{\tan x - \sin x}{x^3} = \frac{\sin x}{\underbrace{x}_{\rightarrow 1}} \cdot \frac{1 - \cos x}{\underbrace{x^2}_{\rightarrow 1/2}} \cdot \frac{1}{\underbrace{\cos x}_{\rightarrow 1}} \rightarrow 1 \cdot \frac{1}{2} \cdot 1 = \frac{1}{2}.$$

Correct method 2 — series (the fast route). With $m = 3$, expand each function to order x^3 :

$$\tan x = x + \frac{x^3}{3} + \cdots, \quad \sin x = x - \frac{x^3}{6} + \cdots.$$

Subtracting, the x terms cancel exactly (the free error check of Procedure C, step 4):

$$\tan x - \sin x = \left(\frac{1}{3} + \frac{1}{6}\right)x^3 + \cdots = \frac{x^3}{2} + \cdots.$$

The coefficient of x^3 is $\frac{1}{2}$, so the limit is $\frac{1}{2}$.

Correct method 3 — L'Hopital (the verification route). The form is $\frac{0}{0}$ at every stage below, which is what makes each step legal.

$$\frac{\tan x - \sin x}{x^3} \rightarrow \frac{\sec^2 x - \cos x}{3x^2} \rightarrow \frac{2\sec^2 x \tan x + \sin x}{6x} \rightarrow \frac{\cos x - 4\sec^2 x + 6\sec^4 x}{6}.$$

At $x = 0$ the last expression is $\frac{1 - 4 + 6}{6} = \frac{3}{6} = \frac{1}{2}$. Three rounds of differentiation for what the series gave in one line — which is exactly why JEE Advanced students reach for the expansion first.

$$\lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x^3} = \boxed{\frac{1}{2}}.$$

Example 4 — [JEE Advanced Level]

Evaluate $\lim_{x \rightarrow 0} \frac{(1+x)^{1/x} - e}{x}$.

Step 1 — Diagnose. From 12_02, $(1+x)^{1/x} \rightarrow e$, so the numerator $\rightarrow 0$ and the denominator $\rightarrow 0$: form $\frac{0}{0}$. This one resists every earlier tool. The 1^∞ rule tells you the *value* e but says nothing about the *rate* at which $(1+x)^{1/x}$ approaches e — and the rate is the whole question. Expansion is the correct instrument.

Step 2 — Convert the power to an exponential. For x near 0 with $x \neq 0$ and $1+x > 0$,

$$(1+x)^{1/x} = \exp\left(\frac{\ln(1+x)}{x}\right).$$

Step 3 — Expand the logarithm. For $|x| < 1$ the table gives $\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots$, so

$$\frac{\ln(1+x)}{x} = 1 - \frac{x}{2} + \frac{x^2}{3} - \cdots.$$

Step 4 — Decide the order needed. The denominator of the original expression is x , so $m = 1$: we need the numerator only to order x . Write

$$\frac{\ln(1+x)}{x} = 1 + u, \quad u = -\frac{x}{2} + \frac{x^2}{3} - \cdots,$$

so that $u \rightarrow 0$ and u has no constant term.

Step 5 — Expand the exponential. Using $e^{1+u} = e \cdot e^u$ and $e^u = 1 + u + \frac{u^2}{2!} + \cdots$:

$$(1+x)^{1/x} = e \left(1 + u + \frac{u^2}{2} + \cdots\right).$$

Since u starts at order x , the term $\frac{u^2}{2}$ starts at order x^2 and is beyond the order we need. Keeping terms up to x :

$$(1+x)^{1/x} = e \left(1 - \frac{x}{2} + (\text{terms of order } x^2 \text{ and higher}) \right) = e - \frac{ex}{2} + \dots$$

Step 6 — Assemble. The constant e cancels — the error check of Procedure C passing, as it must:

$$\frac{(1+x)^{1/x} - e}{x} = \frac{-\frac{ex}{2} + (\text{order } x^2)}{x} = -\frac{e}{2} + (\text{order } x).$$

Step 7 — Read off.

$$\lim_{x \rightarrow 0} \frac{(1+x)^{1/x} - e}{x} = \boxed{-\frac{e}{2}}.$$

Deep point. This is the standard first-order correction to the definition of e : $(1+x)^{1/x} \approx e(1 - \frac{x}{2})$ for small x . It also explains a numerical fact — computing e as $(1 + \frac{1}{n})^n$ converges only about as fast as $\frac{1}{n}$, which is painfully slow. L'Hopital can be forced through this problem, but differentiating $(1+x)^{1/x}$ requires logarithmic differentiation and produces an expression far uglier than the two-line expansion above. When the numerator is a composition of a log inside an exponential, expand.

Section 5: Practice Exercises

1. **[CBSE]** Evaluate $\lim_{x \rightarrow \pi/2} \frac{\cos x}{x - \frac{\pi}{2}}$.
2. **[CBSE]** Evaluate $\lim_{x \rightarrow \pi} \frac{\sin x}{\pi - x}$.
3. **[JEE Main]** Evaluate $\lim_{x \rightarrow \pi/4} \frac{1 - \tan x}{1 - \sqrt{2} \sin x}$.
4. **[JEE Main]** Evaluate $\lim_{x \rightarrow -\infty} \left(\sqrt{x^2 + 3x} + x \right)$.
5. **[JEE Main]** Evaluate $\lim_{x \rightarrow \infty} x^2 \left(1 - \cos \frac{2}{x} \right)$. (A $\infty \times 0$ form — convert it.)
6. **[JEE Main]** Evaluate $\lim_{x \rightarrow 1} (1-x) \tan \frac{\pi x}{2}$. (Both a shift and a $0 \times \infty$ form.)
7. **[JEE Advanced]** Evaluate $\lim_{x \rightarrow 0} \frac{x - \sin x}{x^3}$.
8. **[JEE Advanced]** Evaluate $\lim_{x \rightarrow 0} \frac{\ln(1+x) - x + \frac{x^2}{2}}{x^3}$.

Answers: 1. -1 2. 1 3. 2 4. $-\frac{3}{2}$ 5. 2 6. $\frac{2}{\pi}$ 7. $\frac{1}{6}$ 8. $\frac{1}{3}$

Section 6: Crack the MCQ

Q1. $\lim_{x \rightarrow -\infty} \left(\sqrt{x^2 + x} - x \right)$ equals

- (A) $\frac{1}{2}$ (B) $-\frac{1}{2}$ (C) $+\infty$ (D) 0

Answer: (C). As $x \rightarrow -\infty$, $\sqrt{x^2 + x} \rightarrow +\infty$ and $-x \rightarrow +\infty$ as well, so this is $\infty + \infty$, not an indeterminate form at all — the expression grows without bound. (Substituting $x = -t$ makes it obvious: $\sqrt{t^2 - t} + t \rightarrow \infty$.)

Distractor Analysis. - **(A)** $\frac{1}{2}$ — the answer to the *same expression* at $+\infty$. The student memorised “ $\sqrt{x^2 + bx} - x \rightarrow \frac{b}{2}$ ” without the direction attached to it. This is the single most-punished reflex in the whole family. - **(B)** $-\frac{1}{2}$ — the answer to $\sqrt{x^2 + x} + x$ at $-\infty$. The student correctly remembered that a sign flips somewhere, and flipped the wrong one. - **(D)** 0 — the “ $\infty - \infty = 0$ ” fallacy from 12_05, applied here to an expression that is not even of that form.

Q2. $\lim_{x \rightarrow 0} \frac{x^2 + 3x + 2}{x + 1}$ equals

(A) 2 (B) 3 (C) 0 (D) does not exist

Answer: (A). Direct substitution gives $\frac{0 + 0 + 2}{0 + 1} = 2$. The form is not indeterminate, so no special machinery is needed — and none is permitted.

Distractor Analysis. - **(B)** 3 — the classic illegal L'Hopital. Differentiating top and bottom gives $\frac{2x+3}{1} \rightarrow 3$. The rule was applied to a limit that was never $\frac{0}{0}$, so the “answer” is meaningless. This is why Procedure B step 1 exists. - **(C)** 0 — the student saw $x \rightarrow 0$ and reported the value of x rather than the value of the function. - **(D)** — assumes that any question presented as “evaluate the limit” must be a trick with a non-existent answer.

Q3. $\lim_{x \rightarrow 0} \frac{\tan x - x}{x^3}$ equals

(A) $\frac{1}{3}$ (B) $\frac{1}{6}$ (C) 0 (D) $\frac{1}{2}$

Answer: (A). With $m = 3$, expand $\tan x = x + \frac{x^3}{3} + \dots$. The x terms cancel, leaving $\frac{x^3}{3} + \dots$, so the limit is the coefficient $\frac{1}{3}$.

Distractor Analysis. - **(B)** $\frac{1}{6}$ — the answer to $\frac{x - \sin x}{x^3}$. The student confused the $\tan$ expansion (coefficient $+\frac{1}{3}$) with the $\sin$ expansion (coefficient $-\frac{1}{6}$). Learn the two coefficients as a contrasting pair. - **(C)** 0 — premature truncation: $\tan x \approx x$ makes the numerator “zero”. Exactly the Example 3 trap. - **(D)** $\frac{1}{2}$ — the answer to $\frac{\tan x - \sin x}{x^3}$, misremembered as belonging to this problem.

Q4. $\lim_{x \rightarrow \pi/2} \frac{1 - \sin x}{(x - \frac{\pi}{2})^2}$ equals

(A) $\frac{1}{2}$ (B) 1 (C) 0 (D) $-\frac{1}{2}$

Answer: (A). Shift with $x = \frac{\pi}{2} + h$. Then $\sin x = \cos h$ and $x - \frac{\pi}{2} = h$, so the expression becomes $\frac{1 - \cos h}{h^2} \rightarrow \frac{1}{2}$ by the 12_02 junction.

Distractor Analysis. - **(B)** 1 — the student reached the correct $\frac{1 - \cos h}{h^2}$ but quoted its value as 1, confusing it with $\frac{\sin h}{h} \rightarrow 1$. The half is the whole point of that junction. - **(C)** 0 — the student observed $1 - \sin x \rightarrow 0$ and stopped, ignoring that the denominator also vanishes. -

(D) $-\frac{1}{2}$ — a sign slip from writing $\sin(\frac{\pi}{2} + h) = -\cos h$. Check the shift table: it is $\cos(\frac{\pi}{2} + h)$ that carries the minus sign, not $\sin$.

Section 7: Common Student Pitfalls and Pro-Tips

1. **Substituting $x = a + h$ in only some places.** If a single x survives inside a trigonometric argument, the expression now mixes two variables and is meaningless. **Correction:** rewrite the entire expression in h before you simplify anything, then check that no x remains.
2. **Applying L'Hopital to a form that is not indeterminate.** This manufactures wrong answers with total confidence, because nothing in the algebra complains. **Correction:** substitute first, every single time, and write down which of $\frac{0}{0}$ or $\frac{\infty}{\infty}$ you have before you differentiate.
3. **Differentiating the quotient instead of the two parts.** Writing $(\frac{f}{g})' = \frac{f'g - fg'}{g^2}$ and calling it L'Hopital. **Correction:** L'Hopital forms $\frac{f'}{g'}$ — top and bottom differentiated independently, no cross terms, no g^2 .
4. **Forgetting to re-check the form between successive applications.** After the first differentiation the form may have become $\frac{\text{number}}{0}$, in which case a second application is illegal and the limit is infinite. **Correction:** treat each round as a fresh problem starting at Procedure B step 1.
5. **Concluding “no limit” because $\frac{f'}{g'}$ has no limit.** For $\lim_{x \rightarrow \infty} \frac{x + \sin x}{x}$ the ratio of derivatives is $1 + \cos x$, which oscillates forever — yet the original limit is plainly 1, since $\frac{x + \sin x}{x} = 1 + \frac{\sin x}{x}$ and $|\frac{\sin x}{x}| \leq \frac{1}{x} \rightarrow 0$. **Correction:** when L'Hopital fails to converge, it has told you nothing; switch methods.
6. **Truncating an expansion below the order of the denominator.** The commonest single error in JEE Advanced limit questions. **Correction:** find m first, expand to x^m , and use the cancellation of all lower-order terms as your proof that you did it right.
7. **Expanding about the wrong point.** The table expands about $x = 0$. You cannot write “ $\sin x = x - \frac{x^3}{6}$ ” while $x \rightarrow \pi$. **Correction:** shift with Procedure A first; then the new variable does tend to 0 and the table applies.
8. **Writing $\sqrt{x^2} = x$ when x is negative.** **Correction:** $\sqrt{x^2} = |x|$ always. When $x \rightarrow -\infty$, extract $-x$ from the root, or dodge the issue entirely by substituting $x = -t$.
9. **Treating $0 \times \infty$ as 0.** A vanishing factor and an exploding factor can balance at any value whatsoever. **Correction:** convert to $\frac{0}{0}$ or $\frac{\infty}{\infty}$ and resolve.

Pro-Tip (speed and accuracy). Build a two-second habit: before doing anything, ask *where* the limit is taken. If the answer is not “0 or ∞ ”, your first pen stroke is the substitution line. Second habit: memorise just four expansion coefficients — $\sin x$ gives $-\frac{1}{6}$, $\tan x$ gives $+\frac{1}{3}$, e^x gives $+\frac{1}{6}$, and $\ln(1 + x)$ gives $+\frac{1}{3}$, each at the cubic term. Those four numbers answer a startling share of JEE Advanced limit questions on sight. Third habit: in an MCQ, solve by the syllabus method and *verify* with L'Hopital — two independent routes agreeing is worth more than either route done twice.

Section 8: Cheat Sheet Box

ADVANCED LIMIT TECHNIQUES

- **Shift the point:** $\lim_{x \rightarrow a} f(x) = \lim_{h \rightarrow 0} f(a+h)$. Rewrite every x ; simplify with exact identities.
- **Shift table:** $\sin(\frac{\pi}{2} + h) = \cos h$; $\cos(\frac{\pi}{2} + h) = -\sin h$; $\tan(\frac{\pi}{2} + h) = -\cot h$; $\tan(\pi + h) = \tan h$; $\tan(\frac{\pi}{4} + h) = \frac{1+\tan h}{1-\tan h}$.
- **L'Hopital (Class 12 bridge tool):** for $\frac{0}{0}$ or $\frac{\infty}{\infty}$ **only**, with f, g differentiable near a , $g' \neq 0$, and $\lim_{x \rightarrow a} \frac{f'}{g'}$ existing — then $\lim_{x \rightarrow a} \frac{f}{g} = \lim_{x \rightarrow a} \frac{f'}{g'}$. Differentiate top and bottom **separately**. Re-check the form before every repeat. Never use it to prove a standard limit (circular).
- **Maclaurin table (about $x = 0$):** $e^x = 1 + x + \frac{x^2}{2!} + \dots$ (all x); $\sin x = x - \frac{x^3}{3!} + \dots$ (all x); $\cos x = 1 - \frac{x^2}{2!} + \dots$ (all x); $\tan x = x + \frac{x^3}{3} + \dots$ ($|x| < \frac{\pi}{2}$); $\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots$ ($-1 < x \leq 1$); $(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \dots$ ($|x| < 1$ for non-integer n); $\tan^{-1} x = x - \frac{x^3}{3} + \dots$, $\sin^{-1} x = x + \frac{x^3}{6} + \dots$ ($|x| \leq 1$).
- **Order rule:** denominator vanishes like $x^m \Rightarrow$ expand the numerator to order x^m ; the answer is the ratio of the x^m coefficients. All lower-order terms must cancel.
- **Standard high-order results:** $\frac{x - \sin x}{x^3} \rightarrow \frac{1}{6}$; $\frac{\tan x - x}{x^3} \rightarrow \frac{1}{3}$; $\frac{\tan x - \sin x}{x^3} \rightarrow \frac{1}{2}$; $\frac{e^x - 1 - x}{x^2} \rightarrow \frac{1}{2}$; $\frac{(1+x)^{1/x} - e}{x} \rightarrow -\frac{e}{2}$.
- **At $-\infty$:** put $x = -t$, $t \rightarrow +\infty$. Always $\sqrt{x^2} = |x| = -x$ for $x < 0$. Pair to memorise: $\sqrt{x^2 + bx} - x \rightarrow \frac{b}{2}$ as $x \rightarrow +\infty$; $\sqrt{x^2 + bx} + x \rightarrow -\frac{b}{2}$ as $x \rightarrow -\infty$.
- $0 \times \infty$: rewrite as $\frac{f}{1/g}$ (gives $\frac{0}{0}$) or $\frac{g}{1/f}$ (gives $\frac{\infty}{\infty}$); pick the easier one, then resolve.

Memory aids: - “If it isn’t at zero, move zero to it.” (the shift substitution) - “Check the form, then differentiate — separately.” (L'Hopital discipline) - “Expand as deep as the denominator is low.” (the order rule) - “Root of a square wears a modulus.” ($\sqrt{x^2} = |x|$) - “Zero times infinity is a question, not zero.”

Addendum A — Derivatives from Functional Equations

(extends Subtopic 03 — Derivatives and First Principle)

Exam Relevance: CBSE Boards do not test this. JEE Main asks it as a one-step application — you are handed the functional equation and $f'(0)$, and must produce $f'(x)$. JEE Advanced builds full problems on it: identify f from the equation, then answer a question about its roots, its range or a further derivative.

Why the standard rules are useless here

Every differentiation rule in 12_04 needs a *formula* for f . In a functional-equation problem you are given no formula — only a rule that f obeys, such as $f(x+y) = f(x)f(y)$ for all real

x and y . The rules have nothing to act on.

The first principle, by contrast, needs only the ability to say something about $f(x + h)$. And that is precisely what a functional equation gives you: it lets you **split** $f(x + h)$ **into a piece involving x and a piece involving h** . Once the x -part is factored or added out, the surviving h -limit is a pure number — almost always $f'(0)$ or $f'(1)$ — and the whole derivative collapses.

The master move. Start from

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x + h) - f(x)}{h},$$

substitute the functional equation into $f(x + h)$, and arrange the result so that everything depending on x comes outside the limit.

Procedure: solving a functional-equation derivative problem

1. **Find $f(0)$ (additive-type) or $f(1)$ (multiplicative-type) by substituting convenient values.** Typically $x = y = 0$ or $x = y = 1$. *Why:* the limit you are about to build is a derivative *at that anchor point*, so you must know the function's value there.
2. **Discard the trivial solution.** Equations like $f(0) = f(0)^2$ give $f(0) \in \{0, 1\}$; the branch $f(0) = 0$ usually forces $f \equiv 0$. Say so explicitly and move on. *Why:* examiners award the case discussion, and skipping it leaves the solution logically incomplete.
3. **Write the first principle and substitute the functional equation into $f(x + h)$.** *Why:* this is the only step that uses the given information.
4. **Factor or subtract out everything involving x .** *Why:* what remains is a constant that the problem has told you (or that you can name as $f'(0)$ or $f'(1)$).
5. **Recognise the surviving limit as a derivative at the anchor point**, using $f(0)$ or $f(1)$ from step 1. *Why:* $\lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h}$ is literally $f'(0)$ by definition (12_03).
6. **Solve the resulting relation for f if the question asks for the function itself.** *Why:* the four standard families below have known answers you can quote and then verify by substitution.

The four standard families

Functional equation	Anchor value	Derivative relation	The function
$f(x + y) = f(x) + f(y)$	$f(0) = 0$	$f'(x) = f'(0)$	$f(x) = kx, k = f'(0)$
$f(x + y) = f(x)f(y)$	$f(0) = 1$ (non-trivial)	$f'(x) = f'(0)f(x)$	$f(x) = e^{kx}, k = f'(0)$
$f(xy) = f(x) + f(y),$ $x, y > 0$	$f(1) = 0$	$f'(x) = \frac{f'(1)}{x}$	$f(x) = c \ln x,$ $c = f'(1)$
$f(xy) = f(x)f(y),$ $x, y > 0$	$f(1) = 1$ (non-trivial)	$f'(x) = \frac{f'(1)f(x)}{x}$	$f(x) = x^n, n = f'(1)$

Each row assumes f is differentiable where stated. Learn the middle column, not the right — the derivative relation is what questions actually ask for, and it is what you can rebuild from first principle in twenty seconds if memory fails.

Worked Example A1 — [JEE Main Level]

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfy $f(x+y) = f(x)f(y)$ for all $x, y \in \mathbb{R}$, with $f(x) \neq 0$ for every x , and suppose f is differentiable at $x = 0$ with $f'(0) = 2$. Find $f'(x)$, and hence $f'(3)$.

Step 1 — Anchor value. Put $x = y = 0$: $f(0) = f(0)^2$, so $f(0) = 0$ or $f(0) = 1$. If $f(0) = 0$, then for any x , $f(x) = f(x+0) = f(x)f(0) = 0$, contradicting $f(x) \neq 0$. Hence $f(0) = 1$.

Step 2 — First principle with the equation substituted.

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{f(x)f(h) - f(x)}{h}.$$

Step 3 — Factor out the x -part. $f(x)$ does not depend on h , so it comes outside:

$$f'(x) = f(x) \lim_{h \rightarrow 0} \frac{f(h) - 1}{h}.$$

Step 4 — Recognise the surviving limit. Since $f(0) = 1$,

$$\lim_{h \rightarrow 0} \frac{f(h) - 1}{h} = \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h} = f'(0) = 2.$$

Step 5 — Conclude.

$$f'(x) = 2f(x) \quad \text{for every } x, \quad \text{so } f'(3) = 2f(3).$$

Step 6 — Identify f . The relation $f'(x) = 2f(x)$ with $f(0) = 1$ is satisfied by $f(x) = e^{2x}$ (check: $f(x+y) = e^{2x+2y} = e^{2x}e^{2y} = f(x)f(y)$, and $f'(0) = 2$). Hence

$$f'(3) = 2e^6.$$

Deep point. Read step 3 again. You were told f is differentiable *only* at 0 — nothing was assumed anywhere else. Yet the calculation produced a finite $f'(x)$ at every x . The functional equation propagates differentiability across the whole line. That observation is itself a favourite Advanced-level assertion-reason question.

Worked Example A2 — [JEE Advanced Level]

A differentiable function $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfies

$$f(x+y) = f(x) + f(y) + 2xy - 1 \quad \text{for all } x, y \in \mathbb{R},$$

and $f'(0) = a$. Find $f(x)$, and determine all values of a for which the equation $f(x) = 0$ has real roots.

Step 1 — Anchor value. Put $x = y = 0$: $f(0) = 2f(0) - 1$, hence $f(0) = 1$.

Step 2 — First principle.

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{[f(x) + f(h) + 2xh - 1] - f(x)}{h}.$$

Step 3 — Cancel and split. The $f(x)$ terms cancel:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(h) - 1 + 2xh}{h} = \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h} + \lim_{h \rightarrow 0} 2x = f'(0) + 2x,$$

using $f(0) = 1$ from step 1 and the algebra of limits from 12_01.

Step 4 — Integrate the derivative back. So $f'(x) = 2x + a$. A function whose derivative is $2x + a$ is $x^2 + ax + C$ for a constant C ; the condition $f(0) = 1$ forces $C = 1$. Therefore

$$f(x) = x^2 + ax + 1.$$

Step 5 — Verify against the original equation (never skip this on a constructed answer):

$$f(x) + f(y) + 2xy - 1 = (x^2 + ax + 1) + (y^2 + ay + 1) + 2xy - 1 = x^2 + 2xy + y^2 + a(x + y) + 1,$$

which is exactly $(x + y)^2 + a(x + y) + 1 = f(x + y)$. The identity holds for all x, y .

Step 6 — Answer the root question. $x^2 + ax + 1 = 0$ has real roots precisely when its discriminant is non-negative:

$$a^2 - 4 \geq 0 \iff |a| \geq 2 \iff a \in (-\infty, -2] \cup [2, \infty).$$

Worked Example A3 — [JEE Advanced Level]

Let f be differentiable on $(0, \infty)$ and satisfy $f(xy) = f(x) + f(y)$ for all $x, y > 0$, with $f'(1) = 2$. Find $f'(x)$ and the value of $f(e^2)$.

Step 1 — Anchor value. Put $x = y = 1$: $f(1) = 2f(1)$, so $f(1) = 0$.

Step 2 — Engineer a product inside f . The equation splits *products*, not sums, so rewrite the increment as a multiplicative one. For $x > 0$ and h small,

$$x + h = x \left(1 + \frac{h}{x}\right) \implies f(x + h) = f(x) + f\left(1 + \frac{h}{x}\right).$$

This is the whole trick of the multiplicative families: turn $x + h$ into x times something near 1.

Step 3 — First principle.

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x + h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{f\left(1 + \frac{h}{x}\right)}{\frac{h}{x}}.$$

Step 4 — Match the increment to the denominator. Multiply and divide by x (a constant as far as h is concerned), and use $f(1) = 0$:

$$f'(x) = \frac{1}{x} \lim_{h \rightarrow 0} \frac{f\left(1 + \frac{h}{x}\right) - f(1)}{\frac{h}{x}}.$$

Now set $t = \frac{h}{x}$; as $h \rightarrow 0$ so does t , and the limit is the definition of the derivative of f at 1:

$$f'(x) = \frac{f'(1)}{x} = \frac{2}{x}.$$

Step 5 — Recover f and evaluate. A function on $(0, \infty)$ with derivative $\frac{2}{x}$ is $2\ln x + C$, and $f(1) = 0$ gives $C = 0$. So $f(x) = 2\ln x$ — which indeed satisfies $f(xy) = 2\ln(xy) = 2\ln x + 2\ln y = f(x) + f(y)$. Hence

$$f(e^2) = 2\ln e^2 = 4.$$

Practice Exercises

1. **[JEE Main]** $f(x + y) = f(x) + f(y)$ for all real x, y , and f is differentiable at 0 with $f'(0) = 5$. Find $f(x)$ and $f(3)$.
2. **[JEE Main]** $f(x + y) = f(x)f(y)$ for all real x, y , $f(x) \neq 0$, $f(1) = 3$ and $f'(0) = \ln 3$. Find $f'(2)$.
3. **[JEE Main]** $f(x + y) = f(x) + f(y) + 3xy$ for all real x, y , f differentiable with $f'(0) = 2$. Find $f'(x)$ and $f(2)$.
4. **[JEE Advanced]** $f(xy) = f(x) + f(y)$ for all $x, y > 0$, f differentiable with $f'(1) = -1$. Find $f'(x)$ and $f(\frac{1}{e})$.
5. **[JEE Advanced]** $f(xy) = f(x)f(y)$ for all $x, y > 0$, f differentiable and not identically zero, with $f'(1) = 3$. Find $f(x)$ and $f'(2)$.

Answers: 1. $f(x) = 5x$; $f(3) = 15$. 2. $f(x) = 3^x$, so $f'(2) = 9\ln 3$. 3. $f'(x) = 3x + 2$; $f(x) = \frac{3}{2}x^2 + 2x$, so $f(2) = 10$. 4. $f'(x) = -\frac{1}{x}$; $f(x) = -\ln x$, so $f(\frac{1}{e}) = 1$. 5. $f(x) = x^3$; $f'(2) = 12$.

Pitfalls

- **Forgetting to establish $f(0)$ or $f(1)$ first.** Without it you cannot convert $\lim_{h \rightarrow 0} \frac{f(h)-1}{h}$ into $f'(0)$, and the solution stalls at an unnamed limit.
- **Dropping the trivial branch silently.** Show that $f(0) = 0$ forces $f \equiv 0$ before you discard it.
- **Trying to split $f(x + h)$ with a multiplicative equation, or $f(xh)$ with an additive one.** Match the operation in the functional equation to the operation you create in the increment: additive equations want $x + h$; multiplicative-argument equations want $x(1 + \frac{h}{x})$.
- **Quoting $f(x) = e^{kx}$ without checking it fits the given data.** Always substitute your answer back into the original equation, as in Example A2 step 5. If the given numbers are mutually inconsistent, that check is the only thing that will reveal it.

Addendum B — Finding Constants for Differentiability at a Junction

(extends Subtopic 03 — Derivatives and First Principle)

Exam Relevance: CBSE Boards ask the continuity-only version. JEE Main asks the differentiability version — a piecewise f with unknown constants a and b , two conditions, two unknowns — reliably enough that it is worth drilling to reflex.

The two conditions, and why there are exactly two

Subtopic 01 taught you to find a and b so that a *limit* exists. Differentiability at a junction $x = c$ demands strictly more, and the extra demand is what supplies the second equation.

Condition 1 — continuity at c : $\lim_{x \rightarrow c^-} f(x) = \lim_{x \rightarrow c^+} f(x) = f(c)$.

Condition 2 — matching one-sided derivatives: LHD = RHD, where

$$\text{LHD} = \lim_{h \rightarrow 0^+} \frac{f(c-h) - f(c)}{-h}, \quad \text{RHD} = \lim_{h \rightarrow 0^+} \frac{f(c+h) - f(c)}{h}.$$

Condition 1 is not optional extra work — it is *forced*. From 12_03: differentiable at c implies continuous at c . So a function that fails continuity cannot possibly be differentiable, no matter how nicely the slopes appear to match.

The order matters, and here is why. The convenient shortcut — differentiate each piece separately and equate the two derivative formulas at c — is only valid once you know f is continuous at c . Impose continuity **first**. If you jump straight to matching slopes, you can find constants that make two graph pieces parallel while leaving a vertical gap between them, and a gap is a discontinuity, and a discontinuity is not differentiable. Two conditions, applied in that order, and two unknowns: the system closes.

Figure brief: Three small panels of the same piecewise curve at a junction $x = c$ — panel 1 with a vertical gap (continuity fails), panel 2 joined but with a visible corner (continuity holds, slopes disagree), panel 3 joined smoothly with a single tangent line drawn through the junction (both conditions hold).

Procedure: find a and b for differentiability at $x = c$

1. **Write down the two pieces and identify the junction c .** *Why:* everything is local to that one point; the rest of the domain is irrelevant.
2. **Impose continuity.** Compute $\lim_{x \rightarrow c^-} f(x)$ and $\lim_{x \rightarrow c^+} f(x)$ by substituting c into each piece (legal when the pieces are polynomials, rational with non-vanishing denominator, or standard smooth functions), set them equal to each other and to $f(c)$. This is equation (i). *Why:* differentiability forces it, and it is the cheaper of the two equations.
3. **Compute the LHD and the RHD.** Having secured continuity, you may differentiate each piece and evaluate at c : LHD = (left piece)' at c , RHD = (right piece)' at c . Set them equal — equation (ii). *Why:* this is the definition of $f'(c)$ existing.
4. **Solve (i) and (ii) simultaneously** for a and b .
5. **Verify.** Substitute your constants back and check both that the two pieces agree in value at c and that they agree in slope at c . *Why:* a sign slip in either equation is invisible until you check.

Worked Example B1 — [CBSE / JEE Main Level]

Find a and b so that

$$f(x) = \begin{cases} x^2, & x \leq 1 \\ ax + b, & x > 1 \end{cases}$$

is differentiable at $x = 1$.

Step 1 — Junction. $c = 1$, with $f(1) = 1^2 = 1$.

Step 2 — Continuity (equation (i)).

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} x^2 = 1, \quad \lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} (ax + b) = a + b.$$

Equating, and matching $f(1) = 1$:

$$a + b = 1. \quad (\text{i})$$

Step 3 — Slopes (equation (ii)). The left piece has derivative $2x$, giving $\text{LHD} = 2(1) = 2$. The right piece has derivative a , giving $\text{RHD} = a$. Hence

$$a = 2. \quad (\text{ii})$$

Step 4 — Solve. From (ii), $a = 2$; substituting into (i), $b = 1 - 2 = -1$.

Step 5 — Verify. With $a = 2$, $b = -1$ the right piece is $2x - 1$. At $x = 1$ it takes the value $2(1) - 1 = 1$, matching $1^2 = 1$: continuous. Its slope is 2, matching the left slope $2(1) = 2$: differentiable.

$$\boxed{a = 2, b = -1}$$

Contrast worth noting: if the question had asked only for **continuity**, equation (i) alone would answer it, and there would be infinitely many solutions — every pair with $a + b = 1$. It is condition (ii) that pins the answer down to one pair. Whenever a piecewise problem gives you two unknowns and asks for a unique answer, differentiability is being tested.

Worked Example B2 — [JEE Main Level]

Find a and b so that

$$f(x) = \begin{cases} ax^2 + b, & x < 1 \\ \frac{1}{|x|}, & x \geq 1 \end{cases}$$

is differentiable at $x = 1$.

Step 1 — Junction and a simplification. $c = 1$. Near $x = 1$ we have $x > 0$, so $|x| = x$ and the right piece is simply $\frac{1}{x}$, with $f(1) = 1$. Making that simplification *before* differentiating is the entire difficulty of the problem; the modulus is there to tempt you into a sign error.

Step 2 — Continuity (equation (i)).

$$\lim_{x \rightarrow 1^-} (ax^2 + b) = a + b, \quad \lim_{x \rightarrow 1^+} \frac{1}{x} = 1 = f(1).$$
$$a + b = 1. \quad (\text{i})$$

Step 3 — Slopes (equation (ii)). Left piece: derivative $2ax$, so $\text{LHD} = 2a$. Right piece: $\frac{d}{dx}\left(\frac{1}{x}\right) = -\frac{1}{x^2}$, so $\text{RHD} = -1$.

$$2a = -1 \implies a = -\frac{1}{2}. \quad (\text{ii})$$

Step 4 — Solve. From (i), $b = 1 - a = 1 + \frac{1}{2} = \frac{3}{2}$.

Step 5 — Verify. Left piece: $-\frac{1}{2}x^2 + \frac{3}{2}$. At $x = 1$: $-\frac{1}{2} + \frac{3}{2} = 1$, matching $\frac{1}{1} = 1$. Slope at 1: $2(-\frac{1}{2})(1) = -1$, matching $-\frac{1}{1^2} = -1$. Both conditions hold.

$$\boxed{a = -\frac{1}{2}, b = \frac{3}{2}}$$

Practice Exercises

1. **[CBSE]** Find a and b so that $f(x) = \begin{cases} ax + b, & x < 0 \\ \sin x, & x \geq 0 \end{cases}$ is differentiable at $x = 0$.
2. **[JEE Main]** Find a and b so that $f(x) = \begin{cases} x^2 + 3x + a, & x \leq 1 \\ bx + 2, & x > 1 \end{cases}$ is differentiable at $x = 1$.
3. **[JEE Main]** Find a and b so that $f(x) = \begin{cases} x^2 + a, & x \leq 2 \\ bx + 3, & x > 2 \end{cases}$ is differentiable at $x = 2$.
4. **[JEE Main]** Take the same piecewise function as in Exercise 1 above. Find all pairs (a, b) that make f merely **continuous** at $x = 0$, and explain why the answer is a whole family rather than a single pair.
5. **[JEE Advanced]** Show that no choice of the real constant a makes $f(x) = a|x - 1| + 3$ differentiable at $x = 1$ unless $a = 0$, and identify the LHD and RHD in terms of a .

Answers: 1. $a = 1, b = 0$. 2. $b = 5, a = 3$. 3. $b = 4, a = 7$. 4. Continuity alone requires only $b = 0$, with a arbitrary: a one-parameter family $\{(a, 0) : a \in \mathbb{R}\}$. Two unknowns constrained by one equation leave one degree of freedom; only the slope condition removes it. 5. LHD = $-a$ and RHD = a , so LHD = RHD forces $a = 0$; the modulus contributes a corner at $x = 1$ for every non-zero a .

Pitfalls

- **Matching slopes without first imposing continuity.** Parallel pieces separated by a gap are not differentiable. Continuity is equation (i), always.
- **Substituting the junction value into the wrong piece.** Read the inequalities: if the left piece carries $x \leq c$ then $f(c)$ comes from the left piece, and the right-hand limit must be made to agree with it.
- **Differentiating a modulus without resolving it.** $|x|$, $|x - 1|$ and $\frac{1}{|x|}$ must first be rewritten without the bars on the side you are approaching from. A modulus that survives to the differentiation step is a guaranteed sign error.
- **Assuming two unknowns always need differentiability.** If the question says “continuous”, one equation is all you get, and the honest answer is a family, not a pair.

CHEAT SHEET — DIFFERENTIABILITY AT A JUNCTION

- Differentiable at $c \Rightarrow$ continuous at c . So impose continuity **first**: left value = right value = $f(c)$. That is equation (i).

- Then match one-sided derivatives: $\text{LHD} = \text{RHD}$. That is equation (ii).
- Two equations, two unknowns $\Rightarrow$ a unique (a, b) . Continuity alone $\Rightarrow$ a one-parameter family.
- Resolve every modulus on the correct side *before* differentiating.
- Always verify: same value at c , same slope at c .

Chapter 12: Limits and Derivatives — Round 2

Addendum

Subtopics 01–05 and the Round 1 Supplement (Subtopic 06 plus Addenda A and B) between them close this chapter’s function-limit and derivative machinery. Round 2 found exactly one genuine gap, and it is a structural one rather than a missing technique.

Every limit in the chapter so far has been the limit of a **function of a real variable**: x slides continuously toward a , or continuously out to $\pm\infty$, and existence is settled by comparing two sides. Nowhere does the chapter treat the limit of a **sequence** — the behaviour of an n -th term as $n \rightarrow \infty$ through the integers $1, 2, 3, \dots$ only. NCERT never defines it (its Sandwich Theorem, Theorem 4, is stated for real functions $f(x) \leq g(x) \leq h(x)$ on a common domain, with $x \rightarrow a$), yet JEE draws on it steadily: Vinay Kumar’s *Differential Calculus for JEE* gives it a dedicated section, and the sum-form and recursive-sequence questions built on it are standing JEE Advanced items. The chapter also already quotes $\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e$ in Subtopic 02’s formula table without ever saying what an $n \rightarrow \infty$ limit is.

This addendum therefore adds one full subtopic, continuing the Round 1 numbering.

Item	Topic	Read after
Subtopic 07	Limits of Sequences	12_02 (junctions), 12_05 (limits at infinity), 12_06 (advanced techniques)

Checked and found already closed (no Round 2 action): the “existence of a derivative via continuity-first reasoning” pattern. Round 1’s Addendum B delivers it in full — both conditions, the *order* in which they must be imposed, the reason the order matters (parallel pieces separated by a vertical gap), a five-step procedure, two worked examples, five practice problems and a cheat-sheet box. Nothing further is needed there.

Subtopic 07 — Limits of Sequences

Exam Relevance: CBSE Boards do not test sequence limits under Chapter 12, but they do test the idea in disguise through the infinite-GP sum of Chapter 9, and a Boards-level $\sum$ -form limit is well within a 2–3 mark question. JEE Main asks one question most sessions — usually a $\sum$ -form limit resolved by the Chapter 9 summation formulas, or an n -th root of a sum of powers. JEE Advanced pushes into recursively defined sequences (where you must *prove convergence before* you solve for the limit), the Sandwich Theorem applied to a sum whose number of

terms grows, and assertion-reason questions built on the one-way bridge between function limits and sequence limits. NEET does not test mathematics.

Section 1: Concept Intuition

Every limit you have met so far has been a limit of a function of a continuous variable. When you write $\lim_{x \rightarrow \infty} f(x)$, the variable x sweeps through *every* real number on its way out — it slides. When you write $\lim_{x \rightarrow 2} f(x)$, x can creep in from the left or from the right, and the whole existence question is whether those two approaches agree.

A sequence behaves differently, and the difference is the entire point of this subtopic. A sequence a_n is a function whose input is restricted to the natural numbers. As n grows, it does not slide — **it hops**. There is no value between $n = 7$ and $n = 8$, so there is no “left side” and no “right side”, and there is no finite point to approach at all. The only question a sequence limit can ever ask is: *as you go far enough out along the list, do the terms settle onto a single number?*

Think about it this way... picture the Delhi Metro’s Yellow Line. A car on the road above can stop anywhere at all — outside a shop, mid-block, at a signal. That is x . The train below can only ever be at a station: Kashmere Gate, Chandni Chowk, Chawri Bazar. That is n . Now suppose the stations are getting closer and closer together as you head south, converging on some final terminus. The train never stops between stations, and it approaches the terminus only from one direction — but it still has a perfectly definite destination. **A sequence limit is a destination approached by hopping, from one side only.**

This has a consequence that JEE loves. Because the train visits only the stations, it can miss trouble that the car drives straight through. Take $f(x) = \sin(\pi x)$. The car sees f swinging between -1 and $+1$ forever, so $\lim_{x \rightarrow \infty} \sin(\pi x)$ does not exist. But the train stops only at integers, and $\sin(n\pi) = 0$ for *every* integer n — so the sequence $a_n = \sin(n\pi)$ is the constant sequence $0, 0, 0, \dots$ and $\lim_{n \rightarrow \infty} a_n = 0$ without any drama at all. **A sequence limit can exist where the corresponding function limit does not.**

The implication runs one way only, and this is the single most examinable sentence in the subtopic:

$$\lim_{x \rightarrow \infty} f(x) = L \implies \lim_{n \rightarrow \infty} f(n) = L, \quad \text{but the converse is false.}$$

That one-way arrow is what licenses everything else. It is the reason you may attack $\lim_{n \rightarrow \infty} \frac{2n^2 + 3n + 1}{5n^2 - n}$ with the degree rule from 12_05 without inventing any new machinery: the function version has the limit $\frac{2}{5}$, so the sequence version inherits it. Almost every routine sequence limit is a function limit in disguise, and the bridge theorem is the disguise remover.

Limiting Conditions and Assumptions — where the idea breaks down:

1. **There is no LHL and no RHL.** Do not import the two-sided existence test from 12_01. For a sequence there is only one direction of approach. Existence means the terms settle; failure means they diverge to $\pm\infty$ or oscillate.
2. **The bridge theorem is one-way.** From a function limit you may deduce the sequence

limit. From a sequence limit you may deduce **nothing** about the function limit. Every assertion-reason question in this family lives on that asymmetry.

3. **A term-by-term limit is illegal when the number of terms grows.** The algebra of limits from 12_01 splits a sum of a *fixed* number of pieces. In $a_n = \sum_{k=1}^n c_{n,k}$ the count of pieces is itself n , so the rule does not apply — each term may tend to 0 while the sum tends to something else entirely.
4. **A recursion's fixed point is not automatically its limit.** Solving $L = f(L)$ tells you *what the limit must be if one exists*. It says nothing about whether one exists. Applied to a divergent sequence it returns a confident, completely wrong number.
5. **Oscillation without growth still kills the limit.** $a_n = (-1)^n$ stays bounded between -1 and 1 forever and never settles, so it has no limit — exactly as $\sin \frac{1}{x}$ has none at 0 in 12_01.

Figure brief: Two stacked panels sharing a horizontal axis. Upper panel: the continuous curve $y = \sin(\pi x)$ for $x \in [0, 8]$, oscillating between -1 and 1 , labelled “the car — x slides, no limit”. Lower panel: the same axis showing only the isolated dots at $x = 1, 2, \dots, 8$, all sitting exactly on $y = 0$, joined by no curve, labelled “the train — n hops, limit = 0”. A one-way arrow from the upper panel to the lower one labelled “function limit $\Rightarrow$ sequence limit (never the reverse)”.

Section 2: Key Formulas and Definitions

Note (maths format): no dimensional formulas — every quantity here is a pure number. The payload of this section is four things: the definition and the bridge theorem, the Sandwich Theorem in its sequence form, the standard sequence-limit table with its growth ordering, and the two summation formulas from Chapter 9 that $\sum$ -form limits run on.

Definition. A sequence is a function $a : \mathbb{N} \rightarrow \mathbb{R}$, written a_n . We say $\lim_{n \rightarrow \infty} a_n = L$ if the terms a_n become and remain arbitrarily close to L for all sufficiently large n . The sequence is then called **convergent**. If a_n grows without bound it **diverges** to $+\infty$ or $-\infty$; if it neither settles nor grows without bound it **oscillates**, and in all three of those cases the limit does not exist.

- n = the index, a positive integer, $n \rightarrow \infty$ by jumps;
- a_n = the n -th term;
- L = the (finite) limit value.

The bridge theorem (function limits transfer to sequences). If f is defined for all large real x and $\lim_{x \rightarrow \infty} f(x) = L$, then the sequence $a_n = f(n)$ satisfies $\lim_{n \rightarrow \infty} a_n = L$. The converse is false; the standing counterexample is $f(x) = \sin(\pi x)$, for which $\lim_{n \rightarrow \infty} f(n) = 0$ while $\lim_{x \rightarrow \infty} f(x)$ does not exist.

Algebra of sequence limits. If $\lim a_n = L$ and $\lim b_n = M$, both finite, then $\lim(a_n \pm b_n) = L \pm M$, $\lim(k a_n) = kL$, $\lim(a_n b_n) = LM$, and $\lim \frac{a_n}{b_n} = \frac{L}{M}$ provided $M \neq 0$. These are the 12_01 rules unchanged — **but they split a sum of a fixed number of sequences only**.

Sandwich (Squeeze) Theorem for sequences. If $a_n \leq b_n \leq c_n$ for all $n \geq N$ (some fixed N ; the first few terms are irrelevant), and $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} c_n = L$, then $\lim_{n \rightarrow \infty} b_n = L$.

Corollary (bounded $\times$ null). If $|a_n| \leq M$ for all n (bounded) and $\lim_{n \rightarrow \infty} b_n = 0$ (null), then $\lim_{n \rightarrow \infty} a_n b_n = 0$. This settles $\frac{\sin n}{n} \rightarrow 0$ and $\frac{(-1)^n}{n} \rightarrow 0$ on sight.

Monotone convergence theorem. A sequence that is increasing and bounded above converges; a sequence that is decreasing and bounded below converges. This is the theorem that supplies the missing existence half of a recursion problem.

Standard sequence limits — the table to memorise.

Sequence	Limit	Conditions
$\frac{1}{n^k}$	0	$k > 0$
$a^{1/n}$	1	$a > 0$
$n^{1/n}$	1	—
a^n	0	$ a < 1$
a^n	$+\infty$	$a > 1$
a^n	no limit (oscillates)	$a \leq -1$
$\left(1 + \frac{a}{n}\right)^n$	e^a	$a \in \mathbb{R}$
$\frac{\ln n}{n^k}$	0	$k > 0$
$\frac{n^k}{n^k}$	0	$a > 1$, any fixed k
$\frac{a^n}{a^n}$	0	any fixed a
$\frac{n!}{(n!)^{1/n}}$	$\frac{1}{e}$	— (Advanced)
$\frac{n}{n}$	e	

The growth ordering (each beats everything to its left as $n \rightarrow \infty$):

$$\ln n \ll n^k \ll a^n \ll n! \ll n^n \quad (k > 0, a > 1).$$

Any ratio in which a later item sits in the denominator tends to 0. That single line disposes of the whole $\frac{n^{100}}{2^n}, \frac{2^n}{n!}, \frac{\ln n}{n}$ family without any work.

Chapter 9 summation formulas (chained in, not re-derived) — the engine of Σ -form limits:

$$\sum_{k=1}^n k = \frac{n(n+1)}{2}, \quad \sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}, \quad \sum_{k=1}^n k^3 = \left[\frac{n(n+1)}{2} \right]^2.$$

The ratio-to-root rule (JEE Advanced). If $a_n > 0$ and $\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \ell$, then $\lim_{n \rightarrow \infty} a_n^{1/n} = \ell$ as well. This converts an intractable n -th root into an easy ratio.

Variable glossary: n = the integer index ($n \rightarrow \infty$); k = the summation index; a_n, b_n, c_n = sequences; L, M = finite limit values; N = the index past which an inequality is required to hold; ℓ = the ratio limit in the ratio-to-root rule.

Section 3: Derivations and Key Procedures

Derivation: why the bridge theorem holds, and why its converse fails

1. Suppose $\lim_{x \rightarrow \infty} f(x) = L$. By definition, for any tolerance $\varepsilon > 0$ there is a cut-off X such that $|f(x) - L| < \varepsilon$ for **every** real $x > X$. *Reason:* this is what a limit at infinity asserts.
2. The integers n with $n > X$ are among those real x . *Reason:* $\mathbb{N} \subset \mathbb{R}$ — the stations lie on the road.
3. Therefore $|f(n) - L| < \varepsilon$ for all integers $n > X$, which is exactly the statement $\lim_{n \rightarrow \infty} f(n) = L$. *Reason:* the sequence definition is the function definition restricted to integer inputs.
4. The converse fails because step 2 cannot be reversed: knowing f is well behaved *at the stations* says nothing about its behaviour *between them*. $f(x) = \sin(\pi x)$ is the proof — it vanishes at every integer and reaches ± 1 at every half-integer.

The practical reading of this derivation: **a sequence limit is never harder than the corresponding function limit, and is sometimes easier.** So your first move on any routine sequence limit is to treat n as a continuous variable and reach for 12_05.

Derivation: why a term-by-term limit fails on a growing sum

Consider $S_n = \underbrace{\frac{1}{n} + \frac{1}{n} + \cdots + \frac{1}{n}}_{n \text{ terms}}$.

1. Each individual term tends to 0 as $n \rightarrow \infty$. *Reason:* $\frac{1}{n} \rightarrow 0$.
2. If the algebra of limits applied, the sum of n things each tending to 0 would tend to 0. But $S_n = n \cdot \frac{1}{n} = 1$ for every n , so $S_n \rightarrow 1$. *Reason:* direct computation.
3. The rule $\lim(u_n + v_n) = \lim u_n + \lim v_n$ is proved for **two** sequences, and extends by induction to any **fixed** number. Induction gives you the result for 2, 3, 4, ... terms — it never gives you the result for “ n terms, where n is the same n that is going to infinity”. *Reason:* the induction has no step that reaches a moving target.
4. Therefore a sum whose length grows with n must be handled whole: either summed in closed form (Chapter 9), or trapped between two bounds (Sandwich).

Procedure A: routine sequence limits — cross the bridge

1. Check whether the expression is an ordinary algebraic function of n with no $[]$, no $(-1)^n$ and no factorials. *Why:* those are the cases the bridge theorem hands you free.
2. Replace n by a continuous x and evaluate $\lim_{x \rightarrow \infty}$ using the 12_05 toolkit — the degree rule, divide-by-highest-power, or the conjugate trick for $\infty - \infty$.
3. Transfer the answer back to the sequence by the bridge theorem. *Why:* the arrow runs in the direction you need.

4. Do **not** attempt the reverse. If the function version has no limit, you have learned nothing about the sequence and must go back to a direct argument.

Procedure B: $\sum$ -form limits — sum first, then take the limit

1. Write the sum with an explicit index, $a_n = \sum_{k=1}^n (\text{expression in } k \text{ and } n)$, and confirm the number of terms grows with n . *Why*: this is the flag that forbids term-by-term limits.
2. Pull every factor that does not involve k outside the summation sign. *Why*: as far as $\sum_k$ is concerned, n is a constant.
3. Apply the Chapter 9 formulas for $\sum k$, $\sum k^2$, $\sum k^3$ to obtain a closed-form rational function of n . *Why*: a closed form has a fixed number of terms, so the ordinary rules apply again.
4. Now take $n \rightarrow \infty$ with the degree rule from 12_05. *Why*: the problem has become a Procedure A problem.
5. Sanity check: a sum of n terms each of rough size n^{-p} should land near n^{1-p} . *Why*: it catches an off-by-one-degree slip instantly.

Procedure C: $\sum$ -form limits that will not sum — use the Sandwich

1. Confirm that the summand cannot be summed in closed form. *Why*: Procedure B is always preferable when it is available.
2. Over the range $k = 1$ to $k = n$, find the **largest** and the **smallest** value the summand can take. Usually the summand is monotone in k , so these occur at $k = n$ and $k = 1$. *Why*: replacing every term by the smallest gives a valid lower bound; by the largest, a valid upper bound.
3. Write $n \cdot (\text{smallest term}) \leq a_n \leq n \cdot (\text{largest term})$. *Why*: there are exactly n terms, so both bounds are one multiplication.
4. Take the limit of both bounds. If they agree, that common value is the answer. *Why*: Sandwich Theorem for sequences.
5. If they disagree, the bounds were too crude — tighten them or split the sum.

Procedure D: recursive sequences — prove convergence, then solve

1. Compute a_1, a_2, a_3 numerically. *Why*: three terms reveal whether the sequence is climbing, falling or oscillating, and that decides the whole argument.
2. **Prove monotonicity** by induction: show $a_{n+1} > a_n$ for all n (increasing) or $a_{n+1} < a_n$ for all n (decreasing).
3. **Prove boundedness** by induction: exhibit a bound B with $a_n < B$ for all n (if increasing) or $a_n > B$ (if decreasing). *Why*: steps 2 and 3 together invoke the monotone convergence theorem, which is the only thing that establishes the limit *exists*.
4. **Only now** set $L = \lim a_n$. Since a_{n+1} is the same sequence with its first term removed, $\lim a_{n+1} = L$ too, so passing to the limit in the recursion gives an equation in L .
5. Solve it and **discard the roots the sequence cannot reach** — a sequence of positive terms cannot converge to a negative number, and an increasing sequence cannot converge below its first term.

6. Verify the surviving root against the numerical terms from step 1.

Why step 3 is not optional. Take $a_1 = 1$ and $a_{n+1} = 2a_n + 1$. Passing to the limit blindly gives $L = 2L + 1$, hence $L = -1$: a clean, definite, entirely fictitious answer. The actual terms are 1, 3, 7, 15, 31, ... — at $n = 26$ the term is already 67,108,863 and climbing. The sequence is increasing and **unbounded**, so no limit exists and there was never anything for L to name. The fixed-point equation answers “what would the limit be?”, never “is there one?”.

Section 4: Worked Examples

Example 1 — [CBSE Board Level]

Evaluate $\lim_{n \rightarrow \infty} \frac{1^2 + 2^2 + 3^2 + \dots + n^2}{n^3}$.

Step 1 — Recognise the shape. The numerator is a sum whose **number of terms grows with n** . So you may not send each term to its own limit; you must sum first. This is Procedure B.

Step 2 — Sum in closed form using the Chapter 9 result $\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$:

$$\frac{1^2 + 2^2 + \dots + n^2}{n^3} = \frac{n(n+1)(2n+1)}{6n^3}.$$

Step 3 — Now it is an ordinary limit at infinity. Expand the numerator: $n(n+1)(2n+1) = 2n^3 + 3n^2 + n$. So the expression is

$$\frac{2n^3 + 3n^2 + n}{6n^3} = \frac{1}{3} + \frac{1}{2n} + \frac{1}{6n^2}.$$

Step 4 — Send $n \rightarrow \infty$. Every $\frac{1}{n^k} \rightarrow 0$ (12_05), leaving

$$\lim_{n \rightarrow \infty} \frac{1^2 + 2^2 + \dots + n^2}{n^3} = \boxed{\frac{1}{3}}.$$

Presentation note: write the summation formula down as a separate line before substituting. Boards award a method mark for quoting it correctly, and the equal-degree ratio $\frac{2}{6} = \frac{1}{3}$ is then visible in one glance.

Example 2 — [JEE Main Level]

Evaluate $\lim_{n \rightarrow \infty} (2^n + 3^n + 5^n)^{1/n}$.

Step 1 — Diagnose. The bracket tends to ∞ and the exponent tends to 0, so this is an ∞^0 form — indeterminate, and none of the 12_02 junctions touches it. But notice what happens far out: 5^n utterly dominates 3^n and 2^n , so the bracket is “ 5^n plus smaller stuff”, and the n -th root of 5^n is 5. That intuition is right, and the Sandwich Theorem turns it into a proof.

Step 2 — Build the two bounds. The largest of the three powers is 5^n . Dropping the other two can only decrease the bracket, and replacing all three by the largest can only increase it:

$$5^n \leq 2^n + 3^n + 5^n \leq 3 \cdot 5^n.$$

Step 3 — Take the n -th root. The map $t \mapsto t^{1/n}$ is increasing on $t > 0$, so the inequalities survive intact:

$$5 \leq (2^n + 3^n + 5^n)^{1/n} \leq 3^{1/n} \cdot 5.$$

Step 4 — Squeeze. The left bound is the constant 5. For the right bound, the standard limit $a^{1/n} \rightarrow 1$ with $a = 3$ gives $3^{1/n} \cdot 5 \rightarrow 1 \cdot 5 = 5$. Both bounds tend to 5, so

$$\lim_{n \rightarrow \infty} (2^n + 3^n + 5^n)^{1/n} = \boxed{5}.$$

The general rule worth carrying: for positive constants, $(a_1^n + a_2^n + \cdots + a_m^n)^{1/n} \rightarrow \max\{a_1, \dots, a_m\}$. The largest base swallows the rest. There is nothing to compute — read off the maximum.

Example 3 — [JEE Main — Speed Trap]

Evaluate $\lim_{n \rightarrow \infty} \left(\frac{1}{\sqrt{n^2+1}} + \frac{1}{\sqrt{n^2+2}} + \cdots + \frac{1}{\sqrt{n^2+n}} \right)$.

The trap: taking the limit term by term. Each single term looks harmless — $\frac{1}{\sqrt{n^2+k}} \rightarrow 0$ as $n \rightarrow \infty$, for every fixed k . So the seductive reasoning runs: “a sum of terms that all tend to zero must tend to zero”, and the seductive wrong answer is 0.

Why it fails. The sum does not have a fixed number of terms. It has n of them, and n is the very quantity going to infinity. Each term is roughly $\frac{1}{n}$, and there are n of them, so the total is roughly $n \cdot \frac{1}{n} = 1$ — not 0. The algebra of limits was never licensed here (Section 3’s derivation), and this is exactly the situation Procedure C exists for.

Step 1 — Bound the summand. For k running from 1 to n , the denominator $\sqrt{n^2+k}$ is smallest at $k = 1$ and largest at $k = n$. So each term satisfies

$$\frac{1}{\sqrt{n^2+n}} \leq \frac{1}{\sqrt{n^2+k}} \leq \frac{1}{\sqrt{n^2+1}}.$$

Step 2 — Multiply by the number of terms. There are exactly n terms, so summing the inequality gives

$$\frac{n}{\sqrt{n^2+n}} \leq S_n \leq \frac{n}{\sqrt{n^2+1}}.$$

Step 3 — Take the limit of each bound by dividing inside the root by n^2 (note $n > 0$, so $\sqrt{n^2} = n$ with no sign worry):

$$\frac{n}{\sqrt{n^2+n}} = \frac{1}{\sqrt{1+\frac{1}{n}}} \rightarrow 1, \quad \frac{n}{\sqrt{n^2+1}} = \frac{1}{\sqrt{1+\frac{1}{n^2}}} \rightarrow 1.$$

Step 4 — Squeeze.

$$\lim_{n \rightarrow \infty} S_n = \boxed{1}.$$

Numerical sanity check: at $n = 10$ the bounds are 0.9535 and 0.9950 with $S_{10} = 0.9739$; at $n = 10^4$ they are 0.99995 and 0.999999 with $S_{10^4} = 0.999975$. The bracket closes on 1, exactly as the algebra says — and nowhere near the trap's answer of 0.

Example 4 — [JEE Advanced Level]

The sequence $\{a_n\}$ is defined by $a_1 = 1$ and $a_{n+1} = \sqrt{2 + a_n}$ for $n \geq 1$. Show that $\{a_n\}$ converges and find its limit.

The subtlety: the question says “**show that it converges and find**”, not merely “find”. The two halves need different arguments, and a solution that only produces the number scores roughly half the marks. Existence comes from monotone convergence; the value comes from the fixed-point equation. In that order.

Step 1 — Look at the first few terms. $a_1 = 1$, $a_2 = \sqrt{3} \approx 1.7321$, $a_3 = \sqrt{2 + \sqrt{3}} \approx 1.9319$. The sequence is climbing and appears to be levelling off below 2. That suggests: increasing, bounded above by 2.

Step 2 — Prove $a_n < 2$ for all n , by induction. Base: $a_1 = 1 < 2$. Step: assume $a_n < 2$. Then $2 + a_n < 4$, so $a_{n+1} = \sqrt{2 + a_n} < \sqrt{4} = 2$. Hence $a_n < 2$ for every n , so the sequence is **bounded above**.

Step 3 — Prove the sequence is increasing. All terms are positive, so comparing a_{n+1} with a_n is the same as comparing their squares:

$$a_{n+1}^2 - a_n^2 = (2 + a_n) - a_n^2 = -(a_n^2 - a_n - 2) = -(a_n - 2)(a_n + 1).$$

By Step 2, $a_n - 2 < 0$; and $a_n + 1 > 0$ since every term is positive. The product of a negative and a positive is negative, so $-(a_n - 2)(a_n + 1) > 0$. Hence $a_{n+1}^2 > a_n^2$, and since both are positive, $a_{n+1} > a_n$. The sequence is **increasing**.

Step 4 — Invoke the theorem. Increasing and bounded above, so by the monotone convergence theorem $\{a_n\}$ **converges**. Call the limit L . Existence is now established, and only now may we hunt for the value.

Step 5 — Pass to the limit in the recursion. The sequence $\{a_{n+1}\}$ is $\{a_n\}$ with its first term deleted, so it has the same limit L . Taking $n \rightarrow \infty$ on both sides of $a_{n+1}^2 = 2 + a_n$:

$$L^2 = 2 + L \implies L^2 - L - 2 = 0 \implies (L - 2)(L + 1) = 0 \implies L = 2 \text{ or } L = -1.$$

Step 6 — Discard the impossible root. Every a_n is positive and the sequence is increasing from $a_1 = 1$, so its limit cannot be -1 . Therefore

$$\lim_{n \rightarrow \infty} a_n = \boxed{2}.$$

Deep point. Compare this with the cautionary recursion $a_{n+1} = 2a_n + 1$ from Procedure D, where the identical Step 5 produced $L = -1$ and the identical Step 6 would have had nothing to say — because Steps 2 to 4 fail there: that sequence is increasing but **unbounded**, so no limit exists at all. The fixed-point equation is a perfectly obedient machine that answers whatever you ask it. It is your job to have earned the right to ask.

Section 5: Practice Exercises

1. [CBSE] Evaluate $\lim_{n \rightarrow \infty} \frac{1 + 2 + 3 + \cdots + n}{n^2}$.
2. [CBSE] Evaluate $\lim_{n \rightarrow \infty} \frac{2n^2 + 3n + 1}{5n^2 - n}$.
3. [JEE Main] Evaluate $\lim_{n \rightarrow \infty} \frac{1^3 + 2^3 + \cdots + n^3}{n^4}$.
4. [JEE Main] Evaluate $\lim_{n \rightarrow \infty} (5^n + 7^n)^{1/n}$.
5. [JEE Main] Evaluate $\lim_{n \rightarrow \infty} \frac{\sin n}{n}$, $n \in \mathbb{N}$, and state which theorem you used.
6. [JEE Main] Evaluate $\lim_{n \rightarrow \infty} \left(1 + \frac{3}{n}\right)^{2n}$.
7. [JEE Advanced] The sequence $\{a_n\}$ has $a_1 = 6$ and $a_{n+1} = \sqrt{6 + a_n}$. Prove that it converges and find the limit. (Note that here the sequence is *decreasing* — adapt Example 4's argument.)
8. [JEE Advanced] Evaluate $\lim_{n \rightarrow \infty} \frac{(n!)^{1/n}}{n}$ using the ratio-to-root rule on $a_n = \frac{n!}{n^n}$.

Answers: 1. $\frac{1}{2}$ 2. $\frac{2}{5}$ 3. $\frac{1}{4}$ 4. 7 5. 0, by the bounded $\times$ null corollary of the Sandwich Theorem ($|\sin n| \leq 1$ and $\frac{1}{n} \rightarrow 0$) 6. e^6 7. Decreasing and bounded below by 3, so it converges; $L^2 = 6 + L$ gives $L = 3$ or $L = -2$, and the terms are positive, so $L = 3$ 8. $\frac{1}{e}$ (the ratio is $\frac{a_{n+1}}{a_n} = \frac{1}{(1 + \frac{1}{n})^n} \rightarrow \frac{1}{e}$)

Section 6: Crack the MCQ

Q1. Consider the two statements $P : \lim_{x \rightarrow \infty} \sin(\pi x)$ exists, and $Q : \lim_{n \rightarrow \infty} \sin(n\pi)$ exists, $n \in \mathbb{N}$. Then

- (A) both P and Q are true (B) P is false, Q is true
(C) P is true, Q is false (D) both are false

Answer: (B). As a function of a real variable, $\sin(\pi x)$ oscillates between -1 and $+1$ forever — at $x = \frac{1}{2}, \frac{3}{2}, \frac{5}{2}$ it takes the values $1, -1, 1$ — so P is false. As a sequence, $\sin(n\pi) = 0$ for **every** integer n , so the sequence is identically 0 and Q is true with limit 0.

Distractor Analysis. - **(A)** — the student assumes any expression built from $\sin$ has the same behaviour in both settings, and never notices that restricting the input to integers can remove an oscillation entirely. - **(C)** — the arrow is remembered backwards. The bridge theorem says function limit $\Rightarrow$ sequence limit; this option asserts the reverse implication,

which is exactly the false one. - **(D)** — the student sees $\sin$ of something growing without bound and reflexively answers “oscillates, no limit” for both, without evaluating $\sin(n\pi)$ at a single integer. One substitution, $n = 1$, would have settled it.

Q2. $\lim_{n \rightarrow \infty} \frac{n^{100}}{2^n}$ equals

- (A) $+\infty$ (B) 1 (C) 0 (D) does not exist

Answer: (C). By the growth ordering $n^k \ll a^n$ for $a > 1$, an exponential beats any fixed power, however large the power. At $n = 10^4$ the ratio is already about 5×10^{-2611} .

Distractor Analysis. - **(A)** — the student compares the two at small n , where n^{100} is astronomically bigger, and extrapolates. The crossover is late but it is inevitable: the exponent 100 is *fixed* while the exponent n keeps growing. - **(B)** — a vague “both go to infinity, so the ratio is 1” — the $\frac{\infty}{\infty}$ fallacy from 12_05, which the chapter already flags as indeterminate. - **(D)** — confusing “the limit is not a nice number” with “there is no limit”. The limit is a perfectly ordinary 0.

Q3. A sequence satisfies $a_1 = 1$ and $a_{n+1} = 2a_n + 1$. A student writes: “Let $L = \lim a_n$. Then $L = 2L + 1$, so $L = -1$.” The student’s conclusion is

- (A) correct (B) wrong, because the limit is $+\infty$ and the fixed-point step assumed a finite limit that does not exist (C) wrong, because the correct fixed point is $L = 1$ (D) correct in magnitude but wrong in sign; the limit is $+1$

Answer: (B). The terms are 1, 3, 7, 15, 31, ... — increasing and unbounded (indeed $a_n = 2^n - 1$), so the sequence diverges. Writing $L = \lim a_n$ presupposes a finite limit; with no such L in existence, the equation $L = 2L + 1$ is about nothing at all. Its root is an artefact.

Distractor Analysis. - **(A)** — accepts the algebra without ever checking the hypothesis. Every term of the sequence is positive and growing, so a limit of -1 should have set off an alarm on inspection alone. - **(C)** — an arithmetic slip in solving $L = 2L + 1$; but the deeper error is the same as (A), since no root of that equation is the limit. - **(D)** — patches the sign to make the answer look plausible against the positive terms, which treats a structural failure as a computational one.

Q4. $\lim_{n \rightarrow \infty} \left(\frac{1}{n+1} + \frac{1}{n+2} + \cdots + \frac{1}{n+n} \right)$ is

- (A) 0 (B) a finite number strictly between $\frac{1}{2}$ and 1 (C) 1 (D) $+\infty$

Answer: (B). Bound as in Procedure C: each of the n terms lies between $\frac{1}{2n}$ and $\frac{1}{n+1}$, so the sum lies between $n \cdot \frac{1}{2n} = \frac{1}{2}$ and $\frac{n}{n+1} < 1$. The limit is therefore trapped in $[\frac{1}{2}, 1]$, and since the bounds do not meet, the Sandwich Theorem pins it only to that interval — enough to answer the question. (Its exact value is $\ln 2 \approx 0.693$, obtained by the Riemann-sum method you will meet in Class 12 integration.)

Distractor Analysis. - **(A)** — the Example 3 trap once more: every term tends to 0, so the sum “must” too. The number of terms grows, so the rule does not apply. - **(C)** — the student computes the upper bound $\frac{n}{n+1} \rightarrow 1$ and stops there, reporting a bound as though it were the limit. An upper bound is an upper bound; it is the answer only when the lower bound agrees,

and here it does not. - **(D)** — reasoning that a sum with ever more terms must blow up. It would, if the terms did not shrink; here n terms of size about $\frac{1}{n}$ balance exactly.

Section 7: Common Student Pitfalls and Pro-Tips

1. **Importing LHL and RHL into a sequence problem.** A sequence has no left side. Writing “ $\lim_{n \rightarrow 3^-} a_n$ ” is meaningless — there are no indices between 2 and 3. *Correction:* the only sequence limit is $n \rightarrow \infty$, and it is one-directional.
2. **Running the bridge theorem backwards.** From “ $a_n = f(n)$ converges” you may conclude nothing whatsoever about $\lim_{x \rightarrow \infty} f(x)$. *Correction:* memorise the direction with its counterexample attached — $\sin(\pi x)$ at the integers.
3. **Taking a term-by-term limit on a sum of n terms.** The single most punished error in this family. *Correction:* count the terms. If the count involves n , you must either sum in closed form (Procedure B) or sandwich the whole thing (Procedure C).
4. **Solving $L = f(L)$ without proving convergence.** *Correction:* the fixed-point equation is step 5 of Procedure D, never step 1. Monotone plus bounded comes first; only then does L denote anything.
5. **Keeping every root of the fixed-point equation.** $L^2 = 2 + L$ has roots 2 and -1 , but a sequence of positive increasing terms cannot land on -1 . *Correction:* filter the roots against the sign and monotonicity you already established.
6. **Reporting a bound as the limit.** Having sandwiched a_n between b_n and c_n , the answer is the common limit — and only if the two agree. *Correction:* always compute *both* bounds’ limits and check that they coincide before writing a value.
7. **Mis-ordering the growth hierarchy.** Students routinely believe $n!$ beats n^n , or that a large fixed power like n^{100} beats 2^n . *Correction:* commit $\ln n \ll n^k \ll a^n \ll n! \ll n^n$ to memory as a single line, left to right.
8. **Assuming a bounded sequence must converge.** $(-1)^n$ is bounded and has no limit. *Correction:* bounded is only half the monotone convergence theorem; monotone is the other half, and you need both.

Pro-Tip (speed and accuracy). Build a three-way triage. Ask first: *is this just a function limit wearing an n ?* If the expression is plain algebra in n , cross the bridge and use the 12_05 degree rule — most JEE Main sequence questions die in one line here. Ask second: *is it a sum whose length grows?* Then it is Procedure B if the Chapter 9 formulas apply, Procedure C if they do not. Ask third: *is it defined recursively?* Then it is Procedure D, and your first pen stroke is “ $a_1 =, a_2 =, a_3 =$ ” — three numbers tell you which way it is heading and hand you the bound to prove. And for anything with an n -th root of a sum of powers, do not compute at all: write down the largest base.

Section 8: Cheat Sheet Box

LIMITS OF SEQUENCES

- **Definition:** a_n is a function on $\mathbb{N}$. $n \rightarrow \infty$ **only** — no finite approach point, no LHL, no RHL. Convergent / divergent to $\pm\infty$ / oscillating.

- **Bridge theorem (one-way):** $\lim_{x \rightarrow \infty} f(x) = L \Rightarrow \lim_{n \rightarrow \infty} f(n) = L$. **Converse false** — counterexample $f(x) = \sin(\pi x)$: $\lim_n f(n) = 0$ but $\lim_x f(x)$ does not exist.
- **Sandwich for sequences:** $a_n \leq b_n \leq c_n$ for all $n \geq N$, with $a_n, c_n \rightarrow L \Rightarrow b_n \rightarrow L$.
- **Bounded $\times$ null:** $|a_n| \leq M$ and $b_n \rightarrow 0 \Rightarrow a_n b_n \rightarrow 0$. Kills $\frac{\sin n}{n}$, $\frac{(-1)^n}{n}$ on sight.
- **Monotone convergence:** increasing + bounded above $\Rightarrow$ converges; decreasing + bounded below $\Rightarrow$ converges.
- **Standard limits:** $a^{1/n} \rightarrow 1$ ($a > 0$); $n^{1/n} \rightarrow 1$; $a^n \rightarrow 0$ for $|a| < 1$, $\rightarrow \infty$ for $a > 1$, oscillates for $a \leq -1$; $(1 + \frac{a}{n})^n \rightarrow e^a$; $\frac{(n!)^{1/n}}{n} \rightarrow \frac{1}{e}$.
- **Growth ordering:** $\ln n \ll n^k \ll a^n \ll n! \ll n^n$ ($k > 0$, $a > 1$). Later item in the denominator $\Rightarrow$ limit 0.
- **Max-base rule:** $(a_1^n + \dots + a_m^n)^{1/n} \rightarrow \max\{a_i\}$ for positive a_i .
- **Ratio-to-root:** $a_n > 0$ and $\frac{a_{n+1}}{a_n} \rightarrow \ell \Rightarrow a_n^{1/n} \rightarrow \ell$.
- **Σ -form:** sum first with $\sum k = \frac{n(n+1)}{2}$, $\sum k^2 = \frac{n(n+1)(2n+1)}{6}$, $\sum k^3 = \left[\frac{n(n+1)}{2}\right]^2$, **then** take the limit. Standard results: $\frac{\sum k}{n^2} \rightarrow \frac{1}{2}$, $\frac{\sum k^2}{n^3} \rightarrow \frac{1}{3}$, $\frac{\sum k^3}{n^4} \rightarrow \frac{1}{4}$.
- **Recursion order:** monotone $\rightarrow$ bounded $\rightarrow$ monotone convergence theorem $\rightarrow L = f(L) \rightarrow$ discard impossible roots. Never the fixed point first.

Memory aids:

- “The car slides, the train hops.” (function variable versus sequence index)
- “Function to sequence, one way only.” (the bridge theorem and its false converse)
- “Count the terms before you split the sum.” (no term-by-term limit when the count grows)
- “Log, power, exponential, factorial, n -to-the- n .” (the growth ordering, left to right)
- “Biggest base wins the root.” (the max-base rule)
- “Prove it lands before you say where.” (convergence before the fixed point)

Chapter 13: Statistics

Class 11 Mathematics — Complete Master Content

Drona Ed Tech

June 2026

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Subtopic 01 — Measures of Dispersion and Mean Deviation

Exam Relevance: Statistics carries roughly one guaranteed question in **CBSE Boards** (a 3–5 mark computation of mean deviation or variance from a frequency table). In **JEE Main** it surfaces as a single quick MCQ, usually on mean deviation, variance, or the effect of a transformation on a measure of spread. **JEE Advanced** almost never asks mean deviation as a stand-alone numerical, but the idea that *the median minimises absolute deviation* shows up inside optimisation problems — so the concept matters more than the formula.

Section 1: Concept Intuition

Imagine two vegetable vendors in your local mandi, Ramesh and Suresh. Over a week you note the price of one kilo of tomatoes at each stall. Ramesh quotes ₹28, ₹30, ₹29, ₹31, ₹30. Suresh quotes ₹14, ₹46, ₹12, ₹48, ₹30. Punch both into a calculator and you will find the *same* average: ₹29.6 each. Yet nobody would call these vendors equivalent. Ramesh is boringly reliable; Suresh is a gamble. The average — the centre of the data — has completely hidden the thing that actually distinguishes them: how *scattered* the prices are around that centre.

That scatter is what **dispersion** measures. Central tendency (mean, median, mode) tells you where a data set sits; dispersion tells you how tightly or loosely the observations huddle around that location. Two data sets can share a mean and look nothing alike, and the whole job of this subtopic is to attach a *single number* to “how spread out is this?”

Think about it this way... picture each data set laid out as dots on a number line. Ramesh’s dots cluster into a tight little knot. Suresh’s dots fling themselves to the far corners. Dispersion is just a fair way of summarising “how far, on average, do the dots live from home?”

Visual. Two horizontal number lines stacked vertically, both centred on the same mean marker. Top line (“Low dispersion”) has five dots tightly clustered near the centre. Bottom line (“High dispersion”) has five dots flung far to the left and right of the same centre. A vertical dashed line through both marks the common mean.

There is a hierarchy of ways to capture this. The crudest is the **range** — just the gap between the largest and smallest observation. It is instant to compute and genuinely useful as a first glance (a weather report saying “today’s temperature ranged from 22°C to 41°C” tells you something real). But the range is a bit of a drama queen: it listens *only* to the two most extreme values and ignores everything in between. One freak observation and the range lies about the whole set.

A smarter idea is to measure, for *every* observation, how far it sits from a central value, and then average those distances. That is the **mean deviation**. The subtle

move here is the word “distance.” A value sitting 5 units *below* the mean is just as “spread out” as one sitting 5 units *above* it — so we must throw away the signs and take **absolute values** of the deviations. If we did not, the deviations from the mean would politely cancel to exactly zero every single time (that is a defining property of the mean), and our spread measure would uselessly report 0 for every data set on earth.

Limiting conditions & assumptions. Mean deviation has two honest weaknesses you should carry in your head. First, the absolute-value sign makes it *mathematically clumsy* — you cannot differentiate $|x|$ cleanly at zero, so mean deviation refuses to slot into further algebra (this is precisely why the next subtopic abandons it in favour of variance and standard deviation, which square the deviations instead). Second, mean deviation is only as meaningful as the central value you anchor it to; the *same* data set has a different mean deviation about its mean than about its median. And as we will prove later, the deviation is genuinely *smallest* when taken about the **median** — a fact that quietly powers a lot of clever exam shortcuts.

Section 2: Key Formulas & Definitions

Range of a data set:

$$\text{Range} = x_{\max} - x_{\min}$$

For a grouped (continuous) distribution, the range is taken as the upper boundary of the highest class minus the lower boundary of the lowest class.

Coefficient of range (a unit-free way to compare spread across data sets of different scales):

$$\text{Coefficient of range} = \frac{x_{\max} - x_{\min}}{x_{\max} + x_{\min}}$$

Mean deviation about a central value a (where a is the mean $\bar{x}$, the median M , or the mode):

For *ungrouped* data $x_1, x_2, \dots, x_n$:

$$\text{M.D.}(a) = \frac{1}{n} \sum_{i=1}^n |x_i - a|$$

For a *frequency distribution* (discrete data, or continuous data using class mid-points x_i) with frequencies f_i and $N = \sum f_i$:

$$\text{M.D.}(a) = \frac{1}{N} \sum_{i=1}^n f_i |x_i - a|$$

Coefficient of mean deviation (again, dimensionless):

$$\text{Coefficient of M.D.} = \frac{\text{M.D.}(a)}{a}$$

Key property (memorise this):

$$\sum_i |x_i - a| \text{ is minimum when } a = \text{median} \Rightarrow \text{M.D. is least about the median.}$$

Symbol meanings: x_i = the i -th observation; n = number of observations (ungrouped); f_i = frequency of x_i ; $N = \sum f_i$ = total frequency; $\bar{x}$ = mean; M = median.

Units note (the statistics analogue of dimensional analysis): Range and every mean deviation carry the **same units as the raw data** — if the observations are in rupees, marks, runs, or centimetres, so is the spread. The *coefficient* of range and the *coefficient* of mean deviation are **pure numbers (dimensionless)**, because they divide one same-unit quantity by another. A quick sanity check follows: since each $|x_i - a| \leq \text{Range}$, the mean deviation can never exceed the range. If your M.D. comes out larger than the range, you have made an arithmetic slip.

Section 3: Key Procedures & Logical Framework

Mean deviation is computed, not derived, so here is the rigorous step-by-step framework, with the *reason* behind each step.

(A) Mean deviation of ungrouped data about the mean.

1. **Find the mean** $\bar{x} = \frac{1}{n} \sum x_i$. *Why:* this is the anchor — the central value all distances are measured from.
2. **Find each absolute deviation** $|x_i - \bar{x}|$. *Why:* the absolute value converts a signed “how far above/below” into an unsigned “how far,” so that values on opposite sides of the mean reinforce rather than cancel.
3. **Sum the absolute deviations**, $\sum |x_i - \bar{x}|$. *Why:* we want total scatter before averaging it out.
4. **Divide by n** . *Why:* averaging makes the measure comparable between data sets of different sizes — a data set of 100 points is not “more spread out” merely because it has more terms.

(B) Mean deviation of ungrouped data about the median.

The procedure is identical, except Step 1 finds the **median** (sort the data, then take the middle value for odd n , or the average of the two middle values for even n). *Why bother?* Because the median minimises the total absolute deviation — measuring about it gives the smallest, most “honest” mean deviation the data can offer.

Visual. A number line with observations marked as dots. From a chosen point 'a' on the line, draw short bracket-arrows to each dot labelled with its distance $|x_i - a|$.
Caption: "Mean deviation = average length of these distance-arrows."

(C) Mean deviation of a frequency distribution.

1. For a **continuous** distribution, first replace each class by its **mid-point** $x_i = \frac{\text{lower limit} + \text{upper limit}}{2}$. *Why:* we need a single representative value per class to stand in for every observation in it. (For a discrete distribution, the x_i are already given.)
2. Compute the mean $\bar{x} = \frac{\sum f_i x_i}{N}$, where $N = \sum f_i$.
3. Form the column $|x_i - \bar{x}|$, then the column $f_i |x_i - \bar{x}|$. *Why the f_i weighting:* an observation that occurs ten times must contribute its distance ten times.
4. Mean deviation = $\frac{\sum f_i |x_i - \bar{x}|}{N}$. **Critical:** divide by $N = \sum f_i$, **not** by the number of distinct classes.

(D) Mean deviation about the median of a continuous distribution.

When the central value is the **median** of grouped data, you cannot read it off directly — you must locate the *median class* first.

1. Build the **cumulative frequency** (c.f.) column.
2. The **median class** is the first class whose c.f. reaches $\frac{N}{2}$.
3. Compute the median by interpolation: $M = \ell + \frac{\frac{N}{2} - C}{f} \times h$, where ℓ = lower boundary of the median class, C = c.f. of the class *just before* it, f = frequency of the median class, h = its width. *Why interpolate:* we assume the observations are spread evenly across the median class and step in proportionally to reach the $\frac{N}{2}$ -th value.
4. Then proceed exactly as in (C), using $|x_i - M|$ in place of $|x_i - \bar{x}|$.

Section 4: Worked Examples

Example 1 — [CBSE Board Level] A student's marks in five class tests (out of 40) are 32, 28, 36, 24, 30. Find the range and the mean deviation about the mean.

Given: observations 32, 28, 36, 24, 30; $n = 5$.

Range = $x_{\max} - x_{\min} = 36 - 24 = 12$ marks.

Mean: $\bar{x} = \frac{32 + 28 + 36 + 24 + 30}{5} = \frac{150}{5} = 30$.

Absolute deviations: $|32 - 30| = 2$, $|28 - 30| = 2$, $|36 - 30| = 6$, $|24 - 30| = 6$, $|30 - 30| = 0$.

$$\text{M.D.}(\bar{x}) = \frac{2 + 2 + 6 + 6 + 0}{5} = \frac{16}{5} = \mathbf{3.2} \text{ marks.}$$

Presentation check: answer carries the unit “marks,” and $3.2 \leq 12$, so it is consistent with the range.

Example 2 — [JEE Main Level: Speed / Shortcut] *In a football season a striker scored the following number of goals per match:*

Goals x_i	0	1	2	3	4
Matches f_i	4	6	8	5	2

Find the mean deviation of the number of goals about the mean.

Work the table in one sweep. $N = 4 + 6 + 8 + 5 + 2 = 25$.

$$\sum f_i x_i = (0)(4) + (1)(6) + (2)(8) + (3)(5) + (4)(2) = 0 + 6 + 16 + 15 + 8 = 45.$$

$$\text{Mean } \bar{x} = \frac{45}{25} = 1.8.$$

$$\sum f_i |x_i - 1.8| = 4(1.8) + 6(0.8) + 8(0.2) + 5(1.2) + 2(2.2) = 7.2 + 4.8 + 1.6 + 6.0 + 4.4 = 24.0.$$

$$\text{M.D.} = \frac{24.0}{25} = \mathbf{0.96} \text{ goals.}$$

The trap to dodge fast: the table has *five* rows, and under time pressure students divide 24.0 by 5 and write 4.8. The divisor is $N = \sum f_i = 25$, the number of matches, not the number of distinct goal-counts. Train your eye to grab N from the frequency column total, never from the row count.

Example 3 — [JEE Main Level: Multi-concept] *A data set of monthly electricity bills (in ₹) has range 40 and mean deviation about the mean equal to 12. The provider revises every bill by the rule $y_i = 3x_i - 7$ (a $3\times$ tariff hike minus a ₹7 fixed rebate). Find the new range and the new mean deviation about the mean.*

This links **dispersion** with the **behaviour of a measure under a linear transformation** $x_i \mapsto ax_i + b$.

A transformation shifts and scales. The shift ($b = -7$) moves *every* observation by the same amount, so all the gaps between observations are untouched — **adding a constant cannot change spread**. The scale factor ($a = 3$) stretches every gap by $|a|$. Both range and mean deviation are built purely out of gaps, so each is multiplied by $|a|$ and is blind to b :

$$\text{Range}_{\text{new}} = |3| \times 40 = \mathbf{120}, \quad \text{M.D.}_{\text{new}} = |3| \times 12 = \mathbf{36}.$$

(Had the rule been $y_i = -3x_i - 7$, the answers would be identical — only $|a|$ matters, because flipping a data set about a point does not change how spread out it is.)

Example 4 — [JEE Advanced Level] For the observations 7, 11, 13, 16, 20, 24, consider the function $S(a) = \sum_i |x_i - a|$. Prove that $S(a)$ attains its minimum at the median, find the minimum value, and identify every value of a that achieves it.

Sort (already sorted): 7, 11, 13, 16, 20, 24, with $n = 6$ even.

The pairing argument. Pair the observations symmetrically from the outside in: (7, 24), (11, 20), (13, 16). Take any point a that lies **inside** an outer pair, say between 7 and 24. The sum of that pair's two distances to a is

$$|a - 7| + |24 - a| = (a - 7) + (24 - a) = 17,$$

a *constant* — wherever a sits between them, the two distances trade off exactly. The same holds for each pair, as long as a lies inside all three, i.e. as long as $a \in [13, 16]$, the innermost pair. Within that interval:

$$S(a) = \underbrace{17}_{(7,24)} + \underbrace{9}_{(11,20)} + \underbrace{3}_{(13,16)} = \mathbf{29}.$$

The moment a leaves $[13, 16]$, it falls outside the innermost pair, and that pair's contribution starts to grow (a gain on one side no longer cancels), so $S(a)$ strictly increases. Hence S is minimised, with value 29, for **every** $a \in [13, 16]$ — the entire interval between the two middle observations. The conventional median $M = \frac{13+16}{2} = 14.5$ sits in this interval, confirming $M.D.(M)_{\min} = \frac{29}{6} \approx \mathbf{4.83}$.

The deep takeaway: the median minimises absolute deviation *because* of this outside-in cancellation. For an even-sized data set the minimiser is not a single point but the whole median interval — a subtlety worth a mark if an Advanced problem asks “for how many values of $a...$ ” (answer: infinitely many).

Example 5 — [CBSE Board Level: Continuous distribution, M.D. about mean] The daily wages (in ₹) of 40 workers are distributed as below. Find the mean deviation about the mean.

Wages (₹)	100-200	200-300	300-400	400-500	500-600
Workers f_i	6	10	14	8	2

Replace each class by its mid-point x_i , then build the table. $N = 40$.

x_i	f_i	$f_i x_i$	$ x_i - \bar{x} $	$f_i x_i - \bar{x} $
⁹⁹² 150	6	900	175	1050

x_i	f_i	$f_i x_i$	$ x_i - \bar{x} $	$f_i x_i - \bar{x} $
250	10	2500	75	750
350	14	4900	25	350
450	8	3600	125	1000
550	2	1100	225	450
Total	40	13000		3600

$$\bar{x} = \frac{13000}{40} = 325. \text{ Then}$$

$$\text{M.D.}(\bar{x}) = \frac{\sum f_i |x_i - \bar{x}|}{N} = \frac{3600}{40} = \mathbf{90}.$$

Example 6 — [JEE Main Level: Continuous distribution, M.D. about median] The marks of 40 students are grouped below. Find the mean deviation about the median.

Marks	0-10	10-20	20-30	30-40	40-50
Students f_i	7	8	10	9	6

Step 1 — locate the median class. Cumulative frequencies: 7, 15, 25, 34, 40. Here $\frac{N}{2} = 20$; the first c.f. to reach 20 is 25, so the **median class is 20-30**.

Step 2 — interpolate the median. With $\ell = 20$, $C = 15$ (c.f. before), $f = 10$, $h = 10$:

$$M = 20 + \frac{20 - 15}{10} \times 10 = 20 + 5 = 25.$$

Step 3 — mean deviation about $M = 25$ (mid-points 5, 15, 25, 35, 45):

x_i	f_i	$ x_i - 25 $	$f_i x_i - 25 $
5	7	20	140
15	8	10	80
25	10	0	0
35	9	10	90
45	6	20	120
Total	40		430

$$\text{M.D.}(M) = \frac{430}{40} = \mathbf{10.75} \text{ marks.}$$

Watch-out: the median of grouped data is **not** the middle mid-point — it must be interpolated through the median class, or every downstream number is wrong.

Section 5: Practice Exercises

1. **[CBSE]** Find the range and the mean deviation about the mean of: 14, 18, 22, 26, 30.
2. **[JEE Main]** For the discrete distribution below, find the mean deviation about the mean. $|x_i|$ 5 | 10 | 15 | 20 | 25 | $|f_i|$ 3 | 5 | 8 | 3 | 1 |
3. **[JEE Main]** Find the mean deviation about the median of: 8, 12, 15, 19, 24, 29, 33, 40.
4. **[JEE Main]** A data set has mean deviation about the mean = 7.5 and range = 25. Every observation is replaced by (value $\times 4 - 9$). Find the new mean deviation and the new range.
5. **[JEE Advanced]** For the observations 2, 5, 9, 14, 20, find the value(s) of a that minimise $\sum_i |x_i - a|$, and state the minimum value of the sum.

Answers: 1. Range = 16, M.D. = 4.8. 2. M.D. = 4.3. 3. Median = 21.5, M.D. = 9. 4. New M.D. = 30, new range = 100 (the -9 has no effect). 5. Minimum at $a = 9$ (the median); minimum sum = 27.

Section 6: Crack the MCQ

Q1. Every observation in a data set is increased by 5. Which statement is correct?

- (A) The range increases by 5 (B) The range is unchanged (C) The mean deviation increases by 5 (D) The coefficient of range is unchanged

Correct: (B). Adding a constant shifts $x_{\max}$ and $x_{\min}$ by the same 5, so their *difference* — the range — is untouched; the same translation-invariance leaves the mean deviation unchanged too. *Why the traps bite:* (A) and (C) come from imagining “+5 on each value must add up,” forgetting that a measure of *spread* depends on gaps, which a uniform shift preserves. (D) is the sneakiest: the coefficient of range = $\frac{x_{\max} - x_{\min}}{x_{\max} + x_{\min}}$ keeps the same numerator but a *larger denominator* after the shift, so it actually **does** change — making (D) false.

Q2. The mean deviation about the median of 4, 7, 9, 13, 16 is:

- (A) 3.6 (B) 3.76 (C) 4.0 (D) 18

Correct: (A). Median = 9; absolute deviations 5, 2, 0, 4, 7 sum to 18; M.D. = $18/5 = 3.6$. *Why the traps bite:* (B) 3.76 is what you get if you absent-mindedly measure about the **mean** (9.8) when the question said *median* — the single most common error here. (C) 4.0 is a rounding/arithmetic slip. (D) 18 is the sum of deviations with the final division by n forgotten.

Q3. For the distribution $x_i = 2, 4, 6$ with $f_i = 1, 2, 2$, the mean deviation about the mean is:

- (A) 1.28 (B) 2.13 (C) 6.4 (D) 1.6

Correct: (A). $N = 5$, $\bar{x} = \frac{2+8+12}{5} = 4.4$; $\sum f_i |x_i - \bar{x}| = 1(2.4) + 2(0.4) + 2(1.6) = 6.4$; M.D. = $6.4/5 = 1.28$. *Why the traps bite:* (B) 2.13 divides 6.4 by **3** (the number of classes) instead of $N = 5$ — the frequency-divisor trap. (C) 6.4 skips

the division entirely. (D) 1.6 mistakenly uses the largest deviation rather than the weighted average.

Q4. The mean deviation of a data set is least when it is measured about the:

- (A) mean (B) median (C) mode (D) range

Correct: (B). By the outside-in cancellation argument, $\sum |x_i - a|$ is minimised at the median. *Why the traps bite:* (A) mean is the popular wrong answer — but the mean minimises the sum of *squared* deviations (variance), not absolute deviations. (C) mode minimises nothing of this kind. (D) range is a measure of spread, not a central value, so it cannot be a point we measure deviation *about* — a pure category error planted to catch panicked guessing.

Section 7: Common Student Pitfalls & Pro-Tips

- **Dropping the absolute-value sign.** Computing $\sum (x_i - \bar{x})$ without the modulus gives exactly 0 for every data set (deviations about the mean always cancel), and students then “conclude” M.D. = 0. The modulus is the whole point — it is the difference between measuring *distance* and measuring *signed displacement*.
- **Dividing by the wrong count in frequency tables.** The divisor is always $N = \sum f_i$ (total frequency), never the number of rows/classes. This single slip is the most frequent source of lost marks in Board answers.
- **Using the mean when “least mean deviation” is asked.** If a question wants the *smallest* mean deviation, anchor to the **median**, not the mean. Reading the question for which central value is requested costs two seconds and saves the whole answer.
- **Believing a shift changes the spread.** Adding/subtracting a constant changes neither the range nor any mean deviation; multiplying scales both by $|a|$. Many transformation MCQs are built entirely around testing whether you know this.

Pro-Tip (speed + sanity): For any transformation $x_i \mapsto ax_i + b$, immediately write Range and M.D. $\mapsto |a| \times (\text{old value})$ and ignore b — no recomputation needed. And before you commit an answer, check that your mean deviation is **smaller than the range**; if it isn't, you have an arithmetic error upstream.

Section 8: Cheat Sheet Box

DISPERSION & MEAN DEVIATION — RAPID RECALL

- **Dispersion** = how scattered the data is about its centre. Order of refinement: Range $\rightarrow$ Mean Deviation $\rightarrow$ (next subtopic: Variance & SD).

- **Range** $= x_{\max} - x_{\min}$; **Coeff. of range** $= \frac{x_{\max} - x_{\min}}{x_{\max} + x_{\min}}$.
- **Mean deviation** $M.D.(a) = \frac{1}{n} \sum |x_i - a|$ (ungrouped) $= \frac{1}{N} \sum f_i |x_i - a|$ (frequency, $N = \sum f_i$).
- **Grouped median** (for M.D. about median): $M = \ell + \frac{N/2 - C}{f} \times h$
(median class = first with c.f. $\geq N/2$); then take $|x_i - M|$.
- **M.D. is least about the MEDIAN** — mnemonic: **“Median Minimises Mean-deviation”** (the three M’s).
- **Transformation rule** $x_i \mapsto ax_i + b$: Range and M.D. both $\mapsto |a| \times$ old; the $+b$ does nothing. Mnemonic: **“Add doesn’t matter, Multiply scales.”**
- **Units:** range and M.D. share the data’s units; coefficients are dimensionless. Sanity check: $M.D. \leq \text{Range}$, always.

Subtopic 02 — Variance and Standard Deviation

Exam Relevance: This is the single most heavily examined corner of Statistics. **JEE Main** has carried a variance/SD question in nearly every recent paper — usually the shortcut formula, the transformation property, or combined variance. **CBSE Boards** reliably ask a 3–5 mark computation from a frequency table. **JEE Advanced** leans on the property-based results: combined variance of two groups, variance of the first n natural numbers, or “minimum mean-square deviation” reasoning.

Section 1: Concept Intuition

In the previous subtopic we learned to measure spread by averaging the *distances* of observations from a centre — the mean deviation. We were forced to wrap each deviation in absolute-value bars, otherwise the deviations about the mean cancel to zero. That worked, but the modulus sign is mathematically grumpy: $|x|$ has a sharp corner at zero and refuses to cooperate with calculus or algebra. Statisticians wanted a spread measure that kills the signs **and** stays smooth enough to manipulate. Their trick: instead of taking the *absolute value* of each deviation, **square** it.

Squaring does three jobs at once. It throws away the sign (a square is never negative). It is perfectly smooth — no corners — so it slots into further mathematics effortlessly. And, crucially, it **punishes large deviations far more than small ones**: an observation sitting 2 units from the mean contributes 4, but one sitting 10 units away contributes 100, not 10. The average of these squared deviations is the **variance**, written σ^2 .

Think about it this way... picture each deviation from the mean as the side of a little square drawn on the number line. A point close to the mean draws a tiny square; a far-flung outlier draws an enormous one. Variance is the *average area* of all these squares. One distant outlier can blow up the average area dramatically — which is exactly the sensitivity we want when “how risky / how inconsistent is this?” is the real question.

Visual. A number line with the mean marked. Above several data dots, draw squares whose side equals the dot’s distance from the mean — small squares near the mean, one very large square over a distant outlier. Caption: “Variance = average area of these squares. Distant points dominate.”

There is one inconvenience. If the data is in marks, the squared deviations are in marks², **and so is the variance — an awkward unit nobody can interpret. The cure is simple: take the positive square root of the variance to drag the measure back into the original units. That square root is the standard deviation**^{**}, σ , the single most-used measure of spread in all of statistics.

A real-world payoff. Suppose two batsmen both average 45 runs. Batsman P scores 44, 46, 45, 43, 47 — tight, dependable. Batsman Q scores 5, 90, 12, 88, 30 — same average, wild swings. For a must-win final, you pick P, and the standard deviation is precisely the number that tells you P is the safer bet. Indian SIP investors use the very same idea: two mutual funds may report identical average returns, but the one with the lower standard deviation is the calmer ride.

But what if you want to compare the consistency of two things measured on *different scales* — say, the variability of students' heights versus their weights, or two stocks priced very differently? Standard deviation alone can't, because it carries units. So we normalise it by the mean and express it as a percentage: the **coefficient of variation** (C.V.). Being a pure number, C.V. compares apples to oranges, and the rule is friendly: **lower C.V. means more consistent**.

Limiting conditions & assumptions. Variance is *always* measured about the mean (unlike mean deviation, which can sit on any central value) — and that is not arbitrary: just as the median minimises absolute deviation (Subtopic 01), the **mean minimises the mean of squared deviations**. Variance is hypersensitive to outliers because of the squaring, so for skewed data it can overstate spread. And σ^2 is zero **only** when every single observation equals the mean — a perfectly flat data set.

Section 2: Key Formulas & Definitions

Variance σ^2 = the mean of the squared deviations from the mean. **Standard deviation** $\sigma = +\sqrt{\sigma^2}$ (always the positive root).

Ungrouped data $x_1, \dots, x_n$ with mean $\bar{x}$:

$$\sigma^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 \quad \Longleftrightarrow \quad \boxed{\sigma^2 = \frac{1}{n} \sum x_i^2 - \bar{x}^2}$$

The boxed **shortcut form** is the workhorse — it avoids computing every deviation and is far faster when $\bar{x}$ is not a whole number.

Frequency distribution (f_i , $N = \sum f_i$; for continuous data the x_i are class mid-points):

$$\sigma^2 = \frac{1}{N} \sum f_i (x_i - \bar{x})^2 = \frac{1}{N} \sum f_i x_i^2 - \left(\frac{1}{N} \sum f_i x_i \right)^2$$

Step-deviation form (for tidy hand computation; A = assumed mean, h = class width, $d_i = \frac{x_i - A}{h}$):

$$998 \quad \sigma^2 = h^2 \left[\frac{1}{N} \sum f_i d_i^2 - \left(\frac{1}{N} \sum f_i d_i \right)^2 \right], \quad \sigma = \frac{h}{N} \sqrt{N \sum f_i d_i^2 - \left(\sum f_i d_i \right)^2}$$

Coefficient of variation (dimensionless, in %):

$$\text{C.V.} = \frac{\sigma}{\bar{x}} \times 100 \quad (\text{lower C.V.} \Rightarrow \text{more consistent})$$

Transformation rule (for $y_i = ax_i + b$): variance is **unaffected by the shift b** and **scaled by a^2** :

$$\sigma_y^2 = a^2 \sigma_x^2, \quad \sigma_y = |a| \sigma_x.$$

Combined variance of two groups (sizes n_1, n_2 ; means $\bar{x}_1, \bar{x}_2$; variances σ_1^2, σ_2^2 ; combined mean $\bar{x}$, with $d_1 = \bar{x}_1 - \bar{x}$, $d_2 = \bar{x}_2 - \bar{x}$):

$$\sigma^2 = \frac{n_1(\sigma_1^2 + d_1^2) + n_2(\sigma_2^2 + d_2^2)}{n_1 + n_2}.$$

Standard result: variance of the first n natural numbers $= \frac{n^2 - 1}{12}$.

Symbol meanings: σ^2 variance, σ SD, $\bar{x}$ mean, n/N counts, f_i frequencies, A assumed mean, h class width, d_i step deviation.

Units note (statistics analogue of dimensional analysis): Standard deviation carries the **same units as the data**; **variance carries the squared units** (marks², ₹², cm²) — which is the whole reason SD exists. The coefficient of variation is a **pure number** (a percentage). Sanity checks: $\sigma^2 \geq 0$ always (a negative answer means you subtracted the shortcut terms in the wrong order), and σ can never exceed the range.

Section 3: CBSE Board Derivations / Key Procedures

Derivation of the shortcut formula $\sigma^2 = \frac{1}{n} \sum x_i^2 - \bar{x}^2$.

Start from the definition and expand the square — the reason we expand is to separate the data-dependent sum $\sum x_i^2$ from the constant mean, so we never have to compute individual deviations:

$$\sigma^2 = \frac{1}{n} \sum (x_i - \bar{x})^2 = \frac{1}{n} \sum (x_i^2 - 2\bar{x}x_i + \bar{x}^2).$$

Split the sum into three pieces (legal because summation distributes over addition):

$$\sigma^2 = \frac{1}{n} \sum x_i^2 - \frac{2\bar{x}}{n} \sum x_i + \frac{1}{n} \sum \bar{x}^2.$$

Now use two facts. First, $\frac{1}{n} \sum x_i = \bar{x}$ by definition of the mean, so the middle term becomes $-2\bar{x} \cdot \bar{x} = -2\bar{x}^2$. Second, $\bar{x}^2$ is a constant, so $\sum \bar{x}^2 = n\bar{x}^2$, and

the last term becomes $\bar{x}^2$. Hence

$$\sigma^2 = \frac{1}{n} \sum x_i^2 - 2\bar{x}^2 + \bar{x}^2 = \frac{1}{n} \sum x_i^2 - \bar{x}^2. \quad \blacksquare$$

Why the mean minimises mean-square deviation. In Subtopic 01 we proved the *median* minimises $\sum |x_i - a|$. The parallel result here: $g(a) = \frac{1}{n} \sum (x_i - a)^2$ is minimised at $a = \bar{x}$. Differentiate and set to zero: $g'(a) = \frac{1}{n} \sum 2(x_i - a)(-1) = -\frac{2}{n} \sum (x_i - a) = 0 \Rightarrow \sum x_i = na \Rightarrow a = \bar{x}$. This is *why* variance is always anchored to the mean — no other anchor gives a smaller average squared spread.

Procedure — variance of a continuous frequency distribution (step-deviation method):

1. Replace each class by its mid-point x_i . *Why:* one representative value per class.
2. Choose a convenient assumed mean A (a central mid-point) and class width h ; compute $d_i = (x_i - A)/h$. *Why:* this shrinks the arithmetic to small integers, slashing error.
3. Build columns $f_i d_i$ and $f_i d_i^2$, then their totals.
4. Apply $\sigma^2 = h^2 \left[\frac{\sum f_i d_i^2}{N} - \left(\frac{\sum f_i d_i}{N} \right)^2 \right]$, then $\sigma = \sqrt{\sigma^2}$. *Why the h^2 :* undoing the scaling by h from step 2 — and notice it is h^2 , not h , exactly because variance scales by the square of any factor.

Section 4: Worked Examples

Example 1 — [CBSE Board Level] The number of units a small shop sold on five days was 12, 15, 18, 21, 24. Find the variance and standard deviation.

$$\text{Mean: } \bar{x} = \frac{12 + 15 + 18 + 21 + 24}{5} = \frac{90}{5} = 18.$$

Deviations: $-6, -3, 0, 3, 6$; squared: $36, 9, 0, 9, 36$; sum = 90.

$$\sigma^2 = \frac{90}{5} = \mathbf{18 \text{ (units}^2\text{)}}, \quad \sigma = \sqrt{18} = 3\sqrt{2} \approx \mathbf{4.24 \text{ units}}.$$

Note the units: variance in units^2 , SD in units — and $4.24 \leq \text{range } (12)$, so it is consistent.

Example 2 — [JEE Main Level: Transformation property] The variance of 20 observations is 5. Each observation is transformed by the rule (multiply by 3, then add 4). Find the new variance and the new standard deviation.

¹⁰⁰⁰ The “+4” shifts all data equally and cannot change spread; the “ $\times 3$ ” scales the

spread. For variance the scale factor enters **squared**:

$$\sigma_{\text{new}}^2 = 3^2 \times 5 = \mathbf{45}, \quad \sigma_{\text{new}} = |3| \times \sqrt{5} = 3\sqrt{5} \approx \mathbf{6.71}.$$

The signature trap: students multiply the variance by 3 (giving 15) or square the SD's factor wrongly. Lock it in — *variance scales by a^2 , SD by $|a|$, and the additive constant does nothing*. This single fact is the most common JEE Main statistics question.

Example 3 — [JEE Main Level: Frequency distribution, shortcut] Find the variance of the following distribution.

x_i	4	8	12	16	20
f_i	2	5	8	3	2

$N = 20$. Build the running totals: $\sum f_i x_i = 8 + 40 + 96 + 48 + 40 = 232 \Rightarrow \bar{x} = \frac{232}{20} = 11.6$.

$\sum f_i x_i^2 = 2(16) + 5(64) + 8(144) + 3(256) + 2(400) = 32 + 320 + 1152 + 768 + 800 = 3072$.

$$\sigma^2 = \frac{\sum f_i x_i^2}{N} - \bar{x}^2 = \frac{3072}{20} - (11.6)^2 = 153.6 - 134.56 = \mathbf{19.04}, \quad \sigma \approx \mathbf{4.36}.$$

The shortcut paid off: we never touched a single $(x_i - 11.6)$ deviation.

Example 4 — [JEE Advanced Level: Combined variance] Section A has 30 students with mean marks 60 and SD 8. Section B has 20 students with mean marks 70 and SD 6. Find the standard deviation of all 50 students combined.

This is the classic “you cannot just average two SDs” problem — the gap *between* the group means adds its own spread.

$$\text{Combined mean: } \bar{x} = \frac{30(60) + 20(70)}{50} = \frac{1800 + 1400}{50} = 64.$$

Deviations of group means from the combined mean: $d_1 = 60 - 64 = -4$, $d_2 = 70 - 64 = 6$.

$$\sigma^2 = \frac{n_1(\sigma_1^2 + d_1^2) + n_2(\sigma_2^2 + d_2^2)}{n_1 + n_2} = \frac{30(64 + 16) + 20(36 + 36)}{50} = \frac{2400 + 1440}{50} = 76.8.$$

$$\sigma = \sqrt{76.8} \approx \mathbf{8.76}.$$

Notice the combined SD (8.76) is *larger* than either group's SD (8 and 6) — the between-group terms d_1^2, d_2^2 inflate it. A student who naively averages 8 and 6 to

get 7 misses this entirely.

Example 5 — [CBSE Board Level: Continuous distribution, step-deviation method] Find the variance and standard deviation of the marks of 30 students grouped below.

Marks	0-20	20-40	40-60	60-80	80-100
Students f_i	4	6	10	7	3

Take assumed mean $A = 50$ (a central mid-point) and class width $h = 20$, and code $d_i = \frac{x_i - A}{h}$. The coding turns large mid-points into the tiny integers $-2, \dots, 2$ — far less error-prone.

x_i	f_i	d_i	$f_i d_i$	$f_i d_i^2$
10	4	-2	-8	16
30	6	-1	-6	6
50	10	0	0	0
70	7	1	7	7
90	3	2	6	12
Total	30		-1	41

Apply the step-deviation form (the h^2 undoes the coding — squared, because variance scales by the square of any factor):

$$\sigma^2 = h^2 \left[\frac{\sum f_i d_i^2}{N} - \left(\frac{\sum f_i d_i}{N} \right)^2 \right] = 400 \left[\frac{41}{30} - \left(\frac{-1}{30} \right)^2 \right] = 400 [1.3667 - 0.0011] = \mathbf{546.2}.$$

$$\sigma = \sqrt{546.2} \approx \mathbf{23.4} \text{ marks.}$$

(Mean, as a bonus: $\bar{x} = A + h \cdot \frac{\sum f_i d_i}{N} = 50 + 20 \left(\frac{-1}{30} \right) \approx 49.3$.) Note how small the correction term $\left(\frac{-1}{30} \right)^2$ is — but dropping it is still an error, and graders look for it.

Section 5: Practice Exercises

1. **[CBSE]** Find the variance and standard deviation of: 5, 10, 15, 20, 25.
2. **[JEE Main]** The variance of 15 observations is 12. If each observation is multiplied by 2 and then decreased by 5, find the new variance and new SD.
3. **[JEE Main]** Brand A bulbs: mean life 1200 h, SD 80 h. Brand B bulbs: mean life 1500 h, SD 90 h. Using the coefficient of variation, decide which brand is more consistent.

4. **[JEE Main]** Find the variance of the distribution: $x_i = 1, 2, 3, 4, 5$ with $f_i = 4, 6, 8, 5, 2$.
5. **[JEE Advanced]** Find the variance and standard deviation of the first 10 natural numbers.

Answers: 1. $\sigma^2 = 50$, $\sigma = 5\sqrt{2} \approx 7.07$. 2. new $\sigma^2 = 48$, new $\sigma = 4\sqrt{3} \approx 6.93$ (the -5 has no effect). 3. $\text{C.V.}_A = 6.67\%$, $\text{C.V.}_B = 6\% \Rightarrow$ Brand **B** is more consistent. 4. $\sigma^2 = 1.36$. 5. $\sigma^2 = \frac{10^2 - 1}{12} = 8.25$, $\sigma \approx 2.87$.

Section 6: Crack the MCQ

Q1. The variance of a data set is 9. Each observation is changed to $(\text{value} \times 2 + 7)$. The new standard deviation is:

- (A) 6 (B) 36 (C) 13 (D) 3

Correct: (A). Original SD $= \sqrt{9} = 3$. Under $y = 2x + 7$, the SD scales by $|2|$ and ignores the $+7$: new SD $= 2 \times 3 = 6$. *Why the traps bite:* (B) 36 is the new variance ($2^2 \times 9$) — the student answered the wrong quantity, forgetting the square root. (C) 13 comes from $2 \times 3 + 7$ — illegally adding the shift to a spread measure. (D) 3 assumes nothing changes, forgetting the scale factor.

Q2. Adding a fixed constant k to every observation changes:

- (A) the mean only (B) the variance only (C) both mean and variance (D) neither

Correct: (A). A translation moves the centre (mean $\rightarrow$ mean $+k$) but preserves every gap, so variance is unchanged. *Why the traps bite:* (C) is the popular error — assuming “everything shifts, so spread shifts too,” confusing location with spread. (B) is backwards. (D) forgets that the mean genuinely moves.

Q3. Two data sets have the *same* standard deviation, but set P has a larger mean than set Q. Which is more consistent?

- (A) P (B) Q (C) Equally consistent (D) Cannot be determined

Correct: (A). Consistency is judged by $\text{C.V.} = \sigma/\bar{x}$. With equal σ , the larger mean gives the smaller C.V., so **P** is more consistent. *Why the traps bite:* (C) is the trap of equating consistency with SD alone — but identical SD on a larger mean is *relatively* tighter. (B) inverts the relationship. (D) gives up despite sufficient information.

Q4. For a data set, $\sum x_i = 40$, $\sum x_i^2 = 375$, and $n = 5$. The variance is:

- (A) 11 (B) 64 (C) 75 (D) 36

Correct: (A). $\bar{x} = 40/5 = 8$; $\sigma^2 = \frac{\sum x_i^2}{n} - \bar{x}^2 = \frac{375}{5} - 64 = 75 - 64 = 11$.

Why the traps bite: (C) 75 is $\sum x_i^2/n$ with the student forgetting to subtract $\bar{x}^2$ — stopping one step early. (B) 64 is just $\bar{x}^2$ — the term that should have been

subtracted, reported alone. (D) 36 is a stray arithmetic distractor for panicked guessing.

Section 7: Common Student Pitfalls & Pro-Tips

- **Confusing variance with standard deviation.** They differ by a square root and by units (marks² vs marks). Read the question: if it asks for SD and you stop at variance, you lose the mark even with correct work.
- **Mishandling the transformation rule.** Under $x \mapsto ax + b$: variance scales by a^2 , SD by $|a|$, and b is irrelevant. The classic slip is scaling variance by a (not a^2) or letting the additive b creep in.
- **Stopping the shortcut formula early.** $\sigma^2 = \frac{1}{n} \sum x_i^2 - \bar{x}^2$ — students routinely forget the $-\bar{x}^2$, or subtract $\bar{x}$ instead of $\bar{x}^2$. If your variance comes out *negative*, you have swapped the two terms.
- **Averaging standard deviations of two groups.** Combined SD is **not** the average of the parts' SDs; you must use the combined-variance formula, which adds the between-group terms d_1^2, d_2^2 .

Pro-Tip (speed + sanity): Reach for $\sigma^2 = \frac{1}{n} \sum x_i^2 - \bar{x}^2$ whenever the mean isn't a whole number — it sidesteps ugly decimal deviations. Two instant checks before committing: variance must be ≥ 0 , and SD must be $\leq$ the range. For consistency comparisons, go straight to C.V. (lower wins) rather than comparing SDs that may sit on different means.

Section 8: Cheat Sheet Box

VARIANCE & STANDARD DEVIATION — RAPID RECALL

- **Variance** $\sigma^2 = \frac{1}{n} \sum (x_i - \bar{x})^2 = \frac{1}{n} \sum x_i^2 - \bar{x}^2$; **SD** $\sigma = +\sqrt{\sigma^2}$. Mnemonic: **"Square for Variance, Root for SD."**
- **Frequency form:** $\sigma^2 = \frac{1}{N} \sum f_i x_i^2 - \left(\frac{\sum f_i x_i}{N} \right)^2$, $N = \sum f_i$.
- **Transformation** $x \mapsto ax + b$: $\sigma^2 \mapsto a^2 \sigma^2$, $\sigma \mapsto |a| \sigma$, b does nothing. Mnemonic: **"Shift-proof, Scale-squared."**
- **Coefficient of variation** C.V. = $\frac{\sigma}{\bar{x}} \times 100$ (unit-free %); **lower C.V. = more consistent.**
- **Combined variance:** $\sigma^2 = \frac{n_1(\sigma_1^2 + d_1^2) + n_2(\sigma_2^2 + d_2^2)}{n_1 + n_2}$, $d_i = \bar{x}_i - \bar{x}_{\text{combined}}$ — never just average the SDs.
- **Anchor fact:** the **mean** minimises mean-square deviation (cf. the **median** minimises mean *absolute* deviation). **Std result:** variance of first n naturals = $\frac{n^2 - 1}{12}$.

- **Units:** variance in (data units)², SD in data units, C.V. dimensionless.
Sanity: $\sigma^2 \geq 0$, $\sigma \leq \text{range}$.

Subtopic 03 — Analysis of Frequency Distributions & Advanced Problem Types

Exam Relevance: This is where the chapter pays off. **CBSE Boards** love the coefficient-of-variation comparison (“which is more consistent?”) as a 4–5 mark question, and the *correction* problem (a misread observation) as a 3-mark twist. **JEE Main** turns the reverse problems — finding missing observations from a given mean and variance — into quick MCQs almost every year. **JEE Advanced** combines these ideas: merge two groups, then judge consistency by C.V.

Section 1: Concept Intuition

We can now *measure* spread (range, mean deviation, variance, standard deviation). The final question of the chapter is the practical one: **how do you compare the spread of two different groups?** A coach choosing between two batsmen, a buyer comparing two brands, an investor weighing two funds — all are asking “which one is steadier?”

If the two groups have the **same mean**, the answer is easy: just compare standard deviations — the smaller σ is the steadier group. But life is rarely that tidy. Usually the two groups have *different* means, or even different units (comparing the variability of people’s heights against their weights, say). Standard deviation alone cannot referee that contest, because a spread of 5 kg around a mean of 60 kg is a very different *relative* wobble from a spread of 5 cm around a mean of 170 cm.

Think about it this way... a ₹10 swing matters enormously on a ₹50 item but is invisible on a ₹50,000 item. To compare fairly, you must measure spread *relative to the size of what you’re measuring*. That is exactly what the **coefficient of variation** does: it divides the standard deviation by the mean (and multiplies by 100 to make a percentage). Being a pure ratio, it strips away units and scale, so any two series — rupees vs runs, heights vs marks — can be compared on one level playing field. The verdict is always the same: **lower C.V. = more consistent; higher C.V. = more variable.**

There is a second family of problems this chapter is famous for, and they run the machinery *backwards*. Instead of “here is the data, find the variance,” they ask “here is the mean and the variance — reconstruct the missing data,” or “an observation was recorded wrongly; repair the mean and standard deviation.” These reward you for understanding that the mean is built from $\sum x_i$ and the variance from $\sum x_i^2$ — so if you can track how those two running totals change, you can solve almost anything.

Limiting conditions & assumptions. The coefficient of variation needs a **non-**

zero mean ($\bar{x} \neq 0$) — it is undefined otherwise, and becomes unstable and misleading when the mean is close to zero. It is also meant for **ratio data with a true zero** (heights, wages, marks); applying it to something like temperatures in °C, where zero is arbitrary, gives nonsense. And “more consistent” (lower C.V.) is not the same as “better” — a consistent low scorer is still a low scorer, so always read what the question actually rewards.

Section 2: Key Formulas & Definitions

Coefficient of variation:

$$\text{C.V.} = \frac{\sigma}{\bar{x}} \times 100 \quad (\bar{x} \neq 0); \quad \text{lower C.V.} \Rightarrow \text{more consistent.}$$

Reverse form (recover the mean from a known C.V. and SD):

$$\bar{x} = \frac{\sigma}{\text{C.V.}} \times 100.$$

Same-mean rule: if $\bar{x}_1 = \bar{x}_2$, comparison by C.V. and comparison by σ give the same ranking, so just compare σ directly.

Running-total identities (the engine for reverse problems):

$$\bar{x} = \frac{\sum x_i}{n} \Rightarrow \sum x_i = n\bar{x}; \quad \sigma^2 = \frac{\sum x_i^2}{n} - \bar{x}^2 \Rightarrow \sum x_i^2 = n(\sigma^2 + \bar{x}^2).$$

Correcting a misrecorded value (wrong value w , correct value c):

$$\sum x_i^{\text{correct}} = \sum x_i^{\text{wrong}} - w + c, \quad \sum x_i^{2\text{correct}} = \sum x_i^{2\text{wrong}} - w^2 + c^2,$$

then recompute $\bar{x}$ and σ . (If the value is *omitted* rather than replaced, drop $-w/-w^2$ only and reduce n by 1.)

Combined mean & variance of two groups (recap from Subtopic 02):

$$\bar{x} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2}, \quad \sigma^2 = \frac{n_1(\sigma_1^2 + d_1^2) + n_2(\sigma_2^2 + d_2^2)}{n_1 + n_2}, \quad d_i = \bar{x}_i - \bar{x}.$$

Symbol meanings: σ SD, $\bar{x}$ mean, $\sum x_i$ sum of observations, $\sum x_i^2$ sum of squares, n count.

Units note: C.V. is a **pure number (percentage)** — that unit-free quality is the whole point, since it lets quantities with different units be compared. The repaired mean and SD carry the **original data units**.

Section 3: Key Procedures & Logical Framework

(A) Comparing two distributions for consistency.

1. Compute $\bar{x}$ and σ for each series.
2. If the means are equal, compare σ directly. Otherwise compute $\text{C.V.} = \frac{\sigma}{\bar{x}} \times 100$ for each.
3. The series with the **smaller C.V. is more consistent/stable**; the larger C.V. is **more variable**. *Why divide by the mean:* it converts an absolute wobble into a relative one, so the comparison is fair across different scales.

(B) Finding missing observations from mean and variance.

1. Translate the given mean into $\sum x_i = n\bar{x}$ — this gives one equation in the unknowns.
2. Translate the given variance into $\sum (x_i - \bar{x})^2 = n\sigma^2$ (or use $\sum x_i^2 = n(\sigma^2 + \bar{x}^2)$) — a second equation.
3. Solve the two equations simultaneously. A neat shortcut for two unknowns a, b : from $a + b$ (known) and $(a - \bar{x})^2 + (b - \bar{x})^2$ (known), use $(p + q)^2 = p^2 + q^2 + 2pq$ to get the product, then solve the resulting quadratic. *Why two equations:* two unknowns need two independent constraints, and mean + variance supply exactly that.

(C) Correcting a misrecorded observation.

1. Recover the *wrong* totals: $\sum x_i^{\text{wrong}} = n\bar{x}_{\text{wrong}}$ and $\sum x_i^2{}^{\text{wrong}} = n(\sigma_{\text{wrong}}^2 + \bar{x}_{\text{wrong}}^2)$.
2. Repair each total: subtract the wrong value's contribution, add the correct one's — for the mean use $-w + c$, for the variance use $-w^2 + c^2$. *Why both totals:* the wrong value polluted **both** $\sum x_i$ (affecting the mean) and $\sum x_i^2$ (affecting the variance), so both must be cleaned.
3. Recompute $\bar{x}$ and σ from the repaired totals (using the corrected n if the value was omitted).

Section 4: Worked Examples

Example 1 — [CBSE Board Level: Coefficient of variation] In a Twenty20 series, batsman Arjun averages 50 runs with a standard deviation of 10; batsman Vikram averages 40 runs with a standard deviation of 6. Who is the more consistent batsman? Who would you pick to chase a steep total?

$$\text{C.V.}_{\text{Arjun}} = \frac{10}{50} \times 100 = 20\%, \quad \text{C.V.}_{\text{Vikram}} = \frac{6}{40} \times 100 = 15\%.$$

Vikram's lower C.V. (15 %) makes him the **more consistent** scorer. But consistency isn't everything: Arjun's higher *average* (50 vs 40) makes him the better bet to chase a steep total, where raw run-scoring power matters more than steadiness. *This split verdict is exactly what examiners test — "consistent" and "best"*

are different questions.

Example 2 — [JEE Main Level: Missing observations] The mean of five observations is 6 and their variance is 6.8. If three of the observations are 2, 4, and 9, find the other two.

Let the unknowns be a and b .

From the **mean**: $2 + 4 + 9 + a + b = 5 \times 6 = 30 \Rightarrow a + b = 15$.

From the **variance**: $\sum (x_i - 6)^2 = 5 \times 6.8 = 34$.

$$(2-6)^2 + (4-6)^2 + (9-6)^2 + (a-6)^2 + (b-6)^2 = 34 \Rightarrow 16 + 4 + 9 + (a-6)^2 + (b-6)^2 = 34,$$

so $(a-6)^2 + (b-6)^2 = 5$. Put $p = a - 6$, $q = b - 6$; then $p + q = 15 - 12 = 3$ and $p^2 + q^2 = 5$. Since $(p + q)^2 = p^2 + q^2 + 2pq$, we get $9 = 5 + 2pq \Rightarrow pq = 2$. So p, q are roots of $t^2 - 3t + 2 = 0$, i.e. $t = 1, 2$. Therefore $\{a, b\} = \{7, 8\}$.

Check: the five observations 2, 4, 9, 7, 8 have mean $30/5 = 6$ and variance $\frac{16+4+9+1+4}{5} = \frac{34}{5} = 6.8$.

Example 3 — [JEE Main Level: Correcting an observation] The mean and standard deviation of 20 observations are found to be 30 and 5. On rechecking, an observation recorded as 50 is found to be wrong; its correct value is 30. Find the corrected mean and standard deviation.

Recover the wrong totals:

$$\sum x_i^{\text{wrong}} = 20 \times 30 = 600, \quad \sum x_i^{2\text{wrong}} = 20(5^2 + 30^2) = 20(25 + 900) = 18500.$$

Repair (wrong $w = 50$, correct $c = 30$):

$$\sum x_i^{\text{correct}} = 600 - 50 + 30 = 580, \quad \sum x_i^{2\text{correct}} = 18500 - 50^2 + 30^2 = 18500 - 2500 + 900 = 16900$$

$$\bar{x}_{\text{correct}} = \frac{580}{20} = \mathbf{29}, \quad \sigma^2 = \frac{16900}{20} - 29^2 = 845 - 841 = 4 \Rightarrow \sigma = \mathbf{2}.$$

The single 4-standard-deviation outlier was doing almost all the work — repairing it collapses the SD from 5 to 2.

Example 4 — [JEE Advanced Level: Combine, then compare] Group I has 40 students with mean score 60 and SD 8; Group II has 60 students with mean score 50 and SD 6. Pooling all 100 students, find the combined standard deviation and the combined coefficient of variation.

$$\text{Combined mean: } \bar{x} = \frac{40(60) + 60(50)}{100} = \frac{2400 + 3000}{100} = 54.$$

Group-mean deviations: $d_1 = 60 - 54 = 6$, $d_2 = 50 - 54 = -4$.

$$\sigma^2 = \frac{40(8^2 + 6^2) + 60(6^2 + (-4)^2)}{100} = \frac{40(64 + 36) + 60(36 + 16)}{100} = \frac{4000 + 3120}{100} = 71.2,$$

so $\sigma = \sqrt{71.2} \approx \mathbf{8.44}$, and

$$\text{C.V.} = \frac{8.44}{54} \times 100 \approx \mathbf{15.6\%}.$$

Note the combined SD (8.44) exceeds *both* group SDs (8 and 6): pooling groups with *different means* injects extra spread through the d_1, d_2 terms — a fact a careless “average the SDs” approach would never reveal.

Section 5: Practice Exercises

1. **[CBSE]** Firm A pays a mean monthly wage of ₹4500 with SD ₹360; Firm B pays a mean of ₹5000 with SD ₹450. Using C.V., decide which firm's wages are more consistent.
2. **[JEE Main]** The mean of five observations is 4 and their variance is 5.2. Three observations are 1, 2, and 6. Find the other two.
3. **[JEE Main]** The mean and SD of 5 observations are 6 and 4. An observation recorded as 14 is found to be wrong; its correct value is 6. Find the corrected mean and SD.
4. **[JEE Main]** Two groups of 50 students each have means 40 and 50, both with SD 5. Find the standard deviation of the combined group of 100.
5. **[JEE Advanced]** The coefficients of variation of two series are 25 % and 40 %, and their standard deviations are 5 and 8 respectively. Find their means and state which series is more variable.

Answers: 1. $\text{C.V.}_A = 8\%$, $\text{C.V.}_B = 9\% \Rightarrow$ Firm **A** is more consistent. 2. The other two are **4 and 7**. 3. Corrected mean = 4.4, corrected SD = 0.8. 4. Combined SD = $\sqrt{50} \approx 7.07$. 5. Both means = 20; the second series (C.V. 40 %) is **more variable**.

Section 6: Crack the MCQ

Q1. Two data sets have the **same mean**. Set X has SD 4 and set Y has SD 6. Which is more consistent?

- (A) X (B) Y (C) Equally consistent (D) Cannot decide without the coefficient of variation

Correct: (A). With equal means, the smaller SD wins, so X is more consistent. *Why the traps bite:* (D) is the over-thinking trap — when means are *equal*, C.V. and SD rank the two series identically, so SD alone is sufficient. (B) inverts the rule. (C) ignores the different SDs.

Q2. Every observation in a series is multiplied by a positive constant k . The coefficient of variation:

(A) becomes k C.V. (B) is unchanged (C) becomes $\text{C.V.}/k$ (D) becomes k^2 C.V.

Correct: (B). Both σ and $\bar{x}$ scale by k , so their ratio — and hence the C.V. — is untouched. *Why the traps bite:* (A) and (C) track only the numerator or only the denominator, forgetting both scale together. (D) wrongly imports the k^2 rule for variance; C.V. uses SD, not variance.

Q3. The coefficients of variation of series P and Q are 30 % and 25 % respectively. Which is more consistent?

(A) P (B) Q (C) Equally consistent (D) Cannot say without the means

Correct: (B). Lower C.V. (25 %) means more consistent, so Q wins. *Why the traps bite:* (D) is the trap — C.V. has *already* normalised by the mean, so the comparison is direct; you do **not** need the means again. (A) inverts “lower is more consistent.” (C) ignores the different values.

Q4. The mean of 10 observations is 25. A value recorded as 35 is later found to be wrong; its correct value is 15. The corrected mean is:

(A) 23 (B) 25 (C) 27 (D) 21

Correct: (A). $\sum x_i^{\text{wrong}} = 10 \times 25 = 250$; corrected sum $= 250 - 35 + 15 = 230$; corrected mean $= 230/10 = 23$. *Why the traps bite:* (B) assumes the mean is unaffected. (C) adds the difference instead of subtracting it ($25 + 2$). (D) double-counts the correction. The safe move is always to adjust the *sum*, then re-divide.

Section 7: Common Student Pitfalls & Pro-Tips

- **Comparing different-mean series by SD instead of C.V.** A larger SD does **not** mean “more variable” if the means differ — you must normalise with the coefficient of variation. Only when the means are equal may you compare SDs directly.
- **Forgetting to repair $\sum x_i^2$ in correction problems.** A wrong observation corrupts both the mean (via $\sum x_i$) and the variance (via $\sum x_i^2$). Fixing only the mean is the most common half-mark loss.
- **Feeding variance into the C.V. formula.** C.V. uses the *standard deviation*, not the variance — using σ^2 inflates the answer enormously.
- **Reading “more consistent” as “better.”** Lower C.V. means steadier, not superior. Always answer the question actually asked (consistency vs. performance).

Pro-Tip (speed + sanity): For reverse problems, immediately write the two master totals $\sum x_i = n\bar{x}$ and $\sum x_i^2 = n(\sigma^2 + \bar{x}^2)$ — every missing-observation or correction question is just bookkeeping on these two numbers. Sanity

check: a corrected variance must stay ≥ 0 ; if it goes negative, you have mis-handled $\sum x_i^2$. And recall C.V. is dimensionless — if your “C.V.” carries units, you divided by the wrong thing.

Section 8: Cheat Sheet Box

ANALYSIS OF DISTRIBUTIONS & REVERSE PROBLEMS — RAPID RECALL

- **Coefficient of variation** $\text{C.V.} = \frac{\sigma}{\bar{x}} \times 100$ ($\bar{x} \neq 0$). **Lower C.V. = more consistent; higher = more variable.**
- **Same mean?** Skip C.V. — just compare σ (same ranking). **Reverse:** $\bar{x} = \frac{\sigma}{\text{C.V.}} \times 100$.
- **Master totals for reverse problems:** $\sum x_i = n\bar{x}$ and $\sum x_i^2 = n(\sigma^2 + \bar{x}^2)$.
- **Correction:** $\sum x \rightarrow \sum x - w + c$, $\sum x^2 \rightarrow \sum x^2 - w^2 + c^2$, then recompute. (Omitted value: drop $-w$, $-w^2$ and reduce n .)
- **Missing two obs a, b :** use $a + b$ and $(a - \bar{x})^2 + (b - \bar{x})^2$ with $(p + q)^2 = p^2 + q^2 + 2pq \rightarrow$ quadratic.
- **Combined:** $\sigma^2 = \frac{n_1(\sigma_1^2 + d_1^2) + n_2(\sigma_2^2 + d_2^2)}{n_1 + n_2}$, $d_i = \bar{x}_i - \bar{x}$. Mnemonic: **“Normalise to compare (C.V.), bookkeep to reverse (sums).”**

Chapter 13: Statistics — Supplement

This supplement plugs two holes that Chapter 13 leaves open. The first is a CBSE-examinable case that falls between Procedure (C) and Procedure (D) of Subtopic 01: the **median of a discrete frequency distribution**, which is read off a cumulative frequency column and never interpolated. The second is a JEE Main phrasing that the chapter only ever shows you in grouped-data costume: the **assumed-mean variance identity for raw ungrouped data**. Both are short, both are mechanical once seen, and both cost marks when met cold.

Addendum 13.1 — Mean Deviation about the Median of a Discrete Frequency Distribution

Exam Relevance: This is the exact case NCERT devotes a worked example and several Exercise 13.1 items to, and CBSE Boards has asked it directly as a 3–4 mark numerical. It sits in the blind spot between the two cases Subtopic 01 already covers: you know how to find the median of an *ungrouped list* (sort and take the middle), and you know how to interpolate the median of a *continuous* distribution. A discrete frequency table needs neither — no sorting beyond what the table already gives, and absolutely no interpolation. JEE Main occasionally uses it as a one-step MCQ; the real risk is a Boards student importing the grouped-data formula $M = \ell + \frac{N/2 - C}{f} \times h$ into a table that has no classes and therefore no h .

The Method

Procedure (C2): Mean deviation about the median of a discrete frequency distribution.

You are given distinct values $x_1 < x_2 < \dots < x_k$ with frequencies $f_1, f_2, \dots, f_k$, and $N = \sum f_i$.

1. **Check that the x_i are in ascending order.** If the table is printed out of order, reorder it first, carrying each frequency with its value. *Why:* a cumulative frequency column is meaningless unless the values it accumulates over are sorted.
2. **Build the cumulative frequency (c.f.) column.** The c.f. against x_i is $f_1 + f_2 + \dots + f_i$. *Why:* the c.f. tells you the **serial positions** the value x_i occupies in the sorted list of all N observations. If the c.f. jumps from 12 to 17 at x_i , then x_i is the 13th, 14th, 15th, 16th and 17th observation.
3. **Locate the median by position, not by formula.**

Case	Median	How to read it off
N odd	the $(\frac{N+1}{2})$ -th observation	the first x_i whose c.f. is $\geq \frac{N+1}{2}$
N even	the mean of the $(\frac{N}{2})$ -th and $(\frac{N}{2} + 1)$ -th observations	read both positions off the c.f. column and average the two values

Why: the median is defined as the middle observation of the sorted data, and the c.f. column is a positional index into that sorted data. **There is no interpolation here** — the median of a discrete distribution is always either a data value or the midpoint of two adjacent data values.

- Watch the one subtle sub-case.** When N is even, if some c.f. is *exactly* equal to $\frac{N}{2}$, then the $\frac{N}{2}$ -th and $(\frac{N}{2} + 1)$ -th observations are **different** values and you must genuinely average them. If instead the c.f. leaps straight past $\frac{N}{2}$ (previous c.f. below $\frac{N}{2}$, this one above), both middle observations are the same x_i and the median is simply x_i . Most textbook tables are of the second kind, which is why students never notice the rule has two branches.
- Form the columns $|x_i - M|$ and then $f_i|x_i - M|$.** *Why the weighting:* a value occurring f_i times must contribute its distance f_i times.
- Divide by N :**

$$\text{M.D.}(M) = \frac{1}{N} \sum_{i=1}^k f_i |x_i - M|$$

Divide by $N = \sum f_i$, **never** by k , the number of distinct values.

- If asked, report the coefficient:**

$$\text{Coefficient of M.D.}(M) = \frac{\text{M.D.}(M)}{M}$$

a pure number, useful for comparing spread across data sets measured in different units.

Figure brief: A horizontal strip of N equally spaced tick marks numbered 1 to N , partitioned into coloured blocks whose widths are $f_1, f_2, \dots$ and labelled with $x_1, x_2, \dots$; a vertical arrow drops onto position $\frac{N+1}{2}$ (or onto the pair $\frac{N}{2}, \frac{N}{2} + 1$) with the caption “the c.f. column tells you which block this position lands in — that block’s x value is the median.”

Sanity checks before you write the final answer.

- $\text{M.D.}(M) \leq \text{M.D.}(\bar{x})$ always, because the median minimises $\sum f_i |x_i - a|$ (proved in Subtopic 01). If your M.D. about the median comes out *larger* than the M.D. about the mean, you have made a slip.
- $\text{M.D.}(M) \leq \text{Range}$, since every $|x_i - M|$ is at most the range.
- The answer carries the same units as the data.

Worked Examples

Example S1 — [CBSE Board Level]

A school records the number of days each of 25 students was absent in a term:

Days absent x_i	4	6	8	10	12
Number of students f_i	3	5	8	4	5

Find the mean deviation about the median, and its coefficient.

Solution.

The values are already in ascending order. Build the cumulative frequency column.

x_i	f_i	c.f.	Positions occupied	$ x_i - M $	$f_i x_i - M $
4	3	3	1-3	4	12
6	5	8	4-8	2	10
8	8	16	9-16	0	0
10	4	20	17-20	2	8
12	5	25	21-25	4	20
Total	$N = 25$				50

Here $N = 25$ is **odd**, so the median is the $\frac{N+1}{2} = \frac{26}{2} = 13$ th observation. Running down the c.f. column, the first entry reaching 13 is 16, against $x = 8$ — and indeed $x = 8$ occupies positions 9 through 16, which contains position 13. Hence

$$M = 8 \text{ days.}$$

Now complete the last two columns (already filled above) and divide:

$$\text{M.D.}(M) = \frac{\sum f_i|x_i - M|}{N} = \frac{50}{25} = 2 \text{ days.}$$

$$\text{Coefficient of M.D.}(M) = \frac{2}{8} = 0.25.$$

Verification by the minimisation property. Let us confirm the answer is smaller than the mean deviation about the mean.

$$\sum f_i x_i = 4(3) + 6(5) + 8(8) + 10(4) + 12(5) = 12 + 30 + 64 + 40 + 60 = 206, \quad \bar{x} = \frac{206}{25} = 8.24.$$

$$\sum f_i |x_i - 8.24| = 3(4.24) + 5(2.24) + 8(0.24) + 4(1.76) + 5(3.76) = 12.72 + 11.2 + 1.92 + 7.04 + 18.8 = 51.68,$$

so $\text{M.D.}(\bar{x}) = \frac{51.68}{25} = 2.0672$ days. As promised, $2 \leq 2.0672$: the median gives the tighter measure. Also $2 \leq \text{Range} = 12 - 4 = 8$.

Example S2 — [JEE Main Level — the even- N branch]

Find the mean deviation about the median for the distribution

x_i	5	10	15	20	25
f_i	4	6	5	9	6

Solution.

$$N = 4 + 6 + 5 + 9 + 6 = 30 \quad (\text{even}), \quad \frac{N}{2} = 15.$$

x_i	f_i	c.f.	Positions occupied
5	4	4	1-4
10	6	10	5-10
15	5	15	11-15
20	9	24	16-24
25	6	30	25-30

This is the sub-case Step 4 warned about: a cumulative frequency is **exactly** $\frac{N}{2} = 15$. So the two middle observations are genuinely different values — the 15th observation is 15 (the last position in that block) and the 16th observation is 20 (the first position in the next block). Therefore

$$M = \frac{15 + 20}{2} = 17.5.$$

Now the deviation columns:

x_i	f_i	$ x_i - 17.5 $	$f_i x_i - 17.5 $
5	4	12.5	50
10	6	7.5	45
15	5	2.5	12.5
20	9	2.5	22.5
25	6	7.5	45
Total	30		175

$$\text{M.D.}(M) = \frac{175}{30} = \frac{35}{6} \approx 5.83, \quad \text{Coefficient of M.D.} = \frac{35/6}{17.5} = \frac{35}{6} \cdot \frac{2}{35} = \frac{1}{3}.$$

Verification. In this configuration exactly 15 observations lie at or below 15 and exactly 15 lie at or above 20, so the sum $\sum f_i|x_i - a|$ is *constant* for every a between 15 and 20 — moving a up by a little increases 15 distances and decreases 15 distances, and the two effects cancel exactly. Test it at $a = 15$:

$$4(10) + 6(5) + 5(0) + 9(5) + 6(10) = 40 + 30 + 0 + 45 + 60 = 175,$$

the same total. So the answer $\frac{175}{30}$ is stable, and choosing the conventional midpoint 17.5 costs nothing.

Practice Exercises

1. Find the mean deviation about the median, and the coefficient of mean deviation about the median, for:

x_i	3	5	7	9	11
f_i	2	4	6	5	3

Answer: $N = 20$; the 10th and 11th observations both equal 7, so $M = 7$; $\sum f_i |x_i - M| = 38$; $\text{M.D.}(M) = \frac{38}{20} = 1.9$; coefficient $= \frac{1.9}{7} \approx 0.271$.

2. The daily wages (in hundreds of rupees) of 30 workers at a site are distributed as below. Find the mean deviation about the median.

x_i	10	15	20	25	30	35
f_i	3	5	8	6	4	4

Answer: $N = 30$; the 15th and 16th observations both equal 20 (that value occupies positions 9-16), so $M = 20$; $\sum f_i |x_i - M| = 185$; $\text{M.D.}(M) = \frac{185}{30} = \frac{37}{6} \approx 6.17$ hundred rupees; coefficient $= \frac{37/6}{20} \approx 0.308$.

Pitfalls

- **Do not interpolate.** The formula $M = \ell + \frac{N/2 - C}{f} \times h$ belongs to *continuous* data. A discrete table has no class boundaries ℓ and no width h , so the formula is not merely inaccurate here — it is undefined. Read the median off the c.f. column instead.
- **Do not average the x_i column to get the median.** The median is a positional statistic on all N observations, not on the k distinct values. In Example S1 the “middle” of the value list 4, 6, 8, 10, 12 happens also to be 8, which is a coincidence of that symmetric table; in Example S2 the middle of the value list is 15 while the true median is 17.5.
- **Divide by N , not k .** In Example S2, dividing 175 by 5 instead of 30 gives 35 — an answer larger than the entire range of 20, which the range sanity check catches instantly.
- **Do not compute the mean first out of habit.** Once the question says “about the median,” the mean is irrelevant and computing it wastes ninety seconds you do not have.

Addendum 13.2 — Variance of Ungrouped Data about an Assumed Mean

Exam Relevance: JEE Main asks this in a very specific costume: it hands you $\sum(x_i - A)$ and $\sum(x_i - A)^2$ for some convenient number A , and asks for the mean, the variance or the standard deviation. It is a one-line question if you know the identity and a five-minute reconstruction if you do not. Subtopic 02 already gives you the machinery in its grouped, step-deviation form; what follows is the same identity stripped of classes and class widths, which is the form the raw-data question actually needs. CBSE Boards uses the shortcut version ($A = 0$) routinely; JEE Advanced folds it into transformation and combined-variance problems.

The Method

The identity. For observations $x_1, x_2, \dots, x_n$ and **any** real constant A whatsoever:

$$\bar{x} = A + \frac{1}{n} \sum_{i=1}^n (x_i - A)$$
$$\sigma^2 = \frac{1}{n} \sum_{i=1}^n (x_i - A)^2 - (\bar{x} - A)^2 = \frac{1}{n} \sum_{i=1}^n (x_i - A)^2 - \left[\frac{1}{n} \sum_{i=1}^n (x_i - A) \right]^2$$

- A is the **assumed mean** (also called the reference point or working origin). It is a number you choose, or a number the question chooses for you. It need not be near the data, need not be a data value, and need not be the actual mean.
- n is the number of observations; σ^2 is the variance of the original data x_i ; $\sigma = +\sqrt{\sigma^2}$.
- **Validity:** unrestricted. This holds for every $A \in \mathbb{R}$ and every finite data set.
- Putting $A = 0$ recovers the familiar shortcut $\sigma^2 = \frac{1}{n} \sum x_i^2 - \bar{x}^2$. Putting $A = \bar{x}$ recovers the definition $\sigma^2 = \frac{1}{n} \sum (x_i - \bar{x})^2$. Those two are not different formulas — they are the two ends of one identity.
- The frequency-distribution version is identical with f_i weights and $N = \sum f_i$:

$$\sigma^2 = \frac{1}{N} \sum f_i (x_i - A)^2 - \left[\frac{1}{N} \sum f_i (x_i - A) \right]^2$$

and the step-deviation form in Subtopic 02 is just this with $d_i = \frac{x_i - A}{h}$ and the h^2 pulled back out.

Derivation

Write $d_i = x_i - A$, so $x_i = d_i + A$. This is a **change of origin** — a shift by the constant A , with no scaling.

Averaging both sides of $x_i = d_i + A$ over i gives $\bar{x} = \bar{d} + A$, i.e. $\bar{d} = \bar{x} - A$. That is the mean statement.

For the variance, note that $x_i - \bar{x} = (d_i + A) - (\bar{d} + A) = d_i - \bar{d}$: **the deviations from the mean are untouched by the shift.** Hence

$$\sigma_x^2 = \frac{1}{n} \sum (x_i - \bar{x})^2 = \frac{1}{n} \sum (d_i - \bar{d})^2 = \sigma_d^2,$$

which is exactly the $a = 1, b = -A$ case of the transformation rule $\sigma_y^2 = a^2\sigma_x^2$ already in Subtopic 02. Now apply the ordinary shortcut formula to the d_i :

$$\sigma_d^2 = \frac{1}{n} \sum d_i^2 - \bar{d}^2.$$

Substituting $d_i = x_i - A$ and $\bar{d} = \bar{x} - A$:

$$\sigma^2 = \frac{1}{n} \sum (x_i - A)^2 - (\bar{x} - A)^2. \quad \blacksquare$$

The companion rule you also need. For a general linear transformation $y_i = ax_i + b$:

$$\bar{y} = a\bar{x} + b, \quad \sigma_y^2 = a^2\sigma_x^2, \quad \sigma_y = |a|\sigma_x$$

Chapter 13 states the variance half of this in a formula box but leaves the mean half scattered through solutions. Memorise the pair together: **the mean feels both a and b ; the variance feels only a , and feels it squared.**

Figure brief: A number line with five data dots and the mean marked; below it, the same five dots rigidly slid left by A with the mean sliding by the same amount — caption “shifting the origin moves the mean by A and leaves every gap, hence the variance, unchanged.”

Procedure (E): Reading off mean and variance from deviation sums

1. **Read n and A off the question.** A is the number being subtracted inside the brackets. If two different reference points appear, see Example S4.
2. **Compute $\bar{d} = \frac{1}{n} \sum (x_i - A)$, then $\bar{x} = A + \bar{d}$.** *Why first:* the variance formula needs $\bar{x} - A = \bar{d}$, so this step is not optional bookkeeping.
3. **Compute $\sigma^2 = \frac{1}{n} \sum (x_i - A)^2 - \bar{d}^2$.** *Why subtract:* the first term measures spread about A , which is generally *not* the mean; the correction $\bar{d}^2$ removes the extra distance introduced by anchoring at the wrong point.
4. **Take the positive square root for σ , and check $\sigma^2 \geq 0$.** A negative variance means you either subtracted in the wrong order or squared $\sum (x_i - A)$ where you needed $\sum (x_i - A)^2$.
5. **Apply transformation rules last,** if the question asks about $y_i = ax_i + b$.

Worked Examples

Example S3 — [JEE Main Level — the standard phrasing]

For a set of 9 observations $x_1, x_2, \dots, x_9$ it is given that

$$\sum_{i=1}^9 (x_i - 5) = 9 \quad \text{and} \quad \sum_{i=1}^9 (x_i - 5)^2 = 45.$$

Find the mean and the standard deviation of the observations.

Solution.

Here $n = 9$ and $A = 5$.

Mean. $\bar{d} = \frac{1}{9} \sum (x_i - 5) = \frac{9}{9} = 1$, so

$$\bar{x} = A + \bar{d} = 5 + 1 = 6.$$

Variance.

$$\sigma^2 = \frac{1}{9} \sum (x_i - 5)^2 - (\bar{x} - 5)^2 = \frac{45}{9} - (1)^2 = 5 - 1 = 4.$$

Standard deviation. $\sigma = +\sqrt{4} = 2$.

Verification by a second route. Expand the two given sums in terms of $\sum x_i$ and $\sum x_i^2$ and redo the calculation from the $A = 0$ shortcut.

$$\sum (x_i - 5) = \sum x_i - 45 = 9 \implies \sum x_i = 54 \implies \bar{x} = \frac{54}{9} = 6. \checkmark$$

$$\sum (x_i - 5)^2 = \sum x_i^2 - 10 \sum x_i + 25(9) = \sum x_i^2 - 540 + 225 = 45 \implies \sum x_i^2 = 360.$$

$$\sigma^2 = \frac{\sum x_i^2}{n} - \bar{x}^2 = \frac{360}{9} - 36 = 40 - 36 = 4. \checkmark$$

Both routes give $\sigma = 2$. Note how much longer the second route is — that is precisely why the identity is worth memorising.

Example S4 — [JEE Main Level — two reference points, n unknown]

For a data set $x_1, \dots, x_n$ it is given that

$$\sum (x_i - 2) = 10, \quad \sum (x_i - 5) = -5, \quad \sum (x_i - 2)^2 = 60.$$

Find n , the mean, and the standard deviation.

Solution.

Step 1 — extract n from the two first-order sums. Subtract the second equation from the first. Each term contributes $(x_i - 2) - (x_i - 5) = 3$, and there are n terms:

$$\sum (x_i - 2) - \sum (x_i - 5) = 3n \implies 10 - (-5) = 3n \implies 15 = 3n \implies n = 5.$$

This is the trick the question is built around: the difference of two deviation sums taken about different points kills the data entirely and leaves only n .

Step 2 — mean. With $A = 2$ and $n = 5$: $\bar{d} = \frac{10}{5} = 2$, so

$$\bar{x} = 2 + 2 = 4.$$

Step 3 — variance.

$$\sigma^2 = \frac{1}{5} \sum (x_i - 2)^2 - (\bar{x} - 2)^2 = \frac{60}{5} - (2)^2 = 12 - 4 = 8, \quad \sigma = 2\sqrt{2} \approx 2.83.$$

Verification. Check the second given sum against the answer: $\sum (x_i - 5) = \sum x_i - 5n = 20 - 25 = -5. \checkmark$ And through the $A = 0$ route: $\sum (x_i - 2)^2 = \sum x_i^2 - 4 \sum x_i + 4n = \sum x_i^2 - 80 + 20 = 60$, so $\sum x_i^2 = 120$ and $\sigma^2 = \frac{120}{5} - 4^2 = 24 - 16 = 8. \checkmark$

Practice Exercises

- For 20 observations, $\sum(x_i - 8) = 40$ and $\sum(x_i - 8)^2 = 500$. Find the mean and the standard deviation. **Answer:** $\bar{x} = 8 + \frac{40}{20} = 10$; $\sigma^2 = \frac{500}{20} - 2^2 = 25 - 4 = 21$; $\sigma = \sqrt{21} \approx 4.58$.
- For 100 observations, $\sum(x_i - 50) = -100$ and $\sum(x_i - 50)^2 = 10000$. Find the mean and variance of the x_i , and then the mean and variance of $y_i = 3x_i + 7$. **Answer:** $\bar{x} = 50 + \frac{-100}{100} = 49$; $\sigma_x^2 = \frac{10000}{100} - (-1)^2 = 100 - 1 = 99$. For y : $\bar{y} = 3(49) + 7 = 154$ and $\sigma_y^2 = 3^2(99) = 891$, so $\sigma_y = 3\sqrt{99} = 9\sqrt{11} \approx 29.85$.

Pitfalls

- $(\sum(x_i - A))^2 \neq \sum(x_i - A)^2$. In Example S3 the first is 81 and the second is 45. The correction term in the identity is $\bar{d}^2 = \left(\frac{1}{n} \sum(x_i - A)\right)^2$, i.e. you divide *before* squaring.
- Do not skip the correction when $\bar{d} \neq 0$.** Reporting $\sigma^2 = \frac{45}{9} = 5$ in Example S3 is the single most common wrong answer, and it is exactly what a distractor option will offer. The term $\frac{1}{n} \sum(x_i - A)^2$ is the mean square deviation about A , and it always **overstates** the variance unless $A = \bar{x}$.
- The identity is exact, not approximate.** There is no requirement that A be close to the data. Choose whatever makes the arithmetic clean.
- Standard deviation is the positive root.** $\sigma = |a|\sigma_x$ under $y = ax + b$, not $a\sigma_x$ — a negative multiplier still increases spread.

Bonus — Variance of an Arithmetic Progression

Chapter 13 states, without proof, that the variance of the first n natural numbers is $\frac{n^2-1}{12}$. Here is the proof, and the one-line generalisation JEE Main actually asks for.

Derivation. For $x_i = 1, 2, \dots, n$: $\bar{x} = \frac{n+1}{2}$ and $\sum x_i^2 = \frac{n(n+1)(2n+1)}{6}$. So

$$\sigma^2 = \frac{1}{n} \cdot \frac{n(n+1)(2n+1)}{6} - \left(\frac{n+1}{2}\right)^2 = \frac{(n+1)(2n+1)}{6} - \frac{(n+1)^2}{4} = \frac{(n+1)[2(2n+1) - 3(n+1)]}{12} = \frac{(n+1)(n-1)}{12}$$

Generalisation. An arithmetic progression $a, a + d, a + 2d, \dots, a + (n-1)d$ is the linear image $x_i = (a-d) + di$ of the list $i = 1, 2, \dots, n$. By the transformation rule $\sigma_y^2 = a^2\sigma_x^2$, the shift $(a-d)$ contributes nothing and the multiplier d contributes d^2 :

$$\sigma_{AP}^2 = \frac{d^2(n^2-1)}{12}$$

- First n even numbers $2, 4, \dots, 2n$: $d = 2 \Rightarrow \sigma^2 = \frac{4(n^2-1)}{12} = \frac{n^2-1}{3}$.
- First n odd numbers $1, 3, \dots, 2n-1$: also $d = 2 \Rightarrow \sigma^2 = \frac{n^2-1}{3}$. The two lists differ by a shift of 1, and a shift cannot change variance.

Practice. Find the variance of the first 10 odd natural numbers. **Answer:** $d = 2, n = 10$,

so $\sigma^2 = \frac{4(100-1)}{12} = 33$. (Direct check: mean = 10; deviations $\pm 1, \pm 3, \pm 5, \pm 7, \pm 9$; $\sum (x_i - \bar{x})^2 = 2(1 + 9 + 25 + 49 + 81) = 330$; $\frac{330}{10} = 33$.)

Cheat Sheet Box

Median of a discrete frequency distribution (no interpolation, ever):

- Sort the x_i , build the c.f. column, set $N = \sum f_i$.
- N odd $\Rightarrow M =$ the $\frac{N+1}{2}$ -th observation = the first x_i with c.f. $\geq \frac{N+1}{2}$.
- N even $\Rightarrow M =$ mean of the $\frac{N}{2}$ -th and $(\frac{N}{2} + 1)$ -th observations. If a c.f. equals $\frac{N}{2}$ exactly, these are two different values — average them.
- $\text{M.D.}(M) = \frac{\sum f_i |x_i - M|}{N}$; coefficient = $\frac{\text{M.D.}(M)}{M}$.
- Checks: $\text{M.D.}(M) \leq \text{M.D.}(\bar{x})$; $\text{M.D.} \leq \text{Range}$; units match the data.

Assumed-mean identity (any A , ungrouped):

$$\bar{x} = A + \frac{1}{n} \sum (x_i - A), \quad \sigma^2 = \frac{1}{n} \sum (x_i - A)^2 - (\bar{x} - A)^2$$

- $A = 0$ gives $\sigma^2 = \frac{1}{n} \sum x_i^2 - \bar{x}^2$. Frequency version: replace $\frac{1}{n} \sum$ by $\frac{1}{N} \sum f_i$.
- Two reference points, n unknown: $\sum (x_i - A_1) - \sum (x_i - A_2) = (A_2 - A_1)n$.

Transformation pair for $y_i = ax_i + b$: $\bar{y} = a\bar{x} + b$, $\sigma_y^2 = a^2\sigma_x^2$, $\sigma_y = |a|\sigma_x$.

Standard variances: first n naturals $\frac{n^2 - 1}{12}$; AP with common difference d : $\frac{d^2(n^2 - 1)}{12}$; first n evens or first n odds: $\frac{n^2 - 1}{3}$.

Chapter 13: Statistics — Round 2 Addendum

Round 1 closed the four gaps it found. A second pass over NCERT against the typeset chapter and the Round 1 supplement turns up exactly one more, and it is a procedural one rather than a formula one: every table in Chapter 13 and in the Round 1 supplement is *exclusive* (continuous), so the chapter never has to say what to do when the class limits have gaps between them. NCERT does put such tables in front of you — twice — but only as a bracketed hint at the end of a question, with no explanation anywhere in the running text. This addendum supplies the explanation, and proves the one thing about it that is genuinely non-obvious: the correction changes far less than students fear, and exactly one thing more than they expect.

The combined variance of two groups with different means, flagged as a candidate gap, is already fully closed in the typeset chapter — the formula appears in Subtopic 02 Section 2 with d_1, d_2 defined, is derived in use in Subtopic 02 Example 4, is recapped in Subtopic 03 Section 2, and is combined with the coefficient of variation in Subtopic 03 Example 4. It is also in both cheat sheet boxes. Nothing to add. Weighted mean and weighted variance with arbitrary (non-frequency) weights sit outside the Class 11 syllabus for all three exams; the frequency-weighted forms the chapter already carries are the only version examined.

Addendum 13.3 — Inclusive (Discontinuous) Class Intervals and True Class Boundaries

Exam Relevance: NCERT plants this twice as a bare bracketed hint and never explains it in the body — Exercise 13.1 Q12 (ages given as 16–20, 21–25, ..., with the instruction to subtract 0.5 from each lower limit and add 0.5 to each upper limit) and Exercise 13.2 Q10 (circle diameters 33–36, 37–40, ..., with the same instruction). CBSE Boards can wrap any 3–5 mark dispersion question in an inclusive table, because ages, marks, wages, and counts are naturally recorded as whole numbers. JEE Main uses it as a silent trap inside a median or mean-deviation MCQ: the distractor is precisely the answer you get by treating the printed limits as boundaries. Chapter 13 assumes throughout that limits *are* boundaries — Subtopic 01 Procedure (C) reads the mid-point off “lower limit + upper limit”, and Procedure (D) asks you for “the lower boundary of the median class” without ever telling you how to obtain a boundary when the table has gaps in it. JEE Advanced does not test the bookkeeping, but it will happily hand you an inclusive table inside a longer problem and expect you not to stumble.

The Method

Two kinds of grouped table. A continuous (or *exclusive*) distribution has no gaps: each class ends exactly where the next begins, as in 0–10, 10–20, 20–30. Every table in Chapter 13

is of this kind. An **inclusive** (or *discontinuous*) distribution leaves a gap, as in 16–20, 21–25, 26–30. The name says what it means: in an inclusive class 16–20, both endpoints belong to the class, so a person aged exactly 20 is counted there and a person aged exactly 21 is counted in the next class. Nothing is lost — the gap exists only because the variable was *recorded* in whole years, while the underlying quantity (age) is continuous. Somebody who has lived 20 years and 7 months is recorded as 20 and sits in the first class, and the class therefore really runs up to 20.5.

The correction factor. Let

$$g = (\text{lower limit of a class}) - (\text{upper limit of the previous class})$$

be the gap, and set the correction factor

$$c = \frac{g}{2}.$$

Then the **true class boundaries** are

$$\text{lower boundary} = \text{lower limit} - c, \quad \text{upper boundary} = \text{upper limit} + c.$$

For whole-number data the gap is $g = 1$ and $c = 0.5$, which is the 0.5 in NCERT’s hint. For data recorded to one decimal place — weights as 40.0–44.9, 45.0–49.9 — the gap is $g = 0.1$ and $c = 0.05$, giving boundaries 39.95–44.95, 44.95–49.95. The rule is $g/2$, never a hard-coded 0.5.

Validity. The gap must be the same between every adjacent pair of classes; if a table has unequal gaps it is malformed, not a candidate for this correction. The boundaries produced are continuous by construction — each upper boundary equals the next lower boundary — which is exactly the state Chapter 13’s procedures assume.

What the correction changes, and what it does not. This is the part worth memorising, because it collapses most of the work.

Mid-point invariance. The correction leaves every class mid-point unchanged. For a class with limits l and u , the corrected mid-point is

$$\frac{(l - c) + (u + c)}{2} = \frac{l + u}{2},$$

since the $-c$ and the $+c$ cancel. The class is stretched by c at each end, symmetrically, so its centre does not move.

Every quantity Chapter 13 builds out of mid-points is therefore untouched: the **mean**, the **variance**, the **standard deviation**, the **coefficient of variation**, and the **mean deviation about the mean** are all identical whether you correct the boundaries or not. If a question on an inclusive table asks for any of those, you may ignore the correction entirely.

Two things do change. The first is the **range**, which is defined on boundaries:

$$\text{Range} = (u_k + c) - (l_1 - c) = u_k - l_1 + 2c.$$

The second, and the one that costs marks, is the **interpolated median** — because the median formula

$$M = \ell + \frac{\frac{N}{2} - C}{f} \times h$$

reads both the lower boundary ℓ and the width h off the class, and the correction moves ℓ down by c while stretching h from $u - l$ to $u - l + 2c$. Since the median moves, so does the **mean deviation about the median**.

How far the median moves. Write $t = \frac{N/2 - C}{f}$ for the position fraction inside the median class. Uncorrected you would compute $M_{\text{naive}} = l + t(u - l)$; corrected you get $M = (l - c) + t(u - l + 2c)$. Subtracting,

$$M - M_{\text{naive}} = 2c \left(t - \frac{1}{2} \right).$$

Read what this says. The error is proportional to how far t sits from the middle of the class. If $t = \frac{1}{2}$ exactly — the median lands dead centre in its class — the two routes agree and the correction costs nothing, because a median at the centre of the class *is* the mid-point, and mid-points are invariant. Otherwise the two answers differ, and the error is bounded by c in either direction. It is small, but it is real, and for $g = 1$ it is enough to separate two printed options.

Procedure (F): handling an inclusive distribution.

1. Check whether the table is inclusive at all. Compare the upper limit of any class with the lower limit of the next. Equal means the table is already continuous — do nothing. A gap means you must decide whether you need the correction. *Why first:* running the correction on an already-continuous table is a way to manufacture a wrong answer out of a right table.
2. Read what the question asks for. If it wants the mean, the variance, the standard deviation, the coefficient of variation, or the mean deviation about the **mean**, stop — mid-points are invariant, so build the table straight from $x_i = \frac{l_i + u_i}{2}$ and proceed exactly as in Subtopic 01 Procedure (C) or Subtopic 02. *Why:* you are being asked for a quantity that cannot feel the correction.
3. If the question wants the **median**, the mean deviation about the median, or the **range**, compute g from any adjacent pair and set $c = g/2$. *Why:* these three are the only quantities that read boundaries rather than centres.
4. Rewrite the class column with boundaries $l_i - c$ and $u_i + c$. Confirm the new upper boundary of each class equals the new lower boundary of the next — if it does not, you mis-measured g . *Why:* that equality is the whole point of the correction, and it is a free self-check.

- Build the cumulative frequency column and locate the median class as usual (first class whose c.f. reaches $\frac{N}{2}$). Note that the correction never moves the median *class*, only the numbers inside it, because it does not touch any frequency.
- Interpolate with the corrected ℓ and the corrected $h = u - \ell + 2c$. Then form $|x_i - M|$ and $f_i|x_i - M|$ using the **uncorrected** mid-points, and divide by N .

Figure brief: Two horizontal number-line strips stacked vertically over a shared axis. The top strip shows inclusive classes 16-20, 21-25, 26-30 as three separate blocks with visible white gaps between them, each block's centre marked with a dot labelled 18, 23, 28. The bottom strip shows the same three blocks each grown by 0.5 at both ends so they now abut at 20.5 and 25.5 with no gaps, and the centre dots sit at exactly the same horizontal positions, still labelled 18, 23, 28. A vertical dashed line drops through each pair of dots to show they have not moved. Caption: "The blocks stretch outward; their centres stay put. That is why the mean and variance do not care, and only boundary-based quantities do."

Worked Examples

Example S7 — [CBSE Board Level: the NCERT case]

The ages of 100 persons are distributed as below. Calculate the mean deviation about the median age.

Age (years)	16-20	21-25	26-30	31-35	36-40	41-45	46-50	51-55
Number of persons	5	6	12	14	26	12	16	9

Solution.

Step 1 — spot the gaps. The upper limit 20 is followed by the lower limit 21, so $g = 1$ and $c = 0.5$. The question asks for the mean deviation about the **median**, so the correction is required.

Step 2 — corrected boundaries. 15.5-20.5, 20.5-25.5, 25.5-30.5, 30.5-35.5, 35.5-40.5, 40.5-45.5, 45.5-50.5, 50.5-55.5. Each upper boundary now equals the next lower boundary, and every class has width $h = 5$ (not 4, which is what the printed limits suggest).

Step 3 — mid-points and cumulative frequencies. The mid-points are unchanged by the correction: 18, 23, 28, 33, 38, 43, 48, 53.

x_i	18	23	28	33	38	43	48	53
f_i	5	6	12	14	26	12	16	9
c.f.	5	11	23	37	63	75	91	100

Step 4 — median. $N = 100$, so $\frac{N}{2} = 50$. The first c.f. to reach 50 is 63, so the median class is the one with mid-point 38, i.e. boundaries 35.5-40.5. With $\ell = 35.5$, $C = 37$, $f = 26$, $h = 5$:

$$M = 35.5 + \frac{50 - 37}{26} \times 5 = 35.5 + \frac{13}{26} \times 5 = 35.5 + 2.5 = 38 \text{ years.}$$

Step 5 — mean deviation.

x_i	f_i	$ x_i - 38 $	$f_i x_i - 38 $
18	5	20	100
23	6	15	90
28	12	10	120
33	14	5	70
38	26	0	0
43	12	5	60
48	16	10	160
53	9	15	135
Total	100		735

$$\text{M.D.}(M) = \frac{735}{100} = 7.35 \text{ years.}$$

A word of warning about this particular table. Here $t = \frac{50 - 37}{26} = \frac{1}{2}$ exactly, so by the shift formula $M - M_{\text{naive}} = 2c(t - \frac{1}{2}) = 0$: a student who ignored the correction and used $\ell = 36$, $h = 4$ would have got $36 + \frac{1}{2}(4) = 38$ as well, and would have scored full marks by luck. Do not learn the wrong lesson from that. It happened only because the median landed at the dead centre of its class, where the median coincides with the invariant mid-point 38. Example S8 shows what happens when it does not.

Example S8 — [JEE Main Level: where the trap actually fires]

For the following distribution of scores, find the median and the mean deviation about the median.

Score	1-10	11-20	21-30	31-40	41-50
Frequency	4	10	20	12	4

Solution.

$g = 11 - 10 = 1$, so $c = 0.5$ and the boundaries are 0.5-10.5, 10.5-20.5, 20.5-30.5, 30.5-40.5, 40.5-50.5, each of width $h = 10$.

Mid-points (unchanged): 5.5, 15.5, 25.5, 35.5, 45.5. Cumulative frequencies: 4, 14, 34, 46, 50.

$N = 50$, so $\frac{N}{2} = 25$; the first c.f. to reach 25 is 34, giving the median class 20.5-30.5. With $\ell = 20.5$, $C = 14$, $f = 20$, $h = 10$:

$$M = 20.5 + \frac{25 - 14}{20} \times 10 = 20.5 + \frac{11}{20} \times 10 = 20.5 + 5.5 = 26.$$

x_i	f_i	$ x_i - 26 $	$f_i x_i - 26 $
5.5	4	20.5	82
15.5	10	10.5	105
25.5	20	0.5	10
35.5	12	9.5	114
45.5	4	19.5	78
Total	50		389

$$\text{M.D.}(M) = \frac{389}{50} = 7.78.$$

The seductive wrong answer. Treat the printed limits as boundaries — $\ell = 21$, $h = 9$ — and you get

$$M_{\text{naive}} = 21 + \frac{11}{20} \times 9 = 21 + 4.95 = 25.95, \quad \text{M.D.} = \frac{388.1}{50} = 7.762,$$

and 25.95 and 7.762 are exactly the distractors a paper-setter prints alongside 26 and 7.78. Check the discrepancy against the shift formula: $t = \frac{11}{20}$, so

$$M - M_{\text{naive}} = 2(0.5) \left(\frac{11}{20} - \frac{1}{2} \right) = \frac{1}{20} = 0.05,$$

and indeed $26 - 25.95 = 0.05$. Note also that the naive route did not even use a self-consistent table: its five classes of width 9 with gaps between them account for only 45 of every 50 units of the score axis, so it is measuring the median against a ruler with pieces missing.

The cross-check that costs nothing. The mean of this distribution is

$$\bar{x} = \frac{4(5.5) + 10(15.5) + 20(25.5) + 12(35.5) + 4(45.5)}{50} = \frac{1295}{50} = 25.9,$$

and this value is the same on either convention, because it is built only from mid-points. So is the variance, $\sigma^2 = 107.84$. If you ever find your mean or variance changing when you apply the boundary correction, you have made an arithmetic error — the correction cannot touch them.

Practice Exercises

- For the distribution below, find (i) the range, (ii) the median, and (iii) the mean deviation about the median.

Class	10-19	20-29	30-39	40-49	50-59
Frequency	5	9	15	8	3

Answer: $c = 0.5$; boundaries 9.5-19.5, ..., 49.5-59.5, each of width 10. (i) Range = $59.5 - 9.5 = 50$. (ii) $N = 40$, c.f. = 5, 14, 29, 37, 40; median class 29.5-39.5; $M = 29.5 + \frac{20-14}{15} \times 10 =$

- 33.5. (iii) $\sum f_i |x_i - M| = 342$, so $\text{M.D.}(M) = \frac{342}{40} = 8.55$. (Ignoring the correction gives $M = 33.6$ and $\text{M.D.} = 8.52$ — both wrong.)
2. For the same distribution as Exercise 1, find the mean and the variance, and state in one line why the boundary correction was irrelevant to this question. *Answer:* mid-points 14.5, 24.5, 34.5, 44.5, 54.5; $\bar{x} = \frac{1330}{40} = 33.25$; $\sigma^2 = \frac{1935}{16} = 120.9375$. The correction stretches each class symmetrically, so no mid-point moves, and both quantities are built only from mid-points.
3. The diameters (in mm) of circles in a design are distributed as below. Find the mean diameter and the standard deviation.

Diameter	33-36	37-40	41-44	45-48	49-52
Number of circles	15	17	21	22	25

- Answer:* $\bar{x} = 43.5$ mm, $\sigma^2 = 30.84$ mm², $\sigma \approx 5.55$ mm. NCERT's hint tells you to convert to 32.5-36.5, 36.5-40.5, and so on; do it if you like, but for the mean and the standard deviation it changes nothing, since the mid-points 34.5, 38.5, 42.5, 46.5, 50.5 are the same either way. (Had the question asked for the median, it would have mattered: $43\frac{13}{14} \approx 43.93$ with the correction against $43\frac{4}{7} \approx 43.57$ without.)
4. Weights are recorded to the nearest tenth of a kilogram in the classes 40.0-44.9, 45.0-49.9, 50.0-54.9. Write down the correction factor and the true class boundaries. *Answer:* $g = 45.0 - 44.9 = 0.1$, so $c = 0.05$; boundaries 39.95-44.95, 44.95-49.95, 49.95-54.95, each of width 5. The correction is $g/2$, not a reflexive 0.5.

Pitfalls

- **Hard-coding 0.5.** The correction factor is half the gap, $c = g/2$. Whole-number data happens to give $c = 0.5$, which is all NCERT's two hints ever show you, so students memorise the number instead of the rule and then subtract 0.5 from a table recorded to one decimal place. Measure the gap first, every time.
- **“Correcting” a table that is already continuous.** Classes 0-10, 10-20, 20-30 have $g = 0$ and need nothing. Subtracting 0.5 anyway produces overlapping classes and a wrong median. The test is one subtraction: upper limit of a class minus lower limit of the next.
- **Applying the correction to the mid-points.** The correction moves the *limits*; the mid-point is their average and does not move. Writing $x_i = \frac{15.5+20.5}{2} = 18$ is right, and so is $x_i = \frac{16+20}{2} = 18$ — but a student who “corrects” the mid-point itself to 17.5 or 18.5 has invented a shift out of nothing and will corrupt the mean, the variance, and the mean deviation all at once.
- **Forgetting that h grows too.** The commonest half-correction is to drop ℓ from 36 to 35.5 and leave h at 4. The class was stretched at *both* ends, so h goes from $u - l$ to $u - l + 2c$ — from 4 to 5 for whole-number ages. Half the correction is worse than none, because it breaks the $t = \frac{1}{2}$ coincidence that would otherwise have saved you.
- **Recomputing the mean and variance “to be safe”.** They cannot change. Spending three minutes re-deriving them under corrected boundaries in a Boards paper is three

minutes gone for an identical number.

- **Thinking the median class moves.** It cannot: the correction touches no frequency, so the cumulative frequency column and the class that first reaches $\frac{N}{2}$ are exactly as before. Only ℓ and h inside that class change.

Cheat Sheet Box

Inclusive (discontinuous) classes — rapid recall

- **Spot it:** upper limit of a class $\neq$ lower limit of the next. Gap g ; correction factor $c = \frac{g}{2}$.
- **Fix it:** lower boundary $= l - c$, upper boundary $= u + c$; new width $h = u - l + 2c$. Check: each upper boundary must equal the next lower boundary.
- **Unchanged by the correction** (mid-points are invariant, $\frac{(l-c)+(u+c)}{2} = \frac{l+u}{2}$): mean, variance, standard deviation, C.V., mean deviation about the **mean**. Do not recompute them.
- **Changed by the correction:** range $= u_k - l_1 + 2c$; the interpolated **median**, hence the mean deviation about the median.
- **Size of the median shift:** with $t = \frac{N/2 - C}{f}$,

$$M - M_{\text{naive}} = 2c \left(t - \frac{1}{2} \right),$$

zero exactly when $t = \frac{1}{2}$ (median at the class centre), and never more than c in magnitude.

- **The median class never moves** — no frequency is touched.
- Mnemonic: **“Stretch the edges, not the centre.”**

Class 11 Mathematics

Chapter 14: Probability

Drona Ed Tech — Content Bank

Complete Chapter — Subtopics 01-03

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Subtopic 01 — Sample Space and Events

Exam Relevance: This is the grammar of the entire chapter, so it is everywhere indirectly. CBSE Boards asks it as a direct 1-2 mark “write the sample space / express the event as a set” question. CUET poses it as fast one-liners on event classification. JEE Main rarely asks it alone but assumes it silently in every probability

problem, and JEE Advanced builds set-counting and partition arguments directly on this foundation. Master it now and the rest of the chapter stops feeling like guesswork.

Section 1: Concept Intuition

Imagine you're standing at a railway booking counter, and the clerk is about to tell you which berth you got — Lower, Middle, Upper, Side-Lower, or Side-Upper. Before he speaks, you don't *know* the answer, but you absolutely know the **complete menu of things that could happen**. There are exactly five possibilities, no more, no less. That full menu — the set of every outcome that *could* occur — is the heart of what we're about to build.

In probability, any action whose result we cannot predict with certainty, but whose set of possible results we *can* list in advance, is called a **random experiment**. Tossing a coin, rolling a die, drawing a card, checking whether the 8:10 local is on time — all random experiments. The key word is *random*: if you “experiment” by adding $2 + 3$, you always get 5, so there's nothing random about it. A random experiment needs genuine uncertainty in the result but total clarity about the list of candidates.

Each individual result on that menu is called an **outcome** (or a **sample point**). The complete menu — the set of *all* possible outcomes — is the **sample space**, written S . For a single die, $S = \{1, 2, 3, 4, 5, 6\}$. For one coin, $S = \{H, T\}$. The sample space is the universe of the experiment; everything else we talk about lives *inside* it.

Now, **Think about it this way...** Suppose you've laid out all six faces of a die on the table as six little cards. An **event** is simply you scooping up *some* of those cards — any selection you like — and saying “I care about these.” “The number is even” means you scoop up $\{2, 4, 6\}$. “The number is bigger than 4” means you scoop up $\{5, 6\}$. So an event is nothing more mysterious than a **subset of the sample space**. That single idea — *event = subset* — is the most important sentence in this subtopic. Once you see events as subsets, every later rule becomes set theory you already know from Chapter 1.

An event has **occurred** if the actual outcome of the experiment is one of the cards you scooped up. Roll the die, get a 4 — your “even” event occurred, because $4 \in \{2, 4, 6\}$. Get a 3 — it did not, because $3 \notin \{2, 4, 6\}$.

Two special selections sit at the extremes. If you scoop up *nothing* (the empty set ϕ), no outcome can ever make it occur — that's the **impossible event** (e.g., “the die shows a 7”). If you scoop up *everything* (the whole of S), some outcome must always make it occur — that's the **sure** or **certain event** (e.g., “the die shows a number from 1 to 6”).

Limiting conditions and assumptions. This entire framework rests on two quiet assumptions worth dragging into the light. First, the sample space must be **well-defined and complete** — every outcome listed exactly once, none missing. The moment you mis-specify S , every event built on it is wrong, and this is the single most common silent error in the chapter. Second, for an experiment with **ordered stages** (toss a coin, *then* roll a die), the outcomes are ordered pairs and $\{H, 4\}$ is genuinely different from a later $\{4, H\}$ if the order of stages differs — collapsing order destroys outcomes. Finally, note that “event” and “outcome” are *not* synonyms: an outcome is a single point of S ; an event is a *set* of points (possibly just one, possibly many, possibly none).

Section 2: Key Definitions & The Algebra of Events

Because this subtopic is definitional rather than formula-driven, the “core quantities” here are *set-theoretic objects*, not dimensioned physical quantities. The table below is your working vocabulary.

Term	Symbol	Meaning
Sample space	S	Set of all possible outcomes
Sample point / outcome	ω	A single element of S
Number of sample points	$n(S)$	Count of elements in S
Event	$E, A, B, \dots$	Any subset of S , i.e. $E \subseteq S$
Impossible event	ϕ	The empty subset
Sure (certain) event	S	The whole sample space

Types of events (classified by how many points they hold):

- **Simple / elementary event:** exactly one sample point. A sample space with n outcomes has exactly n simple events.
- **Compound (mixed / composite) event:** more than one sample point.

The algebra of events — combining events with the set operations of Chapter 1. Let $A, B \subseteq S$.

$$A' = \bar{A} = S - A = \{\omega : \omega \in S \text{ and } \omega \notin A\} \quad (\text{complement, the event "not } A\text{")}$$

$$A \cup B = \text{"}A \text{ or } B\text{"} \quad (\text{at least one occurs})$$

$$A \cap B = \text{"}A \text{ and } B\text{"} \quad (\text{both occur simultaneously})$$

$$A - B = A \cap B' = \text{"}A \text{ but not } B\text{"}$$

Relationship definitions (the ones examiners love):

- **Mutually exclusive (disjoint):** $A \cap B = \phi$. The two events cannot occur together.
- **Exhaustive:** $E_1 \cup E_2 \cup \dots \cup E_n = S$. At least one must occur.
- **Mutually exclusive and exhaustive (a partition):** $E_i \cap E_j = \phi$ for all $i \neq j$ and $\bigcup_{i=1}^n E_i = S$. Exactly one occurs, always. *Both conditions are required — neither alone is enough.*
- **Equally likely events:** each is as likely to occur as any other (this is the assumption baked into the word “fair,” and into “at random”).

Note on units: Unlike the Physics bank, none of these quantities carry dimensions or SI units — they are pure set objects. $n(S)$ and $n(A)$ are dimensionless counts (natural numbers). This is the math-v1 analogue of the “dimensional formula” slot.

Section 3: Key Procedures & Logical Framework

Sample Space & Events has no formal *derivation* — it has **procedures**. Here are the three that decide most marks.

Procedure A — Constructing a sample space.

1. Identify what is being *recorded* on each trial. (“Colour AND number,” or just “number”? The sample space changes entirely with what you observe.)
2. If the experiment is **single-stage**, list every distinct outcome once. For a die: $\{1, 2, 3, 4, 5, 6\}$.
3. If it is **multi-stage / sequential**, build a **tree**: branch for stage 1, then sub-branch for stage 2, and read each complete root-to-leaf path as one ordered outcome. *Why a tree?* Because it mechanically guarantees you neither miss a combination nor double-count one — the two failure modes that wreck this question.
4. Watch **with vs. without replacement**: without replacement removes the “repeat” outcomes (you cannot draw the same ball twice), shrinking S .
5. Sanity check $n(S)$: for k independent stages with $m_1, m_2, \dots$ outcomes each, expect $n(S) = m_1 \times m_2 \times \dots$ — if your hand-built list disagrees, you’ve missed or duplicated a branch.

Procedure B — Translating a verbal event into a subset.

1. Write the full sample space first (Procedure A). Never skip this.
2. Read the event’s wording and **scan every outcome of S** , keeping only those that satisfy the condition. The event is exactly that collection.
3. Mind the quantifier words: *at least one* $\neq$ *exactly one* $\neq$ *at most one*. These three phrases produce three different subsets from the same S — and examiners weaponise the difference.

Procedure C — Testing a relationship between events.

1. **Mutually exclusive?** Compute $A \cap B$. If it equals ϕ , yes; otherwise no.
2. **Exhaustive?** Compute $A \cup B \cup \dots$. If it equals S , yes; otherwise no.
3. **Partition (mutually exclusive AND exhaustive)?** Both of the above must hold *together*. Check both — a set can be one without the other.

Section 4: Worked Examples

Example 1 — [CBSE Board Level] A fair spinner is divided into 8 equal sectors numbered 1 to 8. It is spun once. (a) Write the sample space. (b) Let A be the event “the number is a multiple of 3” and B the event “the number is greater than 5.” Express A , B , and $A \cap B$ as subsets.

Solution.

(a) Every sector is a possible outcome, recorded once:

$$S = \{1, 2, 3, 4, 5, 6, 7, 8\}, \quad n(S) = 8.$$

(b) Scan S for multiples of 3: $A = \{3, 6\}$. Scan S for numbers > 5 : $B = \{6, 7, 8\}$. Common to both: $A \cap B = \{6\}$. Since $A \cap B = \{6\} \neq \phi$, the events A and B are **not** mutually exclusive. ■

(Presentation marks come from writing S first, then each event as an explicit set, then the intersection — exactly that order.)

Example 2 — [CUET / Speed Trap] A single card is drawn from a standard 52-card deck. Which one of the following pairs of events is mutually exclusive? (i) “card is a King” and “card is a Heart” (ii) “card is a Spade” and “card is red” (iii) “card is a face card” and “card is a Queen” (iv) “card is even-numbered” and “card is black”

Fast solution — eliminate by overlap, don’t compute. Mutually exclusive means zero overlap.

- (i) The King of Hearts is both $\rightarrow$ overlap exists $\rightarrow$ not exclusive.
- (ii) Spades are black, never red $\rightarrow$ overlap is empty $\rightarrow$ **mutually exclusive.** ✓
- (iii) Every Queen is a face card $\rightarrow$ Queens lie in both $\rightarrow$ not exclusive.
- (iv) The 2 of Spades is even and black $\rightarrow$ overlap exists $\rightarrow$ not exclusive.

Answer: (ii). *Speed tip:* you only need to find a **single** common card to kill an option — hunt for one counterexample, never enumerate the whole set. ■

Example 3 — [JEE Main Level] Two distinguishable dice (one red, one blue) are rolled and the ordered pair (red, blue) is recorded. Let $A =$ “the sum is 7”, $B =$ “the red die shows 6”, and $C =$ “the sum is at least 11.” (a) Find $n(S)$. (b) Write A . (c) Are A and C mutually exclusive? (d) Is $\{A, A'\}$ a partition of S ?

Solution.

- (a) Ordered pairs over two independent stages of 6 outcomes each:

$$n(S) = 6 \times 6 = 36.$$

- (b) Pairs summing to 7:

$$A = \{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\}, \quad n(A) = 6.$$

- (c) $C = \text{sum} \geq 11 = \{(5, 6), (6, 5), (6, 6)\}$. The smallest sum in C is 11, but every pair in A sums to 7, so $A \cap C = \phi$. **Yes, mutually exclusive.**

- (d) For any event, $A \cap A' = \phi$ (mutually exclusive ✓) and $A \cup A' = S$ (exhaustive ✓). Both hold, so $\{A, A'\}$ is **always a partition** of S — this is true for *any* event and its complement, a fact worth memorising. ■

(This problem links sample-space construction (counting) with the partition test (set algebra) — the two-concept pattern JEE Main favours.)

Example 4 — [JEE Advanced Level] A random experiment has a sample space with exactly n distinct outcomes ($n \geq 2$). (a) How many distinct events (subsets, including ϕ and S) does the experiment have? (b) How many of these events are compound? (c) How many unordered pairs $\{A, B\}$ of non-empty events are mutually exclusive (i.e. $A \cap B = \phi$, with $A \neq B$)?

Solution.

- (a) An event is any subset of an n -element set, and a set of size n has 2^n subsets:

$$\text{number of events} = 2^n.$$

- (b) Compound = more than one point. Remove the n simple (singleton) events and the 1 empty event:

$$2^n - n - 1.$$

- (c) Count *ordered* disjoint pairs (A, B) with $A, B \neq \phi$, then halve. Assign each of the n outcomes to one of three bins — “only A ”, “only B ”, or “neither” — giving 3^n assignments. Subtract those that make A empty (2^n , every point in “only B ” or “neither”) and those that make B empty (2^n), then add back the one assignment counted in both (all points in “neither”):

$$\#\{(A, B) : A, B \neq \phi, A \cap B = \phi\} = 3^n - 2 \cdot 2^n + 1.$$

Since the pair is unordered and $A \neq B$ automatically (two disjoint non-empty sets can't be equal), divide by 2:

$$\frac{3^n - 2^{n+1} + 1}{2}.$$

Check $n = 2$, $S = \{a, b\}$: formula gives $\frac{9-8+1}{2} = 1$, and indeed the only disjoint non-empty pair is $\{\{a\}, \{b\}\}$. ✓ ■

Section 5: Practice Exercises

1. **[CBSE]** A coin is tossed and then a die is rolled. Using a tree, write the sample space and state $n(S)$. Express the event “the coin shows Head and the die shows a prime.”
2. **[CUET]** From the set $\{10, 11, 12, \dots, 20\}$ a number is chosen. Are the events “the number is prime” and “the number is even” mutually exclusive? Are they exhaustive?
3. **[JEE Main]** Three coins are tossed and the ordered triple of results is recorded. Write the sample space, then express the events E = “exactly two heads” and F = “at most one head.” Are E and F mutually exclusive?
4. **[JEE Main]** Two balls are drawn in succession *without replacement* from a bag holding 1 green, 1 yellow and 1 black ball, recording the ordered pair. Write S , and the event “the yellow ball is drawn at some stage.”
5. **[JEE Advanced]** A sample space has n outcomes. How many ordered triples (A, B, C) of pairwise-disjoint non-empty events exist? (Hint: four bins per outcome.)

Answers:

1. $n(S) = 12$; event = $\{(H, 2), (H, 3), (H, 5)\}$.
2. Mutually exclusive: **yes** (no even prime in this range — 2 isn't listed). Exhaustive: **no** (e.g. 15 is neither).
3. $n(S) = 8$; $E = \{HHT, HTH, THH\}$, $F = \{HTT, THT, TTH, TTT\}$; **yes**, mutually exclusive ($E \cap F = \phi$).
4. $S = \{(G, Y), (G, B), (Y, G), (Y, B), (B, G), (B, Y)\}$, $n(S) = 6$; event = $\{(G, Y), (Y, G), (Y, B), (B, Y)\}$.
5. $4^n - 3 \cdot 3^n + 3 \cdot 2^n - 1$ (inclusion-exclusion on the three “empty” conditions).

Section 6: Crack the MCQ

Q1. A die is rolled once. The events $A = \{1, 2, 3\}$ and $B = \{3, 4, 5, 6\}$ are:

- (a) mutually exclusive only (b) exhaustive only (c) mutually exclusive and exhaustive (d) neither

Answer: (b). $A \cup B = \{1, 2, 3, 4, 5, 6\} = S \rightarrow$ exhaustive. But $A \cap B = \{3\} \neq \phi \rightarrow$ not mutually exclusive. *Trap (a):* picked by students who eyeball “these look like opposite halves”

without computing the intersection. *Trap (c)*: the classic reflex of assuming exhaustive $\Rightarrow$ mutually exclusive too. *Trap (d)*: missed that the union already fills S .

Q2. For an experiment with sample space S , which statement is **always** true for any event A ?

- (a) $A \cup A' = \phi$ (b) $A \cap A' = S$ (c) $A \cup A' = S$ and $A \cap A' = \phi$ (d) $A = A'$

Answer: (c). An event and its complement always partition S : their union is everything, their intersection is nothing. *Trap (a) and (b)*: swap the roles of $\cup$ and $\cap$ — union *fills*, intersection *empties*, not the reverse. *Trap (d)*: only possible if $A = A'$, impossible for a genuine subset unless S is empty.

Q3. A coin is tossed twice (ordered). How many *compound* events does this experiment have?

- (a) 4 (b) 11 (c) 12 (d) 16

Answer: (b). $S = \{HH, HT, TH, TT\}$, so $n(S) = 4$ and total events $= 2^4 = 16$. Remove 4 simple events and the 1 impossible event: $16 - 4 - 1 = 11$. *Trap (d)*: counted *all* events, forgetting “compound” excludes singletons and ϕ . *Trap (c)*: removed only the empty event ($16 - 4 = 12$), forgetting the simple events. *Trap (a)*: counted only the simple events.

Q4. Two events A and B satisfy $A \subseteq B$. Then $A \cap B$ equals:

- (a) ϕ (b) A (c) B (d) S

Answer: (b). If every point of A is also in B , then the points common to both are exactly the points of A , so $A \cap B = A$. *Trap (c)*: confuses intersection with union — under $A \subseteq B$ it's $A \cup B = B$, not the intersection. *Trap (a)*: assumes “two named events $\Rightarrow$ disjoint,” a frequent reflex. *Trap (d)*: with no reason for either set to be all of S .

Section 7: Common Student Pitfalls & Pro-Tips

- Confusing an outcome with an event.** “Getting a 4” is an *outcome* (ω); “getting an even number” is an *event* (a set $\{2, 4, 6\}$). When a question says “write the event,” it wants a **set**, even if that set has one element — write $\{4\}$, not 4.
- Assuming mutually exclusive $\Rightarrow$ exhaustive (or the reverse).** These are independent conditions. $A = \{1\}$, $B = \{2\}$ on a die are mutually exclusive but nowhere near exhaustive. Always test $A \cap B$ **and** $A \cup B$ separately.
- Mis-built sample space from a multi-stage experiment.** Forgetting “without replacement,” or collapsing ordered pairs, silently changes S and poisons every event after it. Draw the tree — it costs ten seconds and saves the question.
- The mutually-exclusive vs. independent confusion (early warning).** Many students will later, in Class 12, treat “mutually exclusive” and “independent” as the same thing. They are nearly opposites. Plant the flag now: *mutually exclusive* = *cannot happen together*; that's a statement about the **sets** ($A \cap B = \phi$), not about probabilities.

Pro-Tip: To check “mutually exclusive,” you don't need the full intersection — just hunt for **one** outcome sitting in both events. Find one $\Rightarrow$ not mutually exclusive, stop immediately. Find none after scanning $\Rightarrow$ mutually exclusive. This turns a calculation into a one-second search.

Section 8: Cheat Sheet Box

SAMPLE SPACE & EVENTS — RAPID RECALL

- **Random experiment:** uncertain result, but a fully listable set of outcomes.
- **Sample space S :** set of *all* outcomes. **Event E :** any subset, $E \subseteq S$. $\rightarrow$ *Event = subset*.
- **Occurrence:** E occurs iff the actual outcome $\omega \in E$.
- **Impossible** = ϕ ; **Sure** = S ; **Simple** = 1 point; **Compound** = more than 1 point.
- **Algebra:** $A' = S - A$; $A \cup B$ = “or”; $A \cap B$ = “and”; $A - B = A \cap B' =$ “A but not B.”
- **Mutually exclusive:** $A \cap B = \phi$. **Exhaustive:** $\bigcup E_i = S$. **Partition:** *both* together.
- **Counts:** n outcomes $\Rightarrow 2^n$ events, of which n are simple and $2^n - n - 1$ are compound. $\{A, A'\}$ is always a partition.

Mnemonics:

- “*Exclusive = Empty intersection*” (both start with E... the intersection is empty).
- “*Union fills, intersection empties*” — for $A \cup A' = S$ vs. $A \cap A' = \phi$.
- “*At-least / exactly / at-most*” = three different subsets. Never blur them.

Diagram placeholder

Two side-by-side Venn diagrams inside a rectangle labelled S . Left: two overlapping circles A and B with the lens shaded and labelled “ $A \cap B \neq \varnothing$ — NOT mutually exclusive.” Right: two separate non-touching circles A and B labelled “ $A \cap B = \varnothing$ — mutually exclusive.”

Diagram placeholder

A tree diagram for “toss a coin, then roll a die.” First level branches H and T ; from each, six branches labelled 1-6; the twelve leaves read $(H,1) \dots (H,6), (T,1) \dots (T,6)$, illustrating $n(S)=12$ for a two-stage experiment.

Subtopic 02 — Probability of Events and the Axiomatic Approach

Exam Relevance: This is one of the highest-yield subtopics in the chapter. CBSE Boards asks the addition rule and complement law as 2-3 mark numericals almost every year. CUET turns the axioms into fast “is this a valid probability assignment?” one-liners. JEE Main treats $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ and $P(A') = 1 - P(A)$ as everyday tools, and JEE Advanced builds non-uniform (loaded) sample-space problems and inequality bounds directly on the axioms. If Sub-01 gave you the *language*, this gives you the *arithmetic of chance*.

Section 1: Concept Intuition

In Sub-01 we learned to lay out every possible outcome of an experiment and to scoop up the ones we care about into an *event*. But we never answered the obvious next question: **how likely is that event?** Assigning a number — a probability — to each event is the job of this subtopic.

You already met probability informally in earlier classes. Roll a fair die; the chance of an even number is “three good faces out of six,” i.e. $\frac{3}{6} = \frac{1}{2}$. That recipe — *favourable outcomes over total outcomes* — is the **classical definition** of probability. It is wonderfully intuitive and, for fair dice and well-shuffled cards, completely correct.

But it hides two cracks. **Think about it this way...** First, the classical recipe only works when every outcome is *equally likely*. What is the probability that a drawing pin, when tossed, lands point-up? There are two outcomes — point-up, point-down — but they are emphatically *not* equally likely, so “one out of two” is nonsense. Second, the recipe collapses when there are **infinitely many outcomes**: “pick a random natural number” has no finite total to divide by. There is also a third, deeper philosophical crack the NCERT text itself points at: the classical definition quietly *uses the word “equally likely,” which already means “equally probable”* — so we are defining probability in terms of probability. That is a circular definition, and mathematicians dislike circles.

The fix, given by the Russian mathematician **A. N. Kolmogorov in 1933**, was beautifully sneaky. Instead of *defining* what probability “is,” he simply listed the **rules any sensible probability must obey** — and declared anything obeying those rules to be a legitimate probability. This is the **axiomatic approach**. It is the same move geometry makes: Euclid never defines “point” and “line” from scratch; he states axioms they must satisfy and reasons from there.

So picture it like distributing one full kilogram of sand across the outcomes of an experiment. Each outcome gets *some* non-negative amount of sand (possibly zero), and the *total* sand across the whole sample space is exactly one kilogram. The probability of an event is just *how much sand sits on the outcomes inside that event*. A fair die spreads the sand evenly — $\frac{1}{6}$ kg per face. A loaded die piles more onto some faces than others. Both are perfectly valid, because both respect the same two rules: no negative sand, and one kilogram total. That picture is the axiomatic approach, and it instantly handles loaded pins and unfair coins that the classical recipe could not.

Limiting conditions and assumptions. The axioms (in the form you need for Class 11) are stated for **finite sample spaces** and for **mutually exclusive** events — additivity is only guaranteed when events don’t overlap. The familiar $P(E) = \frac{n(E)}{n(S)}$ formula is *not* an axiom; it

is a **special case** that pops out only when all outcomes are equally likely. Forget that caveat and you will confidently apply $\frac{n(E)}{n(S)}$ to a loaded die and get a wrong answer that *looks* right.

Section 2: Key Formulas, Definitions & Validity Conditions

The three axioms of probability (Kolmogorov, finite-sample-space form). For a sample space S and any events $A, B \subseteq S$:

Axiom 1 (Non-negativity): $P(A) \geq 0$

Axiom 2 (Normalization): $P(S) = 1$

Axiom 3 (Additivity): $A \cap B = \phi \Rightarrow P(A \cup B) = P(A) + P(B)$

Probability via sample points. Assign to each sample point $\omega_i \in S$ a number $P(\omega_i)$ with

$$P(\omega_i) \geq 0 \quad \text{and} \quad \sum_{\omega_i \in S} P(\omega_i) = 1.$$

Then for any event E ,

$$P(E) = \sum_{\omega_i \in E} P(\omega_i).$$

Consequences you will use constantly (each derived in §3):

Result	Formula
Range of probability	$0 \leq P(A) \leq 1$
Impossible event	$P(\phi) = 0$
Sure event	$P(S) = 1$
Complement (the “not” law)	$P(A') = 1 - P(A)$
Equally-likely special case	$P(A) = \frac{n(A)}{n(S)}$
Addition rule (general)	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
Addition rule (mutually exclusive)	$P(A \cup B) = P(A) + P(B)$
“A but not B”	$P(A - B) = P(A) - P(A \cap B)$
Exactly one of A, B	$P(A) + P(B) - 2P(A \cap B)$
Neither A nor B (De Morgan)	$P(A' \cap B') = 1 - P(A \cup B)$

Notation. $P(A)$ = probability of event A ; A' = complement; $\cup$ = “or”; $\cap$ = “and”; $n(\cdot)$ = number of sample points. All probabilities are **dimensionless pure numbers** in $[0, 1]$ — the math-v1 analogue of “units”: the only “unit check” available is *does it lie between 0 and 1?* A negative probability or one exceeding 1 is an instant red flag.

Section 3: Derivations from the Axioms

Unlike Sub-01 (which was procedural), this subtopic has genuine derivations — every working formula above is *squeezed out of the three axioms*. Seeing this once removes the need to memorise the table.

Result 1 — The impossible event has probability zero, $P(\phi) = 0$. The sure event S and the impossible event ϕ are mutually exclusive ($S \cap \phi = \phi$), and $S \cup \phi = S$. Apply Axiom 3:

$$P(S) = P(S \cup \phi) = P(S) + P(\phi).$$

Cancelling $P(S)$ from both sides gives $P(\phi) = 0$. *Physical reading: an outcome that can never happen carries zero sand.*

Result 2 — The complement law, $P(A') = 1 - P(A)$. An event and its complement are mutually exclusive ($A \cap A' = \phi$) and together exhaust the sample space ($A \cup A' = S$) — recall this partition fact from Sub-01. Apply Axiom 3, then Axiom 2:

$$P(A) + P(A') = P(A \cup A') = P(S) = 1 \Rightarrow P(A') = 1 - P(A).$$

This is the single most useful trick in the chapter: whenever “at least one” is hard, compute “none” and subtract.

Result 3 — Probability is bounded, $0 \leq P(A) \leq 1$. The lower bound is Axiom 1 directly: $P(A) \geq 0$. For the upper bound, use Result 2: $P(A) = 1 - P(A')$, and since $P(A') \geq 0$ (Axiom 1 applied to A'), we get $P(A) \leq 1$. So every probability is trapped in $[0, 1]$ — no exceptions.

Result 4 — The general addition rule, $P(A \cup B) = P(A) + P(B) - P(A \cap B)$. This is the step students most want to memorise and least understand. The key is to **carve overlapping events into disjoint pieces** so Axiom 3 applies.

Write $A \cup B$ as two disjoint slabs: everything in A , plus the part of B lying *outside* A :

$$A \cup B = A \cup (B - A), \quad A \cap (B - A) = \phi.$$

By Axiom 3:

$$P(A \cup B) = P(A) + P(B - A). \quad (\text{i})$$

Now carve B itself into its overlap with A and its non-overlap:

$$B = (A \cap B) \cup (B - A), \quad (A \cap B) \cap (B - A) = \phi.$$

By Axiom 3 again:

$$P(B) = P(A \cap B) + P(B - A) \Rightarrow P(B - A) = P(B) - P(A \cap B). \quad (\text{ii})$$

Substitute (ii) into (i):

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

Why subtract the overlap? Because $P(A) + P(B)$ counts the sand in $A \cap B$ twice — once in each event — so we remove one copy. When A and B are mutually exclusive, $A \cap B = \phi$, so $P(A \cap B) = 0$ and the rule collapses to $P(A) + P(B)$ — recovering Axiom 3.

Diagram placeholder

Venn diagram of two overlapping circles A and B inside rectangle S , with the lens $A \cap B$ shaded a distinct colour and labelled “counted twice in $P(A) + P(B)$ — subtract once.” Be-

low it, the identity $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

Result 5 — The equally-likely formula, $P(A) = \frac{n(A)}{n(S)}$. Suppose all $n(S)$ outcomes are equally likely, so each sample point gets the same probability p . Their total must be 1 (normalization), so $n(S) \cdot p = 1 \Rightarrow p = \frac{1}{n(S)}$. An event A with $n(A)$ points then carries

$$P(A) = \sum_{\omega_i \in A} p = n(A) \cdot \frac{1}{n(S)} = \frac{n(A)}{n(S)}.$$

This recovers the classical recipe as a *theorem*, not a definition — and makes crystal-clear that it holds **only** under the equally-likely assumption.

Section 4: Worked Examples

Example 1 — [CBSE Board Level] A box contains 4 red, 5 green and 3 blue identical marbles. One marble is drawn at random. Find (a) $P(\text{red})$, (b) $P(\text{not blue})$, (c) $P(\text{red or green})$.

Solution. Total marbles: $n(S) = 4 + 5 + 3 = 12$, all equally likely.

(a) $P(\text{red}) = \frac{n(\text{red})}{n(S)} = \frac{4}{12} = \frac{1}{3}.$

(b) Use the complement law. $P(\text{blue}) = \frac{3}{12} = \frac{1}{4}$, so

$$P(\text{not blue}) = 1 - P(\text{blue}) = 1 - \frac{1}{4} = \frac{3}{4}.$$

(c) “Red or green” are mutually exclusive (a marble can’t be both), so by the additivity axiom:

$$P(\text{red or green}) = \frac{4}{12} + \frac{5}{12} = \frac{9}{12} = \frac{3}{4}.$$

(Sanity check: $P(\text{red or green}) = P(\text{not blue})$, since the only way to be neither red nor green is to be blue — and both came out $\frac{3}{4}$. ✓) ■

Example 2 — [CUET / Speed Trap] For a sample space $S = \{\omega_1, \omega_2, \omega_3, \omega_4\}$, which of the following is a **valid** probability assignment? (i) 0.3, 0.3, 0.3, 0.3 (ii) 0.5, 0.2, -0.1, 0.4 (iii) 0.1, 0.2, 0.3, 0.4 (iv) 0.4, 0.4, 0.1, 0.2

Fast solution — test the two axioms, nothing else. A valid assignment needs *all* values ≥ 0 (Axiom 1) **and** $\text{sum} = 1$ (normalization).

- (i) $\text{Sum} = 1.2 \neq 1$. ✗
- (ii) Contains $-0.1 < 0$. ✗ instantly.
- (iii) All ≥ 0 and $\text{sum} = 0.1 + 0.2 + 0.3 + 0.4 = 1.0$. ✓
- (iv) $\text{Sum} = 1.1 \neq 1$. ✗

Answer: (iii). *Speed tip:* scan for a negative value first (kills an option in one glance), then add — don’t compute anything fancy. ■

Example 3 — [JEE Main Level] For two events A and B , $P(A) = 0.6$, $P(B) = 0.5$, and $P(A \cap B) = 0.3$. Find (a) $P(A \cup B)$, (b) $P(\text{exactly one of } A, B)$, (c) $P(A' \cap B')$.

Solution.

(a) General addition rule:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.6 + 0.5 - 0.3 = 0.8.$$

(b) “Exactly one” excludes the overlap from both sides:

$$P(\text{exactly one}) = P(A) + P(B) - 2P(A \cap B) = 0.6 + 0.5 - 2(0.3) = 0.5.$$

(Equivalently $P(A \cup B) - P(A \cap B) = 0.8 - 0.3 = 0.5$. ✓)

(c) By De Morgan, $A' \cap B' = (A \cup B)'$, so use the complement law:

$$P(A' \cap B') = 1 - P(A \cup B) = 1 - 0.8 = 0.2.$$

(This is the “neither” probability — the part of S outside both circles.) ■

(Two concepts linked: the addition rule feeds the complement-of-union, which is exactly the JEE Main composition pattern.)

Example 4 — [JEE Advanced Level] A die is **loaded** so that the probability of each face is proportional to the number on that face. (a) Find the probability of each face. (b) Find $P(\text{the number is prime})$. (c) Find $P(\text{the number is even OR greater than 4})$. (d) Verify your assignment satisfies the axioms.

Solution.

(a) “Proportional to the face value” means $P(k) = ck$ for $k = 1, \dots, 6$ and some constant c . Normalization (the sand totals one):

$$\sum_{k=1}^6 ck = c(1 + 2 + 3 + 4 + 5 + 6) = 21c = 1 \Rightarrow c = \frac{1}{21}.$$

So $P(k) = \frac{k}{21}$: the faces carry $\frac{1}{21}, \frac{2}{21}, \frac{3}{21}, \frac{4}{21}, \frac{5}{21}, \frac{6}{21}$.

(Notice $P(A) = \frac{n(A)}{n(S)}$ would be wrong here — the outcomes are not equally likely. This is precisely the case the classical recipe cannot touch and the axiomatic approach can.)

(b) Primes on a die: $\{2, 3, 5\}$. Sum their sand:

$$P(\text{prime}) = \frac{2}{21} + \frac{3}{21} + \frac{5}{21} = \frac{10}{21}.$$

(c) Let $E = \text{even} = \{2, 4, 6\}$ and $G = \text{greater than 4} = \{5, 6\}$. Then $E \cap G = \{6\}$. Use the addition rule:

$$P(E) = \frac{2+4+6}{21} = \frac{12}{21}, \quad P(G) = \frac{5+6}{21} = \frac{11}{21}, \quad P(E \cap G) = \frac{6}{21}.$$

$$P(E \cup G) = \frac{12}{21} + \frac{11}{21} - \frac{6}{21} = \frac{17}{21}.$$

(Check directly: $E \cup G = \{2, 4, 5, 6\}$, sand = $\frac{2+4+5+6}{21} = \frac{17}{21}$. ✓)

(d) Axiom 1: every $P(k) = \frac{k}{21} > 0$. ✓ Axiom 2: $\sum P(k) = \frac{21}{21} = 1$. ✓ Additivity holds because each event's probability was computed as a sum over disjoint sample points. ✓

■

Section 5: Practice Exercises

1. **[CBSE]** A card is drawn from a well-shuffled deck of 52. Find $P(\text{a king})$, $P(\text{a red card})$, and $P(\text{not a face card})$.
2. **[CUET]** If $P(A) = 0.45$, is it possible that $P(A') = 0.45$? If $P(A) = 0.7$ and $P(B) = 0.5$, can A and B be mutually exclusive? Answer with one-line reasons.
3. **[JEE Main]** Events A and B satisfy $P(A) = 0.4$, $P(B) = 0.35$, $P(A \cup B) = 0.6$. Find $P(A \cap B)$ and $P(\text{neither } A \text{ nor } B)$.
4. **[JEE Main]** A number is chosen at random from 1 to 30. Find the probability that it is divisible by 3 or by 5.
5. **[JEE Advanced]** A coin is biased so that heads is twice as likely as tails. It is tossed twice, the tosses independent in the everyday sense of “the coin has no memory.” Assign probabilities to the four outcomes $\{HH, HT, TH, TT\}$ consistent with the axioms, and find $P(\text{at least one head})$.

Answers:

1. $\frac{4}{52} = \frac{1}{13}$; $\frac{26}{52} = \frac{1}{2}$; $1 - \frac{12}{52} = \frac{40}{52} = \frac{10}{13}$.
2. Yes — $P(A') = 1 - 0.45 = 0.55$, so 0.45 is impossible for A' (it would need $P(A) = 0.55$). No — mutually exclusive needs $P(A) + P(B) \leq 1$, but $0.7 + 0.5 = 1.2 > 1$.
3. $P(A \cap B) = 0.4 + 0.35 - 0.6 = 0.15$; $P(\text{neither}) = 1 - 0.6 = 0.4$.
4. $P(\text{div } 3) = \frac{10}{30}$, $P(\text{div } 5) = \frac{6}{30}$, $P(\text{div } 15) = \frac{2}{30}$; $P = \frac{10+6-2}{30} = \frac{14}{30} = \frac{7}{15}$.
5. $P(H) = \frac{2}{3}$, $P(T) = \frac{1}{3}$ per toss, so $P(HH) = \frac{4}{9}$, $P(HT) = P(TH) = \frac{2}{9}$, $P(TT) = \frac{1}{9}$ (sum = 1 ✓); $P(\text{at least one head}) = 1 - P(TT) = 1 - \frac{1}{9} = \frac{8}{9}$.

Section 6: Crack the MCQ

Q1. If $P(A) = 0.6$ and $P(A \cup B) = 0.9$, and A, B are mutually exclusive, then $P(B) =$

- (a) 0.3 (b) 0.5 (c) 1.5 (d) 0.9

Answer: (a). Mutually exclusive $\Rightarrow P(A \cup B) = P(A) + P(B)$, so $P(B) = 0.9 - 0.6 = 0.3$. *Trap (c):* added when it should subtract ($0.6 + 0.9$). *Trap (b):* used $1 - 0.6$, confusing union with complement. *Trap (d):* copied $P(A \cup B)$ without using A at all.

Q2. Which value can **never** be a probability of an event?

- (a) 0 (b) $\frac{2}{3}$ (c) 1 (d) $\frac{3}{2}$

Answer: (d). Every probability lies in $[0, 1]$; $\frac{3}{2} > 1$ violates the upper bound. *Traps (a) and (c):* 0 (impossible event) and 1 (sure event) are the legitimate endpoints — students who think probabilities must be strictly between 0 and 1 wrongly reject them.

Q3. For an event A , $P(A) = 0.35$. Then $P(\text{not } A)$ is:

- (a) 0.35 (b) 0.65 (c) 1.35 (d) cannot be determined

Answer: (b). Complement law: $P(A') = 1 - 0.35 = 0.65$. *Trap (a):* assumes “not A ” has the same probability as A . *Trap (d):* believes you need more information, not realising the complement law needs only $P(A)$. *Trap (c):* added instead of subtracting from 1.

Q4. If $P(A) = 0.5$, $P(B) = 0.4$, $P(A \cap B) = 0.2$, then $P(A' \cap B') =$

(a) 0.1 (b) 0.3 (c) 0.7 (d) 0.9

Answer: (b). $P(A \cup B) = 0.5 + 0.4 - 0.2 = 0.7$; by De Morgan, $P(A' \cap B') = 1 - 0.7 = 0.3$. *Trap (c):* stopped at $P(A \cup B) = 0.7$, forgetting the final complement. *Trap (a):* multiplied $P(A')P(B')$ — an *independence* move that is (i) Class 12 and (ii) unjustified here. *Trap (d):* used $1 - P(A \cap B)$, complementing the wrong set.

Section 7: Common Student Pitfalls & Pro-Tips

1. **Applying $\frac{n(A)}{n(S)}$ to a loaded/unfair experiment.** That formula is a *consequence of equal likelihood*, not a definition. For a biased coin or a die loaded by face value, go back to the axioms and sum the actual $P(\omega_i)$.
2. **Forgetting to subtract the overlap.** Writing $P(A \cup B) = P(A) + P(B)$ when A and B overlap double-counts $A \cap B$. Only drop the $-P(A \cap B)$ term when you have *checked* the events are mutually exclusive.
3. **Treating mutually exclusive as independent.** The Class 12 trap, seeded here: for mutually exclusive events $P(A \cap B) = 0$; for independent events $P(A \cap B) = P(A)P(B)$. These are different (and for non-trivial events, incompatible). Don't multiply probabilities in a Class 11 addition-rule problem.
4. **Probabilities outside $[0, 1]$.** If your answer is negative or exceeds 1, you have made an arithmetic error — full stop. Use the range as a free sanity check on every final answer.

Pro-Tip: When a question says “**at least one**,” reach for the complement *first*. $P(\text{at least one}) = 1 - P(\text{none})$ almost always turns a messy union into a single easy product or count. It is the highest-leverage one-liner in the entire chapter.

Section 8: Cheat Sheet Box

AXIOMATIC PROBABILITY — RAPID RECALL

- **Three axioms:** $P(A) \geq 0$; $P(S) = 1$; $A \cap B = \phi \Rightarrow P(A \cup B) = P(A) + P(B)$.
- **Sample points:** $P(\omega_i) \geq 0$, $\sum P(\omega_i) = 1$, and $P(E) = \sum_{\omega_i \in E} P(\omega_i)$.
- **Range:** $0 \leq P(A) \leq 1$. Outside this $\Rightarrow$ error.
- **Complement:** $P(A') = 1 - P(A)$. **Impossible:** $P(\phi) = 0$. **Sure:** $P(S) = 1$.
- **Addition rule:** $P(A \cup B) = P(A) + P(B) - P(A \cap B)$; drop the last term *only* if mutually exclusive.
- **Equally likely (special case only):** $P(A) = \frac{n(A)}{n(S)}$.
- **De Morgan:** $P(A' \cap B') = 1 - P(A \cup B)$ (“neither”).

Mnemonics:

- “*Or = Add, And = subtract once*” — for the addition rule's $+P(B) - P(A \cap B)$.
- “*At least one $\Rightarrow$ flip to none*” — use $1 - P(\text{none})$.
- “*No negatives, totals one*” — the two axiom checks for a valid assignment.

Diagram placeholder

A horizontal probability scale from 0 to 1, with 0 marked “impossible event (φ)”, 1 marked “sure event (S)”, 0.5 marked “even chance”, and a shaded band showing all valid probabilities live on this segment — nothing below 0 or above 1.

Subtopic 03 — Computing Probabilities: Combinatorial Methods, Multi-Event Addition & Odds

Exam Relevance: This is where the marks actually live. Almost every numerical in CBSE Boards, and the overwhelming majority of JEE Main probability questions, are *counting* problems dressed as probability — cards, dice, committees, arrangements. CUET leans on odds and quick “at least one” calculations. JEE Advanced pushes three-event inclusion-exclusion and combinatorial “at least / exactly” counts. Sub-01 taught you to *list* outcomes and Sub-02 gave you the *rules*; this subtopic teaches you to *compute* when listing by hand is hopeless.

Section 1: Concept Intuition

In Sub-02 we met the equally-likely formula $P(E) = \frac{n(E)}{n(S)}$ and used it on tiny experiments — a die, a coin, a handful of marbles — where you could literally write every outcome down. But now imagine being asked: *a poker hand of 5 cards is dealt from a 52-card deck; what is the probability it contains exactly one ace?* The sample space has ${}^{52}C_5 = 2,598,960$ outcomes. You are not going to list those on an exam.

This is exactly the moment Chapter 6 (Permutations and Combinations) comes to the rescue.

Think about it this way... the formula $P(E) = \frac{n(E)}{n(S)}$ never required you to *write* the outcomes — only to *count* them. And counting is precisely what nP_r and nC_r were built for. So computing a probability becomes a two-part counting job: count the favourable outcomes ($n(E)$), count the total outcomes ($n(S)$), and divide. The entire art is deciding *how* to count each.

The single most important decision is: **does order matter?** When you deal a poker hand, the *order* the cards arrive in is irrelevant — a hand is a *set* — so you count with combinations (nC_r). When you arrange people in a queue or fill ranked positions, order *does* matter, so you count with permutations (nP_r). The beautiful part is that as long as you count $n(E)$ and $n(S)$ *the same way* — both ordered or both unordered — the ratio comes out correct either way. Choose whichever is easier and stay consistent.

Two more ideas complete the toolkit. First, **“at least one” is a trap that the complement defuses.** “At least one ace” splits into one ace, two aces, three aces, four aces — four messy cases. But its opposite, “no ace at all,” is a single clean case. So $P(\text{at least one}) = 1 - P(\text{none})$ turns four computations into one. This is the highest-value shortcut in the whole chapter. Second, real exams love the **addition theorem stretched to three events** — “students who play cricket *or* football *or* hockey” — which needs the inclusion-exclusion pattern we will build.

Finally, **odds.** Bookmakers and weather reports rarely say “the probability is 0.6”; they say “the odds are 3 to 2.” Odds are just probability wearing different clothes — a ratio of favourable to unfavourable outcomes instead of favourable to total.

Limiting conditions and assumptions. The counting formula $\frac{n(E)}{n(S)}$ is still the *equally likely* special case — it silently assumes every outcome is equally probable (a fair deck, an unbiased die, “drawn at random”). For a loaded die or a weighted draw it fails, and you must fall back to the axioms. And when outcomes can’t even be assumed equally likely

or counted — “what is the probability this newborn calf survives?” — we abandon counting entirely and use **empirical (statistical) probability**: the long-run relative frequency $P(E) \approx \frac{\text{number of times } E \text{ occurred}}{\text{total number of trials}}$, which stabilises as trials grow large.

Section 2: Key Formulas, Definitions & Validity Conditions

Combinatorial (classical) probability. For an experiment with equally likely outcomes,

$$P(E) = \frac{n(E)}{n(S)},$$

where $n(E), n(S)$ are counted using Chapter 6 tools:

- **Selections (order irrelevant):** choose r from n in ${}^nC_r = \frac{n!}{r!(n-r)!}$ ways.
- **Arrangements (order matters):** arrange r from n in ${}^nP_r = \frac{n!}{(n-r)!}$ ways.

Complement for “at least one”:

$$P(\text{at least one}) = 1 - P(\text{none}).$$

Addition theorem — two events (from Sub-02):

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

Addition theorem — three events (inclusion-exclusion):

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C).$$

Useful derived counts:

$$P(\text{exactly one of } A, B) = P(A) + P(B) - 2P(A \cap B),$$

$$P(\text{exactly one of } A, B, C) = [P(A) + P(B) + P(C)] - 2[P(A \cap B) + P(B \cap C) + P(C \cap A)] + 3P(A \cap B \cap C).$$

Odds. If an event E has $n(E)$ favourable and $n(E')$ unfavourable equally likely outcomes:

$$\text{Odds in favour of } E = n(E) : n(E') = P(E) : P(E'),$$

$$\text{Odds against } E = n(E') : n(E).$$

Converting odds to probability — if odds in favour are $a : b$, then

$$P(E) = \frac{a}{a+b}, \quad P(E') = \frac{b}{a+b}.$$

Empirical probability: $P(E) \approx \frac{f}{N}$, the relative frequency of E over N trials (exact in the limit $N \rightarrow \infty$).

Notation & validity. All probabilities are dimensionless and lie in $[0, 1]$; odds are ratios of positive integers (or reduced fractions). The counting formula requires *equally likely* outcomes — the standing assumption behind “fair,” “unbiased,” and “at random.”

Section 3: Key Procedures & Logical Framework

Procedure A — Solving any combinatorial probability problem.

1. Decide what a single “outcome” is, and whether **order matters**. Hand of cards / committee / group of balls → *combinations*. Queue / ranked positions / sequence → *permutations*.
2. Count $n(S)$ — the total number of ways the experiment can turn out, counted the way you chose in step 1.
3. Count $n(E)$ — the favourable outcomes, counted **the same way**. For “and” conditions across independent groups, *multiply* the per-group counts (e.g. choose men *and* women → ${}^mC_i \times {}^wC_j$).
4. Divide: $P(E) = \frac{n(E)}{n(S)}$, and sanity-check it lies in $[0, 1]$.

Procedure B — “At least one” via the complement.

1. Identify the event as “at least one X .”
2. Write its complement: “no X at all” (zero occurrences).
3. Compute $P(\text{none})$ — usually a single clean count.
4. Answer = $1 - P(\text{none})$. *Reach for this whenever you see “at least.”*

Procedure C — Three-event addition (inclusion-exclusion).

1. Add the three single probabilities (this overcounts every pairwise overlap once).
2. Subtract the three pairwise intersections (this now removes the triple overlap *too many* times).
3. Add back the triple intersection once to repair the count. *The alternating +, −, + pattern is inclusion-exclusion; the reason is the same “don’t double-count the overlap” logic from the two-event rule, applied twice over.*

Procedure D — Odds conversions.

- Odds in favour $a : b \Rightarrow P(E) = \frac{a}{a+b}$.
- Odds against $a : b \Rightarrow P(E) = \frac{b}{a+b}$ (the event’s favourable count is the *second* number).
- Probability → odds: $P(E) = \frac{p}{q}$ (reduced) $\Rightarrow$ odds in favour = $p : (q - p)$.

Section 4: Worked Examples

Example 1 — [CBSE Board Level] A box contains 5 white and 4 black balls, identical apart from colour. Two balls are drawn at random together. Find (a) $P(\text{both white})$, (b) $P(\text{one of each colour})$.

Solution. Drawing two together — order irrelevant — so count with combinations.

$$n(S) = {}^9C_2 = \frac{9 \cdot 8}{2} = 36.$$

(a) Both white: choose 2 of the 5 white balls:

$$n = {}^5C_2 = 10 \quad \Rightarrow \quad P(\text{both white}) = \frac{10}{36} = \frac{5}{18}.$$

(b) One of each: choose 1 white *and* 1 black (multiply the group counts):

$$n = {}^5C_1 \times {}^4C_1 = 5 \times 4 = 20 \Rightarrow P(\text{one of each}) = \frac{20}{36} = \frac{5}{9}.$$

(Check: both black = ${}^4C_2/36 = 6/36$; and $10 + 20 + 6 = 36$ ✓ — the three cases exhaust the sample space.) ■

Example 2 — [CUET / Speed Trap] The odds against a particular horse winning a race are 5 : 2. What is the probability that the horse wins?

Fast solution. “Odds **against** = 5 : 2” means unfavourable : favourable = 5 : 2. So favourable is the **second** number:

$$P(\text{win}) = \frac{2}{5+2} = \frac{2}{7}.$$

Answer: $\frac{2}{7}$. *Speed trap:* the reflex error is to write $\frac{5}{7}$ by grabbing the first number. Anchor it: “**against**” puts the event’s share **second**. ■

Example 3 — [JEE Main Level] A committee of 4 is to be chosen at random from 5 men and 6 women. Find (a) $P(\text{exactly 2 men})$, (b) $P(\text{at least 1 woman})$.

Solution. Committee = unordered selection, so combinations. Total:

$$n(S) = {}^{11}C_4 = 330.$$

(a) Exactly 2 men means 2 men *and* 2 women:

$$n = {}^5C_2 \times {}^6C_2 = 10 \times 15 = 150 \Rightarrow P = \frac{150}{330} = \frac{5}{11}.$$

(b) “At least 1 woman” — use the complement (Procedure B). Its opposite is “no women,” i.e. all 4 from the 5 men:

$$P(\text{no woman}) = \frac{{}^5C_4}{{}^{11}C_4} = \frac{5}{330} = \frac{1}{66}.$$

$$P(\text{at least 1 woman}) = 1 - \frac{1}{66} = \frac{65}{66}.$$

(Two concepts fused: combinatorial counting from Ch.6 and the complement shortcut — the signature JEE Main composition.) ■

Example 4 — [JEE Advanced Level] In a colony, the probability that a household reads newspaper *A* is 0.5, newspaper *B* is 0.4, and newspaper *C* is 0.3. The probabilities of reading any two are $P(A \cap B) = 0.2$, $P(B \cap C) = 0.15$, $P(C \cap A) = 0.1$, and all three is $P(A \cap B \cap C) = 0.05$. Find (a) $P(\text{reads at least one paper})$, (b) $P(\text{reads exactly one paper})$.

Solution.

(a) Three-event inclusion-exclusion (Procedure C):

$$\begin{aligned} P(A \cup B \cup C) &= 0.5 + 0.4 + 0.3 - 0.2 - 0.15 - 0.1 + 0.05. \\ &= 1.2 - 0.45 + 0.05 = 0.80. \end{aligned}$$

(b) “Exactly one” uses the derived identity:

$$\begin{aligned} P(\text{exactly one}) &= [P(A) + P(B) + P(C)] - 2[P(A \cap B) + P(B \cap C) + P(C \cap A)] + 3P(A \cap B \cap C) \\ &= 1.2 - 2(0.45) + 3(0.05) = 1.2 - 0.90 + 0.15 = 0.45. \end{aligned}$$

(Cross-check the structure: at-least-one 0.80 must be $\geq$ exactly-one 0.45, and the gap 0.35 is everyone reading two or more — consistent. ✓) ■

Section 5: Practice Exercises

1. **[CBSE]** A bag holds 4 red and 6 green balls. Three are drawn at random together. Find $P(\text{all three green})$.
2. **[CUET]** The odds in favour of rain tomorrow are 2 : 3. What is the probability that it rains? What is the probability that it does not?
3. **[JEE Main]** A natural number is chosen at random from 1 to 100. Find the probability that it is divisible by 2 or by 5.
4. **[JEE Main]** Five cards are drawn at random from a well-shuffled deck of 52. Find $P(\text{all five are hearts})$ (leave the answer in nC_r form).
5. **[JEE Advanced]** For three events, $P(A) = 0.6$, $P(B) = 0.5$, $P(C) = 0.4$, $P(A \cap B) = 0.3$, $P(B \cap C) = 0.2$, $P(C \cap A) = 0.25$, $P(A \cap B \cap C) = 0.1$. Find $P(\text{at least one of } A, B, C)$.

Answers:

1. $\frac{{}^6C_3}{{}^{10}C_3} = \frac{20}{120} = \frac{1}{6}$.
2. $P(\text{rain}) = \frac{2}{5}$; $P(\text{no rain}) = \frac{3}{5}$.
3. $\frac{50 + 20 - 10}{100} = \frac{60}{100} = \frac{3}{5}$.
4. $\frac{{}^{13}C_5}{{}^{52}C_5} = \frac{1287}{2598960} \approx 4.95 \times 10^{-4}$.
5. $0.6 + 0.5 + 0.4 - 0.3 - 0.2 - 0.25 + 0.1 = 0.85$.

Section 6: Crack the MCQ

Q1. A pair of fair dice is thrown. The probability that the sum is 8 is:

- (a) $\frac{5}{36}$ (b) $\frac{1}{6}$ (c) $\frac{1}{9}$ (d) $\frac{7}{36}$

Answer: (a). Favourable ordered pairs summing to 8: (2, 6), (3, 5), (4, 4), (5, 3), (6, 2) — five of them; $n(S) = 36$. So $P = \frac{5}{36}$. *Trap (b):* counted 6 by wrongly including (4, 4) twice or adding (1, 7). *Trap (d):* confused with the count for sum = 7 (which has 6) or miscounted. *Trap (c):* $\frac{4}{36}$, forgetting one ordered pair.

Q2. The odds against an event are 7 : 5. The probability of the event is:

- (a) $\frac{7}{12}$ (b) $\frac{5}{12}$ (c) $\frac{5}{7}$ (d) $\frac{7}{5}$

Answer: (b). Odds against = unfavourable : favourable = 7 : 5, so $P(E) = \frac{5}{7+5} = \frac{5}{12}$.

Trap (a): read the first number as favourable. *Trap (d):* wrote the raw ratio — impossible, it exceeds 1. *Trap (c):* divided by the wrong total (7 instead of 12).

Q3. Three fair coins are tossed. The probability of getting at least one head is:

- (a) $\frac{1}{8}$ (b) $\frac{3}{8}$ (c) $\frac{7}{8}$ (d) $\frac{1}{2}$

Answer: (c). Complement: $P(\text{no head}) = P(TTT) = \frac{1}{8}$, so $P(\text{at least one head}) = 1 - \frac{1}{8} = \frac{7}{8}$. *Trap (a):* answered $P(\text{no head})$ itself. *Trap (b):* counted only “exactly one head.” *Trap (d):* guessed from a single-coin intuition.

Q4. Two fair dice are rolled. The probability of getting at least one six is:

- (a) $\frac{1}{6}$ (b) $\frac{11}{36}$ (c) $\frac{1}{3}$ (d) $\frac{25}{36}$

Answer: (b). Complement: $P(\text{no six}) = \left(\frac{5}{6}\right)^2 = \frac{25}{36}$, so $P(\text{at least one six}) = 1 - \frac{25}{36} = \frac{11}{36}$. *Trap (d):* stopped at $P(\text{no six})$. *Trap (c):* added $\frac{1}{6} + \frac{1}{6}$, double-counting the (6, 6) outcome. *Trap (a):* used a single die.

Section 7: Common Student Pitfalls & Pro-Tips

- Mixing ordered and unordered counts.** If you count $n(S)$ with combinations, count $n(E)$ with combinations too. Switching midway (e.g. nP_r on top, nC_r below) gives a nonsense ratio. Pick one convention and hold it.
- Brute-forcing “at least one.”** Summing the one-, two-, three-, ... cases is slow and error-prone. The complement $1 - P(\text{none})$ is almost always one short calculation. Train the reflex: *see “at least,” compute “none.”*
- Forgetting the $+P(A \cap B \cap C)$ term.** In three-event problems students subtract the three pairwise overlaps and stop, which removes the triple overlap one time too many. Always add it back.
- Confusing odds with probability.** Odds 3 : 2 is **not** a probability of $\frac{3}{2}$ or $\frac{3}{5}$ “in favour of the loss.” Odds in favour $a : b$ means $P = \frac{a}{a+b}$. And “against” flips which number is the event’s.

Pro-Tip: For “at least one of several independent-looking trials,” the complement collapses the whole thing to a single product: $P(\text{at least one}) = 1 - (\text{chance of none})^{(\text{number of trials})}$. One line, no casework — the biggest time-saver in the chapter.

Section 8: Cheat Sheet Box

COMPUTING PROBABILITIES — RAPID RECALL

- **Classical:** $P(E) = \frac{n(E)}{n(S)}$, count with nC_r (order irrelevant) or nP_r (order matters) — *same convention top and bottom*.
- **“At least one”:** $P = 1 - P(\text{none})$. Single biggest shortcut.
- **Addition (2 events):** $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.
- **Addition (3 events):** $P(A) + P(B) + P(C) - \sum P(\text{pairs}) + P(A \cap B \cap C)$.
- **Exactly one (3 events):** $\sum P - 2 \sum P(\text{pairs}) + 3P(\text{triple})$.
- **Odds in favour** $a : b \Rightarrow P = \frac{a}{a+b}$; **odds against** $a : b \Rightarrow P = \frac{b}{a+b}$.
- **Empirical:** $P(E) \approx \frac{f}{N}$ for large N (when outcomes aren’t equally likely).

Mnemonics:

- “And across groups $\Rightarrow$ multiply” — choosing men AND women is ${}^mC_i \times {}^wC_j$.
- “At least $\Rightarrow$ flip to none.”
- “In-out-in” — the $+$, $-$, $+$ sign pattern of three-event inclusion-exclusion.
- “Against puts the event second” — for odds-against conversions.

Diagram placeholder

A three-circle Venn diagram (A, B, C overlapping) inside rectangle S, with each of the 7 disjoint regions labelled — only-A, only-B, only-C, $A \cap B$ -only, $B \cap C$ -only, $C \cap A$ -only, and the central $A \cap B \cap C$ — annotated to show how inclusion-exclusion adds singles, subtracts pairs, and adds back the centre.

Looking ahead: in Class 12 you will extend this toolkit with conditional probability, the multiplication theorem, independence, and Bayes’ theorem — but those are a separate chapter and are not part of Class 11.

Chapter 14: Probability — Supplement

Chapter 14 teaches you two things it never makes you *use*. The first is ordered counting: nP_r is defined in Subtopic 03, named in Procedure A, and warned about in Pitfall 1 — yet every solved problem in the chapter is a combination, an enumeration or a divisibility count. The second is the family of inequality bounds that Subtopic 02's own Exam Relevance blurb promises to JEE Advanced readers and then never delivers. This supplement closes both.

A note on scope. Everything here stays inside Class 11. Conditional probability $P(A | B)$, independence and the multiplication rule are Class 12 material and the chapter is right to defer them. Nothing below uses them, and one worked example below shows you concretely why reaching for them early produces a wrong number.

Addendum 14.1 — Probability with Permutation Counts

Exam Relevance: JEE Main asks at least one ordered-counting probability almost every session: digits forming a number under a divisibility or size condition, letters of a word arranged with a separation or togetherness condition, people in a queue or in ranked positions. CBSE Boards asks the gentler versions (two particular people together in a row). The counting itself is Chapter 6 material you already own; what is new is the discipline of counting $n(E)$ and $n(S)$ **the same way**, which is exactly what Subtopic 03's Pitfall 1 warns about. JEE Advanced layers a second condition on top and expects you to count without listing.

The Method

Procedure A-P — Ordered-counting probability. This is Procedure A of Subtopic 03 with the “order matters” branch actually taken.

1. **Decide what one outcome is, and fix a counting convention.** An outcome is one complete arrangement — one specific number, one specific word-arrangement, one specific line-up. *Why this matters:* “the numbers formed” and “the arrangements of the letters” are different sample spaces from “the sets of digits chosen,” and they have different sizes.
2. **Count $n(S)$ as an arrangement count.**

- r positions filled from n distinct objects, no repetition: ${}^nP_r = \frac{n!}{(n-r)!}$.
- All n objects in a row, with repeats $p_1, p_2, \dots$ of identical objects: $\frac{n!}{p_1! p_2! \dots}$.
- Digits with a leading-zero restriction: count the whole thing and subtract the arrangements that start with 0, or fill the leading position first.

3. **Count $n(E)$ under the same convention.** Fill the *constrained* positions first. If the condition is on the last digit, choose the last digit before anything else; if it is on the first digit, choose that first; if two conditions collide on the same digit, split into cases.
4. **For separation conditions (“no two X together”), use the gap method.** Arrange the non- X objects first, then choose gaps for the X objects from the (number of non- X + 1) available gaps. *Why:* placing the X ’s into distinct gaps makes adjacency impossible by construction, so no messy subtraction is needed.
5. **Divide and sanity-check:** $P(E) = \frac{n(E)}{n(S)} \in [0, 1]$. Where you can, re-count by a second convention (all objects treated as distinct, or via the complement) and confirm the ratio is unchanged.

The consistency principle, stated plainly. As long as $n(E)$ and $n(S)$ are counted in the same currency — both ordered, or both unordered, or both “all letters distinct” — the ratio is correct. Mixing currencies (nP_r upstairs, nC_r downstairs) is the single most expensive error in this topic, because the answer still looks like a probability.

Figure brief: Four empty boxes in a row labelled “thousands, hundreds, tens, units” with the units box shaded and annotated “constrained — fill this first: 3 even choices”; arrows show the remaining three boxes being filled from the leftover six digits as 6P_3 .

Worked Examples

Example S1 — [JEE Main Level]

Four-digit numbers with no repeated digit are formed using the digits 1, 2, 3, 4, 5, 6, 7. One such number is chosen at random. Find the probability that it is (a) even, (b) divisible by 5, (c) both even and greater than 4000.

Solution.

An outcome is one four-digit number, so order matters and every number is equally likely.

$$n(S) = {}^7P_4 = 7 \times 6 \times 5 \times 4 = 840.$$

(a) Even. A number is even exactly when its units digit is even, so fill the units place first: it must be 2, 4 or 6 — three choices. The remaining three places are then filled from the six unused digits in ${}^6P_3 = 6 \times 5 \times 4 = 120$ ways.

$$n(E) = 3 \times 120 = 360, \quad P(\text{even}) = \frac{360}{840} = \frac{3}{7}.$$

Second method (symmetry). Every one of the seven digits is equally likely to land in the units place, because the digits play symmetric roles in 7P_4 . Three of the seven digits are even, so $P = \frac{3}{7}$ directly. ✓

(b) Divisible by 5. Only 5 is available as a terminal digit, so the units place is forced: 1 choice, then ${}^6P_3 = 120$.

$$P(\text{divisible by } 5) = \frac{120}{840} = \frac{1}{7}.$$

Consistent with the symmetry argument: one favourable digit out of seven.

(c) Even and greater than 4000. Now two positions are constrained at once — the units digit must lie in $\{2, 4, 6\}$ and the thousands digit in $\{4, 5, 6, 7\}$ — and these sets overlap, so split into cases on the units digit.

Units digit	Choices for thousands digit	Middle two places	Count
2	4, 5, 6, 7 — all four available	${}^5P_2 = 20$	$4 \times 20 = 80$
4	5, 6, 7 — 4 is used up	${}^5P_2 = 20$	$3 \times 20 = 60$
6	4, 5, 7 — 6 is used up	${}^5P_2 = 20$	$3 \times 20 = 60$
Total			200

$$P = \frac{200}{840} = \frac{5}{21}.$$

Second method. Count the valid (thousands, units) ordered pairs directly: $4 + 3 + 3 = 10$ pairs, each completed by ${}^5P_2 = 20$ ways of filling the middle two places, giving $10 \times 20 = 200$. ✓

Read this before you move on. Part (c) is not $P(\text{even}) \times P(> 4000)$. That product would be $\frac{3}{7} \times \frac{4}{7} = \frac{12}{49} \approx 0.2449$, whereas the true answer is $\frac{5}{21} \approx 0.2381$. Multiplying probabilities of two events is a Class 12 move that requires a justification this chapter has not given you — and here it would also give the wrong number, because the two conditions compete for the digits 4 and 6. **Count, do not multiply probabilities.**

Example S2 — [JEE Advanced Level]

The letters of the word **ASSASSIN** are arranged at random in a row. Find the probability that (a) no two S's are adjacent, (b) all four S's occur together.

Solution.

ASSASSIN has 8 letters: A twice, S four times, I once, N once.

$$n(S) = \frac{8!}{2!4!} = \frac{40320}{48} = 840.$$

Every distinguishable arrangement is equally likely, since the underlying random shuffle of the eight letter-tiles treats all tiles alike.

(a) No two S's adjacent — the gap method. First arrange the four non-S letters A, A, I, N:

$$\frac{4!}{2!} = 12 \text{ arrangements.}$$

Four letters in a row create 5 gaps (one before, three between, one after). Dropping each S into a **different** gap makes adjacency impossible, and since all four S's are identical we only choose *which* gaps:

$${}^5C_4 = 5 \text{ ways.}$$

$$n(E) = 12 \times 5 = 60, \quad P = \frac{60}{840} = \frac{1}{14}.$$

Second method (treat all eight letters as distinct tiles). Label the tiles $A_1, A_2, S_1, S_2, S_3, S_4, I, N$; then $n(S) = 8! = 40320$. The number of ways to choose 4 pairwise non-adjacent positions out of 8 in a row is ${}^{8-4+1}C_4 = {}^5C_4 = 5$. Place the four S tiles in those positions ($4!$ ways) and the other four tiles in the remaining positions ($4!$ ways):

$$n(E) = 5 \times 24 \times 24 = 2880, \quad P = \frac{2880}{40320} = \frac{1}{14}. \checkmark$$

Two entirely different sample spaces, one answer — which is the consistency principle of Step 5 doing its job.

(b) All four S's together. Glue the four S's into a single block SSSS. You are now arranging 5 objects — the block, A, A, I, N — with A repeated:

$$n(E) = \frac{5!}{2!} = 60, \quad P = \frac{60}{840} = \frac{1}{14}.$$

The S's are identical, so the block has no internal arrangements to count — do **not** multiply by $4!$ here.

Why (a) and (b) came out equal. It is not a fluke of arithmetic: both events reduce to choosing one of exactly 5 position-patterns for the S's (${}^5C_4 = 5$ gap-choices in (a); 5 possible starting positions for a block of four in (b)), each combined with the same 12 arrangements of A, A, I, N.

Sanity check. Both answers lie in $[0, 1]$, and $\frac{1}{14} + \frac{1}{14} = \frac{1}{7} < 1$ — consistent, since “no two together” and “all four together” are mutually exclusive events (with 8 letters and 4 S's, they cannot both hold) and the remaining $\frac{6}{7}$ covers all the partially-clumped arrangements.

Practice Exercises

- Five-digit numbers with no repeated digit are formed using all of the digits 0, 1, 2, 3, 4 (the number may not begin with 0). One is chosen at random. Find the probability that it is even. **Answer:** $n(S) = 4 \times 4! = 96$. Favourable = 60 (units digit 0: $4! = 24$; units digit 2: $3 \times 3! = 18$; units digit 4: 18). $P = \frac{60}{96} = \frac{5}{8}$.
- Four boys and three girls stand in a row in random order. Find the probability that no two girls stand next to each other. **Answer:** $n(S) = 7! = 5040$. Favourable = $4! \times {}^5C_3 \times 3! = 24 \times 10 \times 6 = 1440$. $P = \frac{1440}{5040} = \frac{2}{7}$.

Pitfalls

- Do not mix nP_r and nC_r across the fraction bar.** In Example S1, using ${}^7C_4 = 35$ for $n(S)$ while counting $n(E)$ as ordered arrangements gives a “probability” above 1 — the fastest possible sign that the currencies do not match.

- **Fill the constrained position first.** Filling the thousands place before the units place in Example S1(a) forces you into needless cases.
- **Identical objects have no internal arrangements.** The four S's in Example S2(b) contribute a factor of 1, not 4!. Multiplying by 4! would give $P = \frac{24}{14} > 1$.
- **“Together” and “not together” are complements only for exactly two objects.** With three or more S's, “all together” and “no two together” leave a large middle ground, so $P(\text{no two together}) \neq 1 - P(\text{all together})$.
- **Never convert an “and” condition into a product of probabilities.** Count the joint arrangements, as Example S1(c) does.

Addendum 14.2 — Inequality Bounds from the Axioms

Exam Relevance: Subtopic 02 promised you this and it is time to deliver. JEE Advanced asks it as “find the greatest and least values of $P(A \cap B)$ ”, and as a consistency test: given some probabilities, is the data even possible? JEE Main uses the same bounds as a silent filter in multi-part questions — an option that violates $P(A \cap B) \geq P(A) + P(B) - 1$ can be discarded without any work. CBSE Boards rarely asks the bounds directly but does ask the special case “can A and B be mutually exclusive?”, which is the Bonferroni bound in disguise. Everything below follows from the three axioms alone — no new machinery, no Class 12 concepts.

The Method

Everything rests on two facts you already have from Subtopic 02: **Axiom 1** ($P \geq 0$) and **Result 3** ($P \leq 1$).

Result 6 — Monotonicity. If $A \subseteq B$ then $P(A) \leq P(B)$.

Proof. Carve B into two disjoint slabs, $B = A \cup (B - A)$ with $A \cap (B - A) = \phi$. By Axiom 3, $P(B) = P(A) + P(B - A)$, and $P(B - A) \geq 0$ by Axiom 1. Hence $P(B) \geq P(A)$. ■

Reading it in sand: a bigger heap cannot hold less sand.

Result 7 — Boole’s inequality (the union is never worse than the sum).

$$P(A \cup B) \leq P(A) + P(B)$$

Proof. From the general addition rule, $P(A \cup B) = P(A) + P(B) - P(A \cap B)$, and $P(A \cap B) \geq 0$ by Axiom 1. ■

More generally, for events $A_1, A_2, \dots, A_n$:

$$P(A_1 \cup A_2 \cup \dots \cup A_n) \leq P(A_1) + P(A_2) + \dots + P(A_n)$$

(apply the two-event case repeatedly). Equality holds exactly when the events are pairwise mutually exclusive.

Result 8 — Bonferroni’s inequality (the overlap cannot be too small).

$$P(A \cap B) \geq P(A) + P(B) - 1$$

Proof. Rearrange the addition rule as $P(A \cap B) = P(A) + P(B) - P(A \cup B)$, then use $P(A \cup B) \leq 1$ from Result 3. ■

More generally, $P(A_1 \cap \dots \cap A_n) \geq P(A_1) + \dots + P(A_n) - (n - 1)$, obtained by applying Boole’s inequality to the complements and using De Morgan.

Result 9 — The four bounds, assembled. Given only $P(A)$ and $P(B)$:

$$\max(0, P(A) + P(B) - 1) \leq P(A \cap B) \leq \min(P(A), P(B))$$

$$\max(P(A), P(B)) \leq P(A \cup B) \leq \min(1, P(A) + P(B))$$

Where each piece comes from:

Bound	Source
$P(A \cap B) \geq 0$	Axiom 1
$P(A \cap B) \geq P(A) + P(B) - 1$	Result 8 (Bonferroni)
$P(A \cap B) \leq \min(P(A), P(B))$	Result 6, since $A \cap B \subseteq A$ and $A \cap B \subseteq B$
$P(A \cup B) \geq \max(P(A), P(B))$	Result 6, since $A \subseteq A \cup B$ and $B \subseteq A \cup B$
$P(A \cup B) \leq P(A) + P(B)$	Result 7 (Boole)
$P(A \cup B) \leq 1$	Result 3

The lever that ties the two lines together. The addition rule can be written as

$$P(A \cup B) + P(A \cap B) = P(A) + P(B),$$

a constant. So the union and the intersection are locked in a see-saw: **every unit of probability you push into the overlap comes straight out of the union.** Find the interval for one and you get the interval for the other by subtracting from $P(A) + P(B)$ — and reversing the endpoints.

Attainability. Both endpoints of each interval are genuinely reachable in a suitable sample space, so “greatest value” and “least value” questions have the endpoints as their answers:

- The **maximum** overlap occurs when the smaller event sits entirely inside the larger, i.e. $A \subseteq B$ or $B \subseteq A$.
- The **minimum** overlap occurs when the events are pushed as far apart as the sample space allows — mutually exclusive if $P(A) + P(B) \leq 1$, otherwise overlapping by exactly the forced amount $P(A) + P(B) - 1$ and covering all of S between them.

Figure brief: Two horizontal bars of lengths $P(A)$ and $P(B)$ sliding along a track of total length 1; the left panel shows them pushed apart to the ends of the track with the forced overlap $P(A) + P(B) - 1$ shaded, the right panel shows the shorter bar fully inside the longer with overlap $\min(P(A), P(B))$ shaded.

Worked Examples

Example S3 — [JEE Advanced Level]

For two events A and B , $P(A) = 0.6$ and $P(B) = 0.8$. Find (a) the greatest and least possible values of $P(A \cap B)$, (b) the greatest and least possible values of $P(A \cup B)$, and (c) an explicit sample space realising each extreme.

Solution.

(a) Bounds on $P(A \cap B)$.

Lower bound. By Bonferroni, $P(A \cap B) \geq P(A) + P(B) - 1 = 0.6 + 0.8 - 1 = 0.4$. Since $0.4 > 0$, this beats the trivial bound $P \geq 0$, so

$$P(A \cap B) \geq 0.4.$$

Upper bound. $A \cap B \subseteq A$ and $A \cap B \subseteq B$, so by monotonicity $P(A \cap B) \leq \min(0.6, 0.8) = 0.6$.

$$0.4 \leq P(A \cap B) \leq 0.6$$

Least value 0.4, greatest value 0.6.

(b) Bounds on $P(A \cup B)$. Use the see-saw rather than repeating the argument. Since $P(A \cup B) = P(A) + P(B) - P(A \cap B) = 1.4 - P(A \cap B)$, and $P(A \cap B)$ ranges over $[0.4, 0.6]$:

$$P(A \cup B) \text{ ranges over } [1.4 - 0.6, 1.4 - 0.4] = [0.8, 1.0].$$

Cross-check against the direct bounds: $P(A \cup B) \geq \max(0.6, 0.8) = 0.8$ ✓, and $P(A \cup B) \leq \min(1, 1.4) = 1$ ✓. Note that Boole's bound 1.4 is useless here — the ceiling $P \leq 1$ is the binding constraint.

$$0.8 \leq P(A \cup B) \leq 1$$

(c) Explicit realisations. Take $S = \{1, 2, 3, \dots, 10\}$ with all ten outcomes equally likely, so each carries probability 0.1.

Maximum overlap. Let $A = \{1, 2, 3, 4, 5, 6\}$ and $B = \{1, 2, \dots, 8\}$. Then $P(A) = 0.6$, $P(B) = 0.8$, and since $A \subseteq B$ we get $A \cap B = A$ and $A \cup B = B$:

$$P(A \cap B) = 0.6, \quad P(A \cup B) = 0.8.$$

Both extremes of the “maximum overlap / minimum union” corner are hit simultaneously — as the see-saw guarantees.

Minimum overlap. Let $A = \{1, 2, 3, 4, 5, 6\}$ and $B = \{3, 4, 5, \dots, 10\}$. Then $P(A) = 0.6$, $P(B) = 0.8$, $A \cap B = \{3, 4, 5, 6\}$ and $A \cup B = S$:

$$P(A \cap B) = 0.4, \quad P(A \cup B) = 1.$$

Notice that A and B **cannot** be made mutually exclusive: $0.6 + 0.8 = 1.4 > 1$, so they are forced to share at least 0.4 of the sand.

Example S4 — [JEE Main — Speed Trap: is the data even possible?]

A survey report claims that 80% of students in a coaching batch attend the morning session, 70% attend the evening session, and 40% attend both. A student is picked at random. Is the report consistent?

Solution.

Let A = “attends morning”, B = “attends evening”, so $P(A) = 0.8$, $P(B) = 0.7$, and the report asserts $P(A \cap B) = 0.4$.

Bonferroni gives the floor:

$$P(A \cap B) \geq P(A) + P(B) - 1 = 0.8 + 0.7 - 1 = 0.5.$$

The claimed 0.4 is below that floor, so **the report is inconsistent** — no sample space can produce those three numbers.

The trap. The seductive move is to check only $P(A \cap B) \leq \min(0.8, 0.7) = 0.7$, see that $0.4 \leq 0.7$, and declare the data fine. The upper bound is satisfied; it is the *lower* bound that fails, and the lower bound is the one students never check.

Confirmation from the other side. If $P(A \cap B)$ really were 0.4, then

$$P(A \cup B) = 0.8 + 0.7 - 0.4 = 1.1 > 1,$$

which violates Result 3 outright. The two checks are the same check, seen through the see-saw identity.

Practice Exercises

1. For two events A and B , $P(A) = 0.4$ and $P(B) = 0.75$. Find the least and greatest possible values of (i) $P(A \cap B)$ and (ii) $P(A \cup B)$. **Answer:** (i) least = $\max(0, 0.4 + 0.75 - 1) = 0.15$, greatest = $\min(0.4, 0.75) = 0.4$. (ii) least = $\max(0.4, 0.75) = 0.75$, greatest = $\min(1, 1.15) = 1$. Check: $P(A \cup B) = 1.15 - P(A \cap B)$ maps $[0.15, 0.4]$ onto $[0.75, 1]$.
2. (a) Three events A, B, C each have probability 0.75. Find the least possible value of $P(A \cap B \cap C)$.
(b) Events A and B satisfy $A \subseteq B$, with $P(A) = 0.35$ and $P(B) = 0.62$. Find $P(A \cap B)$, $P(A \cup B)$ and $P(B - A)$. **Answer:** (a) By the n -event Bonferroni inequality, $P(A \cap B \cap C) \geq 0.75 \times 3 - 2 = 0.25$; least value 0.25. (b) $A \subseteq B$ gives $A \cap B = A$ and $A \cup B = B$, so $P(A \cap B) = 0.35$ and $P(A \cup B) = 0.62$; then $P(B - A) = P(B) - P(A \cap B) = 0.62 - 0.35 = 0.27$.

Pitfalls and Pro-Tips

- **Check both ends of the interval.** The upper bound $\min(P(A), P(B))$ is the one students remember and the lower bound is the one exams attack. Whenever $P(A) + P(B) > 1$, the events are *forced* to overlap.
- **$\max(0, \cdot)$ is not decoration.** When $P(A) + P(B) \leq 1$, the Bonferroni floor is negative and therefore useless; the real floor is 0, attained by making A and B mutually exclusive.

- **Boole's inequality can be slack.** In Example S3 it gave $P(A \cup B) \leq 1.4$, which says nothing. Always intersect it with $P \leq 1$.
- **Monotonicity needs a genuine subset.** $P(A) \leq P(B)$ does *not* imply $A \subseteq B$ — the implication runs one way only.
- **Pro-tip — one identity, two answers.** Compute the interval for $P(A \cap B)$, then get the interval for $P(A \cup B)$ for free as $[P(A) + P(B) - \text{upper}, P(A) + P(B) - \text{lower}]$, capped at 1. In an MCQ this halves the work.
- **Pro-tip — bounds as an option filter.** Before computing anything, discard options that violate $0 \leq P \leq 1$, $P(A \cap B) \leq \min(P(A), P(B))$ or $P(A \cup B) \geq \max(P(A), P(B))$. On a four-option question this often leaves one survivor.

Cheat Sheet Box

Ordered counting in probability.

- $n(S)$ and $n(E)$ must be counted in the same currency — both ordered or both unordered.
- r from n , no repetition, order matters: ${}^nP_r = \frac{n!}{(n-r)!}$. All n with repeats: $\frac{n!}{p_1!p_2!\dots}$.
- Constrained position (units digit, leading digit) is filled **first**; overlapping constraints split into cases.
- “No two X together”: arrange the m non- X objects, then choose X -positions from the $m+1$ gaps.
- “ k identical objects together”: glue into one block; identical objects contribute **no** internal $k!$.
- Never replace a joint count by a product of probabilities.

Bounds from the axioms (all provable from $P \geq 0$, $P(S) = 1$, additivity):

Name	Statement
Monotonicity	$A \subseteq B \Rightarrow P(A) \leq P(B)$
Boole	$P(A \cup B) \leq P(A) + P(B)$; in general $P(\bigcup A_i) \leq \sum P(A_i)$
Bonferroni	$P(A \cap B) \geq P(A) + P(B) - 1$; in general $P(\bigcap_{i=1}^n A_i) \geq \sum P(A_i) - (n-1)$
Intersection range	$\max(0, P(A) + P(B) - 1) \leq P(A \cap B) \leq \min(P(A), P(B))$
Union range	$\max(P(A), P(B)) \leq P(A \cup B) \leq \min(1, P(A) + P(B))$
See-saw identity	$P(A \cup B) + P(A \cap B) = P(A) + P(B)$

- Maximum overlap $\Leftrightarrow$ one event contained in the other. Minimum overlap $\Leftrightarrow$ events pushed apart (mutually exclusive when $P(A) + P(B) \leq 1$).
- A and B can be mutually exclusive **only if** $P(A) + P(B) \leq 1$.

- Consistency test for supplied data: reject if $P(A \cap B) < P(A) + P(B) - 1$, or $P(A \cap B) > \min(P(A), P(B))$, or $P(A \cup B) > 1$.

Chapter 14: Probability — Round 2 Addendum

Subtopics 01–03 and the Round 1 Supplement (Addenda 14.1–14.2, Examples S1–S4, Results 6–9) between them now cover this chapter properly, and the chapter is right to be short: NCERT Chapter 14 has only three genuine blocks, and everything past them — conditional probability, the multiplication theorem, independence, Bayes — is Class 12. Round 1 closed the two real holes: ordered counting was defined and never used, and the inequality bounds were promised to JEE Advanced readers and never delivered.

Two question shapes remain that the chapter’s own tools already answer and that neither the chapter nor Round 1 ever demonstrates. Both are grounded in NCERT items the chapter passes over, and both sit strictly inside Class 11.

Addendum	Topic	NCERT source	Read after
14.3	Problems where one outcome is a selection and an arrangement at once (nC_r and nP_r in the same count)	Misc. Examples 15, 17; Misc. Exercise Q9	Subtopic 03, Section 4; Round 1 Addendum 14.1
14.4	The matching problem — inclusion–exclusion past three events, where the count itself mixes nC_k with $(n - k)!$	Misc. Exercise Q6; Solved Example 16	Subtopic 03, Section 3, Procedure C

A note on scope, continued from Round 1. Nothing below uses conditional probability $P(A | B)$, independence, or the multiplication rule. Every count is a direct count of equally likely outcomes, and Addendum 14.4 makes a point of showing an intersection whose probability is *not* the product of the two individual probabilities — which is exactly why inclusion–exclusion, and not multiplication, is the Class 11 tool. Geometric probability was considered for this round and deliberately left out; the reasoning is recorded at the end.

Numbering continues Round 1: Examples S5–S7, Results 10–11.

Addendum 14.3 — Selection and Arrangement in the Same Problem

Exam Relevance: Subtopic 03 counts committees and hands with nC_r ; Round 1’s Addendum 14.1 counts numbers and word-arrangements with nP_r . JEE Main’s

favourite shape is neither — it is the one where a single outcome is *both*: a team is picked **and** given ranked roles, letters are **chosen** from a word and then **arranged**, books are **selected** and then **shelved**. The JEE Advanced probability syllabus names “computation of probability of events using permutations and combinations” in one breath, and this is what that phrase means in practice. CBSE Boards asks the two-letter and two-digit versions for 2–3 marks. The counting is Chapter 6 material you own; what is new is seeing that the two tools live in one multiplication.

The Method

Procedure A-M — mixed selection-and-arrangement probability. This is Procedure A of Subtopic 03 and Procedure A-P of Round 1 run together, one after the other.

1. **Split one outcome into its two acts.** Ask *which objects* (a selection) and then *in what order* (an arrangement). An outcome of “a team of 4 chosen from 9 and given four distinct roles” is a pair: the set of four people, and the assignment of roles to them.
2. **Multiply the two counts.**

$$n(S) = \underbrace{{}^nC_r}_{\text{which}} \times \underbrace{r!}_{\text{in what order}}$$

and when the chosen objects are not all distinct, the arrangement factor is $\frac{r!}{p_1!p_2!\cdots}$ instead of $r!$.

3. **Cross-check with the free identity** ${}^nC_r \times r! = {}^nP_r$. If your two-stage count does not reproduce nP_r , one of the two stages is wrong. This is the cheapest sanity check in the topic.
4. **Count $n(E)$ in the same two acts.** Selection conditions (“exactly two girls”) are applied at stage one; position conditions (“the captain is a girl”, “the three maths books stand together”) are applied at stage two. Apply each at its own stage and multiply — do not try to handle both in one sweep.
5. **If the pool contains repeated objects, the selection splits into cases.** Choosing three letters from a word with doubled letters is *not* a single 8C_3 : the case “all three different” and the case “one letter twice” have different arrangement factors ($3!$ against $\frac{3!}{2!}$), so they must be counted separately and added.
6. **If repetition is allowed, the arrangement stage is n^r , not nP_r .** Round 1’s Procedure A-P covered only the no-repetition case. “Four-digit numbers formed from five digits, repetition allowed” has 5^4 raw arrangements before any restriction — a different sample space from the 5P_4 of the no-repetition version, and NCERT’s Miscellaneous Exercise Q9 asks you for both in one question.

When the roles cancel. If the event’s wording never mentions the positions — “the team has exactly two girls” says nothing about who is captain — then every arrangement of a favourable selection is favourable, the factor $r!$ appears in $n(E)$ and $n(S)$ alike, and it cancels. You may then drop to the pure-selection model and use nC_r throughout. **Recognising the cancellation is worth more marks than performing it**, because it tells you when the harder model was never needed. Example S5 shows both halves of this.

State the experiment before you count. “One of the distinct words that can be formed is chosen at random” and “three letter-tiles are drawn from a bag and laid in a row” are *different experiments* with *different* correct answers, because the second one does not make the distinct words equally likely. Example S6 is built on exactly this.

Figure brief: Two stacked strips. The upper strip shows a pool of nine circles, four shaded (girls) and five unshaded (boys), with a lasso enclosing four of them, captioned “stage 1 — which: 9C_4 ”. The lower strip shows those same four circles dropped into four labelled slots reading Captain, Vice-Captain, Secretary, Treasurer, captioned “stage 2 — in what order: $4!$ ”. A brace joining the two strips is labelled ${}^9C_4 \times 4! = {}^9P_4 = 3024$.

Worked Examples

Example S5 — [JEE Main Level]

A team of 4 is chosen at random from 5 boys and 4 girls, and the four chosen are then assigned the four distinct roles of captain, vice-captain, secretary and treasurer. Find the probability that (a) the team contains exactly two girls, (b) the team contains exactly two girls **and** the captain is a girl.

Solution.

An outcome is a filled role-sheet: four named people in four named posts. Counting the two stages,

$$n(S) = {}^9C_4 \times 4! = 126 \times 24 = 3024,$$

and the identity of Step 3 confirms it: ${}^9P_4 = 9 \times 8 \times 7 \times 6 = 3024$. Every filled role-sheet is equally likely.

(a) Exactly two girls. This is a pure stage-one condition. Choose 2 of the 4 girls and 2 of the 5 boys, then let the roles fall as they may:

$$n(E) = \underbrace{{}^4C_2 \times {}^5C_2}_{\text{which}} \times \underbrace{4!}_{\text{in what order}} = (6 \times 10) \times 24 = 1440,$$

$$P = \frac{1440}{3024} = \frac{10}{21}.$$

Second method (the roles cancel). The condition never mentions a post, so the factor $4!$ sits in both $n(E)$ and $n(S)$ and cancels. Drop to the selection-only model:

$$P = \frac{{}^4C_2 \times {}^5C_2}{{}^9C_4} = \frac{60}{126} = \frac{10}{21}. \checkmark$$

Two sample spaces, one answer — the consistency principle of Round 1, Step 5, doing its job.

(b) Exactly two girls and a girl as captain. Now there is a stage-two condition as well, so handle the stages in order. Stage one is unchanged: ${}^4C_2 \times {}^5C_2 = 60$ favourable teams. Stage two: of the four team members, choose which of the **two girls** takes the captaincy (2 ways), then fill the remaining three posts with the remaining three members ($3! = 6$ ways):

$$n(E) = 60 \times 2 \times 6 = 720, \quad P = \frac{720}{3024} = \frac{5}{21}.$$

Sanity check by symmetry. Every favourable team has 4 members of whom 2 are girls, and the $4!$ role-sheets for that team put each member in the captain's chair equally often. So exactly half of part (a)'s 1440 sheets have a girl as captain, and $\frac{1}{2} \times \frac{10}{21} = \frac{5}{21}$. ✓ This is a counting argument about $4!$ orderings, not a multiplication of two probabilities. ■

Example S6 — [JEE Main — Speed Trap]

All the **distinct** three-letter arrangements (meaningful or not) that can be formed from the letters of the word MATHEMATICS are written out, and one of them is chosen at random. Find the probability that its three letters are all different.

Solution.

MATHEMATICS has 11 letters over 8 distinct symbols $\{M, A, T, H, E, I, C, S\}$, of which M , A and T each occur twice and the other five occur once. No letter occurs three times. By Step 5, the sample space splits into cases.

All three letters different. Choose 3 of the 8 distinct symbols and arrange them:

$${}^8C_3 \times 3! = 56 \times 6 = 336.$$

Exactly two letters the same. The doubled letter must be M , A or T — 3 choices; the third letter is any of the remaining 7 symbols; the arrangement factor is $\frac{3!}{2!} = 3$ because two of the three letters are identical:

$$3 \times 7 \times 3 = 63.$$

All three the same. Impossible — no symbol has three copies.

$$n(S) = 336 + 63 = 399, \quad n(E) = 336, \quad P = \frac{336}{399} = \frac{16}{19}.$$

The trap. The seductive move is to write $n(S) = {}^{11}P_3 = 990$ — “eleven letters, pick three in order” — while leaving $n(E) = 336$, giving $\frac{336}{990} = \frac{56}{165}$. That number answers no question at all. It is Round 1's mixed-currency error in a new costume: 990 counts **tile-sequences** (the two M tiles treated as different objects) while 336 counts **words** (the two M tiles treated as the same symbol).

Both currencies are legal — they just describe different experiments. Suppose instead you put the eleven letter-tiles in a bag, draw three, and lay them in a row. Now $n(S) = {}^{11}P_3 = 990$ equally likely tile-sequences, and you must recount $n(E)$ in tiles too: pick three distinct symbols and one tile for each, then arrange. Splitting on how many of the three symbols come from the doubled set $\{M, A, T\}$,

$$({}^5C_3 \cdot 1) + (3 \cdot {}^5C_2 \cdot 2) + ({}^3C_2 \cdot 5 \cdot 4) + (1 \cdot 8) = 10 + 60 + 60 + 8 = 138,$$

$$n(E) = 138 \times 3! = 828, \quad P = \frac{828}{990} = \frac{46}{55}.$$

And $\frac{16}{19} = 0.8421 \dots$ while $\frac{46}{55} = 0.8363 \dots$ — genuinely different numbers, both right for their own experiment. The reason they differ is worth holding onto: in the bag-of-tiles experiment the 399 words are **not** equally likely. The word MAT arises from $2 \times 2 \times 2 = 8$ tile-sequences,

the word *MAM* from only 4, and the word *HEI* from just 1. Whenever repeated objects are physically distinct, “pick a word at random” and “draw the objects at random” are two different experiments, and the question’s wording — not your preference — decides which one you are in. ■

Practice Exercises

1. **[CBSE]** All the distinct three-letter arrangements formable from the letters of the word LEVEL are listed and one is chosen at random. Find the probability that its three letters are all different.
2. **[JEE Main]** From 6 men and 4 women, three people are chosen and assigned the posts of chairperson, secretary and treasurer. Find the probability that the chairperson is a woman and the panel contains exactly one woman.
3. **[JEE Main]** From 4 mathematics books and 3 physics books (all different), five are chosen at random and arranged in a row on a shelf. Find the probability that exactly three mathematics books are chosen and that they stand together.
4. **[JEE Main — NCERT Miscellaneous Exercise]** Four-digit numbers greater than 5000 are formed at random from the digits 0, 1, 3, 5, 7. Find the probability that such a number is divisible by 5 when (i) repetition of digits is allowed, (ii) repetition is not allowed.

Answers. 1. LEVEL has 5 letters over 3 symbols, *L* and *E* doubled. All different: $3! = 6$. One letter doubled: $2 \times 2 \times \frac{3!}{2!} = 12$. So $n(S) = 18$, $n(E) = 6$, $P = \frac{6}{18} = \frac{1}{3}$. — 2. $n(S) = {}^{10}C_3 \times 3! = {}^{10}P_3 = 720$. Choose the woman chairperson (4 ways), then fill the other two posts from the 6 men in ${}^6P_2 = 30$ ways: $n(E) = 120$, $P = \frac{120}{720} = \frac{1}{6}$. — 3. $n(S) = {}^7C_5 \times 5! = {}^7P_5 = 2520$. Stage one: ${}^4C_3 \times {}^3C_2 = 4 \times 3 = 12$ selections. Stage two: glue the three mathematics books into one block, so you arrange 3 objects (block plus two physics books) in $3! = 6$ ways, and the three different mathematics books inside the block in $3! = 6$ ways — here the block **does** carry an internal $3!$, because these three objects are distinct, unlike the identical S’s of Round 1’s Example S2(b). $n(E) = 12 \times 6 \times 6 = 432$, $P = \frac{432}{2520} = \frac{6}{35}$. — 4. (i) The leading digit must be 5 or 7, giving $2 \times 5^3 = 250$ numbers, but 5000 itself is not greater than 5000, so $n(S) = 249$. Divisible by 5 means the units digit is 0 or 5: $2 \times 5 \times 5 \times 2 = 100$, again less the number 5000, so $n(E) = 99$ and $P = \frac{99}{249} = \frac{33}{83}$. (ii) $n(S) = 2 \times {}^4P_3 = 48$, and no repetition means 5000 never arises. Units digit 0: leading digit 5 or 7 and the middle two from the remaining three digits, $2 \times {}^3P_2 = 12$. Units digit 5: the leading digit is forced to 7, giving $1 \times {}^3P_2 = 6$. So $n(E) = 18$ and $P = \frac{18}{48} = \frac{3}{8}$.

Pitfalls

- **Do not apply a position condition at the selection stage.** “The captain is a girl” is not a statement about which four people were picked. Pick first, then place; the two conditions multiply because they act on different stages of one outcome, not because of any independence argument.
- **Check whether the roles cancel before you build the bigger sample space.** If the event never names a position, the arrangement factor divides out and nC_r alone suffices. Building the nP_r model anyway is not wrong, only slower — but forgetting the factor in $n(E)$ while keeping it in $n(S)$ is wrong.

- **A pool with repeated objects never gives a single nC_r .** Split into cases by how many repeats you took, because each case carries its own $\frac{r!}{p_1! \dots}$.
- **A block of *distinct* objects carries an internal $r!$; a block of *identical* objects does not.** Round 1's Example S2(b) glued four identical S's and multiplied by 1; Practice 3 above glues three different books and multiplies by 3!. The rule is about the objects, not the gluing.
- **Repetition allowed changes the sample space, not just the answer.** 5^4 and 5P_4 are different denominators; read the question for the word "repeated".
- **Read the experiment, not the objects.** "A word is chosen from the list" and "the tiles are drawn from a bag" need different denominators, as Example S6 shows.

Addendum 14.4 — The Matching Problem, and Inclusion-Exclusion Past Three Events

Exam Relevance: Subtopic 03's Procedure C stops at three events, and NCERT's Solved Example 16 proves exactly that case. But NCERT's own Miscellaneous Exercise Q6 — three letters into three addressed envelopes, find $P(\text{at least one is correct})$ — is the first term of a family that JEE Main asks with four or five letters and JEE Advanced asks with n , or asks for *exactly* r correct. CBSE Boards asks the $n = 3$ case verbatim. The whole family needs nothing beyond the addition theorem and Chapter 6 counting, which is why it belongs in Class 11 — and it is the cleanest problem in the chapter in which nC_k and a factorial appear inside a single count.

The Method

Result 10 — Inclusion-exclusion for n events. For any events $A_1, A_2, \dots, A_n$,

$$P\left(\bigcup_{i=1}^n A_i\right) = \sum_i P(A_i) - \sum_{i < j} P(A_i \cap A_j) + \sum_{i < j < k} P(A_i \cap A_j \cap A_k) - \dots + (-1)^{n+1} P(A_1 \cap \dots \cap A_n).$$

Proof (induction on n , from the two-event rule alone). The case $n = 2$ is the general addition rule of Subtopic 02, Result 4. Assume the formula for $n - 1$ events and write $B = A_1 \cup \dots \cup A_{n-1}$. The two-event rule gives

$$P(B \cup A_n) = P(B) + P(A_n) - P(B \cap A_n).$$

By distributivity, $B \cap A_n = \bigcup_{i=1}^{n-1} (A_i \cap A_n)$, which is a union of $n - 1$ events. Apply the induction hypothesis to $P(B)$ and to $P(B \cap A_n)$ separately and collect terms by size: every k -fold intersection appears with sign $(-1)^{k+1}$. This is precisely the route NCERT's Solved Example 16 takes for $n = 3$, run one step further. ■

Nothing here is new machinery — it is the addition rule applied repeatedly, so it rests on the three axioms exactly as Round 1's Results 6–9 do.

Result 11 — The matching probability. Suppose n distinct letters are placed at random into n correspondingly addressed envelopes, one letter per envelope. Then

$$P(\text{at least one letter is in its own envelope}) = \sum_{k=1}^n \frac{(-1)^{k+1}}{k!} = 1 - \frac{1}{2!} + \frac{1}{3!} - \cdots + \frac{(-1)^{n+1}}{n!},$$

$$P(\text{no letter is in its own envelope}) = \sum_{k=0}^n \frac{(-1)^k}{k!}.$$

Proof. An outcome is one of the $n!$ equally likely arrangements, so $n(S) = n!$. Let A_i be the event “letter i lands in envelope i ”. Fix any k of the indices. Requiring those k letters to be correct pins down k envelopes and leaves the other $n - k$ letters free to be arranged in $(n - k)!$ ways, so

$$P(A_{i_1} \cap \cdots \cap A_{i_k}) = \frac{(n - k)!}{n!}.$$

There are nC_k such choices of indices, so the k -th sum in Result 10 is

$${}^nC_k \cdot \frac{(n - k)!}{n!} = \frac{n!}{k!(n - k)!} \cdot \frac{(n - k)!}{n!} = \frac{1}{k!},$$

and Result 10 gives the stated alternating sum. The second line is its complement. ■

This is the mixing of Addendum 14.3 in its purest form: a combination nC_k choosing *which* letters are correct, multiplied by a permutation count $(n - k)!$ arranging the rest, and the whole thing sitting over a permutation sample space $n!$. Notice how cleanly the two cancel into $\frac{1}{k!}$ — that cancellation is the reason this problem has a memorable answer at all.

Attend to this — it is why we are not in Class 12. With $n = 4$ you have $P(A_1) = \frac{3!}{4!} = \frac{1}{4}$ and $P(A_2) = \frac{1}{4}$, but

$$P(A_1 \cap A_2) = \frac{2!}{4!} = \frac{1}{12}, \quad \text{while} \quad P(A_1)P(A_2) = \frac{1}{16}.$$

The events are **not** independent, and $\frac{1}{12} \neq \frac{1}{16}$ would make any multiplication shortcut give a wrong answer. Inclusion–exclusion never asks for the product — it asks you to *count* the intersection, which you just did. This is Round 1’s Example S1(c) lesson in a second setting: **count the joint event, do not multiply the probabilities.**

Figure brief: Four envelopes drawn in a row, addressed 1, 2, 3, 4, with four letter cards above them addressed 1, 2, 3, 4. Curved arrows map letters to envelopes in the derangement 2, 1, 4, 3, with a caption “one of the $D_4 = 9$ arrangements with no match”. To the right, a small bar of 24 cells with 9 shaded, labelled “ $P(\text{no match}) = 9/24 = 3/8$ ”.

Worked Example

Example S7 — [JEE Advanced Level]

Four letters are written and four envelopes are correspondingly addressed. The letters are inserted into the envelopes at random, one letter per envelope. Find (a) the probability that

at least one letter reaches its correct envelope, (b) the probability that no letter does, and (c) the answer to the same question with three letters, so as to recover the NCERT case.

Solution.

(a) With $n(S) = 4! = 24$ equally likely insertions, apply Result 10 to A_1, A_2, A_3, A_4 , computing each sum by counting rather than multiplying.

k	Number of k -subsets	P of each k -fold intersection	Contribution $\frac{{}^4C_k(4-k)!}{24}$
1	${}^4C_1 = 4$	$3!/4! = 6/24$	$+ 24/24$
2	${}^4C_2 = 6$	$2!/4! = 2/24$	$- 12/24$
3	${}^4C_3 = 4$	$1!/4! = 1/24$	$+ 4/24$
4	${}^4C_4 = 1$	$0!/4! = 1/24$	$- 1/24$

$$P(\text{at least one match}) = \frac{24 - 12 + 4 - 1}{24} = \frac{15}{24} = \frac{5}{8}.$$

Equivalently, by Result 11, $1 - \frac{1}{2!} + \frac{1}{3!} - \frac{1}{4!} = \frac{5}{8}$.

(b) By the complement law (Subtopic 02, Result 2),

$$P(\text{no match}) = 1 - \frac{5}{8} = \frac{3}{8},$$

so exactly 9 of the 24 arrangements are match-free. A direct listing confirms 9; these are the *derangements* of four objects, and $D_4 = 9$.

(c) For $n = 3$: $1 - \frac{1}{2!} + \frac{1}{3!} = 1 - \frac{1}{2} + \frac{1}{6} = \frac{2}{3}$, which is NCERT's Miscellaneous Exercise Q6. Checking by hand, $n(S) = 3! = 6$ and the match-free arrangements are just the two cyclic shifts, so $P(\text{at least one}) = 1 - \frac{2}{6} = \frac{2}{3}$. ✓

A remark worth carrying. The answers for $n = 3, 4, 5$ are $\frac{2}{3} = 0.667$, $\frac{5}{8} = 0.625$, $\frac{19}{30} = 0.633$ — they barely move. The alternating series of Result 11 converges to $1 - e^{-1} \approx 0.632$ almost immediately, so with 100 letters the probability that at least one reaches the right envelope is still about 0.632. Adding more letters does not help. ■

Practice Exercises

- [CBSE — NCERT Miscellaneous Exercise]** Three letters are dictated to three persons and an envelope is addressed to each of them. The letters are inserted into the envelopes at random, one per envelope. Find the probability that at least one letter is in its proper envelope.
- [JEE Main]** Five letters are placed at random into five correspondingly addressed envelopes, one per envelope. Find (i) the probability that no letter is in its correct envelope, (ii) the probability that exactly one letter is.
- [JEE Advanced]** Show that the number of arrangements of n letters into n addressed envelopes with **exactly** r correct is ${}^nC_r D_{n-r}$, where D_m is the number of match-free

arrangements of m letters, and use it to find $P(\text{exactly two correct})$ for $n = 5$. Verify that your probabilities over $r = 0, 1, \dots, 5$ sum to 1.

4. **[JEE Main]** Using Result 10 with four events, find the probability that a number chosen at random from 1 to 1000 is divisible by at least one of 2, 3, 5 or 7.

Answers. 1. $1 - \frac{1}{2!} + \frac{1}{3!} = \frac{2}{3}$. — 2. (i) $\sum_{k=0}^5 \frac{(-1)^k}{k!} = 1 - 1 + \frac{1}{2} - \frac{1}{6} + \frac{1}{24} - \frac{1}{120} = \frac{44}{120} = \frac{11}{30}$, so $D_5 = 44$. (ii) Choose the one correct letter in ${}^5C_1 = 5$ ways and derange the other four in $D_4 = 9$ ways: $\frac{5 \times 9}{120} = \frac{45}{120} = \frac{3}{8}$. — 3. Choose which r letters are correct in nC_r ways; the remaining $n - r$ must **all** be wrong, which is D_{n-r} ways — the word “exactly” is what forces the derangement factor, and omitting it (writing ${}^nC_r(n-r)!$) is the standard error, since that counts “at least r ”. For $n = 5, r = 2$: ${}^5C_2 D_3 = 10 \times 2 = 20$, so $P = \frac{20}{120} = \frac{1}{6}$. The full distribution is 44, 45, 20, 10, 0, 1 over 120 — note the 0 at $r = 4$, since four correct forces the fifth — and $44 + 45 + 20 + 10 + 0 + 1 = 120$. ✓ — 4. With A, B, C, D the events “divisible by 2, 3, 5, 7”, the counts are $500 + 333 + 200 + 142 = 1175$ singles; pairs $166 + 100 + 71 + 66 + 47 + 28 = 478$ (divisors 6, 10, 14, 15, 21, 35); triples $33 + 23 + 14 + 9 = 79$ (divisors 30, 42, 70, 105); quadruple 4 (divisor 210). So $n(E) = 1175 - 478 + 79 - 4 = 772$ and $P = \frac{772}{1000} = \frac{193}{250}$.

Pitfalls and Pro-Tips

- **“At least one” here does *not* yield to a single complement count.** Subtopic 03’s Procedure B works when “none” is one clean case. Here “no letter correct” is itself the hard count, so inclusion-exclusion is what produces it — and then the complement gives you the other answer for free. Know which direction is the cheap one.
- **Do not multiply $P(A_i)$ together.** $P(A_1 \cap A_2) = \frac{1}{12}$, not $\frac{1}{16}$. The intersections in Result 10 must be **counted**.
- **An erratum worth marking in Subtopic 03, Section 7.** That subtopic’s closing Pro-Tip reads “ $P(\text{at least one}) = 1 - (\text{chance of none})^{\text{number of trials}}$ ”. As written that is the independence formula, which is Class 12 and is *false* in general — the matching problem is a counterexample sitting inside NCERT’s own exercise set. The two places the chapter uses it are both fine, but only because they are really counts, not products: Subtopic 03’s MCQ Q4 gets $P(\text{no six on two dice}) = \frac{25}{36}$ because 25 of the 36 equally likely ordered pairs avoid a six, and MCQ Q3 gets $P(\text{no head}) = \frac{1}{8}$ because one of the 8 equally likely triples is TTT . **Read that Pro-Tip as a shortcut for counting equally likely outcomes in a product sample space, never as a licence to multiply probabilities of events.** Round 1’s pitfall “never convert an *and* condition into a product of probabilities” governs, and this is where the chapter’s own wording brushes against it.
- **“Exactly r ” needs a derangement, “at least r ” needs a factorial.** ${}^nC_r(n-r)!$ counts arrangements with *at least* r correct, with multiplicity; ${}^nC_r D_{n-r}$ counts exactly r . Mixing them is the standard slip in Practice 3.
- **Pro-tip — the sums shrink fast.** In Result 11 the k -th term is $\frac{1}{k!}$, so by $k = 5$ you are adding $\frac{1}{120}$. For an MCQ, the first three terms already pin the answer to within 1% of $1 - e^{-1} \approx 0.632$, which is usually enough to eliminate three options.

Round 2 Cheat Sheet Additions

- **Mixed selection-and-arrangement:** $n(S) = {}^nC_r \times r! = {}^nP_r$. Apply selection conditions at stage one, position conditions at stage two, and multiply. With repetition allowed the arrangement stage is n^r instead.
- **Roles cancel when the event ignores them.** “Exactly two girls” needs only $\frac{{}^4C_2 {}^5C_2}{{}^9C_4}$; “and the captain is a girl” does not.
- **Pool with repeats $\Rightarrow$ split the selection into cases**, each with its own $\frac{r!}{p_1! p_2! \dots}$. MATH-EMATICS three-letter words: $336 + 63 = 399$, not 8C_3 and not ${}^{11}P_3$.
- **Blocks:** distinct objects glued together carry an internal $r!$; identical objects carry 1.
- **“Word chosen at random” $\neq$ “tiles drawn at random”** when letters repeat — different experiments, different denominators, different correct answers ($\frac{16}{19}$ against $\frac{46}{55}$ for Example S6).
- **Inclusion-exclusion, n events (Result 10):** alternate the sums of k -fold intersections with sign $(-1)^{k+1}$. Provable from the two-event addition rule by induction; no new axioms.
- **Matching problem (Result 11):** $P(\text{at least one match}) = \sum_{k=1}^n \frac{(-1)^{k+1}}{k!}$; $P(\text{no match}) = \sum_{k=0}^n \frac{(-1)^k}{k!}$. Memorise $D_2 = 1$, $D_3 = 2$, $D_4 = 9$, $D_5 = 44$. Exactly r correct: $\frac{{}^nC_r D_{n-r}}{n!}$. The limit is $1 - e^{-1} \approx 0.632$.
- **Intersections get counted, never multiplied.** $P(A_1 \cap A_2) = \frac{2!}{4!} = \frac{1}{12} \neq \frac{1}{16}$ in the four-letter matching problem.

Deliberately out of scope

Recorded so that a later pass does not re-open them.

Geometric probability — “a point is chosen at random inside a square / on a segment; find the probability that ...”, answered by a ratio of lengths, areas or volumes. This was the main candidate for Round 2 and it is being left out on three grounds, not one.

1. **It has no axiom to stand on in this chapter.** Subtopic 02 states Kolmogorov’s axioms in their finite-sample-space form, and says so explicitly in its “Limiting conditions”. A length-ratio or area-ratio probability is a statement about an uncountably infinite sample space in which every individual outcome has probability zero, which the additivity axiom as given here cannot reach. Presenting it would mean asserting a measure the book has not built.
2. **NCERT does not have it, at either level.** Class 11 Chapter 14 goes no further than the coin-tossed-until-a-head sample space $\{H, TH, TTH, \dots\}$, which is infinite but *discrete*, and it never assigns probabilities to it. Its Introduction names infinite sample spaces as a case the classical ratio “cannot be applied to”, and then never returns. The Class 12 chapter is conditional probability, Bayes and distributions, not continuous sample spaces.
3. **Neither examination syllabus lists it.** The JEE Advanced probability syllabus runs “random experiment, sample space, different types of events, addition and multiplica-

tion rules, conditional probability, Bayes theorem, independence, and computation of probability of events using permutations and combinations” — that final clause is what Addenda 14.1, 14.3 and 14.4 serve, and there is no continuous or measure-theoretic clause anywhere in it. JEE Main’s list is a subset of the same. Geometric probability is a genuine topic, but at this level it is coaching-circuit enrichment, not examinable content, and adding it to a master-reference chapter would misrepresent the syllabus.

Conditional probability, independence, the multiplication theorem, Bayes’ theorem, random variables and their distributions, binomial trials — Class 12, exactly as Round 1 held and as Subtopic 03’s closing “Looking ahead” note already signposts. Addendum 14.4 strengthens the boundary rather than crossing it, by exhibiting a natural pair of events whose intersection probability is not the product.

Bertrand-style paradoxes and the ambiguity of “at random” over continuous sets — follows geometric probability out, for the same reason. The finite analogue of that lesson is already taught, and taught honestly, by Example S6’s two experiments.

Errata

This section lists the errors identified in the printed text of the fourteen chapters of the *Class 11 Mathematics Master Reference*. Because the chapters were typeset from a source that is no longer available, these corrections could not be applied in place and are published here instead. Entries are ordered by chapter and then by printed page; the page number is the number printed on the page itself. Each entry names the exact location, quotes what is printed, gives the replacement text, and — where the correction is a mathematical result — shows the working, so that a reader can act on the correction without consulting any other source. Where an exercise is defective beyond repair, a replacement exercise and its answer are supplied. Errors of a purely typographic kind (raw markup, clipped digits, unrendered figures) are listed alongside mathematical errors, since both mislead the reader on the page. Every item below was independently re-verified against the typeset PDF. Defects that exist only in the machine-extracted text of the chapters, and not on the typeset page, are excluded.

Chapter 1: Sets

Page 5 — Subtopic 01, Section 4, Example 2 [JEE Main — Speed Trap]: two options are the empty set

Printed: “Without lengthy computation, identify which of the following is the empty set: (a) $\{x \in \mathbb{R} : x^2 + 1 = 0\}$ (b) $\{x : x \text{ is an even prime}\}$ (c) $\{0\}$ (d) $\{x \in \mathbb{N} : x < 1\}$ ”, with the solution conceding “The trap: (a) and (d) are both empty. If the question allows one answer, the intended one is usually (a).”

Should read: Replace option (d) with a non-empty set, so that exactly one option is empty. Use

(a) $\{x \in \mathbb{R} : x^2 + 1 = 0\}$ (b) $\{x : x \text{ is an even prime}\}$ (c) $\{0\}$ (d) $\{x \in \mathbb{N} : x < 2\}$

and delete the sentence beginning “If the question allows one answer”. The solution should read: (a) is empty because $x^2 = -1$ has no real solution; (b) = $\{2\}$; (c) = $\{0\}$, a singleton; (d) = $\{1\}$, since 1 is the smallest natural number. Answer: (a).

Why: As printed the item has two correct answers — $x^2 + 1 = 0$ has no real root and no natural number is less than 1, so both (a) and (d) denote $\emptyset$. Guidance to guess the examiner’s intent is not a substitute for a well-posed item. Changing the bound in (d) from $x < 1$ to $x < 2$ makes it $\{1\} \neq \emptyset$ while preserving the intended “check the universe first” lesson.

Page 21 — Subtopic 04, opening paragraph: internal build metadata printed as body text

Printed: “Provenance note: this subtopic is [SUPPLIED — grounding waived by founder] — generated from NCERT-aligned curriculum knowledge; no source bundle was retrieved, so it is not RAG-grounded. Accepted into master content by explicit founder decision. All numeric answers are SymPy-verified and all examples are original. It remains flagged as supplied (not [GROUNDED]) for an honest audit trail, and will be upgraded if a source bundle is later audited.”

Should read: The paragraph should be deleted in its entirety. Subtopic 04 begins directly with its Exam Relevance box.

Why: This is production-pipeline metadata, not content. It carries no mathematical information and tells the reader that a quarter of the chapter was not source-checked, which is not a statement a reference text should make in its body.

Chapter 2: Relations and Functions

Page 9 — Subtopic 01, Section 7, first pitfall (“Adding instead of multiplying”): abandoned self-correction left in the text

Printed: “Quick gut check: with $n(A) = n(B) = 2$ you should get 4 pairs, not 4... wait — verify by actually listing once, early in your prep, and the muscle memory sticks.”

Should read: “Quick gut check: with $n(A) = 2$ and $n(B) = 3$ you should get 6 pairs, not 5. Verify by actually listing once, early in your prep, and the muscle memory sticks.”

Why: The intended contrast is between the product $n(A)n(B)$ and the (incorrect) sum $n(A) + n(B)$. With $n(A) = n(B) = 2$ both equal 4, so the check is vacuous and the author’s mid-sentence retraction was left in print. With $n(A) = 2$, $n(B) = 3$ the product is $2 \times 3 = 6$ and the sum is $2 + 3 = 5$, so the check discriminates.

Page 11 — Subtopic 02, Exam Relevance box: reference to a subtopic that does not exist

Printed: “(The deeper machinery — reflexive/symmetric/transitive and equivalence relations — lives in a later subtopic; here we build the foundation everything else stands on.)”

Should read: “(The deeper machinery — reflexive/symmetric/transitive and equivalence relations — is not part of the Class 11 syllabus and is taken up in Class 12, Chapter 1; here we build the foundation everything else stands on.)”

Why: Chapter 2 contains three subtopics only — 01 Cartesian Product of Sets, 02 Relations, 03 Functions — and ends on page 27. No later subtopic treats equivalence relations, so the promise is never fulfilled and no forward pointer is given.

Page 18 — Subtopic 03, Exam Relevance box: reference to a subtopic that does not exist

Printed: “(The mapping-classification layer — one-one, onto, bijective, composition, inverse — is built in a later subtopic; here we install the engine.)”

Should read: “(The mapping-classification layer — one-one, onto, bijective, composition, inverse — is not part of the Class 11 syllabus and is built in Class 12, Chapter 1; here we install the engine.)”

Why: Subtopic 03 is the last subtopic in the chapter. The promised material appears nowhere in it.

Pages 20–21 — Subtopic 03, Section 2: the standard-function table and the graph figure disagree on the count

Printed: The definitions table on page 20 lists seven standard functions (Identity, Constant, Polynomial, Rational, Modulus, Signum, Greatest integer); the sentence introducing the graphs reads “These six are the ones every Class 11 student must be able to sketch on sight”, and the caption on page 21 reads “Figure 2. Graphs of the six standard functions.”

Should read: The introducing sentence should read “These six — every function in the table except the rational function, whose graph varies with p and q — are the ones every Class 11 student must be able to sketch on sight”, and the caption should read “Figure 2. Graphs of six of the seven standard functions (the rational function is omitted).”

Why: The figure shows Identity, Constant, Modulus, Signum, Greatest integer and Polynomial. The rational function is present in the table but absent from the figure, and the text does not say so, leaving the reader unsure how many functions the standard library contains.

Page 21 — Subtopic 03, Section 2, Polynomial bullet: false statement about even-degree polynomials

Printed: “Polynomial $f(x) = x^2$: an upward parabola with vertex at the origin. An even-degree polynomial never reaches negative values, so the range is $[0, \infty)$ — a concrete instance of the ‘even degree $\Rightarrow$ range is a subset of $\mathbb{R}$ ’ rule.”

Should read: “Polynomial $f(x) = x^2$: an upward parabola with vertex at the origin. Its range is $[0, \infty)$ because a square is never negative. More generally, an even-degree polynomial has a range that is bounded below when the leading coefficient is positive, and bounded above when it is negative; it is never all of $\mathbb{R}$.”

Why: Two errors. First, an even-degree polynomial certainly can take negative values: $f(x) = x^2 - 1$ is negative on $(-1, 1)$, and $f(x) = -x^2$ is never positive. What is true is that the range of an even-degree polynomial is of the form $[m, \infty)$ (positive leading coefficient) or $(-\infty, M]$ (negative leading coefficient). Second, “even degree $\Rightarrow$ range is a subset of $\mathbb{R}$ ” is not a rule: the range of every real polynomial is a subset of $\mathbb{R}$, whatever its degree.

Page 21 — Subtopic 03, Section 2: the heading “Algebra of real functions” is run into the preceding paragraph

Printed: “(JEE companion — beyond the Class 11 core: the fractional part $\{x\} = x - [x]$ strips off the integer part of x ; it is a sawtooth with range $[0, 1)$ and is studied alongside $[x]$ in the Class 12 / JEE function toolkit.) **Algebra of real functions.** For real functions f, g and scalar α : ...”

Should read: “Algebra of real functions” should begin a new block, set as a heading in the same style as “Graphs of the standard functions” earlier in the same section, with the parenthetical JEE aside closed off before it.

Why: The bold run-in is appended to the last line of an italic aside, so one of the most examinable topics in the chapter has no visual heading and is easy to skim past.

Chapter 3: Trigonometric Functions

Page 26 — Subtopic 03, Section 6, Practice Exercise 4 and its answer: the identity to be proved is false

Printed: Question 4: “Prove $\tan(90^\circ + \theta) \cdot \tan(90^\circ - \theta) = -1$.” Answer 4: “ $\tan(90^\circ + \theta) = -\cot \theta$, $\tan(90^\circ - \theta) = \cot \theta$; product = $-\cot \theta \cdot \dots$ actually $(-\cot \theta)(\cot \theta) = -\cot^2 \theta$. Hmm — the identity as stated holds only for the special pairing; corrected target: $\tan(90^\circ + \theta) \cot(90^\circ - \theta) \dots$ (Use the reduction values; the intended clean result is that $\tan(90^\circ + \theta) \tan(90^\circ - \theta) = -\cot^2 \theta$.)”

Should read: Question 4: “Prove $\tan(90^\circ + \theta) \cdot \cot(90^\circ - \theta) = -1$.” Answer 4: “ $\tan(90^\circ + \theta) = -\cot \theta$ and $\cot(90^\circ - \theta) = \tan \theta$, so the product is $(-\cot \theta)(\tan \theta) = -1$.”

Why: As printed the statement is false. Since $\tan(90^\circ + \theta) = -\cot \theta$ and $\tan(90^\circ - \theta) = \cot \theta$,

$$\tan(90^\circ + \theta) \tan(90^\circ - \theta) = -\cot^2 \theta,$$

which equals -1 only when $\cot^2 \theta = 1$, i.e. $\theta = 45^\circ + 90^\circ n$. Replacing the second factor by $\cot(90^\circ - \theta) = \tan \theta$ gives $-\cot \theta \cdot \tan \theta = -1$ for every θ at which both sides are defined, which is the intended identity. The printed answer, which argues with itself and addresses the author rather than the reader, must be replaced by the two-line proof above.

Page 33 — Subtopic 04, Section 6, Practice Exercise 6 and answer 6: wrong solution count

Printed: “How many solutions does $\sin x = \frac{x}{10}$ have for $x \geq 0$? (Reason graphically.)” — Answer: “[0, 10] gives 3 solutions (including $x = 0$).”

Should read: “[0, 10] gives 4 solutions (including $x = 0$).”

Why: Solutions require $|x/10| \leq 1$, i.e. $0 \leq x \leq 10$. On $(0, \pi)$ the line $y = x/10$ cuts the first hump once. On $(\pi, 2\pi)$ the sine is negative while the line is positive, so there is no crossing.

On $(2\pi, 3\pi)$ the sine rises to 1 at $x = 3\pi/2 + \pi = 7.854$, where the line is only 0.785, and returns to 0 at $3\pi = 9.425$, where the line is 0.942 — so the line cuts that hump **twice**. On $(3\pi, 10]$ the sine is negative again. Counting $x = 0$, the solutions are

$$x = 0, \quad x \approx 2.8523, \quad x \approx 7.0682, \quad x \approx 8.4232,$$

a total of 4 (equivalently, 3 strictly positive solutions).

Pages 46, 47 and 48 — Subtopic 06, Sections 1, 2 and 4(a): three cross-references point at the wrong subtopic

Printed: Page 46, Section 1: “you use identities (Subtopic 03) to simplify”. Page 47, Section 2: “factorising with sum-to-product (Subtopic 03)”. Page 48, Section 4(a): “Move to one side and apply sum-to-product (Subtopic 03)”.

Should read: In all three places, “Subtopic 03” should read “Subtopic 05”.

Why: Subtopic 03 is “Standard Values, Even-Odd Nature & Allied Angles” (page 21) and contains no compound-angle or sum-to-product material. The identities and the sum-to-product formulas are in Subtopic 05, “Trigonometric Identities and Compound Angle Formulas” (page 35).

Chapter 4: Complex Numbers and Quadratic Equations

Page 19 — Subtopic 03, Section 7, MCQ Q5, analysis of option (D): unresolved self-correction and wrong claim

Printed: “(B) is $-z$; (D) is $1/z$ only when $r = 1$... actually $1/z = \frac{1}{r}(\cos \theta - i \sin \theta)$, so (D) is wrong on both count.”

Should read: “(B) is $-z$. (D) is $\frac{1}{\bar{z}} = \frac{z}{r^2}$: it inverts the modulus but leaves the argument alone, so it is neither $\bar{z}$ nor $1/z$ — note $1/z = \frac{1}{r}(\cos \theta - i \sin \theta)$.”

Why: With $z = r(\cos \theta + i \sin \theta)$, option (D) is $\frac{1}{\bar{z}}(\cos \theta + i \sin \theta) = z/r^2 = 1/\bar{z}$. It is never equal to $1/z$ unless $\sin \theta = 0$, so the printed claim “ $1/z$ only when $r = 1$ ” is itself incorrect; it coincides with z (not with $1/z$) when $r = 1$. The unedited “actually ...” retraction and the ungrammatical “on both count” must both go.

Page 29 — Subtopic 05, Section 6, Practice Exercise 6: retracted half of the question left in the stem

Printed: “Prove $(1 + \omega)^3 = (1 + \omega^2)^3 = -1$... actually show $(1 + \omega)^3 = -1$.” The answer key proves only the first equality.

Should read: “Prove $(1 + \omega)^3 = (1 + \omega^2)^3 = -1$, where ω is a non-real cube root of unity.” Answer: “ $1 + \omega = -\omega^2$, so $(1 + \omega)^3 = -\omega^6 = -(\omega^3)^2 = -1$. Likewise $1 + \omega^2 = -\omega$, so

$$(1 + \omega^2)^3 = -\omega^3 = -1."$$

Why: The retracted half of the claim is true, so the hedge is unnecessary as well as unedited. Since $1 + \omega + \omega^2 = 0$ we have $1 + \omega^2 = -\omega$ and hence $(1 + \omega^2)^3 = -\omega^3 = -1$, using $\omega^3 = 1$.

Page 34 — Subtopic 06, Section 6, answer 6: wrong sign on the centre of the Apollonius circle

Printed: "Apollonius circle ($k = 2 \neq 1$): solving gives centre $\frac{5}{3}$, radius $\frac{4}{3}$ (a circle)."

Should read: "Apollonius circle ($k = 2 \neq 1$): solving gives centre $-\frac{5}{3}$, radius $\frac{4}{3}$ (a circle)."

Why: From $\frac{|z-1|}{|z+1|} = 2$ we get $|z-1|^2 = 4|z+1|^2$, so with $z = x + iy$,

$$(x-1)^2 + y^2 = 4[(x+1)^2 + y^2] \implies 3x^2 + 3y^2 + 10x + 3 = 0 \implies x^2 + y^2 + \frac{10}{3}x + 1 = 0.$$

Hence the centre is $(-\frac{5}{3}, 0)$ and the radius is $\sqrt{\frac{25}{9} - 1} = \frac{4}{3}$. The radius as printed is correct; the sign of the centre is not. The sign is conceptually load-bearing: points satisfying the condition are twice as far from 1 as from -1 , so the circle must lie on the negative side of the origin.

Chapter 5: Linear Inequalities

Page 1 — Title page: three replacement glyphs in place of an em dash

Printed: "Class 11 Mathematics $\diamond\diamond\diamond$ Master Reference" — three solid black diamond replacement glyphs appear between "Mathematics" and "Master".

Should read: "Class 11 Mathematics — Master Reference"

Why: A three-byte UTF-8 em dash was decoded one byte at a time, producing three undefined characters. These are the only such glyphs in the 31-page chapter; the em dash renders correctly everywhere else in the text.

Pages 3, 5, 12, 13, 14 and 29 — All six figures are unrendered placeholders

Printed: Six shaded boxes headed "Figure placeholder", each containing a prose description of the intended artwork instead of the artwork itself, e.g. on page 3: "A horizontal number line with a hollow circle at $x = 2$ and a thick shaded arrow extending right toward $+\infty$, labelled ' $x > 2$ '. Below it, a second number line with a filled circle at $x = 2$ and a thick shaded arrow extending right, labelled ' $x \geq 2$ '."

Should read: Each placeholder should be replaced by the figure it describes: Subtopic 01 Sections 1 and 4 (number lines for strict and non-strict inequalities), Subtopic 02 Sections

3 and 4 (half-plane diagrams, two on Section 4), and Subtopic 04 Section 4 (the wavy-curve sign diagram).

Why: The chapter ships with no graphics at all. Two of the placeholders are load-bearing rather than decorative: the wavy-curve sign diagram in Subtopic 04 Example 4 and the half-plane diagram in the Subtopic 02 board derivation carry reasoning that the surrounding prose only partly restates.

Pages 20 and 22 — Subtopic 03: three cross-references to headings that do not exist

Printed: Page 20, Section 3, step 4: “Solve the inequality or system using the algebra from Subtopics 01–02.” Page 20, Section 3, mixture illustration: “you clear of fractions ... and solve as in Subtopic 01.” Page 22, Section 4: “its solution region (from Subtopic 02) is a bounded polygon.”

Should read: “...from Units 1–2”, “...as in Unit 1”, “...(from Unit 2)” respectively.

Why: This chapter numbers its four major divisions as “1 Linear Inequalities in One Variable”, “2 Linear Inequalities in Two Variables”, “3 Applications & Word Problems of Linear Inequalities” and “4 Non-Linear Inequalities: The Wavy-Curve Method”. There is no heading anywhere in the chapter reading “Subtopic 01” or “Subtopic 02”, so as printed the three references cannot be resolved by the reader.

Pages 22–23 — Unit 3, Section 5, Practice Exercise 3 and its answer: one solution pair omitted

Printed: Question 3: “Find all pairs of consecutive even natural numbers, both larger than 5, whose sum is less than 24.” Answer: “(6, 8), (8, 10) — i.e. smaller number $\in \{6, 8\}$.”

Should read: “(6, 8), (8, 10), (10, 12) — i.e. smaller number $\in \{6, 8, 10\}$.”

Why: Let the smaller number be x , so the pair is $(x, x + 2)$ with x even and $x > 5$. The condition is

$$x + (x + 2) < 24 \implies 2x < 22 \implies x < 11.$$

The even values with $5 < x < 11$ are $x = 6, 8, 10$, giving the pairs (6, 8), (8, 10) and (10, 12). The last is valid: $10 + 12 = 22 < 24$ and both members exceed 5. The printed key is the answer to the version of the problem with the bound 23 rather than 24.

Chapter 6: Permutations and Combinations

Pages 4 and 12 — Raw LaTeX delimiters printed in the running text

Printed: Page 4, Subtopic 01, Section 2, “Conditions of validity”, first bullet: “Domain: $\$n!\$$ exists only for $n \in \{0, 1, 2, 3, \dots\}$.” Page 12, Subtopic 02, Section 1, “Conditions & Assump-

tions”, third bullet: “Circular logic ($(n-1)!$) applies to genuinely rotatable arrangements”.

Should read: “Domain: $n!$ exists only for $n \in \{0, 1, 2, 3, \dots\}$ ” and “Circular logic $((n-1)!)!$ applies to genuinely rotatable arrangements”, with the mathematics set in math italic as it is everywhere else in the chapter.

Why: The inline-math delimiters were not processed, so dollar signs and unstyled roman text appear on the page. Both bullets contain the same symbols that typeset correctly elsewhere.

Page 31 — Subtopic 04, Section 6, Crack the MCQ, Q2: broken and arithmetically false working

Printed: “Answer: (A) 10. $\frac{6!}{3!3!2!} = \frac{720}{6 \cdot 2} \cdot \frac{1}{1} = \frac{720}{36} \cdot \frac{1}{1} \dots$ compute directly: $\frac{6!}{3!3!} = 20$ for labelled groups, then $\div 2!$ for the interchangeable equal groups = 10.”

Should read: “Answer: (A) 10. $\frac{6!}{3!3!2!} = \frac{720}{6 \cdot 6 \cdot 2} = \frac{720}{72} = 10$. Equivalently: $\frac{6!}{3!3!} = 20$ for labelled groups, then $\div 2!$ for the interchangeable equal groups gives 10.”

Why: Both printed denominators are wrong and both printed quotients contradict the stated answer. $3!3!2! = 6 \cdot 6 \cdot 2 = 72$, not $6 \cdot 2 = 12$ and not 36; and $\frac{720}{12} = 60$, $\frac{720}{36} = 20$, neither of which is 10. The final answer (A) 10 and the “compute directly” route are both correct, so only the displayed chain needs replacing.

Chapter 7: Binomial Theorem

Pages 24 and 28 — Subtopic 04, Section 2 “Index/ratio identities” and Section 8 Cheat Sheet Box: false identity

Printed: Under the chapter’s own convention $C_r = {}^nC_r$, both places print

$${}_rC_r = {}_nC_{r-1}.$$

Should read:

$$r {}^nC_r = n {}^{n-1}C_{r-1}.$$

Why: The upper index on the right-hand side must drop by one. Directly:

$$r \cdot \frac{n!}{r!(n-r)!} = \frac{n!}{(r-1)!(n-r)!} = n \cdot \frac{(n-1)!}{(r-1)!((n-1)-(r-1))!} = n {}^{n-1}C_{r-1}.$$

Numerical check with $n = 4$, $r = 2$: the left side is $2 \cdot {}^4C_2 = 2 \cdot 6 = 12$; the corrected right side is $4 \cdot {}^3C_1 = 4 \cdot 3 = 12$; the printed right side is $4 \cdot {}^4C_1 = 16$. This is a typeset fault, not a rendering loss: superscripts on nC_r symbols appear correctly elsewhere on the same lines (for instance ${}^{n+1}C_{r+1}$ in the adjacent identity).

Page 37 — Subtopic 06, Section 5, answer 1: raw LaTeX printed on the page

Printed: “1. $6=5+1 \Rightarrow 6^n-1 = \sum \text{ of terms each divisible by } 5.$ ”

Should read: “1. $6 = 5 + 1$, so $6^n - 1 = (5 + 1)^n - 1 = \sum_{k=1}^n \binom{n}{k} 5^k$, a sum of terms each divisible by 5.”

Why: The dollar delimiters and the caret exponent were not processed. Expanding $(5 + 1)^n$ by the binomial theorem gives $1 + \binom{n}{1}5 + \binom{n}{2}5^2 + \dots$; subtracting 1 leaves only terms carrying a factor of 5.

Page 37 — Subtopic 06, Section 5, answer 2: the answer contradicts its own derivation

Printed: “2. 2 (since $2^3 = 8 \equiv 1$, $2^{50} = 2^{48} \cdot 2^2 \equiv 1 \cdot 4 = 4$) — **remainder 4.**”

Should read: “2. 4 (since $2^3 = 8 \equiv 1 \pmod{7}$, $2^{50} = 2^{48} \cdot 2^2 = (2^3)^{16} \cdot 4 \equiv 1 \cdot 4 = 4$).”

Why: The remainder when 2^{50} is divided by 7 is 4, as the parenthetical derivation and the closing words of the same line both state. The leading “2” is simply wrong and will mislead a reader scanning the answer list.

Chapter 8: Sequences and Series

Page 14 — Unit 2, Section 6, Practice Exercise 8 and its answer: wrong minimising value and stray word

Printed: Question 8: “The first term of an AP is unity and the common difference is $d \neq 0$. For what value of d is $a_1a_3 + a_2a_3$ minimum?” Answer: “ $d = -\frac{2}{3}$ region; minimise the quadratic in d .”

Should read: “ $d = -\frac{5}{4}$, at which the minimum value is $-\frac{9}{8}$.”

Why: With $a_1 = 1$, $a_2 = 1 + d$ and $a_3 = 1 + 2d$,

$$a_1a_3 + a_2a_3 = a_3(a_1 + a_2) = (1 + 2d)(2 + d) = 2d^2 + 5d + 2.$$

This is an upward parabola in d , minimised at $d = -\frac{5}{2 \cdot 2} = -\frac{5}{4}$, where its value is $2\left(\frac{25}{16}\right) - \frac{25}{4} + 2 = -\frac{9}{8}$. The word “region” is a leftover from an unrelated draft and should be deleted.

Page 25 — Unit 3, Section 6, Practice Exercise 7 and its answer: the problem as stated has no valid solution

Printed: Question 7: “Three positive distinct numbers x, y, z are in GP and $x + y, y + z, z + x$ are in AP. Find the common ratio.” Answer: “ $r = 2 - \sqrt{3}$ (or its reciprocal).”

Should read: The exercise is unsalvageable as stated and must be replaced. Substitute:

7. (Adv) Three positive distinct numbers x, y, z are in GP and $x, 2y, z$ are in AP. Find the common ratio.

Answer: $r = 2 - \sqrt{3}$ (or its reciprocal $2 + \sqrt{3}$).

Why: With the printed AP condition, $2(y + z) = (x + y) + (z + x)$ gives $y + z = 2x$; writing $y = xr, z = xr^2$ this becomes

$$r^2 + r - 2 = 0 \implies r = 1 \text{ or } r = -2.$$

But $r = 1$ makes the three numbers equal, contradicting “distinct”, and $r = -2$ makes them alternate in sign, contradicting “positive”. No admissible ratio exists, so the printed answer cannot follow from the printed question. The value $2 - \sqrt{3}$ is the root of $r^2 - 4r + 1 = 0$, which is exactly the condition $x + z = 4y$, i.e. $x, 2y, z$ in AP. The replacement stem restores that condition: $2(2y) = x + z$ gives $4xr = x + xr^2$, hence $r^2 - 4r + 1 = 0$ and $r = 2 \pm \sqrt{3}$, both positive and neither equal to 1.

Page 31 — Unit 4, Section 5, Worked Examples, Example 5: abandoned draft left in the question stem

Printed: “Example 5 — JEE Advanced. If a, b, c are in AP, a, b, d are in GP, and $a, c, e, \dots$ — more directly: if the AM of two numbers is 9 and their HM is 4, find the numbers (using A, G, H).”

Should read: “Example 5 — JEE Advanced. If the AM of two numbers is 9 and their HM is 4, find the numbers (using A, G, H).”

Why: Everything before “more directly” is an unfinished first draft of a different problem and was never deleted; the solution that follows addresses only the second half. That solution is correct and needs no change: $G^2 = AH = 36$ so $G = 6$; then $a + b = 2A = 18$ and $ab = G^2 = 36$, so a and b are the roots of $x^2 - 18x + 36 = 0$, giving $9 \pm 3\sqrt{5}$.

Pages 31-32 — Unit 4, Section 6, Practice Exercise 3 and its answer: only two means inserted, and an endpoint listed as a mean

Printed: Question 3: “Insert 3 HMs between 2 and $\frac{1}{2}$.” Answer: “reciprocal AP $\frac{1}{2}, 1, \frac{3}{2}, 2 \rightarrow$ HMs $1, \frac{2}{3}, \frac{1}{2}$.”

Should read: “reciprocal AP $\frac{1}{2}, \frac{7}{8}, \frac{5}{4}, \frac{13}{8}, 2$ (common difference $D = \frac{3}{8}$) $\rightarrow$ HMs $\frac{8}{7}, \frac{4}{5}, \frac{8}{13}$.”

Why: Inserting 3 harmonic means between 2 and $\frac{1}{2}$ means inserting 3 arithmetic means between their reciprocals $\frac{1}{2}$ and 2, producing a five-term AP. Its common difference is

$$D = \frac{2 - \frac{1}{2}}{3 + 1} = \frac{3}{8},$$

so the reciprocal AP is $\frac{1}{2}, \frac{7}{8}, \frac{5}{4}, \frac{13}{8}, 2$ and the three harmonic means are the reciprocals of the three interior terms: $\frac{8}{7}, \frac{4}{5}, \frac{8}{13}$. The printed reciprocal AP has only four terms, i.e. it inserts

two means, not three; and the third quantity it lists as a “mean”, $\frac{1}{2}$, is the reciprocal of the endpoint 2.

Page 37 — Unit 5, Section 6, Practice Exercise 4 and its answer: wrong numbers

Printed: Question 4: “If the AM of two numbers exceeds their GM by 2 and the numbers differ by 8, find them.” Answer: “16 and 8.”

Should read: “9 and 1.”

Why: For 16 and 8: $A = 12$ and $G = \sqrt{128} \approx 11.314$, so $A - G \approx 0.686$, not 2. Solving properly, let the numbers be $a > b > 0$ with $a - b = 8$ and $A - G = 2$. Then

$$\frac{a+b}{2} - \sqrt{ab} = 2 \implies a+b-2\sqrt{ab} = 4 \implies (\sqrt{a}-\sqrt{b})^2 = 4 \implies \sqrt{a}-\sqrt{b} = 2,$$

and since $(\sqrt{a}-\sqrt{b})(\sqrt{a}+\sqrt{b}) = a-b = 8$, we get $\sqrt{a}+\sqrt{b} = 4$. Hence $\sqrt{a} = 3$, $\sqrt{b} = 1$, so $a = 9$ and $b = 1$. Check: $A = 5$, $G = 3$, $A - G = 2$, $a - b = 8$.

Page 57 — Unit 8, Section 6, Practice Exercise 5 and its answer: the answer does not match the question

Printed: Question 5: “Evaluate $\sum_{r=1}^n \frac{r}{(2r-1)(2r+1)(2r+3)}$ — hint: partial fractions into a telescoper.” Answer: “ $\frac{1}{8} \left[\frac{1}{3} - \frac{1}{(2n+1)(2n+3)} \right]$ (after partial fractions; telescopes in the product-pairs).”

Should read: “ $\frac{n(n+1)}{2(2n+1)(2n+3)}$, which tends to $\frac{1}{8}$ as $n \rightarrow \infty$.”

Why: The partial-fraction decomposition of the printed summand is

$$\frac{r}{(2r-1)(2r+1)(2r+3)} = \frac{1}{16(2r-1)} + \frac{1}{8(2r+1)} - \frac{3}{16(2r+3)},$$

and summing from $r = 1$ to n gives

$$\sum_{r=1}^n \frac{r}{(2r-1)(2r+1)(2r+3)} = \frac{n(n+1)}{2(2n+1)(2n+3)}.$$

Check: $n = 1$ gives $\frac{1 \cdot 2}{2 \cdot 3 \cdot 5} = \frac{1}{15}$, which matches $\frac{1}{1 \cdot 3 \cdot 5}$; $n = 2$ gives $\frac{2 \cdot 3}{2 \cdot 5 \cdot 7} = \frac{3}{35}$, which matches $\frac{1}{15} + \frac{2}{105}$. The printed expression is wrong twice over: it is the closed form for the sum **without** the numerator r — namely $\frac{1}{4} \left[\frac{1}{3} - \frac{1}{(2n+1)(2n+3)} \right] = \frac{n(n+2)}{3(2n+1)(2n+3)}$ — and it carries the coefficient $\frac{1}{8}$ where that form requires $\frac{1}{4}$, so it is exactly half of even that.

Chapter 9: Straight Lines

Page 7 — Subtopic 00 (Coordinate Foundations), Section 5, answer 5: wrong sign in the perpendicular bisector

Printed: “5. Perpendicular bisector $2x - y - 1 = 0$ (i.e. $8x - 4y - 4 = 0 \Rightarrow 2x - y - 1 = 0$).”

Should read: “5. Perpendicular bisector $2x - y + 1 = 0$ (i.e. $-8x + 4y - 4 = 0 \Rightarrow 2x - y + 1 = 0$).”

Why: For $P(x, y)$ equidistant from $A(3, 2)$ and $B(-1, 4)$,

$$(x - 3)^2 + (y - 2)^2 = (x + 1)^2 + (y - 4)^2.$$

Expanding: $x^2 - 6x + 9 + y^2 - 4y + 4 = x^2 + 2x + 1 + y^2 - 8y + 16$, so $-6x - 4y + 13 = 2x - 8y + 17$, i.e. $-8x + 4y - 4 = 0$, i.e. $2x - y + 1 = 0$. Independent check: the midpoint of AB is $(1, 3)$ and must lie on the bisector — it satisfies $2(1) - 3 + 1 = 0$, whereas the printed equation gives $2(1) - 3 - 1 = -2 \neq 0$. Both the intermediate form and the final form carry the sign error.

Page 15 — Subtopic 01 (Slope of a Line and Angles), Section 5, answer 3: wrong value of p

Printed: “3. $p = -3$.”

Should read: “3. $p = \frac{11}{2}$.”

Why: The reference line through $(0, 0)$ and $(7, 14)$ has slope $\frac{14}{7} = 2$. The line through $(p, 4)$ and $(2, -3)$ has slope $\frac{-3 - 4}{2 - p} = \frac{7}{p - 2}$. Parallelism requires

$$\frac{7}{p - 2} = 2 \implies 7 = 2p - 4 \implies p = \frac{11}{2}.$$

The printed value $p = -3$ gives slope $\frac{7}{-5} = -\frac{7}{5}$, which is parallel to nothing in the question.

Page 36 — Subtopic 04 (Point of Intersection, Concurrency & Family of Lines), Section 5, answer 4: wrong value of k

Printed: “4. $k = -3$.”

Should read: “4. $k = 4$.”

Why: The first two lines $x + y - 1 = 0$ and $2x - y + 3 = 0$ meet where $3x = -2$, i.e. at $(-\frac{2}{3}, \frac{5}{3})$. Substituting into $kx + y + 1 = 0$:

$$-\frac{2k}{3} + \frac{5}{3} + 1 = 0 \implies -\frac{2k}{3} + \frac{8}{3} = 0 \implies k = 4.$$

The determinant method taught in this same subtopic agrees:

$$\begin{vmatrix} 1 & 1 & -1 \\ 2 & -1 & 3 \\ k & 1 & 1 \end{vmatrix} = 1(-1-3) - 1(2-3k) + (-1)(2+k) = -4 - 2 + 3k - 2 - k = 2k - 8 = 0,$$

so $k = 4$.

Page 42 — Subtopic 05 (Angle Bisectors of Two Lines), Section 5, answer 3: the conclusion is the opposite of the truth

Printed: “3. $a_1a_2 + b_1b_2 = 36 - 20 = 16 > 0$ (constants already positive) $\Rightarrow$ origin in **obtuse** angle.”

Should read: “3. Normalise the constants first: $3x - 4y + 1 = 0$ already has $c = 1 > 0$, but $12x + 5y - 2 = 0$ has $c = -2 < 0$ and must be rewritten as $-12x - 5y + 2 = 0$. Then $a_1a_2 + b_1b_2 = 3(-12) + (-4)(-5) = -36 + 20 = -16 < 0 \Rightarrow$ origin in the **acute** angle.”

Why: The parenthetical “constants already positive” is false — the second line has $c = -2$ — and the sign test is applied to the unnormalised coefficients. This is precisely the normalisation the subtopic insists on in Section 2, in the Section 3 procedure, in Pitfall 1, in MCQ Q3 and in the cheat sheet. After the required rewrite, $a_1a_2 + b_1b_2 = -16 < 0$, so the origin-containing bisector is the acute bisector. Verified directly: the origin’s bisector is

$$\frac{3x - 4y + 1}{5} = \frac{-12x - 5y + 2}{13} \Rightarrow 33x - 9y + 1 = 0,$$

which makes an angle of 37.87° with each of the two lines, while the angle between the lines themselves is 75.75° — i.e. it bisects the acute angle.

Chapter 10: Conic Sections

Page 5 — Introduction, Section 3: unit-naming scheme mixed on the page

Printed: “This subtopic classifies rather than derives, so the skill is reading a description or an equation and naming the curve with reasoning, not rote.”

Should read: “This unit classifies rather than derives, ...”

Why: This chapter names its five divisions in prose — “Introduction: Sections of a Cone & the Conic Family”, “The Circle”, “The Parabola”, “The Ellipse”, “The Hyperbola” — rather than as “Subtopic NN”. The body text then refers back to them as “subtopics”, mixing both schemes. No content is affected; only the label.

Page 12 — The Circle, Section 5, Practice Exercise 5 and its answer: every reported quantity is wrong

Printed: Question 5: “Find the equation of the circle through $(1, 0)$, $(0, 1)$, and $(2, 1)$, and the length of the tangent to it from the origin.” Answer: “Circle $x^2 + y^2 - 3x - y + 2 = 0$ (centre

$(\frac{3}{2}, \frac{1}{2})$, radius $\sqrt{\frac{1}{2}}$; tangent length from origin $= \sqrt{S_1} = \sqrt{0+0-0-0+2} = \sqrt{2}$."

Should read: "Circle $x^2 + y^2 - 2x - 2y + 1 = 0$ (centre $(1, 1)$, radius 1); tangent length from origin $= \sqrt{S_1} = \sqrt{0+0-0-0+1} = 1$."

Why: The printed circle does not pass through $(0, 1)$: substituting gives $0+1-0-1+2 = 2 \neq 0$. Solving properly with $x^2 + y^2 + 2gx + 2fy + c = 0$:

$$(1, 0) : 1 + 2g + c = 0, \quad (0, 1) : 1 + 2f + c = 0, \quad (2, 1) : 5 + 4g + 2f + c = 0.$$

The first two give $g = f$ and $c = -1 - 2g$; substituting into the third gives $5 + 4g + 2g - 1 - 2g = 0$, so $g = -1$, hence $f = -1$ and $c = 1$. The circle is $x^2 + y^2 - 2x - 2y + 1 = 0$, with centre $(-g, -f) = (1, 1)$ and radius $\sqrt{g^2 + f^2 - c} = \sqrt{1 + 1 - 1} = 1$. The tangent length from the origin is $\sqrt{S_1} = \sqrt{1} = 1$.

Page 12 — The Circle, Section 6, Crack the MCQ, Q4: two options are the same number

Printed: "The length of the tangent from $(0, 0)$ to the circle $x^2 + y^2 - 4x - 6y + 12 = 0$ is (A) $2\sqrt{3}$ (B) $\sqrt{12}$ (C) 2 (D) cannot be drawn", with the explanation "Trap (B): $\sqrt{12}$ is correct but not in simplest form — both (A) and (B) are numerically equal, and exam keys take the simplified surd."

Should read: Option (B) should be replaced by a genuine distractor and its rationale rewritten:

(A) $2\sqrt{3}$ (B) 12 (C) 2 (D) cannot be drawn

"Trap (B): 12 is S_1 itself — the student forgot to take the square root."

Why: $\sqrt{12} = 2\sqrt{3}$, so as printed the item has two correct options and cannot be marked. The mathematics is otherwise sound: $S_1 = 0 + 0 - 0 - 0 + 12 = 12 > 0$, so the origin lies outside and the tangent length is $\sqrt{12} = 2\sqrt{3}$.

Pages 25-26 — The Ellipse, Section 5, Practice Exercise 3 and its answer: the stem is unsolvable and the answer silently rewrites it

Printed: Question 3: "If the latus rectum of an ellipse equals its minor axis, find its eccentricity." Answer: " $\frac{2b^2}{a} = 2b \Rightarrow b = a$? — careful: $\frac{2b^2}{a} = 2b \Rightarrow b = a$, impossible for a true ellipse, so re-read as latus rectum = semi-minor axis b : $\frac{2b^2}{a} = b \Rightarrow b = \frac{a}{2} \Rightarrow e = \sqrt{1 - \frac{1}{4}} = \frac{\sqrt{3}}{2}$."

Should read: Question 3: "If the latus rectum of an ellipse equals its **semi-minor axis**, find its eccentricity." Answer: " $\frac{2b^2}{a} = b \Rightarrow b = \frac{a}{2} \Rightarrow e = \sqrt{1 - \frac{b^2}{a^2}} = \sqrt{1 - \frac{1}{4}} = \frac{\sqrt{3}}{2}$."

Why: As printed, latus rectum = minor axis means $\frac{2b^2}{a} = 2b$, hence $b = a$ — a circle, with $e = 0$ — so the item has no ellipse as a solution. The fix belongs in the stem, not in the answer: the answer key as printed argues with itself in front of the reader and then answers a different question from the one asked.

Page 33 — The Hyperbola, Section 5, Practice Exercise 5 and its answer: unedited draft shipped as an exercise, with no solvable task

Printed: Question 5: “The eccentricity of a hyperbola is the reciprocal of that of the ellipse $\frac{x^2}{25} + \frac{y^2}{16} = 1$... is impossible (explain why), so instead: find the eccentricity of a hyperbola whose conjugate axis equals the distance between its foci.” Answer: “...the ‘impossible’ framing is the trap, the reciprocal route is fine, but the cleaner intended part: conjugate axis = focal distance $\Rightarrow 2b = 2c \Rightarrow b = c$; but $c^2 = a^2 + b^2$ gives $b^2 = a^2 + b^2$, impossible — so no such hyperbola exists... Recognising the impossibility is the answer.”

Should read: The exercise is unsalvageable — an abandoned first attempt and its replacement are both in print, and the replacement is itself impossible. Substitute:

5. [JEE Advanced] Find the eccentricity of a hyperbola whose conjugate axis equals **half** the distance between its foci.

Answer: $e = \frac{2}{\sqrt{3}} = \frac{2\sqrt{3}}{3}$.

Why: The condition “conjugate axis = distance between the foci” reads $2b = 2c$, i.e. $b = c$, which with $c^2 = a^2 + b^2$ forces $a = 0$: no such hyperbola exists, so the exercise ends with nothing for the reader to do. With the corrected condition, $2b = \frac{1}{2}(2c) = c$, so

$$c^2 = a^2 + b^2 \implies 4b^2 = a^2 + b^2 \implies a^2 = 3b^2,$$

and therefore

$$e = \frac{c}{a} = \frac{2b}{\sqrt{3}b} = \frac{2}{\sqrt{3}} = \frac{2\sqrt{3}}{3}.$$

Pages 10, 16, 24 and 31 — Every figure in the chapter is an unrendered placeholder, and the cone diagram is not even requested

Printed: Four shaded boxes headed “Figure to be illustrated”, each containing an art brief instead of artwork: the circle (page 10), the parabola (page 16), the ellipse (page 24) and the hyperbola (page 31). The Introduction unit contains no figure block at all.

Should read: The four placeholders should be replaced by the diagrams they describe, and a fifth figure should be added to the Introduction: a double cone cut by a plane, showing the semi-vertical angle α and the cutting-plane angle β , with the four sections that arise as β varies relative to α .

Why: For this chapter in particular the diagrams are not decorative — focus and directrix, the latus rectum, and the asymptote geometry are close to unteachable in prose alone. The missing cone diagram is the most consequential of the five: the whole Introduction argument turns on comparing β with α , and no figure was even requested for it.

Chapter 11: Introduction to Three-Dimensional Geometry

Page 13 — Subtopic 02, Section 5, Practice Exercise 2, answer (i): wrong y -coordinate

Printed: “2. (i) $\left(-\frac{4}{5}, -\frac{2}{5}, \frac{27}{5}\right)$, (ii) $(-8, 17, 3)$.”

Should read: “2. (i) $\left(-\frac{4}{5}, \frac{1}{5}, \frac{27}{5}\right)$, (ii) $(-8, 17, 3)$.”

Why: Dividing the segment from $A(-2, 3, 5)$ to $B(1, -4, 6)$ internally in the ratio $m : n = 2 : 3$ gives

$$\left(\frac{2(1) + 3(-2)}{5}, \frac{2(-4) + 3(3)}{5}, \frac{2(6) + 3(5)}{5}\right) = \left(\frac{-4}{5}, \frac{-8 + 9}{5}, \frac{27}{5}\right) = \left(-\frac{4}{5}, \frac{1}{5}, \frac{27}{5}\right).$$

The x - and z -coordinates and the external point (ii) are correct as printed; only the y -coordinate is wrong. The error matters disproportionately because this is the chapter’s only numerical instance of the internal section formula.

Page 13 — Subtopic 02, Section 5, Practice Exercise 4 and its answer: wrong ratio, and the division is external

Printed: Question 4: “Show that $A(2, -3, 4)$, $B(-1, 2, 1)$, $C\left(0, \frac{1}{3}, 2\right)$ are collinear, and find the ratio in which B divides AC .” Answer: “ B divides AC in the ratio $1 : 2$ (so collinear).”

Should read: “ B divides AC **externally** in the ratio $3 : 1$. (Equivalently, C divides AB internally in the ratio $2 : 1$.)”

Why: The three points are collinear: $\overrightarrow{AB} = (-3, 5, -3)$ and $\overrightarrow{AC} = \left(-2, \frac{10}{3}, -2\right) = \frac{2}{3}\overrightarrow{AB}$. Since $\overrightarrow{AB} = \frac{3}{2}\overrightarrow{AC}$, the point C lies strictly between A and B , so B lies outside the segment AC and cannot divide it internally. Writing $B = \frac{3C - A}{3 - 1}$ shows that B divides AC externally in the ratio $3 : 1$. The ratio $1 : 2$ is wrong under either convention. Readers should note that the stem is answerable as printed; only the answer requires correction.

Page 16 — Subtopic 03, Exam Relevance box: promised material is not delivered

Printed: “...and rewards the incentre/tetrahedron extensions.”

Should read: “...and rewards the incentre and tetrahedron-centroid formulas, which appear here as formula-block entries only.”

Why: Neither the incentre nor the centroid of a tetrahedron receives a worked example, a practice exercise or an MCQ anywhere in the chapter; both appear only in the formula block and the cheat sheet. The sentence as printed advertises depth the chapter does not contain.

Pages 12 and 20 — Subtopic 02 Section 4 Example 4 and Subtopic 03 Section 5 Question 4 are the same problem, tagged at two difficulty levels

Printed: Page 12, Subtopic 02, Section 4, Example 4 — “[JEE Advanced Level] Find the equation of the set of all points equidistant from $A(1, 2, 3)$ and $B(3, 2, -1)$, and describe the

surface”, solved in full with answer $x - 2z = 0$. Page 20, Subtopic 03, Section 5, Question 4 — “[JEE Main] Find the equation of the set of points equidistant from $A(1, 2, 3)$ and $B(3, 2, -1)$, and name the surface”, answer $x - 2z = 0$, a plane.

Should read: The practice item on page 20 should be replaced with a problem using different points, and its difficulty tag brought into line with the replacement. A suitable substitute:

4. [JEE Main] Find the equation of the set of points equidistant from $A(0, 1, 2)$ and $B(4, 3, 2)$, and name the surface.

Answer: $2x + y - 6 = 0$ — a plane (the perpendicular bisector plane of AB).

Why: As printed, a reader who works the Subtopic 03 practice set is re-solving a fully worked example from eight pages earlier, and the identical problem carries two different difficulty labels. For the replacement: $(x - 0)^2 + (y - 1)^2 + (z - 2)^2 = (x - 4)^2 + (y - 3)^2 + (z - 2)^2$ gives $-2y + 1 = -8x + 16 - 6y + 9$, i.e. $8x + 4y - 24 = 0$, i.e. $2x + y - 6 = 0$. (Check: the midpoint $(2, 2, 2)$ of AB satisfies it.)

Page 20 — Subtopic 03, Section 5, Practice Exercise 5: half the question is malformed and is never answered

Printed: Question 5: “Find the equation of the sphere that is the locus of points at distance 5 from the centre $(2, -1, 3)$, and state its radius and the point on it nearest the origin’s z -axis.”
Answer: “ $(x - 2)^2 + (y + 1)^2 + (z - 3)^2 = 25$; radius 5.”

Should read: The second half of the stem should be replaced and answered:

5. [JEE Advanced] Find the equation of the sphere that is the locus of points at distance 5 from the centre $(2, -1, 3)$, state its radius, and find the points in which it meets the z -axis.

Answer: $(x - 2)^2 + (y + 1)^2 + (z - 3)^2 = 25$; radius 5; the sphere meets the z -axis at $(0, 0, 3 + 2\sqrt{5})$ and $(0, 0, 3 - 2\sqrt{5})$.

Why: The phrase “the origin’s z -axis” is malformed — the z -axis does not belong to the origin — and it is not clear whether “nearest” means nearest to the origin or nearest to the axis; the answer key simply omits the sub-question. The replacement is well defined and answered: putting $x = y = 0$ in the sphere gives

$$(0 - 2)^2 + (0 + 1)^2 + (z - 3)^2 = 25 \implies (z - 3)^2 = 20 \implies z = 3 \pm 2\sqrt{5}.$$

Chapter 12: Limits and Derivatives

Page 24 — Subtopic 04, Section 6 (Crack the MCQ), Q3, analysis of option (D): raw LaTeX on the page, and a rationale that does not produce the option

Printed: “(D) doubles correctly but forgets $u = 1$ and uses $u = 2$ giving $\$3\$4\cdot\$...\$$ — a substitution slip.”

Should read: “(D) 12 arises from substituting $u = 2$ instead of $u(0) = 1$ **and** stopping at $\frac{dy}{du}$: $3u^2 = 3(2)^2 = 12$, with the factor $\frac{du}{dx} = 2$ never applied.”

Why: Two faults in one sentence. First, the inline-math delimiters and the trailing ellipsis were not processed, leaving literal dollar signs on the page — the only ones in the 32-page chapter. Second, the described slip does not yield the option it claims to explain: doubling correctly with $u = 2$ gives $3(2)^2 \cdot 2 = 24$, not 12. The value 12 is $3u^2$ evaluated at $u = 2$ with the chain-rule factor omitted. (The correct working for the item is unaffected: $\frac{dy}{dx} = 3u^2 \cdot 2$ and at $x = 0$, $u = 1$, giving 6, option (B).)

Chapter 13: Statistics

Page 9 — Subtopic 01, Section 5, Practice Exercise 2: raw Markdown table printed inline

Printed: “For the discrete distribution below, find the mean deviation about the mean. | x_i | 5 | 10 | 15 | 20 | 25 | |—|—|—|—|—|—| | f_i | 3 | 5 | 8 | 3 | 1 |”

Should read: The data should be set as a frequency table below the question:

x_i	5	10	15	20	25
f_i	3	5	8	3	1

Why: The table markup was not converted, so pipes and em dashes appear as literal characters inside the numbered list. Every other table in the chapter typesets correctly, so this is an isolated conversion failure. The data and the printed answer (mean deviation = 4.3) are correct: $\sum f_i = 20$, $\sum f_i x_i = 270$, mean = 13.5, and $\sum f_i |x_i - 13.5| = 86$, so the mean deviation is $86/20 = 4.3$.

Page 24 — Subtopic 03, Section 4, Example 3 (correcting a misrecorded observation): a digit is lost off the right margin

Printed: “ $\sum x_i^2$ correct = $18500 - 50^2 + 30^2 = 18500 - 2500 + 900 = 1690$ ” — the line overruns the right margin and the final digit is clipped.

Should read: “ $\sum x_i^2$ correct = $18500 - 50^2 + 30^2 = 18500 - 2500 + 900 = 16900$ ”, set so that it fits within the text block.

Why: $18500 - 2500 + 900 = 16900$. The very next line of the same solution correctly divides 16900 by 20 to get 845, so the printed intermediate contradicts the step that follows it.

Chapter 14: Probability

Page 14 — Subtopic 02, Section 5, answer to Practice Exercise 2: the verdict contradicts its own reasoning

Printed: “2. **Yes** — $P(A') = 1 - 0.45 = 0.55$, so 0.45 is impossible for A' (it would need $P(A) = 0.55$).”

Should read: “2. **No** — $P(A') = 1 - 0.45 = 0.55$, so 0.45 is impossible for A' (it would need $P(A) = 0.55$).”

Why: The question asks whether $P(A') = 0.45$ is possible when $P(A) = 0.45$. The complement law gives $P(A') = 1 - P(A) = 0.55$, so the answer is no — exactly as the printed reasoning shows. The second half of the same answer is correctly verdict-first (“No — mutually exclusive needs $P(A) + P(B) \leq 1...$ ”), which confirms that “Yes” here is an error rather than a different reading of the question. A reader self-checking against the key would conclude that the complement law permits $P(A) = P(A') = 0.45$.

Page 22 — Subtopic 03, Section 7, Pro-Tip: the “at least one” shortcut is stated incorrectly

Printed: “For ‘at least one of several independent-looking trials,’ the complement collapses the whole thing to a single product: $P(\text{at least one}) = 1 - (\text{chance of none})^{(\text{number of trials})}$. One line, no casework — the biggest time-saver in the chapter.”

Should read: “For ‘at least one’ over n repeated draws from the *same* equally-likely sample space — the case every example in this chapter actually uses — the complement collapses the whole thing to a single count:

$$P(\text{at least one}) = 1 - P(\text{none}) = 1 - \frac{(\text{outcomes avoiding it in one draw})^n}{(\text{total equally likely outcomes in one draw})^n}.$$

This is a **counting** shortcut, not a probability-multiplication rule: it works because repeating an equally-likely draw n times produces s^n equally likely ordered outcomes, of which $(s - k)^n$ avoid the target, so the ratio is a power. It is not licence to multiply the probabilities of two *different* events, which needs independence — a Class 12 idea this chapter never establishes.”

Why: As printed, the formula’s exponent base — “chance of none” raised to “number of trials” — reads as the general independence rule $P(\text{none}) = P(A'_1)P(A'_2) \cdots P(A'_n)$, which is Class 12 material with no justification anywhere in this chapter, and which is false in general. It sits in direct tension with Subtopic 02 Pitfall 3 (“Don’t multiply probabilities in a Class 11 addition-rule problem”), the Subtopic 02 MCQ Q4 trap, which calls multiplying $P(A')P(B')$ “a Class

12 move... unjustified here”, and the Round 2 addendum’s Addenda 14.3–14.4 (page 22 area), which name this exact Pro-Tip as an erratum and show that both of the chapter’s own uses of it — $P(\text{no six on two dice}) = (\frac{5}{6})^2$ (Subtopic 03 MCQ Q4) and $P(\text{no head in three tosses}) = (\frac{1}{2})^3$ (Subtopic 03 MCQ Q3) — are correct only because they are counts of equally-likely outcomes over 36 and 8-element sample spaces respectively, not products of probabilities. The revision above keeps the shortcut but grounds it in counting, so it stays inside Class 11 and stops teaching the independence rule by implication. Related passages should be brought into line: in Subtopic 02 Section 5 answer 5 (page 14), the biased-coin calculation $P(HH) = \frac{4}{9}$ is justified only by the hedge “independent in the everyday sense of the coin has no memory” — that hedge should be read the same way, as a counting statement about ordered pairs of tosses, not as an appeal to formal independence.